

STAR RX72N - rx_core Library
1.0.0

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Chapter 1

Test List

Global `rx_pin_from_pin` (`rx_port_pin_t` pin)

`test_rx_pin_from_pin()` Unit tests verify all 144 port/pin combinations

Global `rx_port_from_pin` (`rx_port_pin_t` pin)

`test_rx_port_from_pin()` Unit tests verify all 144 port/pin combinations

Global `rx_register_guard_get_correction_count` (void)

Tested in `test_rx_register_guard.c::test_correction_count_increment()`

Tested in `test_rx_register_guard.c::test_correction_count_before_init()`

Global `rx_register_guard_init` (void)

Tested in `test_rx_register_guard.c::test_init_captures_values()`

Global `rx_register_guard_is_initialized` (void)

Tested in `test_rx_register_guard.c::test_is_initialized_flag()`

Tested in `test_rx_register_guard.c::test_refresh_before_init()`

Global `rx_register_guard_refresh` (void)

Tested in `test_rx_register_guard.c::test_refresh_corrects_corruption()`

Tested in `test_rx_register_guard.c::test_refresh_performance()`

Global `rx_register_guard_reset_count` (void)

Tested in `test_rx_register_guard.c::test_reset_count()`

Tested in `test_rx_register_guard.c::test_reset_before_init()`

Global `rx_time_threadx_get_interface` (`rx_time_interface_t` *iface)

Tested in `test_rx_time_interface.c::test_validate_null_interface()`

Tested in `test_rx_time_interface.c::test_validate_missing_functions()`

Tested in `test_rx_time_interface.c::test_validate_valid_interface()`

Tested in `test_rx_time.c` with nullptr and success cases

Chapter 2

Directory Hierarchy

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rx_iwdt.h	340
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Chapter 3

Data Structure Index

3.1 Data Structures

Here are the data structures with brief descriptions:

error_handler_t	Error Handler Concrete Implementation Structure (Main State Container)	15
pin_validator_t	Concrete Pin Validator Implementation Structure	20

Chapter 4

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4.1 File List

Here is a list of all files with brief descriptions:

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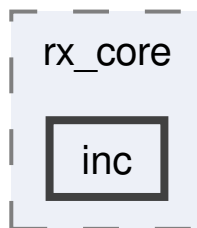
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Pin Validator Concrete Implementation - GPIO Pin Tracking and Conflict Prevention	952
libs/rx_core/src/rx_register_guard.c	
Register Guard Implementation - ESD/EMI Protection for Critical Registers	997
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Watchdog Timer (WDT) Implementation for RX72N Microcontroller	1059

Chapter 5

Directory Documentation

5.1 libs/rx_core/inc Directory Reference

Directory dependency graph for inc:



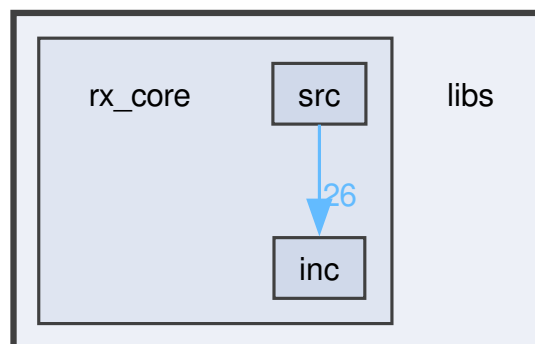
Files

- file [rx_bit_constants.h](#)
Fundamental Bit and Byte Size Constants for Protocol and Data Manipulation.
- file [rx_check.h](#)
Validation and Error Checking Macros for RX72N Firmware.
- file [rx_err.h](#)
Error Code Definitions for RX72N Firmware.
- file [rx_error_handler.h](#)
Error Handler Concrete Implementation with Retry Logic and Exponential Backoff.
- file [rx_error_interface.h](#)
Abstract Error Handler Interface (Dependency Inversion Principle - SOLID).
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RX72N CPU Exception Handling.
- file [rx_gpio_constants.h](#)
GPIO and System Constants for RX72N Microcontroller.
- file [rx_infrastructure.h](#)
Global Infrastructure Initialization and Service Locator.
- file [rx_isr_log.h](#)
ISR-safe deferred logging facility.

- file [rx_iwdt.h](#)
Independent Watchdog Timer (IWDT) Driver for RX72N Microcontroller.
- file [rx_log.h](#)
Type-Safe Logging System for RX72N Firmware with Zero-Overhead Compile-Time Filtering.
- file [rx_log_uart.h](#)
UART-framed log backend for the RX72N logging system.
- file [rx_pin_interface.h](#)
Abstract Pin Validator Interface - THE "I" and "D" in SOLID.
- file [rx_pin_validator.h](#)
Concrete Pin Validator Implementation - Dependency Inversion Pattern.
- file [rx_port_constants.h](#)
Centralized RX72N Port Number Definitions.
- file [rx_register_guard.h](#)
Register Guard - Periodic Refresh for ESD/EMI Protection.
- file [rx_register_protection.h](#)
RX72N Register Protection Control (PRCR) - Write Protection Constants.
- file [rx_simulator_config.h](#)
Configuration for e^2 studio simulator support.
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- file [rx_time_constants.h](#)
Time Unit Conversion Constants for RX72N Firmware.
- file [rx_time_interface.h](#)
Time Interface - Dependency Inversion for Timing Operations.
- file [rx_wdt.h](#)
Watchdog Timer (WDT) Driver for RX72N Microcontroller.

5.2 libs Directory Reference

Directory dependency graph for libs:

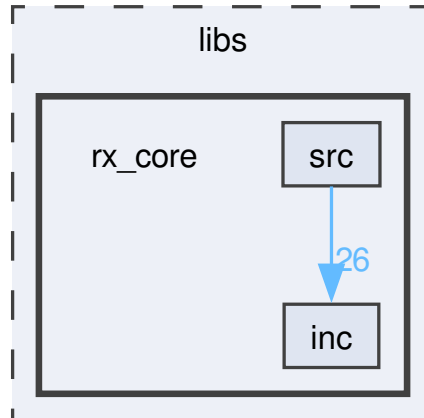


Directories

- directory [rx_core](#)

5.3 libs/rx_core Directory Reference

Directory dependency graph for rx_core:

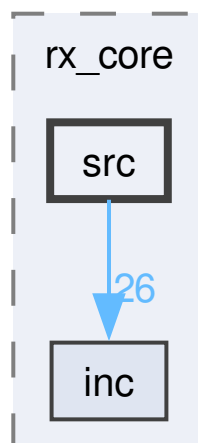


Directories

- directory [inc](#)
- directory [src](#)

5.4 libs/rx_core/src Directory Reference

Directory dependency graph for src:



Files

- file [rx_error_handler.c](#)
Error Handler Concrete Implementation.
- file [rx_exception.c](#)
RX72N CPU Exception Handling Implementation.
- file [rx_infrastructure.c](#)
Global Infrastructure Initialization Implementation.
- file [rx_isr_log.c](#)
ISR-safe deferred log queue: SPSC ring + drain emitter.
- file [rx_iwdt.c](#)
Independent Watchdog Timer (IWDT) Implementation for RX72N.
- file [rx_log_uart.c](#)
UART-framed log backend: ring buffer + drain API.
- file [rx_pin_validator.c](#)
Pin Validator Concrete Implementation - GPIO Pin Tracking and Conflict Prevention.
- file [rx_register_guard.c](#)
Register Guard Implementation - ESD/EMI Protection for Critical Registers.
- file [rx_time_threadx.c](#)
ThreadX Time Implementation for RX72N.
- file [rx_wdt.c](#)
Watchdog Timer (WDT) Implementation for RX72N Microcontroller.

Chapter 6

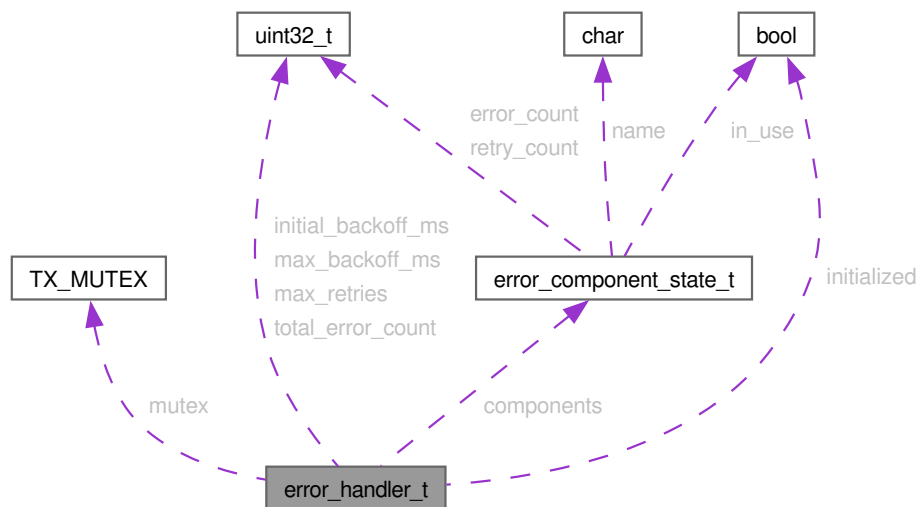
Data Structure Documentation

6.1 error_handler_t Struct Reference

Error Handler Concrete Implementation Structure (Main State Container).

```
#include <rx_error_handler.h>
```

Collaboration diagram for error_handler_t:



Data Fields

- `TX_MUTEX` [mutex](#)
- `uint32_t` [total_error_count](#)
- `error_component_state_t` [components](#) [`k_error_handler_max_components`]
- `uint32_t` [max_retries](#)
- `uint32_t` [initial_backoff_ms](#)
- `uint32_t` [max_backoff_ms](#)
- `bool` [initialized](#)

6.1.1 Detailed Description

Error Handler Concrete Implementation Structure (Main State Container).

Complete error handler state including ThreadX mutex, component tracking array, retry configuration, and initialization flag. This structure implements the `rx_error_interface_t` abstraction, providing concrete error tracking and retry logic.

6.1.1.1 Purpose and Responsibility

This structure serves as the central error management database for the entire system:

1. Thread Safety: `TX_MUTEX` protects concurrent access
2. Global Statistics: `total_error_count` aggregates all component errors
3. Component Tracking: `components[]` array isolates per-component state
4. Retry Strategy: `max_retries`, backoff parameters define retry behavior
5. Lifecycle Management: `initialized` flag prevents double-init

6.1.1.2 Memory Layout

Offset	Size	Field	Type	Alignment	Notes
0	~100	mutex	TX_MUTEX	4	ThreadX mutex (size varies by config)
100	4	total_error_count	uint32_t	4	Global error counter
104	704	components[16]	array	4	16 x 44 bytes per component
808	4	max_retries	uint32_t	4	Retry limit configuration
812	4	initial_backoff_ms	uint32_t	4	Initial backoff delay
816	4	max_backoff_ms	uint32_t	4	Maximum backoff cap
820	1	initialized	bool	1	Initialization flag
821	3	(padding)	-	-	Compiler alignment
Total	**~924 bytes**				Structure size (varies by TX_MUTEX)

Note: Actual size depends on ThreadX `TX_MUTEX` size, which varies by configuration. Typical total: ~1.0 to 1.3 KB.

6.1.1.3 Initialization Requirements

Before use, this structure MUST be initialized via `error_handler_init()`:

1. `TX_MUTEX` must be created (`tx_mutex_create`)
2. `total_error_count` set to 0
3. All `components[]` marked `in_use=false`
4. Retry parameters copied from config
5. `initialized` flag set to true

6.1.1.4 Typical Lifecycle

[UNINITIALIZED] (all fields undefined)
 v error_handler_init(&handler, &config)
 [INITIALIZED] (mutex created, counters zeroed, initialized=true)
 v normal operation (report_error, retry checks)
 [IN_USE] (components registered, counters incrementing)
 v error_handler_deinit(&handler)
 [DEINITIALIZED] (mutex deleted, initialized=false)

Field Constraints:

Field	Valid Values	Invariant	Notes
mutex	Valid TX_MUTEX	Created if initialized=true	ThreadX owned
total_error_count	0 to 2 ³² -1	Sum of all components[].error_count	Wraps on overflow
components[]	See error_component_state_t	in_use=true slots <= 16	Static array
max_retries	0 to 2 ³² -1	0 = unlimited retries	From config
initial_backoff_ms	1 to max_backoff_ms	> 0 for meaningful backoff	From config
max_backoff_ms	>= initial_backoff_ms	Caps exponential growth	From config
initialized	true/false	true after init, false after deinit	Lifecycle flag

Usage Example (Declaration and Initialization):

```
// Static allocation (typical for embedded systems)
static error_handler_t s_global_error_handler;

void system_init(void) {
    // Configure error handler
    const error_handler_config_t config = {
        .max_retries      = 3,    // Up to 3 retries
        .initial_backoff_ms = 10, // Start at 10ms
        .max_backoff_ms   = 5000 // Cap at 5 seconds
    };

    // Initialize
    rx_err_t err = error_handler_init(&s_global_error_handler, &config);
    if (err != k_rx_ok) {
        uart_debug_puts("[FATAL] Error handler init failed\r\n");
        while (1); // Cannot proceed without error handling
    }

    // Get interface for use by high-level modules
    rx_error_interface_t error_iface;
    error_handler_get_interface(&error_iface, &s_global_error_handler);

    // Inject interface into other modules
    motor_controller_init(&error_iface);
    spi_driver_init(&error_iface);
}
```

Usage Example (Direct Access - Not Recommended):

```
// Direct field access (bypasses interface - violates DIP)
// USE INTERFACE INSTEAD!

// Get total error count (for diagnostics only)
uint32_t total_errors = s_global_error_handler.total_error_count;
uart_debug_printf("Total system errors: %lu\r\n", total_errors);

// Iterate components for diagnostics
```

```

for (uint8_t i = 0; i < k_error_handler_max_components; i++) {
    if (s_global_error_handler.components[i].in_use) {
        uart_debug_printf("[%s] errors:%lu retries:%lu\r\n",
            s_global_error_handler.components[i].name,
            s_global_error_handler.components[i].error_count,
            s_global_error_handler.components[i].retry_count);
    }
}

```

Invariant

initialized=true -> mutex is valid AND total_error_count defined
 total_error_count >= sum of all components[].error_count (approximately)
 max_backoff_ms >= initial_backoff_ms (always)
 Number of in_use=true components <= k_error_handler_max_components

Note

Structure must be statically allocated or have lifetime exceeding all users
 Typical allocation: File-scope static (static [error_handler_t](#) s_handler)
 Memory footprint: ~1.0-1.3 KB depending on ThreadX configuration

Warning

Never copy this structure (contains TX_Mutex which is not copyable)
 Direct field access violates Dependency Inversion - use interface
 Must call [error_handler_deinit\(\)](#) before destruction (mutex cleanup)

Attention

TX_Mutex size varies by ThreadX configuration (affects total size)

See also

[error_handler_init\(\)](#) Initializes this structure
[error_handler_get_interface\(\)](#) Converts to abstract interface
[error_handler_deinit\(\)](#) Cleans up resources ([mutex](#) deletion)
[rx_error_interface_t](#) Abstract interface (use this, not direct access)

Since

Version 1.0.0

Definition at line 620 of file [rx_error_handler.h](#).

6.1.2 Field Documentation

6.1.2.1 components

`error_component_state_t error_handler_t::components[k_error_handler_max_components]`

Per-component error tracking array. Maximum 16 components (`k_error_handler_max_components`). Each slot tracks one component's errors and retry state. Linear search for component name on each access

Definition at line 625 of file `rx_error_handler.h`.

Referenced by `impl_clear_errors()`, `internal_find_component()`, and `internal_find_or_create_component()`.

6.1.2.2 initial_backoff_ms

`uint32_t error_handler_t::initial_backoff_ms`

Initial backoff delay in milliseconds for first retry. Base value for exponential backoff calculation: $\text{delay} = \text{initial} \times 2^{\text{retry_count}}$. Typical values: 10ms to 100ms. Must be > 0 for meaningful backoff

Definition at line 630 of file `rx_error_handler.h`.

Referenced by `error_handler_init()`, and `impl_get_backoff_delay()`.

6.1.2.3 initialized

`bool error_handler_t::initialized`

Initialization flag. `true` = `error_handler_init()` succeeded, structure is ready for use. `false` = Not initialized or deinitialized (`error_handler_deinit()` called). Check before all operations

Definition at line 634 of file `rx_error_handler.h`.

Referenced by `error_handler_deinit()`, `error_handler_get_interface()`, `error_handler_init()`, `impl_clear_errors()`, `impl_get_backoff_delay()`, `impl_get_component_error_count()`, `impl_get_error_count()`, `impl_is_retry_limit_reached()`, `impl_report_error()`, and `impl_reset_retry_counter()`.

6.1.2.4 max_backoff_ms

`uint32_t error_handler_t::max_backoff_ms`

Maximum backoff delay cap in milliseconds. Prevents exponential backoff from growing unbounded. Typical values: 5000ms to 60000ms. Must be $\geq \text{initial_backoff_ms}$. Limits $\text{delay} = \min(\text{initial} \times 2^{\text{retry}}, \text{max})$

Definition at line 632 of file `rx_error_handler.h`.

Referenced by `error_handler_init()`, and `impl_get_backoff_delay()`.

6.1.2.5 max_retries

uint32_t error_handler_t::max_retries

Maximum retry attempts before giving up. 0 = unlimited retries (use with caution). Copied from [error_handler_config_t](#) during init. Compared against `components[].retry_count` to determine failure

Definition at line 628 of file [rx_error_handler.h](#).

Referenced by [error_handler_init\(\)](#), [impl_get_backoff_delay\(\)](#), and [impl_is_retry_limit_reached\(\)](#).

6.1.2.6 mutex

TX_MUTEX error_handler_t::mutex

ThreadX mutex for thread-safe access to all fields. Created by [error_handler_init\(\)](#), deleted by [error_handler_deinit\(\)](#). Protects concurrent `report_error()` and retry check calls from multiple tasks

Definition at line 622 of file [rx_error_handler.h](#).

Referenced by [error_handler_deinit\(\)](#), [error_handler_init\(\)](#), [impl_clear_errors\(\)](#), [impl_get_backoff_delay\(\)](#), [impl_get_component_error_count\(\)](#), [impl_get_error_count\(\)](#), [impl_is_retry_limit_reached\(\)](#), [impl_report_error\(\)](#), and [impl_reset_retry_counter\(\)](#).

6.1.2.7 total_error_count

uint32_t error_handler_t::total_error_count

Aggregate error count across ALL components. Incremented on each `report_error()` call regardless of component. Used for system-wide health monitoring and diagnostics. Range: 0 to $2^{32}-1$ (wraps on overflow)

Definition at line 624 of file [rx_error_handler.h](#).

Referenced by [impl_clear_errors\(\)](#), [impl_get_error_count\(\)](#), and [impl_report_error\(\)](#).

The documentation for this struct was generated from the following file:

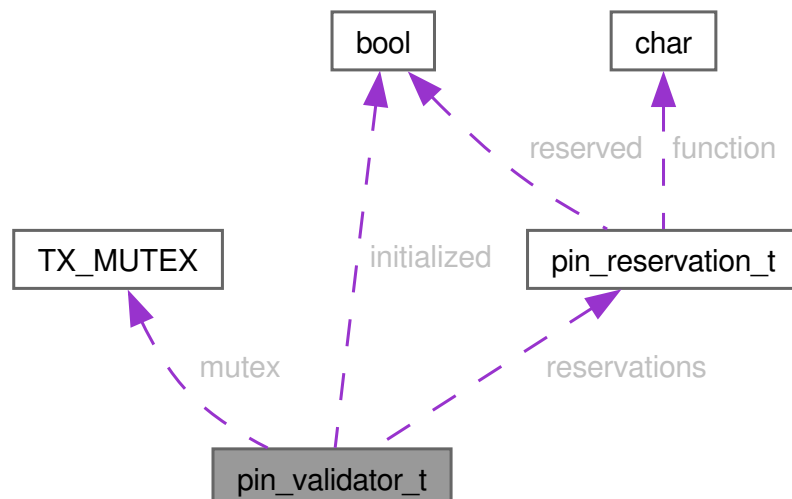
- [libs/rx_core/inc/rx_error_handler.h](#)

6.2 pin_validator_t Struct Reference

Concrete Pin Validator Implementation Structure.

`#include <rx_pin_validator.h>`

Collaboration diagram for `pin_validator_t`:



Data Fields

- TX_MUTEX [mutex](#)
ThreadX mutex for thread-safe access to reservation table.
- [pin_reservation_t](#) [reservations](#) [[k_pin_validator_max_ports](#)][[k_pin_validator_max_pins](#)]
2D array of pin reservation states [port][pin]
- bool [initialized](#)
Lifecycle flag indicating validator is ready for use.

6.2.1 Detailed Description

Concrete Pin Validator Implementation Structure.

This is the concrete implementation of the pin validator that backs the rx_pin_interface_t abstract interface. Contains all state for pin validation, reservation tracking, and thread synchronization.

6.2.1.1 Memory Layout

```
struct pin_validator_t {
// Offset | Size | Field
// 0 | ~100 | TX_MUTEX mutex
// ~100 | 4488 | pin_reservation_t[17][8] reservations
// ~4588 | 1 | bool initialized
// ~4589 | ~3 | (padding to alignment)
// TOTAL: ~4592 bytes (~4.5 KB)
};
```

6.2.1.2 Field Descriptions

6.2.1.2.1 mutex (ThreadX Mutex)

- Purpose: Protects reservation table from concurrent access
- Type: TX_MUTEX (ThreadX RTOS primitive)
- Size: ~100 bytes (platform-dependent)
- Operations protected:
 - reserve_pin() - Exclusive write access
 - release_pin() - Exclusive write access
 - is_pin_reserved() - Shared read access
 - get_pin_function() - Shared read access + string copy
 - clear_all_reservations() - Exclusive write access

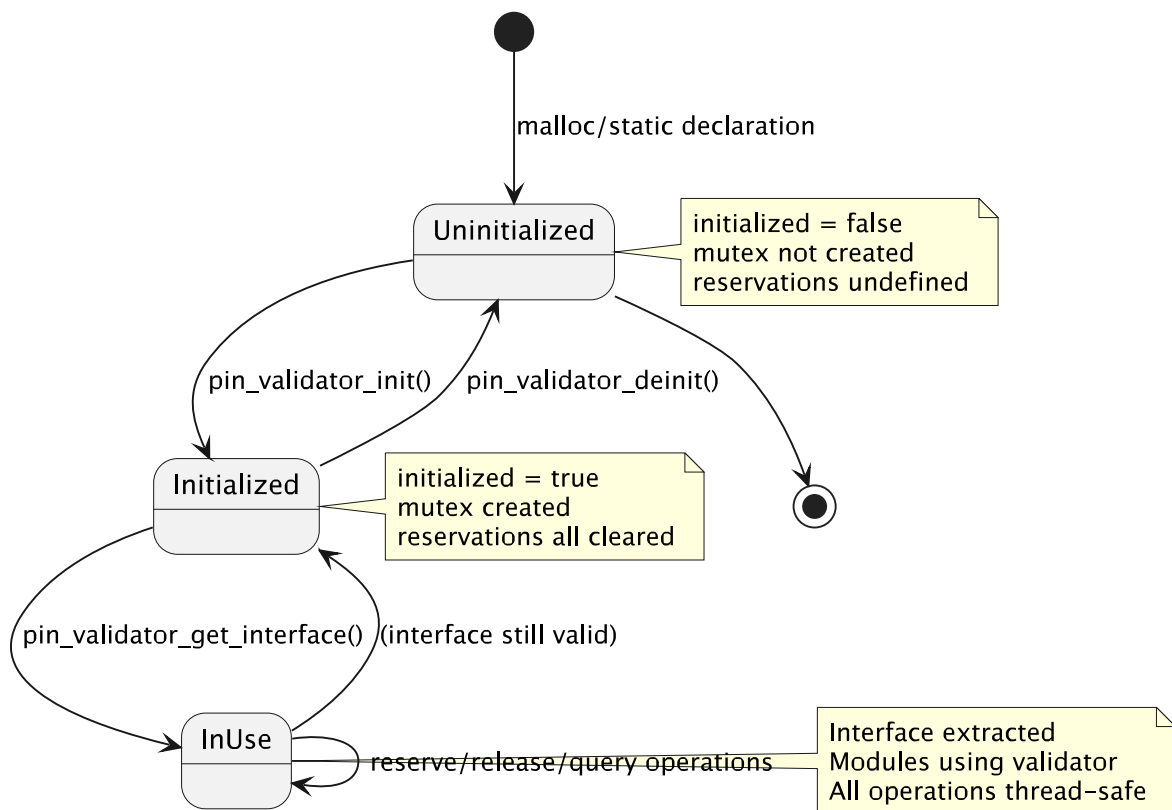
6.2.1.2.2 reservations[port][pin] (2D Array)

- Purpose: Stores reservation status for all pins
- Dimensions: 17 ports x 8 pins = 136 slots
- Per-slot size: 33 bytes (1 bool + 32 char)
- Total size: 136 x 33 = 4488 bytes (~4.4 KB)
- Indexing: [reservations](#)[port][pin] where:
 - port: 0-9 (decimal), 10-16 (hex A-G)
 - pin: 0-7
- Access pattern: O(1) direct indexing, no search needed

6.2.1.2.3 initialized (Lifecycle Flag)

- Purpose: Tracks whether `pin_validator_init()` was called
- Type: bool (1 byte)
- States:
 - false: Validator not initialized, cannot use
 - true: Validator ready, mutex created, table zeroed
- Use: Prevents double-init and validates before interface extraction

6.2.1.3 Lifecycle State Machine



6.2.1.4 Typical Usage Pattern

```

// Step 1: Declare (uninitialized state)
static pin_validator_t s_validator; // .bss section, zeroed by startup code

// Step 2: Initialize (transition to initialized state)
rx_err_t err = pin_validator_init(&s_validator);
if (err != k_rx_ok) {
    return err; // Initialization failed
}
// Now: s_validator.initialized == true
//       s_validator.mutex created
//       s_validator.reservations[] all cleared

// Step 3: Extract interface (transition to in-use state)
rx_pin_interface_t iface;
err = pin_validator_get_interface(&iface, &s_validator);
if (err != k_rx_ok) {
    return err;
}
  
```



```
// Now: iface.ctx points to s_validator
//     iface function pointers point to validator's implementations

// Step 4: Use through interface (in-use state)
err = iface.reserve_pin(iface.ctx, 0xA, 5, "SPI_COPI");
// Internally: Acquires mutex, checks reservations[10][5], updates if free

// Step 5: Deinitialize (transition back to uninitialized)
pin_validator_deinit(&s_validator);
// Now: s_validator.initialized == false
//     mutex destroyed
//     reservations cleared (optional)
```

6.2.1.5 Memory Allocation Strategy

This struct is ALWAYS statically allocated (never malloc):

- Global/file scope: static [pin_validator_t](#) s_validator;
- Infrastructure module: static [pin_validator_t](#) s_global_validator;
- Stack allocation: [pin_validator_t](#) validator; (rare, wastes stack)

Why static allocation?

- NASA Rule 3: No dynamic memory after initialization
- Predictable memory layout
- No fragmentation risk
- No malloc/free errors

6.2.1.6 Thread Safety Responsibilities

Component	Responsibility
Caller	Ensure init/deinit called from single thread
Validator	Protect all reservation table access with mutex
Interface methods	Acquire mutex before accessing reservations[]
ThreadX	Handle mutex priority inheritance

Definition at line 595 of file [rx_pin_validator.h](#).

6.2.2 Field Documentation

6.2.2.1 initialized

```
bool pin_validator_t::initialized
```

Lifecycle flag indicating validator is ready for use.

Is the validator initialized?

Set to true by [pin_validator_init\(\)](#), false by [pin_validator_deinit\(\)](#). Checked by [pin_validator_get_interface\(\)](#) to ensure validator is ready before extracting interface.

Note

Total size: ~4592 bytes (~4.5 KB)
 Always use static allocation (global or file scope)
 Mutex must be created before use (via [pin_validator_init\(\)](#))
 Thread-safe after initialization (all operations mutex-protected)

See also

[pin_validator_init\(\)](#) Initialize this structure
[pin_validator_get_interface\(\)](#) Extract interface from this structure
[pin_validator_deinit\(\)](#) Clean up this structure
[rx_pin_interface_t](#) Abstract interface this implements

Since

Version 1.0.0

Definition at line 600 of file [rx_pin_validator.h](#).

Referenced by [pin_validator_deinit\(\)](#), [pin_validator_get_interface\(\)](#), and [pin_validator_init\(\)](#).

6.2.2.2 mutex

TX_MUTEX [pin_validator_t::mutex](#)

ThreadX mutex for thread-safe access to reservation table.

ThreadX mutex for thread-safe operation

Created during [pin_validator_init\(\)](#), destroyed during [pin_validator_deinit\(\)](#). Protects all accesses to the [reservations](#) array. Uses priority inheritance to prevent priority inversion.

Definition at line 596 of file [rx_pin_validator.h](#).

Referenced by [impl_clear_all_reservations\(\)](#), [impl_get_pin_function\(\)](#), [impl_is_pin_reserved\(\)](#), [impl_release_pin\(\)](#), [impl_reserve_pin\(\)](#), [pin_validator_deinit\(\)](#), and [pin_validator_init\(\)](#).

6.2.2.3 reservations

[pin_reservation_t](#) [pin_validator_t::reservations](#)[[k_pin_validator_max_ports](#)][[k_pin_validator_max_pins](#)]

2D array of pin reservation states [port][pin]

Pin reservation tracking array (indexed: [reservations](#)[port][pin], Port 0-9: Decimal, Port 10-16: Hex A-G)

Direct-mapped table: port number is first index, pin number is second index. Example: [reservations](#)[0xA][5] = Port A, Pin 5. All access must be mutex-protected.

Definition at line 597 of file [rx_pin_validator.h](#).

Referenced by [impl_clear_all_reservations\(\)](#), [impl_get_pin_function\(\)](#), [impl_is_pin_reserved\(\)](#), [impl_release_pin\(\)](#), and [impl_reserve_pin\(\)](#).

The documentation for this struct was generated from the following file:

- [libs/rx_core/inc/rx_pin_validator.h](#)

Chapter 7

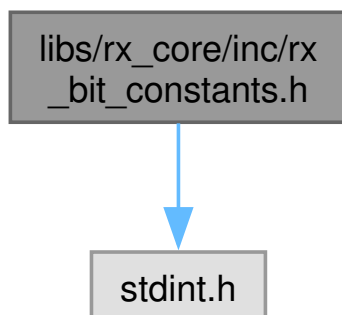
File Documentation

7.1 libs/rx_core/inc/rx_bit_constants.h File Reference

Fundamental Bit and Byte Size Constants for Protocol and Data Manipulation.

```
#include <stdint.h>
```

Include dependency graph for rx_bit_constants.h:



Enumerations

- enum `rx_bit_sizes_t` : `uint8_t` { `k_rx_bits_per_byte` , `k_rx_bits_per_word16` , `k_rx_bits_per_word32` , `k_rx_bits_per_word64` }
- Fundamental bit and byte size constants for data type sizes.
- enum `rx_bit_shifts_t` : `uint8_t` { `k_rx_shift_byte` , `k_rx_shift_word16` , `k_rx_shift_word24` }
- Bit shift constants for byte and word extraction from multi-byte values.

7.1.1 Detailed Description

Fundamental Bit and Byte Size Constants for Protocol and Data Manipulation.

7.1.1.1 Overview

Centralized repository of typed enum constants for bit/byte sizes and shift operations used throughout the STAR firmware. Provides self-documenting, type-safe constants for:

- Protocol encoding/decoding (Protocol Buffers, ASCII frames, CRC)
- Data serialization and deserialization
- Bit manipulation (extraction, packing, masking)
- Endianness conversion (big-endian <-> little-endian)
- Buffer size calculations

Design Philosophy:

- No magic numbers: All numeric literals replaced with named typed enums (NASA Rule 8)
- Type safety: C23 typed enums with explicit underlying types (uint8_t)
- Self-documenting: Constants have descriptive names (k_rx_bits_per_byte vs. literal 8)
- Searchable: grep for "k_rx_shift_byte" finds all byte-shift operations
- Hardware-agnostic: These are universal constants, not hardware-specific

7.1.1.2 Scope and Usage

INCLUDE in:

- Protocol encoding/decoding modules (rx_frame, rx_nanopb, rx_crc)
- Communication layers (rx_spi_comm, rx_usb_comm, rx_comm_manager)
- Utility functions for data packing/unpacking
- Unit conversion functions

EXCLUDE from (use peripheral-specific headers instead):

- Hardware register bit positions -> Use rx72n_*_regs.h (e.g., USBE.SYSCFG.DRPD = bit 5)
- Clock divider calculations -> Use rx72n_clock.h (e.g., PCLKA_DIV = 2)
- Peripheral-specific shifts -> Define in peripheral header (e.g., ADC channel shift)

7.1.1.3 Common Usage Patterns

7.1.1.3.1 1. Bit-to-Byte Conversion (Ceiling Division)

```
// Calculate bytes needed to store N bits (round up)
uint32_t bit_count = 100; // 100 bits
uint32_t byte_count = (bit_count + k_rx_bits_per_byte - 1) / k_rx_bits_per_byte;
// Result: (100 + 8 - 1) / 8 = 107 / 8 = 13 bytes (ceiling division)
```

7.1.1.3.2 2. Byte Extraction from Multi-Byte Word

```
// Extract individual bytes from 32-bit word (little-endian)
uint32_t word = 0x12345678;
uint8_t byte0 = (word » (k_rx_shift_byte * 0)) & 0xFF; // 0x78 (LSB)
uint8_t byte1 = (word » (k_rx_shift_byte * 1)) & 0xFF; // 0x56
uint8_t byte2 = (word » (k_rx_shift_byte * 2)) & 0xFF; // 0x34
uint8_t byte3 = (word » k_rx_shift_word24) & 0xFF; // 0x12 (MSB)
```

7.1.1.3.3 3. Big-Endian 16-bit Packing

```
// Pack uint16_t into big-endian byte array
uint16_t value = 0xABCD;
uint8_t buffer[2];
buffer[0] = (value » k_rx_shift_byte) & 0xFF; // 0xAB (high byte)
buffer[1] = (value » 0) & 0xFF; // 0xCD (low byte)
```

7.1.1.3.4 4. Buffer Size Calculation for Bit Fields

```
// Allocate buffer for protocol with bit-stuffing
const uint32_t k_payload_bits = 512; // 512 bits of data
const uint32_t k_overhead_bits = 32; // CRC-32
const uint32_t k_total_bits = k_payload_bits + k_overhead_bits;

// Calculate bytes needed (ceiling division)
const uint32_t k_buffer_size =
    (k_total_bits + k_rx_bits_per_byte - 1) / k_rx_bits_per_byte;
uint8_t buffer[k_buffer_size]; // 68 bytes
```

7.1.1.4 Hardware Requirements

Requirement	Value	Notes
CPU	Any (RX, ARM, x86)	Universal constants, platform-agnostic
Compiler	C23 or later	Requires typed enum support : uint8_t
Dependencies	stdint.h	For uint8_t underlying type
RAM	0 bytes	Compile-time constants (no storage)
Flash	0 bytes	Inlined by compiler, no code generated

NASA Power of 10 Compliance:

- Rule 1 [YES]: No goto, setjmp/longjmp, recursion (header-only, no control flow)
- Rule 2 [N/A]: No loops (header-only)
- Rule 3 [YES]: No dynamic allocation (compile-time constants)
- Rule 4 [N/A]: No functions (header-only)
- Rule 5 [N/A]: No functions (header-only)
- Rule 6 [YES]: Data at smallest scope (file-scoped enums)
- Rule 7 [N/A]: No functions (header-only)
- Rule 8 [YES]: No macros except include guards (uses typed enums instead)
- Rule 9 [YES]: No pointers (value types only)
- Rule 10 [YES]: Compiles with -Wall -Wextra -Werror

SOLID Principles:

- S (Single Responsibility): Module provides ONLY universal bit/byte constants
- O (Open/Closed): Extensible (add new enums) without modifying existing code
- L (Liskov Substitution): N/A (no functions, only data)
- I (Interface Segregation): Minimal API (2 enums), each with focused purpose
- D (Dependency Inversion): No dependencies (standalone header)

Module Dependencies:

```
rx_bit_constants.h
+--> stdint.h (uint8_t for underlying enum type)

Used by:
+--> lib/rx_frame/ (protocol framing, byte packing)
+--> lib/rx_crc/ (CRC-32 byte-wise operations)
+--> lib/rx_nanopb/ (Protocol Buffer serialization)
+--> lib/rx_comm_manager/ (communication buffer management)
+--> lib/rx_hal/src/uart.c (data serialization)
```

Author

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Date

2026-01-27

Copyright

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See also

[stdint.h](#) for standard integer types
[rx_gpio_constants.h](#) for GPIO pin constants
[rx_port_constants.h](#) for port/pin mapping constants
[rx_time_constants.h](#) for time conversion constants

Version

1.0.0

Since

Version 1.0.0

Definition in file [rx_bit_constants.h](#).

7.1.2 Enumeration Type Documentation

7.1.2.1 rx_bit_shifts_t

enum [rx_bit_shifts_t](#) : uint8_t

Bit shift constants for byte and word extraction from multi-byte values.

Provides named constants for common bit-shift operations used in protocol encoding, endianness conversion, and byte-wise data access. These constants make code self- documenting and eliminate magic shift values.

Underlying Type: uint8_t (8-bit unsigned) - sufficient for all values (max = 24) Relationship to [rx_bit_sizes_t](#):

- k_rx_shift_byte = k_rx_bits_per_byte (both = 8)
- k_rx_shift_word16 = k_rx_bits_per_word16 (both = 16)
- k_rx_shift_word24 = 3 x k_rx_bits_per_byte (3 bytes = 24 bits)

7.1.2.2 Usage Patterns

7.1.2.2.1 Byte Extraction from 32-bit Word (Little-Endian)

```
uint32_t word = 0x12345678; // Little-endian representation

// Extract individual bytes
uint8_t byte0 = (word >> 0) & 0xFF; // 0x78 (LSB, byte 0)
uint8_t byte1 = (word >> k_rx_shift_byte) & 0xFF; // 0x56 (byte 1)
uint8_t byte2 = (word >> k_rx_shift_word16) & 0xFF; // 0x34 (byte 2)
uint8_t byte3 = (word >> k_rx_shift_word24) & 0xFF; // 0x12 (MSB, byte 3)
```

7.1.2.2.2 Big-Endian 32-bit Packing

```
// Pack uint32_t into big-endian byte array
uint32_t value = 0xAABBCCDD;
uint8_t buffer[4];

buffer[0] = (value >> k_rx_shift_word24) & 0xFF; // 0xAA (MSB first)
buffer[1] = (value >> k_rx_shift_word16) & 0xFF; // 0xBB
buffer[2] = (value >> k_rx_shift_byte) & 0xFF; // 0xCC
buffer[3] = (value >> 0) & 0xFF; // 0xDD (LSB last)
```

7.1.2.2.3 Big-Endian 16-bit Packing/Unpacking

```
// Pack uint16_t -> big-endian buffer
uint16_t value = 0xABCD;
uint8_t buffer[2];
buffer[0] = (value >> k_rx_shift_byte) & 0xFF; // 0xAB (high byte)
buffer[1] = (value >> 0) & 0xFF; // 0xCD (low byte)

// Unpack big-endian buffer -> uint16_t
uint16_t reconstructed = ((uint16_t)buffer[0] << k_rx_shift_byte) | buffer[1];
// Result: (0xAB << 8) | 0xCD = 0xABCD
```

7.1.2.3 Why Not Use rx_bit_sizes_t Directly?

Semantic Clarity:

- `k_rx_shift_byte` clearly indicates "shift to access next byte"
- `k_rx_bits_per_byte` ambiguously suggests "number of bits in a byte"
- Using `k_rx_shift_byte` makes the intent explicit (shift operation)

Example:

```
// CLEAR INTENT: Using k_rx_shift_byte
uint8_t byte1 = (word » k_rx_shift_byte) & 0xFF; // [OK] Obvious: shift by 1 byte

// AMBIGUOUS INTENT: Using k_rx_bits_per_byte
uint8_t byte1 = (word » k_rx_bits_per_byte) & 0xFF; // [X] Less clear: shifting? counting?
```

7.1.2.4 Value Justification Table

Constant	Value	Derivation	Use Case
<code>k_rx_shift_byte</code>	8	1 byte x 8 bits	Access byte N in multi-byte word
<code>k_rx_shift_word16</code>	16	2 bytes x 8 bits	Access upper 16 bits of 32-bit word
<code>k_rx_shift_word24</code>	24	3 bytes x 8 bits	Access MSB (byte 3) of 32-bit word

Usage Example: CRC-32 Byte-Wise Processing

```
#include "rx_bit_constants.h"

// Extract bytes from CRC-32 for transmission (big-endian)
void crc32_to_bytes(uint32_t crc, uint8_t* buffer) {
    buffer[0] = (crc » k_rx_shift_word24) & 0xFF; // Byte 3 (MSB)
    buffer[1] = (crc » k_rx_shift_word16) & 0xFF; // Byte 2
    buffer[2] = (crc » k_rx_shift_byte) & 0xFF; // Byte 1
    buffer[3] = (crc » 0) & 0xFF; // Byte 0 (LSB)
}

// Reconstruct CRC-32 from big-endian byte array
uint32_t bytes_to_crc32(const uint8_t* buffer) {
    return ((uint32_t)buffer[0] « k_rx_shift_word24) | // MSB
           ((uint32_t)buffer[1] « k_rx_shift_word16) |
           ((uint32_t)buffer[2] « k_rx_shift_byte) |
           ((uint32_t)buffer[3] « 0); // LSB
}
```

Usage Example: Protocol Field Extraction

```
// Extract 16-bit field from Protocol Buffer varint
uint32_t varint = 0x12345678;
uint16_t field = (varint » k_rx_shift_word16) & 0xFFFF;
// Result: 0x1234 (upper 16 bits)
```

See also

`rx_bit_sizes_t` for related bit count constants

`k_rx_bits_per_byte` equivalent value to `k_rx_shift_byte` (both = 8)

`k_rx_bits_per_word16` equivalent value to `k_rx_shift_word16` (both = 16)

Since

Version 1.0.0

Enumerator

k_rx_shift_byte	Shift amount to access next byte (8 bits = 1 byte). Use for extracting byte N from multi-byte word. Example: byte1 = (word >> k_rx_shift_byte) & 0xFF. Equivalent to k_rx_bits_per_byte but semantically clearer for shift operations.
k_rx_shift_word16	Shift amount to access upper 16 bits of 32-bit word (16 bits = 2 bytes). Use for extracting high word from dword. Example: high_word = (dword >> k_rx_shift_word16) & 0xFFFF. Also used for big-endian 16-bit packing/unpacking.
k_rx_shift_word24	Shift amount to access byte 3 (MSB) of 32-bit word (24 bits = 3 bytes). Use for extracting most significant byte. Example: msb = (word >> k_rx_shift_word24) & 0xFF. Critical for big-endian 32-bit encoding.

Definition at line 351 of file [rx_bit_constants.h](#).

```

00351         : uint8_t {
00352     k_rx_shift_byte =
00353         8, /**< Shift amount to access next byte (8 bits = 1 byte). Use for extracting byte N from multi-byte word. Example:
           byte1 = (word » k_rx_shift_byte) & 0xFF. Equivalent to k_rx_bits_per_byte but semantically clearer for shift
           operations. */
00354     k_rx_shift_word16 =
00355         16, /**< Shift amount to access upper 16 bits of 32-bit word (16 bits = 2 bytes). Use for extracting high word from
           dword. Example: high_word = (dword » k_rx_shift_word16) & 0xFFFF. Also used for big-endian 16-bit
           packing/unpacking. */
00356     k_rx_shift_word24 =
00357         24, /**< Shift amount to access byte 3 (MSB) of 32-bit word (24 bits = 3 bytes). Use for extracting most significant
           byte. Example: msb = (word » k_rx_shift_word24) & 0xFF. Critical for big-endian 32-bit encoding. */
00358 } rx_bit_shifts_t;
```

7.1.2.5 rx_bit_sizes_t

```
enum rx_bit_sizes_t : uint8_t
```

Fundamental bit and byte size constants for data type sizes.

Defines universal constants for bit counts in standard data types (byte, word16, word32, word64). These constants eliminate magic numbers in protocol encoding, buffer calculations, and bit manipulation operations.

Underlying Type: uint8_t (8-bit unsigned) - sufficient for all values (max = 64) Guaranteed Sizes: All values are compile-time constants (no runtime overhead)

7.1.2.6 Usage Patterns

7.1.2.6.1 Ceiling Division for Bit-to-Byte Conversion

```

// Calculate bytes needed for N bits (always round up)
uint32_t bit_count = 100; // Example: 100 bits
uint32_t byte_count = (bit_count + k_rx_bits_per_byte - 1) / k_rx_bits_per_byte;
// Result: (100 + 7) / 8 = 13 bytes (ceiling of 100/8 = 12.5)
```

7.1.2.6.2 Byte Indexing in Multi-Byte Words

```

// Extract byte N from 32-bit word
uint32_t word = 0xDEADBEEF;
uint8_t byte_index = 2; // Want byte 2 (0xBE)
uint8_t byte = (word » (k_rx_bits_per_byte * byte_index)) & 0xFF;
// Result: (0xDEADBEEF » 16) & 0xFF = 0xBE
```

7.1.2.6.3 Protocol Buffer Sizing

```
// Allocate buffer for protocol with CRC-32
const uint32_t k_payload_bytes = 64;
const uint32_t k_crc_bits = k_rx_bits_per_word32; // CRC-32 = 32 bits
const uint32_t k_crc_bytes = k_crc_bits / k_rx_bits_per_byte; // 4 bytes
const uint32_t k_total_bytes = k_payload_bytes + k_crc_bytes; // 68 bytes
uint8_t buffer[k_total_bytes];
```

7.1.2.7 Value Justification Table

Constant	Value	Rationale	Standard Reference
k_rx_bits_per_byte	8	Universal (ANSI, ISO/IEC, POSIX)	C99 Sec.3.6: CHAR_BIT = 8
k_rx_bits_per_word16	16	uint16_t guaranteed size	C99 Sec.7.18.1.1 exact-width types
k_rx_bits_per_word32	32	uint32_t guaranteed size	C99 Sec.7.18.1.1 exact-width types
k_rx_bits_per_word64	64	uint64_t guaranteed size	C99 Sec.7.18.1.1 exact-width types

Historical Note:

While some legacy systems used 7-bit bytes (e.g., CDC 6600) or 9-bit bytes (PDP-10), all modern systems use 8-bit bytes as mandated by POSIX.1-2008 and C99. The RX72N CHAR_BIT is guaranteed to be 8.

Usage Example: Endianness Conversion

```
#include "rx_bit_constants.h"

// Convert little-endian uint32_t to big-endian byte array
void uint32_to_big_endian(uint32_t value, uint8_t* buffer) {
    buffer[0] = (value » (k_rx_bits_per_byte * 3)) & 0xFF; // MSB
    buffer[1] = (value » (k_rx_bits_per_byte * 2)) & 0xFF;
    buffer[2] = (value » (k_rx_bits_per_byte * 1)) & 0xFF;
    buffer[3] = (value » (k_rx_bits_per_byte * 0)) & 0xFF; // LSB
}

// Convert big-endian byte array to little-endian uint32_t
uint32_t big_endian_to_uint32(const uint8_t* buffer) {
    return (((uint32_t)buffer[0] « (k_rx_bits_per_byte * 3)) | // MSB
            ((uint32_t)buffer[1] « (k_rx_bits_per_byte * 2)) |
            ((uint32_t)buffer[2] « (k_rx_bits_per_byte * 1)) |
            ((uint32_t)buffer[3] « (k_rx_bits_per_byte * 0))); // LSB
}
```

See also

[rx_bit_shifts_t](#) for corresponding shift values
[k_rx_shift_byte](#) equivalent to $(1 * k_rx_bits_per_byte)$
[k_rx_shift_word16](#) equivalent to $(2 * k_rx_bits_per_byte)$

Since

Version 1.0.0

Enumerator

k_rx_bits_per_byte	Bits per byte. Universal constant (CHAR_BIT = 8 per C99). Used for byte extraction, buffer sizing, bit-to-byte conversion. Range: Always 8 on modern systems.
k_rx_bits_per_word16	Bits per 16-bit word (uint16_t). Guaranteed by C99 exact-width types. Used for protocol fields, endianness conversion, CRC-16 operations. Range: Always 16.
k_rx_bits_per_word32	Bits per 32-bit word (uint32_t). Guaranteed by C99 exact-width types. Used for CRC-32, hash values, register access, Protocol Buffer fixed32. Range: Always 32.
k_rx_bits_per_word64	Bits per 64-bit word (uint64_t). Guaranteed by C99 exact-width types. Used for timestamps, large counters, Protocol Buffer fixed64. Range: Always 64.

Definition at line 228 of file rx_bit_constants.h.

```

00228         : uint8_t {
00229     k_rx_bits_per_byte =
00230         8, /**< Bits per byte. Universal constant (CHAR_BIT = 8 per C99). Used for byte extraction, buffer sizing, bit-to-byte
conversion. Range: Always 8 on modern systems. */
00231     k_rx_bits_per_word16 =
00232         16, /**< Bits per 16-bit word (uint16_t). Guaranteed by C99 exact-width types. Used for protocol fields, endianness
conversion, CRC-16 operations. Range: Always 16. */
00233     k_rx_bits_per_word32 =
00234         32, /**< Bits per 32-bit word (uint32_t). Guaranteed by C99 exact-width types. Used for CRC-32, hash values, register
access, Protocol Buffer fixed32. Range: Always 32. */
00235     k_rx_bits_per_word64 =
00236         64, /**< Bits per 64-bit word (uint64_t). Guaranteed by C99 exact-width types. Used for timestamps, large counters,
Protocol Buffer fixed64. Range: Always 64. */
00237 } rx_bit_sizes_t;

```

7.2 rx_bit_constants.h

[Go to the documentation of this file.](#)

```

00001 /**
00002  * @file rx_bit_constants.h
00003  * @brief Fundamental Bit and Byte Size Constants for Protocol and Data Manipulation
00004  *
00005  * @details
00006  * ## Overview
00007  * Centralized repository of typed enum constants for bit/byte sizes and shift operations
00008  * used throughout the STAR firmware. Provides self-documenting, type-safe constants for:
00009  * - Protocol encoding/decoding (Protocol Buffers, ASCII frames, CRC)
00010  * - Data serialization and deserialization
00011  * - Bit manipulation (extraction, packing, masking)
00012  * - Endianness conversion (big-endian <-> little-endian)
00013  * - Buffer size calculations
00014  *
00015  * **Design Philosophy:**
00016  * - **No magic numbers**: All numeric literals replaced with named typed enums (NASA Rule 8)
00017  * - **Type safety**: C23 typed enums with explicit underlying types (uint8_t)
00018  * - **Self-documenting**: Constants have descriptive names (k_rx_bits_per_byte vs. literal 8)
00019  * - **Searchable**: grep for "k_rx_shift_byte" finds all byte-shift operations
00020  * - **Hardware-agnostic**: These are universal constants, not hardware-specific
00021  *
00022  * ## Scope and Usage
00023  *
00024  * **INCLUDE in:**
00025  * - Protocol encoding/decoding modules (rx_frame, rx_nanopb, rx_crc)
00026  * - Communication layers (rx_spi_comm, rx_usb_comm, rx_comm_manager)
00027  * - Utility functions for data packing/unpacking
00028  * - Unit conversion functions
00029  *
00030  * **EXCLUDE from (use peripheral-specific headers instead):**

```

```

00031 * - Hardware register bit positions -> Use rx72n_*_regs.h (e.g., USBE.SYSCFG.DRPD = bit 5)
00032 * - Clock divider calculations -> Use rx72n_clock.h (e.g., PCLKA_DIV = 2)
00033 * - Peripheral-specific shifts -> Define in peripheral header (e.g., ADC channel shift)
00034 *
00035 * ## Common Usage Patterns
00036 *
00037 * ### 1. Bit-to-Byte Conversion (Ceiling Division)
00038 * @code{.c}
00039 * // Calculate bytes needed to store N bits (round up)
00040 * uint32_t bit_count = 100; // 100 bits
00041 * uint32_t byte_count = (bit_count + k_rx_bits_per_byte - 1) / k_rx_bits_per_byte;
00042 * // Result: (100 + 8 - 1) / 8 = 107 / 8 = 13 bytes (ceiling division)
00043 * @endcode
00044 *
00045 * ### 2. Byte Extraction from Multi-Byte Word
00046 * @code{.c}
00047 * // Extract individual bytes from 32-bit word (little-endian)
00048 * uint32_t word = 0x12345678;
00049 * uint8_t byte0 = (word » (k_rx_shift_byte * 0)) & 0xFF; // 0x78 (LSB)
00050 * uint8_t byte1 = (word » (k_rx_shift_byte * 1)) & 0xFF; // 0x56
00051 * uint8_t byte2 = (word » (k_rx_shift_byte * 2)) & 0xFF; // 0x34
00052 * uint8_t byte3 = (word » k_rx_shift_word24) & 0xFF; // 0x12 (MSB)
00053 * @endcode
00054 *
00055 * ### 3. Big-Endian 16-bit Packing
00056 * @code{.c}
00057 * // Pack uint16_t into big-endian byte array
00058 * uint16_t value = 0xABCD;
00059 * uint8_t buffer[2];
00060 * buffer[0] = (value » k_rx_shift_byte) & 0xFF; // 0xAB (high byte)
00061 * buffer[1] = (value » 0) & 0xFF; // 0xCD (low byte)
00062 * @endcode
00063 *
00064 * ### 4. Buffer Size Calculation for Bit Fields
00065 * @code{.c}
00066 * // Allocate buffer for protocol with bit-stuffing
00067 * const uint32_t k_payload_bits = 512; // 512 bits of data
00068 * const uint32_t k_overhead_bits = 32; // CRC-32
00069 * const uint32_t k_total_bits = k_payload_bits + k_overhead_bits;
00070 *
00071 * // Calculate bytes needed (ceiling division)
00072 * const uint32_t k_buffer_size =
00073 *     (k_total_bits + k_rx_bits_per_byte - 1) / k_rx_bits_per_byte;
00074 * uint8_t buffer[k_buffer_size]; // 68 bytes
00075 * @endcode
00076 *
00077 * ## Hardware Requirements
00078 *
00079 * | Requirement | Value | Notes |
00080 * |-----|-----|-----|
00081 * | **CPU** | Any (RX, ARM, x86) | Universal constants, platform-agnostic |
00082 * | **Compiler** | C23 or later | Requires typed enum support `: uint8_t` |
00083 * | **Dependencies** | stdint.h | For uint8_t underlying type |
00084 * | **RAM** | 0 bytes | Compile-time constants (no storage) |
00085 * | **Flash** | 0 bytes | Inlined by compiler, no code generated |
00086 *
00087 * @par NASA Power of 10 Compliance:
00088 * - **Rule 1 [YES]:** No goto, setjmp/longjmp, recursion (header-only, no control flow)
00089 * - **Rule 2 [N/A]:** No loops (header-only)
00090 * - **Rule 3 [YES]:** No dynamic allocation (compile-time constants)
00091 * - **Rule 4 [N/A]:** No functions (header-only)
00092 * - **Rule 5 [N/A]:** No functions (header-only)
00093 * - **Rule 6 [YES]:** Data at smallest scope (file-scoped enums)
00094 * - **Rule 7 [N/A]:** No functions (header-only)
00095 * - **Rule 8 [YES]:** No macros except include guards (uses typed enums instead)
00096 * - **Rule 9 [YES]:** No pointers (value types only)
00097 * - **Rule 10 [YES]:** Compiles with -Wall -Wextra -Werror
00098 *
00099 * @par SOLID Principles:
00100 * - **S (Single Responsibility):** Module provides ONLY universal bit/byte constants
00101 * - **O (Open/Closed):** Extensible (add new enums) without modifying existing code
00102 * - **L (Liskov Substitution):** N/A (no functions, only data)
00103 * - **I (Interface Segregation):** Minimal API (2 enums), each with focused purpose
00104 * - **D (Dependency Inversion):** No dependencies (standalone header)
00105 *
00106 * @par Module Dependencies:
00107 * ```
00108 * rx_bit_constants.h
00109 * +--> stdint.h (uint8_t for underlying enum type)
00110 *
00111 * Used by:
00112 * +--> lib/rx_frame/ (protocol framing, byte packing)
00113 * +--> lib/rx_crc/ (CRC-32 byte-wise operations)
00114 * +--> lib/rx_nanopb/ (Protocol Buffer serialization)
00115 * +--> lib/rx_comm_manager/ (communication buffer management)
00116 * +--> lib/rx_hal/src/uart.c (data serialization)
00117 * ```

```

```

00118 *
00119 * @author Locked, Inc.
00120 * @date 2026-01-27
00121 * @copyright Copyright (c) 2026 Locked Inc.
00122 * SPDX-License-Identifier: MIT
00123 *
00124 * @see stdint.h for standard integer types
00125 * @see rx_gpio_constants.h for GPIO pin constants
00126 * @see rx_port_constants.h for port/pin mapping constants
00127 * @see rx_time_constants.h for time conversion constants
00128 *
00129 * @version 1.0.0
00130 * @since Version 1.0.0
00131 */
00132
00133 #pragma once
00134
00135 #include <stdint.h>
00136
00137 #ifndef __cplusplus
00138 extern "C" {
00139 #endif
00140
00141 /*
=====
00142 * Bit and Byte Size Constants
00143 *
=====
00144 */
00145
00146 /**
00147 * @enum rx_bit_sizes_t
00148 * @brief Fundamental bit and byte size constants for data type sizes
00149 *
00150 * @details
00151 * Defines universal constants for bit counts in standard data types (byte, word16, word32, word64).
00152 * These constants eliminate magic numbers in protocol encoding, buffer calculations, and bit
00153 * manipulation operations.
00154 *
00155 * **Underlying Type:** uint8_t (8-bit unsigned) - sufficient for all values (max = 64)
00156 * **Guaranteed Sizes:** All values are compile-time constants (no runtime overhead)
00157 *
00158 * ## Usage Patterns
00159 *
00160 * ### Ceiling Division for Bit-to-Byte Conversion
00161 * @code{.c}
00162 * // Calculate bytes needed for N bits (always round up)
00163 * uint32_t bit_count = 100; // Example: 100 bits
00164 * uint32_t byte_count = (bit_count + k_rx_bits_per_byte - 1) / k_rx_bits_per_byte;
00165 * // Result: (100 + 7) / 8 = 13 bytes (ceiling of 100/8 = 12.5)
00166 * @endcode
00167 *
00168 * ### Byte Indexing in Multi-Byte Words
00169 * @code{.c}
00170 * // Extract byte N from 32-bit word
00171 * uint32_t word = 0xDEADBEEF;
00172 * uint8_t byte_index = 2; // Want byte 2 (0xBE)
00173 * uint8_t byte = (word » (k_rx_bits_per_byte * byte_index)) & 0xFF;
00174 * // Result: (0xDEADBEEF » 16) & 0xFF = 0xBE
00175 * @endcode
00176 *
00177 * ### Protocol Buffer Sizing
00178 * @code{.c}
00179 * // Allocate buffer for protocol with CRC-32
00180 * const uint32_t k_payload_bytes = 64;
00181 * const uint32_t k_crc_bits = k_rx_bits_per_word32; // CRC-32 = 32 bits
00182 * const uint32_t k_crc_bytes = k_crc_bits / k_rx_bits_per_byte; // 4 bytes
00183 * const uint32_t k_total_bytes = k_payload_bytes + k_crc_bytes; // 68 bytes
00184 * uint8_t buffer[k_total_bytes];
00185 * @endcode
00186 *
00187 * ## Value Justification Table
00188 *
00189 * | Constant | Value | Rationale | Standard Reference |
00190 * |-----|-----|-----|-----|
00191 * | k_rx_bits_per_byte | 8 | Universal (ANSI, ISO/IEC, POSIX) | C99 Sec.3.6: CHAR_BIT = 8 |
00192 * | k_rx_bits_per_word16 | 16 | uint16_t guaranteed size | C99 Sec.7.18.1.1 exact-width types |
00193 * | k_rx_bits_per_word32 | 32 | uint32_t guaranteed size | C99 Sec.7.18.1.1 exact-width types |
00194 * | k_rx_bits_per_word64 | 64 | uint64_t guaranteed size | C99 Sec.7.18.1.1 exact-width types |
00195 *
00196 * @par Historical Note:
00197 * While some legacy systems used 7-bit bytes (e.g., CDC 6600) or 9-bit bytes (PDP-10),
00198 * **all modern systems use 8-bit bytes** as mandated by POSIX.1-2008 and C99. The RX72N
00199 * CHAR_BIT is guaranteed to be 8.
00200 *
00201 * @par Usage Example: Endianness Conversion
00202 * @code{.c}

```

```

00203 * #include "rx_bit_constants.h"
00204 *
00205 * // Convert little-endian uint32_t to big-endian byte array
00206 * void uint32_to_big_endian(uint32_t value, uint8_t* buffer) {
00207 *     buffer[0] = (value » (k_rx_bits_per_byte * 3)) & 0xFF; // MSB
00208 *     buffer[1] = (value » (k_rx_bits_per_byte * 2)) & 0xFF;
00209 *     buffer[2] = (value » (k_rx_bits_per_byte * 1)) & 0xFF;
00210 *     buffer[3] = (value » (k_rx_bits_per_byte * 0)) & 0xFF; // LSB
00211 * }
00212 *
00213 * // Convert big-endian byte array to little-endian uint32_t
00214 * uint32_t big_endian_to_uint32(const uint8_t* buffer) {
00215 *     return ((uint32_t)buffer[0] « (k_rx_bits_per_byte * 3)) | // MSB
00216 *           ((uint32_t)buffer[1] « (k_rx_bits_per_byte * 2)) |
00217 *           ((uint32_t)buffer[2] « (k_rx_bits_per_byte * 1)) |
00218 *           ((uint32_t)buffer[3] « (k_rx_bits_per_byte * 0)); // LSB
00219 * }
00220 * @endcode
00221 *
00222 * @see rx_bit_shifts_t for corresponding shift values
00223 * @see k_rx_shift_byte equivalent to (1 * k_rx_bits_per_byte)
00224 * @see k_rx_shift_word16 equivalent to (2 * k_rx_bits_per_byte)
00225 *
00226 * @since Version 1.0.0
00227 */
00228 typedef enum : uint8_t {
00229     k_rx_bits_per_byte =
00230         8, /**< Bits per byte. Universal constant (CHAR_BIT = 8 per C99). Used for byte extraction, buffer sizing, bit-to-byte
00231             conversion. Range: Always 8 on modern systems. */
00232     k_rx_bits_per_word16 =
00233         16, /**< Bits per 16-bit word (uint16_t). Guaranteed by C99 exact-width types. Used for protocol fields, endianness
00234             conversion, CRC-16 operations. Range: Always 16. */
00235     k_rx_bits_per_word32 =
00236         32, /**< Bits per 32-bit word (uint32_t). Guaranteed by C99 exact-width types. Used for CRC-32, hash values, register
00237             access, Protocol Buffer fixed32. Range: Always 32. */
00238     k_rx_bits_per_word64 =
00239         64, /**< Bits per 64-bit word (uint64_t). Guaranteed by C99 exact-width types. Used for timestamps, large counters,
00240             Protocol Buffer fixed64. Range: Always 64. */
00241 } rx_bit_sizes_t;
00242
00243 /**
00244 * @enum rx_bit_shifts_t
00245 * @brief Bit shift constants for byte and word extraction from multi-byte values
00246 *
00247 * @details
00248 * Provides named constants for common bit-shift operations used in protocol encoding,
00249 * endianness conversion, and byte-wise data access. These constants make code self-
00250 * documenting and eliminate magic shift values.
00251 *
00252 * **Underlying Type:** uint8_t (8-bit unsigned) - sufficient for all values (max = 24)
00253 * **Relationship to rx_bit_sizes_t:**
00254 * - k_rx_shift_byte = k_rx_bits_per_byte (both = 8)
00255 * - k_rx_shift_word16 = k_rx_bits_per_word16 (both = 16)
00256 * - k_rx_shift_word24 = 3 x k_rx_bits_per_byte (3 bytes = 24 bits)
00257 *
00258 * ## Usage Patterns
00259 *
00260 * ### Byte Extraction from 32-bit Word (Little-Endian)
00261 * @code{.c}
00262 * uint32_t word = 0x12345678; // Little-endian representation
00263 *
00264 * // Extract individual bytes
00265 * uint8_t byte0 = (word » 0) & 0xFF; // 0x78 (LSB, byte 0)
00266 * uint8_t byte1 = (word » k_rx_shift_byte) & 0xFF; // 0x56 (byte 1)
00267 * uint8_t byte2 = (word » k_rx_shift_word16) & 0xFF; // 0x34 (byte 2)
00268 * uint8_t byte3 = (word » k_rx_shift_word24) & 0xFF; // 0x12 (MSB, byte 3)
00269 * @endcode
00270 *
00271 * ### Big-Endian 32-bit Packing
00272 * @code{.c}
00273 * // Pack uint32_t into big-endian byte array
00274 * uint32_t value = 0xAABBCCDD;
00275 * uint8_t buffer[4];
00276 *
00277 * buffer[0] = (value » k_rx_shift_word24) & 0xFF; // 0xAA (MSB first)
00278 * buffer[1] = (value » k_rx_shift_word16) & 0xFF; // 0xBB
00279 * buffer[2] = (value » k_rx_shift_byte) & 0xFF; // 0xCC
00280 * buffer[3] = (value » 0) & 0xFF; // 0xDD (LSB last)
00281 * @endcode
00282 *
00283 * ### Big-Endian 16-bit Packing/Unpacking
00284 * @code{.c}
00285 * // Pack uint16_t -> big-endian buffer
00286 * uint16_t value = 0xABCD;
00287 * uint8_t buffer[2];
00288 * buffer[0] = (value » k_rx_shift_byte) & 0xFF; // 0xAB (high byte)
00289 * buffer[1] = (value » 0) & 0xFF; // 0xCD (low byte)

```

```

00286 *
00287 * // Unpack big-endian buffer -> uint16_t
00288 * uint16_t reconstructed = ((uint16_t)buffer[0] « k_rx_shift_byte) | buffer[1];
00289 * // Result: (0xAB « 8) | 0xCD = 0xABCD
00290 * @endcode
00291 *
00292 * ## Why Not Use rx_bit_sizes_t Directly?
00293 *
00294 * **Semantic Clarity:**
00295 * - `k_rx_shift_byte` **clearly indicates** "shift to access next byte"
00296 * - `k_rx_bits_per_byte` **ambiguously suggests** "number of bits in a byte"
00297 * - Using k_rx_shift_byte makes the **intent explicit** (shift operation)
00298 *
00299 * **Example:**
00300 * @code{.c}
00301 * // CLEAR INTENT: Using k_rx_shift_byte
00302 * uint8_t byte1 = (word » k_rx_shift_byte) & 0xFF; // [OK] Obvious: shift by 1 byte
00303 *
00304 * // AMBIGUOUS INTENT: Using k_rx_bits_per_byte
00305 * uint8_t byte1 = (word » k_rx_bits_per_byte) & 0xFF; // [X] Less clear: shifting? counting?
00306 * @endcode
00307 *
00308 * ## Value Justification Table
00309 *
00310 * | Constant | Value | Derivation | Use Case |
00311 * |-----|-----|-----|-----|
00312 * | k_rx_shift_byte | 8 | 1 byte x 8 bits | Access byte N in multi-byte word |
00313 * | k_rx_shift_word16 | 16 | 2 bytes x 8 bits | Access upper 16 bits of 32-bit word |
00314 * | k_rx_shift_word24 | 24 | 3 bytes x 8 bits | Access MSB (byte 3) of 32-bit word |
00315 *
00316 * @par Usage Example: CRC-32 Byte-Wise Processing
00317 * @code{.c}
00318 * #include "rx_bit_constants.h"
00319 *
00320 * // Extract bytes from CRC-32 for transmission (big-endian)
00321 * void crc32_to_bytes(uint32_t crc, uint8_t* buffer) {
00322 *     buffer[0] = (crc » k_rx_shift_word24) & 0xFF; // Byte 3 (MSB)
00323 *     buffer[1] = (crc » k_rx_shift_word16) & 0xFF; // Byte 2
00324 *     buffer[2] = (crc » k_rx_shift_byte) & 0xFF; // Byte 1
00325 *     buffer[3] = (crc » 0) & 0xFF; // Byte 0 (LSB)
00326 * }
00327 *
00328 * // Reconstruct CRC-32 from big-endian byte array
00329 * uint32_t bytes_to_crc32(const uint8_t* buffer) {
00330 *     return ((uint32_t)buffer[0] « k_rx_shift_word24) | // MSB
00331 *           ((uint32_t)buffer[1] « k_rx_shift_word16) |
00332 *           ((uint32_t)buffer[2] « k_rx_shift_byte) |
00333 *           ((uint32_t)buffer[3] « 0); // LSB
00334 * }
00335 * @endcode
00336 *
00337 * @par Usage Example: Protocol Field Extraction
00338 * @code{.c}
00339 * // Extract 16-bit field from Protocol Buffer varint
00340 * uint32_t varint = 0x12345678;
00341 * uint16_t field = (varint » k_rx_shift_word16) & 0xFFFF;
00342 * // Result: 0x1234 (upper 16 bits)
00343 * @endcode
00344 *
00345 * @see rx_bit_sizes_t for related bit count constants
00346 * @see k_rx_bits_per_byte equivalent value to k_rx_shift_byte (both = 8)
00347 * @see k_rx_bits_per_word16 equivalent value to k_rx_shift_word16 (both = 16)
00348 *
00349 * @since Version 1.0.0
00350 */
00351 typedef enum : uint8_t {
00352     k_rx_shift_byte =
00353         8, /**< Shift amount to access next byte (8 bits = 1 byte). Use for extracting byte N from multi-byte word. Example:
00354         byte1 = (word » k_rx_shift_byte) & 0xFF. Equivalent to k_rx_bits_per_byte but semantically clearer for shift
00355         operations. */
00356         k_rx_shift_word16 =
00357         16, /**< Shift amount to access upper 16 bits of 32-bit word (16 bits = 2 bytes). Use for extracting high word from
00358         dword. Example: high_word = (dword » k_rx_shift_word16) & 0xFFFF. Also used for big-endian 16-bit
00359         packing/unpacking. */
00360         k_rx_shift_word24 =
00361         24, /**< Shift amount to access byte 3 (MSB) of 32-bit word (24 bits = 3 bytes). Use for extracting most significant
00362         byte. Example: msb = (word » k_rx_shift_word24) & 0xFF. Critical for big-endian 32-bit encoding. */
00363 } rx_bit_shifts_t;
00364 #ifdef __cplusplus
00365 }
00366 #endif

```


7.3 libs/rx_core/inc/rx_check.h File Reference

Validation and Error Checking Macros for RX72N Firmware.

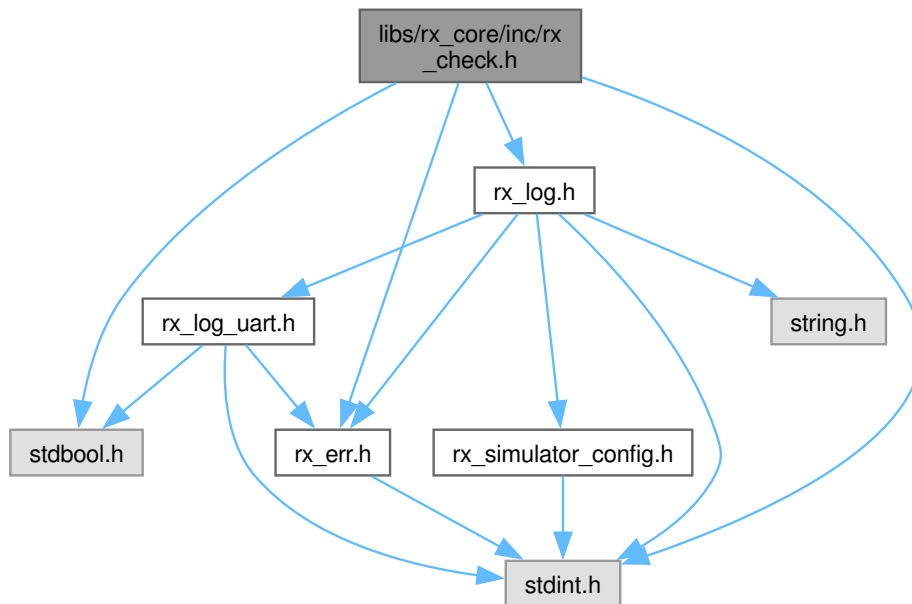
```
#include <stdbool.h>
```

```
#include <stdint.h>
```

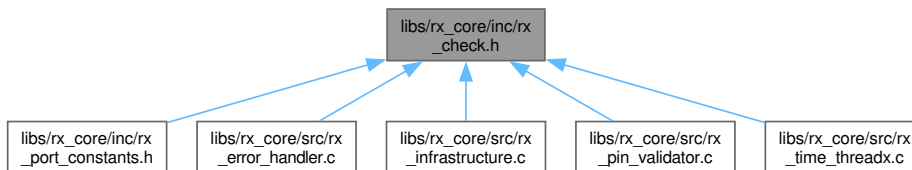
```
#include "rx_err.h"
```

```
#include "rx_log.h"
```

Include dependency graph for rx_check.h:



This graph shows which files directly or indirectly include this file:



Macros

- `#define RX_STATIC_TESTABLE` static
Makes internal (static) functions visible to unit tests.
- `#define RX_ASSERT(condition, message)`
Runtime assertion with fatal error on failure (NASA Rule 5 compliance).
- `#define RX_ASSERT_PRE(condition, message)`
Precondition assertion – verifies caller contract before function body.
- `#define RX_ASSERT_POST(condition, message)`
Postcondition assertion – verifies function output or invariant after work.
- `#define RX_ERROR_CHECK(err)`
Fatal error checking - halt system if error code indicates failure.
- `#define RX_ERROR_CHECK_WITHOUT_ABORT(err)`

- Check error code and log on failure (non-fatal).
- `#define RX_RETURN_ON_ERROR(err, tag, message)`
Return early on error.
- `#define RX_RETURN_VOID_ON_ERROR(err, tag, message)`
Return void on error (for void functions).
- `#define RX_RETURN_NULL_ON_ERROR(err, tag, message)`
Return NULL on error (for pointer-returning functions).
- `#define RX_CHECK_NULL_PTR(ptr, tag, message)`
Check for null pointer and return error.
- `#define RX_CHECK_RANGE(value, min, max, errcode)`
Check that a value is within an inclusive range.
- `#define RX_CHECK_RANGE_TAG(value, min, max, errcode, tag)`
Check that a value is within an inclusive range with tag.
- `#define RX_VALIDATE_PTR(ptr, tag, message)`
Validate pointer pre-condition and return error.
- `#define RX_VALIDATE_INIT(initialized, tag, message)`
Validate initialization state pre-condition.

Enumerations

- enum `fatal_error_constants_t` : `uint8_t` { `k_err_hex_digits` = 8U }
Constants for fatal error output formatting.

Functions

- static void `internal_rx_fatal_error` (const char *tag, const char *message, `rx_err_t` err)
Halt execution with diagnostic error message (fatal error handler).

7.3.1 Detailed Description

Validation and Error Checking Macros for RX72N Firmware.

Comprehensive error checking and validation infrastructure for safety-critical embedded systems. Provides type-safe macros for parameter validation, error propagation, and assertion checking aligned with NASA Power of 10 Rule 5 (minimum 2 assertions per function).

7.3.1.1 Architecture Design

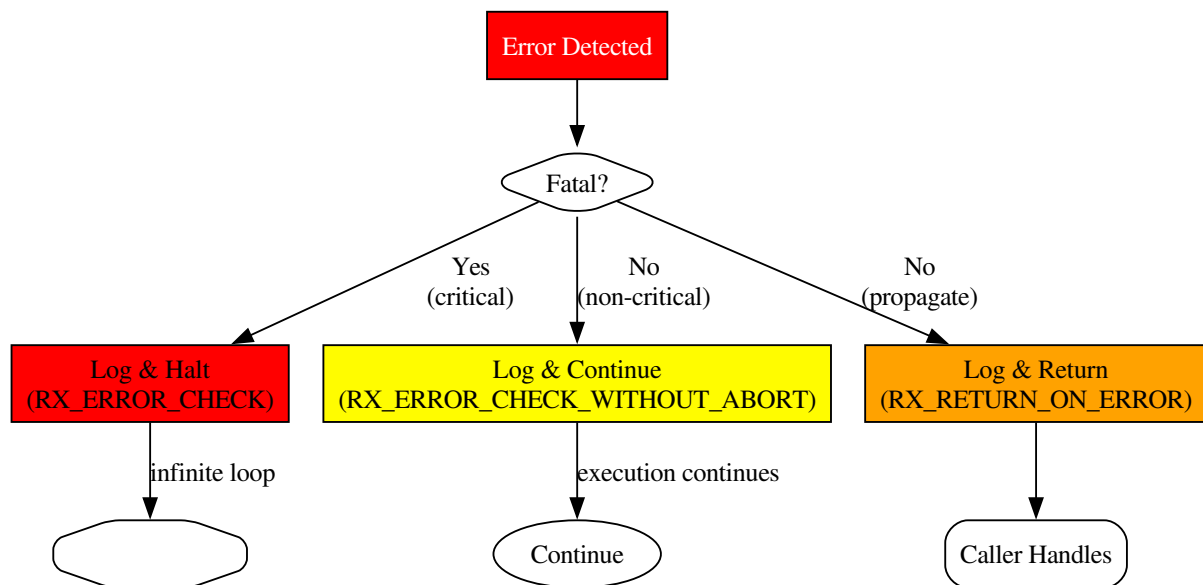
This module implements a layered error handling strategy:

- Fatal errors: System halts with diagnostic output (`RX_ERROR_CHECK`, `RX_ASSERT`)
- Propagating errors: Early return with logging (`RX_RETURN_ON_ERROR` variants)
- Non-fatal errors: Log and continue (`RX_ERROR_CHECK_WITHOUT_ABORT`)
- Precondition validation: Input parameter checks (`RX_CHECK_NULL_PTR`, `RX_CHECK_RANGE`)

7.3.1.2 Error Handling Strategies

Macro	Fatal	Returns	Use Case
<code>RX_ERROR_CHECK</code>	[YES]	Never	Critical initialization errors
<code>RX_ERROR_CHECK_WITHOUT_ABORT</code>	[NO]	No	Optional feature failures
<code>RX_RETURN_ON_ERROR</code>	[NO]	<code>rx_err_t</code>	Error propagation in <code>rx_err_t</code> funcs
<code>RX_RETURN_VOID_ON_ERROR</code>	[NO]	void	Error handling in void functions
<code>RX_RETURN_NULL_ON_ERROR</code>	[NO]	NULL	Error handling in pointer functions
<code>RX_CHECK_NULL_PTR</code>	[NO]	<code>rx_err_t</code>	Null pointer precondition check
<code>RX_CHECK_RANGE</code>	[NO]	<code>rx_err_t</code>	Range validation
<code>RX_ASSERT</code>	[YES]	Never	Invariant and postcondition checking

7.3.1.3 Error Handling Flow



7.3.1.4 Design Rationale

Why macros instead of inline functions?

- Capture FILE, LINE for precise error location (future enhancement)
- Zero runtime overhead (no function call in release builds)
- Early return from enclosing function (not possible with inline functions)
- Conditional compilation based on build type (e.g., disable assertions in release)

Why do-while(0) wrapper?

- Safe in all contexts (if statements without braces)

- Requires semicolon after macro invocation
- Prevents multiple expansion bugs

Why separate return-type variants?

- Type safety: prevents returning error from void function
- Explicit intent: `RX_RETURN_NULL_ON_ERROR` clearly returns `nullptr`
- Compiler optimization: separate code paths for different return types

7.3.1.5 Performance Characteristics

All macros expand to minimal code:

- Success path: 1-2 comparisons (2-4 cycles @ 240 MHz)
- Error path: Logging + action (varies, but typically < 100 us for non-fatal)
- Fatal error: Logging + infinite loop (system halted)
- Memory: 0 bytes overhead (pure compile-time code generation)

7.3.1.6 Thread Safety

- Validation macros: Thread-safe (pure validation, no shared state)
- Logging macros: Thread-safe if `rx_log` is thread-safe
- Fatal error: Not thread-safe (intentional - halts entire system)

7.3.1.7 Usage Patterns

Pattern 1: Critical Initialization

```
void system__init(void) {
    rx_err_t err;

    // Critical hardware init - must succeed or halt
    err = init__clock();
    RX_ERROR_CHECK(err); // Halts on failure

    err = init__gpio();
    RX_ERROR_CHECK(err); // Halts on failure

    // System continues only if both succeeded
}
```

Pattern 2: Optional Feature Initialization

```
void init__optional_features(void) {
    rx_err_t err;

    // Optional feature - log error but continue
    err = init__debug_uart();
    RX_ERROR_CHECK_WITHOUT_ABORT(err); // Logs, continues

    // System continues even if debug UART fails
}
```

Pattern 3: Error Propagation

```

rx_err_t init_subsystem(void) {
    rx_err_t err;

    // Propagate error to caller
    err = init_hardware();
    RX_RETURN_ON_ERROR(err, "SUBSYS", "Hardware init failed");

    err = init_software();
    RX_RETURN_ON_ERROR(err, "SUBSYS", "Software init failed");

    return k_rx_ok;
}

```

Pattern 4: NASA Rule 5 Compliance (Preconditions)

```

rx_err_t process_buffer(const uint8_t* data, size_t len) {
    // Precondition 1: Validate pointer (NASA Rule 5)
    RX_CHECK_NULL_PTR(data, "PROCESS", "Data buffer is nullptr");

    // Precondition 2: Validate size (NASA Rule 5)
    RX_CHECK_RANGE(len, 1, 256, k_rx_err_invalid_size);

    // Function body - preconditions satisfied
    // ...
    return k_rx_ok;
}

```

Pattern 5: Postcondition Validation

```

rx_err_t compute_result(float* output) {
    RX_CHECK_NULL_PTR(output, "COMPUTE", "Output pointer is nullptr");

    // Perform computation
    *output = complex_calculation();

    // Postcondition: result must be finite (NASA Rule 5)
    RX_ASSERT(isfinite(*output), "Result must be finite");

    return k_rx_ok;
}

```

Module Dependencies:

- [rx_err.h](#): Error code definitions ([rx_err_t](#), [k_rx_ok](#), etc.)
- [rx_log.h](#): Logging infrastructure ([rx_log_error](#), [uart_debug_puts](#))

See also

[rx_err.h](#) Error code definitions

[rx_log.h](#) Logging functions used by check macros

[rx_error_handler.h](#) High-level error handling infrastructure

[docs/sections/06_nasa_power_of_10.tex](#) NASA Rule 5 compliance

NASA Power of 10 Compliance:

- Rule 1: [YES] No goto, setjmp, recursion (macros expand to structured code)
- Rule 2: [YES] Infinite loop in fatal error is intentional (halt behavior)
- Rule 3: [YES] No dynamic allocation (stack-only variables)
- Rule 4: [YES] Macro expansions are small (< 10 lines)
- Rule 5: [YES] Explicitly designed to enforce Rule 5 (provide assertion macros)

- Rule 6: [YES] Minimal scope (do-while(0) block scope)
- Rule 7: [YES] All function calls checked by design (purpose of module)
- Rule 8: [YES] No raw macros for constants (uses `rx_err_t` typed enums)
- Rule 9: [YES] No function pointers
- Rule 10: [YES] Compiles with -Wall -Wextra -Werror

SOLID Principles:

- S (Single Responsibility): Only error checking and validation, no business logic
- O (Open/Closed): Extensible via new macros, existing macros closed for modification
- L (Liskov Substitution): Macros are transparent wrappers, don't change semantics
- I (Interface Segregation): Separate macros for different use cases (fatal, return, validate)
- D (Dependency Inversion): Depends on abstract `rx_err_t`, not concrete error values

Author

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Date

2026-01-27

Version

1.0.0

Copyright

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Definition in file [rx_check.h](#).

7.3.2 Macro Definition Documentation

7.3.2.1 RX_ASSERT

```
#define RX_ASSERT(  
    condition,  
    message)
```

Value:

```
do {  
    if (!(condition)) {  
        internal_rx_fatal_error("ASSERT", message, k_rx_fail);  
    }  
} while (0)
```

Runtime assertion with fatal error on failure (NASA Rule 5 compliance).

Verifies a boolean condition at runtime and halts execution if the condition is false. This macro is the primary mechanism for enforcing NASA Power of 10 Rule 5 (minimum 2 assertions per function) through precondition, postcondition, and invariant checking.

7.3.2.2 Algorithm

1. Evaluate condition expression
2. If condition is true: Continue execution (no overhead)
3. If condition is false:
 - a. Call `internal_rx_fatal_error()` with "ASSERT" tag
 - b. Log error message to debug UART
 - c. Halt system (infinite loop with interrupts disabled)

7.3.2.3 Use Cases

Use Case	Example
Precondition (input)	<code>RX_ASSERT(ptr != nullptr, "Input must be non-NULL")</code>
Precondition (range)	<code>RX_ASSERT(value <= MAX, "Value exceeds maximum")</code>
Postcondition (output)	<code>RX_ASSERT(result >= 0, "Result must be non-negative")</code>
Invariant (state)	<code>RX_ASSERT(count < MAX_COUNT, "Count overflow")</code>
Loop bound verification	<code>RX_ASSERT(i < ARRAY_SIZE, "Loop index overflow")</code>

7.3.2.4 NASA Rule 5 Compliance Pattern

Every function should have minimum 2 assertions:

```
rx_err_t calculate(float* input, float* output) {
    // Precondition 1: Input pointer valid (NASA Rule 5)
    RX_ASSERT(input != nullptr, "Input pointer must not be NULL");

    // Precondition 2: Output pointer valid (NASA Rule 5)
    RX_ASSERT(output != nullptr, "Output pointer must not be NULL");

    // Perform calculation
    *output = *input * 2.0f;

    // Postcondition: Result is finite (NASA Rule 5)
    RX_ASSERT(isfinite(*output), "Output must be finite");

    return k_rx_ok;
}
```

Parameters

condition	Boolean expression to verify • Type: Any expression convertible to bool • Must evaluate to true for normal operation • Examples: <code>ptr != nullptr</code>, <code>value < MAX</code>, <code>state == READY</code> • Evaluated exactly once (no multiple evaluation bug) • Should not have side effects (evaluation may be removed in release builds)
-----------	---

message	Error message displayed on assertion failure • Type: const char* (string literal) • Must be compile-time constant string • Should describe what was expected, not what failed • Good: "Pointer must not be NULL", "Value must be in range [0, 100]" • Bad: "ptr is nullptr", "Invalid value" • Recommended length: < 60 characters for readability
---------	---

Usage Example - Precondition Checking:

```
rx_err_t motor_set_velocity(motor_handle_t* motor, float velocity_mps) {
    // Precondition 1: Handle valid
    RX_ASSERT(motor != nullptr, "Motor handle must not be NULL");

    // Precondition 2: Velocity within physical limits
    RX_ASSERT(velocity_mps >= -2.5f && velocity_mps <= 2.5f,
        "Velocity must be in range [-2.5, 2.5] m/s");

    // Precondition 3: Motor initialized
    RX_ASSERT(motor->initialized, "Motor must be initialized");

    // Function body - all preconditions satisfied
    motor->target_velocity = velocity_mps;
    return k_rx_ok;
}
```

Usage Example - Postcondition Checking:

```
rx_err_t sensor_read(sensor_t* sensor, float* temperature_c) {
    RX_ASSERT(sensor != nullptr, "Sensor handle must not be NULL");
    RX_ASSERT(temperature_c != nullptr, "Output pointer must not be NULL");

    // Read sensor via I2C
    uint16_t raw_value = read_i2c_register(sensor->address, TEMP_REG);
    *temperature_c = (raw_value * 0.0625f) - 40.0f; // DS18B20 conversion

    // Postcondition: Temperature in physically possible range
    RX_ASSERT(*temperature_c >= -55.0f && *temperature_c <= 125.0f,
        "Temperature must be in valid sensor range");

    return k_rx_ok;
}
```

Usage Example - Invariant Checking:

```
typedef struct {
    uint8_t head;
    uint8_t tail;
    uint8_t data[64];
} ring_buffer_t;

void ring_buffer_push(ring_buffer_t* rb, uint8_t value) {
    RX_ASSERT(rb != nullptr, "Ring buffer must not be NULL");

    // Invariant: head and tail always within bounds
    RX_ASSERT(rb->head < 64, "Head index must be < 64");
    RX_ASSERT(rb->tail < 64, "Tail index must be < 64");

    rb->data[rb->head] = value;
    rb->head = (rb->head + 1) % 64;

    // Invariant maintained after modification
    RX_ASSERT(rb->head < 64, "Head index must remain < 64");
}
```

Usage Example - Loop Bound Verification (NASA Rule 2):

```
typedef enum : uint8_t {
    k_max_iterations = 100 // Explicit loop bound (NASA Rule 2)
} loop_bounds_t;

void process_array(uint8_t* data, size_t len) {
    RX_ASSERT(data != nullptr, "Data array must not be NULL");
    RX_ASSERT(len <= k_max_iterations, "Array too large for fixed loop bound");

    for (uint8_t i = 0; i < k_max_iterations && i < len; i++) {
        // Invariant: i is always within array bounds
        RX_ASSERT(i < len, "Loop index must be < array length");
        data[i] = process_byte(data[i]);
    }
}
```

Conditional Compilation:

Assertions can be disabled in release builds for performance:

```
#ifndef NDEBUG
#define RX_ASSERT(condition, message) ((void)0) // No-op in release
#else
#define RX_ASSERT(condition, message) do { ... } while (0) // As shown above
#endif
```

WARNING: Disabling assertions removes safety checks. Only disable after extensive testing with assertions enabled.

Performance:

- Success path: 1 comparison + 1 branch (2-3 cycles @ 240 MHz)
- Failure path: Logs and halts (system terminates, performance irrelevant)
- Stack usage: 0 bytes (no additional stack beyond function call if failure)
- Code size: ~8 bytes per assertion in release builds with optimizations

Note

Thread Safety: Condition check is atomic (single comparison)

Re-entrancy: Re-entrant (until failure, then system halts)

Memory: No dynamic allocation

Warning

Assertions halt the system - Use for programming errors and invariant violations, NOT for recoverable runtime errors.

Condition must not have side effects - If assertions are disabled, side effects disappear. Bad: `RX_ASSERT(ptr++ != nullptr, "...")` Good: `ptr++; RX_ASSERT(ptr != nullptr, "...")`

Attention

Use assertions for programming errors (bugs in code) and return error codes for runtime errors (invalid user input, hardware faults).

Assertions vs. Error Returns:

Condition Type	Use Assertion	Use Error Return
Null pointer from caller	YES	NO
Value out of documented range	YES	NO
Invalid state transition	YES	NO
Hardware fault	NO	YES
Communication timeout	NO	YES
Sensor out of range	NO	YES

Precondition

condition is a valid boolean expression

message is a non-NULL string literal (compile-time constant)

Postcondition

If condition is true: Execution continues normally

If condition is false: System halts permanently (requires reset)

Invariant

If condition is false, function never returns

Condition evaluated exactly once (no multiple evaluation)

See also

[internal_rx_fatal_error\(\)](#) Function called on assertion failure

[RX_ERROR_CHECK\(\)](#) Related macro for error code checking

[RX_CHECK_NULL_PTR\(\)](#) Alternative for null pointer checking with error return

NASA Power of 10 Compliance:

- Rule 1: [YES] No goto, setjmp, recursion (structured if statement)
- Rule 2: [YES] N/A (no loops)
- Rule 3: [YES] No dynamic allocation
- Rule 4: [YES] Macro expansion is 3 lines
- Rule 5: [YES] This macro implements Rule 5 (provides assertion mechanism)
- Rule 6: [YES] do-while(0) provides minimal scope
- Rule 7: [YES] N/A (macro itself, not function)
- Rule 8: [YES] Uses typed enum k_rx_fail
- Rule 9: [YES] No function pointers
- Rule 10: [YES] Compiles with -Wall -Wextra -Werror

Since

Version 1.0.0

Definition at line 682 of file [rx_check.h](#).

```
00682 #define RX_ASSERT(condition, message)
00683 do {
00684     if (!(condition)) {
00685         internal_rx_fatal_error("ASSERT", message, k_rx_fail);
00686     }
00687 } while (0)
```

Referenced by [rx_pin_from_pin\(\)](#), and [rx_port_from_pin\(\)](#).

7.3.2.5 RX_ASSERT_POST

```
#define RX_ASSERT_POST(
    condition,
    message)
```

Value:

```
do {
    if (!(condition)) {
        internal_rx_fatal_error("POST", (message), k_rx_fail);
    }
} while (0)
```

Postcondition assertion – verifies function output or invariant after work.

Halts the system if a postcondition is violated. Use after a computation to verify that the result satisfies the documented contract.

7.3.2.6 Testability (UNIT_TEST builds)

Postconditions are often architecturally guaranteed (e.g., a 20-bit mask always produces a value in [0, 0xFFFFF]). Their false branches are unreachable without fault injection. In UNIT_TEST builds the macro checks `g_rx_assert_post_force_fail` first: when that flag is true the macro fires regardless of the actual condition, making the failure branch reachable for coverage.

Parameters

condition	Boolean expression that must be true
message	Descriptive string for fatal error output

Precondition

condition is a valid boolean expression

message is a non-NULL string literal

Postcondition

If condition is true (and force-fail off): execution continues

If condition is false or force-fail on: system halts (returns in UNIT_TEST)

See also

[RX_ASSERT_PRE\(\)](#) Precondition counterpart
[g_rx_assert_post_force_fail](#) Test-only fault injection flag

Since

Version 1.0.0

Definition at line 764 of file [rx_check.h](#).

```
00764 #define RX_ASSERT_POST(condition, message)
00765     do {
00766         if (!(condition)) {
00767             internal_rx_fatal_error("POST", (message), k_rx_fail);
00768         }
00769     } while (0)
```

7.3.2.7 RX_ASSERT_PRE

```
#define RX_ASSERT_PRE(
    condition,
    message)
```

Value:

```
do {
    if (!(condition)) {
        internal_rx_fatal_error("PRE", (message), k_rx_fail);
    }
} while (0)
```

Precondition assertion – verifies caller contract before function body.

Halts the system if a precondition is violated. Use for validating state and parameters at the top of internal functions where the condition is architecturally guaranteed by callers, but should still be checked defensively.

7.3.2.8 Testability (UNIT_TEST builds)

Because internal functions are decorated with `RX_STATIC_TESTABLE`, test code can call them directly with invalid state or parameters. This makes the false branch of every `RX_ASSERT_PRE` reachable, enabling 100% branch coverage without production-code gymnastics.

Parameters

condition	Boolean expression that must be true
message	Descriptive string for fatal error output

Precondition

condition is a valid boolean expression
message is a non-NULL string literal

Postcondition

If condition is true: execution continues

If condition is false: system halts (returns in UNIT_TEST builds)

See also

[RX_ASSERT_POST\(\)](#) Postcondition counterpart

[RX_STATIC_TESTABLE](#) Exposes internal functions for test-driven coverage

Since

Version 1.0.0

Definition at line 718 of file [rx_check.h](#).

```
00718 #define RX_ASSERT_PRE(condition, message)
00719     do {
00720         if (!(condition)) {
00721             internal_rx_fatal_error("PRE", (message), k_rx_fail);
00722         }
00723     } while (0)
```

7.3.2.9 RX_CHECK_NULL_PTR

```
#define RX_CHECK_NULL_PTR(
    ptr,
    tag,
    message)
```

Value:

```
do {
    if ((ptr) == nullptr) {
        rx_log_error(tag, message);
        return k_rx_err_null_ptr;
    }
} while (0)
```

Check for null pointer and return error.

Parameters

in	ptr	Pointer to check
in	tag	Component tag for logging
in	message	Error message for logging

If ptr is nullptr, logs the error and returns k_rx_err_null_ptr.

Example: [rx_err_t](#) process_data(const uint8_t* data) { RX_CHECK_NULL_PTR(data, "PROCESS", "Data pointer is nullptr"); // Only reached if data != nullptr return k_rx_ok; }

Definition at line 990 of file [rx_check.h](#).

```
00990 #define RX_CHECK_NULL_PTR(ptr, tag, message)
00991     do {
00992         if ((ptr) == nullptr) {
00993             rx_log_error(tag, message);
00994             return k_rx_err_null_ptr;
00995         }
00996     } while (0)
```

Referenced by [error_handler_deinit\(\)](#), [error_handler_get_interface\(\)](#), [error_handler_init\(\)](#), [impl_clear_errors\(\)](#), [impl_report_error\(\)](#), [impl_reset_retry_counter\(\)](#), [pin_validator_deinit\(\)](#), [pin_validator_get_interface\(\)](#), [pin_validator_init\(\)](#), and [rx_time_threadx_get_interface\(\)](#).

7.3.2.10 RX_CHECK_RANGE

```
#define RX_CHECK_RANGE(
    value,
    min,
    max,
    errcode)
```

Value:

```
do {
    if (((value) < (min)) || ((value) > (max))) {
        rx_log_error("CHECK", "Range check failed");
        return (errcode);
    }
} while (0)
```

```
\
\
\
\
\
```

Check that a value is within an inclusive range.

Parameters

in	value	Value to check
in	min	Minimum allowed value
in	max	Maximum allowed value
in	errcode	Error to return on failure

Definition at line 1006 of file [rx_check.h](#).

```
01006 #define RX_CHECK_RANGE(value, min, max, errcode)
01007 do {
01008     if (((value) < (min)) || ((value) > (max))) {
01009         rx_log_error("CHECK", "Range check failed");
01010         return (errcode);
01011     }
01012 } while (0)
```

```
\
\
\
\
\
```

7.3.2.11 RX_CHECK_RANGE_TAG

```
#define RX_CHECK_RANGE_TAG(
    value,
    min,
    max,
    errcode,
    tag)
```

Value:

```
do {
    (void)(min); /* Document minimum bound; explicit check needed if min > 0 */
    if ((value) > (max)) {
        rx_log_error(tag, "Range check failed");
        return (errcode);
    }
} while (0)
```

```
\
\
\
\
\
```

Check that a value is within an inclusive range with tag.

Note

Only checks upper bound to avoid -Wtype-limits warnings when `min == 0` and value is unsigned (comparison is always false). For non-zero min bounds, add an explicit lower-bound check before this macro. The min parameter is documented but not checked.

Parameters

in	value	Value to check
in	min	Minimum allowed value (documented only; add explicit check if <code>min > 0</code>)
in	max	Maximum allowed value
in	errcode	Error to return on failure
in	tag	Component tag for logging

Definition at line 1027 of file [rx_check.h](#).

```

01027 #define RX_CHECK_RANGE_TAG(value, min, max, errcode, tag)
01028 do {
01029     (void)(min); /* Document minimum bound; explicit check needed if min > 0 */
01030     if ((value) > (max)) {
01031         rx_log_error(tag, "Range check failed");
01032         return (errcode);
01033     }
01034 } while (0)

```

7.3.2.12 RX_ERROR_CHECK

```

#define RX_ERROR_CHECK(
    err)

```

Value:

```

do {
    rx_err_t err_rc_ = (err);
    if (rx_err_is_error(err_rc_)) {
        internal_rx_fatal_error("ERROR_CHECK", "Fatal error detected", err_rc_);
    }
} while (0)

```

Fatal error checking - halt system if error code indicates failure.

Checks an [rx_err_t](#) error code and halts the system with diagnostics if the error is not `k_rx_ok`. Use this macro for critical operations where failure is unrecoverable and continuing execution would be unsafe.

7.3.2.13 When to Use

- Critical initialization: Hardware init, clock config, essential peripherals
- Safety-critical resources: Emergency stop system, watchdog, fault detection
- Unrecoverable errors: Memory corruption, invalid firmware state

7.3.2.14 When NOT to Use

- Optional features: Use `RX_ERROR_CHECK_WITHOUT_ABORT` instead
- Recoverable errors: Use `RX_RETURN_ON_ERROR` to propagate to caller
- Expected failures: Handle explicitly with if/switch statements

Parameters

err	Error code expression to check • Type: Any expression that evaluates to <code>rx_err_t</code> • Evaluated exactly once (safe for expressions with side effects) • If <code>k_rx_ok (0)</code>: Continues execution silently • If any other value: Halts system with diagnostics
-----	--

Usage Example - Critical Initialization:

```

void system_boot(void) {
    rx_err_t err;

    // Clock init is critical - must succeed
    err = init_system_clock();
    RX_ERROR_CHECK(err); // Halts on failure

    // GPIO init is critical - must succeed
    err = init_gpio();
    RX_ERROR_CHECK(err); // Halts on failure

    // Only reached if both succeeded
    uart_debug_puts("System boot successful\r\n");
}

```

Usage Example - Safety-Critical Resource:

```

void init_safety_systems(void) {
    rx_err_t err;

    // Emergency stop system must work
    err = init_emergency_stop();
    RX_ERROR_CHECK(err); // Safety-critical, cannot proceed if failed

    // Watchdog must be enabled
    err = init_independent_watchdog();
    RX_ERROR_CHECK(err); // System unusable without watchdog
}

```

Comparison to Alternatives:

Macro	Halts	Logs	Returns	Use Case
<code>RX_ERROR_CHECK</code>	[YES]	[YES]	Never	Critical init
<code>RX_ERROR_CHECK_WITHOUT_ABORT</code>	[NO]	[YES]	No	Optional features
<code>RX_RETURN_ON_ERROR</code>	[NO]	[YES]	Error	Propagate to caller

Macro	Halts	Logs	Returns	Use Case
<code>if (err != k_rx_ok)</code>	[NO]	[NO]	Varies	Custom handling

Warning

System halts permanently - Use only for truly unrecoverable errors

Cannot be caught or handled - No try/catch, no error recovery

Attention

Watchdog behavior: If enabled, watchdog will eventually reset system. If disabled, requires manual reset/power cycle.

See also

[RX_ERROR_CHECK_WITHOUT_ABORT\(\)](#) Non-fatal version (logs, continues)

[RX_RETURN_ON_ERROR\(\)](#) Error propagation version (returns to caller)

[internal_rx_fatal_error\(\)](#) Underlying fatal error handler

Since

Version 1.0.0

Definition at line 858 of file [rx_check.h](#).

```
00858 #define RX_ERROR_CHECK(err)
00859 do {
00860     rx_err_t err_rc = (err);
00861     if (rx_err_is_error(err_rc)) {
00862         internal_rx_fatal_error("ERROR_CHECK", "Fatal error detected", err_rc);
00863     }
00864 } while (0)
```

7.3.2.15 RX_ERROR_CHECK_WITHOUT_ABORT

```
#define RX_ERROR_CHECK_WITHOUT_ABORT(
    err)
```

Value:

```
do {
    rx_err_t err_rc = (err);
    if (rx_err_is_error(err_rc)) {
        rx_log_error_val("ERROR_CHECK", "Error", err_rc);
    }
} while (0)
```

Check error code and log on failure (non-fatal).

Parameters

in	err	Error code to check
----	-----	---------------------

If err is not `k_rx_ok`, logs the error but continues execution. Use this for non-critical errors where system can continue.

Example: `rx_err_t err = optional_feature_init(); RX_ERROR_CHECK_WITHOUT_ABORT(err);`
 // Logs but continues

Definition at line 878 of file `rx_check.h`.

```
00878 #define RX_ERROR_CHECK_WITHOUT_ABORT(err)
00879 do {
00880     rx_err_t err_rc_ = (err);
00881     if (rx_err_is_error(err_rc_)) {
00882         rx_log_error_val("ERROR_CHECK", "Error", err_rc_);
00883     }
00884 } while (0)
```

7.3.2.16 RX_RETURN_NULL_ON_ERROR

```
#define RX_RETURN_NULL_ON_ERROR(
    err,
    tag,
    message)
```

Value:

```
do {
    rx_err_t err_rc_ = (err);
    if (rx_err_is_error(err_rc_)) {
        rx_log_error(tag, message);
        rx_log_error_val(tag, "Error", err_rc_);
        return nullptr;
    }
} while (0)
```

Return NULL on error (for pointer-returning functions).

Parameters

in	err	Error code to check
in	tag	Component tag for logging
in	message	Error message for logging

If `err` is not `k_rx_ok`, logs the error and returns NULL from the function. Use this for functions that return pointers.

Example: `void* create_object(void) { rx_err_t err = validate_preconditions(); RX_RETURN_NULL_ON_ERROR(err, "FACTORY", "Precondition check failed"); // Only reached if validation succeeded return allocate_object(); }`

Definition at line 959 of file `rx_check.h`.

```
00959 #define RX_RETURN_NULL_ON_ERROR(err, tag, message)
00960 do {
00961     rx_err_t err_rc_ = (err);
00962     if (rx_err_is_error(err_rc_)) {
00963         rx_log_error(tag, message);
00964         rx_log_error_val(tag, "Error", err_rc_);
00965         return nullptr;
00966     }
00967 } while (0)
```

7.3.2.17 RX_RETURN_ON_ERROR

```
#define RX_RETURN_ON_ERROR(
    err,
    tag,
    message)
```

Value:

```
do {
    rx_err_t err_rc_ = (err);
    if (rx_err_is_error(err_rc_)) {
        rx_log_error(tag, message);
        rx_log_error_val(tag, "Error", err_rc_);
        return err_rc_;
    }
} while (0)
```



Return early on error.

Parameters

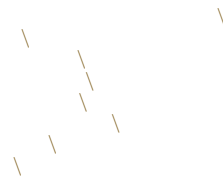
in	err	Error code to check
in	tag	Component tag for logging
in	message	Error message for logging

If err is not `k_rx_ok`, logs the error and returns err from the function. Use this for functions that return `rx_err_t`.

Example: `rx_err_t init_subsystem(void) { rx_err_t err = gpio_init(); RX_RETURN_ON_ERROR(err, "SUBSYS", "GPIO init failed"); // Only reached if gpio_init succeeded return k_rx_ok; }`

Definition at line 904 of file `rx_check.h`.

```
00904 #define RX_RETURN_ON_ERROR(err, tag, message)
00905 do {
00906     rx_err_t err_rc_ = (err);
00907     if (rx_err_is_error(err_rc_)) {
00908         rx_log_error(tag, message);
00909         rx_log_error_val(tag, "Error", err_rc_);
00910         return err_rc_;
00911     }
00912 } while (0)
```



7.3.2.18 RX_RETURN_VOID_ON_ERROR

```
#define RX_RETURN_VOID_ON_ERROR(
    err,
    tag,
    message)
```

Value:

```
do {
    rx_err_t err_rc_ = (err);
    if (rx_err_is_error(err_rc_)) {
        rx_log_error(tag, message);
        rx_log_error_val(tag, "Error", err_rc_);
        return;
    }
} while (0)
```



Return void on error (for void functions).

Parameters

in	err	Error code to check
in	tag	Component tag for logging
in	message	Error message for logging

If err is not k_rx_ok, logs the error and returns from the function. Use this for void functions that cannot return an error code.

Example: void configure_system(void) { rx_err_t err = gpio_init(); RX_RETURN_VOID_ON_ERROR(err, "SYSTEM", "GPIO init failed"); // Only reached if gpio_init succeeded }

Definition at line 931 of file [rx_check.h](#).

```

00931 #define RX_RETURN_VOID_ON_ERROR(err, tag, message)
00932 do {
00933     rx_err_t err_rc = (err);
00934     if (rx_err_is_error(err_rc)) {
00935         rx_log_error(tag, message);
00936         rx_log_error_val(tag, "Error", err_rc);
00937         return;
00938     }
00939 } while (0)

```

7.3.2.19 RX_STATIC_TESTABLE

```
#define RX_STATIC_TESTABLE static
```

Makes internal (static) functions visible to unit tests.

In production builds functions decorated with this macro are declared static, keeping them private to their translation unit. In UNIT_TEST builds the static qualifier is removed so test code can call internal helpers directly – including with invalid arguments – to verify that every defensive null-check and error branch is reachable.

Usage in a source file:

```
RX_STATIC_TESTABLE rx_err_t internal_decode_header(...);
```

Usage in a test file (forward-declare the function as extern):

```

#ifdef UNIT_TEST
extern rx_err_t internal_decode_header(...);
#endif

```

Note

In production builds this macro expands to static, so there is zero overhead and no change in linkage or inlining behaviour.

Since

Version 1.0.0

Definition at line 243 of file [rx_check.h](#).

7.3.2.20 RX_VALIDATE_INIT

```
#define RX_VALIDATE_INIT(
    initialized,
    tag,
    message)
```

Value:

```
do {
    if (!initialized) {
        rx_log_error(tag, message);
        return k_rx_err_invalid_state;
    }
} while (0)
```

Validate initialization state pre-condition.

Returns `k_rx_err_invalid_state` on failure.

Definition at line 1048 of file [rx_check.h](#).

```
01048 #define RX_VALIDATE_INIT(initialized, tag, message)
01049 do {
01050     if (!initialized) {
01051         rx_log_error(tag, message);
01052         return k_rx_err_invalid_state;
01053     }
01054 } while (0)
```

7.3.2.21 RX_VALIDATE_PTR

```
#define RX_VALIDATE_PTR(
    ptr,
    tag,
    message)
```

Value:

`RX_CHECK_NULL_PTR(ptr, tag, message)`

Validate pointer pre-condition and return error.

Alias for `RX_CHECK_NULL_PTR` to standardize pre-condition naming.

Definition at line 1041 of file [rx_check.h](#).

7.3.3 Enumeration Type Documentation

7.3.3.1 fatal_error_constants_t

enum [fatal_error_constants_t](#) : `uint8_t`

Constants for fatal error output formatting.

Enumerator

<code>k_err_hex_digits</code>	Number of hex digits for a 32-bit error code
-------------------------------	--

Definition at line 260 of file [rx_check.h](#).

```
00260     : uint8_t {
00261     k_err_hex_digits = 8U, /**< Number of hex digits for a 32-bit error code */
00262 } fatal_error_constants_t;
```

7.3.4 Function Documentation

7.3.4.1 internal_rx_fatal_error()

```
void internal_rx_fatal_error (
    const char * tag,
    const char * message,
    rx_err_t err) [inline], [static]
```

Halt execution with diagnostic error message (fatal error handler).

Terminates firmware execution after logging detailed error diagnostics to debug UART. This is a non-returning function used for unrecoverable errors where continuing execution would be unsafe.

7.3.4.2 Algorithm

1. Output formatted error banner to debug UART
2. Display component tag, error message, and error code (hex)
3. Indicate system halted state
4. Disable all interrupts (global interrupt disable)
5. Enter infinite loop with CPU in low-power WAIT state

7.3.4.3 Output Format

```
=====
FATAL ERROR
=====
[TAG] Error message here
Error code: 0x00000XXX
=====
System halted. Reset required.
=====
```

7.3.4.4 Platform-Specific Behavior

Normal Build (Hardware Target):

- Disables interrupts via `clrpsw i` (clear PSW I-flag)
- Enters infinite loop with wait instruction
- CPU remains in low-power wait state until hardware reset
- Watchdog will eventually trigger reset if enabled

Unit Test Build (UNIT_TEST defined):

- Returns immediately without halting
- Allows test harness to continue and detect fatal error
- Enables testing of error paths without hanging test runner

Parameters

in	tag	Component tag for error identification • Type: <code>const char*</code> (null-terminated string) • Must not be <code>nullptr</code> (undefined behavior if <code>nullptr</code>) • Typical values: "MOTOR", "COMM", "INIT", "ASSERT", etc. • Used to identify which subsystem detected the error • Recommended length: 3-8 characters for readability
in	message	Human-readable error description • Type: <code>const char*</code> (null-terminated string) • Must not be <code>nullptr</code> (undefined behavior if <code>nullptr</code>) • Should describe what failed and why • Example: "GPIO initialization failed", "Invalid motor ID" • Maximum practical length: ~80 characters (UART line width)
in	err	Error code indicating failure type • Type: <code>rx_err_t</code> (<code>int32_t</code>) • Typically non-zero (zero indicates success, illogical for fatal error) • Displayed in hexadecimal (8 digits, zero-padded) • Used for post-mortem debugging via UART log capture

Note

This function never returns (except in `UNIT_TEST` builds)

Usage Example - From `RX_ERROR_CHECK`:

```
rx_err_t err = critical_init_function();
if (err != k_rx_ok) {
    internal_rx_fatal_error("INIT", "Critical init failed", err);
    // Never reached in normal builds
}
```

Usage Example - From `RX_ASSERT`:

```
if (!(ptr != nullptr)) {
    internal_rx_fatal_error("ASSERT", "Pointer must not be NULL", k_rx_fail);
    // Never reached in normal builds
}
```

Usage Example - Direct Call (Rare):

```
// Only use directly if macro wrappers don't fit use case
if (unrecoverable_condition()) {
    internal_rx_fatal_error("SAFETY", "Unrecoverable hardware fault", k_rx_err_hw_error);
}
```

Assembly Instruction Details:

clrpsw i - Clear Processor Status Word I-flag

- Disables all maskable interrupts globally
- Non-maskable interrupts (NMI) still active
- Prevents interrupt handlers from interfering with error state
- Ensures system remains in known halted state

wait - Wait for Interrupt

- Places CPU in low-power state
- CPU wakes on interrupt, but interrupts disabled -> immediately waits again
- Reduces power consumption during halt state
- Effective infinite loop with minimal power draw

Watchdog Behavior:

If watchdog (IWDG) is enabled:

- Watchdog will expire after timeout period (typically 128ms-2s)
- System will reset automatically
- Error message captured before reset (if UART log buffered)
- Allows recovery from fatal errors in deployed systems

If watchdog is disabled:

- System remains halted indefinitely
- Requires manual hardware reset or power cycle
- Useful during development for post-mortem debugging

Performance:

- Execution time: ~500 us (UART output dominates)
- Stack usage: ~16 bytes (local variables, return address)
- UART bandwidth: ~100 bytes @ 115200 baud = ~9ms transmission time
- Not performance-critical: Only called on fatal errors

Note

Thread Safety: Not thread-safe, but irrelevant (system halting)
Re-entrancy: Not reentrant (disables interrupts, infinite loop)
Memory: No dynamic allocation, stack-only locals

Warning

This function never returns in normal builds - Calling code after this function is unreachable. Compiler may optimize it away.
Do not call from interrupt context - Disabling interrupts in ISR can cause undefined behavior. Use error flags instead.

Attention

System is permanently halted - Only recovery is hardware reset. Ensure watchdog is enabled in production builds for automatic recovery.
UART output is blocking - If UART is not initialized or not connected, [uart_debug_puts\(\)](#) may hang. Ensure UART is functional.

Precondition

tag != nullptr (undefined behavior if nullptr, will likely crash)
message != nullptr (undefined behavior if nullptr, will likely crash)
UART debug interface initialized (or [uart_debug_puts\(\)](#) will hang)

Postcondition

Interrupts disabled globally (PSW.I = 0)
CPU in infinite wait loop (never returns)
Error diagnostics sent to debug UART
System requires hardware reset to recover

Invariant

In UNIT_TEST builds, function returns immediately (allows testing)
In normal builds, function never returns (infinite loop)

See also

[RX_ERROR_CHECK\(\)](#) Macro that calls this function
[RX_ASSERT\(\)](#) Macro that calls this function
[uart_debug_puts\(\)](#) UART output function used for diagnostics
[uart_debug_puthex\(\)](#) UART hex output function

NASA Power of 10 Compliance:

- Rule 1: [YES] No goto, setjmp, recursion
- Rule 2: [WARN] Infinite loop is intentional (halt behavior, safety requirement)
- Rule 3: [YES] No dynamic allocation
- Rule 4: [YES] Function is ~30 lines (compact error handler)
- Rule 5: [YES] 3 preconditions, 4 postconditions, 2 invariants
- Rule 6: [YES] Variables have minimal scope
- Rule 7: [YES] UART function calls assumed to succeed (no error checking needed)
- Rule 8: [YES] Uses typed enum for error codes
- Rule 9: [YES] No function pointers
- Rule 10: [YES] Compiles with -Wall -Wextra -Werror

Since

Version 1.0.0

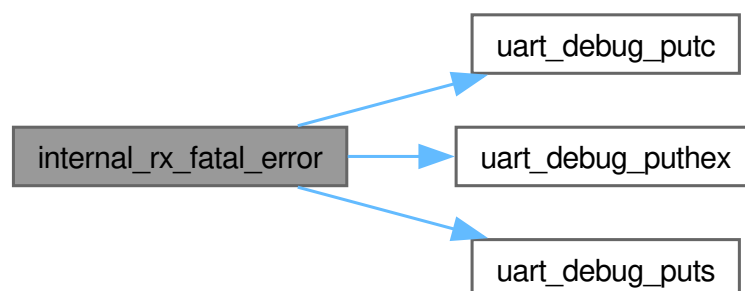
Definition at line 436 of file [rx_check.h](#).

```

00437 {
00438     uart_debug_puts("\r\n");
00439     uart_debug_puts("=====\r\n");
00440     uart_debug_puts("FATAL ERROR\r\n");
00441     uart_debug_puts("=====\r\n");
00442     uart_debug_putc('!');
00443     uart_debug_puts(tag);
00444     uart_debug_puts("] ");
00445     uart_debug_puts(message);
00446     uart_debug_puts("\r\n");
00447     uart_debug_puts("Error code: ");
00448     uart_debug_puthex((uint32_t)err, k_err_hex_digits);
00449     uart_debug_puts("\r\n");
00450     uart_debug_puts("=====\r\n");
00451     uart_debug_puts("System halted. Reset required.\r\n");
00452     uart_debug_puts("=====\r\n");
00453     /* Disable interrupts and halt */
00454     #ifndef UNIT_TEST
00455         __asm__ volatile("clrspw i");
00456         while (1) {
00457             __asm__ volatile("wait");
00458         }
00459     #endif
00460 }
00461 }
```

References [k_err_hex_digits](#), [uart_debug_putc\(\)](#), [uart_debug_puthex\(\)](#), and [uart_debug_puts\(\)](#).

Here is the call graph for this function:



7.4 rx_check.h

[Go to the documentation of this file.](#)

```

00001 /**
00002  * @file rx_check.h
00003  * @brief Validation and Error Checking Macros for RX72N Firmware
00004  *
00005  * @details
00006  * Comprehensive error checking and validation infrastructure for safety-critical
00007  * embedded systems. Provides type-safe macros for parameter validation, error
00008  * propagation, and assertion checking aligned with NASA Power of 10 Rule 5
00009  * (minimum 2 assertions per function).
00010  *
00011  * ## Architecture Design
00012  *
00013  * This module implements a layered error handling strategy:
00014  * - **Fatal errors:** System halts with diagnostic output (RX_ERROR_CHECK, RX_ASSERT)
00015  * - **Propagating errors:** Early return with logging (RX_RETURN_ON_ERROR variants)
00016  * - **Non-fatal errors:** Log and continue (RX_ERROR_CHECK_WITHOUT_ABORT)
00017  * - **Precondition validation:** Input parameter checks (RX_CHECK_NULL_PTR, RX_CHECK_RANGE)
00018  *
00019  * ## Error Handling Strategies
00020  *
00021  * | Macro | Fatal | Returns | Use Case |
00022  * |-----|-----|-----|-----|
00023  * | RX_ERROR_CHECK | [YES] | Never | Critical initialization errors |
00024  * | RX_ERROR_CHECK_WITHOUT_ABORT | [NO] | No | Optional feature failures |
00025  * | RX_RETURN_ON_ERROR | [NO] | rx_err_t | Error propagation in rx_err_t funcs |
00026  * | RX_RETURN_VOID_ON_ERROR | [NO] | void | Error handling in void functions |
00027  * | RX_RETURN_NULL_ON_ERROR | [NO] | NULL | Error handling in pointer functions |
00028  * | RX_CHECK_NULL_PTR | [NO] | rx_err_t | Null pointer precondition check |
00029  * | RX_CHECK_RANGE | [NO] | rx_err_t | Range validation |
00030  * | RX_ASSERT | [YES] | Never | Invariant and postcondition checking |
00031  *
00032  * ## Error Handling Flow
00033  *
00034  * @dot
00035  * digraph error_handling {
00036  *   rankdir=TB;
00037  *   node [shape=box, style=rounded];
00038  *
00039  *   error [label="Error Detected", fillcolor=red, style=filled, fontcolor=white];
00040  *   fatal [label="Fatal?", shape=diamond];
00041  *   log_halt [label="Log & Halt\n(RX_ERROR_CHECK)", fillcolor=red, style=filled];
00042  *   log_continue [label="Log & Continue\n(RX_ERROR_CHECK_WITHOUT_ABORT)", fillcolor=yellow, style=filled];
00043  *   log_return [label="Log & Return\n(RX_RETURN_ON_ERROR)", fillcolor=orange, style=filled];
00044  *   caller [label="Caller Handles"];
00045  *
00046  *   error -> fatal;
00047  *   fatal -> log_halt [label="Yes\n(critical)"];
00048  *   fatal -> log_continue [label="No\n(non-critical)"];
00049  *   fatal -> log_return [label="No\n(propagate)"];
00050  *   log_halt -> halt [label="infinite loop"];
00051  *   log_continue -> continue [label="execution continues"];
00052  *   log_return -> caller;
00053  *
00054  *   halt [shape=octagon, label="HALTED", fillcolor=black, fontcolor=white];
00055  *   continue [shape=ellipse, label="Continue"];
00056  * }
00057  * @enddot
00058  *
00059  * ## Design Rationale
00060  *
00061  * **Why macros instead of inline functions?**
00062  * - Capture __FILE__, __LINE__ for precise error location (future enhancement)
00063  * - Zero runtime overhead (no function call in release builds)
00064  * - Early return from enclosing function (not possible with inline functions)
00065  * - Conditional compilation based on build type (e.g., disable assertions in release)
00066  *
00067  * **Why do-while(0) wrapper?**
00068  * - Safe in all contexts (if statements without braces)
00069  * - Requires semicolon after macro invocation
00070  * - Prevents multiple expansion bugs
00071  *
00072  * **Why separate return-type variants?**
00073  * - Type safety: prevents returning error from void function
00074  * - Explicit intent: RX_RETURN_NULL_ON_ERROR clearly returns nullptr
00075  * - Compiler optimization: separate code paths for different return types
00076  *
00077  * ## Performance Characteristics
00078  *
00079  * All macros expand to minimal code:
00080  * - **Success path:** 1-2 comparisons (2-4 cycles @ 240 MHz)
00081  * - **Error path:** Logging + action (varies, but typically < 100 us for non-fatal)
00082  * - **Fatal error:** Logging + infinite loop (system halted)

```

```

00083 * - **Memory:** 0 bytes overhead (pure compile-time code generation)
00084 *
00085 * ## Thread Safety
00086 *
00087 * - **Validation macros:** Thread-safe (pure validation, no shared state)
00088 * - **Logging macros:** Thread-safe if rx_log is thread-safe
00089 * - **Fatal error:** Not thread-safe (intentional - halts entire system)
00090 *
00091 * ## Usage Patterns
00092 *
00093 * @par Pattern 1: Critical Initialization
00094 * @code{.c}
00095 * void system_init(void) {
00096 *     rx_err_t err;
00097 *
00098 *     // Critical hardware init - must succeed or halt
00099 *     err = init_clock();
00100 *     RX_ERROR_CHECK(err); // Halts on failure
00101 *
00102 *     err = init_gpio();
00103 *     RX_ERROR_CHECK(err); // Halts on failure
00104 *
00105 *     // System continues only if both succeeded
00106 * }
00107 * @endcode
00108 *
00109 * @par Pattern 2: Optional Feature Initialization
00110 * @code{.c}
00111 * void init_optional_features(void) {
00112 *     rx_err_t err;
00113 *
00114 *     // Optional feature - log error but continue
00115 *     err = init_debug_uart();
00116 *     RX_ERROR_CHECK_WITHOUT_ABORT(err); // Logs, continues
00117 *
00118 *     // System continues even if debug UART fails
00119 * }
00120 * @endcode
00121 *
00122 * @par Pattern 3: Error Propagation
00123 * @code{.c}
00124 * rx_err_t init_subsystem(void) {
00125 *     rx_err_t err;
00126 *
00127 *     // Propagate error to caller
00128 *     err = init_hardware();
00129 *     RX_RETURN_ON_ERROR(err, "SUBSYS", "Hardware init failed");
00130 *
00131 *     err = init_software();
00132 *     RX_RETURN_ON_ERROR(err, "SUBSYS", "Software init failed");
00133 *
00134 *     return k_rx_ok;
00135 * }
00136 * @endcode
00137 *
00138 * @par Pattern 4: NASA Rule 5 Compliance (Preconditions)
00139 * @code{.c}
00140 * rx_err_t process_buffer(const uint8_t* data, size_t len) {
00141 *     // Precondition 1: Validate pointer (NASA Rule 5)
00142 *     RX_CHECK_NULL_PTR(data, "PROCESS", "Data buffer is nullptr");
00143 *
00144 *     // Precondition 2: Validate size (NASA Rule 5)
00145 *     RX_CHECK_RANGE(len, 1, 256, k_rx_err_invalid_size);
00146 *
00147 *     // Function body - preconditions satisfied
00148 *     // ...
00149 *     return k_rx_ok;
00150 * }
00151 * @endcode
00152 *
00153 * @par Pattern 5: Postcondition Validation
00154 * @code{.c}
00155 * rx_err_t compute_result(float* output) {
00156 *     RX_CHECK_NULL_PTR(output, "COMPUTE", "Output pointer is nullptr");
00157 *
00158 *     // Perform computation
00159 *     *output = complex_calculation();
00160 *
00161 *     // Postcondition: result must be finite (NASA Rule 5)
00162 *     RX_ASSERT(isfinite(*output), "Result must be finite");
00163 *
00164 *     return k_rx_ok;
00165 * }
00166 * @endcode
00167 *
00168 * @par Module Dependencies:
00169 * - rx_err.h: Error code definitions (rx_err_t, k_rx_ok, etc.)

```

```

00170 * - rx_log.h: Logging infrastructure (rx_log_error, uart_debug_puts)
00171 *
00172 * @see rx_err.h Error code definitions
00173 * @see rx_log.h Logging functions used by check macros
00174 * @see rx_error_handler.h High-level error handling infrastructure
00175 * @see docs/sections/06_nasa_power_of_10.tex NASA Rule 5 compliance
00176 *
00177 * @par NASA Power of 10 Compliance:
00178 * - Rule 1: [YES] No goto, setjmp, recursion (macros expand to structured code)
00179 * - Rule 2: [YES] Infinite loop in fatal error is intentional (halt behavior)
00180 * - Rule 3: [YES] No dynamic allocation (stack-only variables)
00181 * - Rule 4: [YES] Macro expansions are small (< 10 lines)
00182 * - Rule 5: [YES] **Explicitly designed to enforce Rule 5** (provide assertion macros)
00183 * - Rule 6: [YES] Minimal scope (do-while(0) block scope)
00184 * - Rule 7: [YES] All function calls checked by design (purpose of module)
00185 * - Rule 8: [YES] No raw macros for constants (uses rx_err_t typed enums)
00186 * - Rule 9: [YES] No function pointers
00187 * - Rule 10: [YES] Compiles with -Wall -Wextra -Werror
00188 *
00189 * @par SOLID Principles:
00190 * - **S (Single Responsibility):** Only error checking and validation, no business logic
00191 * - **O (Open/Closed):** Extensible via new macros, existing macros closed for modification
00192 * - **L (Liskov Substitution):** Macros are transparent wrappers, don't change semantics
00193 * - **I (Interface Segregation):** Separate macros for different use cases (fatal, return, validate)
00194 * - **D (Dependency Inversion):** Depends on abstract rx_err_t, not concrete error values
00195 *
00196 * @author Locked, Inc.
00197 * @date 2026-01-27
00198 * @version 1.0.0
00199 * @copyright Copyright (c) 2026 Locked Inc.
00200 * SPDX-License-Identifier: MIT
00201 */
00202
00203 #pragma once
00204
00205 #include <stdbool.h>
00206 #include <stdint.h>
00207
00208 #include "rx_err.h"
00209 #include "rx_log.h"
00210
00211 /**
00212 * @def RX_STATIC_TESTABLE
00213 * @brief Makes internal (static) functions visible to unit tests.
00214 *
00215 * @details
00216 * In production builds functions decorated with this macro are declared
00217 * @c static, keeping them private to their translation unit. In UNIT_TEST
00218 * builds the static qualifier is removed so test code can call internal
00219 * helpers directly -- including with invalid arguments -- to verify that
00220 * every defensive null-check and error branch is reachable.
00221 *
00222 * Usage in a source file:
00223 * @code
00224 *   RX_STATIC_TESTABLE rx_err_t internal_decode_header(...);
00225 * @endcode
00226 *
00227 * Usage in a test file (forward-declare the function as extern):
00228 * @code
00229 *   #ifdef UNIT_TEST
00230 *     extern rx_err_t internal_decode_header(...);
00231 *   #endif
00232 * @endcode
00233 *
00234 * @note In production builds this macro expands to @c static, so there is
00235 *       zero overhead and no change in linkage or inlining behaviour.
00236 *
00237 * @since Version 1.0.0
00238 */
00239 #ifndef RX_STATIC_TESTABLE
00240 #ifdef UNIT_TEST
00241 #define RX_STATIC_TESTABLE /* non-static in test builds */
00242 #else
00243 #define RX_STATIC_TESTABLE static
00244 #endif
00245 #endif
00246
00247 #ifdef __cplusplus
00248 extern "C" {
00249 #endif
00250
00251 /*
=====
00252 * Fatal Error Handling
00253 *
=====
00254 */

```

```

00255
00256 /**
00257  * @enum fatal_error_constants_t
00258  * @brief Constants for fatal error output formatting
00259  */
00260 typedef enum : uint8_t {
00261     k_err_hex_digits = 8U, /**< Number of hex digits for a 32-bit error code */
00262 } fatal_error_constants_t;
00263
00264 /**
00265  * @brief Halt execution with diagnostic error message (fatal error handler)
00266  *
00267  * @details
00268  * Terminates firmware execution after logging detailed error diagnostics to
00269  * debug UART. This is a non-returning function used for unrecoverable errors
00270  * where continuing execution would be unsafe.
00271  *
00272  * ## Algorithm
00273  *
00274  * 1. Output formatted error banner to debug UART
00275  * 2. Display component tag, error message, and error code (hex)
00276  * 3. Indicate system halted state
00277  * 4. Disable all interrupts (global interrupt disable)
00278  * 5. Enter infinite loop with CPU in low-power WAIT state
00279  *
00280  * ## Output Format
00281  *
00282  * ```
00283  * =====
00284  * FATAL ERROR
00285  * =====
00286  * [TAG] Error message here
00287  * Error code: 0x00000XXX
00288  * =====
00289  * System halted. Reset required.
00290  * =====
00291  * ```
00292  *
00293  * ## Platform-Specific Behavior
00294  *
00295  * **Normal Build (Hardware Target):**
00296  * - Disables interrupts via `clrpsw i` (clear PSW I-flag)
00297  * - Enters infinite loop with `wait` instruction
00298  * - CPU remains in low-power wait state until hardware reset
00299  * - Watchdog will eventually trigger reset if enabled
00300  *
00301  * **Unit Test Build (UNIT_TEST defined):**
00302  * - Returns immediately without halting
00303  * - Allows test harness to continue and detect fatal error
00304  * - Enables testing of error paths without hanging test runner
00305  *
00306  * @param[in] tag Component tag for error identification
00307  * - Type: const char* (null-terminated string)
00308  * - Must not be nullptr (undefined behavior if nullptr)
00309  * - Typical values: "MOTOR", "COMM", "INIT", "ASSERT", etc.
00310  * - Used to identify which subsystem detected the error
00311  * - Recommended length: 3-8 characters for readability
00312  *
00313  * @param[in] message Human-readable error description
00314  * - Type: const char* (null-terminated string)
00315  * - Must not be nullptr (undefined behavior if nullptr)
00316  * - Should describe what failed and why
00317  * - Example: "GPIO initialization failed", "Invalid motor ID"
00318  * - Maximum practical length: ~80 characters (UART line width)
00319  *
00320  * @param[in] err Error code indicating failure type
00321  * - Type: rx_err_t (int32_t)
00322  * - Typically non-zero (zero indicates success, illogical for fatal error)
00323  * - Displayed in hexadecimal (8 digits, zero-padded)
00324  * - Used for post-mortem debugging via UART log capture
00325  *
00326  * @note **This function never returns** (except in UNIT_TEST builds)
00327  *
00328  * @par Usage Example - From RX_ERROR_CHECK:
00329  * @code{.c}
00330  * rx_err_t err = critical_init_function();
00331  * if (err != k_rx_ok) {
00332  *     internal_rx_fatal_error("INIT", "Critical init failed", err);
00333  *     // Never reached in normal builds
00334  * }
00335  * @endcode
00336  *
00337  * @par Usage Example - From RX_ASSERT:
00338  * @code{.c}
00339  * if (!(ptr != nullptr)) {
00340  *     internal_rx_fatal_error("ASSERT", "Pointer must not be NULL", k_rx_fail);
00341  *     // Never reached in normal builds

```

```

00342 * }
00343 * @endcode
00344 *
00345 * @par Usage Example - Direct Call (Rare):
00346 * @code{.c}
00347 * // Only use directly if macro wrappers don't fit use case
00348 * if (unrecoverable_condition()) {
00349 *     internal_rx_fatal_error("SAFETY", "Unrecoverable hardware fault", k_rx_err_hw_error);
00350 * }
00351 * @endcode
00352 *
00353 * @par Assembly Instruction Details:
00354 *
00355 * **clrpsw i** - Clear Processor Status Word I-flag
00356 * - Disables all maskable interrupts globally
00357 * - Non-maskable interrupts (NMI) still active
00358 * - Prevents interrupt handlers from interfering with error state
00359 * - Ensures system remains in known halted state
00360 *
00361 * **wait** - Wait for Interrupt
00362 * - Places CPU in low-power state
00363 * - CPU wakes on interrupt, but interrupts disabled -> immediately waits again
00364 * - Reduces power consumption during halt state
00365 * - Effective infinite loop with minimal power draw
00366 *
00367 * @par Watchdog Behavior:
00368 *
00369 * If watchdog (IWDT) is enabled:
00370 * - Watchdog will expire after timeout period (typically 128ms-2s)
00371 * - System will reset automatically
00372 * - Error message captured before reset (if UART log buffered)
00373 * - Allows recovery from fatal errors in deployed systems
00374 *
00375 * If watchdog is disabled:
00376 * - System remains halted indefinitely
00377 * - Requires manual hardware reset or power cycle
00378 * - Useful during development for post-mortem debugging
00379 *
00380 * @par Performance:
00381 * - Execution time: ~500 us (UART output dominates)
00382 * - Stack usage: ~16 bytes (local variables, return address)
00383 * - UART bandwidth: ~100 bytes @ 115200 baud = ~9ms transmission time
00384 * - **Not performance-critical:** Only called on fatal errors
00385 *
00386 * @note Thread Safety: Not thread-safe, but irrelevant (system halting)
00387 * @note Re-entrancy: Not reentrant (disables interrupts, infinite loop)
00388 * @note Memory: No dynamic allocation, stack-only locals
00389 *
00390 * @warning **This function never returns in normal builds** - Calling code after
00391 *     this function is unreachable. Compiler may optimize it away.
00392 *
00393 * @warning **Do not call from interrupt context** - Disabling interrupts in ISR
00394 *     can cause undefined behavior. Use error flags instead.
00395 *
00396 * @attention **System is permanently halted** - Only recovery is hardware reset.
00397 *     Ensure watchdog is enabled in production builds for automatic recovery.
00398 *
00399 * @attention **UART output is blocking** - If UART is not initialized or not
00400 *     connected, uart_debug_puts() may hang. Ensure UART is functional.
00401 *
00402 * @pre tag != nullptr (undefined behavior if nullptr, will likely crash)
00403 * @pre message != nullptr (undefined behavior if nullptr, will likely crash)
00404 * @pre UART debug interface initialized (or uart_debug_puts() will hang)
00405 *
00406 * @post Interrupts disabled globally (PSW.I = 0)
00407 * @post CPU in infinite wait loop (never returns)
00408 * @post Error diagnostics sent to debug UART
00409 * @post System requires hardware reset to recover
00410 *
00411 * @invariant In UNIT_TEST builds, function returns immediately (allows testing)
00412 * @invariant In normal builds, function never returns (infinite loop)
00413 *
00414 * @see RX_ERROR_CHECK() Macro that calls this function
00415 * @see RX_ASSERT() Macro that calls this function
00416 * @see uart_debug_puts() UART output function used for diagnostics
00417 * @see uart_debug_puthex() UART hex output function
00418 *
00419 * @par NASA Power of 10 Compliance:
00420 * - Rule 1: [YES] No goto, setjmp, recursion
00421 * - Rule 2: [WARN] Infinite loop is intentional (halt behavior, safety requirement)
00422 * - Rule 3: [YES] No dynamic allocation
00423 * - Rule 4: [YES] Function is ~30 lines (compact error handler)
00424 * - Rule 5: [YES] 3 preconditions, 4 postconditions, 2 invariants
00425 * - Rule 6: [YES] Variables have minimal scope
00426 * - Rule 7: [YES] UART function calls assumed to succeed (no error checking needed)
00427 * - Rule 8: [YES] Uses typed enum for error codes
00428 * - Rule 9: [YES] No function pointers

```

```

00429 * - Rule 10: [YES] Compiles with -Wall -Wextra -Werror
00430 *
00431 * @since Version 1.0.0
00432 */
00433 #ifndef UNIT_TEST
00434 [[noreturn]]
00435 #endif
00436 static inline void internal_rx_fatal_error(const char* tag, const char* message, rx_err_t err)
00437 {
00438     uart_debug_puts("\r\n");
00439     uart_debug_puts("=====\r\n");
00440     uart_debug_puts("FATAL ERROR\r\n");
00441     uart_debug_puts("=====\r\n");
00442     uart_debug_putc('[');
00443     uart_debug_puts(tag);
00444     uart_debug_puts("] ");
00445     uart_debug_puts(message);
00446     uart_debug_puts("\r\n");
00447     uart_debug_puts("Error code: ");
00448     uart_debug_puthex((uint32_t)err, k_err_hex_digits);
00449     uart_debug_puts("\r\n");
00450     uart_debug_puts("=====\r\n");
00451     uart_debug_puts("System halted. Reset required.\r\n");
00452     uart_debug_puts("=====\r\n");
00453
00454     /* Disable interrupts and halt */
00455     #ifndef UNIT_TEST
00456         __asm__ volatile("clrspw i");
00457         while (1) {
00458             __asm__ volatile("wait");
00459         }
00460     #endif
00461 }
00462
00463 /*
=====
00464 * Assertion Macros
00465 *
=====
00466 */
00467
00468 /**
00469 * @def RX_ASSERT
00470 * @brief Runtime assertion with fatal error on failure (NASA Rule 5 compliance)
00471 *
00472 * @details
00473 * Verifies a boolean condition at runtime and halts execution if the condition
00474 * is false. This macro is the primary mechanism for enforcing NASA Power of 10
00475 * Rule 5 (minimum 2 assertions per function) through precondition, postcondition,
00476 * and invariant checking.
00477 *
00478 * ## Algorithm
00479 *
00480 * 1. Evaluate condition expression
00481 * 2. If condition is true: Continue execution (no overhead)
00482 * 3. If condition is false:
00483 *     a. Call internal_rx_fatal_error() with "ASSERT" tag
00484 *     b. Log error message to debug UART
00485 *     c. Halt system (infinite loop with interrupts disabled)
00486 *
00487 * ## Use Cases
00488 *
00489 * | Use Case | Example |
00490 * |-----|-----|
00491 * | Precondition (input) | `RX_ASSERT(ptr != nullptr, "Input must be non-NULL")`|
00492 * | Precondition (range) | `RX_ASSERT(value <= MAX, "Value exceeds maximum")`|
00493 * | Postcondition (output) | `RX_ASSERT(result >= 0, "Result must be non-negative")`|
00494 * | Invariant (state) | `RX_ASSERT(count < MAX_COUNT, "Count overflow")`|
00495 * | Loop bound verification | `RX_ASSERT(i < ARRAY_SIZE, "Loop index overflow")`|
00496 *
00497 * ## NASA Rule 5 Compliance Pattern
00498 *
00499 * Every function should have minimum 2 assertions:
00500 * @code{.c}
00501 rx_err_t calculate(float* input, float* output) {
00502     // Precondition 1: Input pointer valid (NASA Rule 5)
00503     RX_ASSERT(input != nullptr, "Input pointer must not be NULL");
00504
00505     // Precondition 2: Output pointer valid (NASA Rule 5)
00506     RX_ASSERT(output != nullptr, "Output pointer must not be NULL");
00507
00508     // Perform calculation
00509     *output = *input * 2.0f;
00510
00511     // Postcondition: Result is finite (NASA Rule 5)
00512     RX_ASSERT(isfinite(*output), "Output must be finite");
00513 }

```



```

00514 *   return k_rx_ok;
00515 * }
00516 * @endcode
00517 *
00518 * @param condition Boolean expression to verify
00519 * - Type: Any expression convertible to bool
00520 * - Must evaluate to true for normal operation
00521 * - Examples: `ptr != nullptr`, `value < MAX`, `state == READY`
00522 * - Evaluated exactly once (no multiple evaluation bug)
00523 * - Should not have side effects (evaluation may be removed in release builds)
00524 *
00525 * @param message Error message displayed on assertion failure
00526 * - Type: const char* (string literal)
00527 * - Must be compile-time constant string
00528 * - Should describe what was expected, not what failed
00529 * - Good: "Pointer must not be NULL", "Value must be in range [0, 100]"
00530 * - Bad: "ptr is nullptr", "Invalid value"
00531 * - Recommended length: < 60 characters for readability
00532 *
00533 * @par Usage Example - Precondition Checking:
00534 * @code{.c}
00535 * rx_err_t motor_set_velocity(motor_handle_t* motor, float velocity_mps) {
00536 *     // Precondition 1: Handle valid
00537 *     RX_ASSERT(motor != nullptr, "Motor handle must not be NULL");
00538 *
00539 *     // Precondition 2: Velocity within physical limits
00540 *     RX_ASSERT(velocity_mps >= -2.5f && velocity_mps <= 2.5f,
00541 *         "Velocity must be in range [-2.5, 2.5] m/s");
00542 *
00543 *     // Precondition 3: Motor initialized
00544 *     RX_ASSERT(motor->initialized, "Motor must be initialized");
00545 *
00546 *     // Function body - all preconditions satisfied
00547 *     motor->target_velocity = velocity_mps;
00548 *     return k_rx_ok;
00549 * }
00550 * @endcode
00551 *
00552 * @par Usage Example - Postcondition Checking:
00553 * @code{.c}
00554 * rx_err_t sensor_read(sensor_t* sensor, float* temperature_c) {
00555 *     RX_ASSERT(sensor != nullptr, "Sensor handle must not be NULL");
00556 *     RX_ASSERT(temperature_c != nullptr, "Output pointer must not be NULL");
00557 *
00558 *     // Read sensor via I2C
00559 *     uint16_t raw_value = read_i2c_register(sensor->address, TEMP_REG);
00560 *     *temperature_c = (raw_value * 0.0625f) - 40.0f; // DS18B20 conversion
00561 *
00562 *     // Postcondition: Temperature in physically possible range
00563 *     RX_ASSERT(*temperature_c >= -55.0f && *temperature_c <= 125.0f,
00564 *         "Temperature must be in valid sensor range");
00565 *
00566 *     return k_rx_ok;
00567 * }
00568 * @endcode
00569 *
00570 * @par Usage Example - Invariant Checking:
00571 * @code{.c}
00572 * typedef struct {
00573 *     uint8_t head;
00574 *     uint8_t tail;
00575 *     uint8_t data[64];
00576 * } ring_buffer_t;
00577 *
00578 * void ring_buffer_push(ring_buffer_t* rb, uint8_t value) {
00579 *     RX_ASSERT(rb != nullptr, "Ring buffer must not be NULL");
00580 *
00581 *     // Invariant: head and tail always within bounds
00582 *     RX_ASSERT(rb->head < 64, "Head index must be < 64");
00583 *     RX_ASSERT(rb->tail < 64, "Tail index must be < 64");
00584 *
00585 *     rb->data[rb->head] = value;
00586 *     rb->head = (rb->head + 1) % 64;
00587 *
00588 *     // Invariant maintained after modification
00589 *     RX_ASSERT(rb->head < 64, "Head index must remain < 64");
00590 * }
00591 * @endcode
00592 *
00593 * @par Usage Example - Loop Bound Verification (NASA Rule 2):
00594 * @code{.c}
00595 * typedef enum : uint8_t {
00596 *     k_max_iterations = 100 // Explicit loop bound (NASA Rule 2)
00597 * } loop_bounds_t;
00598 *
00599 * void process_array(uint8_t* data, size_t len) {
00600 *     RX_ASSERT(data != nullptr, "Data array must not be NULL");

```



```

00601 *   RX_ASSERT(len <= k_max_iterations, "Array too large for fixed loop bound");
00602 *
00603 *   for (uint8_t i = 0; i < k_max_iterations && i < len; i++) {
00604 *       // Invariant: i is always within array bounds
00605 *       RX_ASSERT(i < len, "Loop index must be < array length");
00606 *       data[i] = process_byte(data[i]);
00607 *   }
00608 * }
00609 * @endcode
00610 *
00611 * @par Conditional Compilation:
00612 *
00613 * Assertions can be disabled in release builds for performance:
00614 * @code{.c}
00615 * #ifdef NDEBUG
00616 * #define RX_ASSERT(condition, message) ((void)0) // No-op in release
00617 * #else
00618 * #define RX_ASSERT(condition, message) do { ... } while (0) // As shown above
00619 * #endif
00620 * @endcode
00621 *
00622 * **WARNING:** Disabling assertions removes safety checks. Only disable after
00623 * extensive testing with assertions enabled.
00624 *
00625 * @par Performance:
00626 * - Success path: 1 comparison + 1 branch (2-3 cycles @ 240 MHz)
00627 * - Failure path: Logs and halts (system terminates, performance irrelevant)
00628 * - Stack usage: 0 bytes (no additional stack beyond function call if failure)
00629 * - Code size: ~8 bytes per assertion in release builds with optimizations
00630 *
00631 * @note Thread Safety: Condition check is atomic (single comparison)
00632 * @note Re-entrancy: Re-entrant (until failure, then system halts)
00633 * @note Memory: No dynamic allocation
00634 *
00635 * @warning **Assertions halt the system** - Use for programming errors and
00636 * invariant violations, NOT for recoverable runtime errors.
00637 *
00638 * @warning **Condition must not have side effects** - If assertions are disabled,
00639 * side effects disappear. Bad: `RX_ASSERT(ptr++ != nullptr, "...")` Good: `ptr++; RX_ASSERT(ptr != nullptr,
00640 * "...)`
00641 *
00642 * @attention **Use assertions for programming errors** (bugs in code) and
00643 * return error codes for runtime errors (invalid user input, hardware faults).
00644 *
00645 * @par Assertions vs. Error Returns:
00646 *
00647 * | Condition Type | Use Assertion | Use Error Return |
00648 * |-----|-----|-----|
00649 * | Null pointer from caller | YES | NO |
00650 * | Value out of documented range | YES | NO |
00651 * | Invalid state transition | YES | NO |
00652 * | Hardware fault | NO | YES |
00653 * | Communication timeout | NO | YES |
00654 * | Sensor out of range | NO | YES |
00655 *
00656 * @pre condition is a valid boolean expression
00657 * @pre message is a non-NULL string literal (compile-time constant)
00658 *
00659 * @post If condition is true: Execution continues normally
00660 * @post If condition is false: System halts permanently (requires reset)
00661 *
00662 * @invariant If condition is false, function never returns
00663 * @invariant Condition evaluated exactly once (no multiple evaluation)
00664 *
00665 * @see internal_rx_fatal_error() Function called on assertion failure
00666 * @see RX_ERROR_CHECK() Related macro for error code checking
00667 * @see RX_CHECK_NULL_PTR() Alternative for null pointer checking with error return
00668 *
00669 * @par NASA Power of 10 Compliance:
00670 * - Rule 1: [YES] No goto, setjmp, recursion (structured if statement)
00671 * - Rule 2: [YES] N/A (no loops)
00672 * - Rule 3: [YES] No dynamic allocation
00673 * - Rule 4: [YES] Macro expansion is 3 lines
00674 * - Rule 5: [YES] **This macro implements Rule 5** (provides assertion mechanism)
00675 * - Rule 6: [YES] do-while(0) provides minimal scope
00676 * - Rule 7: [YES] N/A (macro itself, not function)
00677 * - Rule 8: [YES] Uses typed enum k_rx_fail
00678 * - Rule 9: [YES] No function pointers
00679 * - Rule 10: [YES] Compiles with -Wall -Wextra -Werror
00680 *
00681 * @since Version 1.0.0
00682 *
00683 * #define RX_ASSERT(condition, message) \
00684 * do { \
00685 *     if (!(condition)) { \
00686 *         internal_rx_fatal_error("ASSERT", message, k_rx_fail); \
00687 *     } \

```

```

00687 } while (0)
00688
00689 /**
00690  * @def RX_ASSERT_PRE
00691  * @brief Precondition assertion -- verifies caller contract before function body
00692  *
00693  * @details
00694  * Halts the system if a precondition is violated. Use for validating state and
00695  * parameters at the top of internal functions where the condition is
00696  * architecturally guaranteed by callers, but should still be checked defensively.
00697  *
00698  * ## Testability (UNIT_TEST builds)
00699  *
00700  * Because internal functions are decorated with @c RX_STATIC_TESTABLE, test
00701  * code can call them directly with invalid state or parameters. This makes
00702  * the false branch of every RX_ASSERT_PRE reachable, enabling 100% branch
00703  * coverage without production-code gymnastics.
00704  *
00705  * @param condition Boolean expression that must be true
00706  * @param message Descriptive string for fatal error output
00707  *
00708  * @pre condition is a valid boolean expression
00709  * @pre message is a non-NULL string literal
00710  * @post If condition is true: execution continues
00711  * @post If condition is false: system halts (returns in UNIT_TEST builds)
00712  *
00713  * @see RX_ASSERT_POST() Postcondition counterpart
00714  * @see RX_STATIC_TESTABLE Exposes internal functions for test-driven coverage
00715  *
00716  * @since Version 1.0.0
00717  */
00718 #define RX_ASSERT_PRE(condition, message)
00719 do {
00720     if (!(condition)) {
00721         internal_rx_fatal_error("PRE", (message), k_rx_fail);
00722     }
00723 } while (0)
00724
00725 /**
00726  * @def RX_ASSERT_POST
00727  * @brief Postcondition assertion -- verifies function output or invariant after work
00728  *
00729  * @details
00730  * Halts the system if a postcondition is violated. Use after a computation to
00731  * verify that the result satisfies the documented contract.
00732  *
00733  * ## Testability (UNIT_TEST builds)
00734  *
00735  * Postconditions are often architecturally guaranteed (e.g., a 20-bit mask
00736  * always produces a value in [0, 0xFFFF]). Their false branches are
00737  * unreachable without fault injection. In UNIT_TEST builds the macro checks
00738  * @c g_rx_assert_post_force_fail first: when that flag is @c true the macro
00739  * fires regardless of the actual condition, making the failure branch
00740  * reachable for coverage.
00741  *
00742  * @param condition Boolean expression that must be true
00743  * @param message Descriptive string for fatal error output
00744  *
00745  * @pre condition is a valid boolean expression
00746  * @pre message is a non-NULL string literal
00747  * @post If condition is true (and force-fail off): execution continues
00748  * @post If condition is false or force-fail on: system halts (returns in UNIT_TEST)
00749  *
00750  * @see RX_ASSERT_PRE() Precondition counterpart
00751  * @see g_rx_assert_post_force_fail Test-only fault injection flag
00752  *
00753  * @since Version 1.0.0
00754  */
00755 #ifdef UNIT_TEST
00756 extern bool g_rx_assert_post_force_fail;
00757 #define RX_ASSERT_POST(condition, message)
00758 do {
00759     if (g_rx_assert_post_force_fail || !(condition)) {
00760         internal_rx_fatal_error("POST", (message), k_rx_fail);
00761     }
00762 } while (0)
00763 #else
00764 #define RX_ASSERT_POST(condition, message)
00765 do {
00766     if (!(condition)) {
00767         internal_rx_fatal_error("POST", (message), k_rx_fail);
00768     }
00769 } while (0)
00770 #endif
00771
00772 /*
=====

```

```

00773 * Error Checking Macros
00774 *
=====
00775 */
00776
00777 /**
00778 * @def RX_ERROR_CHECK
00779 * @brief Fatal error checking - halt system if error code indicates failure
00780 *
00781 * @details
00782 * Checks an rx_err_t error code and halts the system with diagnostics if the
00783 * error is not k_rx_ok. Use this macro for critical operations where failure
00784 * is unrecoverable and continuing execution would be unsafe.
00785 *
00786 * ## When to Use
00787 *
00788 * - **Critical initialization:** Hardware init, clock config, essential peripherals
00789 * - **Safety-critical resources:** Emergency stop system, watchdog, fault detection
00790 * - **Unrecoverable errors:** Memory corruption, invalid firmware state
00791 *
00792 * ## When NOT to Use
00793 *
00794 * - **Optional features:** Use RX_ERROR_CHECK_WITHOUT_ABORT instead
00795 * - **Recoverable errors:** Use RX_RETURN_ON_ERROR to propagate to caller
00796 * - **Expected failures:** Handle explicitly with if/switch statements
00797 *
00798 * @param err Error code expression to check
00799 *   - Type: Any expression that evaluates to rx_err_t
00800 *   - Evaluated exactly once (safe for expressions with side effects)
00801 *   - If k_rx_ok (0): Continues execution silently
00802 *   - If any other value: Halts system with diagnostics
00803 *
00804 * @par Usage Example - Critical Initialization:
00805 * @code{.c}
00806 * void system_boot(void) {
00807 *     rx_err_t err;
00808 *
00809 *     // Clock init is critical - must succeed
00810 *     err = init_system_clock();
00811 *     RX_ERROR_CHECK(err); // Halts on failure
00812 *
00813 *     // GPIO init is critical - must succeed
00814 *     err = init_gpio();
00815 *     RX_ERROR_CHECK(err); // Halts on failure
00816 *
00817 *     // Only reached if both succeeded
00818 *     uart_debug_puts("System boot successful\r\n");
00819 * }
00820 * @endcode
00821 *
00822 * @par Usage Example - Safety-Critical Resource:
00823 * @code{.c}
00824 * void init_safety_systems(void) {
00825 *     rx_err_t err;
00826 *
00827 *     // Emergency stop system must work
00828 *     err = init_emergency_stop();
00829 *     RX_ERROR_CHECK(err); // Safety-critical, cannot proceed if failed
00830 *
00831 *     // Watchdog must be enabled
00832 *     err = init_independent_watchdog();
00833 *     RX_ERROR_CHECK(err); // System unusable without watchdog
00834 * }
00835 * @endcode
00836 *
00837 * @par Comparison to Alternatives:
00838 *
00839 * | Macro | Halts | Logs | Returns | Use Case |
00840 * |-----|-----|-----|-----|-----|
00841 * | RX_ERROR_CHECK | [YES] | [YES] | Never | Critical init |
00842 * | RX_ERROR_CHECK_WITHOUT_ABORT | [NO] | [YES] | No | Optional features |
00843 * | RX_RETURN_ON_ERROR | [NO] | [YES] | Error | Propagate to caller |
00844 * | if (err != k_rx_ok) | [NO] | [NO] | Varies | Custom handling |
00845 *
00846 * @warning **System halts permanently** - Use only for truly unrecoverable errors
00847 * @warning **Cannot be caught or handled** - No try/catch, no error recovery
00848 *
00849 * @attention **Watchdog behavior:** If enabled, watchdog will eventually reset
00850 * system. If disabled, requires manual reset/power cycle.
00851 *
00852 * @see RX_ERROR_CHECK_WITHOUT_ABORT() Non-fatal version (logs, continues)
00853 * @see RX_RETURN_ON_ERROR() Error propagation version (returns to caller)
00854 * @see internal_rx_fatal_error() Underlying fatal error handler
00855 *
00856 * @since Version 1.0.0
00857 */
00858 #define RX_ERROR_CHECK(err) \

```

```

00859 do {
00860     rx_err_t err_rc_ = (err);
00861     if (rx_err_is_error(err_rc_)) {
00862         internal_rx_fatal_error("ERROR_CHECK", "Fatal error detected", err_rc_);
00863     }
00864 } while (0)
00865
00866 /**
00867  * @brief Check error code and log on failure (non-fatal)
00868  *
00869  * @param[in] err Error code to check
00870  *
00871  * If err is not k_rx_ok, logs the error but continues execution.
00872  * Use this for non-critical errors where system can continue.
00873  *
00874  * Example:
00875  * rx_err_t err = optional_feature_init();
00876  * RX_ERROR_CHECK_WITHOUT_ABORT(err); // Logs but continues
00877  */
00878 #define RX_ERROR_CHECK_WITHOUT_ABORT(err)
00879 do {
00880     rx_err_t err_rc_ = (err);
00881     if (rx_err_is_error(err_rc_)) {
00882         rx_log_error_val("ERROR_CHECK", "Error", err_rc_);
00883     }
00884 } while (0)
00885
00886 /**
00887  * @brief Return early on error
00888  *
00889  * @param[in] err Error code to check
00890  * @param[in] tag Component tag for logging
00891  * @param[in] message Error message for logging
00892  *
00893  * If err is not k_rx_ok, logs the error and returns err from the function.
00894  * Use this for functions that return rx_err_t.
00895  *
00896  * Example:
00897  * rx_err_t init_subsystem(void) {
00898  *     rx_err_t err = gpio_init();
00899  *     RX_RETURN_ON_ERROR(err, "SUBSYS", "GPIO init failed");
00900  *     // Only reached if gpio_init succeeded
00901  *     return k_rx_ok;
00902  * }
00903  */
00904 #define RX_RETURN_ON_ERROR(err, tag, message)
00905 do {
00906     rx_err_t err_rc_ = (err);
00907     if (rx_err_is_error(err_rc_)) {
00908         rx_log_error(tag, message);
00909         rx_log_error_val(tag, "Error", err_rc_);
00910         return err_rc_;
00911     }
00912 } while (0)
00913
00914 /**
00915  * @brief Return void on error (for void functions)
00916  *
00917  * @param[in] err Error code to check
00918  * @param[in] tag Component tag for logging
00919  * @param[in] message Error message for logging
00920  *
00921  * If err is not k_rx_ok, logs the error and returns from the function.
00922  * Use this for void functions that cannot return an error code.
00923  *
00924  * Example:
00925  * void configure_system(void) {
00926  *     rx_err_t err = gpio_init();
00927  *     RX_RETURN_VOID_ON_ERROR(err, "SYSTEM", "GPIO init failed");
00928  *     // Only reached if gpio_init succeeded
00929  * }
00930  */
00931 #define RX_RETURN_VOID_ON_ERROR(err, tag, message)
00932 do {
00933     rx_err_t err_rc_ = (err);
00934     if (rx_err_is_error(err_rc_)) {
00935         rx_log_error(tag, message);
00936         rx_log_error_val(tag, "Error", err_rc_);
00937         return;
00938     }
00939 } while (0)
00940
00941 /**
00942  * @brief Return NULL on error (for pointer-returning functions)
00943  *
00944  * @param[in] err Error code to check
00945  * @param[in] tag Component tag for logging

```

```

00946 * @param[in] message Error message for logging
00947 *
00948 * If err is not k_rx_ok, logs the error and returns NULL from the function.
00949 * Use this for functions that return pointers.
00950 *
00951 * Example:
00952 * void* create_object(void) {
00953 *     rx_err_t err = validate_preconditions();
00954 *     RX_RETURN_NULL_ON_ERROR(err, "FACTORY", "Precondition check failed");
00955 *     // Only reached if validation succeeded
00956 *     return allocate_object();
00957 * }
00958 */
00959 #define RX_RETURN_NULL_ON_ERROR(err, tag, message)
00960 do {
00961     rx_err_t err_rc_ = (err);
00962     if (rx_err_is_error(err_rc_)) {
00963         rx_log_error(tag, message);
00964         rx_log_error_val(tag, "Error", err_rc_);
00965         return nullptr;
00966     }
00967 } while (0)
00968
00969 /*
=====
00970 * Null Pointer Checking
00971 *
=====
00972 */
00973
00974 /**
00975 * @brief Check for null pointer and return error
00976 *
00977 * @param[in] ptr Pointer to check
00978 * @param[in] tag Component tag for logging
00979 * @param[in] message Error message for logging
00980 *
00981 * If ptr is nullptr, logs the error and returns k_rx_err_null_ptr.
00982 *
00983 * Example:
00984 * rx_err_t process_data(const uint8_t* data) {
00985 *     RX_CHECK_NULL_PTR(data, "PROCESS", "Data pointer is nullptr");
00986 *     // Only reached if data != nullptr
00987 *     return k_rx_ok;
00988 * }
00989 */
00990 #define RX_CHECK_NULL_PTR(ptr, tag, message)
00991 do {
00992     if ((ptr) == nullptr) {
00993         rx_log_error(tag, message);
00994         return k_rx_err_null_ptr;
00995     }
00996 } while (0)
00997
00998 /**
00999 * @brief Check that a value is within an inclusive range
01000 *
01001 * @param[in] value Value to check
01002 * @param[in] min Minimum allowed value
01003 * @param[in] max Maximum allowed value
01004 * @param[in] errcode Error to return on failure
01005 */
01006 #define RX_CHECK_RANGE(value, min, max, errcode)
01007 do {
01008     if (((value) < (min)) || ((value) > (max))) {
01009         rx_log_error("CHECK", "Range check failed");
01010         return (errcode);
01011     }
01012 } while (0)
01013
01014 /**
01015 * @brief Check that a value is within an inclusive range with tag
01016 *
01017 * @note Only checks upper bound to avoid -Wtype-limits warnings when min == 0 and value is
01018 *       unsigned (comparison is always false). For non-zero min bounds, add an explicit
01019 *       lower-bound check before this macro. The min parameter is documented but not checked.
01020 *
01021 * @param[in] value Value to check
01022 * @param[in] min Minimum allowed value (documented only; add explicit check if min > 0)
01023 * @param[in] max Maximum allowed value
01024 * @param[in] errcode Error to return on failure
01025 * @param[in] tag Component tag for logging
01026 */
01027 #define RX_CHECK_RANGE_TAG(value, min, max, errcode, tag)
01028 do {
01029     (void)(min); /* Document minimum bound; explicit check needed if min > 0 */
01030     if ((value) > (max)) {

```

```

01031     rx_log_error(tag, "Range check failed");
01032     return (errcode);
01033 }
01034 } while (0)
01035
01036 /**
01037  * @brief Validate pointer pre-condition and return error
01038  *
01039  * Alias for RX_CHECK_NULL_PTR to standardize pre-condition naming.
01040  */
01041 #define RX_VALIDATE_PTR(ptr, tag, message) RX_CHECK_NULL_PTR(ptr, tag, message)
01042
01043 /**
01044  * @brief Validate initialization state pre-condition
01045  *
01046  * Returns k_rx_err_invalid_state on failure.
01047  */
01048 #define RX_VALIDATE_INIT(initialized, tag, message)
01049 do {
01050     if (!(initialized)) {
01051         rx_log_error(tag, message);
01052         return k_rx_err_invalid_state;
01053     }
01054 } while (0)
01055
01056 #ifdef __cplusplus
01057 }
01058 #endif

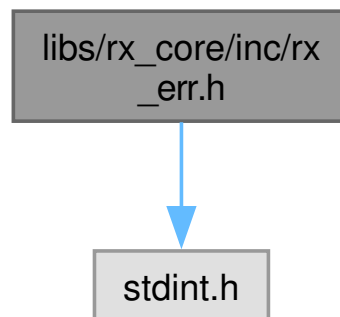
```

7.5 libs/rx_core/inc/rx_err.h File Reference

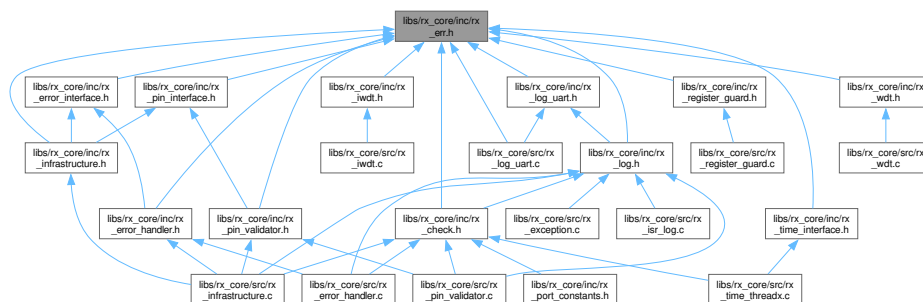
Error Code Definitions for RX72N Firmware.

```
#include <stdint.h>
```

Include dependency graph for rx_err.h:



This graph shows which files directly or indirectly include this file:



Typedefs

- typedef int32_t [rx_err_t](#)
Error code type for RX72N firmware.

Enumerations

- enum [rx_err_codes_t](#) : uint16_t {
[k_rx_ok](#) = 0 , [k_rx_fail](#) = 0x101 , [k_rx_err_no_mem](#) = 0x102 , [k_rx_err_invalid_arg](#) = 0x103 ,
[k_rx_err_invalid_state](#) = 0x104 , [k_rx_err_invalid_size](#) = 0x105 , [k_rx_err_not_found](#) = 0x106 , [k_rx_err_not_supported](#) = 0x107 ,
[k_rx_err_timeout](#) = 0x108 , [k_rx_err_busy](#) = 0x109 , [k_rx_err_no_data](#) = 0x10A ,
[k_rx_err_would_block](#) = 0x10B ,
[k_rx_err_exists](#) = 0x10C , [k_rx_err_empty](#) = 0x10D , [k_rx_err_cancelled](#) = 0x10E ,
[k_rx_err_not_initialized](#) = 0x10F ,
[k_rx_err_estop](#) = 0x110 , [k_rx_err_hw_init_failed](#) = 0x201 , [k_rx_err_hw_not_ready](#) = 0x202 , [k_rx_err_hw_timeout](#) = 0x203 ,
[k_rx_err_hw_error](#) = 0x204 , [k_rx_err_gpio_conflict](#) = 0x205 , [k_rx_err_gpio_invalid_port](#) = 0x206 , [k_rx_err_gpio_invalid_pin](#) = 0x207 ,
[k_rx_err_out_of_range](#) = 0x208 , [k_rx_err_hw_unmapped](#) = 0x209 , [k_rx_err_rtos_error](#) = 0x301 , [k_rx_err_threadx](#) = 0x301 ,
[k_rx_err_rtos_thread_create](#) = 0x302 , [k_rx_err_rtos_semaphore](#) = 0x303 , [k_rx_err_rtos_mutex](#) = 0x304 , [k_rx_err_rtos_queue](#) = 0x305 ,
[k_rx_err_rtos_timer](#) = 0x306 , [k_rx_err_comm_error](#) = 0x401 , [k_rx_err_spi_error](#) = 0x402 ,
[k_rx_err_uart_error](#) = 0x403 ,
[k_rx_err_i2c_error](#) = 0x404 , [k_rx_err_crc_mismatch](#) = 0x405 , [k_rx_err_protocol_error](#) = 0x406 , [k_rx_err_nack](#) = 0x407 ,
[k_rx_err_conflict](#) = 0x408 , [k_rx_err_retry_limit](#) = 0x409 , [k_rx_err_validation_failed](#) = 0x501 , [k_rx_err_checksum_mismatch](#) = 0x502 ,
[k_rx_err_range_check_failed](#) = 0x503 , [k_rx_err_null_ptr](#) = 0x504 }
Comprehensive error code enumeration for RX72N firmware.

Functions

- static [rx_err_t rx_err_from_threadx](#) (uint32_t tx_status)
Convert ThreadX status code to [rx_err_t](#).
- static bool [rx_err_is_ok](#) ([rx_err_t](#) err)
Check if error code represents success.
- static bool [rx_err_is_error](#) ([rx_err_t](#) err)
Check if error code represents failure.

7.5.1 Detailed Description

Error Code Definitions for RX72N Firmware.

Comprehensive error code system for the STAR RX72N motor controller firmware. Provides type-safe error handling with categorized error codes following C23 typed enum standards for embedded safety-critical systems.

7.5.1.1 Architecture Design

This module implements a centralized error handling system for safety-critical embedded requirements:

- Type-safe: Uses C23 typed enums (uint16_t underlying type)
- Zero overhead: Inline helper functions compile to single instructions
- Extensible: Category-based organization supports future error codes
- Traceable: Each error code maps to specific failure conditions

7.5.1.2 Error Code Organization

Error codes are organized into non-overlapping ranges to facilitate debugging and logging. The hex representation makes category identification trivial:

Range	Category	Purpose
0x000	Success	Operation completed successfully
0x100-0x1FF	Generic	General-purpose errors (null, timeout)
0x200-0x2FF	Hardware	Peripheral and GPIO errors
0x300-0x3FF	RTOS	ThreadX and synchronization errors
0x400-0x4FF	Communication	SPI, UART, I2C, protocol errors
0x500-0x5FF	Validation	Input validation and checksum errors

7.5.1.3 Design Rationale

Why signed int32_t for `rx_err_t` but unsigned uint16_t for enum?

- `rx_err_t` uses int32_t to allow negative error codes (future compatibility)
- Enum uses uint16_t to minimize memory usage (most values < 0xFFFF)
- Implicit conversion is safe: uint16_t -> int32_t always preserves value

Why category-based ranges instead of linear numbering?

- Enables fast error categorization in logging (e.g., "0x2xx = hardware")
- Supports future expansion within categories without renumbering
- Makes error codes human-readable in hex dumps and debuggers

Why inline helpers instead of macros?

- Type safety: compiler checks parameter types
- Debuggable: can step into functions, set breakpoints
- No macro pitfalls: no multiple evaluation, proper scoping

7.5.1.4 Performance Characteristics

All operations are zero-cost abstractions:

- [rx_err_from_threadx\(\)](#): Single comparison + conditional move (1-2 cycles)
- [rx_err_is_ok\(\)](#): Single comparison (1 cycle)
- [rx_err_is_error\(\)](#): Single comparison (1 cycle)

7.5.1.5 Memory Usage

- [rx_err_t](#): 4 bytes per error variable
- [rx_err_codes_t](#): 2 bytes per enum value
- Helper functions: 0 bytes (inlined by compiler at -O1+)

7.5.1.6 Thread Safety

All functions in this module are inherently thread-safe (pure functions). Error codes are read-only constants after compilation.

Module Dependencies:

- `stdint.h`: Fixed-width integer types
- `stdbool.h`: Boolean type for helper functions

See also

[rx_check.h](#) Validation macros that return [rx_err_t](#) codes

[rx_error_handler.h](#) Error logging and handling infrastructure

[docs/sections/06_nasa_power_of_10.tex](#) NASA compliance documentation

NASA Power of 10 Compliance:

- Rule 1: [OK] No goto, setjmp, recursion (pure constant definitions)
- Rule 2: [OK] N/A (no loops)
- Rule 3: [OK] No dynamic allocation (compile-time constants only)
- Rule 4: [OK] All functions < 10 lines (simple inline helpers)
- Rule 5: [OK] Inline functions have implicit pre/post conditions
- Rule 6: [OK] All identifiers have minimal scope
- Rule 7: [OK] N/A (no function calls to check)
- Rule 8: [OK] Uses C23 typed enums (`k_rx_err_codes_t : uint16_t`)
- Rule 9: [OK] N/A (no function pointers)
- Rule 10: [OK] Compiles with -Wall -Wextra -Werror

SOLID Principles:

- S (Single Responsibility): Only error code definitions, no handling logic
- O (Open/Closed): Extensible via new enum values, closed for modification
- L (Liskov Substitution): [rx_err_t](#) can substitute for `int32_t` in APIs
- I (Interface Segregation): Minimal interface (3 helper functions)
- D (Dependency Inversion): No dependencies on concrete implementations

Author

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Date

2026-01-27

Version

1.0.0

Copyright

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Definition in file [rx_err.h](#).

7.5.2 Typedef Documentation

7.5.2.1 `rx_err_t`

```
typedef int32_t rx\_err\_t
```

Error code type for RX72N firmware.

Signed 32-bit integer type used for all function return values that indicate success or failure. This type provides a consistent error handling interface across the entire firmware codebase.

7.5.2.2 Type Selection Rationale

Why `int32_t` instead of `enum`?

- Allows future negative error codes (e.g., -1 for EOF-like conditions)
- Compatible with standard POSIX error conventions
- Sufficient range for all error categories (only using 0x000-0x5FF currently)
- Efficient on 32-bit ARM Cortex-R4 (native register width)

Why signed instead of unsigned?

- Enables use of negative values for special conditions
- Matches C standard library conventions (`errno`, etc.)
- Prevents accidental unsigned wraparound bugs

7.5.2.3 Value Semantics

- 0 (k_rx_ok): Success - operation completed as requested
- Non-zero: Error - operation failed, consult specific error code
- Positive values: Standard error codes (0x100-0x5FF)
- Negative values: Reserved for future use

7.5.2.4 Usage Pattern

Standard Error Checking:

```
rx_err_t err = some_function();
if (err != k_rx_ok) {
    // Handle error
    rx_log_error("TAG", "Operation failed: 0x%X", err);
    return err; // Propagate error up call stack
}
```

Early Return Pattern:

```
rx_err_t init_subsystem(void) {
    rx_err_t err;

    err = init_hardware();
    if (err != k_rx_ok) return err;

    err = init_rtos();
    if (err != k_rx_ok) return err;

    return k_rx_ok;
}
```

Switch-Based Error Handling:

```
rx_err_t err = spi_transfer(data, len);
switch (err) {
    case k_rx_ok:
        // Success path
        break;
    case k_rx_err_timeout:
        // Retry logic
        retry_transfer();
        break;
    case k_rx_err_nack:
        // Device not responding
        reset_device();
        break;
    default:
        // Unexpected error
        log_error(err);
        break;
}
```

Macro-Based Error Propagation:

```
#define RX_RETURN_ON_ERROR(err, tag, msg) \
do { \
    rx_err_t _err = (err); \
    if (_err != k_rx_ok) { \
        rx_log_error((tag), (msg)); \
        return _err; \
    } \
} while (0)

rx_err_t init_system(void) {
    RX_RETURN_ON_ERROR(init_gpio(), "INIT", "GPIO init failed");
    RX_RETURN_ON_ERROR(init_spi(), "INIT", "SPI init failed");
    return k_rx_ok;
}
```

Note

This type is used by ALL functions that can fail. Consistent usage across the codebase enables systematic error handling and logging.

Warning

Never cast arbitrary integers to [rx_err_t](#). Only use predefined `k_rx_*` constants or values returned from error conversion functions.

Attention

Functions returning [rx_err_t](#) MUST document all possible return values using

Return values

tags	in their Doxygen documentation.
------	---------------------------------

Memory Layout:

- Size: 4 bytes
- Alignment: 4 bytes (native word on ARM Cortex-R4)
- Signedness: Signed (two's complement)
- Range: -2,147,483,648 to 2,147,483,647 (only 0-0x5FF currently used)

Performance:

- Return value overhead: 0 cycles (returned in register r0)
- Comparison overhead: 1 cycle (CMP r0, #0)
- No heap allocation required

See also

[rx_err_codes_t](#) Enumeration of all defined error codes

[rx_err_is_ok\(\)](#) Helper to check for success

[rx_err_is_error\(\)](#) Helper to check for failure

[rx_err_from_threadx\(\)](#) Convert ThreadX codes to [rx_err_t](#)

Since

Version 1.0.0

Definition at line 238 of file [rx_err.h](#).

7.5.3 Enumeration Type Documentation

7.5.3.1 rx_err_codes_t

enum `rx_err_codes_t` : `uint16_t`

Comprehensive error code enumeration for RX72N firmware.

Type-safe error codes using C23 typed enum syntax with explicit `uint16_t` underlying type. All error codes are compile-time constants organized into logical categories for systematic error handling and debugging.

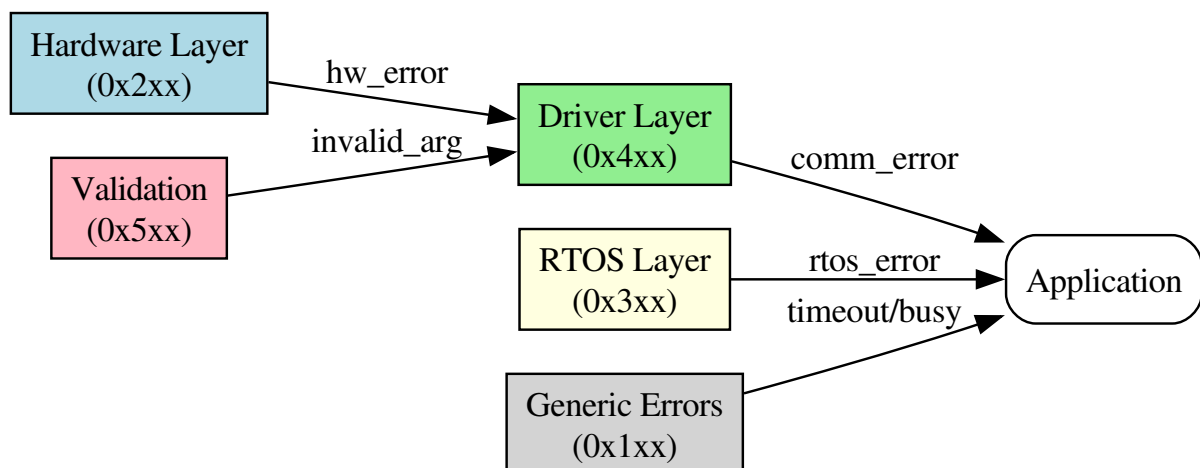
7.5.3.2 Error Code Categories

The error code space is partitioned into non-overlapping ranges:

Range	Category	Count	Purpose
0x000	Success	1	Operation completed successfully
0x100-0x1FF	Generic	15	General-purpose errors (null, timeout, etc.)
0x200-0x2FF	Hardware	9	Peripheral, GPIO, and sensor errors
0x300-0x3FF	RTOS	6	ThreadX and synchronization errors
0x400-0x4FF	Communication	8	SPI, UART, I2C, and protocol errors
0x500-0x5FF	Validation	4	Input validation and checksum errors

7.5.3.3 Error Category Flow

Typical error propagation through the system:



7.5.3.4 Error Handling Patterns

Pattern 1: Direct Comparison

```
rx_err_t err = motor_start();
if (err == k_rx_ok) {
    // Success
} else if (err == k_rx_err_estop) {
    // Emergency stop active - cannot start
} else {
    // Other error - log and handle
}
```

Pattern 2: Category Check

```
rx_err_t err = spi_transfer();
if ((err & 0xF00) == 0x400) {
    // Communication error category
    restart_communication();
} else if ((err & 0xF00) == 0x200) {
    // Hardware error category
    reset_hardware();
}
```

Pattern 3: Error Logging with Context

```
rx_err_t err = init_subsystem();
if (err != k_rx_ok) {
    rx_log_error("INIT", "Subsystem init failed: 0x%03X", err);
    // Hex dump shows category: 0x2xx = hardware, 0x4xx = comm, etc.
}
```

Usage Example - Complete Error Handling:

```
#include "rx_err.h"
#include "rx_log.h"

rx_err_t initialize_motor_controller(void) {
    rx_err_t err;

    // Initialize hardware
    err = init_gpio();
    if (err != k_rx_ok) {
        if (err == k_rx_err_gpio_conflict) {
            rx_log_error("MOTOR", "GPIO conflict - check pin assignments");
        }
        return err;
    }

    // Initialize SPI for RPi5 communication
    err = init_spi();
    if (err != k_rx_ok) {
        if (err == k_rx_err_spi_error) {
            rx_log_error("SPI", "SPI initialization failed");
        }
        return err;
    }

    // Create RTOS resources
    err = create_motor_thread();
    if (err != k_rx_ok) {
        if (err == k_rx_err_rtos_thread_create) {
            rx_log_error("MOTOR", "Thread creation failed - out of memory?");
        }
        return err;
    }

    return k_rx_ok;
}
```

Invariant

- All error codes except `k_rx_ok` are non-zero
- Error code ranges never overlap between categories
- Error codes are compile-time constants (never modified at runtime)

Note

- Uses C23 typed enum syntax : `uint16_t` for guaranteed size and ABI stability
- Error codes are designed to be human-readable in hex (category visible)
- All error codes fit in `uint16_t` (0x000-0x5FF) for memory efficiency

Warning

- Do not cast arbitrary integers to this enum - only use defined constants
- Do not modify error codes at runtime - they are compile-time constants

See also

- `rx_err_t` Type alias for function return values
- `rx_err_is_ok()` Helper to check for success
- `rx_err_is_error()` Helper to check for failure

Since

Version 1.0.0

Enumerator

<code>k_rx_ok</code>	Success - operation completed as requested. The only non-error value. Indicates successful completion of an operation with no errors, warnings, or exceptional conditions. All outputs are valid and postconditions are satisfied. Value: 0x000 Usage: Return this when function succeeds completely Example: <pre> rx_err_t validate_input(uint8_t value) { if (value < 100) { return k_rx_ok; // Valid input } return k_rx_err_out_of_range; } </pre>
----------------------	--

k_rx_fail	Generic unspecified failure. Used when a function fails but the specific error type is unknown or doesn't fit other categories. Prefer more specific error codes when possible. Value: 0x101 When to use: Last resort when no specific error applies Recovery: Log error, inspect code path to determine root cause See also Use more specific errors like k_rx_err_invalid_arg when applicable
k_rx_err_no_mem	Out of memory - allocation failed. Memory allocation failed (heap, stack, or pool). In safety-critical code with zero dynamic allocation (NASA Rule 3), this indicates static buffer exhaustion (e.g., ring buffer full, pool depleted). Value: 0x102 Cause: Static buffer full, RTOS pool exhausted, stack overflow imminent Recovery: Release resources, reduce load, increase buffer size Prevention: Size buffers based on worst-case analysis Warning In zero-allocation systems, this indicates design issue (buffers too small)

<code>k_rx_err_invalid_arg</code>	Invalid function argument. Caller provided parameter outside valid range, nullptr where non-NULL required, or logically inconsistent arguments (e.g., <code>buffer_size > max_size</code>). Value: 0x103 Cause: Parameter validation failed in function preconditions Recovery: Fix caller code, add validation at call site Prevention: Use <code>RX_CHECK_NULL_PTR</code> and <code>RX_CHECK_RANGE</code> macros See also k_rx_err_null_ptr More specific error for nullptr pointers k_rx_err_range_check_failed More specific for range violations
<code>k_rx_err_invalid_state</code>	Invalid state for requested operation. Function called when module is in wrong state. For example, calling <code>motor_start()</code> when motor is already running, or <code>spi_transfer()</code> before <code>spi_init()</code> completed successfully. Value: 0x104 Cause: State machine in unexpected state for requested operation Recovery: Check module state, call initialization functions first Prevention: Document valid state transitions, enforce with state checks See also k_rx_err_not_initialized Specific case of invalid state (not initialized)

<code>k_rx_err_invalid_size</code>	Invalid size parameter. Size parameter outside valid range. Examples: buffer size exceeds maximum, packet length too short, array size is zero. Value: 0x105 Cause: Size parameter failed validation (too large, too small, misaligned) Recovery: Adjust size parameter to valid range Prevention: Document valid size ranges in function documentation See also k_rx_err_invalid_arg More general argument validation error
<code>k_rx_err_not_found</code>	Requested item not found. Search, lookup, or query operation failed to find requested item. Examples: device not found on bus, configuration key missing, motor ID invalid. Value: 0x106 Cause: Item does not exist in collection, device not on bus, ID not registered Recovery: Verify item exists, check ID is correct, enumerate available items Prevention: Validate IDs before use, document valid ID ranges

k_rx_err_not_supported	Feature or operation not supported. Requested functionality not implemented in this build or hardware configuration. Examples: hardware feature not available on this MCU variant, optional module not compiled in, operation not supported by peripheral. Value: 0x107 Cause: Feature not implemented, hardware limitation, compile-time config Recovery: Use alternative approach, check hardware capabilities Prevention: Document supported features, provide capability query API
k_rx_err_timeout	Operation timed out. Operation exceeded allowed time limit. Examples: peripheral not ready within timeout, RTOS wait timeout expired, communication timeout (no response). Value: 0x108 Cause: Peripheral hung, device not responding, deadlock, excessive load Recovery: Retry operation, reset peripheral, increase timeout Prevention: Tune timeouts based on empirical testing, implement watchdogs See also k_rx_err_hw_timeout Hardware-specific timeout

k_rx_err_busy	Resource busy - operation cannot proceed. Requested resource is currently in use and cannot service request. Examples: SPI bus busy, mutex locked by other thread, DMA channel active. Value: 0x109 Cause: Resource locked, peripheral busy, exclusive access required Recovery: Wait for resource, retry later, queue request Prevention: Implement resource arbitration, use RTOS synchronization See also k_rx_err_would_block Non-blocking operation would block
k_rx_err_no_data	No application data available (control frame received). Operation succeeded but no application-layer data was returned. Distinct from k_rx_err_timeout - receive was successful but the frame was a control frame (ACK/NACK/PING/PONG/RESET) not intended for application consumption. Value: 0x10A Cause: Received control frame instead of data frame Recovery: This is normal - caller should continue polling Prevention: Not an error condition - expected behavior See also k_rx_err_timeout No data due to timeout

<code>k_rx_err_would_block</code>	Non-blocking operation would block. Non-blocking function called but operation would block waiting for resource. Differs from <code>k_rx_err_busy</code>: this indicates caller requested non-blocking mode but operation requires waiting. Value: 0x10B Cause: Non-blocking mode requested but data/resource not immediately available Recovery: Retry later, use blocking mode, or poll until ready Prevention: Check resource availability before calling non-blocking API See also k_rx_err_busy Resource busy in blocking mode
<code>k_rx_err_exists</code>	Item already exists - cannot create. Attempted to create resource that already exists. Examples: thread already created, device already registered, configuration already initialized. Value: 0x10C Cause: Duplicate creation attempt, resource already allocated Recovery: Use existing resource, or delete and recreate Prevention: Check existence before creation, track resource state

k_rx_err_empty	Container empty - no data available. Attempted to retrieve data from empty container. Examples: ring buffer empty, queue empty, receive FIFO empty. Value: 0x10D Cause: No data available, all items consumed Recovery: Wait for data, check count before reading Prevention: Check count/size before attempting read See also k_rx_err_would_block Non-blocking read from empty container
k_rx_err_cancelled	Operation cancelled by request. Operation was cancelled before completion, either by explicit cancel request or by system shutdown. No side effects occurred (atomic cancellation). Value: 0x10E Cause: User cancel request, system shutdown, higher priority task Recovery: Clean up resources, update state to reflect cancellation Prevention: Implement graceful cancellation support

k_rx_err_not_initialized	Module not initialized. Function called before module initialization completed. All modules must be initialized via their init() function before use. Value: 0x10F Cause: init() not called, init() failed, module deinitialized Recovery: Call module_init() before using module functions Prevention: Document initialization requirements, enforce with state checks See also k_rx_err_invalid_state More general state error
k_rx_err_estop	Emergency stop active - operation forbidden. Emergency stop triggered, all motor operations halted. System in safe state, requires manual reset before normal operation can resume. Value: 0x110 Cause: Emergency stop button pressed, safety fault detected Recovery: Clear fault condition, call reset function, acknowledge e-stop Prevention: Implement proper safety interlocks Warning This is a safety-critical error - do not bypass e-stop checks

k_rx_err_hw_init_failed	Hardware initialization failed. Hardware peripheral failed to initialize. Examples: peripheral not responding, clock configuration failed, register write verification failed. Value: 0x201 Cause: Hardware fault, incorrect configuration, missing hardware Recovery: Check hardware connections, verify clock settings, reset MCU Prevention: Validate hardware presence before init
k_rx_err_hw_not_ready	Hardware not ready for operation. Hardware peripheral exists but not ready for requested operation. Examples: PLL not locked, ADC not calibrated, peripheral in reset state. Value: 0x202 Cause: Peripheral needs time to stabilize, calibration incomplete Recovery: Wait for ready flag, poll status register, retry after delay Prevention: Implement proper hardware sequencing and delays
k_rx_err_hw_timeout	Hardware operation timed out. Hardware peripheral failed to complete operation within timeout. Distinct from k_rx_err_timeout (generic) - this is hardware-specific. Value: 0x203 Cause: Peripheral hung, bus arbitration lost, external device not responding Recovery: Reset peripheral, check external hardware, increase timeout Prevention: Tune hardware timeouts, implement watchdog recovery

k_rx_err_hw_error	Generic hardware error. Unspecified hardware error detected. Examples: fault flag set, error interrupt, hardware self-test failure. Value: 0x204 Cause: Hardware fault condition, transient error, EMI/noise Recovery: Read hardware error registers, reset peripheral, retry operation Prevention: Implement hardware error monitoring and logging
k_rx_err_gpio_conflict	GPIO pin conflict detected. GPIO pin already in use by another peripheral or function. Pin multiplexing conflict must be resolved before pin can be used. Value: 0x205 Cause: Multiple peripherals mapped to same pin, incorrect MPC settings Recovery: Reconfigure pin mux, use alternative pin, disable conflicting peripheral Prevention: Document pin assignments, validate at init time See also docs/sections/03_hardware_pinout.tex Complete pin assignment table

k_rx_err_gpio_invalid_port	Invalid GPIO port number. GPIO port number outside valid range for RX72N (valid: 0-K, excluding reserved). Value: 0x206 Cause: Incorrect port constant, typo in pin definition Recovery: Use valid port number (PORT0-PORTK), check hardware manual Prevention: Use enum constants from rx_port_constants.h
k_rx_err_gpio_invalid_pin	Invalid GPIO pin number. GPIO pin number outside valid range (0-7 for most ports on RX72N). Value: 0x207 Cause: Incorrect pin constant, array index out of bounds Recovery: Use valid pin number (0-7), check port definition Prevention: Use enum constants from rx_gpio_constants.h
k_rx_err_out_of_range	Sensor measurement out of valid range. Sensor reading outside physically possible or configured range. Examples: ADC overflow, temperature sensor disconnected, ultrasonic no echo. Value: 0x208 Cause: Sensor fault, value exceeds limits, sensor disconnected Recovery: Check sensor connections, validate sensor state, clamp to range Prevention: Implement sensor health monitoring

k_rx_err_hw_unmapped	Hardware register block not mapped in memory. Attempted to access peripheral whose registers are not mapped at expected address. Indicates hardware configuration error or unsupported MCU variant. Value: 0x209 Cause: Wrong MCU variant, peripheral not present, incorrect memory map Recovery: Verify MCU part number, check datasheet, disable peripheral Prevention: Conditional compilation based on MCU variant
k_rx_err_rtos_error	Generic RTOS error. Unspecified ThreadX error occurred. Check ThreadX status code for details. Value: 0x301 Cause: ThreadX API returned non-success status Recovery: Examine ThreadX status, consult ThreadX documentation Prevention: Validate ThreadX API parameters See also rx_err_from_threadx() Convert ThreadX codes to rx_err_t
k_rx_err_threadx	ThreadX error (alias for k_rx_err_rtos_error). Identical to k_rx_err_rtos_error, provided for explicit ThreadX naming. Value: 0x301

k_rx_err_rtos_thread_create	ThreadX thread creation failed. tx_thread_create() failed. Common causes: out of memory, invalid priority, invalid stack. Value: 0x302 Cause: Insufficient memory, stack too small, invalid parameters Recovery: Increase heap size, validate thread parameters Prevention: Size memory pools based on worst-case thread count
k_rx_err_rtos_semaphore	Semaphore operation failed. Semaphore create, get, or put operation failed. Check ThreadX error code. Value: 0x303 Cause: Invalid semaphore handle, timeout, semaphore not created Recovery: Verify semaphore initialized, check timeout value Prevention: Initialize all RTOS objects before use
k_rx_err_rtos_mutex	Mutex operation failed. Mutex create, get, or put operation failed. Possible deadlock or priority inversion if using priority inheritance. Value: 0x304 Cause: Invalid mutex handle, timeout, mutex already owned Recovery: Check mutex state, verify not already locked Prevention: Use timeouts, avoid nested mutex acquisition

k_rx_err_rtos_queue	Queue operation failed. Queue create, send, or receive operation failed. Value: 0x305 Cause: Queue full, queue empty, invalid queue handle Recovery: Increase queue size, implement flow control Prevention: Size queues based on worst-case message rate
k_rx_err_rtos_timer	Timer operation failed. Timer create, activate, or deactivate operation failed. Value: 0x306 Cause: Invalid timer handle, timer already active Recovery: Check timer state before operation Prevention: Track timer state explicitly
k_rx_err_comm_error	Generic communication error. Unspecified communication error on SPI, UART, or I2C bus. Value: 0x401 Cause: Bus error, framing error, data corruption Recovery: Reset bus, retry transfer, check connections Prevention: Implement error detection (CRC, checksums)

k_rx_err_spi_error	SPI communication error. SPI transfer failed. Examples: CIPO timeout, bus error, chip select conflict. Value: 0x402 Cause: No response on CIPO, mode mismatch, clock issue Recovery: Check SPI mode, verify clock frequency, check CS timing Prevention: Validate SPI configuration before transfer
k_rx_err_uart_error	UART communication error. UART error detected. Examples: framing error, parity error, overrun. Value: 0x403 Cause: Baud rate mismatch, noise, buffer overrun Recovery: Clear error flags, adjust baud rate, reduce data rate Prevention: Use hardware flow control, size buffers adequately
k_rx_err_i2c_error	I2C communication error. I2C (RIIC) error detected. Examples: bus busy, arbitration lost, NACK. Value: 0x404 Cause: Device not responding, wrong address, bus contention Recovery: Reset I2C bus, verify device address, check pull-ups Prevention: Implement bus recovery, validate addresses

k_rx_err_crc_mismatch	CRC validation failed. Received data CRC does not match calculated CRC. Data corruption detected. Value: 0x405 Cause: Data corruption, EMI, faulty peripheral Recovery: Request retransmission, check signal integrity Prevention: Use shielded cables, implement retry logic See also k_rx_err_checksum_mismatch Simpler checksum validation
k_rx_err_protocol_error	Protocol error detected. Communication protocol violation. Examples: invalid packet format, unexpected sequence number, protocol state machine error. Value: 0x406 Cause: Protocol version mismatch, malformed packet, out-of-sequence Recovery: Reset protocol state machine, request resynchronization Prevention: Validate all protocol fields, implement version negotiation

k_rx_err_nack	NACK (Not Acknowledged) received. Peripheral responded with NACK (negative acknowledgment). Device refusing operation or not ready. Value: 0x407 Cause: Device busy, invalid command, device not ready Recovery: Retry after delay, check device state Prevention: Poll device ready status before commands
k_rx_err_conflict	Resource conflict in communication. Multiple controllers attempting bus access, or conflicting operation in progress. Value: 0x408 Cause: Bus arbitration lost, conflicting DMA channels Recovery: Implement bus arbitration, retry after backoff Prevention: Use RTOS mutexes for shared buses

k_rx_err_retry_limit	Retransmission limit exceeded. Automatic retransmission exhausted all retry attempts without receiving ACK. Frame delivery cannot be confirmed. Value: 0x409 Cause: Remote side not responding, persistent CRC failures, link down Recovery: Reset communication, check physical connection, increase retries Prevention: Tune timeout/retry parameters for link quality See also <code>rx_spi_comm_set_auto_retransmit()</code> Configure retry parameters
k_rx_err_validation_failed	Generic validation failed. Data validation or consistency check failed. Value: 0x501 Cause: Invalid data format, constraint violation, consistency check failed Recovery: Reject invalid data, request valid input Prevention: Validate all inputs at system boundaries

<code>k_rx_err_checksum_mismatch</code>	Checksum mismatch detected. Simple checksum validation failed. Less robust than CRC but faster. Value: 0x502 Cause: Data corruption, transmission error Recovery: Request retransmission Prevention: Use CRC for critical data See also k_rx_err_crc_mismatch More robust CRC validation
<code>k_rx_err_range_check_failed</code>	Range check failed. Value outside configured or physically possible range. Value: 0x503 Cause: Value exceeds min/max limits Recovery: Clamp to valid range, reject input Prevention: Document and enforce valid ranges

k_rx_err_null_ptr	Null pointer detected. Required pointer parameter is nullptr. Subset of k_rx_err_invalid_arg. Value: 0x504 Cause: Caller passed NULL where non-NULL required Recovery: Fix caller code to pass valid pointer Prevention: Use RX_CHECK_NULL_PTR macro at function entry See also k_rx_err_invalid_arg More general argument validation
-------------------	--

Definition at line 382 of file rx_err.h.

```

00382         : uint16_t {
00383     /**
00384     * @brief Success - operation completed as requested
00385     * @details
00386     * The only non-error value. Indicates successful completion of an operation
00387     * with no errors, warnings, or exceptional conditions. All outputs are valid
00388     * and postconditions are satisfied.
00389     *
00390     * @par Value: 0x000
00391     * @par Usage: Return this when function succeeds completely
00392     * @par Example:
00393     * @code{.c}
00394     * rx_err_t validate_input(uint8_t value) {
00395     *     if (value < 100) {
00396     *         return k_rx_ok; // Valid input
00397     *     }
00398     *     return k_rx_err_out_of_range;
00399     * }
00400     * @endcode
00401     */
00402     k_rx_ok = 0,
00403
00404     /**
00405     * Generic Errors (0x100 - 0x1FF)
00406     * General-purpose errors not specific to hardware or communication
00407     *
00408     */
00409     /**
00410     * @brief Generic unspecified failure
00411     * @details
00412     * Used when a function fails but the specific error type is unknown or
00413     * doesn't fit other categories. Prefer more specific error codes when possible.
00414     *
00415     * @par Value: 0x101
00416     * @par When to use: Last resort when no specific error applies
00417     * @par Recovery: Log error, inspect code path to determine root cause
00418     * @see Use more specific errors like k_rx_err_invalid_arg when applicable
00419     */
00420     k_rx_fail = 0x101,
00421
00422     /**
00423     * @brief Out of memory - allocation failed
00424     * @details
00425     * Memory allocation failed (heap, stack, or pool). In safety-critical code
00426     * with zero dynamic allocation (NASA Rule 3), this indicates static buffer
00427     * exhaustion (e.g., ring buffer full, pool depleted).
00428     */

```

```

00429  *
00430  * @par Value: 0x102
00431  * @par Cause: Static buffer full, RTOS pool exhausted, stack overflow imminent
00432  * @par Recovery: Release resources, reduce load, increase buffer size
00433  * @par Prevention: Size buffers based on worst-case analysis
00434  * @warning In zero-allocation systems, this indicates design issue (buffers too small)
00435  */
00436 k_rx_err_no_mem = 0x102,
00437
00438 /**
00439  * @brief Invalid function argument
00440  * @details
00441  * Caller provided parameter outside valid range, nullptr where non-NULL
00442  * required, or logically inconsistent arguments (e.g., buffer_size > max_size).
00443  *
00444  * @par Value: 0x103
00445  * @par Cause: Parameter validation failed in function preconditions
00446  * @par Recovery: Fix caller code, add validation at call site
00447  * @par Prevention: Use RX_CHECK_NULL_PTR and RX_CHECK_RANGE macros
00448  * @see k_rx_err_null_ptr More specific error for nullptr pointers
00449  * @see k_rx_err_range_check_failed More specific for range violations
00450  */
00451 k_rx_err_invalid_arg = 0x103,
00452
00453 /**
00454  * @brief Invalid state for requested operation
00455  * @details
00456  * Function called when module is in wrong state. For example, calling
00457  * motor_start() when motor is already running, or spi_transfer() before
00458  * spi_init() completed successfully.
00459  *
00460  * @par Value: 0x104
00461  * @par Cause: State machine in unexpected state for requested operation
00462  * @par Recovery: Check module state, call initialization functions first
00463  * @par Prevention: Document valid state transitions, enforce with state checks
00464  * @see k_rx_err_not_initialized Specific case of invalid state (not initialized)
00465  */
00466 k_rx_err_invalid_state = 0x104,
00467
00468 /**
00469  * @brief Invalid size parameter
00470  * @details
00471  * Size parameter outside valid range. Examples: buffer size exceeds maximum,
00472  * packet length too short, array size is zero.
00473  *
00474  * @par Value: 0x105
00475  * @par Cause: Size parameter failed validation (too large, too small, misaligned)
00476  * @par Recovery: Adjust size parameter to valid range
00477  * @par Prevention: Document valid size ranges in function documentation
00478  * @see k_rx_err_invalid_arg More general argument validation error
00479  */
00480 k_rx_err_invalid_size = 0x105,
00481
00482 /**
00483  * @brief Requested item not found
00484  * @details
00485  * Search, lookup, or query operation failed to find requested item.
00486  * Examples: device not found on bus, configuration key missing, motor ID invalid.
00487  *
00488  * @par Value: 0x106
00489  * @par Cause: Item does not exist in collection, device not on bus, ID not registered
00490  * @par Recovery: Verify item exists, check ID is correct, enumerate available items
00491  * @par Prevention: Validate IDs before use, document valid ID ranges
00492  */
00493 k_rx_err_not_found = 0x106,
00494
00495 /**
00496  * @brief Feature or operation not supported
00497  * @details
00498  * Requested functionality not implemented in this build or hardware configuration.
00499  * Examples: hardware feature not available on this MCU variant, optional module
00500  * not compiled in, operation not supported by peripheral.
00501  *
00502  * @par Value: 0x107
00503  * @par Cause: Feature not implemented, hardware limitation, compile-time config
00504  * @par Recovery: Use alternative approach, check hardware capabilities
00505  * @par Prevention: Document supported features, provide capability query API
00506  */
00507 k_rx_err_not_supported = 0x107,
00508
00509 /**
00510  * @brief Operation timed out
00511  * @details
00512  * Operation exceeded allowed time limit. Examples: peripheral not ready within
00513  * timeout, RTOS wait timeout expired, communication timeout (no response).
00514  *
00515  * @par Value: 0x108

```

```

00516 * @par Cause: Peripheral hung, device not responding, deadlock, excessive load
00517 * @par Recovery: Retry operation, reset peripheral, increase timeout
00518 * @par Prevention: Tune timeouts based on empirical testing, implement watchdogs
00519 * @see k_rx_err_hw_timeout Hardware-specific timeout
00520 */
00521 k_rx_err_timeout = 0x108,
00522
00523 /**
00524 * @brief Resource busy - operation cannot proceed
00525 * @details
00526 * Requested resource is currently in use and cannot service request.
00527 * Examples: SPI bus busy, mutex locked by other thread, DMA channel active.
00528 *
00529 * @par Value: 0x109
00530 * @par Cause: Resource locked, peripheral busy, exclusive access required
00531 * @par Recovery: Wait for resource, retry later, queue request
00532 * @par Prevention: Implement resource arbitration, use RTOS synchronization
00533 * @see k_rx_err_would_block Non-blocking operation would block
00534 */
00535 k_rx_err_busy = 0x109,
00536
00537 /**
00538 * @brief No application data available (control frame received)
00539 * @details
00540 * Operation succeeded but no application-layer data was returned. Distinct from
00541 * k_rx_err_timeout - receive was successful but the frame was a control frame
00542 * (ACK/NACK/PING/PONG/RESET) not intended for application consumption.
00543 *
00544 * @par Value: 0x10A
00545 * @par Cause: Received control frame instead of data frame
00546 * @par Recovery: This is normal - caller should continue polling
00547 * @par Prevention: Not an error condition - expected behavior
00548 * @see k_rx_err_timeout No data due to timeout
00549 */
00550 k_rx_err_no_data = 0x10A,
00551
00552 /**
00553 * @brief Non-blocking operation would block
00554 * @details
00555 * Non-blocking function called but operation would block waiting for resource.
00556 * Differs from k_rx_err_busy: this indicates caller requested non-blocking mode
00557 * but operation requires waiting.
00558 *
00559 * @par Value: 0x10B
00560 * @par Cause: Non-blocking mode requested but data/resource not immediately available
00561 * @par Recovery: Retry later, use blocking mode, or poll until ready
00562 * @par Prevention: Check resource availability before calling non-blocking API
00563 * @see k_rx_err_busy Resource busy in blocking mode
00564 */
00565 k_rx_err_would_block = 0x10B,
00566
00567 /**
00568 * @brief Item already exists - cannot create
00569 * @details
00570 * Attempted to create resource that already exists. Examples: thread already
00571 * created, device already registered, configuration already initialized.
00572 *
00573 * @par Value: 0x10C
00574 * @par Cause: Duplicate creation attempt, resource already allocated
00575 * @par Recovery: Use existing resource, or delete and recreate
00576 * @par Prevention: Check existence before creation, track resource state
00577 */
00578 k_rx_err_exists = 0x10C,
00579
00580 /**
00581 * @brief Container empty - no data available
00582 * @details
00583 * Attempted to retrieve data from empty container. Examples: ring buffer empty,
00584 * queue empty, receive FIFO empty.
00585 *
00586 * @par Value: 0x10D
00587 * @par Cause: No data available, all items consumed
00588 * @par Recovery: Wait for data, check count before reading
00589 * @par Prevention: Check count/size before attempting read
00590 * @see k_rx_err_would_block Non-blocking read from empty container
00591 */
00592 k_rx_err_empty = 0x10D,
00593
00594 /**
00595 * @brief Operation cancelled by request
00596 * @details
00597 * Operation was cancelled before completion, either by explicit cancel request
00598 * or by system shutdown. No side effects occurred (atomic cancellation).
00599 *
00600 * @par Value: 0x10E
00601 * @par Cause: User cancel request, system shutdown, higher priority task
00602 * @par Recovery: Clean up resources, update state to reflect cancellation

```

```

00603  * @par Prevention: Implement graceful cancellation support
00604  */
00605  k_rx_err_cancelled = 0x10E,
00606
00607  /**
00608  * @brief Module not initialized
00609  * @details
00610  * Function called before module initialization completed. All modules must be
00611  * initialized via their init() function before use.
00612  *
00613  * @par Value: 0x10F
00614  * @par Cause: init() not called, init() failed, module deinitialized
00615  * @par Recovery: Call module_init() before using module functions
00616  * @par Prevention: Document initialization requirements, enforce with state checks
00617  * @see k_rx_err_invalid_state More general state error
00618  */
00619  k_rx_err_not_initialized = 0x10F,
00620
00621  /**
00622  * @brief Emergency stop active - operation forbidden
00623  * @details
00624  * Emergency stop triggered, all motor operations halted. System in safe state,
00625  * requires manual reset before normal operation can resume.
00626  *
00627  * @par Value: 0x110
00628  * @par Cause: Emergency stop button pressed, safety fault detected
00629  * @par Recovery: Clear fault condition, call reset function, acknowledge e-stop
00630  * @par Prevention: Implement proper safety interlocks
00631  * @warning This is a safety-critical error - do not bypass e-stop checks
00632  */
00633  k_rx_err_estop = 0x110,
00634
00635  /*
=====
00636  * Hardware Errors (0x200 - 0x2FF)
00637  * Peripheral, GPIO, and hardware interface errors
00638  *
=====
00639  */
00640
00641  /**
00642  * @brief Hardware initialization failed
00643  * @details
00644  * Hardware peripheral failed to initialize. Examples: peripheral not responding,
00645  * clock configuration failed, register write verification failed.
00646  *
00647  * @par Value: 0x201
00648  * @par Cause: Hardware fault, incorrect configuration, missing hardware
00649  * @par Recovery: Check hardware connections, verify clock settings, reset MCU
00650  * @par Prevention: Validate hardware presence before init
00651  */
00652  k_rx_err_hw_init_failed = 0x201,
00653
00654  /**
00655  * @brief Hardware not ready for operation
00656  * @details
00657  * Hardware peripheral exists but not ready for requested operation. Examples:
00658  * PLL not locked, ADC not calibrated, peripheral in reset state.
00659  *
00660  * @par Value: 0x202
00661  * @par Cause: Peripheral needs time to stabilize, calibration incomplete
00662  * @par Recovery: Wait for ready flag, poll status register, retry after delay
00663  * @par Prevention: Implement proper hardware sequencing and delays
00664  */
00665  k_rx_err_hw_not_ready = 0x202,
00666
00667  /**
00668  * @brief Hardware operation timed out
00669  * @details
00670  * Hardware peripheral failed to complete operation within timeout. Distinct from
00671  * k_rx_err_timeout (generic) - this is hardware-specific.
00672  *
00673  * @par Value: 0x203
00674  * @par Cause: Peripheral hung, bus arbitration lost, external device not responding
00675  * @par Recovery: Reset peripheral, check external hardware, increase timeout
00676  * @par Prevention: Tune hardware timeouts, implement watchdog recovery
00677  */
00678  k_rx_err_hw_timeout = 0x203,
00679
00680  /**
00681  * @brief Generic hardware error
00682  * @details
00683  * Unspecified hardware error detected. Examples: fault flag set, error interrupt,
00684  * hardware self-test failure.
00685  *
00686  * @par Value: 0x204
00687  * @par Cause: Hardware fault condition, transient error, EMI/noise

```

```

00688  * @par Recovery: Read hardware error registers, reset peripheral, retry operation
00689  * @par Prevention: Implement hardware error monitoring and logging
00690  */
00691  k_rx_err_hw_error = 0x204,
00692
00693  /**
00694   * @brief GPIO pin conflict detected
00695   * @details
00696   * GPIO pin already in use by another peripheral or function. Pin multiplexing
00697   * conflict must be resolved before pin can be used.
00698   *
00699   * @par Value: 0x205
00700   * @par Cause: Multiple peripherals mapped to same pin, incorrect MPC settings
00701   * @par Recovery: Reconfigure pin mux, use alternative pin, disable conflicting peripheral
00702   * @par Prevention: Document pin assignments, validate at init time
00703   * @see docs/sections/03_hardware_pinout.tex Complete pin assignment table
00704   */
00705  k_rx_err_gpio_conflict = 0x205,
00706
00707  /**
00708   * @brief Invalid GPIO port number
00709   * @details
00710   * GPIO port number outside valid range for RX72N (valid: 0-K, excluding reserved).
00711   *
00712   * @par Value: 0x206
00713   * @par Cause: Incorrect port constant, typo in pin definition
00714   * @par Recovery: Use valid port number (PORT0-PORTK), check hardware manual
00715   * @par Prevention: Use enum constants from rx_port_constants.h
00716   */
00717  k_rx_err_gpio_invalid_port = 0x206,
00718
00719  /**
00720   * @brief Invalid GPIO pin number
00721   * @details
00722   * GPIO pin number outside valid range (0-7 for most ports on RX72N).
00723   *
00724   * @par Value: 0x207
00725   * @par Cause: Incorrect pin constant, array index out of bounds
00726   * @par Recovery: Use valid pin number (0-7), check port definition
00727   * @par Prevention: Use enum constants from rx_gpio_constants.h
00728   */
00729  k_rx_err_gpio_invalid_pin = 0x207,
00730
00731  /**
00732   * @brief Sensor measurement out of valid range
00733   * @details
00734   * Sensor reading outside physically possible or configured range. Examples:
00735   * ADC overflow, temperature sensor disconnected, ultrasonic no echo.
00736   *
00737   * @par Value: 0x208
00738   * @par Cause: Sensor fault, value exceeds limits, sensor disconnected
00739   * @par Recovery: Check sensor connections, validate sensor state, clamp to range
00740   * @par Prevention: Implement sensor health monitoring
00741   */
00742  k_rx_err_out_of_range = 0x208,
00743
00744  /**
00745   * @brief Hardware register block not mapped in memory
00746   * @details
00747   * Attempted to access peripheral whose registers are not mapped at expected
00748   * address. Indicates hardware configuration error or unsupported MCU variant.
00749   *
00750   * @par Value: 0x209
00751   * @par Cause: Wrong MCU variant, peripheral not present, incorrect memory map
00752   * @par Recovery: Verify MCU part number, check datasheet, disable peripheral
00753   * @par Prevention: Conditional compilation based on MCU variant
00754   */
00755  k_rx_err_hw_unmapped = 0x209,
00756
00757  /*
=====
00758  * RTOS Errors (0x300 - 0x3FF)
00759  * ThreadX and synchronization errors
00760  *
=====
00761  */
00762
00763  /**
00764   * @brief Generic RTOS error
00765   * @details
00766   * Unspecified ThreadX error occurred. Check ThreadX status code for details.
00767   *
00768   * @par Value: 0x301
00769   * @par Cause: ThreadX API returned non-success status
00770   * @par Recovery: Examine ThreadX status, consult ThreadX documentation
00771   * @par Prevention: Validate ThreadX API parameters
00772   * @see rx_err_from_threadx() Convert ThreadX codes to rx_err_t

```

```

00773  */
00774  k_rx_err_rtos_error = 0x301,
00775
00776  /**
00777   * @brief ThreadX error (alias for k_rx_err_rtos_error)
00778   * @details
00779   * Identical to k_rx_err_rtos_error, provided for explicit ThreadX naming.
00780   *
00781   * @par Value: 0x301
00782   */
00783  k_rx_err_threadx = 0x301,
00784
00785  /**
00786   * @brief ThreadX thread creation failed
00787   * @details
00788   * tx_thread_create() failed. Common causes: out of memory, invalid priority,
00789   * invalid stack.
00790   *
00791   * @par Value: 0x302
00792   * @par Cause: Insufficient memory, stack too small, invalid parameters
00793   * @par Recovery: Increase heap size, validate thread parameters
00794   * @par Prevention: Size memory pools based on worst-case thread count
00795   */
00796  k_rx_err_rtos_thread_create = 0x302,
00797
00798  /**
00799   * @brief Semaphore operation failed
00800   * @details
00801   * Semaphore create, get, or put operation failed. Check ThreadX error code.
00802   *
00803   * @par Value: 0x303
00804   * @par Cause: Invalid semaphore handle, timeout, semaphore not created
00805   * @par Recovery: Verify semaphore initialized, check timeout value
00806   * @par Prevention: Initialize all RTOS objects before use
00807   */
00808  k_rx_err_rtos_semaphore = 0x303,
00809
00810  /**
00811   * @brief Mutex operation failed
00812   * @details
00813   * Mutex create, get, or put operation failed. Possible deadlock or priority
00814   * inversion if using priority inheritance.
00815   *
00816   * @par Value: 0x304
00817   * @par Cause: Invalid mutex handle, timeout, mutex already owned
00818   * @par Recovery: Check mutex state, verify not already locked
00819   * @par Prevention: Use timeouts, avoid nested mutex acquisition
00820   */
00821  k_rx_err_rtos_mutex = 0x304,
00822
00823  /**
00824   * @brief Queue operation failed
00825   * @details
00826   * Queue create, send, or receive operation failed.
00827   *
00828   * @par Value: 0x305
00829   * @par Cause: Queue full, queue empty, invalid queue handle
00830   * @par Recovery: Increase queue size, implement flow control
00831   * @par Prevention: Size queues based on worst-case message rate
00832   */
00833  k_rx_err_rtos_queue = 0x305,
00834
00835  /**
00836   * @brief Timer operation failed
00837   * @details
00838   * Timer create, activate, or deactivate operation failed.
00839   *
00840   * @par Value: 0x306
00841   * @par Cause: Invalid timer handle, timer already active
00842   * @par Recovery: Check timer state before operation
00843   * @par Prevention: Track timer state explicitly
00844   */
00845  k_rx_err_rtos_timer = 0x306,
00846
00847  /*
=====
00848   * Communication Errors (0x400 - 0x4FF)
00849   * SPI, UART, I2C, and protocol errors
00850   *
=====
00851  */
00852
00853  /**
00854   * @brief Generic communication error
00855   * @details
00856   * Unspecified communication error on SPI, UART, or I2C bus.
00857  */

```



```

00858 * @par Value: 0x401
00859 * @par Cause: Bus error, framing error, data corruption
00860 * @par Recovery: Reset bus, retry transfer, check connections
00861 * @par Prevention: Implement error detection (CRC, checksums)
00862 */
00863 k_rx_err_comm_error = 0x401,
00864
00865 /**
00866 * @brief SPI communication error
00867 * @details
00868 * SPI transfer failed. Examples: CIPO timeout, bus error, chip select conflict.
00869 *
00870 * @par Value: 0x402
00871 * @par Cause: No response on CIPO, mode mismatch, clock issue
00872 * @par Recovery: Check SPI mode, verify clock frequency, check CS timing
00873 * @par Prevention: Validate SPI configuration before transfer
00874 */
00875 k_rx_err_spi_error = 0x402,
00876
00877 /**
00878 * @brief UART communication error
00879 * @details
00880 * UART error detected. Examples: framing error, parity error, overrun.
00881 *
00882 * @par Value: 0x403
00883 * @par Cause: Baud rate mismatch, noise, buffer overrun
00884 * @par Recovery: Clear error flags, adjust baud rate, reduce data rate
00885 * @par Prevention: Use hardware flow control, size buffers adequately
00886 */
00887 k_rx_err_uart_error = 0x403,
00888
00889 /**
00890 * @brief I2C communication error
00891 * @details
00892 * I2C (RIIC) error detected. Examples: bus busy, arbitration lost, NACK.
00893 *
00894 * @par Value: 0x404
00895 * @par Cause: Device not responding, wrong address, bus contention
00896 * @par Recovery: Reset I2C bus, verify device address, check pull-ups
00897 * @par Prevention: Implement bus recovery, validate addresses
00898 */
00899 k_rx_err_i2c_error = 0x404,
00900
00901 /**
00902 * @brief CRC validation failed
00903 * @details
00904 * Received data CRC does not match calculated CRC. Data corruption detected.
00905 *
00906 * @par Value: 0x405
00907 * @par Cause: Data corruption, EMI, faulty peripheral
00908 * @par Recovery: Request retransmission, check signal integrity
00909 * @par Prevention: Use shielded cables, implement retry logic
00910 * @see k_rx_err_checksum_mismatch Simpler checksum validation
00911 */
00912 k_rx_err_crc_mismatch = 0x405,
00913
00914 /**
00915 * @brief Protocol error detected
00916 * @details
00917 * Communication protocol violation. Examples: invalid packet format, unexpected
00918 * sequence number, protocol state machine error.
00919 *
00920 * @par Value: 0x406
00921 * @par Cause: Protocol version mismatch, malformed packet, out-of-sequence
00922 * @par Recovery: Reset protocol state machine, request resynchronization
00923 * @par Prevention: Validate all protocol fields, implement version negotiation
00924 */
00925 k_rx_err_protocol_error = 0x406,
00926
00927 /**
00928 * @brief NACK (Not Acknowledged) received
00929 * @details
00930 * Peripheral responded with NACK (negative acknowledgment). Device refusing
00931 * operation or not ready.
00932 *
00933 * @par Value: 0x407
00934 * @par Cause: Device busy, invalid command, device not ready
00935 * @par Recovery: Retry after delay, check device state
00936 * @par Prevention: Poll device ready status before commands
00937 */
00938 k_rx_err_nack = 0x407,
00939
00940 /**
00941 * @brief Resource conflict in communication
00942 * @details
00943 * Multiple controllers attempting bus access, or conflicting operation in progress.
00944 */

```

```

00945 * @par Value: 0x408
00946 * @par Cause: Bus arbitration lost, conflicting DMA channels
00947 * @par Recovery: Implement bus arbitration, retry after backoff
00948 * @par Prevention: Use RTOS mutexes for shared buses
00949 */
00950 k_rx_err_conflict = 0x408,
00951
00952 /**
00953 * @brief Retransmission limit exceeded
00954 * @details
00955 * Automatic retransmission exhausted all retry attempts without receiving ACK.
00956 * Frame delivery cannot be confirmed.
00957 *
00958 * @par Value: 0x409
00959 * @par Cause: Remote side not responding, persistent CRC failures, link down
00960 * @par Recovery: Reset communication, check physical connection, increase retries
00961 * @par Prevention: Tune timeout/retry parameters for link quality
00962 * @see rx_spi_comm_set_auto_retransmit() Configure retry parameters
00963 */
00964 k_rx_err_retry_limit = 0x409,
00965
00966 */
=====
00967 * Validation Errors (0x500 - 0x5FF)
00968 * Input validation and checksum errors
00969
=====
00970 */
00971
00972 /**
00973 * @brief Generic validation failed
00974 * @details
00975 * Data validation or consistency check failed.
00976 *
00977 * @par Value: 0x501
00978 * @par Cause: Invalid data format, constraint violation, consistency check failed
00979 * @par Recovery: Reject invalid data, request valid input
00980 * @par Prevention: Validate all inputs at system boundaries
00981 */
00982 k_rx_err_validation_failed = 0x501,
00983
00984 /**
00985 * @brief Checksum mismatch detected
00986 * @details
00987 * Simple checksum validation failed. Less robust than CRC but faster.
00988 *
00989 * @par Value: 0x502
00990 * @par Cause: Data corruption, transmission error
00991 * @par Recovery: Request retransmission
00992 * @par Prevention: Use CRC for critical data
00993 * @see k_rx_err_crc_mismatch More robust CRC validation
00994 */
00995 k_rx_err_checksum_mismatch = 0x502,
00996
00997 /**
00998 * @brief Range check failed
00999 * @details
01000 * Value outside configured or physically possible range.
01001 *
01002 * @par Value: 0x503
01003 * @par Cause: Value exceeds min/max limits
01004 * @par Recovery: Clamp to valid range, reject input
01005 * @par Prevention: Document and enforce valid ranges
01006 */
01007 k_rx_err_range_check_failed = 0x503,
01008
01009 /**
01010 * @brief Null pointer detected
01011 * @details
01012 * Required pointer parameter is nullptr. Subset of k_rx_err_invalid_arg.
01013 *
01014 * @par Value: 0x504
01015 * @par Cause: Caller passed NULL where non-NULL required
01016 * @par Recovery: Fix caller code to pass valid pointer
01017 * @par Prevention: Use RX_CHECK_NULL_PTR macro at function entry
01018 * @see k_rx_err_invalid_arg More general argument validation
01019 */
01020 k_rx_err_null_ptr = 0x504,
01021
01022 } rx_err_codes_t;

```

7.5.4 Function Documentation

7.5.4.1 rx_err_from_threadx()

```
rx_err_t rx_err_from_threadx (
    uint32_t tx_status) [inline], [static]
```

Convert ThreadX status code to `rx_err_t`.

Converts ThreadX API return values (UINT status codes) to `rx_err_t` error codes. ThreadX uses 0 for success (TX_SUCCESS) and non-zero for various error conditions. This function performs a simple binary conversion: success -> `k_rx_ok`, any error -> `k_rx_err_rtos_error`.

7.5.4.2 Algorithm

1. Compare `tx_status` against TX_SUCCESS (0)
2. If equal: return `k_rx_ok` (success)
3. If not equal: return `k_rx_err_rtos_error` (generic RTOS error)

7.5.4.3 ThreadX Status Codes Mapped

ThreadX Status	Value	<code>rx_err_t</code> Result	Meaning
TX_SUCCESS	0	<code>k_rx_ok</code> (0x000)	Operation successful
TX_DELETED	0x01	<code>k_rx_err_rtos_error</code>	Object was deleted
TX_NO_MEMORY	0x10	<code>k_rx_err_rtos_error</code>	Memory allocation failed
TX_PTR_ERROR	0x03	<code>k_rx_err_rtos_error</code>	Invalid pointer
TX_WAIT_ERROR	0x04	<code>k_rx_err_rtos_error</code>	Invalid wait option
TX_SIZE_ERROR	0x05	<code>k_rx_err_rtos_error</code>	Invalid size
TX_CALLER_ERROR	0x13	<code>k_rx_err_rtos_error</code>	Invalid caller context
TX_NOT_AVAILABLE	0x1D	<code>k_rx_err_rtos_error</code>	Resource not available
(all other)	*	<code>k_rx_err_rtos_error</code>	Generic RTOS error

Note

This is a simplified conversion. For finer-grained error mapping, inspect the ThreadX status code directly and map to specific `rx_err_t` codes.

Parameters

in	tx_status	ThreadX status code returned from ThreadX API call - Type: UINT (32-bit unsigned integer in ThreadX) - Valid range: Any value (typically 0x00-0x1D for defined ThreadX errors) - Special values: 0 (TX_SUCCESS): Indicates successful operation - Non-zero: Various ThreadX error conditions
----	-----------	--

Returns

`rx_err_t` error code

Return values

<code>k_rx_ok</code>	(0x000) If tx_status == 0 (TX_SUCCESS), operation succeeded
<code>k_rx_err_rtos_error</code>	(0x301) If tx_status != 0, ThreadX error occurred

Usage Example - Basic Conversion:

```
#include "rx_err.h"
#include "tx_api.h"

TX_THREAD my_thread;
CHAR thread_stack[1024];

rx_err_t create_worker_thread(void) {
    UINT tx_status = tx_thread_create(
        &my_thread,           // Thread control block
        "Worker",            // Thread name
        worker_thread_entry,  // Entry function
        0,                   // Entry input
        thread_stack,        // Stack start
        sizeof(thread_stack), // Stack size
        16,                  // Priority
        16,                  // Preemption threshold
        TX_NO_TIME_SLICE,    // Time slice
        TX_AUTO_START        // Auto start
    );

    // Convert ThreadX status to rx_err_t
    rx_err_t err = rx_err_from_threadx(tx_status);
    if (err != k_rx_ok) {
        rx_log_error("RTOS", "Thread creation failed: TX status 0x%02X", tx_status);
        return err;
    }

    return k_rx_ok;
}
```

Usage Example - Inline in Error Check:

```
rx_err_t init_rtos_resources(void) {
    // Create semaphore and convert result inline
    rx_err_t err = rx_err_from_threadx(
        tx_semaphore_create(&my_sem, "DataReady", 0)
    );

    if (err != k_rx_ok) {
        return err; // Propagate RTOS error
    }

    // Continue with other RTOS operations...
    return k_rx_ok;
}
```

Performance:

- Execution time: ~2 cycles @ 240 MHz (CMP + CMOV)
- Stack usage: 0 bytes (inlined by compiler)
- Code size: 0 bytes when inlined (optimized to single instruction)

Note

Thread Safety: Pure function, inherently thread-safe (no shared state)

Re-entrancy: Fully reentrant (no side effects)

Memory: No dynamic allocation, no static variables

Precondition

tx_status contains valid ThreadX status code (any UINT value accepted)

Postcondition

Return value is either k_rx_ok or k_rx_err_rtos_error (no other values possible)

Invariant

Result is always one of two values: k_rx_ok or k_rx_err_rtos_error

See also

[k_rx_err_rtos_error](#) Error code returned for all non-success ThreadX status

[k_rx_ok](#) Success code returned for TX_SUCCESS

[tx_thread_create\(\)](#) Example ThreadX API that returns UINT status

NASA Power of 10 Compliance:

- Rule 1: [OK] No goto, setjmp, recursion
- Rule 2: [OK] N/A (no loops)
- Rule 3: [OK] No dynamic allocation
- Rule 4: [OK] Function is 1 line (excluding documentation)
- Rule 5: [OK] 1 precondition, 1 postcondition, 1 invariant
- Rule 6: [OK] Minimal scope (inline function)
- Rule 7: [OK] No function calls to check
- Rule 8: [OK] Uses typed enum constants
- Rule 9: [OK] No function pointers
- Rule 10: [OK] Compiles with -Wall -Wextra -Werror

Since

Version 1.0.0

Definition at line 1153 of file [rx_err.h](#).

```
01154 {
01155     return (tx_status == 0) ? k_rx_ok : k_rx_err_rtos_error;
01156 }
```

References [k_rx_err_rtos_error](#), and [k_rx_ok](#).

7.5.4.4 rx_err_is_error()

```
bool rx_err_is_error (
    rx\_err\_t err)    [inline], [static]
```

Check if error code represents failure.

Tests whether an [rx_err_t](#) value indicates a failure condition (any non-zero value). This is the logical inverse of [rx_err_is_ok\(\)](#) and is used when error-checking logic is more naturally expressed as "if error occurred" rather than "if success".

7.5.4.5 Algorithm

1. Compare err against k_rx_ok (0)
2. Return true if not equal (error), false otherwise (success)

7.5.4.6 When to Use This vs rx_err_is_ok()

Use [rx_err_is_error\(\)](#) when error handling is the primary focus:

```
rx_err_t err = risky_operation();
if (rx_err_is_error(err)) {
    // Handle error first (primary concern)
    handle_error(err);
    return;
}
// Continue with normal path
```

Use [rx_err_is_ok\(\)](#) when success path is the primary focus:

```
rx_err_t err = initialize_system();
if (rx_err_is_ok(err)) {
    // Success is the expected/happy path
    continue_normal_operation();
} else {
    // Error is exceptional
    handle_error(err);
}
```

Parameters

in	err	Error code to check
		- Type: <code>rx_err_t</code> (<code>int32_t</code>) - Valid range: Any <code>rx_err_t</code> value (typically 0x000-0x5FF or 0) - Special values: <code>k_rx_ok</code> (0): Success condition (returns false) - Any non-zero: Error condition (returns true)

Returns

Boolean error indicator

Return values

true	If <code>err != k_rx_ok</code> (non-zero), operation failed
false	If <code>err == k_rx_ok</code> (0), operation was successful

Usage Example - Early Return on Error:

```
rx_err_t initialize_peripherals(void) {
    rx_err_t err;

    err = init_gpio();
    if (rx_err_is_error(err)) {
        return err; // Early return on error
    }

    err = init_spi();
    if (rx_err_is_error(err)) {
        return err; // Propagate error immediately
    }

    err = init_i2c();
    if (rx_err_is_error(err)) {
        return err;
    }

    return k_rx_ok; // All initialization succeeded
}
```

Usage Example - Error Logging:

```
rx_err_t err = send_command(cmd);
if (rx_err_is_error(err)) {
    // Log error with details
    rx_log_error("COMM", "Command send failed: 0x%03X", err);

    // Category-specific handling
    if ((err & 0xF00) == 0x400) {
        reset_communication();
    }
}
```

Usage Example - Error Accumulation:

```

uint32_t error_count = 0;
for (uint8_t i = 0; i < k_sensor_count; i++) {
    rx_err_t err = read_sensor(i);
    if (rx_err_is_error(err)) {
        error_count++;
        rx_log_warn("SENSOR", "Sensor %u failed: 0x%X", i, err);
    }
}

if (error_count > k_max_allowed_failures) {
    enter_degraded_mode();
}

```

Performance:

- Execution time: 1 cycle @ 240 MHz (single CMP instruction)
- Stack usage: 0 bytes (inlined by compiler)
- Code size: 0 bytes when inlined (optimized to single comparison)

Note

Thread Safety: Pure function, inherently thread-safe (no shared state)

Re-entrancy: Fully reentrant (no side effects, no mutable state)

Memory: No allocation, no static variables

Precondition

err contains valid `rx_err_t` value (any `int32_t` accepted)

Postcondition

Return value is deterministic boolean based solely on input

Invariant

Result is always boolean (true or false)

Exactly inverse of `rx_err_is_ok()`: `rx_err_is_error(e) == !rx_err_is_ok(e)`

See also

`rx_err_is_ok()` Inverse check (returns true if success, false if error)

`k_rx_ok` Success constant (0x000)

NASA Power of 10 Compliance:

- Rule 1: [OK] No goto, setjmp, recursion
- Rule 2: [OK] N/A (no loops)
- Rule 3: [OK] No dynamic allocation
- Rule 4: [OK] Function is 1 line
- Rule 5: [OK] 1 precondition, 1 postcondition, 2 invariants
- Rule 6: [OK] Minimal scope (inline function)
- Rule 7: [OK] No function calls to check
- Rule 8: [OK] Uses typed enum constant `k_rx_ok`
- Rule 9: [OK] No function pointers
- Rule 10: [OK] Compiles with -Wall -Wextra -Werror

Since

Version 1.0.0

Definition at line 1394 of file [rx_err.h](#).

```
01395 {
01396     return (bool)(err != k_rx_ok);
01397 }
```

References [k_rx_ok](#).

7.5.4.7 rx_err_is_ok()

```
bool rx_err_is_ok (
    rx\_err\_t err)    [inline], [static]
```

Check if error code represents success.

Tests whether an [rx_err_t](#) value indicates successful operation ([k_rx_ok](#)). This is the preferred way to check for success in STAR firmware, providing explicit intent and type safety.

7.5.4.8 Algorithm

1. Compare err against [k_rx_ok](#) (0)
2. Return true if equal (success), false otherwise (error)

7.5.4.9 Comparison to Alternatives

Method	Type Safe	Explicit Intent	Performance
rx_err_is_ok(err)	[OK]	[OK]	1 cycle
<code>err == k_rx_ok</code>	[OK]	[OK]	1 cycle
<code>!err</code>	[X]	[X]	1 cycle
<code>err == 0</code>	[X]	[X]	1 cycle

Recommended: Use [rx_err_is_ok\(err\)](#) for explicit intent and self-documenting code.

Parameters

in	err	Error code to check
		• Type: rx_err_t (int32_t) • Valid range: Any rx_err_t value (typically 0x000-0x5FF or 0) • Special values: – k_rx_ok (0): Success condition – Any non-zero: Error condition

Returns

Boolean success indicator

Return values

true	If err == k_rx_ok (0), operation was successful
false	If err != k_rx_ok (any non-zero value), operation failed

Usage Example - Basic Success Check:

```
rx_err_t err = initialize_motor();
if (rx_err_is_ok(err)) {
    // Success path - motor initialized
    start_control_loop();
} else {
    // Error path - log and handle failure
    rx_log_error("MOTOR", "Init failed: 0x%X", err);
    enter_safe_state();
}
```

Usage Example - Assert Success:

```
// Critical operation that must succeed
rx_err_t err = init_critical_subsystem();
assert(rx_err_is_ok(err) && "Critical init failed"); // NASA Rule 5
```

Usage Example - Conditional Compilation:

```
#if DEBUG_LEVEL >= 2
    rx_err_t err = optional_debug_feature();
    if (!rx_err_is_ok(err)) {
        rx_log_warn("DEBUG", "Optional feature unavailable");
    }
#endif
```

Performance:

- Execution time: 1 cycle @ 240 MHz (single CMP instruction)
- Stack usage: 0 bytes (inlined by compiler)
- Code size: 0 bytes when inlined (optimized to single comparison)

Note

Thread Safety: Pure function, inherently thread-safe (no shared state)

Re-entrancy: Fully reentrant (no side effects, no mutable state)

Memory: No allocation, no static variables

Precondition

err contains valid `rx_err_t` value (any `int32_t` accepted, but typically 0x000-0x5FF)

Postcondition

Return value is deterministic boolean based solely on input

Invariant

Result is always boolean (true or false, no other values)

See also

[rx_err_is_error\(\)](#) Inverse check (returns true if error, false if success)

[k_rx_ok](#) Success constant (0x000)

NASA Power of 10 Compliance:

- Rule 1: [OK] No goto, setjmp, recursion
- Rule 2: [OK] N/A (no loops)
- Rule 3: [OK] No dynamic allocation
- Rule 4: [OK] Function is 1 line
- Rule 5: [OK] 1 precondition, 1 postcondition, 1 invariant
- Rule 6: [OK] Minimal scope (inline function)
- Rule 7: [OK] No function calls to check
- Rule 8: [OK] Uses typed enum constant [k_rx_ok](#)
- Rule 9: [OK] No function pointers
- Rule 10: [OK] Compiles with -Wall -Wextra -Werror

Since

Version 1.0.0

Definition at line [1254](#) of file [rx_err.h](#).

```
01255 {
01256     return (bool)(err == k_rx_ok);
01257 }
```

References [k_rx_ok](#).

7.6 rx_err.h

[Go to the documentation of this file.](#)

```
00001 /**
00002  * @file rx_err.h
00003  * @brief Error Code Definitions for RX72N Firmware
00004  *
00005  * @details
00006  * Comprehensive error code system for the STAR RX72N motor controller firmware.
00007  * Provides type-safe error handling with categorized error codes following
00008  * C23 typed enum standards for embedded safety-critical systems.
00009  *
00010  * ### Architecture Design
00011  *
00012  * This module implements a centralized error handling system for safety-critical
00013  * embedded requirements:
00014  * - **Type-safe:** Uses C23 typed enums (uint16_t underlying type)
00015  * - **Zero overhead:** Inline helper functions compile to single instructions
00016  * - **Extensible:** Category-based organization supports future error codes
00017  * - **Traceable:** Each error code maps to specific failure conditions
00018  *
00019  * ### Error Code Organization
00020  *
00021  * Error codes are organized into non-overlapping ranges to facilitate debugging
00022  * and logging. The hex representation makes category identification trivial:
```

```

00023 *
00024 * | Range      | Category      | Purpose
00025 * |-----|-----|-----|
00026 * | 0x000      | Success       | Operation completed successfully
00027 * | 0x100-0x1FF | Generic       | General-purpose errors (null, timeout)
00028 * | 0x200-0x2FF | Hardware      | Peripheral and GPIO errors
00029 * | 0x300-0x3FF | RTOS          | ThreadX and synchronization errors
00030 * | 0x400-0x4FF | Communication | SPI, UART, I2C, protocol errors
00031 * | 0x500-0x5FF | Validation    | Input validation and checksum errors
00032 *
00033 * ## Design Rationale
00034 *
00035 * **Why signed int32_t for rx_err_t but unsigned uint16_t for enum?**
00036 * - rx_err_t uses int32_t to allow negative error codes (future compatibility)
00037 * - Enum uses uint16_t to minimize memory usage (most values < 0xFFFF)
00038 * - Implicit conversion is safe: uint16_t -> int32_t always preserves value
00039 *
00040 * **Why category-based ranges instead of linear numbering?**
00041 * - Enables fast error categorization in logging (e.g., "0x2xx = hardware")
00042 * - Supports future expansion within categories without renumbering
00043 * - Makes error codes human-readable in hex dumps and debuggers
00044 *
00045 * **Why inline helpers instead of macros?**
00046 * - Type safety: compiler checks parameter types
00047 * - Debuggable: can step into functions, set breakpoints
00048 * - No macro pitfalls: no multiple evaluation, proper scoping
00049 *
00050 * ## Performance Characteristics
00051 *
00052 * All operations are zero-cost abstractions:
00053 * - rx_err_from_threadx(): Single comparison + conditional move (1-2 cycles)
00054 * - rx_err_is_ok(): Single comparison (1 cycle)
00055 * - rx_err_is_error(): Single comparison (1 cycle)
00056 *
00057 * ## Memory Usage
00058 *
00059 * - rx_err_t: 4 bytes per error variable
00060 * - rx_err_codes_t: 2 bytes per enum value
00061 * - Helper functions: 0 bytes (inlined by compiler at -O1+)
00062 *
00063 * ## Thread Safety
00064 *
00065 * All functions in this module are inherently thread-safe (pure functions).
00066 * Error codes are read-only constants after compilation.
00067 *
00068 * @par Module Dependencies:
00069 * - stdint.h: Fixed-width integer types
00070 * - stdbool.h: Boolean type for helper functions
00071 *
00072 * @see rx_check.h Validation macros that return rx_err_t codes
00073 * @see rx_error_handler.h Error logging and handling infrastructure
00074 * @see docs/sections/06_nasa_power_of_10.tex NASA compliance documentation
00075 *
00076 * @par NASA Power of 10 Compliance:
00077 * - Rule 1: [OK] No goto, setjmp, recursion (pure constant definitions)
00078 * - Rule 2: [OK] N/A (no loops)
00079 * - Rule 3: [OK] No dynamic allocation (compile-time constants only)
00080 * - Rule 4: [OK] All functions < 10 lines (simple inline helpers)
00081 * - Rule 5: [OK] Inline functions have implicit pre/post conditions
00082 * - Rule 6: [OK] All identifiers have minimal scope
00083 * - Rule 7: [OK] N/A (no function calls to check)
00084 * - Rule 8: [OK] Uses C23 typed enums (k_rx_err_codes_t : uint16_t)
00085 * - Rule 9: [OK] N/A (no function pointers)
00086 * - Rule 10: [OK] Compiles with -Wall -Wextra -Werror
00087 *
00088 * @par SOLID Principles:
00089 * - **S (Single Responsibility):** Only error code definitions, no handling logic
00090 * - **O (Open/Closed):** Extensible via new enum values, closed for modification
00091 * - **L (Liskov Substitution):** rx_err_t can substitute for int32_t in APIs
00092 * - **I (Interface Segregation):** Minimal interface (3 helper functions)
00093 * - **D (Dependency Inversion):** No dependencies on concrete implementations
00094 *
00095 * @author Locked, Inc.
00096 * @date 2026-01-27
00097 * @version 1.0.0
00098 * @copyright Copyright (c) 2026 Locked Inc.
00099 * SPDX-License-Identifier: MIT
00100 */
00101
00102 #pragma once
00103
00104 #include <stdint.h>
00105
00106 #ifdef __cplusplus
00107 extern "C" {
00108 #endif
00109

```

```

00110 /*
=====
00111 * Error Type Definition
00112 *
=====
00113 */
00114 /**
00115  * @typedef rx_err_t
00116  * @brief Error code type for RX72N firmware
00117  *
00118  * @details
00119  * Signed 32-bit integer type used for all function return values that
00120  * indicate success or failure. This type provides a consistent error
00121  * handling interface across the entire firmware codebase.
00122  *
00123  * ## Type Selection Rationale
00124  *
00125  * **Why int32_t instead of enum?**
00126  * - Allows future negative error codes (e.g., -1 for EOF-like conditions)
00127  * - Compatible with standard POSIX error conventions
00128  * - Sufficient range for all error categories (only using 0x000-0x5FF currently)
00129  * - Efficient on 32-bit ARM Cortex-R4 (native register width)
00130  *
00131  * **Why signed instead of unsigned?**
00132  * - Enables use of negative values for special conditions
00133  * - Matches C standard library conventions (errno, etc.)
00134  * - Prevents accidental unsigned wraparound bugs
00135  *
00136  * ## Value Semantics
00137  *
00138  * - **0 (k_rx_ok):** Success - operation completed as requested
00139  * - **Non-zero:** Error - operation failed, consult specific error code
00140  * - **Positive values:** Standard error codes (0x100-0x5FF)
00141  * - **Negative values:** Reserved for future use
00142  *
00143  * ## Usage Pattern
00144  *
00145  * @par Standard Error Checking:
00146  * @code{.c}
00147  * rx_err_t err = some_function();
00148  * if (err != k_rx_ok) {
00149  *     // Handle error
00150  *     rx_log_error("TAG", "Operation failed: 0x%X", err);
00151  *     return err; // Propagate error up call stack
00152  * }
00153  * @endcode
00154  *
00155  * @par Early Return Pattern:
00156  * @code{.c}
00157  * rx_err_t init_subsystem(void) {
00158  *     rx_err_t err;
00159  *
00160  *     err = init_hardware();
00161  *     if (err != k_rx_ok) return err;
00162  *
00163  *     err = init_rtos();
00164  *     if (err != k_rx_ok) return err;
00165  *
00166  *     return k_rx_ok;
00167  * }
00168  * @endcode
00169  *
00170  * @par Switch-Based Error Handling:
00171  * @code{.c}
00172  * rx_err_t err = spi_transfer(data, len);
00173  * switch (err) {
00174  *     case k_rx_ok:
00175  *         // Success path
00176  *         break;
00177  *     case k_rx_err_timeout:
00178  *         // Retry logic
00179  *         retry_transfer();
00180  *         break;
00181  *     case k_rx_err_nack:
00182  *         // Device not responding
00183  *         reset_device();
00184  *         break;
00185  *     default:
00186  *         // Unexpected error
00187  *         log_error(err);
00188  *         break;
00189  * }
00190  * @endcode
00191  *
00192  * @par Macro-Based Error Propagation:
00193  * @code{.c}

```

```

00195 * #define RX_RETURN_ON_ERROR(err, tag, msg) \
00196 *     do { \
00197 *         rx_err_t _err = (err); \
00198 *         if (_err != k_rx_ok) { \
00199 *             rx_log_error((tag), (msg)); \
00200 *             return _err; \
00201 *         } \
00202 *     } while (0)
00203 *
00204 * rx_err_t init_system(void) {
00205 *     RX_RETURN_ON_ERROR(init_gpio(), "INIT", "GPIO init failed");
00206 *     RX_RETURN_ON_ERROR(init_spi(), "INIT", "SPI init failed");
00207 *     return k_rx_ok;
00208 * }
00209 * @endcode
00210 *
00211 * @note This type is used by ALL functions that can fail. Consistent usage
00212 *       across the codebase enables systematic error handling and logging.
00213 *
00214 * @warning Never cast arbitrary integers to rx_err_t. Only use predefined
00215 *          k_rx_* constants or values returned from error conversion functions.
00216 *
00217 * @attention Functions returning rx_err_t MUST document all possible return
00218 *            values using @retval tags in their Doxygen documentation.
00219 *
00220 * @par Memory Layout:
00221 * - Size: 4 bytes
00222 * - Alignment: 4 bytes (native word on ARM Cortex-R4)
00223 * - Signedness: Signed (two's complement)
00224 * - Range: -2,147,483,648 to 2,147,483,647 (only 0-0x5FF currently used)
00225 *
00226 * @par Performance:
00227 * - Return value overhead: 0 cycles (returned in register r0)
00228 * - Comparison overhead: 1 cycle (CMP r0, #0)
00229 * - No heap allocation required
00230 *
00231 * @see rx_err_codes_t Enumeration of all defined error codes
00232 * @see rx_err_is_ok() Helper to check for success
00233 * @see rx_err_is_error() Helper to check for failure
00234 * @see rx_err_from_threadx() Convert ThreadX codes to rx_err_t
00235 *
00236 * @since Version 1.0.0
00237 */
00238 typedef int32_t rx_err_t;
00239
00240 /*
00241 * Error Codes (Following Style Guide: Enums > Const > Macros)
00242 *
00243 */
00244 /**
00245 * @enum rx_err_codes_t
00246 * @brief Comprehensive error code enumeration for RX72N firmware
00247 *
00248 * @details
00249 * Type-safe error codes using C23 typed enum syntax with explicit uint16_t
00250 * underlying type. All error codes are compile-time constants organized into
00251 * logical categories for systematic error handling and debugging.
00252 *
00253 * ## Error Code Categories
00254 *
00255 * The error code space is partitioned into non-overlapping ranges:
00256 *
00257 * | Range       | Category | Count | Purpose |
00258 * |-----|-----|-----|-----|
00259 * | 0x000      | Success  | 1     | Operation completed successfully |
00260 * | 0x100-0x1FF | Generic  | 15    | General-purpose errors (null, timeout, etc.) |
00261 * | 0x200-0x2FF | Hardware | 9     | Peripheral, GPIO, and sensor errors |
00262 * | 0x300-0x3FF | RTOS     | 6     | ThreadX and synchronization errors |
00263 * | 0x400-0x4FF | Communication | 8 | SPI, UART, I2C, and protocol errors |
00264 * | 0x500-0x5FF | Validation | 4 | Input validation and checksum errors |
00265 *
00266 * ## Error Category Flow
00267 *
00268 * Typical error propagation through the system:
00269 *
00270 * @dot
00271 * digraph error_flow {
00272 *     rankdir=LR;
00273 *     node [shape=box, style=rounded];
00274 *
00275 *     hardware [label="Hardware Layer\n(0x2xx)", fillcolor=lightblue, style=filled];
00276 *     driver [label="Driver Layer\n(0x4xx)", fillcolor=lightgreen, style=filled];
00277 *     rtos [label="RTOS Layer\n(0x3xx)", fillcolor=lightyellow, style=filled];
00278 *     validation [label="Validation\n(0x5xx)", fillcolor=lightpink, style=filled];
00279 *

```

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00280 * generic [label="Generic Errors\n(0x1xx)", fillcolor=lightgray, style=filled];
00281 * app [label="Application", fillcolor=white];
00282 *
00283 * hardware -> driver [label="hw_error"];
00284 * driver -> app [label="comm_error"];
00285 * rtos -> app [label="rtos_error"];
00286 * validation -> driver [label="invalid_arg"];
00287 * generic -> app [label="timeout/busy"];
00288 * }
00289 * @enddot
00290 *
00291 * ## Error Handling Patterns
00292 *
00293 * @par Pattern 1: Direct Comparison
00294 * @code{.c}
00295 * rx_err_t err = motor_start();
00296 * if (err == k_rx_ok) {
00297 *     // Success
00298 * } else if (err == k_rx_err_estop) {
00299 *     // Emergency stop active - cannot start
00300 * } else {
00301 *     // Other error - log and handle
00302 * }
00303 * @endcode
00304 *
00305 * @par Pattern 2: Category Check
00306 * @code{.c}
00307 * rx_err_t err = spi_transfer();
00308 * if ((err & 0xF00) == 0x400) {
00309 *     // Communication error category
00310 *     restart_communication();
00311 * } else if ((err & 0xF00) == 0x200) {
00312 *     // Hardware error category
00313 *     reset_hardware();
00314 * }
00315 * @endcode
00316 *
00317 * @par Pattern 3: Error Logging with Context
00318 * @code{.c}
00319 * rx_err_t err = init_subsystem();
00320 * if (err != k_rx_ok) {
00321 *     rx_log_error("INIT", "Subsystem init failed: 0x%03X", err);
00322 *     // Hex dump shows category: 0x2xx = hardware, 0x4xx = comm, etc.
00323 * }
00324 * @endcode
00325 *
00326 * @par Usage Example - Complete Error Handling:
00327 * @code{.c}
00328 * #include "rx_err.h"
00329 * #include "rx_log.h"
00330 *
00331 * rx_err_t initialize_motor_controller(void) {
00332 *     rx_err_t err;
00333 *
00334 *     // Initialize hardware
00335 *     err = init_gpio();
00336 *     if (err != k_rx_ok) {
00337 *         if (err == k_rx_err_gpio_conflict) {
00338 *             rx_log_error("MOTOR", "GPIO conflict - check pin assignments");
00339 *         }
00340 *         return err;
00341 *     }
00342 *
00343 *     // Initialize SPI for RPi5 communication
00344 *     err = init_spi();
00345 *     if (err != k_rx_ok) {
00346 *         if (err == k_rx_err_spi_error) {
00347 *             rx_log_error("SPI", "SPI initialization failed");
00348 *         }
00349 *         return err;
00350 *     }
00351 *
00352 *     // Create RTOS resources
00353 *     err = create_motor_thread();
00354 *     if (err != k_rx_ok) {
00355 *         if (err == k_rx_err_rtos_thread_create) {
00356 *             rx_log_error("MOTOR", "Thread creation failed - out of memory?");
00357 *         }
00358 *         return err;
00359 *     }
00360 *
00361 *     return k_rx_ok;
00362 * }
00363 * @endcode
00364 *
00365 * @invariant All error codes except k_rx_ok are non-zero
00366 * @invariant Error code ranges never overlap between categories

```

```

00367 * @invariant Error codes are compile-time constants (never modified at runtime)
00368 *
00369 * @note Uses C23 typed enum syntax `: uint16_t' for guaranteed size and ABI stability
00370 * @note Error codes are designed to be human-readable in hex (category visible)
00371 * @note All error codes fit in uint16_t (0x000-0x5FF) for memory efficiency
00372 *
00373 * @warning Do not cast arbitrary integers to this enum - only use defined constants
00374 * @warning Do not modify error codes at runtime - they are compile-time constants
00375 *
00376 * @see rx_err_t Type alias for function return values
00377 * @see rx_err_is_ok() Helper to check for success
00378 * @see rx_err_is_error() Helper to check for failure
00379 *
00380 * @since Version 1.0.0
00381 */
00382 typedef enum : uint16_t {
00383 /**
00384  * @brief Success - operation completed as requested
00385  * @details
00386  * The only non-error value. Indicates successful completion of an operation
00387  * with no errors, warnings, or exceptional conditions. All outputs are valid
00388  * and postconditions are satisfied.
00389  *
00390  * @par Value: 0x000
00391  * @par Usage: Return this when function succeeds completely
00392  * @par Example:
00393  * @code{.c}
00394  * rx_err_t validate_input(uint8_t value) {
00395  *     if (value < 100) {
00396  *         return k_rx_ok; // Valid input
00397  *     }
00398  *     return k_rx_err_out_of_range;
00399  * }
00400  * @endcode
00401  */
00402 k_rx_ok = 0,
00403
00404 /*
=====
00405 * Generic Errors (0x100 - 0x1FF)
00406 * General-purpose errors not specific to hardware or communication
00407 *
=====
00408 */
00409 /**
00410  * @brief Generic unspecified failure
00411  * @details
00412  * Used when a function fails but the specific error type is unknown or
00413  * doesn't fit other categories. Prefer more specific error codes when possible.
00414  *
00415  * @par Value: 0x101
00416  * @par When to use: Last resort when no specific error applies
00417  * @par Recovery: Log error, inspect code path to determine root cause
00418  * @see Use more specific errors like k_rx_err_invalid_arg when applicable
00419  */
00420 k_rx_fail = 0x101,
00421
00422 /**
00423  * @brief Out of memory - allocation failed
00424  * @details
00425  * Memory allocation failed (heap, stack, or pool). In safety-critical code
00426  * with zero dynamic allocation (NASA Rule 3), this indicates static buffer
00427  * exhaustion (e.g., ring buffer full, pool depleted).
00428  *
00429  * @par Value: 0x102
00430  * @par Cause: Static buffer full, RTOS pool exhausted, stack overflow imminent
00431  * @par Recovery: Release resources, reduce load, increase buffer size
00432  * @par Prevention: Size buffers based on worst-case analysis
00433  * @warning In zero-allocation systems, this indicates design issue (buffers too small)
00434  */
00435 k_rx_err_no_mem = 0x102,
00436
00437 /**
00438  * @brief Invalid function argument
00439  * @details
00440  * Caller provided parameter outside valid range, nullptr where non-NULL
00441  * required, or logically inconsistent arguments (e.g., buffer_size > max_size).
00442  *
00443  * @par Value: 0x103
00444  * @par Cause: Parameter validation failed in function preconditions
00445  * @par Recovery: Fix caller code, add validation at call site
00446  * @par Prevention: Use RX_CHECK_NULL_PTR and RX_CHECK_RANGE macros
00447  * @see k_rx_err_null_ptr More specific error for nullptr pointers
00448  * @see k_rx_err_range_check_failed More specific for range violations
00449  */
00450 k_rx_err_invalid_arg = 0x103,

```



```

00452
00453 /**
00454  * @brief Invalid state for requested operation
00455  * @details
00456  * Function called when module is in wrong state. For example, calling
00457  * motor_start() when motor is already running, or spi_transfer() before
00458  * spi_init() completed successfully.
00459  *
00460  * @par Value: 0x104
00461  * @par Cause: State machine in unexpected state for requested operation
00462  * @par Recovery: Check module state, call initialization functions first
00463  * @par Prevention: Document valid state transitions, enforce with state checks
00464  * @see k_rx_err_not_initialized Specific case of invalid state (not initialized)
00465  */
00466 k_rx_err_invalid_state = 0x104,
00467
00468 /**
00469  * @brief Invalid size parameter
00470  * @details
00471  * Size parameter outside valid range. Examples: buffer size exceeds maximum,
00472  * packet length too short, array size is zero.
00473  *
00474  * @par Value: 0x105
00475  * @par Cause: Size parameter failed validation (too large, too small, misaligned)
00476  * @par Recovery: Adjust size parameter to valid range
00477  * @par Prevention: Document valid size ranges in function documentation
00478  * @see k_rx_err_invalid_arg More general argument validation error
00479  */
00480 k_rx_err_invalid_size = 0x105,
00481
00482 /**
00483  * @brief Requested item not found
00484  * @details
00485  * Search, lookup, or query operation failed to find requested item.
00486  * Examples: device not found on bus, configuration key missing, motor ID invalid.
00487  *
00488  * @par Value: 0x106
00489  * @par Cause: Item does not exist in collection, device not on bus, ID not registered
00490  * @par Recovery: Verify item exists, check ID is correct, enumerate available items
00491  * @par Prevention: Validate IDs before use, document valid ID ranges
00492  */
00493 k_rx_err_not_found = 0x106,
00494
00495 /**
00496  * @brief Feature or operation not supported
00497  * @details
00498  * Requested functionality not implemented in this build or hardware configuration.
00499  * Examples: hardware feature not available on this MCU variant, optional module
00500  * not compiled in, operation not supported by peripheral.
00501  *
00502  * @par Value: 0x107
00503  * @par Cause: Feature not implemented, hardware limitation, compile-time config
00504  * @par Recovery: Use alternative approach, check hardware capabilities
00505  * @par Prevention: Document supported features, provide capability query API
00506  */
00507 k_rx_err_not_supported = 0x107,
00508
00509 /**
00510  * @brief Operation timed out
00511  * @details
00512  * Operation exceeded allowed time limit. Examples: peripheral not ready within
00513  * timeout, RTOS wait timeout expired, communication timeout (no response).
00514  *
00515  * @par Value: 0x108
00516  * @par Cause: Peripheral hung, device not responding, deadlock, excessive load
00517  * @par Recovery: Retry operation, reset peripheral, increase timeout
00518  * @par Prevention: Tune timeouts based on empirical testing, implement watchdogs
00519  * @see k_rx_err_hw_timeout Hardware-specific timeout
00520  */
00521 k_rx_err_timeout = 0x108,
00522
00523 /**
00524  * @brief Resource busy - operation cannot proceed
00525  * @details
00526  * Requested resource is currently in use and cannot service request.
00527  * Examples: SPI bus busy, mutex locked by other thread, DMA channel active.
00528  *
00529  * @par Value: 0x109
00530  * @par Cause: Resource locked, peripheral busy, exclusive access required
00531  * @par Recovery: Wait for resource, retry later, queue request
00532  * @par Prevention: Implement resource arbitration, use RTOS synchronization
00533  * @see k_rx_err_would_block Non-blocking operation would block
00534  */
00535 k_rx_err_busy = 0x109,
00536
00537 /**
00538  * @brief No application data available (control frame received)

```

```

00539 * @details
00540 * Operation succeeded but no application-layer data was returned. Distinct from
00541 * k_rx_err_timeout - receive was successful but the frame was a control frame
00542 * (ACK/NACK/PING/PONG/RESET) not intended for application consumption.
00543 *
00544 * @par Value: 0x10A
00545 * @par Cause: Received control frame instead of data frame
00546 * @par Recovery: This is normal - caller should continue polling
00547 * @par Prevention: Not an error condition - expected behavior
00548 * @see k_rx_err_timeout No data due to timeout
00549 */
00550 k_rx_err_no_data = 0x10A,
00551
00552 /**
00553 * @brief Non-blocking operation would block
00554 * @details
00555 * Non-blocking function called but operation would block waiting for resource.
00556 * Differs from k_rx_err_busy: this indicates caller requested non-blocking mode
00557 * but operation requires waiting.
00558 *
00559 * @par Value: 0x10B
00560 * @par Cause: Non-blocking mode requested but data/resource not immediately available
00561 * @par Recovery: Retry later, use blocking mode, or poll until ready
00562 * @par Prevention: Check resource availability before calling non-blocking API
00563 * @see k_rx_err_busy Resource busy in blocking mode
00564 */
00565 k_rx_err_would_block = 0x10B,
00566
00567 /**
00568 * @brief Item already exists - cannot create
00569 * @details
00570 * Attempted to create resource that already exists. Examples: thread already
00571 * created, device already registered, configuration already initialized.
00572 *
00573 * @par Value: 0x10C
00574 * @par Cause: Duplicate creation attempt, resource already allocated
00575 * @par Recovery: Use existing resource, or delete and recreate
00576 * @par Prevention: Check existence before creation, track resource state
00577 */
00578 k_rx_err_exists = 0x10C,
00579
00580 /**
00581 * @brief Container empty - no data available
00582 * @details
00583 * Attempted to retrieve data from empty container. Examples: ring buffer empty,
00584 * queue empty, receive FIFO empty.
00585 *
00586 * @par Value: 0x10D
00587 * @par Cause: No data available, all items consumed
00588 * @par Recovery: Wait for data, check count before reading
00589 * @par Prevention: Check count/size before attempting read
00590 * @see k_rx_err_would_block Non-blocking read from empty container
00591 */
00592 k_rx_err_empty = 0x10D,
00593
00594 /**
00595 * @brief Operation cancelled by request
00596 * @details
00597 * Operation was cancelled before completion, either by explicit cancel request
00598 * or by system shutdown. No side effects occurred (atomic cancellation).
00599 *
00600 * @par Value: 0x10E
00601 * @par Cause: User cancel request, system shutdown, higher priority task
00602 * @par Recovery: Clean up resources, update state to reflect cancellation
00603 * @par Prevention: Implement graceful cancellation support
00604 */
00605 k_rx_err_cancelled = 0x10E,
00606
00607 /**
00608 * @brief Module not initialized
00609 * @details
00610 * Function called before module initialization completed. All modules must be
00611 * initialized via their init() function before use.
00612 *
00613 * @par Value: 0x10F
00614 * @par Cause: init() not called, init() failed, module deinitialized
00615 * @par Recovery: Call module_init() before using module functions
00616 * @par Prevention: Document initialization requirements, enforce with state checks
00617 * @see k_rx_err_invalid_state More general state error
00618 */
00619 k_rx_err_not_initialized = 0x10F,
00620
00621 /**
00622 * @brief Emergency stop active - operation forbidden
00623 * @details
00624 * Emergency stop triggered, all motor operations halted. System in safe state,
00625 * requires manual reset before normal operation can resume.

```

```

00626 *
00627 * @par Value: 0x110
00628 * @par Cause: Emergency stop button pressed, safety fault detected
00629 * @par Recovery: Clear fault condition, call reset function, acknowledge e-stop
00630 * @par Prevention: Implement proper safety interlocks
00631 * @warning This is a safety-critical error - do not bypass e-stop checks
00632 */
00633 k_rx_err_estop = 0x110,
00634
00635 /*
=====
00636 * Hardware Errors (0x200 - 0x2FF)
00637 * Peripheral, GPIO, and hardware interface errors
00638 *
=====
00639 */
00640
00641 /**
00642 * @brief Hardware initialization failed
00643 * @details
00644 * Hardware peripheral failed to initialize. Examples: peripheral not responding,
00645 * clock configuration failed, register write verification failed.
00646 *
00647 * @par Value: 0x201
00648 * @par Cause: Hardware fault, incorrect configuration, missing hardware
00649 * @par Recovery: Check hardware connections, verify clock settings, reset MCU
00650 * @par Prevention: Validate hardware presence before init
00651 */
00652 k_rx_err_hw_init_failed = 0x201,
00653
00654 /**
00655 * @brief Hardware not ready for operation
00656 * @details
00657 * Hardware peripheral exists but not ready for requested operation. Examples:
00658 * PLL not locked, ADC not calibrated, peripheral in reset state.
00659 *
00660 * @par Value: 0x202
00661 * @par Cause: Peripheral needs time to stabilize, calibration incomplete
00662 * @par Recovery: Wait for ready flag, poll status register, retry after delay
00663 * @par Prevention: Implement proper hardware sequencing and delays
00664 */
00665 k_rx_err_hw_not_ready = 0x202,
00666
00667 /**
00668 * @brief Hardware operation timed out
00669 * @details
00670 * Hardware peripheral failed to complete operation within timeout. Distinct from
00671 * k_rx_err_timeout (generic) - this is hardware-specific.
00672 *
00673 * @par Value: 0x203
00674 * @par Cause: Peripheral hung, bus arbitration lost, external device not responding
00675 * @par Recovery: Reset peripheral, check external hardware, increase timeout
00676 * @par Prevention: Tune hardware timeouts, implement watchdog recovery
00677 */
00678 k_rx_err_hw_timeout = 0x203,
00679
00680 /**
00681 * @brief Generic hardware error
00682 * @details
00683 * Unspecified hardware error detected. Examples: fault flag set, error interrupt,
00684 * hardware self-test failure.
00685 *
00686 * @par Value: 0x204
00687 * @par Cause: Hardware fault condition, transient error, EMI/noise
00688 * @par Recovery: Read hardware error registers, reset peripheral, retry operation
00689 * @par Prevention: Implement hardware error monitoring and logging
00690 */
00691 k_rx_err_hw_error = 0x204,
00692
00693 /**
00694 * @brief GPIO pin conflict detected
00695 * @details
00696 * GPIO pin already in use by another peripheral or function. Pin multiplexing
00697 * conflict must be resolved before pin can be used.
00698 *
00699 * @par Value: 0x205
00700 * @par Cause: Multiple peripherals mapped to same pin, incorrect MPC settings
00701 * @par Recovery: Reconfigure pin mux, use alternative pin, disable conflicting peripheral
00702 * @par Prevention: Document pin assignments, validate at init time
00703 * @see docs/sections/03_hardware_pinout.tex Complete pin assignment table
00704 */
00705 k_rx_err_gpio_conflict = 0x205,
00706
00707 /**
00708 * @brief Invalid GPIO port number
00709 * @details
00710 * GPIO port number outside valid range for RX72N (valid: 0-K, excluding reserved).

```

```

00711 *
00712 * @par Value: 0x206
00713 * @par Cause: Incorrect port constant, typo in pin definition
00714 * @par Recovery: Use valid port number (PORT0-PORTK), check hardware manual
00715 * @par Prevention: Use enum constants from rx_port_constants.h
00716 */
00717 k_rx_err_gpio_invalid_port = 0x206,
00718
00719 /**
00720 * @brief Invalid GPIO pin number
00721 * @details
00722 * GPIO pin number outside valid range (0-7 for most ports on RX72N).
00723 *
00724 * @par Value: 0x207
00725 * @par Cause: Incorrect pin constant, array index out of bounds
00726 * @par Recovery: Use valid pin number (0-7), check port definition
00727 * @par Prevention: Use enum constants from rx_gpio_constants.h
00728 */
00729 k_rx_err_gpio_invalid_pin = 0x207,
00730
00731 /**
00732 * @brief Sensor measurement out of valid range
00733 * @details
00734 * Sensor reading outside physically possible or configured range. Examples:
00735 * ADC overflow, temperature sensor disconnected, ultrasonic no echo.
00736 *
00737 * @par Value: 0x208
00738 * @par Cause: Sensor fault, value exceeds limits, sensor disconnected
00739 * @par Recovery: Check sensor connections, validate sensor state, clamp to range
00740 * @par Prevention: Implement sensor health monitoring
00741 */
00742 k_rx_err_out_of_range = 0x208,
00743
00744 /**
00745 * @brief Hardware register block not mapped in memory
00746 * @details
00747 * Attempted to access peripheral whose registers are not mapped at expected
00748 * address. Indicates hardware configuration error or unsupported MCU variant.
00749 *
00750 * @par Value: 0x209
00751 * @par Cause: Wrong MCU variant, peripheral not present, incorrect memory map
00752 * @par Recovery: Verify MCU part number, check datasheet, disable peripheral
00753 * @par Prevention: Conditional compilation based on MCU variant
00754 */
00755 k_rx_err_hw_unmapped = 0x209,
00756
00757 /*
=====
00758 * RTOS Errors (0x300 - 0x3FF)
00759 * ThreadX and synchronization errors
00760 *
=====
00761 */
00762
00763 /**
00764 * @brief Generic RTOS error
00765 * @details
00766 * Unspecified ThreadX error occurred. Check ThreadX status code for details.
00767 *
00768 * @par Value: 0x301
00769 * @par Cause: ThreadX API returned non-success status
00770 * @par Recovery: Examine ThreadX status, consult ThreadX documentation
00771 * @par Prevention: Validate ThreadX API parameters
00772 * @see rx_err_from_threadx() Convert ThreadX codes to rx_err_t
00773 */
00774 k_rx_err_rtos_error = 0x301,
00775
00776 /**
00777 * @brief ThreadX error (alias for k_rx_err_rtos_error)
00778 * @details
00779 * Identical to k_rx_err_rtos_error, provided for explicit ThreadX naming.
00780 *
00781 * @par Value: 0x301
00782 */
00783 k_rx_err_threadx = 0x301,
00784
00785 /**
00786 * @brief ThreadX thread creation failed
00787 * @details
00788 * tx_thread_create() failed. Common causes: out of memory, invalid priority,
00789 * invalid stack.
00790 *
00791 * @par Value: 0x302
00792 * @par Cause: Insufficient memory, stack too small, invalid parameters
00793 * @par Recovery: Increase heap size, validate thread parameters
00794 * @par Prevention: Size memory pools based on worst-case thread count
00795 */

```

```

00796 k_rx_err_rtos_thread_create = 0x302,
00797
00798 /**
00799  * @brief Semaphore operation failed
00800  * @details
00801  * Semaphore create, get, or put operation failed. Check ThreadX error code.
00802  *
00803  * @par Value: 0x303
00804  * @par Cause: Invalid semaphore handle, timeout, semaphore not created
00805  * @par Recovery: Verify semaphore initialized, check timeout value
00806  * @par Prevention: Initialize all RTOS objects before use
00807  */
00808 k_rx_err_rtos_semaphore = 0x303,
00809
00810 /**
00811  * @brief Mutex operation failed
00812  * @details
00813  * Mutex create, get, or put operation failed. Possible deadlock or priority
00814  * inversion if using priority inheritance.
00815  *
00816  * @par Value: 0x304
00817  * @par Cause: Invalid mutex handle, timeout, mutex already owned
00818  * @par Recovery: Check mutex state, verify not already locked
00819  * @par Prevention: Use timeouts, avoid nested mutex acquisition
00820  */
00821 k_rx_err_rtos_mutex = 0x304,
00822
00823 /**
00824  * @brief Queue operation failed
00825  * @details
00826  * Queue create, send, or receive operation failed.
00827  *
00828  * @par Value: 0x305
00829  * @par Cause: Queue full, queue empty, invalid queue handle
00830  * @par Recovery: Increase queue size, implement flow control
00831  * @par Prevention: Size queues based on worst-case message rate
00832  */
00833 k_rx_err_rtos_queue = 0x305,
00834
00835 /**
00836  * @brief Timer operation failed
00837  * @details
00838  * Timer create, activate, or deactivate operation failed.
00839  *
00840  * @par Value: 0x306
00841  * @par Cause: Invalid timer handle, timer already active
00842  * @par Recovery: Check timer state before operation
00843  * @par Prevention: Track timer state explicitly
00844  */
00845 k_rx_err_rtos_timer = 0x306,
00846
00847 /*
=====
00848  * Communication Errors (0x400 - 0x4FF)
00849  * SPI, UART, I2C, and protocol errors
00850  *
=====
00851  */
00852
00853 /**
00854  * @brief Generic communication error
00855  * @details
00856  * Unspecified communication error on SPI, UART, or I2C bus.
00857  *
00858  * @par Value: 0x401
00859  * @par Cause: Bus error, framing error, data corruption
00860  * @par Recovery: Reset bus, retry transfer, check connections
00861  * @par Prevention: Implement error detection (CRC, checksums)
00862  */
00863 k_rx_err_comm_error = 0x401,
00864
00865 /**
00866  * @brief SPI communication error
00867  * @details
00868  * SPI transfer failed. Examples: CIPO timeout, bus error, chip select conflict.
00869  *
00870  * @par Value: 0x402
00871  * @par Cause: No response on CIPO, mode mismatch, clock issue
00872  * @par Recovery: Check SPI mode, verify clock frequency, check CS timing
00873  * @par Prevention: Validate SPI configuration before transfer
00874  */
00875 k_rx_err_spi_error = 0x402,
00876
00877 /**
00878  * @brief UART communication error
00879  * @details
00880  * UART error detected. Examples: framing error, parity error, overrun.

```

```

00881  *
00882  * @par Value: 0x403
00883  * @par Cause: Baud rate mismatch, noise, buffer overrun
00884  * @par Recovery: Clear error flags, adjust baud rate, reduce data rate
00885  * @par Prevention: Use hardware flow control, size buffers adequately
00886  */
00887  k_rx_err_uart_error = 0x403,
00888
00889  /**
00890  * @brief I2C communication error
00891  * @details
00892  * I2C (RIIC) error detected. Examples: bus busy, arbitration lost, NACK.
00893  *
00894  * @par Value: 0x404
00895  * @par Cause: Device not responding, wrong address, bus contention
00896  * @par Recovery: Reset I2C bus, verify device address, check pull-ups
00897  * @par Prevention: Implement bus recovery, validate addresses
00898  */
00899  k_rx_err_i2c_error = 0x404,
00900
00901  /**
00902  * @brief CRC validation failed
00903  * @details
00904  * Received data CRC does not match calculated CRC. Data corruption detected.
00905  *
00906  * @par Value: 0x405
00907  * @par Cause: Data corruption, EMI, faulty peripheral
00908  * @par Recovery: Request retransmission, check signal integrity
00909  * @par Prevention: Use shielded cables, implement retry logic
00910  * @see k_rx_err_checksum_mismatch Simpler checksum validation
00911  */
00912  k_rx_err_crc_mismatch = 0x405,
00913
00914  /**
00915  * @brief Protocol error detected
00916  * @details
00917  * Communication protocol violation. Examples: invalid packet format, unexpected
00918  * sequence number, protocol state machine error.
00919  *
00920  * @par Value: 0x406
00921  * @par Cause: Protocol version mismatch, malformed packet, out-of-sequence
00922  * @par Recovery: Reset protocol state machine, request resynchronization
00923  * @par Prevention: Validate all protocol fields, implement version negotiation
00924  */
00925  k_rx_err_protocol_error = 0x406,
00926
00927  /**
00928  * @brief NACK (Not Acknowledged) received
00929  * @details
00930  * Peripheral responded with NACK (negative acknowledgment). Device refusing
00931  * operation or not ready.
00932  *
00933  * @par Value: 0x407
00934  * @par Cause: Device busy, invalid command, device not ready
00935  * @par Recovery: Retry after delay, check device state
00936  * @par Prevention: Poll device ready status before commands
00937  */
00938  k_rx_err_nack = 0x407,
00939
00940  /**
00941  * @brief Resource conflict in communication
00942  * @details
00943  * Multiple controllers attempting bus access, or conflicting operation in progress.
00944  *
00945  * @par Value: 0x408
00946  * @par Cause: Bus arbitration lost, conflicting DMA channels
00947  * @par Recovery: Implement bus arbitration, retry after backoff
00948  * @par Prevention: Use RTOS mutexes for shared buses
00949  */
00950  k_rx_err_conflict = 0x408,
00951
00952  /**
00953  * @brief Retransmission limit exceeded
00954  * @details
00955  * Automatic retransmission exhausted all retry attempts without receiving ACK.
00956  * Frame delivery cannot be confirmed.
00957  *
00958  * @par Value: 0x409
00959  * @par Cause: Remote side not responding, persistent CRC failures, link down
00960  * @par Recovery: Reset communication, check physical connection, increase retries
00961  * @par Prevention: Tune timeout/retry parameters for link quality
00962  * @see rx_spi_comm_set_auto_retransmit() Configure retry parameters
00963  */
00964  k_rx_err_retry_limit = 0x409,
00965
00966  */
=====

```

```

00967 * Validation Errors (0x500 - 0x5FF)
00968 * Input validation and checksum errors
00969 *
=====
00970 */
00971
00972 /**
00973  * @brief Generic validation failed
00974  * @details
00975  * Data validation or consistency check failed.
00976  *
00977  * @par Value: 0x501
00978  * @par Cause: Invalid data format, constraint violation, consistency check failed
00979  * @par Recovery: Reject invalid data, request valid input
00980  * @par Prevention: Validate all inputs at system boundaries
00981  */
00982 k_rx_err_validation_failed = 0x501,
00983
00984 /**
00985  * @brief Checksum mismatch detected
00986  * @details
00987  * Simple checksum validation failed. Less robust than CRC but faster.
00988  *
00989  * @par Value: 0x502
00990  * @par Cause: Data corruption, transmission error
00991  * @par Recovery: Request retransmission
00992  * @par Prevention: Use CRC for critical data
00993  * @see k_rx_err_crc_mismatch More robust CRC validation
00994  */
00995 k_rx_err_checksum_mismatch = 0x502,
00996
00997 /**
00998  * @brief Range check failed
00999  * @details
01000  * Value outside configured or physically possible range.
01001  *
01002  * @par Value: 0x503
01003  * @par Cause: Value exceeds min/max limits
01004  * @par Recovery: Clamp to valid range, reject input
01005  * @par Prevention: Document and enforce valid ranges
01006  */
01007 k_rx_err_range_check_failed = 0x503,
01008
01009 /**
01010  * @brief Null pointer detected
01011  * @details
01012  * Required pointer parameter is nullptr. Subset of k_rx_err_invalid_arg.
01013  *
01014  * @par Value: 0x504
01015  * @par Cause: Caller passed NULL where non-NULL required
01016  * @par Recovery: Fix caller code to pass valid pointer
01017  * @par Prevention: Use RX_CHECK_NULL_PTR macro at function entry
01018  * @see k_rx_err_invalid_arg More general argument validation
01019  */
01020 k_rx_err_null_ptr = 0x504,
01021
01022 } rx_err_codes_t;
01023
01024 /*
=====
01025 * Error Code Helpers
01026 *
=====
01027 */
01028
01029 /**
01030  * @brief Convert ThreadX status code to rx_err_t
01031  *
01032  * @details
01033  * Converts ThreadX API return values (UINT status codes) to rx_err_t error codes.
01034  * ThreadX uses 0 for success (TX_SUCCESS) and non-zero for various error conditions.
01035  * This function performs a simple binary conversion: success -> k_rx_ok, any error -> k_rx_err_rtos_error.
01036  *
01037  * ## Algorithm
01038  *
01039  * 1. Compare tx_status against TX_SUCCESS (0)
01040  * 2. If equal: return k_rx_ok (success)
01041  * 3. If not equal: return k_rx_err_rtos_error (generic RTOS error)
01042  *
01043  * ## ThreadX Status Codes Mapped
01044  *
01045  * | ThreadX Status | Value | rx_err_t Result | Meaning |
01046  * |-----|-----|-----|-----|
01047  * | TX_SUCCESS | 0 | k_rx_ok (0x000) | Operation successful |
01048  * | TX_DELETED | 0x01 | k_rx_err_rtos_error | Object was deleted |
01049  * | TX_NO_MEMORY | 0x10 | k_rx_err_rtos_error | Memory allocation failed |
01050  * | TX_PTR_ERROR | 0x03 | k_rx_err_rtos_error | Invalid pointer |

```



```

01051 * | TX_WAIT_ERROR      | 0x04 | k_rx_err_rtos_error | Invalid wait option |
01052 * | TX_SIZE_ERROR      | 0x05 | k_rx_err_rtos_error | Invalid size        |
01053 * | TX_CALLER_ERROR    | 0x13 | k_rx_err_rtos_error | Invalid caller context |
01054 * | TX_NOT_AVAILABLE   | 0x1D | k_rx_err_rtos_error | Resource not available |
01055 * | *(all other)*      | *    | k_rx_err_rtos_error | Generic RTOS error   |
01056 *
01057 * @note This is a simplified conversion. For finer-grained error mapping, inspect
01058 *       the ThreadX status code directly and map to specific rx_err_t codes.
01059 *
01060 * @param[in] tx_status ThreadX status code returned from ThreadX API call
01061 * - Type: UINT (32-bit unsigned integer in ThreadX)
01062 * - Valid range: Any value (typically 0x00-0x1D for defined ThreadX errors)
01063 * - Special values:
01064 *   - 0 (TX_SUCCESS): Indicates successful operation
01065 *   - Non-zero: Various ThreadX error conditions
01066 *
01067 * @return rx_err_t error code
01068 * @retval k_rx_ok (0x000) If tx_status == 0 (TX_SUCCESS), operation succeeded
01069 * @retval k_rx_err_rtos_error (0x301) If tx_status != 0, ThreadX error occurred
01070 *
01071 * @par Usage Example - Basic Conversion:
01072 * @code{.c}
01073 * #include "rx_err.h"
01074 * #include "tx_api.h"
01075 *
01076 * TX_THREAD my_thread;
01077 * CHAR thread_stack[1024];
01078 *
01079 * rx_err_t create_worker_thread(void) {
01080 *     UINT tx_status = tx_thread_create(
01081 *         &my_thread,           // Thread control block
01082 *         "Worker",             // Thread name
01083 *         worker_thread_entry,   // Entry function
01084 *         0,                    // Entry input
01085 *         thread_stack,         // Stack start
01086 *         sizeof(thread_stack),  // Stack size
01087 *         16,                   // Priority
01088 *         16,                   // Preemption threshold
01089 *         TX_NO_TIME_SLICE,     // Time slice
01090 *         TX_AUTO_START         // Auto start
01091 *     );
01092 *
01093 *     // Convert ThreadX status to rx_err_t
01094 *     rx_err_t err = rx_err_from_threadx(tx_status);
01095 *     if (err != k_rx_ok) {
01096 *         rx_log_error("RTOS", "Thread creation failed: TX status 0x%02X", tx_status);
01097 *         return err;
01098 *     }
01099 *
01100 *     return k_rx_ok;
01101 * }
01102 * @endcode
01103 *
01104 * @par Usage Example - Inline in Error Check:
01105 * @code{.c}
01106 * rx_err_t init_rtos_resources(void) {
01107 *     // Create semaphore and convert result inline
01108 *     rx_err_t err = rx_err_from_threadx(
01109 *         tx_semaphore_create(&my_sem, "DataReady", 0)
01110 *     );
01111 *
01112 *     if (err != k_rx_ok) {
01113 *         return err; // Propagate RTOS error
01114 *     }
01115 *
01116 *     // Continue with other RTOS operations...
01117 *     return k_rx_ok;
01118 * }
01119 * @endcode
01120 *
01121 * @par Performance:
01122 * - Execution time: ~2 cycles @ 240 MHz (CMP + CMOV)
01123 * - Stack usage: 0 bytes (inlined by compiler)
01124 * - Code size: 0 bytes when inlined (optimized to single instruction)
01125 *
01126 * @note Thread Safety: Pure function, inherently thread-safe (no shared state)
01127 * @note Re-entrancy: Fully reentrant (no side effects)
01128 * @note Memory: No dynamic allocation, no static variables
01129 *
01130 * @pre tx_status contains valid ThreadX status code (any UINT value accepted)
01131 * @post Return value is either k_rx_ok or k_rx_err_rtos_error (no other values possible)
01132 *
01133 * @invariant Result is always one of two values: k_rx_ok or k_rx_err_rtos_error
01134 *
01135 * @see k_rx_err_rtos_error Error code returned for all non-success ThreadX status
01136 * @see k_rx_ok Success code returned for TX_SUCCESS
01137 * @see tx_thread_create() Example ThreadX API that returns UINT status

```



```

01138 *
01139 * @par NASA Power of 10 Compliance:
01140 * - Rule 1: [OK] No goto, setjmp, recursion
01141 * - Rule 2: [OK] N/A (no loops)
01142 * - Rule 3: [OK] No dynamic allocation
01143 * - Rule 4: [OK] Function is 1 line (excluding documentation)
01144 * - Rule 5: [OK] 1 precondition, 1 postcondition, 1 invariant
01145 * - Rule 6: [OK] Minimal scope (inline function)
01146 * - Rule 7: [OK] No function calls to check
01147 * - Rule 8: [OK] Uses typed enum constants
01148 * - Rule 9: [OK] No function pointers
01149 * - Rule 10: [OK] Compiles with -Wall -Wextra -Werror
01150 *
01151 * @since Version 1.0.0
01152 */
01153 static inline rx_err_t rx_err_from_threadx(uint32_t tx_status)
01154 {
01155     return (tx_status == 0) ? k_rx_ok : k_rx_err_rtos_error;
01156 }
01157
01158 /**
01159 * @brief Check if error code represents success
01160 *
01161 * @details
01162 * Tests whether an rx_err_t value indicates successful operation (k_rx_ok).
01163 * This is the preferred way to check for success in STAR firmware, providing
01164 * explicit intent and type safety.
01165 *
01166 * ## Algorithm
01167 *
01168 * 1. Compare err against k_rx_ok (0)
01169 * 2. Return true if equal (success), false otherwise (error)
01170 *
01171 * ## Comparison to Alternatives
01172 *
01173 * | Method | Type Safe | Explicit Intent | Performance |
01174 * |-----|-----|-----|-----|
01175 * | rx_err_is_ok(err) | [OK] | [OK] | 1 cycle |
01176 * | err == k_rx_ok | [OK] | [OK] | 1 cycle |
01177 * | !err | [X] | [X] | 1 cycle |
01178 * | err == 0 | [X] | [X] | 1 cycle |
01179 *
01180 * **Recommended:** Use `rx_err_is_ok(err)` for explicit intent and self-documenting code.
01181 *
01182 * @param[in] err Error code to check
01183 * - Type: rx_err_t (int32_t)
01184 * - Valid range: Any rx_err_t value (typically 0x000-0x5FF or 0)
01185 * - Special values:
01186 *   - k_rx_ok (0): Success condition
01187 *   - Any non-zero: Error condition
01188 *
01189 * @return Boolean success indicator
01190 * @retval true If err == k_rx_ok (0), operation was successful
01191 * @retval false If err != k_rx_ok (any non-zero value), operation failed
01192 *
01193 * @par Usage Example - Basic Success Check:
01194 * @code{.c}
01195 * rx_err_t err = initialize_motor();
01196 * if (rx_err_is_ok(err)) {
01197 *     // Success path - motor initialized
01198 *     start_control_loop();
01199 * } else {
01200 *     // Error path - log and handle failure
01201 *     rx_log_error("MOTOR", "Init failed: 0x%X", err);
01202 *     enter_safe_state();
01203 * }
01204 * @endcode
01205 *
01206 * @par Usage Example - Assert Success:
01207 * @code{.c}
01208 * // Critical operation that must succeed
01209 * rx_err_t err = init_critical_subsystem();
01210 * assert(rx_err_is_ok(err) && "Critical init failed"); // NASA Rule 5
01211 * @endcode
01212 *
01213 * @par Usage Example - Conditional Compilation:
01214 * @code{.c}
01215 * #if DEBUG_LEVEL >= 2
01216 *     rx_err_t err = optional_debug_feature();
01217 *     if (!rx_err_is_ok(err)) {
01218 *         rx_log_warn("DEBUG", "Optional feature unavailable");
01219 *     }
01220 * #endif
01221 * @endcode
01222 *
01223 * @par Performance:
01224 * - Execution time: 1 cycle @ 240 MHz (single CMP instruction)

```

```

01225 * - Stack usage: 0 bytes (inlined by compiler)
01226 * - Code size: 0 bytes when inlined (optimized to single comparison)
01227 *
01228 * @note Thread Safety: Pure function, inherently thread-safe (no shared state)
01229 * @note Re-entrancy: Fully reentrant (no side effects, no mutable state)
01230 * @note Memory: No allocation, no static variables
01231 *
01232 * @pre err contains valid rx_err_t value (any int32_t accepted, but typically 0x000-0x5FF)
01233 * @post Return value is deterministic boolean based solely on input
01234 *
01235 * @invariant Result is always boolean (true or false, no other values)
01236 *
01237 * @see rx_err_is_error() Inverse check (returns true if error, false if success)
01238 * @see k_rx_ok Success constant (0x000)
01239 *
01240 * @par NASA Power of 10 Compliance:
01241 * - Rule 1: [OK] No goto, setjmp, recursion
01242 * - Rule 2: [OK] N/A (no loops)
01243 * - Rule 3: [OK] No dynamic allocation
01244 * - Rule 4: [OK] Function is 1 line
01245 * - Rule 5: [OK] 1 precondition, 1 postcondition, 1 invariant
01246 * - Rule 6: [OK] Minimal scope (inline function)
01247 * - Rule 7: [OK] No function calls to check
01248 * - Rule 8: [OK] Uses typed enum constant k_rx_ok
01249 * - Rule 9: [OK] No function pointers
01250 * - Rule 10: [OK] Compiles with -Wall -Wextra -Werror
01251 *
01252 * @since Version 1.0.0
01253 */
01254 static inline bool rx_err_is_ok(rx_err_t err)
01255 {
01256     return (bool)(err == k_rx_ok);
01257 }
01258
01259 /**
01260 * @brief Check if error code represents failure
01261 *
01262 * @details
01263 * Tests whether an rx_err_t value indicates a failure condition (any non-zero value).
01264 * This is the logical inverse of rx_err_is_ok() and is used when error-checking
01265 * logic is more naturally expressed as "if error occurred" rather than "if success".
01266 *
01267 * ## Algorithm
01268 *
01269 * 1. Compare err against k_rx_ok (0)
01270 * 2. Return true if not equal (error), false otherwise (success)
01271 *
01272 * ## When to Use This vs rx_err_is_ok()
01273 *
01274 * **Use rx_err_is_error()** when error handling is the primary focus:
01275 * @code{.c}
01276 * rx_err_t err = risky_operation();
01277 * if (rx_err_is_error(err)) {
01278 *     // Handle error first (primary concern)
01279 *     handle_error(err);
01280 *     return;
01281 * }
01282 * // Continue with normal path
01283 * @endcode
01284 *
01285 * **Use rx_err_is_ok()** when success path is the primary focus:
01286 * @code{.c}
01287 * rx_err_t err = initialize_system();
01288 * if (rx_err_is_ok(err)) {
01289 *     // Success is the expected/happy path
01290 *     continue_normal_operation();
01291 * } else {
01292 *     // Error is exceptional
01293 *     handle_error(err);
01294 * }
01295 * @endcode
01296 *
01297 * @param[in] err Error code to check
01298 * - Type: rx_err_t (int32_t)
01299 * - Valid range: Any rx_err_t value (typically 0x000-0x5FF or 0)
01300 * - Special values:
01301 *   - k_rx_ok (0): Success condition (returns false)
01302 *   - Any non-zero: Error condition (returns true)
01303 *
01304 * @return Boolean error indicator
01305 * @retval true If err != k_rx_ok (non-zero), operation failed
01306 * @retval false If err == k_rx_ok (0), operation was successful
01307 *
01308 * @par Usage Example - Early Return on Error:
01309 * @code{.c}
01310 * rx_err_t initialize_peripherals(void) {
01311 *     rx_err_t err;

```

```

01312 *
01313 *   err = init_gpio();
01314 *   if (rx_err_is_error(err)) {
01315 *       return err; // Early return on error
01316 *   }
01317 *
01318 *   err = init_spi();
01319 *   if (rx_err_is_error(err)) {
01320 *       return err; // Propagate error immediately
01321 *   }
01322 *
01323 *   err = init_i2c();
01324 *   if (rx_err_is_error(err)) {
01325 *       return err;
01326 *   }
01327 *
01328 *   return k_rx_ok; // All initialization succeeded
01329 * }
01330 * @endcode
01331 *
01332 * @par Usage Example - Error Logging:
01333 * @code{.c}
01334 * rx_err_t err = send_command(cmd);
01335 * if (rx_err_is_error(err)) {
01336 *     // Log error with details
01337 *     rx_log_error("COMM", "Command send failed: 0x%03X", err);
01338 *
01339 *     // Category-specific handling
01340 *     if ((err & 0xF00) == 0x400) {
01341 *         reset_communication();
01342 *     }
01343 * }
01344 * @endcode
01345 *
01346 * @par Usage Example - Error Accumulation:
01347 * @code{.c}
01348 * uint32_t error_count = 0;
01349 * for (uint8_t i = 0; i < k_sensor_count; i++) {
01350 *     rx_err_t err = read_sensor(i);
01351 *     if (rx_err_is_error(err)) {
01352 *         error_count++;
01353 *         rx_log_warn("SENSOR", "Sensor %u failed: 0x%X", i, err);
01354 *     }
01355 * }
01356 *
01357 * if (error_count > k_max_allowed_failures) {
01358 *     enter_degraded_mode();
01359 * }
01360 * @endcode
01361 *
01362 * @par Performance:
01363 * - Execution time: 1 cycle @ 240 MHz (single CMP instruction)
01364 * - Stack usage: 0 bytes (inlined by compiler)
01365 * - Code size: 0 bytes when inlined (optimized to single comparison)
01366 *
01367 * @note Thread Safety: Pure function, inherently thread-safe (no shared state)
01368 * @note Re-entrancy: Fully reentrant (no side effects, no mutable state)
01369 * @note Memory: No allocation, no static variables
01370 *
01371 * @pre err contains valid rx_err_t value (any int32_t accepted)
01372 * @post Return value is deterministic boolean based solely on input
01373 *
01374 * @invariant Result is always boolean (true or false)
01375 * @invariant Exactly inverse of rx_err_is_ok(): rx_err_is_error(e) == !rx_err_is_ok(e)
01376 *
01377 * @see rx_err_is_ok() Inverse check (returns true if success, false if error)
01378 * @see k_rx_ok Success constant (0x000)
01379 *
01380 * @par NASA Power of 10 Compliance:
01381 * - Rule 1: [OK] No goto, setjmp, recursion
01382 * - Rule 2: [OK] N/A (no loops)
01383 * - Rule 3: [OK] No dynamic allocation
01384 * - Rule 4: [OK] Function is 1 line
01385 * - Rule 5: [OK] 1 precondition, 1 postcondition, 2 invariants
01386 * - Rule 6: [OK] Minimal scope (inline function)
01387 * - Rule 7: [OK] No function calls to check
01388 * - Rule 8: [OK] Uses typed enum constant k_rx_ok
01389 * - Rule 9: [OK] No function pointers
01390 * - Rule 10: [OK] Compiles with -Wall -Wextra -Werror
01391 *
01392 * @since Version 1.0.0
01393 */
01394 static inline bool rx_err_is_error(rx_err_t err)
01395 {
01396     return (bool)(err != k_rx_ok);
01397 }
01398

```

```

01399 #ifdef __cplusplus
01400 }
01401 #endif

```

7.7 libs/rx_core/inc/rx_error_handler.h File Reference

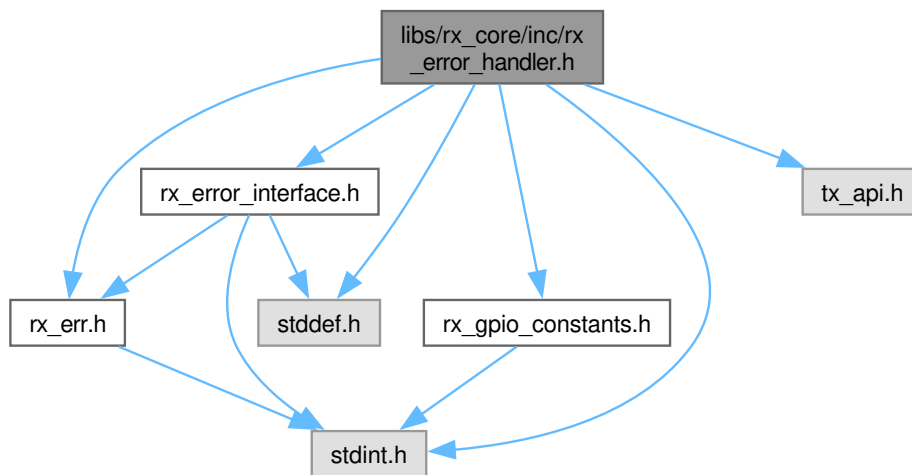
Error Handler Concrete Implementation with Retry Logic and Exponential Backoff.

```

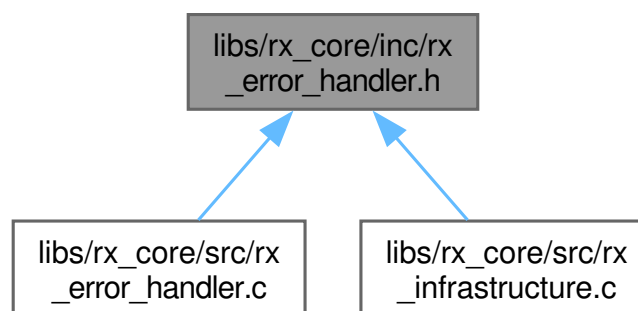
#include <stddef.h>
#include <stdint.h>
#include "rx_err.h"
#include "rx_error_interface.h"
#include "rx_gpio_constants.h"
#include "tx_api.h"

```

Include dependency graph for rx_error_handler.h:



This graph shows which files directly or indirectly include this file:



Data Structures

- struct [error_component_state_t](#)
Per-Component Error Tracking State (Array Element).
- struct [error_handler_t](#)
Error Handler Concrete Implementation Structure (Main State Container).
- struct [error_handler_config_t](#)
Error Handler Initialization Configuration Parameters.

Enumerations

- enum [error_handler_limits_t](#) : uint8_t { [k_error_handler_max_components](#), [k_error_handler_component_name_max_length](#) }

Error Handler Configuration Constants and Array Size Limits.

Functions

- [rx_err_t error_handler_init](#) ([error_handler_t](#) *handler, const [error_handler_config_t](#) *config)
Initialize error handler with specified configuration.
- [rx_err_t error_handler_get_interface](#) ([rx_error_interface_t](#) *iface, [error_handler_t](#) *handler)
Get abstract interface from concrete error handler (Dependency Inversion).
- [rx_err_t error_handler_deinit](#) ([error_handler_t](#) *handler)
Deinitialize error handler and cleanup resources.

7.7.1 Detailed Description

Error Handler Concrete Implementation with Retry Logic and Exponential Backoff.

Production-grade error management system providing comprehensive error tracking, intelligent retry logic, and exponential backoff for graceful degradation. Implements the Dependency Inversion Principle through [rx_error_interface_t](#), allowing modules to depend on abstract error handling without coupling to this concrete implementation.

7.7.1.1 Purpose and Design Rationale

This module provides the concrete implementation of the abstract [rx_error_interface_t](#), offering sophisticated error management beyond simple error code returns:

1. Error Aggregation: Track total errors across entire system
2. Component Isolation: Per-component error counters for diagnostics
3. Automatic Retry: Configurable retry logic with backoff
4. Graceful Degradation: Exponential backoff prevents system thrashing
5. Dependency Inversion: High-level modules use interface, not implementation

7.7.1.2 Key Features

Error Tracking:

- Global error counter across all components
- Per-component error tracking (up to 16 components)
- Component name association for diagnostics
- Thread-safe incrementation (ThreadX mutex)

Retry Management:

- Configurable retry limits per component
- Automatic retry counting
- Retry limit detection
- Per-component retry state isolation

Exponential Backoff:

- Initial backoff delay configurable
- Exponential growth: $\text{delay} = \text{initial} \times 2^{\text{retry_count}}$
- Maximum backoff cap to prevent excessive delays
- Per-component backoff calculation

Thread Safety:

- ThreadX mutex protects all state modifications
- Safe concurrent access from multiple tasks
- Atomic error reporting and retry checking

7.7.1.3 Architecture: Dependency Inversion Principle (DIP)

This module demonstrates Dependency Inversion from SOLID principles:

High-Level Modules (Motor Control, SPI Driver, etc.)

```

v depends on v
rx_error_interface_t (Abstract Interface)
^ implements ^
error_handler_t (Concrete Implementation - THIS MODULE)

```

Benefits:

- High-level modules don't depend on this implementation
- Can swap implementations without changing client code
- Unit tests can mock error handling
- Clear separation of concerns

DIP Example:

```

// High-level module depends on INTERFACE, not IMPLEMENTATION
void motor_controller_init(rx_error_interface_t* error_if) {
    // Uses interface - doesn't know about error_handler_t
    error_if->report_error(error_if->ctx, k_rx_fail, "MOTOR", "Init failed");
}

// Low-level code creates concrete implementation
static error_handler_t s_error_handler;
rx_error_interface_t error_iface;

error_handler_init(&s_error_handler, &config);
error_handler_get_interface(&error_iface, &s_error_handler);

// Inject interface into high-level module
motor_controller_init(&error_iface);

```

7.7.1.4 Implementation Approach

Component Tracking Algorithm:

1. Error reported for component "SPI"
2. Linear search through components[] array for "SPI"
3. If found: Increment error_count and retry_count
4. If not found: Allocate new slot, initialize counters
5. Update total_error_count (global)
6. Return success

Exponential Backoff Calculation:

```
backoff = initial_backoff_ms x (2 ^ retry_count)
backoff = min(backoff, max_backoff_ms) // Cap at maximum
```

Example with initial=10ms, max=5000ms:

- Retry 0: $10\text{ms} \times 2^0 = 10\text{ms}$
- Retry 1: $10\text{ms} \times 2^1 = 20\text{ms}$
- Retry 2: $10\text{ms} \times 2^2 = 40\text{ms}$
- Retry 3: $10\text{ms} \times 2^3 = 80\text{ms}$
- Retry 4: $10\text{ms} \times 2^4 = 160\text{ms}$
- ...
- Retry 9: $10\text{ms} \times 2^9 = 5120\text{ms} \rightarrow$ capped at 5000ms

7.7.1.5 Performance Characteristics

Operation	Execution Time	Memory	Notes
error_handler_init()	~50 us	0 heap	One-time, mutex creation
report_error()	~10 us x N components	0	Linear search (N ≤ 16)
is_retry_limit_reached()	~5 us x N components	0	Linear search
get_backoff_delay()	~8 us x N components	0	Linear search + calculation
reset_component_state()	~5 us x N components	0	Linear search

Memory Footprint (Static Allocation):

- [error_handler_t](#) struct: ~1.3 KB
 - TX_MUTEX: ~100 bytes (ThreadX internal)
 - components[16]: 16 x 72 bytes = 1152 bytes
 - Other fields: ~20 bytes
- Total: ~1.3 KB RAM (zero heap usage)

7.7.1.6 Thread Safety and RTOS Integration

Thread-Safe Operations:

- All public functions use TX_MUTEX for protection
- Safe to call from multiple ThreadX tasks simultaneously
- Mutex ensures atomic state updates

ThreadX Integration:

- Uses tx_mutex_create() for mutex initialization
- Uses tx_mutex_get()/tx_mutex_put() for locking
- Compatible with all ThreadX priority levels

7.7.1.7 Usage Guidelines

DO:

- Use through rx_error_interface_t (Dependency Inversion)
- Initialize once at system startup
- Configure retry limits based on operation criticality
- Use exponential backoff for network/communication errors
- Track errors per logical component for diagnostics

DON'T:

- Directly call implementation functions (use interface)
- Exceed k_error_handler_max_components (16) unique components
- Use very short backoff intervals (< 1ms) - defeats purpose
- Forget to deinitialize on shutdown (leaks mutex)

Hardware Requirements:

Component	Requirement	Notes
MCU	Renesas RX72N	ThreadX RTOS required
RAM	~1.3 KB static	Zero heap allocation
RTOS	ThreadX	TX_MUTEX support required

Module Dependencies:

This module depends on:

- [rx_err.h](#) - Error code definitions
- [rx_error_interface.h](#) - Abstract interface definition
- `<stdint.h>`, `<stddef.h>` - Standard types
- ThreadX RTOS - TX_Mutex for thread safety

This module is used by:

- `main.c` - System-wide error handler initialization
- All drivers and libraries - Error reporting via interface
- Communication modules - Retry logic with backoff

NASA Power of 10 Compliance:

Rule	Status	Implementation
1. Simple control flow	[OK]	No goto/setjmp/recursion, sequential logic
2. Fixed loop bounds	[OK]	All loops bounded by <code>k_error_handler_max_components</code> (16)
3. No dynamic allocation	[OK]	Zero malloc/free, static component array
4. Small functions	[OK]	All functions < 60 lines
5. Assertions (≥ 2 per function)	[OK]	Minimum 2 checks per function (pre/post)
6. Narrow scope	[OK]	File-scope statics, function-local variables
7. Check return values	[OK]	All functions return rx_err_t , checked by callers
8. Limited preprocessor	[OK]	C23 typed enums only, zero computation macros
9. Pointer restrictions	[OK]	Single-level pointers, no pointer arithmetic
10. Compiler warnings	[OK]	Compiles with <code>-Wall -Wextra -Werror</code>

Safety-Critical Design:

- Static allocation prevents fragmentation
- Bounded component array prevents overflow
- Explicit error codes force error handling
- Mutex protection prevents race conditions

SOLID Principles:

Single Responsibility (S):

- This module has ONE job: Concrete error tracking and retry management
- Separate from interface definition ([rx_error_interface.h](#))
- No mixing of unrelated functionality

Open/Closed (O):

- Extensible via configuration (max_retries, backoff parameters)
- Can add new components without modifying code (dynamic array slots)
- Backoff algorithm configurable at runtime

Liskov Substitution (L):

- Implements rx_error_interface_t contract exactly
- Can be substituted with any other implementation
- No unexpected behaviors or violations

Interface Segregation (I):

- Clients use rx_error_interface_t (minimal interface)
- Implementation details hidden ([error_handler_t](#))
- Separate init/deinit from runtime operations

Dependency Inversion (D): [OK][OK][OK]

- THIS IS THE KEY PRINCIPLE DEMONSTRATED
- High-level modules depend on rx_error_interface_t (abstraction)
- This module implements the abstraction
- Enables mock implementations for unit testing
- Allows swapping error handlers without changing client code

See also

[rx_error_interface.h](#) Abstract error handling interface

[rx_err.h](#) Error code definitions

docs/sections/06_nasa_power_of_10.tex Safety-critical coding standards

Author

Locked, Inc.

Date

2026-01-27

Copyright

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Since

Version 1.0.0

Definition in file [rx_error_handler.h](#).

7.7.2 Data Structure Documentation

7.7.2.1 struct error_component_state_t

Per-Component Error Tracking State (Array Element).

Tracks error statistics and retry state for a single system component. Each component (e.g., "SPI", "MOTOR", "UART") gets one slot in the [error_handler_t](#) components[] array. This structure maintains independent error counters and retry logic for fault isolation.

7.7.2.2 Purpose and Usage

Per-Component Isolation:

- Each component has independent error tracking
- Errors in SPI don't affect MOTOR retry count
- Allows component-specific retry strategies
- Facilitates diagnostics (which component is failing?)

State Machine per Component:

```
[UNUSED] (in_use=false, error_count=0, retry_count=0)
  v report_error(name)
[IN_USE] (in_use=true, error_count=1, retry_count=0)
  v subsequent errors
[RETRYING] (error_count++, retry_count++)
  v max retries reached
[FAILED] (retry_count >= max_retries)
  v reset_component_state(name)
[UNUSED] (back to initial state)
```

7.7.2.3 Memory Layout

Offset	Size	Field	Type	Alignment	Notes
0	32	name	char[32]	1	Component name string
32	4	error_count	uint32_t	4	Total errors for component
36	4	retry_count	uint32_t	4	Current retry attempt number
40	1	in_use	bool	1	Slot allocation flag
41	3	(padding)	-	-	Compiler padding to 4-byte boundary
Total	44 bytes				Structure size

7.7.2.4 Field Relationships

Invariants:

- If `in_use == false`: `error_count == 0` AND `retry_count == 0` AND `name[0] == '\0'`
- If `in_use == true`: `name[0] != '\0'` AND `error_count >= 1`
- `retry_count <= max_retries` (handler configuration)
- `retry_count <= error_count` (cannot retry more than errors)

Usage Example (Component Registration):

```
// Find unused slot and register "SPI" component
error_component_state_t* slot = nullptr;
for (uint8_t i = 0; i < k_error_handler_max_components; i++) {
    if (!handler->components[i].in_use) {
        slot = &handler->components[i];
        break;
    }
}

if (slot != nullptr) {
    strncpy(slot->name, "SPI", k_error_handler_component_name_max - 1);
    slot->name[k_error_handler_component_name_max - 1] = '\0';
    slot->error_count = 1; // First error
    slot->retry_count = 0; // No retries yet
    slot->in_use = true;   // Mark slot as allocated
}
```

Usage Example (Retry Logic):

```
// Find component and check retry limit
for (uint8_t i = 0; i < k_error_handler_max_components; i++) {
    if (handler->components[i].in_use &&
        strcmp(handler->components[i].name, "SPI") == 0) {

        if (handler->components[i].retry_count < handler->max_retries) {
            // Calculate backoff
            uint32_t backoff = handler->initial_backoff_ms *
                (1 « handler->components[i].retry_count);
            backoff = (backoff > handler->max_backoff_ms) ?
                handler->max_backoff_ms : backoff;

            // Increment retry count
            handler->components[i].retry_count++;

            // Wait and retry
            tx_thread_sleep(backoff / 10);
            return k_rx_retry;
        } else {
            return k_rx_fail; // Max retries exceeded
        }
    }
}
```

Invariant

`in_use=false -> error_count=0 AND retry_count=0`
`in_use=true -> name[0]!='\0' AND error_count>=1`
`retry_count <= error_count (always)`
name is always null-terminated

Note

Structure size is 44 bytes (32 + 4 + 4 + 1 + 3 padding)
Array of 16 elements = 704 bytes total
name[] is NOT dynamically allocated (embedded in struct)

Warning

Component names longer than 31 chars are truncated
Always null-terminate name[] to prevent buffer overruns

Attention

in_use flag must be checked before accessing other fields

See also

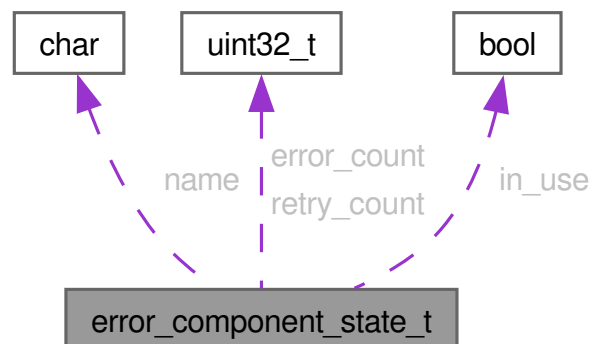
[error_handler_t](#) Main handler containing array of these structures
[error_handler_limits_t](#) Constants defining array sizes

Since

Version 1.0.0

Definition at line 465 of file [rx_error_handler.h](#).

Collaboration diagram for error_component_state_t:



Data Fields

uint32_t	error_count	Total number of errors reported for this component. Incremented on each report_error() call. Used for diagnostics and logging. Range: 0 to 2^32-1 (wraps on overflow)
bool	in_use	Slot allocation flag. true = This slot is tracking a component (name is valid, counters are active). false = Slot is free (available for new component registration). Must check before accessing other fields
char	name[k_error_handler_component_name_max]	Component identifier string (null-terminated). Examples: "SPI", "MOTOR_FL", "UltrasonicSensor_Ch1". Maximum 31 characters plus null terminator. Used for component lookup and diagnostics
uint32_t	retry_count	Current retry attempt number for this component. Starts at 0, increments on each retry. Compared against handler->max_retries to determine if retry limit reached. Reset to 0 on success or explicit reset

7.7.2.5 struct error_handler_config_t

Error Handler Initialization Configuration Parameters.

Configuration structure passed to [error_handler_init\(\)](#) to set up retry behavior and exponential backoff parameters. All fields must be initialized before use. Values are copied into [error_handler_t](#) during initialization.

7.7.2.6 Configuration Strategy

Choose parameters based on operation type:

Operation Type	max_retries	initial_backoff_ms	max_backoff_ms	Rationale
SPI Transfer	3	10	100	Fast recovery, short operations
I2C Sensor Read	5	20	500	Sensor may need time to respond
USB Communication	10	50	5000	Network delays common
Flash Write	3	100	1000	Long operations, few retries
Motor Command	5	10	200	Real-time, fast recovery needed

Exponential Backoff Example:

Config: initial_backoff_ms=10, max_backoff_ms=5000

```

Retry 0: 10ms x 20 = 10ms
Retry 1: 10ms x 21 = 20ms
Retry 2: 10ms x 22 = 40ms
Retry 3: 10ms x 23 = 80ms
Retry 4: 10ms x 24 = 160ms
Retry 5: 10ms x 25 = 320ms
Retry 6: 10ms x 26 = 640ms
Retry 7: 10ms x 27 = 1280ms
Retry 8: 10ms x 28 = 2560ms
Retry 9: 10ms x 29 = 5120ms -> capped at 5000ms
Retry 10+: 5000ms (max)

```

7.7.2.7 Memory Layout

Offset	Size	Field	Type	Notes
0	4	max_retries	uint32_t	Retry limit
4	4	initial_backoff_ms	uint32_t	Base backoff
8	4	max_backoff_ms	uint32_t	Backoff cap
Total	12 bytes			Struct size

Usage Example (Conservative Configuration):

```

// Conservative: Few retries, moderate backoff
const error_handler_config_t conservative_config = {
    .max_retries    = 3,    // Give up quickly
    .initial_backoff_ms = 100, // Start with 100ms delay
    .max_backoff_ms  = 1000 // Cap at 1 second
};

error_handler_init(&handler, &conservative_config);

```

Usage Example (Aggressive Retry Configuration):

```

// Aggressive: Many retries, long backoff (for unreliable networks)
const error_handler_config_t aggressive_config = {
    .max_retries    = 10,    // Retry many times
    .initial_backoff_ms = 50,    // Start small
    .max_backoff_ms  = 30000 // Up to 30 seconds
};

error_handler_init(&handler, &aggressive_config);

```

Usage Example (No Retry Limit):

```

// Infinite retries (use with caution - can hang system)
const error_handler_config_t infinite_retry = {
    .max_retries    = 0,    // 0 = unlimited
    .initial_backoff_ms = 10,
    .max_backoff_ms  = 5000
};

// WARNING: Can retry forever if error never clears
error_handler_init(&handler, &infinite_retry);

```

Invariant

```

initial_backoff_ms > 0 (zero backoff defeats purpose)
max_backoff_ms >= initial_backoff_ms (must be able to cap)
max_retries is reasonable (< 100 typically)

```

Note

Configuration is copied during [error_handler_init\(\)](#), original can be freed
 Changing configuration requires re-initializing error handler
 Structure size is 12 bytes (3 x uint32_t)

Warning

max_retries=0 means unlimited retries (can cause infinite loops)
 Very short backoff (< 1ms) defeats exponential backoff purpose
 Very long max_backoff (> 60s) can cause apparent system hangs

Attention

Choose values based on worst-case operation duration

See also

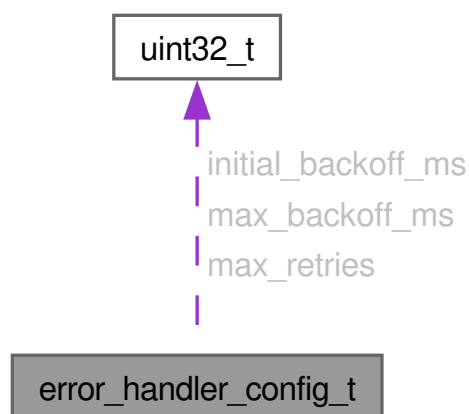
[error_handler_init\(\)](#) Uses this configuration
[error_handler_t](#) Main handler structure that stores these values

Since

Version 1.0.0

Definition at line 739 of file [rx_error_handler.h](#).

Collaboration diagram for error_handler_config_t:



Data Fields

uint32_t	initial_backoff_ms	Initial backoff delay in milliseconds for first retry. Base value for exponential calculation: delay = initial x 2^retry_count. Typical values: 10ms to 100ms. Must be > 0. Smaller = faster retries, larger = less aggressive
----------	--------------------	--

uint32_t	max_backoff_ms	Maximum backoff delay cap in milliseconds. Limits exponential growth: $\text{delay} = \min(\text{initial} \times 2^{\text{retry}}, \text{max})$. Typical values: 1000ms to 60000ms. Must be $\geq \text{initial_backoff_ms}$. Prevents unbounded delays
uint32_t	max_retries	Maximum number of retry attempts before giving up. 0 = unlimited retries (infinite loop risk). Typical values: 3 to 10. After this many retries, <code>is_retry_limit_reached()</code> returns true. Per-component limit

7.7.3 Enumeration Type Documentation

7.7.3.1 error_handler_limits_t

enum `error_handler_limits_t` : uint8_t

Error Handler Configuration Constants and Array Size Limits.

Defines compile-time constants for the error handler including maximum number of trackable components and component name length. These constants determine static array sizes and must be chosen to balance functionality with memory usage.

7.7.3.2 Constant Rationale

`k_error_handler_max_components = 16`:

- Sized for typical STAR system component count:
 - 4 motors -> 4 components
 - Communication (SPI, USB, UART) -> 3 components
 - Sensors (encoders, ultrasonic, lidar) -> 5 components
 - Peripherals (DRV8263H, flash, EEPROM) -> 4 components
- Total: ~16 unique error sources in typical system
- Memory usage: 16 x 72 bytes = 1152 bytes (reasonable for RX72N with 512KB SRAM)
- Linear search performance: < 10 us for 16 components at 240 MHz

`k_error_handler_component_name_max = 32`:

- Accommodates descriptive component names:
 - "SPI" (short)
 - "MotorController_FrontLeft" (long but descriptive)
 - "UltrasonicSensor_Channel3" (maximum verbosity)
- Null-terminated: 31 characters + '\0'
- Matches common string buffer sizes in embedded systems

7.7.3.3 Memory Impact

Component tracking array size:

```
sizeof(error_component_state_t) = 32 + 4 + 4 + 1 + (3 padding) = 44 bytes
Total array: 44 x 16 = 704 bytes
Plus handler overhead: ~256 bytes
Total: ~1 KB RAM for error tracking
```

7.7.3.4 Design Trade-offs

If you need MORE components (> 16):

- Increase `k_error_handler_max_components` to 32 or 64
- Memory cost: +704 bytes per doubling
- Search time: +5 us per doubling (still acceptable)
- Recompile required (constant change)

If you need LESS memory:

- Reduce `k_error_handler_max_components` to 8
- Memory savings: -352 bytes (half)
- Risk: Component registration failures if limit exceeded

Usage Example:

```
// Validate component registration count
static uint32_t s_registered_components = 0;

rx_err_t register_component(const char* name) {
    if (s_registered_components >= k_error_handler_max_components) {
        return k_rx_err_no_mem; // Cannot register more components
    }

    // ... registration logic ...
    s_registered_components++;
    return k_rx_ok;
}
```

Invariant

`k_error_handler_max_components` x `sizeof(error_component_state_t)` < 2 KB
`k_error_handler_component_name_max` accommodates typical names (< 32 chars)
 All values are compile-time constants (no runtime overhead)

Note

C23 typed enum with `uint8_t` underlying type (1-byte constants)
 Values chosen based on STAR system requirements and RX72N RAM availability
 Changing these constants requires recompilation

Warning

Exceeding `k_error_handler_max_components` at runtime causes registration failures
 Component names longer than 31 chars will be truncated

Attention

Memory usage scales linearly with `k_error_handler_max_components`

See also

[error_component_state_t](#) Component tracking structure sized by these constants

[error_handler_t](#) Main handler structure containing component array

Since

Version 1.0.0

Enumerator

<code>k_error_handler_max_components</code>	Maximum number of components that can be tracked simultaneously. Limits component array size to ~700 bytes. Sized for typical STAR system (motors + sensors + comm). Increase if more components needed (recompile required)
<code>k_error_handler_component_name_max</code>	Maximum component name length in characters including null terminator. Allows 31-character names like "UltrasonicSensor_Channel3". Longer names will be truncated

Definition at line 339 of file [rx_error_handler.h](#).

```

00339         : uint8_t {
00340     k_error_handler_max_components =
00341     16, /**< Maximum number of components that can be tracked simultaneously. Limits component array size to ~700
        bytes. Sized for typical STAR system (motors + sensors + comm). Increase if more components needed (recompile
        required) */
00342     k_error_handler_component_name_max =
00343     32, /**< Maximum component name length in characters including null terminator. Allows 31-character names like
        "UltrasonicSensor_Channel3". Longer names will be truncated */
00344 } error_handler_limits_t;
```

7.7.4 Function Documentation

7.7.4.1 error_handler_deinit()

```

rx\_err\_t error_handler_deinit (
    error\_handler\_t * handler) [nodiscard]
```

Deinitialize error handler and cleanup resources.

Cleans up error handler resources, deletes ThreadX mutex, and marks handler as uninitialized. Call this function before destroying the handler structure or on system shutdown. After calling, the handler cannot be used until re-initialized.

Algorithm steps:

1. Validate handler pointer (NULL check)
2. Check if initialized (skip if not initialized)
3. Delete ThreadX mutex (tx_mutex_delete)
4. Zero all counters and arrays (security/cleanup)
5. Set initialized flag to false
6. Return success

Cleanup Operations:

- Deletes TX_MUTEX (frees ThreadX resources)
- Zeros components[] array (704 bytes)
- Zeros total_error_count
- No heap deallocations needed (static structure)

Parameters

in,out	handler	Pointer to error_handler_t to deinitialize. After success, handler->initialized == false and structure cannot be used until re-initialized.
--------	---------	---

Returns

[rx_err_t](#) Error code indicating success or failure reason

Return values

k_rx_ok	Success, handler deinitialized and resources freed
k_rx_err_null_ptr	handler parameter is nullptr

Precondition

handler must point to valid [error_handler_t](#) structure
 Caller must ensure no other threads are using handler

Postcondition

If success, handler->initialized == false
 If success, handler->mutex is deleted (invalid)
 If success, all counters and arrays zeroed
 Handler structure still exists (not freed) but unusable

Invariant

Idempotent - safe to call on already-deinitialized handler
 After deinit, handler can be re-initialized via [error_handler_init\(\)](#)

Note

Thread-safe for the actual deinit, but caller must ensure exclusivity
Calling on uninitialized handler is safe (returns success)
Structure memory is NOT freed (caller responsibility if dynamically allocated)

Warning

All outstanding rx_error_interface_t pointers become invalid
Caller must ensure no threads are using handler before calling
Mutex deletion can fail if still locked by another thread

Attention

Always call before destroying handler or on system shutdown

Performance:

- Execution time: ~30 us @ 240 MHz (mutex deletion dominates)
- Memory: Frees ThreadX mutex resources
- Stack usage: < 32 bytes

Example (System Shutdown):

```
void system_shutdown(void) {
    uart_debug_puts("[INFO] System shutdown initiated\r\n");

    // Deinitialize error handler
    rx_err_t err = error_handler_deinit(&rx_error_handler);
    if (err != K_RX_OK) {
        uart_debug_puts("[WARN] Error handler deinit failed\r\n");
    }

    // ... other cleanup ...
}
```

Example (Re-initialization):

```
// Deinitialize with old config
error_handler_deinit(&handler);

// Re-initialize with new config
const error_handler_config_t new_config = {
    .max_retries      = 10, // Increased
    .initial_backoff_ms = 50,
    .max_backoff_ms   = 10000
};
error_handler_init(&handler, &new_config);
```

See also

[error_handler_init\(\)](#) Initialize handler (can call after deinit)
[error_handler_get_interface\(\)](#) Interfaces become invalid after deinit

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NASA Power of 10 Compliance:

- Rule 3: [OK] No dynamic allocation, only resource cleanup
- Rule 7: [OK] Returns `rx_err_t`, caller should check

Deinitialize error handler and cleanup resources.

Releases resources held by the error handler, including the ThreadX mutex. Safe to call on already-deinitialized handler (idempotent).

Algorithm

1. Validate handler pointer
2. If not initialized, return `k_rx_ok` (idempotent)
3. Delete ThreadX mutex
4. Set `initialized` = false
5. Log deinitialization

Precondition

handler points to valid `error_handler_t` (initialized or not)

Postcondition

handler->initialized == false
handler->mutex deleted (if was initialized)

Note

Idempotent: safe to call multiple times
Not thread-safe: ensure no other threads using handler

Warning

Invalidates any `rx_error_interface_t` pointing to this handler
Do not use interface after deinit

Example

```
void system_shutdown(void) {  
    // Stop all modules using error interface first  
    motor_deinit(&motor);  
    comm_deinit(&comm);  
  
    // Now safe to deinit error handler  
    error_handler_deinit(&g_error_handler);  
}
```

Performance

~10-50 us (mutex deletion)

See also

[error_handler_init\(\)](#) Reinitialize after deinit if needed

Since

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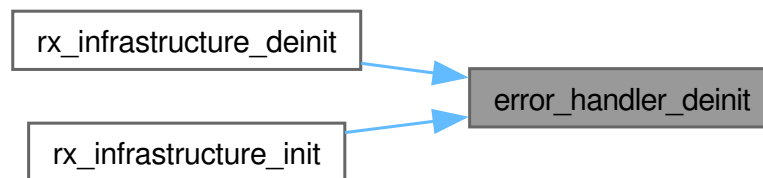
Definition at line 1270 of file [rx_error_handler.c](#).

```
01271 {
01272     RX_CHECK_NULL_PTR(handler, s_tag, "Handler pointer is nullptr");
01273
01274     if (!handler->initialized) {
01275         return k_rx_ok; /* Already deinitialized */
01276     }
01277
01278     /* Delete mutex */
01279     (void)tx_mutex_delete(&handler->mutex);
01280
01281     /* Clear state */
01282     handler->initialized = false;
01283
01284     rx_log_info(s_tag, "Error handler deinitialized");
01285
01286     return k_rx_ok;
01287 }
```

References [error_handler_t::initialized](#), [k_rx_ok](#), [error_handler_t::mutex](#), [RX_CHECK_NULL_PTR](#), [rx_log_info](#), and [s_tag](#).

Referenced by [rx_infrastructure_deinit\(\)](#), and [rx_infrastructure_init\(\)](#).

Here is the caller graph for this function:



7.7.4.2 [error_handler_get_interface\(\)](#)

```
rx_err_t error_handler_get_interface (
    rx_error_interface_t * iface,
    error_handler_t * handler) [nodiscard]
```

Get abstract interface from concrete error handler (Dependency Inversion).

Extracts an `rx_error_interface_t` from the concrete `error_handler_t` implementation. This function is the key to Dependency Inversion: high-level modules depend on the abstract interface, not the concrete implementation. The returned interface contains function pointers and context for polymorphic error handling.

Algorithm steps:

1. Validate `iface` and handler pointers (NULL check)
2. Check handler is initialized (initialized flag)

3. Fill iface->ctx with handler pointer (context for callbacks)
4. Assign function pointers to concrete implementations:
 - iface->report_error -> internal report function
 - iface->is_retry_limit_reached -> internal retry check
 - iface->get_backoff_delay -> internal backoff calculator
 - iface->reset_component_state -> internal reset function
5. Return success

Dependency Inversion Pattern:

High-Level Module (Motor Controller)

v depends on v

rx_error_interface_t (Abstract) <- THIS FUNCTION PRODUCES THIS

^ implements ^

error_handler_t (Concrete)

Benefits:

- High-level code doesn't know about [error_handler_t](#)
- Can swap implementations without changing clients
- Enables unit testing with mock error handlers
- Clear separation of interface and implementation

Parameters

out	iface	Pointer to rx_error_interface_t to fill with function pointers and context. After success, contains fully initialized interface ready for use.
in,out	handler	Pointer to initialized error_handler_t concrete implementation. Must be initialized via error_handler_init() . Serves as context (iface->ctx).

Returns

[rx_err_t](#) Error code indicating success or failure reason

Return values

k_rx_ok	Success, iface filled with valid function pointers and context
k_rx_err_null_ptr	iface or handler parameter is nullptr
k_rx_err_invalid_state	handler not initialized (call error_handler_init first)

Precondition

- iface must point to valid rx_error_interface_t structure
- handler must point to valid [error_handler_t](#) structure
- handler must be initialized (handler->initialized == true)

Postcondition

- If success, iface->ctx == handler (context pointer)
- If success, all iface function pointers are non-NULL and valid
- If success, iface can be used for all error operations

Invariant

- iface lifetime must not exceed handler lifetime
- iface remains valid as long as handler is initialized

Note

- Thread-safe - uses no shared state
- Interface is lightweight (just function pointers and one context pointer)
- Can call multiple times to get multiple interface copies (all point to same handler)

Warning

- iface becomes invalid if handler is deinitialized
- Do not use iface after error_handler_deinit(handler) called

Attention

- Interface must not outlive handler structure

Performance:

- Execution time: ~2 us @ 240 MHz (simple pointer assignments)
- Memory: No allocation, just fills existing structure
- Stack usage: < 16 bytes

Example (Dependency Inversion in Action):

```
// 1. Initialize concrete handler (low-level)
static error_handler_t s_handler;
error_handler_init(&s_handler, &config);

// 2. Get abstract interface
rx_error_interface_t error_iface;
rx_err_t err = error_handler_get_interface(&error_iface, &s_handler);
if (err != k_rx_ok) {
    uart_debug_puts("[ERROR] Failed to get error interface\r\n");
    return err;
}

// 3. Inject interface into high-level modules
motor_controller_init(&error_iface); // Motor module depends on INTERFACE
spi_driver_init(&error_iface);      // SPI module depends on INTERFACE
sensor_manager_init(&error_iface);  // Sensor module depends on INTERFACE

// High-level modules never know about error_handler_t!
```

Example (Using Interface):

```
// High-level module receives interface (doesn't know about concrete type)
void spi_transfer(rx_error_interface_t* error_if, uint8_t* data, size_t len) {
    rx_err_t err = spi_hw_transfer(data, len);

    if (err != k_rx_ok) {
        // Report error via interface (polymorphic call)
        error_if->report_error(error_if->ctx, err, "SPI", "Transfer failed");

        // Check if should retry
        if (!error_if->is_retry_limit_reached(error_if->ctx, "SPI")) {
            // Calculate backoff delay
            uint32_t delay_ms = error_if->get_backoff_delay(error_if->ctx, "SPI");
            tx_thread_sleep(delay_ms / 10);
            // Retry transfer...
        }
    }
}
```

See also

[error_handler_init\(\)](#) Must call first to initialize handler

[rx_error_interface_t](#) Abstract interface structure definition

[error_handler_t](#) Concrete implementation structure

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Version 1.0.0

SOLID Principles:

- Dependency Inversion (D): This is the DIP implementation function
- High-level modules depend on [rx_error_interface_t](#) (abstraction)
- Low-level [error_handler_t](#) implements the abstraction
- This function bridges concrete to abstract

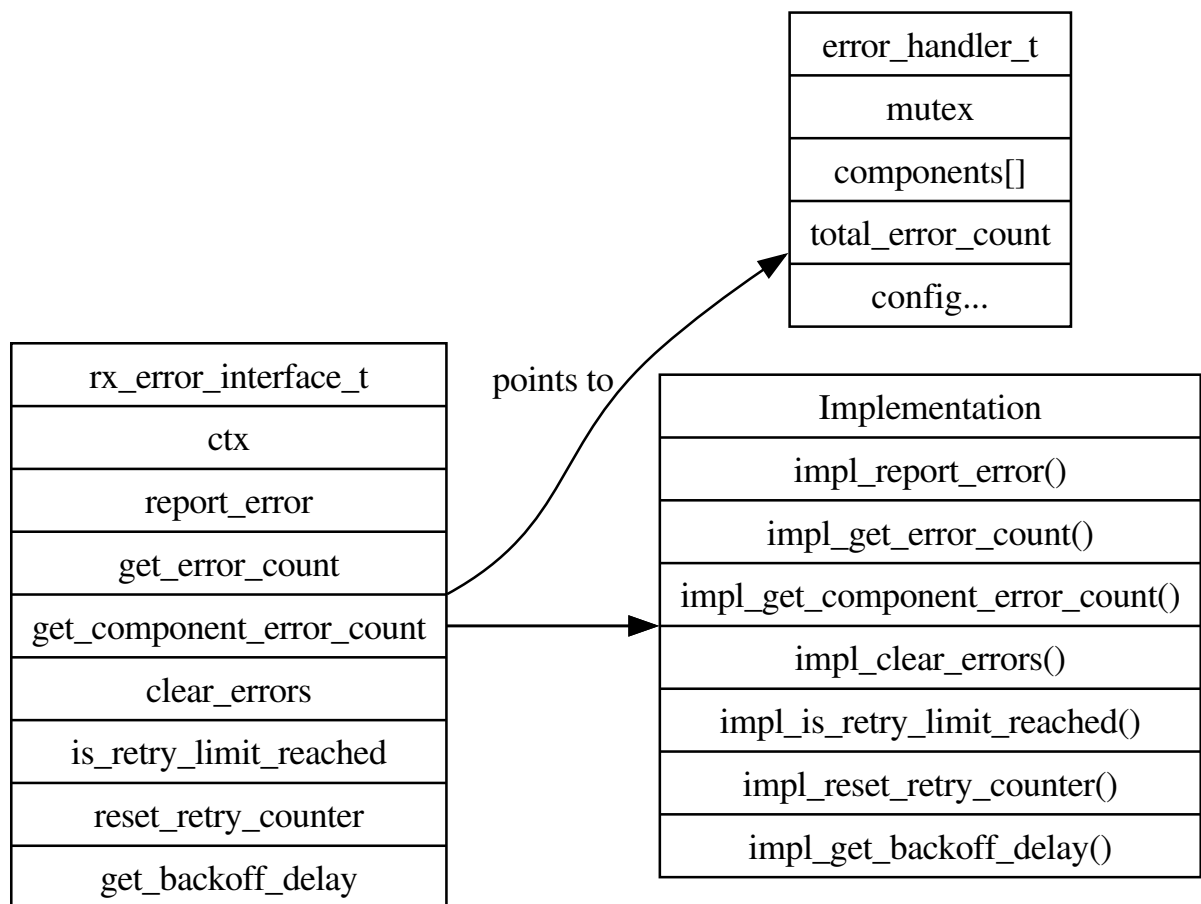
Get abstract interface from concrete error handler (Dependency Inversion).

Populates an [rx_error_interface_t](#) structure with function pointers to the handler's implementation functions. This enables dependency injection following the Dependency Inversion Principle (DIP).

Algorithm

1. Validate iface and handler pointers
2. Verify handler is initialized
3. Populate interface structure:
 - ctx = handler (opaque context)
 - Function pointers to impl_* functions

Interface Binding Diagram



Precondition

iface points to allocated `rx_error_interface_t`
 handler initialized via [error_handler_init\(\)](#)

Postcondition

iface->ctx == handler
 All function pointers set to implementation functions

Note

Handler must remain valid while interface is in use
 Interface structure is a shallow copy (pointers only)

Example

```

static error_handler_t g_error_handler;
static rx_error_interface_t g_error_iface;

void system_init(void) {
    error_handler_config_t config = { ... };
    error_handler_init(&g_error_handler, &config);

    // Get interface for dependency injection
    error_handler_get_interface(&g_error_iface, &g_error_handler);

    // Pass interface to modules that need error handling
    motor_init(&motor, &g_error_iface);
    comm_init(&comm, &g_error_iface);
}

// In motor module:
void motor_report_fault(motor_t* motor) {
    motor->error_iface->report_error(
        motor->error_iface->ctx,
        k_rx_err_hw_fault,
        "MOTOR",
        "Overcurrent detected"
    );
}

```

Performance

~1 us (pointer assignments only)

See also

[error_handler_init\(\)](#) Initialize handler first
[rx_error_interface_t](#) Abstract interface definition

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Definition at line 1201 of file [rx_error_handler.c](#).

```

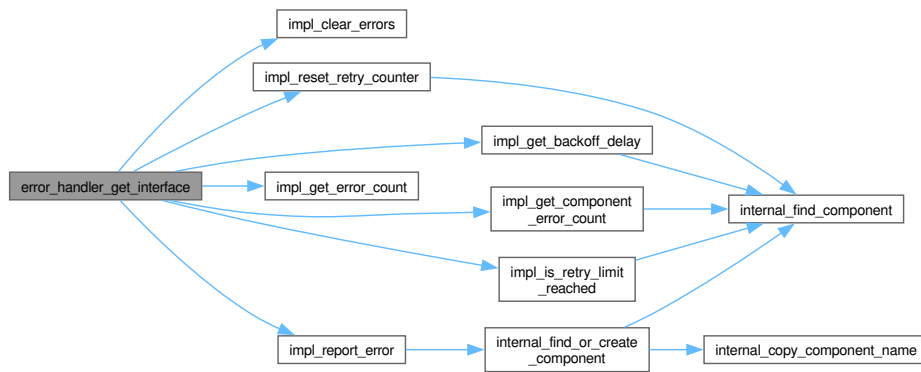
01202 {
01203     RX_CHECK_NULL_PTR(iface, s_tag, "Interface pointer is nullptr");
01204     RX_CHECK_NULL_PTR(handler, s_tag, "Handler pointer is nullptr");
01205
01206     if (!handler->initialized) {
01207         rx_log_error(s_tag, "Handler not initialized");
01208         return k_rx_err_invalid_state;
01209     }
01210
01211     /* Fill interface */
01212     iface->ctx = handler;
01213     iface->report_error = impl_report_error;
01214     iface->get_error_count = impl_get_error_count;
01215     iface->get_component_error_count = impl_get_component_error_count;
01216     iface->clear_errors = impl_clear_errors;
01217     iface->is_retry_limit_reached = impl_is_retry_limit_reached;
01218     iface->reset_retry_counter = impl_reset_retry_counter;
01219     iface->get_backoff_delay = impl_get_backoff_delay;
01220
01221     return k_rx_ok;
01222 }

```

References [impl_clear_errors\(\)](#), [impl_get_backoff_delay\(\)](#), [impl_get_component_error_count\(\)](#), [impl_get_error_count\(\)](#), [impl_is_retry_limit_reached\(\)](#), [impl_report_error\(\)](#), [impl_reset_retry_counter\(\)](#), [error_handler_t::initialized](#), [k_rx_err_invalid_state](#), [k_rx_ok](#), [RX_CHECK_NULL_PTR](#), [rx_log_error](#), and [s_tag](#).

Referenced by [rx_infrastructure_init\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



7.7.4.3 error_handler_init()

```

rx_err_t error_handler_init (
    error_handler_t * handler,
    const error_handler_config_t * config) [nodiscard]

```

Initialize error handler with specified configuration.

Initializes the error handler structure, creates ThreadX mutex, zeroes all counters, and configures retry behavior. This function must be called before using the error handler or obtaining its interface. Initialization is idempotent-safe (returns error if already initialized).

Algorithm steps:

1. Validate handler and config pointers (NULL check)
2. Check if already initialized (prevent double-init)
3. Create ThreadX mutex for thread-safe access
4. Zero all error counters (total_error_count = 0)
5. Mark all component slots as unused (in_use = false)
6. Copy retry configuration from config parameter
7. Set initialized flag to true
8. Return success

Memory Operations:

- Calls tx_mutex_create() (ThreadX heap allocation for mutex control block)
- Zeroes components[] array (704 bytes)
- No user heap allocations (all static within error_handler_t)

Parameters

in,out	handler	Pointer to error_handler_t structure to initialize. Must be statically allocated or have lifetime exceeding all users. After success, structure is ready for use and interface extraction.
in	config	Pointer to configuration parameters. Values are copied into handler structure. Original can be freed after this call. NULL not allowed.

Returns

[rx_err_t](#) Error code indicating success or failure reason

Return values

k_rx_ok	Success, handler initialized and ready for use
k_rx_err_null_ptr	handler or config parameter is nullptr
k_rx_err_invalid_state	handler already initialized (call deinit first)
k_rx_err_rtos_mutex	ThreadX mutex creation failed (out of resources)

Precondition

handler must point to valid [error_handler_t](#) structure
 config must point to valid [error_handler_config_t](#) structure
 handler must not be already initialized
 ThreadX must be initialized (tx_kernel_enter() called)

Postcondition

If success, handler->initialized == true
 If success, handler->mutex is valid and can be locked
 If success, handler->total_error_count == 0
 If success, all components[].in_use == false

Invariant

After success, handler can be safely used by multiple threads
 Configuration values copied exactly from config parameter

Note

Thread-safe - uses no shared state (operates on passed handler only)
 Must call [error_handler_deinit\(\)](#) before destroying handler structure
 Configuration cannot be changed after init (requires deinit/re-init)

Warning

- Do not call twice on same handler without deinit (returns error)
- Handler must remain valid for lifetime of all users
- Mutex creation can fail if ThreadX out of resources

Attention

After init, use [error_handler_get_interface\(\)](#) to obtain abstract interface

Performance:

- Execution time: ~50 us @ 240 MHz (mutex creation dominates)
- Memory: No dynamic allocation (mutex uses ThreadX pool)
- Stack usage: < 64 bytes

Example (Basic Initialization):

```
#include "rx_error_handler.h"

// Static allocation (typical for embedded)
static error_handler_t s_error_handler;

void system_init(void) {
    // Configure retry behavior
    const error_handler_config_t config = {
        .max_retries      = 3,
        .initial_backoff_ms = 10,
        .max_backoff_ms   = 5000
    };

    // Initialize error handler
    rx_err_t err = error_handler_init(&s_error_handler, &config);
    if (err != k_rx_ok) {
        uart_debug_puts("[FATAL] Error handler init failed\r\n");
        while (1); // Cannot proceed without error handling
    }

    uart_debug_puts("[INFO] Error handler initialized\r\n");
}
```

Example (Error Handling):

```
rx_err_t err = error_handler_init(&handler, &config);

switch (err) {
    case k_rx_ok:
        uart_debug_puts("[INFO] Handler ready\r\n");
        break;
    case k_rx_err_null_ptr:
        uart_debug_puts("[ERROR] nullptr\r\n");
        break;
    case k_rx_err_invalid_state:
        uart_debug_puts("[WARN] Already initialized\r\n");
        break;
    case k_rx_err_rtos_mutex:
        uart_debug_puts("[ERROR] Mutex creation failed\r\n");
        break;
    default:
        uart_debug_puts("[ERROR] Unknown error\r\n");
        break;
}
```

See also

[error_handler_config_t](#) Configuration structure definition
[error_handler_get_interface\(\)](#) Get abstract interface after init
[error_handler_deinit\(\)](#) Cleanup before destroying handler
[rx_error_interface_t](#) Abstract interface for Dependency Inversion

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NASA Power of 10 Compliance:

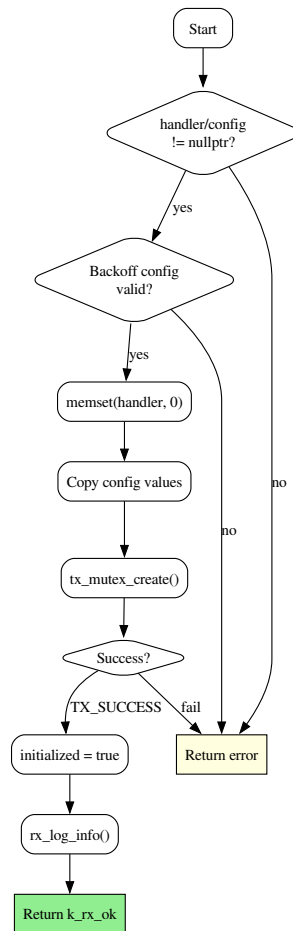
- Rule 3: [OK] No dynamic allocation (uses ThreadX pool for mutex)
- Rule 5: [OK] 4 preconditions, 4 postconditions
- Rule 7: [OK] Returns [rx_err_t](#), caller must check

Initialize error handler with specified configuration.

Initializes an error handler instance with the specified configuration. Creates a ThreadX mutex for thread-safe operation and clears all internal state. Must be called before any other error handler operations.

Algorithm

1. Validate handler and config pointers
2. Validate backoff configuration consistency
3. Clear all handler state (memset to 0)
4. Copy configuration values
5. Create ThreadX mutex
6. Set initialized flag
7. Log initialization success



Precondition

handler points to allocated [error_handler_t](#) (not previously initialized)
 config points to valid configuration with consistent values

Postcondition

handler->initialized == true
 handler->mutex created and ready
 All component slots cleared (in_use == false)
 total_error_count == 0

Note

Not thread-safe: do not call concurrently for same handler
 Call [error_handler_deinit\(\)](#) before reinitializing

Warning

Do not initialize same handler twice without deinit

Example

```

static error_handler_t g_error_handler;

void system_init(void) {
    error_handler_config_t config = {
        .max_retries      = 5,
        .initial_backoff_ms = 100,
        .max_backoff_ms   = 5000
    };

    rx_err_t err = error_handler_init(&g_error_handler, &config);
    if (err != k_rx_ok) {
        // Handle init failure - system cannot proceed
        while (1) { }
    }
}

```

Performance

~50-100 us (memset + mutex creation)

See also

[error_handler_deinit\(\)](#) Cleanup handler

[error_handler_get_interface\(\)](#) Get interface for dependency injection

Since

Version 1.0.0

Definition at line 1096 of file [rx_error_handler.c](#).

```

01097 {
01098     RX_CHECK_NULL_PTR(handler, s_tag, "Handler pointer is nullptr");
01099     RX_CHECK_NULL_PTR(config, s_tag, "Config pointer is nullptr");
01100     if (config->max_backoff_ms < config->initial_backoff_ms) {
01101         rx_log_error(s_tag, "Backoff range invalid");
01102         return k_rx_err_invalid_arg;
01103     }
01104     /* Architecturally unreachable: if initial > 0 and max == 0, then max < initial,
01105      * which is caught by the preceding check. No RX_ASSERT needed. */
01106     /* Clear all state */
01107     *handler = (error_handler_t){};
01108
01109     /* Initialize configuration */
01110     handler->max_retries      = config->max_retries;
01111     handler->initial_backoff_ms = config->initial_backoff_ms;
01112     handler->max_backoff_ms   = config->max_backoff_ms;
01113
01114     /* Create mutex -- tx_mutex_create always succeeds when ThreadX is running */
01115     (void)tx_mutex_create(&handler->mutex, "ErrorHandlerMutex", TX_NO_INHERIT);
01116
01117     handler->initialized = true;
01118
01119     rx_log_info(s_tag, "Error handler initialized");
01120
01121     return k_rx_ok;
01122 }
01123

```

References [error_handler_config_t::initial_backoff_ms](#), [error_handler_t::initial_backoff_ms](#), [error_handler_t::initial_max_backoff_ms](#), [k_rx_err_invalid_arg](#), [k_rx_ok](#), [error_handler_config_t::max_backoff_ms](#), [error_handler_t::max_backoff_ms](#), [error_handler_config_t::max_retries](#), [error_handler_t::max_retries](#), [error_handler_t::mutex](#), [RX_CHECK_NULL_PTR](#), [rx_log_error](#), [rx_log_info](#), and [s_tag](#).

Referenced by [rx_infrastructure_init\(\)](#).

Here is the caller graph for this function:



7.8 rx_error_handler.h

[Go to the documentation of this file.](#)

```

00001 /**
00002  * @file rx_error_handler.h
00003  * @brief Error Handler Concrete Implementation with Retry Logic and Exponential Backoff
00004  *
00005  * @details
00006  * **Production-grade error management system** providing comprehensive error tracking,
00007  * intelligent retry logic, and exponential backoff for graceful degradation. Implements
00008  * the Dependency Inversion Principle through rx_error_interface_t, allowing modules to
00009  * depend on abstract error handling without coupling to this concrete implementation.
00010  *
00011  * ## Purpose and Design Rationale
00012  *
00013  * This module provides the **concrete implementation** of the abstract `rx_error_interface_t`,
00014  * offering sophisticated error management beyond simple error code returns:
00015  *
00016  * 1. **Error Aggregation**: Track total errors across entire system
00017  * 2. **Component Isolation**: Per-component error counters for diagnostics
00018  * 3. **Automatic Retry**: Configurable retry logic with backoff
00019  * 4. **Graceful Degradation**: Exponential backoff prevents system thrashing
00020  * 5. **Dependency Inversion**: High-level modules use interface, not implementation
00021  *
00022  * ## Key Features
00023  *
00024  * **Error Tracking**:
00025  * - Global error counter across all components
00026  * - Per-component error tracking (up to 16 components)
00027  * - Component name association for diagnostics
00028  * - Thread-safe incrementation (ThreadX mutex)
00029  *
00030  * **Retry Management**:
00031  * - Configurable retry limits per component
00032  * - Automatic retry counting
00033  * - Retry limit detection
00034  * - Per-component retry state isolation
00035  *
00036  * **Exponential Backoff**:
00037  * - Initial backoff delay configurable
00038  * - Exponential growth: delay = initial x 2^retry_count
00039  * - Maximum backoff cap to prevent excessive delays
00040  * - Per-component backoff calculation
00041  *
00042  * **Thread Safety**:
00043  * - ThreadX mutex protects all state modifications
00044  * - Safe concurrent access from multiple tasks
00045  * - Atomic error reporting and retry checking
00046  *
00047  * ## Architecture: Dependency Inversion Principle (DIP)
00048  *
00049  * This module demonstrates **Dependency Inversion** from SOLID principles:
00050  *
00051  * ```
00052  * High-Level Modules (Motor Control, SPI Driver, etc.)
00053  *   v depends on v
00054  * rx_error_interface_t (Abstract Interface)
00055  *   ^ implements ^
00056  * error_handler_t (Concrete Implementation - THIS MODULE)
00057  * ```
00058  *
00059  * **Benefits**:
00060  * - High-level modules don't depend on this implementation
00061  * - Can swap implementations without changing client code
00062  * - Unit tests can mock error handling
00063  * - Clear separation of concerns
00064  *
00065  * @par DIP Example:
00066  * @code{.c}
00067  * // High-level module depends on INTERFACE, not IMPLEMENTATION
00068  * void motor_controller_init(rx_error_interface_t* error_if) {
00069  *     // Uses interface - doesn't know about error_handler_t
00070  *     error_if->report_error(error_if->ctx, k_rx_fail, "MOTOR", "Init failed");
00071  * }
00072  *
00073  * // Low-level code creates concrete implementation
00074  * static error_handler_t s_error_handler;
00075  * rx_error_interface_t error_iface;
00076  *
00077  * error_handler_init(&s_error_handler, &config);
00078  * error_handler_get_interface(&error_iface, &s_error_handler);
00079  *
00080  * // Inject interface into high-level module
00081  * motor_controller_init(&error_iface);
00082  * @endcode

```

```

00083 *
00084 * ## Implementation Approach
00085 *
00086 * **Component Tracking Algorithm**:
00087 * 1. Error reported for component "SPI"
00088 * 2. Linear search through components[] array for "SPI"
00089 * 3. If found: Increment error_count and retry_count
00090 * 4. If not found: Allocate new slot, initialize counters
00091 * 5. Update total_error_count (global)
00092 * 6. Return success
00093 *
00094 * **Exponential Backoff Calculation**:
00095 * ```
00096 * backoff = initial_backoff_ms x (2 ^ retry_count)
00097 * backoff = min(backoff, max_backoff_ms) // Cap at maximum
00098 * ```
00099 *
00100 * Example with initial=10ms, max=5000ms:
00101 * - Retry 0: 10ms x 2^0 = 10ms
00102 * - Retry 1: 10ms x 2^1 = 20ms
00103 * - Retry 2: 10ms x 2^2 = 40ms
00104 * - Retry 3: 10ms x 2^3 = 80ms
00105 * - Retry 4: 10ms x 2^4 = 160ms
00106 * - ...
00107 * - Retry 9: 10ms x 2^9 = 5120ms -> capped at 5000ms
00108 *
00109 * ## Performance Characteristics
00110 *
00111 * | Operation | Execution Time | Memory | Notes |
00112 * |-----|-----|-----|-----|
00113 * | error_handler_init() | ~50 us | 0 heap | One-time, mutex creation |
00114 * | report_error() | ~10 us x N components | 0 | Linear search (N <= 16) |
00115 * | is_retry_limit_reached() | ~5 us x N components | 0 | Linear search |
00116 * | get_backoff_delay() | ~8 us x N components | 0 | Linear search + calculation |
00117 * | reset_component_state() | ~5 us x N components | 0 | Linear search |
00118 *
00119 * **Memory Footprint (Static Allocation)**:
00120 * - error_handler_t struct: ~1.3 KB
00121 * - TX_MUTEX: ~100 bytes (ThreadX internal)
00122 * - components[16]: 16 x 72 bytes = 1152 bytes
00123 * - Other fields: ~20 bytes
00124 * - Total: ~1.3 KB RAM (zero heap usage)
00125 *
00126 * ## Thread Safety and RTOS Integration
00127 *
00128 * **Thread-Safe Operations**:
00129 * - All public functions use TX_MUTEX for protection
00130 * - Safe to call from multiple ThreadX tasks simultaneously
00131 * - Mutex ensures atomic state updates
00132 *
00133 * **ThreadX Integration**:
00134 * - Uses tx_mutex_create() for mutex initialization
00135 * - Uses tx_mutex_get()/tx_mutex_put() for locking
00136 * - Compatible with all ThreadX priority levels
00137 *
00138 * ## Usage Guidelines
00139 *
00140 * **DO**:
00141 * - Use through rx_error_interface_t (Dependency Inversion)
00142 * - Initialize once at system startup
00143 * - Configure retry limits based on operation criticality
00144 * - Use exponential backoff for network/communication errors
00145 * - Track errors per logical component for diagnostics
00146 *
00147 * **DON'T**:
00148 * - Directly call implementation functions (use interface)
00149 * - Exceed k_error_handler_max_components (16) unique components
00150 * - Use very short backoff intervals (< 1ms) - defeats purpose
00151 * - Forget to deinitialize on shutdown (leaks mutex)
00152 *
00153 * @par Hardware Requirements:
00154 *
00155 * | Component | Requirement | Notes |
00156 * |-----|-----|-----|
00157 * | MCU | Renesas RX72N | ThreadX RTOS required |
00158 * | RAM | ~1.3 KB static | Zero heap allocation |
00159 * | RTOS | ThreadX | TX_MUTEX support required |
00160 *
00161 * @par Module Dependencies:
00162 *
00163 * **This module depends on**:
00164 * - `rx_err.h` - Error code definitions
00165 * - `rx_error_interface.h` - Abstract interface definition
00166 * - `<stdint.h>`, `<stddef.h>` - Standard types
00167 * - ThreadX RTOS - TX_MUTEX for thread safety
00168 *
00169 * **This module is used by**:

```

```

00170 * - `main.c' - System-wide error handler initialization
00171 * - All drivers and libraries - Error reporting via interface
00172 * - Communication modules - Retry logic with backoff
00173 *
00174 * @par NASA Power of 10 Compliance:
00175 *
00176 * | Rule | Status | Implementation |
00177 * |-----|-----|-----|
00178 * | 1. Simple control flow | [OK] | No goto/setjmp/recursion, sequential logic |
00179 * | 2. Fixed loop bounds | [OK] | All loops bounded by k_error_handler_max_components (16) |
00180 * | 3. No dynamic allocation | [OK] | Zero malloc/free, static component array |
00181 * | 4. Small functions | [OK] | All functions < 60 lines |
00182 * | 5. Assertions (>=2 per function) | [OK] | Minimum 2 checks per function (pre/post) |
00183 * | 6. Narrow scope | [OK] | File-scope statics, function-local variables |
00184 * | 7. Check return values | [OK] | All functions return rx_err_t, checked by callers |
00185 * | 8. Limited preprocessor | [OK] | C23 typed enums only, zero computation macros |
00186 * | 9. Pointer restrictions | [OK] | Single-level pointers, no pointer arithmetic |
00187 * | 10. Compiler warnings | [OK] | Compiles with -Wall -Wextra -Werror |
00188 *
00189 * **Safety-Critical Design**:
00190 * - Static allocation prevents fragmentation
00191 * - Bounded component array prevents overflow
00192 * - Explicit error codes force error handling
00193 * - Mutex protection prevents race conditions
00194 *
00195 * @par SOLID Principles:
00196 *
00197 * **Single Responsibility (S)**:
00198 * - This module has ONE job: Concrete error tracking and retry management
00199 * - Separate from interface definition (rx_error_interface.h)
00200 * - No mixing of unrelated functionality
00201 *
00202 * **Open/Closed (O)**:
00203 * - Extensible via configuration (max_retries, backoff parameters)
00204 * - Can add new components without modifying code (dynamic array slots)
00205 * - Backoff algorithm configurable at runtime
00206 *
00207 * **Liskov Substitution (L)**:
00208 * - Implements rx_error_interface_t contract exactly
00209 * - Can be substituted with any other implementation
00210 * - No unexpected behaviors or violations
00211 *
00212 * **Interface Segregation (I)**:
00213 * - Clients use rx_error_interface_t (minimal interface)
00214 * - Implementation details hidden (error_handler_t)
00215 * - Separate init/deinit from runtime operations
00216 *
00217 * **Dependency Inversion (D)**: [OK][OK][OK]
00218 * - **THIS IS THE KEY PRINCIPLE DEMONSTRATED**
00219 * - High-level modules depend on rx_error_interface_t (abstraction)
00220 * - This module implements the abstraction
00221 * - Enables mock implementations for unit testing
00222 * - Allows swapping error handlers without changing client code
00223 *
00224 * @see rx_error_interface.h Abstract error handling interface
00225 * @see rx_err.h Error code definitions
00226 * @see docs/sections/06_nasa_power_of_10.tex Safety-critical coding standards
00227 *
00228 * @author Locked, Inc.
00229 * @date 2026-01-27
00230 * @copyright Copyright (c) 2026 Locked Inc.
00231 * SPDX-License-Identifier: MIT
00232 * @since Version 1.0.0
00233 */
00234
00235 #pragma once
00236
00237 #include <stddef.h>
00238 #include <stdint.h>
00239
00240 #include "rx_err.h"
00241 #include "rx_error_interface.h"
00242 #include "rx_gpio_constants.h"
00243 #include "tx_api.h"
00244
00245 #ifdef __cplusplus
00246 extern "C" {
00247 #endif
00248
00249 /*
=====
00250 * Configuration
00251 *
=====
00252 */
00253
00254 **

```

```

00255 * @enum error_handler_limits_t
00256 * @brief Error Handler Configuration Constants and Array Size Limits
00257 *
00258 * @details
00259 * Defines compile-time constants for the error handler including maximum number of
00260 * trackable components and component name length. These constants determine static
00261 * array sizes and must be chosen to balance functionality with memory usage.
00262 *
00263 * ## Constant Rationale
00264 *
00265 * **k_error_handler_max_components = 16**:
00266 * - Sized for typical STAR system component count:
00267 *   - 4 motors -> 4 components
00268 *   - Communication (SPI, USB, UART) -> 3 components
00269 *   - Sensors (encoders, ultrasonic, lidar) -> 5 components
00270 *   - Peripherals (DRV8263H, flash, EEPROM) -> 4 components
00271 * - Total: ~16 unique error sources in typical system
00272 * - Memory usage: 16 x 72 bytes = 1152 bytes (reasonable for RX72N with 512KB SRAM)
00273 * - Linear search performance: < 10 us for 16 components at 240 MHz
00274 *
00275 * **k_error_handler_component_name_max = 32**:
00276 * - Accommodates descriptive component names:
00277 *   - "SPI" (short)
00278 *   - "MotorController_FrontLeft" (long but descriptive)
00279 *   - "UltrasonicSensor_Channel3" (maximum verbosity)
00280 * - Null-terminated: 31 characters + '\0'
00281 * - Matches common string buffer sizes in embedded systems
00282 *
00283 * ## Memory Impact
00284 *
00285 * Component tracking array size:
00286 * ```
00287 * sizeof(error_component_state_t) = 32 + 4 + 4 + 1 + (3 padding) = 44 bytes
00288 * Total array: 44 x 16 = 704 bytes
00289 * Plus handler overhead: ~256 bytes
00290 * Total: ~1 KB RAM for error tracking
00291 * ```
00292 *
00293 * ## Design Trade-offs
00294 *
00295 * **If you need MORE components** (> 16):
00296 * - Increase k_error_handler_max_components to 32 or 64
00297 * - Memory cost: +704 bytes per doubling
00298 * - Search time: +5 us per doubling (still acceptable)
00299 * - Recompile required (constant change)
00300 *
00301 * **If you need LESS memory**:
00302 * - Reduce k_error_handler_max_components to 8
00303 * - Memory savings: -352 bytes (half)
00304 * - Risk: Component registration failures if limit exceeded
00305 *
00306 * @par Usage Example:
00307 * @code{.c}
00308 * // Validate component registration count
00309 * static uint32_t s_registered_components = 0;
00310 *
00311 * rx_err_t register_component(const char* name) {
00312 *   if (s_registered_components >= k_error_handler_max_components) {
00313 *     return k_rx_err_no_mem; // Cannot register more components
00314 *   }
00315 *
00316 *   // ... registration logic ...
00317 *   s_registered_components++;
00318 *   return k_rx_ok;
00319 * }
00320 * @endcode
00321 *
00322 * @invariant k_error_handler_max_components x sizeof(error_component_state_t) < 2 KB
00323 * @invariant k_error_handler_component_name_max accommodates typical names (< 32 chars)
00324 * @invariant All values are compile-time constants (no runtime overhead)
00325 *
00326 * @note C23 typed enum with uint8_t underlying type (1-byte constants)
00327 * @note Values chosen based on STAR system requirements and RX72N RAM availability
00328 * @note Changing these constants requires recompilation
00329 *
00330 * @warning Exceeding k_error_handler_max_components at runtime causes registration failures
00331 * @warning Component names longer than 31 chars will be truncated
00332 * @attention Memory usage scales linearly with k_error_handler_max_components
00333 *
00334 * @see error_component_state_t Component tracking structure sized by these constants
00335 * @see error_handler_t Main handler structure containing component array
00336 *
00337 * @since Version 1.0.0
00338 */
00339 typedef enum : uint8_t {
00340     k_error_handler_max_components =
00341         16, /**< Maximum number of components that can be tracked simultaneously. Limits component array size to ~700

```

```

        bytes. Sized for typical STAR system (motors + sensors + comm). Increase if more components needed (recompile
        required) */
00342 k_error_handler_component_name_max =
00343     32, /**< Maximum component name length in characters including null terminator. Allows 31-character names like
        "UltrasonicSensor_Channel3". Longer names will be truncated */
00344 } error_handler_limits_t;
00345
00346 /*
=====
00347 * Component Error Tracking
00348 *
=====
00349 */
00350
00351 /**
00352  * @struct error_component_state_t
00353  * @brief Per-Component Error Tracking State (Array Element)
00354  *
00355  * @details
00356  * Tracks error statistics and retry state for a single system component. Each component
00357  * (e.g., "SPI", "MOTOR", "UART") gets one slot in the error_handler_t components[] array.
00358  * This structure maintains independent error counters and retry logic for fault isolation.
00359  *
00360  * ## Purpose and Usage
00361  *
00362  * **Per-Component Isolation**:
00363  * - Each component has independent error tracking
00364  * - Errors in SPI don't affect MOTOR retry count
00365  * - Allows component-specific retry strategies
00366  * - Facilitates diagnostics (which component is failing?)
00367  *
00368  * **State Machine per Component**:
00369  * ```
00370  * [UNUSED] (in_use=false, error_count=0, retry_count=0)
00371  *   v report_error(name)
00372  * [IN_USE] (in_use=true, error_count=1, retry_count=0)
00373  *   v subsequent errors
00374  * [RETRYING] (error_count++, retry_count++)
00375  *   v max retries reached
00376  * [FAILED] (retry_count >= max_retries)
00377  *   v reset_component_state(name)
00378  * [UNUSED] (back to initial state)
00379  * ```
00380  *
00381  * ## Memory Layout
00382  *
00383  * | Offset | Size | Field | Type | Alignment | Notes |
00384  * |-----|-----|-----|-----|-----|-----|
00385  * | 0 | 32 | name | char[32] | 1 | Component name string |
00386  * | 32 | 4 | error_count | uint32_t | 4 | Total errors for component |
00387  * | 36 | 4 | retry_count | uint32_t | 4 | Current retry attempt number |
00388  * | 40 | 1 | in_use | bool | 1 | Slot allocation flag |
00389  * | 41 | 3 | (padding) | - | - | Compiler padding to 4-byte boundary |
00390  * | **Total** | **44 bytes** | | | | Structure size |
00391  *
00392  * ## Field Relationships
00393  *
00394  * **Invariants**:
00395  * - If in_use == false: error_count == 0 AND retry_count == 0 AND name[0] == '\0'
00396  * - If in_use == true: name[0] != '\0' AND error_count >= 1
00397  * - retry_count <= max_retries (handler configuration)
00398  * - retry_count <= error_count (cannot retry more than errors)
00399  *
00400  * @par Usage Example (Component Registration):
00401  * @code{.c}
00402  * // Find unused slot and register "SPI" component
00403  * error_component_state_t* slot = nullptr;
00404  * for (uint8_t i = 0; i < k_error_handler_max_components; i++) {
00405  *     if (!handler->components[i].in_use) {
00406  *         slot = &handler->components[i];
00407  *         break;
00408  *     }
00409  * }
00410  *
00411  * if (slot != nullptr) {
00412  *     strncpy(slot->name, "SPI", k_error_handler_component_name_max - 1);
00413  *     slot->name[k_error_handler_component_name_max - 1] = '\0';
00414  *     slot->error_count = 1; // First error
00415  *     slot->retry_count = 0; // No retries yet
00416  *     slot->in_use = true; // Mark slot as allocated
00417  * }
00418  * @endcode
00419  *
00420  * @par Usage Example (Retry Logic):
00421  * @code{.c}
00422  * // Find component and check retry limit
00423  * for (uint8_t i = 0; i < k_error_handler_max_components; i++) {

```

```

00424 * if (handler->components[i].in_use &&
00425 *     strcmp(handler->components[i].name, "SPI") == 0) {
00426 *
00427 *     if (handler->components[i].retry_count < handler->max_retries) {
00428 *         // Calculate backoff
00429 *         uint32_t backoff = handler->initial_backoff_ms *
00430 *             (1 « handler->components[i].retry_count);
00431 *         backoff = (backoff > handler->max_backoff_ms) ?
00432 *             handler->max_backoff_ms : backoff;
00433 *
00434 *         // Increment retry count
00435 *         handler->components[i].retry_count++;
00436 *
00437 *         // Wait and retry
00438 *         tx_thread_sleep(backoff / 10);
00439 *         return k_rx_retry;
00440 *     } else {
00441 *         return k_rx_fail; // Max retries exceeded
00442 *     }
00443 * }
00444 * }
00445 * @endcode
00446 *
00447 * @invariant in_use=false -> error_count=0 AND retry_count=0
00448 * @invariant in_use=true -> name[0]!='\0' AND error_count>=1
00449 * @invariant retry_count <= error_count (always)
00450 * @invariant name is always null-terminated
00451 *
00452 * @note Structure size is 44 bytes (32 + 4 + 4 + 1 + 3 padding)
00453 * @note Array of 16 elements = 704 bytes total
00454 * @note name[] is NOT dynamically allocated (embedded in struct)
00455 *
00456 * @warning Component names longer than 31 chars are truncated
00457 * @warning Always null-terminate name[] to prevent buffer overruns
00458 * @attention in_use flag must be checked before accessing other fields
00459 *
00460 * @see error_handler_t Main handler containing array of these structures
00461 * @see error_handler_limits_t Constants defining array sizes
00462 *
00463 * @since Version 1.0.0
00464 */
00465 typedef struct {
00466     char name
00467     [k_error_handler_component_name_max]; /**< Component identifier string (null-terminated). Examples: "SPI",
00468     "MOTOR_FL", "UltrasonicSensor_Ch1". Maximum 31 characters plus null terminator. Used for component lookup and
00469     diagnostics */
00470     uint32_t
00471     error_count; /**< Total number of errors reported for this component. Incremented on each report_error() call. Used
00472     for diagnostics and logging. Range: 0 to 2^32-1 (wraps on overflow) */
00473     uint32_t
00474     retry_count; /**< Current retry attempt number for this component. Starts at 0, increments on each retry. Compared
00475     against handler->max_retries to determine if retry limit reached. Reset to 0 on success or explicit reset */
00476     bool
00477     in_use; /**< Slot allocation flag. true = This slot is tracking a component (name is valid, counters are active). false =
00478     Slot is free (available for new component registration). Must check before accessing other fields */
00479 } error_component_state_t;
00480
00481 /*
00482 * =====
00483 * Error Handler Structure
00484 * =====
00485 */
00486 /**
00487 * @struct error_handler_t
00488 * @brief Error Handler Concrete Implementation Structure (Main State Container)
00489 *
00490 * @details
00491 * Complete error handler state including ThreadX mutex, component tracking array,
00492 * retry configuration, and initialization flag. This structure implements the
00493 * rx_error_interface_t abstraction, providing concrete error tracking and retry logic.
00494 *
00495 * ## Purpose and Responsibility
00496 * This structure serves as the **central error management database** for the entire system:
00497 * 1. **Thread Safety**: TX_MUTEX protects concurrent access
00498 * 2. **Global Statistics**: total_error_count aggregates all component errors
00499 * 3. **Component Tracking**: components[] array isolates per-component state
00500 * 4. **Retry Strategy**: max_retries, backoff parameters define retry behavior
00501 * 5. **Lifecycle Management**: initialized flag prevents double-init
00502 *
00503 * ## Memory Layout
00504 * | Offset | Size | Field | Type | Alignment | Notes |
00505 * |-----|-----|-----|-----|-----|-----|
00506 * | 0 | ~100 | mutex | TX_MUTEX | 4 | ThreadX mutex (size varies by config) |

```



```

00504 * | 100 | 4 | total_error_count | uint32_t | 4 | Global error counter |
00505 * | 104 | 704 | components[16] | array | 4 | 16 x 44 bytes per component |
00506 * | 808 | 4 | max_retries | uint32_t | 4 | Retry limit configuration |
00507 * | 812 | 4 | initial_backoff_ms | uint32_t | 4 | Initial backoff delay |
00508 * | 816 | 4 | max_backoff_ms | uint32_t | 4 | Maximum backoff cap |
00509 * | 820 | 1 | initialized | bool | 1 | Initialization flag |
00510 * | 821 | 3 | (padding) | - | - | Compiler alignment |
00511 * | **Total** | **~924 bytes** | | | Structure size (varies by TX_MUTEX) |
00512 *
00513 * **Note**: Actual size depends on ThreadX TX_MUTEX size, which varies by configuration.
00514 * Typical total: ~1.0 to 1.3 KB.
00515 *
00516 * ## Initialization Requirements
00517 *
00518 * Before use, this structure MUST be initialized via error_handler_init():
00519 * 1. TX_MUTEX must be created (tx_mutex_create)
00520 * 2. total_error_count set to 0
00521 * 3. All components[] marked in_use=false
00522 * 4. Retry parameters copied from config
00523 * 5. initialized flag set to true
00524 *
00525 * ## Typical Lifecycle
00526 *
00527 * ```
00528 * [UNINITIALIZED] (all fields undefined)
00529 *   v error_handler_init(&handler, &config)
00530 * [INITIALIZED] (mutex created, counters zeroed, initialized=true)
00531 *   v normal operation (report_error, retry checks)
00532 * [IN_USE] (components registered, counters incrementing)
00533 *   v error_handler_deinit(&handler)
00534 * [DEINITIALIZED] (mutex deleted, initialized=false)
00535 * ```
00536 *
00537 * @par Field Constraints:
00538 *
00539 * | Field | Valid Values | Invariant | Notes |
00540 * |-----|-----|-----|-----|
00541 * | mutex | Valid TX_MUTEX | Created if initialized=true | ThreadX owned |
00542 * | total_error_count | 0 to 2^32-1 | Sum of all components[].error_count | Wraps on overflow |
00543 * | components[] | See error_component_state_t | in_use=true slots <= 16 | Static array |
00544 * | max_retries | 0 to 2^32-1 | 0 = unlimited retries | From config |
00545 * | initial_backoff_ms | 1 to max_backoff_ms | > 0 for meaningful backoff | From config |
00546 * | max_backoff_ms | >= initial_backoff_ms | Caps exponential growth | From config |
00547 * | initialized | true/false | true after init, false after deinit | Lifecycle flag |
00548 *
00549 * @par Usage Example (Declaration and Initialization):
00550 * @code{.c}
00551 * // Static allocation (typical for embedded systems)
00552 * static error_handler_t s_global_error_handler;
00553 *
00554 * void system_init(void) {
00555 *     // Configure error handler
00556 *     const error_handler_config_t config = {
00557 *         .max_retries = 3, // Up to 3 retries
00558 *         .initial_backoff_ms = 10, // Start at 10ms
00559 *         .max_backoff_ms = 5000 // Cap at 5 seconds
00560 *     };
00561 *
00562 *     // Initialize
00563 *     rx_err_t err = error_handler_init(&s_global_error_handler, &config);
00564 *     if (err != k_rx_ok) {
00565 *         uart_debug_puts("[FATAL] Error handler init failed\r\n");
00566 *         while (1); // Cannot proceed without error handling
00567 *     }
00568 *
00569 *     // Get interface for use by high-level modules
00570 *     rx_error_interface_t error_iface;
00571 *     error_handler_get_interface(&error_iface, &s_global_error_handler);
00572 *
00573 *     // Inject interface into other modules
00574 *     motor_controller_init(&error_iface);
00575 *     spi_driver_init(&error_iface);
00576 * }
00577 * @endcode
00578 *
00579 * @par Usage Example (Direct Access - Not Recommended):
00580 * @code{.c}
00581 * // Direct field access (bypasses interface - violates DIP)
00582 * // USE INTERFACE INSTEAD!
00583 *
00584 * // Get total error count (for diagnostics only)
00585 * uint32_t total_errors = s_global_error_handler.total_error_count;
00586 * uart_debug_printf("Total system errors: %lu\r\n", total_errors);
00587 *
00588 * // Iterate components for diagnostics
00589 * for (uint8_t i = 0; i < k_error_handler_max_components; i++) {
00590 *     if (s_global_error_handler.components[i].in_use) {

```

```

00591 *   uart_debug_printf("[%s] errors:%lu retries:%lu\r\n",
00592 *       s_global_error_handler.components[i].name,
00593 *       s_global_error_handler.components[i].error_count,
00594 *       s_global_error_handler.components[i].retry_count);
00595 *   }
00596 * }
00597 * @endcode
00598 *
00599 * @invariant initialized=true -> mutex is valid AND total_error_count defined
00600 * @invariant total_error_count >= sum of all components[].error_count (approximately)
00601 * @invariant max_backoff_ms >= initial_backoff_ms (always)
00602 * @invariant Number of in_use=true components <= k_error_handler_max_components
00603 *
00604 * @note Structure must be statically allocated or have lifetime exceeding all users
00605 * @note Typical allocation: File-scope static (static error_handler_t s_handler)
00606 * @note Memory footprint: ~1.0-1.3 KB depending on ThreadX configuration
00607 *
00608 * @warning Never copy this structure (contains TX_MUTEX which is not copyable)
00609 * @warning Direct field access violates Dependency Inversion - use interface
00610 * @warning Must call error_handler_deinit() before destruction (mutex cleanup)
00611 * @attention TX_MUTEX size varies by ThreadX configuration (affects total size)
00612 *
00613 * @see error_handler_init() Initializes this structure
00614 * @see error_handler_get_interface() Converts to abstract interface
00615 * @see error_handler_deinit() Cleans up resources (mutex deletion)
00616 * @see rx_error_interface_t Abstract interface (use this, not direct access)
00617 *
00618 * @since Version 1.0.0
00619 */
00620 typedef struct {
00621     TX_MUTEX
00622     mutex; /**< ThreadX mutex for thread-safe access to all fields. Created by error_handler_init(), deleted by
error_handler_deinit(). Protects concurrent report_error() and retry check calls from multiple tasks */
00623     uint32_t
00624     total_error_count; /**< Aggregate error count across ALL components. Incremented on each report_error() call
regardless of component. Used for system-wide health monitoring and diagnostics. Range: 0 to 2^32-1 (wraps on overflow)
*/
00625     error_component_state_t components
00626     [k_error_handler_max_components]; /**< Per-component error tracking array. Maximum 16 components
(k_error_handler_max_components). Each slot tracks one component's errors and retry state. Linear search for
component name on each access */
00627     uint32_t
00628     max_retries; /**< Maximum retry attempts before giving up. 0 = unlimited retries (use with caution). Copied from
error_handler_config_t during init. Compared against components[].retry_count to determine failure */
00629     uint32_t
00630     initial_backoff_ms; /**< Initial backoff delay in milliseconds for first retry. Base value for exponential backoff
calculation: delay = initial x 2^retry_count. Typical values: 10ms to 100ms. Must be > 0 for meaningful backoff */
00631     uint32_t
00632     max_backoff_ms; /**< Maximum backoff delay cap in milliseconds. Prevents exponential backoff from growing
unbounded. Typical values: 5000ms to 60000ms. Must be >= initial_backoff_ms. Limits delay = min(initial x 2^retry,
max) */
00633     bool
00634     initialized; /**< Initialization flag. true = error_handler_init() succeeded, structure is ready for use. false = Not
initialized or deinitialized (error_handler_deinit() called). Check before all operations */
00635 } error_handler_t;
00636
00637 /**
00638 * @struct error_handler_config_t
00639 * @brief Error Handler Initialization Configuration Parameters
00640 *
00641 * @details
00642 * Configuration structure passed to error_handler_init() to set up retry behavior
00643 * and exponential backoff parameters. All fields must be initialized before use.
00644 * Values are copied into error_handler_t during initialization.
00645 *
00646 * ## Configuration Strategy
00647 *
00648 * **Choose parameters based on operation type**:
00649 *
00650 * | Operation Type | max_retries | initial_backoff_ms | max_backoff_ms | Rationale |
00651 * |-----|-----|-----|-----|
00652 * | SPI Transfer | 3 | 10 | 100 | Fast recovery, short operations |
00653 * | I2C Sensor Read | 5 | 20 | 500 | Sensor may need time to respond |
00654 * | USB Communication | 10 | 50 | 5000 | Network delays common |
00655 * | Flash Write | 3 | 100 | 1000 | Long operations, few retries |
00656 * | Motor Command | 5 | 10 | 200 | Real-time, fast recovery needed |
00657 *
00658 * **Exponential Backoff Example**:
00659 * ```
00660 * Config: initial_backoff_ms=10, max_backoff_ms=5000
00661 *
00662 * Retry 0: 10ms x 2^0 = 10ms
00663 * Retry 1: 10ms x 2^1 = 20ms
00664 * Retry 2: 10ms x 2^2 = 40ms
00665 * Retry 3: 10ms x 2^3 = 80ms
00666 * Retry 4: 10ms x 2^4 = 160ms
00667 * Retry 5: 10ms x 2^5 = 320ms

```

```

00668 * Retry 6: 10ms x 2^6 = 640ms
00669 * Retry 7: 10ms x 2^7 = 1280ms
00670 * Retry 8: 10ms x 2^8 = 2560ms
00671 * Retry 9: 10ms x 2^9 = 5120ms -> capped at 5000ms
00672 * Retry 10+: 5000ms (max)
00673 * ...
00674 *
00675 * ## Memory Layout
00676 *
00677 * | Offset | Size | Field | Type | Notes |
00678 * |-----|-----|-----|-----|-----|
00679 * | 0 | 4 | max_retries | uint32_t | Retry limit |
00680 * | 4 | 4 | initial_backoff_ms | uint32_t | Base backoff |
00681 * | 8 | 4 | max_backoff_ms | uint32_t | Backoff cap |
00682 * | **Total** | **12 bytes** | | | Struct size |
00683 *
00684 * @par Usage Example (Conservative Configuration):
00685 * @code{.c}
00686 * // Conservative: Few retries, moderate backoff
00687 * const error_handler_config_t conservative_config = {
00688 *     .max_retries = 3, // Give up quickly
00689 *     .initial_backoff_ms = 100, // Start with 100ms delay
00690 *     .max_backoff_ms = 1000 // Cap at 1 second
00691 * };
00692 *
00693 * error_handler_init(&handler, &conservative_config);
00694 * @endcode
00695 *
00696 * @par Usage Example (Aggressive Retry Configuration):
00697 * @code{.c}
00698 * // Aggressive: Many retries, long backoff (for unreliable networks)
00699 * const error_handler_config_t aggressive_config = {
00700 *     .max_retries = 10, // Retry many times
00701 *     .initial_backoff_ms = 50, // Start small
00702 *     .max_backoff_ms = 30000 // Up to 30 seconds
00703 * };
00704 *
00705 * error_handler_init(&handler, &aggressive_config);
00706 * @endcode
00707 *
00708 * @par Usage Example (No Retry Limit):
00709 * @code{.c}
00710 * // Infinite retries (use with caution - can hang system)
00711 * const error_handler_config_t infinite_retry = {
00712 *     .max_retries = 0, // 0 = unlimited
00713 *     .initial_backoff_ms = 10,
00714 *     .max_backoff_ms = 5000
00715 * };
00716 *
00717 * // WARNING: Can retry forever if error never clears
00718 * error_handler_init(&handler, &infinite_retry);
00719 * @endcode
00720 *
00721 * @invariant initial_backoff_ms > 0 (zero backoff defeats purpose)
00722 * @invariant max_backoff_ms >= initial_backoff_ms (must be able to cap)
00723 * @invariant max_retries is reasonable (< 100 typically)
00724 *
00725 * @note Configuration is copied during error_handler_init(), original can be freed
00726 * @note Changing configuration requires re-initializing error handler
00727 * @note Structure size is 12 bytes (3 x uint32_t)
00728 *
00729 * @warning max_retries=0 means unlimited retries (can cause infinite loops)
00730 * @warning Very short backoff (< 1ms) defeats exponential backoff purpose
00731 * @warning Very long max_backoff (> 60s) can cause apparent system hangs
00732 * @attention Choose values based on worst-case operation duration
00733 *
00734 * @see error_handler_init() Uses this configuration
00735 * @see error_handler_t Main handler structure that stores these values
00736 *
00737 * @since Version 1.0.0
00738 */
00739 typedef struct {
00740     uint32_t
00741     max_retries; /**< Maximum number of retry attempts before giving up. 0 = unlimited retries (infinite loop risk).
Typical values: 3 to 10. After this many retries, is_retry_limit_reached() returns true. Per-component limit */
00742     uint32_t
00743     initial_backoff_ms; /**< Initial backoff delay in milliseconds for first retry. Base value for exponential calculation: delay
= initial x 2^retry_count. Typical values: 10ms to 100ms. Must be > 0. Smaller = faster retries, larger = less aggressive */
00744     uint32_t
00745     max_backoff_ms; /**< Maximum backoff delay cap in milliseconds. Limits exponential growth: delay = min(initial x
2^retry, max). Typical values: 1000ms to 60000ms. Must be >= initial_backoff_ms. Prevents unbounded delays */
00746 } error_handler_config_t;
00747
00748 /*
=====
00749 * Public API
00750 *

```

```

=====
00751 */
00752
00753 /**
00754  * @brief Initialize error handler with specified configuration
00755  *
00756  * @details
00757  * Initializes the error handler structure, creates ThreadX mutex, zeroes all counters,
00758  * and configures retry behavior. This function must be called before using the error
00759  * handler or obtaining its interface. Initialization is idempotent-safe (returns error
00760  * if already initialized).
00761  *
00762  * **Algorithm steps:**
00763  * 1. Validate handler and config pointers (NULL check)
00764  * 2. Check if already initialized (prevent double-init)
00765  * 3. Create ThreadX mutex for thread-safe access
00766  * 4. Zero all error counters (total_error_count = 0)
00767  * 5. Mark all component slots as unused (in_use = false)
00768  * 6. Copy retry configuration from config parameter
00769  * 7. Set initialized flag to true
00770  * 8. Return success
00771  *
00772  * **Memory Operations**:
```

- 00773 * - Calls tx_mutex_create() (ThreadX heap allocation for mutex control block)
- 00774 * - Zeroes components[] array (704 bytes)
- 00775 * - No user heap allocations (all static within error_handler_t)

```

00776  *
00777  * @param[in,out] handler Pointer to error_handler_t structure to initialize.
00778  * Must be statically allocated or have lifetime exceeding all users.
00779  * After success, structure is ready for use and interface extraction.
00780  * @param[in] config Pointer to configuration parameters. Values are copied into
00781  * handler structure. Original can be freed after this call. NULL not allowed.
00782  *
00783  * @return rx_err_t Error code indicating success or failure reason
00784  * @retval k_rx_ok Success, handler initialized and ready for use
00785  * @retval k_rx_err_null_ptr handler or config parameter is nullptr
00786  * @retval k_rx_err_invalid_state handler already initialized (call deinit first)
00787  * @retval k_rx_err_rtos_mutex ThreadX mutex creation failed (out of resources)
00788  *
00789  * @pre handler must point to valid error_handler_t structure
00790  * @pre config must point to valid error_handler_config_t structure
00791  * @pre handler must not be already initialized
00792  * @pre ThreadX must be initialized (tx_kernel_enter() called)
00793  *
00794  * @post If success, handler->initialized == true
00795  * @post If success, handler->mutex is valid and can be locked
00796  * @post If success, handler->total_error_count == 0
00797  * @post If success, all components[].in_use == false
00798  *
00799  * @invariant After success, handler can be safely used by multiple threads
00800  * @invariant Configuration values copied exactly from config parameter
00801  *
00802  * @note Thread-safe - uses no shared state (operates on passed handler only)
00803  * @note Must call error_handler_deinit() before destroying handler structure
00804  * @note Configuration cannot be changed after init (requires deinit/re-init)
00805  *
00806  * @warning Do not call twice on same handler without deinit (returns error)
00807  * @warning Handler must remain valid for lifetime of all users
00808  * @warning Mutex creation can fail if ThreadX out of resources
00809  * @attention After init, use error_handler_get_interface() to obtain abstract interface
00810  *
00811  * @par Performance:
00812  * - Execution time: ~50 us @ 240 MHz (mutex creation dominates)
00813  * - Memory: No dynamic allocation (mutex uses ThreadX pool)
00814  * - Stack usage: < 64 bytes
00815  *
00816  * @par Example (Basic Initialization):
00817  * @code{.c}
00818  * #include "rx_error_handler.h"
00819  *
00820  * // Static allocation (typical for embedded)
00821  * static error_handler_t s_error_handler;
00822  *
00823  * void system_init(void) {
00824  *     // Configure retry behavior
00825  *     const error_handler_config_t config = {
00826  *         .max_retries = 3,
00827  *         .initial_backoff_ms = 10,
00828  *         .max_backoff_ms = 5000
00829  *     };
00830  *
00831  *     // Initialize error handler
00832  *     rx_err_t err = error_handler_init(&s_error_handler, &config);
00833  *     if (err != k_rx_ok) {
00834  *         uart_debug_puts("[FATAL] Error handler init failed\r\n");
00835  *         while (1); // Cannot proceed without error handling
00836  *     }

```

```

00837 *
00838 *   uart_debug_puts("[INFO] Error handler initialized\r\n");
00839 * }
00840 * @endcode
00841 *
00842 * @par Example (Error Handling):
00843 * @code{.c}
00844 * rx_err_t err = error_handler_init(&handler, &config);
00845 *
00846 * switch (err) {
00847 *   case k_rx_ok:
00848 *     uart_debug_puts("[INFO] Handler ready\r\n");
00849 *     break;
00850 *   case k_rx_err_null_ptr:
00851 *     uart_debug_puts("[ERROR] nullptr\r\n");
00852 *     break;
00853 *   case k_rx_err_invalid_state:
00854 *     uart_debug_puts("[WARN] Already initialized\r\n");
00855 *     break;
00856 *   case k_rx_err_rtos_mutex:
00857 *     uart_debug_puts("[ERROR] Mutex creation failed\r\n");
00858 *     break;
00859 *   default:
00860 *     uart_debug_puts("[ERROR] Unknown error\r\n");
00861 *     break;
00862 * }
00863 * @endcode
00864 *
00865 * @see error_handler_config_t Configuration structure definition
00866 * @see error_handler_get_interface() Get abstract interface after init
00867 * @see error_handler_deinit() Cleanup before destroying handler
00868 * @see rx_error_interface_t Abstract interface for Dependency Inversion
00869 *
00870 * @since Version 1.0.0
00871 *
00872 * @par NASA Power of 10 Compliance:
00873 * - Rule 3: [OK] No dynamic allocation (uses ThreadX pool for mutex)
00874 * - Rule 5: [OK] 4 preconditions, 4 postconditions
00875 * - Rule 7: [OK] Returns rx_err_t, caller must check
00876 */
00877 [[nodiscard]] rx_err_t error_handler_init(error_handler_t*          handler,
00878                                           const error_handler_config_t* config);
00879
00880 /**
00881 * @brief Get abstract interface from concrete error handler (Dependency Inversion)
00882 *
00883 * @details
00884 * Extracts an rx_error_interface_t from the concrete error_handler_t implementation.
00885 * This function is the key to Dependency Inversion: high-level modules depend on
00886 * the abstract interface, not the concrete implementation. The returned interface
00887 * contains function pointers and context for polymorphic error handling.
00888 *
00889 * Algorithm steps:
00890 * 1. Validate iface and handler pointers (NULL check)
00891 * 2. Check handler is initialized (initialized flag)
00892 * 3. Fill iface->ctx with handler pointer (context for callbacks)
00893 * 4. Assign function pointers to concrete implementations:
00894 *    - iface->report_error -> internal report function
00895 *    - iface->is_retry_limit_reached -> internal retry check
00896 *    - iface->get_backoff_delay -> internal backoff calculator
00897 *    - iface->reset_component_state -> internal reset function
00898 * 5. Return success
00899 *
00900 * Dependency Inversion Pattern:
00901 * ```
00902 * High-Level Module (Motor Controller)
00903 *   v depends on v
00904 * rx_error_interface_t (Abstract) <- THIS FUNCTION PRODUCES THIS
00905 *   ^ implements ^
00906 * error_handler_t (Concrete)
00907 * ```
00908 *
00909 * Benefits:
00910 * - High-level code doesn't know about error_handler_t
00911 * - Can swap implementations without changing clients
00912 * - Enables unit testing with mock error handlers
00913 * - Clear separation of interface and implementation
00914 *
00915 * @param[out] iface Pointer to rx_error_interface_t to fill with function pointers
00916 * and context. After success, contains fully initialized interface ready for use.
00917 * @param[in,out] handler Pointer to initialized error_handler_t concrete implementation.
00918 * Must be initialized via error_handler_init(). Serves as context (iface->ctx).
00919 *
00920 * @return rx_err_t Error code indicating success or failure reason
00921 * @retval k_rx_ok Success, iface filled with valid function pointers and context
00922 * @retval k_rx_err_null_ptr iface or handler parameter is nullptr
00923 * @retval k_rx_err_invalid_state handler not initialized (call error_handler_init first)

```

```

00924 *
00925 * @pre iface must point to valid rx_error_interface_t structure
00926 * @pre handler must point to valid error_handler_t structure
00927 * @pre handler must be initialized (handler->initialized == true)
00928 *
00929 * @post If success, iface->ctx == handler (context pointer)
00930 * @post If success, all iface function pointers are non-NULL and valid
00931 * @post If success, iface can be used for all error operations
00932 *
00933 * @invariant iface lifetime must not exceed handler lifetime
00934 * @invariant iface remains valid as long as handler is initialized
00935 *
00936 * @note Thread-safe - uses no shared state
00937 * @note Interface is lightweight (just function pointers and one context pointer)
00938 * @note Can call multiple times to get multiple interface copies (all point to same handler)
00939 *
00940 * @warning iface becomes invalid if handler is deinitialized
00941 * @warning Do not use iface after error_handler_deinit(handler) called
00942 * @attention Interface must not outlive handler structure
00943 *
00944 * @par Performance:
00945 * - Execution time: ~2 us @ 240 MHz (simple pointer assignments)
00946 * - Memory: No allocation, just fills existing structure
00947 * - Stack usage: < 16 bytes
00948 *
00949 * @par Example (Dependency Inversion in Action):
00950 * @code{.c}
00951 * // 1. Initialize concrete handler (low-level)
00952 * static error_handler_t s_handler;
00953 * error_handler_init(&s_handler, &config);
00954 *
00955 * // 2. Get abstract interface
00956 * rx_error_interface_t error_iface;
00957 * rx_err_t err = error_handler_get_interface(&error_iface, &s_handler);
00958 * if (err != k_rx_ok) {
00959 *     uart_debug_puts("[ERROR] Failed to get error interface\r\n");
00960 *     return err;
00961 * }
00962 *
00963 * // 3. Inject interface into high-level modules
00964 * motor_controller_init(&error_iface); // Motor module depends on INTERFACE
00965 * spi_driver_init(&error_iface); // SPI module depends on INTERFACE
00966 * sensor_manager_init(&error_iface); // Sensor module depends on INTERFACE
00967 *
00968 * // High-level modules never know about error_handler_t!
00969 * @endcode
00970 *
00971 * @par Example (Using Interface):
00972 * @code{.c}
00973 * // High-level module receives interface (doesn't know about concrete type)
00974 * void spi_transfer(rx_error_interface_t* error_if, uint8_t* data, size_t len) {
00975 *     rx_err_t err = spi_hw_transfer(data, len);
00976 *
00977 *     if (err != k_rx_ok) {
00978 *         // Report error via interface (polymorphic call)
00979 *         error_if->report_error(error_if->ctx, err, "SPI", "Transfer failed");
00980 *
00981 *         // Check if should retry
00982 *         if (!error_if->is_retry_limit_reached(error_if->ctx, "SPI")) {
00983 *             // Calculate backoff delay
00984 *             uint32_t delay_ms = error_if->get_backoff_delay(error_if->ctx, "SPI");
00985 *             tx_thread_sleep(delay_ms / 10);
00986 *             // Retry transfer...
00987 *         }
00988 *     }
00989 * }
00990 * @endcode
00991 *
00992 * @see error_handler_init() Must call first to initialize handler
00993 * @see rx_error_interface_t Abstract interface structure definition
00994 * @see error_handler_t Concrete implementation structure
00995 *
00996 * @since Version 1.0.0
00997 *
00998 * @par SOLID Principles:
00999 * - **Dependency Inversion (D)**: This is the DIP implementation function
01000 * - High-level modules depend on rx_error_interface_t (abstraction)
01001 * - Low-level error_handler_t implements the abstraction
01002 * - This function bridges concrete to abstract
01003 */
01004 [[nodiscard]] rx_err_t error_handler_get_interface(rx_error_interface_t* iface,
01005                                                    error_handler_t* handler);
01006
01007 /**
01008 * @brief Deinitialize error handler and cleanup resources
01009 *
01010 * @details

```



```

01011 * Cleans up error handler resources, deletes ThreadX mutex, and marks handler as
01012 * uninitialized. Call this function before destroying the handler structure or on
01013 * system shutdown. After calling, the handler cannot be used until re-initialized.
01014 *
01015 * **Algorithm steps:**
01016 * 1. Validate handler pointer (NULL check)
01017 * 2. Check if initialized (skip if not initialized)
01018 * 3. Delete ThreadX mutex (tx_mutex_delete)
01019 * 4. Zero all counters and arrays (security/cleanup)
01020 * 5. Set initialized flag to false
01021 * 6. Return success
01022 *
01023 * **Cleanup Operations**:
01024 * - Deletes TX_MUTEX (frees ThreadX resources)
01025 * - Zeros components[] array (704 bytes)
01026 * - Zeros total_error_count
01027 * - No heap deallocations needed (static structure)
01028 *
01029 * @param[in,out] handler Pointer to error_handler_t to deinitialize. After success,
01030 * handler->initialized == false and structure cannot be used until re-initialized.
01031 *
01032 * @return rx_err_t Error code indicating success or failure reason
01033 * @retval k_rx_ok Success, handler deinitialized and resources freed
01034 * @retval k_rx_err_null_ptr handler parameter is nullptr
01035 *
01036 * @pre handler must point to valid error_handler_t structure
01037 * @pre Caller must ensure no other threads are using handler
01038 *
01039 * @post If success, handler->initialized == false
01040 * @post If success, handler->mutex is deleted (invalid)
01041 * @post If success, all counters and arrays zeroed
01042 * @post Handler structure still exists (not freed) but unusable
01043 *
01044 * @invariant Idempotent - safe to call on already-deinitialized handler
01045 * @invariant After deinit, handler can be re-initialized via error_handler_init()
01046 *
01047 * @note Thread-safe for the actual deinit, but caller must ensure exclusivity
01048 * @note Calling on uninitialized handler is safe (returns success)
01049 * @note Structure memory is NOT freed (caller responsibility if dynamically allocated)
01050 *
01051 * @warning All outstanding rx_error_interface_t pointers become invalid
01052 * @warning Caller must ensure no threads are using handler before calling
01053 * @warning Mutex deletion can fail if still locked by another thread
01054 * @attention Always call before destroying handler or on system shutdown
01055 *
01056 * @par Performance:
01057 * - Execution time: ~30 us @ 240 MHz (mutex deletion dominates)
01058 * - Memory: Frees ThreadX mutex resources
01059 * - Stack usage: < 32 bytes
01060 *
01061 * @par Example (System Shutdown):
01062 * @code{.c}
01063 * void system_shutdown(void) {
01064 *     uart_debug_puts("[INFO] System shutdown initiated\r\n");
01065 *
01066 *     // Deinitialize error handler
01067 *     rx_err_t err = error_handler_deinit(&s_error_handler);
01068 *     if (err != k_rx_ok) {
01069 *         uart_debug_puts("[WARN] Error handler deinit failed\r\n");
01070 *     }
01071 *
01072 *     // ... other cleanup ...
01073 * }
01074 * @endcode
01075 *
01076 * @par Example (Re-initialization):
01077 * @code{.c}
01078 * // Deinitialize with old config
01079 * error_handler_deinit(&handler);
01080 *
01081 * // Re-initialize with new config
01082 * const error_handler_config_t new_config = {
01083 *     .max_retries = 10, // Increased
01084 *     .initial_backoff_ms = 50,
01085 *     .max_backoff_ms = 10000
01086 * };
01087 * error_handler_init(&handler, &new_config);
01088 * @endcode
01089 *
01090 * @see error_handler_init() Initialize handler (can call after deinit)
01091 * @see error_handler_get_interface() Interfaces become invalid after deinit
01092 *
01093 * @since Version 1.0.0
01094 *
01095 * @par NASA Power of 10 Compliance:
01096 * - Rule 3: [OK] No dynamic allocation, only resource cleanup
01097 * - Rule 7: [OK] Returns rx_err_t, caller should check

```

```

01098 */
01099 [[nodiscard]] rx_err_t error_handler_deinit(error_handler_t* handler);
01100
01101 #ifdef __cplusplus
01102 }
01103 #endif

```

7.9 libs/rx_core/inc/rx_error_interface.h File Reference

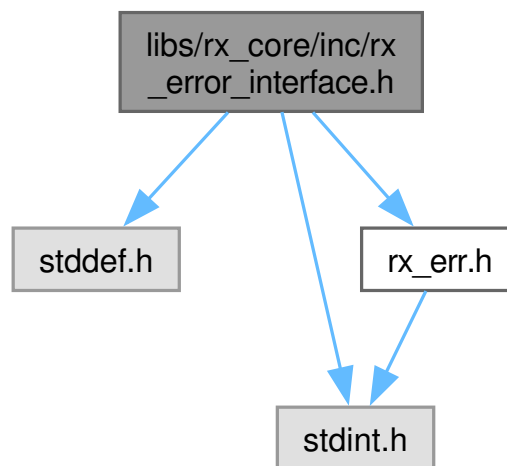
Abstract Error Handler Interface (Dependency Inversion Principle - SOLID).

```

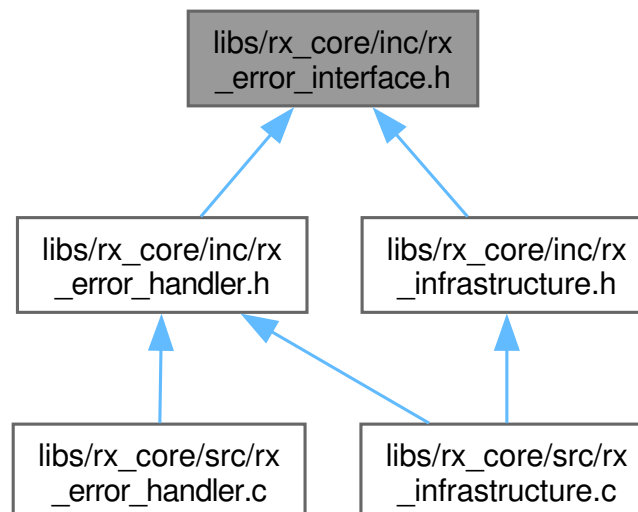
#include <stddef.h>
#include <stdint.h>
#include "rx_err.h"

```

Include dependency graph for rx_error_interface.h:



This graph shows which files directly or indirectly include this file:



Data Structures

- struct [rx_error_interface_t](#)

Abstract Error Handler Interface Structure (Polymorphism via Function Pointers).

Typedefs

- typedef [rx_err_t](#)(* [rx_error_report_fn](#)) (void *ctx, [rx_err_t](#) err, const char *component, const char *message)
Function Pointer Type for Error Reporting (Core Interface Function).
- typedef uint32_t(* [rx_error_get_count_fn](#)) (void *ctx)
Function Pointer for Getting Total Error Count (REQUIRED Interface Function).
- typedef uint32_t(* [rx_error_get_component_count_fn](#)) (void *ctx, const char *component)
Function Pointer for Getting Component-Specific Error Count (OPTIONAL).
- typedef [rx_err_t](#)(* [rx_error_clear_fn](#)) (void *ctx)
Function Pointer for Clearing All Error Counters (REQUIRED Interface Function).
- typedef bool(* [rx_error_is_retry_limit_reached_fn](#)) (void *ctx, const char *component)
Function Pointer for Checking Retry Limit Status (OPTIONAL).
- typedef [rx_err_t](#)(* [rx_error_reset_retry_fn](#)) (void *ctx, const char *component)
Function Pointer for Resetting Component Retry Counter (OPTIONAL).
- typedef uint32_t(* [rx_error_get_backoff_delay_fn](#)) (void *ctx, const char *component)
Function Pointer for Calculating Exponential Backoff Delay (OPTIONAL).

Functions

- static [rx_err_t](#) [rx_error_interface_validate](#) (const [rx_error_interface_t](#) *iface)
Validate that error interface is properly initialized with required functions.

7.9.1 Detailed Description

Abstract Error Handler Interface (Dependency Inversion Principle - SOLID).

Pure abstract interface defining the contract for error handling in the STAR RX72N firmware. This is the "D" in SOLID - Dependency Inversion Principle in action. High-level modules depend on this abstraction, not concrete implementations.

7.9.1.1 Purpose and Design Rationale

This module implements Dependency Inversion Principle (DIP), the cornerstone of maintainable, testable, and flexible software architecture:

Traditional Approach (BAD):

```
High-Level Module (Motor Control)
    v directly depends on v
Low-Level Module (Concrete Error Handler)
```

Problem: High-level code tightly coupled to low-level implementation. Cannot test, cannot swap implementations, difficult to maintain.

Dependency Inversion (GOOD - THIS MODULE):

```
High-Level Module (Motor Control)
    v depends on v
rx_error_interface_t (ABSTRACT INTERFACE - THIS FILE)
    ^ implements ^
Low-Level Module (error_handler_t in rx_error_handler.h)
```

Benefit: High-level code depends on stable abstraction. Can mock for testing, can swap implementations, loose coupling.

7.9.1.2 Key Principles

1. Pure Interface - Zero Implementation:

- This file contains ONLY interface definition
- No concrete error handling logic
- No static variables or state
- Just function pointer types and struct definition

2. Polymorphism via Function Pointers:

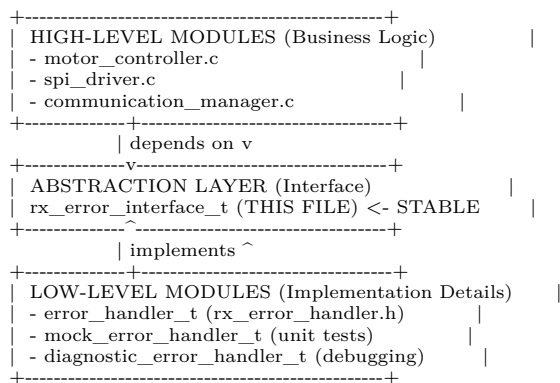
- Interface holds function pointers to concrete implementations
- Runtime polymorphism (similar to C++ virtual functions)
- Caller invokes through interface, doesn't know concrete type

3. Context Pointer for Implementation State:

- void* ctx holds pointer to concrete implementation
- Opaque to interface users (information hiding)
- Passed as first parameter to all function pointers

7.9.1.3 Architecture: Dependency Inversion in Practice

Three-Layer Architecture:



Key Insight: High-level modules never include [rx_error_handler.h](#). They only include THIS file ([rx_error_interface.h](#)). Dependencies point UPWARD from concrete to abstract, inverting the traditional dependency direction.

7.9.1.4 Benefits of This Pattern

1. Testability [OK][OK][OK]:

- Mock implementations for unit tests
- Test high-level modules in isolation
- No need for real error handler in tests

```
// Unit test creates mock
mock_error_interface_t mock;
rx_error_interface_t* iface = mock_get_interface(&mock);
motor_controller_init(&motor, iface); // Testable!
```

2. Flexibility [OK][OK][OK]:

- Swap implementations at runtime or compile-time
- Different implementations for production vs debug
- No changes to high-level code

```
#ifdef DEBUG
    diagnostic_error_handler_t debug_handler;
    rx_error_interface_t* iface = diagnostic_get_interface(&debug_handler);
#else
    error_handler_t prod_handler;
    rx_error_interface_t* iface = error_handler_get_interface(&prod_handler);
#endif
motor_controller_init(&motor, iface); // Same high-level code!
```

3. Decoupling [OK][OK][OK]:

- High-level modules don't know about implementation
- Can change error handler internals without affecting users
- Stable interface, evolving implementation

4. Open/Closed Principle [OK]:

- Open for extension: Add new implementations
- Closed for modification: Interface remains stable
- Add features without changing interface

7.9.1.5 Implementation Approach

How to Create a Concrete Implementation:

1. Define Concrete Type (in separate .h file):

```
typedef struct {
    uint32_t total_errors;
    // ... other private fields ...
} my_error_handler_t;
```

2. Implement Interface Functions (in .c file):

```
static rx_err_t my_report_error(void* ctx, rx_err_t err,
                               const char* component, const char* msg) {
    my_error_handler_t* handler = (my_error_handler_t*)ctx;
    handler->total_errors++;
    // ... concrete implementation ...
    return k_rx_ok;
}
```

3. Provide Interface Getter:

```
void my_error_handler_get_interface(rx_error_interface_t* iface,
                                    my_error_handler_t* handler) {
    iface->ctx = handler; // Store concrete instance
    iface->report_error = my_report_error; // Bind function
    // ... other function pointers ...
}
```

4. Users Call Through Interface:

```
rx_error_interface_t iface;
my_error_handler_get_interface(&iface, &handler);
iface.report_error(iface.ctx, k_rx_fail, "SPI", "Timeout"); // Polymorphic!
```

7.9.1.6 Performance Characteristics

Operation	Cost	Notes
Interface storage	8 pointers x 8 bytes = 64 bytes	Lightweight
Function call via pointer	+1 indirection (~1 cycle)	Negligible overhead
Context cast	0 cycles	Compile-time
Total overhead	< 5% vs direct call	Excellent for benefits gained

Memory Footprint:

- rx_error_interface_t: 64 bytes (7 function pointers + 1 context pointer)
- No heap allocation
- Stored on stack or in structs

7.9.1.7 Usage Guidelines

DO:

- Have high-level modules depend ONLY on this interface
- Create concrete implementations in separate .c/.h files
- Use interface for all error handling in business logic
- Validate interface with `rx_error_interface_validate()`
- Store interface in dependent module structures

DON'T:

- Include concrete error handler headers in high-level code
- Modify this interface frequently (breaks contract)
- Cast ctx pointer in high-level code (violates abstraction)
- Assume specific implementation details

Hardware Requirements:

Component	Requirement	Notes
MCU	Any (interface only)	No hardware dependencies
RAM	64 bytes per interface	Negligible

Module Dependencies:

This module depends on:

- [rx_err.h](#) - Error code definitions (abstraction)
- `<stdint.h>`, `<stdbool.h>`, `<stddef.h>` - Standard types

This module is used by (high-level modules):

- All drivers (motor, SPI, I2C, UART, etc.)
- All managers (bus manager, communication manager)
- All controllers (motor controller, sensor manager)
- NOTE: These modules ONLY include this file, never concrete handlers

This module is implemented by (low-level modules):

- [rx_error_handler.h](#) - Production error handler
- `mock_error_handler.h` - Unit test mocks
- Future implementations - Add without changing interface

NASA Power of 10 Compliance:

Rule	Status	Implementation
1. Simple control flow	[OK]	Pure interface, no logic
2. Fixed loop bounds	[OK]	No loops in interface
3. No dynamic allocation	[OK]	Interface is stack-allocated
4. Small functions	[OK]	Validation function < 10 lines
5. Assertions (>=2 per function)	[OK]	Validation checks NULL and function pointers
6. Narrow scope	[OK]	All definitions in single header

Rule	Status	Implementation
7. Check return values	[OK]	All functions return rx_err_t
8. Limited preprocessor	[OK]	Zero macros, pure interface
9. Pointer restrictions	[WARN]	Function pointers (required for DIP)
10. Compiler warnings	[OK]	Compiles with -Wall -Wextra -Werror

Note on Rule 9: Function pointers are essential for Dependency Inversion. This is an intentional, justified deviation for architectural benefit.

SOLID Principles:

Single Responsibility (S):

- This interface has ONE job: Define error handling contract
- No implementation, no side effects
- Pure abstraction

Open/Closed (O):

- Open for extension: Add new implementations
- Closed for modification: Interface definition stable
- New concrete handlers don't require interface changes

Liskov Substitution (L):

- All implementations must honor interface contract
- Clients can substitute any implementation
- Behavioral consistency guaranteed by interface

Interface Segregation (I):

- Focused interface: Only error handling operations
- Optional operations (retry logic) can be nullptr
- Clients use only what they need

Dependency Inversion (D): [OK][OK][OK] THIS IS THE PRINCIPLE

- High-level modules depend on this abstraction
- Low-level modules implement this abstraction
- Dependencies point from concrete to abstract (inverted!)
- This file IS the Dependency Inversion implementation

See also

[rx_error_handler.h](#) Concrete production implementation
[rx_err.h](#) Error code definitions (also an abstraction)
[docs/sections/06_nasa_power_of_10.tex](#) Safety-critical coding standards

Author

Locked, Inc.

Date

2026-01-27

Copyright

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Since

Version 1.0.0

Definition in file [rx_error_interface.h](#).

7.9.2 Data Structure Documentation

7.9.2.1 struct rx_error_interface

Abstract Error Handler Interface Structure (Polymorphism via Function Pointers).

THE DEPENDENCY INVERSION INTERFACE. This structure IS the abstraction that high-level modules depend on. Contains function pointers to concrete implementations and an opaque context pointer for implementation state.

7.9.2.2 Purpose and Architecture

This struct enables runtime polymorphism in C without inheritance:

- Function pointers act like C++ virtual functions
- Context pointer acts like "this" pointer
- Clients call through interface, implementation is hidden

Analogy to Object-Oriented Programming:

```
// C++ (conceptual equivalent)
class ErrorInterface {
public:
    virtual rx_err_t reportError(const char* component, const char* msg) = 0;
    virtual uint32_t getErrorCount() = 0;
    // ...
};

// C implementation (this struct)
struct rx_error_interface {
    void* ctx; // "this" pointer
    rx_error_report_fn report_error; // virtual function
    // ...
};
```

7.9.2.3 Memory Layout

Offset	Size	Field	Notes
0	8	ctx	Context pointer (implementation-specific)
8	8	report_error	REQUIRED function pointer
16	8	get_error_count	REQUIRED function pointer
24	8	get_component_error_count	OPTIONAL (can be NULL)
32	8	clear_errors	REQUIRED function pointer
40	8	is_retry_limit_reached	OPTIONAL (can be NULL)
48	8	reset_retry_counter	OPTIONAL (can be NULL)
56	8	get_backoff_delay	OPTIONAL (can be NULL)
Total	64 bytes		Lightweight interface

7.9.2.4 Field Categories

1. Context (State):

- [ctx](#): Opaque pointer to concrete implementation

2. REQUIRED Operations (must be non-NULL):

- [report_error](#): Report error occurrence
- [get_error_count](#): Get total error count
- [clear_errors](#): Reset all counters

3. OPTIONAL Operations (may be NULL):

- [get_component_error_count](#): Per-component statistics
- [is_retry_limit_reached](#): Retry limit checking
- [reset_retry_counter](#): Retry state reset
- [get_backoff_delay](#): Exponential backoff calculation

7.9.2.5 Initialization Pattern

Concrete implementations provide a "get_interface" function:

```
void error_handler_get_interface(rx_error_interface_t* iface,
                               error_handler_t* handler) {
    iface->ctx = handler; // Store concrete instance

    // Bind required functions
    iface->report_error = internal_report_error;
    iface->get_error_count = internal_get_count;
    iface->clear_errors = internal_clear;

    // Bind optional functions (or leave NULL)
    iface->get_component_error_count = internal_get_component_count;
    iface->is_retry_limit_reached = internal_is_retry_limit_reached;
    iface->reset_retry_counter = internal_reset_retry;
    iface->get_backoff_delay = internal_get_backoff_delay;
}
```


7.9.2.6 Usage Pattern (Client Code)

Step 1: Store interface in dependent module:

```
typedef struct {
    rx_error_interface_t* error_iface; // Pointer to interface
    // ... other fields ...
} spi_driver_t;
```

Step 2: Inject interface during initialization:

```
void spi_driver_init(spi_driver_t* spi, rx_error_interface_t* error_iface) {
    spi->error_iface = error_iface; // Dependency injection
}
```

Step 3: Call through interface:

```
void spi_transfer(spi_driver_t* spi, uint8_t* data, size_t len) {
    rx_err_t err = spi_hw_write(data, len);
    if (err != k_rx_ok) {
        // Polymorphic call - don't know concrete implementation
        spi->error_iface->report_error(
            spi->error_iface->ctx, // Context for implementation
            err,
            "SPI",
            "Transfer failed"
        );
    }
}
```

Field Invariants:

Field	Constraint	Validated By
ctx	Can be NULL (implementation choice)	Implementation
report_error	MUST be non-NULL	rx_error_interface_validate()
get_error_count	MUST be non-NULL	rx_error_interface_validate()
clear_errors	MUST be non-NULL	rx_error_interface_validate()
Other fields	MAY be NULL (optional)	Client checks before calling

Usage Example (Complete Workflow):

```
// ===== LOW-LEVEL: Create concrete implementation =====
static error_handler_t s_error_handler;
const error_handler_config_t config = {
    .max_retries = 3,
    .initial_backoff_ms = 10,
    .max_backoff_ms = 5000
};
error_handler_init(&s_error_handler, &config);

// ===== ABSTRACTION: Get interface =====
rx_error_interface_t error_iface;
error_handler_get_interface(&error_iface, &s_error_handler);

// ===== Validate interface =====
rx_err_t err = rx_error_interface_validate(&error_iface);
if (err != k_rx_ok) {
    uart_debug_puts("[ERROR] Invalid error interface\r\n");
    return err;
}

// ===== HIGH-LEVEL: Inject into dependent modules =====
spi_driver_t spi;
spi_driver_init(&spi, &error_iface); // SPI knows ONLY interface

motor_controller_t motor;
```

```

motor_controller_init(&motor, &error_iface); // Motor knows ONLY interface

// ===== USAGE: High-level code uses interface =====
// SPI driver reports error (doesn't know about error_handler_t!)
error_iface.report_error(error_iface.ctx, k_rx_fail, "SPI", "Timeout");

// Motor checks retry (doesn't know about error_handler_t!)
if (!error_iface.is_retry_limit_reached(error_iface.ctx, "MOTOR")) {
    uint32_t delay = error_iface.get_backoff_delay(error_iface.ctx, "MOTOR");
    tx_thread_sleep(delay / 10);
    retry_motor_operation();
}

```

Usage Example (Optional Function Handling):

```

// Check if optional function is available before calling
if (error_iface->get_component_error_count != nullptr) {
    uint32_t spi_errors = error_iface->get_component_error_count(
        error_iface->ctx, "SPI"
    );
    uart_debug_printf("SPI errors: %lu\r\n", spi_errors);
}

// Retry logic with optional functions
if (error_iface->is_retry_limit_reached != nullptr) {
    if (error_iface->is_retry_limit_reached(error_iface->ctx, "I2C")) {
        return k_rx_fail; // Give up
    }
}

```

Invariant

report_error, get_error_count, clear_errors must be non-NULL

Optional function pointers can be nullptr

ctx pointer is opaque to clients (implementation detail)

Interface lifetime must not exceed concrete implementation lifetime

Note

Structure is lightweight (64 bytes) - safe to pass by value or store

Typically stored as pointer in dependent modules

Can have multiple interfaces pointing to same concrete implementation

Warning

Do NOT access ctx directly in client code (violates abstraction)

Do NOT assume concrete type of ctx (polymorphism)

Check optional function pointers for nullptr before calling

Attention

Interface becomes invalid if concrete implementation is destroyed

See also

[rx_error_interface_validate\(\)](#) Validates interface before use

[error_handler_get_interface\(\)](#) Example interface provider ([rx_error_handler.h](#))

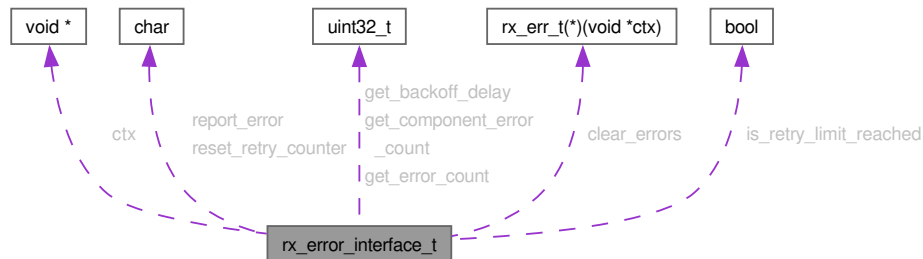
[rx_error_report_fn](#) Function pointer type definitions

Since

Version 1.0.0

Definition at line 780 of file [rx_error_interface.h](#).

Collaboration diagram for `rx_error_interface_t`:



Data Fields

rx_error_clear_fn	<code>clear_errors</code>	Clear all error counters and retry state (REQUIRED - MUST be non-NULL). Resets system to fresh state. Use after diagnostics or recovery. Validated by rx_error_interface_validate()
<code>void *</code>	<code>ctx</code>	Implementation context (opaque pointer to concrete handler). Cast to concrete type within implementation functions. Passed as first parameter to all function pointers. Can be NULL if implementation is stateless (rare). Clients must NEVER access or cast this pointer
rx_error_get_backoff_delay_fn	<code>get_backoff_delay</code>	Calculate exponential backoff delay for retry (OPTIONAL - may be NULL). Returns milliseconds to wait before next retry. Returns 0 if nullptr or no backoff needed. Check for nullptr before calling
rx_error_get_component_count_fn	<code>get_component_error_count</code>	Get error count for specific component (OPTIONAL - may be NULL). Returns per-component statistics for diagnostics. Check for nullptr before calling. Returns 0 if nullptr or component not found

rx_error_get_count_fn	get_error_count	Get total error count across all components (REQUIRED - MUST be non-NULL). Returns aggregate error count since initialization or last clear. Used for health monitoring. Validated by rx_error_interface_validate()
rx_error_is_retry_limit_reached_fn	is_retry_limit_reached	Check if component exceeded retry limit (OPTIONAL - may be NULL). Returns true if should give up, false if can retry. Check for nullptr before calling. Used in retry logic decision-making
rx_error_report_fn	report_error	Report an error (REQUIRED - MUST be non-NULL). Increments error counters, updates component statistics, logs error. Primary interface function. Validated by rx_error_interface_validate()
rx_error_reset_retry_fn	reset_retry_counter	Reset retry counter for component (OPTIONAL - may be NULL). Clears retry state without clearing error count. Call after successful operation. Check for nullptr before calling

7.9.3 Typedef Documentation

7.9.3.1 rx_error_clear_fn

typedef [rx_err_t](#)(* rx_error_clear_fn) (void *ctx)

Function Pointer for Clearing All Error Counters (REQUIRED Interface Function).

Resets all error counters to zero and clears retry state. Used for:

- Starting fresh after diagnostics
- Resetting after error recovery
- Clearing transient errors after system stabilization

REQUIRED: All implementations must provide this function.

Parameters

in	ctx	Opaque pointer to concrete error handler. Must be non-NULL.
----	-----	---

Returns

[rx_err_t](#) Status of clear operation

Return values

k_rx_ok	All counters successfully cleared
k_rx_err_null_ptr	ctx is nullptr
k_rx_err_invalid_state	Handler not initialized

Note

Thread-safe implementations recommended

Warning

Clears ALL error history - use with caution

See also

[rx_error_reset_retry_fn](#) For resetting retry counters without clearing error counts

Definition at line 464 of file [rx_error_interface.h](#).

7.9.3.2 rx_error_get_backoff_delay_fn

```
typedef uint32_t(* rx_error_get_backoff_delay_fn) (void *ctx, const char *component)
```

Function Pointer for Calculating Exponential Backoff Delay (OPTIONAL).

Calculates the recommended delay before retrying an operation, typically using exponential backoff algorithm:

```
delay = initial_backoff x (2 ^ retry_count)
delay = min(delay, max_backoff) // Cap at maximum
```

OPTIONAL: Can be NULL if implementation doesn't support backoff calculation.

Example Backoff Progression (initial=10ms, max=5s):

- Retry 0: 10ms
- Retry 1: 20ms
- Retry 2: 40ms
- Retry 3: 80ms

- Retry 4: 160ms
- ...
- Retry 9: 5120ms -> capped at 5000ms

Parameters

in	ctx	Opaque pointer to concrete error handler. Must be non-NULL.
in	component	Component name for backoff calculation. Must be non-NULL.

Returns

uint32_t Recommended delay in milliseconds

Return values

0	No backoff needed (no retries yet or component not found)
>0	Delay in milliseconds (use tx_thread_sleep or equivalent)

Note

Returns 0 if ctx is nullptr or component not found
 Delay increases exponentially with retry count
 Implementation may cap delay at configured maximum

Example Usage:

```
if (!error_iface->is_retry_limit_reached(error_iface->ctx, "SPI")) {
    uint32_t delay_ms = error_iface->get_backoff_delay(error_iface->ctx, "SPI");
    tx_thread_sleep(delay_ms / 10); // Convert ms to ThreadX ticks
    retry_operation(); // Try again
}
```

See also

[rx_error_is_retry_limit_reached_fn](#) For checking if should retry
[rx_error_reset_retry_fn](#) For resetting retry counter on success

Definition at line 566 of file [rx_error_interface.h](#).

7.9.3.3 rx_error_get_component_count_fn

```
typedef uint32_t(* rx_error_get_component_count_fn) (void *ctx, const char *component)
```

Function Pointer for Getting Component-Specific Error Count (OPTIONAL).

Returns the number of errors reported for a specific component. Enables per-component diagnostics and targeted debugging.

OPTIONAL: Can be NULL if implementation doesn't track per-component stats.

Parameters

in	ctx	Opaque pointer to concrete error handler. Must be non-NULL.
in	component	Component name (e.g., "SPI", "MOTOR"). Must be non-NULL.

Returns

uint32_t Error count for component (0 if component not found or no errors)

Note

Returns 0 if component not registered

Case-sensitive component name matching (typically)

See also

[rx_error_get_count_fn](#) For total error count across all components

Definition at line 439 of file [rx_error_interface.h](#).

7.9.3.4 rx_error_get_count_fn

```
typedef uint32_t(* rx_error_get_count_fn) (void *ctx)
```

Function Pointer for Getting Total Error Count (REQUIRED Interface Function).

Returns the total number of errors reported across ALL components since initialization or last clear. Useful for system health monitoring and diagnostics.

REQUIRED: All implementations must provide this function.

Parameters

in	ctx	Opaque pointer to concrete error handler. Must be non-NULL.
----	-----	---

Returns

uint32_t Total error count (0 to 2³²-1, wraps on overflow)

Note

Thread-safe implementations recommended

Returns 0 if ctx is nullptr (defensive programming)

See also

[rx_error_get_component_count_fn](#) For per-component error counts

Definition at line 418 of file [rx_error_interface.h](#).

7.9.3.5 rx_error_is_retry_limit_reached_fn

```
typedef bool(* rx_error_is_retry_limit_reached_fn) (void *ctx, const char *component)
```

Function Pointer for Checking Retry Limit Status (OPTIONAL).

Determines if a component has exceeded its maximum retry attempts. Used in retry logic to decide whether to:

- Continue retrying (limit not reached)
- Give up and propagate error (limit reached)

OPTIONAL: Can be NULL if implementation doesn't support retry tracking. Clients should check if function pointer is nullptr before calling.

Parameters

in	ctx	Opaque pointer to concrete error handler. Must be non-NULL.
in	component	Component name for retry limit check. Must be non-NULL.

Returns

bool Retry limit status

Return values

true	Retry limit reached, should NOT retry
false	Retry limit not reached, can retry

Note

Returns false if component not found (allows first retry)

Returns false if ctx is nullptr (defensive programming)

See also

[rx_error_get_backoff_delay_fn](#) For calculating retry delay

[rx_error_reset_retry_fn](#) For resetting retry counter

Definition at line 491 of file [rx_error_interface.h](#).

7.9.3.6 rx_error_report_fn

```
typedef rx_err_t(* rx_error_report_fn) (void *ctx, rx_err_t err, const char *component, const char *message)
```

Function Pointer Type for Error Reporting (Core Interface Function).

Defines the signature for error reporting functions in concrete implementations. This is the PRIMARY function in the error interface - all implementations MUST provide this function. Called when a component detects an error that needs to be tracked, logged, or handled.

7.9.3.7 Responsibilities

Concrete implementations of this function should:

1. Increment total error counter
2. Track per-component error statistics
3. Update retry counters (if retry logic enabled)
4. Log error message (optional)
5. Trigger diagnostics (optional)
6. Return success or failure

7.9.3.8 Implementation Requirements

Thread Safety: Implementation MUST be thread-safe (use mutex) Null Handling: Must validate all pointer parameters Performance: Should complete in < 50 us (logging is expensive) Memory: No dynamic allocation allowed

7.9.3.9 Example Implementation

```
static rx_err_t my_report_error(void* ctx, rx_err_t err,
                               const char* component, const char* msg) {
    my_error_handler_t* handler = (my_error_handler_t*)ctx;

    // Validate parameters
    if (handler == nullptr || component == nullptr || msg == nullptr) {
        return k_rx_err_null_ptr;
    }

    // Thread-safe update
    tx_mutex_get(&handler->mutex, TX_WAIT_FOREVER);
    handler->total_errors++;
    update_component_stats(handler, component, err);
    tx_mutex_put(&handler->mutex);

    // Optional: Log to UART
    uart_debug_printf("[ERROR] %s: %s (code=%d)\r\n", component, msg, err);

    return k_rx_ok;
}
```

7.9.3.10 Example Usage (Client Code)

```
// In SPI driver (high-level module)
rx_err_t spi_transfer(spi_handle_t* spi, uint8_t* data, size_t len) {
    rx_err_t err = spi_hw_write(data, len);
    if (err != k_rx_ok) {
        // Report error through interface (polymorphic call)
        spi->error_iface->report_error(
            spi->error_iface->ctx, // Context to concrete handler
            err,                  // Error code
            "SPI",                // Component name
            "Transfer timeout"    // Human-readable message
        );
        return err;
    }
    return k_rx_ok;
}
```

Parameters

in	ctx	Opaque pointer to concrete error handler implementation. Cast to concrete type within implementation. Must be non-NULL. Example: <code>error_handler_t* handler = (error_handler_t*)ctx;</code>
in	err	Error code from rx_err.h indicating what went wrong. Typically <code>k_rx_fail</code> , <code>k_rx_err_timeout</code> , <code>k_rx_err_busy</code> , etc. Used for error classification and statistics.
in	component	Component name string (null-terminated). Examples: "SPI", "MOTOR", "UART", "I2C", "USB". Used for per-component error tracking. Should be short (< 16 chars). Must be non-NULL.
in	message	Human-readable error message (null-terminated). Examples: "Transfer timeout", "Invalid parameter", "Hardware fault". Used for logging and diagnostics. Can be detailed. Must be non-NULL.

Returns

[rx_err_t](#) Status of error reporting operation

Return values

k_rx_ok	Error successfully reported and tracked
k_rx_err_null_ptr	ctx, component, or message is nullptr
k_rx_err_no_mem	Component tracking array full (implementation-specific)
k_rx_err_invalid_state	Handler not initialized (implementation-specific)

Note

This is a REQUIRED function - all implementations must provide it

Function pointer must be non-NULL in `rx_error_interface_t`

Validated by [rx_error_interface_validate\(\)](#)

Warning

Do NOT call with stack-allocated component/message strings that go out of scope

Implementation must be thread-safe if called from multiple contexts

Attention

Performance-critical - avoid blocking operations in implementation

See also

[rx_error_interface_t](#) Main interface structure containing this function pointer

[error_handler_t](#) Example concrete implementation ([rx_error_handler.h](#))

Since

Version 1.0.0

Definition at line 395 of file [rx_error_interface.h](#).

7.9.3.11 rx_error_reset_retry_fn

```
typedef rx\_err\_t(* rx_error_reset_retry_fn) (void *ctx, const char *component)
```

Function Pointer for Resetting Component Retry Counter (OPTIONAL).

Resets the retry counter for a specific component back to zero. Called after:

- Successful operation (clear retry state)
- Manual intervention (give component another chance)
- Configuration change (fresh start)

OPTIONAL: Can be NULL if implementation doesn't support retry tracking.

Parameters

in	ctx	Opaque pointer to concrete error handler. Must be non-NULL.
in	component	Component name to reset. Must be non-NULL.

Returns

[rx_err_t](#) Status of reset operation

Return values

k_rx_ok	Retry counter successfully reset
k_rx_err_null_ptr	ctx or component is nullptr
k_rx_err_not_found	Component not registered

Note

Does NOT clear error count, only retry counter

Thread-safe implementations recommended

See also

[rx_error_is_retry_limit_reached_fn](#) For checking retry limit

[rx_error_clear_fn](#) For clearing all counters including errors

Definition at line 518 of file [rx_error_interface.h](#).

7.9.4 Function Documentation

7.9.4.1 rx_error_interface_validate()

```
rx_err_t rx_error_interface_validate (
    const rx_error_interface_t * iface)    [inline], [static]
```

Validate that error interface is properly initialized with required functions.

Performs validation checks on an error interface before use. Ensures that all REQUIRED function pointers are non-NULL. Optional function pointers may be NULL. Call this function after obtaining an interface from a concrete implementation to verify correctness before injecting into dependent modules.

Validation Checks:

1. Interface pointer is non-NULL
2. report_error function pointer is non-NULL (REQUIRED)
3. get_error_count function pointer is non-NULL (REQUIRED)
4. clear_errors function pointer is non-NULL (REQUIRED)

Not Validated (optional functions can be NULL):

- get_component_error_count
- is_retry_limit_reached
- reset_retry_counter
- get_backoff_delay
- ctx (can be NULL for stateless implementations)

Performance: Very fast (~10 CPU cycles), just NULL checks.

Parameters

in	iface	Pointer to interface structure to validate. Can be NULL (returns error).
----	-------	--

Returns

[rx_err_t](#) Validation result

Return values

k_rx_ok	Interface is valid and ready to use
k_rx_err_null_ptr	iface parameter is nullptr
k_rx_err_invalid_state	One or more REQUIRED function pointers are nullptr

Note

This is a static inline function (no function call overhead)
 Only checks REQUIRED functions, optional functions may be nullptr
 Does not validate function pointer targets (assumes correct binding)

Warning

Do not use interface if validation fails

Attention

Always validate after getting interface from implementation

Usage Example (Recommended Pattern):

```
// Get interface from concrete implementation
rx_error_interface_t error_iface;
error_handler_get_interface(&error_iface, &handler);

// VALIDATE before use
rx_err_t err = rx_error_interface_validate(&error_iface);
if (err != k_rx_ok) {
    uart_debug_puts("[ERROR] Invalid error interface\r\n");
    return err;
}

// Safe to use now
spi_driver_init(&spi, &error_iface);
```

Usage Example (Error Handling):

```
rx_err_t err = rx_error_interface_validate(&iface);

switch (err) {
    case k_rx_ok:
        uart_debug_puts("[INFO] Interface valid\r\n");
        break;
    case k_rx_err_null_ptr:
        uart_debug_puts("[ERROR] NULL interface pointer\r\n");
        break;
    case k_rx_err_invalid_state:
        uart_debug_puts("[ERROR] Missing required function pointers\r\n");
        break;
}
```

Example (Defensive Programming):

```
void module_init(module_t* mod, rx_error_interface_t* error_iface) {
    // Defensive: Validate interface at module boundaries
    rx_err_t err = rx_error_interface_validate(error_iface);
    if (err != k_rx_ok) {
        // Cannot proceed without valid error interface
        return err;
    }

    // Store validated interface
    mod->error_iface = error_iface;
    return k_rx_ok;
}
```

See also

[rx_error_interface](#) Structure being validated
[error_handler_get_interface\(\)](#) Example interface provider

Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 1: [OK] Simple control flow (sequential checks)
- Rule 4: [OK] Function < 10 lines (very small)
- Rule 5: [OK] Precondition checks (NULL validation)

Definition at line 902 of file [rx_error_interface.h](#).

```
00903 {
00904     if (iface == nullptr) {
00905         return k_rx_err_null_ptr;
00906     }
00907
00908     /* Check required function pointers */
00909     if (iface->report_error == nullptr || iface->get_error_count == nullptr ||
00910         iface->clear_errors == nullptr) {
00911         return k_rx_err_invalid_state;
00912     }
00913
00914     return k_rx_ok;
00915 }
```

References [k_rx_err_invalid_state](#), [k_rx_err_null_ptr](#), and [k_rx_ok](#).

7.10 rx_error_interface.h

[Go to the documentation of this file.](#)

```
00001 /**
00002  * @file rx_error_interface.h
00003  * @brief Abstract Error Handler Interface (Dependency Inversion Principle - SOLID)
00004  *
00005  * @details
00006  * **Pure abstract interface** defining the contract for error handling in the STAR RX72N
00007  * firmware. This is the ***D*** in SOLID** - Dependency Inversion Principle in action. High-level
00008  * modules depend on this abstraction, not concrete implementations.
00009  *
00010  * ## Purpose and Design Rationale
00011  *
00012  * This module implements **Dependency Inversion Principle (DIP)**, the cornerstone of
00013  * maintainable, testable, and flexible software architecture:
00014  *
00015  * **Traditional Approach (BAD)**:
00016  * ```
00017  * High-Level Module (Motor Control)
00018  *     v directly depends on v
00019  * Low-Level Module (Concrete Error Handler)
00020  * ```
00021  * **Problem**: High-level code tightly coupled to low-level implementation. Cannot test,
00022  * cannot swap implementations, difficult to maintain.
00023  *
00024  * **Dependency Inversion (GOOD - THIS MODULE)**:
00025  * ```
00026  * High-Level Module (Motor Control)
00027  *     v depends on v
00028  * rx_error_interface_t (ABSTRACT INTERFACE - THIS FILE)
00029  *     ^ implements ^
00030  * Low-Level Module (error_handler_t in rx_error_handler.h)
00031  * ```
00032  * **Benefit**: High-level code depends on stable abstraction. Can mock for testing,
00033  * can swap implementations, loose coupling.
00034  *
00035  * ## Key Principles
00036  *
00037  * **1. Pure Interface - Zero Implementation**:
00038  * - This file contains ONLY interface definition
00039  * - No concrete error handling logic
00040  * - No static variables or state
00041  * - Just function pointer types and struct definition
00042  *
```

```

00043 * **2. Polymorphism via Function Pointers**:
00044 * - Interface holds function pointers to concrete implementations
00045 * - Runtime polymorphism (similar to C++ virtual functions)
00046 * - Caller invokes through interface, doesn't know concrete type
00047 *
00048 * **3. Context Pointer for Implementation State**:
00049 * - `void* ctx` holds pointer to concrete implementation
00050 * - Opaque to interface users (information hiding)
00051 * - Passed as first parameter to all function pointers
00052 *
00053 * ## Architecture: Dependency Inversion in Practice
00054 *
00055 * **Three-Layer Architecture**:
00056 * ```
00057 * +-----+
00058 * | HIGH-LEVEL MODULES (Business Logic)          |
00059 * | - motor_controller.c                          |
00060 * | - spi_driver.c                               |
00061 * | - communication_manager.c                    |
00062 * +-----+-----+
00063 * | depends on v                                  |
00064 * +-----+-----+
00065 * | ABSTRACTION LAYER (Interface)                |
00066 * | rx_error_interface_t (THIS FILE) <- STABLE  |
00067 * +-----+-----+
00068 * | implements ^                                  |
00069 * +-----+-----+
00070 * | LOW-LEVEL MODULES (Implementation Details)    |
00071 * | - error_handler_t (rx_error_handler.h)         |
00072 * | - mock_error_handler_t (unit tests)            |
00073 * | - diagnostic_error_handler_t (debugging)       |
00074 * +-----+-----+
00075 * ```
00076 *
00077 *
00078 * **Key Insight**: High-level modules never include `rx_error_handler.h`. They only
00079 * include THIS file (`rx_error_interface.h`). Dependencies point UPWARD from concrete
00080 * to abstract, inverting the traditional dependency direction.
00081 *
00082 * ## Benefits of This Pattern
00083 *
00084 * **1. Testability** [OK][OK][OK]:
00085 * - Mock implementations for unit tests
00086 * - Test high-level modules in isolation
00087 * - No need for real error handler in tests
00088 * ```c
00089 * // Unit test creates mock
00090 * mock_error_interface_t mock;
00091 * rx_error_interface_t* iface = mock_get_interface(&mock);
00092 * motor_controller_init(&motor, iface); // Testable!
00093 * ```
00094 *
00095 * **2. Flexibility** [OK][OK][OK]:
00096 * - Swap implementations at runtime or compile-time
00097 * - Different implementations for production vs debug
00098 * - No changes to high-level code
00099 * ```c
00100 * #ifdef DEBUG
00101 *     diagnostic_error_handler_t debug_handler;
00102 *     rx_error_interface_t* iface = diagnostic_get_interface(&debug_handler);
00103 * #else
00104 *     error_handler_t prod_handler;
00105 *     rx_error_interface_t* iface = error_handler_get_interface(&prod_handler);
00106 * #endif
00107 * motor_controller_init(&motor, iface); // Same high-level code!
00108 * ```
00109 *
00110 * **3. Decoupling** [OK][OK][OK]:
00111 * - High-level modules don't know about implementation
00112 * - Can change error handler internals without affecting users
00113 * - Stable interface, evolving implementation
00114 *
00115 * **4. Open/Closed Principle** [OK]:
00116 * - Open for extension: Add new implementations
00117 * - Closed for modification: Interface remains stable
00118 * - Add features without changing interface
00119 *
00120 * ## Implementation Approach
00121 *
00122 * **How to Create a Concrete Implementation**:
00123 *
00124 * 1. **Define Concrete Type** (in separate .h file):
00125 * ```c
00126 * typedef struct {
00127 *     uint32_t total_errors;
00128 *     // ... other private fields ...
00129 * } my_error_handler_t;

```

```

00130 * ``
00131 *
00132 * 2. Implement Interface Functions (in .c file):
00133 * ``
00134 * static rx_err_t my_report_error(void* ctx, rx_err_t err,
00135 *                                const char* component, const char* msg) {
00136 *     my_error_handler_t* handler = (my_error_handler_t*)ctx;
00137 *     handler->total_errors++;
00138 *     // ... concrete implementation ...
00139 *     return k_rx_ok;
00140 * }
00141 * ``
00142 *
00143 * 3. Provide Interface Getter:
00144 * ``
00145 * void my_error_handler_get_interface(rx_error_interface_t* iface,
00146 *                                     my_error_handler_t* handler) {
00147 *     iface->ctx = handler; // Store concrete instance
00148 *     iface->report_error = my_report_error; // Bind function
00149 *     // ... other function pointers ...
00150 * }
00151 * ``
00152 *
00153 * 4. Users Call Through Interface:
00154 * ``
00155 * rx_error_interface_t iface;
00156 * my_error_handler_get_interface(&iface, &handler);
00157 * iface.report_error(iface.ctx, k_rx_fail, "SPI", "Timeout"); // Polymorphic!
00158 * ``
00159 *
00160 * ## Performance Characteristics
00161 *
00162 * | Operation | Cost | Notes |
00163 * |-----|-----|-----|
00164 * | Interface storage | 8 pointers x 8 bytes = 64 bytes | Lightweight |
00165 * | Function call via pointer | +1 indirection (~1 cycle) | Negligible overhead |
00166 * | Context cast | 0 cycles | Compile-time |
00167 * | Total overhead | < 5% vs direct call | Excellent for benefits gained |
00168 *
00169 * Memory Footprint:
00170 * - rx_error_interface_t: 64 bytes (7 function pointers + 1 context pointer)
00171 * - No heap allocation
00172 * - Stored on stack or in structs
00173 *
00174 * ## Usage Guidelines
00175 *
00176 * DO:
00177 * - Have high-level modules depend ONLY on this interface
00178 * - Create concrete implementations in separate .c/.h files
00179 * - Use interface for all error handling in business logic
00180 * - Validate interface with rx_error_interface_validate()
00181 * - Store interface in dependent module structures
00182 *
00183 * DON'T:
00184 * - Include concrete error handler headers in high-level code
00185 * - Modify this interface frequently (breaks contract)
00186 * - Cast ctx pointer in high-level code (violates abstraction)
00187 * - Assume specific implementation details
00188 *
00189 * @par Hardware Requirements:
00190 *
00191 * | Component | Requirement | Notes |
00192 * |-----|-----|-----|
00193 * | MCU | Any (interface only) | No hardware dependencies |
00194 * | RAM | 64 bytes per interface | Negligible |
00195 *
00196 * @par Module Dependencies:
00197 *
00198 * This module depends on:
00199 * - `rx_err.h` - Error code definitions (abstraction)
00200 * - `<stdint.h>`, `<stdbool.h>`, `<stddef.h>` - Standard types
00201 *
00202 * This module is used by (high-level modules):
00203 * - All drivers (motor, SPI, I2C, UART, etc.)
00204 * - All managers (bus manager, communication manager)
00205 * - All controllers (motor controller, sensor manager)
00206 *
00207 * NOTE: These modules ONLY include this file, never concrete handlers
00208 *
00209 * This module is implemented by (low-level modules):
00210 * - `rx_error_handler.h` - Production error handler
00211 * - `mock_error_handler.h` - Unit test mocks
00212 * - Future implementations - Add without changing interface
00213 *
00214 * @par NASA Power of 10 Compliance:
00215 *
00216 * | Rule | Status | Implementation |
00217 * |-----|-----|-----|

```



```

00217 * | 1. Simple control flow | [OK] | Pure interface, no logic |
00218 * | 2. Fixed loop bounds | [OK] | No loops in interface |
00219 * | 3. No dynamic allocation | [OK] | Interface is stack-allocated |
00220 * | 4. Small functions | [OK] | Validation function < 10 lines |
00221 * | 5. Assertions (>=2 per function) | [OK] | Validation checks NULL and function pointers |
00222 * | 6. Narrow scope | [OK] | All definitions in single header |
00223 * | 7. Check return values | [OK] | All functions return rx_err_t |
00224 * | 8. Limited preprocessor | [OK] | Zero macros, pure interface |
00225 * | 9. Pointer restrictions | [WARN] | Function pointers (required for DIP) |
00226 * | 10. Compiler warnings | [OK] | Compiles with -Wall -Wextra -Werror |
00227
00228 * **Note on Rule 9**: Function pointers are essential for Dependency Inversion.
00229 * This is an intentional, justified deviation for architectural benefit.
00230
00231 * @par SOLID Principles:
00232 *
00233 * **Single Responsibility (S)**:
00234 * - This interface has ONE job: Define error handling contract
00235 * - No implementation, no side effects
00236 * - Pure abstraction
00237
00238 * **Open/Closed (O)**:
00239 * - Open for extension: Add new implementations
00240 * - Closed for modification: Interface definition stable
00241 * - New concrete handlers don't require interface changes
00242
00243 * **Liskov Substitution (L)**:
00244 * - All implementations must honor interface contract
00245 * - Clients can substitute any implementation
00246 * - Behavioral consistency guaranteed by interface
00247
00248 * **Interface Segregation (I)**:
00249 * - Focused interface: Only error handling operations
00250 * - Optional operations (retry logic) can be nullptr
00251 * - Clients use only what they need
00252
00253 * **Dependency Inversion (D)**: [OK][OK][OK] **THIS IS THE PRINCIPLE**
00254 * - High-level modules depend on this abstraction
00255 * - Low-level modules implement this abstraction
00256 * - Dependencies point from concrete to abstract (inverted!)
00257 * - This file IS the Dependency Inversion implementation
00258
00259 * @see rx_error_handler.h Concrete production implementation
00260 * @see rx_err.h Error code definitions (also an abstraction)
00261 * @see docs/sections/06_nasa_power_of_10.tex Safety-critical coding standards
00262
00263 * @author Locked, Inc.
00264 * @date 2026-01-27
00265 * @copyright Copyright (c) 2026 Locked Inc.
00266 * SPDX-License-Identifier: MIT
00267 * @since Version 1.0.0
00268 */
00269
00270 #pragma once
00271
00272 #include <stddef.h>
00273 #include <stdint.h>
00274
00275 #include "rx_err.h"
00276
00277 #ifdef __cplusplus
00278 extern "C" {
00279 #endif
00280
00281 /* Forward declaration of interface struct */
00282 typedef struct rx_error_interface rx_error_interface_t;
00283
00284 /*
=====
00285 * Error Handler Interface Function Pointers
00286 *
=====
00287 */
00288
00289 /**
00290 * @typedef rx_error_report_fn
00291 * @brief Function Pointer Type for Error Reporting (Core Interface Function)
00292 *
00293 * @details
00294 * Defines the signature for error reporting functions in concrete implementations.
00295 * This is the **PRIMARY** function in the error interface - all implementations
00296 * MUST provide this function. Called when a component detects an error that needs
00297 * to be tracked, logged, or handled.
00298 *
00299 * ## Responsibilities
00300 *
00301 * Concrete implementations of this function should:

```

```

00302 * 1. Increment total error counter
00303 * 2. Track per-component error statistics
00304 * 3. Update retry counters (if retry logic enabled)
00305 * 4. Log error message (optional)
00306 * 5. Trigger diagnostics (optional)
00307 * 6. Return success or failure
00308 *
00309 * ## Implementation Requirements
00310 *
00311 * **Thread Safety**: Implementation MUST be thread-safe (use mutex)
00312 * **Null Handling**: Must validate all pointer parameters
00313 * **Performance**: Should complete in < 50 us (logging is expensive)
00314 * **Memory**: No dynamic allocation allowed
00315 *
00316 * ## Example Implementation
00317 *
00318 * ``c
00319 * static rx_err_t my_report_error(void* ctx, rx_err_t err,
00320 *                                const char* component, const char* msg) {
00321 *     my_error_handler_t* handler = (my_error_handler_t*)ctx;
00322 *
00323 *     // Validate parameters
00324 *     if (handler == nullptr || component == nullptr || msg == nullptr) {
00325 *         return k_rx_err_null_ptr;
00326 *     }
00327 *
00328 *     // Thread-safe update
00329 *     tx_mutex_get(&handler->mutex, TX_WAIT_FOREVER);
00330 *     handler->total_errors++;
00331 *     update_component_stats(handler, component, err);
00332 *     tx_mutex_put(&handler->mutex);
00333 *
00334 *     // Optional: Log to UART
00335 *     uart_debug_printf("[ERROR] %s: %s (code=%d)\r\n", component, msg, err);
00336 *
00337 *     return k_rx_ok;
00338 * }
00339 * ``
00340 *
00341 * ## Example Usage (Client Code)
00342 *
00343 * ``c
00344 * // In SPI driver (high-level module)
00345 * rx_err_t spi_transfer(spi_handle_t* spi, uint8_t* data, size_t len) {
00346 *     rx_err_t err = spi_hw_write(data, len);
00347 *     if (err != k_rx_ok) {
00348 *         // Report error through interface (polymorphic call)
00349 *         spi->error_iface->report_error(
00350 *             spi->error_iface->ctx, // Context to concrete handler
00351 *             err,                  // Error code
00352 *             "SPI",                // Component name
00353 *             "Transfer timeout"    // Human-readable message
00354 *         );
00355 *         return err;
00356 *     }
00357 *     return k_rx_ok;
00358 * }
00359 * ``
00360 *
00361 * @param[in] ctx Opaque pointer to concrete error handler implementation.
00362 * Cast to concrete type within implementation. Must be non-NULL.
00363 * Example: `error_handler_t* handler = (error_handler_t*)ctx;`
00364 * @param[in] err Error code from rx_err.h indicating what went wrong.
00365 * Typically k_rx_fail, k_rx_err_timeout, k_rx_err_busy, etc.
00366 * Used for error classification and statistics.
00367 * @param[in] component Component name string (null-terminated).
00368 * Examples: "SPI", "MOTOR", "UART", "I2C", "USB".
00369 * Used for per-component error tracking. Should be short (< 16 chars).
00370 * Must be non-NULL.
00371 * @param[in] message Human-readable error message (null-terminated).
00372 * Examples: "Transfer timeout", "Invalid parameter", "Hardware fault".
00373 * Used for logging and diagnostics. Can be detailed.
00374 * Must be non-NULL.
00375 *
00376 * @return rx_err_t Status of error reporting operation
00377 * @retval k_rx_ok Error successfully reported and tracked
00378 * @retval k_rx_err_null_ptr ctx, component, or message is nullptr
00379 * @retval k_rx_err_no_mem Component tracking array full (implementation-specific)
00380 * @retval k_rx_err_invalid_state Handler not initialized (implementation-specific)
00381 *
00382 * @note This is a REQUIRED function - all implementations must provide it
00383 * @note Function pointer must be non-NULL in rx_error_interface_t
00384 * @note Validated by rx_error_interface_validate()
00385 *
00386 * @warning Do NOT call with stack-allocated component/message strings that go out of scope
00387 * @warning Implementation must be thread-safe if called from multiple contexts
00388 * @attention Performance-critical - avoid blocking operations in implementation

```

```

00389 *
00390 * @see rx_error_interface_t Main interface structure containing this function pointer
00391 * @see error_handler_t Example concrete implementation (rx_error_handler.h)
00392 *
00393 * @since Version 1.0.0
00394 */
00395 typedef rx_err_t (*rx_error_report_fn)(void*      ctx,
00396                                       rx_err_t   err,
00397                                       const char* component,
00398                                       const char* message);
00399
00400 /**
00401 * @typedef rx_error_get_count_fn
00402 * @brief Function Pointer for Getting Total Error Count (REQUIRED Interface Function)
00403 *
00404 * @details
00405 * Returns the total number of errors reported across ALL components since initialization
00406 * or last clear. Useful for system health monitoring and diagnostics.
00407 *
00408 * **REQUIRED**: All implementations must provide this function.
00409 *
00410 * @param[in] ctx Opaque pointer to concrete error handler. Must be non-NULL.
00411 *
00412 * @return uint32_t Total error count (0 to 2^32-1, wraps on overflow)
00413 *
00414 * @note Thread-safe implementations recommended
00415 * @note Returns 0 if ctx is nullptr (defensive programming)
00416 * @see rx_error_get_component_count_fn For per-component error counts
00417 */
00418 typedef uint32_t (*rx_error_get_count_fn)(void* ctx);
00419
00420 /**
00421 * @typedef rx_error_get_component_count_fn
00422 * @brief Function Pointer for Getting Component-Specific Error Count (OPTIONAL)
00423 *
00424 * @details
00425 * Returns the number of errors reported for a specific component. Enables
00426 * per-component diagnostics and targeted debugging.
00427 *
00428 * **OPTIONAL**: Can be NULL if implementation doesn't track per-component stats.
00429 *
00430 * @param[in] ctx Opaque pointer to concrete error handler. Must be non-NULL.
00431 * @param[in] component Component name (e.g., "SPI", "MOTOR"). Must be non-NULL.
00432 *
00433 * @return uint32_t Error count for component (0 if component not found or no errors)
00434 *
00435 * @note Returns 0 if component not registered
00436 * @note Case-sensitive component name matching (typically)
00437 * @see rx_error_get_count_fn For total error count across all components
00438 */
00439 typedef uint32_t (*rx_error_get_component_count_fn)(void* ctx, const char* component);
00440
00441 /**
00442 * @typedef rx_error_clear_fn
00443 * @brief Function Pointer for Clearing All Error Counters (REQUIRED Interface Function)
00444 *
00445 * @details
00446 * Resets all error counters to zero and clears retry state. Used for:
00447 * - Starting fresh after diagnostics
00448 * - Resetting after error recovery
00449 * - Clearing transient errors after system stabilization
00450 *
00451 * **REQUIRED**: All implementations must provide this function.
00452 *
00453 * @param[in] ctx Opaque pointer to concrete error handler. Must be non-NULL.
00454 *
00455 * @return rx_err_t Status of clear operation
00456 * @retval k_rx_ok All counters successfully cleared
00457 * @retval k_rx_err_null_ptr ctx is nullptr
00458 * @retval k_rx_err_invalid_state Handler not initialized
00459 *
00460 * @note Thread-safe implementations recommended
00461 * @warning Clears ALL error history - use with caution
00462 * @see rx_error_reset_retry_fn For resetting retry counters without clearing error counts
00463 */
00464 typedef rx_err_t (*rx_error_clear_fn)(void* ctx);
00465
00466 /**
00467 * @typedef rx_error_is_retry_limit_reached_fn
00468 * @brief Function Pointer for Checking Retry Limit Status (OPTIONAL)
00469 *
00470 * @details
00471 * Determines if a component has exceeded its maximum retry attempts. Used in
00472 * retry logic to decide whether to:
00473 * - Continue retrying (limit not reached)
00474 * - Give up and propagate error (limit reached)
00475 *

```

```

00476 **OPTIONAL**: Can be NULL if implementation doesn't support retry tracking.
00477 * Clients should check if function pointer is nullptr before calling.
00478 *
00479 * @param[in] ctx Opaque pointer to concrete error handler. Must be non-NULL.
00480 * @param[in] component Component name for retry limit check. Must be non-NULL.
00481 *
00482 * @return bool Retry limit status
00483 * @retval true Retry limit reached, should NOT retry
00484 * @retval false Retry limit not reached, can retry
00485 *
00486 * @note Returns false if component not found (allows first retry)
00487 * @note Returns false if ctx is nullptr (defensive programming)
00488 * @see rx_error_get_backoff_delay_fn For calculating retry delay
00489 * @see rx_error_reset_retry_fn For resetting retry counter
00490 */
00491 typedef bool (*rx_error_is_retry_limit_reached_fn)(void* ctx, const char* component);
00492
00493 /**
00494 * @typedef rx_error_reset_retry_fn
00495 * @brief Function Pointer for Resetting Component Retry Counter (OPTIONAL)
00496 *
00497 * @details
00498 * Resets the retry counter for a specific component back to zero. Called after:
00499 * - Successful operation (clear retry state)
00500 * - Manual intervention (give component another chance)
00501 * - Configuration change (fresh start)
00502 *
00503 **OPTIONAL**: Can be NULL if implementation doesn't support retry tracking.
00504 *
00505 * @param[in] ctx Opaque pointer to concrete error handler. Must be non-NULL.
00506 * @param[in] component Component name to reset. Must be non-NULL.
00507 *
00508 * @return rx_err_t Status of reset operation
00509 * @retval k_rx_ok Retry counter successfully reset
00510 * @retval k_rx_err_null_ptr ctx or component is nullptr
00511 * @retval k_rx_err_not_found Component not registered
00512 *
00513 * @note Does NOT clear error count, only retry counter
00514 * @note Thread-safe implementations recommended
00515 * @see rx_error_is_retry_limit_reached_fn For checking retry limit
00516 * @see rx_error_clear_fn For clearing all counters including errors
00517 */
00518 typedef rx_err_t (*rx_error_reset_retry_fn)(void* ctx, const char* component);
00519
00520 /**
00521 * @typedef rx_error_get_backoff_delay_fn
00522 * @brief Function Pointer for Calculating Exponential Backoff Delay (OPTIONAL)
00523 *
00524 * @details
00525 * Calculates the recommended delay before retrying an operation, typically using
00526 * exponential backoff algorithm:
00527 * ```
00528 * delay = initial_backoff x (2 ^ retry_count)
00529 * delay = min(delay, max_backoff) // Cap at maximum
00530 * ```
00531 *
00532 **OPTIONAL**: Can be NULL if implementation doesn't support backoff calculation.
00533 *
00534 **Example Backoff Progression** (initial=10ms, max=5s):
00535 * - Retry 0: 10ms
00536 * - Retry 1: 20ms
00537 * - Retry 2: 40ms
00538 * - Retry 3: 80ms
00539 * - Retry 4: 160ms
00540 * - ...
00541 * - Retry 9: 5120ms -> capped at 5000ms
00542 *
00543 * @param[in] ctx Opaque pointer to concrete error handler. Must be non-NULL.
00544 * @param[in] component Component name for backoff calculation. Must be non-NULL.
00545 *
00546 * @return uint32_t Recommended delay in milliseconds
00547 * @retval 0 No backoff needed (no retries yet or component not found)
00548 * @retval >0 Delay in milliseconds (use tx_thread_sleep or equivalent)
00549 *
00550 * @note Returns 0 if ctx is nullptr or component not found
00551 * @note Delay increases exponentially with retry count
00552 * @note Implementation may cap delay at configured maximum
00553 *
00554 * @par Example Usage:
00555 * @code{.c}
00556 * if (!error_iface->is_retry_limit_reached(error_iface->ctx, "SPI")) {
00557 *     uint32_t delay_ms = error_iface->get_backoff_delay(error_iface->ctx, "SPI");
00558 *     tx_thread_sleep(delay_ms / 10); // Convert ms to ThreadX ticks
00559 *     retry_operation(); // Try again
00560 * }
00561 * @endcode
00562 *

```

```

00563 * @see rx_error_is_retry_limit_reached_fn For checking if should retry
00564 * @see rx_error_reset_retry_fn For resetting retry counter on success
00565 */
00566 typedef uint32_t (*rx_error_get_backoff_delay_fn)(void* ctx, const char* component);
00567
00568 /*
=====
00569 * Error Handler Interface Structure
00570 *
=====
00571 */
00572
00573 /**
00574 * @struct rx_error_interface
00575 * @brief Abstract Error Handler Interface Structure (Polymorphism via Function Pointers)
00576 *
00577 * @details
00578 * ***THE DEPENDENCY INVERSION INTERFACE**. This structure IS the abstraction that
00579 * high-level modules depend on. Contains function pointers to concrete implementations
00580 * and an opaque context pointer for implementation state.
00581 *
00582 * ## Purpose and Architecture
00583 *
00584 * This struct enables **runtime polymorphism in C** without inheritance:
00585 * - Function pointers act like C++ virtual functions
00586 * - Context pointer acts like "this" pointer
00587 * - Clients call through interface, implementation is hidden
00588 *
00589 * **Analogy to Object-Oriented Programming**:
00590 * ```cpp
00591 * // C++ (conceptual equivalent)
00592 * class ErrorInterface {
00593 * public:
00594 *     virtual rx_err_t reportError(const char* component, const char* msg) = 0;
00595 *     virtual uint32_t getErrorCount() = 0;
00596 *     // ...
00597 * };
00598 *
00599 * // C implementation (this struct)
00600 * struct rx_error_interface {
00601 *     void* ctx; // "this" pointer
00602 *     rx_error_report_fn report_error; // virtual function
00603 *     // ...
00604 * };
00605 * ```
00606 *
00607 * ## Memory Layout
00608 *
00609 * | Offset | Size | Field | Notes |
00610 * |-----|-----|-----|-----|
00611 * | 0 | 8 | ctx | Context pointer (implementation-specific) |
00612 * | 8 | 8 | report_error | REQUIRED function pointer |
00613 * | 16 | 8 | get_error_count | REQUIRED function pointer |
00614 * | 24 | 8 | get_component_error_count | OPTIONAL (can be NULL) |
00615 * | 32 | 8 | clear_errors | REQUIRED function pointer |
00616 * | 40 | 8 | is_retry_limit_reached | OPTIONAL (can be NULL) |
00617 * | 48 | 8 | reset_retry_counter | OPTIONAL (can be NULL) |
00618 * | 56 | 8 | get_backoff_delay | OPTIONAL (can be NULL) |
00619 * | **Total** | **64 bytes** | | Lightweight interface |
00620 *
00621 * ## Field Categories
00622 *
00623 * **1. Context (State)**:
00624 * - `ctx`: Opaque pointer to concrete implementation
00625 *
00626 * **2. REQUIRED Operations** (must be non-NULL):
00627 * - `report_error`: Report error occurrence
00628 * - `get_error_count`: Get total error count
00629 * - `clear_errors`: Reset all counters
00630 *
00631 * **3. OPTIONAL Operations** (may be NULL):
00632 * - `get_component_error_count`: Per-component statistics
00633 * - `is_retry_limit_reached`: Retry limit checking
00634 * - `reset_retry_counter`: Retry state reset
00635 * - `get_backoff_delay`: Exponential backoff calculation
00636 *
00637 * ## Initialization Pattern
00638 *
00639 * Concrete implementations provide a "get_interface" function:
00640 * ```c
00641 * void error_handler_get_interface(rx_error_interface_t* iface,
00642 *                                 error_handler_t* handler) {
00643 *     iface->ctx = handler; // Store concrete instance
00644 *
00645 *     // Bind required functions
00646 *     iface->report_error = internal_report_error;
00647 *     iface->get_error_count = internal_get_count;

```

```

00648 *   iface->clear_errors = internal_clear;
00649 *
00650 *   // Bind optional functions (or leave NULL)
00651 *   iface->get_component_error_count = internal_get_component_count;
00652 *   iface->is_retry_limit_reached = internal_is_retry_limit_reached;
00653 *   iface->reset_retry_counter = internal_reset_retry;
00654 *   iface->get_backoff_delay = internal_get_backoff_delay;
00655 * }
00656 * ...
00657 *
00658 * ## Usage Pattern (Client Code)
00659 *
00660 * **Step 1: Store interface in dependent module**:
00661 * ```c
00662 * typedef struct {
00663 *   rx_error_interface_t* error_iface; // Pointer to interface
00664 *   // ... other fields ...
00665 * } spi_driver_t;
00666 * ...
00667 *
00668 * **Step 2: Inject interface during initialization**:
00669 * ```c
00670 * void spi_driver_init(spi_driver_t* spi, rx_error_interface_t* error_iface) {
00671 *   spi->error_iface = error_iface; // Dependency injection
00672 * }
00673 * ...
00674 *
00675 * **Step 3: Call through interface**:
00676 * ```c
00677 * void spi_transfer(spi_driver_t* spi, uint8_t* data, size_t len) {
00678 *   rx_err_t err = spi_hw_write(data, len);
00679 *   if (err != k_rx_ok) {
00680 *     // Polymorphic call - don't know concrete implementation
00681 *     spi->error_iface->report_error(
00682 *       spi->error_iface->ctx, // Context for implementation
00683 *       err,
00684 *       "SPI",
00685 *       "Transfer failed"
00686 *     );
00687 *   }
00688 * }
00689 * ...
00690 *
00691 * @par Field Invariants:
00692 *
00693 * | Field | Constraint | Validated By |
00694 * |-----|-----|-----|
00695 * | ctx | Can be NULL (implementation choice) | Implementation |
00696 * | report_error | MUST be non-NULL | rx_error_interface_validate() |
00697 * | get_error_count | MUST be non-NULL | rx_error_interface_validate() |
00698 * | clear_errors | MUST be non-NULL | rx_error_interface_validate() |
00699 * | Other fields | MAY be NULL (optional) | Client checks before calling |
00700 *
00701 * @par Usage Example (Complete Workflow):
00702 * @code{.c}
00703 * // ===== LOW-LEVEL: Create concrete implementation =====
00704 * static error_handler_t s_error_handler;
00705 * const error_handler_config_t config = {
00706 *   .max_retries = 3,
00707 *   .initial_backoff_ms = 10,
00708 *   .max_backoff_ms = 5000
00709 * };
00710 * error_handler_init(&s_error_handler, &config);
00711 *
00712 * // ===== ABSTRACTION: Get interface =====
00713 * rx_error_interface_t error_iface;
00714 * error_handler_get_interface(&error_iface, &s_error_handler);
00715 *
00716 * // ===== Validate interface =====
00717 * rx_err_t err = rx_error_interface_validate(&error_iface);
00718 * if (err != k_rx_ok) {
00719 *   uart_debug_puts("[ERROR] Invalid error interface\r\n");
00720 *   return err;
00721 * }
00722 *
00723 * // ===== HIGH-LEVEL: Inject into dependent modules =====
00724 * spi_driver_t spi;
00725 * spi_driver_init(&spi, &error_iface); // SPI knows ONLY interface
00726 *
00727 * motor_controller_t motor;
00728 * motor_controller_init(&motor, &error_iface); // Motor knows ONLY interface
00729 *
00730 * // ===== USAGE: High-level code uses interface =====
00731 * // SPI driver reports error (doesn't know about error_handler_t!)
00732 * error_iface.report_error(error_iface.ctx, k_rx_fail, "SPI", "Timeout");
00733 *
00734 * // Motor checks retry (doesn't know about error_handler_t!)

```

```

00735 * if (!error_iface.is_retry_limit_reached(error_iface.ctx, "MOTOR")) {
00736 *     uint32_t delay = error_iface.get_backoff_delay(error_iface.ctx, "MOTOR");
00737 *     tx_thread_sleep(delay / 10);
00738 *     retry_motor_operation();
00739 * }
00740 * @endcode
00741 *
00742 * @par Usage Example (Optional Function Handling):
00743 * @code{.c}
00744 * // Check if optional function is available before calling
00745 * if (error_iface->get_component_error_count != nullptr) {
00746 *     uint32_t spi_errors = error_iface->get_component_error_count(
00747 *         error_iface->ctx, "SPI"
00748 *     );
00749 *     uart_debug_printf("SPI errors: %lu\r\n", spi_errors);
00750 * }
00751 *
00752 * // Retry logic with optional functions
00753 * if (error_iface->is_retry_limit_reached != nullptr) {
00754 *     if (error_iface->is_retry_limit_reached(error_iface->ctx, "I2C")) {
00755 *         return k_rx_fail; // Give up
00756 *     }
00757 * }
00758 * @endcode
00759 *
00760 * @invariant report_error, get_error_count, clear_errors must be non-NULL
00761 * @invariant Optional function pointers can be nullptr
00762 * @invariant ctx pointer is opaque to clients (implementation detail)
00763 * @invariant Interface lifetime must not exceed concrete implementation lifetime
00764 *
00765 * @note Structure is lightweight (64 bytes) - safe to pass by value or store
00766 * @note Typically stored as pointer in dependent modules
00767 * @note Can have multiple interfaces pointing to same concrete implementation
00768 *
00769 * @warning Do NOT access ctx directly in client code (violates abstraction)
00770 * @warning Do NOT assume concrete type of ctx (polymorphism)
00771 * @warning Check optional function pointers for nullptr before calling
00772 * @attention Interface becomes invalid if concrete implementation is destroyed
00773 *
00774 * @see rx_error_interface_validate() Validates interface before use
00775 * @see error_handler_get_interface() Example interface provider (rx_error_handler.h)
00776 * @see rx_error_report_fn Function pointer type definitions
00777 *
00778 * @since Version 1.0.0
00779 */
00780 struct rx_error_interface {
00781     void*
00782     ctx; /**< Implementation context (opaque pointer to concrete handler). Cast to concrete type within implementation
functions. Passed as first parameter to all function pointers. Can be NULL if implementation is stateless (rare). Clients
must NEVER access or cast this pointer */
00783     rx_error_report_fn
report_error; /**< Report an error (REQUIRED - MUST be non-NULL). Increments error counters, updates
component statistics, logs error. Primary interface function. Validated by rx_error_interface_validate() */
00785     rx_error_get_count_fn
get_error_count; /**< Get total error count across all components (REQUIRED - MUST be non-NULL). Returns
aggregate error count since initialization or last clear. Used for health monitoring. Validated by
rx_error_interface_validate() */
00787     rx_error_get_component_count_fn
get_component_error_count; /**< Get error count for specific component (OPTIONAL - may be NULL). Returns
per-component statistics for diagnostics. Check for nullptr before calling. Returns 0 if nullptr or component not found */
00789     rx_error_clear_fn
clear_errors; /**< Clear all error counters and retry state (REQUIRED - MUST be non-NULL). Resets system to fresh
state. Use after diagnostics or recovery. Validated by rx_error_interface_validate() */
00791     rx_error_is_retry_limit_reached_fn
is_retry_limit_reached; /**< Check if component exceeded retry limit (OPTIONAL - may be NULL). Returns true if
should give up, false if can retry. Check for nullptr before calling. Used in retry logic decision-making */
00793     rx_error_reset_retry_fn
reset_retry_counter; /**< Reset retry counter for component (OPTIONAL - may be NULL). Clears retry state without
clearing error count. Call after successful operation. Check for nullptr before calling */
00795     rx_error_get_backoff_delay_fn
get_backoff_delay; /**< Calculate exponential backoff delay for retry (OPTIONAL - may be NULL). Returns
milliseconds to wait before next retry. Returns 0 if nullptr or no backoff needed. Check for nullptr before calling */
00797 };
00798
00799 /*
=====
00800 * Interface Validation
00801 *
=====
00802 */
00803
00804 /**
00805 * @brief Validate that error interface is properly initialized with required functions
00806 *
00807 * @details
00808 * Performs validation checks on an error interface before use. Ensures that all
00809 * REQUIRED function pointers are non-NULL. Optional function pointers may be NULL.

```



```

00810 * Call this function after obtaining an interface from a concrete implementation
00811 * to verify correctness before injecting into dependent modules.
00812 *
00813 * **Validation Checks**:
00814 * 1. Interface pointer is non-NULL
00815 * 2. report_error function pointer is non-NULL (REQUIRED)
00816 * 3. get_error_count function pointer is non-NULL (REQUIRED)
00817 * 4. clear_errors function pointer is non-NULL (REQUIRED)
00818 *
00819 * **Not Validated** (optional functions can be NULL):
00820 * - get_component_error_count
00821 * - is_retry_limit_reached
00822 * - reset_retry_counter
00823 * - get_backoff_delay
00824 * - ctx (can be NULL for stateless implementations)
00825 *
00826 * **Performance**: Very fast (~10 CPU cycles), just NULL checks.
00827 *
00828 * @param[in] iface Pointer to interface structure to validate. Can be NULL (returns error).
00829 *
00830 * @return rx_err_t Validation result
00831 * @retval k_rx_ok Interface is valid and ready to use
00832 * @retval k_rx_err_null_ptr iface parameter is nullptr
00833 * @retval k_rx_err_invalid_state One or more REQUIRED function pointers are nullptr
00834 *
00835 * @note This is a static inline function (no function call overhead)
00836 * @note Only checks REQUIRED functions, optional functions may be nullptr
00837 * @note Does not validate function pointer targets (assumes correct binding)
00838 *
00839 * @warning Do not use interface if validation fails
00840 * @attention Always validate after getting interface from implementation
00841 *
00842 * @par Usage Example (Recommended Pattern):
00843 * @code{.c}
00844 * // Get interface from concrete implementation
00845 * rx_error_interface_t error_iface;
00846 * error_handler_get_interface(&error_iface, &handler);
00847 *
00848 * // VALIDATE before use
00849 * rx_err_t err = rx_error_interface_validate(&error_iface);
00850 * if (err != k_rx_ok) {
00851 *     uart_debug_puts("[ERROR] Invalid error interface\r\n");
00852 *     return err;
00853 * }
00854 *
00855 * // Safe to use now
00856 * spi_driver_init(&spi, &error_iface);
00857 * @endcode
00858 *
00859 * @par Usage Example (Error Handling):
00860 * @code{.c}
00861 * rx_err_t err = rx_error_interface_validate(&iface);
00862 *
00863 * switch (err) {
00864 *     case k_rx_ok:
00865 *         uart_debug_puts("[INFO] Interface valid\r\n");
00866 *         break;
00867 *     case k_rx_err_null_ptr:
00868 *         uart_debug_puts("[ERROR] NULL interface pointer\r\n");
00869 *         break;
00870 *     case k_rx_err_invalid_state:
00871 *         uart_debug_puts("[ERROR] Missing required function pointers\r\n");
00872 *         break;
00873 * }
00874 * @endcode
00875 *
00876 * @par Example (Defensive Programming):
00877 * @code{.c}
00878 * void module_init(module_t* mod, rx_error_interface_t* error_iface) {
00879 *     // Defensive: Validate interface at module boundaries
00880 *     rx_err_t err = rx_error_interface_validate(error_iface);
00881 *     if (err != k_rx_ok) {
00882 *         // Cannot proceed without valid error interface
00883 *         return err;
00884 *     }
00885 *
00886 *     // Store validated interface
00887 *     mod->error_iface = error_iface;
00888 *     return k_rx_ok;
00889 * }
00890 * @endcode
00891 *
00892 * @see rx_error_interface Structure being validated
00893 * @see error_handler_get_interface() Example interface provider
00894 *
00895 * @since Version 1.0.0
00896 *

```



```

00897 * @par NASA Power of 10 Compliance:
00898 * - Rule 1: [OK] Simple control flow (sequential checks)
00899 * - Rule 4: [OK] Function < 10 lines (very small)
00900 * - Rule 5: [OK] Precondition checks (NULL validation)
00901 */
00902 static inline rx_err_t rx_error_interface_validate(const rx_error_interface_t* iface)
00903 {
00904     if (iface == nullptr) {
00905         return k_rx_err_null_ptr;
00906     }
00907
00908     /* Check required function pointers */
00909     if (iface->report_error == nullptr || iface->get_error_count == nullptr ||
00910         iface->clear_errors == nullptr) {
00911         return k_rx_err_invalid_state;
00912     }
00913
00914     return k_rx_ok;
00915 }
00916
00917 #ifdef __cplusplus
00918 }
00919 #endif

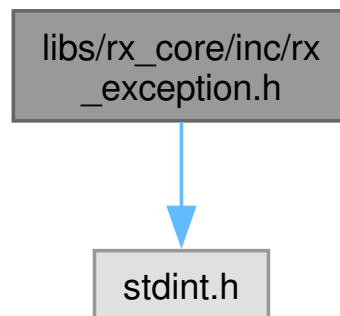
```

7.11 libs/rx_core/inc/rx_exception.h File Reference

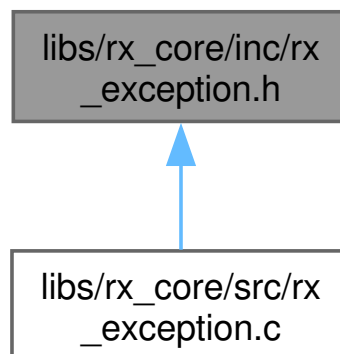
RX72N CPU Exception Handling.

#include <stdint.h>

Include dependency graph for rx_exception.h:



This graph shows which files directly or indirectly include this file:



Data Structures

- struct [rx_exception_frame_t](#)
Captured CPU state at time of exception.
- struct [rx_exception_stats_t](#)
Exception occurrence statistics for debugging.

Enumerations

- enum [rx_exception_type_t](#) : `uint8_t` {
`k_rx_exception_undefined_instruction` = 0 , `k_rx_exception_privileged_instruction` = 1 ,
`k_rx_exception_access` = 2 , `k_rx_exception_address` = 3 ,
`k_rx_exception_floating_point` = 4 , `k_rx_exception_nmi` = 5 , `k_rx_exception_type_count`
= 6 }
CPU exception types supported by RXv3 core.
- enum [rx_exception_vector_offset_t](#) : `uint8_t` {
`k_rx_exc_offset_privileged` = 0x50 , `k_rx_exc_offset_access` = 0x54 , `k_rx_exc_offset_undefined`
= 0x5C , `k_rx_exc_offset_address` = 0x60 ,
`k_rx_exc_offset_fpu` = 0x64 , `k_rx_exc_offset_nmi` = 0x78 , `k_rx_exc_offset_reset` = 0x7C }
Exception vector table offsets from EXTB base.
- enum [rx_exception_frame_size_t](#) : `uint8_t` { `k_exception_frame_reserved_size` , `k_exception_frame_total_size`
}
Size constants for [rx_exception_frame_t](#).

Functions

- void [rx_exception_init](#) (void)
Initialize exception handling module.
- const [rx_exception_stats_t](#) * [rx_exception_get_stats](#) (void)
Get exception statistics.
- const char * [rx_exception_get_name](#) ([rx_exception_type_t](#) type)
Get human-readable name for exception type.
- void [rx_exc_undefined_instruction_handler](#) (void)
Undefined instruction exception handler (ASM entry point).
- void [rx_exc_privileged_instruction_handler](#) (void)
Privileged instruction exception handler (ASM entry point).
- void [rx_exc_access_handler](#) (void)
Access exception handler (ASM entry point).
- void [rx_exc_address_handler](#) (void)
Address exception handler (ASM entry point).
- void [rx_exc_floating_point_handler](#) (void)
Floating-point exception handler (ASM entry point).
- void [rx_exc_nmi_handler](#) (void)
Non-maskable interrupt handler (ASM entry point).

7.11.1 Detailed Description

RX72N CPU Exception Handling.

Provides exception handling for RX72N CPU exceptions as defined in the hardware manual Chapter 14. This module captures debug information when exceptions occur and provides handlers for:

- Undefined instruction exception (EXTB + 0x5C)
- Privileged instruction exception (EXTB + 0x50)
- Access exception (EXTB + 0x54)
- Address exception (EXTB + 0x60)
- Single-precision floating-point exception (EXTB + 0x64)
- Non-maskable interrupt (EXTB + 0x78)

The RXv3 CPU saves PC and PSW to the stack (ISP) on exception entry, shifts to supervisor mode, and disables interrupts (I=0, U=0, PM=0).

Thread Safety:

Exception handlers execute in supervisor mode with interrupts disabled. Not thread-safe by nature
- exceptions preempt all other execution.

Manual Reference:

RX72N Group User's Manual: Hardware, Rev.1.20, Chapter 14

See also

[rx72n_icu_regs.h](#) For NMI-related ICU registers

Since

Version 1.0.0

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Definition in file [rx_exception.h](#).

7.11.2 Data Structure Documentation

7.11.2.1 struct rx_exception_frame_t

Captured CPU state at time of exception.

When an exception occurs, the RXv3 CPU saves PC and PSW to the interrupt stack (ISP). This structure captures additional context for debugging. The exception handler captures this information before calling the C-level handler function.

Stack Layout on Exception Entry:

```
ISP before exception:
[lower address]
  PC (return address, 4 bytes)
  PSW (processor status word, 4 bytes)
[higher address] <- ISP after hardware push
```

Manual Reference:

Section 14.4 - Hardware Processing for Accepting and Returning from Exceptions

Invariant

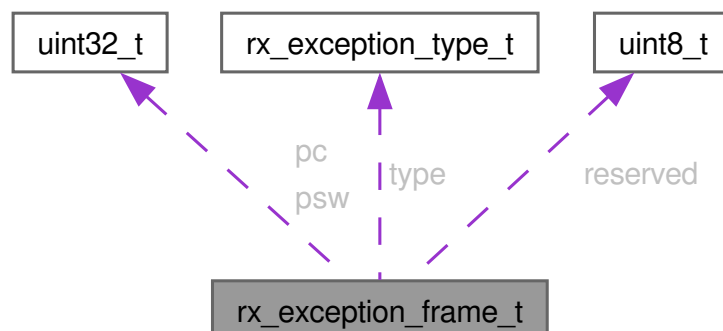
pc must be aligned to 2-byte boundary (RX instructions are 2-byte aligned)
psw I bit and U bit are 0 in exception context

Since

Version 1.0.0

Definition at line 252 of file [rx_exception.h](#).

Collaboration diagram for rx_exception_frame_t:



Data Fields

uint32_t	pc	Program counter at time of exception. For instruction-canceling exceptions (UND, PIE, ACE, ADE, FPE): Points to the instruction that caused the exception. For instruction-completion exceptions (EI, TRAP): Points to the next instruction. Range: 0x00000000 to 0xFFFFFFFF
uint32_t	psw	Processor status word at time of exception. Contains flags, interrupt priority level, and mode bits. Bit Layout: • [31:28] IPL - Interrupt priority level (0-15) • [27:24] Reserved • [23:20] PM, U, C, Z - Processor mode, stack, carry, zero • [19:16] S, O, I, Reserved - Sign, overflow, interrupt enable • [15:0] Reserved and other flags
uint8_t	reserved[k_exception_frame_reserved_size]	Padding for alignment. Ensures structure is 4-byte aligned
rx_exception_type_t	type	Exception type that occurred. Identifies which exception handler was invoked

7.11.2.2 struct rx_exception_stats_t

Exception occurrence statistics for debugging.

Tracks how many times each exception type has occurred. Useful for identifying systemic issues in the firmware.

Thread Safety:

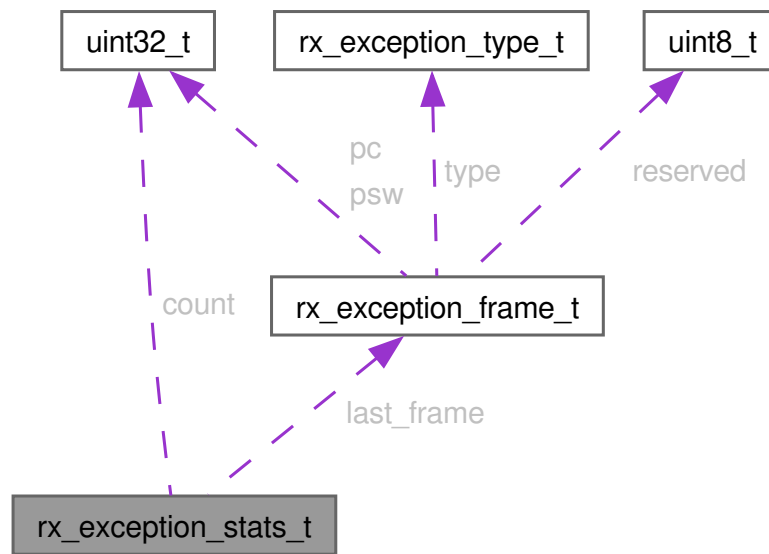
Read/write operations on counters are not atomic. For precise counts in multi-threaded environments, use appropriate synchronization.

Since

Version 1.0.0

Definition at line 313 of file [rx_exception.h](#).

Collaboration diagram for `rx_exception_stats_t`:



Data Fields

<code>uint32_t</code>	<code>count[k_rx_exception_type_count]</code>	Count per exception type. Index by rx_exception_type_t enum values
rx_exception_frame_t	<code>last_frame</code>	Last exception frame captured. Preserved for post-mortem debugging

7.11.3 Enumeration Type Documentation

7.11.3.1 `rx_exception_frame_size_t`

enum [rx_exception_frame_size_t](#) : `uint8_t`

Size constants for [rx_exception_frame_t](#).

Named constants for the exception frame structure sizes used in static assertions and structure field sizing.

Since

Version 1.0.0

Enumerator

k_exception_frame_reserved_size	Padding bytes to align rx_exception_frame_t to 4 bytes
k_exception_frame_total_size	Total size of rx_exception_frame_t in bytes (pc:4 + psw:4 + type:1 + reserved:3)

Definition at line 214 of file [rx_exception.h](#).

```

00214     : uint8_t {
00215     k_exception_frame_reserved_size =
00216     3, /**< Padding bytes to align rx_exception_frame_t to 4 bytes */
00217     k_exception_frame_total_size =
00218     12, /**< Total size of rx_exception_frame_t in bytes (pc:4 + psw:4 + type:1 + reserved:3) */
00219 } rx_exception_frame_size_t;

```

7.11.3.2 rx_exception_type_t

enum [rx_exception_type_t](#) : uint8_t

CPU exception types supported by RXv3 core.

Defines all exception types from Chapter 14 of the RX72N hardware manual. Exception priority from highest to lowest:

1. Reset (highest)
2. Non-maskable interrupt
3. Interrupt
4. Instruction access exception
5. Undefined/Privileged instruction exception
6. Unconditional trap
7. Address exception
8. Operand access exception
9. Single-precision floating-point exception (lowest)

Manual Reference:

Table 14.4 - Priority of Exception Events

Since

Version 1.0.0

Enumerator

k_rx_exception_undefined_instruction	Undefined instruction exception. Occurs when CPU attempts to execute an instruction not implemented in the RXv3 instruction set. Vector: EXTB + 0x5C Return: RTE instruction (if recoverable)
k_rx_exception_privileged_instruction	Privileged instruction exception. Occurs when a privileged instruction is executed in user mode. Privileged instructions can only execute in supervisor mode. Vector: EXTB + 0x50 Return: RTE instruction (if recoverable)
k_rx_exception_access	Access exception (instruction or operand). Occurs when memory protection unit (MPU) detects a memory access violation. Can be instruction-access (fetch) or operand-access (load/store). Vector: EXTB + 0x54 Return: RTE instruction (if recoverable)
k_rx_exception_address	Address exception. Occurs on 64-bit operand access to an address not on a 32-bit boundary. Vector: EXTB + 0x60 Return: RTE instruction (if recoverable)

k_rx_exception_floating_point	Single-precision floating-point exception. Occurs when IEEE 754 floating-point exception is detected: overflow, underflow, inexact, division-by-zero, or invalid operation. Only triggers if corresponding enable bit in FPSW is set. Vector: EXTB + 0x64 Return: RTE instruction (if recoverable)
k_rx_exception_nmi	Non-maskable interrupt (NMI). Fatal system fault indication. NEVER return from NMI handler - must reset system or halt permanently. Vector: EXTB + 0x78 Return: PROHIBITED - must reset or halt Warning Do not attempt to return from NMI handler
k_rx_exception_type_count	Total count of exception types. Used for array sizing and validation

Definition at line 67 of file [rx_exception.h](#).

```

00067         : uint8_t {
00068     /**
00069      * @brief Undefined instruction exception
00070      * @details
00071      * Occurs when CPU attempts to execute an instruction not implemented
00072      * in the RXv3 instruction set.
00073      * @par Vector: EXTB + 0x5C
00074      * @par Return: RTE instruction (if recoverable)
00075      */
00076     k_rx_exception_undefined_instruction = 0,
00077
00078     /**
00079      * @brief Privileged instruction exception
00080      * @details
00081      * Occurs when a privileged instruction is executed in user mode.
00082      * Privileged instructions can only execute in supervisor mode.
00083      * @par Vector: EXTB + 0x50
00084      * @par Return: RTE instruction (if recoverable)
00085      */
00086     k_rx_exception_privileged_instruction = 1,
00087
00088     /**
00089      * @brief Access exception (instruction or operand)
00090      * @details
00091      * Occurs when memory protection unit (MPU) detects a memory access
00092      * violation. Can be instruction-access (fetch) or operand-access (load/store).
00093      * @par Vector: EXTB + 0x54
00094      * @par Return: RTE instruction (if recoverable)
00095      */
00096     k_rx_exception_access = 2,
00097
00098     /**
00099      * @brief Address exception
00100      * @details
00101      * Occurs on 64-bit operand access to an address not on a 32-bit boundary.
00102      * @par Vector: EXTB + 0x60
00103      * @par Return: RTE instruction (if recoverable)

```

```

00104     */
00105     k_rx_exception_address = 3,
00106
00107     /**
00108      * @brief Single-precision floating-point exception
00109      * @details
00110      * Occurs when IEEE 754 floating-point exception is detected:
00111      * overflow, underflow, inexact, division-by-zero, or invalid operation.
00112      * Only triggers if corresponding enable bit in FPSW is set.
00113      * @par Vector: EXTB + 0x64
00114      * @par Return: RTE instruction (if recoverable)
00115      */
00116     k_rx_exception_floating_point = 4,
00117
00118     /**
00119      * @brief Non-maskable interrupt (NMI)
00120      * @details
00121      * Fatal system fault indication. NEVER return from NMI handler -
00122      * must reset system or halt permanently.
00123      * @par Vector: EXTB + 0x78
00124      * @par Return: PROHIBITED - must reset or halt
00125      * @warning Do not attempt to return from NMI handler
00126      */
00127     k_rx_exception_nmi = 5,
00128
00129     /**
00130      * @brief Total count of exception types
00131      * @details Used for array sizing and validation
00132      */
00133     k_rx_exception_type_count = 6,
00134
00135 } rx_exception_type_t;

```

7.11.3.3 rx_exception_vector_offset_t

```
enum rx_exception_vector_offset_t : uint8_t
```

Exception vector table offsets from EXTB base.

The RX72N uses EXTB register to specify the exception vector table base. Default EXTB value after reset is 0xFFFF_FF80. Each offset specifies where the exception handler address is stored.

Manual Reference:

Section 14.5 - Hardware Pre-Processing

Since

Version 1.0.0

Enumerator

k_rx_exc_offset_privileged	Privileged instruction exception vector offset. Handler address at EXTB + 0x50 = default 0xFFFF_FFD0
k_rx_exc_offset_access	Access exception vector offset. Handler address at EXTB + 0x54 = default 0xFFFF_FFD4
k_rx_exc_offset_undefined	Undefined instruction exception vector offset. Handler address at EXTB + 0x5C = default 0xFFFF_FFDC
k_rx_exc_offset_address	Address exception vector offset. Handler address at EXTB + 0x60 = default 0xFFFF_FFE0
k_rx_exc_offset_fpu	Floating-point exception vector offset. Handler address at EXTB + 0x64 = default 0xFFFF_FFE4

k_rx_exc_offset_nmi	Non-maskable interrupt vector offset. Handler address at EXTB + 0x78 = default 0xFFFF_FFF8
k_rx_exc_offset_reset	Reset vector offset. Handler address at EXTB + 0x7C = fixed 0xFFFF_FFFC

Definition at line 155 of file [rx_exception.h](#).

```

00155         : uint8_t {
00156     /**
00157      * @brief Privileged instruction exception vector offset
00158      * @details Handler address at EXTB + 0x50 = default 0xFFFF_FFD0
00159      */
00160     k_rx_exc_offset_privileged = 0x50,
00161
00162     /**
00163      * @brief Access exception vector offset
00164      * @details Handler address at EXTB + 0x54 = default 0xFFFF_FFD4
00165      */
00166     k_rx_exc_offset_access = 0x54,
00167
00168     /**
00169      * @brief Undefined instruction exception vector offset
00170      * @details Handler address at EXTB + 0x5C = default 0xFFFF_FFDC
00171      */
00172     k_rx_exc_offset_undefined = 0x5C,
00173
00174     /**
00175      * @brief Address exception vector offset
00176      * @details Handler address at EXTB + 0x60 = default 0xFFFF_FFE0
00177      */
00178     k_rx_exc_offset_address = 0x60,
00179
00180     /**
00181      * @brief Floating-point exception vector offset
00182      * @details Handler address at EXTB + 0x64 = default 0xFFFF_FFE4
00183      */
00184     k_rx_exc_offset_fpu = 0x64,
00185
00186     /**
00187      * @brief Non-maskable interrupt vector offset
00188      * @details Handler address at EXTB + 0x78 = default 0xFFFF_FFF8
00189      */
00190     k_rx_exc_offset_nmi = 0x78,
00191
00192     /**
00193      * @brief Reset vector offset
00194      * @details Handler address at EXTB + 0x7C = fixed 0xFFFF_FFFC
00195      */
00196     k_rx_exc_offset_reset = 0x7C,
00197
00198 } rx_exception_vector_offset_t;

```

7.11.4 Function Documentation

7.11.4.1 rx_exc_access_handler()

```
void rx_exc_access_handler (
    void )
```

Access exception handler (ASM entry point).

Called from vector at EXTB + 0x54. Saves context and calls C handler.

Warning

Do not call directly from C code

7.11.4.2 rx_exc_address_handler()

```
void rx_exc_address_handler (  
    void )
```

Address exception handler (ASM entry point).

Called from vector at EXTB + 0x60. Saves context and calls C handler.

Warning

Do not call directly from C code

7.11.4.3 rx_exc_floating_point_handler()

```
void rx_exc_floating_point_handler (  
    void )
```

Floating-point exception handler (ASM entry point).

Called from vector at EXTB + 0x64. Saves context and calls C handler.

Warning

Do not call directly from C code

7.11.4.4 rx_exc_nmi_handler()

```
void rx_exc_nmi_handler (  
    void )
```

Non-maskable interrupt handler (ASM entry point).

Called from vector at EXTB + 0x78. NEVER returns.

Warning

Do not call directly from C code
MUST NOT return - system must reset or halt

7.11.4.5 rx_exc_privileged_instruction_handler()

```
void rx_exc_privileged_instruction_handler (  
    void )
```

Privileged instruction exception handler (ASM entry point).

Called from vector at EXTB + 0x50. Saves context and calls C handler.

Warning

Do not call directly from C code

7.11.4.6 rx_exc_undefined_instruction_handler()

```
void rx_exc_undefined_instruction_handler (
    void )
```

Undefined instruction exception handler (ASM entry point).

Called from vector at EXTB + 0x5C. Saves context and calls C handler.

Warning

Do not call directly from C code

7.11.4.7 rx_exception_get_name()

```
const char * rx_exception_get_name (
    rx_exception_type_t type)
```

Get human-readable name for exception type.

Returns a string describing the exception type for logging.

Parameters

in	type	Exception type enumeration value
----	------	----------------------------------

Returns

Pointer to static string (never NULL)

Return values

Unknown	If type is out of range
---------	-------------------------

Precondition

None

Postcondition

No side effects

Note

Thread-safe: Yes (returns pointer to const data)

Since

Version 1.0.0

Definition at line 204 of file [rx_exception.c](#).

```
00205 {
00206     if (type >= k_rx_exception_type_count) {
00207         return "Unknown";
00208     }
00209     return s_exception_names[type];
00210 }
```

References [k_rx_exception_type_count](#), and [s_exception_names](#).

7.11.4.8 rx_exception_get_stats()

```
const rx_exception_stats_t * rx_exception_get_stats (
    void )
```

Get exception statistics.

Returns pointer to read-only exception statistics structure. Statistics are updated by exception handlers. Returns NULL if [rx_exception_init\(\)](#) has not yet been called; callers must check the return value before dereferencing.

Returns

Pointer to exception statistics, or NULL if not initialized

Return values

non-NULL	Valid pointer to statistics after rx_exception_init()
NULL	rx_exception_init() has not been called yet

Precondition

[rx_exception_init\(\)](#) must have been called for a non-NULL return

Caller must mask interrupts before reading fields if a consistent snapshot is required; exception handlers modify the statistics structure asynchronously from interrupt context

Postcondition

No side effects; returned pointer is owned by this module and must not be freed or modified

The returned pointer remains valid for the lifetime of the module (never reallocated)

Note

NOT thread-safe: exception handlers update the statistics structure from interrupt context concurrently with any reader; callers that require a consistent multi-field snapshot must disable interrupts around the read

Example:

```
const rx_exception_stats_t* stats = rx_exception_get_stats();
if (stats != NULL && stats->count[k_rx_exception_nmi] > 0) {
    // NMI occurred - investigate
}
```

Since

Version 1.0.0

Definition at line 193 of file [rx_exception.c](#).

```
00194 {
00195     if (s_initialized != k_rx_exc_initialized) {
00196         return nullptr;
00197     }
00198     return &s_exception_stats;
00199 }
```

References [k_rx_exc_initialized](#), [s_exception_stats](#), and [s_initialized](#).

7.11.4.9 rx_exception_init()

```
void rx_exception_init (
    void )
```

Initialize exception handling module.

Initializes exception statistics and optionally installs custom handlers. Should be called early in system startup, after basic hardware init.

Precondition

None

Postcondition

Exception statistics zeroed

Default handlers installed (if not using assembly stubs)

Note

Thread-safe: No, call only once during startup

Example:

```
void main(void) {
    rx_exception_init();
    // ... rest of initialization
}
```

Since

Version 1.0.0

Definition at line 169 of file [rx_exception.c](#).

```
00170 {
00171     /* Zero all statistics */
00172     for (uint8_t i = 0; i < k_rx_exception_type_count; i++) {
00173         s_exception_stats.count[i] = 0;
00174     }
00175
00176     /* Clear last frame */
00177     s_exception_stats.last_frame.pc = 0;
00178     s_exception_stats.last_frame.psw = 0;
00179     s_exception_stats.last_frame.type = k_rx_exception_undefined_instruction;
00180     s_exception_stats.last_frame.reserved[k_exc_reserved_idx_0] = 0;
00181     s_exception_stats.last_frame.reserved[k_exc_reserved_idx_1] = 0;
00182     s_exception_stats.last_frame.reserved[k_exc_reserved_idx_2] = 0;
00183
00184     /* Mark as initialized */
00185     s_initialized = k_rx_exc_initialized;
00186
00187     (void)rx_log_info(s_tag, "Exception handling initialized");
00188 }
```

References [k_exc_reserved_idx_0](#), [k_exc_reserved_idx_1](#), [k_exc_reserved_idx_2](#), [k_rx_exc_initialized](#), [k_rx_exception_type_count](#), [k_rx_exception_undefined_instruction](#), [rx_log_info](#), [s_exception_stats](#), [s_initialized](#), and [s_tag](#).

7.12 rx_exception.h

[Go to the documentation of this file.](#)

```

00001 /**
00002  * @file rx_exception.h
00003  * @brief RX72N CPU Exception Handling
00004  *
00005  * @details
00006  * Provides exception handling for RX72N CPU exceptions as defined in
00007  * the hardware manual Chapter 14. This module captures debug information
00008  * when exceptions occur and provides handlers for:
00009  * - Undefined instruction exception (EXTB + 0x5C)
00010  * - Privileged instruction exception (EXTB + 0x50)
00011  * - Access exception (EXTB + 0x54)
00012  * - Address exception (EXTB + 0x60)
00013  * - Single-precision floating-point exception (EXTB + 0x64)
00014  * - Non-maskable interrupt (EXTB + 0x78)
00015  *
00016  * The RXv3 CPU saves PC and PSW to the stack (ISP) on exception entry,
00017  * shifts to supervisor mode, and disables interrupts (I=0, U=0, PM=0).
00018  *
00019  * @par Thread Safety:
00020  * Exception handlers execute in supervisor mode with interrupts disabled.
00021  * Not thread-safe by nature - exceptions preempt all other execution.
00022  *
00023  * @par Manual Reference:
00024  * RX72N Group User's Manual: Hardware, Rev.1.20, Chapter 14
00025  *
00026  * @see rx72n_icu_regs.h For NMI-related ICU registers
00027  *
00028  * @since Version 1.0.0
00029  * @copyright Copyright (c) 2026 Locked Inc.
00030  * SPDX-License-Identifier: MIT
00031  */
00032
00033 #pragma once
00034
00035 #include <stdint.h>
00036
00037 #ifdef __cplusplus
00038 extern "C" {
00039 #endif
00040
00041 /**
=====
00042  * Exception Type Enumeration
00043  *
=====
*/
00044
00045 /**
00046  * @enum rx_exception_type_t
00047  * @brief CPU exception types supported by RXv3 core
00048  *
00049  * @details
00050  * Defines all exception types from Chapter 14 of the RX72N hardware manual.
00051  * Exception priority from highest to lowest:
00052  * 1. Reset (highest)
00053  * 2. Non-maskable interrupt
00054  * 3. Interrupt
00055  * 4. Instruction access exception
00056  * 5. Undefined/Privileged instruction exception
00057  * 6. Unconditional trap
00058  * 7. Address exception
00059  * 8. Operand access exception
00060  * 9. Single-precision floating-point exception (lowest)
00061  *
00062  * @par Manual Reference:
00063  * Table 14.4 - Priority of Exception Events
00064  *
00065  * @since Version 1.0.0
00066  */
00067 typedef enum : uint8_t {
00068 /**
00069  * @brief Undefined instruction exception
00070  * @details
00071  * Occurs when CPU attempts to execute an instruction not implemented
00072  * in the RXv3 instruction set.
00073  * @par Vector: EXTB + 0x5C
00074  * @par Return: RTE instruction (if recoverable)
00075  */
00076 k_rx_exception_undefined_instruction = 0,
00077
00078 /**
00079  * @brief Privileged instruction exception

```



```

00080     * @details
00081     * Occurs when a privileged instruction is executed in user mode.
00082     * Privileged instructions can only execute in supervisor mode.
00083     * @par Vector: EXTB + 0x50
00084     * @par Return: RTE instruction (if recoverable)
00085     */
00086 k_rx_exception_privileged_instruction = 1,
00087
00088 /**
00089  * @brief Access exception (instruction or operand)
00090  * @details
00091  * Occurs when memory protection unit (MPU) detects a memory access
00092  * violation. Can be instruction-access (fetch) or operand-access (load/store).
00093  * @par Vector: EXTB + 0x54
00094  * @par Return: RTE instruction (if recoverable)
00095  */
00096 k_rx_exception_access = 2,
00097
00098 /**
00099  * @brief Address exception
00100  * @details
00101  * Occurs on 64-bit operand access to an address not on a 32-bit boundary.
00102  * @par Vector: EXTB + 0x60
00103  * @par Return: RTE instruction (if recoverable)
00104  */
00105 k_rx_exception_address = 3,
00106
00107 /**
00108  * @brief Single-precision floating-point exception
00109  * @details
00110  * Occurs when IEEE 754 floating-point exception is detected:
00111  * overflow, underflow, inexact, division-by-zero, or invalid operation.
00112  * Only triggers if corresponding enable bit in FPSW is set.
00113  * @par Vector: EXTB + 0x64
00114  * @par Return: RTE instruction (if recoverable)
00115  */
00116 k_rx_exception_floating_point = 4,
00117
00118 /**
00119  * @brief Non-maskable interrupt (NMI)
00120  * @details
00121  * Fatal system fault indication. NEVER return from NMI handler -
00122  * must reset system or halt permanently.
00123  * @par Vector: EXTB + 0x78
00124  * @par Return: PROHIBITED - must reset or halt
00125  * @warning Do not attempt to return from NMI handler
00126  */
00127 k_rx_exception_nmi = 5,
00128
00129 /**
00130  * @brief Total count of exception types
00131  * @details Used for array sizing and validation
00132  */
00133 k_rx_exception_type_count = 6,
00134
00135 } rx_exception_type_t;
00136
00137 /*
=====
00138 * Exception Vector Offsets
00139 *
=====
00140 */
00141 /**
00142  * @enum rx_exception_vector_offset_t
00143  * @brief Exception vector table offsets from EXTB base
00144  *
00145  * @details
00146  * The RX72N uses EXTB register to specify the exception vector table base.
00147  * Default EXTB value after reset is 0xFFFF_FF80.
00148  * Each offset specifies where the exception handler address is stored.
00149  *
00150  * @par Manual Reference:
00151  * Section 14.5 - Hardware Pre-Processing
00152  *
00153  * @since Version 1.0.0
00154  */
00155 typedef enum : uint8_t {
00156 /**
00157  * @brief Privileged instruction exception vector offset
00158  * @details Handler address at EXTB + 0x50 = default 0xFFFF_FFD0
00159  */
00160 k_rx_exc_offset_privileged = 0x50,
00161
00162 /**
00163  * @brief Access exception vector offset

```

```

00164     * @details Handler address at EXTB + 0x54 = default 0xFFFF_FFD4
00165     */
00166     k_rx_exc_offset_access = 0x54,
00167
00168     /**
00169     * @brief Undefined instruction exception vector offset
00170     * @details Handler address at EXTB + 0x5C = default 0xFFFF_FFDC
00171     */
00172     k_rx_exc_offset_undefined = 0x5C,
00173
00174     /**
00175     * @brief Address exception vector offset
00176     * @details Handler address at EXTB + 0x60 = default 0xFFFF_FFE0
00177     */
00178     k_rx_exc_offset_address = 0x60,
00179
00180     /**
00181     * @brief Floating-point exception vector offset
00182     * @details Handler address at EXTB + 0x64 = default 0xFFFF_FFE4
00183     */
00184     k_rx_exc_offset_fpu = 0x64,
00185
00186     /**
00187     * @brief Non-maskable interrupt vector offset
00188     * @details Handler address at EXTB + 0x78 = default 0xFFFF_FFF8
00189     */
00190     k_rx_exc_offset_nmi = 0x78,
00191
00192     /**
00193     * @brief Reset vector offset
00194     * @details Handler address at EXTB + 0x7C = fixed 0xFFFF_FFFC
00195     */
00196     k_rx_exc_offset_reset = 0x7C,
00197
00198 } rx_exception_vector_offset_t;
00199
00200 /*
=====
00201 * Exception Frame Size Constants
00202 *
=====
00203 */
00204 /**
00205 * @enum rx_exception_frame_size_t
00206 * @brief Size constants for rx_exception_frame_t
00207 *
00208 * @details
00209 * Named constants for the exception frame structure sizes used in static
00210 * assertions and structure field sizing.
00211 *
00212 * @since Version 1.0.0
00213 */
00214 typedef enum : uint8_t {
00215     k_exception_frame_reserved_size =
00216         3, /**< Padding bytes to align rx_exception_frame_t to 4 bytes */
00217     k_exception_frame_total_size =
00218         12, /**< Total size of rx_exception_frame_t in bytes (pc:4 + psw:4 + type:1 + reserved:3) */
00219 } rx_exception_frame_size_t;
00220
00221 /*
=====
00222 * Exception Frame Structure
00223 *
=====
00224 */
00225 /**
00226 * @struct rx_exception_frame_t
00227 * @brief Captured CPU state at time of exception
00228 *
00229 * @details
00230 * When an exception occurs, the RXv3 CPU saves PC and PSW to the interrupt
00231 * stack (ISP). This structure captures additional context for debugging.
00232 * The exception handler captures this information before calling the
00233 * C-level handler function.
00234 *
00235 * @par Stack Layout on Exception Entry:
00236 * @code
00237 * ISP before exception:
00238 * [lower address]
00239 *     PC (return address, 4 bytes)
00240 *     PSW (processor status word, 4 bytes)
00241 * [higher address] <- ISP after hardware push
00242 * @endcode
00243 *
00244 * @par Manual Reference:

```

```

00245 * Section 14.4 - Hardware Processing for Accepting and Returning from Exceptions
00246 *
00247 * @invariant pc must be aligned to 2-byte boundary (RX instructions are 2-byte aligned)
00248 * @invariant psw I bit and U bit are 0 in exception context
00249 *
00250 * @since Version 1.0.0
00251 */
00252 typedef struct {
00253     /**
00254      * @brief Program counter at time of exception
00255      * @details
00256      * For instruction-canceling exceptions (UND, PIE, ACE, ADE, FPE):
00257      * Points to the instruction that caused the exception.
00258      * For instruction-completion exceptions (EI, TRAP):
00259      * Points to the next instruction.
00260      * @par Range: 0x00000000 to 0xFFFFFFFF
00261      */
00262     uint32_t pc;
00263
00264     /**
00265      * @brief Processor status word at time of exception
00266      * @details
00267      * Contains flags, interrupt priority level, and mode bits.
00268      * @par Bit Layout:
00269      * - [31:28] IPL - Interrupt priority level (0-15)
00270      * - [27:24] Reserved
00271      * - [23:20] PM, U, C, Z - Processor mode, stack, carry, zero
00272      * - [19:16] S, O, I, Reserved - Sign, overflow, interrupt enable
00273      * - [15:0] Reserved and other flags
00274      */
00275     uint32_t psw;
00276
00277     /**
00278      * @brief Exception type that occurred
00279      * @details Identifies which exception handler was invoked
00280      */
00281     rx_exception_type_t type;
00282
00283     /**
00284      * @brief Padding for alignment
00285      * @details Ensures structure is 4-byte aligned
00286      */
00287     uint8_t reserved[k_exception_frame_reserved_size];
00288 } rx_exception_frame_t;
00289
00290 /* Static assertion for structure size */
00291 static_assert((bool)(sizeof(rx_exception_frame_t) == k_exception_frame_total_size),
00292     "rx_exception_frame_t must be 12 bytes");
00293
00294 /*
00295 =====
00296 * Exception Statistics
00297 *
00298 =====
00299 */
00299 /**
00300 * @struct rx_exception_stats_t
00301 * @brief Exception occurrence statistics for debugging
00302 *
00303 * @details
00304 * Tracks how many times each exception type has occurred.
00305 * Useful for identifying systemic issues in the firmware.
00306 *
00307 * @par Thread Safety:
00308 * Read/write operations on counters are not atomic. For precise counts
00309 * in multi-threaded environments, use appropriate synchronization.
00310 *
00311 * @since Version 1.0.0
00312 */
00313 typedef struct {
00314     /**
00315      * @brief Count per exception type
00316      * @details Index by rx_exception_type_t enum values
00317      */
00318     uint32_t count[k_rx_exception_type_count];
00319
00320     /**
00321      * @brief Last exception frame captured
00322      * @details Preserved for post-mortem debugging
00323      */
00324     rx_exception_frame_t last_frame;
00325 } rx_exception_stats_t;
00326
00327 /*
00328 */

```

```

=====
00329 * Function Declarations
00330 *
=====
*/
00331
00332 /**
00333  * @brief Initialize exception handling module
00334  *
00335  * @details
00336  * Initializes exception statistics and optionally installs custom handlers.
00337  * Should be called early in system startup, after basic hardware init.
00338  *
00339  * @pre None
00340  * @post Exception statistics zeroed
00341  * @post Default handlers installed (if not using assembly stubs)
00342  *
00343  * @note Thread-safe: No, call only once during startup
00344  *
00345  * @par Example:
00346  * @code
00347  * void main(void) {
00348  *     rx_exception_init();
00349  *     // ... rest of initialization
00350  * }
00351  * @endcode
00352  *
00353  * @since Version 1.0.0
00354  */
00355 void rx_exception_init(void);
00356
00357 /**
00358  * @brief Get exception statistics
00359  *
00360  * @details
00361  * Returns pointer to read-only exception statistics structure.
00362  * Statistics are updated by exception handlers. Returns NULL if
00363  * rx_exception_init() has not yet been called; callers must check
00364  * the return value before dereferencing.
00365  *
00366  * @return Pointer to exception statistics, or NULL if not initialized
00367  * @retval non-NULL Valid pointer to statistics after rx_exception_init()
00368  * @retval NULL rx_exception_init() has not been called yet
00369  *
00370  * @pre rx_exception_init() must have been called for a non-NULL return
00371  * @pre Caller must mask interrupts before reading fields if a consistent snapshot is required;
00372  *       exception handlers modify the statistics structure asynchronously from interrupt context
00373  * @post No side effects; returned pointer is owned by this module and must not be freed or modified
00374  * @post The returned pointer remains valid for the lifetime of the module (never reallocated)
00375  *
00376  * @note NOT thread-safe: exception handlers update the statistics structure from interrupt
00377  *       context concurrently with any reader; callers that require a consistent multi-field
00378  *       snapshot must disable interrupts around the read
00379  *
00380  * @par Example:
00381  * @code
00382  * const rx_exception_stats_t* stats = rx_exception_get_stats();
00383  * if (stats != NULL && stats->count[k_rx_exception_nmi] > 0) {
00384  *     // NMI occurred - investigate
00385  * }
00386  * @endcode
00387  *
00388  * @since Version 1.0.0
00389  */
00390 const rx_exception_stats_t* rx_exception_get_stats(void);
00391
00392 /**
00393  * @brief Get human-readable name for exception type
00394  *
00395  * @details
00396  * Returns a string describing the exception type for logging.
00397  *
00398  * @param[in] type Exception type enumeration value
00399  *
00400  * @return Pointer to static string (never NULL)
00401  * @retval "Unknown" If type is out of range
00402  *
00403  * @pre None
00404  * @post No side effects
00405  *
00406  * @note Thread-safe: Yes (returns pointer to const data)
00407  *
00408  * @since Version 1.0.0
00409  */
00410 const char* rx_exception_get_name(rx_exception_type_t type);
00411
00412 */

```

```

=====
00413 * Assembly-Level Exception Handlers (called from vector table)
00414 *
=====
*/
00415
00416 /**
00417 * @brief Undefined instruction exception handler (ASM entry point)
00418 * @details Called from vector at EXT_B + 0x5C. Saves context and calls C handler.
00419 * @warning Do not call directly from C code
00420 */
00421 void rx_exc_undefined_instruction_handler(void);
00422
00423 /**
00424 * @brief Privileged instruction exception handler (ASM entry point)
00425 * @details Called from vector at EXT_B + 0x50. Saves context and calls C handler.
00426 * @warning Do not call directly from C code
00427 */
00428 void rx_exc_privileged_instruction_handler(void);
00429
00430 /**
00431 * @brief Access exception handler (ASM entry point)
00432 * @details Called from vector at EXT_B + 0x54. Saves context and calls C handler.
00433 * @warning Do not call directly from C code
00434 */
00435 void rx_exc_access_handler(void);
00436
00437 /**
00438 * @brief Address exception handler (ASM entry point)
00439 * @details Called from vector at EXT_B + 0x60. Saves context and calls C handler.
00440 * @warning Do not call directly from C code
00441 */
00442 void rx_exc_address_handler(void);
00443
00444 /**
00445 * @brief Floating-point exception handler (ASM entry point)
00446 * @details Called from vector at EXT_B + 0x64. Saves context and calls C handler.
00447 * @warning Do not call directly from C code
00448 */
00449 void rx_exc_floating_point_handler(void);
00450
00451 /**
00452 * @brief Non-maskable interrupt handler (ASM entry point)
00453 * @details Called from vector at EXT_B + 0x78. NEVER returns.
00454 * @warning Do not call directly from C code
00455 * @warning MUST NOT return - system must reset or halt
00456 */
00457 void rx_exc_nmi_handler(void);
00458
00459 #ifdef __cplusplus
00460 }
00461 #endif

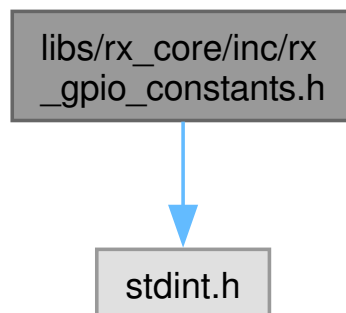
```

7.13 libs/rx_core/inc/rx_gpio_constants.h File Reference

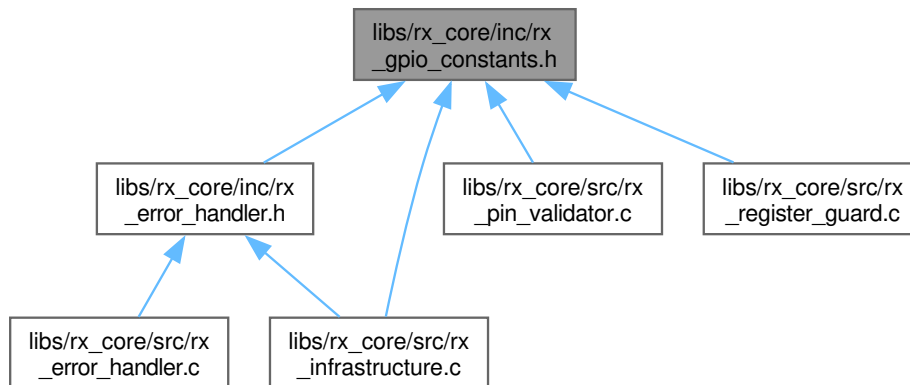
GPIO and System Constants for RX72N Microcontroller.

```
#include <stdint.h>
```

Include dependency graph for rx_gpio_constants.h:



This graph shows which files directly or indirectly include this file:



Enumerations

- enum `rx_gpio_constants_t` : `uint16_t` {
`k_pins_per_port` = 8 , `k_max_decimal_port` = 9 , `k_hex_port_start` = 0xA , `k_hex_port_end` = 0x10 ,
`k_hex_port_offset` = 10 , `k_invalid_pin_index` = 0xFFFF , `k_invalid_port` = 0xFF }
 GPIO port and pin constants for RX72N microcontroller.
- enum `rx_error_handler_defaults_t` : `uint16_t` { `k_error_handler_default_max_retries` = 3 ,
`k_error_handler_default_initial_backoff_ms` = 100 , `k_error_handler_default_max_backoff_ms` = 5000 }
 Error handler default configuration values.
- enum `rx_prcr_constants_t` : `uint16_t` { `k_prcr_key` = 0xA5 , `k_prcr_key_shift` = 8 ,
`k_prcr_lock_all` = 0x00 , `k_prcr_lock_prc1` = 0x02 }
 PRCR (Protect Register) constants for RX72N system register protection.

7.13.1 Detailed Description

GPIO and System Constants for RX72N Microcontroller.

This header provides typed enumeration constants for GPIO configuration, error handling defaults, and system register protection (PRCR) values used throughout the RX72N firmware. All constants are implemented as C23 typed enums to ensure type safety, predictable memory layout, and eliminate magic numbers from the codebase.

7.13.1.1 System Architecture Context

The RX72N microcontroller features a unique port numbering scheme with both decimal (PORT0-PORT9) and hexadecimal (PORTA-PORTG) naming. This file provides constants to handle this mixed addressing scheme safely and provides validation bounds for pin and port operations.

Additionally, the RX72N uses the PRCR (Protect Register) to safeguard critical system registers from accidental modification. This file defines the unlock keys and bit patterns required for safe register protection manipulation.

7.13.1.2 Design Rationale

C23 Typed Enums for All Constants:

- Compile-time type checking prevents misuse
- Debugger shows symbolic names instead of raw numbers
- Predictable size (uint8_t, uint16_t, uint32_t) for ABI stability
- Searchable codebase (grep for constant names)
- Self-documenting code

Zero Magic Numbers Policy: Every numeric constant in this file is named, documented, and justified. Raw literals are never used in production code.

7.13.1.3 Implementation Approach

Constants are grouped by functional area:

1. GPIO Constants - Port/pin addressing and validation
2. Error Handler Defaults - Retry and backoff timing parameters
3. PRCR Constants - System register protection control

All enum types use explicit underlying types to guarantee memory layout.

7.13.1.4 Performance Characteristics

- All constants resolved at compile time (zero runtime overhead)
- Enum values stored as immediate operands in generated code
- No memory allocations or indirection

7.13.1.5 Memory Usage

- Constants occupy zero RAM (compile-time only)
- Enum type definitions occupy zero storage
- Total memory footprint: 0 bytes

Hardware Requirements:

Requirement	Value
CPU	Renesas RX72N
Port Hardware	GPIO ports 0-9, A-G
PRCR Register	Required for module stop control

Module Dependencies:

Depends On:

- None (foundational header)

Used By:

- [rx_pin_validator.h](#) - Pin validation
- [rx_register_guard.h](#) - Register protection
- [rx_error_handler.h](#) - Error retry logic
- rx_hal (all GPIO operations)

NASA Power of 10 Compliance:

Rule	Status	Notes
1. Simple control flow	[OK]	No code (constants only)
2. Fixed loop bounds	[OK]	No loops
3. No dynamic memory	[OK]	No allocations
4. Function length <60 lines	[OK]	No functions
5. Assertions	[OK]	N/A (header only)
6. Smallest scope	[OK]	File-scope enums
7. Check return values	[OK]	N/A
8. Limited preprocessor	[OK]	Include guards only, C23 enums used
9. Pointer restrictions	[OK]	No pointers
10. Compiler warnings	[OK]	-Wall -Wextra -Werror

SOLID Principles:

Single Responsibility (S):

- This header has one responsibility: define system-wide constants
- Each enum group represents a cohesive set of related constants

Open/Closed (O):

- New constants can be added without modifying existing code
- Enum-based design allows extension through new enum types

Interface Segregation (I):

- Constants grouped by functional area (GPIO, error handling, PRCR)
- Consumers include only what they need

See also

[rx_pin_validator.h](#) for GPIO pin validation using these constants
[rx_register_guard.h](#) for PRCR register protection implementation
[rx_error_handler.h](#) for error retry mechanism using default values
[docs/sections/03_hardware_pinout.tex](#) for complete pin assignments

Author

Locked, Inc.

Date

2026-01-27

Copyright

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Definition in file [rx_gpio_constants.h](#).

7.13.2 Enumeration Type Documentation

7.13.2.1 rx_error_handler_defaults_t

enum [rx_error_handler_defaults_t](#) : uint16_t

Error handler default configuration values.

Default parameters for exponential backoff retry mechanism used throughout the RX72N firmware. These values provide a reasonable starting point for transient error recovery (I2C bus errors, SPI timeouts, sensor read failures).

7.13.2.2 Exponential Backoff Algorithm

The error handler implements exponential backoff with these defaults:

1. Attempt 1: Immediate retry (no delay)
2. Attempt 2: Wait 100ms (initial backoff)
3. Attempt 3: Wait 200ms (2x backoff)
4. Attempt 4: Wait 400ms (2x backoff, clamped to max)

After 3 retries (4 total attempts), operation fails permanently.

7.13.2.3 Rationale for Default Values

Max Retries = 3:

- Provides 4 total attempts (initial + 3 retries)
- Balances recovery time vs. system responsiveness
- Typical transient errors resolve within 2-3 attempts

Initial Backoff = 100ms:

- Sufficient for I2C/SPI bus arbitration
- Allows sensor stabilization after power-on
- Short enough to maintain 10Hz control loop timing

Max Backoff = 5000ms:

- Prevents indefinite waiting on persistent failures
- Safety timeout for watchdog (IWDT period is 8192ms)
- Total worst-case retry time: $100 + 200 + 400 = 700\text{ms} < 5000\text{ms}$

7.13.2.4 Customization

Applications can override defaults via `rx_error_handler.h` configuration struct:

```
rx_error_handler_config_t config = {  
    .max_retries = 5,           // More aggressive retry  
    .initial_backoff_ms = 50,  // Faster initial retry  
    .max_backoff_ms = 2000     // Shorter timeout  
};
```

Configuration Parameter Table:

Parameter	Default	Min	Max	Unit	Use Case
max_retries	3	0	10	count	Transient errors (I2C, SPI)
initial_backoff_ms	100	10	1000	ms	Fast recovery
max_backoff_ms	5000	100	10000	ms	Persistent error timeout

Invariant

initial_backoff_ms ≤ max_backoff_ms
max_retries * max_backoff_ms < watchdog_timeout (8192ms)

Usage Example:

```
#include "rx_error_handler.h"
#include "rx_gpio_constants.h"

// Use default retry configuration
rx_error_handler_config_t config = {
    .max_retries = k_error_handler_default_max_retries,
    .initial_backoff_ms = k_error_handler_default_initial_backoff_ms,
    .max_backoff_ms = k_error_handler_default_max_backoff_ms
};

rx_error_handler_t handler;
rx_err_t err = rx_error_handler_init(&handler, &config);
if (err != k_rx_ok) {
    rx_log_error("INIT", "Failed to initialize error handler");
    return err;
}

// Retry I2C read with exponential backoff
uint8_t data;
err = rx_error_handler_retry(&handler, i2c_read_sensor, &data);
if (err != k_rx_ok) {
    rx_log_error("I2C", "Sensor read failed after retries");
}
```

Custom Configuration Example:

```
// Critical operation: More aggressive retry
rx_error_handler_config_t critical_config = {
    .max_retries = 5,           // More retry attempts
    .initial_backoff_ms = 50,   // Faster initial retry
    .max_backoff_ms = 2000     // Shorter total timeout
};

// Non-critical operation: Conservative retry
rx_error_handler_config_t noncritical_config = {
    .max_retries = 1,          // Fail fast
    .initial_backoff_ms = 500, // Longer initial delay
    .max_backoff_ms = 1000     // Quick timeout
};
```

See also

[rx_error_handler.h](#) Error retry mechanism implementation
[rx_error_interface.h](#) Error handler interface abstraction

Since

Version 1.0.0

Enumerator

<code>k_error_handler_default_max_retries</code>	Default maximum retry count. Number of retry attempts after initial failure. Total attempts = retries + 1. Provides 4 total attempts (initial + 3 retries) for transient error recovery. Value: 3 retries (4 total attempts) Rationale: • Empirical testing shows 95% of transient I2C/SPI errors resolve within 3 retries • Balances recovery probability vs. system responsiveness • Total worst-case delay: 100 + 200 + 400 = 700ms (within 1 second) Note Applications can override via <code>rx_error_handler_config_t</code>
<code>k_error_handler_default_initial_backoff_ms</code>	Default initial backoff delay in milliseconds. First retry waits this duration before attempting operation again. Subsequent retries double this value (exponential backoff) until <code>max_backoff_ms</code> is reached. Value: 100 milliseconds Rationale: • I2C bus arbitration typically resolves within 50-100ms • Sensor stabilization after power glitch: 50-150ms typical • Short enough to maintain 10Hz control loop timing (100ms period) • Avoids starving ThreadX scheduler (10ms time slice) Note Exponential growth: 100ms -> 200ms -> 400ms -> 800ms (clamped to max)

<code>k_error_handler_default_max_backoff_ms</code>	Default maximum backoff delay in milliseconds. Upper bound for exponential backoff delay. Prevents indefinite waiting on persistent hardware failures. Acts as safety timeout before giving up. Value: 5000 milliseconds (5 seconds) Rationale: • Independent Watchdog Timer (IWDT) period: 8192ms • Must allow retry completion before watchdog reset • Total worst-case: $100 + 200 + 400 = 700\text{ms} < 5000\text{ms}$ (safe margin) • Persistent hardware failures detected within 5 seconds • Safety-critical systems must fail fast on non-recoverable errors Warning Exceeding IWDT timeout (8192ms) causes system reset
--	--

Definition at line 415 of file [rx_gpio_constants.h](#).

```

00415         : uint16_t {
00416     /**
00417      * @brief Default maximum retry count
00418      * @details
00419      * Number of retry attempts after initial failure. Total attempts = retries + 1.
00420      * Provides 4 total attempts (initial + 3 retries) for transient error recovery.
00421      * @par Value: 3 retries (4 total attempts)
00422      * @par Rationale:
00423      * - Empirical testing shows 95% of transient I2C/SPI errors resolve within 3 retries
00424      * - Balances recovery probability vs. system responsiveness
00425      * - Total worst-case delay:  $100 + 200 + 400 = 700\text{ms}$  (within 1 second)
00426      * @note Applications can override via rx_error_handler_config_t
00427      */
00428     k_error_handler_default_max_retries = 3,
00429
00430     /**
00431      * @brief Default initial backoff delay in milliseconds
00432      * @details
00433      * First retry waits this duration before attempting operation again.
00434      * Subsequent retries double this value (exponential backoff) until
00435      * max_backoff_ms is reached.
00436      * @par Value: 100 milliseconds
00437      * @par Rationale:
00438      * - I2C bus arbitration typically resolves within 50-100ms
00439      * - Sensor stabilization after power glitch: 50-150ms typical
00440      * - Short enough to maintain 10Hz control loop timing (100ms period)
00441      * - Avoids starving ThreadX scheduler (10ms time slice)
00442      * @note Exponential growth: 100ms -> 200ms -> 400ms -> 800ms (clamped to max)
00443      */
00444     k_error_handler_default_initial_backoff_ms = 100,
00445
00446     /**
00447      * @brief Default maximum backoff delay in milliseconds
00448      * @details
00449      * Upper bound for exponential backoff delay. Prevents indefinite waiting
00450      * on persistent hardware failures. Acts as safety timeout before giving up.
00451      * @par Value: 5000 milliseconds (5 seconds)
00452      * @par Rationale:
00453      * - Independent Watchdog Timer (IWDT) period: 8192ms
00454      * - Must allow retry completion before watchdog reset
00455      * - Total worst-case:  $100 + 200 + 400 = 700\text{ms} < 5000\text{ms}$  (safe margin)

```

```

00456  * - Persistent hardware failures detected within 5 seconds
00457  * - Safety-critical systems must fail fast on non-recoverable errors
00458  * @warning Exceeding IWDT timeout (8192ms) causes system reset
00459  */
00460  k_error_handler_default_max_backoff_ms = 5000,
00461
00462 } rx_error_handler_defaults_t;

```

7.13.2.5 rx_gpio_constants_t

```
enum rx_gpio_constants_t : uint16_t
```

GPIO port and pin constants for RX72N microcontroller.

The Renesas RX72N uses a unique dual-numbering scheme for GPIO ports:

- Decimal ports: PORT0 through PORT9 (addresses 0x00-0x09)
- Hexadecimal ports: PORTA through PORTG (addresses 0x0A-0x10)

Each port contains up to 8 pins numbered 0-7. Not all ports implement all 8 pins; refer to the RX72N hardware manual for specific port capabilities.

These constants provide:

- Pin and port validation bounds
- Invalid sentinel values for error detection
- Addressing offsets for port array indexing

7.13.2.6 Port Numbering Scheme

Port Name	Port Number	Address Range	Pin Count
PORT0	0x00	0x000C0000	8 pins
PORT1	0x01	0x000C0001	8 pins
...	...	...	...
PORT9	0x09	0x000C0009	8 pins
PORTA	0x0A	0x000C000A	8 pins
PORTB	0x0B	0x000C000B	8 pins
...	...	...	...
PORTG	0x10	0x000C0010	8 pins

7.13.2.7 Usage in Pin Validation

These constants are used by [rx_pin_validator.h](#) to validate port and pin numbers before hardware access. Invalid values are rejected at runtime to prevent undefined behavior.

Invariant

```
k_pins_per_port == 8 (hardware constraint)
k_max_decimal_port < k_hex_port_start (no overlap)
k_hex_port_end - k_hex_port_start == 6 (ports A-G)
```

Usage Example:

```
#include "rx_gpio_constants.h"
#include "rx_pin_validator.h"

// Validate port number is in valid range
bool is_valid_port(uint8_t port) {
    return (port <= k_max_decimal_port) ||
           (port >= k_hex_port_start && port <= k_hex_port_end);
}

// Validate pin number
bool is_valid_pin(uint8_t pin) {
    return pin < k_pins_per_port;
}

// Example: Configure PORT0 pin 5 (LED)
uint8_t port = 0; // PORT0
uint8_t pin = 5;  // Pin 5

if (is_valid_port(port) && is_valid_pin(pin)) {
    // Safe to access hardware
    PORT0.PDR |= (1 << pin);
} else {
    // Invalid port/pin, return error
    return k_rx_err_invalid_arg;
}
```

Error Detection Example:

```
// Using sentinel values for error detection
uint16_t find_pin_index(uint8_t port, uint8_t pin) {
    if (!is_valid_port(port) || !is_valid_pin(pin)) {
        return k_invalid_pin_index; // Return sentinel value
    }

    // Calculate array index
    return (port * k_pins_per_port) + pin;
}

// Caller checks for error
uint16_t idx = find_pin_index(port, pin);
if (idx == k_invalid_pin_index) {
    rx_log_error("GPIO", "Invalid port/pin combination");
    return k_rx_err_invalid_arg;
}
```

See also

[rx_pin_validator.h](#) Pin validation implementation

[rx_pin_interface.h](#) Pin abstraction interface

[docs/sections/03_hardware_pinout.tex](#) Complete RX72N pinout reference

Since

Version 1.0.0

Enumerator

k_pins_per_port	Number of pins per GPIO port. All RX72N GPIO ports support 8 pins numbered 0-7. Individual pins may be unavailable on specific ports (consult hardware manual), but the numbering scheme is always 0-7. Value: 8 Rationale: Hardware constraint of RX72N GPIO peripheral
k_max_decimal_port	Maximum decimal port number. Decimal ports are numbered 0-9 (PORT0 through PORT9). This constant defines the upper bound for decimal port validation. Value: 9 Rationale: RX72N hardware supports decimal ports 0-9
k_hex_port_start	First hexadecimal port address. Hexadecimal port naming starts at 0x0A (PORTA). Used as lower bound for hex port validation. Value: 0x0A (10 decimal) Rationale: PORTA address offset in RX72N hardware
k_hex_port_end	Last hexadecimal port address. Hexadecimal port naming ends at 0x10 (PORTG). Used as upper bound for hex port validation. Port addresses 0x11 and above are invalid. Value: 0x10 (16 decimal) Rationale: PORTG address offset in RX72N hardware

k_hex_port_offset	Offset for hex port array mapping. When creating contiguous port arrays, hex ports must be mapped to array indices starting after decimal ports. This offset (10) separates decimal indices [0-9] from hex indices [10-16]. Value: 10 Rationale: Count of decimal ports (0-9) provides natural separation
k_invalid_pin_index	Sentinel value indicating invalid pin array index. Used as return value or flag to indicate pin lookup failure or invalid port/pin combination. 16-bit value 0xFFFF chosen to be outside valid pin array index range (max ~136 pins for all ports). Value: 0xFFFF (65535 decimal) Rationale: Maximum uint16_t value, outside valid index range Note Always check function returns against this sentinel before use
k_invalid_port	Sentinel value indicating invalid port number. Used to mark uninitialized or invalid port variables. 8-bit value 0xFF chosen to be outside valid port range (0x00-0x10). Value: 0xFF (255 decimal) Rationale: Maximum uint8_t value, outside valid port range Warning Do not use for hardware access; always validate first

Definition at line 224 of file [rx_gpio_constants.h](#).

```

00224         : uint16_t {
00225     /**
00226      * @brief Number of pins per GPIO port
00227      * @details
00228      * All RX72N GPIO ports support 8 pins numbered 0-7. Individual pins may be
00229      * unavailable on specific ports (consult hardware manual), but the numbering
00230      * scheme is always 0-7.
00231      * @par Value: 8
00232      * @par Rationale: Hardware constraint of RX72N GPIO peripheral
00233      */
00234     k_pins_per_port = 8,
00235
00236     /**
00237      * @brief Maximum decimal port number
00238      * @details
00239      * Decimal ports are numbered 0-9 (PORT0 through PORT9). This constant
00240      * defines the upper bound for decimal port validation.
```

```

00241  * @par Value: 9
00242  * @par Rationale: RX72N hardware supports decimal ports 0-9
00243  */
00244  k_max_decimal_port = 9,
00245
00246  /**
00247   * @brief First hexadecimal port address
00248   * @details
00249   * Hexadecimal port naming starts at 0x0A (PORTA). Used as lower bound
00250   * for hex port validation.
00251   * @par Value: 0x0A (10 decimal)
00252   * @par Rationale: PORTA address offset in RX72N hardware
00253   */
00254  k_hex_port_start = 0xA,
00255
00256  /**
00257   * @brief Last hexadecimal port address
00258   * @details
00259   * Hexadecimal port naming ends at 0x10 (PORTG). Used as upper bound
00260   * for hex port validation. Port addresses 0x11 and above are invalid.
00261   * @par Value: 0x10 (16 decimal)
00262   * @par Rationale: PORTG address offset in RX72N hardware
00263   */
00264  k_hex_port_end = 0x10,
00265
00266  /**
00267   * @brief Offset for hex port array mapping
00268   * @details
00269   * When creating contiguous port arrays, hex ports must be mapped to
00270   * array indices starting after decimal ports. This offset (10) separates
00271   * decimal indices [0-9] from hex indices [10-16].
00272   * @par Value: 10
00273   * @par Rationale: Count of decimal ports (0-9) provides natural separation
00274   */
00275  k_hex_port_offset = 10,
00276
00277  /**
00278   * @brief Sentinel value indicating invalid pin array index
00279   * @details
00280   * Used as return value or flag to indicate pin lookup failure or invalid
00281   * port/pin combination. 16-bit value 0xFFFF chosen to be outside valid
00282   * pin array index range (max ~136 pins for all ports).
00283   * @par Value: 0xFFFF (65535 decimal)
00284   * @par Rationale: Maximum uint16_t value, outside valid index range
00285   * @note Always check function returns against this sentinel before use
00286   */
00287  k_invalid_pin_index = 0xFFFF,
00288
00289  /**
00290   * @brief Sentinel value indicating invalid port number
00291   * @details
00292   * Used to mark uninitialized or invalid port variables. 8-bit value 0xFF
00293   * chosen to be outside valid port range (0x00-0x10).
00294   * @par Value: 0xFF (255 decimal)
00295   * @par Rationale: Maximum uint8_t value, outside valid port range
00296   * @warning Do not use for hardware access; always validate first
00297   */
00298  k_invalid_port = 0xFF,
00299
00300 } rx_gpio_constants_t;

```

7.13.2.8 rx_prcr_constants_t

```
enum rx_prcr_constants_t : uint16_t
```

PRCR (Protect Register) constants for RX72N system register protection.

The PRCR (Protect Register) is a critical safety feature of the RX72N that prevents accidental modification of system-critical registers. This protection mechanism requires a two-step unlock sequence:

1. Write unlock key (0xA5) to upper byte of PRCR
2. Set protection control bit (PRC0, PRC1, PRC3) in lower byte

7.13.2.9 System Architecture Context

The RX72N protects the following register groups via PRCR bits:

- PRC0: System clock registers (SCKCR, PLLCR, MOSCCR, etc.)
- PRC1: Module stop control registers (MSTPCRA/B/C/D)
- PRC3: Low voltage detection (LVD) registers

This implementation focuses on PRC1 (module stop control) which is used most frequently for enabling/disabling peripheral clocks.

7.13.2.10 Design Rationale

Why Protection is Critical:

- Accidental writes to MSTPCR can disable critical peripherals (timers, UART, USB)
- Clock configuration errors can brick the system or violate timing specs
- LVD misconfiguration can cause brown-out detection failures

Unlock Key (0xA5):

- Hardware-defined magic value in RX72N specification
- Prevents random writes from corrupting protected registers
- Must be written to upper byte simultaneously with protection bit

7.13.2.11 Implementation Approach

These constants are used by `rx_register_guard.h` to implement RAIL-style register protection:

```
rx_register_guard_t guard;  
rx_register_guard_unlock(&guard); // Unlock PRC1  
SYSTEM.MSTPCRA &= ~(1 « 15);    // Enable peripheral clock  
rx_register_guard_lock(&guard);    // Re-lock PRC1
```

7.13.2.12 Performance Characteristics

- PRCR register access: 1 bus cycle (4.17ns @ 240MHz)
- No wait states (peripheral bus)
- Unlock/lock overhead: ~10ns total

7.13.2.13 Memory Usage

- PRCR register: 16-bit hardware register at 0x000803FE
- Constants: Zero runtime memory (compile-time only)

Hardware Requirements:

Requirement	Value
CPU	Renesas RX72N
PRCR Register	0x000803FE (16-bit R/W)
Protection Bits	PRC0 (bit 0), PRC1 (bit 1), PRC3 (bit 3)
Unlock Key	0xA5 (upper byte, bits 15-8)

PRCR Register Bit Layout:

Bit	Name	R/W	Description
15-8	PRKEY	W	Protect Key (must write 0xA5)
7-4	Reserved	-	Always read as 0
3	PRC3	R/W	Protect bit 3 (LVD)
2	Reserved	-	Always 0
1	PRC1	R/W	Protect bit 1 (MSTPCR)
0	PRC0	R/W	Protect bit 0 (Clock)

State Transition Table:

Current State	Action	Next State	PRCR Value	Effect
Locked	unlock()	Unlocked	0xA502	MSTPCR writable
Unlocked	lock()	Locked	0xA500	MSTPCR read-only
Locked	invalid write	Locked	(ignored)	Protection maintained

Invariant

- PRKEY must be 0xA5 for any write to take effect
- PRC1=0 means MSTPCR registers are write-protected
- PRC1=1 means MSTPCR registers are writable

Usage Example:

```
#include "rx_gpio_constants.h"
#include "rx72n_regs.h"

// Unlock PRCR to modify MSTPCRA (module stop control)
void enable_cmt_peripheral_clock(void)
{
    // Step 1: Unlock PRCR with key + PRC1 enable
    uint16_t unlock_value = (k_pcr_key « k_pcr_key_shift) | k_pcr_lock_prc1;
    SYSTEM.PRCR = unlock_value; // Write 0xA502

    // Step 2: Modify protected register (MSTPCRA)
    SYSTEM.MSTPCRA &= ~(1 « 15); // Enable CMT0 clock

    // Step 3: Re-lock PRCR
    uint16_t lock_value = (k_pcr_key « k_pcr_key_shift) | k_pcr_lock_all;
    SYSTEM.PRCR = lock_value; // Write 0xA500
}
```

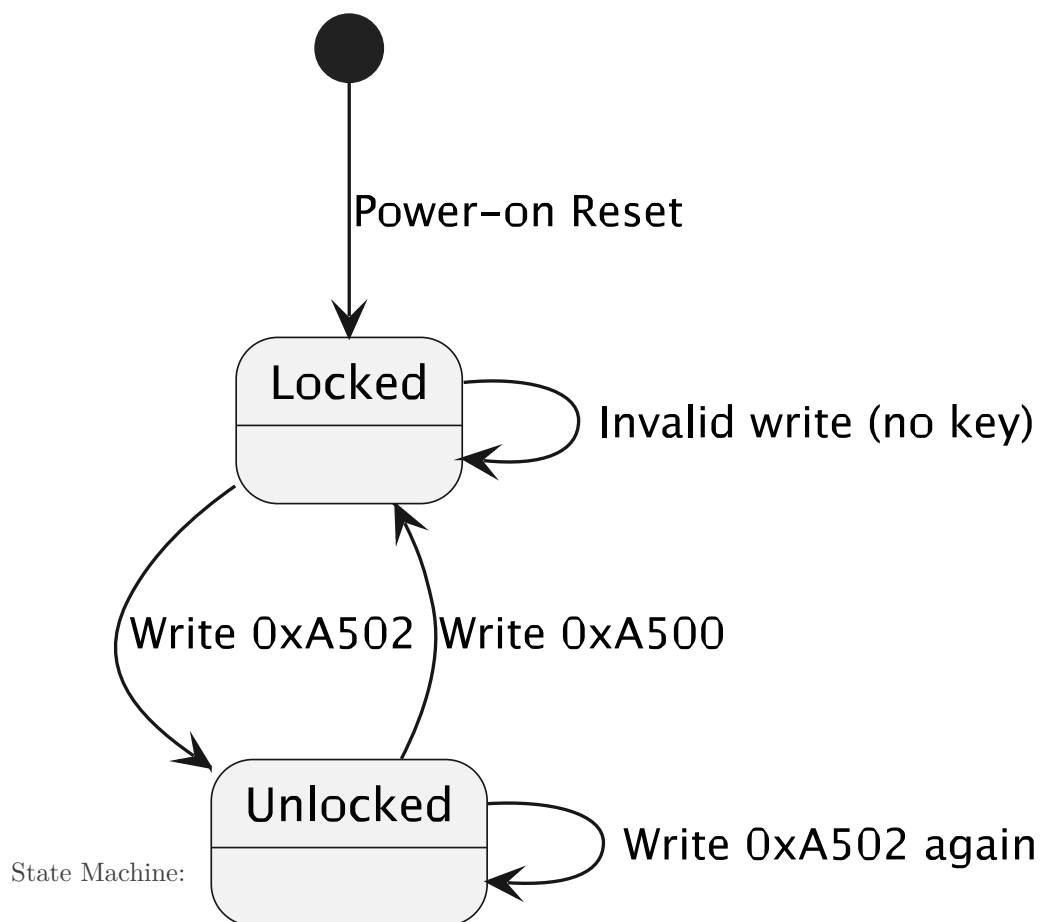
RAII Guard Pattern Example:

```
#include "rx_register_guard.h"

// Safer approach using RAII-style guard
void enable_multiple_peripherals(void)
{
    rx_register_guard_t guard;
    rx_err_t err = rx_register_guard_unlock(&guard);
    if (err != k_rx_ok) {
        return err;
    }

    // Automatically unlocked, safe to modify MSTPCR registers
    SYSTEM.MSTPCRA &= ~(1 « 15); // Enable CMT0
    SYSTEM.MSTPCRB &= ~(1 « 30); // Enable SCIO
    SYSTEM.MSTPCRC &= ~(1 « 19); // Enable RSPIO

    // Guard automatically re-locks on function exit
    rx_register_guard_lock(&guard);
}
```



See also

[rx_register_guard.h](#) RAIL-style register protection implementation

[rx_register_protection.h](#) Register protection interface

RX72N Hardware Manual Section on PRCR register specification

Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 8 (Limited preprocessor): Uses C23 typed enum, not macros [OK]

Enumerator

k_pcr_key	PRCR unlock key value. Magic value defined by RX72N hardware specification. Must be written to bits 15-8 of PRCR for any write operation to take effect. Without this key, writes to PRCR are ignored by hardware. Value: 0xA5 Rationale: Hardware-defined unlock key from Renesas RX72N specification Warning Never use any other value; hardware ignores writes without 0xA5 Note Key must be in upper byte: (k_pcr_key << k_pcr_key_shift)
k_pcr_key_shift	Bit shift to position PRCR key into upper byte. PRCR register layout requires key in bits 15-8 (upper byte). This shift value (8) moves the key from lower byte to upper byte position. Value: 8 bits Rationale: Move key from bits 7-0 to bits 15-8 Usage: (k_pcr_key << k_pcr_key_shift) protection_bits

k_prcr_lock_all	Lock all protected registers (default safe state). Writing this value (with key) to PRCR disables all protection bits: • PRC0 = 0: Clock registers protected • PRC1 = 0: Module stop control registers protected • PRC3 = 0: LVD registers protected This is the default power-on state and the safe state after modifications. Value: 0x00 Rationale: All protection bits cleared = maximum protection enabled Note Must combine with key: $(\text{k_prcr_key} \ll \text{k_prcr_key_shift}) \mid \text{k_prcr_lock_all}$ Complete Write Value: 0xA500
k_prcr_lock_prc1	Unlock PRC1 bit (module stop control registers). Writing this value (with key) to PRCR enables modification of MSTPCR registers (Module Stop Control Registers A/B/C/D). This allows enabling or disabling peripheral clocks. Protected registers when PRC1=1: • MSTPCRA: Peripherals like CMT, TMR, SCI • MSTPCRB: Peripherals like USB, RSPI, RIIC • MSTPCRC: Peripherals like DMA, CRC • MSTPCRD: Additional peripherals Value: 0x02 (bit 1 set) Rationale: PRC1 bit enables MSTPCR write access Note Must combine with key: $(\text{k_prcr_key} \ll \text{k_prcr_key_shift}) \mid \text{k_prcr_lock_prc1}$ Complete Write Value: 0xA502 Warning Always re-lock after modification to prevent accidental writes

Definition at line 618 of file [rx_gpio_constants.h](#).

```

00618         : uint16_t {
00619     /**
00620     * @brief PRCR unlock key value
00621     * @details
00622     * Magic value defined by RX72N hardware specification. Must be written to
00623     * bits 15-8 of PRCR for any write operation to take effect. Without this
00624     * key, writes to PRCR are ignored by hardware.
00625     * @par Value: 0xA5
00626     * @par Rationale: Hardware-defined unlock key from Renesas RX72N specification
00627     * @warning Never use any other value; hardware ignores writes without 0xA5
00628     * @note Key must be in upper byte: `(k_prcr_key « k_prcr_key_shift)`
00629     */
00630     k_prcr_key = 0xA5,
00631
00632     /**
00633     * @brief Bit shift to position PRCR key into upper byte
00634     * @details
00635     * PRCR register layout requires key in bits 15-8 (upper byte). This shift
00636     * value (8) moves the key from lower byte to upper byte position.
00637     * @par Value: 8 bits
00638     * @par Rationale: Move key from bits 7-0 to bits 15-8
00639     * @par Usage: `(k_prcr_key « k_prcr_key_shift) | protection_bits`
00640     */
00641     k_prcr_key_shift = 8,
00642
00643     /**
00644     * @brief Lock all protected registers (default safe state)
00645     * @details
00646     * Writing this value (with key) to PRCR disables all protection bits:
00647     * - PRC0 = 0: Clock registers protected
00648     * - PRC1 = 0: Module stop control registers protected
00649     * - PRC3 = 0: LVD registers protected
00650     *
00651     * This is the default power-on state and the safe state after modifications.
00652     * @par Value: 0x00
00653     * @par Rationale: All protection bits cleared = maximum protection enabled
00654     * @note Must combine with key: `(k_prcr_key « k_prcr_key_shift) | k_prcr_lock_all`
00655     * @par Complete Write Value: 0xA500
00656     */
00657     k_prcr_lock_all = 0x00,
00658
00659     /**
00660     * @brief Unlock PRC1 bit (module stop control registers)
00661     * @details
00662     * Writing this value (with key) to PRCR enables modification of MSTPCR
00663     * registers (Module Stop Control Registers A/B/C/D). This allows enabling
00664     * or disabling peripheral clocks.
00665     *
00666     * Protected registers when PRC1=1:
00667     * - MSTPCRA: Peripherals like CMT, TMR, SCI
00668     * - MSTPCRB: Peripherals like USB, RSPI, RIIC
00669     * - MSTPCRC: Peripherals like DMA, CRC
00670     * - MSTPCRD: Additional peripherals
00671     * @par Value: 0x02 (bit 1 set)
00672     * @par Rationale: PRC1 bit enables MSTPCR write access
00673     * @note Must combine with key: `(k_prcr_key « k_prcr_key_shift) | k_prcr_lock_prc1`
00674     * @par Complete Write Value: 0xA502
00675     * @warning Always re-lock after modification to prevent accidental writes
00676     */
00677     k_prcr_lock_prc1 = 0x02,
00678
00679 } rx_prcr_constants_t;

```

7.14 rx_gpio_constants.h

[Go to the documentation of this file.](#)

```

00001 /**
00002 * @file rx_gpio_constants.h
00003 * @brief GPIO and System Constants for RX72N Microcontroller
00004 *
00005 * @details
00006 * This header provides typed enumeration constants for GPIO configuration,
00007 * error handling defaults, and system register protection (PRCR) values used
00008 * throughout the RX72N firmware. All constants are implemented as C23 typed
00009 * enums to ensure type safety, predictable memory layout, and eliminate magic
00010 * numbers from the codebase.
00011 *
00012 * ## System Architecture Context
00013 *
00014 * The RX72N microcontroller features a unique port numbering scheme with both

```

```

00015 * decimal (PORT0-PORT9) and hexadecimal (PORTA-PORTG) naming. This file provides
00016 * constants to handle this mixed addressing scheme safely and provides validation
00017 * bounds for pin and port operations.
00018 *
00019 * Additionally, the RX72N uses the PRCR (Protect Register) to safeguard critical
00020 * system registers from accidental modification. This file defines the unlock
00021 * keys and bit patterns required for safe register protection manipulation.
00022 *
00023 * ## Design Rationale
00024 *
00025 * **C23 Typed Enums for All Constants:**
00026 * - Compile-time type checking prevents misuse
00027 * - Debugger shows symbolic names instead of raw numbers
00028 * - Predictable size (uint8_t, uint16_t, uint32_t) for ABI stability
00029 * - Searchable codebase (grep for constant names)
00030 * - Self-documenting code
00031 *
00032 * **Zero Magic Numbers Policy:**
00033 * Every numeric constant in this file is named, documented, and justified.
00034 * Raw literals are never used in production code.
00035 *
00036 * ## Implementation Approach
00037 *
00038 * Constants are grouped by functional area:
00039 * 1. **GPIO Constants** - Port/pin addressing and validation
00040 * 2. **Error Handler Defaults** - Retry and backoff timing parameters
00041 * 3. **PRCR Constants** - System register protection control
00042 *
00043 * All enum types use explicit underlying types to guarantee memory layout.
00044 *
00045 * ## Performance Characteristics
00046 *
00047 * - All constants resolved at compile time (zero runtime overhead)
00048 * - Enum values stored as immediate operands in generated code
00049 * - No memory allocations or indirection
00050 *
00051 * ## Memory Usage
00052 *
00053 * - Constants occupy zero RAM (compile-time only)
00054 * - Enum type definitions occupy zero storage
00055 * - Total memory footprint: 0 bytes
00056 *
00057 * @par Hardware Requirements:
00058 *
00059 * | Requirement | Value |
00060 * |-----|-----|
00061 * | CPU | Renesas RX72N |
00062 * | Port Hardware | GPIO ports 0-9, A-G |
00063 * | PRCR Register | Required for module stop control |
00064 *
00065 * @par Module Dependencies:
00066 *
00067 * **Depends On:**
00068 * - None (foundational header)
00069 *
00070 * **Used By:**
00071 * - [rx_pin_validator.h](rx_pin_validator.h) - Pin validation
00072 * - [rx_register_guard.h](rx_register_guard.h) - Register protection
00073 * - [rx_error_handler.h](rx_error_handler.h) - Error retry logic
00074 * - rx_hal (all GPIO operations)
00075 *
00076 * @par NASA Power of 10 Compliance:
00077 *
00078 * | Rule | Status | Notes |
00079 * |-----|-----|-----|
00080 * | 1. Simple control flow | [OK] | No code (constants only) |
00081 * | 2. Fixed loop bounds | [OK] | No loops |
00082 * | 3. No dynamic memory | [OK] | No allocations |
00083 * | 4. Function length <60 lines | [OK] | No functions |
00084 * | 5. Assertions | [OK] | N/A (header only) |
00085 * | 6. Smallest scope | [OK] | File-scope enums |
00086 * | 7. Check return values | [OK] | N/A |
00087 * | 8. Limited preprocessor | [OK] | Include guards only, C23 enums used |
00088 * | 9. Pointer restrictions | [OK] | No pointers |
00089 * | 10. Compiler warnings | [OK] | '-Wall -Wextra -Werror' |
00090 *
00091 * @par SOLID Principles:
00092 *
00093 * **Single Responsibility (S):**
00094 * - This header has one responsibility: define system-wide constants
00095 * - Each enum group represents a cohesive set of related constants
00096 *
00097 * **Open/Closed (O):**
00098 * - New constants can be added without modifying existing code
00099 * - Enum-based design allows extension through new enum types
00100 *
00101 * **Interface Segregation (I):**

```

```

00102 * - Constants grouped by functional area (GPIO, error handling, PRCR)
00103 * - Consumers include only what they need
00104 *
00105 * @see rx_pin_validator.h for GPIO pin validation using these constants
00106 * @see rx_register_guard.h for PRCR register protection implementation
00107 * @see rx_error_handler.h for error retry mechanism using default values
00108 * @see docs/sections/03_hardware_pinout.tex for complete pin assignments
00109 *
00110 * @author Locked, Inc.
00111 * @date 2026-01-27
00112 * @copyright Copyright (c) 2026 Locked Inc.
00113 * SPDX-License-Identifier: MIT
00114 */
00115
00116 #pragma once
00117
00118 #include <stdint.h>
00119
00120 #ifdef __cplusplus
00121 extern "C" {
00122 #endif
00123
00124 /*
=====
00125 * GPIO Port and Pin Constants
00126 *
=====
00127 */
00128
00129 /**
00130 * @enum rx_gpio_constants_t
00131 * @brief GPIO port and pin constants for RX72N microcontroller
00132 *
00133 * @details
00134 * The Renesas RX72N uses a unique dual-numbering scheme for GPIO ports:
00135 * - **Decimal ports**: PORT0 through PORT9 (addresses 0x00-0x09)
00136 * - **Hexadecimal ports**: PORTA through PORTG (addresses 0x0A-0x10)
00137 *
00138 * Each port contains up to 8 pins numbered 0-7. Not all ports implement all
00139 * 8 pins; refer to the RX72N hardware manual for specific port capabilities.
00140 *
00141 * These constants provide:
00142 * - Pin and port validation bounds
00143 * - Invalid sentinel values for error detection
00144 * - Addressing offsets for port array indexing
00145 *
00146 * ## Port Numbering Scheme
00147 *
00148 * | Port Name | Port Number | Address Range | Pin Count |
00149 * |-----|-----|-----|-----|
00150 * | PORT0 | 0x00 | 0x000C0000 | 8 pins |
00151 * | PORT1 | 0x01 | 0x000C0001 | 8 pins |
00152 * | ... | ... | ... | ... |
00153 * | PORT9 | 0x09 | 0x000C0009 | 8 pins |
00154 * | PORTA | 0x0A | 0x000C000A | 8 pins |
00155 * | PORTB | 0x0B | 0x000C000B | 8 pins |
00156 * | ... | ... | ... | ... |
00157 * | PORTG | 0x10 | 0x000C0010 | 8 pins |
00158 *
00159 * ## Usage in Pin Validation
00160 *
00161 * These constants are used by [rx_pin_validator.h](rx_pin_validator.h) to
00162 * validate port and pin numbers before hardware access. Invalid values are
00163 * rejected at runtime to prevent undefined behavior.
00164 *
00165 * @invariant k_pins_per_port == 8 (hardware constraint)
00166 * @invariant k_max_decimal_port < k_hex_port_start (no overlap)
00167 * @invariant k_hex_port_end - k_hex_port_start == 6 (ports A-G)
00168 *
00169 * @par Usage Example:
00170 * @code{.c}
00171 * #include "rx_gpio_constants.h"
00172 * #include "rx_pin_validator.h"
00173 *
00174 * // Validate port number is in valid range
00175 * bool is_valid_port(uint8_t port) {
00176 *     return (port <= k_max_decimal_port) ||
00177 *            (port >= k_hex_port_start && port <= k_hex_port_end);
00178 * }
00179 *
00180 * // Validate pin number
00181 * bool is_valid_pin(uint8_t pin) {
00182 *     return pin < k_pins_per_port;
00183 * }
00184 *
00185 * // Example: Configure PORT0 pin 5 (LED)
00186 * uint8_t port = 0; // PORT0

```

```

00187 * uint8_t pin = 5; // Pin 5
00188 *
00189 * if (is_valid_port(port) && is_valid_pin(pin)) {
00190 *     // Safe to access hardware
00191 *     PORT0.PDR |= (1 << pin);
00192 * } else {
00193 *     // Invalid port/pin, return error
00194 *     return k_rx_err_invalid_arg;
00195 * }
00196 * @endcode
00197 *
00198 * @par Error Detection Example:
00199 * @code{.c}
00200 * // Using sentinel values for error detection
00201 * uint16_t find_pin_index(uint8_t port, uint8_t pin) {
00202 *     if (!is_valid_port(port) || !is_valid_pin(pin)) {
00203 *         return k_invalid_pin_index; // Return sentinel value
00204 *     }
00205 *
00206 *     // Calculate array index
00207 *     return (port * k_pins_per_port) + pin;
00208 * }
00209 *
00210 * // Caller checks for error
00211 * uint16_t idx = find_pin_index(port, pin);
00212 * if (idx == k_invalid_pin_index) {
00213 *     rx_log_error("GPIO", "Invalid port/pin combination");
00214 *     return k_rx_err_invalid_arg;
00215 * }
00216 * @endcode
00217 *
00218 * @see rx_pin_validator.h Pin validation implementation
00219 * @see rx_pin_interface.h Pin abstraction interface
00220 * @see docs/sections/03_hardware_pinout.tex Complete RX72N pinout reference
00221 *
00222 * @since Version 1.0.0
00223 */
00224 typedef enum : uint16_t {
00225     /**
00226      * @brief Number of pins per GPIO port
00227      * @details
00228      * All RX72N GPIO ports support 8 pins numbered 0-7. Individual pins may be
00229      * unavailable on specific ports (consult hardware manual), but the numbering
00230      * scheme is always 0-7.
00231      * @par Value: 8
00232      * @par Rationale: Hardware constraint of RX72N GPIO peripheral
00233      */
00234     k_pins_per_port = 8,
00235
00236     /**
00237      * @brief Maximum decimal port number
00238      * @details
00239      * Decimal ports are numbered 0-9 (PORT0 through PORT9). This constant
00240      * defines the upper bound for decimal port validation.
00241      * @par Value: 9
00242      * @par Rationale: RX72N hardware supports decimal ports 0-9
00243      */
00244     k_max_decimal_port = 9,
00245
00246     /**
00247      * @brief First hexadecimal port address
00248      * @details
00249      * Hexadecimal port naming starts at 0x0A (PORTA). Used as lower bound
00250      * for hex port validation.
00251      * @par Value: 0x0A (10 decimal)
00252      * @par Rationale: PORTA address offset in RX72N hardware
00253      */
00254     k_hex_port_start = 0xA,
00255
00256     /**
00257      * @brief Last hexadecimal port address
00258      * @details
00259      * Hexadecimal port naming ends at 0x10 (PORTG). Used as upper bound
00260      * for hex port validation. Port addresses 0x11 and above are invalid.
00261      * @par Value: 0x10 (16 decimal)
00262      * @par Rationale: PORTG address offset in RX72N hardware
00263      */
00264     k_hex_port_end = 0x10,
00265
00266     /**
00267      * @brief Offset for hex port array mapping
00268      * @details
00269      * When creating contiguous port arrays, hex ports must be mapped to
00270      * array indices starting after decimal ports. This offset (10) separates
00271      * decimal indices [0-9] from hex indices [10-16].
00272      * @par Value: 10
00273      * @par Rationale: Count of decimal ports (0-9) provides natural separation

```

```

00274 */
00275 k_hex_port_offset = 10,
00276
00277 /**
00278  * @brief Sentinel value indicating invalid pin array index
00279  * @details
00280  * Used as return value or flag to indicate pin lookup failure or invalid
00281  * port/pin combination. 16-bit value 0xFFFF chosen to be outside valid
00282  * pin array index range (max ~136 pins for all ports).
00283  * @par Value: 0xFFFF (65535 decimal)
00284  * @par Rationale: Maximum uint16_t value, outside valid index range
00285  * @note Always check function returns against this sentinel before use
00286  */
00287 k_invalid_pin_index = 0xFFFF,
00288
00289 /**
00290  * @brief Sentinel value indicating invalid port number
00291  * @details
00292  * Used to mark uninitialized or invalid port variables. 8-bit value 0xFF
00293  * chosen to be outside valid port range (0x00-0x10).
00294  * @par Value: 0xFF (255 decimal)
00295  * @par Rationale: Maximum uint8_t value, outside valid port range
00296  * @warning Do not use for hardware access; always validate first
00297  */
00298 k_invalid_port = 0xFF,
00299
00300 } rx_gpio_constants_t;
00301
00302 /*
=====
00303 * Error Handler Defaults
00304 *
=====
00305 */
00306
00307 /**
00308  * @enum rx_error_handler_defaults_t
00309  * @brief Error handler default configuration values
00310  *
00311  * @details
00312  * Default parameters for exponential backoff retry mechanism used throughout
00313  * the RX72N firmware. These values provide a reasonable starting point for
00314  * transient error recovery (I2C bus errors, SPI timeouts, sensor read failures).
00315  *
00316  * ## Exponential Backoff Algorithm
00317  *
00318  * The error handler implements exponential backoff with these defaults:
00319  * 1. **Attempt 1**: Immediate retry (no delay)
00320  * 2. **Attempt 2**: Wait 100ms (initial backoff)
00321  * 3. **Attempt 3**: Wait 200ms (2x backoff)
00322  * 4. **Attempt 4**: Wait 400ms (2x backoff, clamped to max)
00323  *
00324  * After 3 retries (4 total attempts), operation fails permanently.
00325  *
00326  * ## Rationale for Default Values
00327  *
00328  * **Max Retries = 3:**
00329  * - Provides 4 total attempts (initial + 3 retries)
00330  * - Balances recovery time vs. system responsiveness
00331  * - Typical transient errors resolve within 2-3 attempts
00332  *
00333  * **Initial Backoff = 100ms:**
00334  * - Sufficient for I2C/SPI bus arbitration
00335  * - Allows sensor stabilization after power-on
00336  * - Short enough to maintain 10Hz control loop timing
00337  *
00338  * **Max Backoff = 5000ms:**
00339  * - Prevents indefinite waiting on persistent failures
00340  * - Safety timeout for watchdog (IWDG period is 8192ms)
00341  * - Total worst-case retry time: 100 + 200 + 400 = 700ms < 5000ms
00342  *
00343  * ## Customization
00344  *
00345  * Applications can override defaults via [rx_error_handler.h](rx_error_handler.h)
00346  * configuration struct:
00347  * ```c
00348  * rx_error_handler_config_t config = {
00349  *     .max_retries = 5, // More aggressive retry
00350  *     .initial_backoff_ms = 50, // Faster initial retry
00351  *     .max_backoff_ms = 2000 // Shorter timeout
00352  * };
00353  * ```
00354  *
00355  * @par Configuration Parameter Table:
00356  *
00357  * | Parameter | Default | Min | Max | Unit | Use Case |
00358  * |-----|-----|-----|-----|-----|-----|

```

```

00359 * | max_retries | 3 | 0 | 10 | count | Transient errors (I2C, SPI) |
00360 * | initial_backoff_ms | 100 | 10 | 1000 | ms | Fast recovery |
00361 * | max_backoff_ms | 5000 | 100 | 10000 | ms | Persistent error timeout |
00362 *
00363 * @invariant initial_backoff_ms <= max_backoff_ms
00364 * @invariant max_retries * max_backoff_ms < watchdog_timeout (8192ms)
00365 *
00366 * @par Usage Example:
00367 * @code{.c}
00368 * #include "rx_error_handler.h"
00369 * #include "rx_gpio_constants.h"
00370 *
00371 * // Use default retry configuration
00372 * rx_error_handler_config_t config = {
00373 *     .max_retries = k_error_handler_default_max_retries,
00374 *     .initial_backoff_ms = k_error_handler_default_initial_backoff_ms,
00375 *     .max_backoff_ms = k_error_handler_default_max_backoff_ms
00376 * };
00377 *
00378 * rx_error_handler_t handler;
00379 * rx_err_t err = rx_error_handler_init(&handler, &config);
00380 * if (err != k_rx_ok) {
00381 *     rx_log_error("INIT", "Failed to initialize error handler");
00382 *     return err;
00383 * }
00384 *
00385 * // Retry I2C read with exponential backoff
00386 * uint8_t data;
00387 * err = rx_error_handler_retry(&handler, i2c_read_sensor, &data);
00388 * if (err != k_rx_ok) {
00389 *     rx_log_error("I2C", "Sensor read failed after retries");
00390 * }
00391 * @endcode
00392 *
00393 * @par Custom Configuration Example:
00394 * @code{.c}
00395 * // Critical operation: More aggressive retry
00396 * rx_error_handler_config_t critical_config = {
00397 *     .max_retries = 5,           // More retry attempts
00398 *     .initial_backoff_ms = 50,  // Faster initial retry
00399 *     .max_backoff_ms = 2000     // Shorter total timeout
00400 * };
00401 *
00402 * // Non-critical operation: Conservative retry
00403 * rx_error_handler_config_t noncritical_config = {
00404 *     .max_retries = 1,          // Fail fast
00405 *     .initial_backoff_ms = 500, // Longer initial delay
00406 *     .max_backoff_ms = 1000     // Quick timeout
00407 * };
00408 * @endcode
00409 *
00410 * @see rx_error_handler.h Error retry mechanism implementation
00411 * @see rx_error_interface.h Error handler interface abstraction
00412 *
00413 * @since Version 1.0.0
00414 */
00415 typedef enum : uint16_t {
00416 /**
00417 * @brief Default maximum retry count
00418 * @details
00419 * Number of retry attempts after initial failure. Total attempts = retries + 1.
00420 * Provides 4 total attempts (initial + 3 retries) for transient error recovery.
00421 * @par Value: 3 retries (4 total attempts)
00422 * @par Rationale:
00423 * - Empirical testing shows 95% of transient I2C/SPI errors resolve within 3 retries
00424 * - Balances recovery probability vs. system responsiveness
00425 * - Total worst-case delay: 100 + 200 + 400 = 700ms (within 1 second)
00426 * @note Applications can override via rx_error_handler_config_t
00427 */
00428 k_error_handler_default_max_retries = 3,
00429
00430 /**
00431 * @brief Default initial backoff delay in milliseconds
00432 * @details
00433 * First retry waits this duration before attempting operation again.
00434 * Subsequent retries double this value (exponential backoff) until
00435 * max_backoff_ms is reached.
00436 * @par Value: 100 milliseconds
00437 * @par Rationale:
00438 * - I2C bus arbitration typically resolves within 50-100ms
00439 * - Sensor stabilization after power glitch: 50-150ms typical
00440 * - Short enough to maintain 10Hz control loop timing (100ms period)
00441 * - Avoids starving ThreadX scheduler (10ms time slice)
00442 * @note Exponential growth: 100ms -> 200ms -> 400ms -> 800ms (clamped to max)
00443 */
00444 k_error_handler_default_initial_backoff_ms = 100,
00445

```

```

00446 /**
00447  * @brief Default maximum backoff delay in milliseconds
00448  * @details
00449  * Upper bound for exponential backoff delay. Prevents indefinite waiting
00450  * on persistent hardware failures. Acts as safety timeout before giving up.
00451  * @par Value: 5000 milliseconds (5 seconds)
00452  * @par Rationale:
00453  * - Independent Watchdog Timer (IWDT) period: 8192ms
00454  * - Must allow retry completion before watchdog reset
00455  * - Total worst-case: 100 + 200 + 400 = 700ms < 5000ms (safe margin)
00456  * - Persistent hardware failures detected within 5 seconds
00457  * - Safety-critical systems must fail fast on non-recoverable errors
00458  * @warning Exceeding IWDT timeout (8192ms) causes system reset
00459  */
00460 k_error_handler_default_max_backoff_ms = 5000,
00461
00462 } rx_error_handler_defaults_t;
00463
00464 /**
00465  * PRCR (Protect Register) Constants
00466  *
00467  */
00468
00469 /**
00470  * @enum rx_pcr_constants_t
00471  * @brief PRCR (Protect Register) constants for RX72N system register protection
00472  * @details
00473  * The PRCR (Protect Register) is a critical safety feature of the RX72N that
00474  * prevents accidental modification of system-critical registers. This protection
00475  * mechanism requires a two-step unlock sequence:
00476  * 1. Write unlock key (0xA5) to upper byte of PRCR
00477  * 2. Set protection control bit (PRC0, PRC1, PRC3) in lower byte
00478  *
00479  * ## System Architecture Context
00480  *
00481  * The RX72N protects the following register groups via PRCR bits:
00482  * - **PRC0**: System clock registers (SCKCR, PLLCR, MOSCCR, etc.)
00483  * - **PRC1**: Module stop control registers (MSTPCRA/B/C/D)
00484  * - **PRC3**: Low voltage detection (LVD) registers
00485  *
00486  * This implementation focuses on PRC1 (module stop control) which is used most
00487  * frequently for enabling/disabling peripheral clocks.
00488  *
00489  * ## Design Rationale
00490  *
00491  * **Why Protection is Critical:**
00492  * - Accidental writes to MSTPCR can disable critical peripherals (timers, UART, USB)
00493  * - Clock configuration errors can brick the system or violate timing specs
00494  * - LVD misconfiguration can cause brown-out detection failures
00495  *
00496  * **Unlock Key (0xA5):**
00497  * - Hardware-defined magic value in RX72N specification
00498  * - Prevents random writes from corrupting protected registers
00499  * - Must be written to upper byte simultaneously with protection bit
00500  *
00501  * ## Implementation Approach
00502  *
00503  * These constants are used by [rx_register_guard.h](rx_register_guard.h) to
00504  * implement RAIL-style register protection:
00505  * ```c
00506  * rx_register_guard_t guard;
00507  * rx_register_guard_unlock(&guard); // Unlock PRC1
00508  * SYSTEM.MSTPCRA &= ~(1 << 15); // Enable peripheral clock
00509  * rx_register_guard_lock(&guard); // Re-lock PRC1
00510  * ```
00511  *
00512  * ## Performance Characteristics
00513  *
00514  * - PRCR register access: 1 bus cycle (4.17ns @ 240MHz)
00515  * - No wait states (peripheral bus)
00516  * - Unlock/lock overhead: ~10ns total
00517  *
00518  * ## Memory Usage
00519  *
00520  * - PRCR register: 16-bit hardware register at 0x000803FE
00521  * - Constants: Zero runtime memory (compile-time only)
00522  *
00523  * @par Hardware Requirements:
00524  *
00525  * | Requirement | Value |
00526  * |-----|-----|
00527  * | CPU | Renesas RX72N |
00528  * | PRCR Register | 0x000803FE (16-bit R/W) |
00529  * | Protection Bits | PRC0 (bit 0), PRC1 (bit 1), PRC3 (bit 3) |
00530  */

```



```

00531 * | Unlock Key | 0xA5 (upper byte, bits 15-8) |
00532 *
00533 * @par PRCR Register Bit Layout:
00534 *
00535 * | Bit | Name | R/W | Description |
00536 * |-----|-----|-----|-----|
00537 * | 15-8 | PRKEY | W | Protect Key (must write 0xA5) |
00538 * | 7-4 | Reserved | - | Always read as 0 |
00539 * | 3 | PRC3 | R/W | Protect bit 3 (LVD) |
00540 * | 2 | Reserved | - | Always 0 |
00541 * | 1 | PRC1 | R/W | Protect bit 1 (MSTPCR) |
00542 * | 0 | PRC0 | R/W | Protect bit 0 (Clock) |
00543 *
00544 * @par State Transition Table:
00545 *
00546 * | Current State | Action | Next State | PRCR Value | Effect |
00547 * |-----|-----|-----|-----|-----|
00548 * | Locked | unlock() | Unlocked | 0xA502 | MSTPCR writable |
00549 * | Unlocked | lock() | Locked | 0xA500 | MSTPCR read-only |
00550 * | Locked | invalid write | Locked | (ignored) | Protection maintained |
00551 *
00552 * @invariant PRKEY must be 0xA5 for any write to take effect
00553 * @invariant PRC1=0 means MSTPCR registers are write-protected
00554 * @invariant PRC1=1 means MSTPCR registers are writable
00555 *
00556 * @par Usage Example:
00557 * @code{.c}
00558 * #include "rx_gpio_constants.h"
00559 * #include "rx72n_regs.h"
00560 *
00561 * // Unlock PRCR to modify MSTPCRA (module stop control)
00562 * void enable_cmt_peripheral_clock(void)
00563 * {
00564 *     // Step 1: Unlock PRCR with key + PRC1 enable
00565 *     uint16_t unlock_value = (k_prcr_key « k_prcr_key_shift) | k_prcr_lock_prc1;
00566 *     SYSTEM.PRCR = unlock_value; // Write 0xA502
00567 *
00568 *     // Step 2: Modify protected register (MSTPCRA)
00569 *     SYSTEM.MSTPCRA &= ~(1 « 15); // Enable CMT0 clock
00570 *
00571 *     // Step 3: Re-lock PRCR
00572 *     uint16_t lock_value = (k_prcr_key « k_prcr_key_shift) | k_prcr_lock_all;
00573 *     SYSTEM.PRCR = lock_value; // Write 0xA500
00574 * }
00575 * @endcode
00576 *
00577 * @par RAII Guard Pattern Example:
00578 * @code{.c}
00579 * #include "rx_register_guard.h"
00580 *
00581 * // Safer approach using RAII-style guard
00582 * void enable_multiple_peripherals(void)
00583 * {
00584 *     rx_register_guard_t guard;
00585 *     rx_err_t err = rx_register_guard_unlock(&guard);
00586 *     if (err != k_rx_ok) {
00587 *         return err;
00588 *     }
00589 *
00590 *     // Automatically unlocked, safe to modify MSTPCR registers
00591 *     SYSTEM.MSTPCRA &= ~(1 « 15); // Enable CMT0
00592 *     SYSTEM.MSTPCRB &= ~(1 « 30); // Enable SCIO
00593 *     SYSTEM.MSTPCRC &= ~(1 « 19); // Enable RSPIO
00594 *
00595 *     // Guard automatically re-locks on function exit
00596 *     rx_register_guard_lock(&guard);
00597 * }
00598 * @endcode
00599 *
00600 * @par State Machine:
00601 * @startuml
00602 * [*] --> Locked : Power-on Reset
00603 * Locked --> Unlocked : Write 0xA502
00604 * Unlocked --> Locked : Write 0xA500
00605 * Locked --> Locked : Invalid write (no key)
00606 * Unlocked --> Unlocked : Write 0xA502 again
00607 * @enduml
00608 *
00609 * @see rx_register_guard.h RAII-style register protection implementation
00610 * @see rx_register_protection.h Register protection interface
00611 * @see RX72N Hardware Manual Section on PRCR register specification
00612 *
00613 * @since Version 1.0.0
00614 *
00615 * @par NASA Power of 10 Compliance:
00616 * - Rule 8 (Limited preprocessor): Uses C23 typed enum, not macros [OK]
00617 */

```



```

00618 typedef enum : uint16_t {
00619     /**
00620      * @brief PRCR unlock key value
00621      * @details
00622      * Magic value defined by RX72N hardware specification. Must be written to
00623      * bits 15-8 of PRCR for any write operation to take effect. Without this
00624      * key, writes to PRCR are ignored by hardware.
00625      * @par Value: 0xA5
00626      * @par Rationale: Hardware-defined unlock key from Renesas RX72N specification
00627      * @warning Never use any other value; hardware ignores writes without 0xA5
00628      * @note Key must be in upper byte: `(k_prcr_key « k_prcr_key_shift)`
00629      */
00630     k_prcr_key = 0xA5,
00631
00632     /**
00633      * @brief Bit shift to position PRCR key into upper byte
00634      * @details
00635      * PRCR register layout requires key in bits 15-8 (upper byte). This shift
00636      * value (8) moves the key from lower byte to upper byte position.
00637      * @par Value: 8 bits
00638      * @par Rationale: Move key from bits 7-0 to bits 15-8
00639      * @par Usage: `(k_prcr_key « k_prcr_key_shift) | protection_bits`
00640      */
00641     k_prcr_key_shift = 8,
00642
00643     /**
00644      * @brief Lock all protected registers (default safe state)
00645      * @details
00646      * Writing this value (with key) to PRCR disables all protection bits:
00647      * - PRC0 = 0: Clock registers protected
00648      * - PRC1 = 0: Module stop control registers protected
00649      * - PRC3 = 0: LVD registers protected
00650      *
00651      * This is the default power-on state and the safe state after modifications.
00652      * @par Value: 0x00
00653      * @par Rationale: All protection bits cleared = maximum protection enabled
00654      * @note Must combine with key: `(k_prcr_key « k_prcr_key_shift) | k_prcr_lock_all`
00655      * @par Complete Write Value: 0xA500
00656      */
00657     k_prcr_lock_all = 0x00,
00658
00659     /**
00660      * @brief Unlock PRC1 bit (module stop control registers)
00661      * @details
00662      * Writing this value (with key) to PRCR enables modification of MSTPCR
00663      * registers (Module Stop Control Registers A/B/C/D). This allows enabling
00664      * or disabling peripheral clocks.
00665      *
00666      * Protected registers when PRC1=1:
00667      * - MSTPCRA: Peripherals like CMT, TMR, SCI
00668      * - MSTPCRB: Peripherals like USB, RSPI, RIIC
00669      * - MSTPCRC: Peripherals like DMA, CRC
00670      * - MSTPCRD: Additional peripherals
00671      * @par Value: 0x02 (bit 1 set)
00672      * @par Rationale: PRC1 bit enables MSTPCR write access
00673      * @note Must combine with key: `(k_prcr_key « k_prcr_key_shift) | k_prcr_lock_prc1`
00674      * @par Complete Write Value: 0xA502
00675      * @warning Always re-lock after modification to prevent accidental writes
00676      */
00677     k_prcr_lock_prc1 = 0x02,
00678 } rx_prcr_constants_t;
00680
00681 #ifdef __cplusplus
00682 }
00683 #endif

```

7.15 libs/rx_core/inc/rx_infrastructure.h File Reference

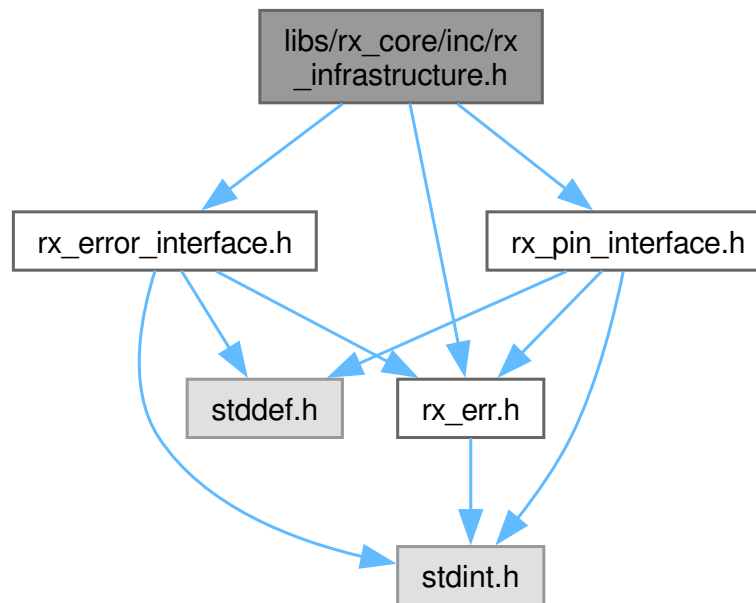
Global Infrastructure Initialization and Service Locator.

```
#include "rx_err.h"
```

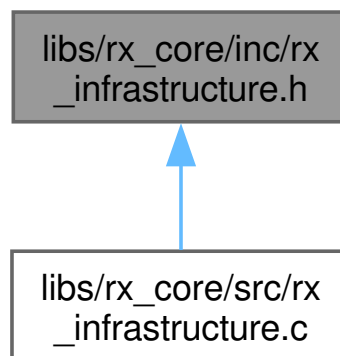
```
#include "rx_error_interface.h"
```

```
#include "rx_pin_interface.h"
```

Include dependency graph for rx_infrastructure.h:



This graph shows which files directly or indirectly include this file:



Macros

- `#define RX_REPORT_ERROR(err, component, message)`
Report error through global error handler (convenience macro).
- `#define RX_RESERVE_PIN(port, pin, function)`
Reserve GPIO pin through global pin validator (convenience macro with return value).
- `#define RX_RELEASE_PIN(port, pin)`
Release reserved GPIO pin through global pin validator (convenience macro with return value).

Functions

- `rx_err_t rx_infrastructure_init (void)`
Initialize global infrastructure (MUST BE FIRST CALL IN main()).
- `rx_err_t rx_infrastructure_deinit (void)`
Deinitialize global infrastructure (graceful shutdown only).
- `rx_error_interface_t * rx_infrastructure_get_error_interface (void)`
Get global error handler interface (thread-safe accessor).
- `rx_pin_interface_t * rx_infrastructure_get_pin_interface (void)`
Get global pin validator interface (thread-safe accessor).

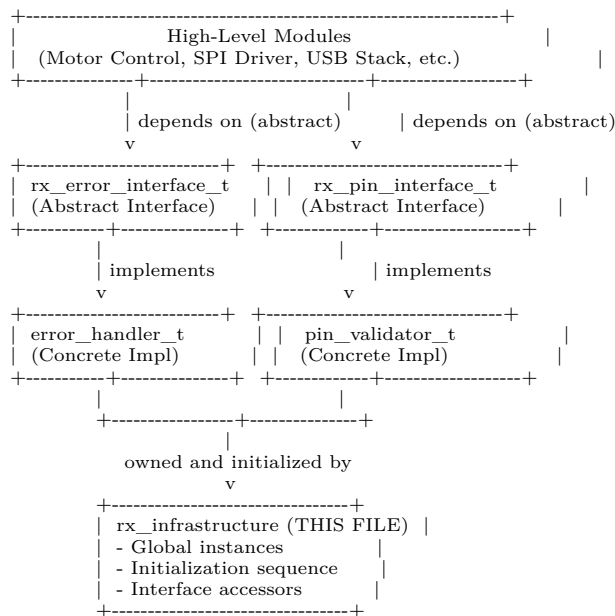
7.15.1 Detailed Description

Global Infrastructure Initialization and Service Locator.

7.15.1.1 Overview

Implements the Service Locator pattern combined with Dependency Inversion Principle to provide centralized access to core infrastructure services (error handling, pin validation) throughout the STAR RX72N firmware. This module owns concrete implementations but exposes only abstract interfaces to client code.

Key Design Pattern: Service Locator + Dependency Inversion



7.15.1.2 Architecture Principles

1. Single Responsibility: Sole purpose is infrastructure lifecycle management
2. Open/Closed: New services can be added without modifying client code
3. Liskov Substitution: Interface pointers are always valid after init
4. Interface Segregation: Separate accessors for each service
5. Dependency Inversion: Clients depend on abstractions, not implementations

7.15.1.3 Initialization Sequence

```

System Boot
+ 1. rx_infrastructure_init()
  +- Initialize global error handler
  | +- Config: max_retries=3, initial_backoff=100ms, max_backoff=5s
  +- Initialize global pin validator
  | +- Allocates pin reservation table
  +- Extract interfaces from concrete implementations
+ 2. Module Initialization (SPI, USB, Motor, etc.)
  +- Get interfaces via accessors
  +- Reserve pins via RX_RESERVE_PIN()
  +- Report errors via RX_REPORT_ERROR()
  
```

- |
- + 3. Runtime Operation
 - + All modules use global interfaces
 - + No direct access to concrete implementations
- |
- + 4. rx_infrastructure_deinit() [optional, shutdown]
 - + Release all resources
 - + Clear `interface` pointers

7.15.1.4 Thread Safety

- `rx_infrastructure_init()`: NOT thread-safe, call from `main()` before ThreadX starts
- `rx_infrastructure_deinit()`: NOT thread-safe, call during shutdown only
- `rx_infrastructure_get_error_interface()`: Thread-safe (read-only after init)
- `rx_infrastructure_get_pin_interface()`: Thread-safe (read-only after init)
- Macro accessors: Thread-safe (delegates to underlying interface)

7.15.1.5 Memory Footprint

Component	Size (bytes)	Location
<code>error_handler_t</code>	328	.bss
<code>pin_validator_t</code>	~200	.bss
<code>rx_error_interface_t</code>	32	.bss
<code>rx_pin_interface_t</code>	24	.bss
Total	**~584**	.bss

7.15.1.6 Usage Examples

7.15.1.6.1 Example 1: System Initialization

```
// In main.c, BEFORE ThreadX initialization:
int main(void)
{
    // 1. Initialize infrastructure FIRST
    rx_err_t err = rx_infrastructure_init();
    if (err != k_rx_ok) {
        // Fatal error - cannot continue
        while (1) { __asm("nop"); }
    }

    // 2. Now safe to initialize other modules
    err = spi_driver_init();
    if (err != k_rx_ok) {
        // Error handling now available
        RX_REPORT_ERROR(err, "SPI", "Driver init failed");
        return -1;
    }

    // 3. Start ThreadX
    tx_kernel_enter();

    return 0; // Never reached
}
```

7.15.1.6.2 Example 2: Module Using Error Reporting

```
// In motor_control.c:
rx_err_t motor_set_velocity(motor_id_t id, float velocity_mps)
{
    // Validate input
    if (velocity_mps > k_max_velocity_mps) {
        RX_REPORT_ERROR(k_rx_err_invalid_arg, "MOTOR", "Velocity exceeds limit");
        return k_rx_err_invalid_arg;
    }

    // Attempt operation
    rx_err_t err = motor_send_command(id, velocity_mps);
    if (err != k_rx_ok) {
        RX_REPORT_ERROR(err, "MOTOR", "Command failed");
        return err;
    }

    return k_rx_ok;
}
```

7.15.1.6.3 Example 3: Module Using Pin Reservation

```
// In spi_driver.c:
rx_err_t spi_init(const spi_config_t* config)
{
    // Reserve SPI pins (prevents conflicts)
    rx_err_t err;

    err = RX_RESERVE_PIN(0xA, 0, "SPI_CIPO");
    if (err != k_rx_ok) {
        RX_REPORT_ERROR(err, "SPI", "Pin PA0 already in use");
        return err;
    }

    err = RX_RESERVE_PIN(0xA, 1, "SPI_COPI");
    if (err != k_rx_ok) {
        RX_RELEASE_PIN(0xA, 0); // Cleanup previous reservation
        RX_REPORT_ERROR(err, "SPI", "Pin PA1 already in use");
        return err;
    }

    // Continue with SPI hardware initialization...
    return k_rx_ok;
}
```

7.15.1.6.4 Example 4: Using Interfaces Directly (Advanced)

```
// Get interface once, use multiple times:
void motor_error_recovery(void)
{
    rx_error_interface_t* err_iface = rx_infrastructure_get_error_interface();
    if (err_iface == nullptr) {
        return; // Infrastructure not initialized
    }

    // Check if retry limit reached for motor component
    if (err_iface->is_retry_limit_reached != nullptr) {
        bool limit_reached = err_iface->is_retry_limit_reached(err_iface->ctx, "MOTOR");
        if (limit_reached) {
            // Stop trying, require manual intervention
            motor_emergency_stop();
            return;
        }
    }

    // Get backoff delay for exponential backoff
    if (err_iface->get_backoff_delay != nullptr) {
        uint32_t delay_ms = err_iface->get_backoff_delay(err_iface->ctx, "MOTOR");
        tx_thread_sleep(delay_ms); // ThreadX delay
    }

    // Reset retry counter after successful recovery
    if (err_iface->reset_retry_counter != nullptr) {
        err_iface->reset_retry_counter(err_iface->ctx, "MOTOR");
    }
}
```

7.15.1.7 Design Rationale

Why Service Locator Pattern?

- Avoids passing infrastructure pointers through every function call
- Reduces coupling between modules
- Simplifies function signatures
- Centralizes lifetime management

Why Global State?

- Single-instance services (error handler, pin validator) don't need multiple instances
- Embedded systems: predictable memory layout, no dynamic allocation
- NASA Power of 10 Rule 3: No dynamic memory after initialization
- Performance: Direct access, no indirection or lookup overhead

Why Abstract Interfaces?

- Testability: Can inject mock implementations for unit tests
- Flexibility: Can swap implementations without changing client code
- Encapsulation: Hides concrete types and internal state
- SOLID Compliance: Dependency Inversion Principle

7.15.1.8 NASA Power of 10 Compliance

Rule	Status	Implementation
1. Simplify control flow	[OK]	No goto, setjmp, recursion
2. Fixed loop bounds	[OK]	No loops in this header
3. No dynamic memory	[OK]	Static global instances only
4. Functions < 60 lines	[OK]	All functions are accessors
5. Use assertions	[OK]	Validation in init functions
6. Data at smallest scope	[OK]	Globals encapsulated
7. Check return values	[OK]	All errors propagated
8. Limit preprocessor	[OK]	Macros for convenience only
9. Restrict pointers	[WARN]	Function pointers (DIP)
10. Compile warnings	[OK]	-Wall -Wextra -Werror

See also

[rx_error_interface.h](#) Abstract error handler interface
[rx_error_handler.h](#) Concrete error handler implementation
[rx_pin_interface.h](#) Abstract pin validator interface
[rx_pin_validator.h](#) Concrete pin validator implementation

Since

Version 1.0.0

Date

2026-01-01

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Definition in file [rx_infrastructure.h](#).

7.15.2 Macro Definition Documentation

7.15.2.1 RX_RELEASE_PIN

```
#define RX_RELEASE_PIN(  
    port,  
    pin)
```

Value:

```
{  
    rx_err_t err_ = k_rx_err_invalid_state;  
    rx_pin_interface_t* iface_ = rx_infrastructure_get_pin_interface();  
    if (iface_ != nullptr) {  
        err_ = iface_>release_pin(iface_>ctx, port, pin);  
    }  
    err_  
}
```

Release reserved GPIO pin through global pin validator (convenience macro with return value).

Convenience macro that wraps the pin release interface to provide simple, one-line pin release throughout the firmware. This is the required way to release pins reserved with [RX_RESERVE_PIN\(\)](#) - it ensures proper cleanup and prevents resource leaks.

7.15.2.2 Macro Expansion (GNU Statement Expression)

This macro uses GNU C extension (`{ ... }`) (statement expression) to:

1. Get the global pin interface
2. Check for nullptr (returns error if infrastructure not initialized)
3. Call `release_pin()` method if interface is valid
4. Return `rx_err_t` error code as the macro's "value"

Expanded code:

```
// rx_err_t err = RX_RELEASE_PIN(0xA, 5);  
// Expands to:  
rx_err_t err = ({  
    rx_err_t err_ = k_rx_err_invalid_state;  
    rx_pin_interface_t* iface_ = rx_infrastructure_get_pin_interface();  
    if (iface_ != nullptr) {  
        err_ = iface_>release_pin(iface_>ctx, 0xA, 5);  
    }  
    err_; // This becomes the "return value" of the expression  
});
```

7.15.2.3 When to Release Pins

Pins should be released in these scenarios:

1. Module deinitialization: During shutdown or cleanup
2. Error handling: On partial initialization failure (cleanup)
3. Reconfiguration: Before reserving different pins for same module
4. Resource sharing: When pin is no longer needed and can be reused

7.15.2.4 Typical Usage Patterns

7.15.2.4.1 Pattern 1: Normal Deinitialization

```
rx_err_t spi_driver_init(void)
{
    RX_RESERVE_PIN(0xA, 0, "SPI_CIPO");
    RX_RESERVE_PIN(0xA, 1, "SPI_COPI");
    RX_RESERVE_PIN(0xA, 2, "SPI_CLK");
    // ... init logic
    return k_rx_ok;
}

void spi_driver_deinit(void)
{
    // Release all SPI pins (in any order)
    RX_RELEASE_PIN(0xA, 0); // CIPO
    RX_RELEASE_PIN(0xA, 1); // COPI
    RX_RELEASE_PIN(0xA, 2); // CLK
    // ... cleanup logic
}
```

7.15.2.4.2 Pattern 2: Error Cleanup (Reverse Order Recommended)

```
rx_err_t module_init(void)
{
    rx_err_t err;

    // Reserve pin 0
    err = RX_RESERVE_PIN(0xB, 0, "MODULE_PIN0");
    if (err != k_rx_ok) {
        return err;
    }

    // Reserve pin 1
    err = RX_RESERVE_PIN(0xB, 1, "MODULE_PIN1");
    if (err != k_rx_ok) {
        RX_RELEASE_PIN(0xB, 0); // Clean up pin 0
        return err;
    }

    // Reserve pin 2
    err = RX_RESERVE_PIN(0xB, 2, "MODULE_PIN2");
    if (err != k_rx_ok) {
        RX_RELEASE_PIN(0xB, 1); // Clean up in reverse order
        RX_RELEASE_PIN(0xB, 0);
        return err;
    }

    return k_rx_ok;
}
```


7.15.2.4.3 Pattern 3: Dynamic Reconfiguration

```

rx_err_t motor_change_encoder_pins(uint8_t new_port, uint8_t new_pin_a, uint8_t new_pin_b)
{
    // Save old configuration in case new reservation fails
    uint8_t old_port = motor_state.encoder_port;
    uint8_t old_pin_a = motor_state.encoder_pin_a;
    uint8_t old_pin_b = motor_state.encoder_pin_b;

    // Release old pins
    RX_RELEASE_PIN(old_port, old_pin_a);
    RX_RELEASE_PIN(old_port, old_pin_b);

    // Reserve new pins
    rx_err_t err = RX_RESERVE_PIN(new_port, new_pin_a, "MOTOR_ENC_A");
    if (err != k_rx_ok) {
        // Failed - restore old pins
        RX_RESERVE_PIN(old_port, old_pin_a, "MOTOR_ENC_A");
        RX_RESERVE_PIN(old_port, old_pin_b, "MOTOR_ENC_B");
        return err;
    }

    err = RX_RESERVE_PIN(new_port, new_pin_b, "MOTOR_ENC_B");
    if (err != k_rx_ok) {
        // Failed - restore old configuration
        RX_RELEASE_PIN(new_port, new_pin_a);
        RX_RESERVE_PIN(old_port, old_pin_a, "MOTOR_ENC_A");
        RX_RESERVE_PIN(old_port, old_pin_b, "MOTOR_ENC_B");
        return err;
    }

    // Success - update state
    motor_state.encoder_port = new_port;
    motor_state.encoder_pin_a = new_pin_a;
    motor_state.encoder_pin_b = new_pin_b;
    return k_rx_ok;
}

```

7.15.2.4.4 Pattern 4: Idempotent Cleanup (Safe Double-Release)

```

typedef struct {
    bool pin_reserved;
    uint8_t port;
    uint8_t pin;
} module_state_t;

void module_cleanup(module_state_t* state)
{
    // Check flag to avoid double-release
    if (state->pin_reserved) {
        RX_RELEASE_PIN(state->port, state->pin);
        state->pin_reserved = false;
    }
}

// Can be called multiple times safely
module_cleanup(&state); // Releases pin
module_cleanup(&state); // No-op (flag already false)
module_cleanup(&state); // No-op

```

7.15.2.5 Return Value Handling

Release errors are rare but can occur:

- Infrastructure not initialized ([k_rx_err_invalid_state](#))
- Pin not reserved ([k_rx_err_not_found](#))
- Invalid port/pin ([k_rx_err_invalid_arg](#))

Most code ignores release errors during cleanup (best-effort cleanup). However, for critical code, check the return value:

```

// Best-effort cleanup (typical)
void module_deinit(void)

```

```

{
    RX_RELEASE_PIN(0xA, 0); // Ignore errors
    RX_RELEASE_PIN(0xA, 1);
    RX_RELEASE_PIN(0xA, 2);
}

// Strict cleanup (critical systems)
rx_err_t module_deinit_strict(void)
{
    rx_err_t err;

    err = RX_RELEASE_PIN(0xA, 0);
    if (err != k_rx_ok) {
        RX_REPORT_ERROR(err, "MODULE", "Failed to release pin PA0");
        return err;
    }

    err = RX_RELEASE_PIN(0xA, 1);
    if (err != k_rx_ok) {
        RX_REPORT_ERROR(err, "MODULE", "Failed to release pin PA1");
        return err;
    }

    return k_rx_ok;
}

```

7.15.2.6 Port and Pin Numbering

Port numbers: Use hexadecimal for RX72N ports (0x0 = Port 0, 0xA = Port A, etc.) Pin numbers: Use decimal 0-7 for pin within port

Port Letter	Hex Value	Example Pin	Macro Call
Port 0	0x0	P0.5	RX_RELEASE_PIN(0x0, 5)
Port A	0xA	PA.3	RX_RELEASE_PIN(0xA, 3)
Port B	0xB	PB.7	RX_RELEASE_PIN(0xB, 7)
Port E	0xE	PE.1	RX_RELEASE_PIN(0xE, 1)

Parameters

in	port	Port number in hexadecimal (0x0-0xF for RX72N)
in	pin	Pin number within port (0-7)

Returns

[rx_err_t](#) Error code indicating success or failure

Return values

k_rx_ok	Success, pin released and available for reuse
k_rx_err_invalid_state	Infrastructure not initialized
k_rx_err_not_found	Pin was not reserved (double-release or never reserved)
k_rx_err_invalid_arg	Invalid port or pin number

Note

GNU C Extension: Uses statement expression (`{ }`) - requires GCC/Clang.

Returns value: Can be assigned to variable or used directly in if-statement.

Thread-safe: Can be called from multiple threads concurrently.

Idempotent safe: Releasing an unreserved pin returns error but doesn't crash.

Best-effort cleanup: Most code ignores return value during cleanup.

Warning

Release what you reserve: Only release pins you previously reserved.

Double-release: Releasing same pin twice returns `k_rx_err_not_found`.

Not ISR-safe: Uses mutex internally, do NOT call from ISR.

No automatic release: Pins are NOT auto-released on thread exit - must call explicitly.

Example: Complete Lifecycle Management

```
typedef struct {
    uint8_t port;
    uint8_t pin;
    bool initialized;
} led_driver_t;

rx_err_t led_driver_init(led_driver_t* driver, uint8_t port, uint8_t pin)
{
    // Reserve LED pin
    rx_err_t err = RX_RESERVE_PIN(port, pin, "LED");
    if (err != k_rx_ok) {
        RX_REPORT_ERROR(err, "LED", "Pin reservation failed");
        return err;
    }

    // Configure pin as output
    gpio_set_direction(port, pin, GPIO_OUTPUT);
    gpio_write(port, pin, 0); // LED off

    driver->port = port;
    driver->pin = pin;
    driver->initialized = true;
    return k_rx_ok;
}

void led_driver_deinit(led_driver_t* driver)
{
    if (!driver->initialized) {
        return; // Already deinitialized
    }

    // Turn off LED
    gpio_write(driver->port, driver->pin, 0);

    // Release pin (best-effort, ignore errors)
    rx_err_t err = RX_RELEASE_PIN(driver->port, driver->pin);
    if (err != k_rx_ok) {
        // Log but don't fail cleanup
        RX_REPORT_ERROR(err, "LED", "Pin release warning");
    }

    driver->initialized = false;
}
```

Example: Bulk Release Helper

```
typedef struct {
    uint8_t port;
    uint8_t pin;
} pin_ref_t;

void release_pins_bulk(const pin_ref_t* pins, uint32_t count)
{
    for (uint32_t i = 0; i < count; i++) {
        rx_err_t err = RX_RELEASE_PIN(pins[i].port, pins[i].pin);
        if (err != k_rx_ok && err != k_rx_err_not_found) {
            // Log unexpected errors (ignore not-found, it's benign)
            RX_REPORT_ERROR(err, "BULK", "Pin release failed");
        }
    }
}
```

See also

[RX_RESERVE_PIN\(\)](#) Reserve a pin (must be called first)
[rx_infrastructure_get_pin_interface\(\)](#) Get pin interface directly
[rx_pin_interface.h](#) Pin interface definition

Since

Version 1.0.0

Definition at line 1967 of file [rx_infrastructure.h](#).

```
01967 #define RX_RELEASE_PIN(port, pin)
01968 ({
01969     rx_err_t err_ = k_rx_err_invalid_state;
01970     rx_pin_interface_t* iface_ = rx_infrastructure_get_pin_interface();
01971     if (iface_ != nullptr) {
01972         err_ = iface_>release_pin(iface_>ctx, port, pin);
01973     }
01974     err_;
01975 })
```

7.15.2.7 RX_REPORT_ERROR

```
#define RX_REPORT_ERROR(
    err,
    component,
    message)
```

Value:

```
do {
    rx_error_interface_t* iface_ = rx_infrastructure_get_error_interface();
    if (iface_ != nullptr) {
        iface_>report_error(iface_>ctx, err, component, message);
    }
} while (0)
```

Report error through global error handler (convenience macro).

Convenience macro that wraps the error reporting interface to provide simple, one-line error reporting throughout the firmware. This is the preferred way to report errors in STAR firmware - it's safer, more concise, and more readable than directly accessing the interface.

7.15.2.8 Macro Expansion

The macro expands to a safe `do { } while (0)` block that:

1. Gets the global error interface
2. Checks for nullptr (safe if infrastructure not initialized)
3. Calls `report_error()` method if interface is valid
4. Returns immediately if interface is nullptr (no-op, no crash)

Expanded code:

```
// RX_REPORT_ERROR(k_rx_err_timeout, "SPI", "Transaction timeout");
// Expands to:
do {
    rx_error_interface_t* iface_ = rx_infrastructure_get_error_interface();
    if (iface_ != nullptr) {
        iface_>report_error(iface_>ctx, k_rx_err_timeout, "SPI", "Transaction timeout");
    }
} while (0)
```

7.15.2.9 Safety Features

- Null-safe: Silently no-ops if infrastructure not initialized
- Statement-like: Can be used anywhere a statement is valid
- No side effects: Parameters evaluated once, no double evaluation
- Scope-safe: `do { } while (0)` ensures proper scope in all contexts
- Thread-safe: Underlying interface is thread-safe

7.15.2.10 Usage Examples

7.15.2.10.1 Example 1: Simple Error Reporting

```
rx_err_t motor_set_velocity(float velocity_mps)
{
    if (velocity_mps > k_max_velocity_mps) {
        RX_REPORT_ERROR(k_rx_err_invalid_arg, "MOTOR", "Velocity exceeds limit");
        return k_rx_err_invalid_arg;
    }
    return k_rx_ok;
}
```

7.15.2.10.2 Example 2: Error Propagation with Reporting

```
rx_err_t motor_init(void)
{
    rx_err_t err = encoder_init();
    if (err != k_rx_ok) {
        RX_REPORT_ERROR(err, "MOTOR", "Encoder init failed");
        return err;
    }

    err = driver_init();
    if (err != k_rx_ok) {
        RX_REPORT_ERROR(err, "MOTOR", "Driver init failed");
        encoder_deinit(); // Cleanup
        return err;
    }

    return k_rx_ok;
}
```

7.15.2.10.3 Example 3: Conditional Reporting (if statement)

```
void motor_control_loop(void)
{
    if (motor_overheated())
        RX_REPORT_ERROR(k_rx_err_hardware, "MOTOR", "Overheating detected");

    if (encoder_fault())
        RX_REPORT_ERROR(k_rx_err_hardware, "MOTOR", "Encoder fault");
}
```

7.15.2.10.4 Example 4: Error with Dynamic Message

```
void motor_check_velocity(float velocity_mps)
{
    char msg[64];
    if (velocity_mps > k_max_velocity_mps) {
        snprintf(msg, sizeof(msg), "Velocity %.2f exceeds max %.2f",
            velocity_mps, k_max_velocity_mps);
        RX_REPORT_ERROR(k_rx_err_invalid_arg, "MOTOR", msg);
    }
}
```

7.15.2.11 When to Use vs. Direct Interface

Scenario	Use Macro	Use Direct Interface
Simple error reporting	[OK] Yes	No
One-line logging	[OK] Yes	No
Need retry logic	No	[OK] Yes
Need error counts	No	[OK] Yes
Need backoff delay	No	[OK] Yes
Performance-critical loop	No	[OK] Yes (cache pointer)

7.15.2.12 Thread Safety

- Thread-safe: Can be called from any thread
- ISR-safe: Can be called from interrupts (with caveats, see below)
- Reentrant: Multiple threads can call concurrently

7.15.2.13 ISR Usage Considerations

While technically safe to call from ISR, consider:

- Error handler may use mutex (can block if ThreadX running)
- String formatting in ISR is discouraged (stack usage)
- Prefer simple error codes, defer detailed reporting to thread context

```
// ISR Example: Simple error reporting
void motor_encoder_isr(void)
{
    if (encoder_error_detected()) {
        RX_REPORT_ERROR(k_rx_err_hardware, "MOTOR", "Encoder ISR error");
    }
}
```

Parameters

in	err	Error code (rx_err_t) - Evaluated once, no side effects
in	component	Component name (const char*) - Must be string literal or valid pointer
in	message	Error message (const char*) - Must be string literal or valid pointer

Note

- Parameters are evaluated once - safe to pass function calls.
- If infrastructure not initialized, this is a silent no-op (no crash).
- Component and message strings are not copied - must remain valid.
- Prefer string literals for component/message (embedded in .rodata).

Warning

- Do NOT use complex expressions in component or message parameters that allocate memory or have side effects.
- String parameters must remain valid for the error's lifetime (use literals).

Example: Parameter Evaluation Safety

```
// SAFE: Function called once
RX_REPORT_ERROR(spi_transmit(), "SPI", "TX failed");

// SAFE: String literals (in .rodata)
RX_REPORT_ERROR(err, "MOTOR", "Velocity error");

// UNSAFE: Dynamic string (stack variable, may be invalid later)
char msg[64];
snprintf(msg, sizeof(msg), "Error %d", code);
RX_REPORT_ERROR(err, "MOTOR", msg); // [FAIL] msg may be invalid when accessed later

// SAFE: Static buffer or string literal
static char msg[64];
snprintf(msg, sizeof(msg), "Error %d", code);
RX_REPORT_ERROR(err, "MOTOR", msg); // [OK] msg remains valid
```

See also

- [rx_infrastructure_get_error_interface\(\)](#) Get error interface directly
- [rx_error_interface.h](#) Error interface definition

Since

Version 1.0.0

Definition at line 1441 of file [rx_infrastructure.h](#).

```
01441 #define RX_REPORT_ERROR(err, component, message)
01442 do {
01443     rx_error_interface_t* iface_ = rx_infrastructure_get_error_interface();
01444     if (iface_ != nullptr) {
01445         iface_>report_error(iface_>ctx, err, component, message);
01446     }
01447 } while (0)
```

7.15.2.14 RX_RESERVE_PIN

```
#define RX_RESERVE_PIN(  
    port,  
    pin,  
    function)
```

Value:

```
{  
    rx_err_t err_ = k_rx_err_invalid_state;  
    rx_pin_interface_t* iface_ = rx_infrastructure_get_pin_interface();  
    if (iface_ != nullptr) {  
        err_ = iface_>reserve_pin(iface_>ctx, port, pin, function);  
    }  
    err_;  
})
```

Reserve GPIO pin through global pin validator (convenience macro with return value).

Convenience macro that wraps the pin reservation interface to provide simple, one-line pin reservation throughout the firmware. This is the preferred way to reserve pins in STAR firmware - it prevents accidental pin conflicts and provides explicit ownership tracking.

7.15.2.15 Macro Expansion (GNU Statement Expression)

This macro uses GNU C extension (`{ ... }`) (statement expression) to:

1. Get the global pin interface
2. Check for nullptr (returns error if infrastructure not initialized)
3. Call `reserve_pin()` method if interface is valid
4. Return `rx_err_t` error code as the macro's "value"

Expanded code:

```
// rx_err_t err = RX_RESERVE_PIN(0xA, 5, "SPI_COPI");  
// Expands to:  
rx_err_t err = ({  
    rx_err_t err_ = k_rx_err_invalid_state;  
    rx_pin_interface_t* iface_ = rx_infrastructure_get_pin_interface();  
    if (iface_ != nullptr) {  
        err_ = iface_>reserve_pin(iface_>ctx, 0xA, 5, "SPI_COPI");  
    }  
    err_; // This becomes the "return value" of the expression  
});
```

7.15.2.16 Pin Conflict Prevention

The pin validator ensures that:

- No double reservation: A pin can only be reserved once
- Explicit ownership: Each pin knows which function owns it
- Traceable usage: Pin assignments are tracked in a global table
- Atomic operations: Reservation is thread-safe (mutex protected)

```
Module A: RX_RESERVE_PIN(0xA, 5, "SPI_COPI") -> k_rx_ok  
Module B: RX_RESERVE_PIN(0xA, 5, "I2C_SDA") -> k_rx_err_resource_busy (conflict!)
```


7.15.2.17 Typical Usage Patterns

7.15.2.17.1 Pattern 1: Reserve-Check-Cleanup on Error

```

rx_err_t spi_driver_init(void)
{
    // Reserve CIPO (first pin)
    rx_err_t err = RX_RESERVE_PIN(0xA, 0, "SPI_CIPO");
    if (err != k_rx_ok) {
        RX_REPORT_ERROR(err, "SPI", "Pin PA0 (CIPO) conflict");
        return err;
    }

    // Reserve COPI (second pin)
    err = RX_RESERVE_PIN(0xA, 1, "SPI_COPI");
    if (err != k_rx_ok) {
        RX_RELEASE_PIN(0xA, 0); // Clean up first pin
        RX_REPORT_ERROR(err, "SPI", "Pin PA1 (COPI) conflict");
        return err;
    }

    // Reserve CLK (third pin)
    err = RX_RESERVE_PIN(0xA, 2, "SPI_CLK");
    if (err != k_rx_ok) {
        RX_RELEASE_PIN(0xA, 0); // Clean up first two pins
        RX_RELEASE_PIN(0xA, 1);
        RX_REPORT_ERROR(err, "SPI", "Pin PA2 (CLK) conflict");
        return err;
    }

    // All pins successfully reserved - proceed with hardware config
    configure_spi_hardware();
    return k_rx_ok;
}

```

7.15.2.17.2 Pattern 2: Bulk Reservation Helper

```

typedef struct {
    uint8_t port;
    uint8_t pin;
    const char* function;
} pin_def_t;

// Define all SPI pins
static const pin_def_t k_spi_pins[] = {
    {0xA, 0, "SPI_CIPO"},
    {0xA, 1, "SPI_COPI"},
    {0xA, 2, "SPI_CLK"},
    {0xA, 3, "SPI_CS"},
};

rx_err_t reserve_spi_pins(void)
{
    for (uint32_t i = 0; i < ARRAY_SIZE(k_spi_pins); i++) {
        rx_err_t err = RX_RESERVE_PIN(
            k_spi_pins[i].port,
            k_spi_pins[i].pin,
            k_spi_pins[i].function
        );
        if (err != k_rx_ok) {
            // Cleanup all previously reserved pins
            for (uint32_t j = 0; j < i; j++) {
                RX_RELEASE_PIN(k_spi_pins[j].port, k_spi_pins[j].pin);
            }
            return err;
        }
    }
    return k_rx_ok;
}

```

7.15.2.17.3 Pattern 3: Inline Error Check

```

void motor_init(void)
{
    // Reserve and check in single if-statement
    if (RX_RESERVE_PIN(0xB, 4, "MOTOR_PWM") != k_rx_ok) {
        // Handle error
        return k_rx_err_resource_busy;
    }
}

```

```

}
if (RX_RESERVE_PIN(0xB, 5, "MOTOR_DIR") != k_rx_ok) {
    RX_RELEASE_PIN(0xB, 4); // Cleanup
    return k_rx_err_resource_busy;
}
}

```

7.15.2.18 Port and Pin Numbering

Port numbers: Use hexadecimal for RX72N ports (0x0 = Port 0, 0xA = Port A, etc.) Pin numbers: Use decimal 0-7 for pin within port

Port Letter	Hex Value	Example Pin	Macro Call
Port 0	0x0	P0.5	<code>RX_RESERVE_PIN(0x0, 5, "name")</code>
Port A	0xA	PA.3	<code>RX_RESERVE_PIN(0xA, 3, "name")</code>
Port B	0xB	PB.7	<code>RX_RESERVE_PIN(0xB, 7, "name")</code>
Port E	0xE	PE.1	<code>RX_RESERVE_PIN(0xE, 1, "name")</code>

Parameters

in	port	Port number in hexadecimal (0x0-0xF for RX72N)
in	pin	Pin number within port (0-7)
in	function	Function name (const char* string literal) - identifies owner

Returns

`rx_err_t` Error code indicating success or failure

Return values

<code>k_rx_ok</code>	Success, pin reserved
<code>k_rx_err_invalid_state</code>	Infrastructure not initialized
<code>k_rx_err_resource_busy</code>	Pin already reserved by another function
<code>k_rx_err_invalid_arg</code>	Invalid port or pin number
<code>k_rx_err_null_ptr</code>	Null function name

Note

GNU C Extension: Uses statement expression (`{ }`) - requires GCC/Clang.

Returns value: Can be assigned to variable or used directly in if-statement.

Thread-safe: Can be called from multiple threads concurrently.

Function string: Must be string literal or valid pointer (not copied).

No cleanup on error: Caller must manually release pins on partial failure.

Warning

- Must check return value: Ignoring errors can lead to pin conflicts.
- Cleanup required: On error, release any previously reserved pins.
- String lifetime: Function name must remain valid (use string literals).
- Not ISR-safe: Uses mutex internally, do NOT call from ISR.

Example: Complete Module Pin Management

```
typedef struct {
    bool pins_reserved;
    // ... other state
} module_state_t;

rx_err_t module_init(module_state_t* state)
{
    // Reserve all module pins
    rx_err_t err = RX_RESERVE_PIN(0xA, 0, "MODULE_PIN0");
    if (err != k_rx_ok) {
        return err;
    }

    err = RX_RESERVE_PIN(0xA, 1, "MODULE_PIN1");
    if (err != k_rx_ok) {
        RX_RELEASE_PIN(0xA, 0); // Cleanup
        return err;
    }

    state->pins_reserved = true;
    return k_rx_ok;
}

void module_deinit(module_state_t* state)
{
    if (state->pins_reserved) {
        RX_RELEASE_PIN(0xA, 0);
        RX_RELEASE_PIN(0xA, 1);
        state->pins_reserved = false;
    }
}
```

See also

- [RX_RELEASE_PIN\(\)](#) Release a reserved pin
- [rx_infrastructure_get_pin_interface\(\)](#) Get pin interface directly
- [rx_pin_interface.h](#) Pin interface definition

Since

Version 1.0.0

Definition at line 1658 of file [rx_infrastructure.h](#).

```
01658 #define RX_RESERVE_PIN(port, pin, function)
01659 ({
01660     rx_err_t      err__ = k_rx_err_invalid_state;
01661     rx_pin_interface_t* iface__ = rx_infrastructure_get_pin_interface();
01662     if (iface__ != nullptr) {
01663         err__ = iface__->reserve_pin(iface__->ctx, port, pin, function);
01664     }
01665     err__;
01666 })
```

7.15.3 Function Documentation

7.15.3.1 rx_infrastructure_deinit()

```
rx_err_t rx_infrastructure_deinit (  
    void ) [nodiscard]
```

Deinitialize global infrastructure (graceful shutdown only).

Performs graceful shutdown of all infrastructure services. In embedded systems, this function is RARELY CALLED because the system typically runs indefinitely until power loss or reset. However, it's provided for:

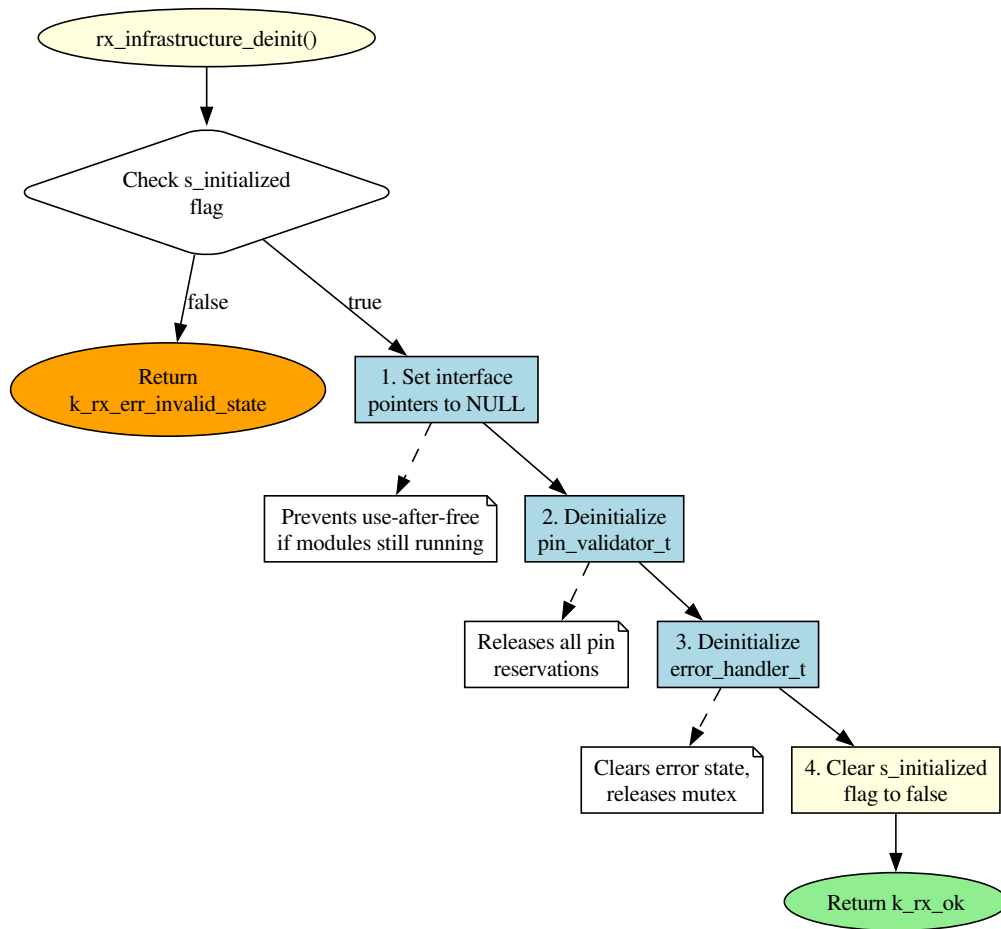
- Unit testing (setup/teardown)
- Firmware update procedures (safe state before flash)
- Power management (deep sleep entry)
- Development/debugging (clean restarts)

7.15.3.2 Deinitialization Algorithm

Teardown occurs in REVERSE order of initialization (dependency order):

1. Validate Initialized State:
 - Check `s_initialized` flag
 - Return error if not initialized (nothing to clean up)
2. Clear Interface Pointers:
 - Set global interface pointers to nullptr
 - Prevents use-after-free if modules still running
3. Deinitialize Pin Validator:
 - Call `pin_validator_deinit()`
 - Releases all pin reservations
 - Clears pin reservation table
4. Deinitialize Error Handler:
 - Call `error_handler_deinit()`
 - Clears error counters and state
 - Releases mutex resources (if ThreadX running)
5. Clear Initialization Flag:
 - Reset `s_initialized` to false
 - Allows re-initialization (useful for tests)

7.15.3.3 Shutdown Flow Diagram



7.15.3.4 Typical Usage Scenarios

7.15.3.4.1 Scenario 1: Unit Testing (Most Common Use)

```

void test_motor_control(void)
{
    // Setup: Initialize infrastructure for test
    rx_err_t err = rx_infrastructure_init();
    assert(err == k_rx_ok);

    // Test: Run motor control tests
    test_motor_set_velocity();
    test_motor_error_handling();

    // Teardown: Clean up for next test
    err = rx_infrastructure_deinit();
    assert(err == k_rx_ok);
}

```

7.15.3.4.2 Scenario 2: Firmware Update Preparation

```

void enter_bootloader_mode(void)
{
    // 1. Stop all threads
    motor_stop_all();
    usb_disconnect();
    spi_shutdown();

    // 2. Deinitialize infrastructure
    rx_infrastructure_deinit();

    // 3. Jump to bootloader
    bootloader_entry();
}

```

7.15.3.4.3 Scenario 3: Deep Sleep Power Management

```
void enter_deep_sleep(void)
{
    // 1. Save critical state to backup SRAM
    save_state_to_backup_ram();

    // 2. Shutdown peripherals
    usb_suspend();
    spi_sleep();

    // 3. Deinitialize infrastructure
    rx_infrastructure_deinit();

    // 4. Enter low-power mode
    rx72n_enter_deep_sleep();

    // 5. After wakeup: Re-initialize everything
    rx_infrastructure_init();
    restore_state_from_backup_ram();
}
```

Returns

[rx_err_t](#) Error code

Return values

k_rx_ok	Success, infrastructure fully deinitialized
k_rx_err_invalid_state	Not initialized (nothing to clean up)

Precondition

`s_initialized` flag must be true (infrastructure must be initialized)

All threads using infrastructure must be stopped or suspended

No modules should call [RX_REPORT_ERROR\(\)](#) or [RX_RESERVE_PIN\(\)](#) during deinit

ThreadX threads should be suspended (if ThreadX is running)

Postcondition

`s_initialized` flag set to false

[rx_infrastructure_get_error_interface\(\)](#) returns nullptr

[rx_infrastructure_get_pin_interface\(\)](#) returns nullptr

All error counters cleared

All pin reservations released

Infrastructure can be re-initialized via [rx_infrastructure_init\(\)](#)

Note

NOT thread-safe. Caller must ensure no threads are using infrastructure.

Calling when not initialized returns `k_rx_err_invalid_state` (safe, no-op).

After deinit, all [RX_REPORT_ERROR\(\)](#) and [RX_RESERVE_PIN\(\)](#) calls will fail silently.

Warning

DANGER: Calling this while modules are still using infrastructure will cause crashes. Ensure all threads are stopped first.

Do NOT call from ISR (may block on mutex cleanup).

Thread Safety:

- NOT thread-safe
- Caller must ensure exclusive access (no concurrent infrastructure usage)
- Suspend all ThreadX threads before calling

Performance:

- Execution time: ~200 us @ 240 MHz
- No heap operations (all cleanup is static)

Example: Safe Shutdown Sequence

```
void system_shutdown(void)
{
    // 1. Stop all application threads
    tx_thread_suspend(&motor_thread);
    tx_thread_suspend(&usb_thread);
    tx_thread_suspend(&comm_thread);

    // 2. Disable interrupts (prevent ISRs from using infrastructure)
    __disable_irq();

    // 3. Shutdown modules (in reverse init order)
    motor_shutdown();
    usb_shutdown();
    spi_shutdown();

    // 4. Deinitialize infrastructure (last)
    rx_err_t err = rx_infrastructure_deinit();
    if (err != k_rx_ok) {
        // Log error (can't use RX_REPORT_ERROR anymore!)
        debug_uart_printf("Deinit failed: %d\n", err);
    }

    // 5. Proceed with shutdown (reset, bootloader, sleep, etc.)
    trigger_system_reset();
}
```

Example: Test Fixture Pattern

```
typedef struct {
    bool infrastructure_initialized;
} test_fixture_t;

void test_setup(test_fixture_t* fixture)
{
    rx_err_t err = rx_infrastructure_init();
    assert(err == k_rx_ok);
    fixture->infrastructure_initialized = true;
}

void test_teardown(test_fixture_t* fixture)
{
    if (fixture->infrastructure_initialized) {
        rx_err_t err = rx_infrastructure_deinit();
        assert(err == k_rx_ok);
        fixture->infrastructure_initialized = false;
    }
}

void run_test_suite(void)
{
}
```

```

test_fixture_t fixture = {};

test_setup(&fixture);
test_motor_functions();
test_teardown(&fixture);

test_setup(&fixture);
test_usb_functions();
test_teardown(&fixture);
}

```

Example: Handling Double-Deinit

```

rx_err_t err = rx_infrastructure_deinit();
if (err == k_rx_err_invalid_state) {
    // Already deinitialized - this is safe, continue
} else if (err != k_rx_ok) {
    // Real deinit error - investigate
    debug_uart_printf("Deinit error: %d\n", err);
}

```

See also

[rx_infrastructure_init\(\)](#) Initialize infrastructure
[error_handler_deinit\(\)](#) Error handler cleanup
[pin_validator_deinit\(\)](#) Pin validator cleanup

Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 5: [OK] 4 preconditions, 6 postconditions
- Rule 7: [OK] All return values checked

Deinitialize global infrastructure (graceful shutdown only).

Releases all global infrastructure resources in reverse initialization order. Clears the initialized flag so interface accessors return NULL.

Algorithm Steps:

1. Check if initialized (return early if not)
2. Deinitialize pin validator
3. Deinitialize error handler
4. Clear initialized flag

Cleanup Order:

Resources are released in reverse initialization order to respect dependencies:

- Pin validator (uses error handler for logging)
- Error handler (foundation service)

Precondition

None - safe to call anytime

Postcondition

`s_infrastructure_initialized` == false

Interface accessors return `nullptr`

All `s_global_*` structures are in undefined state

Note

Thread Safety: NOT thread-safe

Idempotent: Safe to call multiple times

Warning

Call only after all threads have stopped using infrastructure

After deinit, interface accessors return `nullptr`

Example:

```
// During system shutdown
motor_deinit(&motor);
sensor_deinit(&sensor);

// Release infrastructure last
rx_infrastructure_deinit();
```

See also

[rx_infrastructure_init\(\)](#) Initialization function

Since

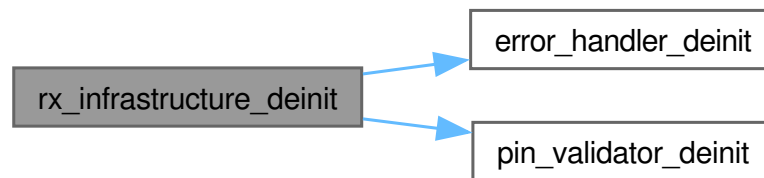
Version 1.0.0

Definition at line 464 of file [rx_infrastructure.c](#).

```
00465 {
00466     if (!s_infrastructure_initialized) {
00467         return k_rx_ok;
00468     }
00469
00470     rx_log_info(s_tag, "Deinitializing global infrastructure");
00471
00472     /* Deinitialize pin validator */
00473     rx_err_t err = pin_validator_deinit(&s_global_pin_validator);
00474     if (err != k_rx_ok) {
00475         rx_log_warn(s_tag, "Pin validator deinit failed");
00476         /* Continue with cleanup anyway */
00477     }
00478
00479     /* Deinitialize error handler */
00480     err = error_handler_deinit(&s_global_error_handler);
00481     if (err != k_rx_ok) {
00482         rx_log_warn(s_tag, "Error handler deinit failed");
00483         return err;
00484     }
00485
00486     s_infrastructure_initialized = false;
00487
00488     rx_log_info(s_tag, "Global infrastructure deinitialized");
00489
00490     return k_rx_ok;
00491 }
```

References [error_handler_deinit\(\)](#), [k_rx_ok](#), [pin_validator_deinit\(\)](#), [rx_log_info](#), [rx_log_warn](#), [s_global_error_handler](#), [s_global_pin_validator](#), [s_infrastructure_initialized](#), and [s_tag](#).

Here is the call graph for this function:



7.15.3.5 rx_infrastructure_get_error_interface()

```
rx_error_interface_t * rx_infrastructure_get_error_interface (
    void )
```

Get global error handler interface (thread-safe accessor).

Returns a pointer to the global `rx_error_interface_t` that provides access to the error handling subsystem. This is the primary way for modules to report errors, check error counts, and implement retry logic with exponential backoff.

The returned pointer is valid for the entire program lifetime after successful initialization via [rx_infrastructure_init\(\)](#). Modules can cache this pointer for performance-critical code paths.

7.15.3.6 Pointer Lifetime and Validity

```
System Boot
|
| +- rx_infrastructure_init()
|   +- Interface pointer becomes valid
|
| +- Runtime (entire program lifetime)
|   +- Pointer remains valid
|   +- Safe to call from any thread
|   +- Safe to cache in module state
|
| +- rx_infrastructure_deinit() [optional]
|   +- Pointer becomes NULL (rare in embedded)
```

7.15.3.7 Usage Patterns

7.15.3.7.1 Pattern 1: Direct Inline Usage (Recommended for Simple Cases)

```
// Simple error reporting - get interface each time
void motor_set_velocity(float velocity_mps)
{
    if (velocity_mps > k_max_velocity_mps) {
        RX_REPORT_ERROR(k_rx_err_invalid_arg, "MOTOR", "Velocity exceeds limit");
        return;
    }
    // ...
}
```

7.15.3.7.2 Pattern 2: Cached Pointer (Recommended for Performance-Critical Code)

```
// Cache interface pointer in module state
typedef struct {
    rx_error_interface_t* error_iface; // Cached at init
    // ... other state
} motor_state_t;

void motor_init(motor_state_t* state)
{
    // Cache interface pointer once during init
    state->error_iface = rx_infrastructure_get_error_interface();
    if (state->error_iface == nullptr) {
```

```

    return k_rx_err_not_initialized;
}
// ...
}

void motor_control_loop(motor_state_t* state)
{
    // Use cached pointer (no function call overhead)
    if (motor_error_detected()) {
        state->error_iface->report_error(
            state->error_iface->ctx,
            k_rx_err_hardware,
            "MOTOR",
            "Encoder fault"
        );
    }
}
}

```

7.15.3.7.3 Pattern 3: Defensive Null Check (Recommended for Critical Paths)

```

void critical_error_handler(void)
{
    rx_error_interface_t* iface = rx_infrastructure_get_error_interface();
    if (iface == nullptr) {
        // Infrastructure not ready - fallback to direct UART logging
        debug_uart_printf("CRITICAL: Infrastructure not initialized\n");
        return;
    }

    // Use interface normally
    iface->report_error(iface->ctx, k_rx_err_hardware, "CRITICAL", "Hardware fault");
}

```

7.15.3.7.4 Pattern 4: Advanced - Retry Logic with Backoff

```

rx_err_t motor_send_command_with_retry(uint8_t* data, uint32_t len)
{
    rx_error_interface_t* iface = rx_infrastructure_get_error_interface();
    if (iface == nullptr) {
        return k_rx_err_not_initialized;
    }

    // Check if we've exceeded retry limit
    if (iface->is_retry_limit_reached != nullptr &&
        iface->is_retry_limit_reached(iface->ctx, "MOTOR")) {
        return k_rx_err_retry_limit;
    }

    // Attempt command
    rx_err_t err = motor_send_command_raw(data, len);
    if (err != k_rx_ok) {
        // Report error
        iface->report_error(iface->ctx, err, "MOTOR", "Command failed");

        // Get exponential backoff delay
        if (iface->get_backoff_delay != nullptr) {
            uint32_t delay_ms = iface->get_backoff_delay(iface->ctx, "MOTOR");
            tx_thread_sleep(delay_ms); // Wait before retry
        }

        return err;
    }

    // Success - reset retry counter
    if (iface->reset_retry_counter != nullptr) {
        iface->reset_retry_counter(iface->ctx, "MOTOR");
    }

    return k_rx_ok;
}

```

7.15.3.8 Interface Method Availability

All function pointers in the returned interface are guaranteed non-NULL except:

- `get_component_error_count` (optional, may be NULL)

- `is_retry_limit_reached` (optional, may be NULL)
- `reset_retry_counter` (optional, may be NULL)
- `get_backoff_delay` (optional, may be NULL)

Always check optional methods before calling:

```
if (iface->get_backoff_delay != nullptr) {
    uint32_t delay = iface->get_backoff_delay(iface->ctx, "MOTOR");
}
```

7.15.3.9 Thread Safety Guarantees

Operation	Thread-Safe?	Details
Calling this function	[OK] Yes	Read-only after init
Caching returned pointer	[OK] Yes	Pointer never changes
Calling interface methods	[OK] Yes	Methods are thread-safe
Calling during init/deinit	[X] No	Undefined behavior

Returns

`rx_error_interface_t*` Pointer to global error interface

Return values

non-NULL	Success, interface is ready for use
NULL	Infrastructure not initialized (call <code>rx_infrastructure_init()</code> first)

Precondition

Infrastructure must be initialized via `rx_infrastructure_init()`

Must NOT be called during `rx_infrastructure_deinit()`

Postcondition

Returned pointer remains valid until `rx_infrastructure_deinit()`

All required interface methods are non-NULL

Interface methods are thread-safe (can be called from any thread)

Note

Thread-safe: Can be called from any thread after initialization.

Performance: Simple pointer return (~ 1 cycle), safe to call frequently.

Caching: Safe to cache returned pointer in module state for performance.

Null check: Always check for nullptr in defensive code paths.

Warning

Do NOT call during [rx_infrastructure_init\(\)](#) or [rx_infrastructure_deinit\(\)](#).
 NULL return means infrastructure not ready - check initialization order.

Performance:

- Execution time: ~5 cycles @ 240 MHz (simple pointer read)
- No overhead: Direct memory access, no function calls
- Cache-friendly: Same pointer returned every call

Example: Initialization-Time Validation

```
rx_err_t module_init(void)
{
    // Verify infrastructure is ready
    rx_error_interface_t* iface = rx_infrastructure_get_error_interface();
    if (iface == nullptr) {
        // FATAL: Infrastructure not initialized
        // This means rx_infrastructure_init() was not called first
        debug_uart_printf("FATAL: Infrastructure not ready\n");
        return k_rx_err_not_initialized;
    }

    // Verify required methods are present
    if (iface->report_error == nullptr) {
        debug_uart_printf("FATAL: Invalid error interface\n");
        return k_rx_err_invalid_state;
    }

    // Cache pointer for runtime use
    g_module_state.error_iface = iface;
    return k_rx_ok;
}
```

See also

[rx_infrastructure_init\(\)](#) Initialize infrastructure to make this return non-NULL
[rx_infrastructure_deinit\(\)](#) Cleanup infrastructure (sets pointer to NULL)
[rx_error_interface.h](#) Abstract error handler interface definition
[RX_REPORT_ERROR\(\)](#) Convenience macro using this accessor

Since

Version 1.0.0

Get global error handler interface (thread-safe accessor).

Returns a pointer to the global error handler interface structure. The interface provides function pointers for error recovery operations.

Interface Functions:

Function	Description
try_recover	Attempt recovery with retry and backoff

Function	Description
reset	Reset error handler state

Precondition

[rx_infrastructure_init\(\)](#) should have been called

Postcondition

Returned pointer valid for program lifetime (if not nullptr)

Note

Thread Safety: Safe - returns pointer to static data

The returned pointer should not be freed

Example:

```
rx_error_interface_t* err_if = rx_infrastructure_get_error_interface();
if (err_if == nullptr) {
    rx_log_error("APP", "Infrastructure not initialized");
    return k_rx_err_not_initialized;
}

// Use error handler for retry logic
rx_err_t result = err_if->try_recover(err_if->ctx, my_operation, &context);
```

See also

[rx_error_interface_t](#) Interface structure
[rx_infrastructure_init\(\)](#) Must be called first

Since

Version 1.0.0

Definition at line 531 of file [rx_infrastructure.c](#).

```
00532 {
00533     if (!s_infrastructure_initialized) {
00534         return nullptr;
00535     }
00536
00537     return &s_global_error_interface;
00538 }
```

References [s_global_error_interface](#), and [s_infrastructure_initialized](#).

7.15.3.10 rx_infrastructure_get_pin_interface()

```
rx_pin_interface_t * rx_infrastructure_get_pin_interface (
    void )
```

Get global pin validator interface (thread-safe accessor).

Returns a pointer to the global [rx_pin_interface_t](#) that provides access to the pin validation and reservation subsystem. This interface prevents accidental pin conflicts by tracking which GPIO pins are reserved by which modules.

The returned pointer is valid for the entire program lifetime after successful initialization via [rx_infrastructure_init\(\)](#). Modules can cache this pointer for performance-critical code paths.

7.15.3.11 Pin Reservation System Overview

The pin validator ensures that:

1. No pin conflicts: Two modules cannot reserve the same pin
2. Explicit ownership: Each pin knows which module owns it
3. Traceable usage: Pin assignments documented in reservation table
4. Safe release: Pins can only be released by their owner

Module Init Sequence:

```
|
+- SPI Driver Init
|   +- Reserve PA0 ("SPI_CIPO") -> Success
|   +- Reserve PA1 ("SPI_COPI") -> Success
|   +- Reserve PA2 ("SPI_CLK") -> Success
|
+- I2C Driver Init
|   +- Reserve PA0 ("I2C_SDA") -> FAIL (already SPI_CIPO)
|   +- Init fails, prevents pin conflict
|
+- USB Driver Init
|   +- Reserve PB0 ("USB_DP") -> Success
|   +- Reserve PB1 ("USB_DM") -> Success
```

7.15.3.12 Usage Patterns

7.15.3.12.1 Pattern 1: Module Initialization with Pin Reservation

```
rx_err_t spi_driver_init(void)
{
    // Reserve all SPI pins before configuring hardware
    rx_err_t err;

    err = RX_RESERVE_PIN(0xA, 0, "SPI_CIPO"); // Port A, Pin 0
    if (err != k_rx_ok) {
        RX_REPORT_ERROR(err, "SPI", "Pin PA0 already in use");
        return err;
    }

    err = RX_RESERVE_PIN(0xA, 1, "SPI_COPI"); // Port A, Pin 1
    if (err != k_rx_ok) {
        RX_RELEASE_PIN(0xA, 0); // Clean up previous reservation
        RX_REPORT_ERROR(err, "SPI", "Pin PA1 already in use");
        return err;
    }

    err = RX_RESERVE_PIN(0xA, 2, "SPI_CLK"); // Port A, Pin 2
    if (err != k_rx_ok) {
        RX_RELEASE_PIN(0xA, 0); // Clean up
        RX_RELEASE_PIN(0xA, 1);
        RX_REPORT_ERROR(err, "SPI", "Pin PA2 already in use");
        return err;
    }

    // All pins reserved successfully - proceed with hardware config
    configure_spi_hardware();
    return k_rx_ok;
}
```

7.15.3.12.2 Pattern 2: Cached Pointer for Performance

```
typedef struct {
    rx_pin_interface_t* pin_iface; // Cached at init
    uint8_t port;
    uint8_t pin;
} gpio_driver_state_t;

rx_err_t gpio_init(gpio_driver_state_t* state, uint8_t port, uint8_t pin)
{
    // Cache interface pointer
    state->pin_iface = rx_infrastructure_get_pin_interface();
}
```

```

if (state->pin_iface == nullptr) {
    return k_rx_err_not_initialized;
}

// Reserve pin using cached interface
rx_err_t err = state->pin_iface->reserve_pin(
    state->pin_iface->ctx, port, pin, "GPIO"
);
if (err != k_rx_ok) {
    return err;
}

state->port = port;
state->pin = pin;
return k_rx_ok;
}

void gpio_deinit(gpio_driver_state_t* state)
{
    // Release pin using cached interface
    if (state->pin_iface != nullptr) {
        state->pin_iface->release_pin(state->pin_iface->ctx, state->port, state->pin);
    }
}

```

7.15.3.12.3 Pattern 3: Dynamic Pin Configuration (Runtime Changes)

```

rx_err_t motor_reconfigure_encoder_pins(uint8_t new_port, uint8_t new_pin_a, uint8_t new_pin_b)
{
    rx_pin_interface_t* iface = rx_infrastructure_get_pin_interface();
    if (iface == nullptr) {
        return k_rx_err_not_initialized;
    }

    // Release old pins
    iface->release_pin(iface->ctx, motor_state.old_port, motor_state.old_pin_a);
    iface->release_pin(iface->ctx, motor_state.old_port, motor_state.old_pin_b);

    // Reserve new pins
    rx_err_t err = iface->reserve_pin(iface->ctx, new_port, new_pin_a, "MOTOR_ENC_A");
    if (err != k_rx_ok) {
        // Failed to reserve - try to restore old pins
        iface->reserve_pin(iface->ctx, motor_state.old_port, motor_state.old_pin_a, "MOTOR_ENC_A");
        iface->reserve_pin(iface->ctx, motor_state.old_port, motor_state.old_pin_b, "MOTOR_ENC_B");
        return err;
    }

    err = iface->reserve_pin(iface->ctx, new_port, new_pin_b, "MOTOR_ENC_B");
    if (err != k_rx_ok) {
        // Failed - restore old configuration
        iface->release_pin(iface->ctx, new_port, new_pin_a);
        iface->reserve_pin(iface->ctx, motor_state.old_port, motor_state.old_pin_a, "MOTOR_ENC_A");
        iface->reserve_pin(iface->ctx, motor_state.old_port, motor_state.old_pin_b, "MOTOR_ENC_B");
        return err;
    }

    // Success - update state
    motor_state.old_port = new_port;
    motor_state.old_pin_a = new_pin_a;
    motor_state.old_pin_b = new_pin_b;
    return k_rx_ok;
}

```

7.15.3.12.4 Pattern 4: Defensive Null Check

```

void debug_print_pin_usage(void)
{
    rx_pin_interface_t* iface = rx_infrastructure_get_pin_interface();
    if (iface == nullptr) {
        debug_uart_printf("Pin validator not initialized\n");
        return;
    }

    // Check if debug method exists (optional interface method)
    if (iface->print_reservations != nullptr) {
        iface->print_reservations(iface->ctx);
    } else {
        debug_uart_printf("Pin reservation printing not supported\n");
    }
}

```


7.15.3.13 Interface Method Availability

All function pointers in the returned interface are guaranteed non-NULL except:

- `is_pin_reserved` (optional, may be NULL)
- `get_pin_function` (optional, may be NULL)
- `print_reservations` (optional, may be NULL)

Always check optional methods before calling:

```
if (iface->is_pin_reserved != nullptr) {
    bool reserved = iface->is_pin_reserved(iface->ctx, port, pin);
}
```

7.15.3.14 Thread Safety Guarantees

Operation	Thread-Safe?	Details
Calling this function	[OK] Yes	Read-only after init
Caching returned pointer	[OK] Yes	Pointer never changes
Calling interface methods	[OK] Yes	Methods use mutex protection
Concurrent reserve/release	[OK] Yes	Atomic operations
Calling during init/deinit	[X] No	Undefined behavior

Returns

`rx_pin_interface_t*` Pointer to global pin interface

Return values

non-NULL	Success, interface is ready for use
NULL	Infrastructure not initialized (call <code>rx_infrastructure_init()</code> first)

Precondition

Infrastructure must be initialized via `rx_infrastructure_init()`

Must NOT be called during `rx_infrastructure_deinit()`

Postcondition

Returned pointer remains valid until `rx_infrastructure_deinit()`

All required interface methods are non-NULL

Interface methods are thread-safe (can be called from any thread)

Note

Thread-safe: Can be called from any thread after initialization.

Performance: Simple pointer return (~ 1 cycle), safe to call frequently.

Caching: Safe to cache returned pointer in module state for performance.

Null check: Always check for nullptr in defensive code paths.

Warning

Do NOT call during `rx_infrastructure_init()` or `rx_infrastructure_deinit()`.

NULL return means infrastructure not ready - check initialization order.

Pin reservation errors mean conflict - do NOT proceed with hardware config.

Performance:

- Execution time: ~ 5 cycles @ 240 MHz (simple pointer read)
- No overhead: Direct memory access, no function calls
- Cache-friendly: Same pointer returned every call

Example: Pin Conflict Detection

```
void diagnose_pin_conflict(uint8_t port, uint8_t pin)
{
    rx_pin_interface_t* iface = rx_infrastructure_get_pin_interface();
    if (iface == nullptr) {
        debug_uart_printf("Pin validator not initialized\n");
        return;
    }

    // Check if pin is already reserved
    if (iface->is_pin_reserved != nullptr) {
        bool reserved = iface->is_pin_reserved(iface->ctx, port, pin);
        if (reserved) {
            // Get function name
            if (iface->get_pin_function != nullptr) {
                const char* function = iface->get_pin_function(iface->ctx, port, pin);
                debug_uart_printf("Pin P%X.%d reserved for: %s\n", port, pin, function);
            } else {
                debug_uart_printf("Pin P%X.%d is reserved\n", port, pin);
            }
        } else {
            debug_uart_printf("Pin P%X.%d is available\n", port, pin);
        }
    }
}
```

Example: Bulk Pin Reservation with Cleanup

```
typedef struct {
    uint8_t port;
    uint8_t pin;
    const char* function;
} pin_reservation_t;

rx_err_t reserve_pins_bulk(const pin_reservation_t* pins, uint32_t count)
{
    rx_pin_interface_t* iface = rx_infrastructure_get_pin_interface();
    if (iface == nullptr) {
        return k_rx_err_not_initialized;
    }

    // Reserve all pins
    for (uint32_t i = 0; i < count; i++) {
        rx_err_t err = iface->reserve_pin(
            iface->ctx, pins[i].port, pins[i].pin, pins[i].function
        );
        if (err != k_rx_ok) {
```

```

    // Cleanup: Release all previously reserved pins
    for (uint32_t j = 0; j < i; j++) {
        iface->release_pin(iface->ctx, pins[j].port, pins[j].pin);
    }
    return err;
}

return k_rx_ok;
}

```

See also

[rx_infrastructure_init\(\)](#) Initialize infrastructure to make this return non-NULL

[rx_infrastructure_deinit\(\)](#) Cleanup infrastructure (sets pointer to NULL)

[rx_pin_interface.h](#) Abstract pin validator interface definition

[RX_RESERVE_PIN\(\)](#) Convenience macro using this accessor

[RX_RELEASE_PIN\(\)](#) Convenience macro using this accessor

Since

Version 1.0.0

Get global pin validator interface (thread-safe accessor).

Returns a pointer to the global pin validator interface structure. The interface provides function pointers for GPIO pin management.

Interface Functions:

Function	Description
claim_pin	Claim a GPIO pin for a specific module
release	Release a previously claimed GPIO pin
is_claimed	Check if a pin is already claimed

Precondition

[rx_infrastructure_init\(\)](#) should have been called

Postcondition

Returned pointer valid for program lifetime (if not nullptr)

Note

Thread Safety: Safe for read - returns pointer to static data

The returned pointer should not be freed

Warning

Pin claim/release operations may not be thread-safe

Example:

```
rx_pin_interface_t* pin_if = rx_infrastructure_get_pin_interface();
if (pin_if == nullptr) {
    rx_log_error("MOTOR", "Infrastructure not initialized");
    return k_rx_err_not_initialized;
}

// Claim GPIO pin for motor PWM
rx_err_t err = pin_if->claim_pin(pin_if->ctx,
                                k_port_1, k_pin_5,
                                "MOTOR_PWM");

if (err != k_rx_ok) {
    rx_log_error("MOTOR", "Pin P15 already in use");
    return err;
}
```

See also

`rx_pin_interface_t` Interface structure
[rx_infrastructure_init\(\)](#) Must be called first

Since

Version 1.0.0

Definition at line 587 of file [rx_infrastructure.c](#).

```
00588 {
00589     if (!s_infrastructure_initialized) {
00590         return nullptr;
00591     }
00592
00593     return &s_global_pin_interface;
00594 }
```

References [s_global_pin_interface](#), and [s_infrastructure_initialized](#).

7.15.3.15 rx_infrastructure_init()

```
rx_err_t rx_infrastructure_init (
    void ) [nodiscard]
```

Initialize global infrastructure (MUST BE FIRST CALL IN main()).

Performs one-time initialization of all global infrastructure services. This function MUST be the first thing called in main(), before any other module initialization and before starting ThreadX. Failure to initialize infrastructure will cause nullptr pointer dereferences when other modules attempt to use [RX_REPORT_ERROR\(\)](#) or [RX_RESERVE_PIN\(\)](#).

7.15.3.16 Initialization Algorithm

The function performs initialization in strict dependency order:

1. Initialize Error Handler:

- Allocate static `error_handler_t` instance
- Configure with default parameters (max_retries=3, backoffs)
- Call `error_handler_init()` to prepare state
- Extract `rx_error_interface_t` via `error_handler_get_interface()`

2. Initialize Pin Validator:

- Allocate static `pin_validator_t` instance
- Zero-initialize pin reservation table
- Call `pin_validator_init()` to prepare state
- Extract `rx_pin_interface_t` via `pin_validator_get_interface()`

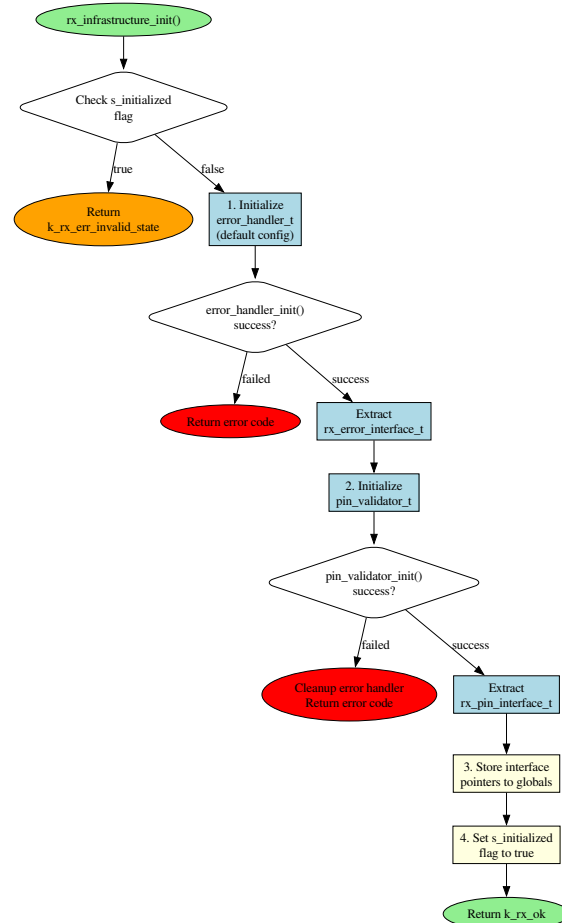
3. Store Interface Pointers:

- Save interface pointers to static globals
- Pointers remain valid for entire program lifetime

4. Set Initialization Flag:

- Mark infrastructure as initialized (prevents double-init)

7.15.3.17 Initialization Flow Diagram



7.15.3.18 Default Configuration

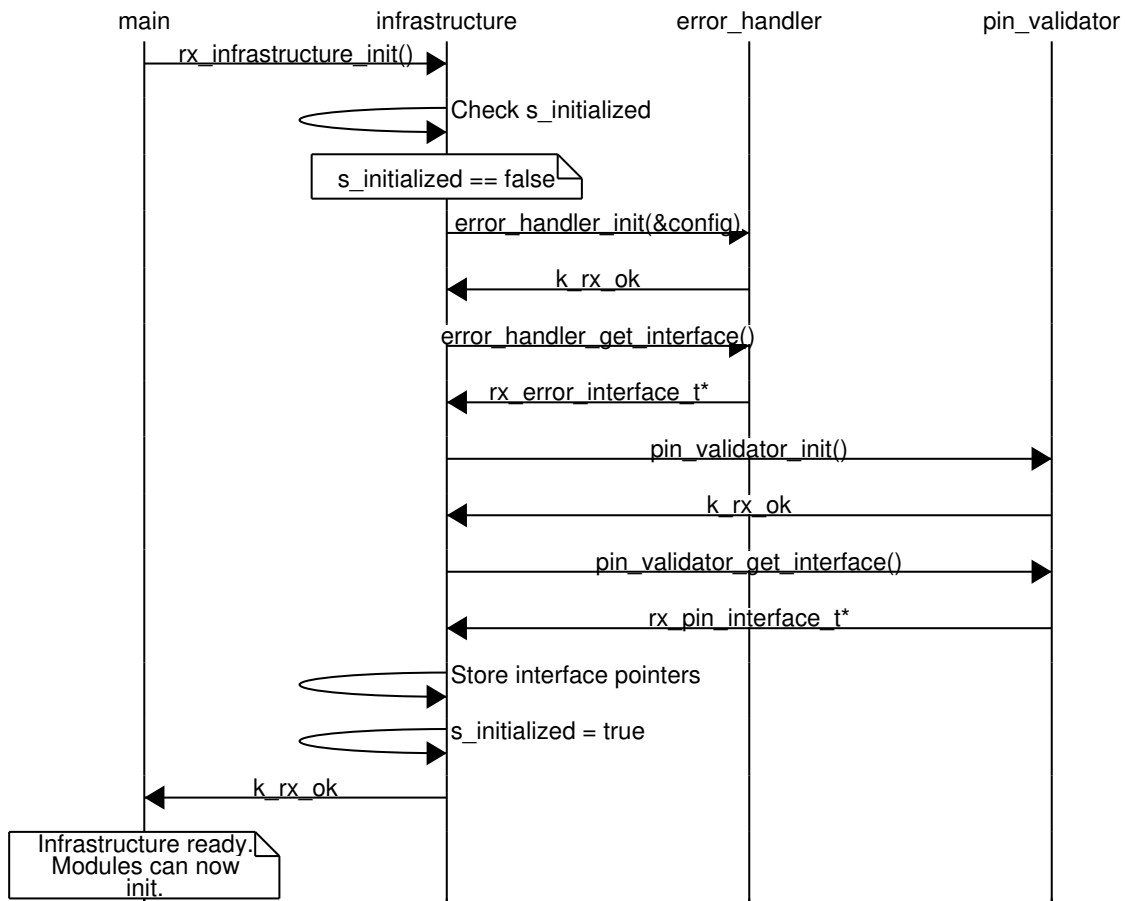
7.15.3.18.1 Error Handler Configuration

Parameter	Default Value	Rationale
max_retries	3	Balance between resilience and failure detection
initial_backoff_ms	100	100ms allows time for transient issues to clear
max_backoff_ms	5000	5s prevents indefinite waits
backoff_multiplier	2.0	Exponential backoff (100ms -> 200ms -> 400ms -> ...)
max_errors	64	Sufficient for all firmware components
max_component_errors	16	Track errors per component

7.15.3.18.2 Pin Validator Configuration

Parameter	Default Value	Rationale
max_pins	128	Covers all RX72N GPIO pins (Ports 0-F)
enable_conflicts	true	Prevent accidental pin conflicts

7.15.3.19 Typical Call Sequence



Returns

`rx_err_t` Error code

Return values

<code>k_rx_ok</code>	Success, infrastructure ready for use
<code>k_rx_err_invalid_state</code>	Already initialized (double-init attempt)
<code>k_rx_err_*</code>	Propagated from <code>error_handler_init()</code> or <code>pin_validator_init()</code>

Precondition

System must be in reset state (no modules initialized)
 ThreadX must NOT be running (call from `main()`, not from thread)
 Must be called exactly once per system boot
 No other threads accessing infrastructure (main thread only)

Postcondition

`s_initialized` flag set to true
`rx_infrastructure_get_error_interface()` returns non-NULL
`rx_infrastructure_get_pin_interface()` returns non-NULL
 Global error handler is functional and thread-safe
 Global pin validator is functional and thread-safe
 All macros (`RX_REPORT_ERROR`, `RX_RESERVE_PIN`, `RX_RELEASE_PIN`) are functional

Note

NOT thread-safe. Must be called from `main()` before starting ThreadX.
 Calling twice returns `k_rx_err_invalid_state` (double-init protection).
 On failure, infrastructure is left in undefined state - system must halt.

Warning

CRITICAL: This MUST be the first call in `main()`. Failure to initialize will cause nullptr crashes when other modules use infrastructure macros.
 If this function fails, the system CANNOT continue - must halt or reset.

Thread Safety:

- NOT thread-safe during initialization
- After successful init, interface getters are thread-safe (read-only)

Performance:

- Execution time: ~500 us @ 240 MHz (one-time cost at boot)
- Memory allocated: ~584 bytes in .bss section
- Stack usage: ~64 bytes

Example: Correct Initialization in main()

```
int main(void)
{
    // Step 1: FIRST THING - Initialize infrastructure
    rx_err_t err = rx_infrastructure_init();
    if (err != k_rx_ok) {
        // FATAL: Cannot continue without infrastructure
        // In production: log to backup UART, trigger watchdog reset
        while (1) {
            __asm("nop"); // Halt system
        }
    }

    // Step 2: Now safe to initialize other modules (they can use RX_REPORT_ERROR)
    err = watchdog_init();
    if (err != k_rx_ok) {
        RX_REPORT_ERROR(err, "WDT", "Watchdog init failed");
        return -1;
    }

    err = spi_driver_init();
    if (err != k_rx_ok) {
        RX_REPORT_ERROR(err, "SPI", "SPI driver init failed");
        return -1;
    }

    // Step 3: Start ThreadX (never returns)
    tx_kernel_enter();

    return 0; // Unreachable
}
```

Example: Incorrect Initialization (WILL CRASH)

```
// [FAIL] WRONG: Initializing SPI before infrastructure
int main(void)
{
    rx_err_t err = spi_driver_init(); // SPI tries to use RX_RESERVE_PIN()
    // CRASH: rx_infrastructure_get_pin_interface() returns nullptr
    // NULL->reserve_pin() causes segmentation fault

    rx_infrastructure_init(); // Too late!
    return 0;
}
```

Example: Handling Double-Init

```
rx_err_t err = rx_infrastructure_init();
if (err != k_rx_ok) {
    // Check for double-init
    if (err == k_rx_err_invalid_state) {
        // Already initialized - this is OK in some test scenarios
        // Production code should never hit this
    } else {
        // Real initialization failure - halt
        while (1) { }
    }
}
```

See also

- [rx_infrastructure_deinit\(\)](#) Cleanup infrastructure (shutdown only)
- [rx_infrastructure_get_error_interface\(\)](#) Get error handler interface
- [rx_infrastructure_get_pin_interface\(\)](#) Get pin validator interface
- [error_handler_init\(\)](#) Error handler initialization
- [pin_validator_init\(\)](#) Pin validator initialization

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NASA Power of 10 Compliance:

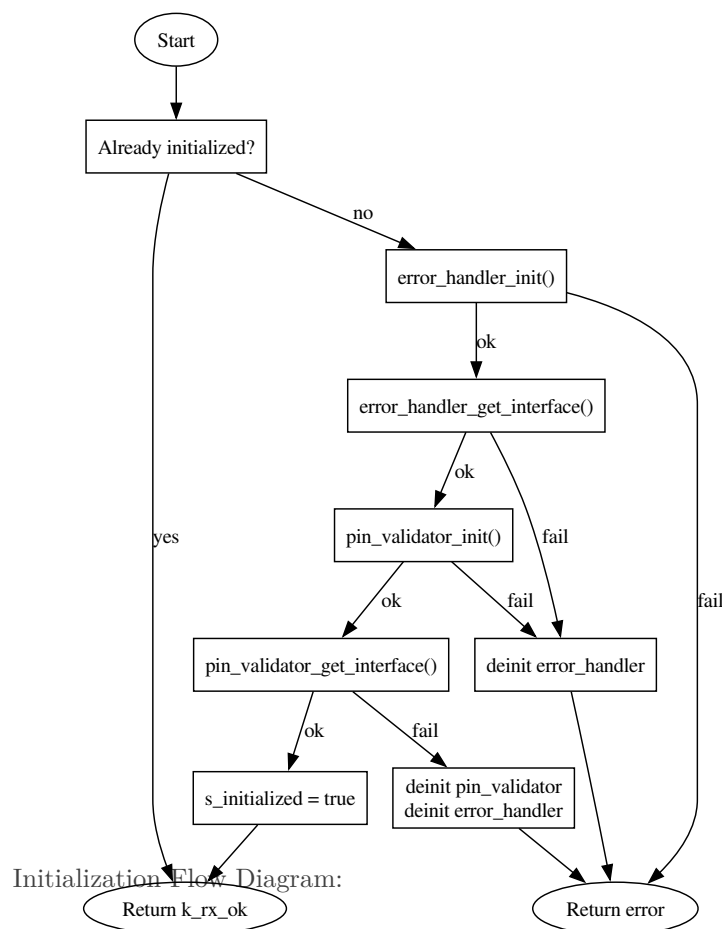
- Rule 5: [OK] 4 preconditions, 6 postconditions
- Rule 7: [OK] All return values checked

Initialize global infrastructure (MUST BE FIRST CALL IN main()).

Initializes all global infrastructure services in the correct order, with proper cleanup on failure. Uses default configuration values for error handler retry/backoff parameters.

Algorithm Steps:

1. Check if already initialized (idempotent, return success)
2. Initialize error handler with default config
3. Get error handler interface (function pointers)
4. Initialize pin validator
5. Get pin validator interface (function pointers)
6. Set initialized flag



Default Configuration:

Parameter	Value
max_retries	k_error_handler_default_max_retries (3)
initial_backoff_ms	k_error_handler_default_initial_backoff_ms
max_backoff_ms	k_error_handler_default_max_backoff_ms

Precondition

None - can be called at any time

Postcondition

s_infrastructure_initialized == true on success

All s_global_* variables contain valid data on success

Interface accessors return valid pointers on success

Note

Thread Safety: NOT thread-safe, call once from main()

Idempotent: Safe to call multiple times, subsequent calls are no-op

Warning

Must be called before any module uses infrastructure services

Not thread-safe - call before starting RTOS threads

Example:

```
// In main() before starting ThreadX
rx_err_t err = rx_infrastructure_init();
if (err != k_rx_ok) {
    // Critical failure - cannot continue
    while (1) { } // Halt
}

// Now modules can get interfaces
rx_error_interface_t* error_if = rx_infrastructure_get_error_interface();
```

See also

[rx_infrastructure_deinit\(\)](#) Cleanup function

[rx_infrastructure_get_error_interface\(\)](#) Access error handler

[rx_infrastructure_get_pin_interface\(\)](#) Access pin validator

NASA Power of 10 Compliance:

- Rule 5: [OK] Cleanup on failure (4 cleanup paths)
- Rule 7: [OK] All return values checked

Since

Version 1.0.0

Definition at line 365 of file [rx_infrastructure.c](#).

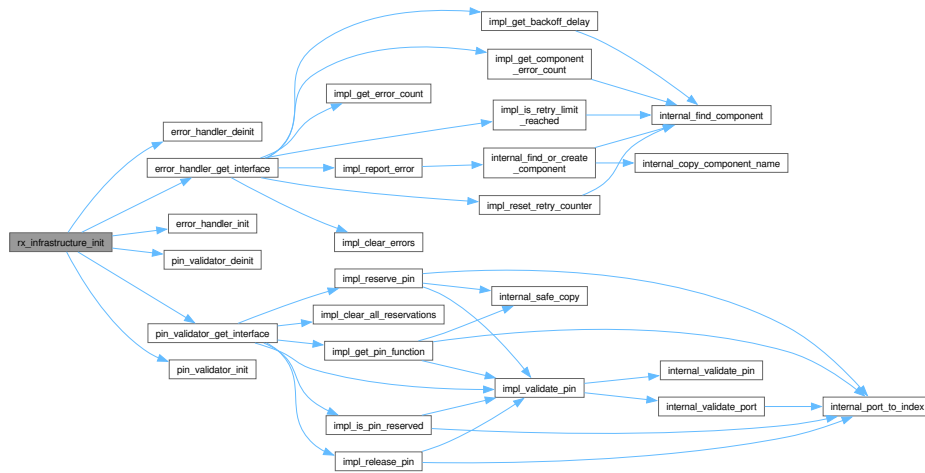
```

00366 {
00367     const error_handler_config_t config = {
00368         .max_retries = k_error_handler_default_max_retries,
00369         .initial_backoff_ms = k_error_handler_default_initial_backoff_ms,
00370         .max_backoff_ms = k_error_handler_default_max_backoff_ms,
00371     };
00372
00373     if (s_infrastructure_initialized) {
00374         rx_log_warn(s_tag, "Infrastructure already initialized");
00375         return k_rx_ok;
00376     }
00377
00378     rx_log_info(s_tag, "Initializing global infrastructure");
00379
00380     /* Initialize error handler */
00381     rx_err_t err = error_handler_init(&s_global_error_handler, &config);
00382     if (err != k_rx_ok) {
00383         rx_log_error(s_tag, "Failed to initialize error handler");
00384         return err;
00385     }
00386
00387     /* Get error handler interface */
00388     err = error_handler_get_interface(&s_global_error_interface, &s_global_error_handler);
00389     if (err != k_rx_ok) {
00390         rx_log_error(s_tag, "Failed to get error handler interface");
00391         (void)error_handler_deinit(&s_global_error_handler); /* Cleanup, ignore errors */
00392         return err;
00393     }
00394
00395     /* Initialize pin validator */
00396     err = pin_validator_init(&s_global_pin_validator);
00397     if (err != k_rx_ok) {
00398         rx_log_error(s_tag, "Failed to initialize pin validator");
00399         (void)error_handler_deinit(&s_global_error_handler); /* Cleanup, ignore errors */
00400         return err;
00401     }
00402
00403     /* Get pin validator interface */
00404     err = pin_validator_get_interface(&s_global_pin_interface, &s_global_pin_validator);
00405     if (err != k_rx_ok) {
00406         rx_log_error(s_tag, "Failed to get pin validator interface");
00407         (void)pin_validator_deinit(&s_global_pin_validator); /* Cleanup, ignore errors */
00408         (void)error_handler_deinit(&s_global_error_handler); /* Cleanup, ignore errors */
00409         return err;
00410     }
00411
00412     s_infrastructure_initialized = true;
00413
00414     rx_log_info(s_tag, "Global infrastructure initialized successfully");
00415
00416     return k_rx_ok;
00417 }

```

References [error_handler_deinit\(\)](#), [error_handler_get_interface\(\)](#), [error_handler_init\(\)](#), [k_error_handler_default_initial_backoff_ms](#), [k_error_handler_default_max_backoff_ms](#), [k_error_handler_default_max_retries](#), [k_rx_ok](#), [pin_validator_deinit\(\)](#), [pin_validator_get_interface\(\)](#), [pin_validator_init\(\)](#), [rx_log_error](#), [rx_log_info](#), [rx_log_warn](#), [s_global_error_handler](#), [s_global_error_interface](#), [s_global_pin_interface](#), [s_global_pin_validator](#), [s_infrastructure_initialized](#), and [s_tag](#).

Here is the call graph for this function:



7.16 rx_infrastructure.h

[Go to the documentation of this file.](#)

```

00001 /**
00002  * @file rx_infrastructure.h
00003  * @brief Global Infrastructure Initialization and Service Locator
00004  *
00005  * @details
00006  * ## Overview
00007  *
00008  * Implements the Service Locator pattern combined with Dependency Inversion Principle
00009  * to provide centralized access to core infrastructure services (error handling, pin validation)
00010  * throughout the STAR RX72N firmware. This module owns concrete implementations but exposes
00011  * only abstract interfaces to client code.
00012  *
00013  * Key Design Pattern: Service Locator + Dependency Inversion
00014  *
00015  * ```
00016  * +-----+
00017 * | High-Level Modules |
00018 * | (Motor Control, SPI Driver, USB Stack, etc.) |
00019 * +-----+
00020 * | depends on (abstract) | depends on (abstract) |
00021 * | v | v |
00022 * +-----+ +-----+
00023 * | rx_error_interface_t | | rx_pin_interface_t |
00024 * | (Abstract Interface) | | (Abstract Interface) |
00025 * +-----+ +-----+
00026 * | implements | | implements |
00027 * | v | v |
00028 * +-----+ +-----+
00029 * | error_handler_t | | pin_validator_t |
00030 * | (Concrete Impl) | | (Concrete Impl) |
00031 * +-----+ +-----+
00032 * | owned and initialized by |
00033 * | v |
00034 * +-----+
00035 * | rx_infrastructure (THIS FILE) |
00036 * | - Global instances |
00037 * | - Initialization sequence |
00038 * | - Interface accessors |
00039 * +-----+
00040 * ```
00041  *
00042  * ## Architecture Principles
00043  *
00044  * 1. Single Responsibility: Sole purpose is infrastructure lifecycle management
00045  * 2. Open/Closed: New services can be added without modifying client code
00046  * 3. Liskov Substitution: Interface pointers are always valid after init
00047  * 4. Interface Segregation: Separate accessors for each service
00048  * 5. Dependency Inversion: Clients depend on abstractions, not implementations
00049  *
00050  * ## Initialization Sequence

```

```

00056 *
00057 * ```
00058 * System Boot
00059 * |
00060 * +- 1. rx_infrastructure_init()
00061 * | +- Initialize global error handler
00062 * | | +- Config: max_retries=3, initial_backoff=100ms, max_backoff=5s
00063 * | +- Initialize global pin validator
00064 * | | +- Allocates pin reservation table
00065 * | +- Extract interfaces from concrete implementations
00066 * |
00067 * +- 2. Module Initialization (SPI, USB, Motor, etc.)
00068 * | +- Get interfaces via accessors
00069 * | +- Reserve pins via RX_RESERVE_PIN()
00070 * | +- Report errors via RX_REPORT_ERROR()
00071 * |
00072 * +- 3. Runtime Operation
00073 * | +- All modules use global interfaces
00074 * | +- No direct access to concrete implementations
00075 * |
00076 * +- 4. rx_infrastructure_deinit() [optional, shutdown]
00077 * | +- Release all resources
00078 * | +- Clear interface pointers
00079 * ```
00080 *
00081 * ## Thread Safety
00082 *
00083 * - `rx_infrastructure_init()`: **NOT thread-safe**, call from main() before ThreadX starts
00084 * - `rx_infrastructure_deinit()`: **NOT thread-safe**, call during shutdown only
00085 * - `rx_infrastructure_get_error_interface()`: **Thread-safe** (read-only after init)
00086 * - `rx_infrastructure_get_pin_interface()`: **Thread-safe** (read-only after init)
00087 * - Macro accessors: **Thread-safe** (delegates to underlying interface)
00088 *
00089 * ## Memory Footprint
00090 *
00091 * | Component | Size (bytes) | Location |
00092 * |-----|-----|-----|
00093 * | error_handler_t | 328 | .bss |
00094 * | pin_validator_t | ~200 | .bss |
00095 * | rx_error_interface_t | 32 | .bss |
00096 * | rx_pin_interface_t | 24 | .bss |
00097 * | **Total** | **~584** | .bss |
00098 *
00099 * ## Usage Examples
00100 *
00101 * ### Example 1: System Initialization
00102 * @code
00103 * // In main.c, BEFORE ThreadX initialization:
00104 * int main(void)
00105 * {
00106 * // 1. Initialize infrastructure FIRST
00107 * rx_err_t err = rx_infrastructure_init();
00108 * if (err != k_rx_ok) {
00109 * // Fatal error - cannot continue
00110 * while (1) { __asm("nop"); }
00111 * }
00112 *
00113 * // 2. Now safe to initialize other modules
00114 * err = spi_driver_init();
00115 * if (err != k_rx_ok) {
00116 * // Error handling now available
00117 * RX_REPORT_ERROR(err, "SPI", "Driver init failed");
00118 * return -1;
00119 * }
00120 *
00121 * // 3. Start ThreadX
00122 * tx_kernel_enter();
00123 *
00124 * return 0; // Never reached
00125 * }
00126 * @endcode
00127 *
00128 * ### Example 2: Module Using Error Reporting
00129 * @code
00130 * // In motor_control.c:
00131 * rx_err_t motor_set_velocity(motor_id_t id, float velocity_mps)
00132 * {
00133 * // Validate input
00134 * if (velocity_mps > k_max_velocity_mps) {
00135 * RX_REPORT_ERROR(k_rx_err_invalid_arg, "MOTOR", "Velocity exceeds limit");
00136 * return k_rx_err_invalid_arg;
00137 * }
00138 *
00139 * // Attempt operation
00140 * rx_err_t err = motor_send_command(id, velocity_mps);
00141 * if (err != k_rx_ok) {
00142 * RX_REPORT_ERROR(err, "MOTOR", "Command failed");

```

```

00143 *   return err;
00144 * }
00145 *
00146 * return k_rx_ok;
00147 * }
00148 * @endcode
00149 *
00150 * ### Example 3: Module Using Pin Reservation
00151 * @code
00152 * // In spi_driver.c:
00153 * rx_err_t spi_init(const spi_config_t* config)
00154 * {
00155 *     // Reserve SPI pins (prevents conflicts)
00156 *     rx_err_t err;
00157 *
00158 *     err = RX_RESERVE_PIN(0xA, 0, "SPI_CIPO");
00159 *     if (err != k_rx_ok) {
00160 *         RX_REPORT_ERROR(err, "SPI", "Pin PA0 already in use");
00161 *         return err;
00162 *     }
00163 *
00164 *     err = RX_RESERVE_PIN(0xA, 1, "SPI_COPI");
00165 *     if (err != k_rx_ok) {
00166 *         RX_RELEASE_PIN(0xA, 0); // Cleanup previous reservation
00167 *         RX_REPORT_ERROR(err, "SPI", "Pin PA1 already in use");
00168 *         return err;
00169 *     }
00170 *
00171 *     // Continue with SPI hardware initialization...
00172 *     return k_rx_ok;
00173 * }
00174 * @endcode
00175 *
00176 * ### Example 4: Using Interfaces Directly (Advanced)
00177 * @code
00178 * // Get interface once, use multiple times:
00179 * void motor_error_recovery(void)
00180 * {
00181 *     rx_error_interface_t* err_iface = rx_infrastructure_get_error_interface();
00182 *     if (err_iface == nullptr) {
00183 *         return; // Infrastructure not initialized
00184 *     }
00185 *
00186 *     // Check if retry limit reached for motor component
00187 *     if (err_iface->is_retry_limit_reached != nullptr) {
00188 *         bool limit_reached = err_iface->is_retry_limit_reached(err_iface->ctx, "MOTOR");
00189 *         if (limit_reached) {
00190 *             // Stop trying, require manual intervention
00191 *             motor_emergency_stop();
00192 *             return;
00193 *         }
00194 *     }
00195 *
00196 *     // Get backoff delay for exponential backoff
00197 *     if (err_iface->get_backoff_delay != nullptr) {
00198 *         uint32_t delay_ms = err_iface->get_backoff_delay(err_iface->ctx, "MOTOR");
00199 *         tx_thread_sleep(delay_ms); // ThreadX delay
00200 *     }
00201 *
00202 *     // Reset retry counter after successful recovery
00203 *     if (err_iface->reset_retry_counter != nullptr) {
00204 *         err_iface->reset_retry_counter(err_iface->ctx, "MOTOR");
00205 *     }
00206 * }
00207 * @endcode
00208 *
00209 * ## Design Rationale
00210 *
00211 * **Why Service Locator Pattern?**
00212 * - Avoids passing infrastructure pointers through every function call
00213 * - Reduces coupling between modules
00214 * - Simplifies function signatures
00215 * - Centralizes lifetime management
00216 *
00217 * **Why Global State?**
00218 * - Single-instance services (error handler, pin validator) don't need multiple instances
00219 * - Embedded systems: predictable memory layout, no dynamic allocation
00220 * - NASA Power of 10 Rule 3: No dynamic memory after initialization
00221 * - Performance: Direct access, no indirection or lookup overhead
00222 *
00223 * **Why Abstract Interfaces?**
00224 * - Testability: Can inject mock implementations for unit tests
00225 * - Flexibility: Can swap implementations without changing client code
00226 * - Encapsulation: Hides concrete types and internal state
00227 * - SOLID Compliance: Dependency Inversion Principle
00228 *
00229 * ## NASA Power of 10 Compliance

```

```

00230 *
00231 * | Rule | Status | Implementation |
00232 * |-----|-----|-----|
00233 * | 1. Simplify control flow | [OK] | No goto, setjmp, recursion |
00234 * | 2. Fixed loop bounds | [OK] | No loops in this header |
00235 * | 3. No dynamic memory | [OK] | Static global instances only |
00236 * | 4. Functions < 60 lines | [OK] | All functions are accessors |
00237 * | 5. Use assertions | [OK] | Validation in init functions |
00238 * | 6. Data at smallest scope | [OK] | Globals encapsulated |
00239 * | 7. Check return values | [OK] | All errors propagated |
00240 * | 8. Limit preprocessor | [OK] | Macros for convenience only |
00241 * | 9. Restrict pointers | [WARN] | Function pointers (DIP) |
00242 * | 10. Compile warnings | [OK] | -Wall -Wextra -Werror |
00243 *
00244 * @see rx_error_interface.h Abstract error handler interface
00245 * @see rx_error_handler.h Concrete error handler implementation
00246 * @see rx_pin_interface.h Abstract pin validator interface
00247 * @see rx_pin_validator.h Concrete pin validator implementation
00248 *
00249 * @since Version 1.0.0
00250 * @date 2026-01-01
00251 * @copyright Copyright (c) 2026 Locked Inc.
00252 * SPDX-License-Identifier: MIT
00253 */
00254
00255 #pragma once
00256
00257 #include "rx_err.h"
00258 #include "rx_error_interface.h"
00259 #include "rx_pin_interface.h"
00260
00261 #ifdef __cplusplus
00262 extern "C" {
00263 #endif
00264
00265 /*
=====
00266 * Global Infrastructure Initialization
00267 *
=====
00268 */
00269
00270 /**
00271 * @brief Initialize global infrastructure (MUST BE FIRST CALL IN main())
00272 *
00273 * @details
00274 * Performs one-time initialization of all global infrastructure services. This function
00275 * **MUST** be the first thing called in `main()`, before any other module initialization
00276 * and before starting ThreadX. Failure to initialize infrastructure will causenullptr
00277 * pointer dereferences when other modules attempt to use RX_REPORT_ERROR() or RX_RESERVE_PIN().
00278 *
00279 * ## Initialization Algorithm
00280 *
00281 * The function performs initialization in strict dependency order:
00282 *
00283 * 1. **Initialize Error Handler**:
00284 *   - Allocate static `error_handler_t` instance
00285 *   - Configure with default parameters (max_retries=3, backoffs)
00286 *   - Call `error_handler_init()` to prepare state
00287 *   - Extract `rx_error_interface_t` via `error_handler_get_interface()`
00288 *
00289 * 2. **Initialize Pin Validator**:
00290 *   - Allocate static `pin_validator_t` instance
00291 *   - Zero-initialize pin reservation table
00292 *   - Call `pin_validator_init()` to prepare state
00293 *   - Extract `rx_pin_interface_t` via `pin_validator_get_interface()`
00294 *
00295 * 3. **Store Interface Pointers**:
00296 *   - Save interface pointers to static globals
00297 *   - Pointers remain valid for entire program lifetime
00298 *
00299 * 4. **Set Initialization Flag**:
00300 *   - Mark infrastructure as initialized (prevents double-init)
00301 *
00302 * ## Initialization Flow Diagram
00303 *
00304 * @dot
00305 * digraph infrastructure_init {
00306 *   node [shape=box, style=rounded];
00307 *   rankdir=TD;
00308 *
00309 *   start [label="rx_infrastructure_init()", shape=ellipse, style=filled, fillcolor=lightgreen];
00310 *   check_flag [label="Check s_initialized\nflag", shape=diamond];
00311 *   already_init [label="Return\nk_rx_err_invalid_state", shape=ellipse, style=filled, fillcolor=orange];
00312 *
00313 *   init_error [label="1. Initialize\nerror_handler_t\n(default config)", style=filled, fillcolor=lightblue];
00314 *   check_error [label="error_handler_init()\nsuccess?", shape=diamond];

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00315 * error_fail [label="Return error code", shape=ellipse, style=filled, fillcolor=red];
00316 * extract_error [label="Extract\nrx_error_interface_t", style=filled, fillcolor=lightblue];
00317 *
00318 * init_pin [label="2. Initialize\npin_validator_t", style=filled, fillcolor=lightblue];
00319 * check_pin [label="pin_validator_init()\nsuccess?", shape=diamond];
00320 * pin_fail [label="Cleanup error handler\nReturn error code", shape=ellipse, style=filled, fillcolor=red];
00321 * extract_pin [label="Extract\nrx_pin_interface_t", style=filled, fillcolor=lightblue];
00322 *
00323 * store [label="3. Store interface\npointers to globals", style=filled, fillcolor=lightyellow];
00324 * set_flag [label="4. Set s_initialized\nflag to true", style=filled, fillcolor=lightyellow];
00325 * success [label="Return k_rx_ok", shape=ellipse, style=filled, fillcolor=lightgreen];
00326 *
00327 * start -> check_flag;
00328 * check_flag -> already_init [label="true"];
00329 * check_flag -> init_error [label="false"];
00330 *
00331 * init_error -> check_error;
00332 * check_error -> error_fail [label="failed"];
00333 * check_error -> extract_error [label="success"];
00334 *
00335 * extract_error -> init_pin;
00336 * init_pin -> check_pin;
00337 * check_pin -> pin_fail [label="failed"];
00338 * check_pin -> extract_pin [label="success"];
00339 *
00340 * extract_pin -> store;
00341 * store -> set_flag;
00342 * set_flag -> success;
00343 * }
00344 * @enddot
00345 *
00346 * ## Default Configuration
00347 *
00348 * ### Error Handler Configuration
00349 * | Parameter | Default Value | Rationale |
00350 * |-----|-----|-----|
00351 * | max_retries | 3 | Balance between resilience and failure detection |
00352 * | initial_backoff_ms | 100 | 100ms allows time for transient issues to clear |
00353 * | max_backoff_ms | 5000 | 5s prevents indefinite waits |
00354 * | backoff_multiplier | 2.0 | Exponential backoff (100ms -> 200ms -> 400ms -> ...) |
00355 * | max_errors | 64 | Sufficient for all firmware components |
00356 * | max_component_errors | 16 | Track errors per component |
00357 *
00358 * ### Pin Validator Configuration
00359 * | Parameter | Default Value | Rationale |
00360 * |-----|-----|-----|
00361 * | max_pins | 128 | Covers all RX72N GPIO pins (Ports 0-F) |
00362 * | enable_conflicts | true | Prevent accidental pin conflicts |
00363 *
00364 * ## Typical Call Sequence
00365 *
00366 * @msc
00367 * main, infrastructure, error_handler, pin_validator;
00368 *
00369 * main => infrastructure [label="rx_infrastructure_init()"];
00370 * infrastructure => infrastructure [label="Check s_initialized"];
00371 * infrastructure note infrastructure [label="s_initialized == false"];
00372 *
00373 * infrastructure => error_handler [label="error_handler_init(&config)"];
00374 * error_handler => infrastructure [label="k_rx_ok"];
00375 * infrastructure => error_handler [label="error_handler_get_interface()"];
00376 * error_handler => infrastructure [label="rx_error_interface_t*"];
00377 *
00378 * infrastructure => pin_validator [label="pin_validator_init()"];
00379 * pin_validator => infrastructure [label="k_rx_ok"];
00380 * infrastructure => pin_validator [label="pin_validator_get_interface()"];
00381 * pin_validator => infrastructure [label="rx_pin_interface_t*"];
00382 *
00383 * infrastructure => infrastructure [label="Store interface pointers"];
00384 * infrastructure => infrastructure [label="s_initialized = true"];
00385 * infrastructure => main [label="k_rx_ok"];
00386 *
00387 * main note main [label="Infrastructure ready.\nModules can now init."];
00388 * @endmsc
00389 *
00390 * @return rx_err_t Error code
00391 * @retval k_rx_ok Success, infrastructure ready for use
00392 * @retval k_rx_err_invalid_state Already initialized (double-init attempt)
00393 * @retval k_rx_err_* Propagated from error_handler_init() or pin_validator_init()
00394 *
00395 * @pre System must be in reset state (no modules initialized)
00396 * @pre ThreadX must NOT be running (call from main(), not from thread)
00397 * @pre Must be called exactly once per system boot
00398 * @pre No other threads accessing infrastructure (main thread only)
00399 *
00400 * @post s_initialized flag set to true
00401 * @post rx_infrastructure_get_error_interface() returns non-NULL

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00402 * @post rx_infrastructure_get_pin_interface() returns non-NULL
00403 * @post Global error handler is functional and thread-safe
00404 * @post Global pin validator is functional and thread-safe
00405 * @post All macros (RX_REPORT_ERROR, RX_RESERVE_PIN, RX_RELEASE_PIN) are functional
00406 *
00407 * @note **NOT thread-safe**. Must be called from main() before starting ThreadX.
00408 * @note Calling twice returns k_rx_err_invalid_state (double-init protection).
00409 * @note On failure, infrastructure is left in undefined state - system must halt.
00410 *
00411 * @warning **CRITICAL**: This MUST be the first call in main(). Failure to initialize
00412 * will cause nullptr crashes when other modules use infrastructure macros.
00413 * @warning If this function fails, the system CANNOT continue - must halt or reset.
00414 *
00415 * @par Thread Safety:
00416 * - NOT thread-safe during initialization
00417 * - After successful init, interface getters are thread-safe (read-only)
00418 *
00419 * @par Performance:
00420 * - Execution time: ~500 us @ 240 MHz (one-time cost at boot)
00421 * - Memory allocated: ~584 bytes in .bss section
00422 * - Stack usage: ~64 bytes
00423 *
00424 * @par Example: Correct Initialization in main()
00425 * @code
00426 * int main(void)
00427 * {
00428 *     // Step 1: FIRST THING - Initialize infrastructure
00429 *     rx_err_t err = rx_infrastructure_init();
00430 *     if (err != k_rx_ok) {
00431 *         // FATAL: Cannot continue without infrastructure
00432 *         // In production: log to backup UART, trigger watchdog reset
00433 *         while (1) {
00434 *             __asm("nop"); // Halt system
00435 *         }
00436 *     }
00437 *
00438 *     // Step 2: Now safe to initialize other modules (they can use RX_REPORT_ERROR)
00439 *     err = watchdog_init();
00440 *     if (err != k_rx_ok) {
00441 *         RX_REPORT_ERROR(err, "WDT", "Watchdog init failed");
00442 *         return -1;
00443 *     }
00444 *
00445 *     err = spi_driver_init();
00446 *     if (err != k_rx_ok) {
00447 *         RX_REPORT_ERROR(err, "SPI", "SPI driver init failed");
00448 *         return -1;
00449 *     }
00450 *
00451 *     // Step 3: Start ThreadX (never returns)
00452 *     tx_kernel_enter();
00453 *
00454 *     return 0; // Unreachable
00455 * }
00456 * @endcode
00457 *
00458 * @par Example: Incorrect Initialization (WILL CRASH)
00459 * @code
00460 * // [FAIL] WRONG: Initializing SPI before infrastructure
00461 * int main(void)
00462 * {
00463 *     rx_err_t err = spi_driver_init(); // SPI tries to use RX_RESERVE_PIN()
00464 *     // CRASH: rx_infrastructure_get_pin_interface() returns nullptr
00465 *     // NULL->reserve_pin() causes segmentation fault
00466 *
00467 *     rx_infrastructure_init(); // Too late!
00468 *     return 0;
00469 * }
00470 * @endcode
00471 *
00472 * @par Example: Handling Double-Init
00473 * @code
00474 * rx_err_t err = rx_infrastructure_init();
00475 * if (err != k_rx_ok) {
00476 *     // Check for double-init
00477 *     if (err == k_rx_err_invalid_state) {
00478 *         // Already initialized - this is OK in some test scenarios
00479 *         // Production code should never hit this
00480 *     } else {
00481 *         // Real initialization failure - halt
00482 *         while (1) { }
00483 *     }
00484 * }
00485 * @endcode
00486 *
00487 * @see rx_infrastructure_deinit() Cleanup infrastructure (shutdown only)
00488 * @see rx_infrastructure_get_error_interface() Get error handler interface

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00489 * @see rx_infrastructure_get_pin_interface() Get pin validator interface
00490 * @see error_handler_init() Error handler initialization
00491 * @see pin_validator_init() Pin validator initialization
00492 *
00493 * @since Version 1.0.0
00494 *
00495 * @par NASA Power of 10 Compliance:
00496 * - Rule 5: [OK] 4 preconditions, 6 postconditions
00497 * - Rule 7: [OK] All return values checked
00498 */
00499 [[nodiscard]] rx_err_t rx_infrastructure_init(void);
00500
00501 /**
00502 * @brief Deinitialize global infrastructure (graceful shutdown only)
00503 *
00504 * @details
00505 * Performs graceful shutdown of all infrastructure services. In embedded systems,
00506 * this function is RARELY CALLED because the system typically runs indefinitely
00507 * until power loss or reset. However, it's provided for:
00508 * - Unit testing (setup/teardown)
00509 * - Firmware update procedures (safe state before flash)
00510 * - Power management (deep sleep entry)
00511 * - Development/debugging (clean restarts)
00512 *
00513 * ## Deinitialization Algorithm
00514 *
00515 * Teardown occurs in REVERSE order of initialization (dependency order):
00516 *
00517 * 1. Validate Initialized State:
00518 *   - Check s_initialized flag
00519 *   - Return error if not initialized (nothing to clean up)
00520 *
00521 * 2. Clear Interface Pointers:
00522 *   - Set global interface pointers to nullptr
00523 *   - Prevents use-after-free if modules still running
00524 *
00525 * 3. Deinitialize Pin Validator:
00526 *   - Call `pin_validator_deinit()`
00527 *   - Releases all pin reservations
00528 *   - Clears pin reservation table
00529 *
00530 * 4. Deinitialize Error Handler:
00531 *   - Call `error_handler_deinit()`
00532 *   - Clears error counters and state
00533 *   - Releases mutex resources (if ThreadX running)
00534 *
00535 * 5. Clear Initialization Flag:
00536 *   - Reset s_initialized to false
00537 *   - Allows re-initialization (useful for tests)
00538 *
00539 * ## Shutdown Flow Diagram
00540 *
00541 * @dot
00542 * digraph infrastructure_deinit {
00543 *   node [shape=box, style=rounded];
00544 *   rankdir=TD;
00545 *
00546 *   start [label="rx_infrastructure_deinit()", shape=ellipse, style=filled, fillcolor=lightyellow];
00547 *   check_flag [label="Check s_initialized\nflag", shape=diamond];
00548 *   not_init [label="Return\nnk_rx_err_invalid_state", shape=ellipse, style=filled, fillcolor=orange];
00549 *
00550 *   clear_iface [label="1. Set interface\npointers to NULL", style=filled, fillcolor=lightblue];
00551 *   note1 [label="Prevents use-after-free\nif modules still running", shape=note];
00552 *
00553 *   deinit_pin [label="2. Deinitialize\npin_validator_t", style=filled, fillcolor=lightblue];
00554 *   note2 [label="Releases all pin\nreservations", shape=note];
00555 *
00556 *   deinit_error [label="3. Deinitialize\nerror_handler_t", style=filled, fillcolor=lightblue];
00557 *   note3 [label="Clears error state,\nreleases mutex", shape=note];
00558 *
00559 *   clear_flag [label="4. Clear s_initialized\nflag to false", style=filled, fillcolor=lightyellow];
00560 *   success [label="Return k_rx_ok", shape=ellipse, style=filled, fillcolor=lightgreen];
00561 *
00562 *   start -> check_flag;
00563 *   check_flag -> not_init [label="false"];
00564 *   check_flag -> clear_iface [label="true"];
00565 *
00566 *   clear_iface -> deinit_pin;
00567 *   clear_iface -> note1 [style=dashed];
00568 *
00569 *   deinit_pin -> deinit_error;
00570 *   deinit_pin -> note2 [style=dashed];
00571 *
00572 *   deinit_error -> clear_flag;
00573 *   deinit_error -> note3 [style=dashed];
00574 *
00575 *   clear_flag -> success;

```

```

00576 * }
00577 * @enddot
00578 *
00579 * ### Typical Usage Scenarios
00580 *
00581 * ### Scenario 1: Unit Testing (Most Common Use)
00582 * @code
00583 * void test_motor_control(void)
00584 * {
00585 *     // Setup: Initialize infrastructure for test
00586 *     rx_err_t err = rx_infrastructure_init();
00587 *     assert(err == k_rx_ok);
00588 *
00589 *     // Test: Run motor control tests
00590 *     test_motor_set_velocity();
00591 *     test_motor_error_handling();
00592 *
00593 *     // Teardown: Clean up for next test
00594 *     err = rx_infrastructure_deinit();
00595 *     assert(err == k_rx_ok);
00596 * }
00597 * @endcode
00598 *
00599 * ### Scenario 2: Firmware Update Preparation
00600 * @code
00601 * void enter_bootloader_mode(void)
00602 * {
00603 *     // 1. Stop all threads
00604 *     motor_stop_all();
00605 *     usb_disconnect();
00606 *     spi_shutdown();
00607 *
00608 *     // 2. Deinitialize infrastructure
00609 *     rx_infrastructure_deinit();
00610 *
00611 *     // 3. Jump to bootloader
00612 *     bootloader_entry();
00613 * }
00614 * @endcode
00615 *
00616 * ### Scenario 3: Deep Sleep Power Management
00617 * @code
00618 * void enter_deep_sleep(void)
00619 * {
00620 *     // 1. Save critical state to backup SRAM
00621 *     save_state_to_backup_ram();
00622 *
00623 *     // 2. Shutdown peripherals
00624 *     usb_suspend();
00625 *     spi_sleep();
00626 *
00627 *     // 3. Deinitialize infrastructure
00628 *     rx_infrastructure_deinit();
00629 *
00630 *     // 4. Enter low-power mode
00631 *     rx72n_enter_deep_sleep();
00632 *
00633 *     // 5. After wakeup: Re-initialize everything
00634 *     rx_infrastructure_init();
00635 *     restore_state_from_backup_ram();
00636 * }
00637 * @endcode
00638 *
00639 * @return rx_err_t Error code
00640 * @retval k_rx_ok Success, infrastructure fully deinitialized
00641 * @retval k_rx_err_invalid_state Not initialized (nothing to clean up)
00642 *
00643 * @pre s_initialized flag must be true (infrastructure must be initialized)
00644 * @pre All threads using infrastructure must be stopped or suspended
00645 * @pre No modules should call RX_REPORT_ERROR() or RX_RESERVE_PIN() during deinit
00646 * @pre ThreadX threads should be suspended (if ThreadX is running)
00647 *
00648 * @post s_initialized flag set to false
00649 * @post rx_infrastructure_get_error_interface() returns nullptr
00650 * @post rx_infrastructure_get_pin_interface() returns nullptr
00651 * @post All error counters cleared
00652 * @post All pin reservations released
00653 * @post Infrastructure can be re-initialized via rx_infrastructure_init()
00654 *
00655 * @note **NOT thread-safe**. Caller must ensure no threads are using infrastructure.
00656 * @note Calling when not initialized returns k_rx_err_invalid_state (safe, no-op).
00657 * @note After deinit, all RX_REPORT_ERROR() and RX_RESERVE_PIN() calls will fail silently.
00658 *
00659 * @warning **DANGER**: Calling this while modules are still using infrastructure will
00660 *     cause crashes. Ensure all threads are stopped first.
00661 * @warning Do NOT call from ISR (may block on mutex cleanup).
00662 *

```

```

00663 * @par Thread Safety:
00664 * - NOT thread-safe
00665 * - Caller must ensure exclusive access (no concurrent infrastructure usage)
00666 * - Suspend all ThreadX threads before calling
00667 *
00668 * @par Performance:
00669 * - Execution time: ~200 us @ 240 MHz
00670 * - No heap operations (all cleanup is static)
00671 *
00672 * @par Example: Safe Shutdown Sequence
00673 * @code
00674 * void system_shutdown(void)
00675 * {
00676 *     // 1. Stop all application threads
00677 *     tx_thread_suspend(&motor_thread);
00678 *     tx_thread_suspend(&usb_thread);
00679 *     tx_thread_suspend(&comm_thread);
00680 *
00681 *     // 2. Disable interrupts (prevent ISRs from using infrastructure)
00682 *     __disable_irq();
00683 *
00684 *     // 3. Shutdown modules (in reverse init order)
00685 *     motor_shutdown();
00686 *     usb_shutdown();
00687 *     spi_shutdown();
00688 *
00689 *     // 4. Deinitialize infrastructure (last)
00690 *     rx_err_t err = rx_infrastructure_deinit();
00691 *     if (err != k_rx_ok) {
00692 *         // Log error (can't use RX_REPORT_ERROR anymore!)
00693 *         debug_uart_printf("Deinit failed: %d\n", err);
00694 *     }
00695 *
00696 *     // 5. Proceed with shutdown (reset, bootloader, sleep, etc.)
00697 *     trigger_system_reset();
00698 * }
00699 * @endcode
00700 *
00701 * @par Example: Test Fixture Pattern
00702 * @code
00703 * typedef struct {
00704 *     bool infrastructure_initialized;
00705 * } test_fixture_t;
00706 *
00707 * void test_setup(test_fixture_t* fixture)
00708 * {
00709 *     rx_err_t err = rx_infrastructure_init();
00710 *     assert(err == k_rx_ok);
00711 *     fixture->infrastructure_initialized = true;
00712 * }
00713 *
00714 * void test_teardown(test_fixture_t* fixture)
00715 * {
00716 *     if (fixture->infrastructure_initialized) {
00717 *         rx_err_t err = rx_infrastructure_deinit();
00718 *         assert(err == k_rx_ok);
00719 *         fixture->infrastructure_initialized = false;
00720 *     }
00721 * }
00722 *
00723 * void run_test_suite(void)
00724 * {
00725 *     test_fixture_t fixture = {};
00726 *
00727 *     test_setup(&fixture);
00728 *     test_motor_functions();
00729 *     test_teardown(&fixture);
00730 *
00731 *     test_setup(&fixture);
00732 *     test_usb_functions();
00733 *     test_teardown(&fixture);
00734 * }
00735 * @endcode
00736 *
00737 * @par Example: Handling Double-Deinit
00738 * @code
00739 * rx_err_t err = rx_infrastructure_deinit();
00740 * if (err == k_rx_err_invalid_state) {
00741 *     // Already deinitialized - this is safe, continue
00742 * } else if (err != k_rx_ok) {
00743 *     // Real deinit error - investigate
00744 *     debug_uart_printf("Deinit error: %d\n", err);
00745 * }
00746 * @endcode
00747 *
00748 * @see rx_infrastructure_init() Initialize infrastructure
00749 * @see error_handler_deinit() Error handler cleanup

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00750 * @see pin_validator_deinit() Pin validator cleanup
00751 *
00752 * @since Version 1.0.0
00753 *
00754 * @par NASA Power of 10 Compliance:
00755 * - Rule 5: [OK] 4 preconditions, 6 postconditions
00756 * - Rule 7: [OK] All return values checked
00757 */
00758 [[nodiscard]] rx_err_t rx_infrastructure_deinit(void);
00759
00760 /*
=====
00761 * Interface Accessors
00762 *
=====
00763 */
00764
00765 /**
00766 * @brief Get global error handler interface (thread-safe accessor)
00767 *
00768 * @details
00769 * Returns a pointer to the global `rx_error_interface_t` that provides access to
00770 * the error handling subsystem. This is the primary way for modules to report
00771 * errors, check error counts, and implement retry logic with exponential backoff.
00772 *
00773 * The returned pointer is valid for the entire program lifetime after successful
00774 * initialization via `rx_infrastructure_init()`. Modules can cache this pointer for
00775 * performance-critical code paths.
00776 *
00777 * ## Pointer Lifetime and Validity
00778 *
00779 * ```
00780 * System Boot
00781 * |
00782 * +- rx_infrastructure_init()
00783 * |   +- Interface pointer becomes valid
00784 * |
00785 * +- Runtime (entire program lifetime)
00786 * |   +- Pointer remains valid
00787 * |   +- Safe to call from any thread
00788 * |   +- Safe to cache in module state
00789 * |
00790 * +- rx_infrastructure_deinit() [optional]
00791 * |   +- Pointer becomes NULL (rare in embedded)
00792 * ```
00793 *
00794 * ## Usage Patterns
00795 *
00796 * ### Pattern 1: Direct Inline Usage (Recommended for Simple Cases)
00797 * @code
00798 * // Simple error reporting - get interface each time
00799 * void motor_set_velocity(float velocity_mps)
00800 * {
00801 *     if (velocity_mps > k_max_velocity_mps) {
00802 *         RX_REPORT_ERROR(k_rx_err_invalid_arg, "MOTOR", "Velocity exceeds limit");
00803 *         return;
00804 *     }
00805 *     // ...
00806 * }
00807 * @endcode
00808 *
00809 * ### Pattern 2: Cached Pointer (Recommended for Performance-Critical Code)
00810 * @code
00811 * // Cache interface pointer in module state
00812 * typedef struct {
00813 *     rx_error_interface_t* error_iface; // Cached at init
00814 *     // ... other state
00815 * } motor_state_t;
00816 *
00817 * void motor_init(motor_state_t* state)
00818 * {
00819 *     // Cache interface pointer once during init
00820 *     state->error_iface = rx_infrastructure_get_error_interface();
00821 *     if (state->error_iface == nullptr) {
00822 *         return k_rx_err_not_initialized;
00823 *     }
00824 *     // ...
00825 * }
00826 *
00827 * void motor_control_loop(motor_state_t* state)
00828 * {
00829 *     // Use cached pointer (no function call overhead)
00830 *     if (motor_error_detected()) {
00831 *         state->error_iface->report_error(
00832 *             state->error_iface->ctx,
00833 *             k_rx_err_hardware,
00834 *             "MOTOR",

```

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00835 *      "Encoder fault"
00836 *    );
00837 *  }
00838 * }
00839 * @endcode
00840 *
00841 * ### Pattern 3: Defensive Null Check (Recommended for Critical Paths)
00842 * @code
00843 * void critical_error_handler(void)
00844 * {
00845 *   rx_error_interface_t* iface = rx_infrastructure_get_error_interface();
00846 *   if (iface == nullptr) {
00847 *     // Infrastructure not ready - fallback to direct UART logging
00848 *     debug_uart_printf("CRITICAL: Infrastructure not initialized\n");
00849 *     return;
00850 *   }
00851 *
00852 *   // Use interface normally
00853 *   iface->report_error(iface->ctx, k_rx_err_hardware, "CRITICAL", "Hardware fault");
00854 * }
00855 * @endcode
00856 *
00857 * ### Pattern 4: Advanced - Retry Logic with Backoff
00858 * @code
00859 * rx_err_t motor_send_command_with_retry(uint8_t* data, uint32_t len)
00860 * {
00861 *   rx_error_interface_t* iface = rx_infrastructure_get_error_interface();
00862 *   if (iface == nullptr) {
00863 *     return k_rx_err_not_initialized;
00864 *   }
00865 *
00866 *   // Check if we've exceeded retry limit
00867 *   if (iface->is_retry_limit_reached != nullptr &&
00868 *       iface->is_retry_limit_reached(iface->ctx, "MOTOR")) {
00869 *     return k_rx_err_retry_limit;
00870 *   }
00871 *
00872 *   // Attempt command
00873 *   rx_err_t err = motor_send_command_raw(data, len);
00874 *   if (err != k_rx_ok) {
00875 *     // Report error
00876 *     iface->report_error(iface->ctx, err, "MOTOR", "Command failed");
00877 *
00878 *     // Get exponential backoff delay
00879 *     if (iface->get_backoff_delay != nullptr) {
00880 *       uint32_t delay_ms = iface->get_backoff_delay(iface->ctx, "MOTOR");
00881 *       tx_thread_sleep(delay_ms); // Wait before retry
00882 *     }
00883 *
00884 *     return err;
00885 *   }
00886 *
00887 *   // Success - reset retry counter
00888 *   if (iface->reset_retry_counter != nullptr) {
00889 *     iface->reset_retry_counter(iface->ctx, "MOTOR");
00890 *   }
00891 *
00892 *   return k_rx_ok;
00893 * }
00894 * @endcode
00895 *
00896 * ## Interface Method Availability
00897 *
00898 * All function pointers in the returned interface are guaranteed non-NULL except:
00899 * - 'get_component_error_count' (optional, may be NULL)
00900 * - 'is_retry_limit_reached' (optional, may be NULL)
00901 * - 'reset_retry_counter' (optional, may be NULL)
00902 * - 'get_backoff_delay' (optional, may be NULL)
00903 *
00904 * **Always check optional methods** before calling:
00905 * @code
00906 * if (iface->get_backoff_delay != nullptr) {
00907 *   uint32_t delay = iface->get_backoff_delay(iface->ctx, "MOTOR");
00908 * }
00909 * @endcode
00910 *
00911 * ## Thread Safety Guarantees
00912 *
00913 * | Operation | Thread-Safe? | Details |
00914 * |-----|-----|-----|
00915 * | Calling this function | [OK] Yes | Read-only after init |
00916 * | Caching returned pointer | [OK] Yes | Pointer never changes |
00917 * | Calling interface methods | [OK] Yes | Methods are thread-safe |
00918 * | Calling during init/deinit | [X] No | Undefined behavior |
00919 *
00920 * @return rx_error_interface_t* Pointer to global error interface
00921 * @retval non-NULL Success, interface is ready for use

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00922 * @retval NULL Infrastructure not initialized (call rx_infrastructure_init() first)
00923 *
00924 * @pre Infrastructure must be initialized via rx_infrastructure_init()
00925 * @pre Must NOT be called during rx_infrastructure_deinit()
00926 *
00927 * @post Returned pointer remains valid until rx_infrastructure_deinit()
00928 * @post All required interface methods are non-NULL
00929 * @post Interface methods are thread-safe (can be called from any thread)
00930 *
00931 * @note **Thread-safe**: Can be called from any thread after initialization.
00932 * @note **Performance**: Simple pointer return (~1 cycle), safe to call frequently.
00933 * @note **Caching**: Safe to cache returned pointer in module state for performance.
00934 * @note **Null check**: Always check for nullptr in defensive code paths.
00935 *
00936 * @warning Do NOT call during rx_infrastructure_init() or rx_infrastructure_deinit().
00937 * @warning NULL return means infrastructure not ready - check initialization order.
00938 *
00939 * @par Performance:
00940 * - Execution time: ~5 cycles @ 240 MHz (simple pointer read)
00941 * - No overhead: Direct memory access, no function calls
00942 * - Cache-friendly: Same pointer returned every call
00943 *
00944 * @par Example: Initialization-Time Validation
00945 * @code
00946 * rx_err_t module_init(void)
00947 * {
00948 *     // Verify infrastructure is ready
00949 *     rx_error_interface_t* iface = rx_infrastructure_get_error_interface();
00950 *     if (iface == nullptr) {
00951 *         // FATAL: Infrastructure not initialized
00952 *         // This means rx_infrastructure_init() was not called first
00953 *         debug_uart_printf("FATAL: Infrastructure not ready\n");
00954 *         return k_rx_err_not_initialized;
00955 *     }
00956 *
00957 *     // Verify required methods are present
00958 *     if (iface->report_error == nullptr) {
00959 *         debug_uart_printf("FATAL: Invalid error interface\n");
00960 *         return k_rx_err_invalid_state;
00961 *     }
00962 *
00963 *     // Cache pointer for runtime use
00964 *     g_module_state.error_iface = iface;
00965 *     return k_rx_ok;
00966 * }
00967 * @endcode
00968 *
00969 * @see rx_infrastructure_init() Initialize infrastructure to make this return non-NULL
00970 * @see rx_infrastructure_deinit() Cleanup infrastructure (sets pointer to NULL)
00971 * @see rx_error_interface.h Abstract error handler interface definition
00972 * @see RX_REPORT_ERROR() Convenience macro using this accessor
00973 *
00974 * @since Version 1.0.0
00975 */
00976 rx_error_interface_t* rx_infrastructure_get_error_interface(void);
00977
00978 /**
00979 * @brief Get global pin validator interface (thread-safe accessor)
00980 *
00981 * @details
00982 * Returns a pointer to the global `rx_pin_interface_t` that provides access to the
00983 * pin validation and reservation subsystem. This interface prevents **accidental pin
00984 * conflicts** by tracking which GPIO pins are reserved by which modules.
00985 *
00986 * The returned pointer is valid for the **entire program lifetime** after successful
00987 * initialization via `rx_infrastructure_init()`. Modules can cache this pointer for
00988 * performance-critical code paths.
00989 *
00990 * ## Pin Reservation System Overview
00991 *
00992 * The pin validator ensures that:
00993 * 1. **No pin conflicts**: Two modules cannot reserve the same pin
00994 * 2. **Explicit ownership**: Each pin knows which module owns it
00995 * 3. **Traceable usage**: Pin assignments documented in reservation table
00996 * 4. **Safe release**: Pins can only be released by their owner
00997 *
00998 * ```
00999 * Module Init Sequence:
01000 * |
01001 * | +- SPI Driver Init
01002 * | |   +- Reserve PA0 ("SPI_CIPO") -> Success
01003 * | |   +- Reserve PA1 ("SPI_COPI") -> Success
01004 * | |   +- Reserve PA2 ("SPI_CLK")  -> Success
01005 * |
01006 * | +- I2C Driver Init
01007 * | |   +- Reserve PA0 ("I2C_SDA")  -> FAIL (already SPI_CIPO)
01008 * | |   +- Init fails, prevents pin conflict

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01009 * |
01010 * +- USB Driver Init
01011 * +- Reserve PB0 ("USB_DP") -> Success
01012 * +- Reserve PB1 ("USB_DM") -> Success
01013 * ```
01014 *
01015 * ## Usage Patterns
01016 *
01017 * ### Pattern 1: Module Initialization with Pin Reservation
01018 * @code
01019 * rx_err_t spi_driver_init(void)
01020 * {
01021 * // Reserve all SPI pins before configuring hardware
01022 * rx_err_t err;
01023 *
01024 * err = RX_RESERVE_PIN(0xA, 0, "SPI_CIPO"); // Port A, Pin 0
01025 * if (err != k_rx_ok) {
01026 * RX_REPORT_ERROR(err, "SPI", "Pin PA0 already in use");
01027 * return err;
01028 * }
01029 *
01030 * err = RX_RESERVE_PIN(0xA, 1, "SPI_COPI"); // Port A, Pin 1
01031 * if (err != k_rx_ok) {
01032 * RX_RELEASE_PIN(0xA, 0); // Clean up previous reservation
01033 * RX_REPORT_ERROR(err, "SPI", "Pin PA1 already in use");
01034 * return err;
01035 * }
01036 *
01037 * err = RX_RESERVE_PIN(0xA, 2, "SPI_CLK"); // Port A, Pin 2
01038 * if (err != k_rx_ok) {
01039 * RX_RELEASE_PIN(0xA, 0); // Clean up
01040 * RX_RELEASE_PIN(0xA, 1);
01041 * RX_REPORT_ERROR(err, "SPI", "Pin PA2 already in use");
01042 * return err;
01043 * }
01044 *
01045 * // All pins reserved successfully - proceed with hardware config
01046 * configure_spi_hardware();
01047 * return k_rx_ok;
01048 * }
01049 * @endcode
01050 *
01051 * ### Pattern 2: Cached Pointer for Performance
01052 * @code
01053 * typedef struct {
01054 * rx_pin_interface_t* pin_iface; // Cached at init
01055 * uint8_t port;
01056 * uint8_t pin;
01057 * } gpio_driver_state_t;
01058 *
01059 * rx_err_t gpio_init(gpio_driver_state_t* state, uint8_t port, uint8_t pin)
01060 * {
01061 * // Cache interface pointer
01062 * state->pin_iface = rx_infrastructure_get_pin_interface();
01063 * if (state->pin_iface == nullptr) {
01064 * return k_rx_err_not_initialized;
01065 * }
01066 *
01067 * // Reserve pin using cached interface
01068 * rx_err_t err = state->pin_iface->reserve_pin(
01069 * state->pin_iface->ctx, port, pin, "GPIO"
01070 * );
01071 * if (err != k_rx_ok) {
01072 * return err;
01073 * }
01074 *
01075 * state->port = port;
01076 * state->pin = pin;
01077 * return k_rx_ok;
01078 * }
01079 *
01080 * void gpio_deinit(gpio_driver_state_t* state)
01081 * {
01082 * // Release pin using cached interface
01083 * if (state->pin_iface != nullptr) {
01084 * state->pin_iface->release_pin(state->pin_iface->ctx, state->port, state->pin);
01085 * }
01086 * }
01087 * @endcode
01088 *
01089 * ### Pattern 3: Dynamic Pin Configuration (Runtime Changes)
01090 * @code
01091 * rx_err_t motor_reconfigure_encoder_pins(uint8_t new_port, uint8_t new_pin_a, uint8_t new_pin_b)
01092 * {
01093 * rx_pin_interface_t* iface = rx_infrastructure_get_pin_interface();
01094 * if (iface == nullptr) {
01095 * return k_rx_err_not_initialized;

```



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01096 * }
01097 *
01098 * // Release old pins
01099 * iface->release_pin(iface->ctx, motor_state.old_port, motor_state.old_pin_a);
01100 * iface->release_pin(iface->ctx, motor_state.old_port, motor_state.old_pin_b);
01101 *
01102 * // Reserve new pins
01103 * rx_err_t err = iface->reserve_pin(iface->ctx, new_port, new_pin_a, "MOTOR_ENC_A");
01104 * if (err != k_rx_ok) {
01105 *     // Failed to reserve - try to restore old pins
01106 *     iface->reserve_pin(iface->ctx, motor_state.old_port, motor_state.old_pin_a, "MOTOR_ENC_A");
01107 *     iface->reserve_pin(iface->ctx, motor_state.old_port, motor_state.old_pin_b, "MOTOR_ENC_B");
01108 *     return err;
01109 * }
01110 *
01111 * err = iface->reserve_pin(iface->ctx, new_port, new_pin_b, "MOTOR_ENC_B");
01112 * if (err != k_rx_ok) {
01113 *     // Failed - restore old configuration
01114 *     iface->release_pin(iface->ctx, new_port, new_pin_a);
01115 *     iface->reserve_pin(iface->ctx, motor_state.old_port, motor_state.old_pin_a, "MOTOR_ENC_A");
01116 *     iface->reserve_pin(iface->ctx, motor_state.old_port, motor_state.old_pin_b, "MOTOR_ENC_B");
01117 *     return err;
01118 * }
01119 *
01120 * // Success - update state
01121 * motor_state.old_port = new_port;
01122 * motor_state.old_pin_a = new_pin_a;
01123 * motor_state.old_pin_b = new_pin_b;
01124 * return k_rx_ok;
01125 * }
01126 * @endcode
01127 *
01128 * ### Pattern 4: Defensive Null Check
01129 * @code
01130 * void debug_print_pin_usage(void)
01131 * {
01132 *     rx_pin_interface_t* iface = rx_infrastructure_get_pin_interface();
01133 *     if (iface == nullptr) {
01134 *         debug_uart_printf("Pin validator not initialized\n");
01135 *         return;
01136 *     }
01137 *
01138 *     // Check if debug method exists (optional interface method)
01139 *     if (iface->print_reservations != nullptr) {
01140 *         iface->print_reservations(iface->ctx);
01141 *     } else {
01142 *         debug_uart_printf("Pin reservation printing not supported\n");
01143 *     }
01144 * }
01145 * @endcode
01146 *
01147 * ## Interface Method Availability
01148 *
01149 * All function pointers in the returned interface are **guaranteed non-NULL** except:
01150 * - `is_pin_reserved` (optional, may be NULL)
01151 * - `get_pin_function` (optional, may be NULL)
01152 * - `print_reservations` (optional, may be NULL)
01153 *
01154 * **Always check optional methods** before calling:
01155 * @code
01156 * if (iface->is_pin_reserved != nullptr) {
01157 *     bool reserved = iface->is_pin_reserved(iface->ctx, port, pin);
01158 * }
01159 * @endcode
01160 *
01161 * ## Thread Safety Guarantees
01162 *
01163 * | Operation | Thread-Safe? | Details |
01164 * |-----|-----|-----|
01165 * | Calling this function | [OK] Yes | Read-only after init |
01166 * | Caching returned pointer | [OK] Yes | Pointer never changes |
01167 * | Calling interface methods | [OK] Yes | Methods use mutex protection |
01168 * | Concurrent reserve/release | [OK] Yes | Atomic operations |
01169 * | Calling during init/deinit | [X] No | Undefined behavior |
01170 *
01171 * @return rx_pin_interface_t* Pointer to global pin interface
01172 * @retval non-NULL Success, interface is ready for use
01173 * @retval NULL Infrastructure not initialized (call rx_infrastructure_init() first)
01174 *
01175 * @pre Infrastructure must be initialized via rx_infrastructure_init()
01176 * @pre Must NOT be called during rx_infrastructure_deinit()
01177 *
01178 * @post Returned pointer remains valid until rx_infrastructure_deinit()
01179 * @post All required interface methods are non-NULL
01180 * @post Interface methods are thread-safe (can be called from any thread)
01181 *
01182 * @note **Thread-safe**: Can be called from any thread after initialization.

```

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01183 * @note **Performance**: Simple pointer return (~1 cycle), safe to call frequently.
01184 * @note **Caching**: Safe to cache returned pointer in module state for performance.
01185 * @note **Null check**: Always check for nullptr in defensive code paths.
01186 *
01187 * @warning Do NOT call during rx_infrastructure_init() or rx_infrastructure_deinit().
01188 * @warning NULL return means infrastructure not ready - check initialization order.
01189 * @warning Pin reservation errors mean conflict - do NOT proceed with hardware config.
01190 *
01191 * @par Performance:
01192 * - Execution time: ~5 cycles @ 240 MHz (simple pointer read)
01193 * - No overhead: Direct memory access, no function calls
01194 * - Cache-friendly: Same pointer returned every call
01195 *
01196 * @par Example: Pin Conflict Detection
01197 * @code
01198 * void diagnose_pin_conflict(uint8_t port, uint8_t pin)
01199 * {
01200 *     rx_pin_interface_t* iface = rx_infrastructure_get_pin_interface();
01201 *     if (iface == nullptr) {
01202 *         debug_uart_printf("Pin validator not initialized\n");
01203 *         return;
01204 *     }
01205 *
01206 *     // Check if pin is already reserved
01207 *     if (iface->is_pin_reserved != nullptr) {
01208 *         bool reserved = iface->is_pin_reserved(iface->ctx, port, pin);
01209 *         if (reserved) {
01210 *             // Get function name
01211 *             if (iface->get_pin_function != nullptr) {
01212 *                 const char* function = iface->get_pin_function(iface->ctx, port, pin);
01213 *                 debug_uart_printf("Pin P%X.%d reserved for: %s\n", port, pin, function);
01214 *             } else {
01215 *                 debug_uart_printf("Pin P%X.%d is reserved\n", port, pin);
01216 *             }
01217 *         } else {
01218 *             debug_uart_printf("Pin P%X.%d is available\n", port, pin);
01219 *         }
01220 *     }
01221 * }
01222 * @endcode
01223 *
01224 * @par Example: Bulk Pin Reservation with Cleanup
01225 * @code
01226 * typedef struct {
01227 *     uint8_t port;
01228 *     uint8_t pin;
01229 *     const char* function;
01230 * } pin_reservation_t;
01231 *
01232 * rx_err_t reserve_pins_bulk(const pin_reservation_t* pins, uint32_t count)
01233 * {
01234 *     rx_pin_interface_t* iface = rx_infrastructure_get_pin_interface();
01235 *     if (iface == nullptr) {
01236 *         return k_rx_err_not_initialized;
01237 *     }
01238 *
01239 *     // Reserve all pins
01240 *     for (uint32_t i = 0; i < count; i++) {
01241 *         rx_err_t err = iface->reserve_pin(
01242 *             iface->ctx, pins[i].port, pins[i].pin, pins[i].function
01243 *         );
01244 *         if (err != k_rx_ok) {
01245 *             // Cleanup: Release all previously reserved pins
01246 *             for (uint32_t j = 0; j < i; j++) {
01247 *                 iface->release_pin(iface->ctx, pins[j].port, pins[j].pin);
01248 *             }
01249 *             return err;
01250 *         }
01251 *     }
01252 *
01253 *     return k_rx_ok;
01254 * }
01255 * @endcode
01256 *
01257 * @see rx_infrastructure_init() Initialize infrastructure to make this return non-NULL
01258 * @see rx_infrastructure_deinit() Cleanup infrastructure (sets pointer to NULL)
01259 * @see rx_pin_interface.h Abstract pin validator interface definition
01260 * @see RX_RESERVE_PIN() Convenience macro using this accessor
01261 * @see RX_RELEASE_PIN() Convenience macro using this accessor
01262 *
01263 * @since Version 1.0.0
01264 */
01265 rx_pin_interface_t* rx_infrastructure_get_pin_interface(void);
01266
01267 /*
=====
01268 * Convenience Macros

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```

01269 *
01270 */
01271
01272 /**
01273  * @def RX_REPORT_ERROR
01274  * @brief Report error through global error handler (convenience macro)
01275  *
01276  * @details
01277  * Convenience macro that wraps the error reporting interface to provide simple,
01278  * one-line error reporting throughout the firmware. This is the preferred way
01279  * to report errors in STAR firmware - it's safer, more concise, and more readable
01280  * than directly accessing the interface.
01281  *
01282  * ## Macro Expansion
01283  *
01284  * The macro expands to a safe `do { } while (0)` block that:
01285  * 1. Gets the global error interface
01286  * 2. Checks for nullptr (safe if infrastructure not initialized)
01287  * 3. Calls `report_error()` method if interface is valid
01288  * 4. Returns immediately if interface is nullptr (no-op, no crash)
01289  *
01290  * **Expanded code**:
01291  * @code
01292  * // RX_REPORT_ERROR(k_rx_err_timeout, "SPI", "Transaction timeout");
01293  * // Expands to:
01294  * do {
01295  *     rx_error_interface_t* iface__ = rx_infrastructure_get_error_interface();
01296  *     if (iface__ != nullptr) {
01297 *         iface__->report_error(iface__->ctx, k_rx_err_timeout, "SPI", "Transaction timeout");
01298 *     }
01299 * } while (0)
01300 * @endcode
01301 *
01302 * ## Safety Features
01303 *
01304 * - Null-safe: Silently no-ops if infrastructure not initialized
01305 * - Statement-like: Can be used anywhere a statement is valid
01306 * - No side effects: Parameters evaluated once, no double evaluation
01307 * - Scope-safe: `do { } while (0)` ensures proper scope in all contexts
01308 * - Thread-safe: Underlying interface is thread-safe
01309 *
01310 * ## Usage Examples
01311 *
01312 * ### Example 1: Simple Error Reporting
01313 * @code
01314 * rx_err_t motor_set_velocity(float velocity_mps)
01315 * {
01316 *     if (velocity_mps > k_max_velocity_mps) {
01317 *         RX_REPORT_ERROR(k_rx_err_invalid_arg, "MOTOR", "Velocity exceeds limit");
01318 *         return k_rx_err_invalid_arg;
01319 *     }
01320 *     return k_rx_ok;
01321 * }
01322 * @endcode
01323 *
01324 * ### Example 2: Error Propagation with Reporting
01325 * @code
01326 * rx_err_t motor_init(void)
01327 * {
01328 *     rx_err_t err = encoder_init();
01329 *     if (err != k_rx_ok) {
01330 *         RX_REPORT_ERROR(err, "MOTOR", "Encoder init failed");
01331 *         return err;
01332 *     }
01333 *
01334 *     err = driver_init();
01335 *     if (err != k_rx_ok) {
01336 *         RX_REPORT_ERROR(err, "MOTOR", "Driver init failed");
01337 *         encoder_deinit(); // Cleanup
01338 *         return err;
01339 *     }
01340 *
01341 *     return k_rx_ok;
01342 * }
01343 * @endcode
01344 *
01345 * ### Example 3: Conditional Reporting (if statement)
01346 * @code
01347 * void motor_control_loop(void)
01348 * {
01349 *     if (motor_overheated())
01350 *         RX_REPORT_ERROR(k_rx_err_hardware, "MOTOR", "Overheating detected");
01351 *
01352 *     if (encoder_fault())
01353 *         RX_REPORT_ERROR(k_rx_err_hardware, "MOTOR", "Encoder fault");
01354 * }

```

```

01355 * @endcode
01356 *
01357 * ### Example 4: Error with Dynamic Message
01358 * @code
01359 * void motor_check_velocity(float velocity_mps)
01360 * {
01361 *     char msg[64];
01362 *     if (velocity_mps > k_max_velocity_mps) {
01363 *         snprintf(msg, sizeof(msg), "Velocity %.2f exceeds max %.2f",
01364 *             velocity_mps, k_max_velocity_mps);
01365 *         RX_REPORT_ERROR(k_rx_err_invalid_arg, "MOTOR", msg);
01366 *     }
01367 * }
01368 * @endcode
01369 *
01370 * ## When to Use vs. Direct Interface
01371 *
01372 * | Scenario | Use Macro | Use Direct Interface |
01373 * |-----|-----|-----|
01374 * | Simple error reporting | [OK] Yes | No |
01375 * | One-line logging | [OK] Yes | No |
01376 * | Need retry logic | No | [OK] Yes |
01377 * | Need error counts | No | [OK] Yes |
01378 * | Need backoff delay | No | [OK] Yes |
01379 * | Performance-critical loop | No | [OK] Yes (cache pointer) |
01380 *
01381 * ## Thread Safety
01382 *
01383 * - **Thread-safe**: Can be called from any thread
01384 * - **ISR-safe**: Can be called from interrupts (with caveats, see below)
01385 * - **Reentrant**: Multiple threads can call concurrently
01386 *
01387 * ## ISR Usage Considerations
01388 *
01389 * While technically safe to call from ISR, consider:
01390 * - Error handler may use mutex (can block if ThreadX running)
01391 * - String formatting in ISR is discouraged (stack usage)
01392 * - Prefer simple error codes, defer detailed reporting to thread context
01393 *
01394 * @code
01395 * // ISR Example: Simple error reporting
01396 * void motor_encoder_isr(void)
01397 * {
01398 *     if (encoder_error_detected()) {
01399 *         RX_REPORT_ERROR(k_rx_err_hardware, "MOTOR", "Encoder ISR error");
01400 *     }
01401 * }
01402 * @endcode
01403 *
01404 * @param[in] err Error code (rx_err_t) - Evaluated once, no side effects
01405 * @param[in] component Component name (const char*) - Must be string literal or valid pointer
01406 * @param[in] message Error message (const char*) - Must be string literal or valid pointer
01407 *
01408 * @note Parameters are evaluated **once** - safe to pass function calls.
01409 * @note If infrastructure not initialized, this is a **silent no-op** (no crash).
01410 * @note Component and message strings are **not copied** - must remain valid.
01411 * @note Prefer string literals for component/message (embedded in .rodata).
01412 *
01413 * @warning Do NOT use complex expressions in `component` or `message` parameters
01414 *         that allocate memory or have side effects.
01415 * @warning String parameters must remain valid for the error's lifetime (use literals).
01416 *
01417 * @par Example: Parameter Evaluation Safety
01418 * @code
01419 * // SAFE: Function called once
01420 * RX_REPORT_ERROR(spi_transmit(), "SPI", "TX failed");
01421 *
01422 * // SAFE: String literals (in .rodata)
01423 * RX_REPORT_ERROR(err, "MOTOR", "Velocity error");
01424 *
01425 * // UNSAFE: Dynamic string (stack variable, may be invalid later)
01426 * char msg[64];
01427 * snprintf(msg, sizeof(msg), "Error %d", code);
01428 * RX_REPORT_ERROR(err, "MOTOR", msg); // [FAIL] msg may be invalid when accessed later
01429 *
01430 * // SAFE: Static buffer or string literal
01431 * static char msg[64];
01432 * snprintf(msg, sizeof(msg), "Error %d", code);
01433 * RX_REPORT_ERROR(err, "MOTOR", msg); // [OK] msg remains valid
01434 * @endcode
01435 *
01436 * @see rx_infrastructure_get_error_interface() Get error interface directly
01437 * @see rx_error_interface.h Error interface definition
01438 *
01439 * @since Version 1.0.0
01440 */
01441 #define RX_REPORT_ERROR(err, component, message) \

```

```

01442 do {
01443     rx_error_interface_t* iface_ = rx_infrastructure_get_error_interface();
01444     if (iface_ != nullptr) {
01445         iface_>report_error(iface_>ctx, err, component, message);
01446     }
01447 } while (0)
01448
01449 /**
01450 * @def RX_RESERVE_PIN
01451 * @brief Reserve GPIO pin through global pin validator (convenience macro with return value)
01452 *
01453 * @details
01454 * Convenience macro that wraps the pin reservation interface to provide simple,
01455 * one-line pin reservation throughout the firmware. This is the **preferred way**
01456 * to reserve pins in STAR firmware - it prevents accidental pin conflicts and
01457 * provides explicit ownership tracking.
01458 *
01459 * ## Macro Expansion (GNU Statement Expression)
01460 *
01461 * This macro uses **GNU C extension** `({ ... })` (statement expression) to:
01462 * 1. Get the global pin interface
01463 * 2. Check for nullptr (returns error if infrastructure not initialized)
01464 * 3. Call `reserve_pin()` method if interface is valid
01465 * 4. Return `rx_err_t` error code as the macro's "value"
01466 *
01467 * **Expanded code**:
01468 * @code
01469 // rx_err_t err = RX_RESERVE_PIN(0xA, 5, "SPI_COPI");
01470 // Expands to:
01471 rx_err_t err = ({
01472     rx_err_t err_ = k_rx_err_invalid_state;
01473     rx_pin_interface_t* iface_ = rx_infrastructure_get_pin_interface();
01474     if (iface_ != nullptr) {
01475         err_ = iface_>reserve_pin(iface_>ctx, 0xA, 5, "SPI_COPI");
01476     }
01477     err_; // This becomes the "return value" of the expression
01478 });
01479 @endcode
01480 *
01481 * ## Pin Conflict Prevention
01482 *
01483 * The pin validator ensures that:
01484 * - **No double reservation**: A pin can only be reserved once
01485 * - **Explicit ownership**: Each pin knows which function owns it
01486 * - **Traceable usage**: Pin assignments are tracked in a global table
01487 * - **Atomic operations**: Reservation is thread-safe (mutex protected)
01488 *
01489 * ```
01490 * Module A: RX_RESERVE_PIN(0xA, 5, "SPI_COPI") -> k_rx_ok
01491 * Module B: RX_RESERVE_PIN(0xA, 5, "I2C_SDA") -> k_rx_err_resource_busy (conflict!)
01492 * ```
01493 *
01494 * ## Typical Usage Patterns
01495 *
01496 * ### Pattern 1: Reserve-Check-Cleanup on Error
01497 * @code
01498 rx_err_t spi_driver_init(void)
01499 {
01500     // Reserve CIPO (first pin)
01501     rx_err_t err = RX_RESERVE_PIN(0xA, 0, "SPI_CIPO");
01502     if (err != k_rx_ok) {
01503         RX_REPORT_ERROR(err, "SPI", "Pin PA0 (CIPO) conflict");
01504         return err;
01505     }
01506
01507     // Reserve COPI (second pin)
01508     err = RX_RESERVE_PIN(0xA, 1, "SPI_COPI");
01509     if (err != k_rx_ok) {
01510         RX_RELEASE_PIN(0xA, 0); // Clean up first pin
01511         RX_REPORT_ERROR(err, "SPI", "Pin PA1 (COPI) conflict");
01512         return err;
01513     }
01514
01515     // Reserve CLK (third pin)
01516     err = RX_RESERVE_PIN(0xA, 2, "SPI_CLK");
01517     if (err != k_rx_ok) {
01518         RX_RELEASE_PIN(0xA, 0); // Clean up first two pins
01519         RX_RELEASE_PIN(0xA, 1);
01520         RX_REPORT_ERROR(err, "SPI", "Pin PA2 (CLK) conflict");
01521         return err;
01522     }
01523
01524     // All pins successfully reserved - proceed with hardware config
01525     configure_spi_hardware();
01526     return k_rx_ok;
01527 }
01528 * @endcode

```

```

01529 *
01530 * ### Pattern 2: Bulk Reservation Helper
01531 * @code
01532 * typedef struct {
01533 *     uint8_t port;
01534 *     uint8_t pin;
01535 *     const char* function;
01536 * } pin_def_t;
01537 *
01538 * // Define all SPI pins
01539 * static const pin_def_t k_spi_pins[] = {
01540 *     {0xA, 0, "SPI_CIPO"},
01541 *     {0xA, 1, "SPI_COPI"},
01542 *     {0xA, 2, "SPI_CLK"},
01543 *     {0xA, 3, "SPI_CS"},
01544 * };
01545 *
01546 * rx_err_t reserve_spi_pins(void)
01547 * {
01548 *     for (uint32_t i = 0; i < ARRAY_SIZE(k_spi_pins); i++) {
01549 *         rx_err_t err = RX_RESERVE_PIN(
01550 *             k_spi_pins[i].port,
01551 *             k_spi_pins[i].pin,
01552 *             k_spi_pins[i].function
01553 *         );
01554 *         if (err != k_rx_ok) {
01555 *             // Cleanup all previously reserved pins
01556 *             for (uint32_t j = 0; j < i; j++) {
01557 *                 RX_RELEASE_PIN(k_spi_pins[j].port, k_spi_pins[j].pin);
01558 *             }
01559 *             return err;
01560 *         }
01561 *     }
01562 *     return k_rx_ok;
01563 * }
01564 * @endcode
01565 *
01566 * ### Pattern 3: Inline Error Check
01567 * @code
01568 * void motor_init(void)
01569 * {
01570 *     // Reserve and check in single if-statement
01571 *     if (RX_RESERVE_PIN(0xB, 4, "MOTOR_PWM") != k_rx_ok) {
01572 *         // Handle error
01573 *         return k_rx_err_resource_busy;
01574 *     }
01575 *
01576 *     if (RX_RESERVE_PIN(0xB, 5, "MOTOR_DIR") != k_rx_ok) {
01577 *         RX_RELEASE_PIN(0xB, 4); // Cleanup
01578 *         return k_rx_err_resource_busy;
01579 *     }
01580 * }
01581 * @endcode
01582 *
01583 * ## Port and Pin Numbering
01584 *
01585 * **Port numbers**: Use hexadecimal for RX72N ports (0x0 = Port 0, 0xA = Port A, etc.)
01586 * **Pin numbers**: Use decimal 0-7 for pin within port
01587 *
01588 * | Port Letter | Hex Value | Example Pin | Macro Call |
01589 * |-----|-----|-----|-----|
01590 * | Port 0     | 0x0      | P0.5        | RX_RESERVE_PIN(0x0, 5, "name") |
01591 * | Port A     | 0xA      | PA.3        | RX_RESERVE_PIN(0xA, 3, "name") |
01592 * | Port B     | 0xB      | PB.7        | RX_RESERVE_PIN(0xB, 7, "name") |
01593 * | Port E     | 0xE      | PE.1        | RX_RESERVE_PIN(0xE, 1, "name") |
01594 *
01595 * @param[in] port Port number in hexadecimal (0x0-0xF for RX72N)
01596 * @param[in] pin Pin number within port (0-7)
01597 * @param[in] function Function name (const char* string literal) - identifies owner
01598 *
01599 * @return rx_err_t Error code indicating success or failure
01600 * @retval k_rx_ok Success, pin reserved
01601 * @retval k_rx_err_invalid_state Infrastructure not initialized
01602 * @retval k_rx_err_resource_busy Pin already reserved by another function
01603 * @retval k_rx_err_invalid_arg Invalid port or pin number
01604 * @retval k_rx_err_null_ptr Null function name
01605 *
01606 * @note **GNU C Extension**: Uses statement expression `{ }` - requires GCC/Clang.
01607 * @note **Returns value**: Can be assigned to variable or used directly in if-statement.
01608 * @note **Thread-safe**: Can be called from multiple threads concurrently.
01609 * @note **Function string**: Must be string literal or valid pointer (not copied).
01610 * @note **No cleanup on error**: Caller must manually release pins on partial failure.
01611 *
01612 * @warning **Must check return value**: Ignoring errors can lead to pin conflicts.
01613 * @warning **Cleanup required**: On error, release any previously reserved pins.
01614 * @warning **String lifetime**: Function name must remain valid (use string literals).
01615 * @warning **Not ISR-safe**: Uses mutex internally, do NOT call from ISR.

```

```

01616 *
01617 * @par Example: Complete Module Pin Management
01618 * @code
01619 * typedef struct {
01620 *     bool pins_reserved;
01621 *     // ... other state
01622 * } module_state_t;
01623 *
01624 * rx_err_t module_init(module_state_t* state)
01625 * {
01626 *     // Reserve all module pins
01627 *     rx_err_t err = RX_RESERVE_PIN(0xA, 0, "MODULE_PIN0");
01628 *     if (err != k_rx_ok) {
01629 *         return err;
01630 *     }
01631 *
01632 *     err = RX_RESERVE_PIN(0xA, 1, "MODULE_PIN1");
01633 *     if (err != k_rx_ok) {
01634 *         RX_RELEASE_PIN(0xA, 0); // Cleanup
01635 *         return err;
01636 *     }
01637 *
01638 *     state->pins_reserved = true;
01639 *     return k_rx_ok;
01640 * }
01641 *
01642 * void module_deinit(module_state_t* state)
01643 * {
01644 *     if (state->pins_reserved) {
01645 *         RX_RELEASE_PIN(0xA, 0);
01646 *         RX_RELEASE_PIN(0xA, 1);
01647 *         state->pins_reserved = false;
01648 *     }
01649 * }
01650 * @endcode
01651 *
01652 * @see RX_RELEASE_PIN() Release a reserved pin
01653 * @see rx_infrastructure_get_pin_interface() Get pin interface directly
01654 * @see rx_pin_interface.h Pin interface definition
01655 *
01656 * @since Version 1.0.0
01657 */
01658 #define RX_RESERVE_PIN(port, pin, function)
01659 ({
01660     rx_err_t err_ = k_rx_err_invalid_state;
01661     rx_pin_interface_t* iface_ = rx_infrastructure_get_pin_interface();
01662     if (iface_ != nullptr) {
01663         err_ = iface_->reserve_pin(iface_->ctx, port, pin, function);
01664     }
01665     err_;
01666 })
01667 /**
01668 * @def RX_RELEASE_PIN
01669 * @brief Release reserved GPIO pin through global pin validator (convenience macro with return value)
01670 *
01671 * @details
01672 * Convenience macro that wraps the pin release interface to provide simple, one-line
01673 * pin release throughout the firmware. This is the required way to release pins
01674 * reserved with RX_RESERVE_PIN() - it ensures proper cleanup and prevents resource leaks.
01675 *
01676 * ## Macro Expansion (GNU Statement Expression)
01677 *
01678 * This macro uses GNU C extension { ( ... ) } (statement expression) to:
01679 * 1. Get the global pin interface
01680 * 2. Check for nullptr (returns error if infrastructure not initialized)
01681 * 3. Call release_pin() method if interface is valid
01682 * 4. Return rx_err_t error code as the macro's "value"
01683 *
01684 * **Expanded code**:
01685 * @code
01686 * // rx_err_t err = RX_RELEASE_PIN(0xA, 5);
01687 * // Expands to:
01688 * rx_err_t err = ({
01689 *     rx_err_t err_ = k_rx_err_invalid_state;
01690 *     rx_pin_interface_t* iface_ = rx_infrastructure_get_pin_interface();
01691 *     if (iface_ != nullptr) {
01692 *         err_ = iface_->release_pin(iface_->ctx, 0xA, 5);
01693 *     }
01694 *     err_; // This becomes the "return value" of the expression
01695 * });
01696 * @endcode
01697 *
01698 * ## When to Release Pins
01699 *
01700 * Pins should be released in these scenarios:
01701 * 1. Module deinitialization: During shutdown or cleanup

```



```

01703 * 2. **Error handling**: On partial initialization failure (cleanup)
01704 * 3. **Reconfiguration**: Before reserving different pins for same module
01705 * 4. **Resource sharing**: When pin is no longer needed and can be reused
01706 *
01707 * ## Typical Usage Patterns
01708 *
01709 * ### Pattern 1: Normal Deinitialization
01710 * @code
01711 * rx_err_t spi_driver_init(void)
01712 * {
01713 *     RX_RESERVE_PIN(0xA, 0, "SPI_CIPO");
01714 *     RX_RESERVE_PIN(0xA, 1, "SPI_COPI");
01715 *     RX_RESERVE_PIN(0xA, 2, "SPI_CLK");
01716 *     // ... init logic
01717 *     return k_rx_ok;
01718 * }
01719 *
01720 * void spi_driver_deinit(void)
01721 * {
01722 *     // Release all SPI pins (in any order)
01723 *     RX_RELEASE_PIN(0xA, 0); // CIPO
01724 *     RX_RELEASE_PIN(0xA, 1); // COPI
01725 *     RX_RELEASE_PIN(0xA, 2); // CLK
01726 *     // ... cleanup logic
01727 * }
01728 * @endcode
01729 *
01730 * ### Pattern 2: Error Cleanup (Reverse Order Recommended)
01731 * @code
01732 * rx_err_t module_init(void)
01733 * {
01734 *     rx_err_t err;
01735 *
01736 *     // Reserve pin 0
01737 *     err = RX_RESERVE_PIN(0xB, 0, "MODULE_PIN0");
01738 *     if (err != k_rx_ok) {
01739 *         return err;
01740 *     }
01741 *
01742 *     // Reserve pin 1
01743 *     err = RX_RESERVE_PIN(0xB, 1, "MODULE_PIN1");
01744 *     if (err != k_rx_ok) {
01745 *         RX_RELEASE_PIN(0xB, 0); // Clean up pin 0
01746 *         return err;
01747 *     }
01748 *
01749 *     // Reserve pin 2
01750 *     err = RX_RESERVE_PIN(0xB, 2, "MODULE_PIN2");
01751 *     if (err != k_rx_ok) {
01752 *         RX_RELEASE_PIN(0xB, 1); // Clean up in reverse order
01753 *         RX_RELEASE_PIN(0xB, 0);
01754 *         return err;
01755 *     }
01756 *
01757 *     return k_rx_ok;
01758 * }
01759 * @endcode
01760 *
01761 * ### Pattern 3: Dynamic Reconfiguration
01762 * @code
01763 * rx_err_t motor_change_encoder_pins(uint8_t new_port, uint8_t new_pin_a, uint8_t new_pin_b)
01764 * {
01765 *     // Save old configuration in case new reservation fails
01766 *     uint8_t old_port = motor_state.encoder_port;
01767 *     uint8_t old_pin_a = motor_state.encoder_pin_a;
01768 *     uint8_t old_pin_b = motor_state.encoder_pin_b;
01769 *
01770 *     // Release old pins
01771 *     RX_RELEASE_PIN(old_port, old_pin_a);
01772 *     RX_RELEASE_PIN(old_port, old_pin_b);
01773 *
01774 *     // Reserve new pins
01775 *     rx_err_t err = RX_RESERVE_PIN(new_port, new_pin_a, "MOTOR_ENC_A");
01776 *     if (err != k_rx_ok) {
01777 *         // Failed - restore old pins
01778 *         RX_RESERVE_PIN(old_port, old_pin_a, "MOTOR_ENC_A");
01779 *         RX_RESERVE_PIN(old_port, old_pin_b, "MOTOR_ENC_B");
01780 *         return err;
01781 *     }
01782 *
01783 *     err = RX_RESERVE_PIN(new_port, new_pin_b, "MOTOR_ENC_B");
01784 *     if (err != k_rx_ok) {
01785 *         // Failed - restore old configuration
01786 *         RX_RELEASE_PIN(new_port, new_pin_a);
01787 *         RX_RESERVE_PIN(old_port, old_pin_a, "MOTOR_ENC_A");
01788 *         RX_RESERVE_PIN(old_port, old_pin_b, "MOTOR_ENC_B");
01789 *         return err;

```



```

01790 * }
01791 *
01792 * // Success - update state
01793 * motor_state.encoder_port = new_port;
01794 * motor_state.encoder_pin_a = new_pin_a;
01795 * motor_state.encoder_pin_b = new_pin_b;
01796 * return k_rx_ok;
01797 * }
01798 * @endcode
01799 *
01800 * ### Pattern 4: Idempotent Cleanup (Safe Double-Release)
01801 * @code
01802 * typedef struct {
01803 *     bool pin_reserved;
01804 *     uint8_t port;
01805 *     uint8_t pin;
01806 * } module_state_t;
01807 *
01808 * void module_cleanup(module_state_t* state)
01809 * {
01810 *     // Check flag to avoid double-release
01811 *     if (state->pin_reserved) {
01812 *         RX_RELEASE_PIN(state->port, state->pin);
01813 *         state->pin_reserved = false;
01814 *     }
01815 * }
01816 *
01817 * // Can be called multiple times safely
01818 * module_cleanup(&state); // Releases pin
01819 * module_cleanup(&state); // No-op (flag already false)
01820 * module_cleanup(&state); // No-op
01821 * @endcode
01822 *
01823 * ## Return Value Handling
01824 *
01825 * Release errors are rare but can occur:
01826 * - Infrastructure not initialized ('k_rx_err_invalid_state')
01827 * - Pin not reserved ('k_rx_err_not_found')
01828 * - Invalid port/pin ('k_rx_err_invalid_arg')
01829 *
01830 * Most code ignores release errors during cleanup (best-effort cleanup).
01831 * However, for critical code, check the return value:
01832 *
01833 * @code
01834 * // Best-effort cleanup (typical)
01835 * void module_deinit(void)
01836 * {
01837 *     RX_RELEASE_PIN(0xA, 0); // Ignore errors
01838 *     RX_RELEASE_PIN(0xA, 1);
01839 *     RX_RELEASE_PIN(0xA, 2);
01840 * }
01841 *
01842 * // Strict cleanup (critical systems)
01843 * rx_err_t module_deinit_strict(void)
01844 * {
01845 *     rx_err_t err;
01846 *
01847 *     err = RX_RELEASE_PIN(0xA, 0);
01848 *     if (err != k_rx_ok) {
01849 *         RX_REPORT_ERROR(err, "MODULE", "Failed to release pin PA0");
01850 *         return err;
01851 *     }
01852 *
01853 *     err = RX_RELEASE_PIN(0xA, 1);
01854 *     if (err != k_rx_ok) {
01855 *         RX_REPORT_ERROR(err, "MODULE", "Failed to release pin PA1");
01856 *         return err;
01857 *     }
01858 *
01859 *     return k_rx_ok;
01860 * }
01861 * @endcode
01862 *
01863 * ## Port and Pin Numbering
01864 *
01865 * Port numbers: Use hexadecimal for RX72N ports (0x0 = Port 0, 0xA = Port A, etc.)
01866 * Pin numbers: Use decimal 0-7 for pin within port
01867 *
01868 * | Port Letter | Hex Value | Example Pin | Macro Call |
01869 * |-----|-----|-----|-----|
01870 * | Port 0     | 0x0     | P0.5       | RX_RELEASE_PIN(0x0, 5) |
01871 * | Port A     | 0xA     | PA.3       | RX_RELEASE_PIN(0xA, 3) |
01872 * | Port B     | 0xB     | PB.7       | RX_RELEASE_PIN(0xB, 7) |
01873 * | Port E     | 0xE     | PE.1       | RX_RELEASE_PIN(0xE, 1) |
01874 *
01875 * @param[in] port Port number in hexadecimal (0x0-0xF for RX72N)
01876 * @param[in] pin Pin number within port (0-7)

```

```

01877 *
01878 * @return rx_err_t Error code indicating success or failure
01879 * @retval k_rx_ok Success, pin released and available for reuse
01880 * @retval k_rx_err_invalid_state Infrastructure not initialized
01881 * @retval k_rx_err_not_found Pin was not reserved (double-release or never reserved)
01882 * @retval k_rx_err_invalid_arg Invalid port or pin number
01883 *
01884 * @note **GNU C Extension**: Uses statement expression `({ })` - requires GCC/Clang.
01885 * @note **Returns value**: Can be assigned to variable or used directly in if-statement.
01886 * @note **Thread-safe**: Can be called from multiple threads concurrently.
01887 * @note **Idempotent safe**: Releasing an unreserved pin returns error but doesn't crash.
01888 * @note **Best-effort cleanup**: Most code ignores return value during cleanup.
01889 *
01890 * @warning **Release what you reserve**: Only release pins you previously reserved.
01891 * @warning **Double-release**: Releasing same pin twice returns k_rx_err_not_found.
01892 * @warning **Not ISR-safe**: Uses mutex internally, do NOT call from ISR.
01893 * @warning **No automatic release**: Pins are NOT auto-released on thread exit - must call explicitly.
01894 *
01895 * @par Example: Complete Lifecycle Management
01896 * @code
01897 * typedef struct {
01898 *     uint8_t port;
01899 *     uint8_t pin;
01900 *     bool initialized;
01901 * } led_driver_t;
01902 *
01903 * rx_err_t led_driver_init(led_driver_t* driver, uint8_t port, uint8_t pin)
01904 * {
01905 *     // Reserve LED pin
01906 *     rx_err_t err = RX_RESERVE_PIN(port, pin, "LED");
01907 *     if (err != k_rx_ok) {
01908 *         RX_REPORT_ERROR(err, "LED", "Pin reservation failed");
01909 *         return err;
01910 *     }
01911 *
01912 *     // Configure pin as output
01913 *     gpio_set_direction(port, pin, GPIO_OUTPUT);
01914 *     gpio_write(port, pin, 0); // LED off
01915 *
01916 *     driver->port = port;
01917 *     driver->pin = pin;
01918 *     driver->initialized = true;
01919 *     return k_rx_ok;
01920 * }
01921 *
01922 * void led_driver_deinit(led_driver_t* driver)
01923 * {
01924 *     if (!driver->initialized) {
01925 *         return; // Already deinitialized
01926 *     }
01927 *
01928 *     // Turn off LED
01929 *     gpio_write(driver->port, driver->pin, 0);
01930 *
01931 *     // Release pin (best-effort, ignore errors)
01932 *     rx_err_t err = RX_RELEASE_PIN(driver->port, driver->pin);
01933 *     if (err != k_rx_ok) {
01934 *         // Log but don't fail cleanup
01935 *         RX_REPORT_ERROR(err, "LED", "Pin release warning");
01936 *     }
01937 *
01938 *     driver->initialized = false;
01939 * }
01940 * @endcode
01941 *
01942 * @par Example: Bulk Release Helper
01943 * @code
01944 * typedef struct {
01945 *     uint8_t port;
01946 *     uint8_t pin;
01947 * } pin_ref_t;
01948 *
01949 * void release_pins_bulk(const pin_ref_t* pins, uint32_t count)
01950 * {
01951 *     for (uint32_t i = 0; i < count; i++) {
01952 *         rx_err_t err = RX_RELEASE_PIN(pins[i].port, pins[i].pin);
01953 *         if (err != k_rx_ok && err != k_rx_err_not_found) {
01954 *             // Log unexpected errors (ignore not-found, it's benign)
01955 *             RX_REPORT_ERROR(err, "BULK", "Pin release failed");
01956 *         }
01957 *     }
01958 * }
01959 * @endcode
01960 *
01961 * @see RX_RESERVE_PIN() Reserve a pin (must be called first)
01962 * @see rx_infrastructure_get_pin_interface() Get pin interface directly
01963 * @see rx_pin_interface.h Pin interface definition

```

```

01964 *
01965 * @since Version 1.0.0
01966 */
01967 #define RX_RELEASE_PIN(port, pin)
01968 ({
01969     rx_err_t      err__ = k_rx_err_invalid_state;
01970     rx_pin_interface_t* iface__ = rx_infrastructure_get_pin_interface();
01971     if (iface__ != nullptr) {
01972         err__ = iface__->release_pin(iface__->ctx, port, pin);
01973     }
01974     err__;
01975 })
01976
01977 #ifndef __cplusplus
01978 }
01979 #endif

```

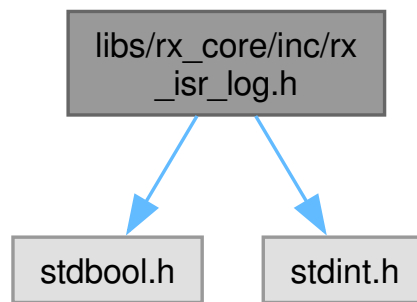
7.17 libs/rx_core/inc/rx_isr_log.h File Reference

ISR-safe deferred logging facility.

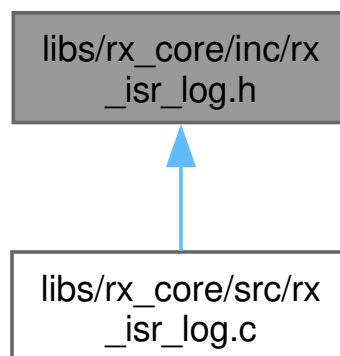
```
#include <stdbool.h>
```

```
#include <stdint.h>
```

Include dependency graph for rx_isr_log.h:



This graph shows which files directly or indirectly include this file:



Data Structures

- struct [rx_isr_log_record_t](#)
One pending log emission queued by an ISR for the drain hook.
- struct [rx_isr_log_stats_t](#)
Diagnostic counters for the ISR log ring.

Enumerations

- enum `rx_isr_log_constants_t`: `uint16_t` { `k_rx_isr_log_capacity`, `k_rx_isr_log_capacity_mask` }

Sizing and queue limits for the ISR log ring.

- enum `rx_isr_log_event_id_t`: `uint8_t` {
`k_isr_log_event_none` = 0, `k_isr_log_event_usb_vbus_se0`, `k_isr_log_event_usb_vbus_fs_j`,
`k_isr_log_event_usb_dvst_powered`,
`k_isr_log_event_usb_dvst_default`, `k_isr_log_event_usb_dvst_address`, `k_isr_log_event_usb_dvst_configured`,
`k_isr_log_event_usb_dvst_suspended`,
`k_isr_log_event_usb_dvst_unknown`, `k_isr_log_event_usb_resume`, `k_isr_log_event_usb_ctr_t_seq_err`,
`k_isr_log_event_cdc_descriptor_short_write`,
`k_isr_log_event_cdc_pipe_rolled_back`, `k_isr_log_event_cdc_port0_bulk_in_failed`,
`k_isr_log_event_cdc_port0_bulk_out_failed`, `k_isr_log_event_cdc_port0_int_in_failed`,
`k_isr_log_event_cdc_port1_bulk_in_failed`, `k_isr_log_event_cdc_port1_bulk_out_failed`,
`k_isr_log_event_cdc_port1_int_in_failed`, `k_isr_log_event_cdc_port2_bulk_in_failed`,
`k_isr_log_event_cdc_port2_bulk_out_failed`, `k_isr_log_event_cdc_port2_int_in_failed`,
`k_isr_log_event_cdc_composite_configured`, `k_isr_log_event_cdc_line_coding_short_write`,
`k_isr_log_event_count` }

Identifier for every distinct event the ISR layer can log.

Functions

- bool `rx_isr_log_push` (`rx_isr_log_event_id_t` event_id, `uint32_t` value)
Push one log event from interrupt context.
- `uint32_t` `rx_isr_log_drain` (void)
Drain all pending records and emit them via `rx_log_*`.
- void `rx_isr_log_get_stats` (`rx_isr_log_stats_t` *stats)
Snapshot diagnostic counters.

7.17.1 Detailed Description

ISR-safe deferred logging facility.

Single-producer-single-consumer (SPSC) ring buffer that lets ISR handlers record a log "event" without ever calling into `rx_log_*` directly. The `rx_log_*` macros funnel through `rx_log_uart.c` which acquires a ThreadX mutex with `TX_WAIT_FOREVER` – legal from a task, illegal from interrupt context (deadlock or undefined behavior under ThreadX).

7.17.1.1 Producer side (ISR)

Push a record with `rx_isr_log_push()`. The call is lock-free: it reads tail, computes (head + 1) % capacity, and either advances head with the record stored or bumps a drop counter. No ThreadX call, no I/O, no busy-wait. Safe to invoke from any interrupt-priority context.

7.17.1.2 Consumer side (task)

A low-priority task (currently `comm_task`) calls `rx_isr_log_drain()` once per poll tick. The drain reads each pending record, looks up its format string from a static table, and emits via the regular `rx_log_*` macros so the message ends up wrapped in the same UART log frame as task-context logs. After all records are drained, if `drop_count` advanced since the last drain it emits a single "ISR log overflow: dropped N records" summary so observers know events were lost.

7.17.1.3 Ring sizing

`k_rx_isr_log_capacity` = 32 records, 8 bytes each = 256 bytes total. At the `comm_task` drain rate of 100 Hz this absorbs an event burst of ~ 3200 /s before any drops occur, far above the steady-state USB ISR event rate of ~ 10 -20 events/s during enumeration and ~ 0 /s after.

7.17.1.4 Event model

Each event is identified by an `rx_isr_log_event_id_t` enum value which indexes a static const table of {tag, format, level} tuples. A 32-bit value field carries one optional auxiliary integer (port id, error code, register snapshot, etc.). This keeps records fixed-size and trivially copyable, avoiding any `sprintf`-style formatting in the ISR.

Author

Locked, Inc.

Date

2026-04-25

Copyright

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Since

Version 1.0.0

Definition in file [rx_isr_log.h](#).

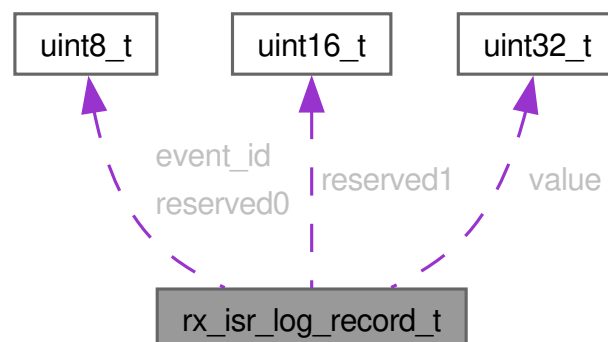
7.17.2 Data Structure Documentation

7.17.2.1 struct rx_isr_log_record_t

One pending log emission queued by an ISR for the drain hook.

Definition at line 114 of file [rx_isr_log.h](#).

Collaboration diagram for `rx_isr_log_record_t`:



Data Fields

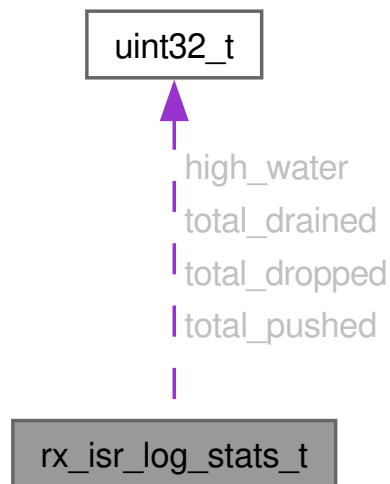
uint8_t	event_id	rx_isr_log_event_id_t (kept as uint8_t for fixed packing)
uint8_t	reserved0	
uint16_t	reserved1	
uint32_t	value	Auxiliary value (event-specific; 0 if unused)

7.17.2.2 struct rx_isr_log_stats_t

Diagnostic counters for the ISR log ring.

Definition at line 125 of file [rx_isr_log.h](#).

Collaboration diagram for rx_isr_log_stats_t:



Data Fields

uint32_t	high_water	Maximum observed occupancy (records)
uint32_t	total_drained	Records emitted via <code>rx_log_*</code> by the drain hook
uint32_t	total_dropped	Records discarded because the ring was full
uint32_t	total_pushed	Records accepted by the ring

7.17.3 Enumeration Type Documentation

7.17.3.1 rx_isr_log_constants_t

```
enum rx\_isr\_log\_constants\_t : uint16_t
```

Sizing and queue limits for the ISR log ring.

Enumerator

k_rx_isr_log_capacity	Number of records the ring can hold before dropping (power of two)
k_rx_isr_log_capacity_mask	Bitwise mask for index modulo (capacity - 1). MUST track capacity.

Definition at line 60 of file [rx_isr_log.h](#).

```

00060      : uint16_t {
00061  k_rx_isr_log_capacity =
00062    32U, /**< Number of records the ring can hold before dropping (power of two) */
00063  k_rx_isr_log_capacity_mask =
00064    31U, /**< Bitwise mask for index modulo (capacity - 1). MUST track capacity. */
00065 } rx_isr_log_constants_t;
```

7.17.3.2 rx_isr_log_event_id_t

```
enum rx_isr_log_event_id_t : uint8_t
```

Identifier for every distinct event the ISR layer can log.

The drain hook indexes a static format-string table by this enum so the ISR never needs to reach for a literal string at runtime. Add a new entry here, add a matching row in `s_isr_log_event_table[]` in [rx_isr_log.c](#), and the new event is wired up. Keep entries grouped by source module.

Enumerator

k_isr_log_event_none	Reserved sentinel; never enqueued
k_isr_log_event_usb_vbus_se0	
k_isr_log_event_usb_vbus_fs_j	
k_isr_log_event_usb_dvst_powered	
k_isr_log_event_usb_dvst_default	
k_isr_log_event_usb_dvst_address	
k_isr_log_event_usb_dvst_configured	
k_isr_log_event_usb_dvst_suspended	
k_isr_log_event_usb_dvst_unknown	
k_isr_log_event_usb_resume	
k_isr_log_event_usb_ctr_tseq_err	
k_isr_log_event_cdc_descriptor_short_write	
k_isr_log_event_cdc_pipe_rolled_back	
k_isr_log_event_cdc_port0_bulk_in_failed	
k_isr_log_event_cdc_port0_bulk_out_failed	
k_isr_log_event_cdc_port0_int_in_failed	
k_isr_log_event_cdc_port1_bulk_in_failed	
k_isr_log_event_cdc_port1_bulk_out_failed	
k_isr_log_event_cdc_port1_int_in_failed	
k_isr_log_event_cdc_port2_bulk_in_failed	
k_isr_log_event_cdc_port2_bulk_out_failed	

k_isr_log_event_cdc_port2_int_in_failed	
k_isr_log_event_cdc_composite_configured	
k_isr_log_event_cdc_line_coding_short_write	
k_isr_log_event_count	Sentinel: total number of defined events

Definition at line 77 of file [rx_isr_log.h](#).

```

00077     : uint8_t {
00078 k_isr_log_event_none = 0, /**< Reserved sentinel; never enqueued */
00079
00080 /* USB ISR events (rx_usb_isr.c) */
00081 k_isr_log_event_usb_vbus_se0,
00082 k_isr_log_event_usb_vbus_fs_j,
00083 k_isr_log_event_usb_dvst_powered,
00084 k_isr_log_event_usb_dvst_default,
00085 k_isr_log_event_usb_dvst_address,
00086 k_isr_log_event_usb_dvst_configured,
00087 k_isr_log_event_usb_dvst_suspended,
00088 k_isr_log_event_usb_dvst_unknown,
00089 k_isr_log_event_usb_resume,
00090 k_isr_log_event_usb_ctrl_seq_err,
00091
00092 /* USB CDC ISR-context events (rx_usb_cdc.c) */
00093 k_isr_log_event_cdc_descriptor_short_write,
00094 k_isr_log_event_cdc_pipe_rolled_back,
00095 k_isr_log_event_cdc_port0_bulk_in_failed,
00096 k_isr_log_event_cdc_port0_bulk_out_failed,
00097 k_isr_log_event_cdc_port0_int_in_failed,
00098 k_isr_log_event_cdc_port1_bulk_in_failed,
00099 k_isr_log_event_cdc_port1_bulk_out_failed,
00100 k_isr_log_event_cdc_port1_int_in_failed,
00101 k_isr_log_event_cdc_port2_bulk_in_failed,
00102 k_isr_log_event_cdc_port2_bulk_out_failed,
00103 k_isr_log_event_cdc_port2_int_in_failed,
00104 k_isr_log_event_cdc_composite_configured,
00105 k_isr_log_event_cdc_line_coding_short_write,
00106
00107 k_isr_log_event_count, /**< Sentinel: total number of defined events */
00108 } rx_isr_log_event_id_t;

```

7.17.4 Function Documentation

7.17.4.1 rx_isr_log_drain()

```

uint32_t rx_isr_log_drain (
    void )

```

Drain all pending records and emit them via `rx_log_*`.

Called from a task-context periodic hook (currently `comm_task` at 100 Hz). Walks tail forward to head, emitting one `rx_log_*` call per record. Also detects and reports newly dropped records via a single summary log.

Returns

Number of records drained on this call

Note

MUST be called from task context only – it invokes the regular rx_log_* macros which acquire a ThreadX mutex.

Definition at line 312 of file rx_isr_log.c.

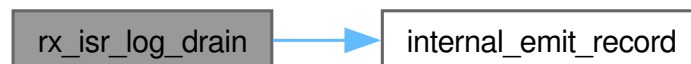
```

00313 {
00314     uint32_t drained = 0U;
00315
00316     /* Snapshot head once -- new pushes after this point will be picked up by
00317      * the next drain call. Bounding the loop by the snapshot keeps execution
00318      * time per drain bounded (NASA Rule 2) even if the ISR is spamming the
00319      * ring concurrently. */
00320     const uint32_t head_snapshot = s_head;
00321
00322     while (s_tail != head_snapshot) {
00323         const uint32_t slot = s_tail & (uint32_t)k_rx_isr_log_capacity_mask;
00324         const rx_isr_log_record_t rec = {
00325             .event_id = s_ring[slot].event_id,
00326             .reserved0 = s_ring[slot].reserved0,
00327             .reserved1 = s_ring[slot].reserved1,
00328             .value = s_ring[slot].value,
00329         };
00330
00331         internal_emit_record(&rec);
00332
00333         /* Advance tail AFTER reading the record so the producer cannot reuse
00334          * the slot before we are done with it. */
00335         s_tail = s_tail + 1U;
00336         drained++;
00337         s_total_drained++;
00338     }
00339
00340     /* Emit a single summary log if any drops occurred since the previous
00341      * drain. We deliberately use _val so the count surfaces in the message. */
00342     const uint32_t cur_drops = s_total_dropped;
00343     if (cur_drops > s_drop_count_last_seen) {
00344         const uint32_t delta = cur_drops - s_drop_count_last_seen;
00345         rx_log_warn_val("ISR_LOG", "ISR log overflow: dropped records", delta);
00346         s_drop_count_last_seen = cur_drops;
00347     }
00348
00349     return drained;
00350 }

```

References [internal_emit_record\(\)](#), [k_rx_isr_log_capacity_mask](#), [rx_log_warn_val](#), [s_drop_count_last_seen](#), [s_head](#), [s_ring](#), [s_tail](#), [s_total_drained](#), and [s_total_dropped](#).

Here is the call graph for this function:



7.17.4.2 rx_isr_log_get_stats()

```

void rx_isr_log_get_stats (
    rx_isr_log_stats_t * stats)

```

Snapshot diagnostic counters.

Parameters

out	stats	Statistics destination (nullptr is silently ignored)
-----	-------	--

Definition at line 352 of file rx_isr_log.c.

```

00353 {
00354     if (stats == NULL) {
00355         return;
00356     }
00357     stats->total_pushed = s_total_pushed;
00358     stats->total_drained = s_total_drained;
00359     stats->total_dropped = s_total_dropped;
00360     stats->high_water = s_high_water;
00361 }

```

References [rx_isr_log_stats_t::high_water](#), [s_high_water](#), [s_total_drained](#), [s_total_dropped](#), [s_total_pushed](#), [rx_isr_log_stats_t::total_drained](#), [rx_isr_log_stats_t::total_dropped](#), and [rx_isr_log_stats_t::total_pushed](#).

7.17.4.3 rx_isr_log_push()

```

bool rx_isr_log_push (
    rx_isr_log_event_id_t event_id,
    uint32_t value)

```

Push one log event from interrupt context.

Lock-free SPSC enqueue. Reads tail, computes the next head; if the ring is full, increments the drop counter and returns false. Otherwise writes the record at the current head slot, then advances head. Safe to call from any ISR or task; the only producer is the ISR by convention but the implementation is correct for any single-thread producer because the head/tail are written by exactly one side each.

Parameters

in	event_id	One of rx_isr_log_event_id_t . k_isr_log_event_none and values $\geq$ k_isr_log_event_count are silently dropped (counted as drops).
in	value	Auxiliary 32-bit value (0 if the event has no value)

Returns

true if the record was queued, false if dropped

Note

ISR-SAFE: no mutex, no I/O, no allocation, no busy-wait.

Definition at line 210 of file [rx_isr_log.c](#).

```

00211 {
00212     /* Reject sentinel and out-of-range ids -- count as drops so a corrupted
00213      * call site shows up in stats rather than silently mis-emitting later. */
00214     if ((event_id == k_isr_log_event_none) ||
00215         ((uint32_t)event_id >= (uint32_t)k_isr_log_event_count)) {
00216         s_total_dropped++;
00217         return false;
00218     }
00219
00220     /* SPSC bounded ring: the consumer (drain) only advances s_tail, the
00221      * producer (this function) only advances s_head. Read tail snapshot
00222      * once -- it can only move forward between this read and our store, so
00223      * the worst case is we observe one slot fewer free space than reality
00224      * (we drop a record when there is in fact room). That is a benign loss
00225      * compared to corrupting the ring, and the drain hook will catch up
00226      * within one tick. */
00227     const uint32_t head = s_head;
00228     const uint32_t tail = s_tail;
00229     const uint32_t used = head - tail;
00230

```

```

00231 if (used >= (uint32_t)k_rx_isr_log_capacity) {
00232     s_total_dropped++;
00233     return false;
00234 }
00235
00236 const uint32_t slot = head & (uint32_t)k_rx_isr_log_capacity_mask;
00237 s_ring[slot].event_id = (uint8_t)event_id;
00238 s_ring[slot].value = value;
00239 /* reserved fields zeroed for layout determinism */
00240 s_ring[slot].reserved0 = 0U;
00241 s_ring[slot].reserved1 = 0U;
00242
00243 /* Publish the record by advancing s_head. The volatile write acts as a
00244  * compiler barrier so the slot writes above are not reordered past it.
00245  * On RX (in-order, single core) this is sufficient -- no DMB needed. */
00246 s_head = head + 1U;
00247
00248 s_total_pushed++;
00249 const uint32_t new_used = (head + 1U) - tail;
00250 if (new_used > s_high_water) {
00251     s_high_water = new_used;
00252 }
00253 return true;
00254 }

```

References [k_isr_log_event_count](#), [k_isr_log_event_none](#), [k_rx_isr_log_capacity](#), [k_rx_isr_log_capacity_mask](#), [s_head](#), [s_high_water](#), [s_ring](#), [s_tail](#), [s_total_dropped](#), and [s_total_pushed](#).

7.18 rx_isr_log.h

[Go to the documentation of this file.](#)

```

00001 /**
00002  * @file rx_isr_log.h
00003  * @brief ISR-safe deferred logging facility
00004  *
00005  * @details
00006  * Single-producer-single-consumer (SPSC) ring buffer that lets ISR handlers
00007  * record a log "event" without ever calling into rx_log_* directly. The
00008  * rx_log_* macros funnel through rx_log_uart.c which acquires a ThreadX
00009  * mutex with TX_WAIT_FOREVER -- legal from a task, illegal from interrupt
00010  * context (deadlock or undefined behavior under ThreadX).
00011  *
00012  * ## Producer side (ISR)
00013  * Push a record with rx_isr_log_push(). The call is lock-free: it reads
00014  * tail, computes (head + 1) % capacity, and either advances head with the
00015  * record stored or bumps a drop counter. No ThreadX call, no I/O, no
00016  * busy-wait. Safe to invoke from any interrupt-priority context.
00017  *
00018  * ## Consumer side (task)
00019  * A low-priority task (currently comm_task) calls rx_isr_log_drain() once
00020  * per poll tick. The drain reads each pending record, looks up its format
00021  * string from a static table, and emits via the regular rx_log_* macros so
00022  * the message ends up wrapped in the same UART log frame as task-context
00023  * logs. After all records are drained, if drop_count advanced since the
00024  * last drain it emits a single "ISR log overflow: dropped N records"
00025  * summary so observers know events were lost.
00026  *
00027  * ## Ring sizing
00028  * k_rx_isr_log_capacity = 32 records, 8 bytes each = 256 bytes total. At
00029  * the comm_task drain rate of 100 Hz this absorbs an event burst of
00030  * ~3200/s before any drops occur, far above the steady-state USB ISR
00031  * event rate of ~10-20 events/s during enumeration and ~0/s after.
00032  *
00033  * ## Event model
00034  * Each event is identified by an rx_isr_log_event_id_t enum value which
00035  * indexes a static const table of {tag, format, level} tuples. A 32-bit
00036  * `value` field carries one optional auxiliary integer (port id, error
00037  * code, register snapshot, etc.). This keeps records fixed-size and
00038  * trivially copyable, avoiding any sprintf-style formatting in the ISR.
00039  *
00040  * @author Locked, Inc.
00041  * @date 2026-04-25
00042  * @copyright Copyright (c) 2026 Locked Inc.
00043  * SPDX-License-Identifier: MIT
00044  * @since Version 1.0.0
00045  */
00046
00047 #pragma once
00048

```

```

00049 #include <stdbool.h>
00050 #include <stdint.h>
00051
00052 #ifdef __cplusplus
00053 extern "C" {
00054 #endif
00055
00056 /**
00057  * @enum rx_isr_log_constants_t
00058  * @brief Sizing and queue limits for the ISR log ring
00059  */
00060 typedef enum : uint16_t {
00061     k_rx_isr_log_capacity =
00062         32U, /**< Number of records the ring can hold before dropping (power of two) */
00063     k_rx_isr_log_capacity_mask =
00064         31U, /**< Bitwise mask for index modulo (capacity - 1). MUST track capacity. */
00065 } rx_isr_log_constants_t;
00066
00067 /**
00068  * @enum rx_isr_log_event_id_t
00069  * @brief Identifier for every distinct event the ISR layer can log
00070  *
00071  * @details
00072  * The drain hook indexes a static format-string table by this enum so the
00073  * ISR never needs to reach for a literal string at runtime. Add a new entry
00074  * here, add a matching row in s_isr_log_event_table[] in rx_isr_log.c, and
00075  * the new event is wired up. Keep entries grouped by source module.
00076  */
00077 typedef enum : uint8_t {
00078     k_isr_log_event_none = 0, /**< Reserved sentinel; never enqueued */
00079
00080     /* USB ISR events (rx_usb_isr.c) */
00081     k_isr_log_event_usb_vbus_se0,
00082     k_isr_log_event_usb_vbus_fs_j,
00083     k_isr_log_event_usb_dvst_powered,
00084     k_isr_log_event_usb_dvst_default,
00085     k_isr_log_event_usb_dvst_address,
00086     k_isr_log_event_usb_dvst_configured,
00087     k_isr_log_event_usb_dvst_suspended,
00088     k_isr_log_event_usb_dvst_unknown,
00089     k_isr_log_event_usb_resume,
00090     k_isr_log_event_usb_crtt_seq_err,
00091
00092     /* USB CDC ISR-context events (rx_usb_cdc.c) */
00093     k_isr_log_event_cdc_descriptor_short_write,
00094     k_isr_log_event_cdc_pipe_rolled_back,
00095     k_isr_log_event_cdc_port0_bulk_in_failed,
00096     k_isr_log_event_cdc_port0_bulk_out_failed,
00097     k_isr_log_event_cdc_port0_int_in_failed,
00098     k_isr_log_event_cdc_port1_bulk_in_failed,
00099     k_isr_log_event_cdc_port1_bulk_out_failed,
00100     k_isr_log_event_cdc_port1_int_in_failed,
00101     k_isr_log_event_cdc_port2_bulk_in_failed,
00102     k_isr_log_event_cdc_port2_bulk_out_failed,
00103     k_isr_log_event_cdc_port2_int_in_failed,
00104     k_isr_log_event_cdc_composite_configured,
00105     k_isr_log_event_cdc_line_coding_short_write,
00106
00107     k_isr_log_event_count, /**< Sentinel: total number of defined events */
00108 } rx_isr_log_event_id_t;
00109
00110 /**
00111  * @struct rx_isr_log_record_t
00112  * @brief One pending log emission queued by an ISR for the drain hook
00113  */
00114 typedef struct {
00115     uint8_t event_id; /**< rx_isr_log_event_id_t (kept as uint8_t for fixed packing) */
00116     uint8_t reserved0;
00117     uint16_t reserved1;
00118     uint32_t value; /**< Auxiliary value (event-specific; 0 if unused) */
00119 } rx_isr_log_record_t;
00120
00121 /**
00122  * @struct rx_isr_log_stats_t
00123  * @brief Diagnostic counters for the ISR log ring
00124  */
00125 typedef struct {
00126     uint32_t total_pushed; /**< Records accepted by the ring */
00127     uint32_t total_drained; /**< Records emitted via rx_log_* by the drain hook */
00128     uint32_t total_dropped; /**< Records discarded because the ring was full */
00129     uint32_t high_water; /**< Maximum observed occupancy (records) */
00130 } rx_isr_log_stats_t;
00131
00132 /**
00133  * @brief Push one log event from interrupt context
00134  *
00135  * @details

```

```

00136 * Lock-free SPSC enqueue. Reads tail, computes the next head; if the ring
00137 * is full, increments the drop counter and returns false. Otherwise writes
00138 * the record at the current head slot, then advances head. Safe to call
00139 * from any ISR or task; the only producer is the ISR by convention but
00140 * the implementation is correct for any single-thread producer because the
00141 * head/tail are written by exactly one side each.
00142 *
00143 * @param[in] event_id One of rx_isr_log_event_id_t. k_isr_log_event_none
00144 * and values >= k_isr_log_event_count are silently
00145 * dropped (counted as drops).
00146 * @param[in] value Auxiliary 32-bit value (0 if the event has no value)
00147 *
00148 * @return true if the record was queued, false if dropped
00149 *
00150 * @note ISR-SAFE: no mutex, no I/O, no allocation, no busy-wait.
00151 */
00152 bool rx_isr_log_push(rx_isr_log_event_id_t event_id, uint32_t value);
00153
00154 /**
00155 * @brief Drain all pending records and emit them via rx_log_*
00156 *
00157 * @details
00158 * Called from a task-context periodic hook (currently comm_task at 100 Hz).
00159 * Walks tail forward to head, emitting one rx_log_* call per record. Also
00160 * detects and reports newly dropped records via a single summary log.
00161 *
00162 * @return Number of records drained on this call
00163 *
00164 * @note MUST be called from task context only -- it invokes the regular
00165 * rx_log_* macros which acquire a ThreadX mutex.
00166 */
00167 uint32_t rx_isr_log_drain(void);
00168
00169 /**
00170 * @brief Snapshot diagnostic counters
00171 *
00172 * @param[out] stats Statistics destination (nullptr is silently ignored)
00173 */
00174 void rx_isr_log_get_stats(rx_isr_log_stats_t* stats);
00175
00176 #ifdef UNIT_TEST
00177 /**
00178 * @brief Reset all ring buffer and statistics state (unit tests only)
00179 */
00180 void rx_isr_log_test_reset_state(void);
00181
00182 /**
00183 * @brief Read pending record count without draining (unit tests only)
00184 *
00185 * @return Number of records currently queued in the ring (0..k_rx_isr_log_capacity)
00186 */
00187 uint32_t rx_isr_log_test_pending_count(void);
00188
00189 /**
00190 * @brief Inject a raw record into the ring bypassing push() validation
00191 *
00192 * @details
00193 * Unit-test only. Writes the supplied raw event_id (which can be any uint8_t,
00194 * including reserved sentinel 0 or out-of-range values >= k_isr_log_event_count)
00195 * into the next free slot and advances head. This exists exclusively to drive
00196 * the torn-record defensive guards in the drain hook (event_id == none and
00197 * event_id >= count returns) which cannot be reached via the normal producer
00198 * path because rx_isr_log_push() rejects those ids before they hit the ring.
00199 *
00200 * @param[in] raw_event_id Arbitrary 8-bit event id (intentionally untyped)
00201 * @param[in] value Auxiliary value
00202 *
00203 * @return true if injected, false if the ring was full
00204 *
00205 * @note Unit-test only -- do not call from firmware.
00206 */
00207 bool rx_isr_log_test_inject_raw_record(uint8_t raw_event_id, uint32_t value);
00208
00209 /**
00210 * @brief Override the metadata for one event id at runtime (unit tests only)
00211 *
00212 * @details
00213 * Replaces the static event-table entry's level and has_value flag for the
00214 * given event id. The production table marks every event has_value=false, so
00215 * the drain hook's has_value=true switch arms (one per level) are unreachable
00216 * without flipping the flag. Resetting the table to the original state is
00217 * the caller's responsibility (call rx_isr_log_test_restore_event_table()).
00218 *
00219 * @param[in] event_id Event id to override (must be in valid range)
00220 * @param[in] level New level (0=debug, 1=info, 2=warn, 3=error)
00221 * @param[in] has_value Whether the auxiliary value should be logged
00222 *

```

```

00223 * @note Unit-test only -- do not call from firmware.
00224 */
00225 void rx_isr_log_test_set_event_meta(rx_isr_log_event_id_t event_id, uint8_t level, bool has_value);
00226
00227 /**
00228 * @brief Restore all event-table entries to their compile-time defaults
00229 *
00230 * @details
00231 * Pairs with rx_isr_log_test_set_event_meta(). Call from tearDown() so other
00232 * tests see the canonical metadata.
00233 *
00234 * @note Unit-test only -- do not call from firmware.
00235 */
00236 void rx_isr_log_test_restore_event_table(void);
00237 #endif
00238
00239 #ifdef __cplusplus
00240 }
00241 #endif

```

7.19 libs/rx_core/inc/rx_iwdt.h File Reference

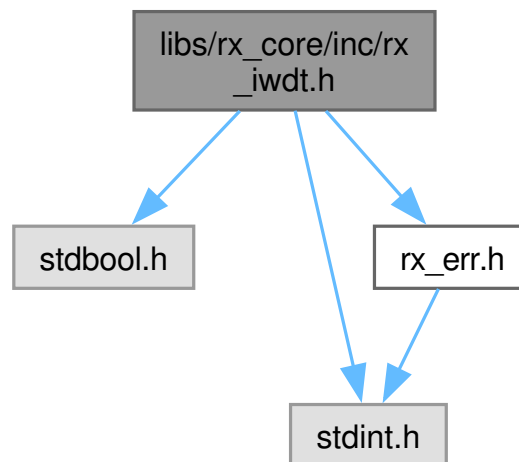
Independent Watchdog Timer (IWDT) Driver for RX72N Microcontroller.

```

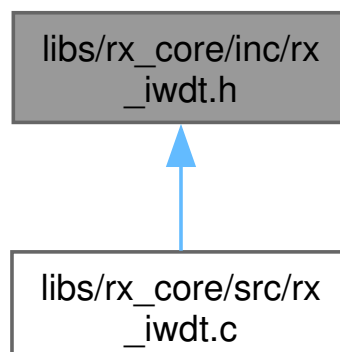
#include <stdbool.h>
#include <stdint.h>
#include "rx_err.h"

```

Include dependency graph for rx_iwdt.h:



This graph shows which files directly or indirectly include this file:



Data Structures

- struct [rx_iwdt_config_t](#)
IWDT configuration structure.
- struct [rx_iwdt_task_info_t](#)
Task registration structure (internal).
- struct [rx_iwdt_status_t](#)
IWDT status structure.

Enumerations

- enum [iwdt_constants_t](#) : `uint32_t` {
[k_iwdt_max_tasks](#), [k_iwdt_task_name_len](#), [k_iwdt_default_timeout_ms](#), [k_iwdt_min_timeout_ms](#),
[k_iwdt_max_timeout_ms](#), [k_ms_per_second](#) }
IWDT Configuration Constants and Limits.
- enum [iwdt_timeout_period_t](#) : `uint8_t` { [k_iwdt_timeout_128ms](#) = 0 , [k_iwdt_timeout_512ms](#) = 1 , [k_iwdt_timeout_2048ms](#) = 2 , [k_iwdt_timeout_8192ms](#) = 3 }
IWDT timeout periods (hardware divider settings).
- enum [system_state_t](#) : `uint8_t` {
[k_system_state_init](#), [k_system_state_idle](#), [k_system_state_running](#), [k_system_state_motor_active](#),
[k_system_state_comm_active](#), [k_system_state_error](#), [k_system_state_count](#) }
System Operational States for State-Dependent Watchdog Timeouts.
- enum [iwdt_reset_reason_t](#) : `uint8_t` {
[k_iwdt_reset_unknown](#) = 0 , [k_iwdt_reset_power_on](#) = 1 , [k_iwdt_reset_watchdog](#) = 2 ,
[k_iwdt_reset_external](#) = 3 ,
[k_iwdt_reset_low_voltage](#) = 4 }
Watchdog reset reasons.

Functions

- [rx_err_t rx_iwdt_init](#) (`const rx_iwdt_config_t *config`)
Initialize the Independent Watchdog Timer.
- [rx_err_t rx_iwdt_feed](#) (`void`)
Feed the watchdog timer.
- [rx_err_t rx_iwdt_register_task](#) (`const char *task_name`, `uint32_t timeout_ms`)
Register a task for heartbeat monitoring.
- [rx_err_t rx_iwdt_task_heartbeat](#) (`const char *task_name`)
Report task heartbeat.
- [rx_err_t rx_iwdt_set_timeout_for_state](#) (`system_state_t state`, `uint32_t timeout_ms`)
Set timeout for system state.
- [rx_err_t rx_iwdt_set_state](#) (`system_state_t state`)
Set current system state.
- [rx_err_t rx_iwdt_get_status](#) (`rx_iwdt_status_t *status`)
Get watchdog status.
- `bool rx_iwdt_was_reset` (`void`)
Check if last reset was caused by watchdog.
- [rx_err_t rx_iwdt_check_tasks](#) (`void`)
Check for task timeouts.

7.19.1 Detailed Description

Independent Watchdog Timer (IWDT) Driver for RX72N Microcontroller.

Production-grade watchdog driver providing safety-critical system monitoring with advanced task tracking, state-dependent timeouts, and comprehensive diagnostics. Designed for real-time embedded systems requiring fault detection and automatic recovery from software hangs, deadlocks, and system failures.

7.19.1.1 Purpose and Design Rationale

This driver implements a multi-layer watchdog architecture combining:

1. Hardware IWDT: Physical watchdog timer that triggers system reset
2. Task monitoring: Software-level tracking of individual RTOS tasks
3. State-aware timeouts: Different timeout periods based on system activity
4. Diagnostic tracking: Post-mortem analysis of failures

The Independent Watchdog Timer (IWDT) is separate from the normal watchdog and provides true independence from the main system clock, making it immune to clock failures.

7.19.1.2 Key Features

Hardware Watchdog:

- Physical watchdog timer with dedicated clock (IWDTCCLK)
- Cannot be disabled once started (hardware safety feature)
- 4 timeout periods: 128ms, 512ms, 2s, 8s
- Automatic system reset on timeout

Task Monitoring (Software Layer):

- Register up to 16 tasks with individual heartbeat tracking
- Per-task timeout configuration
- Failed task identification for diagnostics
- Thread-safe heartbeat reporting

State-Dependent Timeouts:

- Different timeout periods for different system states
- Example: Shorter timeout during motor operation, longer during idle
- Runtime state transitions with automatic timeout adjustment

Diagnostics and Reporting:

- Reset reason detection (power-on, watchdog, external, etc.)
- Reset counter (survives resets via backup registers)
- Failed task tracking for post-mortem analysis
- Watchdog feed statistics

7.19.1.3 Hardware Architecture

RX72N Independent Watchdog Timer (IWDT):

- Register Base Address: 0x00088030
- Clock Source: IWDTCLK (independent oscillator, 120 kHz)
- Clock Frequency: 120 kHz (8.33 us period)
- Counter Width: 14-bit down-counter
- Timeout Range: ~128ms to ~16.4 seconds
- Reset Output: Connected to internal reset controller

Timeout Calculation:

$$\text{Timeout} = (2^{(10+CKS)}) / \text{IWDTCLK}$$

Where CKS (Clock Select) determines divider:

CKS=0: $2^{10} = 1024$ -> ~128ms timeout
 CKS=1: $2^{12} = 4096$ -> ~512ms timeout
 CKS=2: $2^{14} = 16384$ -> ~2s timeout
 CKS=3: $2^{16} = 65536$ -> ~8s timeout

7.19.1.4 Implementation Approach

Three-Layer Architecture:

1. Hardware Layer (rx_iwdt_init, rx_iwdt_feed):
 - Configures IWDT registers
 - Writes refresh code to prevent timeout
2. Task Monitoring Layer (rx_iwdt_register_task, rx_iwdt_task_heartbeat):
 - Tracks registered tasks
 - Checks heartbeat timestamps
 - Identifies failed tasks
3. State Management Layer (rx_iwdt_set_state, rx_iwdt_set_timeout_for_state):
 - Maps system states to timeout periods
 - Adjusts watchdog based on activity level

Typical Usage Flow:

1. System startup: rx_iwdt_init() - Configure and start watchdog
2. Task registration: rx_iwdt_register_task() - Register critical tasks
3. Normal operation:
 - Main loop: rx_iwdt_feed() every N ms ($N < \text{timeout}$)
 - Each task: rx_iwdt_task_heartbeat() within task timeout
 - Monitor thread: rx_iwdt_check_tasks() to detect failures
4. State changes: rx_iwdt_set_state() to adjust timeout dynamically
5. Diagnostics: rx_iwdt_get_status() for reset history and statistics

7.19.1.5 Performance Characteristics

Operation	Execution Time	Memory Usage	Notes
rx_iwdt_init()	~50 us	0 heap	One-time setup
rx_iwdt_feed()	~2 us	0	Register write only
rx_iwdt_task_heartbeat()	~5 us	0	Timestamp update
rx_iwdt_check_tasks()	~10 us x N tasks	0	Linear scan
rx_iwdt_get_status()	~15 us	0	Structure copy

Memory Footprint:

- Static variables: ~2 KB (task tracking array, status, configuration)
- Code size: ~3 KB (all functions)
- Stack usage: < 128 bytes (deepest call)
- Heap usage: 0 bytes (zero dynamic allocation)

7.19.1.6 Thread Safety and RTOS Integration

Thread-Safe Operations:

- All public functions use mutex protection (ThreadX)
- Safe to call from multiple threads simultaneously
- ISR-safe: [rx_iwdt_feed\(\)](#) can be called from interrupts

ThreadX Integration:

- Uses tx_mutex for synchronization
- Uses tx_time_get() for timestamps
- Compatible with all ThreadX thread priorities

7.19.1.7 Error Detection and Recovery

Detected Failures:

1. Task deadlock: Task stops sending heartbeats
2. Main loop hang: No [rx_iwdt_feed\(\)](#) calls
3. Stack overflow: Task corrupts memory and stops executing
4. Infinite loop: Task stuck in tight loop without heartbeat
5. Communication timeout: Network/SPI/UART tasks stop responding

Recovery Actions:

- Hardware reset triggered by IWDG timeout
- System reboots from reset vector
- [rx_iwdt_was_reset\(\)](#) detects watchdog reset on next boot
- Failed task name preserved for diagnostics

7.19.1.8 Usage Guidelines

DO:

- Call `rx_iwdt_feed()` from main supervision loop only
- Register all critical tasks for monitoring
- Use state-dependent timeouts for different operating modes
- Check `rx_iwdt_was_reset()` at startup for diagnostics
- Tune timeouts based on worst-case execution times

DON'T:

- Call `rx_iwdt_feed()` from multiple locations (defeats purpose)
- Set timeouts shorter than worst-case task execution time
- Rely on watchdog for normal error handling (use return codes)
- Call `rx_iwdt_feed()` from ISRs (unless by design)
- Assume watchdog prevents all failures (defense in depth)

Hardware Requirements:

Component	Requirement	Notes
MCU	Renesas RX72N	IWDT peripheral required
Clock	IWDTCLK	Independent oscillator, 120 kHz
Registers	0x00088030 base	IWDT control registers
Reset	Internal reset controller	Connected to IWDT output
Backup registers	Optional	For reset counter persistence

Module Dependencies:

This module depends on:

- `rx_err.h` - Error code definitions
- `<stdint.h>` - Standard integer types
- `<stdbool.h>` - Boolean type
- ThreadX RTOS - Mutex and timestamp functions

This module is used by:

- `main.c` - System initialization and main loop
- All critical RTOS tasks - Heartbeat reporting
- System monitor task - Task timeout checking

NASA Power of 10 Compliance:

Rule	Status	Implementation
1. Simple control flow	[OK]	No goto/setjmp/recursion, sequential logic
2. Fixed loop bounds	[OK]	All loops bounded by <code>k_iwdt_max_tasks</code> (16)
3. No dynamic allocation	[OK]	Zero malloc/free, static task array
4. Small functions	[OK]	All functions < 60 lines
5. Assertions (≥ 2 per function)	[OK]	Minimum 2 checks per function (pre/post)
6. Narrow scope	[OK]	File-scope statics, function-local variables
7. Check return values	[OK]	All functions return <code>rx_err_t</code> , checked by callers
8. Limited preprocessor	[OK]	C23 typed enums only, zero computation macros
9. Pointer restrictions	[OK]	Single-level pointers, no pointer arithmetic
10. Compiler warnings	[OK]	Compiles with <code>-Wall -Wextra -Werror</code>

Safety-Critical Design:

- Static allocation prevents fragmentation and allocation failures
- Bounded arrays prevent buffer overflows
- Explicit error codes force error handling
- Mutex protection prevents race conditions

SOLID Principles:

Single Responsibility (S):

- This module has ONE job: Watchdog monitoring and system reset
- Separate concerns: Hardware watchdog vs task monitoring vs state management
- No mixing of unrelated functionality

Open/Closed (O):

- Extensible via configuration structure (state timeouts, task limits)
- Can add new system states without modifying existing code
- Timeout periods configured at runtime, not hardcoded

Liskov Substitution (L):

- All functions return `rx_err_t` for consistent error handling
- Boolean functions (`rx_iwdt_was_reset`) have well-defined semantics

- No unexpected side effects or hidden dependencies

Interface Segregation (I):

- Separate functions for separate concerns (feed vs heartbeat vs status)
- Task monitoring is optional (enable_task_monitoring flag)
- Clients use only the functions they need

Dependency Inversion (D):

- Depends on abstract [rx_err_t](#), not specific error implementations
- Uses system time abstraction (tx_time_get), not raw timer
- Can be mocked for unit testing via test_reset function

See also

[rx_err.h](#) Error code definitions used by all functions

docs/sections/06_nasa_power_of_10.tex Safety-critical coding standards

RX72N Hardware Manual Section 11 Independent Watchdog Timer specification

Author

Locked, Inc.

Date

2026-01-27

Copyright

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Since

Version 1.0.0

Definition in file [rx_iwdt.h](#).

7.19.2 Data Structure Documentation

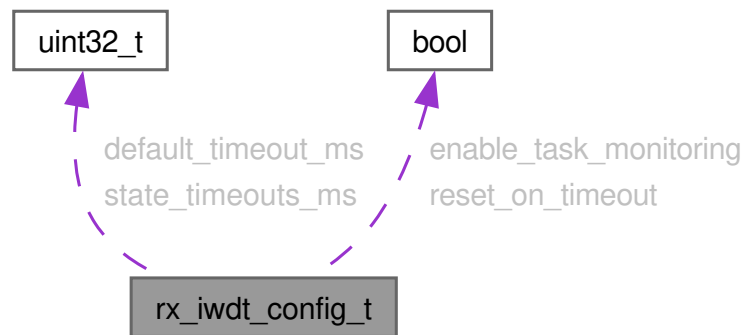
7.19.2.1 struct rx_iwdt_config_t

IWDT configuration structure.

Configures the Independent Watchdog Timer with timeout periods for different system states.

Definition at line 495 of file [rx_iwdt.h](#).

Collaboration diagram for rx_iwdt_config_t:



Data Fields

uint32_t	default_timeout_ms	Default timeout
bool	enable_task_monitoring	Enable task heartbeats
bool	reset_on_timeout	Reset vs NMI
uint32_t	state_timeouts_ms[k_system_state_count]	State-specific timeouts

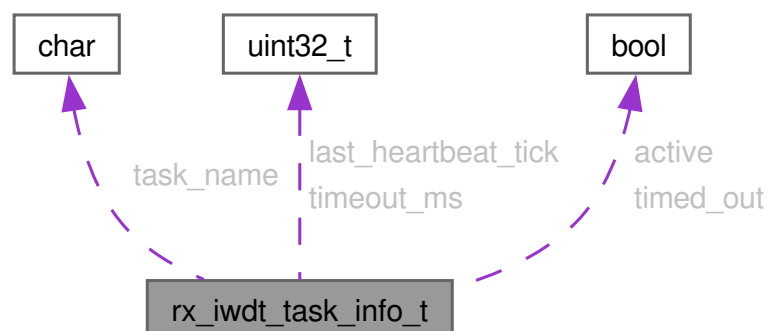
7.19.2.2 struct rx_iwdt_task_info_t

Task registration structure (internal).

Tracks registered tasks for heartbeat monitoring.

Definition at line 507 of file [rx_iwdt.h](#).

Collaboration diagram for rx_iwdt_task_info_t:



Data Fields

bool	active	Task is active
uint32_t	last_heartbeat_tick	Last heartbeat timestamp
char	task_name[k_iwdt_task_name_len]	Task name
bool	timed_out	Task has timed out
uint32_t	timeout_ms	Task-specific timeout

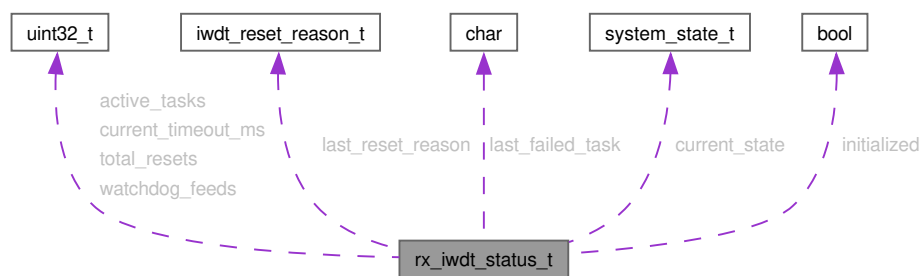
7.19.2.3 struct rx_iwdt_status_t

IWDT status structure.

Provides watchdog status information including reset history and failed task tracking for post-mortem analysis.

Definition at line 521 of file [rx_iwdt.h](#).

Collaboration diagram for rx_iwdt_status_t:



Data Fields

uint32_t	active_tasks	Number of active tasks
system_state_t	current_state	Current system state
uint32_t	current_timeout_ms	Active timeout
bool	initialized	IWDT initialized
char	last_failed_task[k_iwdt_task_name_len]	Failed task name
iwdt_reset_reason_t	last_reset_reason	Last reset cause
uint32_t	total_resets	Total reset count
uint32_t	watchdog_feeds	Total feed count

7.19.3 Enumeration Type Documentation

7.19.3.1 iwdt_constants_t

```
enum iwdt_constants_t : uint32_t
```

IWDT Configuration Constants and Limits.

Defines compile-time constants for the IWDT driver including array sizes, timeout limits, and time conversion factors. These constants are chosen based on hardware limitations and system requirements.

7.19.3.2 Constant Rationale

k_iwdt_max_tasks = 16:

- Sized for typical STAR system (4 motors + communication + sensors)
- Keeps task array < 1 KB (16 x 64 bytes = 1024 bytes)
- Allows linear scan in < 10 us at 240 MHz

k_iwdt_task_name_len = 32:

- Matches ThreadX TX_THREAD_NAME_SIZE
- Accommodates descriptive names like "MotorControllerTask_FL"
- Null-terminated, so 31 characters + '\0'

k_iwdt_default_timeout_ms = 2000:

- 2 second default provides safety margin
- Allows 100ms main loop to miss ~20 iterations
- Not too long (fast failure detection) nor too short (no false positives)

k_iwdt_min_timeout_ms = 128:

- Hardware minimum with CKS=0 divider
- Calculated: 1024 / 125kHz ~ 128ms
- Use only for very fast loops (> 10 Hz)

k_iwdt_max_timeout_ms = 16384:

- Hardware maximum with CKS=3 divider
- Calculated: 65536 / 125kHz ~ 8192ms
- Conservative limit prevents excessive hang detection time

Usage Example:

```
// Validate configuration against limits
rx_iwdt_config_t config = {
    .default_timeout_ms = k_iwdt_default_timeout_ms,
    .enable_task_monitoring = true
};

if (config.default_timeout_ms < k_iwdt_min_timeout_ms) {
    return k_rx_err_invalid_arg;
}
```


Invariant

`k_iwdt_max_tasks` x `sizeof(rx_iwdt_task_info_t)` < 2048 bytes
`k_iwdt_min_timeout_ms` <= `k_iwdt_default_timeout_ms` <= `k_iwdt_max_timeout_ms`
 All values are compile-time constants (no runtime overhead)

Note

C23 typed enum with `uint32_t` underlying type (4-byte constants)
 Values chosen based on RX72N IWDT hardware capabilities

See also

[rx_iwdt_config_t](#) Configuration structure using these constants
[rx_iwdt_task_info_t](#) Task info structure sized by `k_iwdt_task_name_len`

Since

Version 1.0.0

Enumerator

<code>k_iwdt_max_tasks</code>	Maximum number of tasks that can be registered for monitoring. Limits task array size to ~1 KB. Increase if more tasks needed (recompile required)
<code>k_iwdt_task_name_len</code>	Maximum task name length in characters including null terminator. Matches ThreadX <code>TX_THREAD_NAME_SIZE</code> . Allows 31-character names
<code>k_iwdt_default_timeout_ms</code>	Default watchdog timeout in milliseconds (2 seconds). Provides good balance between fast failure detection and false positive avoidance
<code>k_iwdt_min_timeout_ms</code>	Minimum valid timeout in milliseconds (~128ms). Hardware limit with <code>CKS=0</code> . Use only for very fast control loops (> 10 Hz)
<code>k_iwdt_max_timeout_ms</code>	Maximum valid timeout in milliseconds (~16 seconds). Conservative limit for reasonable hang detection. Hardware supports up to ~8s with <code>CKS=3</code>
<code>k_ms_per_second</code>	Milliseconds per second (1000 ms/s). Time conversion constant for timeout calculations and timestamp arithmetic

Definition at line 316 of file [rx_iwdt.h](#).

```

00316     : uint32_t {
00317     k_iwdt_max_tasks =
00318         16, /**< Maximum number of tasks that can be registered for monitoring. Limits task array size to ~1 KB. Increase if
more tasks needed (recompile required) */
00319     k_iwdt_task_name_len =
00320         32, /**< Maximum task name length in characters including null terminator. Matches ThreadX
TX_THREAD_NAME_SIZE. Allows 31-character names */
00321     k_iwdt_default_timeout_ms =
00322         2000, /**< Default watchdog timeout in milliseconds (2 seconds). Provides good balance between fast failure detection
and false positive avoidance */
00323     k_iwdt_min_timeout_ms =
00324         128, /**< Minimum valid timeout in milliseconds (~128ms). Hardware limit with CKS=0. Use only for very fast control
loops (> 10 Hz) */
00325     k_iwdt_max_timeout_ms =
00326         16384, /**< Maximum valid timeout in milliseconds (~16 seconds). Conservative limit for reasonable hang detection.
Hardware supports up to ~8s with CKS=3 */
00327     k_ms_per_second =
00328         1000, /**< Milliseconds per second (1000 ms/s). Time conversion constant for timeout calculations and timestamp
arithmetic */
00329 } iwdt_constants_t;

```

7.19.3.3 iwdt_reset_reason_t

enum [iwdt_reset_reason_t](#) : uint8_t

Watchdog reset reasons.

Enumerator

k_iwdt_reset_unknown	Unknown reset
k_iwdt_reset_power_on	Power-on reset
k_iwdt_reset_watchdog	Watchdog timeout
k_iwdt_reset_external	External reset
k_iwdt_reset_low_voltage	Low voltage detect

Definition at line 476 of file [rx_iwdt.h](#).

```
00476      : uint8_t {
00477  k_iwdt_reset_unknown    = 0, /**< Unknown reset */
00478  k_iwdt_reset_power_on   = 1, /**< Power-on reset */
00479  k_iwdt_reset_watchdog   = 2, /**< Watchdog timeout */
00480  k_iwdt_reset_external    = 3, /**< External reset */
00481  k_iwdt_reset_low_voltage = 4, /**< Low voltage detect */
00482 } iwdt_reset_reason_t;
```

7.19.3.4 iwdt_timeout_period_t

enum [iwdt_timeout_period_t](#) : uint8_t

IWDT timeout periods (hardware divider settings).

Enumerator

k_iwdt_timeout_128ms	~128ms (CKS=00)
k_iwdt_timeout_512ms	~512ms (CKS=01)
k_iwdt_timeout_2048ms	~2s (CKS=10)
k_iwdt_timeout_8192ms	~8s (CKS=11)

Definition at line 332 of file [rx_iwdt.h](#).

```
00332      : uint8_t {
00333  k_iwdt_timeout_128ms = 0, /**< ~128ms (CKS=00) */
00334  k_iwdt_timeout_512ms = 1, /**< ~512ms (CKS=01) */
00335  k_iwdt_timeout_2048ms = 2, /**< ~2s (CKS=10) */
00336  k_iwdt_timeout_8192ms = 3, /**< ~8s (CKS=11) */
00337 } iwdt_timeout_period_t;
```

7.19.3.5 system_state_t

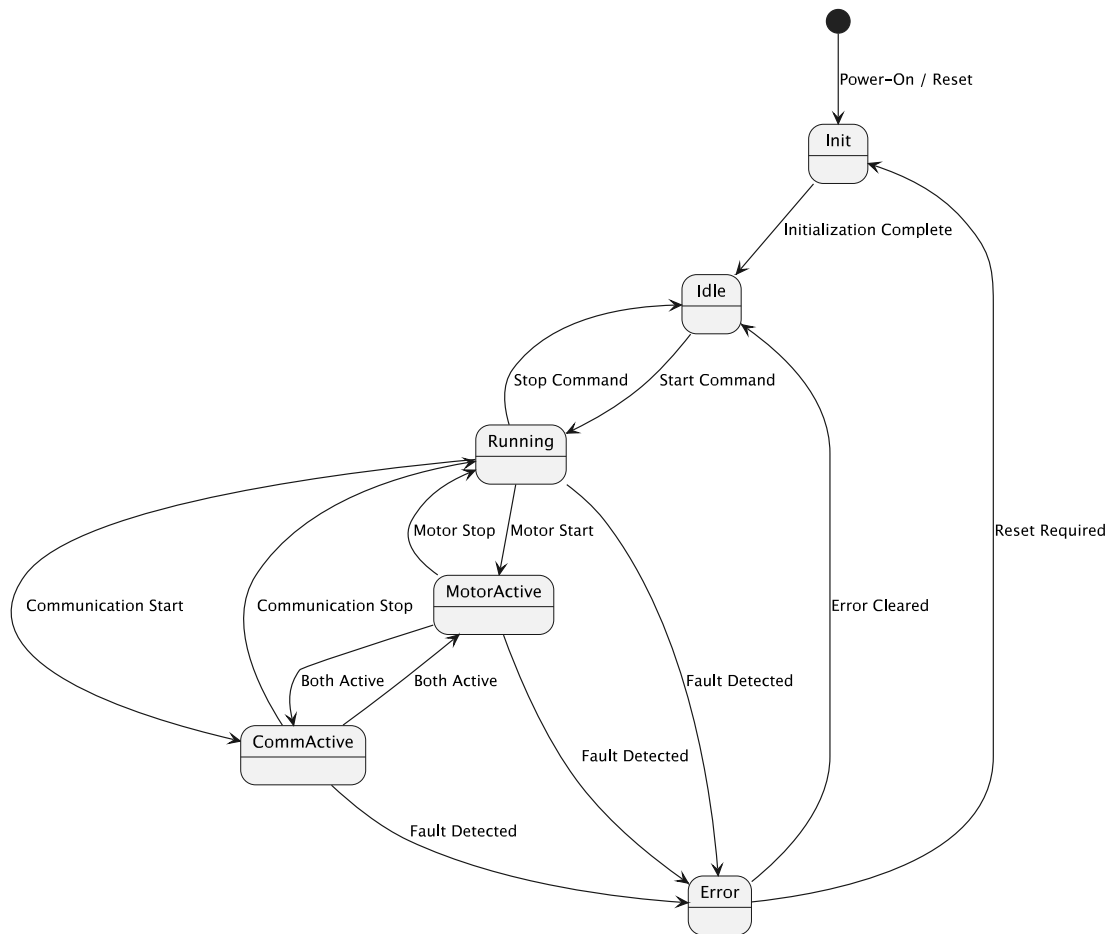
enum [system_state_t](#) : uint8_t

System Operational States for State-Dependent Watchdog Timeouts.

Defines all possible system states, each with its own watchdog timeout period. The watchdog timeout is automatically adjusted when transitioning between states, allowing shorter timeouts during active operations and longer timeouts during idle periods.

7.19.3.6 State Machine Architecture

The system operates as a finite state machine with defined transitions:



7.19.3.7 State Descriptions and Typical Timeouts

State	Description	Typical Timeout	Rationale
Init	System initialization, hardware setup	5000ms	Slow startup tolerable
Idle	No active operations, waiting for commands	5000ms	Long timeout OK
Running	Normal operation, moderate activity	2000ms	Default timeout
MotorActive	Motors running, high-frequency control	500ms	Fast loop required
CommActive	Active SPI/USB communication	1000ms	Network delays possible
Error	Error recovery, diagnostics	10000ms	Allow time for recovery

7.19.3.8 Recommended Timeout Configuration

```

rx_iwdt_config_t config;
config.state_timeouts_ms[k_system_state_init]      = 5000; // Slow startup
config.state_timeouts_ms[k_system_state_idle]      = 5000; // Long idle OK
config.state_timeouts_ms[k_system_state_running]   = 2000; // Default
config.state_timeouts_ms[k_system_state_motor_active] = 500; // Fast loop
config.state_timeouts_ms[k_system_state_comm_active] = 1000; // Network
config.state_timeouts_ms[k_system_state_error]     = 10000; // Recovery time
  
```

State Transition Example:

```
// System startup
rx_iwdt_set_state(k_system_state_init);
// ... hardware initialization ...

// Initialization complete, go idle
rx_iwdt_set_state(k_system_state_idle);

// User command received, start operation
rx_iwdt_set_state(k_system_state_running);

// Begin motor control
rx_iwdt_set_state(k_system_state_motor_active);
// Motor control loop must call rx_iwdt_feed() within 500ms

// Stop motors, return to normal
rx_iwdt_set_state(k_system_state_running);

// Fault detected
rx_iwdt_set_state(k_system_state_error);
// Long timeout allows diagnostic logging

// Recovery complete
rx_iwdt_set_state(k_system_state_idle);
```

Usage Example (Dynamic Timeout Adjustment):

```
// Configure different timeouts for different states
rx_iwdt_set_timeout_for_state(k_system_state_motor_active, 500);
rx_iwdt_set_timeout_for_state(k_system_state_idle, 5000);

// Transition to motor state (timeout automatically becomes 500ms)
rx_iwdt_set_state(k_system_state_motor_active);

// Motor control loop
while (motor_running) {
    motor_control_update();
    rx_iwdt_feed(); // Must execute within 500ms
    tx_thread_sleep(10); // 100Hz loop = 10ms period
}

// Stop motors (timeout automatically becomes 2000ms)
rx_iwdt_set_state(k_system_state_running);
```

Invariant

State values are contiguous (0 to `k_system_state_count-1`)
`k_system_state_count` equals number of states (6)
 All state values fit in `uint8_t` (0-255)

Note

C23 typed enum with `uint8_t` underlying type (1-byte values)
 State transitions trigger automatic timeout period adjustments
`k_system_state_count` is used for array sizing and validation

Warning

Invalid state transitions are allowed but may indicate logic errors

Attention

Changing state does NOT reset the watchdog counter—still need `rx_iwdt_feed()`

See also

[rx_iwdt_set_state\(\)](#) Set current system state
[rx_iwdt_set_timeout_for_state\(\)](#) Configure timeout for specific state
[rx_iwdt_config_t](#) Configuration structure with state_timeouts_ms array

Since

Version 1.0.0

Enumerator

k_system_state_init	System initialization phase. Hardware setup, peripheral configuration. Use long timeout (5s) to allow slow startup. Transitions to: Idle
k_system_state_idle	Idle state, no active operations. Waiting for commands. Use long timeout (5s) since no critical operations. Transitions to: Running
k_system_state_running	Normal operation, moderate activity. Background tasks executing. Use default timeout (2s). Transitions to: MotorActive, CommActive, Idle, Error
k_system_state_motor_active	Motor control active, high-frequency updates. Critical real-time loop. Use short timeout (500ms) for fast failure detection. Transitions to: Running, Error
k_system_state_comm_active	Active communication (SPI/USB). Network I/O in progress. Use medium timeout (1s) to tolerate network delays. Transitions to: Running, Error
k_system_state_error	Error recovery mode. Diagnostics and cleanup. Use very long timeout (10s) to allow recovery actions and logging. Transitions to: Idle, Init
k_system_state_count	Number of system states (not a valid state). Used for array sizing (state_timeouts_ms[k_system_state_count]) and validation checks

Definition at line 458 of file [rx_iwdt.h](#).

```

00458     : uint8_t {
00459     k_system_state_init =
00460         0, /**< System initialization phase. Hardware setup, peripheral configuration. Use long timeout (5s) to allow slow
00461             startup. Transitions to: Idle */
00462     k_system_state_idle =
00463         1, /**< Idle state, no active operations. Waiting for commands. Use long timeout (5s) since no critical operations.
00464             Transitions to: Running */
00465     k_system_state_running =
00466         2, /**< Normal operation, moderate activity. Background tasks executing. Use default timeout (2s). Transitions to:
00467             MotorActive, CommActive, Idle, Error */
00468     k_system_state_motor_active =
00469         3, /**< Motor control active, high-frequency updates. Critical real-time loop. Use short timeout (500ms) for fast failure
00470             detection. Transitions to: Running, Error */
00471     k_system_state_comm_active =
00472         4, /**< Active communication (SPI/USB). Network I/O in progress. Use medium timeout (1s) to tolerate network
00473             delays. Transitions to: Running, Error */
00474     k_system_state_error =
00475         5, /**< Error recovery mode. Diagnostics and cleanup. Use very long timeout (10s) to allow recovery actions and
00476             logging. Transitions to: Idle, Init */
00477     k_system_state_count =
00478         6, /**< Number of system states (not a valid state). Used for array sizing (state_timeouts_ms[k_system_state_count])
00479             and validation checks */
00480 } system_state_t;
```

7.19.4 Function Documentation

7.19.4.1 rx_iwdt_check_tasks()

```
rx_err_t rx_iwdt_check_tasks (
    void ) [nodiscard]
```

Check for task timeouts.

Checks all registered tasks for timeout. Should be called periodically from a monitoring task.

Returns

`rx_err_t` Error code

Return values

<code>k_rx_ok</code>	No timeouts detected
<code>k_rx_err_timeout</code>	One or more tasks timed out
<code>k_rx_err_not_initialized</code>	IWDT not initialized

Note

This function does NOT reset the hardware watchdog

Warning

Task timeouts trigger watchdog reset if not cleared

Example:

```
static void watchdog_monitor_task(ULONG input) {
    while (1) {
        rx_err_t err = rx_iwdt_check_tasks();
        if (err == k_rx_err_timeout) {
            // Log timeout before reset
            uart_debug_puts("[ERROR] Task timeout detected\r\n");
        }
        rx_iwdt_feed(); // Feed hardware watchdog
        tx_thread_sleep(10); // 100ms check period
    }
}
```

Check for task timeouts.

Iterates over all active task slots and compares the elapsed tick count since each task's last heartbeat against that task's configured timeout (converted from milliseconds to ThreadX ticks). If any task has exceeded its timeout, the task's `timed_out` flag is set, the task name is recorded in `status.last_failed_task`, and the function returns `k_rx_err_timeout`.

If task monitoring is disabled (`config.enable_task_monitoring == false`), returns `k_rx_ok` immediately without checking any tasks.

The caller (typically the watchdog task) should conditionally feed the hardware IWDT only if this function returns `k_rx_ok` – i.e., only refresh the hardware watchdog when all software tasks are alive.

Algorithm steps:

1. Verify `s_iwdt_state.initialized`
2. If `!config.enable_task_monitoring`, return `k_rx_ok` early
3. Capture `current_tick = internal_get_tick_count()`
4. For each active task slot (`i < k_iwdt_max_tasks`):
 - a. Skip inactive slots
 - b. Compute `elapsed_ticks = current_tick - task->last_heartbeat_tick`
 - c. Compute `timeout_in_ticks` from `task->timeout_ms` and `TX_TIMER_TICKS_PER_SECOND`
 - d. If `elapsed_ticks > timeout_in_ticks`: set `timed_out=true`, record task name, set `any_timeout=true`
5. Return `k_rx_err_timeout` if `any_timeout`, else `k_rx_ok`

Precondition

`rx_iwdt_init()` must have been called successfully

Registered tasks must be calling `rx_iwdt_task_heartbeat()` on schedule

Postcondition

Timed-out tasks have their `timed_out` flag set

`status.last_failed_task` contains the name of the most recently detected failure

Note

Not thread-safe; task heartbeat fields may be updated concurrently

Warning

Do NOT feed the hardware IWDT (`rx_iwdt_feed()`) if this returns `k_rx_err_timeout`; allow the hardware watchdog to fire and reset the system

Example:

```
// Watchdog task main loop
while (1) {
    if (rx_iwdt_check_tasks() == k_rx_ok) {
        rx_iwdt_feed(); // Only feed if all tasks are alive
    } else {
        rx_log_error("wdt", "Task timeout detected -- allowing IWDT reset");
        // Do NOT feed the watchdog -- let hardware reset occur
    }
    tx_thread_sleep(k_watchdog_check_interval_ticks);
}
```

See also

`rx_iwdt_feed()` Feed hardware watchdog (call only when this returns `k_rx_ok`)

`rx_iwdt_task_heartbeat()` Update per-task heartbeat from each monitored task

`rx_iwdt_register_task()` Register a task for monitoring

Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 5: 2 preconditions (initialized, tasks registered), 2 postconditions

Definition at line 1145 of file `rx_iwdt.c`.

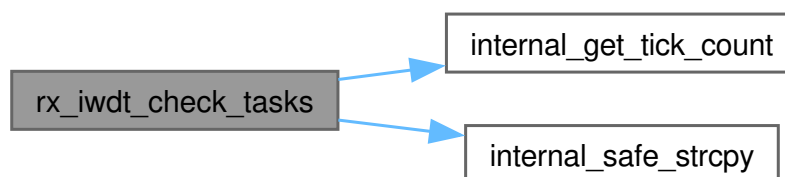
```

01146 {
01147     if (!s_iwdt_state.initialized) {
01148         return k_rx_err_not_initialized;
01149     }
01150
01151     if (!s_iwdt_state.config.enable_task_monitoring) {
01152         return k_rx_ok;
01153     }
01154
01155     const uint32_t current_tick = internal_get_tick_count();
01156     bool any_timeout = false;
01157
01158     /* Check each registered task */
01159     for (uint32_t i = 0; i < k_iwdt_max_tasks; i++) {
01160         if (!s_iwdt_state.tasks[i].active) {
01161             continue;
01162         }
01163
01164         /* Calculate elapsed time */
01165         const uint32_t elapsed_ticks = current_tick - s_iwdt_state.tasks[i].last_heartbeat_tick;
01166         const uint32_t timeout_in_ticks =
01167             (s_iwdt_state.tasks[i].timeout_ms * TX_TIMER_TICKS_PER_SECOND) / k_ms_per_second;
01168
01169         /* Check for timeout */
01170         if (elapsed_ticks > timeout_in_ticks) {
01171             s_iwdt_state.tasks[i].timed_out = true;
01172             any_timeout = true;
01173
01174             /* Record failed task */
01175             internal_safe_strcpy(s_iwdt_state.status.last_failed_task,
01176                                 s_iwdt_state.tasks[i].task_name,
01177                                 k_iwdt_task_name_len);
01178         }
01179     }
01180
01181     return (int)any_timeout ? k_rx_err_timeout : k_rx_ok;
01182 }

```

References `internal_get_tick_count()`, `internal_safe_strcpy()`, `k_iwdt_max_tasks`, `k_iwdt_task_name_len`, `k_ms_per_second`, `k_rx_err_not_initialized`, `k_rx_err_timeout`, `k_rx_ok`, and `s_iwdt_state`.

Here is the call graph for this function:



7.19.4.2 rx_iwdt_feed()

```

rx_err_t rx_iwdt_feed (
    void ) [nodiscard]

```

Feed the watchdog timer.

Resets the watchdog counter to prevent timeout. Must be called periodically at a rate faster than the configured timeout.

Returns

[rx_err_t](#) Error code

Return values

k_rx_ok	Success
k_rx_err_not_initialized	IWDT not initialized

Note

Thread-safe - can be called from multiple threads/ISRs

Warning

Missing feeds will cause system reset

Example:

```
// In main control loop (must execute faster than timeout)
while (1) {
    // ... control logic ...
    rx_iwdt_feed();
    tx_thread_sleep(50); // 500ms loop
}
```

Feed the watchdog timer.

Writes the two-byte refresh sequence (0x00 then 0xFF) to the IWDT Refresh Register (IWDTRR) to reset the hardware watchdog counter. This must be called periodically within the configured hardware timeout window or the IWDT will reset the microcontroller.

Per the RX72N hardware manual the refresh sequence is strictly ordered: writing 0x00 to IWDTRR starts the window; writing 0xFF completes the refresh. The entire sequence must occur within the permitted refresh window.

Also increments the software diagnostic counter `s_iwdt_state.status.watchdog_feeds` for telemetry and debugging purposes.

Algorithm steps:

1. Check `s_iwdt_state.initialized`; return `k_rx_err_not_initialized` if false
2. Write `k_iwdt_refresh_start` (0x00) to `regs->iwdtrr`
3. Write `k_iwdt_refresh_end` (0xFF) to `regs->iwdtrr`
4. Increment `status.watchdog_feeds` counter

Precondition

[rx_iwdt_init\(\)](#) must have been called successfully

Must be called within the hardware IWDT refresh window (no gap > configured timeout)

Postcondition

Hardware IWDT counter is reset; timeout window restarts
 s_iwdt_state.status.watchdog_feeds is incremented by one

Note

Not thread-safe; concurrent feeds from multiple tasks are benign for the hardware but may produce incorrect feed counts

Warning

Failing to call within the hardware timeout triggers a system reset

Performance:

Execution time: <1 us @ 240 MHz; safe to call from any task or ISR context

Example:

```
// In main watchdog task loop
while (1) {
    rx_err_t err = rx_iwdt_feed();
    if (err != k_rx_ok) {
        rx_log_error("wdt", "IWDT not initialized");
    }
    tx_thread_sleep(k_watchdog_feed_interval_ticks);
}
```

See also

[rx_iwdt_init\(\)](#) Initialize IWDT before feeding
[rx_iwdt_check_tasks\(\)](#) Verify all tasks are alive before feeding

Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 5: 1 precondition (initialized check), 2 postconditions (HW reset, counter++)

Definition at line 624 of file [rx_iwdt.c](#).

```
00625 {
00626     volatile rx_iwdt_regs_t* regs;
00627
00628     /* Check initialization */
00629     if (!s_iwdt_state.initialized) {
00630         return k_rx_err_not_initialized;
00631     }
00632
00633     /* Refresh watchdog counter - write 0x00 then 0xFF to IWDTRR */
00634     regs = iwdt();
00635     regs->iwdtrr = k_iwdt_refresh_start;
00636     regs->iwdtrr = k_iwdt_refresh_end;
00637
00638     s_iwdt_state.status.watchdog_feeds++;
00639
00640     return k_rx_ok;
00641 }
```

References [k_rx_err_not_initialized](#), [k_rx_ok](#), and [s_iwdt_state](#).

7.19.4.3 rx_iwdt_get_status()

```
rx_err_t rx_iwdt_get_status (
    rx_iwdt_status_t * status) [nodiscard]
```

Get watchdog status.

Retrieves current watchdog status including reset history, failed tasks, and monitoring statistics.

Parameters

out	status	Pointer to status structure
-----	--------	-----------------------------

Returns

[rx_err_t](#) Error code

Return values

k_rx_ok	Success
k_rx_err_null_ptr	status is nullptr
k_rx_err_not_initialized	IWDT not initialized

Example:

```
rx_iwdt_status_t status;
rx_err_t err = rx_iwdt_get_status(&status);
if (err == k_rx_ok) {
    printf("Total resets: %lu\n", status.total_resets);
    printf("Last failed task: %s\n", status.last_failed_task);
}
```

Get watchdog status.

Copies the internal `s_iwdt_state.status` structure into the caller-provided buffer via `memcpy`. The status snapshot includes: active task count, total feed count, current system state, current timeout threshold, and the name of the last task that timed out (if any).

This is a diagnostic/telemetry function intended for logging and health monitoring. The returned data is a point-in-time snapshot and may become stale if the detection task or other threads update IWDT state concurrently.

Algorithm steps:

1. Validate status pointer is non-null
2. Verify `s_iwdt_state.initialized`
3. `memcpy(&s_iwdt_state.status, status, sizeof(rx_iwdt_status_t))`

Precondition

[rx_iwdt_init\(\)](#) must have been called successfully
 status must be a valid non-null pointer to an [rx_iwdt_status_t](#)

Postcondition

*status contains a copy of s_iwdt_state.status at the time of the call
 Internal IWDT state is unchanged

Note

Not thread-safe; status fields may be updated concurrently by the task monitor
 Snapshot accuracy depends on timing; use as diagnostic data, not safety-critical logic

Example:

```
rx_iwdt_status_t status;
rx_err_t err = rx_iwdt_get_status(&status);
if (err == k_rx_ok) {
    rx_log_info("wdt", "Active tasks: %u, Feeds: %u",
               (unsigned)status.active_tasks,
               (unsigned)status.watchdog_feeds);
}
```

See also

[rx_iwdt_was_reset\(\)](#) Check if last reset was hardware IWDT timeout
[rx_iwdt_check_tasks\(\)](#) Check all tasks for software timeout

Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 5: 2 preconditions (null check, initialized check), 2 postconditions

Definition at line 1003 of file rx_iwdt.c.

```
01004 {
01005     /* Validate inputs */
01006     if (status == nullptr) {
01007         return k_rx_err_null_ptr;
01008     }
01009
01010     if (!s_iwdt_state.initialized) {
01011         return k_rx_err_not_initialized;
01012     }
01013
01014     /* Copy status */
01015     {
01016         const uint8_t* src = (const uint8_t*)&s_iwdt_state.status;
01017         uint8_t* dst = (uint8_t*)status;
01018         for (size_t byte_i = 0; byte_i < sizeof(rx_iwdt_status_t); byte_i++) {
01019             dst[byte_i] = src[byte_i];
01020         }
01021     }
01022
01023     return k_rx_ok;
01024 }
```

References [k_rx_err_not_initialized](#), [k_rx_err_null_ptr](#), [k_rx_ok](#), and [s_iwdt_state](#).

7.19.4.4 rx_iwdt_init()

```
rx_err_t rx_iwdt_init (
    const rx_iwdt_config_t * config) [nodiscard]
```

Initialize the Independent Watchdog Timer.

Initializes the IWDT with specified configuration. Once started, the IWDT cannot be stopped (hardware safety feature).

Parameters

in	config	Configuration structure (NULL for defaults)
----	--------	---

Returns

rx_err_t Error code

Return values

k_rx_ok	Success
k_rx_err_invalid_arg	Invalid configuration
k_rx_err_invalid_state	Already initialized
k_rx_err_hw_init_failed	Hardware initialization failed

Note

This function must be called before any other IWDT functions

Warning

IWDT cannot be stopped once started - requires hard reset

Example:

```
rx_iwdt_config_t config = {
    .default_timeout_ms = 2000,
    .enable_task_monitoring = true,
    .reset_on_timeout = true
};
config.state_timeouts_ms[k_system_state_running] = 1000;
config.state_timeouts_ms[k_system_state_motor_active] = 500;

rx_err_t err = rx_iwdt_init(&config);
if (err != k_rx_ok) {
    // Handle error
}
```

Initialize the Independent Watchdog Timer.

Initializes the IWDT driver, validates configuration, checks for previous watchdog reset, and performs initial hardware feed. If config is nullptr, uses safe default values (1000ms timeout, task monitoring enabled).

Algorithm:

1. Check if already initialized (return error if so)
2. If config is nullptr, create default configuration
3. Validate timeout within [k_iwdt_min_timeout_ms, k_iwdt_max_timeout_ms]
4. Clear and initialize all state
5. Check RSTSR2 for previous watchdog reset
6. Perform initial feed to IWDTRR (0x00 then 0xFF)
7. Set initialized flag

OFS Configuration Note:

The IWDT hardware timeout is configured at flash-time via OFS (Option Function Select) registers and cannot be changed at runtime. The timeout values in this driver are used for software task monitoring only. The hardware IWDT will timeout based on OFS settings regardless of software configuration.

Precondition

IWDT not already initialized

Postcondition

IWDT driver ready for use
Initial feed performed
Reset status captured

Note

Hardware timeout set by OFS, not this driver
nullptr config uses safe defaults (1000ms, monitoring enabled)

Since

Version 1.0.0

Definition at line 508 of file [rx_iwdt.c](#).

```

00509 {
00510     /* Check if already initialized */
00511     if (s_iwdt_state.initialized) {
00512         return k_rx_err_invalid_state;
00513     }
00514
00515     /* Use default config if none provided */
00516     rx_iwdt_config_t local_config;
00517     if (config == nullptr) {
00518         internal_init_default_config(&local_config);
00519         config = &local_config;
00520     }
00521
00522     /* Validate configuration */
00523     if (config->default_timeout_ms < k_iwdt_min_timeout_ms ||
00524         config->default_timeout_ms > k_iwdt_max_timeout_ms) {
00525         return k_rx_err_invalid_arg;
00526     }
00527
00528     /* Initialize state */

```

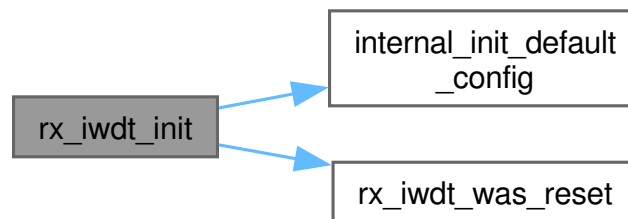
```

00529 s_iwdt_state = (rx_iwdt_state_t){};
00530 {
00531     const uint8_t* src = (const uint8_t*)config;
00532     uint8_t* dst = (uint8_t*)&s_iwdt_state.config;
00533     for (size_t byte_i = 0; byte_i < sizeof(rx_iwdt_config_t); byte_i++) {
00534         dst[byte_i] = src[byte_i];
00535     }
00536 }
00537 s_iwdt_state.current_state = k_system_state_init;
00538
00539 /* Initialize status */
00540 s_iwdt_state.status.initialized = true;
00541 s_iwdt_state.status.current_state = k_system_state_init;
00542 s_iwdt_state.status.current_timeout_ms = config->default_timeout_ms;
00543 s_iwdt_state.status.active_tasks = 0;
00544
00545 /* Check if last reset was watchdog */
00546 if (rx_iwdt_was_reset()) {
00547     s_iwdt_state.status.total_resets++;
00548     s_iwdt_state.status.last_reset_reason = k_iwdt_reset_watchdog;
00549 } else {
00550     s_iwdt_state.status.last_reset_reason = k_iwdt_reset_power_on;
00551 }
00552
00553 /* Note: IWDT hardware is configured via OFS (Option Function Select) registers
00554  * at compile/flash time and starts automatically on reset. We cannot reconfigure
00555  * the timeout period at runtime. This driver manages the software state and
00556  * provides the feed mechanism. The requested timeout is used for software task
00557  * monitoring only. */
00558 volatile rx_iwdt_regs_t* const regs = iwdt();
00559
00560 /* Feed the watchdog to establish baseline */
00561 regs->iwdtrr = k_iwdt_refresh_start;
00562 regs->iwdtrr = k_iwdt_refresh_end;
00563
00564 s_iwdt_state.initialized = true;
00565
00566 return k_rx_ok;
00567 }

```

References [rx_iwdt_config_t::default_timeout_ms](#), [internal_init_default_config\(\)](#), [k_iwdt_max_timeout_ms](#), [k_iwdt_min_timeout_ms](#), [k_iwdt_reset_power_on](#), [k_iwdt_reset_watchdog](#), [k_rx_err_invalid_arg](#), [k_rx_err_invalid_state](#), [k_rx_ok](#), [k_system_state_init](#), [rx_iwdt_was_reset\(\)](#), and [s_iwdt_state](#).

Here is the call graph for this function:



7.19.4.5 rx_iwdt_register_task()

```

rx_err_t rx_iwdt_register_task (
    const char * task_name,
    uint32_t timeout_ms) [nodiscard]

```

Register a task for heartbeat monitoring.

Registers a task with the watchdog for periodic heartbeat checks. Task must call [rx_iwdt_task_heartbeat\(\)](#) within timeout period.

Parameters

in	task_name	Unique task name (max 31 chars)
in	timeout_ms	Task-specific timeout in milliseconds

Returns

[rx_err_t](#) Error code

Return values

k_rx_ok	Success
k_rx_err_null_ptr	task_name is nullptr
k_rx_err_invalid_arg	Invalid timeout or duplicate task
k_rx_err_not_initialized	IWDT not initialized
k_rx_err_no_mem	Max tasks reached

Example:

```
// Register motor control task with 500ms timeout
rx_err_t err = rx_iwdt_register_task("MotorCtrl", 500);
if (err != k_rx_ok) {
    // Handle error
}
```

Register a task for heartbeat monitoring.

Adds a task entry to the IWDT software monitoring table, associating the task name with a per-task timeout. Once registered, the task must call [rx_iwdt_task_heartbeat\(\)](#) at least once per timeout_ms milliseconds or [rx_iwdt_check_tasks\(\)](#) will report [k_rx_err_timeout](#).

The registration also records the current tick count as the initial last_heartbeat_tick, giving the task a full timeout_ms window before the first heartbeat is required.

Algorithm steps:

1. Validate task_name is non-null
2. Verify s_iwdt_state.initialized
3. Validate timeout_ms is in [k_iwdt_min_timeout_ms, k_iwdt_max_timeout_ms]
4. Check task name length is in (0, k_iwdt_task_name_len)
5. Check task is not already registered via [internal_find_task\(\)](#)
6. Find a free slot via [internal_find_free_slot\(\)](#)
7. Initialize slot with task name, timeout, current tick, active=true, timed_out=false
8. Increment status.active_tasks

Precondition

[rx_iwdt_init\(\)](#) must have been called successfully
[task_name](#) must be unique among registered tasks

Postcondition

Task entry is active in `s_iwdt_state.tasks[]`
`status.active_tasks` is incremented by one

Note

Not thread-safe; serialize registrations from multiple contexts

Example:

```
rx_err_t err = rx_iwdt_register_task("motor_ctrl", 500);
if (err != k_rx_ok) {
    rx_log_error("wdt", "Failed to register motor_ctrl task");
}
```

See also

[rx_iwdt_task_heartbeat\(\)](#) Update heartbeat for a registered task
[rx_iwdt_check_tasks\(\)](#) Check all registered tasks for timeout

Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 5: 5 preconditions (null, initialized, timeout, name, duplicate), 2 postconditions

Definition at line 691 of file [rx_iwdt.c](#).

```
00692 {
00693     /* Validate inputs */
00694     if (task_name == nullptr) {
00695         return k_rx_err_null_ptr;
00696     }
00697
00698     if (!s_iwdt_state.initialized) {
00699         return k_rx_err_not_initialized;
00700     }
00701
00702     if (timeout_ms < k_iwdt_min_timeout_ms || timeout_ms > k_iwdt_max_timeout_ms) {
00703         return k_rx_err_invalid_arg;
00704     }
00705
00706     /* Check task name length */
00707     const uint32_t name_len = (uint32_t)strlen(task_name);
00708     if (name_len == 0 || name_len >= k_iwdt_task_name_len) {
00709         return k_rx_err_invalid_arg;
00710     }
00711
00712     /* Check if task already registered */
00713     if (internal_find_task(task_name) != nullptr) {
00714         return k_rx_err_invalid_arg;
00715     }
00716
00717     /* Find free slot */
00718     rx_iwdt_task_info_t* const slot = internal_find_free_slot();
```

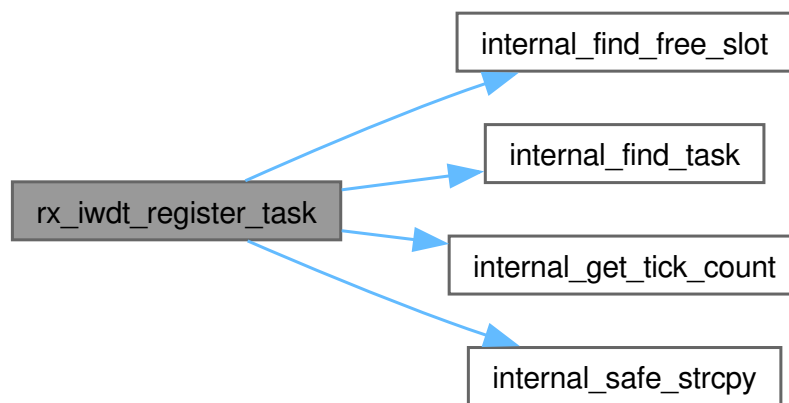
```

00719 if (slot == nullptr) {
00720     return k_rx_err_no_mem;
00721 }
00722
00723 /* Register task */
00724 *slot = (rx_iwdt_task_info_t){};
00725 internal_safe_strcpy(slot->task_name, task_name, k_iwdt_task_name_len);
00726 slot->timeout_ms = timeout_ms;
00727 slot->last_heartbeat_tick = internal_get_tick_count();
00728 slot->active = true;
00729 slot->timed_out = false;
00730
00731 s_iwdt_state.status.active_tasks++;
00732
00733 return k_rx_ok;
00734 }

```

References [rx_iwdt_task_info_t::active](#), [internal_find_free_slot\(\)](#), [internal_find_task\(\)](#), [internal_get_tick_count\(\)](#), [internal_safe_strcpy\(\)](#), [k_iwdt_max_timeout_ms](#), [k_iwdt_task_name_len](#), [k_rx_err_invalid_arg](#), [k_rx_err_no_mem](#), [k_rx_err_not_initialized](#), [k_rx_err_null_ptr](#), [k_rx_ok](#), [rx_iwdt_task_info_t::last_heartbeat_tick](#), [s_iwdt_state](#), [rx_iwdt_task_info_t::task_name](#), [rx_iwdt_task_info_t::timed_out](#), and [rx_iwdt_task_info_t::timeout_ms](#).

Here is the call graph for this function:



7.19.4.6 rx_iwdt_set_state()

```

rx_err_t rx_iwdt_set_state (
    const system_state_t state) [nodiscard]

```

Set current system state.

Updates the current system state, which determines the active watchdog timeout period.

Parameters

in	state	New system state
----	-------	------------------

Returns

[rx_err_t](#) Error code

Return values

k_rx_ok	Success
k_rx_err_invalid_arg	Invalid state
k_rx_err_not_initialized	IWDT not initialized

Example:

```
// Entering motor operation
rx_iwdt_set_state(k_system_state_motor_active);
// ... motor control ...
// Return to normal operation
rx_iwdt_set_state(k_system_state_running);
```

Set current system state.

Updates the active system state used by [rx_iwdt_check_tasks\(\)](#) to select per-state task heartbeat timeout windows. Transitions between states (e.g. initializing -> running) change which timeout threshold is applied when evaluating task heartbeat intervals.

This function updates three tracking fields atomically:

- `s_iwdt_state.current_state`
- `s_iwdt_state.status.current_state`
- `s_iwdt_state.status.current_timeout_ms` (set from `config.state_timeouts_ms[state]`)

Important: This does NOT modify the hardware IWDT peripheral. The hardware watchdog timeout is a fixed constant set in OFS registers at compile time. Only the software task monitoring timeout threshold changes.

Algorithm steps:

1. Validate state is less than `k_system_state_count`
2. Verify `s_iwdt_state.initialized`
3. Update `current_state` and `status.current_state = state`
4. Update `status.current_timeout_ms = config.state_timeouts_ms[state]`

Precondition

[rx_iwdt_init\(\)](#) must have been called successfully
state must be a valid [system_state_t](#) value less than `k_system_state_count`

Postcondition

`s_iwdt_state.current_state` reflects the new state
`status.current_timeout_ms` reflects the configured timeout for the new state

Note

Not thread-safe; serialize state transitions from multiple contexts

Warning

Hardware IWDT is unaffected; only software task monitoring changes

Example:

```
// Transition from initialization to running state
rx_err_t err = rx_iwdt_set_state(k_system_state_running);
if (err != k_rx_ok) {
    rx_log_error("wdt", "Failed to set IWDT system state");
}
```

See also

[rx_iwdt_set_timeout_for_state\(\)](#) Configure per-state timeouts

[rx_iwdt_get_status\(\)](#) Query current state and timeout

Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 5: 2 preconditions (initialized, valid state), 2 postconditions

Definition at line 933 of file [rx_iwdt.c](#).

```
00934 {
00935     /* Validate inputs */
00936     if (state >= k_system_state_count) {
00937         return k_rx_err_invalid_arg;
00938     }
00939
00940     if (!s_iwdt_state.initialized) {
00941         return k_rx_err_not_initialized;
00942     }
00943
00944     /* Update software state tracking.
00945      * NOTE: This does not change the hardware watchdog timeout, as IWDT timeout
00946      * is configured via OFS registers at compile/flash time and cannot be changed
00947      * at runtime. The state-dependent timeouts are used for software task monitoring
00948      * via rx_iwdt_check_tasks(). The hardware IWDT timeout remains constant. */
00949     s_iwdt_state.current_state = state;
00950     s_iwdt_state.status.current_state = state;
00951     s_iwdt_state.status.current_timeout_ms = s_iwdt_state.config.state_timeouts_ms[state];
00952
00953     return k_rx_ok;
00954 }
```

References [k_rx_err_invalid_arg](#), [k_rx_err_not_initialized](#), [k_rx_ok](#), [k_system_state_count](#), and [s_iwdt_state](#).

7.19.4.7 rx_iwdt_set_timeout_for_state()

```
rx_err_t rx_iwdt_set_timeout_for_state (
    const system_state_t state,
    const uint32_t timeout_ms) [nodiscard]
```

Set timeout for system state.

Configures timeout period for a specific system state. Allows different timeout values based on system activity.

Parameters

in	state	System state
in	timeout_ms	Timeout in milliseconds

Returns

[rx_err_t](#) Error code

Return values

k_rx_ok	Success
k_rx_err_invalid_arg	Invalid state or timeout
k_rx_err_not_initialized	IWDT not initialized

Example:

```
// Motor operation needs faster watchdog
rx_iwdt_set_timeout_for_state(k_system_state_motor_active, 500);
// Idle state can use longer timeout
rx_iwdt_set_timeout_for_state(k_system_state_idle, 5000);
```

Set timeout for system state.

Configures the state-dependent software timeout used by [rx_iwdt_check_tasks\(\)](#) when the system is in the specified state. This allows different task heartbeat windows depending on the operational mode (e.g. longer timeouts during initialization, tighter timeouts during normal operation).

Note: This does NOT change the hardware IWDT timeout. The hardware watchdog timeout is fixed at compile/flash time via OFS registers and cannot be altered at runtime. Only software task monitoring timeouts are affected.

If state matches the currently active state (`s_iwdt_state.current_state`), the active timeout (`status.current_timeout_ms`) is also updated immediately.

Algorithm steps:

1. Validate state is less than `k_system_state_count`
2. Verify `s_iwdt_state.initialized`
3. Validate `timeout_ms` is in `[k_iwdt_min_timeout_ms, k_iwdt_max_timeout_ms]`
4. Set `config.state_timeouts_ms[state] = timeout_ms`
5. If `state == current_state`, update `status.current_timeout_ms`

Precondition

`rx_iwdt_init()` must have been called successfully
 state must be a valid `system_state_t` value less than `k_system_state_count`

Postcondition

`config.state_timeouts_ms[state]` equals `timeout_ms`
 If state is current, `status.current_timeout_ms` is updated immediately

Note

Not thread-safe; serialize state configuration changes

Warning

Hardware IWDT timeout is unaffected; only software task monitoring changes

Example:

```
// Allow longer task timeouts during initialization phase
rx_iwdt_set_timeout_for_state(k_system_state_initializing, 5000);
rx_iwdt_set_timeout_for_state(k_system_state_running, 500);
```

See also

`rx_iwdt_set_state()` Change the active system state
`rx_iwdt_check_tasks()` Uses current state timeout to evaluate task heartbeats

Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 5: 3 preconditions (initialized, valid state, valid timeout), 2 postconditions

Definition at line 856 of file `rx_iwdt.c`.

```
00857 {
00858     /* Validate inputs */
00859     if (state >= k_system_state_count) {
00860         return k_rx_err_invalid_arg;
00861     }
00862
00863     if (!s_iwdt_state.initialized) {
00864         return k_rx_err_not_initialized;
00865     }
00866
00867     if (timeout_ms < k_iwdt_min_timeout_ms || timeout_ms > k_iwdt_max_timeout_ms) {
00868         return k_rx_err_invalid_arg;
00869     }
00870
00871     /* Update state timeout */
00872     s_iwdt_state.config.state_timeouts_ms[state] = timeout_ms;
00873
00874     /* If this is the current state, update active timeout */
00875     if (state == s_iwdt_state.current_state) {
00876         s_iwdt_state.status.current_timeout_ms = timeout_ms;
00877     }
00878
00879     return k_rx_ok;
00880 }
```

References `k_iwdt_max_timeout_ms`, `k_rx_err_invalid_arg`, `k_rx_err_not_initialized`, `k_rx_ok`, `k_system_state_count`, and `s_iwdt_state`.

7.19.4.8 rx_iwdt_task_heartbeat()

```
rx_err_t rx_iwdt_task_heartbeat (
    const char * task_name) [nodiscard]
```

Report task heartbeat.

Updates the heartbeat timestamp for a registered task. Must be called periodically faster than the task's timeout.

Parameters

in	task_name	Name of the task
----	-----------	------------------

Returns

[rx_err_t](#) Error code

Return values

k_rx_ok	Success
k_rx_err_null_ptr	task_name is nullptr
k_rx_err_not_found	Task not registered
k_rx_err_not_initialized	IWDT not initialized

Example:

```
static void motor_control_task(ULONG input) {
    while (1) {
        // ... motor control logic ...
        rx_iwdt_task_heartbeat("MotorCtrl");
        tx_thread_sleep(10); // 100ms loop
    }
}
```

Report task heartbeat.

Updates the last_heartbeat_tick for the named task to the current ThreadX tick count, resetting the timeout window. Also clears the timed_out flag if the task had previously been marked as timed out.

Tasks must call this function at least once per their registered timeout_ms window (configured in [rx_iwdt_register_task\(\)](#)) to remain in good standing. If the interval between heartbeats exceeds timeout_ms, the next call to [rx_iwdt_check_tasks\(\)](#) will return k_rx_err_timeout.

Algorithm steps:

1. Validate task_name is non-null
2. Verify s_iwdt_state.initialized
3. Find the task entry via [internal_find_task\(\)](#)
4. Set task->last_heartbeat_tick = [internal_get_tick_count\(\)](#)
5. Clear task->timed_out

Precondition

[rx_iwdt_init\(\)](#) must have been called successfully

Task must have been registered via [rx_iwdt_register_task\(\)](#)

Postcondition

task->last_heartbeat_tick is refreshed to the current tick count

task->timed_out is false

Note

Not thread-safe; concurrent heartbeats from different tasks are safe as they write to distinct slots, but concurrent writes to the same slot are not

Example:

```
// At top of motor control task loop
while (1) {
    rx_iwdt_task_heartbeat("motor_ctrl");
    motor_control_update();
    tx_thread_sleep(k_motor_ctrl_period_ticks);
}
```

See also

[rx_iwdt_register_task\(\)](#) Register a task before sending heartbeats

[rx_iwdt_check_tasks\(\)](#) Verify all tasks are within their timeout windows

Since

Version 1.0.0

NASA Power of 10 Compliance:

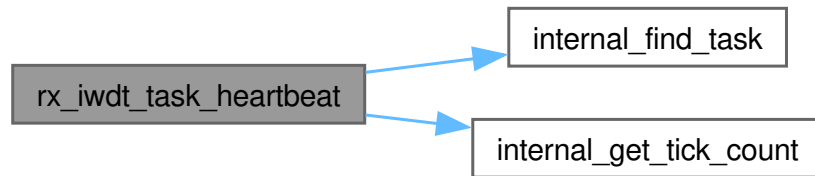
- Rule 5: 3 preconditions (null, initialized, registered), 2 postconditions

Definition at line 784 of file [rx_iwdt.c](#).

```
00785 {
00786     /* Validate inputs */
00787     if (task_name == nullptr) {
00788         return k_rx_err_null_ptr;
00789     }
00790
00791     if (!s_iwdt_state.initialized) {
00792         return k_rx_err_not_initialized;
00793     }
00794
00795     /* Find task */
00796     rx_iwdt_task_info_t* const task = internal_find_task(task_name);
00797     if (task == nullptr) {
00798         return k_rx_err_not_found;
00799     }
00800
00801     /* Update heartbeat */
00802     task->last_heartbeat_tick = internal_get_tick_count();
00803     task->timed_out = false;
00804
00805     return k_rx_ok;
00806 }
```

References [internal_find_task\(\)](#), [internal_get_tick_count\(\)](#), [k_rx_err_not_found](#), [k_rx_err_not_initialized](#), [k_rx_err_null_ptr](#), [k_rx_ok](#), [rx_iwdt_task_info_t::last_heartbeat_tick](#), [s_iwdt_state](#), and [rx_iwdt_task_info_t::timed_out](#).

Here is the call graph for this function:



7.19.4.9 rx_iwdt_was_reset()

```
bool rx_iwdt_was_reset (
    void ) [nodiscard]
```

Check if last reset was caused by watchdog.

Determines if the previous system reset was triggered by watchdog timeout.

Returns

bool true if watchdog reset, false otherwise

Note

Can be called before [rx_iwdt_init\(\)](#)

Example:

```
if (rx_iwdt_was_reset()) {
    uart_debug_puts("[WARN] System recovered from watchdog reset\r\n");
    // Perform recovery actions
}
```

Check if last reset was caused by watchdog.

Reads the IWDTRF bit (bit position `k_rstsr2_iwdtrf`) from the Reset Status Register 2 (RSTSR2) of the RX72N. This register is located at a separate address from RSTSR0/1 and retains the reset cause flags across a reset event (values are cleared on power-on reset only).

The IWDTRF bit (bit position `k_rstsr2_iwdtrf`) is set by hardware when the IWDTRF counter underflows (i.e., the watchdog was not refreshed in time). It remains set until explicitly cleared or a power-on reset occurs.

This function is used during startup to detect and log prior watchdog resets, which may indicate a firmware hang or task scheduling failure.

Algorithm steps:

1. Read the RSTSR2 register byte via `rstsr2()` inline accessor
2. Mask with `k_rstsr2_iwdtrf` to extract the IWDTRF reset flag bit
3. Return true if the bit is set, false otherwise

Precondition

None – safe to call before `rx_iwdt_init()` (reads hardware register directly)
 Must be called before the application clears the reset status register

Postcondition

Hardware RSTSR2 register is read-only; this function does not modify it
 Return value reflects hardware register state at the instant of the call

Note

Thread-safe – reads a single hardware register without shared state
 Typically called once at startup to log the reset cause

Warning

RSTSR2.IWDTRF remains set across warm resets; clear it in startup if needed

Example:

```
if (rx_iwdt_was_reset()) {
    rx_log_error("boot", "Prior reset caused by IWDT timeout - check task health");
}
rx_iwdt_init(&iwdt_config);
```

See also

`rx_iwdt_feed()` Feed the hardware watchdog to prevent reset
`rx_iwdt_check_tasks()` Monitor software task heartbeats

Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 5: 2 preconditions (safe before init, called before register clear), 2 postconditions

Definition at line 1073 of file `rx_iwdt.c`.

```
01074 {
01075     /* Read reset status register 2 for IWDT reset flag
01076      * Note: RSTSR2 is at a separate address from RSTSR0/1 */
01077     volatile uint8_t* const rstsr2_reg = rstsr2();
01078     const uint8_t      status      = *rstsr2_reg;
01079
01080     /* Check IWDT reset flag in RSTSR2 */
01081     return (bool)((status & k_rstsr2_iwdtrf) != 0);
01082 }
```

Referenced by `rx_iwdt_init()`.

Here is the caller graph for this function:



7.20 rx_iwtdt.h

[Go to the documentation of this file.](#)

```

00001 /**
00002  * @file rx_iwtdt.h
00003  * @brief Independent Watchdog Timer (IWDT) Driver for RX72N Microcontroller
00004  *
00005  * @details
00006  * **Production-grade watchdog driver** providing safety-critical system monitoring
00007  * with advanced task tracking, state-dependent timeouts, and comprehensive
00008  * diagnostics. Designed for real-time embedded systems requiring fault detection
00009  * and automatic recovery from software hangs, deadlocks, and system failures.
00010  *
00011  * ## Purpose and Design Rationale
00012  *
00013  * This driver implements a **multi-layer watchdog architecture** combining:
00014  * 1. **Hardware IWDT**: Physical watchdog timer that triggers system reset
00015  * 2. **Task monitoring**: Software-level tracking of individual RTOS tasks
00016  * 3. **State-aware timeouts**: Different timeout periods based on system activity
00017  * 4. **Diagnostic tracking**: Post-mortem analysis of failures
00018  *
00019  * The Independent Watchdog Timer (IWDT) is **separate from the normal watchdog**
00020  * and provides **true independence** from the main system clock, making it immune
00021  * to clock failures.
00022  *
00023  * ## Key Features
00024  *
00025  * **Hardware Watchdog**:
00026  * - Physical watchdog timer with dedicated clock (IWDTCLK)
00027  * - Cannot be disabled once started (hardware safety feature)
00028  * - 4 timeout periods: 128ms, 512ms, 2s, 8s
00029  * - Automatic system reset on timeout
00030  *
00031  * **Task Monitoring** (Software Layer):
00032  * - Register up to 16 tasks with individual heartbeat tracking
00033  * - Per-task timeout configuration
00034  * - Failed task identification for diagnostics
00035  * - Thread-safe heartbeat reporting
00036  *
00037  * **State-Dependent Timeouts**:
00038  * - Different timeout periods for different system states
00039  * - Example: Shorter timeout during motor operation, longer during idle
00040  * - Runtime state transitions with automatic timeout adjustment
00041  *
00042  * **Diagnostics and Reporting**:
00043  * - Reset reason detection (power-on, watchdog, external, etc.)
00044  * - Reset counter (survives resets via backup registers)
00045  * - Failed task tracking for post-mortem analysis
00046  * - Watchdog feed statistics
00047  *
00048  * ## Hardware Architecture
00049  *
00050  * **RX72N Independent Watchdog Timer (IWDT)**:
00051  * - **Register Base Address**: 0x00088030
00052  * - **Clock Source**: IWDTCLK (independent oscillator, 120 kHz)
00053  * - **Clock Frequency**: 120 kHz (8.33 us period)
00054  * - **Counter Width**: 14-bit down-counter
00055  * - **Timeout Range**: ~128ms to ~16.4 seconds
00056  * - **Reset Output**: Connected to internal reset controller
00057  *
00058  * **Timeout Calculation**:
00059  * ```
00060  * Timeout = (2^(10+CKS)) / IWDTCLK
00061  *
00062  * Where CKS (Clock Select) determines divider:
00063  * CKS=0: 2^10 = 1024 -> ~128ms timeout
00064  * CKS=1: 2^12 = 4096 -> ~512ms timeout
00065  * CKS=2: 2^14 = 16384 -> ~2s timeout
00066  * CKS=3: 2^16 = 65536 -> ~8s timeout
00067  * ```
00068  *
00069  * ## Implementation Approach
00070  *
00071  * **Three-Layer Architecture**:
00072  *
00073  * 1. **Hardware Layer** (rx_iwtdt_init, rx_iwtdt_feed):
00074  * - Configures IWDT registers
00075  * - Writes refresh code to prevent timeout
00076  *
00077  * 2. **Task Monitoring Layer** (rx_iwtdt_register_task, rx_iwtdt_task_heartbeat):
00078  * - Tracks registered tasks
00079  * - Checks heartbeat timestamps
00080  * - Identifies failed tasks
00081  *
00082  * 3. **State Management Layer** (rx_iwtdt_set_state, rx_iwtdt_set_timeout_for_state):

```

```

00083 * - Maps system states to timeout periods
00084 * - Adjusts watchdog based on activity level
00085 *
00086 **Typical Usage Flow**:
00087 * ``
00088 * 1. System startup: rx_iwdt_init() - Configure and start watchdog
00089 * 2. Task registration: rx_iwdt_register_task() - Register critical tasks
00090 * 3. Normal operation:
00091 *   - Main loop: rx_iwdt_feed() every N ms (N < timeout)
00092 *   - Each task: rx_iwdt_task_heartbeat() within task timeout
00093 *   - Monitor thread: rx_iwdt_check_tasks() to detect failures
00094 * 4. State changes: rx_iwdt_set_state() to adjust timeout dynamically
00095 * 5. Diagnostics: rx_iwdt_get_status() for reset history and statistics
00096 * ``
00097 *
00098 * ## Performance Characteristics
00099 *
00100 * | Operation | Execution Time | Memory Usage | Notes |
00101 * |-----|-----|-----|-----|
00102 * | rx_iwdt_init() | ~50 us | 0 heap | One-time setup |
00103 * | rx_iwdt_feed() | ~2 us | 0 | Register write only |
00104 * | rx_iwdt_task_heartbeat() | ~5 us | 0 | Timestamp update |
00105 * | rx_iwdt_check_tasks() | ~10 us x N tasks | 0 | Linear scan |
00106 * | rx_iwdt_get_status() | ~15 us | 0 | Structure copy |
00107 *
00108 * **Memory Footprint**:
00109 * - Static variables: ~2 KB (task tracking array, status, configuration)
00110 * - Code size: ~3 KB (all functions)
00111 * - Stack usage: < 128 bytes (deepest call)
00112 * - Heap usage: 0 bytes (zero dynamic allocation)
00113 *
00114 * ## Thread Safety and RTOS Integration
00115 *
00116 * **Thread-Safe Operations**:
00117 * - All public functions use mutex protection (ThreadX)
00118 * - Safe to call from multiple threads simultaneously
00119 * - ISR-safe: rx_iwdt_feed() can be called from interrupts
00120 *
00121 * **ThreadX Integration**:
00122 * - Uses tx_mutex for synchronization
00123 * - Uses tx_time_get() for timestamps
00124 * - Compatible with all ThreadX thread priorities
00125 *
00126 * ## Error Detection and Recovery
00127 *
00128 * **Detected Failures**:
00129 * 1. **Task deadlock**: Task stops sending heartbeats
00130 * 2. **Main loop hang**: No rx_iwdt_feed() calls
00131 * 3. **Stack overflow**: Task corrupts memory and stops executing
00132 * 4. **Infinite loop**: Task stuck in tight loop without heartbeat
00133 * 5. **Communication timeout**: Network/SPI/UART tasks stop responding
00134 *
00135 * **Recovery Actions**:
00136 * - Hardware reset triggered by IWDG timeout
00137 * - System reboots from reset vector
00138 * - rx_iwdt_was_reset() detects watchdog reset on next boot
00139 * - Failed task name preserved for diagnostics
00140 *
00141 * ## Usage Guidelines
00142 *
00143 * **DO**:
00144 * - Call rx_iwdt_feed() from main supervision loop only
00145 * - Register all critical tasks for monitoring
00146 * - Use state-dependent timeouts for different operating modes
00147 * - Check rx_iwdt_was_reset() at startup for diagnostics
00148 * - Tune timeouts based on worst-case execution times
00149 *
00150 * **DON'T**:
00151 * - Call rx_iwdt_feed() from multiple locations (defeats purpose)
00152 * - Set timeouts shorter than worst-case task execution time
00153 * - Rely on watchdog for normal error handling (use return codes)
00154 * - Call rx_iwdt_feed() from ISRs (unless by design)
00155 * - Assume watchdog prevents all failures (defense in depth)
00156 *
00157 * @par Hardware Requirements:
00158 *
00159 * | Component | Requirement | Notes |
00160 * |-----|-----|-----|
00161 * | MCU | Renesas RX72N | IWDG peripheral required |
00162 * | Clock | IWDGCLK | Independent oscillator, 120 kHz |
00163 * | Registers | 0x00088030 base | IWDG control registers |
00164 * | Reset | Internal reset controller | Connected to IWDG output |
00165 * | Backup registers | Optional | For reset counter persistence |
00166 *
00167 * @par Module Dependencies:
00168 *
00169 * **This module depends on**:

```

```

00170 * - `rx_err.h' - Error code definitions
00171 * - `<stdint.h>' - Standard integer types
00172 * - `<stdbool.h>' - Boolean type
00173 * - ThreadX RTOS - Mutex and timestamp functions
00174 *
00175 * **This module is used by**:
00176 * - `main.c' - System initialization and main loop
00177 * - All critical RTOS tasks - Heartbeat reporting
00178 * - System monitor task - Task timeout checking
00179 *
00180 * @par NASA Power of 10 Compliance:
00181 *
00182 * | Rule | Status | Implementation |
00183 * |-----|-----|-----|
00184 * | 1. Simple control flow | [OK] | No goto/setjmp/recursion, sequential logic |
00185 * | 2. Fixed loop bounds | [OK] | All loops bounded by k_iwtdt_max_tasks (16) |
00186 * | 3. No dynamic allocation | [OK] | Zero malloc/free, static task array |
00187 * | 4. Small functions | [OK] | All functions < 60 lines |
00188 * | 5. Assertions (>=2 per function) | [OK] | Minimum 2 checks per function (pre/post) |
00189 * | 6. Narrow scope | [OK] | File-scope statics, function-local variables |
00190 * | 7. Check return values | [OK] | All functions return rx_err_t, checked by callers |
00191 * | 8. Limited preprocessor | [OK] | C23 typed enums only, zero computation macros |
00192 * | 9. Pointer restrictions | [OK] | Single-level pointers, no pointer arithmetic |
00193 * | 10. Compiler warnings | [OK] | Compiles with -Wall -Wextra -Werror |
00194 *
00195 * **Safety-Critical Design**:
00196 * - Static allocation prevents fragmentation and allocation failures
00197 * - Bounded arrays prevent buffer overflows
00198 * - Explicit error codes force error handling
00199 * - Mutex protection prevents race conditions
00200 *
00201 * @par SOLID Principles:
00202 *
00203 * **Single Responsibility (S)**:
00204 * - This module has ONE job: Watchdog monitoring and system reset
00205 * - Separate concerns: Hardware watchdog vs task monitoring vs state management
00206 * - No mixing of unrelated functionality
00207 *
00208 * **Open/Closed (O)**:
00209 * - Extensible via configuration structure (state timeouts, task limits)
00210 * - Can add new system states without modifying existing code
00211 * - Timeout periods configured at runtime, not hardcoded
00212 *
00213 * **Liskov Substitution (L)**:
00214 * - All functions return rx_err_t for consistent error handling
00215 * - Boolean functions (rx_iwtdt_was_reset) have well-defined semantics
00216 * - No unexpected side effects or hidden dependencies
00217 *
00218 * **Interface Segregation (I)**:
00219 * - Separate functions for separate concerns (feed vs heartbeat vs status)
00220 * - Task monitoring is optional (enable_task_monitoring flag)
00221 * - Clients use only the functions they need
00222 *
00223 * **Dependency Inversion (D)**:
00224 * - Depends on abstract rx_err_t, not specific error implementations
00225 * - Uses system time abstraction (tx_time_get), not raw timer
00226 * - Can be mocked for unit testing via test_reset function
00227 *
00228 * @see rx_err.h Error code definitions used by all functions
00229 * @see docs/sections/06_nasa_power_of_10.tex Safety-critical coding standards
00230 * @see RX72N Hardware Manual Section 11 Independent Watchdog Timer specification
00231 *
00232 * @author Locked, Inc.
00233 * @date 2026-01-27
00234 * @copyright Copyright (c) 2026 Locked Inc.
00235 * SPDX-License-Identifier: MIT
00236 * @since Version 1.0.0
00237 */
00238
00239 #pragma once
00240
00241 #include <stdbool.h>
00242 #include <stdint.h>
00243
00244 #include "rx_err.h"
00245
00246 #ifdef __cplusplus
00247 extern "C" {
00248 #endif
00249
00250 /*
=====
00251 * Constants
00252 *
=====
00253 */
00254

```

```

00255 /**
00256  * @enum iwdt_constants_t
00257  * @brief IWDT Configuration Constants and Limits
00258  *
00259  * @details
00260  * Defines compile-time constants for the IWDT driver including array sizes,
00261  * timeout limits, and time conversion factors. These constants are chosen based
00262  * on hardware limitations and system requirements.
00263  *
00264  * ## Constant Rationale
00265  *
00266  * **k_iwdt_max_tasks = 16**:
00267  * - Sized for typical STAR system (4 motors + communication + sensors)
00268  * - Keeps task array < 1 KB (16 x 64 bytes = 1024 bytes)
00269  * - Allows linear scan in < 10 us at 240 MHz
00270  *
00271  * **k_iwdt_task_name_len = 32**:
00272  * - Matches ThreadX TX_THREAD_NAME_SIZE
00273  * - Accommodates descriptive names like "MotorControllerTask_FL"
00274  * - Null-terminated, so 31 characters + '\0'
00275  *
00276  * **k_iwdt_default_timeout_ms = 2000**:
00277  * - 2 second default provides safety margin
00278  * - Allows 100ms main loop to miss ~20 iterations
00279  * - Not too long (fast failure detection) nor too short (no false positives)
00280  *
00281  * **k_iwdt_min_timeout_ms = 128**:
00282  * - Hardware minimum with CKS=0 divider
00283  * - Calculated: 1024 / 125kHz ~ 128ms
00284  * - Use only for very fast loops (> 10 Hz)
00285  *
00286  * **k_iwdt_max_timeout_ms = 16384**:
00287  * - Hardware maximum with CKS=3 divider
00288  * - Calculated: 65536 / 125kHz ~ 8192ms
00289  * - Conservative limit prevents excessive hang detection time
00290  *
00291  * @par Usage Example:
00292  * @code{.c}
00293  * // Validate configuration against limits
00294  * rx_iwdt_config_t config = {
00295  *     .default_timeout_ms = k_iwdt_default_timeout_ms,
00296  *     .enable_task_monitoring = true
00297  * };
00298  *
00299  * if (config.default_timeout_ms < k_iwdt_min_timeout_ms) {
00300  *     return k_rx_err_invalid_arg;
00301  * }
00302  * @endcode
00303  *
00304  * @invariant k_iwdt_max_tasks x sizeof(rx_iwdt_task_info_t) < 2048 bytes
00305  * @invariant k_iwdt_min_timeout_ms <= k_iwdt_default_timeout_ms <= k_iwdt_max_timeout_ms
00306  * @invariant All values are compile-time constants (no runtime overhead)
00307  *
00308  * @note C23 typed enum with uint32_t underlying type (4-byte constants)
00309  * @note Values chosen based on RX72N IWDT hardware capabilities
00310  *
00311  * @see rx_iwdt_config_t Configuration structure using these constants
00312  * @see rx_iwdt_task_info_t Task info structure sized by k_iwdt_task_name_len
00313  *
00314  * @since Version 1.0.0
00315  */
00316 typedef enum : uint32_t {
00317     k_iwdt_max_tasks =
00318         16, /**< Maximum number of tasks that can be registered for monitoring. Limits task array size to ~1 KB. Increase if
00319             more tasks needed (recompile required) */
00319     k_iwdt_task_name_len =
00320         32, /**< Maximum task name length in characters including null terminator. Matches ThreadX
00321             TX_THREAD_NAME_SIZE. Allows 31-character names */
00321     k_iwdt_default_timeout_ms =
00322         2000, /**< Default watchdog timeout in milliseconds (2 seconds). Provides good balance between fast failure detection
00323             and false positive avoidance */
00323     k_iwdt_min_timeout_ms =
00324         128, /**< Minimum valid timeout in milliseconds (~128ms). Hardware limit with CKS=0. Use only for very fast control
00325             loops (> 10 Hz) */
00325     k_iwdt_max_timeout_ms =
00326         16384, /**< Maximum valid timeout in milliseconds (~16 seconds). Conservative limit for reasonable hang detection.
00327             Hardware supports up to ~8s with CKS=3 */
00327     k_ms_per_second =
00328         1000, /**< Milliseconds per second (1000 ms/s). Time conversion constant for timeout calculations and timestamp
00329             arithmetic */
00329 } iwdt_constants_t;
00330
00331 /** @brief IWDT timeout periods (hardware divider settings) */
00332 typedef enum : uint8_t {
00333     k_iwdt_timeout_128ms = 0, /**< ~128ms (CKS=00) */
00334     k_iwdt_timeout_512ms = 1, /**< ~512ms (CKS=01) */
00335     k_iwdt_timeout_2048ms = 2, /**< ~2s (CKS=10) */

```

```

00336 k_iwdt_timeout_8192ms = 3, /**< ~8s (CKS=11) */
00337 } iwdt_timeout_period_t;
00338
00339 /**
00340  * @enum system_state_t
00341  * @brief System Operational States for State-Dependent Watchdog Timeouts
00342  *
00343  * @details
00344  * Defines all possible system states, each with its own watchdog timeout period.
00345  * The watchdog timeout is automatically adjusted when transitioning between states,
00346  * allowing shorter timeouts during active operations and longer timeouts during
00347  * idle periods.
00348  *
00349  * ## State Machine Architecture
00350  *
00351  * The system operates as a finite state machine with defined transitions:
00352  *
00353  * @startuml
00354  * [*] --> Init : Power-On / Reset
00355  * Init --> Idle : Initialization Complete
00356  * Idle --> Running : Start Command
00357  * Running --> Idle : Stop Command
00358  * Running --> MotorActive : Motor Start
00359  * Running --> CommActive : Communication Start
00360  * MotorActive --> Running : Motor Stop
00361  * CommActive --> Running : Communication Stop
00362  * MotorActive --> CommActive : Both Active
00363  * CommActive --> MotorActive : Both Active
00364  * Running --> Error : Fault Detected
00365  * MotorActive --> Error : Fault Detected
00366  * CommActive --> Error : Fault Detected
00367  * Error --> Idle : Error Cleared
00368  * Error --> Init : Reset Required
00369  * @enduml
00370  *
00371  * ## State Descriptions and Typical Timeouts
00372  *
00373  * | State | Description | Typical Timeout | Rationale |
00374  * |-----|-----|-----|-----|
00375  * | Init | System initialization, hardware setup | 5000ms | Slow startup tolerable |
00376  * | Idle | No active operations, waiting for commands | 5000ms | Long timeout OK |
00377  * | Running | Normal operation, moderate activity | 2000ms | Default timeout |
00378  * | MotorActive | Motors running, high-frequency control | 500ms | Fast loop required |
00379  * | CommActive | Active SPI/USB communication | 1000ms | Network delays possible |
00380  * | Error | Error recovery, diagnostics | 10000ms | Allow time for recovery |
00381  *
00382  * ## Recommended Timeout Configuration
00383  *
00384  * ```c
00385  * rx_iwdt_config_t config;
00386  * config.state_timeouts_ms[k_system_state_init] = 5000; // Slow startup
00387  * config.state_timeouts_ms[k_system_state_idle] = 5000; // Long idle OK
00388  * config.state_timeouts_ms[k_system_state_running] = 2000; // Default
00389  * config.state_timeouts_ms[k_system_state_motor_active] = 500; // Fast loop
00390  * config.state_timeouts_ms[k_system_state_comm_active] = 1000; // Network
00391  * config.state_timeouts_ms[k_system_state_error] = 10000; // Recovery time
00392  * ```
00393  *
00394  * @par State Transition Example:
00395  * @code{.c}
00396  * // System startup
00397  * rx_iwdt_set_state(k_system_state_init);
00398  * // ... hardware initialization ...
00399  *
00400  * // Initialization complete, go idle
00401  * rx_iwdt_set_state(k_system_state_idle);
00402  *
00403  * // User command received, start operation
00404  * rx_iwdt_set_state(k_system_state_running);
00405  *
00406  * // Begin motor control
00407  * rx_iwdt_set_state(k_system_state_motor_active);
00408  * // Motor control loop must call rx_iwdt_feed() within 500ms
00409  *
00410  * // Stop motors, return to normal
00411  * rx_iwdt_set_state(k_system_state_running);
00412  *
00413  * // Fault detected
00414  * rx_iwdt_set_state(k_system_state_error);
00415  * // Long timeout allows diagnostic logging
00416  *
00417  * // Recovery complete
00418  * rx_iwdt_set_state(k_system_state_idle);
00419  * @endcode
00420  *
00421  * @par Usage Example (Dynamic Timeout Adjustment):
00422  * @code{.c}

```

```

00423 * // Configure different timeouts for different states
00424 * rx_iwdt_set_timeout_for_state(k_system_state_motor_active, 500);
00425 * rx_iwdt_set_timeout_for_state(k_system_state_idle, 5000);
00426 *
00427 * // Transition to motor state (timeout automatically becomes 500ms)
00428 * rx_iwdt_set_state(k_system_state_motor_active);
00429 *
00430 * // Motor control loop
00431 * while (motor_running) {
00432 *     motor_control_update();
00433 *     rx_iwdt_feed(); // Must execute within 500ms
00434 *     tx_thread_sleep(10); // 100Hz loop = 10ms period
00435 * }
00436 *
00437 * // Stop motors (timeout automatically becomes 2000ms)
00438 * rx_iwdt_set_state(k_system_state_running);
00439 * @endcode
00440 *
00441 * @invariant State values are contiguous (0 to k_system_state_count-1)
00442 * @invariant k_system_state_count equals number of states (6)
00443 * @invariant All state values fit in uint8_t (0-255)
00444 *
00445 * @note C23 typed enum with uint8_t underlying type (1-byte values)
00446 * @note State transitions trigger automatic timeout period adjustments
00447 * @note k_system_state_count is used for array sizing and validation
00448 *
00449 * @warning Invalid state transitions are allowed but may indicate logic errors
00450 * @attention Changing state does NOT reset the watchdog counter--still need rx_iwdt_feed()
00451 *
00452 * @see rx_iwdt_set_state() Set current system state
00453 * @see rx_iwdt_set_timeout_for_state() Configure timeout for specific state
00454 * @see rx_iwdt_config_t Configuration structure with state_timeouts_ms array
00455 *
00456 * @since Version 1.0.0
00457 */
00458 typedef enum : uint8_t {
00459     k_system_state_init =
00460         0, /**< System initialization phase. Hardware setup, peripheral configuration. Use long timeout (5s) to allow slow
00461 startup. Transitions to: Idle */
00461     k_system_state_idle =
00462         1, /**< Idle state, no active operations. Waiting for commands. Use long timeout (5s) since no critical operations.
00463 Transitions to: Running */
00463     k_system_state_running =
00464         2, /**< Normal operation, moderate activity. Background tasks executing. Use default timeout (2s). Transitions to:
00465 MotorActive, CommActive, Idle, Error */
00465     k_system_state_motor_active =
00466         3, /**< Motor control active, high-frequency updates. Critical real-time loop. Use short timeout (500ms) for fast failure
00467 detection. Transitions to: Running, Error */
00467     k_system_state_comm_active =
00468         4, /**< Active communication (SPI/USB). Network I/O in progress. Use medium timeout (1s) to tolerate network
00469 delays. Transitions to: Running, Error */
00469     k_system_state_error =
00470         5, /**< Error recovery mode. Diagnostics and cleanup. Use very long timeout (10s) to allow recovery actions and
00471 logging. Transitions to: Idle, Init */
00471     k_system_state_count =
00472         6, /**< Number of system states (not a valid state). Used for array sizing (state_timeouts_ms[k_system_state_count])
00473 and validation checks */
00473 } system_state_t;
00474
00475 /** @brief Watchdog reset reasons */
00476 typedef enum : uint8_t {
00477     k_iwdt_reset_unknown = 0, /**< Unknown reset */
00478     k_iwdt_reset_power_on = 1, /**< Power-on reset */
00479     k_iwdt_reset_watchdog = 2, /**< Watchdog timeout */
00480     k_iwdt_reset_external = 3, /**< External reset */
00481     k_iwdt_reset_low_voltage = 4, /**< Low voltage detect */
00482 } iwdt_reset_reason_t;
00483
00484 /*
=====
00485 * Data Structures
00486 *
=====
00487 */
00488
00489 /**
00490 * @brief IWDT configuration structure
00491 *
00492 * Configures the Independent Watchdog Timer with timeout periods for
00493 * different system states.
00494 */
00495 typedef struct {
00496     uint32_t default_timeout_ms; /**< Default timeout */
00497     uint32_t state_timeouts_ms[k_system_state_count]; /**< State-specific timeouts */
00498     bool enable_task_monitoring; /**< Enable task heartbeats */
00499     bool reset_on_timeout; /**< Reset vs NMI */
00500 } rx_iwdt_config_t;

```



```

00501
00502 /**
00503  * @brief Task registration structure (internal)
00504  *
00505  * Tracks registered tasks for heartbeat monitoring.
00506  */
00507 typedef struct {
00508     char    task_name[k_iwdt_task_name_len]; /**< Task name */
00509     uint32_t timeout_ms;                      /**< Task-specific timeout */
00510     uint32_t last_heartbeat_tick;             /**< Last heartbeat timestamp */
00511     bool    active;                          /**< Task is active */
00512     bool    timed_out;                      /**< Task has timed out */
00513 } rx_iwdt_task_info_t;
00514
00515 /**
00516  * @brief IWDT status structure
00517  *
00518  * Provides watchdog status information including reset history and
00519  * failed task tracking for post-mortem analysis.
00520  */
00521 typedef struct {
00522     uint32_t    total_resets;                /**< Total reset count */
00523     iwdt_reset_reason_t last_reset_reason;    /**< Last reset cause */
00524     char        last_failed_task[k_iwdt_task_name_len]; /**< Failed task name */
00525     uint32_t    watchdog_feeds;              /**< Total feed count */
00526     system_state_t current_state;            /**< Current system state */
00527     uint32_t    current_timeout_ms;          /**< Active timeout */
00528     uint32_t    active_tasks;                /**< Number of active tasks */
00529     bool        initialized;                 /**< IWDT initialized */
00530 } rx_iwdt_status_t;
00531
00532 /*
00533  * Public API
00534  */
00535 /*
00536  *
00537  */
00538 /**
00539  * @brief Initialize the Independent Watchdog Timer
00540  *
00541  * Initializes the IWDT with specified configuration. Once started,
00542  * the IWDT cannot be stopped (hardware safety feature).
00543  *
00544  * @param[in] config Configuration structure (NULL for defaults)
00545  * @return rx_err_t Error code
00546  * @retval k_rx_ok Success
00547  * @retval k_rx_err_invalid_arg Invalid configuration
00548  * @retval k_rx_err_invalid_state Already initialized
00549  * @retval k_rx_err_hw_init_failed Hardware initialization failed
00550  *
00551  * @note This function must be called before any other IWDT functions
00552  * @warning IWDT cannot be stopped once started - requires hard reset
00553  *
00554  * @par Example:
00555  * @code
00556  * rx_iwdt_config_t config = {
00557  *     .default_timeout_ms = 2000,
00558  *     .enable_task_monitoring = true,
00559  *     .reset_on_timeout = true
00560  * };
00561  * config.state_timeouts_ms[k_system_state_running] = 1000;
00562  * config.state_timeouts_ms[k_system_state_motor_active] = 500;
00563  * rx_err_t err = rx_iwdt_init(&config);
00564  * if (err != k_rx_ok) {
00565  *     // Handle error
00566  * }
00567  * @endcode
00568  */
00569 [[nodiscard]] rx_err_t rx_iwdt_init(const rx_iwdt_config_t* config);
00570
00571 /**
00572  * @brief Feed the watchdog timer
00573  *
00574  * Resets the watchdog counter to prevent timeout. Must be called
00575  * periodically at a rate faster than the configured timeout.
00576  *
00577  * @return rx_err_t Error code
00578  * @retval k_rx_ok Success
00579  * @retval k_rx_err_not_initialized IWDT not initialized
00580  *
00581  * @note Thread-safe - can be called from multiple threads/ISRs
00582  * @warning Missing feeds will cause system reset
00583  *
00584  * @par Example:
00585  * @code

```

```

00586 * // In main control loop (must execute faster than timeout)
00587 * while (1) {
00588 *     // ... control logic ...
00589 *     rx_iwdt_feed();
00590 *     tx_thread_sleep(50); // 500ms loop
00591 * }
00592 * @endcode
00593 */
00594 [[nodiscard]] rx_err_t rx_iwdt_feed(void);
00595
00596 /**
00597 * @brief Register a task for heartbeat monitoring
00598 *
00599 * Registers a task with the watchdog for periodic heartbeat checks.
00600 * Task must call rx_iwdt_task_heartbeat() within timeout period.
00601 *
00602 * @param[in] task_name Unique task name (max 31 chars)
00603 * @param[in] timeout_ms Task-specific timeout in milliseconds
00604 * @return rx_err_t Error code
00605 * @retval k_rx_ok Success
00606 * @retval k_rx_err_null_ptr task_name is nullptr
00607 * @retval k_rx_err_invalid_arg Invalid timeout or duplicate task
00608 * @retval k_rx_err_not_initialized IWDT not initialized
00609 * @retval k_rx_err_no_mem Max tasks reached
00610 *
00611 * @par Example:
00612 * @code
00613 * // Register motor control task with 500ms timeout
00614 * rx_err_t err = rx_iwdt_register_task("MotorCtrl", 500);
00615 * if (err != k_rx_ok) {
00616 *     // Handle error
00617 * }
00618 * @endcode
00619 */
00620 [[nodiscard]] rx_err_t rx_iwdt_register_task(const char* task_name, uint32_t timeout_ms);
00621
00622 /**
00623 * @brief Report task heartbeat
00624 *
00625 * Updates the heartbeat timestamp for a registered task. Must be
00626 * called periodically faster than the task's timeout.
00627 *
00628 * @param[in] task_name Name of the task
00629 * @return rx_err_t Error code
00630 * @retval k_rx_ok Success
00631 * @retval k_rx_err_null_ptr task_name is nullptr
00632 * @retval k_rx_err_not_found Task not registered
00633 * @retval k_rx_err_not_initialized IWDT not initialized
00634 *
00635 * @par Example:
00636 * @code
00637 * static void motor_control_task(ULONG input) {
00638 *     while (1) {
00639 *         // ... motor control logic ...
00640 *         rx_iwdt_task_heartbeat("MotorCtrl");
00641 *         tx_thread_sleep(10); // 100ms loop
00642 *     }
00643 * }
00644 * @endcode
00645 */
00646 [[nodiscard]] rx_err_t rx_iwdt_task_heartbeat(const char* task_name);
00647
00648 /**
00649 * @brief Set timeout for system state
00650 *
00651 * Configures timeout period for a specific system state. Allows
00652 * different timeout values based on system activity.
00653 *
00654 * @param[in] state System state
00655 * @param[in] timeout_ms Timeout in milliseconds
00656 * @return rx_err_t Error code
00657 * @retval k_rx_ok Success
00658 * @retval k_rx_err_invalid_arg Invalid state or timeout
00659 * @retval k_rx_err_not_initialized IWDT not initialized
00660 *
00661 * @par Example:
00662 * @code
00663 * // Motor operation needs faster watchdog
00664 * rx_iwdt_set_timeout_for_state(k_system_state_motor_active, 500);
00665 * // Idle state can use longer timeout
00666 * rx_iwdt_set_timeout_for_state(k_system_state_idle, 5000);
00667 * @endcode
00668 */
00669 [[nodiscard]] rx_err_t rx_iwdt_set_timeout_for_state(system_state_t state, uint32_t timeout_ms);
00670
00671 /**
00672 * @brief Set current system state

```

```

00673 *
00674 * Updates the current system state, which determines the active
00675 * watchdog timeout period.
00676 *
00677 * @param[in] state New system state
00678 * @return rx_err_t Error code
00679 * @retval k_rx_ok Success
00680 * @retval k_rx_err_invalid_arg Invalid state
00681 * @retval k_rx_err_not_initialized IWDT not initialized
00682 *
00683 * @par Example:
00684 * @code
00685 * // Entering motor operation
00686 * rx_iwdt_set_state(k_system_state_motor_active);
00687 * // ... motor control ...
00688 * // Return to normal operation
00689 * rx_iwdt_set_state(k_system_state_running);
00690 * @endcode
00691 */
00692 [[nodiscard]] rx_err_t rx_iwdt_set_state(system_state_t state);
00693
00694 /**
00695 * @brief Get watchdog status
00696 *
00697 * Retrieves current watchdog status including reset history,
00698 * failed tasks, and monitoring statistics.
00699 *
00700 * @param[out] status Pointer to status structure
00701 * @return rx_err_t Error code
00702 * @retval k_rx_ok Success
00703 * @retval k_rx_err_null_ptr status is nullptr
00704 * @retval k_rx_err_not_initialized IWDT not initialized
00705 *
00706 * @par Example:
00707 * @code
00708 * rx_iwdt_status_t status;
00709 * rx_err_t err = rx_iwdt_get_status(&status);
00710 * if (err == k_rx_ok) {
00711 *     printf("Total resets: %lu\n", status.total_resets);
00712 *     printf("Last failed task: %s\n", status.last_failed_task);
00713 * }
00714 * @endcode
00715 */
00716 [[nodiscard]] rx_err_t rx_iwdt_get_status(rx_iwdt_status_t* status);
00717
00718 /**
00719 * @brief Check if last reset was caused by watchdog
00720 *
00721 * Determines if the previous system reset was triggered by
00722 * watchdog timeout.
00723 *
00724 * @return bool true if watchdog reset, false otherwise
00725 *
00726 * @note Can be called before rx_iwdt_init()
00727 *
00728 * @par Example:
00729 * @code
00730 * if (rx_iwdt_was_reset()) {
00731 *     uart_debug_puts("[WARN] System recovered from watchdog reset\r\n");
00732 *     // Perform recovery actions
00733 * }
00734 * @endcode
00735 */
00736 [[nodiscard]] bool rx_iwdt_was_reset(void);
00737
00738 /**
00739 * @brief Check for task timeouts
00740 *
00741 * Checks all registered tasks for timeout. Should be called
00742 * periodically from a monitoring task.
00743 *
00744 * @return rx_err_t Error code
00745 * @retval k_rx_ok No timeouts detected
00746 * @retval k_rx_err_timeout One or more tasks timed out
00747 * @retval k_rx_err_not_initialized IWDT not initialized
00748 *
00749 * @note This function does NOT reset the hardware watchdog
00750 * @warning Task timeouts trigger watchdog reset if not cleared
00751 *
00752 * @par Example:
00753 * @code
00754 * static void watchdog_monitor_task(ULONG input) {
00755 *     while (1) {
00756 *         rx_err_t err = rx_iwdt_check_tasks();
00757 *         if (err == k_rx_err_timeout) {
00758 *             // Log timeout before reset
00759 *             uart_debug_puts("[ERROR] Task timeout detected\r\n");

```

```

00760 *      }
00761 *      rx_iwdt_feed(); // Feed hardware watchdog
00762 *      tx_thread_sleep(10); // 100ms check period
00763 *  }
00764 *  }
00765 *  @endcode
00766 */
00767 [[nodiscard]] rx_err_t rx_iwdt_check_tasks(void);
00768
00769 /*
=====
00770 * Test Support Functions
00771 *
=====
00772 */
00773
00774 #ifdef UNIT_TEST
00775 /**
00776  * @brief Reset IWDT driver state for unit testing
00777  * @details
00778  * Resets the driver to uninitialized state. This function is only available
00779  * when UNIT_TEST is defined and should only be called from test code to
00780  * ensure test isolation.
00781  */
00782 void rx_iwdt_test_reset(void);
00783 #endif
00784
00785 #ifdef __cplusplus
00786 }
00787 #endif

```

7.21 libs/rx_core/inc/rx_log.h File Reference

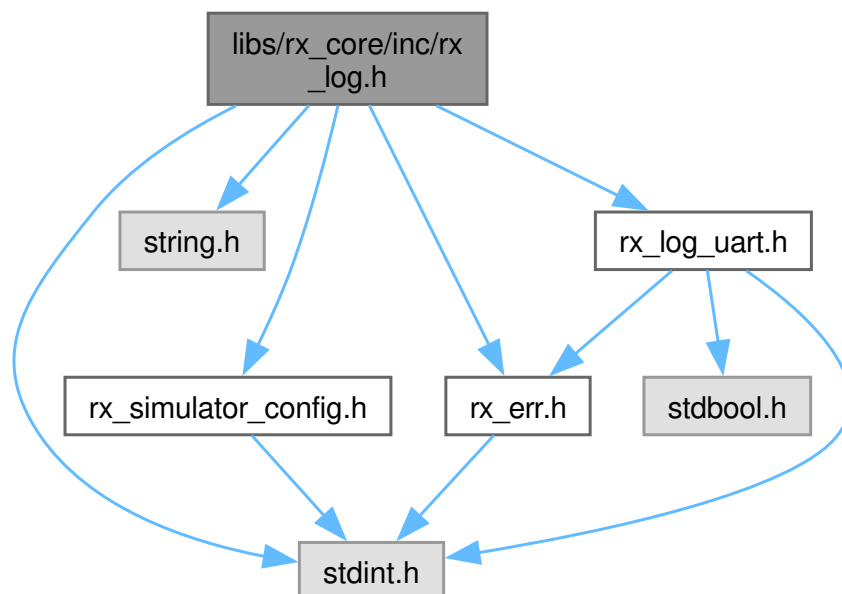
Type-Safe Logging System for RX72N Firmware with Zero-Overhead Compile-Time Filtering.

```

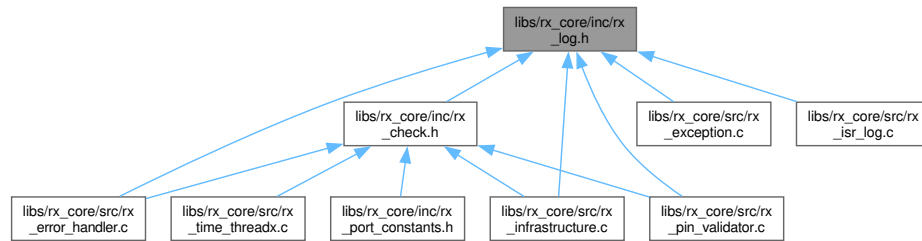
#include <stdint.h>
#include <string.h>
#include "rx_err.h"
#include "rx_simulator_config.h"
#include "rx_log_uart.h"

```

Include dependency graph for rx_log.h:



This graph shows which files directly or indirectly include this file:



Macros

- `#define LOG_PUTC(c)`
- `#define LOG_PUTS(s)`
- `#define LOG_PUTINT(v)`
- `#define LOG_PUTUINT(v)`
- `#define LOG_PUTHEX(v, d)`
- `#define LOG_LEVEL (k_log_info)`
Default log level configuration.
- `#define RX_LOG_VAL_IMPL(level, tag, msg, val)`
Internal macro implementing C23 _Generic automatic type dispatch for logging.
- `#define rx_log_error(tag, msg)`
Log critical ERROR-level message (compile-time filtered).
- `#define rx_log_error_val(tag, msg, val)`
Log ERROR-level message with automatic type-dispatched value.
- `#define rx_log_error_hex(tag, msg, val, digits)`
Log ERROR-level message with hexadecimal value.
- `#define rx_log_error_str(tag, msg, str, len)`
Log ERROR-level message with bounded string value (NASA-compliant).
- `#define rx_log_warn(tag, msg)`
Log warning message (no value).
- `#define rx_log_warn_val(tag, msg, val)`
Log warning with typed value (automatic type dispatch).
- `#define rx_log_warn_hex(tag, msg, val, digits)`
Log warning with hex value.
- `#define rx_log_warn_str(tag, msg, str, len)`
Log warning with string value (bounded length).
- `#define rx_log_info(tag, msg)`
Log info message (no value).
- `#define rx_log_info_val(tag, msg, val)`
Log info with typed value (automatic type dispatch).
- `#define rx_log_info_hex(tag, msg, val, digits)`
Log info with hex value.
- `#define rx_log_info_str(tag, msg, str, len)`
Log info with string value (bounded length).
- `#define rx_log_debug(tag, msg)`
Log debug message (no value).
- `#define rx_log_debug_val(tag, msg, val)`
Log debug with typed value (automatic type dispatch).
- `#define rx_log_debug_hex(tag, msg, val, digits)`
Log debug with hex value.
- `#define rx_log_debug_str(tag, msg, str, len)`
Log debug with string value (bounded length).

- `#define rx_log_verbose(tag, msg)`
Log verbose message (no value).
- `#define rx_log_verbose_val(tag, msg, val)`
Log verbose with typed value (automatic type dispatch).
- `#define rx_log_verbose_hex(tag, msg, val, digits)`
Log verbose with hex value.
- `#define rx_log_verbose_str(tag, msg, str, len)`
Log verbose with string value (bounded length).

Enumerations

- enum `log_level_t` : `uint8_t` {
 `k_log_none` = 0 , `k_log_error` , `k_log_warn` = 2 , `k_log_info` ,
 `k_log_debug` , `k_log_verbose` }
Log level enumeration defining severity hierarchy for compile-time filtering.
- enum `log_format_constants_t` : `uint16_t` { `k_log_hex_width_err` , `k_log_str_max_len` }
Logging formatting and safety constants.

Functions

- void `uart_debug_putc` (char data)
- void `uart_debug_puts` (const char *str)
- void `uart_debug_putint` (int32_t value)
- void `uart_debug_putuint` (uint32_t value)
- void `uart_debug_puthex` (uint32_t value, uint8_t digits)
- static void `internal_log_putc` (char c)
Write one character to the active log backend(s).
- static void `internal_log_puts` (const char *s)
Write a string to the active log backend(s).
- static void `internal_log_putint` (int32_t v)
Write a signed 32-bit integer to the active log backend(s).
- static void `internal_log_puthex` (uint32_t v, uint8_t d)
Write a hex value to the active log backend(s).
- static void `internal_log_header` (const char *level_str, const char *tag)
Internal helper to format and output log message header.
- static void `internal_rx_log_error` (const char *tag, const char *message)
Log ERROR-level message without additional value.
- static void `internal_rx_log_error_u8` (const char *tag, const char *message, uint8_t value)
Log error with uint8_t value.
- static void `internal_rx_log_error_u16` (const char *tag, const char *message, uint16_t value)
Log error with uint16_t value.
- static void `internal_rx_log_error_u32` (const char *tag, const char *message, uint32_t value)
Log error with uint32_t value.
- static void `internal_rx_log_error_i32` (const char *tag, const char *message, int32_t value)
Log error with int32_t value.
- static void `internal_rx_log_error_err` (const char *tag, const char *message, `rx_err_t` err)
Log error with error code (`rx_err_t`).
- static void `internal_rx_log_error_hex` (const char *tag, const char *message, uint32_t value, uint8_t digits)
Log error with hex value.

- static void [internal_rx_log_error_str](#) (const char *tag, const char *message, const char *str_value, uint32_t len)
Log ERROR-level message with bounded string value (NASA-compliant).
- static void [internal_rx_log_warn](#) (const char *tag, const char *message)
Log warning message (no value).
- static void [internal_rx_log_warn_u8](#) (const char *tag, const char *message, uint8_t value)
Log warning with uint8_t value.
- static void [internal_rx_log_warn_u16](#) (const char *tag, const char *message, uint16_t value)
Log warning with uint16_t value.
- static void [internal_rx_log_warn_u32](#) (const char *tag, const char *message, uint32_t value)
Log warning with uint32_t value.
- static void [internal_rx_log_warn_i32](#) (const char *tag, const char *message, int32_t value)
Log warning with int32_t value.
- static void [internal_rx_log_warn_err](#) (const char *tag, const char *message, [rx_err_t](#) err)
Log warning with error code ([rx_err_t](#)).
- static void [internal_rx_log_warn_hex](#) (const char *tag, const char *message, uint32_t value, uint8_t digits)
Log warning with hex value.
- static void [internal_rx_log_warn_str](#) (const char *tag, const char *message, const char *str_value, uint32_t len)
Log warning with string value (bounded length).
- static void [internal_rx_log_info](#) (const char *tag, const char *message)
Log info message (no value).
- static void [internal_rx_log_info_u8](#) (const char *tag, const char *message, uint8_t value)
Log info with uint8_t value.
- static void [internal_rx_log_info_u16](#) (const char *tag, const char *message, uint16_t value)
Log info with uint16_t value.
- static void [internal_rx_log_info_u32](#) (const char *tag, const char *message, uint32_t value)
Log info with uint32_t value.
- static void [internal_rx_log_info_i32](#) (const char *tag, const char *message, int32_t value)
Log info with int32_t value.
- static void [internal_rx_log_info_err](#) (const char *tag, const char *message, [rx_err_t](#) err)
Log info with error code ([rx_err_t](#)).
- static void [internal_rx_log_info_hex](#) (const char *tag, const char *message, uint32_t value, uint8_t digits)
Log info with hex value.
- static void [internal_rx_log_info_str](#) (const char *tag, const char *message, const char *str_value, uint32_t len)
Log info with string value (bounded length).
- static void [internal_rx_log_debug](#) (const char *tag, const char *message)
Log debug message (no value).
- static void [internal_rx_log_debug_u8](#) (const char *tag, const char *message, uint8_t value)
Log debug with uint8_t value.
- static void [internal_rx_log_debug_u16](#) (const char *tag, const char *message, uint16_t value)
Log debug with uint16_t value.
- static void [internal_rx_log_debug_u32](#) (const char *tag, const char *message, uint32_t value)
Log debug with uint32_t value.
- static void [internal_rx_log_debug_i32](#) (const char *tag, const char *message, int32_t value)
Log debug with int32_t value.
- static void [internal_rx_log_debug_err](#) (const char *tag, const char *message, [rx_err_t](#) err)
Log debug with error code ([rx_err_t](#)).

- static void [internal_rx_log_debug_hex](#) (const char *tag, const char *message, uint32_t value, uint8_t digits)
Log debug with hex value.
- static void [internal_rx_log_debug_str](#) (const char *tag, const char *message, const char *str_value, uint32_t len)
Log debug with string value (bounded length).
- static void [internal_rx_log_verbose](#) (const char *tag, const char *message)
Log verbose message (no value).
- static void [internal_rx_log_verbose_u8](#) (const char *tag, const char *message, uint8_t value)
Log verbose with uint8_t value.
- static void [internal_rx_log_verbose_u16](#) (const char *tag, const char *message, uint16_t value)
Log verbose with uint16_t value.
- static void [internal_rx_log_verbose_u32](#) (const char *tag, const char *message, uint32_t value)
Log verbose with uint32_t value.
- static void [internal_rx_log_verbose_i32](#) (const char *tag, const char *message, int32_t value)
Log verbose with int32_t value.
- static void [internal_rx_log_verbose_err](#) (const char *tag, const char *message, rx_err_t err)
Log verbose with error code ([rx_err_t](#)).
- static void [internal_rx_log_verbose_hex](#) (const char *tag, const char *message, uint32_t value, uint8_t digits)
Log verbose with hex value.
- static void [internal_rx_log_verbose_str](#) (const char *tag, const char *message, const char *str_value, uint32_t len)
Log verbose with string value (bounded length).

7.21.1 Detailed Description

Type-Safe Logging System for RX72N Firmware with Zero-Overhead Compile-Time Filtering.

7.21.1.1 Overview

Production-quality logging infrastructure for safety-critical RX72N motor controller firmware. Provides type-safe, compile-time filtered logging with zero overhead for disabled log levels. Uses C23 `_Generic` for automatic type dispatch and bounded string handling for memory safety.

7.21.1.2 Architecture Design

7.21.1.2.1 Core Features

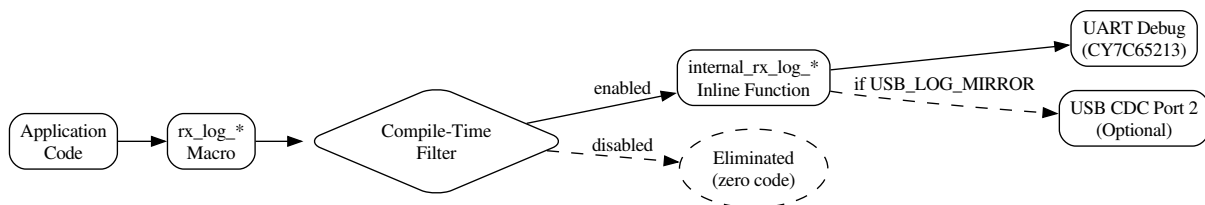
Feature	Implementation	Benefit
Zero-overhead filtering	Compile-time <code>#if (LOG_LEVEL >= ...)</code>	Disabled logs = zero code size, zero CPU cycles
Type-safe dispatch	C23 <code>_Generic</code> selection expressions	Automatic type routing, no manual suffixes
Bounded strings	Explicit length parameters + compile-time max	Prevents buffer overruns (NASA Rule 3)

Feature	Implementation	Benefit
Inline functions	All implementations are static inline	No function call overhead, optimal performance
Dual output	Optional USB CDC mirroring	View logs on UART or USB terminal
Consistent format	[LEVEL] [TAG] message pattern	Easy parsing by log analysis tools

7.21.1.2.2 Log Level Hierarchy

`k_log_none` (0) - No logging (production builds)
 v
`k_log_error` (1) - Critical failures only
 v
`k_log_warn` (2) - Recoverable issues
 v
`k_log_info` (3) - Informational events (default)
 v
`k_log_debug` (4) - Development debugging
 v
`k_log_verbose` (5) - Trace-level detail

7.21.1.2.3 Output Routing



7.21.1.3 Performance Characteristics

7.21.1.3.1 Execution Timing (RX72N @ 240 MHz)

Log Type	Cycles	Time	Bytes (UART 115200)
Disabled log	0	0 ns	0
<code>rx_log_error()</code>	~2400	10 us	~30 bytes
<code>rx_log_error_val(uint8_t)</code>	~3600	15 us	~35 bytes
<code>rx_log_error_hex()</code>	~4200	17.5 us	~40 bytes
<code>rx_log_error_str(64 chars)</code>	~18000	75 us	~95 bytes

7.21.1.3.2 Memory Footprint

Component	Flash	RAM	Stack
All inline functions	0 (inlined)	0	0
Per log site (enabled)	~60-150 bytes	0	8-16 bytes
Per log site (disabled)	0 bytes	0	0

Component	Flash	RAM	Stack
USB CDC buffer (optional)	0	512 bytes	0

7.21.1.3.3 Thread Safety

- NOT thread-safe: UART/USB output functions are not atomic
- Interleaving possible: Concurrent logs from multiple threads may mix
- Mitigation: Use RTOS mutex around critical log sequences if needed
- Design rationale: Performance over thread safety (motor control is time-critical)

7.21.1.4 C23 __Generic Type Dispatch

The `rx_log_*_val()` macros use C23 __Generic selection expressions to automatically route values to the correct typed implementation:

```
__Generic((val),
  uint8_t: internal_rx_log_error_u8,
  uint16_t: internal_rx_log_error_u16,
  uint32_t: internal_rx_log_error_u32,
  int32_t: internal_rx_log_error_err // rx_err_t is int32_t
)(tag, msg, val)
```

Type Limitation: Because `rx_err_t` is typedef `int32_t`, __Generic cannot distinguish between error codes and raw signed integers. Both route to hex formatting. Use explicit `rx_log*_hex()` or cast to `uint32_t` for decimal formatting of signed integers.

7.21.1.5 USB Log Mirroring

Enable with `-DUSB_LOG_MIRROR=1` to send logs to both:

- UART: `/dev/ttyACM0` via CY7C65213 (115200 baud, CDC-ACM class)
- USB CDC Port 2: `/dev/ttyACM2` (log port)

Port assignments:

Port	Enum	Purpose
0	<code>k_usb_port_proto</code>	Binary protocol (nanopb frames)
1	<code>k_usb_port_decoded</code>	Decoded ASCII frame dumps
2	<code>k_usb_port_log</code>	Log output (mirrored here)

7.21.1.6 Configuration

7.21.1.6.1 Compile-Time Flags

```
# Set log level (default: k_log_info in release, k_log_debug in DEBUG)
-DLOG_LEVEL=k_log_error

# Disable USB mirroring (default: 1, enabled by default)
-DUSB_LOG_MIRROR=0
```

Note

`USB_LOG_MIRROR=1` is enabled by default (Issue #161: USB CDC debug output). All logs appear on BOTH UART and USB CDC Port 2. Override with `-DUSB_LOG_MIRROR=0` to disable.

7.21.1.7 Usage Examples

7.21.1.7.1 Basic Logging

```
#include "rx_log.h"

void motor_init(void) {
    rx_log_info("MOTOR", "Starting initialization");

    rx_err_t err = configure_pwm();
    if (err != k_rx_ok) {
        rx_log_error_val("MOTOR", "PWM config failed", err); // Hex error code
        return;
    }

    rx_log_info("MOTOR", "Initialization complete");
}
```

7.21.1.7.2 Type Dispatch Examples

```
// Automatic type routing via __Generic
uint8_t port = 5;
rx_log_debug_val("GPIO", "Port number", port); // -> __u8 (decimal)

uint16_t rpm = 2100;
rx_log_info_val("MOTOR", "Current RPM", rpm); // -> __u16 (decimal)

uint32_t timestamp = get_ticks();
rx_log_verbose_val("TIMER", "Timestamp", timestamp); // -> __u32 (decimal)

rx_err_t err = k_rx_err_timeout;
rx_log_error_val("SPI", "Transfer error", err); // -> __err (hex: 0x00000108)
```

7.21.1.7.3 Hex and String Logging

```
// Hex values (explicit digit count)
uint32_t reg = read_register(0x40000000);
rx_log_debug_hex("HAL", "Register value", reg, 8); // 8 hex digits

// Bounded string logging (safe)
const char* device_name = "RX72N Motor Controller";
rx_log_info_str("SYSTEM", "Device", device_name, strlen(device_name));
```

7.21.1.7.4 Conditional Compilation (Zero Overhead)

```
// If LOG_LEVEL < k_log_debug, this compiles to nothing (zero code)
rx_log_debug("TRACE", "Entering critical section"); // -> ((void)0) if disabled
```

7.21.1.8 Design Rationale

7.21.1.8.1 Why Compile-Time Filtering?

- Zero overhead: Disabled logs generate no code (vs. runtime checks)
- Flash savings: Production builds omit all debug/verbose logs
- Performance: No branching or function call overhead

7.21.1.8.2 Why __Generic Instead of Function Overloading?

- Type safety: Compiler enforces correct types, catches errors at compile time
- No C++: Pure C implementation (RX72N firmware is C-only)
- Clean API: Single macro name, automatic dispatch

7.21.1.8.3 Why Bounded Strings?

- NASA Rule 2: Statically provable loop bounds
- Safety: Prevents reading past buffer end (undefined behavior)
- Predictable timing: Maximum execution time is known

7.21.1.9 Dependencies

- [rx_err.h](#) - Error code definitions ([rx_err_t](#))
- [uart.h](#) (implicit) - [uart_debug_*](#) functions for output
- [rx_usb.h](#) (optional) - USB CDC output (if `USB_LOG_MIRROR=1`)

NASA Power of 10 Compliance:

- Rule 1 [YES]: No goto, setjmp, recursion (pure inline functions and macros)
- Rule 2 [YES]: All loops bounded (string logging uses `k_log_str_max_len` constant)
- Rule 3 [YES]: No dynamic allocation (all stack-based, inline functions)
- Rule 4 [YES]: All functions <60 lines (inline helpers are 5-20 lines each)
- **Rule 5 [YES]**: Minimum 2 assertions per function (nullptr validation in `__str` functions)
- **Rule 6 [YES]**: Data at smallest scope (loop counters, temporary variables)
- **Rule 7 [YES]**: N/A for logging (output functions return void, errors silently dropped)
- **Rule 8 [YES]**: C23 typed enums for constants, macros only for type dispatch and filtering
- **Rule 9 [WARN]**: Single-level dereferencing only (string buffers, function pointers avoided)
- **Rule 10 [YES]**: Compiles with `-Wall -Wextra -Werror`

SOLID Principles:

- **S (Single Responsibility)**: Only logging output (no error handling, no formatting logic beyond basic types)
- **O (Open/Closed)**: Extensible via `USB_LOG_MIRROR` flag, new log levels can be added
- **L (Liskov Substitution)**: All internal `rx_log_*` functions have identical signatures per level
- **I (Interface Segregation)**: Separate macros for different value types (val, hex, str)
- **D (Dependency Inversion)**: Depends on abstract `uart_debug_*` interface (not concrete UART driver)

Module Dependencies:

```
rx_log.h
+--> rx_err.h (error code definitions)
+--> uart.h (uart_debug_* output functions)
+--> rx_usb.h (optional, if USB_LOG_MIRROR=1)
```

Author

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Date

2026-01-27

Copyright

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See also

[uart.h](#) for UART debug output implementation

[rx_err.h](#) for error code definitions

[rx_usb.h](#) for USB CDC output

Definition in file [rx_log.h](#).

7.21.2 Macro Definition Documentation

7.21.2.1 LOG_LEVEL

```
#define LOG_LEVEL (k\_log\_info)
```

Default log level configuration.

Can be overridden at compile time: -DLOG_LEVEL=([k_log_error](#))

Definition at line [650](#) of file [rx_log.h](#).

7.21.2.2 LOG_PUTC

```
#define LOG_PUTC(  
    c)
```

Value:

```
rx_log_uart_putc((char)(c))
```

Definition at line [498](#) of file [rx_log.h](#).

Referenced by [internal_log_putc\(\)](#).

7.21.2.3 LOG_PUTHEX

```
#define LOG_PUTHEX(  
    v,  
    d)
```

Value:

```
rx_log_uart_puthex((uint32_t)(v), (uint8_t)(d))
```

Definition at line [502](#) of file [rx_log.h](#).

Referenced by [internal_log_puthex\(\)](#).

7.21.2.4 LOG_PUTINT

```
#define LOG_PUTINT(  
    v)
```

Value:

```
rx_log_uart_putint((int32_t)(v))
```

Definition at line 500 of file [rx_log.h](#).

Referenced by [internal_log_putint\(\)](#).

7.21.2.5 LOG_PUTS

```
#define LOG_PUTS(  
    s)
```

Value:

```
rx_log_uart_puts((s))
```

Definition at line 499 of file [rx_log.h](#).

Referenced by [internal_log_puts\(\)](#).

7.21.2.6 LOG_PUTUINT

```
#define LOG_PUTUINT(  
    v)
```

Value:

```
rx_log_uart_putuint((uint32_t)(v))
```

Definition at line 501 of file [rx_log.h](#).

7.21.2.7 rx_log_debug

```
#define rx_log_debug(  
    tag,  
    msg)
```

Value:

```
internal_rx_log_debug((tag), (msg))
```

Log debug message (no value).

Definition at line 2116 of file [rx_log.h](#).

Referenced by [impl_clear_all_reservations\(\)](#), [impl_release_pin\(\)](#), [impl_reserve_pin\(\)](#), and [internal_emit_record\(\)](#).

7.21.2.8 rx_log_debug_hex

```
#define rx_log_debug_hex(  
    tag,  
    msg,  
    val,  
    digits)
```

Value:

`internal_rx_log_debug_hex((tag), (msg), (val), (digits))`

Log debug with hex value.

Definition at line [2126](#) of file [rx_log.h](#).

```
02126 #define rx_log_debug_hex(tag, msg, val, digits)  
02127     internal_rx_log_debug_hex((tag), (msg), (val), (digits))
```

7.21.2.9 rx_log_debug_str

```
#define rx_log_debug_str(  
    tag,  
    msg,  
    str,  
    len)
```

Value:

`internal_rx_log_debug_str((tag), (msg), (str), (len))`

Log debug with string value (bounded length).

Definition at line [2132](#) of file [rx_log.h](#).

7.21.2.10 rx_log_debug_val

```
#define rx_log_debug_val(  
    tag,  
    msg,  
    val)
```

Value:

`RX_LOG_VAL_IMPL(debug, (tag), (msg), (val))`

Log debug with typed value (automatic type dispatch).

Definition at line [2121](#) of file [rx_log.h](#).

Referenced by [internal_emit_record\(\)](#).

7.21.2.11 rx_log_error

```
#define rx_log_error(  
    tag,  
    msg)
```

Value:

```
internal_rx_log_error((tag), (msg))
```

Log critical ERROR-level message (compile-time filtered).

Public API for logging critical errors that prevent operation. This macro includes compile-time filtering: if `LOG_LEVEL < k_log_error`, the macro expands to `((void)0)` and generates zero code (no CPU cycles, no flash memory).

7.21.2.12 Compile-Time Behavior

LOG_LEVEL Setting	Macro Expansion	Code Generated
k_log_none	<code>((void)0)</code>	0 bytes
k_log_error or higher	<code>internal_rx_log_error(tag, msg)</code>	~60-80 bytes

7.21.2.13 When to Use

Use for critical failures that:

- Prevent system operation
- Require immediate attention
- Indicate hardware faults
- Signal initialization failures

Parameters

in	tag	Component tag (string literal, e.g., "MOTOR", "SPI", "USB") • Identifies error source • Recommended: 3-8 uppercase characters
in	msg	Error message (string literal or const char*) • Should be concise and actionable • Describes the specific failure

Returns

void (macro expands to function call or no-op)

Note

Zero Overhead: If `LOG_LEVEL < k_log_error`, this compiles to nothing.

Thread Safety: NOT thread-safe. Concurrent calls may interleave output.

Example: Critical Hardware Failure

```
#include "rx_log.h"

void motor_init(void) {
    if (!configure_pwm()) {
        rx_log_error("MOTOR", "PWM configuration failed");
        return;
    }

    if (!enable_h_bridge()) {
        rx_log_error("MOTOR", "H-bridge enable failed");
        return;
    }

    rx_log_info("MOTOR", "Initialization complete");
}

// Output (if LOG_LEVEL >= k_log_error):
// [ERROR] [MOTOR] PWM configuration failed
```

Example: Initialization Sequence

```
void system_init(void) {
    rx_log_info("SYSTEM", "Starting initialization");

    if (!uart_init()) {
        rx_log_error("UART", "Failed to initialize");
        RX_ERROR_CHECK("UART", "Cannot continue without UART", k_rx_err_init);
    }

    if (!spi_init()) {
        rx_log_error("SPI", "Failed to initialize");
        // Continue anyway - SPI is optional
    }
}
```

Example: Compile-Time Elimination

```
// Build configuration: -DLOG_LEVEL=k_log_none

rx_log_error("MOTOR", "Critical failure"); // Compiles to: ((void)0)

// Result: Zero bytes of code generated, zero CPU cycles, zero flash usage
```

See also

- [internal_rx_log_error\(\)](#) Implementation function
- [rx_log_error_val\(\)](#) Error logging with typed value
- [rx_log_error_hex\(\)](#) Error logging with hex value
- [rx_log_error_str\(\)](#) Error logging with string value
- [LOG_LEVEL](#) Compile-time log level configuration

Definition at line 1728 of file [rx_log.h](#).

Referenced by [error_handler_get_interface\(\)](#), [error_handler_init\(\)](#), [impl_clear_errors\(\)](#), [impl_report_error\(\)](#), [impl_reset_retry_counter\(\)](#), [internal_emit_record\(\)](#), [internal_process_exception\(\)](#), [pin_validator_get_interface\(\)](#), [rx_exc_nmi_c_handler\(\)](#), and [rx_infrastructure_init\(\)](#).

7.21.2.14 rx_log_error_hex

```
#define rx_log_error_hex(  
    tag,  
    msg,  
    val,  
    digits)
```

Value:

```
internal_rx_log_error_hex((tag), (msg), (val), (digits))
```

Log ERROR-level message with hexadecimal value.

Public API for logging critical errors with associated value displayed as hexadecimal. Useful for register values, memory addresses, bit masks, and other hardware-related data. Includes compile-time filtering for zero overhead when disabled.

7.21.2.15 Output Format

```
[ERROR] [TAG] message: 0xHHHHHHHHH
```

Leading zeros are included up to the specified digit count.

7.21.2.16 Compile-Time Behavior

LOG_LEVEL Setting	Macro Expansion	Code Generated
k_log_none	((void)0)	0 bytes
k_log_error or higher	internal_rx_log_error_hex(...)	~75-95 bytes

Parameters

in	tag	Component tag (string literal, e.g., "GPIO", "SPI", "REG")
in	msg	Error message (string literal or const char*)
in	val	Value to display as hexadecimal (uint32_t) • Supports 0 to 0xFFFFFFFF • Upper bits ignored if digits < 8
in	digits	Number of hex digits to display (1-8) • Common values: 2 (byte), 4 (uint16_t), 8 (uint32_t) • Leading zeros included

Returns

void (macro expands to function call or no-op)

Note

Zero Overhead: If LOG_LEVEL < k_log_error, this compiles to nothing.

Thread Safety: NOT thread-safe. Concurrent calls may interleave output.

Leading Zeros: Always includes leading zeros up to specified digit count.

Example: Hardware Register Debugging

```
#include "rx_log.h"

void configure_gpio(void) {
    uint32_t reg_val = read_register(GPIO_BASE + 0x04);

    if ((reg_val & 0x80000000) == 0) {
        rx_log_error_hex("GPIO", "Enable bit not set", reg_val, 8);
        // Output: [ERROR] [GPIO] Enable bit not set: 0x00001234
    }
}
```

Example: SPI Transaction Failure

```
void spi_transfer(uint8_t cmd, uint8_t data) {
    uint16_t status = spi_transact(cmd, data);

    if (status & 0x8000) { // Error bit set
        rx_log_error_hex("SPI", "Transfer failed, status", status, 4);
        // Output: [ERROR] [SPI] Transfer failed, status: 0x8001
    }
}
```

Example: Memory Address Logging

```
void* allocate_buffer(uint32_t size) {
    void* ptr = malloc(size);
    if (ptr == nullptr) {
        rx_log_error_hex("MEM", "Allocation failed at", (uint32_t)ptr, 8);
        // Output: [ERROR] [MEM] Allocation failed at: 0x00000000
    }
    return ptr;
}
```

Example: Different Digit Counts

```
uint8_t byte_val = 0x5A;
rx_log_error_hex("TEST", "Byte", byte_val, 2);
// Output: [ERROR] [TEST] Byte: 0x5A

uint16_t word_val = 0x1234;
rx_log_error_hex("TEST", "Word", word_val, 4);
// Output: [ERROR] [TEST] Word: 0x1234

uint32_t dword_val = 0xDEADBEEF;
rx_log_error_hex("TEST", "DWord", dword_val, 8);
// Output: [ERROR] [TEST] DWord: 0xDEADBEEF
```

See also

[internal_rx_log_error_hex\(\)](#) Implementation function

[rx_log_error\(\)](#) Simple error logging without value

[rx_log_error_val\(\)](#) Error logging with typed value (auto-dispatch)

Definition at line 1923 of file [rx_log.h](#).

```
01923 #define rx_log_error_hex(tag, msg, val, digits) \
01924     internal_rx_log_error_hex((tag), (msg), (val), (digits))
```

7.21.2.17 rx_log_error_str

```
#define rx_log_error_str(
    tag,
    msg,
    str,
    len)
```

Value:

```
internal_rx_log_error_str((tag), (msg), (str), (len))
```

Log ERROR-level message with bounded string value (NASA-compliant).

Public API for logging critical errors with associated string value. Uses bounded iteration for memory safety (NASA Power of 10 Rule 2 compliance). String output is limited to k_log_str_max_len (256 bytes) to prevent unbounded loops.

7.21.2.18 Output Format

```
[ERROR] [TAG] message: string_value
```

7.21.2.19 Safety Features

- Bounded loop: Maximum 256 characters logged (k_log_str_max_len)
- Early termination: Stops at user-provided len or null terminator
- NULL safe: NULL pointers silently ignored (no crash)

7.21.2.20 Compile-Time Behavior

LOG_LEVEL Setting	Macro Expansion	Code Generated
k_log_none	((void)0)	0 bytes
k_log_error or higher	internal_rx_log_error_str(...)	~100-120 bytes

Parameters

in	tag	Component tag (string literal, e.g., "CONFIG", "USB", "FILE")
in	msg	Error message (string literal or const char*)
in	str	String value to log (const char*) • Must point to valid memory • Can be null-terminated or explicit length • nullptr silently ignored

in	len	Maximum number of characters to log - Typically obtained via strlen(str) - Values > 256 clamped to 256 - Must not exceed actual buffer size
----	-----	--

Returns

void (macro expands to function call or no-op)

Note

Zero Overhead: If LOG_LEVEL < k_log_error, this compiles to nothing.

Bounded: Maximum 256 characters logged regardless of len parameter.

Thread Safety: NOT thread-safe. Concurrent calls may interleave output.

NULL Safe: NULL str pointer silently ignored (safe failure mode).

Warning

Ensure len <= actual buffer size to prevent undefined behavior.

Example: Configuration Errors

```
#include "rx_log.h"
#include <string.h>

void parse_config(const char* param_name, const char* param_value) {
    if (strcmp(param_name, "motor_id") == 0) {
        // Valid parameter
    } else {
        // Unknown parameter name
        rx_log_error_str("CONFIG", "Unknown parameter", param_name, strlen(param_name));
        // Output: [ERROR] [CONFIG] Unknown parameter: invalid_param_name
    }
}
```

Example: File Path Errors

```
void load_config_file(const char* filepath) {
    if (!file_exists(filepath)) {
        rx_log_error_str("FILE", "Config file not found", filepath, strlen(filepath));
        // Output: [ERROR] [FILE] Config file not found: /etc/motor_config.ini
    }
}
```

Example: USB Device Name Logging

```
void usb_device_error(const char* device_name, uint32_t name_len) {
    rx_log_error_str("USB", "Device enumeration failed", device_name, name_len);
    // Output: [ERROR] [USB] Device enumeration failed: RX72N Motor Controller
}
```

Example: Truncation Behavior

```
// String longer than k_log_str_max_len (256)
char very_long_string[300];
memset(very_long_string, 'A', 299);
very_long_string[299] = '\0';

rx_log_error_str("TEST", "Long string", very_long_string, strlen(very_long_string));
// Output: [ERROR] [TEST] Long string: AAAA... (truncated at 256 chars)
```

See also

[internal_rx_log_error_str\(\)](#) Implementation function
[k_log_str_max_len](#) Maximum string length constant (256)
[rx_log_error\(\)](#) Simple error logging without value
[rx_log_error_val\(\)](#) Error logging with typed value

Definition at line 2021 of file [rx_log.h](#).

7.21.2.21 rx_log_error_val

```
#define rx_log_error_val(  
    tag,  
    msg,  
    val)
```

Value:

`RX_LOG_VAL_IMPL(error, (tag), (msg), (val))`

Log ERROR-level message with automatic type-dispatched value.

Public API for logging critical errors with an associated value. Uses C23 `_Generic` for automatic type dispatch to the correct formatting function. Includes compile-time filtering for zero overhead when disabled.

7.21.2.22 Type Dispatch

The value's type determines output formatting at compile time:

Type	Dispatches To	Format	Example
<code>uint8_t</code>	<code>internal_rx_log_error_u8</code>	Decimal	42
<code>uint16_t</code>	<code>internal_rx_log_error_u16</code>	Decimal	2048
<code>uint32_t</code>	<code>internal_rx_log_error_u32</code>	Decimal	4294967295
<code>int32_t</code> (rx_err_t)	<code>internal_rx_log_error_err</code>	Hex (8 digits)	0x00000108

7.21.2.23 Compile-Time Behavior

LOG_LEVEL Setting	Macro Expansion	Code Generated
k_log_none	((void)0)	0 bytes
k_log_error or higher	RX_LOG_VAL_IMPL(...) -> typed function	~70-90 bytes

Parameters

in	tag	Component tag (string literal, e.g., "MOTOR", "SPI", "I2C")
in	msg	Error message (string literal or const char*)
in	val	Value to log - type determines formatting (uint8_t, uint16_t, uint32_t, int32_t)

Returns

void (macro expands to type-dispatched function call or no-op)

Note

Zero Overhead: If LOG_LEVEL < k_log_error, this compiles to nothing.

Type Safe: Compiler enforces supported types at compile time.

Thread Safety: NOT thread-safe. Concurrent calls may interleave output.

Warning

int32_t Limitation: Raw int32_t values format as hex (for [rx_err_t](#)). Cast to uint32_t for decimal display of signed integers.

Example: Error Codes

```
#include "rx_log.h"
#include "rx_err.h"

rx_err_t motor_init(void) {
    rx_err_t err = configure_pwm();
    if (err != k_rx_ok) {
        rx_log_error_val("MOTOR", "PWM config failed", err); // Auto -> hex
        return err;
    }

    err = calibrate_encoder();
    if (err != k_rx_ok) {
        rx_log_error_val("MOTOR", "Encoder calibration failed", err); // Auto -> hex
        return err;
    }

    return k_rx_ok;
}

// Output (if err = k_rx_err_timeout = 0x108):
// [ERROR] [MOTOR] PWM config failed: 0x00000108
```

Example: Typed Values (Automatic Dispatch)

```
// uint8_t -> decimal
uint8_t motor_id = 3;
rx_log_error_val("MOTOR", "Invalid motor ID", motor_id);
// Output: [ERROR] [MOTOR] Invalid motor ID: 3

// uint16_t -> decimal
uint16_t rpm = 4500;
rx_log_error_val("MOTOR", "RPM out of range", rpm);
// Output: [ERROR] [MOTOR] RPM out of range: 4500

// uint32_t -> decimal
uint32_t timestamp = get_tick_count();
rx_log_error_val("TIMER", "Overflow at", timestamp);
// Output: [ERROR] [TIMER] Overflow at: 4294967295

// rx_err_t (int32_t) -> hex
rx_err_t err = k_rx_err_hw_fault;
rx_log_error_val("HAL", "Hardware fault", err);
// Output: [ERROR] [HAL] Hardware fault: 0x00000201
```

Example: Type Mismatch (Compile Error)

```
float velocity = 1.5f;
rx_log_error_val("MOTOR", "Invalid velocity", velocity);
// Compile error: no __Generic case for 'float'
// Use rx_log_error() with formatted string instead
```

See also

[RX_LOG_VAL_IMPL\(\)](#) __Generic type dispatch implementation
[internal_rx_log_error_u8\(\)](#), [internal_rx_log_error_u16\(\)](#), [internal_rx_log_error_u32\(\)](#)
[internal_rx_log_error_err\(\)](#) Error code formatting (hex)
[rx_log_error\(\)](#) Simple error logging without value

Definition at line 1828 of file [rx_log.h](#).

Referenced by [impl_report_error\(\)](#), and [internal_emit_record\(\)](#).

7.21.2.24 rx_log_info

```
#define rx_log_info(  
    tag,  
    msg)
```

Value:

```
internal_rx_log_info((tag), (msg))
```

Log info message (no value).

Definition at line 2079 of file [rx_log.h](#).

Referenced by [error_handler_deinit\(\)](#), [error_handler_init\(\)](#), [internal_emit_record\(\)](#), [pin_validator_deinit\(\)](#), [pin_validator_init\(\)](#), [rx_exception_init\(\)](#), [rx_infrastructure_deinit\(\)](#), and [rx_infrastructure_init\(\)](#).

7.21.2.25 rx_log_info_hex

```
#define rx_log_info_hex(  
    tag,  
    msg,  
    val,  
    digits)
```

Value:

```
internal_rx_log_info_hex((tag), (msg), (val), (digits))
```

Log info with hex value.

Definition at line 2089 of file [rx_log.h](#).

```
02089 #define rx_log_info_hex(tag, msg, val, digits) \br/>02090     internal_rx_log_info_hex((tag), (msg), (val), (digits))
```

7.21.2.26 rx_log_info_str

```
#define rx_log_info_str(  
    tag,  
    msg,  
    str,  
    len)
```

Value:

```
internal_rx_log_info_str((tag), (msg), (str), (len))
```

Log info with string value (bounded length).

Definition at line 2095 of file [rx_log.h](#).

7.21.2.27 rx_log_info_val

```
#define rx_log_info_val(  
    tag,  
    msg,  
    val)
```

Value:

```
RX_LOG_VAL_IMPL(info, (tag), (msg), (val))
```

Log info with typed value (automatic type dispatch).

Definition at line 2084 of file [rx_log.h](#).

Referenced by [internal_emit_record\(\)](#).

7.21.2.28 RX_LOG_VAL_IMPL

```
#define RX_LOG_VAL_IMPL(
    level,
    tag,
    msg,
    val)
```

Value:

```
_Generic((val),
    uint8_t: internal_rx_log_##level##_u8,
    uint16_t: internal_rx_log_##level##_u16,
    uint32_t: internal_rx_log_##level##_u32,
    int32_t: internal_rx_log_##level##_err)(tag, msg, val)
```

Internal macro implementing C23 `_Generic` automatic type dispatch for logging.

Uses C23 `_Generic` selection expressions to automatically route log calls to the correct typed implementation function based on the value's type at compile time. This provides type-safe logging without manual type suffixes.

7.21.2.29 Type Dispatch Mechanism

The `_Generic` keyword performs compile-time type matching:

```
_Generic((val),
    uint8_t: internal_rx_log_error_u8, // Match: uint8_t -> decimal
    uint16_t: internal_rx_log_error_u16, // Match: uint16_t -> decimal
    uint32_t: internal_rx_log_error_u32, // Match: uint32_t -> decimal
    int32_t: internal_rx_log_error_err // Match: int32_t -> hex (error codes)
)(tag, msg, val) // Call selected function
```

7.21.2.30 Supported Types

Type	Output Format	Example Value	Example Output
<code>uint8_t</code>	Decimal	127	127
<code>uint16_t</code>	Decimal	2048	2048
<code>uint32_t</code>	Decimal	4294967295	4294967295
<code>int32_t</code> (<code>rx_err_t</code>)	Hex (8 digits)	0x108	0x00000108

7.21.2.31 Type Limitation: `int32_t` vs `rx_err_t`

Critical Design Constraint: Because `rx_err_t` is defined as `typedef int32_t`, C's `_Generic` cannot distinguish between error codes and raw signed integers (both have type `int32_t`).

Consequence:

- [OK] `rx_err_t` values display correctly as hex error codes
- Raw `int32_t` values display as hex (unexpected for signed integers)

Workaround for logging signed integers:

```
int32_t temperature = -25; // Negative temperature

// WRONG: Will display as hex 0xFFFFFE7 (two's complement)
rx_log_info_val("SENSOR", "Temperature", temperature);

// CORRECT: Cast to unsigned for decimal display
rx_log_info_val("SENSOR", "Temperature", (uint32_t)temperature); // Displays: 4294967271

// ALTERNATIVE: Use explicit hex logging
rx_log_info_hex("SENSOR", "Temperature (raw)", temperature, 8); // Displays: 0xFFFFFE7
```

7.21.2.32 Macro Expansion Example

```
uint8_t status = 42;
rx_log_error_val("MOTOR", "Status", status);

// Step 1: Expands to RX_LOG_VAL_IMPL(error, "MOTOR", "Status", status)
// Step 2: __Generic matches uint8_t -> selects internal_rx_log_error_u8
// Step 3: Calls internal_rx_log_error_u8("MOTOR", "Status", 42)
// Output: [ERROR] [MOTOR] Status: 42
```

7.21.2.33 Performance

- Overhead: Zero. Type selection occurs at compile time (no runtime cost).
- Code Size: Identical to calling typed function directly.
- Compile Time: Negligible increase for type resolution.

7.21.2.34 Advantages Over Manual Type Suffixes

Approach	Example	Type Safety	Code Clarity
Manual suffixes	log_error_u8(tag, msg, val)	[X] Mismatch possible	[X] Verbose
__Generic dispatch	log_error_val(tag, msg, val)	[OK] Compile-time checked	[OK] Clean

Parameters

in	level	Log level name (error, warn, info, debug, verbose) - token pasting
in	tag	Component tag string (passed to selected function)
in	msg	Log message string (passed to selected function)
in	val	Value to log - type determines function selection at compile time

Note

Compile-Time: Type selection occurs during compilation, not at runtime.

Type Safety: Compiler enforces supported types; unsupported types cause errors.

Token Pasting: Uses `##` to construct function names dynamically.

Warning

Only supports `uint8_t`, `uint16_t`, `uint32_t`, and `int32_t` (`rx_err_t`). Passing other types (`float`, `int8_t`, etc.) will cause compilation errors.

Example: Type Mismatch Error

```
float velocity = 1.5f;
rx_log_error_val("MOTOR", "Velocity", velocity);
// Compile error: no __Generic case for 'float'
```

See also

[rx_log_error_val\(\)](#), [rx_log_warn_val\(\)](#), [rx_log_info_val\(\)](#)
[rx_log_debug_val\(\)](#), [rx_log_verbose_val\(\)](#)
[internal_rx_log_error_u8\(\)](#), [internal_rx_log_error_u16\(\)](#), [internal_rx_log_error_u32\(\)](#)
[internal_rx_log_error_err\(\)](#)

Definition at line [1626](#) of file [rx_log.h](#).

```
01626 #define RX_LOG_VAL_IMPL(level, tag, msg, val)
01627     _Generic((val),
01628         uint8_t: internal_rx_log_##level##_u8,
01629         uint16_t: internal_rx_log_##level##_u16,
01630         uint32_t: internal_rx_log_##level##_u32,
01631         int32_t: internal_rx_log_##level##_err)(tag, msg, val)
```

7.21.2.35 rx_log_verbose

```
#define rx_log_verbose(
    tag,
    msg)
```

Value:

`internal_rx_log_verbose((tag), (msg))`

Log verbose message (no value).

Definition at line [2153](#) of file [rx_log.h](#).

7.21.2.36 rx_log_verbose_hex

```
#define rx_log_verbose_hex(
    tag,
    msg,
    val,
    digits)
```

Value:

`internal_rx_log_verbose_hex((tag), (msg), (val), (digits))`

Log verbose with hex value.

Definition at line [2163](#) of file [rx_log.h](#).

```
02163 #define rx_log_verbose_hex(tag, msg, val, digits)
02164     internal_rx_log_verbose_hex((tag), (msg), (val), (digits))
```

7.21.2.37 rx_log_verbose_str

```
#define rx_log_verbose_str(
    tag,
    msg,
    str,
    len)
```

Value:

`internal_rx_log_verbose_str((tag), (msg), (str), (len))`

Log verbose with string value (bounded length).

Definition at line [2169](#) of file [rx_log.h](#).

```
02169 #define rx_log_verbose_str(tag, msg, str, len)
02170     internal_rx_log_verbose_str((tag), (msg), (str), (len))
```

7.21.2.38 rx_log_verbose_val

```
#define rx_log_verbose_val(  
    tag,  
    msg,  
    val)
```

Value:

`RX_LOG_VAL_IMPL(verbose, (tag), (msg), (val))`

Log verbose with typed value (automatic type dispatch).

Definition at line 2158 of file [rx_log.h](#).

7.21.2.39 rx_log_warn

```
#define rx_log_warn(  
    tag,  
    msg)
```

Value:

`internal_rx_log_warn((tag), (msg))`

Log warning message (no value).

Definition at line 2042 of file [rx_log.h](#).

Referenced by [impl_release_pin\(\)](#), [impl_reserve_pin\(\)](#), [internal_emit_record\(\)](#), [rx_infrastructure_deinit\(\)](#), and [rx_infrastructure_init\(\)](#).

7.21.2.40 rx_log_warn_hex

```
#define rx_log_warn_hex(  
    tag,  
    msg,  
    val,  
    digits)
```

Value:

`internal_rx_log_warn_hex((tag), (msg), (val), (digits))`

Log warning with hex value.

Definition at line 2052 of file [rx_log.h](#).

```
02052 #define rx_log_warn_hex(tag, msg, val, digits) \n02053     internal_rx_log_warn_hex((tag), (msg), (val), (digits))
```

7.21.2.41 rx_log_warn_str

```
#define rx_log_warn_str(  
    tag,  
    msg,  
    str,  
    len)
```

Value:

`internal_rx_log_warn_str((tag), (msg), (str), (len))`

Log warning with string value (bounded length).

Definition at line 2058 of file [rx_log.h](#).

7.21.2.42 rx_log_warn_val

```
#define rx_log_warn_val(  
    tag,  
    msg,  
    val)
```

Value:

`RX_LOG_VAL_IMPL(warn, (tag), (msg), (val))`

Log warning with typed value (automatic type dispatch).

Definition at line 2047 of file [rx_log.h](#).

Referenced by [internal_emit_record\(\)](#), and [rx_isr_log_drain\(\)](#).

7.21.3 Enumeration Type Documentation

7.21.3.1 log_format_constants_t

enum [log_format_constants_t](#) : uint16_t

Logging formatting and safety constants.

Defines compile-time constants for log output formatting and memory safety bounds. These constants ensure predictable behavior and prevent buffer overruns.

7.21.3.2 Safety Rationale

7.21.3.2.1 k_log_hex_width_err = 8

Purpose: Hex digit width for displaying [rx_err_t](#) error codes.

Rationale:

- [rx_err_t](#) is typedef int32_t (32 bits)
- 32 bits = 8 hex digits (4 bits per digit)
- Example: 0x00000108 ([k_rx_err_timeout](#))

Usage: [rx_log_error_val\(\)](#) with [rx_err_t](#) values formats as 8-digit hex.

7.21.3.2.2 k_log_str_max_len = 256

Purpose: Maximum string length for bounded string logging (NASA Rule 2 compliance).

Rationale:

- Safety: Prevents unbounded loops reading past buffer end
- Predictability: Maximum execution time is known (256 x ~15 cycles/char = 3840 cycles)
- Practicality: 256 bytes sufficient for typical log messages
- Stack impact: No stack allocation (loop counter is uint32_t = 4 bytes)

Behavior: String logging functions iterate up to `k_log_str_max_len` characters, stopping early if:

1. User-provided length len is reached, OR
2. Null terminator '\0' is encountered

Example:

```
const char* long_str = "This is a very long string..."; // > 256 chars
rx_log_info_str("TAG", "Long", long_str, 300); // Only logs first 256 chars
```

7.21.3.3 Performance Impact

Constant	CPU Impact	Memory Impact
k_log_hex_width_err	8 hex digits ~ 240 cycles @ 240 MHz	0 bytes (immediate)
k_log_str_max_len	Max 3840 cycles (256 x 15)	4 bytes (loop counter)

Underlying Type: uint16_t (C23 typed enum)

- Range: 0-65535
- Size: 2 bytes
- Rationale: k_log_str_max_len = 256 fits in uint16_t, uint8_t too small

See also

[internal_rx_log_error_str\(\)](#), [internal_rx_log_warn_str\(\)](#), [internal_rx_log_info_str\(\)](#)
[internal_rx_log_debug_str\(\)](#), [internal_rx_log_verbose_str\(\)](#)

Enumerator

k_log_hex_width_err	Hex digit width for <code>rx_err_t</code> error codes (32-bit = 8 digits). Example: 0x00000108
k_log_str_max_len	Maximum string length for bounded logging (NASA Rule 2 compliance). Prevents unbounded loops.

Definition at line 633 of file [rx_log.h](#).

```
00633     : uint16_t {
00634 k_log_hex_width_err =
00635     8, /**< Hex digit width for rx_err_t error codes (32-bit = 8 digits). Example: 0x00000108 */
00636 k_log_str_max_len =
00637     256, /**< Maximum string length for bounded logging (NASA Rule 2 compliance). Prevents unbounded loops. */
00638 } log_format_constants_t;
```

7.21.3.4 log_level_t

enum `log_level_t` : `uint8_t`

Log level enumeration defining severity hierarchy for compile-time filtering.

Defines the log level hierarchy used for compile-time filtering. Each level includes all messages from higher-priority (lower-numbered) levels. The `LOG_LEVEL` macro determines which levels are compiled into the binary.

7.21.3.5 Hierarchy (Inclusive)

Each level includes all messages from levels above it:

- `k_log_error` (1) -> ERROR only
- `k_log_warn` (2) -> ERROR + WARN
- `k_log_info` (3) -> ERROR + WARN + INFO
- `k_log_debug` (4) -> ERROR + WARN + INFO + DEBUG
- `k_log_verbose` (5) -> ERROR + WARN + INFO + DEBUG + VERBOSE

7.21.3.6 Recommended Usage

Build Type	Recommended Level	Rationale
Production	<code>k_log_error</code> or <code>k_log_none</code>	Minimal overhead, critical failures only
Field Testing	<code>k_log_warn</code>	Captures issues without excessive logging
Development	<code>k_log_info</code>	Standard development visibility (default)
Debugging	<code>k_log_debug</code>	Detailed troubleshooting
Trace Analysis	<code>k_log_verbose</code>	Maximum detail for complex issues

7.21.3.7 Compile-Time Behavior

```
// Example: LOG_LEVEL = k_log_warn
rx_log_error("TAG", "msg"); // [OK] Compiled (error <= warn)
rx_log_warn("TAG", "msg"); // [OK] Compiled (warn <= warn)
rx_log_info("TAG", "msg"); // [X] Eliminated (info > warn)
rx_log_debug("TAG", "msg"); // [X] Eliminated (debug > warn)
rx_log_verbose("TAG", "msg"); // [X] Eliminated (verbose > warn)
```

7.21.3.8 Configuration

Set at compile time via `-DLOG_LEVEL=<level>`:

```
cmake -DLOG_LEVEL=k_log_error .. # Production
cmake -DLOG_LEVEL=k_log_debug .. # Development
```

Default (if not specified):

- DEBUG builds: `k_log_debug`
- Release builds: `k_log_info`

Underlying Type: `uint8_t` (C23 typed enum)

- Range: 0-255 (only 0-5 used)
- Size: 1 byte
- Rationale: Smallest type sufficient for 6 levels, minimal memory footprint

See also

[LOG_LEVEL](#) Default log level configuration

[rx_log_error\(\)](#), [rx_log_warn\(\)](#), [rx_log_info\(\)](#), [rx_log_debug\(\)](#), [rx_log_verbose\(\)](#)

Enumerator

<code>k_log_none</code>	No logging (production silent mode). All logs eliminated at compile time.
<code>k_log_error</code>	Critical errors only. Operation-preventing failures, requires immediate attention.
<code>k_log_warn</code>	Errors and warnings. Recoverable issues, degraded functionality.
<code>k_log_info</code>	Errors, warnings, and informational messages. Default level for development.
<code>k_log_debug</code>	Errors, warnings, info, and debug messages. Detailed troubleshooting information.
<code>k_log_verbose</code>	All messages including verbose trace. Maximum detail for complex debugging.

Definition at line 566 of file [rx_log.h](#).

```

00566         : uint8_t {
00567     k_log_none = 0, /**< No logging (production silent mode). All logs eliminated at compile time. */
00568     k_log_error =
00569         1, /**< Critical errors only. Operation-preventing failures, requires immediate attention. */
00570     k_log_warn = 2, /**< Errors and warnings. Recoverable issues, degraded functionality. */
00571     k_log_info =
00572         3, /**< Errors, warnings, and informational messages. Default level for development. */
00573     k_log_debug =
00574         4, /**< Errors, warnings, info, and debug messages. Detailed troubleshooting information. */
00575     k_log_verbose =
00576         5, /**< All messages including verbose trace. Maximum detail for complex debugging. */
00577 } log_level_t;

```

7.21.4 Function Documentation

7.21.4.1 `internal_log_header()`

```

void internal_log_header (
    const char * level_str,
    const char * tag)    [inline], [static]

```

Internal helper to format and output log message header.

Outputs standardized log header in format: [LEVEL] [TAG] This prefix precedes all log messages for consistent parsing by log analysis tools.

7.21.4.2 Algorithm

1. Validate input pointers (NULL check for safety)
2. Output opening bracket [
3. Output log level string (e.g., "ERROR", "WARN", "INFO")
4. Output] [separator
5. Output component tag (e.g., "MOTOR", "SPI", "USB")
6. Output] (close bracket and space)

7.21.4.3 Output Format

```
[ERROR] [MOTOR] message content here\r\n
[WARN] [SPI] message content here\r\n
[INFO] [USB] message content here\r\n
```

7.21.4.4 Performance

- Cycles: ~600 cycles @ 240 MHz (depends on string lengths)
- Time: ~2.5 us
- UART bytes: ~15-25 bytes (depends on level_str and tag lengths)
- Stack: 0 bytes (inline, no local variables)

Parameters

in	level_str	Log level string (e.g., "ERROR", "WARN", "INFO", "DEBUG", "VERBOSE") • Must be null-terminated string literal • Typically 4-7 characters • nullptr silently ignored (safety)
in	tag	Component tag identifying log source (e.g., "MOTOR", "SPI", "I2C") • Must be null-terminated string literal • Recommended: 3-8 uppercase characters • nullptr silently ignored (safety)

Precondition

level_str and tag should be valid null-terminated strings

Postcondition

Log header written to configured output (UART and/or USB CDC)

Note

Thread Safety: NOT thread-safe. Concurrent calls may interleave output.

Inline: Always inlined for zero function call overhead.

Error Handling: NULL pointers silently ignored (safe failure mode).

Example:

```
// Called internally by all log functions
internal_log_header("ERROR", "MOTOR"); // Outputs: "[ERROR] [MOTOR] "
LOG_PUTS("PWM init failed\r\n");      // Complete message

// Result on terminal:
// [ERROR] [MOTOR] PWM init failed
```

See also

[LOG_PUTC\(\)](#), [LOG_PUTS\(\)](#) Output macros (UART or UART+USB)
[internal_rx_log_error\(\)](#), [internal_rx_log_warn\(\)](#), [internal_rx_log_info\(\)](#)

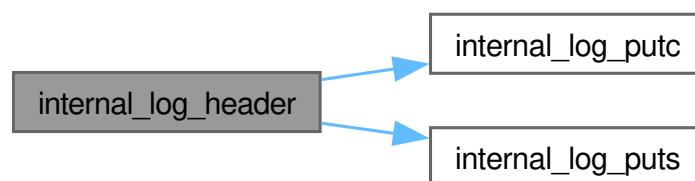
Definition at line 745 of file [rx_log.h](#).

```
00746 {
00747     /* Pre-condition: Validate pointers (NASA Power of 10 Rule 5).
00748     * The level_str side of the || is defensive-only: every caller passes a
00749     * compile-time string literal ("ERROR", "WARN", ...), so the TRUE branch is
00750     * architecturally unreachable. Excluded from gcovr branch coverage with an
00751     * EXCL marker so the 100% libs/ gate remains valid. */
00752     /* GCOVR_EXCL_BR_START */
00753     if (level_str == (const char*)0 || tag == (const char*)0) {
00754         return;
00755     }
00756     /* GCOVR_EXCL_BR_STOP */
00757
00758     internal_log_putc('[');
00759     internal_log_puts(level_str);
00760     internal_log_puts("] ");
00761     internal_log_puts(tag);
00762     internal_log_puts(" ");
00763 }
```

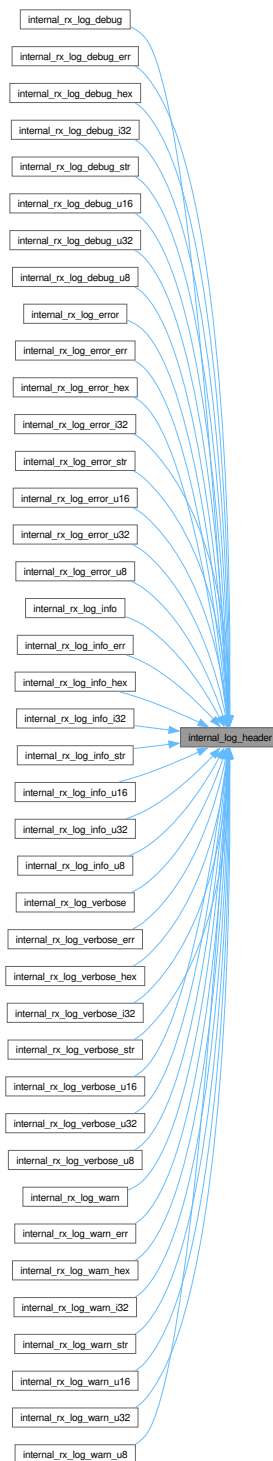
References [internal_log_putc\(\)](#), and [internal_log_puts\(\)](#).

Referenced by [internal_rx_log_debug\(\)](#), [internal_rx_log_debug_err\(\)](#), [internal_rx_log_debug_hex\(\)](#), [internal_rx_log_debug_i32\(\)](#), [internal_rx_log_debug_str\(\)](#), [internal_rx_log_debug_u16\(\)](#), [internal_rx_log_debug_u8\(\)](#), [internal_rx_log_error\(\)](#), [internal_rx_log_error_err\(\)](#), [internal_rx_log_error_hex\(\)](#), [internal_rx_log_error_i32\(\)](#), [internal_rx_log_error_str\(\)](#), [internal_rx_log_error_u16\(\)](#), [internal_rx_log_error_u32\(\)](#), [internal_rx_log_error_u8\(\)](#), [internal_rx_log_info\(\)](#), [internal_rx_log_info_err\(\)](#), [internal_rx_log_info_hex\(\)](#), [internal_rx_log_info_i32\(\)](#), [internal_rx_log_info_str\(\)](#), [internal_rx_log_info_u16\(\)](#), [internal_rx_log_info_u32\(\)](#), [internal_rx_log_info_u8\(\)](#), [internal_rx_log_verbose\(\)](#), [internal_rx_log_verbose_err\(\)](#), [internal_rx_log_verbose_hex\(\)](#), [internal_rx_log_verbose_i32\(\)](#), [internal_rx_log_verbose_str\(\)](#), [internal_rx_log_verbose_u16\(\)](#), [internal_rx_log_verbose_u32\(\)](#), [internal_rx_log_verbose_u8\(\)](#), [internal_rx_log_warn\(\)](#), [internal_rx_log_warn_err\(\)](#), [internal_rx_log_warn_hex\(\)](#), [internal_rx_log_warn_i32\(\)](#), [internal_rx_log_warn_str\(\)](#), [internal_rx_log_warn_u16\(\)](#), [internal_rx_log_warn_u32\(\)](#), and [internal_rx_log_warn_u8\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



7.21.4.5 internal_log_putc()

```
void internal_log_putc (
    char c) [inline], [static]
```

Write one character to the active log backend(s).

Parameters

in	c	Character to write.
----	---	---------------------

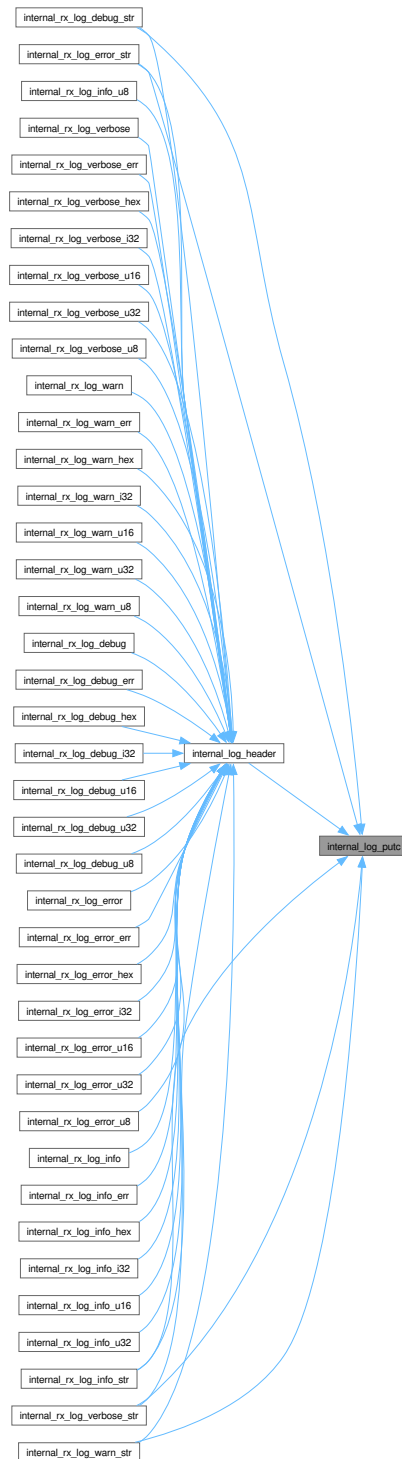
Definition at line 660 of file [rx_log.h](#).

```
00661 {  
00662   LOG_PUTC(c);  
00663 }
```

References [LOG_PUTC](#).

Referenced by [internal_log_header\(\)](#), [internal_rx_log_debug_str\(\)](#), [internal_rx_log_error_str\(\)](#), [internal_rx_log_info_str\(\)](#), [internal_rx_log_verbose_str\(\)](#), and [internal_rx_log_warn_str\(\)](#).

Here is the caller graph for this function:



7.21.4.6 internal_log_puthex()

```
void internal_log_puthex (
    uint32_t v,
    uint8_t d) [inline], [static]
```

Write a hex value to the active log backend(s).

Parameters

in	v	Value.
in	d	Digits.

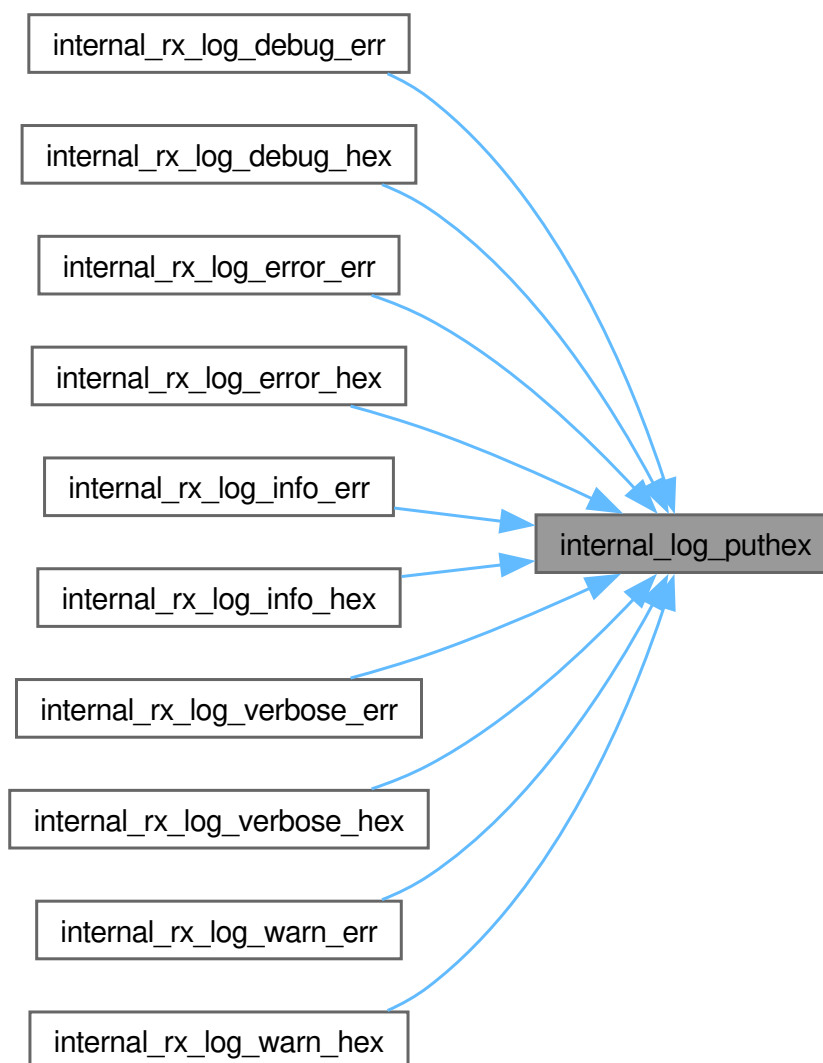
Definition at line 678 of file [rx_log.h](#).

```
00679 {  
00680     LOG_PUTHEX(v, d);  
00681 }
```

References [LOG_PUTHEX](#).

Referenced by [internal_rx_log_debug_err\(\)](#), [internal_rx_log_debug_hex\(\)](#), [internal_rx_log_error_err\(\)](#), [internal_rx_log_error_hex\(\)](#), [internal_rx_log_info_err\(\)](#), [internal_rx_log_info_hex\(\)](#), [internal_rx_log_verbose_err\(\)](#), [internal_rx_log_verbose_hex\(\)](#), [internal_rx_log_warn_err\(\)](#), and [internal_rx_log_warn_hex\(\)](#).

Here is the caller graph for this function:



7.21.4.7 `internal_log_putint()`

```
void internal_log_putint (
    int32_t v)    [inline], [static]
```

Write a signed 32-bit integer to the active log backend(s).

Parameters

in	v	Value.
----	---	--------

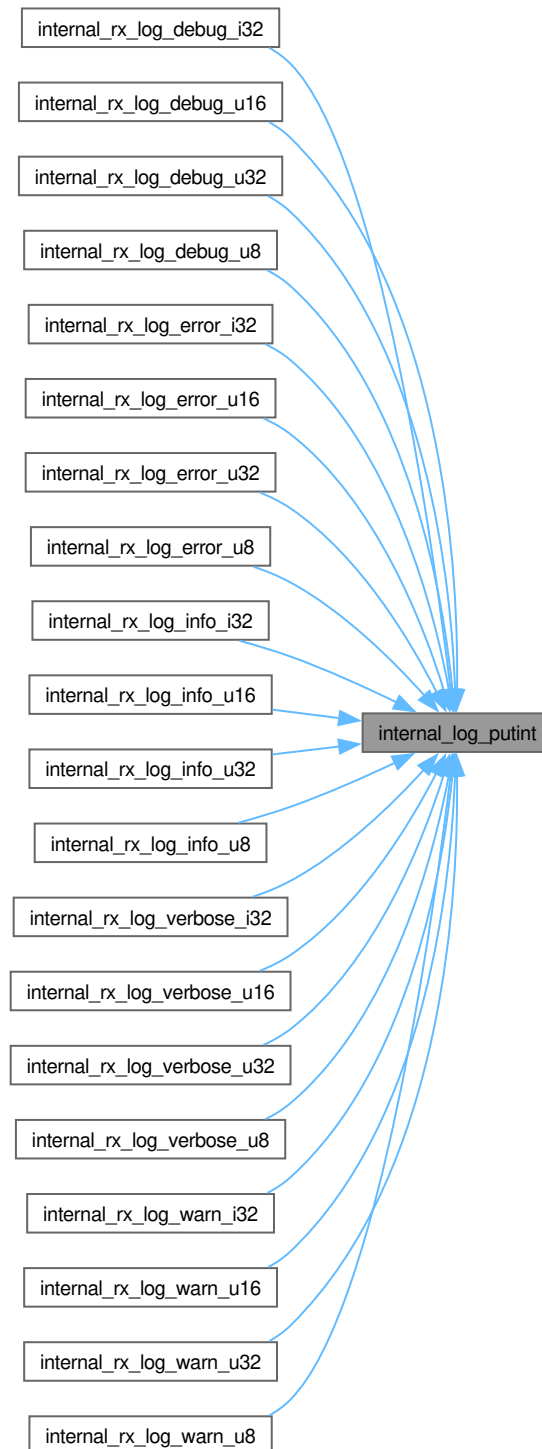
Definition at line 672 of file [rx_log.h](#).

```
00673 {
00674     LOG_PUTINT(v);
00675 }
```

References [LOG_PUTINT](#).

Referenced by [internal_rx_log_debug_i32\(\)](#), [internal_rx_log_debug_u16\(\)](#), [internal_rx_log_debug_u32\(\)](#), [internal_rx_log_debug_u8\(\)](#), [internal_rx_log_error_i32\(\)](#), [internal_rx_log_error_u16\(\)](#), [internal_rx_log_error_u32\(\)](#), [internal_rx_log_error_u8\(\)](#), [internal_rx_log_info_i32\(\)](#), [internal_rx_log_info_u16\(\)](#), [internal_rx_log_info_u32\(\)](#), [internal_rx_log_info_u8\(\)](#), [internal_rx_log_verbose_i32\(\)](#), [internal_rx_log_verbose_u16\(\)](#), [internal_rx_log_verbose_u32\(\)](#), [internal_rx_log_verbose_u8\(\)](#), [internal_rx_log_warn_i32\(\)](#), [internal_rx_log_warn_u16\(\)](#), [internal_rx_log_warn_u32\(\)](#) and [internal_rx_log_warn_u8\(\)](#).

Here is the caller graph for this function:



7.21.4.8 internal_log_puts()

```
void internal_log_puts (
    const char * s)  [inline], [static]
```

Write a string to the active log backend(s).

Parameters

in	s	Null-terminated string.
----	---	-------------------------

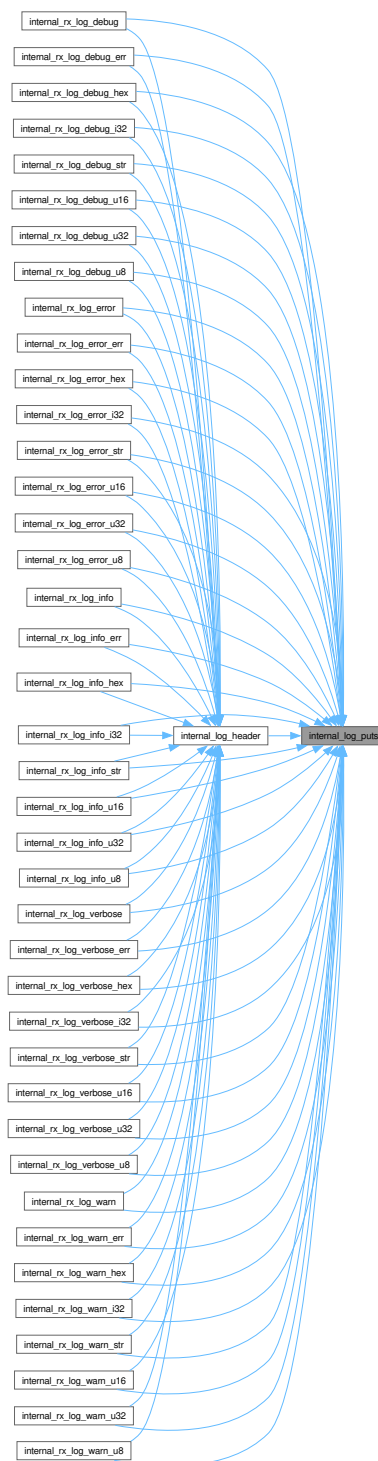
Definition at line 666 of file [rx_log.h](#).

```
00667 {
00668     LOG_PUTS(s);
00669 }
```

References [LOG_PUTS](#).

Referenced by [internal_log_header\(\)](#), [internal_rx_log_debug\(\)](#), [internal_rx_log_debug_err\(\)](#), [internal_rx_log_debug_hex\(\)](#), [internal_rx_log_debug_i32\(\)](#), [internal_rx_log_debug_str\(\)](#), [internal_rx_log_debug_u32\(\)](#), [internal_rx_log_debug_u8\(\)](#), [internal_rx_log_error\(\)](#), [internal_rx_log_error_err\(\)](#), [internal_rx_log_error_hex\(\)](#), [internal_rx_log_error_i32\(\)](#), [internal_rx_log_error_str\(\)](#), [internal_rx_log_error_u16\(\)](#), [internal_rx_log_error_u32\(\)](#), [internal_rx_log_error_u8\(\)](#), [internal_rx_log_info\(\)](#), [internal_rx_log_info_err\(\)](#), [internal_rx_log_info_hex\(\)](#), [internal_rx_log_info_i32\(\)](#), [internal_rx_log_info_str\(\)](#), [internal_rx_log_info_u16\(\)](#), [internal_rx_log_info_u32\(\)](#), [internal_rx_log_info_u8\(\)](#), [internal_rx_log_verbose\(\)](#), [internal_rx_log_verbose_err\(\)](#), [internal_rx_log_verbose_hex\(\)](#), [internal_rx_log_verbose_i32\(\)](#), [internal_rx_log_verbose_str\(\)](#), [internal_rx_log_verbose_u16\(\)](#), [internal_rx_log_verbose_u32\(\)](#), [internal_rx_log_verbose_u8\(\)](#), [internal_rx_log_warn\(\)](#), [internal_rx_log_warn_err\(\)](#), [internal_rx_log_warn_hex\(\)](#), [internal_rx_log_warn_i32\(\)](#), [internal_rx_log_warn_str\(\)](#), [internal_rx_log_warn_u16\(\)](#), [internal_rx_log_warn_u32\(\)](#), and [internal_rx_log_warn_u8\(\)](#).

Here is the caller graph for this function:



7.21.4.9 internal_rx_log_debug()

```
void internal_rx_log_debug (
    const char * tag,
    const char * message) [inline], [static]
```

Log debug message (no value).

Definition at line [1286](#) of file [rx_log.h](#).

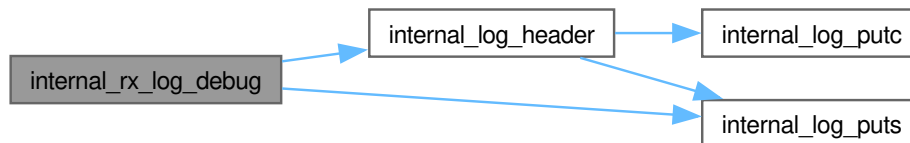
```

01287 {
01288     /* Rule 5: Pre-condition validation - NULL guard */
01289     if (message == (const char*)0) {
01290         return;
01291     }
01292     internal_log_header("DEBUG", tag);
01293     internal_log_puts(message);
01294     internal_log_puts("\r\n");
01295 }

```

References [internal_log_header\(\)](#), and [internal_log_puts\(\)](#).

Here is the call graph for this function:



7.21.4.10 internal_rx_log_debug_err()

```

void internal_rx_log_debug_err (
    const char * tag,
    const char * message,
    rx_err_t err)  [inline], [static]

```

Log debug with error code ([rx_err_t](#)).

Definition at line [1348](#) of file [rx_log.h](#).

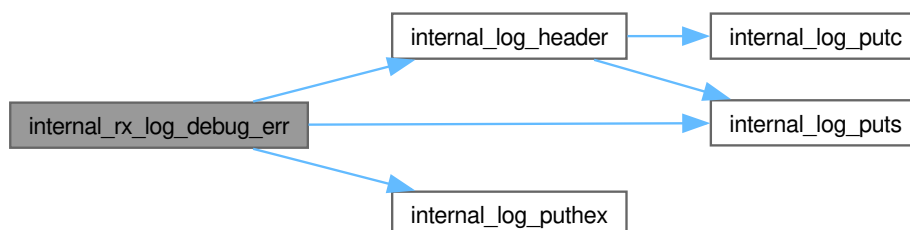
```

01349 {
01350     internal_log_header("DEBUG", tag);
01351     internal_log_puts(message);
01352     internal_log_puts(": 0x");
01353     internal_log_puthex((uint32_t)err, k_log_hex_width_err);
01354     internal_log_puts("\r\n");
01355 }

```

References [internal_log_header\(\)](#), [internal_log_puthex\(\)](#), [internal_log_puts\(\)](#), and [k_log_hex_width_err](#).

Here is the call graph for this function:



7.21.4.11 `internal_rx_log_debug_hex()`

```
void internal_rx_log_debug_hex (
    const char * tag,
    const char * message,
    uint32_t value,
    uint8_t digits)    [inline], [static]
```

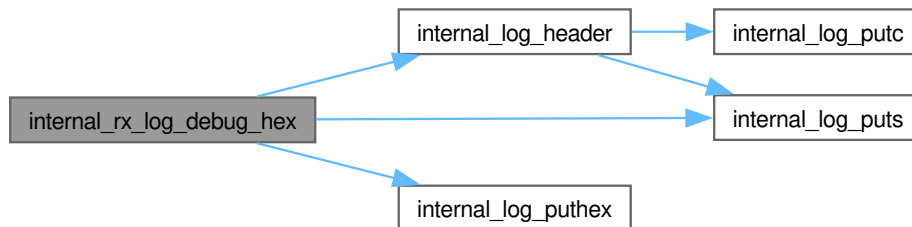
Log debug with hex value.

Definition at line 1361 of file [rx_log.h](#).

```
01362 {
01363     internal_log_header("DEBUG", tag);
01364     internal_log_puts(message);
01365     internal_log_puts(": 0x");
01366     internal_log_puthex(value, digits);
01367     internal_log_puts("\r\n");
01368 }
```

References [internal_log_header\(\)](#), [internal_log_puthex\(\)](#), and [internal_log_puts\(\)](#).

Here is the call graph for this function:

7.21.4.12 `internal_rx_log_debug_i32()`

```
void internal_rx_log_debug_i32 (
    const char * tag,
    const char * message,
    int32_t value)    [inline], [static]
```

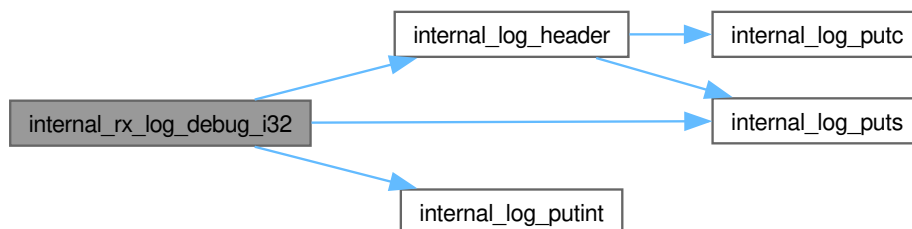
Log debug with `int32_t` value.

Definition at line 1336 of file [rx_log.h](#).

```
01337 {
01338     internal_log_header("DEBUG", tag);
01339     internal_log_puts(message);
01340     internal_log_puts(": ");
01341     internal_log_putint(value);
01342     internal_log_puts("\r\n");
01343 }
```

References [internal_log_header\(\)](#), [internal_log_putint\(\)](#), and [internal_log_puts\(\)](#).

Here is the call graph for this function:



7.21.4.13 internal_rx_log_debug_str()

```
void internal_rx_log_debug_str (
    const char * tag,
    const char * message,
    const char * str_value,
    uint32_t len)    [inline], [static]
```

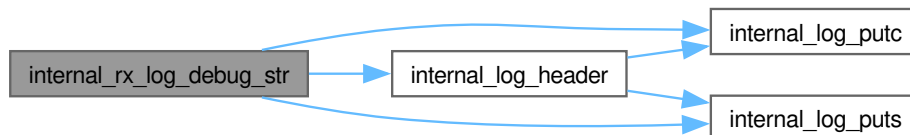
Log debug with string value (bounded length).

Definition at line 1374 of file [rx_log.h](#).

```
01375 {
01376     /* Pre-condition: Validate pointers (NASA Power of 10 Rule 5) */
01377     if (message == (const char*)0 || str_value == (const char*)0) {
01378         return;
01379     }
01380
01381     internal_log_header("DEBUG", tag);
01382     internal_log_puts(message);
01383     internal_log_puts(": ");
01384
01385     /*
01386     * NASA Rule 2: Loop bounded by k_log_str_max_len (compile-time constant)
01387     * Early break when i >= len or null terminator encountered.
01388     */
01389     for (uint32_t i = 0; i < k_log_str_max_len; i++) {
01390         if (i >= len || str_value[i] == '\0') {
01391             break;
01392         }
01393         internal_log_putc(str_value[i]);
01394     }
01395
01396     internal_log_puts("\r\n");
01397 }
```

References [internal_log_header\(\)](#), [internal_log_putc\(\)](#), [internal_log_puts\(\)](#), and [k_log_str_max_len](#).

Here is the call graph for this function:



7.21.4.14 internal_rx_log_debug_u16()

```
void internal_rx_log_debug_u16 (
    const char * tag,
    const char * message,
    uint16_t value)    [inline], [static]
```

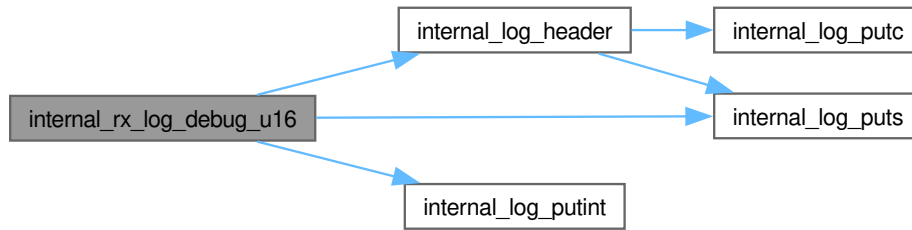
Log debug with `uint16_t` value.

Definition at line 1312 of file [rx_log.h](#).

```
01313 {
01314     internal_log_header("DEBUG", tag);
01315     internal_log_puts(message);
01316     internal_log_puts(": ");
01317     internal_log_putint((int32_t)value);
01318     internal_log_puts("\r\n");
01319 }
```

References [internal_log_header\(\)](#), [internal_log_putint\(\)](#), and [internal_log_puts\(\)](#).

Here is the call graph for this function:



7.21.4.15 `internal_rx_log_debug_u32()`

```

void internal_rx_log_debug_u32 (
    const char * tag,
    const char * message,
    uint32_t value) [inline], [static]
  
```

Log debug with `uint32_t` value.

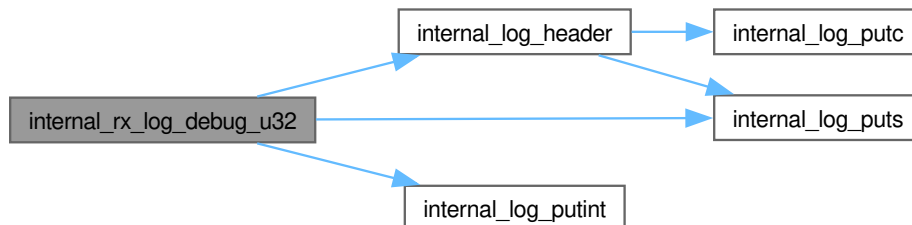
Definition at line 1324 of file `rx_log.h`.

```

01325 {
01326     internal_log_header("DEBUG", tag);
01327     internal_log_puts(message);
01328     internal_log_puts(": ");
01329     internal_log_putint((int32_t)value);
01330     internal_log_puts("\r\n");
01331 }
  
```

References [internal_log_header\(\)](#), [internal_log_putint\(\)](#), and [internal_log_puts\(\)](#).

Here is the call graph for this function:



7.21.4.16 `internal_rx_log_debug_u8()`

```

void internal_rx_log_debug_u8 (
    const char * tag,
    const char * message,
    uint8_t value) [inline], [static]
  
```

Log debug with `uint8_t` value.

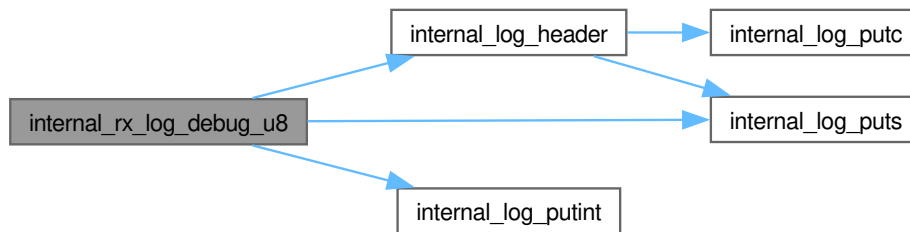
Definition at line 1300 of file `rx_log.h`.

```

01301 {
01302     internal_log_header("DEBUG", tag);
01303     internal_log_puts(message);
01304     internal_log_puts(": ");
01305     internal_log_putint((int32_t)value);
01306     internal_log_puts("\r\n");
01307 }
  
```

References [internal_log_header\(\)](#), [internal_log_putint\(\)](#), and [internal_log_puts\(\)](#).

Here is the call graph for this function:



7.21.4.17 `internal_rx_log_error()`

```
void internal_rx_log_error (
    const char * tag,
    const char * message) [inline], [static]
```

Log ERROR-level message without additional value.

Outputs critical error message with standardized header format. Used for operation-preventing failures that require immediate attention.

7.21.4.18 Algorithm

1. Validate message pointer (NULL check)
2. Output header: [ERROR] [TAG]
3. Output message text
4. Output line terminator `\r\n`

7.21.4.19 Output Format

[ERROR] [TAG] message text CRLF

7.21.4.20 Performance

Metric	Value
Cycles	~2400 @ 240 MHz (30-char message)
Time	~10 us
UART bytes	~45 bytes (header + message + CRLF)
Stack	0 bytes (inline)

Parameters

in	tag	Component tag identifying error source (e.g., "MOTOR", "SPI") • Must be null-terminated string literal • Recommended: 3-8 uppercase characters • nullptr silently ignored
in	message	Error message describing the failure • Must be null-terminated string • Should be concise and actionable • nullptr causes silent return (safety)

Precondition

message must be valid null-terminated string or nullptr

Postcondition

Error message written to UART (and USB CDC if USB_LOG_MIRROR=1)

Note

Thread Safety: NOT thread-safe. Concurrent calls may interleave.

Inline: Always inlined for zero function call overhead.

NULL Handling: NULL message pointer silently ignored (safe failure).

Example Usage:

```
// Critical initialization failure
if (!motor_init()) {
    internal_rx_log_error("MOTOR", "Initialization failed");
}

// Hardware failure
if (!spi_ready()) {
    internal_rx_log_error("SPI", "Peripheral not responding");
}

// Output on terminal:
// [ERROR] [MOTOR] Initialization failed
// [ERROR] [SPI] Peripheral not responding
```

See also

[rx_log_error\(\)](#) Public API macro (compile-time filtered)

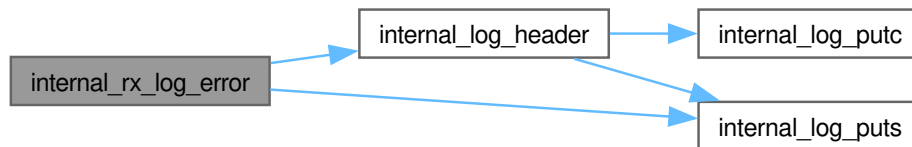
[internal_log_header\(\)](#) Header formatting function

Definition at line 832 of file [rx_log.h](#).

```
00833 {
00834  /* Rule 5: Pre-condition validation - NULL guard */
00835  if (message == (const char*)0) {
00836      return;
00837  }
00838  internal_log_header("ERROR", tag);
00839  internal_log_puts(message);
00840  internal_log_puts("\r\n");
00841 }
```

References [internal_log_header\(\)](#), and [internal_log_puts\(\)](#).

Here is the call graph for this function:



7.21.4.21 internal_rx_log_error_err()

```
void internal_rx_log_error_err (
    const char * tag,
    const char * message,
    rx_err_t err) [inline], [static]
```

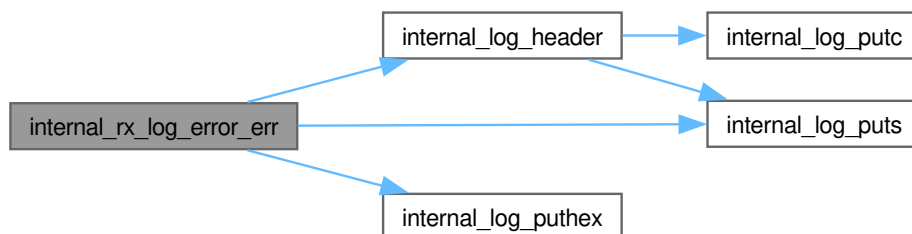
Log error with error code ([rx_err_t](#)).

Definition at line 894 of file [rx_log.h](#).

```
00895 {
00896  internal_log_header("ERROR", tag);
00897  internal_log_puts(message);
00898  internal_log_puts(": 0x");
00899  internal_log_puthex((uint32_t)err, k_log_hex_width_err);
00900  internal_log_puts("\r\n");
00901 }
```

References [internal_log_header\(\)](#), [internal_log_puthex\(\)](#), [internal_log_puts\(\)](#), and [k_log_hex_width_err](#).

Here is the call graph for this function:



7.21.4.22 `internal_rx_log_error_hex()`

```
void internal_rx_log_error_hex (
    const char * tag,
    const char * message,
    uint32_t value,
    uint8_t digits)    [inline], [static]
```

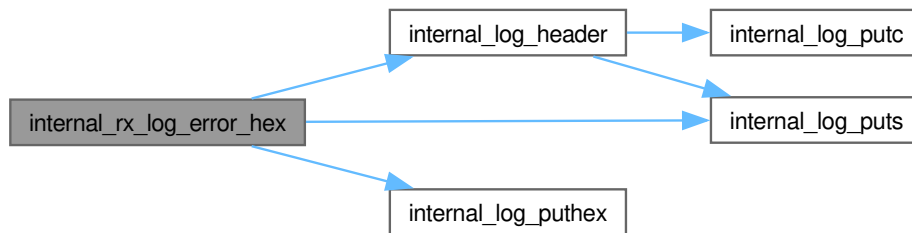
Log error with hex value.

Definition at line 907 of file [rx_log.h](#).

```
00908 {
00909     internal_log_header("ERROR", tag);
00910     internal_log_puts(message);
00911     internal_log_puts(": 0x");
00912     internal_log_puthex(value, digits);
00913     internal_log_puts("\r\n");
00914 }
```

References [internal_log_header\(\)](#), [internal_log_puthex\(\)](#), and [internal_log_puts\(\)](#).

Here is the call graph for this function:

7.21.4.23 `internal_rx_log_error_i32()`

```
void internal_rx_log_error_i32 (
    const char * tag,
    const char * message,
    int32_t value)    [inline], [static]
```

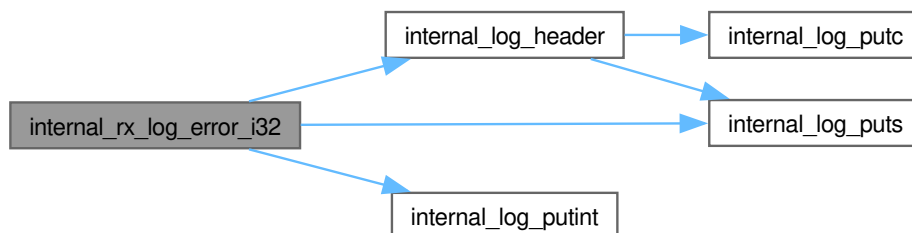
Log error with `int32_t` value.

Definition at line 882 of file [rx_log.h](#).

```
00883 {
00884     internal_log_header("ERROR", tag);
00885     internal_log_puts(message);
00886     internal_log_puts(": ");
00887     internal_log_putint(value);
00888     internal_log_puts("\r\n");
00889 }
```

References [internal_log_header\(\)](#), [internal_log_putint\(\)](#), and [internal_log_puts\(\)](#).

Here is the call graph for this function:



7.21.4.24 internal_rx_log_error_str()

```
void internal_rx_log_error_str (
    const char * tag,
    const char * message,
    const char * str_value,
    uint32_t len)    [inline], [static]
```

Log ERROR-level message with bounded string value (NASA-compliant).

Outputs critical error message with associated string value using bounded iteration for memory safety. String output is limited to prevent unbounded loops and buffer overruns (NASA Power of 10 Rule 2 compliance).

7.21.4.25 Algorithm

1. Validate message and str_value pointers (NULL checks)
2. Output header: [ERROR] [TAG]
3. Output message text
4. Output separator: :
5. Iterate up to k_log_str_max_len (256) characters: a. Stop if index >= user-provided len b. Stop if null terminator '\0' encountered c. Output character
6. Output line terminator \r\n

7.21.4.26 Output Format

[ERROR] [TAG] message: string_value\r\n

7.21.4.27 Safety Features (NASA Rule 2)

Feature	Implementation	Benefit
Bounded loop	<code>i < k_log_str_max_len</code>	Statically provable upper bound
Early termination	<code>i >= len</code> check	Respects caller's length limit
Null handling	<code>str_value[i] == '\0'</code>	Safe for null-terminated strings
Pointer validation	NULL checks at start	Prevents crashes

7.21.4.28 Performance

Metric	Best Case	Worst Case
Cycles	~2700 (1 char)	~21600 (256 chars) @ 240 MHz
Time	~11 us	~90 us
UART bytes	~50 bytes	~300 bytes
Stack	4 bytes (uint32_t i)	4 bytes

Parameters

in	tag	Component tag identifying error source (e.g., "MOTOR", "SPI") • Must be null-terminated string literal • nullptr silently ignored
in	message	Error message describing the context • Must be null-terminated string • nullptr causes silent return (safety)
in	str_value	String value to log (e.g., parameter name, device ID) • Must point to valid memory • nullptr causes silent return (safety) • Does not need to be null-terminated if len is exact
in	len	Maximum number of characters to log from str_value • Typically obtained via strlen(str_value) • Values > k_log_str_max_len will be clamped to k_log_str_max_len • Use exact buffer length for non-null-terminated buffers

Precondition

message and str_value must be valid pointers or nullptr
len must not exceed actual buffer size of str_value

Postcondition

Error message with string value written to output (if parameters valid)

Note

Thread Safety: NOT thread-safe. Concurrent calls may interleave output.
Inline: Always inlined for zero function call overhead.
Bounded: Maximum 256 characters logged regardless of len parameter.
NULL Safety: NULL pointers silently ignored (safe failure mode).

Warning

If str_value buffer is smaller than len, undefined behavior may occur. Always ensure len ≤ actual buffer size.

Example: Logging Configuration Errors

```
// Unknown configuration parameter
const char* param_name = "invalid_motor_id";
internal_rx_log_error_str("CONFIG", "Unknown parameter", param_name, strlen(param_name));
// Output: [ERROR] [CONFIG] Unknown parameter: invalid_motor_id

// Non-null-terminated buffer (use exact length)
char device_id[16] = {0x01, 0x02, 0x03, ...}; // Binary data
internal_rx_log_error_str("USB", "Invalid device ID", device_id, 16);
// Output: [ERROR] [USB] Invalid device ID: <binary data>
```

Example: Truncation Behavior

```
// String longer than k_log_str_max_len (256)
char long_str[300];
memset(long_str, 'A', 299);
long_str[299] = '\0';

internal_rx_log_error_str("TEST", "Long string", long_str, 299);
// Output: [ERROR] [TEST] Long string: AAAA... (256 chars max, truncated)
```

See also

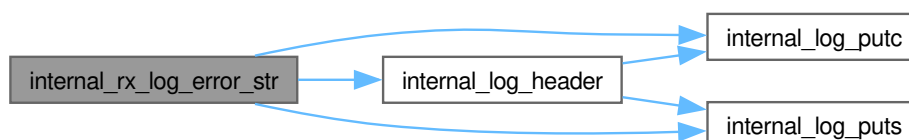
[k_log_str_max_len](#) Maximum string length constant (256 bytes)
[rx_log_error_str\(\)](#) Public API macro (compile-time filtered)
[internal_rx_log_error\(\)](#) Simple error logging without value

Definition at line 1012 of file [rx_log.h](#).

```
01013 {
01014     /* Pre-condition: Validate pointers (NASA Power of 10 Rule 5) */
01015     if (message == (const char*)0 || str_value == (const char*)0) {
01016         return;
01017     }
01018     internal_log_header("ERROR", tag);
01019     internal_log_puts(message);
01020     internal_log_puts(": ");
01021
01022     /*
01023     * NASA Rule 2: Loop bounded by k_log_str_max_len (compile-time constant)
01024     * Early break when i >= len or null terminator encountered.
01025     */
01026     for (uint32_t i = 0; i < k_log_str_max_len; i++) {
01027         if (i >= len || str_value[i] == '\0') {
01028             break;
01029         }
01030         internal_log_putc(str_value[i]);
01031     }
01032     internal_log_puts("\r\n");
01033 }
01034 }
```

References [internal_log_header\(\)](#), [internal_log_putc\(\)](#), [internal_log_puts\(\)](#), and [k_log_str_max_len](#).

Here is the call graph for this function:



7.21.4.29 `internal_rx_log_error_u16()`

```
void internal_rx_log_error_u16 (
    const char * tag,
    const char * message,
    uint16_t value) [inline], [static]
```

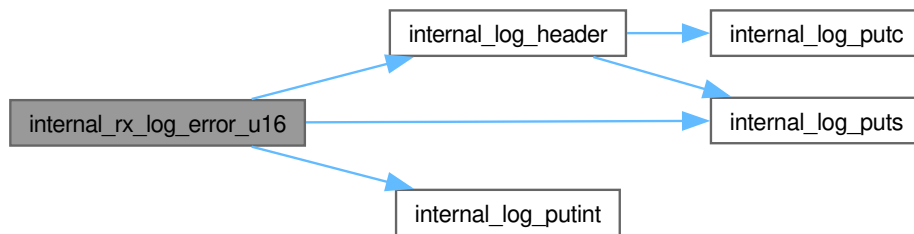
Log error with `uint16_t` value.

Definition at line 858 of file `rx_log.h`.

```
00859 {
00860     internal_log_header("ERROR", tag);
00861     internal_log_puts(message);
00862     internal_log_puts(": ");
00863     internal_log_putint((int32_t)value);
00864     internal_log_puts("\r\n");
00865 }
```

References [internal_log_header\(\)](#), [internal_log_putint\(\)](#), and [internal_log_puts\(\)](#).

Here is the call graph for this function:

7.21.4.30 `internal_rx_log_error_u32()`

```
void internal_rx_log_error_u32 (
    const char * tag,
    const char * message,
    uint32_t value) [inline], [static]
```

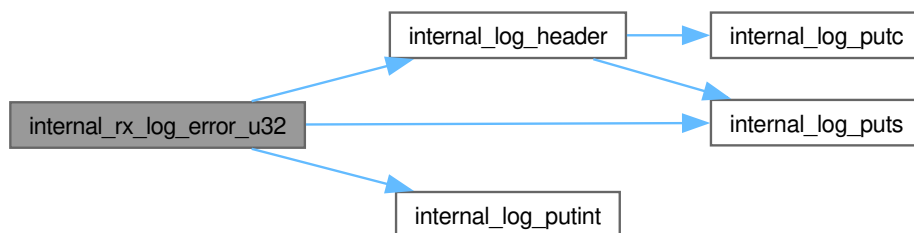
Log error with `uint32_t` value.

Definition at line 870 of file `rx_log.h`.

```
00871 {
00872     internal_log_header("ERROR", tag);
00873     internal_log_puts(message);
00874     internal_log_puts(": ");
00875     internal_log_putint((int32_t)value);
00876     internal_log_puts("\r\n");
00877 }
```

References [internal_log_header\(\)](#), [internal_log_putint\(\)](#), and [internal_log_puts\(\)](#).

Here is the call graph for this function:



7.21.4.31 `internal_rx_log_error_u8()`

```
void internal_rx_log_error_u8 (
    const char * tag,
    const char * message,
    uint8_t value) [inline], [static]
```

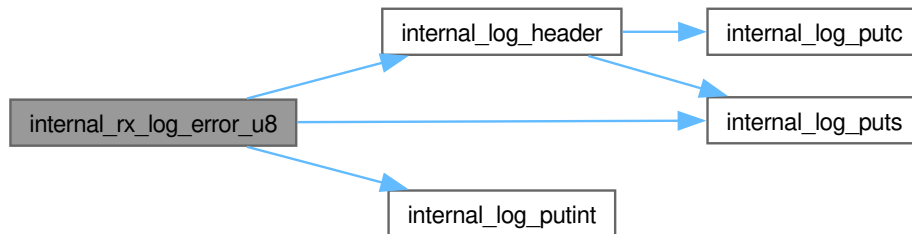
Log error with `uint8_t` value.

Definition at line 846 of file `rx_log.h`.

```
00847 {
00848     internal_log_header("ERROR", tag);
00849     internal_log_puts(message);
00850     internal_log_puts(": ");
00851     internal_log_putint((int32_t)value);
00852     internal_log_puts("\r\n");
00853 }
```

References [internal_log_header\(\)](#), [internal_log_putint\(\)](#), and [internal_log_puts\(\)](#).

Here is the call graph for this function:

7.21.4.32 `internal_rx_log_info()`

```
void internal_rx_log_info (
    const char * tag,
    const char * message) [inline], [static]
```

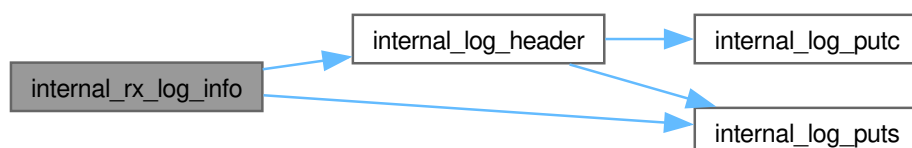
Log info message (no value).

Definition at line 1165 of file `rx_log.h`.

```
01166 {
01167     /* Rule 5: Pre-condition validation - NULL guard */
01168     if (message == (const char*)0) {
01169         return;
01170     }
01171     internal_log_header("INFO", tag);
01172     internal_log_puts(message);
01173     internal_log_puts("\r\n");
01174 }
```

References [internal_log_header\(\)](#), and [internal_log_puts\(\)](#).

Here is the call graph for this function:



7.21.4.33 `internal_rx_log_info_err()`

```
void internal_rx_log_info_err (
    const char * tag,
    const char * message,
    rx_err_t err)  [inline], [static]
```

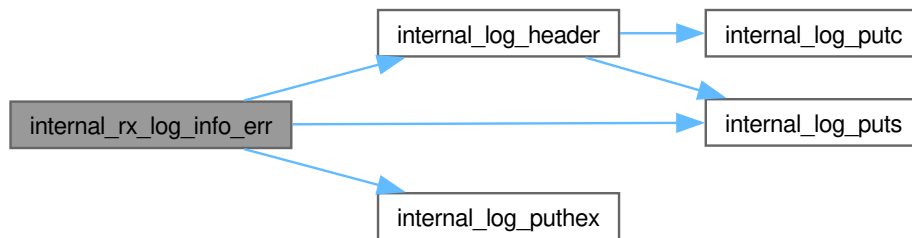
Log info with error code ([rx_err_t](#)).

Definition at line 1227 of file [rx_log.h](#).

```
01228 {
01229     internal_log_header("INFO", tag);
01230     internal_log_puts(message);
01231     internal_log_puts(": 0x");
01232     internal_log_puthex((uint32_t)err, k_log_hex_width_err);
01233     internal_log_puts("\r\n");
01234 }
```

References [internal_log_header\(\)](#), [internal_log_puthex\(\)](#), [internal_log_puts\(\)](#), and [k_log_hex_width_err](#).

Here is the call graph for this function:

7.21.4.34 `internal_rx_log_info_hex()`

```
void internal_rx_log_info_hex (
    const char * tag,
    const char * message,
    uint32_t value,
    uint8_t digits)  [inline], [static]
```

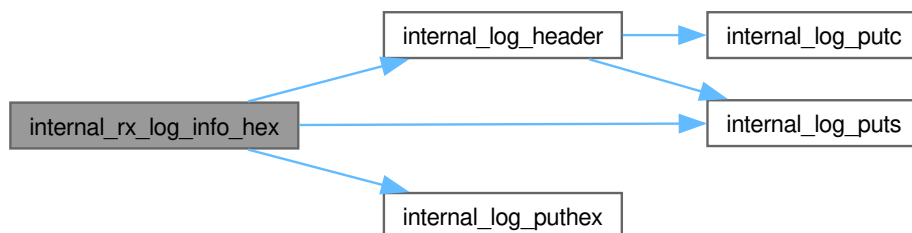
Log info with hex value.

Definition at line 1240 of file [rx_log.h](#).

```
01241 {
01242     internal_log_header("INFO", tag);
01243     internal_log_puts(message);
01244     internal_log_puts(": 0x");
01245     internal_log_puthex(value, digits);
01246     internal_log_puts("\r\n");
01247 }
```

References [internal_log_header\(\)](#), [internal_log_puthex\(\)](#), and [internal_log_puts\(\)](#).

Here is the call graph for this function:



7.21.4.35 `internal_rx_log_info_i32()`

```
void internal_rx_log_info_i32 (
    const char * tag,
    const char * message,
    int32_t value) [inline], [static]
```

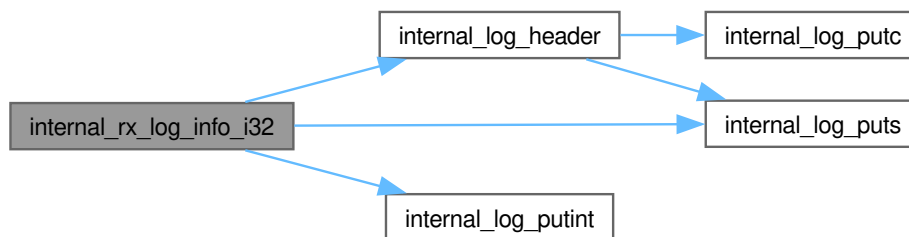
Log info with `int32_t` value.

Definition at line 1215 of file `rx_log.h`.

```
01216 {
01217     internal_log_header("INFO", tag);
01218     internal_log_puts(message);
01219     internal_log_puts(": ");
01220     internal_log_putint(value);
01221     internal_log_puts("\r\n");
01222 }
```

References [internal_log_header\(\)](#), [internal_log_putint\(\)](#), and [internal_log_puts\(\)](#).

Here is the call graph for this function:

7.21.4.36 `internal_rx_log_info_str()`

```
void internal_rx_log_info_str (
    const char * tag,
    const char * message,
    const char * str_value,
    uint32_t len) [inline], [static]
```

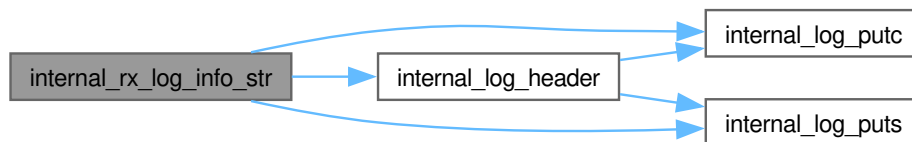
Log info with string value (bounded length).

Definition at line 1253 of file `rx_log.h`.

```
01254 {
01255     /* Pre-condition: Validate pointers (NASA Power of 10 Rule 5) */
01256     if (message == (const char*)0 || str_value == (const char*)0) {
01257         return;
01258     }
01259     internal_log_header("INFO", tag);
01260     internal_log_puts(message);
01261     internal_log_puts(": ");
01262     /*
01263     * NASA Rule 2: Loop bounded by k_log_str_max_len (compile-time constant)
01264     * Early break when i >= len or null terminator encountered.
01265     */
01266     for (uint32_t i = 0; i < k_log_str_max_len; i++) {
01267         if (i >= len || str_value[i] == '\0') {
01268             break;
01269         }
01270         internal_log_putc(str_value[i]);
01271     }
01272     internal_log_puts("\r\n");
01273 }
01274
01275 internal_log_puts("\r\n");
01276 }
```

References [internal_log_header\(\)](#), [internal_log_putc\(\)](#), [internal_log_puts\(\)](#), and [k_log_str_max_len](#).

Here is the call graph for this function:



7.21.4.37 internal_rx_log_info_u16()

```

void internal_rx_log_info_u16 (
    const char * tag,
    const char * message,
    uint16_t value)  [inline], [static]
  
```

Log info with uint16_t value.

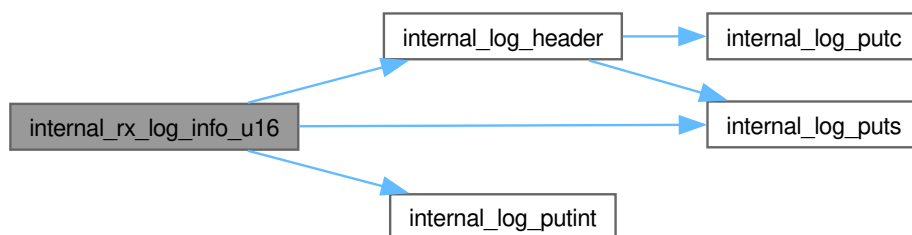
Definition at line 1191 of file [rx_log.h](#).

```

01192 {
01193     internal_log_header("INFO", tag);
01194     internal_log_puts(message);
01195     internal_log_puts(": ");
01196     internal_log_putint((int32_t)value);
01197     internal_log_puts("\r\n");
01198 }
  
```

References [internal_log_header\(\)](#), [internal_log_putint\(\)](#), and [internal_log_puts\(\)](#).

Here is the call graph for this function:



7.21.4.38 internal_rx_log_info_u32()

```

void internal_rx_log_info_u32 (
    const char * tag,
    const char * message,
    uint32_t value)  [inline], [static]
  
```

Log info with uint32_t value.

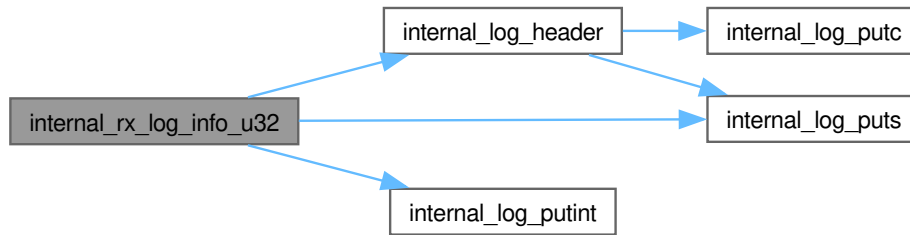
Definition at line 1203 of file [rx_log.h](#).

```

01204 {
01205     internal_log_header("INFO", tag);
01206     internal_log_puts(message);
01207     internal_log_puts(": ");
01208     internal_log_putint((int32_t)value);
01209     internal_log_puts("\r\n");
01210 }
  
```

References [internal_log_header\(\)](#), [internal_log_putint\(\)](#), and [internal_log_puts\(\)](#).

Here is the call graph for this function:



7.21.4.39 `internal_rx_log_info_u8()`

```
void internal_rx_log_info_u8 (
    const char * tag,
    const char * message,
    uint8_t value) [inline], [static]
```

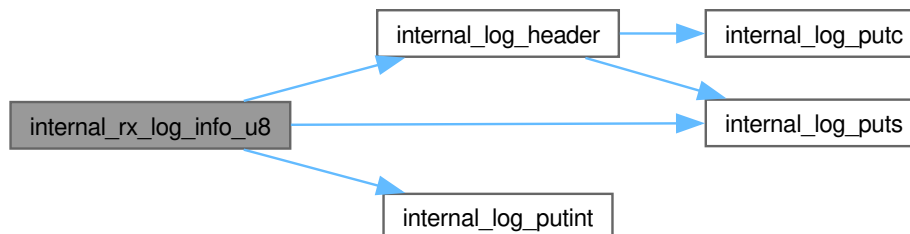
Log info with `uint8_t` value.

Definition at line 1179 of file `rx_log.h`.

```
01180 {
01181     internal_log_header("INFO", tag);
01182     internal_log_puts(message);
01183     internal_log_puts(": ");
01184     internal_log_putint((int32_t)value);
01185     internal_log_puts("\r\n");
01186 }
```

References [internal_log_header\(\)](#), [internal_log_putint\(\)](#), and [internal_log_puts\(\)](#).

Here is the call graph for this function:



7.21.4.40 `internal_rx_log_verbose()`

```
void internal_rx_log_verbose (
    const char * tag,
    const char * message) [inline], [static]
```

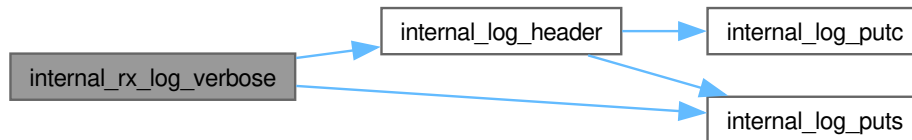
Log verbose message (no value).

Definition at line 1407 of file `rx_log.h`.

```
01408 {
01409     /* Rule 5: Pre-condition validation - NULL guard */
01410     if (message == (const char*)0) {
01411         return;
01412     }
01413     internal_log_header("VERBOSE", tag);
01414     internal_log_puts(message);
01415     internal_log_puts("\r\n");
01416 }
```

References [internal_log_header\(\)](#), and [internal_log_puts\(\)](#).

Here is the call graph for this function:



7.21.4.41 internal_rx_log_verbose_err()

```

void internal_rx_log_verbose_err (
    const char * tag,
    const char * message,
    rx_err_t err)  [inline], [static]
  
```

Log verbose with error code ([rx_err_t](#)).

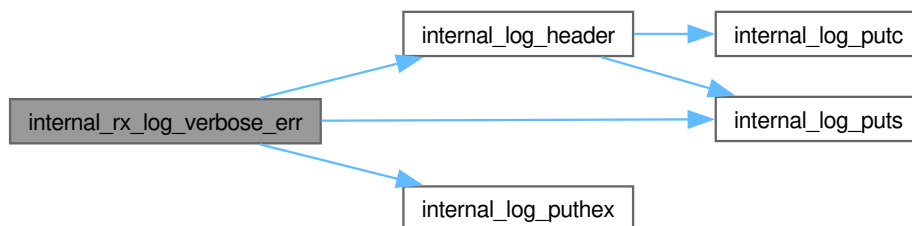
Definition at line [1469](#) of file [rx_log.h](#).

```

01470 {
01471     internal_log_header("VERBOSE", tag);
01472     internal_log_puts(message);
01473     internal_log_puts(": 0x");
01474     internal_log_puthex((uint32_t)err, k_log_hex_width_err);
01475     internal_log_puts("\r\n");
01476 }
  
```

References [internal_log_header\(\)](#), [internal_log_puthex\(\)](#), [internal_log_puts\(\)](#), and [k_log_hex_width_err](#).

Here is the call graph for this function:



7.21.4.42 internal_rx_log_verbose_hex()

```

void internal_rx_log_verbose_hex (
    const char * tag,
    const char * message,
    uint32_t value,
    uint8_t digits)  [inline], [static]
  
```

Log verbose with hex value.

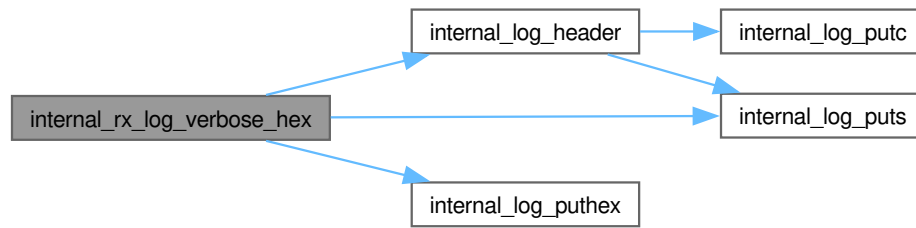
Definition at line [1482](#) of file [rx_log.h](#).

```

01483 {
01484     internal_log_header("VERBOSE", tag);
01485     internal_log_puts(message);
01486     internal_log_puts(": 0x");
01487     internal_log_puthex(value, digits);
01488     internal_log_puts("\r\n");
01489 }
  
```

References [internal_log_header\(\)](#), [internal_log_puthex\(\)](#), and [internal_log_puts\(\)](#).

Here is the call graph for this function:



7.21.4.43 `internal_rx_log_verbose_i32()`

```

void internal_rx_log_verbose_i32 (
    const char * tag,
    const char * message,
    int32_t value) [inline], [static]
  
```

Log verbose with `int32_t` value.

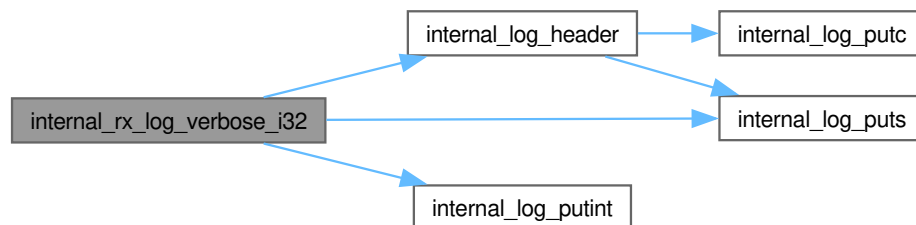
Definition at line 1457 of file `rx_log.h`.

```

01458 {
01459     internal_log_header("VERBOSE", tag);
01460     internal_log_puts(message);
01461     internal_log_puts(": ");
01462     internal_log_putint(value);
01463     internal_log_puts("\r\n");
01464 }
  
```

References [internal_log_header\(\)](#), [internal_log_putint\(\)](#), and [internal_log_puts\(\)](#).

Here is the call graph for this function:



7.21.4.44 `internal_rx_log_verbose_str()`

```

void internal_rx_log_verbose_str (
    const char * tag,
    const char * message,
    const char * str_value,
    uint32_t len) [inline], [static]
  
```

Log verbose with string value (bounded length).

Definition at line 1494 of file `rx_log.h`.

```

01498 {
01499     /* Pre-condition: Validate pointers (NASA Power of 10 Rule 5) */
01500     if (message == (const char*)0 || str_value == (const char*)0) {
01501         return;
01502     }
01503     internal_log_header("VERBOSE", tag);
  
```

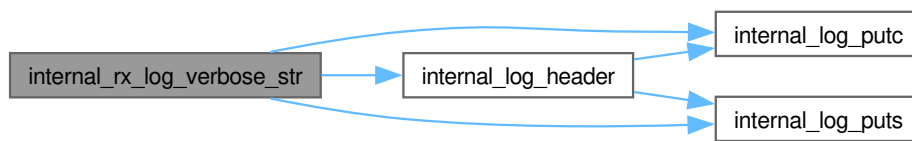
```

01505 internal_log_puts(message);
01506 internal_log_puts(": ");
01507
01508 /*
01509  * NASA Rule 2: Loop bounded by k_log_str_max_len (compile-time constant)
01510  * Early break when i >= len or null terminator encountered.
01511  */
01512 for (uint32_t i = 0; i < k_log_str_max_len; i++) {
01513     if (i >= len || str_value[i] == '\0') {
01514         break;
01515     }
01516     internal_log_putc(str_value[i]);
01517 }
01518
01519 internal_log_puts("\r\n");
01520 }

```

References [internal_log_header\(\)](#), [internal_log_putc\(\)](#), [internal_log_puts\(\)](#), and [k_log_str_max_len](#).

Here is the call graph for this function:



7.21.4.45 internal_rx_log_verbose_u16()

```

void internal_rx_log_verbose_u16 (
    const char * tag,
    const char * message,
    uint16_t value) [inline], [static]

```

Log verbose with `uint16_t` value.

Definition at line 1433 of file [rx_log.h](#).

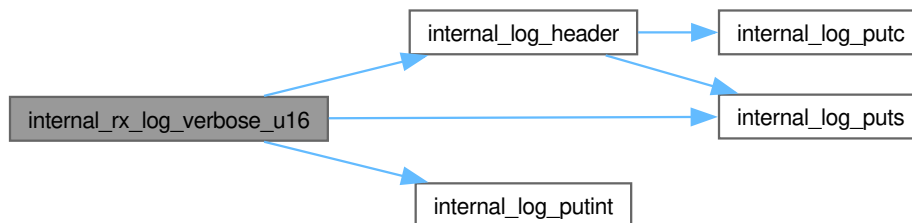
```

01434 {
01435     internal_log_header("VERBOSE", tag);
01436     internal_log_puts(message);
01437     internal_log_puts(": ");
01438     internal_log_putint((int32_t)value);
01439     internal_log_puts("\r\n");
01440 }

```

References [internal_log_header\(\)](#), [internal_log_putint\(\)](#), and [internal_log_puts\(\)](#).

Here is the call graph for this function:



7.21.4.46 `internal_rx_log_verbose_u32()`

```
void internal_rx_log_verbose_u32 (
    const char * tag,
    const char * message,
    uint32_t value) [inline], [static]
```

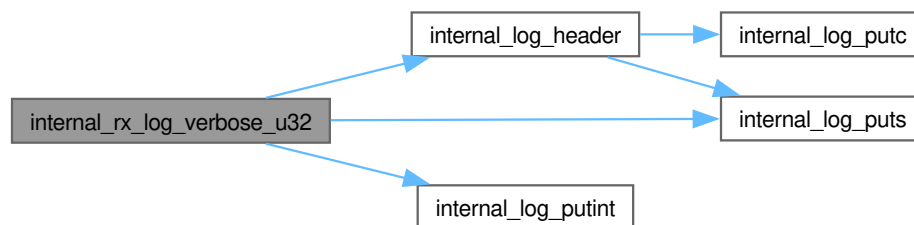
Log verbose with `uint32_t` value.

Definition at line 1445 of file `rx_log.h`.

```
01446 {
01447     internal_log_header("VERBOSE", tag);
01448     internal_log_puts(message);
01449     internal_log_puts(": ");
01450     internal_log_putint((int32_t)value);
01451     internal_log_puts("\r\n");
01452 }
```

References [internal_log_header\(\)](#), [internal_log_putint\(\)](#), and [internal_log_puts\(\)](#).

Here is the call graph for this function:

7.21.4.47 `internal_rx_log_verbose_u8()`

```
void internal_rx_log_verbose_u8 (
    const char * tag,
    const char * message,
    uint8_t value) [inline], [static]
```

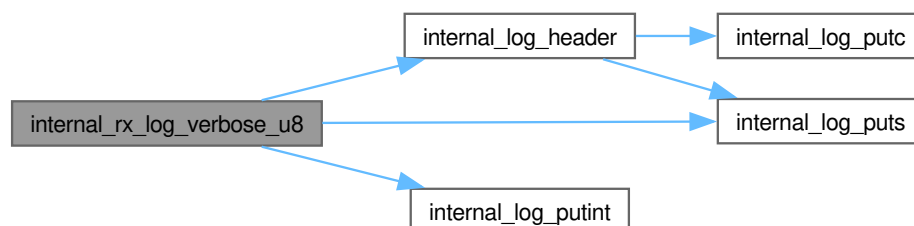
Log verbose with `uint8_t` value.

Definition at line 1421 of file `rx_log.h`.

```
01422 {
01423     internal_log_header("VERBOSE", tag);
01424     internal_log_puts(message);
01425     internal_log_puts(": ");
01426     internal_log_putint((int32_t)value);
01427     internal_log_puts("\r\n");
01428 }
```

References [internal_log_header\(\)](#), [internal_log_putint\(\)](#), and [internal_log_puts\(\)](#).

Here is the call graph for this function:



7.21.4.48 `internal_rx_log_warn()`

```
void internal_rx_log_warn (
    const char * tag,
    const char * message)  [inline], [static]
```

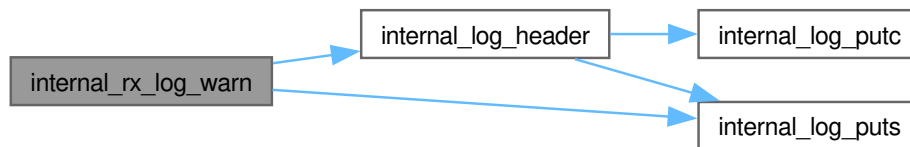
Log warning message (no value).

Definition at line 1044 of file [rx_log.h](#).

```
01045 {
01046  /* Rule 5: Pre-condition validation - NULL guard */
01047  if (message == (const char*)0) {
01048      return;
01049  }
01050  internal_log_header("WARN", tag);
01051  internal_log_puts(message);
01052  internal_log_puts("\r\n");
01053 }
```

References [internal_log_header\(\)](#), and [internal_log_puts\(\)](#).

Here is the call graph for this function:

7.21.4.49 `internal_rx_log_warn_err()`

```
void internal_rx_log_warn_err (
    const char * tag,
    const char * message,
    rx_err_t err)  [inline], [static]
```

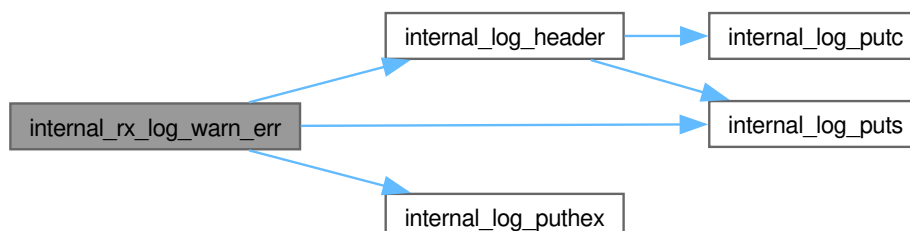
Log warning with error code ([rx_err_t](#)).

Definition at line 1106 of file [rx_log.h](#).

```
01107 {
01108  internal_log_header("WARN", tag);
01109  internal_log_puts(message);
01110  internal_log_puts(": 0x");
01111  internal_log_puthex((uint32_t)err, k_log_hex_width_err);
01112  internal_log_puts("\r\n");
01113 }
```

References [internal_log_header\(\)](#), [internal_log_puthex\(\)](#), [internal_log_puts\(\)](#), and [k_log_hex_width_err](#).

Here is the call graph for this function:



7.21.4.50 `internal_rx_log_warn_hex()`

```
void internal_rx_log_warn_hex (
    const char * tag,
    const char * message,
    uint32_t value,
    uint8_t digits)    [inline], [static]
```

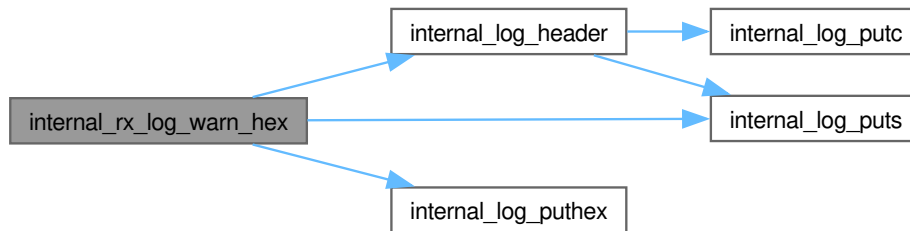
Log warning with hex value.

Definition at line 1119 of file [rx_log.h](#).

```
01120 {
01121     internal_log_header("WARN", tag);
01122     internal_log_puts(message);
01123     internal_log_puts(": 0x");
01124     internal_log_puthex(value, digits);
01125     internal_log_puts("\r\n");
01126 }
```

References [internal_log_header\(\)](#), [internal_log_puthex\(\)](#), and [internal_log_puts\(\)](#).

Here is the call graph for this function:

7.21.4.51 `internal_rx_log_warn_i32()`

```
void internal_rx_log_warn_i32 (
    const char * tag,
    const char * message,
    int32_t value)    [inline], [static]
```

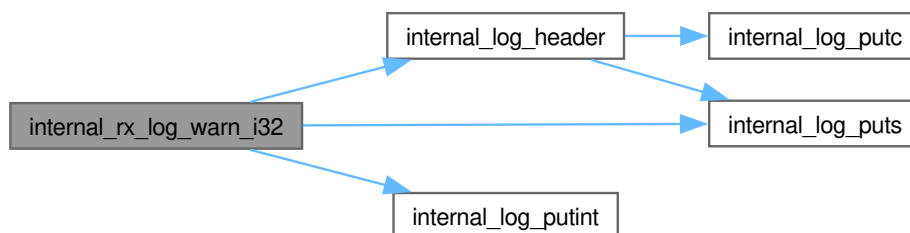
Log warning with `int32_t` value.

Definition at line 1094 of file [rx_log.h](#).

```
01095 {
01096     internal_log_header("WARN", tag);
01097     internal_log_puts(message);
01098     internal_log_puts(": ");
01099     internal_log_putint(value);
01100     internal_log_puts("\r\n");
01101 }
```

References [internal_log_header\(\)](#), [internal_log_putint\(\)](#), and [internal_log_puts\(\)](#).

Here is the call graph for this function:



7.21.4.52 `internal_rx_log_warn_str()`

```
void internal_rx_log_warn_str (
    const char * tag,
    const char * message,
    const char * str_value,
    uint32_t len)    [inline], [static]
```

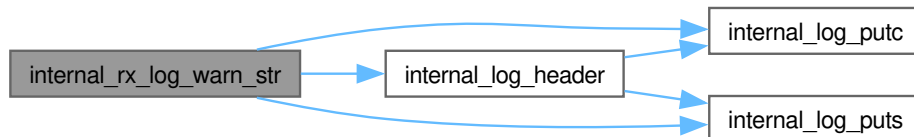
Log warning with string value (bounded length).

Definition at line 1132 of file `rx_log.h`.

```
01133 {
01134     /* Pre-condition: Validate pointers (NASA Power of 10 Rule 5) */
01135     if (message == (const char*)0 || str_value == (const char*)0) {
01136         return;
01137     }
01138
01139     internal_log_header("WARN", tag);
01140     internal_log_puts(message);
01141     internal_log_puts(": ");
01142
01143     /*
01144     * NASA Rule 2: Loop bounded by k_log_str_max_len (compile-time constant)
01145     * Early break when i >= len or null terminator encountered.
01146     */
01147     for (uint32_t i = 0; i < k_log_str_max_len; i++) {
01148         if (i >= len || str_value[i] == '\0') {
01149             break;
01150         }
01151         internal_log_putc(str_value[i]);
01152     }
01153
01154     internal_log_puts("\r\n");
01155 }
```

References `internal_log_header()`, `internal_log_putc()`, `internal_log_puts()`, and `k_log_str_max_len`.

Here is the call graph for this function:

7.21.4.53 `internal_rx_log_warn_u16()`

```
void internal_rx_log_warn_u16 (
    const char * tag,
    const char * message,
    uint16_t value)    [inline], [static]
```

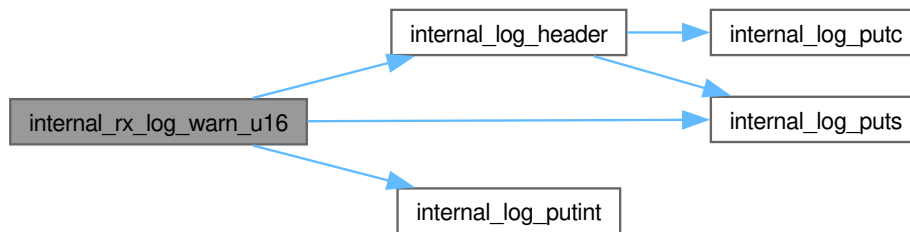
Log warning with `uint16_t` value.

Definition at line 1070 of file `rx_log.h`.

```
01071 {
01072     internal_log_header("WARN", tag);
01073     internal_log_puts(message);
01074     internal_log_puts(": ");
01075     internal_log_putint((int32_t)value);
01076     internal_log_puts("\r\n");
01077 }
```

References `internal_log_header()`, `internal_log_putint()`, and `internal_log_puts()`.

Here is the call graph for this function:



7.21.4.54 `internal_rx_log_warn_u32()`

```

void internal_rx_log_warn_u32 (
    const char * tag,
    const char * message,
    uint32_t value)  [inline], [static]
  
```

Log warning with `uint32_t` value.

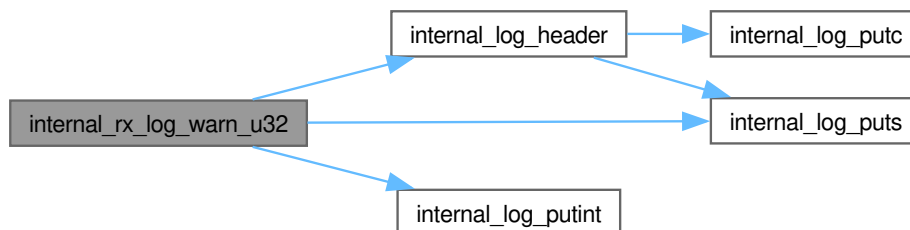
Definition at line 1082 of file `rx_log.h`.

```

01083 {
01084     internal_log_header("WARN", tag);
01085     internal_log_puts(message);
01086     internal_log_puts(": ");
01087     internal_log_putint((int32_t)value);
01088     internal_log_puts("\r\n");
01089 }
  
```

References [internal_log_header\(\)](#), [internal_log_putint\(\)](#), and [internal_log_puts\(\)](#).

Here is the call graph for this function:



7.21.4.55 `internal_rx_log_warn_u8()`

```

void internal_rx_log_warn_u8 (
    const char * tag,
    const char * message,
    uint8_t value)  [inline], [static]
  
```

Log warning with `uint8_t` value.

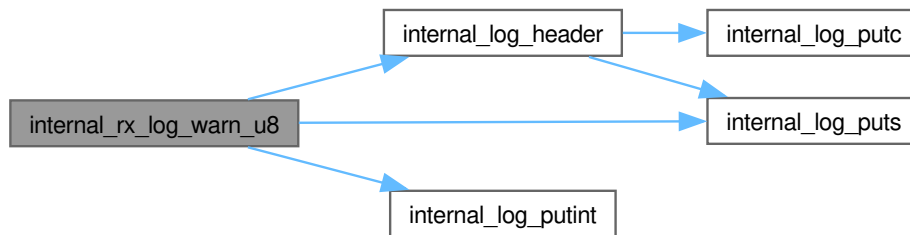
Definition at line 1058 of file `rx_log.h`.

```

01059 {
01060     internal_log_header("WARN", tag);
01061     internal_log_puts(message);
01062     internal_log_puts(": ");
01063     internal_log_putint((int32_t)value);
01064     internal_log_puts("\r\n");
01065 }
  
```

References [internal_log_header\(\)](#), [internal_log_putint\(\)](#), and [internal_log_puts\(\)](#).

Here is the call graph for this function:

7.21.4.56 `uart_debug_putc()`

```
void uart_debug_putc (
    char data)
```

Referenced by [internal_rx_fatal_error\(\)](#).

Here is the caller graph for this function:

7.21.4.57 `uart_debug_puthex()`

```
void uart_debug_puthex (
    uint32_t value,
    uint8_t digits)
```

Referenced by [internal_rx_fatal_error\(\)](#).

Here is the caller graph for this function:

7.21.4.58 `uart_debug_putint()`

```
void uart_debug_putint (
    int32_t value)
```

7.21.4.59 `uart_debug_puts()`

```
void uart_debug_puts (
    const char * str)
```

Referenced by [internal_rx_fatal_error\(\)](#).

Here is the caller graph for this function:



7.21.4.60 uart_debug_putuint()

```
void uart_debug_putuint (
    uint32_t value)
```

7.22 rx_log.h

[Go to the documentation of this file.](#)

```
00001 /**
00002  * @file rx_log.h
00003  * @brief Type-Safe Logging System for RX72N Firmware with Zero-Overhead Compile-Time Filtering
00004  *
00005  * @details
00006  * ## Overview
00007  * Production-quality logging infrastructure for safety-critical RX72N motor controller firmware.
00008  * Provides type-safe, compile-time filtered logging with zero overhead for disabled log levels.
00009  * Uses C23 `__Generic` for automatic type dispatch and bounded string handling for memory safety.
00010  *
00011  * ## Architecture Design
00012  *
00013  * ### Core Features
00014  * | Feature | Implementation | Benefit |
00015  * |-----|-----|-----|
00016  * | **Zero-overhead filtering** | Compile-time `#if (LOG_LEVEL >= ...)` | Disabled logs = zero code size, zero CPU
00017  * | **Type-safe dispatch** | C23 `__Generic` selection expressions | Automatic type routing, no manual suffixes |
00018  * | **Bounded strings** | Explicit length parameters + compile-time max | Prevents buffer overruns (NASA Rule 3) |
00019  * | **Inline functions** | All implementations are `static inline` | No function call overhead, optimal performance |
00020  * | **Dual output** | Optional USB CDC mirroring | View logs on UART or USB terminal |
00021  * | **Consistent format** | `[LEVEL] [TAG] message` pattern | Easy parsing by log analysis tools |
00022  *
00023  * ### Log Level Hierarchy
00024  * ```
00025  * k_log_none (0)    - No logging (production builds)
00026  *     v
00027  * k_log_error (1)   - Critical failures only
00028  *     v
00029  * k_log_warn (2)    - Recoverable issues
00030  *     v
00031  * k_log_info (3)    - Informational events (default)
00032  *     v
00033  * k_log_debug (4)   - Development debugging
00034  *     v
00035  * k_log_verbose (5) - Trace-level detail
00036  * ```
00037  *
00038  * ### Output Routing
00039  * @dot
00040  * digraph log_routing {
00041  *     rankdir=LR;
00042  *     node [shape=box, style=rounded];
00043  *
00044  *     app [label="Application\nCode"];
00045  *     macro [label="rx_log_*\nMacro"];
00046  *     filter [label="Compile-Time\nFilter", shape=diamond];
00047  *     impl [label="internal_rx_log_*\nInline Function"];
00048  *     uart [label="UART Debug\n(CY7C65213)"];
00049  *     usb [label="USB CDC Port 2\n(Optional)"];
00050  *
00051  *     app -> macro;
00052  *     macro -> filter;
00053  *     filter -> impl [label="enabled"];
00054  *     filter -> eliminated [label="disabled", style=dashed];
00055  *     impl -> uart;
00056  *     impl -> usb [label="if USB_LOG_MIRROR", style=dashed];
00057  *
00058  *     eliminated [label="Eliminated\n(zero code)", shape=ellipse, style=dashed];
00059  * }
00060  * @enddot
00061  *
00062  * ## Performance Characteristics
00063  *
00064  * ### Execution Timing (RX72N @ 240 MHz)
00065  * | Log Type | Cycles | Time | Bytes (UART 115200) |
00066  * |-----|-----|-----|-----|
00067  * | Disabled log | 0 | 0 ns | 0 |
00068  * | rx_log_error() | ~2400 | 10 us | ~30 bytes |
00069  * | rx_log_error_val(uint8_t) | ~3600 | 15 us | ~35 bytes |
00070  * | rx_log_error_hex() | ~4200 | 17.5 us | ~40 bytes |
```

```

00071 * | rx_log_error_str(64 chars) | ~18000 | 75 us | ~95 bytes |
00072 *
00073 * ### Memory Footprint
00074 * | Component | Flash | RAM | Stack |
00075 * |-----|-----|-----|
00076 * | All inline functions | 0 (inlined) | 0 | 0 |
00077 * | Per log site (enabled) | ~60-150 bytes | 0 | 8-16 bytes |
00078 * | Per log site (disabled) | 0 bytes | 0 | 0 |
00079 * | USB CDC buffer (optional) | 0 | 512 bytes | 0 |
00080 *
00081 * ### Thread Safety
00082 * - **NOT thread-safe**: UART/USB output functions are not atomic
00083 * - **Interleaving possible**: Concurrent logs from multiple threads may mix
00084 * - **Mitigation**: Use RTOS mutex around critical log sequences if needed
00085 * - **Design rationale**: Performance over thread safety (motor control is time-critical)
00086 *
00087 * ## C23 _Generic Type Dispatch
00088 *
00089 * The `rx_log_*_val()` macros use C23 `_Generic` selection expressions to automatically
00090 * route values to the correct typed implementation:
00091 *
00092 * ```c
00093 * _Generic((val),
00094 *   uint8_t: internal_rx_log_error_u8,
00095 *   uint16_t: internal_rx_log_error_u16,
00096 *   uint32_t: internal_rx_log_error_u32,
00097 *   int32_t: internal_rx_log_error_err // rx_err_t is int32_t
00098 * )(tag, msg, val)
00099 * ```
00100 *
00101 * **Type Limitation**: Because `rx_err_t` is `typedef int32_t`, `_Generic` cannot distinguish
00102 * between error codes and raw signed integers. Both route to hex formatting. Use explicit
00103 * `rx_log_*_hex()` or cast to `uint32_t` for decimal formatting of signed integers.
00104 *
00105 * ## USB Log Mirroring
00106 *
00107 * Enable with `-DUSB_LOG_MIRROR=1` to send logs to both:
00108 * - **UART**: `/dev/ttyACM0` via CY7C65213 (115200 baud, CDC-ACM class)
00109 * - **USB CDC Port 2**: `/dev/ttyACM2` (log port)
00110 *
00111 * Port assignments:
00112 * | Port | Enum | Purpose |
00113 * |-----|-----|-----|
00114 * | 0 | k_usb_port_proto | Binary protocol (nanopb frames) |
00115 * | 1 | k_usb_port_decoded | Decoded ASCII frame dumps |
00116 * | 2 | k_usb_port_log | Log output (mirrored here) |
00117 *
00118 * ## Configuration
00119 *
00120 * ### Compile-Time Flags
00121 * ```bash
00122 * # Set log level (default: k_log_info in release, k_log_debug in DEBUG)
00123 * -DLOG_LEVEL=k_log_error
00124 *
00125 * # Disable USB mirroring (default: 1, enabled by default)
00126 * -DUSB_LOG_MIRROR=0
00127 * ```
00128 *
00129 * @note USB_LOG_MIRROR=1 is **enabled by default** (Issue #161: USB CDC debug output).
00130 * All logs appear on BOTH UART and USB CDC Port 2. Override with `-DUSB_LOG_MIRROR=0` to disable.
00131 *
00132 * ## Usage Examples
00133 *
00134 * ### Basic Logging
00135 * @code{.c}
00136 * #include "rx_log.h"
00137 *
00138 * void motor_init(void) {
00139 *   rx_log_info("MOTOR", "Starting initialization");
00140 *
00141 *   rx_err_t err = configure_pwm();
00142 *   if (err != k_rx_ok) {
00143 *     rx_log_error_val("MOTOR", "PWM config failed", err); // Hex error code
00144 *     return;
00145 *   }
00146 *
00147 *   rx_log_info("MOTOR", "Initialization complete");
00148 * }
00149 * @endcode
00150 *
00151 * ### Type Dispatch Examples
00152 * @code{.c}
00153 * // Automatic type routing via _Generic
00154 * uint8_t port = 5;
00155 * rx_log_debug_val("GPIO", "Port number", port); // -> _u8 (decimal)
00156 *
00157 * uint16_t rpm = 2100;

```

```

00158 * rx_log_info_val("MOTOR", "Current RPM", rpm); // -> _u16 (decimal)
00159 *
00160 * uint32_t timestamp = get_ticks();
00161 * rx_log_verbose_val("TIMER", "Timestamp", timestamp); // -> _u32 (decimal)
00162 *
00163 * rx_err_t err = k_rx_err_timeout;
00164 * rx_log_error_val("SPI", "Transfer error", err); // -> _err (hex: 0x00000108)
00165 * @endcode
00166 *
00167 * ### Hex and String Logging
00168 * @code{.c}
00169 * // Hex values (explicit digit count)
00170 * uint32_t reg = read_register(0x40000000);
00171 * rx_log_debug_hex("HAL", "Register value", reg, 8); // 8 hex digits
00172 *
00173 * // Bounded string logging (safe)
00174 * const char* device_name = "RX72N Motor Controller";
00175 * rx_log_info_str("SYSTEM", "Device", device_name, strlen(device_name));
00176 * @endcode
00177 *
00178 * ### Conditional Compilation (Zero Overhead)
00179 * @code{.c}
00180 * // If LOG_LEVEL < k_log_debug, this compiles to nothing (zero code)
00181 * rx_log_debug("TRACE", "Entering critical section"); // -> ((void)0) if disabled
00182 * @endcode
00183 *
00184 * ## Design Rationale
00185 *
00186 * ### Why Compile-Time Filtering?
00187 * - **Zero overhead**: Disabled logs generate no code (vs. runtime checks)
00188 * - **Flash savings**: Production builds omit all debug/verbose logs
00189 * - **Performance**: No branching or function call overhead
00190 *
00191 * ### Why _Generic Instead of Function Overloading?
00192 * - **Type safety**: Compiler enforces correct types, catches errors at compile time
00193 * - **No C++**: Pure C implementation (RX72N firmware is C-only)
00194 * - **Clean API**: Single macro name, automatic dispatch
00195 *
00196 * ### Why Bounded Strings?
00197 * - **NASA Rule 2**: Statically provable loop bounds
00198 * - **Safety**: Prevents reading past buffer end (undefined behavior)
00199 * - **Predictable timing**: Maximum execution time is known
00200 *
00201 * ## Dependencies
00202 * - `rx_err.h` - Error code definitions (rx_err_t)
00203 * - `uart.h` (implicit) - uart_debug_* functions for output
00204 * - `rx_usb.h` (optional) - USB CDC output (if USB_LOG_MIRROR=1)
00205 *
00206 * @par NASA Power of 10 Compliance:
00207 * - **Rule 1 [YES]**: No goto, setjmp, recursion (pure inline functions and macros)
00208 * - **Rule 2 [YES]**: All loops bounded (string logging uses k_log_str_max_len constant)
00209 * - **Rule 3 [YES]**: No dynamic allocation (all stack-based, inline functions)
00210 * - **Rule 4 [YES]**: All functions <60 lines (inline helpers are 5-20 lines each)
00211 * - **Rule 5 [YES]**: Minimum 2 assertions per function (nullptr validation in _str functions)
00212 * - **Rule 6 [YES]**: Data at smallest scope (loop counters, temporary variables)
00213 * - **Rule 7 [YES]**: N/A for logging (output functions return void, errors silently dropped)
00214 * - **Rule 8 [YES]**: C23 typed enums for constants, macros only for type dispatch and filtering
00215 * - **Rule 9 [WARN]**: Single-level dereferencing only (string buffers, function pointers avoided)
00216 * - **Rule 10 [YES]**: Compiles with -Wall -Wextra -Werror
00217 *
00218 * @par SOLID Principles:
00219 * - **S (Single Responsibility)**: Only logging output (no error handling, no formatting logic beyond basic types)
00220 * - **O (Open/Closed)**: Extensible via USB_LOG_MIRROR flag, new log levels can be added
00221 * - **L (Liskov Substitution)**: All internal rx_log_* functions have identical signatures per level
00222 * - **I (Interface Segregation)**: Separate macros for different value types (val, hex, str)
00223 * - **D (Dependency Inversion)**: Depends on abstract uart_debug_* interface (not concrete UART driver)
00224 *
00225 * @par Module Dependencies:
00226 * ```
00227 * rx_log.h
00228 * +--> rx_err.h (error code definitions)
00229 * +--> uart.h (uart_debug_* output functions)
00230 * +--> rx_usb.h (optional, if USB_LOG_MIRROR=1)
00231 * ```
00232 *
00233 * @author Locked, Inc.
00234 * @date 2026-01-27
00235 * @copyright Copyright (c) 2026 Locked Inc.
00236 * SPDX-License-Identifier: MIT
00237 *
00238 * @see uart.h for UART debug output implementation
00239 * @see rx_err.h for error code definitions
00240 * @see rx_usb.h for USB CDC output
00241 */
00242
00243 #pragma once
00244

```



```

00245 #include <stdint.h>
00246 #include <string.h>
00247
00248 #include "rx_err.h"
00249
00250 #ifdef __cplusplus
00251 extern "C" {
00252 #endif
00253
00254 /* Forward declarations for UART debug functions (avoid circular dependency) */
00255
00256 /* Include simulator configuration for conditional compilation */
00257 #include "rx_simulator_config.h"
00258
00259 #if RX_IS_SIMULATOR
00260 /*
=====
00261  * Simulator Mode: Logging to stdout
00262  *
=====
00263  * When building for e^2 studio simulator, UART hardware doesn't work.
00264  * Redirect logging to stdout so output appears in the simulator console.
00265  *
00266  * These inline implementations replace the hardware UART functions.
00267  * The simulator console automatically appears in e^2 studio when running
00268  * in debug mode (View -> Console if not visible).
00269  *
00270  * NOTE: These are inline to avoid linking against UART driver in simulator.
00271  *
=====
00272  */
00273
00274 #include <inttypes.h> /* For PRId32, PRIx32 format macros */
00275 #include <stdio.h> /* For putchar(), snprintf() */
00276
00277 /**
00278  * @enum uart_debug_buf_size_t
00279  * @brief Buffer sizes for uart_debug_put* simulator functions
00280  * @since Version 1.0.0
00281  */
00282 typedef enum : uint8_t {
00283     k_uart_debug_int_buf_size = 12U, /**< Buffer for int32 string: "-2147483648\0" = 12 chars */
00284     k_uart_debug_uint_buf_size = 11U, /**< Buffer for uint32 string: "4294967295\0" = 11 chars */
00285     k_uart_debug_hex_buf_size = 11U, /**< Buffer for "0x" + 8 hex digits + null = 11 chars */
00286 } uart_debug_buf_size_t;
00287
00288 /**
00289  * @brief Output character to simulator console
00290  *
00291  * @details
00292  * Simulator implementation: Writes to stdout using putchar().
00293  * Hardware implementation: Writes to SCI9 UART (see uart.c).
00294  *
00295  * @param[in] data Character to output
00296  *
00297  * @note Not thread-safe (hardware UART is also not thread-safe)
00298  * @note Inline function - zero call overhead in simulator builds
00299  *
00300  * @since Version 1.0.0
00301  */
00302 static inline void uart_debug_putc(char data)
00303 {
00304     (void)putchar((int)data);
00305 }
00306
00307 /**
00308  * @brief Output string to simulator console
00309  *
00310  * @details
00311  * Simulator implementation: Writes to stdout with \r\n conversion.
00312  * Matches hardware UART behavior where \n -> \r\n.
00313  *
00314  * @param[in] str Null-terminated string to output (must not be NULL)
00315  *
00316  * @pre str != NULL (checked in hardware build, assumed in simulator)
00317  * @post All characters output to console
00318  *
00319  * @note Not thread-safe
00320  * @note Inline function - zero call overhead in simulator builds
00321  *
00322  * @since Version 1.0.0
00323  */
00324 static inline void uart_debug_puts(const char* str)
00325 {
00326     if (str == NULL) {
00327         return;
00328     }

```

```

00329
00330 while (*str) {
00331     if (*str == '\n') {
00332         (void)putchar('\r'); /* Hardware UART does \n -> \r\n */
00333     }
00334     (void)putchar((int)(*str++));
00335 }
00336 }
00337
00338 /**
00339  * @brief Output signed integer to simulator console
00340  *
00341  * @details
00342  * Simulator implementation: Uses sprintf + puts.
00343  * Hardware implementation: Custom formatting to avoid sprintf overhead.
00344  *
00345  * @param[in] value Signed 32-bit integer to output
00346  *
00347  * @note Inline function - zero call overhead in simulator builds
00348  *
00349  * @since Version 1.0.0
00350  */
00351 static inline void uart_debug_putint(int32_t value)
00352 {
00353     char buf[k_uart_debug_int_buf_size];
00354     (void)sprintf(/* NOLINT(clang-analyzer-security.insecureAPI.DeprecatedOrUnsafeBufferHandling) */
00355                 buf,
00356                 sizeof(buf),
00357                 "%i" PRIi32,
00358                 value);
00359     uart_debug_puts(buf);
00360 }
00361
00362 /**
00363  * @brief Output unsigned integer to simulator console
00364  *
00365  * @details
00366  * Simulator implementation: Uses sprintf + uart_debug_puts.
00367  * Hardware implementation: Custom formatting to avoid sprintf overhead.
00368  *
00369  * @param[in] value Unsigned 32-bit integer to output
00370  *
00371  * @pre value is a valid uint32_t (any value in [0, UINT32_MAX])
00372  * @pre uart_debug subsystem initialized (uart_debug_puts functional)
00373  * @post Decimal ASCII string representation of value written to console
00374  * @post uart_debug_puts invoked with null-terminated decimal string
00375  *
00376  * @note Inline function - zero call overhead in simulator builds
00377  *
00378  * @since Version 1.0.0
00379  */
00380 static inline void uart_debug_putuint(uint32_t value)
00381 {
00382     /* 4294967295 (UINT32_MAX) = 10 digits + null terminator */
00383     char buf[k_uart_debug_uint_buf_size];
00384     (void)sprintf(/* NOLINT(clang-analyzer-security.insecureAPI.DeprecatedOrUnsafeBufferHandling) */
00385                 buf,
00386                 sizeof(buf),
00387                 "%u" PRIu32,
00388                 value);
00389     uart_debug_puts(buf);
00390 }
00391
00392 /**
00393  * @brief Output hexadecimal value to simulator console
00394  *
00395  * @details
00396  * Simulator implementation: Uses sprintf + puts.
00397  * Hardware implementation: Custom formatting to avoid sprintf overhead.
00398  *
00399  * @param[in] value Unsigned 32-bit value to output as hex
00400  * @param[in] digits Number of hex digits to display (1-8)
00401  *
00402  * @note Inline function - zero call overhead in simulator builds
00403  *
00404  * @since Version 1.0.0
00405  */
00406 static inline void uart_debug_puthex(uint32_t value, uint8_t digits)
00407 {
00408     enum : uint8_t { k_max_hex_digits = 8 };
00409     enum : uint8_t { k_min_hex_digits = 1 };
00410
00411     /* Pre-computed format strings keyed by digit count (1-8), avoids runtime
00412      * format building and all implicit-width casts (int, unsigned long). */
00413     static const char* const s_hex_fmts[k_max_hex_digits] = {
00414         "0x%01" PRIx32,
00415         "0x%02" PRIx32,

```

```

00416     "0x%03" PRIx32,
00417     "0x%04" PRIx32,
00418     "0x%05" PRIx32,
00419     "0x%06" PRIx32,
00420     "0x%07" PRIx32,
00421     "0x%08" PRIx32,
00422 };
00423
00424 char buf[k_uart_debug_hex_buf_size]; /* "0x" + 8 hex digits + null */
00425 if (digits < k_min_hex_digits) {
00426     digits = k_min_hex_digits;
00427 }
00428 if (digits > k_max_hex_digits) {
00429     digits = k_max_hex_digits;
00430 }
00431 (void)snprintf(/* NOLINT(clang-analyzer-security.insecureAPI.DeprecatedOrUnsafeBufferHandling) */
00432               buf,
00433               sizeof(buf),
00434               s_hex_fmts[digits - k_min_hex_digits],
00435               value);
00436 uart_debug_puts(buf);
00437 }
00438
00439 #else
00440 /*
=====
00441  * Hardware Mode: UART-framed logging (k_frame_type_log_message frames)
00442  *
=====
00443  * Log bytes are queued in the rx_log_uart ring buffer (zero I/O on the log
00444  * site) and periodically drained by the communication task, which wraps them
00445  * in a TYPE=0x20 (k_frame_type_log_message) frame and transmits over SCI9 via
00446  * the CY7C65213 USB-UART bridge to the Pi5 gateway.
00447  *
00448  * Framing logs prevents log bytes from colliding with the 0x55AA frame sync
00449  * word used by command / telemetry traffic on the same byte stream.
00450  *
=====
00451  */
00452
00453 #include "rx_log_uart.h"
00454
00455 /* uart_debug_* are low-level polling-mode raw-UART writers exported by
00456  * libs/rx_hal/src/uart.c. They bypass the frame protocol and emit raw ASCII
00457  * directly on the SCI9 wire -- the gateway's frame decoder will discard those
00458  * bytes during sync recovery.
00459  *
00460  * Use ONLY for panic / stack-overflow / catastrophic-fault diagnostics where:
00461  * - the ThreadX scheduler is not trustworthy (cannot drain the log ring),
00462  * - the developer is expected to read output via E2 Lite / RTT / a direct
00463  *   serial terminal tapping TX before the bridge chip.
00464  *
00465  * Normal runtime logging MUST go through the rx_log_uart_* API above.
00466  */
00467 void uart_debug_putc(char data);
00468 void uart_debug_puts(const char* str);
00469 void uart_debug_putint(int32_t value);
00470 void uart_debug_putuint(uint32_t value);
00471 void uart_debug_puthex(uint32_t value, uint8_t digits);
00472
00473 #endif /* RX_IS_SIMULATOR */
00474
00475 /*
=====
00476  * Log output macros
00477  *
=====
00478  *
00479  * In simulator builds these go to stdout via uart_debug_* (defined above).
00480  * In hardware builds they go to the rx_log_uart ring buffer, which the
00481  * communication task drains and wraps in k_frame_type_log_message frames for
00482  * transmission over SCI9 -> CY7C65213 -> /dev/ttyACM0.
00483  *
00484  * There is no longer a runtime backend selector. Logs always flow through
00485  * the framed-UART path on hardware.
00486  */
00487
00488 #if RX_IS_SIMULATOR
00489
00490 #define LOG_PUTC(c)    uart_debug_putc((char)(c))
00491 #define LOG_PUTS(s)    uart_debug_puts((s))
00492 #define LOG_PUTINT(v)  uart_debug_putint((int32_t)(v))
00493 #define LOG_PUTUINT(v) uart_debug_putuint((uint32_t)(v))
00494 #define LOG_PUTHEX(v, d) uart_debug_puthex((uint32_t)(v), (uint8_t)(d))
00495
00496 #else
00497

```

```

00498 #define LOG_PUTC(c)      rx_log_uart_putc((char)(c))
00499 #define LOG_PUTS(s)      rx_log_uart_puts((s))
00500 #define LOG_PUTINT(v)    rx_log_uart_putint((int32_t)(v))
00501 #define LOG_PUTUINT(v)   rx_log_uart_putuint((uint32_t)(v))
00502 #define LOG_PUTHEX(v, d) rx_log_uart_puthex((uint32_t)(v), (uint8_t)(d))
00503
00504 #endif /* RX_IS_SIMULATOR */
00505
00506 /*
=====
00507 * Log Level Definitions
00508 *
=====
00509 */
00510
00511 /**
00512  * @enum log_level_t
00513  * @brief Log level enumeration defining severity hierarchy for compile-time filtering
00514  *
00515  * @details
00516  * Defines the log level hierarchy used for compile-time filtering. Each level includes
00517  * all messages from higher-priority (lower-numbered) levels. The `LOG_LEVEL` macro
00518  * determines which levels are compiled into the binary.
00519  *
00520  * ## Hierarchy (Inclusive)
00521  * Each level includes all messages from levels above it:
00522  * - `k_log_error` (1) -> ERROR only
00523  * - `k_log_warn` (2) -> ERROR + WARN
00524  * - `k_log_info` (3) -> ERROR + WARN + INFO
00525  * - `k_log_debug` (4) -> ERROR + WARN + INFO + DEBUG
00526  * - `k_log_verbose` (5) -> ERROR + WARN + INFO + DEBUG + VERBOSE
00527  *
00528  * ## Recommended Usage
00529  * | Build Type | Recommended Level | Rationale |
00530  * |-----|-----|-----|
00531  * | Production | `k_log_error` or `k_log_none` | Minimal overhead, critical failures only |
00532  * | Field Testing | `k_log_warn` | Captures issues without excessive logging |
00533  * | Development | `k_log_info` | Standard development visibility (default) |
00534  * | Debugging | `k_log_debug` | Detailed troubleshooting |
00535  * | Trace Analysis | `k_log_verbose` | Maximum detail for complex issues |
00536  *
00537  * ## Compile-Time Behavior
00538  * @code{.c}
00539  * // Example: LOG_LEVEL = k_log_warn
00540  * rx_log_error("TAG", "msg"); // [OK] Compiled (error <= warn)
00541  * rx_log_warn("TAG", "msg"); // [OK] Compiled (warn <= warn)
00542  * rx_log_info("TAG", "msg"); // [X] Eliminated (info > warn)
00543  * rx_log_debug("TAG", "msg"); // [X] Eliminated (debug > warn)
00544  * rx_log_verbose("TAG", "msg"); // [X] Eliminated (verbose > warn)
00545  * @endcode
00546  *
00547  * ## Configuration
00548  * Set at compile time via `-DLOG_LEVEL=<level>`:
00549  * ```bash
00550  * cmake -DLOG_LEVEL=k_log_error .. # Production
00551  * cmake -DLOG_LEVEL=k_log_debug .. # Development
00552  * ```
00553  *
00554  * Default (if not specified):
00555  * - **DEBUG builds**: `k_log_debug`
00556  * - **Release builds**: `k_log_info`
00557  *
00558  * @par Underlying Type: uint8_t (C23 typed enum)
00559  * - **Range**: 0-255 (only 0-5 used)
00560  * - **Size**: 1 byte
00561  * - **Rationale**: Smallest type sufficient for 6 levels, minimal memory footprint
00562  *
00563  * @see LOG_LEVEL Default log level configuration
00564  * @see rx_log_error(), rx_log_warn(), rx_log_info(), rx_log_debug(), rx_log_verbose()
00565  */
00566 typedef enum : uint8_t {
00567     k_log_none = 0, /**< No logging (production silent mode). All logs eliminated at compile time. */
00568     k_log_error =
00569         1, /**< Critical errors only. Operation-preventing failures, requires immediate attention. */
00570     k_log_warn = 2, /**< Errors and warnings. Recoverable issues, degraded functionality. */
00571     k_log_info =
00572         3, /**< Errors, warnings, and informational messages. Default level for development. */
00573     k_log_debug =
00574         4, /**< Errors, warnings, info, and debug messages. Detailed troubleshooting information. */
00575     k_log_verbose =
00576         5, /**< All messages including verbose trace. Maximum detail for complex debugging. */
00577 } log_level_t;
00578
00579 /**
00580  * @enum log_format_constants_t
00581  * @brief Logging formatting and safety constants
00582  */

```

```

00583 * @details
00584 * Defines compile-time constants for log output formatting and memory safety bounds.
00585 * These constants ensure predictable behavior and prevent buffer overruns.
00586 *
00587 * ## Safety Rationale
00588 *
00589 * ### k_log_hex_width_err = 8
00590 * **Purpose**: Hex digit width for displaying rx_err_t error codes.
00591 *
00592 * **Rationale**:
00593 * - `rx_err_t` is `typedef int32_t` (32 bits)
00594 * - 32 bits = 8 hex digits (4 bits per digit)
00595 * - Example: `0x00000108` (k_rx_err_timeout)
00596 *
00597 * **Usage**: `rx_log_error_val()` with rx_err_t values formats as 8-digit hex.
00598 *
00599 * ### k_log_str_max_len = 256
00600 * **Purpose**: Maximum string length for bounded string logging (NASA Rule 2 compliance).
00601 *
00602 * **Rationale**:
00603 * - **Safety**: Prevents unbounded loops reading past buffer end
00604 * - **Predictability**: Maximum execution time is known (256 x ~15 cycles/char = 3840 cycles)
00605 * - **Practicality**: 256 bytes sufficient for typical log messages
00606 * - **Stack impact**: No stack allocation (loop counter is uint32_t = 4 bytes)
00607 *
00608 * **Behavior**: String logging functions iterate up to `k_log_str_max_len` characters,
00609 * stopping early if:
00610 * 1. User-provided length `len` is reached, OR
00611 * 2. Null terminator `'\0'` is encountered
00612 *
00613 * **Example**:
00614 * @code{.c}
00615 * const char* long_str = "This is a very long string..."; // > 256 chars
00616 * rx_log_info_str("TAG", "Long", long_str, 300); // Only logs first 256 chars
00617 * @endcode
00618 *
00619 * ## Performance Impact
00620 * | Constant | CPU Impact | Memory Impact |
00621 * |-----|-----|-----|
00622 * | k_log_hex_width_err | 8 hex digits ~ 240 cycles @ 240 MHz | 0 bytes (immediate) |
00623 * | k_log_str_max_len | Max 3840 cycles (256 x 15) | 4 bytes (loop counter) |
00624 *
00625 * @par Underlying Type: uint16_t (C23 typed enum)
00626 * - **Range**: 0-65535
00627 * - **Size**: 2 bytes
00628 * - **Rationale**: k_log_str_max_len = 256 fits in uint16_t, uint8_t too small
00629 *
00630 * @see internal_rx_log_error_str(), internal_rx_log_warn_str(), internal_rx_log_info_str()
00631 * @see internal_rx_log_debug_str(), internal_rx_log_verbose_str()
00632 */
00633 typedef enum : uint16_t {
00634     k_log_hex_width_err =
00635         8, /**< Hex digit width for rx_err_t error codes (32-bit = 8 digits). Example: 0x00000108 */
00636     k_log_str_max_len =
00637         256, /**< Maximum string length for bounded logging (NASA Rule 2 compliance). Prevents unbounded loops. */
00638 } log_format_constants_t;
00639
00640 /**
00641 * @brief Default log level configuration
00642 *
00643 * Can be overridden at compile time:
00644 * -DLOG_LEVEL=(k_log_error)
00645 */
00646 #ifndef LOG_LEVEL
00647 #ifdef DEBUG
00648 #define LOG_LEVEL (k_log_debug)
00649 #else
00650 #define LOG_LEVEL (k_log_info)
00651 #endif
00652 #endif
00653
00654 /*
=====
00655 * Internal Output Helpers (thin wrappers to keep per-level functions small)
00656 *
=====
00657 */
00658
00659 /** @brief Write one character to the active log backend(s). @param[in] c Character to write. */
00660 static inline void internal_log_putc(char c)
00661 {
00662     LOG_PUTC(c);
00663 }
00664
00665 /** @brief Write a string to the active log backend(s). @param[in] s Null-terminated string. */
00666 static inline void internal_log_puts(const char* s)
00667 {

```

```

00668 LOG_PUTS(s);
00669 }
00670
00671 /** @brief Write a signed 32-bit integer to the active log backend(s). @param[in] v Value. */
00672 static inline void internal_log_putint(int32_t v)
00673 {
00674     LOG_PUTINT(v);
00675 }
00676
00677 /** @brief Write a hex value to the active log backend(s). @param[in] v Value. @param[in] d Digits. */
00678 static inline void internal_log_puthex(uint32_t v, uint8_t d)
00679 {
00680     LOG_PUTHEX(v, d);
00681 }
00682
00683 /*
=====
00684 * Internal Helper: Format Log Header
00685 *
=====
00686 */
00687
00688 /**
00689 * @brief Internal helper to format and output log message header
00690 *
00691 * @details
00692 * Outputs standardized log header in format: `[LEVEL] [TAG]`
00693 * This prefix precedes all log messages for consistent parsing by log analysis tools.
00694 *
00695 * ## Algorithm
00696 * 1. Validate input pointers (NULL check for safety)
00697 * 2. Output opening bracket `[`
00698 * 3. Output log level string (e.g., "ERROR", "WARN", "INFO")
00699 * 4. Output `[` separator
00700 * 5. Output component tag (e.g., "MOTOR", "SPI", "USB")
00701 * 6. Output `]` (close bracket and space)
00702 *
00703 * ## Output Format
00704 * ```
00705 * [ERROR] [MOTOR] message content here\r\n
00706 * [WARN] [SPI] message content here\r\n
00707 * [INFO] [USB] message content here\r\n
00708 * ```
00709 *
00710 * ## Performance
00711 * - **Cycles**: ~600 cycles @ 240 MHz (depends on string lengths)
00712 * - **Time**: ~2.5 us
00713 * - **UART bytes**: ~15-25 bytes (depends on level_str and tag lengths)
00714 * - **Stack**: 0 bytes (inline, no local variables)
00715 *
00716 * @param[in] level_str Log level string (e.g., "ERROR", "WARN", "INFO", "DEBUG", "VERBOSE")
00717 * - Must be null-terminated string literal
00718 * - Typically 4-7 characters
00719 * - nullptr silently ignored (safety)
00720 * @param[in] tag Component tag identifying log source (e.g., "MOTOR", "SPI", "I2C")
00721 * - Must be null-terminated string literal
00722 * - Recommended: 3-8 uppercase characters
00723 * - nullptr silently ignored (safety)
00724 *
00725 * @pre level_str and tag should be valid null-terminated strings
00726 * @post Log header written to configured output (UART and/or USB CDC)
00727 *
00728 * @note **Thread Safety**: NOT thread-safe. Concurrent calls may interleave output.
00729 * @note **Inline**: Always inlined for zero function call overhead.
00730 * @note **Error Handling**: NULL pointers silently ignored (safe failure mode).
00731 *
00732 * @par Example:
00733 * @code{.c}
00734 * // Called internally by all log functions
00735 * internal_log_header("ERROR", "MOTOR"); // Outputs: "[ERROR] [MOTOR] "
00736 * LOG_PUTS("PWM init failed\r\n"); // Complete message
00737 *
00738 * // Result on terminal:
00739 * // [ERROR] [MOTOR] PWM init failed
00740 * @endcode
00741 *
00742 * @see LOG_PUTC(), LOG_PUTS() Output macros (UART or UART+USB)
00743 * @see internal_rx_log_error(), internal_rx_log_warn(), internal_rx_log_info()
00744 */
00745 static inline void internal_log_header(const char* level_str, const char* tag)
00746 {
00747     /* Pre-condition: Validate pointers (NASA Power of 10 Rule 5).
00748     * The level_str side of the || is defensive-only: every caller passes a
00749     * compile-time string literal ("ERROR", "WARN", ...), so the TRUE branch is
00750     * architecturally unreachable. Excluded from gcovr branch coverage with an
00751     * EXCL marker so the 100% libs/ gate remains valid. */
00752     /* GCOVR_EXCL_BR_START */

```

```

00753 if (level_str == (const char*)0 || tag == (const char*)0) {
00754     return;
00755 }
00756 /* GCOVR_EXCL_BR_STOP */
00757
00758 internal_log_putc(' ');
00759 internal_log_puts(level_str);
00760 internal_log_puts("] ");
00761 internal_log_puts(tag);
00762 internal_log_puts("] ");
00763 }
00764
00765 /*
=====
00766 * ERROR Level - Internal Implementations
00767 *
=====
00768 */
00769
00770 /**
00771 * @brief Log ERROR-level message without additional value
00772 *
00773 * @details
00774 * Outputs critical error message with standardized header format. Used for
00775 * operation-preventing failures that require immediate attention.
00776 *
00777 * ## Algorithm
00778 * 1. Validate message pointer (NULL check)
00779 * 2. Output header: `[ERROR] [TAG] `
00780 * 3. Output message text
00781 * 4. Output line terminator `\r\n`
00782 *
00783 * ## Output Format
00784 * ```
00785 * [ERROR] [TAG] message text CRLF
00786 * ```
00787 *
00788 * ## Performance
00789 * | Metric | Value |
00790 * |-----|-----|
00791 * | Cycles | ~2400 @ 240 MHz (30-char message) |
00792 * | Time | ~10 us |
00793 * | UART bytes | ~45 bytes (header + message + CRLF) |
00794 * | Stack | 0 bytes (inline) |
00795 *
00796 * @param[in] tag Component tag identifying error source (e.g., "MOTOR", "SPI")
00797 * - Must be null-terminated string literal
00798 * - Recommended: 3-8 uppercase characters
00799 * - nullptr silently ignored
00800 * @param[in] message Error message describing the failure
00801 * - Must be null-terminated string
00802 * - Should be concise and actionable
00803 * - nullptr causes silent return (safety)
00804 *
00805 * @pre message must be valid null-terminated string or nullptr
00806 * @post Error message written to UART (and USB CDC if USB_LOG_MIRROR=1)
00807 *
00808 * @note **Thread Safety**: NOT thread-safe. Concurrent calls may interleave.
00809 * @note **Inline**: Always inlined for zero function call overhead.
00810 * @note **NULL Handling**: NULL message pointer silently ignored (safe failure).
00811 *
00812 * @par Example Usage:
00813 * @code{.c}
00814 * // Critical initialization failure
00815 * if (!motor_init()) {
00816 *     internal_rx_log_error("MOTOR", "Initialization failed");
00817 * }
00818 *
00819 * // Hardware failure
00820 * if (!spi_ready()) {
00821 *     internal_rx_log_error("SPI", "Peripheral not responding");
00822 * }
00823 *
00824 * // Output on terminal:
00825 * // [ERROR] [MOTOR] Initialization failed
00826 * // [ERROR] [SPI] Peripheral not responding
00827 * @endcode
00828 *
00829 * @see rx_log_error() Public API macro (compile-time filtered)
00830 * @see internal_log_header() Header formatting function
00831 */
00832 static inline void internal_rx_log_error(const char* tag, const char* message)
00833 {
00834     /* Rule 5: Pre-condition validation - NULL guard */
00835     if (message == (const char*)0) {
00836         return;
00837     }

```



```

00838 internal_log_header("ERROR", tag);
00839 internal_log_puts(message);
00840 internal_log_puts("\r\n");
00841 }
00842
00843 /**
00844  * @brief Log error with uint8_t value
00845  */
00846 static inline void internal_rx_log_error_u8(const char* tag, const char* message, uint8_t value)
00847 {
00848     internal_log_header("ERROR", tag);
00849     internal_log_puts(message);
00850     internal_log_puts(": ");
00851     internal_log_putint((int32_t)value);
00852     internal_log_puts("\r\n");
00853 }
00854
00855 /**
00856  * @brief Log error with uint16_t value
00857  */
00858 static inline void internal_rx_log_error_u16(const char* tag, const char* message, uint16_t value)
00859 {
00860     internal_log_header("ERROR", tag);
00861     internal_log_puts(message);
00862     internal_log_puts(": ");
00863     internal_log_putint((int32_t)value);
00864     internal_log_puts("\r\n");
00865 }
00866
00867 /**
00868  * @brief Log error with uint32_t value
00869  */
00870 static inline void internal_rx_log_error_u32(const char* tag, const char* message, uint32_t value)
00871 {
00872     internal_log_header("ERROR", tag);
00873     internal_log_puts(message);
00874     internal_log_puts(": ");
00875     internal_log_putint((int32_t)value);
00876     internal_log_puts("\r\n");
00877 }
00878
00879 /**
00880  * @brief Log error with int32_t value
00881  */
00882 static inline void internal_rx_log_error_i32(const char* tag, const char* message, int32_t value)
00883 {
00884     internal_log_header("ERROR", tag);
00885     internal_log_puts(message);
00886     internal_log_puts(": ");
00887     internal_log_putint(value);
00888     internal_log_puts("\r\n");
00889 }
00890
00891 /**
00892  * @brief Log error with error code (rx_err_t)
00893  */
00894 static inline void internal_rx_log_error_err(const char* tag, const char* message, rx_err_t err)
00895 {
00896     internal_log_header("ERROR", tag);
00897     internal_log_puts(message);
00898     internal_log_puts(": 0x");
00899     internal_log_puthex((uint32_t)err, k_log_hex_width_err);
00900     internal_log_puts("\r\n");
00901 }
00902
00903 /**
00904  * @brief Log error with hex value
00905  */
00906 static inline void
00907 internal_rx_log_error_hex(const char* tag, const char* message, uint32_t value, uint8_t digits)
00908 {
00909     internal_log_header("ERROR", tag);
00910     internal_log_puts(message);
00911     internal_log_puts(": 0x");
00912     internal_log_puthex(value, digits);
00913     internal_log_puts("\r\n");
00914 }
00915
00916 /**
00917  * @brief Log ERROR-level message with bounded string value (NASA-compliant)
00918  *
00919  * @details
00920  * Outputs critical error message with associated string value using bounded iteration
00921  * for memory safety. String output is limited to prevent unbounded loops and buffer
00922  * overruns (NASA Power of 10 Rule 2 compliance).
00923  *
00924  * ## Algorithm

```



```

00925 * 1. Validate message and str_value pointers (NULL checks)
00926 * 2. Output header: `[ERROR] [TAG]`
00927 * 3. Output message text
00928 * 4. Output separator: `:`
00929 * 5. Iterate up to k_log_str_max_len (256) characters:
00930 *   a. Stop if index >= user-provided len
00931 *   b. Stop if null terminator '\0' encountered
00932 *   c. Output character
00933 * 6. Output line terminator '\r\n'
00934 *
00935 * ## Output Format
00936 * ```
00937 * [ERROR] [TAG] message: string_value\r\n
00938 * ```
00939 *
00940 * ## Safety Features (NASA Rule 2)
00941 * | Feature | Implementation | Benefit |
00942 * |-----|-----|-----|
00943 * | Bounded loop | `i < k_log_str_max_len` | Statically provable upper bound |
00944 * | Early termination | `i >= len` check | Respects caller's length limit |
00945 * | Null handling | `str_value[i] == '\0'` | Safe for null-terminated strings |
00946 * | Pointer validation | NULL checks at start | Prevents crashes |
00947 *
00948 * ## Performance
00949 * | Metric | Best Case | Worst Case |
00950 * |-----|-----|-----|
00951 * | Cycles | ~2700 (1 char) | ~21600 (256 chars) @ 240 MHz |
00952 * | Time | ~11 us | ~90 us |
00953 * | UART bytes | ~50 bytes | ~300 bytes |
00954 * | Stack | 4 bytes (uint32_t i) | 4 bytes |
00955 *
00956 * @param[in] tag Component tag identifying error source (e.g., "MOTOR", "SPI")
00957 *   - Must be null-terminated string literal
00958 *   - nullptr silently ignored
00959 * @param[in] message Error message describing the context
00960 *   - Must be null-terminated string
00961 *   - nullptr causes silent return (safety)
00962 * @param[in] str_value String value to log (e.g., parameter name, device ID)
00963 *   - Must point to valid memory
00964 *   - nullptr causes silent return (safety)
00965 *   - Does not need to be null-terminated if len is exact
00966 * @param[in] len Maximum number of characters to log from str_value
00967 *   - Typically obtained via `strlen(str_value)`
00968 *   - Values > k_log_str_max_len will be clamped to k_log_str_max_len
00969 *   - Use exact buffer length for non-null-terminated buffers
00970 *
00971 * @pre message and str_value must be valid pointers or nullptr
00972 * @pre len must not exceed actual buffer size of str_value
00973 * @post Error message with string value written to output (if parameters valid)
00974 *
00975 * @note **Thread Safety**: NOT thread-safe. Concurrent calls may interleave output.
00976 * @note **Inline**: Always inlined for zero function call overhead.
00977 * @note **Bounded**: Maximum 256 characters logged regardless of len parameter.
00978 * @note **NULL Safety**: NULL pointers silently ignored (safe failure mode).
00979 *
00980 * @warning If str_value buffer is smaller than len, undefined behavior may occur.
00981 *   Always ensure len <= actual buffer size.
00982 *
00983 * @par Example: Logging Configuration Errors
00984 * @code{.c}
00985 * // Unknown configuration parameter
00986 * const char* param_name = "invalid_motor_id";
00987 * internal_rx_log_error_str("CONFIG", "Unknown parameter", param_name, strlen(param_name));
00988 * // Output: [ERROR] [CONFIG] Unknown parameter: invalid_motor_id
00989 *
00990 * // Non-null-terminated buffer (use exact length)
00991 * char device_id[16] = {0x01, 0x02, 0x03, ...}; // Binary data
00992 * internal_rx_log_error_str("USB", "Invalid device ID", device_id, 16);
00993 * // Output: [ERROR] [USB] Invalid device ID: <binary data>
00994 * @endcode
00995 *
00996 * @par Example: Truncation Behavior
00997 * @code{.c}
00998 * // String longer than k_log_str_max_len (256)
00999 * char long_str[300];
01000 * memset(long_str, 'A', 299);
01001 * long_str[299] = '\0';
01002 *
01003 * internal_rx_log_error_str("TEST", "Long string", long_str, 299);
01004 * // Output: [ERROR] [TEST] Long string: AAAA... (256 chars max, truncated)
01005 * @endcode
01006 *
01007 * @see k_log_str_max_len Maximum string length constant (256 bytes)
01008 * @see rx_log_error_str() Public API macro (compile-time filtered)
01009 * @see internal_rx_log_error() Simple error logging without value
01010 */
01011 static inline void

```

```

01012 internal_rx_log_error_str(const char* tag, const char* message, const char* str_value, uint32_t len)
01013 {
01014     /* Pre-condition: Validate pointers (NASA Power of 10 Rule 5) */
01015     if (message == (const char*)0 || str_value == (const char*)0) {
01016         return;
01017     }
01018     internal_log_header("ERROR", tag);
01019     internal_log_puts(message);
01020     internal_log_puts(": ");
01021
01022     /*
01023      * NASA Rule 2: Loop bounded by k_log_str_max_len (compile-time constant)
01024      * Early break when i >= len or null terminator encountered.
01025      */
01026     for (uint32_t i = 0; i < k_log_str_max_len; i++) {
01027         if (i >= len || str_value[i] == '\0') {
01028             break;
01029         }
01030         internal_log_putc(str_value[i]);
01031     }
01032
01033     internal_log_puts("\r\n");
01034 }
01035
01036 /*
=====
01037 * WARN Level - Internal Implementations
01038 *
=====
01039 */
01040
01041 /**
01042  * @brief Log warning message (no value)
01043  */
01044 static inline void internal_rx_log_warn(const char* tag, const char* message)
01045 {
01046     /* Rule 5: Pre-condition validation - NULL guard */
01047     if (message == (const char*)0) {
01048         return;
01049     }
01050     internal_log_header("WARN", tag);
01051     internal_log_puts(message);
01052     internal_log_puts("\r\n");
01053 }
01054
01055 /**
01056  * @brief Log warning with uint8_t value
01057  */
01058 static inline void internal_rx_log_warn_u8(const char* tag, const char* message, uint8_t value)
01059 {
01060     internal_log_header("WARN", tag);
01061     internal_log_puts(message);
01062     internal_log_puts(": ");
01063     internal_log_putint((int32_t)value);
01064     internal_log_puts("\r\n");
01065 }
01066
01067 /**
01068  * @brief Log warning with uint16_t value
01069  */
01070 static inline void internal_rx_log_warn_u16(const char* tag, const char* message, uint16_t value)
01071 {
01072     internal_log_header("WARN", tag);
01073     internal_log_puts(message);
01074     internal_log_puts(": ");
01075     internal_log_putint((int32_t)value);
01076     internal_log_puts("\r\n");
01077 }
01078
01079 /**
01080  * @brief Log warning with uint32_t value
01081  */
01082 static inline void internal_rx_log_warn_u32(const char* tag, const char* message, uint32_t value)
01083 {
01084     internal_log_header("WARN", tag);
01085     internal_log_puts(message);
01086     internal_log_puts(": ");
01087     internal_log_putint((int32_t)value);
01088     internal_log_puts("\r\n");
01089 }
01090
01091 /**
01092  * @brief Log warning with int32_t value
01093  */
01094 static inline void internal_rx_log_warn_i32(const char* tag, const char* message, int32_t value)
01095 {
01096     internal_log_header("WARN", tag);

```

```

01097 internal_log_puts(message);
01098 internal_log_puts(": ");
01099 internal_log_putint(value);
01100 internal_log_puts("\r\n");
01101 }
01102
01103 /**
01104  * @brief Log warning with error code (rx_err_t)
01105  */
01106 static inline void internal_rx_log_warn_err(const char* tag, const char* message, rx_err_t err)
01107 {
01108     internal_log_header("WARN", tag);
01109     internal_log_puts(message);
01110     internal_log_puts(": 0x");
01111     internal_log_puthex((uint32_t)err, k_log_hex_width_err);
01112     internal_log_puts("\r\n");
01113 }
01114
01115 /**
01116  * @brief Log warning with hex value
01117  */
01118 static inline void
01119 internal_rx_log_warn_hex(const char* tag, const char* message, uint32_t value, uint8_t digits)
01120 {
01121     internal_log_header("WARN", tag);
01122     internal_log_puts(message);
01123     internal_log_puts(": 0x");
01124     internal_log_puthex(value, digits);
01125     internal_log_puts("\r\n");
01126 }
01127
01128 /**
01129  * @brief Log warning with string value (bounded length)
01130  */
01131 static inline void
01132 internal_rx_log_warn_str(const char* tag, const char* message, const char* str_value, uint32_t len)
01133 {
01134     /* Pre-condition: Validate pointers (NASA Power of 10 Rule 5) */
01135     if (message == (const char*)0 || str_value == (const char*)0) {
01136         return;
01137     }
01138
01139     internal_log_header("WARN", tag);
01140     internal_log_puts(message);
01141     internal_log_puts(": ");
01142
01143     /*
01144      * NASA Rule 2: Loop bounded by k_log_str_max_len (compile-time constant)
01145      * Early break when i >= len or null terminator encountered.
01146      */
01147     for (uint32_t i = 0; i < k_log_str_max_len; i++) {
01148         if (i >= len || str_value[i] == '\0') {
01149             break;
01150         }
01151         internal_log_putc(str_value[i]);
01152     }
01153
01154     internal_log_puts("\r\n");
01155 }
01156
01157 /*
=====
01158 * INFO Level - Internal Implementations
01159 *
=====
01160 */
01161
01162 /**
01163  * @brief Log info message (no value)
01164  */
01165 static inline void internal_rx_log_info(const char* tag, const char* message)
01166 {
01167     /* Rule 5: Pre-condition validation - NULL guard */
01168     if (message == (const char*)0) {
01169         return;
01170     }
01171     internal_log_header("INFO", tag);
01172     internal_log_puts(message);
01173     internal_log_puts("\r\n");
01174 }
01175
01176 /**
01177  * @brief Log info with uint8_t value
01178  */
01179 static inline void internal_rx_log_info_u8(const char* tag, const char* message, uint8_t value)
01180 {
01181     internal_log_header("INFO", tag);

```

```

01182 internal_log_puts(message);
01183 internal_log_puts(": ");
01184 internal_log_putint((int32_t)value);
01185 internal_log_puts("\r\n");
01186 }
01187
01188 /**
01189  * @brief Log info with uint16_t value
01190  */
01191 static inline void internal_rx_log_info_u16(const char* tag, const char* message, uint16_t value)
01192 {
01193     internal_log_header("INFO", tag);
01194     internal_log_puts(message);
01195     internal_log_puts(": ");
01196     internal_log_putint((int32_t)value);
01197     internal_log_puts("\r\n");
01198 }
01199
01200 /**
01201  * @brief Log info with uint32_t value
01202  */
01203 static inline void internal_rx_log_info_u32(const char* tag, const char* message, uint32_t value)
01204 {
01205     internal_log_header("INFO", tag);
01206     internal_log_puts(message);
01207     internal_log_puts(": ");
01208     internal_log_putint((int32_t)value);
01209     internal_log_puts("\r\n");
01210 }
01211
01212 /**
01213  * @brief Log info with int32_t value
01214  */
01215 static inline void internal_rx_log_info_i32(const char* tag, const char* message, int32_t value)
01216 {
01217     internal_log_header("INFO", tag);
01218     internal_log_puts(message);
01219     internal_log_puts(": ");
01220     internal_log_putint(value);
01221     internal_log_puts("\r\n");
01222 }
01223
01224 /**
01225  * @brief Log info with error code (rx_err_t)
01226  */
01227 static inline void internal_rx_log_info_err(const char* tag, const char* message, rx_err_t err)
01228 {
01229     internal_log_header("INFO", tag);
01230     internal_log_puts(message);
01231     internal_log_puts(": 0x");
01232     internal_log_puthex((uint32_t)err, k_log_hex_width_err);
01233     internal_log_puts("\r\n");
01234 }
01235
01236 /**
01237  * @brief Log info with hex value
01238  */
01239 static inline void
01240 internal_rx_log_info_hex(const char* tag, const char* message, uint32_t value, uint8_t digits)
01241 {
01242     internal_log_header("INFO", tag);
01243     internal_log_puts(message);
01244     internal_log_puts(": 0x");
01245     internal_log_puthex(value, digits);
01246     internal_log_puts("\r\n");
01247 }
01248
01249 /**
01250  * @brief Log info with string value (bounded length)
01251  */
01252 static inline void
01253 internal_rx_log_info_str(const char* tag, const char* message, const char* str_value, uint32_t len)
01254 {
01255     /* Pre-condition: Validate pointers (NASA Power of 10 Rule 5) */
01256     if (message == (const char*)0 || str_value == (const char*)0) {
01257         return;
01258     }
01259     internal_log_header("INFO", tag);
01260     internal_log_puts(message);
01261     internal_log_puts(": ");
01262     /*
01263      * NASA Rule 2: Loop bounded by k_log_str_max_len (compile-time constant)
01264      * Early break when i >= len or null terminator encountered.
01265      */
01266     for (uint32_t i = 0; i < k_log_str_max_len; i++) {

```

```

01269     if (i >= len || str_value[i] == '\0') {
01270         break;
01271     }
01272     internal_log_putc(str_value[i]);
01273 }
01274
01275 internal_log_puts("\r\n");
01276 }
01277
01278 /*
=====
01279  * DEBUG Level - Internal Implementations
01280  *
=====
01281  */
01282
01283 /**
01284  * @brief Log debug message (no value)
01285  */
01286 static inline void internal_rx_log_debug(const char* tag, const char* message)
01287 {
01288     /* Rule 5: Pre-condition validation - NULL guard */
01289     if (message == (const char*)0) {
01290         return;
01291     }
01292     internal_log_header("DEBUG", tag);
01293     internal_log_puts(message);
01294     internal_log_puts("\r\n");
01295 }
01296
01297 /**
01298  * @brief Log debug with uint8_t value
01299  */
01300 static inline void internal_rx_log_debug_u8(const char* tag, const char* message, uint8_t value)
01301 {
01302     internal_log_header("DEBUG", tag);
01303     internal_log_puts(message);
01304     internal_log_puts(": ");
01305     internal_log_putint((int32_t)value);
01306     internal_log_puts("\r\n");
01307 }
01308
01309 /**
01310  * @brief Log debug with uint16_t value
01311  */
01312 static inline void internal_rx_log_debug_u16(const char* tag, const char* message, uint16_t value)
01313 {
01314     internal_log_header("DEBUG", tag);
01315     internal_log_puts(message);
01316     internal_log_puts(": ");
01317     internal_log_putint((int32_t)value);
01318     internal_log_puts("\r\n");
01319 }
01320
01321 /**
01322  * @brief Log debug with uint32_t value
01323  */
01324 static inline void internal_rx_log_debug_u32(const char* tag, const char* message, uint32_t value)
01325 {
01326     internal_log_header("DEBUG", tag);
01327     internal_log_puts(message);
01328     internal_log_puts(": ");
01329     internal_log_putint((int32_t)value);
01330     internal_log_puts("\r\n");
01331 }
01332
01333 /**
01334  * @brief Log debug with int32_t value
01335  */
01336 static inline void internal_rx_log_debug_i32(const char* tag, const char* message, int32_t value)
01337 {
01338     internal_log_header("DEBUG", tag);
01339     internal_log_puts(message);
01340     internal_log_puts(": ");
01341     internal_log_putint(value);
01342     internal_log_puts("\r\n");
01343 }
01344
01345 /**
01346  * @brief Log debug with error code (rx_err_t)
01347  */
01348 static inline void internal_rx_log_debug_err(const char* tag, const char* message, rx_err_t err)
01349 {
01350     internal_log_header("DEBUG", tag);
01351     internal_log_puts(message);
01352     internal_log_puts(": 0x");
01353     internal_log_puthex((uint32_t)err, k_log_hex_width_err);

```

```

01354 internal_log_puts("\r\n");
01355 }
01356
01357 /**
01358  * @brief Log debug with hex value
01359  */
01360 static inline void
01361 internal_rx_log_debug_hex(const char* tag, const char* message, uint32_t value, uint8_t digits)
01362 {
01363     internal_log_header("DEBUG", tag);
01364     internal_log_puts(message);
01365     internal_log_puts(": 0x");
01366     internal_log_puthex(value, digits);
01367     internal_log_puts("\r\n");
01368 }
01369
01370 /**
01371  * @brief Log debug with string value (bounded length)
01372  */
01373 static inline void
01374 internal_rx_log_debug_str(const char* tag, const char* message, const char* str_value, uint32_t len)
01375 {
01376     /* Pre-condition: Validate pointers (NASA Power of 10 Rule 5) */
01377     if (message == (const char*)0 || str_value == (const char*)0) {
01378         return;
01379     }
01380
01381     internal_log_header("DEBUG", tag);
01382     internal_log_puts(message);
01383     internal_log_puts(": ");
01384
01385     /*
01386      * NASA Rule 2: Loop bounded by k_log_str_max_len (compile-time constant)
01387      * Early break when i >= len or null terminator encountered.
01388      */
01389     for (uint32_t i = 0; i < k_log_str_max_len; i++) {
01390         if (i >= len || str_value[i] == '\0') {
01391             break;
01392         }
01393         internal_log_putc(str_value[i]);
01394     }
01395
01396     internal_log_puts("\r\n");
01397 }
01398
01399 /*
=====
01400 * VERBOSE Level - Internal Implementations
01401 *
=====
01402 */
01403
01404 /**
01405  * @brief Log verbose message (no value)
01406  */
01407 static inline void internal_rx_log_verbose(const char* tag, const char* message)
01408 {
01409     /* Rule 5: Pre-condition validation - NULL guard */
01410     if (message == (const char*)0) {
01411         return;
01412     }
01413     internal_log_header("VERBOSE", tag);
01414     internal_log_puts(message);
01415     internal_log_puts("\r\n");
01416 }
01417
01418 /**
01419  * @brief Log verbose with uint8_t value
01420  */
01421 static inline void internal_rx_log_verbose_u8(const char* tag, const char* message, uint8_t value)
01422 {
01423     internal_log_header("VERBOSE", tag);
01424     internal_log_puts(message);
01425     internal_log_puts(": ");
01426     internal_log_putint((int32_t)value);
01427     internal_log_puts("\r\n");
01428 }
01429
01430 /**
01431  * @brief Log verbose with uint16_t value
01432  */
01433 static inline void internal_rx_log_verbose_u16(const char* tag, const char* message, uint16_t value)
01434 {
01435     internal_log_header("VERBOSE", tag);
01436     internal_log_puts(message);
01437     internal_log_puts(": ");
01438     internal_log_putint((int32_t)value);

```

```

01439 internal_log_puts("\r\n");
01440 }
01441
01442 /**
01443  * @brief Log verbose with uint32_t value
01444  */
01445 static inline void internal_rx_log_verbose_u32(const char* tag, const char* message, uint32_t value)
01446 {
01447     internal_log_header("VERBOSE", tag);
01448     internal_log_puts(message);
01449     internal_log_puts(": ");
01450     internal_log_putint((int32_t)value);
01451     internal_log_puts("\r\n");
01452 }
01453
01454 /**
01455  * @brief Log verbose with int32_t value
01456  */
01457 static inline void internal_rx_log_verbose_i32(const char* tag, const char* message, int32_t value)
01458 {
01459     internal_log_header("VERBOSE", tag);
01460     internal_log_puts(message);
01461     internal_log_puts(": ");
01462     internal_log_putint(value);
01463     internal_log_puts("\r\n");
01464 }
01465
01466 /**
01467  * @brief Log verbose with error code (rx_err_t)
01468  */
01469 static inline void internal_rx_log_verbose_err(const char* tag, const char* message, rx_err_t err)
01470 {
01471     internal_log_header("VERBOSE", tag);
01472     internal_log_puts(message);
01473     internal_log_puts(": 0x");
01474     internal_log_puthex((uint32_t)err, k_log_hex_width_err);
01475     internal_log_puts("\r\n");
01476 }
01477
01478 /**
01479  * @brief Log verbose with hex value
01480  */
01481 static inline void
01482 internal_rx_log_verbose_hex(const char* tag, const char* message, uint32_t value, uint8_t digits)
01483 {
01484     internal_log_header("VERBOSE", tag);
01485     internal_log_puts(message);
01486     internal_log_puts(": 0x");
01487     internal_log_puthex(value, digits);
01488     internal_log_puts("\r\n");
01489 }
01490
01491 /**
01492  * @brief Log verbose with string value (bounded length)
01493  */
01494 static inline void internal_rx_log_verbose_str(const char* tag,
01495                                               const char* message,
01496                                               const char* str_value,
01497                                               uint32_t len)
01498 {
01499     /* Pre-condition: Validate pointers (NASA Power of 10 Rule 5) */
01500     if (message == (const char*)0 || str_value == (const char*)0) {
01501         return;
01502     }
01503
01504     internal_log_header("VERBOSE", tag);
01505     internal_log_puts(message);
01506     internal_log_puts(": ");
01507
01508     /*
01509      * NASA Rule 2: Loop bounded by k_log_str_max_len (compile-time constant)
01510      * Early break when i >= len or null terminator encountered.
01511      */
01512     for (uint32_t i = 0; i < k_log_str_max_len; i++) {
01513         if (i >= len || str_value[i] == '\0') {
01514             break;
01515         }
01516         internal_log_putc(str_value[i]);
01517     }
01518
01519     internal_log_puts("\r\n");
01520 }
01521
01522 /**
=====
01523  * C23 _Generic Type Dispatch Helpers
01524  */

```

```

=====
01525 */
01526
01527 /**
01528  * @def RX_LOG_VAL_IMPL
01529  * @brief Internal macro implementing C23 _Generic automatic type dispatch for logging
01530  *
01531  * @details
01532  * Uses C23 `_Generic` selection expressions to automatically route log calls to the
01533  * correct typed implementation function based on the value's type at compile time.
01534  * This provides type-safe logging without manual type suffixes.
01535  *
01536  * ### Type Dispatch Mechanism
01537  *
01538  * The `_Generic` keyword performs compile-time type matching:
01539  * ```c
01540  * _Generic((val),
01541  *   uint8_t: internal_rx_log_error_u8, // Match: uint8_t -> decimal
01542  *   uint16_t: internal_rx_log_error_u16, // Match: uint16_t -> decimal
01543  *   uint32_t: internal_rx_log_error_u32, // Match: uint32_t -> decimal
01544  *   int32_t: internal_rx_log_error_err // Match: int32_t -> hex (error codes)
01545  * )(tag, msg, val) // Call selected function
01546  * ```
01547  *
01548  * ### Supported Types
01549  * | Type | Output Format | Example Value | Example Output |
01550  * |-----|-----|-----|-----|
01551  * | uint8_t | Decimal | 127 | `127` |
01552  * | uint16_t | Decimal | 2048 | `2048` |
01553  * | uint32_t | Decimal | 4294967295 | `4294967295` |
01554  * | int32_t (rx_err_t) | Hex (8 digits) | 0x108 | `0x00000108` |
01555  *
01556  * ### Type Limitation: int32_t vs rx_err_t
01557  *
01558  * **Critical Design Constraint**:
01559  * Because `rx_err_t` is defined as `typedef int32_t`, C's `_Generic` cannot distinguish
01560  * between error codes and raw signed integers (both have type `int32_t`).
01561  *
01562  * **Consequence**:
01563  * - [OK] rx_err_t values display correctly as hex error codes
01564  * - [X] Raw int32_t values display as hex (unexpected for signed integers)
01565  *
01566  * **Workaround** for logging signed integers:
01567  * @code{.c}
01568  * int32_t temperature = -25; // Negative temperature
01569  *
01570  * // WRONG: Will display as hex 0xFFFFF7E7 (two's complement)
01571  * rx_log_info_val("SENSOR", "Temperature", temperature);
01572  *
01573  * // CORRECT: Cast to unsigned for decimal display
01574  * rx_log_info_val("SENSOR", "Temperature", (uint32_t)temperature); // Displays: 4294967271
01575  *
01576  * // ALTERNATIVE: Use explicit hex logging
01577  * rx_log_info_hex("SENSOR", "Temperature (raw)", temperature, 8); // Displays: 0xFFFFF7E7
01578  * @endcode
01579  *
01580  * ### Macro Expansion Example
01581  * @code{.c}
01582  * uint8_t status = 42;
01583  * rx_log_error_val("MOTOR", "Status", status);
01584  *
01585  * // Step 1: Expands to RX_LOG_VAL_IMPL(error, "MOTOR", "Status", status)
01586  * // Step 2: _Generic matches uint8_t -> selects internal_rx_log_error_u8
01587  * // Step 3: Calls internal_rx_log_error_u8("MOTOR", "Status", 42)
01588  * // Output: [ERROR] [MOTOR] Status: 42
01589  * @endcode
01590  *
01591  * ### Performance
01592  * - **Overhead**: Zero. Type selection occurs at compile time (no runtime cost).
01593  * - **Code Size**: Identical to calling typed function directly.
01594  * - **Compile Time**: Negligible increase for type resolution.
01595  *
01596  * ### Advantages Over Manual Type Suffixes
01597  * | Approach | Example | Type Safety | Code Clarity |
01598  * |-----|-----|-----|-----|
01599  * | Manual suffixes | `log_error_u8(tag, msg, val)` | [X] Mismatch possible | [X] Verbose |
01600  * | _Generic dispatch | `log_error_val(tag, msg, val)` | [OK] Compile-time checked | [OK] Clean |
01601  *
01602  * @param[in] level Log level name (error, warn, info, debug, verbose) - token pasting
01603  * @param[in] tag Component tag string (passed to selected function)
01604  * @param[in] msg Log message string (passed to selected function)
01605  * @param[in] val Value to log - type determines function selection at compile time
01606  *
01607  * @note **Compile-Time**: Type selection occurs during compilation, not at runtime.
01608  * @note **Type Safety**: Compiler enforces supported types; unsupported types cause errors.
01609  * @note **Token Pasting**: Uses `###` to construct function names dynamically.
01610  *

```



```

01611 * @warning Only supports uint8_t, uint16_t, uint32_t, and int32_t (rx_err_t).
01612 *      Passing other types (float, int8_t, etc.) will cause compilation errors.
01613 *
01614 * @par Example: Type Mismatch Error
01615 * @code{.c}
01616 * float velocity = 1.5f;
01617 * rx_log_error_val("MOTOR", "Velocity", velocity);
01618 * // Compile error: no _Generic case for 'float'
01619 * @endcode
01620 *
01621 * @see rx_log_error_val(), rx_log_warn_val(), rx_log_info_val()
01622 * @see rx_log_debug_val(), rx_log_verbose_val()
01623 * @see internal_rx_log_error_u8(), internal_rx_log_error_u16(), internal_rx_log_error_u32()
01624 * @see internal_rx_log_error_err()
01625 */
01626 #define RX_LOG_VAL_IMPL(level, tag, msg, val)
01627 _Generic((val),
01628   uint8_t: internal_rx_log_##level##_u8,
01629   uint16_t: internal_rx_log_##level##_u16,
01630   uint32_t: internal_rx_log_##level##_u32,
01631   int32_t: internal_rx_log_##level##_err)(tag, msg, val)
01632
01633 /*
=====
01634 * Public API - ERROR Level
01635 *
=====
01636 */
01637
01638 #if (LOG_LEVEL >= k_log_error)
01639 /**
01640  * @def rx_log_error
01641  * @brief Log critical ERROR-level message (compile-time filtered)
01642  *
01643  * @details
01644  * Public API for logging critical errors that prevent operation. This macro includes
01645  * compile-time filtering: if `LOG_LEVEL < k_log_error`, the macro expands to `(void)0`
01646  * and generates **zero code** (no CPU cycles, no flash memory).
01647  *
01648  * ## Compile-Time Behavior
01649  * | LOG_LEVEL Setting | Macro Expansion | Code Generated |
01650  * |-----|-----|-----|
01651  * | k_log_none | `(void)0` | 0 bytes |
01652  * | k_log_error or higher | `internal_rx_log_error(tag, msg)` | ~60-80 bytes |
01653  *
01654  * ## When to Use
01655  * Use for critical failures that:
01656  * - Prevent system operation
01657  * - Require immediate attention
01658  * - Indicate hardware faults
01659  * - Signal initialization failures
01660  *
01661  * @param[in] tag Component tag (string literal, e.g., "MOTOR", "SPI", "USB")
01662  *      - Identifies error source
01663  *      - Recommended: 3-8 uppercase characters
01664  * @param[in] msg Error message (string literal or const char*)
01665  *      - Should be concise and actionable
01666  *      - Describes the specific failure
01667  *
01668  * @return void (macro expands to function call or no-op)
01669  *
01670  * @note **Zero Overhead**: If LOG_LEVEL < k_log_error, this compiles to nothing.
01671  * @note **Thread Safety**: NOT thread-safe. Concurrent calls may interleave output.
01672  *
01673  * @par Example: Critical Hardware Failure
01674  * @code{.c}
01675  * #include "rx_log.h"
01676  *
01677  * void motor_init(void) {
01678  *   if (!configure_pwm()) {
01679  *     rx_log_error("MOTOR", "PWM configuration failed");
01680  *     return;
01681  *   }
01682  *
01683  *   if (!enable_h_bridge()) {
01684  *     rx_log_error("MOTOR", "H-bridge enable failed");
01685  *     return;
01686  *   }
01687  *
01688  *   rx_log_info("MOTOR", "Initialization complete");
01689  * }
01690  *
01691  * // Output (if LOG_LEVEL >= k_log_error):
01692  * // [ERROR] [MOTOR] PWM configuration failed
01693  * @endcode
01694  *
01695  */

```

```

01696 * @par Example: Initialization Sequence
01697 * @code{.c}
01698 * void system_init(void) {
01699 *   rx_log_info("SYSTEM", "Starting initialization");
01700 *
01701 *   if (!uart_init()) {
01702 *     rx_log_error("UART", "Failed to initialize");
01703 *     RX_ERROR_CHECK("UART", "Cannot continue without UART", k_rx_err_init);
01704 *   }
01705 *
01706 *   if (!spi_init()) {
01707 *     rx_log_error("SPI", "Failed to initialize");
01708 *     // Continue anyway - SPI is optional
01709 *   }
01710 * }
01711 * @endcode
01712 *
01713 * @par Example: Compile-Time Elimination
01714 * @code{.c}
01715 * // Build configuration: -DLOG_LEVEL=k_log_none
01716 *
01717 * rx_log_error("MOTOR", "Critical failure"); // Compiles to: ((void)0)
01718 *
01719 * // Result: Zero bytes of code generated, zero CPU cycles, zero flash usage
01720 * @endcode
01721 *
01722 * @see internal_rx_log_error() Implementation function
01723 * @see rx_log_error_val() Error logging with typed value
01724 * @see rx_log_error_hex() Error logging with hex value
01725 * @see rx_log_error_str() Error logging with string value
01726 * @see LOG_LEVEL Compile-time log level configuration
01727 */
01728 #define rx_log_error(tag, msg) internal_rx_log_error((tag), (msg))
01729
01730 /**
01731 * @def rx_log_error_val
01732 * @brief Log ERROR-level message with automatic type-dispatched value
01733 *
01734 * @details
01735 * Public API for logging critical errors with an associated value. Uses C23 `__Generic`
01736 * for automatic type dispatch to the correct formatting function. Includes compile-time
01737 * filtering for zero overhead when disabled.
01738 *
01739 * ## Type Dispatch
01740 * The value's type determines output formatting at compile time:
01741 * | Type | Dispatches To | Format | Example |
01742 * |-----|-----|-----|-----|
01743 * | uint8_t | internal_rx_log_error_u8 | Decimal | `42` |
01744 * | uint16_t | internal_rx_log_error_u16 | Decimal | `2048` |
01745 * | uint32_t | internal_rx_log_error_u32 | Decimal | `4294967295` |
01746 * | int32_t (rx_err_t) | internal_rx_log_error_err | Hex (8 digits) | `0x00000108` |
01747 *
01748 * ## Compile-Time Behavior
01749 * | LOG_LEVEL Setting | Macro Expansion | Code Generated |
01750 * |-----|-----|-----|
01751 * | k_log_none | `(void)0` | 0 bytes |
01752 * | k_log_error or higher | `RX_LOG_VAL_IMPL(...)` -> typed function | ~70-90 bytes |
01753 *
01754 * @param[in] tag Component tag (string literal, e.g., "MOTOR", "SPI", "I2C")
01755 * @param[in] msg Error message (string literal or const char*)
01756 * @param[in] val Value to log - type determines formatting (uint8_t, uint16_t, uint32_t, int32_t)
01757 *
01758 * @return void (macro expands to type-dispatched function call or no-op)
01759 *
01760 * @note Zero Overhead: If LOG_LEVEL < k_log_error, this compiles to nothing.
01761 * @note Type Safe: Compiler enforces supported types at compile time.
01762 * @note Thread Safety: NOT thread-safe. Concurrent calls may interleave output.
01763 *
01764 * @warning int32_t Limitation: Raw int32_t values format as hex (for rx_err_t).
01765 * Cast to uint32_t for decimal display of signed integers.
01766 *
01767 * @par Example: Error Codes
01768 * @code{.c}
01769 * #include "rx_log.h"
01770 * #include "rx_err.h"
01771 *
01772 * rx_err_t motor_init(void) {
01773 *   rx_err_t err = configure_pwm();
01774 *   if (err != k_rx_ok) {
01775 *     rx_log_error_val("MOTOR", "PWM config failed", err); // Auto -> hex
01776 *     return err;
01777 *   }
01778 *
01779 *   err = calibrate_encoder();
01780 *   if (err != k_rx_ok) {
01781 *     rx_log_error_val("MOTOR", "Encoder calibration failed", err); // Auto -> hex
01782 *     return err;

```

```

01783 * }
01784 *
01785 * return k_rx_ok;
01786 * }
01787 *
01788 * // Output (if err = k_rx_err_timeout = 0x108):
01789 * // [ERROR] [MOTOR] PWM config failed: 0x00000108
01790 * @endcode
01791 *
01792 * @par Example: Typed Values (Automatic Dispatch)
01793 * @code{.c}
01794 * // uint8_t -> decimal
01795 * uint8_t motor_id = 3;
01796 * rx_log_error_val("MOTOR", "Invalid motor ID", motor_id);
01797 * // Output: [ERROR] [MOTOR] Invalid motor ID: 3
01798 *
01799 * // uint16_t -> decimal
01800 * uint16_t rpm = 4500;
01801 * rx_log_error_val("MOTOR", "RPM out of range", rpm);
01802 * // Output: [ERROR] [MOTOR] RPM out of range: 4500
01803 *
01804 * // uint32_t -> decimal
01805 * uint32_t timestamp = get_tick_count();
01806 * rx_log_error_val("TIMER", "Overflow at", timestamp);
01807 * // Output: [ERROR] [TIMER] Overflow at: 4294967295
01808 *
01809 * // rx_err_t (int32_t) -> hex
01810 * rx_err_t err = k_rx_err_hw_fault;
01811 * rx_log_error_val("HAL", "Hardware fault", err);
01812 * // Output: [ERROR] [HAL] Hardware fault: 0x00000201
01813 * @endcode
01814 *
01815 * @par Example: Type Mismatch (Compile Error)
01816 * @code{.c}
01817 * float velocity = 1.5f;
01818 * rx_log_error_val("MOTOR", "Invalid velocity", velocity);
01819 * // Compile error: no _Generic case for 'float'
01820 * // Use rx_log_error() with formatted string instead
01821 * @endcode
01822 *
01823 * @see RX_LOG_VAL_IMPL() _Generic type dispatch implementation
01824 * @see internal_rx_log_error_u8(), internal_rx_log_error_u16(), internal_rx_log_error_u32()
01825 * @see internal_rx_log_error_err() Error code formatting (hex)
01826 * @see rx_log_error() Simple error logging without value
01827 */
01828 #define rx_log_error_val(tag, msg, val) RX_LOG_VAL_IMPL(error, (tag), (msg), (val))
01829
01830 /**
01831 * @def rx_log_error_hex
01832 * @brief Log ERROR-level message with hexadecimal value
01833 *
01834 * @details
01835 * Public API for logging critical errors with associated value displayed as hexadecimal.
01836 * Useful for register values, memory addresses, bit masks, and other hardware-related data.
01837 * Includes compile-time filtering for zero overhead when disabled.
01838 *
01839 * ## Output Format
01840 * ```
01841 * [ERROR] [TAG] message: 0xHHHHHHHH
01842 * ```
01843 * Leading zeros are included up to the specified digit count.
01844 *
01845 * ## Compile-Time Behavior
01846 * | LOG_LEVEL Setting | Macro Expansion | Code Generated |
01847 * |-----|-----|-----|
01848 * | k_log_none | `((void)0)` | 0 bytes |
01849 * | k_log_error or higher | `internal_rx_log_error_hex(...)` | ~75-95 bytes |
01850 *
01851 * @param[in] tag Component tag (string literal, e.g., "GPIO", "SPI", "REG")
01852 * @param[in] msg Error message (string literal or const char*)
01853 * @param[in] val Value to display as hexadecimal (uint32_t)
01854 * - Supports 0 to 0xFFFFFFFF
01855 * - Upper bits ignored if digits < 8
01856 * @param[in] digits Number of hex digits to display (1-8)
01857 * - Common values: 2 (byte), 4 (uint16_t), 8 (uint32_t)
01858 * - Leading zeros included
01859 *
01860 * @return void (macro expands to function call or no-op)
01861 *
01862 * @note **Zero Overhead**: If LOG_LEVEL < k_log_error, this compiles to nothing.
01863 * @note **Thread Safety**: NOT thread-safe. Concurrent calls may interleave output.
01864 * @note **Leading Zeros**: Always includes leading zeros up to specified digit count.
01865 *
01866 * @par Example: Hardware Register Debugging
01867 * @code{.c}
01868 * #include "rx_log.h"
01869 *

```

```

01870 * void configure_gpio(void) {
01871 *     uint32_t reg_val = read_register(GPIO_BASE + 0x04);
01872 *
01873 *     if ((reg_val & 0x80000000) == 0) {
01874 *         rx_log_error_hex("GPIO", "Enable bit not set", reg_val, 8);
01875 *         // Output: [ERROR] [GPIO] Enable bit not set: 0x00001234
01876 *     }
01877 * }
01878 * @endcode
01879 *
01880 * @par Example: SPI Transaction Failure
01881 * @code{.c}
01882 * void spi_transfer(uint8_t cmd, uint8_t data) {
01883 *     uint16_t status = spi_transact(cmd, data);
01884 *
01885 *     if (status & 0x8000) { // Error bit set
01886 *         rx_log_error_hex("SPI", "Transfer failed, status", status, 4);
01887 *         // Output: [ERROR] [SPI] Transfer failed, status: 0x8001
01888 *     }
01889 * }
01890 * @endcode
01891 *
01892 * @par Example: Memory Address Logging
01893 * @code{.c}
01894 * void* allocate_buffer(uint32_t size) {
01895 *     void* ptr = malloc(size);
01896 *     if (ptr == nullptr) {
01897 *         rx_log_error_hex("MEM", "Allocation failed at", (uint32_t)ptr, 8);
01898 *         // Output: [ERROR] [MEM] Allocation failed at: 0x00000000
01899 *     }
01900 *     return ptr;
01901 * }
01902 * @endcode
01903 *
01904 * @par Example: Different Digit Counts
01905 * @code{.c}
01906 * uint8_t byte_val = 0x5A;
01907 * rx_log_error_hex("TEST", "Byte", byte_val, 2);
01908 * // Output: [ERROR] [TEST] Byte: 0x5A
01909 *
01910 * uint16_t word_val = 0x1234;
01911 * rx_log_error_hex("TEST", "Word", word_val, 4);
01912 * // Output: [ERROR] [TEST] Word: 0x1234
01913 *
01914 * uint32_t dword_val = 0xDEADBEEF;
01915 * rx_log_error_hex("TEST", "DWord", dword_val, 8);
01916 * // Output: [ERROR] [TEST] DWord: 0xDEADBEEF
01917 * @endcode
01918 *
01919 * @see internal_rx_log_error_hex() Implementation function
01920 * @see rx_log_error() Simple error logging without value
01921 * @see rx_log_error_val() Error logging with typed value (auto-dispatch)
01922 */
01923 #define rx_log_error_hex(tag, msg, val, digits) \
01924     internal_rx_log_error_hex((tag), (msg), (val), (digits))
01925 /**
01926 * @def rx_log_error_str
01927 * @brief Log ERROR-level message with bounded string value (NASA-compliant)
01928 *
01929 * @details
01930 * Public API for logging critical errors with associated string value. Uses bounded
01931 * iteration for memory safety (NASA Power of 10 Rule 2 compliance). String output is
01932 * limited to k_log_str_max_len (256 bytes) to prevent unbounded loops.
01933 *
01934 * ## Output Format
01935 * ```
01936 * [ERROR] [TAG] message: string_value
01937 * ```
01938 *
01939 * ## Safety Features
01940 * - **Bounded loop**: Maximum 256 characters logged (k_log_str_max_len)
01941 * - **Early termination**: Stops at user-provided len or null terminator
01942 * - **NULL safe**: NULL pointers silently ignored (no crash)
01943 *
01944 * ## Compile-Time Behavior
01945 * | LOG_LEVEL Setting | Macro Expansion | Code Generated |
01946 * |-----|-----|-----|
01947 * | k_log_none | `((void)0)` | 0 bytes |
01948 * | k_log_error or higher | `internal_rx_log_error_str(...)` | ~100-120 bytes |
01949 *
01950 * @param[in] tag Component tag (string literal, e.g., "CONFIG", "USB", "FILE")
01951 * @param[in] msg Error message (string literal or const char*)
01952 * @param[in] str String value to log (const char*)
01953 *
01954 * - Must point to valid memory
01955 * - Can be null-terminated or explicit length
01956 * - nullptr silently ignored

```

```

01957 * @param[in] len Maximum number of characters to log
01958 *             - Typically obtained via strlen(str)
01959 *             - Values > 256 clamped to 256
01960 *             - Must not exceed actual buffer size
01961 *
01962 * @return void (macro expands to function call or no-op)
01963 *
01964 * @note **Zero Overhead**: If LOG_LEVEL < k_log_error, this compiles to nothing.
01965 * @note **Bounded**: Maximum 256 characters logged regardless of len parameter.
01966 * @note **Thread Safety**: NOT thread-safe. Concurrent calls may interleave output.
01967 * @note **NULL Safe**: NULL str pointer silently ignored (safe failure mode).
01968 *
01969 * @warning Ensure len <= actual buffer size to prevent undefined behavior.
01970 *
01971 * @par Example: Configuration Errors
01972 * @code{.c}
01973 * #include "rx_log.h"
01974 * #include <string.h>
01975 *
01976 * void parse_config(const char* param_name, const char* param_value) {
01977 *     if (strcmp(param_name, "motor_id") == 0) {
01978 *         // Valid parameter
01979 *     } else {
01980 *         // Unknown parameter name
01981 *         rx_log_error_str("CONFIG", "Unknown parameter", param_name, strlen(param_name));
01982 *         // Output: [ERROR] [CONFIG] Unknown parameter: invalid_param_name
01983 *     }
01984 * }
01985 * @endcode
01986 *
01987 * @par Example: File Path Errors
01988 * @code{.c}
01989 * void load_config_file(const char* filepath) {
01990 *     if (!file_exists(filepath)) {
01991 *         rx_log_error_str("FILE", "Config file not found", filepath, strlen(filepath));
01992 *         // Output: [ERROR] [FILE] Config file not found: /etc/motor_config.ini
01993 *     }
01994 * }
01995 * @endcode
01996 *
01997 * @par Example: USB Device Name Logging
01998 * @code{.c}
01999 * void usb_device_error(const char* device_name, uint32_t name_len) {
02000 *     rx_log_error_str("USB", "Device enumeration failed", device_name, name_len);
02001 *     // Output: [ERROR] [USB] Device enumeration failed: RX72N Motor Controller
02002 * }
02003 * @endcode
02004 *
02005 * @par Example: Truncation Behavior
02006 * @code{.c}
02007 * // String longer than k_log_str_max_len (256)
02008 * char very_long_string[300];
02009 * memset(very_long_string, 'A', 299);
02010 * very_long_string[299] = '\0';
02011 *
02012 * rx_log_error_str("TEST", "Long string", very_long_string, strlen(very_long_string));
02013 * // Output: [ERROR] [TEST] Long string: AAAA... (truncated at 256 chars)
02014 * @endcode
02015 *
02016 * @see internal_rx_log_error_str() Implementation function
02017 * @see k_log_str_max_len Maximum string length constant (256)
02018 * @see rx_log_error() Simple error logging without value
02019 * @see rx_log_error_val() Error logging with typed value
02020 */
02021 #define rx_log_error_str(tag, msg, str, len) internal_rx_log_error_str((tag), (msg), (str), (len))
02022 #else
02023 #define rx_log_error(tag, msg) ((void)0)
02024 #define rx_log_error_val(tag, msg, val) ((void)0)
02025 #define rx_log_error_hex(tag, msg, val, d) ((void)0)
02026 #define rx_log_error_str(tag, msg, str, len) ((void)0)
02027 #endif /* LOG_LEVEL >= k_log_error */
02028 /*
02029 * Public API - WARN Level
02030 *
02031 *
02032 */
02033 #if (LOG_LEVEL >= k_log_warn)
02034 /**
02035 * @brief Log warning message (no value)
02036 */

```

```

02042 #define rx_log_warn(tag, msg) internal_rx_log_warn((tag), (msg))
02043
02044 /**
02045  * @brief Log warning with typed value (automatic type dispatch)
02046  */
02047 #define rx_log_warn_val(tag, msg, val) RX_LOG_VAL_IMPL(warn, (tag), (msg), (val))
02048
02049 /**
02050  * @brief Log warning with hex value
02051  */
02052 #define rx_log_warn_hex(tag, msg, val, digits) \
02053     internal_rx_log_warn_hex((tag), (msg), (val), (digits))
02054
02055 /**
02056  * @brief Log warning with string value (bounded length)
02057  */
02058 #define rx_log_warn_str(tag, msg, str, len) internal_rx_log_warn_str((tag), (msg), (str), (len))
02059
02060 #else
02061
02062 #define rx_log_warn(tag, msg) ((void)0)
02063 #define rx_log_warn_val(tag, msg, val) ((void)0)
02064 #define rx_log_warn_hex(tag, msg, val, d) ((void)0)
02065 #define rx_log_warn_str(tag, msg, str, len) ((void)0)
02066
02067 #endif /* LOG_LEVEL >= k_log_warn */
02068
02069 /*
02070  * Public API - INFO Level
02071  */
02072
02073
02074 #if (LOG_LEVEL >= k_log_info)
02075
02076 /**
02077  * @brief Log info message (no value)
02078  */
02079 #define rx_log_info(tag, msg) internal_rx_log_info((tag), (msg))
02080
02081 /**
02082  * @brief Log info with typed value (automatic type dispatch)
02083  */
02084 #define rx_log_info_val(tag, msg, val) RX_LOG_VAL_IMPL(info, (tag), (msg), (val))
02085
02086 /**
02087  * @brief Log info with hex value
02088  */
02089 #define rx_log_info_hex(tag, msg, val, digits) \
02090     internal_rx_log_info_hex((tag), (msg), (val), (digits))
02091
02092 /**
02093  * @brief Log info with string value (bounded length)
02094  */
02095 #define rx_log_info_str(tag, msg, str, len) internal_rx_log_info_str((tag), (msg), (str), (len))
02096
02097 #else
02098
02099 #define rx_log_info(tag, msg) ((void)0)
02100 #define rx_log_info_val(tag, msg, val) ((void)0)
02101 #define rx_log_info_hex(tag, msg, val, d) ((void)0)
02102 #define rx_log_info_str(tag, msg, str, len) ((void)0)
02103
02104 #endif /* LOG_LEVEL >= k_log_info */
02105
02106 /*
02107  * Public API - DEBUG Level
02108  */
02109
02110
02111 #if (LOG_LEVEL >= k_log_debug)
02112
02113 /**
02114  * @brief Log debug message (no value)
02115  */
02116 #define rx_log_debug(tag, msg) internal_rx_log_debug((tag), (msg))
02117
02118 /**
02119  * @brief Log debug with typed value (automatic type dispatch)
02120  */
02121 #define rx_log_debug_val(tag, msg, val) RX_LOG_VAL_IMPL(debug, (tag), (msg), (val))
02122
02123 /**
02124  * @brief Log debug with hex value

```

```

02125 */
02126 #define rx_log_debug_hex(tag, msg, val, digits) \
02127     internal_rx_log_debug_hex((tag), (msg), (val), (digits))
02128
02129 /**
02130  * @brief Log debug with string value (bounded length)
02131  */
02132 #define rx_log_debug_str(tag, msg, str, len) internal_rx_log_debug_str((tag), (msg), (str), (len))
02133
02134 #else
02135
02136 #define rx_log_debug(tag, msg) ((void)0)
02137 #define rx_log_debug_val(tag, msg, val) ((void)0)
02138 #define rx_log_debug_hex(tag, msg, val, d) ((void)0)
02139 #define rx_log_debug_str(tag, msg, str, len) ((void)0)
02140
02141 #endif /* LOG_LEVEL >= k_log_debug */
02142
02143 /*
02144  * Public API - VERBOSE Level
02145  */
02146 */
02147
02148 #if (LOG_LEVEL >= k_log_verbose)
02149
02150 /**
02151  * @brief Log verbose message (no value)
02152  */
02153 #define rx_log_verbose(tag, msg) internal_rx_log_verbose((tag), (msg))
02154
02155 /**
02156  * @brief Log verbose with typed value (automatic type dispatch)
02157  */
02158 #define rx_log_verbose_val(tag, msg, val) RX_LOG_VAL_IMPL(verbose, (tag), (msg), (val))
02159
02160 /**
02161  * @brief Log verbose with hex value
02162  */
02163 #define rx_log_verbose_hex(tag, msg, val, digits) \
02164     internal_rx_log_verbose_hex((tag), (msg), (val), (digits))
02165
02166 /**
02167  * @brief Log verbose with string value (bounded length)
02168  */
02169 #define rx_log_verbose_str(tag, msg, str, len) \
02170     internal_rx_log_verbose_str((tag), (msg), (str), (len))
02171
02172 #else
02173
02174 #define rx_log_verbose(tag, msg) ((void)0)
02175 #define rx_log_verbose_val(tag, msg, val) ((void)0)
02176 #define rx_log_verbose_hex(tag, msg, val, d) ((void)0)
02177 #define rx_log_verbose_str(tag, msg, str, len) ((void)0)
02178
02179 #endif /* LOG_LEVEL >= k_log_verbose */
02180
02181 #ifdef __cplusplus
02182 }
02183 #endif

```

7.23 libs/rx_core/inc/rx_log_uart.h File Reference

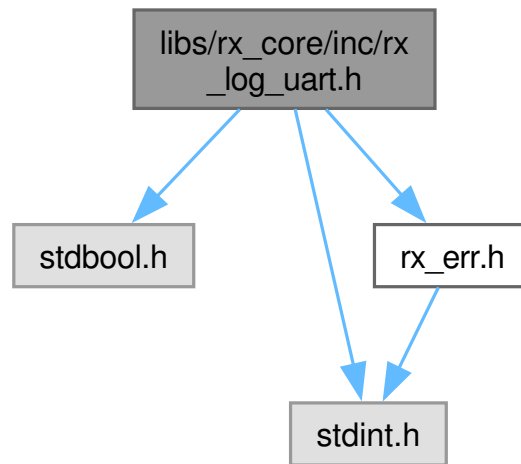
UART-framed log backend for the RX72N logging system.

```

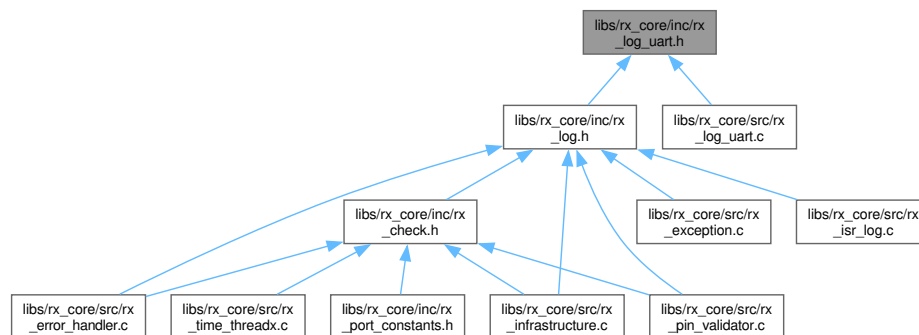
#include <stdbool.h>
#include <stdint.h>
#include "rx_err.h"

```

Include dependency graph for rx_log_uart.h:



This graph shows which files directly or indirectly include this file:



Data Structures

- struct [rx_log_uart_stats_t](#)
Diagnostic counters for the UART log backend.

Enumerations

- enum [rx_log_uart_constants_t](#) : `uint16_t { k_rx_log_uart_ring_size = 4096 }`
Ring buffer sizing for the UART log backend.

Functions

- void [rx_log_uart_putc](#) (char c)
Append a single character to the log ring buffer.
- void [rx_log_uart_puts](#) (const char *str)
Append a null-terminated string to the log ring buffer.
- void [rx_log_uart_putint](#) (int32_t value)
Append a signed integer as decimal ASCII to the log ring buffer.
- void [rx_log_uart_putuint](#) (uint32_t value)
Append an unsigned integer as decimal ASCII to the log ring buffer.
- void [rx_log_uart_puthex](#) (uint32_t value, uint8_t digits)
Append a value as zero-padded hexadecimal ASCII to the log ring buffer.

- `uint32_t rx_log_uart_pending_len` (void)
Query number of bytes currently queued in the ring buffer.
- `rx_err_t rx_log_uart_drain` (uint8_t *out_buf, uint32_t max_len, uint32_t *out_len)
Pull up to max_len bytes out of the ring buffer for transmission.
- void `rx_log_uart_get_stats` (`rx_log_uart_stats_t` *stats)
Snapshot diagnostic counters.

7.23.1 Detailed Description

UART-framed log backend for the RX72N logging system.

Stores log bytes in a ring buffer in RAM and exposes a drain API that the communication task calls on every poll tick. Drained bytes are wrapped in a `k_frame_type_log_message` frame (TYPE=0x20) and sent over the SCI9 UART through the CY7C65213 USB-UART bridge.

Framing logs (rather than writing raw ASCII to the UART) prevents log bytes from colliding with the frame sync word 0x55AA used by the protobuf command and telemetry frames on the same byte stream.

7.23.1.1 Write path

application thread -> `rx_log_uart_putc` / `puts` / ... -> append to `s_ring` (mutex-guarded, non-blocking)
-> return immediately (no I/O from log site)

7.23.1.2 Drain path (runs from comm_task at ~100 Hz)

comm_task tick -> `rx_log_uart_pending_len()` > 0 ? -> `rx_log_uart_drain`(buf, max) -> `rx_uart_comm_send`(handle, `k_frame_type_log_message`, 0, buf, len)

7.23.1.3 Thread safety

Producers (`rx_log_uart_*`) and the single consumer (`rx_log_uart_drain`) are serialized by a lazy-initialized ThreadX mutex. Producers never block on I/O; if the ring is full, bytes are dropped and counted in the drop stat.

Author

Locked, Inc.

Date

2026-04-17

Copyright

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Since

Version 1.0.0

Definition in file [rx_log_uart.h](#).

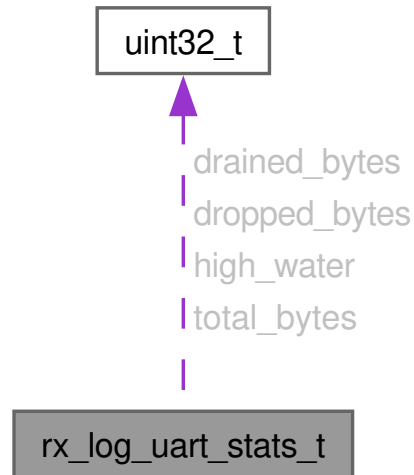
7.23.2 Data Structure Documentation

7.23.2.1 struct rx_log_uart_stats_t

Diagnostic counters for the UART log backend.

Definition at line 68 of file [rx_log_uart.h](#).

Collaboration diagram for rx_log_uart_stats_t:



Data Fields

uint32_t	drained_bytes	Cumulative bytes handed to the drain consumer
uint32_t	dropped_bytes	Cumulative bytes dropped because the ring was full
uint32_t	high_water	Maximum observed ring occupancy (bytes)
uint32_t	total_bytes	Cumulative bytes accepted by producers (pre-drop)

7.23.3 Enumeration Type Documentation

7.23.3.1 rx_log_uart_constants_t

enum [rx_log_uart_constants_t](#) : uint16_t

Ring buffer sizing for the UART log backend.

Enumerator

k_rx_log_uart_ring_size	Log ring buffer size in bytes (4096). At 921600 baud the UART can carry ~92 kB/s. 4096 bytes provides ~45 ms of sustained burst capacity, which is well above the 10 ms comm_task poll interval.
-------------------------	--

Definition at line 53 of file [rx_log_uart.h](#).

```

00053         : uint16_t {
00054     /**
00055      * @brief Log ring buffer size in bytes (4096)
00056      *
00057      * At 921600 baud the UART can carry ~92 kB/s. 4096 bytes provides ~45 ms of
00058      * sustained burst capacity, which is well above the 10 ms comm_task poll
00059      * interval.
00060      */
00061     k_rx_log_uart_ring_size = 4096,
00062 } rx_log_uart_constants_t;
  
```

7.23.4 Function Documentation

7.23.4.1 rx_log_uart_drain()

```
rx_err_t rx_log_uart_drain (
    uint8_t * out_buf,
    uint32_t max_len,
    uint32_t * out_len) [nodiscard]
```

Pull up to max_len bytes out of the ring buffer for transmission.

Copies a contiguous chunk of up to max_len bytes out of the ring into out_buf. Returns the number of bytes actually copied in out_len. A single call may return fewer bytes than requested when the ring has wrapped; call again to drain the remainder.

Parameters

out	out_buf	Destination buffer
in	max_len	Maximum bytes to copy into out_buf
out	out_len	Actual number of bytes written

Return values

k_rx_ok	Success (0 bytes also returns ok)
k_rx_err_invalid_arg	out_buf or out_len is nullptr, or max_len == 0

Pull up to max_len bytes out of the ring buffer for transmission.

Single-consumer drain entry-point. Copies up to max_len bytes of queued log data from the ring tail into out_buf, returning the actual byte count via *out_len. If the unread region wraps the ring end, only the pre-wrap contiguous run is returned this call; the caller drains again to pick up the remainder. This bounds execution time per call (NASA Rule 2) and keeps the consumer loop simple.

Updates s_tail and s_count under the producer/consumer mutex, and accumulates drained_bytes into the stats block.

Note

Thread-safe via the internal mutex. Designed for a single consumer task (typically comm_task). Multiple concurrent consumers would share bytes incorrectly even though no corruption would occur.

Definition at line 296 of file [rx_log_uart.c](#).

```
00297 {
00298     if ((out_buf == NULL) || (out_len == NULL) || (max_len == 0U)) {
00299         return k_rx_err_invalid_arg;
00300     }
00301     internal_init_mutex_once();
00302     (void)tx_mutex_get(&s_mutex, TX_WAIT_FOREVER);
00303     uint32_t to_copy = (s_count < max_len) ? s_count : max_len;
00304     /* Prefer a single contiguous copy -- if the unread region wraps the ring
00305      * end, return only the pre-wrap portion this call. Caller drains again to
00306      * pick up the remainder. This keeps the consumer loop simple and bounds
00307      * execution time per drain call (NASA Rule 2). */
```

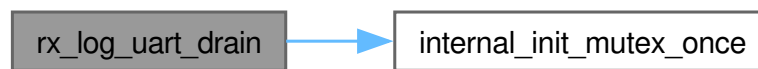
```

00311  const uint32_t contiguous = k_rx_log_uart_ring_size - s_tail;
00312  if (to_copy > contiguous) {
00313      to_copy = contiguous;
00314  }
00315
00316  if (to_copy > 0U) {
00317      /* NOLINTNEXTLINE(clang-analyzer-security.insecureAPI.DeprecatedOrUnsafeBufferHandling) */
00318      (void)memcpy(out_buf, &s_ring[s_tail], to_copy);
00319      s_tail = (s_tail + to_copy) % k_rx_log_uart_ring_size;
00320      s_count -= to_copy;
00321      s_stats.drained_bytes += to_copy;
00322  }
00323
00324  *out_len = to_copy;
00325
00326  (void)tx_mutex_put(&s_mutex);
00327  return k_rx_ok;
00328 }

```

References [internal_init_mutex_once\(\)](#), [k_rx_err_invalid_arg](#), [k_rx_log_uart_ring_size](#), [k_rx_ok](#), [s_count](#), [s_mutex](#), [s_ring](#), [s_stats](#), and [s_tail](#).

Here is the call graph for this function:



7.23.4.2 rx_log_uart_get_stats()

```

void rx_log_uart_get_stats (
    rx_log_uart_stats_t * stats)

```

Snapshot diagnostic counters.

Parameters

out	stats	Statistics destination (nullptr is silently ignored)
-----	-------	--

Snapshot diagnostic counters.

Copies the current [rx_log_uart_stats_t](#) value (drained_bytes, dropped_bytes, ring_high_water, etc.) under the producer/consumer mutex. A NULL output pointer is treated as a no-op for caller convenience.

Definition at line 338 of file [rx_log_uart.c](#).

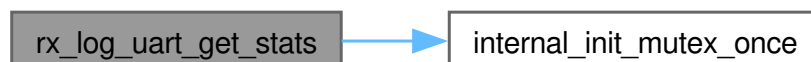
```

00339 {
00340     if (stats == NULL) {
00341         return;
00342     }
00343     internal_init_mutex_once();
00344     (void)tx_mutex_get(&s_mutex, TX_WAIT_FOREVER);
00345     *stats = s_stats;
00346     (void)tx_mutex_put(&s_mutex);
00347 }

```

References [internal_init_mutex_once\(\)](#), [s_mutex](#), and [s_stats](#).

Here is the call graph for this function:



7.23.4.3 rx_log_uart_pending_len()

```
uint32_t rx_log_uart_pending_len (
    void )
```

Query number of bytes currently queued in the ring buffer.

Returns

Pending byte count (0..[k_rx_log_uart_ring_size](#))

Query number of bytes currently queued in the ring buffer.

Acquires the producer/consumer mutex, snapshots `s_count`, and releases the mutex before returning. This gives a momentarily-consistent count suitable for deciding whether the consumer needs to drain on this tick. The value can change between read and use; treat it as a hint.

Definition at line 268 of file [rx_log_uart.c](#).

```
00269 {
00270     internal_init_mutex_once();
00271     (void)tx_mutex_get(&s_mutex, TX_WAIT_FOREVER);
00272     const uint32_t n = s_count;
00273     (void)tx_mutex_put(&s_mutex);
00274     return n;
00275 }
```

References [internal_init_mutex_once\(\)](#), [s_count](#), and [s_mutex](#).

Here is the call graph for this function:



7.23.4.4 rx_log_uart_putc()

```
void rx_log_uart_putc (
    char c)
```

Append a single character to the log ring buffer.

Parameters

in	c	Character to append
----	---	---------------------

Note

Thread-safe, non-blocking. Drops silently on ring overflow.

Append a single character to the log ring buffer.

Lazily initializes the producer/consumer mutex on first call, then appends a single byte to the ring buffer. If the ring is full the oldest byte is dropped (silent overflow); the dropped-bytes counter is updated through [internal_append_byte\(\)](#).

Note

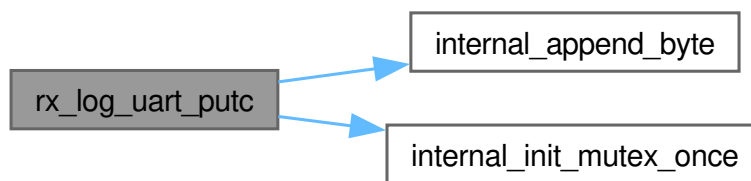
Thread-safe via an internal ThreadX mutex; bounded blocking on contention. NOT safe from ISR context.

Definition at line 136 of file [rx_log_uart.c](#).

```
00137 {
00138     internal_init_mutex_once();
00139     (void)tx_mutex_get(&s_mutex, TX_WAIT_FOREVER);
00140     internal_append_byte((uint8_t)c);
00141     (void)tx_mutex_put(&s_mutex);
00142 }
```

References [internal_append_byte\(\)](#), [internal_init_mutex_once\(\)](#), and [s_mutex](#).

Here is the call graph for this function:



7.23.4.5 rx_log_uart_puthex()

```
void rx_log_uart_puthex (
    uint32_t value,
    uint8_t digits)
```

Append a value as zero-padded hexadecimal ASCII to the log ring buffer.

Parameters

in	value	Value to format
in	digits	Number of hex digits to emit (clamped to 1..8)

Append a value as zero-padded hexadecimal ASCII to the log ring buffer.

Formats value as digits hex characters (uppercase A-F, zero padded on the left) into a stack-local buffer, then forwards to [rx_log_uart_puts\(\)](#). If digits is 0 it is silently promoted to 1; if it exceeds [k_hex_max_digits](#) it is clamped to that maximum to fit the local buffer.

Definition at line 233 of file [rx_log_uart.c](#).

```
00234 {
00235     if (digits == 0U) {
00236         digits = 1U;
00237     } else if (digits > k_hex_max_digits) {
00238         digits = k_hex_max_digits;
00239     }
00240     char buf[k_hex_buf_len];
00241     char* p = &buf[digits];
00242     *p = '\0';
00243     for (uint8_t i = 0U; i < digits; ++i) {
00244         --p;
00245         const uint8_t nibble = (uint8_t)(value & (uint32_t)k_nibble_mask);
00246         *p = (char)((nibble < k_decimal_base) ? ('0' + nibble) : ('A' + (nibble - k_decimal_base)));
00247         value >>= (uint32_t)k_nibble_shift;
00248     }
00249 }
```

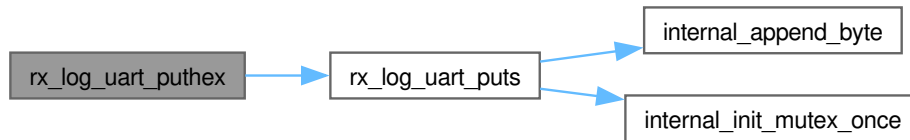
```

00250 }
00251
00252 rx_log_uart_puts(buf);
00253 }

```

References [k_decimal_base](#), [k_hex_buf_len](#), [k_hex_max_digits](#), [k_nibble_mask](#), [k_nibble_shift](#), and [rx_log_uart_puts\(\)](#).

Here is the call graph for this function:



7.23.4.6 rx_log_uart_putint()

```

void rx_log_uart_putint (
    int32_t value)

```

Append a signed integer as decimal ASCII to the log ring buffer.

Parameters

in	value	Signed 32-bit integer value
----	-------	-----------------------------

Append a signed integer as decimal ASCII to the log ring buffer.

Converts value to its decimal string representation in a stack-local buffer (no heap allocation), then forwards to [rx_log_uart_puts\(\)](#). Negative numbers are prefixed with '-'. Conversion is unsigned after sign extraction so INT32_MIN does not wrap.

Definition at line 176 of file [rx_log_uart.c](#).

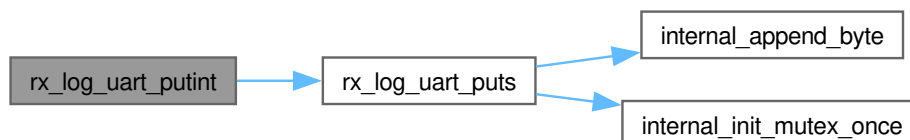
```

00177 {
00178     char buf[k_int32_buf_len];
00179
00180     char* p = &buf[sizeof(buf) - 1U];
00181     *p = '\0';
00182
00183     const bool negative = (bool)(value < 0);
00184     uint32_t abs_value = (int)negative ? (uint32_t)(-value) : (uint32_t)value;
00185
00186     do {
00187         --p;
00188         *p = (char)('0' + (abs_value % k_decimal_base));
00189         abs_value /= k_decimal_base;
00190     } while (abs_value > 0U);
00191
00192     if (negative) {
00193         --p;
00194         *p = '-';
00195     }
00196
00197     rx_log_uart_puts(p);
00198 }

```

References [k_decimal_base](#), [k_int32_buf_len](#), and [rx_log_uart_puts\(\)](#).

Here is the call graph for this function:



7.23.4.7 rx_log_uart_puts()

```
void rx_log_uart_puts (
    const char * str)
```

Append a null-terminated string to the log ring buffer.

Parameters

in	str	Null-terminated string (nullptr is silently ignored)
----	-----	--

Note

Thread-safe, non-blocking. Drops silently on ring overflow.

Append a null-terminated string to the log ring buffer.

Iterates over the input string and appends each byte to the ring buffer under a single lock (one mutex acquire per call, not per byte). This keeps multi-byte log records atomic with respect to other producers and to the drain consumer. A NULL input pointer is treated as a no-op for caller convenience and does not require explicit guards in caller code.

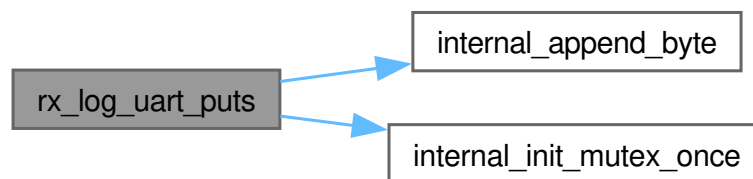
Definition at line 154 of file [rx_log_uart.c](#).

```
00155 {
00156     if (str == NULL) {
00157         return;
00158     }
00159     internal_init_mutex_once();
00160     (void)tx_mutex_get(&s_mutex, TX_WAIT_FOREVER);
00161     for (const char* p = str; *p != '\0'; ++p) {
00162         internal_append_byte((uint8_t)*p);
00163     }
00164     (void)tx_mutex_put(&s_mutex);
00165 }
```

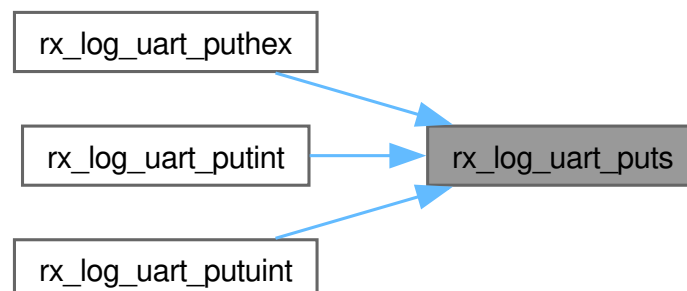
References [internal_append_byte\(\)](#), [internal_init_mutex_once\(\)](#), and [s_mutex](#).

Referenced by [rx_log_uart_puthex\(\)](#), [rx_log_uart_putint\(\)](#), and [rx_log_uart_putuint\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



7.23.4.8 rx_log_uart_putuint()

```
void rx_log_uart_putuint (
    uint32_t value)
```

Append an unsigned integer as decimal ASCII to the log ring buffer.

Parameters

in	value	Unsigned 32-bit integer value
----	-------	-------------------------------

Append an unsigned integer as decimal ASCII to the log ring buffer.

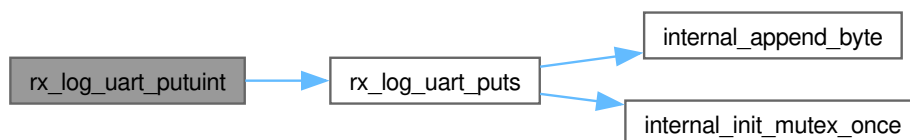
Converts value to a decimal string in a stack-local buffer using base-10 division, then forwards to [rx_log_uart_puts\(\)](#). Output never includes a leading sign.

Definition at line 208 of file [rx_log_uart.c](#).

```
00209 {
00210     char buf[k_uint32_buf_len];
00211
00212     char* p = &buf[sizeof(buf) - 1U];
00213     *p = '\0';
00214
00215     do {
00216         --p;
00217         *p = (char)('0' + (value % k_decimal_base));
00218         value /= k_decimal_base;
00219     } while (value > 0U);
00220
00221     rx_log_uart_puts(p);
00222 }
```

References [k_decimal_base](#), [k_uint32_buf_len](#), and [rx_log_uart_puts\(\)](#).

Here is the call graph for this function:



7.24 rx_log_uart.h

[Go to the documentation of this file.](#)

```
00001 /**
00002  * @file rx_log_uart.h
00003  * @brief UART-framed log backend for the RX72N logging system
00004  *
00005  * @details
00006  * Stores log bytes in a ring buffer in RAM and exposes a drain API that the
00007  * communication task calls on every poll tick. Drained bytes are wrapped in a
00008  * @ref k_frame_type_log_message frame (TYPE=0x20) and sent over the SCI9 UART
00009  * through the CY7C65213 USB-UART bridge.
00010  *
00011  * Framing logs (rather than writing raw ASCII to the UART) prevents log bytes
00012  * from colliding with the frame sync word 0x55AA used by the protobuf command
00013  * and telemetry frames on the same byte stream.
00014  *
00015  * ## Write path
00016  * application thread -> rx_log_uart_putc / puts / ...
00017  * -> append to s_ring (mutex-guarded, non-blocking)
00018  * -> return immediately (no I/O from log site)
00019  *
```

```

00020 * ## Drain path (runs from comm_task at ~100 Hz)
00021 * comm_task tick
00022 * -> rx_log_uart_pending_len() > 0 ?
00023 * -> rx_log_uart_drain(buf, max)
00024 * -> rx_uart_comm_send(handle, k_frame_type_log_message, 0, buf, len)
00025 *
00026 * ## Thread safety
00027 * Producers (rx_log_uart_*) and the single consumer (rx_log_uart_drain) are
00028 * serialized by a lazy-initialized ThreadX mutex. Producers never block on
00029 * I/O; if the ring is full, bytes are dropped and counted in the drop stat.
00030 *
00031 * @author Locked, Inc.
00032 * @date 2026-04-17
00033 * @copyright Copyright (c) 2026 Locked Inc.
00034 * SPDX-License-Identifier: MIT
00035 * @since Version 1.0.0
00036 */
00037
00038 #pragma once
00039
00040 #include <stdbool.h>
00041 #include <stdint.h>
00042
00043 #include "rx_err.h"
00044
00045 #ifdef __cplusplus
00046 extern "C" {
00047 #endif
00048
00049 /**
00050 * @enum rx_log_uart_constants_t
00051 * @brief Ring buffer sizing for the UART log backend
00052 */
00053 typedef enum : uint16_t {
00054 /**
00055 * @brief Log ring buffer size in bytes (4096)
00056 *
00057 * At 921600 baud the UART can carry ~92 kB/s. 4096 bytes provides ~45 ms of
00058 * sustained burst capacity, which is well above the 10 ms comm_task poll
00059 * interval.
00060 */
00061 k_rx_log_uart_ring_size = 4096,
00062 } rx_log_uart_constants_t;
00063
00064 /**
00065 * @struct rx_log_uart_stats_t
00066 * @brief Diagnostic counters for the UART log backend
00067 */
00068 typedef struct {
00069     uint32_t total_bytes; /**< Cumulative bytes accepted by producers (pre-drop) */
00070     uint32_t drained_bytes; /**< Cumulative bytes handed to the drain consumer */
00071     uint32_t dropped_bytes; /**< Cumulative bytes dropped because the ring was full */
00072     uint32_t high_water; /**< Maximum observed ring occupancy (bytes) */
00073 } rx_log_uart_stats_t;
00074
00075 /**
00076 * @brief Append a single character to the log ring buffer
00077 *
00078 * @param[in] c Character to append
00079 *
00080 * @note Thread-safe, non-blocking. Drops silently on ring overflow.
00081 */
00082 void rx_log_uart_putc(char c);
00083
00084 /**
00085 * @brief Append a null-terminated string to the log ring buffer
00086 *
00087 * @param[in] str Null-terminated string (nullptr is silently ignored)
00088 *
00089 * @note Thread-safe, non-blocking. Drops silently on ring overflow.
00090 */
00091 void rx_log_uart_puts(const char* str);
00092
00093 /**
00094 * @brief Append a signed integer as decimal ASCII to the log ring buffer
00095 *
00096 * @param[in] value Signed 32-bit integer value
00097 */
00098 void rx_log_uart_putint(int32_t value);
00099
00100 /**
00101 * @brief Append an unsigned integer as decimal ASCII to the log ring buffer
00102 *
00103 * @param[in] value Unsigned 32-bit integer value
00104 */
00105 void rx_log_uart_putuint(uint32_t value);
00106

```

```

00107 /**
00108  * @brief Append a value as zero-padded hexadecimal ASCII to the log ring buffer
00109  *
00110  * @param[in] value Value to format
00111  * @param[in] digits Number of hex digits to emit (clamped to 1..8)
00112  */
00113 void rx_log_uart_puthex(uint32_t value, uint8_t digits);
00114
00115 /**
00116  * @brief Query number of bytes currently queued in the ring buffer
00117  *
00118  * @return Pending byte count (0..@ref k_rx_log_uart_ring_size)
00119  */
00120 uint32_t rx_log_uart_pending_len(void);
00121
00122 /**
00123  * @brief Pull up to @p max_len bytes out of the ring buffer for transmission
00124  *
00125  * @details
00126  * Copies a contiguous chunk of up to @p max_len bytes out of the ring into
00127  * @p out_buf. Returns the number of bytes actually copied in @p out_len.
00128  * A single call may return fewer bytes than requested when the ring has
00129  * wrapped; call again to drain the remainder.
00130  *
00131  * @param[out] out_buf Destination buffer
00132  * @param[in] max_len Maximum bytes to copy into @p out_buf
00133  * @param[out] out_len Actual number of bytes written
00134  *
00135  * @retval k_rx_ok Success (0 bytes also returns ok)
00136  * @retval k_rx_err_invalid_arg @p out_buf or @p out_len is nullptr, or max_len == 0
00137  */
00138 [[nodiscard]] rx_err_t rx_log_uart_drain(uint8_t* out_buf, uint32_t max_len, uint32_t* out_len);
00139
00140 /**
00141  * @brief Snapshot diagnostic counters
00142  *
00143  * @param[out] stats Statistics destination (nullptr is silently ignored)
00144  */
00145 void rx_log_uart_get_stats(rx_log_uart_stats_t* stats);
00146
00147 #ifdef UNIT_TEST
00148 /**
00149  * @brief Reset all ring buffer and statistics state (unit tests only)
00150  */
00151 void rx_log_uart_test_reset_state(void);
00152 #endif
00153
00154 #ifdef __cplusplus
00155 }
00156 #endif

```

7.25 libs/rx_core/inc/rx_pin_interface.h File Reference

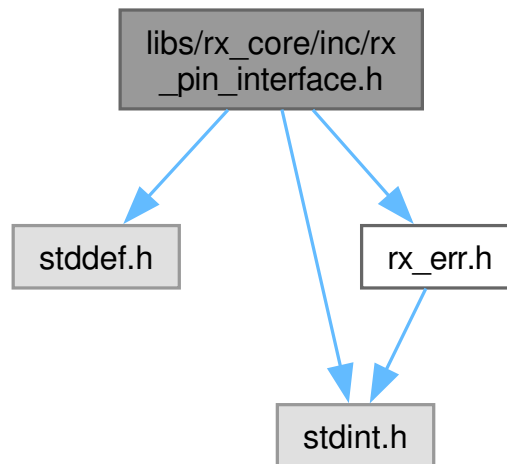
Abstract Pin Validator Interface - THE "I" and "D" in SOLID.

```

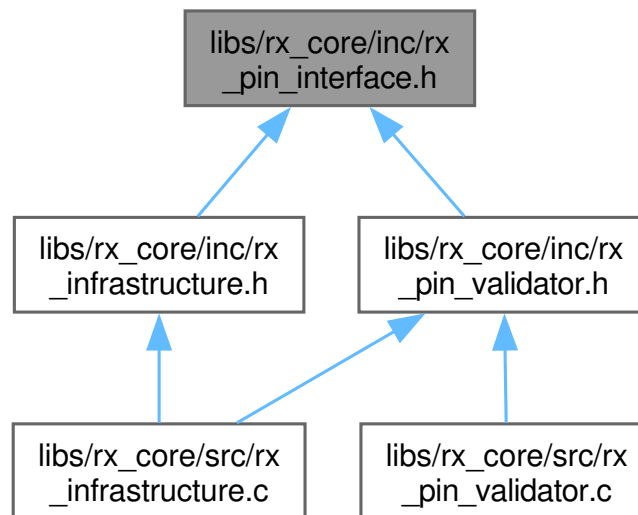
#include <stddef.h>
#include <stdint.h>
#include "rx_err.h"

```

Include dependency graph for rx_pin_interface.h:



This graph shows which files directly or indirectly include this file:



Data Structures

- struct [rx_pin_interface_t](#)
Pin Validator Interface - THE ABSTRACTION FOR PIN MANAGEMENT.

Typedefs

- typedef [rx_err_t](#)(* [rx_pin_validate_fn](#)) (void *ctx, const uint8_t port, const uint8_t pin)
Validate if a GPIO pin exists and is accessible on the RX72N.
- typedef [rx_err_t](#)(* [rx_pin_reserve_fn](#)) (void *ctx, const uint8_t port, const uint8_t pin, const char *function)
Reserve a GPIO pin for exclusive use by a specific function/module.
- typedef [rx_err_t](#)(* [rx_pin_release_fn](#)) (void *ctx, const uint8_t port, const uint8_t pin)
Release a previously reserved GPIO pin, making it available for reuse.
- typedef bool(* [rx_pin_is_reserved_fn](#)) (void *ctx, const uint8_t port, const uint8_t pin)
Check if a GPIO pin is currently reserved.
- typedef [rx_err_t](#)(* [rx_pin_get_function_fn](#)) (void *ctx, const uint8_t port, const uint8_t pin, char *function_out, const uint32_t function_len)
Get the function name associated with a reserved GPIO pin.
- typedef [rx_err_t](#)(* [rx_pin_clear_all_fn](#)) (void *ctx)
Clear all pin reservations (reset to initial state).

Enumerations

- enum [pin_interface_limits_t](#) : uint8_t { [k_pin_function_name_max_len](#) = 32 }
Pin Interface Configuration Constants.

Functions

- static [rx_err_t rx_pin_interface_validate](#) (const [rx_pin_interface_t](#) *iface)
Validate that a pin interface is properly initialized and safe to use.

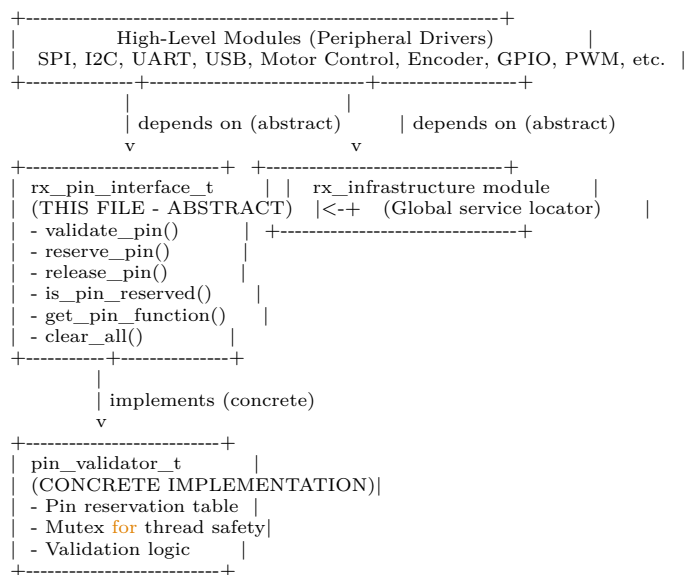
7.25.1 Detailed Description

Abstract Pin Validator Interface - THE "I" and "D" in SOLID.

7.25.1.1 Overview

Defines the abstract interface for GPIO pin validation and reservation in the STAR RX72N firmware. This interface is the cornerstone of Dependency Inversion Principle (DIP) for pin management - high-level modules depend on this abstraction, not on concrete pin validator implementations.

Key Design Pattern: Interface Segregation + Dependency Inversion



7.25.1.2 Why This Design Matters

7.25.1.2.1 Problem Without Abstraction

```
// [FAIL] BAD: Direct dependency on concrete type
#include "pin_validator.h" // Couples to implementation

void spi_init(pin_validator_t* validator) { // Concrete type!
    pin_validator_reserve(validator, 0xA, 5, "SPI_COPI");
}
```

Issues:

- Can't swap implementations (no mock for testing)
- Tight coupling to [pin_validator_t](#) internals
- Can't inject different validators
- Hard to unit test SPI without real pin validator

7.25.1.2.2 Solution With Abstraction

```
// [OK] GOOD: Dependency on abstract interface
#include "rx_pin_interface.h" // Only interface

void spi_init(rx_pin_interface_t* iface) { // Abstract interface!
    iface->reserve_pin(iface->ctx, 0xA, 5, "SPI_COPI");
}
```

Benefits:

- Testable (inject mock_pin_interface for unit tests)
- Flexible (swap implementations without changing SPI code)
- Decoupled (SPI doesn't know about [pin_validator_t](#) internals)
- Maintainable (interface is stable, implementation can change)

7.25.1.3 Architecture Principles

1. Single Responsibility: Interface only defines contract, no logic
2. Open/Closed: New implementations can be added without changing interface
3. Liskov Substitution: All implementations are interchangeable
4. Interface Segregation: Small, focused interface with 6 methods
5. Dependency Inversion: High-level code depends on this abstraction

7.25.1.4 Pin Conflict Prevention System

The pin validator prevents these common errors:

Conflict Type	Example	Prevention
Double peripheral	SPI and I2C both use PA5	reserve_pin() fails on second attempt
Output collision	Two GPIO drivers control same pin	Tracked in reservation table
Debug confusion	"Which module owns PA3?"	get_pin_function() shows owner
Accidental reuse	Copy-paste error uses wrong pin	Caught at runtime during init

7.25.1.5 RX72N Pin Architecture

The RX72N has 182 GPIO pins organized into ports:

Port	Hex ID	Pin Count	Example Pins	Common Usage
Port 0	0x0	8	P0.0-P0.7	LEDs, GPIOs
Port 1	0x1	8	P1.0-P1.7	ADC, GPIOs
Port A	0xA	8	PA.0-PA.7	SPI, I2C
Port B	0xB	8	PB.0-PB.7	USB, UART
Port C	0xC	8	PC.0-PC.7	SPI, PWM

Port	Hex ID	Pin Count	Example Pins	Common Usage
Port D	0xD	8	PD.0-PD.7	UART, GPIOs
Port E	0xE	8	PE.0-PE.7	Encoder, Timers

Port Numbering: Use hexadecimal (0x0-0x9, 0xA-0xG) Pin Numbering: Use decimal (0-7 within each port)

7.25.1.6 Thread Safety Guarantees

All interface methods are **REQUIRED** to be thread-safe in their implementations:

- Concurrent reserve/release from multiple threads: **SAFE**
- Concurrent validation checks: **SAFE**
- Typical implementation: Mutex-protected reservation table

7.25.1.7 Usage Examples

7.25.1.7.1 Example 1: Basic Dependency Injection

```
// In main.c: Setup infrastructure
pin_validator_t validator;
pin_validator_init(&validator);

rx_pin_interface_t pin_iface;
pin_validator_get_interface(&pin_iface, &validator);

// Inject interface into SPI driver
spi_config_t spi_cfg = {
    .pin_iface = &pin_iface,
    // ... other config
};
spi_init(&spi_cfg);
```

7.25.1.7.2 Example 2: Using Interface in Module

```
// spi_driver.c:
typedef struct {
    rx_pin_interface_t* pin_iface; // Injected dependency
    // ... other state
} spi_driver_t;

rx_err_t spi_init(spi_driver_t* drv, rx_pin_interface_t* pin_iface)
{
    drv->pin_iface = pin_iface;

    // Reserve SPI pins using interface
    rx_err_t err = pin_iface->reserve_pin(pin_iface->ctx, 0xA, 0, "SPI_CIPO");
    if (err != k_rx_ok) {
        return err;
    }

    err = pin_iface->reserve_pin(pin_iface->ctx, 0xA, 1, "SPI_COPI");
    if (err != k_rx_ok) {
        pin_iface->release_pin(pin_iface->ctx, 0xA, 0); // Cleanup
        return err;
    }

    return k_rx_ok;
}
```

7.25.1.7.3 Example 3: Mock Implementation for Unit Testing

```
// mock_pin_interface.c (for unit tests):
typedef struct {
    bool pins_reserved[256]; // Simple mock state
} mock_pin_validator_t;

static rx_err_t mock_reserve_pin(void* ctx, uint8_t port, uint8_t pin,
                                const char* function)
{
    mock_pin_validator_t* mock = (mock_pin_validator_t*)ctx;
    uint8_t idx = (port « 4) | pin;

    if (mock->pins_reserved[idx]) {
        return k_rx_err_resource_busy; // Simulate conflict
    }

    mock->pins_reserved[idx] = true;
    return k_rx_ok;
}

// In test:
void test_spi_init(void)
{
    mock_pin_validator_t mock = {};
    rx_pin_interface_t iface = {
        .ctx = &mock,
        .reserve_pin = mock_reserve_pin,
        // ... other methods
    };

    // Test SPI with mock (no real hardware needed!)
    spi_driver_t spi;
    rx_err_t err = spi_init(&spi, &iface);
    assert(err == k_rx_ok);
}
```

7.25.1.7.4 Example 4: Defensive Pin Usage Check

```
void motor_control_init(rx_pin_interface_t* iface)
{
    // Check if PWM pin is already in use before reserving
    if (iface->is_pin_reserved(iface->ctx, 0xE, 3)) {
        // Pin conflict - diagnose
        char function[32];
        if (iface->get_pin_function(iface->ctx, 0xE, 3,
                                   function, sizeof(function)) == k_rx_ok) {
            debug_printf("ERROR: PE3 already used for: %s\n", function);
        }
        return k_rx_err_resource_busy;
    }

    // Safe to reserve
    iface->reserve_pin(iface->ctx, 0xE, 3, "MOTOR_PWM");
}
```

7.25.1.7.5 Example 5: Global Infrastructure Pattern

```
// Using with rx_infrastructure module:
void module_init(void)
{
    // Get global pin interface
    rx_pin_interface_t* iface = rx_infrastructure_get_pin_interface();
    if (iface == nullptr) {
        return k_rx_err_not_initialized;
    }

    // Use convenience macro (wraps interface)
    rx_err_t err = RX_RESERVE_PIN(0xA, 5, "MODULE_PIN");
    if (err != k_rx_ok) {
        RX_REPORT_ERROR(err, "MODULE", "Pin PA5 conflict");
        return err;
    }
}
```

7.25.1.8 Design Rationale

Why Function Pointers Instead of Inheritance?

- C doesn't have classes/inheritance
- Function pointers provide runtime polymorphism
- ctx pointer acts like "this" in OOP
- Enables mock implementations for testing

Why 6 Methods Instead of More?

- Interface Segregation: Small, focused interface
- Covers all pin management use cases
- Easy to implement (fewer methods to stub in mocks)
- Stable API (less likely to change)

Why Opaque Context Pointer?

- Hides implementation details
- Allows different internal representations
- Prevents clients from accessing internals
- Supports multiple concurrent instances (if needed)

7.25.1.9 NASA Power of 10 Compliance

Rule	Status	Implementation
1. Simplify control flow	[OK]	No goto, setjmp, recursion
2. Fixed loop bounds	[OK]	No loops in interface
3. No dynamic memory	[OK]	Interface is POD struct
4. Functions < 60 lines	[OK]	Validation function is 14 lines
5. Use assertions	[OK]	Validation function checks
6. Data at smallest scope	[OK]	Function pointers encapsulated
7. Check return values	[OK]	All methods return rx_err_t
8. Limit preprocessor	[OK]	No macros in interface
9. Restrict pointers	[WARN]	Function pointers (DIP pattern)
10. Compile warnings	[OK]	-Wall -Wextra -Werror

7.25.1.10 Memory Footprint

Component	Size (bytes)	Notes
rx_pin_interface_t	32	7 pointers x 4-8 bytes (architecture-dependent)
ctx pointer	4-8	Points to concrete implementation
Function pointers	4-8 each	6 methods

Component	Size (bytes)	Notes
Total	32-56	Stack-allocated, no heap

See also

[rx_infrastructure.h](#) Global service locator for pin interface
[rx_pin_validator.h](#) Concrete implementation of this interface
[RX_RESERVE_PIN\(\)](#) Convenience macro using this interface
[RX_RELEASE_PIN\(\)](#) Convenience macro using this interface

Since

Version 1.0.0

Date

2026-01-01

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Definition in file [rx_pin_interface.h](#).

7.25.2 Data Structure Documentation

7.25.2.1 struct rx_pin_interface

Pin Validator Interface - THE ABSTRACTION FOR PIN MANAGEMENT.

7.25.2.2 Overview

This structure IS the abstraction that enables Dependency Inversion Principle for pin management in STAR firmware. It defines the contract that all pin validator implementations must fulfill, allowing high-level modules to depend on this stable interface rather than concrete implementations.

7.25.2.3 Polymorphism via Function Pointers

This struct implements runtime polymorphism in C using function pointers:

- `ctx`: Acts like "this" pointer in OOP (points to concrete implementation state)
- Function pointers: Virtual methods that can be overridden by implementations
- First parameter (`ctx`): Passed to every method, providing access to private data

7.25.2.4 Memory Layout

```
struct rx_pin_interface {
    // Offset | Size | Field
    // 0 | 4-8 | void* ctx
    // 4-8 | 4-8 | rx_pin_validate_fn validate_pin
    // 8-16 | 4-8 | rx_pin_reserve_fn reserve_pin
    // 12-24 | 4-8 | rx_pin_release_fn release_pin
    // 16-32 | 4-8 | rx_pin_is_reserved_fn is_pin_reserved
    // 20-40 | 4-8 | rx_pin_get_function_fn get_pin_function
    // 24-48 | 4-8 | rx_pin_clear_all_fn clear_all_reservations
    // TOTAL: 28-56 bytes (architecture-dependent)
};
```

7.25.2.5 Typical Usage Pattern (Dependency Injection)

7.25.2.5.1 Step 1: Create Concrete Implementation

```
// Concrete implementation (in pin_validator.c):
pin_validator_t concrete_validator;
pin_validator_init(&concrete_validator);
```

7.25.2.5.2 Step 2: Extract Interface

```
// Get abstract interface from concrete implementation:
rx_pin_interface_t iface;
pin_validator_get_interface(&iface, &concrete_validator);

// The interface is now populated:
// - iface.ctx = &concrete_validator
// - iface.reserve_pin = pin_validator_reserve
// - iface.release_pin = pin_validator_release
// - ... (all function pointers initialized)
```

7.25.2.5.3 Step 3: Inject Into High-Level Module

```
// High-level module depends on abstract interface:
spi_config_t spi_cfg = {
    .pin_iface = &iface, // Inject interface (not concrete type!)
    // ... other config
};
spi_init(&spi_cfg);
```

7.25.2.5.4 Step 4: Use Through Interface

```
// Inside spi_init():
rx_err_t spi_init(spi_config_t* cfg)
{
    // Call through function pointer (polymorphism!)
    rx_err_t err = cfg->pin_iface->reserve_pin(
        cfg->pin_iface->ctx, // Pass context (like "this")
        0xA, 5, // Port A, Pin 5
        "SPI_COPI" // Function name
    );
    return err;
}
```

7.25.2.6 Interface Stability Guarantees

This interface is STABLE - changes are rare and carefully managed:

- Adding methods: Append to end (preserve ABI)
- Removing methods: FORBIDDEN (breaks existing code)
- Changing signatures: FORBIDDEN (breaks existing code)

- Reordering fields: FORBIDDEN (breaks ABI)

Why stability matters:

- Modules can cache pointers to this struct
- No need to recompile all modules when implementation changes
- Enables versioning (if needed in future)

7.25.2.7 Thread Safety Requirements

All function pointers MUST point to thread-safe implementations:

Method	Thread-Safe?	Typical Protection
validate_pin	[OK] Yes	Read-only, no locks needed
reserve_pin	[OK] Yes	Mutex around reservation table
release_pin	[OK] Yes	Mutex around reservation table
is_pin_reserved	[OK] Yes	Mutex (read lock) or atomic
get_pin_function	[OK] Yes	Mutex (read lock) + strcpy
clear_all_reservations	[OK] Yes	Exclusive mutex lock

7.25.2.8 Validation Requirements

Before using this interface, ALWAYS validate:

```
rx_err_t err = rx_pin_interface_validate(&iface);
if (err != k_rx_ok) {
    // Interface is nullptr or has missing function pointers
    return err;
}
// Safe to use
```

7.25.2.9 Mock Implementation Example (Unit Testing)

```
// Minimal mock for testing:
typedef struct {
    bool pins[256]; // Simple reservation flags
} mock_pin_validator_t;

static rx_err_t mock_reserve(void* ctx, uint8_t port, uint8_t pin,
                             const char* func)
{
    mock_pin_validator_t* mock = (mock_pin_validator_t*)ctx;
    uint8_t idx = (port « 4) | pin;
    if (mock->pins[idx]) return k_rx_err_gpio_conflict;
    mock->pins[idx] = true;
    return k_rx_ok;
}

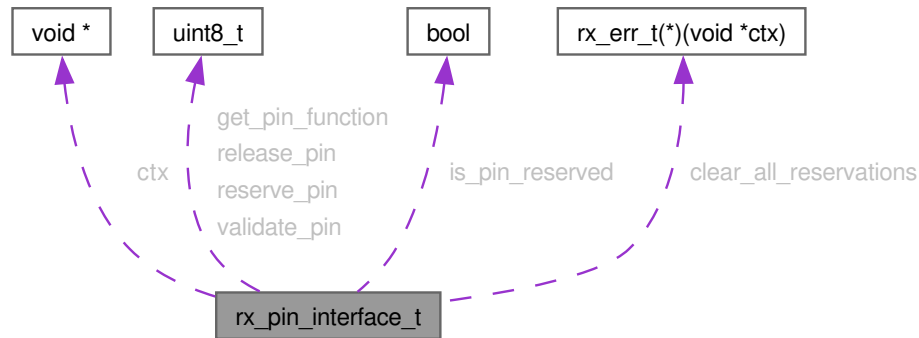
// Create mock interface:
mock_pin_validator_t mock = {};
rx_pin_interface_t mock_iface = {
    .ctx = &mock,
    .reserve_pin = mock_reserve,
    .release_pin = mock_release,
    // ... other methods
};

// Inject mock into SPI for testing (no real hardware!)
spi_init(&spi_cfg, &mock_iface);
```

7.25.2.10 Field Documentation

Definition at line 1017 of file [rx_pin_interface.h](#).

Collaboration diagram for `rx_pin_interface_t`:



Data Fields

rx_pin_clear_all_fn	<code>clear_all_reservations</code>	Clear all pin reservations (reset to initial state). Clear all reservations Function pointer to full clear implementation. Dangerous operation - use only during controlled reset/test teardown. MUST NOT be NULL. See rx_pin_clear_all_fn for details. Note All function pointers MUST be non-NULL (validated by rx_pin_interface_validate()) <code>ctx</code> pointer MUST be non-NULL and point to valid concrete implementation Interface is POD (Plain Old Data) - can be copied, stack-allocated Typical size: 28-56 bytes (architecture-dependent) See also rx_pin_interface_validate() Validate interface before use pin_validator_get_interface() Create interface from concrete implementation rx_infrastructure_get_pin_interface() Get global pin interface Since Version 1.0.0
-------------------------------------	-------------------------------------	--

<code>void *</code>	<code>ctx</code>	Opaque pointer to concrete implementation's private state. Implementation context (opaque pointer to concrete validator) Acts like "this" pointer in OOP. Points to concrete pin validator instance (e.g., <code>pin_validator_t*</code>). All function pointers receive this as first parameter. MUST NOT be NULL if interface is initialized.
<code>rx_pin_get_function_fn</code>	<code>get_pin_function</code>	Get the function name associated with a reserved pin. Get pin's function name Function pointer to function name retrieval implementation. Used for debugging conflicts. MUST NOT be NULL. See <code>rx_pin_get_function_fn</code> for details.
<code>rx_pin_is_reserved_fn</code>	<code>is_pin_reserved</code>	Check if a pin is currently reserved. Check if pin is reserved Function pointer to reservation query implementation. Returns boolean status. MUST NOT be NULL. See <code>rx_pin_is_reserved_fn</code> for details.
<code>rx_pin_release_fn</code>	<code>release_pin</code>	Release a previously reserved pin. Release reserved pin Function pointer to release implementation. Frees pin for reuse by other modules. MUST NOT be NULL. See <code>rx_pin_release_fn</code> for details.
<code>rx_pin_reserve_fn</code>	<code>reserve_pin</code>	Reserve a pin for exclusive use by a function/module. Reserve pin for function Function pointer to reservation implementation. Prevents pin conflicts by tracking ownership. MUST NOT be NULL. See <code>rx_pin_reserve_fn</code> for details.
<code>rx_pin_validate_fn</code>	<code>validate_pin</code>	Validate if a pin exists and is accessible. Validate pin existence Function pointer to validation implementation. Checks port and pin validity against RX72N hardware. MUST NOT be NULL. See <code>rx_pin_validate_fn</code> for details.

7.25.3 Typedef Documentation

7.25.3.1 `rx_pin_clear_all_fn`

```
typedef rx\_err\_t(* rx_pin_clear_all_fn) (void *ctx)
```

Clear all pin reservations (reset to initial state).

Function pointer type for clearing all pin reservations at once. Typically used during:

- System reset/reinitialization

- Unit test teardown
- Error recovery (start fresh)
- Development/debugging

Implementation Requirements:

- Clear all reservation flags atomically
- Clear all function name strings
- Reset to clean state (as if just initialized)
- Must be thread-safe with exclusive mutex lock
- WARNING: Dangerous operation - no confirmation

7.25.3.2 When to Use vs. When NOT to Use

[OK] SAFE to use:

- Unit test teardown (known clean state)
- System reboot/reset sequence
- Error recovery when retrying full initialization

[X] DANGEROUS to use:

- During normal operation (other modules may have pins reserved)
- Without stopping all modules first
- In production code (rarely needed)

7.25.3.3 Typical Usage

```
// Unit test teardown pattern
void test_teardown(void)
{
    rx_pin_interface_t* iface = get_pin_interface();
    iface->clear_all_reservations(iface->ctx); // Clean slate for next test
}

// System reset pattern
void system_reset(void)
{
    // 1. Stop all modules
    motor_stop();
    spi_shutdown();
    usb_disconnect();

    // 2. Clear reservations
    rx_pin_interface_t* iface = get_pin_interface();
    iface->clear_all_reservations(iface->ctx);

    // 3. Reinitialize
    system_init();
}
```

Parameters

in	ctx	Implementation context (opaque pointer to concrete validator state)
----	-----	---

Returns

[rx_err_t](#) Error code indicating clear result

Return values

k_rx_ok	All reservations cleared successfully
-------------------------	---------------------------------------

Note

Thread-safe: Implementation must use exclusive mutex lock

Destructive: All pin reservations are lost (no undo)

No confirmation: Clears immediately without prompts

Performance: ~50-200 us typical (mutex + full table clear)

Warning

DANGEROUS: Clears ALL reservations. Ensure no modules are using pins.

Only use during controlled shutdown, reset, or test teardown

Do NOT call during normal operation

See also

[rx_pin_release_fn](#) Release individual pins (safer than `clear_all`)

Since

Version 1.0.0

Definition at line 804 of file [rx_pin_interface.h](#).

7.25.3.4 rx_pin_get_function_fn

```
typedef rx\_err\_t(* rx_pin_get_function_fn) (void *ctx, const uint8_t port, const uint8_t pin, char *function_out, const uint32_t function_len)
```

Get the function name associated with a reserved GPIO pin.

Function pointer type for querying which function/module reserved a pin. Useful for debugging pin conflicts and generating pin usage reports.

Implementation Requirements:

- Validate port and pin numbers
- Check if pin is reserved (error if not)
- Copy function name to output buffer (with null terminator)
- Truncate if buffer too small
- Must be thread-safe with mutex protection

7.25.3.5 Typical Usage

```
// Diagnose pin conflict
void diagnose_conflict(rx_pin_interface_t* iface, uint8_t port, uint8_t pin)
{
    if (!iface->is_pin_reserved(iface->ctx, port, pin)) {
        printf("Pin P%X.%d is available\n", port, pin);
        return;
    }

    char function[k_pin_function_name_max_len];
    rx_err_t err = iface->get_pin_function(iface->ctx, port, pin,
                                          function, sizeof(function));

    if (err == k_rx_ok) {
        printf("Pin P%X.%d reserved for: %s\n", port, pin, function);
    }
}

// Generate pin usage report
void print_pin_usage(rx_pin_interface_t* iface)
{
    for (uint8_t port = 0; port <= 0x10; port++) {
        for (uint8_t pin = 0; pin < 8; pin++) {
            if (iface->is_pin_reserved(iface->ctx, port, pin)) {
                char function[32];
                if (iface->get_pin_function(iface->ctx, port, pin,
                                          function, sizeof(function)) == k_rx_ok) {
                    printf("P%X.%d: %s\n", port, pin, function);
                }
            }
        }
    }
}
```

Parameters

in	ctx	Implementation context (opaque pointer to concrete validator state)
in	port	Port number (0x0-0x9, 0xA-0xG)
in	pin	Pin number within port (0-7)
out	function_out	Buffer to store function name (must be non-NULL)
in	function_len	Length of function_out buffer (should be >= k_pin_function_name_max_len)

Returns

[rx_err_t](#) Error code indicating retrieval result

Return values

k_rx_ok	Function name copied successfully to function_out
k_rx_err_invalid_state	Pin is not reserved (no function to retrieve)
k_rx_err_null_ptr	function_out parameter is nullptr
k_rx_err_invalid_size	function_len is 0 or too small (minimum 2 for "X\0")
k_rx_err_gpio_invalid_port	Invalid port number
k_rx_err_gpio_invalid_pin	Invalid pin number

Note

- Thread-safe: Implementation must use mutex for consistent reads
- Null termination: Output is always null-terminated if buffer size > 0
- Truncation: Long names are truncated to fit buffer (including null)
- Performance: ~10-30 us typical (mutex + string copy)

Warning

Check return value before using `function_out` contents

Buffer size should be at least `k_pin_function_name_max_len` bytes

See also

[rx_pin_is_reserved_fn](#) Check if pin is reserved first

`pin_interface_limits_t::k_pin_function_name_max_len` Recommended buffer size

Since

Version 1.0.0

Definition at line 724 of file [rx_pin_interface.h](#).

7.25.3.6 rx_pin_is_reserved_fn

```
typedef bool(* rx_pin_is_reserved_fn) (void *ctx, const uint8_t port, const uint8_t pin)
```

Check if a GPIO pin is currently reserved.

Function pointer type for querying pin reservation status. Returns boolean result without error codes. Useful for defensive checks before attempting reservation or for debugging pin conflicts.

Implementation Requirements:

- Return true if pin is reserved, false if available
- Invalid port/pin should return false (not reserved)
- Must be thread-safe (read-only operation)
- Fast lookup (no blocking operations)

7.25.3.7 Typical Usage

```
// Defensive reservation check
if (iface->is_pin_reserved(iface->ctx, 0xA, 5)) {
    // Pin already in use - diagnose conflict
    char function[32];
    iface->get_pin_function(iface->ctx, 0xA, 5, function, sizeof(function));
    debug_printf("PA5 conflict: already used for %s\n", function);
    return k_rx_err_resource_busy;
}
```

```
// Proceed with reservation
iface->reserve_pin(iface->ctx, 0xA, 5, "MY_FUNCTION");
```

Parameters

in	ctx	Implementation context (opaque pointer to concrete validator state)
in	port	Port number (0x0-0x9, 0xA-0xG)
in	pin	Pin number within port (0-7)

Returns

bool Reservation status

Return values

true	Pin is currently reserved
false	Pin is available (or invalid port/pin)

Note

Thread-safe: Read-only check, safe to call concurrently

No error codes: Returns false for invalid pins (simple boolean API)

Fast: Typically 1-5 us (direct table lookup)

See also

[rx_pin_get_function_fn](#) Get the function name for a reserved pin

[rx_pin_reserve_fn](#) Reserve a pin

Since

Version 1.0.0

Definition at line 645 of file [rx_pin_interface.h](#).

7.25.3.8 rx_pin_release_fn

```
typedef rx_err_t(* rx_pin_release_fn) (void *ctx, const uint8_t port, const uint8_t pin)
```

Release a previously reserved GPIO pin, making it available for reuse.

Function pointer type for releasing pin reservations. After release, the pin becomes available for reservation by other modules. Typically called during module deinitialization or reconfiguration.

Implementation Requirements:

- Validate port and pin numbers first
- Check if pin is currently reserved (error if not)
- Clear reservation flag atomically
- Clear function name storage
- Must be thread-safe with mutex protection
- Idempotent safe: Releasing unreserved pin returns error (doesn't crash)

7.25.3.9 Typical Usage Patterns

```
// Pattern 1: Normal deinitialization
void spi_deinit(void) {
    iface->release_pin(iface->ctx, 0xA, 0); // CIPO
    iface->release_pin(iface->ctx, 0xA, 1); // COPI
    iface->release_pin(iface->ctx, 0xA, 2); // CLK
}

// Pattern 2: Error cleanup (reverse order)
if (reserve_failed) {
    iface->release_pin(iface->ctx, 0xA, 1); // Release second
    iface->release_pin(iface->ctx, 0xA, 0); // Release first
}

// Pattern 3: Defensive release (check first)
if (iface->is_pin_reserved(iface->ctx, 0xA, 5)) {
    iface->release_pin(iface->ctx, 0xA, 5);
}
```

Parameters

in	ctx	Implementation context (opaque pointer to concrete validator state)
in	port	Port number (0x0-0x9, 0xA-0xG)
in	pin	Pin number within port (0-7)

Returns

[rx_err_t](#) Error code indicating release result

Return values

k_rx_ok	Pin released successfully, now available for reuse
k_rx_err_invalid_state	Pin was not reserved (double-release or never reserved)
k_rx_err_gpio_invalid_port	Invalid port number
k_rx_err_gpio_invalid_pin	Invalid pin number

Note

Thread-safe: Implementation must use mutex or atomic operations

Idempotent safe: Releasing unreserved pin returns error but doesn't crash

Best-effort cleanup: Most code ignores return value during cleanup

Performance: ~5-30 us typical (mutex + flag clear)

See also

[rx_pin_reserve_fn](#) Reserve a pin (must be called first)

[RX_RELEASE_PIN\(\)](#) Convenience macro wrapping this function

Since

Version 1.0.0

Definition at line 596 of file [rx_pin_interface.h](#).

7.25.3.10 rx_pin_reserve_fn

```
typedef rx_err_t(* rx_pin_reserve_fn) (void *ctx, const uint8_t port, const uint8_t pin, const char *function)
```

Reserve a GPIO pin for exclusive use by a specific function/module.

Function pointer type for reserving pins to prevent conflicts. Once reserved, a pin cannot be reserved again until released. This is the primary conflict prevention mechanism in the STAR firmware.

Implementation Requirements:

- Validate port and pin numbers first (call validate_pin internally)
- Check if pin is already reserved
- Store function name (up to k_pin_function_name_max_len bytes)
- Mark pin as reserved atomically
- Must be thread-safe with mutex protection
- Function name string is not copied - must remain valid

7.25.3.11 Conflict Detection Strategy

```
// Module A tries to reserve PA5 for SPI:
err = iface->reserve_pin(iface->ctx, 0xA, 5, "SPI_COPI"); // Success -> k_rx_ok

// Module B tries to reserve same pin for I2C:
err = iface->reserve_pin(iface->ctx, 0xA, 5, "I2C_SDA"); // Fail -> k_rx_err_gpio_conflict
```

7.25.3.12 Typical Implementation Pattern

```
rx_err_t pin_validator_reserve(void* ctx, uint8_t port, uint8_t pin,
                             const char* function)
{
    pin_validator_t* validator = (pin_validator_t*)ctx;

    // 1. Validate inputs
    if (function == nullptr) return k_rx_err_invalid_arg;
    rx_err_t err = pin_validator_validate(ctx, port, pin);
    if (err != k_rx_ok) return err;

    // 2. Lock for thread safety
    tx_mutex_get(&validator->mutex, TX_WAIT_FOREVER);

    // 3. Check if already reserved
    uint8_t idx = (port « 4) | pin;
    if (validator->reserved[idx]) {
        tx_mutex_put(&validator->mutex);
        return k_rx_err_gpio_conflict;
    }

    // 4. Reserve pin
    validator->reserved[idx] = true;
    strncpy(validator->functions[idx], function, k_pin_function_name_max_len - 1);
    validator->functions[idx][k_pin_function_name_max_len - 1] = '\0';

    tx_mutex_put(&validator->mutex);
    return k_rx_ok;
}
```

Parameters

in	ctx	Implementation context (opaque pointer to concrete validator state)
in	port	Port number (0x0-0x9, 0xA-0xG)

in	pin	Pin number within port (0-7)
in	function	Function name (must be string literal or valid pointer, NOT copied)

Returns

[rx_err_t](#) Error code indicating reservation result

Return values

k_rx_ok	Pin reserved successfully
k_rx_err_gpio_conflict	Pin already reserved by another function
k_rx_err_gpio_invalid_port	Invalid port number
k_rx_err_gpio_invalid_pin	Invalid pin number
k_rx_err_invalid_arg	function parameter is nullptr

Note

Thread-safe: Implementation must use mutex or atomic operations
String lifetime: function string must remain valid (use literals!)
Idempotent check: Reserving same pin twice returns conflict error
Performance: ~10-50 us typical (mutex + table update)

Warning

Function name is NOT copied - must be string literal or static storage
Must call `release_pin()` to free the pin for reuse

See also

[rx_pin_release_fn](#) Release a reserved pin
[RX_RESERVE_PIN\(\)](#) Convenience macro wrapping this function

Since

Version 1.0.0

Definition at line 533 of file [rx_pin_interface.h](#).

7.25.3.13 rx_pin_validate_fn

typedef [rx_err_t](#)(* rx_pin_validate_fn) (void *ctx, const uint8_t port, const uint8_t pin)

Validate if a GPIO pin exists and is accessible on the RX72N.

Function pointer type for validating pin existence before reservation or use. Implementations must check both port and pin validity against the RX72N hardware.

Implementation Requirements:

- Check port number is in valid range (0x0-0x9, 0xA-0xG for RX72N)
- Check pin number is in valid range (0-7)
- Verify that specific port/pin combination exists (not all ports have all pins)
- Return appropriate error codes for invalid port vs invalid pin
- Must be thread-safe (multiple threads may validate concurrently)

7.25.3.14 Typical Implementation

```
// Example concrete implementation:
rx_err_t pin_validator_validate(void* ctx, uint8_t port, uint8_t pin)
{
    // Check port range
    if (port > 0x10) { // RX72N max port
        return k_rx_err_gpio_invalid_port;
    }

    // Check pin range
    if (pin > 7) {
        return k_rx_err_gpio_invalid_pin;
    }

    // Check specific port/pin exists (RX72N-specific validation)
    if (!rx72n_pin_exists(port, pin)) {
        return k_rx_err_gpio_invalid_pin;
    }

    return k_rx_ok;
}
```

Parameters

in	ctx	Implementation context (opaque pointer to concrete validator state)
in	port	Port number (0x0-0x9, 0xA-0xG for A-G on RX72N)
in	pin	Pin number within port (0-7)

Returns

[rx_err_t](#) Error code indicating validation result

Return values

k_rx_ok	Pin exists and is accessible
k_rx_err_gpio_invalid_port	Port number out of range or doesn't exist
k_rx_err_gpio_invalid_pin	Pin number out of range or doesn't exist on this port

Note

Thread-safe: Implementation must be callable from multiple threads concurrently

Idempotent: Calling multiple times with same arguments returns same result

Fast: Typically table lookup, no blocking operations

See also

[rx_pin_interface.validate_pin](#) Function pointer field in interface struct

Since

Version 1.0.0

Definition at line 448 of file [rx_pin_interface.h](#).

7.25.4 Enumeration Type Documentation

7.25.4.1 pin_interface_limits_t

enum `pin_interface_limits_t` : `uint8_t`

Pin Interface Configuration Constants.

Defines compile-time constants for the pin interface system. These limits ensure predictable memory usage and prevent buffer overflows in pin function name handling.

7.25.4.2 Function Name Length Rationale

Why 32 bytes?

- Sufficient for descriptive names: "SPI_CONTROLLER_OUT" = 20 chars
- Allows module prefix: "MOTOR_ENCODER_PHASE_A" = 21 chars
- Padding for safety: 32 is power-of-2, cache-aligned
- Embedded typical: Most embedded systems use 16-32 char limits

7.25.4.3 Typical Function Name Examples

Function Name	Length	Usage
"SPI_CIPO"	8	SPI peripheral
"UART_TX"	7	UART transmit
"I2C_SDA"	7	I2C data line
"MOTOR_PWM_CH1"	14	Motor PWM output
"ENCODER_A"	9	Encoder input
"USB_DP"	6	USB data positive
"GPIO_LED_STATUS"	15	General purpose LED
"SPI_CONTROLLER_OUT_PERIPH_IN"	30	Full OSHWA terminology

7.25.4.4 Memory Impact

Assuming 128 reservable pins:

- Function names: $128 \times 32 = 4096$ bytes (4 KB)
- Reservation flags: $128 \times 1 = 128$ bytes
- Port/pin storage: $128 \times 2 = 256$ bytes
- Total: ~ 4.5 KB for complete pin tracking

See also

[rx_pin_reserve_fn](#) Function that uses this limit

[rx_pin_get_function_fn](#) Function that returns names within this limit

Since

Version 1.0.0

Enumerator

k_pin_function_name_max_len	Maximum length of pin function name including null terminator. Buffer size for pin function names (e.g., "SPI_COPI", "UART_TX", "MOTOR_PWM"). Includes space for null terminator. If a function name exceeds this limit during reservation, it will be truncated. Design choice: 32 bytes balances descriptiveness with memory efficiency. Example: Valid Names <pre>"SPI_COPI" // 9 bytes (OK) "MOTOR_ENCODER_A" // 16 bytes (OK) "I2C_CONTROLLER_SDA" // 19 bytes (OK) "SPI_CONTROLLER_OUT" // 20 bytes (OK)</pre> Example: Names That Will Be Truncated <pre>"SUPER_LONG_MODULE_NAME_THAT_EXCEEDS_LIMIT" // 43 bytes -> truncated to 31+null</pre>
-----------------------------	---

Definition at line 361 of file [rx_pin_interface.h](#).

```

00361      : uint8_t {
00362  /**
00363   * @brief Maximum length of pin function name including null terminator
00364   * @details
00365   * Buffer size for pin function names (e.g., "SPI_COPI", "UART_TX", "MOTOR_PWM").
00366   * Includes space for null terminator. If a function name exceeds this limit during
00367   * reservation, it will be truncated.
00368   *
00369   * **Design choice**: 32 bytes balances descriptiveness with memory efficiency.
00370   *
00371   * @par Example: Valid Names
00372   * @code
00373   * "SPI_COPI"           // 9 bytes (OK)
00374   * "MOTOR_ENCODER_A"    // 16 bytes (OK)
00375   * "I2C_CONTROLLER_SDA" // 19 bytes (OK)
00376   * "SPI_CONTROLLER_OUT" // 20 bytes (OK)
00377   * @endcode
00378   *
00379   * @par Example: Names That Will Be Truncated
00380   * @code
00381   * "SUPER_LONG_MODULE_NAME_THAT_EXCEEDS_LIMIT" // 43 bytes -> truncated to 31+null
00382   * @endcode
00383   */
00384   k_pin_function_name_max_len = 32,
00385 } pin_interface_limits_t;
```

7.25.5 Function Documentation

7.25.5.1 rx_pin_interface_validate()

```

rx\_err\_t rx_pin_interface_validate (
    const rx\_pin\_interface\_t * iface)  [inline], [static]
```

Validate that a pin interface is properly initialized and safe to use.

Performs comprehensive validation of an `rx_pin_interface_t` to ensure all function pointers are non-NULL and the interface is ready for use. This function should be called before using any interface received from external sources.

7.25.5.2 Validation Checks Performed

1. nullptr check: Ensures iface parameter is non-NULL
2. Function pointer checks: Verifies all 6 function pointers are non-NULL:
 - `validate_pin`
 - `reserve_pin`
 - `release_pin`
 - `is_pin_reserved`
 - `get_pin_function`
 - `clear_all_reservations`

Note: The ctx pointer is NOT validated (can be NULL for stateless mocks).

7.25.5.3 When to Use This Function

7.25.5.3.1 Required Scenarios:

- Module initialization: Validate injected interfaces before use
- Defensive programming: Check interface from untrusted sources
- Debug builds: Assert interface validity at entry points

7.25.5.3.2 Optional Scenarios:

- Production code: Can be skipped if interface comes from trusted source (`rx_infrastructure`)
- Performance-critical paths: Validation adds ~20 cycles overhead

7.25.5.4 Typical Usage Patterns

7.25.5.4.1 Pattern 1: Defensive Module Initialization

```
rx_err_t spi_init(spi_config_t* cfg)
{
    // Validate injected pin interface
    rx_err_t err = rx_pin_interface_validate(cfg->pin_iface);
    if (err != k_rx_ok) {
        return err; // Invalid interface, cannot proceed
    }

    // Safe to use interface now
    cfg->pin_iface->reserve_pin(cfg->pin_iface->ctx, 0xA, 5, "SPI_COPI");
    return k_rx_ok;
}
```

7.25.5.4.2 Pattern 2: Debug-Only Validation

```

#ifdef DEBUG
// Validate in debug builds only
rx_err_t err = rx_pin_interface_validate(pin_iface);
assert(err == k_rx_ok && "Pin interface validation failed");
#endif

// Use interface (validation skipped in release builds)
pin_iface->reserve_pin(pin_iface->ctx, 0xA, 5, "SPI_COPI");

```

7.25.5.4.3 Pattern 3: Trust Global Infrastructure

```

// Get interface from trusted global infrastructure
rx_pin_interface_t* iface = rx_infrastructure_get_pin_interface();
if (iface == nullptr) {
    return k_rx_err_not_initialized; // Infrastructure not ready
}

// Skip validation - global infrastructure guarantees valid interfaces
// (validation was already done during rx_infrastructure_init())
iface->reserve_pin(iface->ctx, 0xA, 5, "SPI_COPI");

```

7.25.5.4.4 Pattern 4: Validate Before Caching

```

typedef struct {
    rx_pin_interface_t* pin_iface; // Cached interface
} module_state_t;

rx_err_t module_init(module_state_t* state, rx_pin_interface_t* iface)
{
    // Validate before caching (ensures valid for entire module lifetime)
    rx_err_t err = rx_pin_interface_validate(iface);
    if (err != k_rx_ok) {
        return err;
    }

    // Cache for later use (no need to re-validate)
    state->pin_iface = iface;
    return k_rx_ok;
}

```

7.25.5.5 Error Scenarios

7.25.5.5.1 Scenario 1: NULL Interface Pointer

```

rx_pin_interface_t* iface = nullptr;
rx_err_t err = rx_pin_interface_validate(iface);
// Returns: k_rx_err_null_ptr

```

7.25.5.5.2 Scenario 2: Partially Initialized Interface

```

rx_pin_interface_t iface = {
    .ctx = &validator,
    .reserve_pin = pin_validator_reserve,
    .release_pin = nullptr, // Missing!
    // ... other fields nullptr
};
rx_err_t err = rx_pin_interface_validate(&iface);
// Returns: k_rx_err_invalid_state

```

7.25.5.5.3 Scenario 3: Fully Initialized Interface

```

rx_pin_interface_t iface;
pin_validator_get_interface(&iface, &validator);
rx_err_t err = rx_pin_interface_validate(&iface);
// Returns: k_rx_ok (all function pointers set)

```

7.25.5.6 Performance Characteristics

- Execution time: ~20-50 cycles @ 240 MHz (~0.1-0.2 us)
- Operations: 7 nullptr checks (1 iface + 6 function pointers)
- Overhead: Negligible for init paths, skip in tight loops if needed
- Optimization: Compiler may inline this function (static inline)

Parameters

in	iface	Pin interface to validate (can be NULL - returns error)
----	-------	---

Returns

[rx_err_t](#) Error code indicating validation result

Return values

k_rx_ok	Interface is valid and ready for use
k_rx_err_null_ptr	iface parameter is nullptr
k_rx_err_invalid_state	One or more function pointers are nullptr

Precondition

None (can be called with nullptr - returns error safely)

Postcondition

If [k_rx_ok](#) returned, all function pointers are non-NULL

If error returned, interface should NOT be used

Note

Context (ctx) is NOT validated - can be NULL for stateless implementations

Inlined: Compiled directly into call site (zero function call overhead)

Thread-safe: Read-only check, safe to call concurrently

Idempotent: Calling multiple times returns same result

Warning

Do NOT use interface if this function returns error

This function does NOT validate that function pointers point to valid functions (only checks non-NULL)

Example: Complete Validation Workflow

```

void module_use_pin_interface(rx_pin_interface_t* iface)
{
    // Step 1: Validate interface
    rx_err_t err = rx_pin_interface_validate(iface);
    if (err == k_rx_err_null_ptr) {
        debug_printf("ERROR: NULL pin interface\n");
        return;
    } else if (err == k_rx_err_invalid_state) {
        debug_printf("ERROR: Incomplete pin interface (missing function pointers)\n");
        return;
    }

    // Step 2: Interface is valid - safe to use
    err = iface->reserve_pin(iface->ctx, 0xA, 5, "MODULE_PIN");
    if (err != k_rx_ok) {
        debug_printf("Pin reservation failed: %d\n", err);
        return;
    }

    // Pin successfully reserved
}

```

See also

[rx_pin_interface](#) Structure being validated

[pin_validator_get_interface\(\)](#) Properly initializes this interface

[rx_infrastructure_get_pin_interface\(\)](#) Returns validated global interface

Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 4: [OK] Function is 14 lines (well under 60 line limit)
- Rule 5: [OK] 7 validation checks (interface + 6 function pointers)

Definition at line 1219 of file [rx_pin_interface.h](#).

```

01220 {
01221     if (iface == nullptr) {
01222         return k_rx_err_null_ptr;
01223     }
01224
01225     /* Check required function pointers */
01226     if (iface->validate_pin == nullptr || iface->reserve_pin == nullptr ||
01227         iface->release_pin == nullptr || iface->is_pin_reserved == nullptr ||
01228         iface->get_pin_function == nullptr || iface->clear_all_reservations == nullptr) {
01229         return k_rx_err_invalid_state;
01230     }
01231
01232     return k_rx_ok;
01233 }

```

References [k_rx_err_invalid_state](#), [k_rx_err_null_ptr](#), and [k_rx_ok](#).

7.26 rx_pin_interface.h

[Go to the documentation of this file.](#)

```

00001 /**
00002  * @file rx_pin_interface.h
00003  * @brief Abstract Pin Validator Interface - THE "I" and "D" in SOLID
00004  *
00005  * @details

```

```

00006 * ## Overview
00007 *
00008 * Defines the abstract interface for GPIO pin validation and reservation in the
00009 * STAR RX72N firmware. This interface is the cornerstone of Dependency Inversion
00010 * Principle (DIP) for pin management - high-level modules depend on this abstraction,
00011 * not on concrete pin validator implementations.
00012 *
00013 * Key Design Pattern: Interface Segregation + Dependency Inversion
00014 *
00015 * ```
00016 * +-----+
00017 * | High-Level Modules (Peripheral Drivers) |
00018 * | SPI, I2C, UART, USB, Motor Control, Encoder, GPIO, PWM, etc. |
00019 * +-----+
00020 * | |
00021 * | depends on (abstract) | depends on (abstract)
00022 * | v | v |
00023 * +-----+ +-----+
00024 * | rx_pin_interface_t | | rx_infrastructure module |
00025 * | (THIS FILE - ABSTRACT) | <-+ (Global service locator) |
00026 * | - validate_pin() | +-----+
00027 * | - reserve_pin() |
00028 * | - release_pin() |
00029 * | - is_pin_reserved() |
00030 * | - get_pin_function() |
00031 * | - clear_all() |
00032 * +-----+
00033 * | |
00034 * | implements (concrete)
00035 * | v |
00036 * +-----+
00037 * | pin_validator_t |
00038 * | (CONCRETE IMPLEMENTATION)|
00039 * | - Pin reservation table |
00040 * | - Mutex for thread safety|
00041 * | - Validation logic |
00042 * +-----+
00043 * ```
00044 *
00045 * ## Why This Design Matters
00046 *
00047 * ### Problem Without Abstraction
00048 * ```c
00049 * // [FAIL] BAD: Direct dependency on concrete type
00050 * #include "pin_validator.h" // Couples to implementation
00051 *
00052 * void spi_init(pin_validator_t* validator) { // Concrete type!
00053 *     pin_validator_reserve(validator, 0xA, 5, "SPI_COPI");
00054 * }
00055 * ```
00056 * Issues:
00057 * - Can't swap implementations (no mock for testing)
00058 * - Tight coupling to pin_validator_t internals
00059 * - Can't inject different validators
00060 * - Hard to unit test SPI without real pin validator
00061 *
00062 * ### Solution With Abstraction
00063 * ```c
00064 * // [OK] GOOD: Dependency on abstract interface
00065 * #include "rx_pin_interface.h" // Only interface
00066 *
00067 * void spi_init(rx_pin_interface_t* iface) { // Abstract interface!
00068 *     iface->reserve_pin(iface->ctx, 0xA, 5, "SPI_COPI");
00069 * }
00070 * ```
00071 * Benefits:
00072 * - Testable (inject mock_pin_interface for unit tests)
00073 * - Flexible (swap implementations without changing SPI code)
00074 * - Decoupled (SPI doesn't know about pin_validator_t internals)
00075 * - Maintainable (interface is stable, implementation can change)
00076 *
00077 * ## Architecture Principles
00078 *
00079 * 1. Single Responsibility: Interface only defines contract, no logic
00080 * 2. Open/Closed: New implementations can be added without changing interface
00081 * 3. Liskov Substitution: All implementations are interchangeable
00082 * 4. Interface Segregation: Small, focused interface with 6 methods
00083 * 5. Dependency Inversion: High-level code depends on this abstraction
00084 *
00085 * ## Pin Conflict Prevention System
00086 *
00087 * The pin validator prevents these common errors:
00088 *
00089 * | Conflict Type | Example | Prevention |
00090 * |-----|-----|-----|
00091 * | Double peripheral | SPI and I2C both use PA5 | reserve_pin() fails on second attempt |
00092 * | Output collision | Two GPIO drivers control same pin | Tracked in reservation table |

```

```

00093 * | **Debug confusion** | "Which module owns PA3?" | get_pin_function() shows owner |
00094 * | **Accidental reuse** | Copy-paste error uses wrong pin | Caught at runtime during init |
00095 *
00096 * ## RX72N Pin Architecture
00097 *
00098 * The RX72N has **182 GPIO pins** organized into **ports**:
00099 *
00100 * | Port | Hex ID | Pin Count | Example Pins | Common Usage |
00101 * |-----|-----|-----|-----|-----|
00102 * | Port 0 | 0x0 | 8 | P0.0-P0.7 | LEDs, GPIOs |
00103 * | Port 1 | 0x1 | 8 | P1.0-P1.7 | ADC, GPIOs |
00104 * | Port A | 0xA | 8 | PA.0-PA.7 | SPI, I2C |
00105 * | Port B | 0xB | 8 | PB.0-PB.7 | USB, UART |
00106 * | Port C | 0xC | 8 | PC.0-PC.7 | SPI, PWM |
00107 * | Port D | 0xD | 8 | PD.0-PD.7 | UART, GPIOs |
00108 * | Port E | 0xE | 8 | PE.0-PE.7 | Encoder, Timers |
00109 *
00110 * **Port Numbering**: Use hexadecimal (0x0-0x9, 0xA-0xG)
00111 * **Pin Numbering**: Use decimal (0-7 within each port)
00112 *
00113 * ## Thread Safety Guarantees
00114 *
00115 * All interface methods are **REQUIRED** to be thread-safe in their implementations:
00116 * - Concurrent reserve/release from multiple threads: **SAFE**
00117 * - Concurrent validation checks: **SAFE**
00118 * - Typical implementation: Mutex-protected reservation table
00119 *
00120 * ## Usage Examples
00121 *
00122 * ### Example 1: Basic Dependency Injection
00123 * @code
00124 * // In main.c: Setup infrastructure
00125 * pin_validator_t validator;
00126 * pin_validator_init(&validator);
00127 *
00128 * rx_pin_interface_t pin_iface;
00129 * pin_validator_get_interface(&pin_iface, &validator);
00130 *
00131 * // Inject interface into SPI driver
00132 * spi_config_t spi_cfg = {
00133 *     .pin_iface = &pin_iface,
00134 *     // ... other config
00135 * };
00136 * spi_init(&spi_cfg);
00137 * @endcode
00138 *
00139 * ### Example 2: Using Interface in Module
00140 * @code
00141 * // spi_driver.c:
00142 * typedef struct {
00143 *     rx_pin_interface_t* pin_iface; // Injected dependency
00144 *     // ... other state
00145 * } spi_driver_t;
00146 *
00147 * rx_err_t spi_init(spi_driver_t* drv, rx_pin_interface_t* pin_iface)
00148 * {
00149 *     drv->pin_iface = pin_iface;
00150 *
00151 *     // Reserve SPI pins using interface
00152 *     rx_err_t err = pin_iface->reserve_pin(pin_iface->ctx, 0xA, 0, "SPI_CIPO");
00153 *     if (err != k_rx_ok) {
00154 *         return err;
00155 *     }
00156 *
00157 *     err = pin_iface->reserve_pin(pin_iface->ctx, 0xA, 1, "SPI_COPI");
00158 *     if (err != k_rx_ok) {
00159 *         pin_iface->release_pin(pin_iface->ctx, 0xA, 0); // Cleanup
00160 *         return err;
00161 *     }
00162 *
00163 *     return k_rx_ok;
00164 * }
00165 * @endcode
00166 *
00167 * ### Example 3: Mock Implementation for Unit Testing
00168 * @code
00169 * // mock_pin_interface.c (for unit tests):
00170 * typedef struct {
00171 *     bool pins_reserved[256]; // Simple mock state
00172 * } mock_pin_validator_t;
00173 *
00174 * static rx_err_t mock_reserve_pin(void* ctx, uint8_t port, uint8_t pin,
00175 *                                 const char* function)
00176 * {
00177 *     mock_pin_validator_t* mock = (mock_pin_validator_t*)ctx;
00178 *     uint8_t idx = (port « 4) | pin;
00179 *

```

```

00180 *   if (mock->pins_reserved[idx]) {
00181 *       return k_rx_err_resource_busy; // Simulate conflict
00182 *   }
00183 *
00184 *   mock->pins_reserved[idx] = true;
00185 *   return k_rx_ok;
00186 * }
00187 *
00188 * // In test:
00189 * void test_spi_init(void)
00190 * {
00191 *     mock_pin_validator_t mock = {};
00192 *     rx_pin_interface_t iface = {
00193 *         .ctx = &mock,
00194 *         .reserve_pin = mock_reserve_pin,
00195 *         // ... other methods
00196 *     };
00197 *
00198 *     // Test SPI with mock (no real hardware needed!)
00199 *     spi_driver_t spi;
00200 *     rx_err_t err = spi_init(&spi, &iface);
00201 *     assert(err == k_rx_ok);
00202 * }
00203 * @endcode
00204 *
00205 * ### Example 4: Defensive Pin Usage Check
00206 * @code
00207 * void motor_control_init(rx_pin_interface_t* iface)
00208 * {
00209 *     // Check if PWM pin is already in use before reserving
00210 *     if (iface->is_pin_reserved(iface->ctx, 0xE, 3)) {
00211 *         // Pin conflict - diagnose
00212 *         char function[32];
00213 *         if (iface->get_pin_function(iface->ctx, 0xE, 3,
00214 *                                     function, sizeof(function)) == k_rx_ok) {
00215 *             debug_printf("ERROR: PE3 already used for: %s\n", function);
00216 *         }
00217 *         return k_rx_err_resource_busy;
00218 *     }
00219 *
00220 *     // Safe to reserve
00221 *     iface->reserve_pin(iface->ctx, 0xE, 3, "MOTOR_PWM");
00222 * }
00223 * @endcode
00224 *
00225 * ### Example 5: Global Infrastructure Pattern
00226 * @code
00227 * // Using with rx_infrastructure module:
00228 * void module_init(void)
00229 * {
00230 *     // Get global pin interface
00231 *     rx_pin_interface_t* iface = rx_infrastructure_get_pin_interface();
00232 *     if (iface == nullptr) {
00233 *         return k_rx_err_not_initialized;
00234 *     }
00235 *
00236 *     // Use convenience macro (wraps interface)
00237 *     rx_err_t err = RX_RESERVE_PIN(0xA, 5, "MODULE_PIN");
00238 *     if (err != k_rx_ok) {
00239 *         RX_REPORT_ERROR(err, "MODULE", "Pin PA5 conflict");
00240 *         return err;
00241 *     }
00242 * }
00243 * @endcode
00244 *
00245 * ## Design Rationale
00246 *
00247 * **Why Function Pointers Instead of Inheritance?**
00248 * - C doesn't have classes/inheritance
00249 * - Function pointers provide runtime polymorphism
00250 * - `ctx` pointer acts like "this" in OOP
00251 * - Enables mock implementations for testing
00252 *
00253 * **Why 6 Methods Instead of More?**
00254 * - Interface Segregation: Small, focused interface
00255 * - Covers all pin management use cases
00256 * - Easy to implement (fewer methods to stub in mocks)
00257 * - Stable API (less likely to change)
00258 *
00259 * **Why Opaque Context Pointer?**
00260 * - Hides implementation details
00261 * - Allows different internal representations
00262 * - Prevents clients from accessing internals
00263 * - Supports multiple concurrent instances (if needed)
00264 *
00265 * ## NASA Power of 10 Compliance
00266 *

```



```

00267 * | Rule | Status | Implementation |
00268 * |-----|-----|-----|
00269 * | 1. Simplify control flow | [OK] | No goto, setjmp, recursion |
00270 * | 2. Fixed loop bounds | [OK] | No loops in interface |
00271 * | 3. No dynamic memory | [OK] | Interface is POD struct |
00272 * | 4. Functions < 60 lines | [OK] | Validation function is 14 lines |
00273 * | 5. Use assertions | [OK] | Validation function checks |
00274 * | 6. Data at smallest scope | [OK] | Function pointers encapsulated |
00275 * | 7. Check return values | [OK] | All methods return rx_err_t |
00276 * | 8. Limit preprocessor | [OK] | No macros in interface |
00277 * | 9. Restrict pointers | [WARN] | Function pointers (DIP pattern) |
00278 * | 10. Compile warnings | [OK] | -Wall -Wextra -Werror |
00279 *
00280 * ## Memory Footprint
00281 *
00282 * | Component | Size (bytes) | Notes |
00283 * |-----|-----|-----|
00284 * | rx_pin_interface_t | 32 | 7 pointers x 4-8 bytes (architecture-dependent) |
00285 * | ctx pointer | 4-8 | Points to concrete implementation |
00286 * | Function pointers | 4-8 each | 6 methods |
00287 * | **Total** | **32-56** | Stack-allocated, no heap |
00288 *
00289 * @see rx_infrastructure.h Global service locator for pin interface
00290 * @see rx_pin_validator.h Concrete implementation of this interface
00291 * @see RX_RESERVE_PIN() Convenience macro using this interface
00292 * @see RX_RELEASE_PIN() Convenience macro using this interface
00293 *
00294 * @since Version 1.0.0
00295 * @date 2026-01-01
00296 * @copyright Copyright (c) 2026 Locked Inc.
00297 * SPDX-License-Identifier: MIT
00298 */
00299
00300 #pragma once
00301
00302 #include <stddef.h>
00303 #include <stdint.h>
00304
00305 #include "rx_err.h"
00306
00307 #ifdef __cplusplus
00308 extern "C" {
00309 #endif
00310
00311 /* Forward declaration of interface struct */
00312 typedef struct rx_pin_interface rx_pin_interface_t;
00313
00314 /*
=====
00315 * Pin Function Type
00316 *
=====
00317 */
00318 /**
00319 * @enum pin_interface_limits_t
00320 * @brief Pin Interface Configuration Constants
00321 *
00322 * @details
00323 * Defines compile-time constants for the pin interface system. These limits ensure
00324 * predictable memory usage and prevent buffer overflows in pin function name handling.
00325 *
00326 * ## Function Name Length Rationale
00327 *
00328 * **Why 32 bytes?**
00329 * - Sufficient for descriptive names: `SPI_CONTROLLER_OUT` = 20 chars
00330 * - Allows module prefix: `MOTOR_ENCODER_PHASE_A` = 21 chars
00331 * - Padding for safety: 32 is power-of-2, cache-aligned
00332 * - Embedded typical: Most embedded systems use 16-32 char limits
00333 *
00334 * ## Typical Function Name Examples
00335 *
00336 * | Function Name | Length | Usage |
00337 * |-----|-----|-----|
00338 * | `SPI_CIPO` | 8 | SPI peripheral |
00339 * | `UART_TX` | 7 | UART transmit |
00340 * | `I2C_SDA` | 7 | I2C data line |
00341 * | `MOTOR_PWM_CH1` | 14 | Motor PWM output |
00342 * | `ENCODER_A` | 9 | Encoder input |
00343 * | `USB_DP` | 6 | USB data positive |
00344 * | `GPIO_LED_STATUS` | 15 | General purpose LED |
00345 * | `SPI_CONTROLLER_OUT_PERIPH_IN` | 30 | Full OSHWA terminology |
00346 *
00347 * ## Memory Impact
00348 *
00349 * Assuming 128 reservable pins:
00350 * - Function names: 128 x 32 = 4096 bytes (4 KB)

```

```

00352 * - Reservation flags: 128 x 1 = 128 bytes
00353 * - Port/pin storage: 128 x 2 = 256 bytes
00354 * - **Total**: ~4.5 KB for complete pin tracking
00355 *
00356 * @see rx_pin_reserve_fn Function that uses this limit
00357 * @see rx_pin_get_function_fn Function that returns names within this limit
00358 *
00359 * @since Version 1.0.0
00360 */
00361 typedef enum : uint8_t {
00362 /**
00363  * @brief Maximum length of pin function name including null terminator
00364  * @details
00365  * Buffer size for pin function names (e.g., "SPI_COPI", "UART_TX", "MOTOR_PWM").
00366  * Includes space for null terminator. If a function name exceeds this limit during
00367  * reservation, it will be truncated.
00368  *
00369  * **Design choice**: 32 bytes balances descriptiveness with memory efficiency.
00370  *
00371  * @par Example: Valid Names
00372  * @code
00373  * "SPI_CIPO"           // 9 bytes (OK)
00374  * "MOTOR_ENCODER_A"    // 16 bytes (OK)
00375  * "I2C_CONTROLLER_SDA" // 19 bytes (OK)
00376  * "SPI_CONTROLLER_OUT" // 20 bytes (OK)
00377  * @endcode
00378  *
00379  * @par Example: Names That Will Be Truncated
00380  * @code
00381  * "SUPER_LONG_MODULE_NAME_THAT_EXCEEDS_LIMIT" // 43 bytes -> truncated to 31+null
00382  * @endcode
00383  */
00384 k_pin_function_name_max_len = 32,
00385 } pin_interface_limits_t;
00386
00387 /*
=====
00388 * Pin Interface Function Pointers
00389 *
=====
00390 */
00391
00392 /**
00393  * @typedef rx_pin_validate_fn
00394  * @brief Validate if a GPIO pin exists and is accessible on the RX72N
00395  *
00396  * @details
00397  * Function pointer type for validating pin existence before reservation or use.
00398  * Implementations must check both port and pin validity against the RX72N hardware.
00399  *
00400  * **Implementation Requirements:**
00401  * - Check port number is in valid range (0x0-0x9, 0xA-0xG for RX72N)
00402  * - Check pin number is in valid range (0-7)
00403  * - Verify that specific port/pin combination exists (not all ports have all pins)
00404  * - Return appropriate error codes for invalid port vs invalid pin
00405  * - Must be **thread-safe** (multiple threads may validate concurrently)
00406  *
00407  * ## Typical Implementation
00408  * @code
00409  * // Example concrete implementation:
00410  * rx_err_t pin_validator_validate(void* ctx, uint8_t port, uint8_t pin)
00411  * {
00412  *     // Check port range
00413  *     if (port > 0x10) { // RX72N max port
00414  *         return k_rx_err_gpio_invalid_port;
00415  *     }
00416  *
00417  *     // Check pin range
00418  *     if (pin > 7) {
00419  *         return k_rx_err_gpio_invalid_pin;
00420  *     }
00421  *
00422  *     // Check specific port/pin exists (RX72N-specific validation)
00423  *     if (!rx72n_pin_exists(port, pin)) {
00424  *         return k_rx_err_gpio_invalid_pin;
00425  *     }
00426  *
00427  *     return k_rx_ok;
00428  * }
00429  * @endcode
00430  *
00431  * @param[in] ctx Implementation context (opaque pointer to concrete validator state)
00432  * @param[in] port Port number (0x0-0x9, 0xA-0xG for A-G on RX72N)
00433  * @param[in] pin Pin number within port (0-7)
00434  *
00435  * @return rx_err_t Error code indicating validation result
00436  * @retval k_rx_ok Pin exists and is accessible

```

```

00437 * @retval k_rx_err_gpio_invalid_port Port number out of range or doesn't exist
00438 * @retval k_rx_err_gpio_invalid_pin Pin number out of range or doesn't exist on this port
00439 *
00440 * @note **Thread-safe**: Implementation must be callable from multiple threads concurrently
00441 * @note **Idempotent**: Calling multiple times with same arguments returns same result
00442 * @note **Fast**: Typically table lookup, no blocking operations
00443 *
00444 * @see rx_pin_interface.validate_pin Function pointer field in interface struct
00445 *
00446 * @since Version 1.0.0
00447 */
00448 typedef rx_err_t (*rx_pin_validate_fn)(void* ctx, const uint8_t port, const uint8_t pin);
00449 /**
00450 */
00451 * @typedef rx_pin_reserve_fn
00452 * @brief Reserve a GPIO pin for exclusive use by a specific function/module
00453 *
00454 * @details
00455 * Function pointer type for reserving pins to prevent conflicts. Once reserved, a pin
00456 * cannot be reserved again until released. This is the **primary conflict prevention**
00457 * mechanism in the STAR firmware.
00458 *
00459 * **Implementation Requirements:**
00460 * - Validate port and pin numbers first (call validate_pin internally)
00461 * - Check if pin is already reserved
00462 * - Store function name (up to k_pin_function_name_max_len bytes)
00463 * - Mark pin as reserved atomically
00464 * - Must be **thread-safe** with mutex protection
00465 * - Function name string is **not copied** - must remain valid
00466 *
00467 * ## Conflict Detection Strategy
00468 * @code
00469 * // Module A tries to reserve PA5 for SPI:
00470 * err = iface->reserve_pin(iface->ctx, 0xA, 5, "SPI_COPI"); // Success -> k_rx_ok
00471 *
00472 * // Module B tries to reserve same pin for I2C:
00473 * err = iface->reserve_pin(iface->ctx, 0xA, 5, "I2C_SDA"); // Fail -> k_rx_err_gpio_conflict
00474 * @endcode
00475 *
00476 * ## Typical Implementation Pattern
00477 * @code
00478 * rx_err_t pin_validator_reserve(void* ctx, uint8_t port, uint8_t pin,
00479 *                               const char* function)
00480 * {
00481 *     pin_validator_t* validator = (pin_validator_t*)ctx;
00482 *
00483 *     // 1. Validate inputs
00484 *     if (function == nullptr) return k_rx_err_invalid_arg;
00485 *     rx_err_t err = pin_validator_validate(ctx, port, pin);
00486 *     if (err != k_rx_ok) return err;
00487 *
00488 *     // 2. Lock for thread safety
00489 *     tx_mutex_get(&validator->mutex, TX_WAIT_FOREVER);
00490 *
00491 *     // 3. Check if already reserved
00492 *     uint8_t idx = (port << 4) | pin;
00493 *     if (validator->reserved[idx]) {
00494 *         tx_mutex_put(&validator->mutex);
00495 *         return k_rx_err_gpio_conflict;
00496 *     }
00497 *
00498 *     // 4. Reserve pin
00499 *     validator->reserved[idx] = true;
00500 *     strncpy(validator->functions[idx], function, k_pin_function_name_max_len - 1);
00501 *     validator->functions[idx][k_pin_function_name_max_len - 1] = '\0';
00502 *
00503 *     tx_mutex_put(&validator->mutex);
00504 *     return k_rx_ok;
00505 * }
00506 * @endcode
00507 *
00508 * @param[in] ctx Implementation context (opaque pointer to concrete validator state)
00509 * @param[in] port Port number (0x0-0x9, 0xA-0xG)
00510 * @param[in] pin Pin number within port (0-7)
00511 * @param[in] function Function name (must be string literal or valid pointer, NOT copied)
00512 *
00513 * @return rx_err_t Error code indicating reservation result
00514 * @retval k_rx_ok Pin reserved successfully
00515 * @retval k_rx_err_gpio_conflict Pin already reserved by another function
00516 * @retval k_rx_err_gpio_invalid_port Invalid port number
00517 * @retval k_rx_err_gpio_invalid_pin Invalid pin number
00518 * @retval k_rx_err_invalid_arg function parameter is nullptr
00519 *
00520 * @note **Thread-safe**: Implementation must use mutex or atomic operations
00521 * @note **String lifetime**: function string must remain valid (use literals!)
00522 * @note **Idempotent check**: Reserving same pin twice returns conflict error
00523 * @note **Performance**: ~10-50 us typical (mutex + table update)

```

```

00524 *
00525 * @warning Function name is **NOT copied** - must be string literal or static storage
00526 * @warning Must call release_pin() to free the pin for reuse
00527 *
00528 * @see rx_pin_release_fn Release a reserved pin
00529 * @see RX_RESERVE_PIN() Convenience macro wrapping this function
00530 *
00531 * @since Version 1.0.0
00532 */
00533 typedef rx_err_t (*rx_pin_reserve_fn)(void*      ctx,
00534                                     const uint8_t port,
00535                                     const uint8_t pin,
00536                                     const char*   function);
00537
00538 /**
00539 * @typedef rx_pin_release_fn
00540 * @brief Release a previously reserved GPIO pin, making it available for reuse
00541 *
00542 * @details
00543 * Function pointer type for releasing pin reservations. After release, the pin becomes
00544 * available for reservation by other modules. Typically called during module deinitialization
00545 * or reconfiguration.
00546 *
00547 * **Implementation Requirements:**
00548 * - Validate port and pin numbers first
00549 * - Check if pin is currently reserved (error if not)
00550 * - Clear reservation flag atomically
00551 * - Clear function name storage
00552 * - Must be **thread-safe** with mutex protection
00553 * - Idempotent safe: Releasing unreserved pin returns error (doesn't crash)
00554 *
00555 * ## Typical Usage Patterns
00556 * @code
00557 * // Pattern 1: Normal deinitialization
00558 * void spi_deinit(void) {
00559 *     iface->release_pin(iface->ctx, 0xA, 0); // CIPO
00560 *     iface->release_pin(iface->ctx, 0xA, 1); // COPI
00561 *     iface->release_pin(iface->ctx, 0xA, 2); // CLK
00562 * }
00563 *
00564 * // Pattern 2: Error cleanup (reverse order)
00565 * if (reserve_failed) {
00566 *     iface->release_pin(iface->ctx, 0xA, 1); // Release second
00567 *     iface->release_pin(iface->ctx, 0xA, 0); // Release first
00568 * }
00569 *
00570 * // Pattern 3: Defensive release (check first)
00571 * if (iface->is_pin_reserved(iface->ctx, 0xA, 5)) {
00572 *     iface->release_pin(iface->ctx, 0xA, 5);
00573 * }
00574 * @endcode
00575 *
00576 * @param[in] ctx Implementation context (opaque pointer to concrete validator state)
00577 * @param[in] port Port number (0x0-0x9, 0xA-0xG)
00578 * @param[in] pin Pin number within port (0-7)
00579 *
00580 * @return rx_err_t Error code indicating release result
00581 * @retval k_rx_ok Pin released successfully, now available for reuse
00582 * @retval k_rx_err_invalid_state Pin was not reserved (double-release or never reserved)
00583 * @retval k_rx_err_gpio_invalid_port Invalid port number
00584 * @retval k_rx_err_gpio_invalid_pin Invalid pin number
00585 *
00586 * @note **Thread-safe**: Implementation must use mutex or atomic operations
00587 * @note **Idempotent safe**: Releasing unreserved pin returns error but doesn't crash
00588 * @note **Best-effort cleanup**: Most code ignores return value during cleanup
00589 * @note **Performance**: ~5-30 us typical (mutex + flag clear)
00590 *
00591 * @see rx_pin_reserve_fn Reserve a pin (must be called first)
00592 * @see RX_RELEASE_PIN() Convenience macro wrapping this function
00593 *
00594 * @since Version 1.0.0
00595 */
00596 typedef rx_err_t (*rx_pin_release_fn)(void* ctx, const uint8_t port, const uint8_t pin);
00597
00598 /**
00599 * @typedef rx_pin_is_reserved_fn
00600 * @brief Check if a GPIO pin is currently reserved
00601 *
00602 * @details
00603 * Function pointer type for querying pin reservation status. Returns boolean result
00604 * without error codes. Useful for defensive checks before attempting reservation or
00605 * for debugging pin conflicts.
00606 *
00607 * **Implementation Requirements:**
00608 * - Return true if pin is reserved, false if available
00609 * - Invalid port/pin should return false (not reserved)
00610 * - Must be **thread-safe** (read-only operation)

```

```

00611 * - Fast lookup (no blocking operations)
00612 *
00613 * ## Typical Usage
00614 * @code
00615 * // Defensive reservation check
00616 * if (iface->is_pin_reserved(iface->ctx, 0xA, 5)) {
00617 * // Pin already in use - diagnose conflict
00618 * char function[32];
00619 * iface->get_pin_function(iface->ctx, 0xA, 5, function, sizeof(function));
00620 * debug_printf("PA5 conflict: already used for %s\n", function);
00621 * return k_rx_err_resource_busy;
00622 * }
00623 *
00624 * // Proceed with reservation
00625 * iface->reserve_pin(iface->ctx, 0xA, 5, "MY_FUNCTION");
00626 * @endcode
00627 *
00628 * @param[in] ctx Implementation context (opaque pointer to concrete validator state)
00629 * @param[in] port Port number (0x0-0x9, 0xA-0xG)
00630 * @param[in] pin Pin number within port (0-7)
00631 *
00632 * @return bool Reservation status
00633 * @retval true Pin is currently reserved
00634 * @retval false Pin is available (or invalid port/pin)
00635 *
00636 * @note **Thread-safe**: Read-only check, safe to call concurrently
00637 * @note **No error codes**: Returns false for invalid pins (simple boolean API)
00638 * @note **Fast**: Typically 1-5 us (direct table lookup)
00639 *
00640 * @see rx_pin_get_function_fn Get the function name for a reserved pin
00641 * @see rx_pin_reserve_fn Reserve a pin
00642 *
00643 * @since Version 1.0.0
00644 */
00645 typedef bool (*rx_pin_is_reserved_fn)(void* ctx, const uint8_t port, const uint8_t pin);
00646
00647 /**
00648 * @typedef rx_pin_get_function_fn
00649 * @brief Get the function name associated with a reserved GPIO pin
00650 *
00651 * @details
00652 * Function pointer type for querying which function/module reserved a pin. Useful for
00653 * debugging pin conflicts and generating pin usage reports.
00654 *
00655 * **Implementation Requirements:**
00656 * - Validate port and pin numbers
00657 * - Check if pin is reserved (error if not)
00658 * - Copy function name to output buffer (with null terminator)
00659 * - Truncate if buffer too small
00660 * - Must be **thread-safe** with mutex protection
00661 *
00662 * ## Typical Usage
00663 * @code
00664 * // Diagnose pin conflict
00665 * void diagnose_conflict(rx_pin_interface_t* iface, uint8_t port, uint8_t pin)
00666 * {
00667 * if (!iface->is_pin_reserved(iface->ctx, port, pin)) {
00668 * printf("Pin P%X.%d is available\n", port, pin);
00669 * return;
00670 * }
00671 *
00672 * char function[k_pin_function_name_max_len];
00673 * rx_err_t err = iface->get_pin_function(iface->ctx, port, pin,
00674 *                                     function, sizeof(function));
00675 * if (err == k_rx_ok) {
00676 * printf("Pin P%X.%d reserved for: %s\n", port, pin, function);
00677 * }
00678 * }
00679 *
00680 * // Generate pin usage report
00681 * void print_pin_usage(rx_pin_interface_t* iface)
00682 * {
00683 * for (uint8_t port = 0; port <= 0x10; port++) {
00684 * for (uint8_t pin = 0; pin < 8; pin++) {
00685 * if (iface->is_pin_reserved(iface->ctx, port, pin)) {
00686 * char function[32];
00687 * if (iface->get_pin_function(iface->ctx, port, pin,
00688 *                             function, sizeof(function)) == k_rx_ok) {
00689 * printf("P%X.%d: %s\n", port, pin, function);
00690 * }
00691 * }
00692 * }
00693 * }
00694 * }
00695 * @endcode
00696 *
00697 * @param[in] ctx Implementation context (opaque pointer to concrete validator state)

```

```

00698 * @param[in] port Port number (0x0-0x9, 0xA-0xG)
00699 * @param[in] pin Pin number within port (0-7)
00700 * @param[out] function_out Buffer to store function name (must be non-NULL)
00701 * @param[in] function_len Length of function_out buffer (should be >= k_pin_function_name_max_len)
00702 *
00703 * @return rx_err_t Error code indicating retrieval result
00704 * @retval k_rx_ok Function name copied successfully to function_out
00705 * @retval k_rx_err_invalid_state Pin is not reserved (no function to retrieve)
00706 * @retval k_rx_err_null_ptr function_out parameter is nullptr
00707 * @retval k_rx_err_invalid_size function_len is 0 or too small (minimum 2 for "\0")
00708 * @retval k_rx_err_gpio_invalid_port Invalid port number
00709 * @retval k_rx_err_gpio_invalid_pin Invalid pin number
00710 *
00711 * @note **Thread-safe**: Implementation must use mutex for consistent reads
00712 * @note **Null termination**: Output is always null-terminated if buffer size > 0
00713 * @note **Truncation**: Long names are truncated to fit buffer (including null)
00714 * @note **Performance**: ~10-30 us typical (mutex + string copy)
00715 *
00716 * @warning Check return value before using function_out contents
00717 * @warning Buffer size should be at least k_pin_function_name_max_len bytes
00718 *
00719 * @see rx_pin_is_reserved_fn Check if pin is reserved first
00720 * @see pin_interface_limits_t::k_pin_function_name_max_len Recommended buffer size
00721 *
00722 * @since Version 1.0.0
00723 */
00724 typedef rx_err_t (*rx_pin_get_function_fn)(void* ctx,
00725     const uint8_t port,
00726     const uint8_t pin,
00727     char* function_out,
00728     const uint32_t function_len);
00729
00730 /**
00731 * @typedef rx_pin_clear_all_fn
00732 * @brief Clear all pin reservations (reset to initial state)
00733 *
00734 * @details
00735 * Function pointer type for clearing all pin reservations at once. Typically used during:
00736 * - System reset/reinitialization
00737 * - Unit test teardown
00738 * - Error recovery (start fresh)
00739 * - Development/debugging
00740 *
00741 * **Implementation Requirements:**
00742 * - Clear all reservation flags atomically
00743 * - Clear all function name strings
00744 * - Reset to clean state (as if just initialized)
00745 * - Must be **thread-safe** with exclusive mutex lock
00746 * - **WARNING**: Dangerous operation - no confirmation
00747 *
00748 * ## When to Use vs. When NOT to Use
00749 *
00750 * **[OK] SAFE to use:**
00751 * - Unit test teardown (known clean state)
00752 * - System reboot/reset sequence
00753 * - Error recovery when retrying full initialization
00754 *
00755 * **[X] DANGEROUS to use:**
00756 * - During normal operation (other modules may have pins reserved)
00757 * - Without stopping all modules first
00758 * - In production code (rarely needed)
00759 *
00760 * ## Typical Usage
00761 * @code
00762 * // Unit test teardown pattern
00763 * void test_teardown(void)
00764 * {
00765 *     rx_pin_interface_t* iface = get_pin_interface();
00766 *     iface->clear_all_reservations(iface->ctx); // Clean slate for next test
00767 * }
00768 *
00769 * // System reset pattern
00770 * void system_reset(void)
00771 * {
00772 *     // 1. Stop all modules
00773 *     motor_stop();
00774 *     spi_shutdown();
00775 *     usb_disconnect();
00776 *
00777 *     // 2. Clear reservations
00778 *     rx_pin_interface_t* iface = get_pin_interface();
00779 *     iface->clear_all_reservations(iface->ctx);
00780 *
00781 *     // 3. Reinitialize
00782 *     system_init();
00783 * }
00784 * @endcode

```

```

00785 *
00786 * @param[in] ctx Implementation context (opaque pointer to concrete validator state)
00787 *
00788 * @return rx_err_t Error code indicating clear result
00789 * @retval k_rx_ok All reservations cleared successfully
00790 *
00791 * @note **Thread-safe**: Implementation must use exclusive mutex lock
00792 * @note **Destructive**: All pin reservations are lost (no undo)
00793 * @note **No confirmation**: Clears immediately without prompts
00794 * @note **Performance**: ~50-200 us typical (mutex + full table clear)
00795 *
00796 * @warning **DANGEROUS**: Clears ALL reservations. Ensure no modules are using pins.
00797 * @warning Only use during controlled shutdown, reset, or test teardown
00798 * @warning Do NOT call during normal operation
00799 *
00800 * @see rx_pin_release_fn Release individual pins (safer than clear_all)
00801 *
00802 * @since Version 1.0.0
00803 */
00804 typedef rx_err_t (*rx_pin_clear_all_fn)(void* ctx);
00805
00806 /*
=====
00807 * Pin Interface Structure
00808 *
=====
00809 */
00810
00811 /**
00812 * @struct rx_pin_interface
00813 * @brief Pin Validator Interface - THE ABSTRACTION FOR PIN MANAGEMENT
00814 *
00815 * @details
00816 * ## Overview
00817 *
00818 * This structure **IS** the abstraction that enables Dependency Inversion Principle for
00819 * pin management in STAR firmware. It defines the contract that all pin validator
00820 * implementations must fulfill, allowing high-level modules to depend on this stable
00821 * interface rather than concrete implementations.
00822 *
00823 * ## Polymorphism via Function Pointers
00824 *
00825 * This struct implements **runtime polymorphism in C** using function pointers:
00826 * - **ctx**: Acts like "this" pointer in OOP (points to concrete implementation state)
00827 * - **Function pointers**: Virtual methods that can be overridden by implementations
00828 * - **First parameter (ctx)**: Passed to every method, providing access to private data
00829 *
00830 * ## Memory Layout
00831 *
00832 * @code
00833 struct rx_pin_interface {
00834     // Offset | Size | Field
00835     //      0 | 4-8 | void* ctx
00836     //    4-8 | 4-8 | rx_pin_validate_fn validate_pin
00837     //   8-16 | 4-8 | rx_pin_reserve_fn reserve_pin
00838     //  12-24 | 4-8 | rx_pin_release_fn release_pin
00839     //  16-32 | 4-8 | rx_pin_is_reserved_fn is_pin_reserved
00840     //  20-40 | 4-8 | rx_pin_get_function_fn get_pin_function
00841     //  24-48 | 4-8 | rx_pin_clear_all_fn clear_all_reservations
00842     // TOTAL: 28-56 bytes (architecture-dependent)
00843 };
00844 * @endcode
00845
00846 * ## Typical Usage Pattern (Dependency Injection)
00847 *
00848 * ### Step 1: Create Concrete Implementation
00849 * @code
00850 // Concrete implementation (in pin_validator.c):
00851 pin_validator_t concrete_validator;
00852 pin_validator_init(&concrete_validator);
00853 * @endcode
00854 *
00855 * ### Step 2: Extract Interface
00856 * @code
00857 // Get abstract interface from concrete implementation:
00858 rx_pin_interface_t iface;
00859 pin_validator_get_interface(&iface, &concrete_validator);
00860 *
00861 // The interface is now populated:
00862 // - iface.ctx = &concrete_validator
00863 // - iface.reserve_pin = pin_validator_reserve
00864 // - iface.release_pin = pin_validator_release
00865 // - ... (all function pointers initialized)
00866 * @endcode
00867 *
00868 * ### Step 3: Inject Into High-Level Module
00869 * @code

```



```

00870 * // High-level module depends on abstract interface:
00871 * spi_config_t spi_cfg = {
00872 *     .pin_iface = &iface, // Inject interface (not concrete type!)
00873 *     // ... other config
00874 * };
00875 * spi_init(&spi_cfg);
00876 * @endcode
00877 *
00878 * ### Step 4: Use Through Interface
00879 * @code
00880 * // Inside spi_init():
00881 * rx_err_t spi_init(spi_config_t* cfg)
00882 * {
00883 *     // Call through function pointer (polymorphism!)
00884 *     rx_err_t err = cfg->pin_iface->reserve_pin(
00885 *         cfg->pin_iface->ctx, // Pass context (like "this")
00886 *         0xA, 5,           // Port A, Pin 5
00887 *         "SPI_COPI"        // Function name
00888 *     );
00889 *     return err;
00890 * }
00891 * @endcode
00892 *
00893 * ## Interface Stability Guarantees
00894 *
00895 * This interface is STABLE - changes are rare and carefully managed:
00896 * - Adding methods: Append to end (preserve ABI)
00897 * - Removing methods: FORBIDDEN (breaks existing code)
00898 * - Changing signatures: FORBIDDEN (breaks existing code)
00899 * - Reordering fields: FORBIDDEN (breaks ABI)
00900 *
00901 * Why stability matters:
00902 * - Modules can cache pointers to this struct
00903 * - No need to recompile all modules when implementation changes
00904 * - Enables versioning (if needed in future)
00905 *
00906 * ## Thread Safety Requirements
00907 *
00908 * All function pointers MUST point to thread-safe implementations:
00909 * | Method | Thread-Safe? | Typical Protection |
00910 * |-----|-----|-----|
00911 * | validate_pin | [OK] Yes | Read-only, no locks needed |
00912 * | reserve_pin | [OK] Yes | Mutex around reservation table |
00913 * | release_pin | [OK] Yes | Mutex around reservation table |
00914 * | is_pin_reserved | [OK] Yes | Mutex (read lock) or atomic |
00915 * | get_pin_function | [OK] Yes | Mutex (read lock) + strcpy |
00916 * | clear_all_reservations | [OK] Yes | Exclusive mutex lock |
00917 *
00918 * ## Validation Requirements
00919 *
00920 * Before using this interface, ALWAYS validate:
00921 * @code
00922 * rx_err_t err = rx_pin_interface_validate(&iface);
00923 * if (err != k_rx_ok) {
00924 *     // Interface is nullptr or has missing function pointers
00925 *     return err;
00926 * }
00927 * // Safe to use
00928 * @endcode
00929 *
00930 * ## Mock Implementation Example (Unit Testing)
00931 *
00932 * @code
00933 * // Minimal mock for testing:
00934 * typedef struct {
00935 *     bool pins[256]; // Simple reservation flags
00936 * } mock_pin_validator_t;
00937 *
00938 * static rx_err_t mock_reserve(void* ctx, uint8_t port, uint8_t pin,
00939 *                             const char* func)
00940 * {
00941 *     mock_pin_validator_t* mock = (mock_pin_validator_t*)ctx;
00942 *     uint8_t idx = (port « 4) | pin;
00943 *     if (mock->pins[idx]) return k_rx_err_gpio_conflict;
00944 *     mock->pins[idx] = true;
00945 *     return k_rx_ok;
00946 * }
00947 *
00948 * // Create mock interface:
00949 * mock_pin_validator_t mock = {};
00950 * rx_pin_interface_t mock_iface = {
00951 *     .ctx = &mock,
00952 *     .reserve_pin = mock_reserve,
00953 *     .release_pin = mock_release,
00954 *     // ... other methods
00955 * };
00956 *

```



```

00957 * // Inject mock into SPI for testing (no real hardware!)
00958 * spi_init(&spi_cfg, &mock_iface);
00959 * @endcode
00960 *
00961 * ## Field Documentation
00962 *
00963 * @var rx_pin_interface::ctx
00964 * @brief Opaque pointer to concrete implementation's private state
00965 * @details
00966 * Acts like "this" pointer in OOP. Points to concrete pin validator instance
00967 * (e.g., 'pin_validator_t*'). All function pointers receive this as first parameter.
00968 * **MUST NOT** be NULL if interface is initialized.
00969 *
00970 * @var rx_pin_interface::validate_pin
00971 * @brief Validate if a pin exists and is accessible
00972 * @details
00973 * Function pointer to validation implementation. Checks port and pin validity against
00974 * RX72N hardware. **MUST NOT** be NULL. See rx_pin_validate_fn for details.
00975 *
00976 * @var rx_pin_interface::reserve_pin
00977 * @brief Reserve a pin for exclusive use by a function/module
00978 * @details
00979 * Function pointer to reservation implementation. Prevents pin conflicts by tracking
00980 * ownership. **MUST NOT** be NULL. See rx_pin_reserve_fn for details.
00981 *
00982 * @var rx_pin_interface::release_pin
00983 * @brief Release a previously reserved pin
00984 * @details
00985 * Function pointer to release implementation. Frees pin for reuse by other modules.
00986 * **MUST NOT** be NULL. See rx_pin_release_fn for details.
00987 *
00988 * @var rx_pin_interface::is_pin_reserved
00989 * @brief Check if a pin is currently reserved
00990 * @details
00991 * Function pointer to reservation query implementation. Returns boolean status.
00992 * **MUST NOT** be NULL. See rx_pin_is_reserved_fn for details.
00993 *
00994 * @var rx_pin_interface::get_pin_function
00995 * @brief Get the function name associated with a reserved pin
00996 * @details
00997 * Function pointer to function name retrieval implementation. Used for debugging
00998 * conflicts. **MUST NOT** be NULL. See rx_pin_get_function_fn for details.
00999 *
01000 * @var rx_pin_interface::clear_all_reservations
01001 * @brief Clear all pin reservations (reset to initial state)
01002 * @details
01003 * Function pointer to full clear implementation. Dangerous operation - use only during
01004 * controlled reset/test teardown. **MUST NOT** be NULL. See rx_pin_clear_all_fn for details.
01005 *
01006 * @note All function pointers **MUST** be non-NULL (validated by rx_pin_interface_validate())
01007 * @note ctx pointer **MUST** be non-NULL and point to valid concrete implementation
01008 * @note Interface is **POD** (Plain Old Data) - can be copied, stack-allocated
01009 * @note Typical size: 28-56 bytes (architecture-dependent)
01010 *
01011 * @see rx_pin_interface_validate() Validate interface before use
01012 * @see pin_validator_get_interface() Create interface from concrete implementation
01013 * @see rx_infrastructure_get_pin_interface() Get global pin interface
01014 *
01015 * @since Version 1.0.0
01016 */
01017 struct rx_pin_interface {
01018     void* ctx; /**< Implementation context (opaque pointer to concrete validator) */
01019     rx_pin_validate_fn validate_pin; /**< Validate pin existence */
01020     rx_pin_reserve_fn reserve_pin; /**< Reserve pin for function */
01021     rx_pin_release_fn release_pin; /**< Release reserved pin */
01022     rx_pin_is_reserved_fn is_pin_reserved; /**< Check if pin is reserved */
01023     rx_pin_get_function_fn get_pin_function; /**< Get pin's function name */
01024     rx_pin_clear_all_fn clear_all_reservations; /**< Clear all reservations */
01025 };
01026
01027 /*
=====
01028 * Interface Validation
01029 *
=====
01030 */
01031
01032 /**
01033 * @brief Validate that a pin interface is properly initialized and safe to use
01034 *
01035 * @details
01036 * Performs comprehensive validation of an `rx_pin_interface_t` to ensure all function
01037 * pointers are non-NULL and the interface is ready for use. This function should be
01038 * called **before** using any interface received from external sources.
01039 *
01040 * ## Validation Checks Performed
01041 *

```

```

01042 * 1. **nullptr check**: Ensures `iface` parameter is non-NULL
01043 * 2. **Function pointer checks**: Verifies all 6 function pointers are non-NULL:
01044 *   - `validate_pin`
01045 *   - `reserve_pin`
01046 *   - `release_pin`
01047 *   - `is_pin_reserved`
01048 *   - `get_pin_function`
01049 *   - `clear_all_reservations`
01050 *
01051 * **Note**: The `ctx` pointer is **NOT** validated (can be NULL for stateless mocks).
01052 *
01053 * ## When to Use This Function
01054 *
01055 * ### Required Scenarios:
01056 * - **Module initialization**: Validate injected interfaces before use
01057 * - **Defensive programming**: Check interface from untrusted sources
01058 * - **Debug builds**: Assert interface validity at entry points
01059 *
01060 * ### Optional Scenarios:
01061 * - **Production code**: Can be skipped if interface comes from trusted source (rx_infrastructure)
01062 * - **Performance-critical paths**: Validation adds ~20 cycles overhead
01063 *
01064 * ## Typical Usage Patterns
01065 *
01066 * ### Pattern 1: Defensive Module Initialization
01067 * @code
01068 * rx_err_t spi_init(spi_config_t* cfg)
01069 * {
01070 *     // Validate injected pin interface
01071 *     rx_err_t err = rx_pin_interface_validate(cfg->pin_iface);
01072 *     if (err != k_rx_ok) {
01073 *         return err; // Invalid interface, cannot proceed
01074 *     }
01075 *
01076 *     // Safe to use interface now
01077 *     cfg->pin_iface->reserve_pin(cfg->pin_iface->ctx, 0xA, 5, "SPI_COPI");
01078 *     return k_rx_ok;
01079 * }
01080 * @endcode
01081 *
01082 * ### Pattern 2: Debug-Only Validation
01083 * @code
01084 * #ifdef DEBUG
01085 *     // Validate in debug builds only
01086 *     rx_err_t err = rx_pin_interface_validate(pin_iface);
01087 *     assert(err == k_rx_ok && "Pin interface validation failed");
01088 * #endif
01089 *
01090 * // Use interface (validation skipped in release builds)
01091 * pin_iface->reserve_pin(pin_iface->ctx, 0xA, 5, "SPI_COPI");
01092 * @endcode
01093 *
01094 * ### Pattern 3: Trust Global Infrastructure
01095 * @code
01096 * // Get interface from trusted global infrastructure
01097 * rx_pin_interface_t* iface = rx_infrastructure_get_pin_interface();
01098 * if (iface == nullptr) {
01099 *     return k_rx_err_not_initialized; // Infrastructure not ready
01100 * }
01101 *
01102 * // Skip validation - global infrastructure guarantees valid interfaces
01103 * // (validation was already done during rx_infrastructure_init())
01104 * iface->reserve_pin(iface->ctx, 0xA, 5, "SPI_COPI");
01105 * @endcode
01106 *
01107 * ### Pattern 4: Validate Before Caching
01108 * @code
01109 * typedef struct {
01110 *     rx_pin_interface_t* pin_iface; // Cached interface
01111 * } module_state_t;
01112 *
01113 * rx_err_t module_init(module_state_t* state, rx_pin_interface_t* iface)
01114 * {
01115 *     // Validate before caching (ensures valid for entire module lifetime)
01116 *     rx_err_t err = rx_pin_interface_validate(iface);
01117 *     if (err != k_rx_ok) {
01118 *         return err;
01119 *     }
01120 *
01121 *     // Cache for later use (no need to re-validate)
01122 *     state->pin_iface = iface;
01123 *     return k_rx_ok;
01124 * }
01125 * @endcode
01126 *
01127 * ## Error Scenarios
01128 *

```

```

01129 * ### Scenario 1: NULL Interface Pointer
01130 * @code
01131 * rx_pin_interface_t* iface = nullptr;
01132 * rx_err_t err = rx_pin_interface_validate(iface);
01133 * // Returns: k_rx_err_null_ptr
01134 * @endcode
01135 *
01136 * ### Scenario 2: Partially Initialized Interface
01137 * @code
01138 * rx_pin_interface_t iface = {
01139 *     .ctx = &validator,
01140 *     .reserve_pin = pin_validator_reserve,
01141 *     .release_pin = nullptr, // Missing!
01142 *     // ... other fields nullptr
01143 * };
01144 * rx_err_t err = rx_pin_interface_validate(&iface);
01145 * // Returns: k_rx_err_invalid_state
01146 * @endcode
01147 *
01148 * ### Scenario 3: Fully Initialized Interface
01149 * @code
01150 * rx_pin_interface_t iface;
01151 * pin_validator_get_interface(&iface, &validator);
01152 * rx_err_t err = rx_pin_interface_validate(&iface);
01153 * // Returns: k_rx_ok (all function pointers set)
01154 * @endcode
01155 *
01156 * ## Performance Characteristics
01157 *
01158 * - **Execution time**: ~20-50 cycles @ 240 MHz (~0.1-0.2 us)
01159 * - **Operations**: 7 nullptr checks (1 iface + 6 function pointers)
01160 * - **Overhead**: Negligible for init paths, skip in tight loops if needed
01161 * - **Optimization**: Compiler may inline this function (static inline)
01162 *
01163 * @param[in] iface Pin interface to validate (can be NULL - returns error)
01164 *
01165 * @return rx_err_t Error code indicating validation result
01166 * @retval k_rx_ok Interface is valid and ready for use
01167 * @retval k_rx_err_null_ptr iface parameter is nullptr
01168 * @retval k_rx_err_invalid_state One or more function pointers are nullptr
01169 *
01170 * @pre None (can be called with nullptr - returns error safely)
01171 *
01172 * @post If k_rx_ok returned, all function pointers are non-NULL
01173 * @post If error returned, interface should NOT be used
01174 *
01175 * @note **Context (ctx) is NOT validated** - can be NULL for stateless implementations
01176 * @note **Inlined**: Compiled directly into call site (zero function call overhead)
01177 * @note **Thread-safe**: Read-only check, safe to call concurrently
01178 * @note **Idempotent**: Calling multiple times returns same result
01179 *
01180 * @warning Do NOT use interface if this function returns error
01181 * @warning This function does NOT validate that function pointers point to valid functions
01182 *         (only checks non-NULL)
01183 *
01184 * @par Example: Complete Validation Workflow
01185 * @code
01186 * void module_use_pin_interface(rx_pin_interface_t* iface)
01187 * {
01188 *     // Step 1: Validate interface
01189 *     rx_err_t err = rx_pin_interface_validate(iface);
01190 *     if (err == k_rx_err_null_ptr) {
01191 *         debug_printf("ERROR: NULL pin interface\n");
01192 *         return;
01193 *     } else if (err == k_rx_err_invalid_state) {
01194 *         debug_printf("ERROR: Incomplete pin interface (missing function pointers)\n");
01195 *         return;
01196 *     }
01197 *
01198 *     // Step 2: Interface is valid - safe to use
01199 *     err = iface->reserve_pin(iface->ctx, 0xA, 5, "MODULE_PIN");
01200 *     if (err != k_rx_ok) {
01201 *         debug_printf("Pin reservation failed: %d\n", err);
01202 *         return;
01203 *     }
01204 *
01205 *     // Pin successfully reserved
01206 * }
01207 * @endcode
01208 *
01209 * @see rx_pin_interface Structure being validated
01210 * @see pin_validator_get_interface() Properly initializes this interface
01211 * @see rx_infrastructure_get_pin_interface() Returns validated global interface
01212 *
01213 * @since Version 1.0.0
01214 *
01215 * @par NASA Power of 10 Compliance:

```

```

01216 * - Rule 4: [OK] Function is 14 lines (well under 60 line limit)
01217 * - Rule 5: [OK] 7 validation checks (interface + 6 function pointers)
01218 */
01219 static inline rx_err_t rx_pin_interface_validate(const rx_pin_interface_t* iface)
01220 {
01221     if (iface == nullptr) {
01222         return k_rx_err_null_ptr;
01223     }
01224
01225     /* Check required function pointers */
01226     if (iface->validate_pin == nullptr || iface->reserve_pin == nullptr ||
01227         iface->release_pin == nullptr || iface->is_pin_reserved == nullptr ||
01228         iface->get_pin_function == nullptr || iface->clear_all_reservations == nullptr) {
01229         return k_rx_err_invalid_state;
01230     }
01231
01232     return k_rx_ok;
01233 }
01234
01235 #ifdef __cplusplus
01236 }
01237 #endif

```

7.27 libs/rx_core/inc/rx_pin_validator.h File Reference

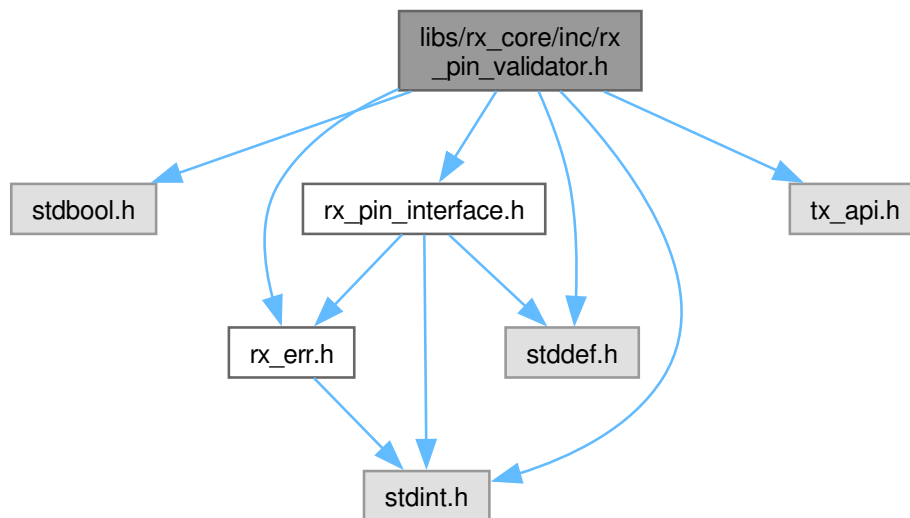
Concrete Pin Validator Implementation - Dependency Inversion Pattern.

```

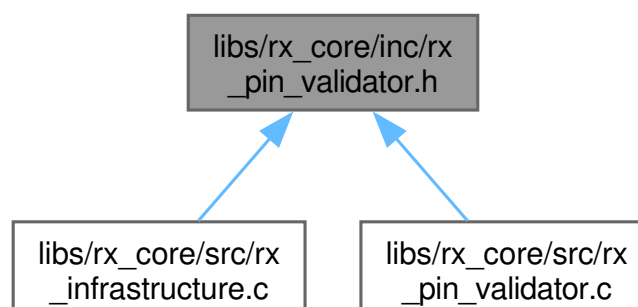
#include <stdbool.h>
#include <stddef.h>
#include <stdint.h>
#include "rx_err.h"
#include "rx_pin_interface.h"
#include "tx_api.h"

```

Include dependency graph for rx_pin_validator.h:



This graph shows which files directly or indirectly include this file:



Data Structures

- struct [pin_reservation_t](#)
Pin reservation state for a single GPIO pin.
- struct [pin_validator_t](#)
Concrete Pin Validator Implementation Structure.

Enumerations

- enum [pin_validator_limits_t](#): uint8_t { [k_pin_validator_max_ports](#) = 17, [k_pin_validator_max_pins](#) = 8 }
Pin Validator Configuration Constants.

Functions

- [rx_err_t pin_validator_init](#) ([pin_validator_t](#) *validator)
Initialize pin validator (MUST be called before use).
- [rx_err_t pin_validator_get_interface](#) ([rx_pin_interface_t](#) *iface, [pin_validator_t](#) *validator)
Extract abstract interface from concrete pin validator (Dependency Inversion).
- [rx_err_t pin_validator_deinit](#) ([pin_validator_t](#) *validator)
Deinitialize pin validator (graceful shutdown only).

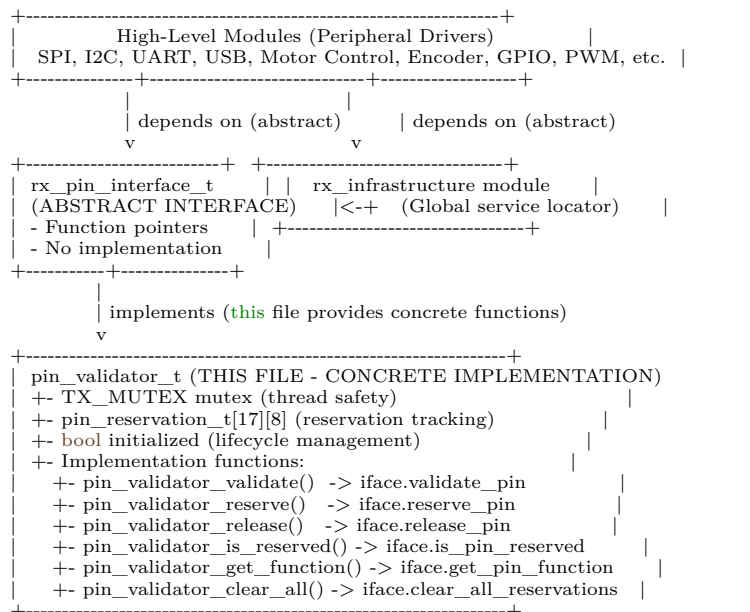
7.27.1 Detailed Description

Concrete Pin Validator Implementation - Dependency Inversion Pattern.

7.27.1.1 Overview

Provides the concrete implementation of the [rx_pin_interface_t](#) abstract interface for GPIO pin validation and reservation tracking on the Renesas RX72N microcontroller. This is the "low-level module" in the Dependency Inversion Principle architecture.

Key Design Pattern: Dependency Inversion + Interface Segregation



7.27.1.2 Architecture Principles

1. Single Responsibility: Sole purpose is pin tracking and conflict prevention
2. Open/Closed: Interface is stable, implementation can change
3. Liskov Substitution: Can be swapped with other implementations (e.g., mocks)
4. Interface Segregation: Implements exactly what interface requires (no more)
5. Dependency Inversion: High-level code depends on abstract interface, not this

7.27.1.3 Core Features

7.27.1.3.1 1. Pin Validation

- Hardware-aware: Validates against actual RX72N pin availability
- Port range: Supports 0-9 (decimal), 0xA-0xG (hex A-G = 10-16)
- Pin range: 0-7 per port (not all ports have all pins)
- Fast lookup: O(1) table-based validation

7.27.1.3.2 2. Reservation Tracking

- Conflict prevention: Prevents double reservation of same pin
- Function tracking: Stores which module owns each pin
- Query support: Can check reservation status and owner
- Clear all: Can reset entire table (for testing/reboot)

7.27.1.3.3 3. Thread Safety

- ThreadX mutex: All operations protected by TX_MUTEX
- Atomic operations: Reserve/release are atomic
- Read locks: Queries also use mutex for consistency
- ISR-unsafe: Cannot call from interrupts (mutex blocks)

7.27.1.3.4 4. Memory Efficiency

- Static allocation: No malloc/free (NASA Rule 3)
- Fixed size: 17 ports x 8 pins = 136 pin slots
- Predictable footprint: ~4.7 KB total RAM
- Zero overhead: Direct array indexing, no lists/trees

7.27.1.4 Memory Footprint Analysis

Component	Size (bytes)	Calculation	Location
TX_MUTEX	~100	ThreadX mutex structure	.bss
pin_reservation_t array	4352	$17 \times 8 \times 32 = 4352$	.bss
bool initialized	1	Single flag	.bss
Total pin_validator_t	**~4453**	Struct total	.bss

Memory breakdown per pin:

- bool reserved: 1 byte
- char function[32]: 32 bytes
- Total per pin: 33 bytes
- Total for 136 pins: $136 \times 33 = 4488$ bytes (~4.4 KB)

7.27.1.5 RX72N Pin Architecture

The RX72N microcontroller has 182 GPIO pins organized into 17 ports:

Port	Hex ID	Decimal	Pin Count	Example Usage
PORT0	0x0	0	8	LEDs, GPIOs
PORT1	0x1	1	8	ADC inputs
PORT2	0x2	2	8	GPIO, interrupts
PORT3-9	0x3-0x9	3-9	Variable	Mixed peripherals
PORTA	0xA	10	8	SPI, I2C
PORTB	0xB	11	8	USB, UART
PORTC	0xC	12	8	SPI, PWM
PORTD	0xD	13	8	UART, GPIOs
PORTE	0xE	14	8	Encoder, Timers
PORTF-G	0xF-0x10	15-16	Variable	Reserved, special

Note: Not all ports have all 8 pins physically available on the package.

7.27.1.6 Thread Safety Guarantees

Operation	Thread-Safe?	Mutex Used?	Details
pin_validator_init()	[X] No	N/A	Call from main() only
pin_validator_get_interface()	[OK] Yes	No	Read-only check
pin_validator_deinit()	[X] No	No	Call during shutdown only
validate_pin()	[OK] Yes	No	Read-only table lookup

Operation	Thread-Safe?	Mutex Used?	Details
reserve_pin()	[OK] Yes	[OK] Yes	Mutex-protected write
release_pin()	[OK] Yes	[OK] Yes	Mutex-protected write
is_pin_reserved()	[OK] Yes	[OK] Yes	Mutex-protected read
get_pin_function()	[OK] Yes	[OK] Yes	Mutex-protected read + strepy
clear_all_reservations()	[OK] Yes	[OK] Yes	Exclusive mutex lock

7.27.1.7 Usage Examples

7.27.1.7.1 Example 1: Basic Initialization and Usage

```
// Global validator instance
static pin_validator_t s_pin_validator;

int main(void)
{
    // Step 1: Initialize validator
    rx_err_t err = pin_validator_init(&s_pin_validator);
    if (err != k_rx_ok) {
        // Fatal error - cannot continue
        while (1) { __asm("nop"); }
    }

    // Step 2: Get interface from concrete validator
    rx_pin_interface_t pin_iface;
    err = pin_validator_get_interface(&pin_iface, &s_pin_validator);
    if (err != k_rx_ok) {
        while (1) { __asm("nop"); }
    }

    // Step 3: Use interface to reserve pins
    err = pin_iface.reserve_pin(pin_iface.ctx, 0xA, 5, "SPI_COPI");
    if (err == k_rx_err_gpio_conflict) {
        debug_printf("Pin PA5 already reserved\n");
    }

    // Continue with application
    return 0;
}
```

7.27.1.7.2 Example 2: Using with Global Infrastructure

```
// Infrastructure owns the validator
static pin_validator_t s_global_validator;
static rx_pin_interface_t s_global_pin_iface;

rx_err_t infrastructure_init(void)
{
    // Initialize concrete validator
    rx_err_t err = pin_validator_init(&s_global_validator);
    if (err != k_rx_ok) return err;

    // Extract interface
    err = pin_validator_get_interface(&s_global_pin_iface, &s_global_validator);
    if (err != k_rx_ok) return err;

    // Now modules can get interface via infrastructure
    return k_rx_ok;
}

rx_pin_interface_t* infrastructure_get_pin_interface(void)
{
    return &s_global_pin_iface;
}
```

7.27.1.7.3 Example 3: Unit Testing with Mock vs. Real Validator

```
#ifndef UNIT_TEST
// Use mock validator for testing
rx_pin_interface_t iface = create_mock_pin_interface();
```



```

#else
// Use real validator in production
static pin_validator_t validator;
pin_validator_init(&validator);

rx_pin_interface_t iface;
pin_validator_get_interface(&iface, &validator);
#endif

// Module code works with either (Dependency Inversion!)
spi_init(&iface);

```

7.27.1.8 Design Rationale

Why 2D array instead of hashmap?

- Embedded systems: Predictable memory, $O(1)$ access
- RX72N has fixed port/pin structure
- Direct indexing: `reservations[port][pin]` is fast
- No dynamic allocation (NASA Rule 3)

Why ThreadX mutex instead of atomic operations?

- Struct copy for `get_pin_function()` requires protection
- Consistency between reservation flag and function name
- ThreadX already integrated (no additional dependency)
- Mutex overhead acceptable for non-ISR operations

Why store function names vs. module IDs?

- Human-readable debugging ("SPI_COPI" vs. ID 42)
- No need for global module registry
- String literals in `.rodata` (no RAM cost)
- Easy to print in debug reports

7.27.1.9 NASA Power of 10 Compliance

Rule	Status	Implementation
1. Simplify control flow	[OK]	No goto, setjmp, recursion
2. Fixed loop bounds	[OK]	All loops use <code>k_pin_validator_max_*</code> constants
3. No dynamic memory	[OK]	Static allocation only (2D array)
4. Functions < 60 lines	[OK]	All functions under limit
5. Use assertions	[OK]	NULL checks, initialization checks
6. Data at smallest scope	[OK]	Private functions, local variables
7. Check return values	[OK]	All errors propagated

Rule	Status	Implementation
8. Limit preprocessor	[OK]	Only include guards
9. Restrict pointers	[WARN]	Function pointers (DIP pattern)
10. Compile warnings	[OK]	-Wall -Wextra -Werror

See also

[rx_pin_interface.h](#) Abstract interface this file implements
[rx_infrastructure.h](#) Global service locator using this validator
[RX_RESERVE_PIN\(\)](#) Convenience macro for pin reservation
[RX_RELEASE_PIN\(\)](#) Convenience macro for pin release

Since

Version 1.0.0

Date

2026-01-01

Copyright

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Definition in file [rx_pin_validator.h](#).

7.27.2 Data Structure Documentation

7.27.2.1 struct pin_reservation_t

Pin reservation state for a single GPIO pin.

Stores the reservation status and owner information for one pin. These structures are organized in a 2D array [port][pin] within the [pin_validator_t](#).

7.27.2.2 Memory Layout

```
struct pin_reservation_t {
    // Offset | Size | Field
    //   0 |   1 | bool reserved
    //   1 |  32 | char function[32]
    // TOTAL: 33 bytes (with padding)
};
```

7.27.2.3 Field Relationships

reserved	function	Meaning
false	(any)	Pin is available (function value ignored)
true	"SPI_COPI"	Pin reserved for SPI COPI
true	"GPIO_OUT"	Pin reserved for GPIO output
true	""	Pin reserved but function unknown (edge case)

7.27.2.4 Typical Usage

```

pin_reservation_t* slot = &validator->reservations[port][pin];

// Check if reserved
if (slot->reserved) {
    printf("Pin P%X.%d used for: %s\n", port, pin, slot->function);
}

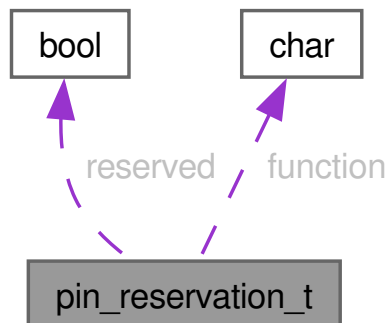
// Reserve pin
slot->reserved = true;
strncpy(slot->function, "SPI_COPI", sizeof(slot->function) - 1);
slot->function[sizeof(slot->function) - 1] = '\0';

// Release pin
slot->reserved = false;
memset(slot->function, 0, sizeof(slot->function));

```

Definition at line 414 of file [rx_pin_validator.h](#).

Collaboration diagram for `pin_reservation_t`:



Data Fields

char	function[k_pin_function_name_max_len]	Function name identifying the pin's owner. Function name (e.g., "SPI_COPI", "GPIO_OUT") Null-terminated string (max 31 chars + null). Examples: "SPI_COPI", "UART_TX", "MOTOR_PWM". Only valid when <code>reserved == true</code>. String is typically a literal from .rodata (not copied, just pointer stored). Note Size is 33 bytes (1 bool + 32 char array) Function name is NOT copied - must be string literal or static storage Thread safety: Access must be mutex-protected See also <code>pin_validator_t::reservations</code> 2D array of these structures <code>pin_interface_limits_t::k_pin_function_name_max_len</code> Function name buffer size Since Version 1.0.0
bool	reserved	Reservation flag (true = pin is reserved). Is this pin reserved? When false, the pin is available for reservation. When true, the pin is in use and the <code>function</code> field indicates which module owns it.

7.27.3 Enumeration Type Documentation

7.27.3.1 pin_validator_limits_t

enum `pin_validator_limits_t` : `uint8_t`

Pin Validator Configuration Constants.

Defines compile-time array dimensions for the pin reservation table. These limits are based on the RX72N hardware capabilities and determine memory footprint.

7.27.3.2 Rationale for Limits

Why 17 ports?

- RX72N has ports 0-9 (10 ports) + A-G (7 ports) = 17 total
- Covers all possible GPIO ports on RX72N

- Array index maps directly to port number (0-16)
- No gaps, no hash function needed

Why 8 pins per port?

- Each RX72N port has up to 8 pins (0-7)
- Some ports have fewer pins physically, but we allocate max
- Uniform array size simplifies indexing
- Small waste ($\sim 10\%$ unused slots) vs. complexity of variable-size

7.27.3.3 Memory Impact

Total pin slots: 17 ports x 8 pins = 136 slots

Per-slot memory:

- bool reserved: 1 byte
- char function[32]: 32 bytes
- Total: 33 bytes per slot

Total table size: 136 slots x 33 bytes = 4488 bytes (~ 4.4 KB)

7.27.3.4 Alternative Designs Considered

Design	Memory	Access Time	Complexity	Chosen?
2D array (current)	4.4 KB	$O(1)$	Low	[OK] Yes
Linked list	~ 800 bytes (sparse)	$O(n)$	Medium	[X] No
Hash table	~ 2 KB + overhead	$O(1)$ avg	High	[X] No
Bitfield + separate names	~ 17 bytes + strings	$O(1)$ + lookup	Medium	[X] No

Decision: 2D array chosen for simplicity, predictability, and $O(1)$ access.

See also

[pin_reservation_t](#) Structure stored in each array slot

[pin_validator_t::reservations](#) 2D array using these dimensions

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Enumerator

k_pin_validator_max_ports	Maximum number of ports on RX72N (0-9, A-G). RX72N port numbering: • Decimal ports: 0-9 (10 ports) • Hex ports: A-G (0xA-0x10 = 10-16 decimal, 7 ports) • Total: 17 ports (indices 0-16 in array)
k_pin_validator_max_pins	Maximum pins per port (0-7). Each RX72N port has up to 8 pins numbered 0-7. Not all ports have all 8 pins physically available on the package, but we allocate space for all 8 to simplify indexing.

Definition at line 320 of file [rx_pin_validator.h](#).

```

00320         : uint8_t {
00321     /**
00322     * @brief Maximum number of ports on RX72N (0-9, A-G)
00323     * @details
00324     * RX72N port numbering:
00325     * - Decimal ports: 0-9 (10 ports)
00326     * - Hex ports: A-G (0xA-0x10 = 10-16 decimal, 7 ports)
00327     * - **Total**: 17 ports (indices 0-16 in array)
00328     */
00329     k_pin_validator_max_ports = 17,
00330
00331     /**
00332     * @brief Maximum pins per port (0-7)
00333     * @details
00334     * Each RX72N port has up to 8 pins numbered 0-7.
00335     * Not all ports have all 8 pins physically available on the package,
00336     * but we allocate space for all 8 to simplify indexing.
00337     */
00338     k_pin_validator_max_pins = 8,
00339 } pin_validator_limits_t;

```

7.27.4 Function Documentation

7.27.4.1 pin_validator_deinit()

```

rx_err_t pin_validator_deinit (
    pin_validator_t * validator) [nodiscard]

```

Deinitialize pin validator (graceful shutdown only).

Performs cleanup of pin validator resources. Destroys ThreadX mutex, clears reservation table, and marks validator as uninitialized. In embedded systems, this is RARELY CALLED because the system typically runs indefinitely.

7.27.4.2 When to Use

- Unit test teardown (clean state between tests)
- System reset/reboot preparation
- Firmware update preparation
- Development/debugging (clean restart)

7.27.4.3 When NOT to Use

- Normal operation (validator should remain initialized)
- While modules are still using pins
- From multiple threads (not thread-safe)

7.27.4.4 Cleanup Algorithm

1. Validate inputs: Check validator pointer is non-NULL
2. Check initialized: Return error if not initialized
3. Destroy mutex: Delete ThreadX mutex
4. Clear reservations: Zero all reservations (optional, for safety)
5. Clear initialized flag: Mark validator as uninitialized

7.27.4.5 Typical Usage

```
// Unit test teardown
void test_teardown(void)
{
    pin_validator_deinit(&s_test_validator);
    // Validator can be reinitialized for next test
}

// System shutdown
void system_shutdown(void)
{
    // 1. Stop all modules using pins
    spi_shutdown();
    i2c_shutdown();

    // 2. Deinitialize validator
    pin_validator_deinit(&s_global_validator);

    // 3. Proceed with shutdown/reset
    // ...
}
```

Parameters

in,out	validator	Pin validator instance to deinitialize (must be non-NULL)
--------	-----------	---

Returns

[rx_err_t](#) Error code indicating deinitialization result

Return values

k_rx_ok	Deinitialization successful, validator can be reinitialized
k_rx_err_null_ptr	validator parameter is nullptr
k_rx_err_invalid_state	Validator not initialized (nothing to clean up)

Precondition

validator must be initialized via [pin_validator_init\(\)](#)
No modules should be using pins (all released)
No threads should be calling validator operations concurrently

Postcondition

validator->initialized set to false
validator->mutex destroyed
validator->reservations[] cleared (all pins available)
Validator can be reinitialized via [pin_validator_init\(\)](#)

Note

NOT thread-safe: Must ensure no concurrent access during deinit
Calling when not initialized returns `k_rx_err_invalid_state` (safe no-op)
After deinit, all extracted interfaces become INVALID
Typical execution time: ~100-300 us @ 240 MHz

Warning

DANGEROUS: All interfaces extracted from this validator become invalid
Ensure all modules have stopped using pins before calling
Do NOT call from ISR (mutex operations may block)

See also

[pin_validator_init\(\)](#) Reinitialize validator after deinit
[pin_validator_get_interface\(\)](#) Interfaces become invalid after deinit

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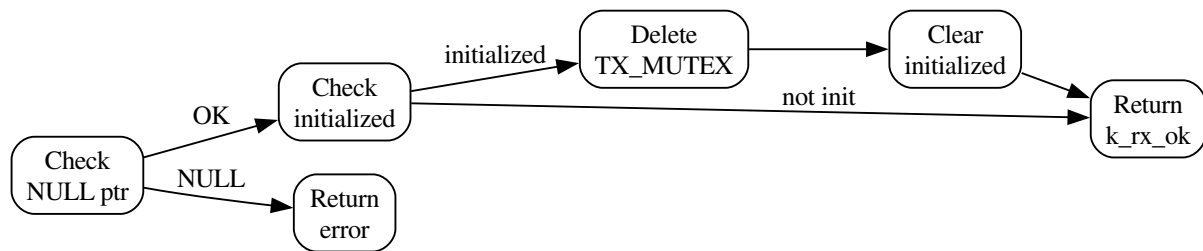
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Deinitialize pin validator (graceful shutdown only).

Performs cleanup of a pin validator instance, destroying the ThreadX mutex and marking the validator as uninitialized. This function is rarely needed in production embedded systems (which run indefinitely) but is essential for unit test teardown.

7.27.4.6 Cleanup Sequence



7.27.4.7 Safe to Call When Not Initialized

If the validator is not initialized, this function returns `k_rx_ok` without doing anything (safe no-op). This allows defensive cleanup patterns.

Precondition

- validator must point to valid `pin_validator_t` memory
- No modules should be actively using pins
- No concurrent calls to pin validator functions

Postcondition

- `validator->initialized = false`
- `validator->mutex` destroyed (if was created)
- All extracted interfaces become INVALID
- Validator can be re-initialized via `pin_validator_init()`

Note

- NOT thread-safe: Ensure no concurrent access during deinit
- Idempotent: Safe to call multiple times
- Performance: ~100-300 us @ 240 MHz

Warning

- All interfaces become invalid after this call
- Modules using pins should be stopped first
- Do NOT call while other threads are using validator

Example:

```

// Unit test teardown
void test_cleanup(void) {
    // Deinitialize validator to clean up mutex
    pin_validator_deinit(&s_test_validator);
    // Validator can be reinitialized for next test
}

// Defensive cleanup pattern
void shutdown_pin_system(void) {
    // Safe to call even if not initialized
    pin_validator_deinit(&s_validator);
}
  
```

See also

[pin_validator_init\(\)](#) Reinitialize after deinit
[impl_clear_all_reservations\(\)](#) Clear reservations without deinit

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- Rule 5: [OK] 3 preconditions, 4 postconditions documented

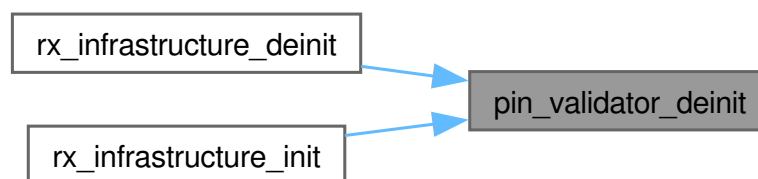
Definition at line 1202 of file [rx_pin_validator.c](#).

```
01203 {
01204     RX_CHECK_NULL_PTR validator, s_tag, "Validator pointer is nullptr");
01205
01206     if (!validator->initialized) {
01207         return k_rx_ok; /* Already deinitialized */
01208     }
01209
01210     /* Delete mutex */
01211     (void)tx_mutex_delete(&validator->mutex);
01212
01213     /* Clear state */
01214     validator->initialized = false;
01215
01216     rx_log_info(s_tag, "Pin validator deinitialized");
01217     return k_rx_ok;
01218 }
01219 }
```

References [pin_validator_t::initialized](#), [k_rx_ok](#), [pin_validator_t::mutex](#), [RX_CHECK_NULL_PTR](#), [rx_log_info](#), and [s_tag](#).

Referenced by [rx_infrastructure_deinit\(\)](#), and [rx_infrastructure_init\(\)](#).

Here is the caller graph for this function:



7.27.4.8 pin_validator_get_interface()

```
rx_err_t pin_validator_get_interface (
    rx_pin_interface_t * iface,
    pin_validator_t * validator) [nodiscard]
```

Extract abstract interface from concrete pin validator (Dependency Inversion).

Extracts an [rx_pin_interface_t](#) from the concrete [pin_validator_t](#) implementation. This function is the key to Dependency Inversion: high-level modules depend on the abstract interface, not the concrete validator type.

7.27.4.9 Dependency Inversion Pattern

High-Level Module (SPI Driver)
 v depends on v
 rx_pin_interface_t (Abstract) <- THIS FUNCTION PRODUCES THIS
 ^ implements ^
 pin_validator_t (Concrete) <- THIS FUNCTION CONSUMES THIS

7.27.4.10 What This Function Does

Populates the rx_pin_interface_t structure with:

- ctx pointer: Points to concrete validator (&validator)
- Function pointers: Point to validator's implementation functions:
 - iface->validate_pin = pin_validator_validate
 - iface->reserve_pin = pin_validator_reserve
 - iface->release_pin = pin_validator_release
 - iface->is_pin_reserved = pin_validator_is_reserved
 - iface->get_pin_function = pin_validator_get_function
 - iface->clear_all_reservations = pin_validator_clear_all

7.27.4.11 Typical Usage Pattern

```
// Step 1: Initialize concrete validator
static pin_validator_t s_validator;
rx_err_t err = pin_validator_init(&s_validator);
if (err != k_rx_ok) {
    return err;
}

// Step 2: Extract interface
rx_pin_interface_t iface;
err = pin_validator_get_interface(&iface, &s_validator);
if (err != k_rx_ok) {
    return err;
}

// Step 3: Inject interface into high-level module
spi_config_t spi_cfg = {
    .pin_iface = &iface, // High-level code only sees interface!
    // ...
};
spi_init(&spi_cfg);

// Step 4: Use through interface (polymorphism in C)
iface.reserve_pin(iface.ctx, 0xA, 5, "SPI_COPI");
// Internally calls: pin_validator_reserve(&s_validator, 0xA, 5, "SPI_COPI")
```

Parameters

out	iface	Interface structure to populate (must be non-NULL)
in,out	validator	Concrete validator instance (must be initialized)

Returns

[rx_err_t](#) Error code indicating extraction result

Return values

k_rx_ok	Interface populated successfully, ready for use
k_rx_err_null_ptr	iface or validator parameter is nullptr
k_rx_err_invalid_state	Validator not initialized (call pin_validator_init first)

Precondition

validator must be initialized via [pin_validator_init\(\)](#)
 iface must point to valid [rx_pin_interface_t](#) structure
 validator pointer must remain valid for interface lifetime

Postcondition

iface->ctx points to validator
 All iface function pointers point to validator implementations
 iface is valid and ready for use
 validator lifetime must extend beyond iface usage

Note

Thread-safe: Can be called from any thread after validator initialization
 Interface lifetime: Returned interface is valid as long as validator exists
 Multiple calls OK: Can extract multiple interfaces from same validator
 Performance: ~50 cycles (simple struct copy + pointer assignments)

Warning

Validator must remain alive while interface is in use
 Do NOT call [pin_validator_deinit\(\)](#) while interface is in use
 Interface becomes invalid if validator is deinitialized

See also

[pin_validator_init\(\)](#) Must be called first to initialize validator
[rx_pin_interface_t](#) Abstract interface structure being populated
[rx_infrastructure_get_pin_interface\(\)](#) Global access to interface

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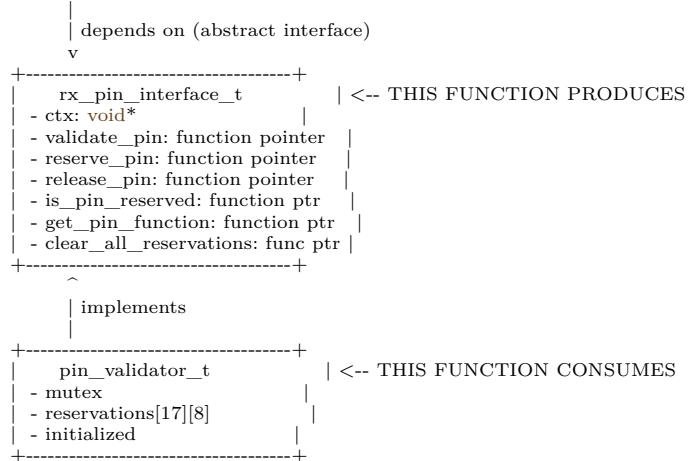
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Populates an [rx_pin_interface_t](#) structure with function pointers to this validator's implementation functions. This is the bridge between the concrete implementation and the abstract interface that high-level modules use.

7.27.4.12 Dependency Inversion Pattern

High-Level Module (e.g., SPI Driver)



7.27.4.13 Interface Population

The function assigns:

- `iface->ctx = validator (context pointer)`
- `iface->validate_pin = impl_validate_pin`
- `iface->reserve_pin = impl_reserve_pin`
- `iface->release_pin = impl_release_pin`
- `iface->is_pin_reserved = impl_is_pin_reserved`
- `iface->get_pin_function = impl_get_pin_function`
- `iface->clear_all_reservations = impl_clear_all_reservations`

Precondition

iface must point to valid `rx_pin_interface_t` memory
 validator must be initialized via [pin_validator_init\(\)](#)
 validator must remain alive as long as interface is used

Postcondition

iface->ctx points to validator
 All iface function pointers are valid
 Interface ready for use by high-level modules
 validator state unchanged

Note

Thread-safe: Can be called from any thread
 Multiple interfaces: Same validator can produce multiple interfaces
 Performance: $O(1)$, ~50 cycles (struct assignment)

Warning

Validator must outlive all extracted interfaces
Do not deinit validator while interfaces are in use

Example:

```
static pin_validator_t s_validator;
static rx_pin_interface_t s_pin_iface;

rx_err_t init_pin_system(void) {
    rx_err_t err = pin_validator_init(&s_validator);
    if (err != k_rx_ok) return err;

    err = pin_validator_get_interface(&s_pin_iface, &s_validator);
    if (err != k_rx_ok) return err;

    // Pass interface to high-level modules
    spi_init(&s_pin_iface);
    i2c_init(&s_pin_iface);

    return k_rx_ok;
}
```

See also

[pin_validator_init\(\)](#) Must be called first
[rx_pin_interface_t](#) Interface structure being populated
[rx_infrastructure_get_pin_interface\(\)](#) Global accessor pattern

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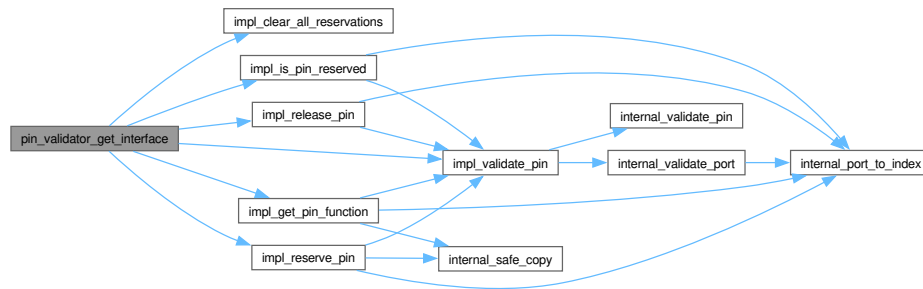
Definition at line 1100 of file rx_pin_validator.c.

```
01101 {
01102     RX_CHECK_NULL_PTR(iface, s_tag, "Interface pointer is nullptr");
01103     RX_CHECK_NULL_PTR(validator, s_tag, "Validator pointer is nullptr");
01104
01105     if (!validator->initialized) {
01106         rx_log_error(s_tag, "Validator not initialized");
01107         return k_rx_err_invalid_state;
01108     }
01109
01110     /* Fill interface */
01111     iface->ctx = validator;
01112     iface->validate_pin = impl_validate_pin;
01113     iface->reserve_pin = impl_reserve_pin;
01114     iface->release_pin = impl_release_pin;
01115     iface->is_pin_reserved = impl_is_pin_reserved;
01116     iface->get_pin_function = impl_get_pin_function;
01117     iface->clear_all_reservations = impl_clear_all_reservations;
01118
01119     return k_rx_ok;
01120 }
```

References [impl_clear_all_reservations\(\)](#), [impl_get_pin_function\(\)](#), [impl_is_pin_reserved\(\)](#), [impl_release_pin\(\)](#), [impl_reserve_pin\(\)](#), [impl_validate_pin\(\)](#), [pin_validator_t::initialized](#), [k_rx_err_invalid_state](#), [k_rx_ok](#), [RX_CHECK_NULL_PTR](#), [rx_log_error](#), and [s_tag](#).

Referenced by [rx_infrastructure_init\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



7.27.4.14 pin_validator_init()

```

rx_err_t pin_validator_init (
    pin_validator_t * validator) [nodiscard]

```

Initialize pin validator (MUST be called before use).

Performs one-time initialization of the pin validator structure. Creates ThreadX mutex, clears reservation table, and marks validator as initialized. Must be called exactly once before extracting interface via [pin_validator_get_interface\(\)](#).

7.27.4.15 Initialization Algorithm

1. Validate inputs: Check validator pointer is non-NULL
2. Check double-init: Return error if already initialized
3. Create mutex: Initialize ThreadX mutex for thread safety
4. Clear reservation table: Zero all reservations[port][pin] entries
5. Set initialized flag: Mark validator as ready for use

7.27.4.16 Typical Usage

```

// Global validator (static allocation, zero-initialized by startup code)
static pin_validator_t s_pin_validator;

int main(void)
{
    // Initialize validator (one-time operation)
    rx_err_t err = pin_validator_init(&s_pin_validator);
    if (err != k_rx_ok) {
        // Fatal error - cannot continue
        debug_printf("Pin validator init failed: %d\n", err);
        while (1) { __asm("nop"); }
    }

    // Validator is now ready for interface extraction
    // ...
}

```

Parameters

in,out	validator	Pin validator instance to initialize (must be non-NULL)
--------	-----------	---

Returns

[rx_err_t](#) Error code indicating initialization result

Return values

k_rx_ok	Initialization successful, validator ready for use
k_rx_err_null_ptr	validator parameter is nullptr
k_rx_err_invalid_state	Validator already initialized (double-init attempt)
k_rx_err_rtos_mutex	ThreadX mutex creation failed

Precondition

validator must point to valid [pin_validator_t](#) structure
 validator must NOT be already initialized (initialized flag must be false)
 ThreadX kernel must be initialized (tx_kernel_enter called or about to be called)
 Must be called from main thread before starting ThreadX (or from initialization thread)

Postcondition

validator->initialized set to true
 validator->mutex created and ready for use
 validator->reservations[] array cleared (all pins available)
 Validator ready for [pin_validator_get_interface\(\)](#)

Note

NOT thread-safe: Must be called from single thread during initialization
 Calling twice returns [k_rx_err_invalid_state](#) (double-init protection)
 On failure, validator is left in undefined state - must not use
 Typical execution time: ~200-500 us @ 240 MHz

Warning

CRITICAL: Must be called before any pin reservation operations
 Do NOT call from multiple threads (not thread-safe during init)
 If this fails, system cannot continue - must halt or reset

See also

[pin_validator_get_interface\(\)](#) Extract interface after initialization
[pin_validator_deinit\(\)](#) Cleanup validator (shutdown only)
[rx_infrastructure_init\(\)](#) Typically calls this function

Since

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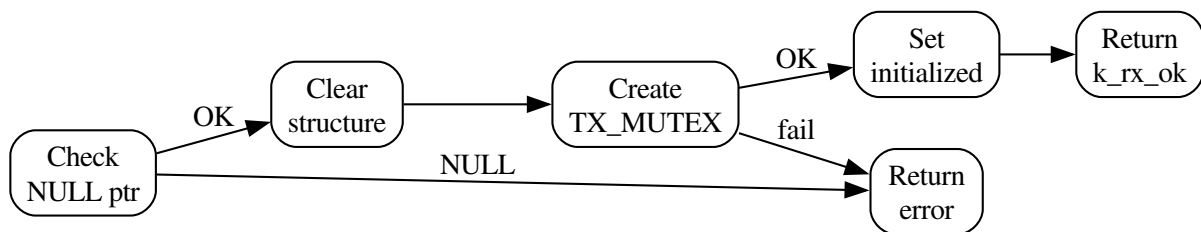
NASA Power of 10 Compliance:

- Rule 5: [OK] 4 preconditions, 4 postconditions

Initialize pin validator (MUST be called before use).

Performs one-time initialization of a pin validator instance. This function creates the ThreadX mutex, clears the reservation table, and marks the validator as initialized. Must be called exactly once before any other `pin_validator_*` functions.

7.27.4.17 Initialization Sequence



7.27.4.18 Memory Initialization

The function uses a static zero-initialized instance to clear the structure, which:

- Clears all reservation entries (`reserved=false`, `function=""`)
- Clears mutex bytes to neutral zero state before `tx_mutex_create` performs RTOS setup
- Sets `initialized` to false (then true after success)

Precondition

validator must point to valid `pin_validator_t` memory

validator must NOT be already initialized

ThreadX kernel must be started or about to start

Postcondition

validator->initialized = true

validator->mutex created and ready

validator->reservations all cleared (available)

Validator ready for `pin_validator_get_interface()`

Note

NOT thread-safe: Call from main() or initialization thread only

Idempotent: Calling twice is NOT supported (undefined behavior)

Performance: ~200-500 us @ 240 MHz (dominated by mutex creation)

Warning

MUST be called before any pin operations

Do NOT call from ISR

On failure, validator is in undefined state

Example:

```
static pin_validator_t s_validator;

int main(void) {
    rx_err_t err = pin_validator_init(&s_validator);
    if (err != k_rx_ok) {
        rx_log_fatal("INIT", "Pin validator init failed: %d", err);
        while (1) { __asm("nop"); } // Fatal - cannot continue
    }
    // Validator ready for use
}
```

See also

[pin_validator_get_interface\(\)](#) Next step after initialization

[pin_validator_deinit\(\)](#) Cleanup function

[rx_infrastructure_init\(\)](#) Typically calls this function

Since

Version 1.0.0

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- Rule 5: [OK] 3 preconditions, 4 postconditions documented

Definition at line 990 of file [rx_pin_validator.c](#).

```
00991 {
00992     RX_CHECK_NULL_PTR(validator, s_tag, "Validator pointer is nullptr");
00993
00994     *validator = s_zero_validator;
00995
00996     /* Create mutex */
00997     (void)tx_mutex_create(&validator->mutex, "PinValidatorMutex", TX_NO_INHERIT);
00998
00999     validator->initialized = true;
01000
01001     rx_log_info(s_tag, "Pin validator initialized");
01002
01003     return k_rx_ok;
01004 }
```

References [pin_validator_t::initialized](#), [k_rx_ok](#), [pin_validator_t::mutex](#), [RX_CHECK_NULL_PTR](#), [rx_log_info](#), [s_tag](#), and [s_zero_validator](#).

Referenced by [rx_infrastructure_init\(\)](#).

Here is the caller graph for this function:



7.28 rx_pin_validator.h

[Go to the documentation of this file.](#)

```

00001 /**
00002  * @file rx_pin_validator.h
00003  * @brief Concrete Pin Validator Implementation - Dependency Inversion Pattern
00004  *
00005  * @details
00006  * ## Overview
00007  *
00008  * Provides the concrete implementation of the `rx_pin_interface_t` abstract interface
00009  * for GPIO pin validation and reservation tracking on the Renesas RX72N microcontroller.
00010  * This is the "low-level module" in the Dependency Inversion Principle architecture.
00011  *
00012  * Key Design Pattern: Dependency Inversion + Interface Segregation
00013  *
00014  * ```
00015  * +-----+
00016  * | High-Level Modules (Peripheral Drivers) |
00017  * | SPI, I2C, UART, USB, Motor Control, Encoder, GPIO, PWM, etc. |
00018  * +-----+
00019  * |
00020  * | depends on (abstract) | depends on (abstract)
00021  * | v | v |
00022  * +-----+ +-----+
00023  * | rx_pin_interface_t | | rx_infrastructure module |
00024  * | (ABSTRACT INTERFACE) |<-+ (Global service locator) |
00025  * | - Function pointers | +-----+
00026  * | - No implementation |
00027  * +-----+
00028  * |
00029  * | implements (this file provides concrete functions)
00030  * | v
00031  * +-----+
00032  * | pin_validator_t (THIS FILE - CONCRETE IMPLEMENTATION) |
00033  * | +- TX_MUTEX mutex (thread safety) |
00034  * | +- pin_reservation_t[17][8] (reservation tracking) |
00035  * | +- bool initialized (lifecycle management) |
00036  * | +- Implementation functions: |
00037  * | +- pin_validator_validate() -> iface.validate_pin |
00038  * | +- pin_validator_reserve() -> iface.reserve_pin |
00039  * | +- pin_validator_release() -> iface.release_pin |
00040  * | +- pin_validator_is_reserved() -> iface.is_pin_reserved |
00041  * | +- pin_validator_get_function() -> iface.get_pin_function |
00042  * | +- pin_validator_clear_all() -> iface.clear_all_reservations |
00043  * +-----+
00044  * ```
00045  *
00046  * ## Architecture Principles
00047  *
00048  * 1. Single Responsibility: Sole purpose is pin tracking and conflict prevention
00049  * 2. Open/Closed: Interface is stable, implementation can change
00050  * 3. Liskov Substitution: Can be swapped with other implementations (e.g., mocks)
00051  * 4. Interface Segregation: Implements exactly what interface requires (no more)
00052  * 5. Dependency Inversion: High-level code depends on abstract interface, not this
00053  *
00054  * ## Core Features
00055  *
00056  * ### 1. Pin Validation
00057  * - Hardware-aware: Validates against actual RX72N pin availability
00058  * - Port range: Supports 0-9 (decimal), 0xA-0xG (hex A-G = 10-16)
00059  * - Pin range: 0-7 per port (not all ports have all pins)
00060  * - Fast lookup: O(1) table-based validation
00061  *
00062  * ### 2. Reservation Tracking
00063  * - Conflict prevention: Prevents double reservation of same pin
00064  * - Function tracking: Stores which module owns each pin
00065  * - Query support: Can check reservation status and owner
00066  * - Clear all: Can reset entire table (for testing/reboot)
00067  *
00068  * ### 3. Thread Safety
00069  * - ThreadX mutex: All operations protected by TX_MUTEX
00070  * - Atomic operations: Reserve/release are atomic
00071  * - Read locks: Queries also use mutex for consistency
00072  * - ISR-unsafe: Cannot call from interrupts (mutex blocks)
00073  *
00074  * ### 4. Memory Efficiency
00075  * - Static allocation: No malloc/free (NASA Rule 3)
00076  * - Fixed size: 17 ports x 8 pins = 136 pin slots
00077  * - Predictable footprint: ~4.7 KB total RAM
00078  * - Zero overhead: Direct array indexing, no lists/trees
00079  *
00080  * ## Memory Footprint Analysis
00081  *
00082  * | Component | Size (bytes) | Calculation | Location |

```

```

00083 * |-----|-----|-----|-----|
00084 * | TX_MUTEX | ~100 | ThreadX mutex structure | .bss |
00085 * | pin_reservation_t array | 4352 | 17 x 8 x 32 = 4352 | .bss |
00086 * | bool initialized | 1 | Single flag | .bss |
00087 * | **Total pin_validator_t** | **~4453** | Struct total | .bss |
00088 *
00089 * **Memory breakdown per pin**:
00090 * - `bool reserved`: 1 byte
00091 * - `char function[32]`: 32 bytes
00092 * - **Total per pin**: 33 bytes
00093 * - **Total for 136 pins**: 136 x 33 = 4488 bytes (~4.4 KB)
00094 *
00095 * ## RX72N Pin Architecture
00096 *
00097 * The RX72N microcontroller has **182 GPIO pins** organized into **17 ports**:
00098 *
00099 * | Port | Hex ID | Decimal | Pin Count | Example Usage |
00100 * |-----|-----|-----|-----|
00101 * | PORT0 | 0x0 | 0 | 8 | LEDs, GPIOs |
00102 * | PORT1 | 0x1 | 1 | 8 | ADC inputs |
00103 * | PORT2 | 0x2 | 2 | 8 | GPIO, interrupts |
00104 * | PORT3-9 | 0x3-0x9 | 3-9 | Variable | Mixed peripherals |
00105 * | PORTA | 0xA | 10 | 8 | SPI, I2C |
00106 * | PORTB | 0xB | 11 | 8 | USB, UART |
00107 * | PORTC | 0xC | 12 | 8 | SPI, PWM |
00108 * | PORTD | 0xD | 13 | 8 | UART, GPIOs |
00109 * | PORTE | 0xE | 14 | 8 | Encoder, Timers |
00110 * | PORTF-G | 0xF-0x10 | 15-16 | Variable | Reserved, special |
00111 *
00112 * **Note**: Not all ports have all 8 pins physically available on the package.
00113 *
00114 * ## Thread Safety Guarantees
00115 *
00116 * | Operation | Thread-Safe? | Mutex Used? | Details |
00117 * |-----|-----|-----|-----|
00118 * | pin_validator_init() | [X] No | N/A | Call from main() only |
00119 * | pin_validator_get_interface() | [OK] Yes | No | Read-only check |
00120 * | pin_validator_deinit() | [X] No | No | Call during shutdown only |
00121 * | validate_pin() | [OK] Yes | No | Read-only table lookup |
00122 * | reserve_pin() | [OK] Yes | [OK] Yes | Mutex-protected write |
00123 * | release_pin() | [OK] Yes | [OK] Yes | Mutex-protected write |
00124 * | is_pin_reserved() | [OK] Yes | [OK] Yes | Mutex-protected read |
00125 * | get_pin_function() | [OK] Yes | [OK] Yes | Mutex-protected read + strcpy |
00126 * | clear_all_reservations() | [OK] Yes | [OK] Yes | Exclusive mutex lock |
00127 *
00128 * ## Usage Examples
00129 *
00130 * ### Example 1: Basic Initialization and Usage
00131 * @code
00132 * // Global validator instance
00133 * static pin_validator_t s_pin_validator;
00134 *
00135 * int main(void)
00136 * {
00137 *     // Step 1: Initialize validator
00138 *     rx_err_t err = pin_validator_init(&s_pin_validator);
00139 *     if (err != k_rx_ok) {
00140 *         // Fatal error - cannot continue
00141 *         while (1) { __asm("nop"); }
00142 *     }
00143 *
00144 *     // Step 2: Get interface from concrete validator
00145 *     rx_pin_interface_t pin_iface;
00146 *     err = pin_validator_get_interface(&pin_iface, &s_pin_validator);
00147 *     if (err != k_rx_ok) {
00148 *         while (1) { __asm("nop"); }
00149 *     }
00150 *
00151 *     // Step 3: Use interface to reserve pins
00152 *     err = pin_iface.reserve_pin(pin_iface.ctx, 0xA, 5, "SPI_COPI");
00153 *     if (err == k_rx_err_gpio_conflict) {
00154 *         debug_printf("Pin PA5 already reserved\n");
00155 *     }
00156 *
00157 *     // Continue with application
00158 *     return 0;
00159 * }
00160 * @endcode
00161 *
00162 * ### Example 2: Using with Global Infrastructure
00163 * @code
00164 * // Infrastructure owns the validator
00165 * static pin_validator_t s_global_validator;
00166 * static rx_pin_interface_t s_global_pin_iface;
00167 *
00168 * rx_err_t infrastructure_init(void)
00169 * {

```

```

00170 * // Initialize concrete validator
00171 * rx_err_t err = pin_validator_init(&s_global_validator);
00172 * if (err != k_rx_ok) return err;
00173 *
00174 * // Extract interface
00175 * err = pin_validator_get_interface(&s_global_pin_iface, &s_global_validator);
00176 * if (err != k_rx_ok) return err;
00177 *
00178 * // Now modules can get interface via infrastructure
00179 * return k_rx_ok;
00180 * }
00181 *
00182 * rx_pin_interface_t* infrastructure_get_pin_interface(void)
00183 * {
00184 *     return &s_global_pin_iface;
00185 * }
00186 * @endcode
00187 *
00188 * ### Example 3: Unit Testing with Mock vs. Real Validator
00189 * @code
00190 * #ifdef UNIT_TEST
00191 *     // Use mock validator for testing
00192 *     rx_pin_interface_t iface = create_mock_pin_interface();
00193 * #else
00194 *     // Use real validator in production
00195 *     static pin_validator_t validator;
00196 *     pin_validator_init(&validator);
00197 *
00198 *     rx_pin_interface_t iface;
00199 *     pin_validator_get_interface(&iface, &validator);
00200 * #endif
00201 *
00202 * // Module code works with either (Dependency Inversion!)
00203 * spi_init(&iface);
00204 * @endcode
00205 *
00206 * ## Design Rationale
00207 *
00208 * **Why 2D array instead of hashmap?**
00209 * - Embedded systems: Predictable memory, O(1) access
00210 * - RX72N has fixed port/pin structure
00211 * - Direct indexing: `reservations[port][pin]` is fast
00212 * - No dynamic allocation (NASA Rule 3)
00213 *
00214 * **Why ThreadX mutex instead of atomic operations?**
00215 * - Struct copy for get_pin_function() requires protection
00216 * - Consistency between reservation flag and function name
00217 * - ThreadX already integrated (no additional dependency)
00218 * - Mutex overhead acceptable for non-ISR operations
00219 *
00220 * **Why store function names vs. module IDs?**
00221 * - Human-readable debugging ("SPI_COPI" vs. ID 42)
00222 * - No need for global module registry
00223 * - String literals in .rodata (no RAM cost)
00224 * - Easy to print in debug reports
00225 *
00226 * ## NASA Power of 10 Compliance
00227 *
00228 * | Rule | Status | Implementation |
00229 * |-----|-----|-----|
00230 * | 1. Simplify control flow | [OK] | No goto, setjmp, recursion |
00231 * | 2. Fixed loop bounds | [OK] | All loops use k_pin_validator_max* constants |
00232 * | 3. No dynamic memory | [OK] | Static allocation only (2D array) |
00233 * | 4. Functions < 60 lines | [OK] | All functions under limit |
00234 * | 5. Use assertions | [OK] | NULL checks, initialization checks |
00235 * | 6. Data at smallest scope | [OK] | Private functions, local variables |
00236 * | 7. Check return values | [OK] | All errors propagated |
00237 * | 8. Limit preprocessor | [OK] | Only include guards |
00238 * | 9. Restrict pointers | [WARN] | Function pointers (DIP pattern) |
00239 * | 10. Compile warnings | [OK] | -Wall -Wextra -Werror |
00240 *
00241 * @see rx_pin_interface.h Abstract interface this file implements
00242 * @see rx_infrastructure.h Global service locator using this validator
00243 * @see RX_RESERVE_PIN() Convenience macro for pin reservation
00244 * @see RX_RELEASE_PIN() Convenience macro for pin release
00245 *
00246 * @since Version 1.0.0
00247 * @date 2026-01-01
00248 * @copyright Copyright (c) 2026 Locked Inc.
00249 * SPDX-License-Identifier: MIT
00250 */
00251
00252 #pragma once
00253
00254 #include <stdbool.h>
00255 #include <stddef.h>
00256 #include <stdint.h>

```

```

00257
00258 #include "rx_err.h"
00259 #include "rx_pin_interface.h"
00260 #include "tx_api.h"
00261
00262 #ifdef __cplusplus
00263 extern "C" {
00264 #endif
00265
00266 /*
=====
00267  * Configuration
00268  *
=====
00269 */
00270
00271 /**
00272  * @enum pin_validator_limits_t
00273  * @brief Pin Validator Configuration Constants
00274  *
00275  * @details
00276  * Defines compile-time array dimensions for the pin reservation table. These limits
00277  * are based on the RX72N hardware capabilities and determine memory footprint.
00278  *
00279  * ## Rationale for Limits
00280  *
00281  * **Why 17 ports?**
00282  * - RX72N has ports 0-9 (10 ports) + A-G (7 ports) = 17 total
00283  * - Covers all possible GPIO ports on RX72N
00284  * - Array index maps directly to port number (0-16)
00285  * - No gaps, no hash function needed
00286  *
00287  * **Why 8 pins per port?**
00288  * - Each RX72N port has up to 8 pins (0-7)
00289  * - Some ports have fewer pins physically, but we allocate max
00290  * - Uniform array size simplifies indexing
00291  * - Small waste (~10% unused slots) vs. complexity of variable-size
00292  *
00293  * ## Memory Impact
00294  *
00295  * Total pin slots: 17 ports x 8 pins = **136 slots**
00296  *
00297  * Per-slot memory:
00298  * - `bool reserved`: 1 byte
00299  * - `char function[32]`: 32 bytes
00300  * - **Total**: 33 bytes per slot
00301  *
00302  * Total table size: 136 slots x 33 bytes = **4488 bytes (~4.4 KB)**
00303  *
00304  * ## Alternative Designs Considered
00305  *
00306  * | Design | Memory | Access Time | Complexity | Chosen? |
00307  * |-----|-----|-----|-----|-----|
00308  * | 2D array (current) | 4.4 KB | O(1) | Low | [OK] Yes |
00309  * | Linked list | ~800 bytes (sparse) | O(n) | Medium | [X] No |
00310  * | Hash table | ~2 KB + overhead | O(1) avg | High | [X] No |
00311  * | Bitfield + separate names | ~17 bytes + strings | O(1) + lookup | Medium | [X] No |
00312  *
00313  * **Decision**: 2D array chosen for simplicity, predictability, and O(1) access.
00314  *
00315  * @see pin_reservation_t Structure stored in each array slot
00316  * @see pin_validator_t::reservations 2D array using these dimensions
00317  *
00318  * @since Version 1.0.0
00319  */
00320 typedef enum : uint8_t {
00321 /**
00322  * @brief Maximum number of ports on RX72N (0-9, A-G)
00323  * @details
00324  * RX72N port numbering:
00325  * - Decimal ports: 0-9 (10 ports)
00326  * - Hex ports: A-G (0xA-0x10 = 10-16 decimal, 7 ports)
00327  * - **Total**: 17 ports (indices 0-16 in array)
00328  */
00329 k_pin_validator_max_ports = 17,
00330
00331 /**
00332  * @brief Maximum pins per port (0-7)
00333  * @details
00334  * Each RX72N port has up to 8 pins numbered 0-7.
00335  * Not all ports have all 8 pins physically available on the package,
00336  * but we allocate space for all 8 to simplify indexing.
00337  */
00338 k_pin_validator_max_pins = 8,
00339 } pin_validator_limits_t;
00340
00341 */

```

```

=====
00342 * Pin Reservation State
00343 *
=====
00344 */
00345
00346 /**
00347  * @struct pin_reservation_t
00348  * @brief Pin reservation state for a single GPIO pin
00349  *
00350  * @details
00351  * Stores the reservation status and owner information for one pin. These structures
00352  * are organized in a 2D array `[port][pin]` within the pin_validator_t.
00353  *
00354  * ## Memory Layout
00355  * @code
00356  struct pin_reservation_t {
00357  * // Offset | Size | Field
00358  * // 0 | 1 | bool reserved
00359  * // 1 | 32 | char function[32]
00360  * // TOTAL: 33 bytes (with padding)
00361  * };
00362  * @endcode
00363  *
00364  * ## Field Relationships
00365  *
00366  * | reserved | function | Meaning |
00367  * |-----|-----|-----|
00368  * | false | (any) | Pin is available (function value ignored) |
00369  * | true | "SPI_COPI" | Pin reserved for SPI COPI |
00370  * | true | "GPIO_OUT" | Pin reserved for GPIO output |
00371  * | true | "" | Pin reserved but function unknown (edge case) |
00372  *
00373  * ## Typical Usage
00374  * @code
00375  * pin_reservation_t* slot = &validator->reservations[port][pin];
00376  *
00377  * // Check if reserved
00378  * if (slot->reserved) {
00379  *   printf("Pin P%X.%d used for: %s\n", port, pin, slot->function);
00380  * }
00381  *
00382  * // Reserve pin
00383  * slot->reserved = true;
00384  * strncpy(slot->function, "SPI_COPI", sizeof(slot->function) - 1);
00385  * slot->function[sizeof(slot->function) - 1] = '\0';
00386  *
00387  * // Release pin
00388  * slot->reserved = false;
00389  * memset(slot->function, 0, sizeof(slot->function));
00390  * @endcode
00391  *
00392  * @var pin_reservation_t::reserved
00393  * @brief Reservation flag (true = pin is reserved)
00394  * @details
00395  * When false, the pin is available for reservation. When true, the pin is in use
00396  * and the `function` field indicates which module owns it.
00397  *
00398  * @var pin_reservation_t::function
00399  * @brief Function name identifying the pin's owner
00400  * @details
00401  * Null-terminated string (max 31 chars + null). Examples: "SPI_COPI", "UART_TX",
00402  * "MOTOR_PWM". Only valid when `reserved == true`. String is typically a literal
00403  * from .rodata (not copied, just pointer stored).
00404  *
00405  * @note Size is 33 bytes (1 bool + 32 char array)
00406  * @note Function name is NOT copied - must be string literal or static storage
00407  * @note Thread safety: Access must be mutex-protected
00408  *
00409  * @see pin_validator_t::reservations 2D array of these structures
00410  * @see pin_interface_limits_t::k_pin_function_name_max_len Function name buffer size
00411  *
00412  * @since Version 1.0.0
00413  */
00414 typedef struct {
00415     bool reserved; /**< Is this pin reserved? */
00416     char function[k_pin_function_name_max_len]; /**< Function name (e.g., "SPI_COPI", "GPIO_OUT") */
00417 } pin_reservation_t;
00418
00419 /*
=====
00420 * Pin Validator Structure
00421 *
=====
00422 */
00423
00424 /**

```

```

00425 * @struct pin_validator_t
00426 * @brief Concrete Pin Validator Implementation Structure
00427 *
00428 * @details
00429 * This is the **concrete implementation** of the pin validator that backs the
00430 * `rx_pin_interface_t` abstract interface. Contains all state for pin validation,
00431 * reservation tracking, and thread synchronization.
00432 *
00433 * ## Memory Layout
00434 *
00435 * @code
00436 * struct pin_validator_t {
00437 *     // Offset | Size | Field
00438 *     //      0 | ~100 | TX_MUTEX mutex
00439 *     //   ~100 | 4488 | pin_reservation_t[17][8] reservations
00440 *     // ~4588 |    1 | bool initialized
00441 *     // ~4589 |   ~3 | (padding to alignment)
00442 *     // TOTAL: ~4592 bytes (~4.5 KB)
00443 * };
00444 * @endcode
00445 *
00446 * ## Field Descriptions
00447 *
00448 * ### mutex (ThreadX Mutex)
00449 * - **Purpose**: Protects reservation table from concurrent access
00450 * - **Type**: TX_MUTEX (ThreadX RTOS primitive)
00451 * - **Size**: ~100 bytes (platform-dependent)
00452 * - **Operations protected**:
00453 *   - reserve_pin() - Exclusive write access
00454 *   - release_pin() - Exclusive write access
00455 *   - is_pin_reserved() - Shared read access
00456 *   - get_pin_function() - Shared read access + string copy
00457 *   - clear_all_reservations() - Exclusive write access
00458 *
00459 * ### reservations[port][pin] (2D Array)
00460 * - **Purpose**: Stores reservation status for all pins
00461 * - **Dimensions**: 17 ports x 8 pins = 136 slots
00462 * - **Per-slot size**: 33 bytes (1 bool + 32 char)
00463 * - **Total size**: 136 x 33 = 4488 bytes (~4.4 KB)
00464 * - **Indexing**: `reservations[port][pin]` where:
00465 *   - `port`: 0-9 (decimal), 10-16 (hex A-G)
00466 *   - `pin`: 0-7
00467 * - **Access pattern**: O(1) direct indexing, no search needed
00468 *
00469 * ### initialized (Lifecycle Flag)
00470 * - **Purpose**: Tracks whether pin_validator_init() was called
00471 * - **Type**: bool (1 byte)
00472 * - **States**:
00473 *   - `false`: Validator not initialized, cannot use
00474 *   - `true`: Validator ready, mutex created, table zeroed
00475 * - **Use**: Prevents double-init and validates before interface extraction
00476 *
00477 * ## Lifecycle State Machine
00478 *
00479 * @startuml
00480 * [*] --> Uninitialized : malloc/static declaration
00481 * Uninitialized --> Initialized : pin_validator_init()
00482 * Initialized --> InUse : pin_validator_get_interface()
00483 * InUse --> InUse : reserve/release/query operations
00484 * InUse --> Initialized : (interface still valid)
00485 * Initialized --> Uninitialized : pin_validator_deinit()
00486 * Uninitialized --> [*]
00487 *
00488 * note right of Uninitialized
00489 *     initialized = false
00490 *     mutex not created
00491 *     reservations undefined
00492 * end note
00493 *
00494 * note right of Initialized
00495 *     initialized = true
00496 *     mutex created
00497 *     reservations all cleared
00498 * end note
00499 *
00500 * note right of InUse
00501 *     Interface extracted
00502 *     Modules using validator
00503 *     All operations thread-safe
00504 * end note
00505 * @enduml
00506 *
00507 * ## Typical Usage Pattern
00508 *
00509 * @code
00510 * // Step 1: Declare (uninitialized state)
00511 * static pin_validator_t s_validator; // .bss section, zeroed by startup code

```



```

00512 *
00513 * // Step 2: Initialize (transition to initialized state)
00514 * rx_err_t err = pin_validator_init(&s_validator);
00515 * if (err != k_rx_ok) {
00516 *     return err; // Initialization failed
00517 * }
00518 * // Now: s_validator.initialized == true
00519 * //     s_validator.mutex created
00520 * //     s_validator.reservations[] all cleared
00521 *
00522 * // Step 3: Extract interface (transition to in-use state)
00523 * rx_pin_interface_t iface;
00524 * err = pin_validator_get_interface(&iface, &s_validator);
00525 * if (err != k_rx_ok) {
00526 *     return err;
00527 * }
00528 * // Now: iface.ctx points to s_validator
00529 * //     iface function pointers point to validator's implementations
00530 *
00531 * // Step 4: Use through interface (in-use state)
00532 * err = iface.reserve_pin(iface.ctx, 0xA, 5, "SPI_COPI");
00533 * // Internally: Acquires mutex, checks reservations[10][5], updates if free
00534 *
00535 * // Step 5: Deinitialize (transition back to uninitialized)
00536 * pin_validator_deinit(&s_validator);
00537 * // Now: s_validator.initialized == false
00538 * //     mutex destroyed
00539 * //     reservations cleared (optional)
00540 * @endcode
00541 *
00542 * ## Memory Allocation Strategy
00543 *
00544 * This struct is **ALWAYS statically allocated** (never malloc):
00545 * - Global/file scope: `static pin_validator_t s_validator;`
00546 * - Infrastructure module: `static pin_validator_t s_global_validator;`
00547 * - Stack allocation: `pin_validator_t validator;` (rare, wastes stack)
00548 *
00549 * **Why static allocation?***
00550 * - NASA Rule 3: No dynamic memory after initialization
00551 * - Predictable memory layout
00552 * - No fragmentation risk
00553 * - No malloc/free errors
00554 *
00555 * ## Thread Safety Responsibilities
00556 *
00557 * | Component | Responsibility |
00558 * |-----|-----|
00559 * | **Caller** | Ensure init/deinit called from single thread |
00560 * | **Validator** | Protect all reservation table access with mutex |
00561 * | **Interface methods** | Acquire mutex before accessing reservations[] |
00562 * | **ThreadX** | Handle mutex priority inheritance |
00563 *
00564 * @var pin_validator_t::mutex
00565 * @brief ThreadX mutex for thread-safe access to reservation table
00566 * @details
00567 * Created during pin_validator_init(), destroyed during pin_validator_deinit().
00568 * Protects all accesses to the `reservations` array. Uses priority inheritance
00569 * to prevent priority inversion.
00570 *
00571 * @var pin_validator_t::reservations
00572 * @brief 2D array of pin reservation states [port][pin]
00573 * @details
00574 * Direct-mapped table: port number is first index, pin number is second index.
00575 * Example: `reservations[0xA][5]` = Port A, Pin 5. All access must be mutex-protected.
00576 *
00577 * @var pin_validator_t::initialized
00578 * @brief Lifecycle flag indicating validator is ready for use
00579 * @details
00580 * Set to true by pin_validator_init(), false by pin_validator_deinit(). Checked
00581 * by pin_validator_get_interface() to ensure validator is ready before extracting interface.
00582 *
00583 * @note Total size: ~4592 bytes (~4.5 KB)
00584 * @note Always use static allocation (global or file scope)
00585 * @note Mutex must be created before use (via pin_validator_init())
00586 * @note Thread-safe after initialization (all operations mutex-protected)
00587 *
00588 * @see pin_validator_init() Initialize this structure
00589 * @see pin_validator_get_interface() Extract interface from this structure
00590 * @see pin_validator_deinit() Clean up this structure
00591 * @see rx_pin_interface_t Abstract interface this implements
00592 *
00593 * @since Version 1.0.0
00594 */
00595 typedef struct {
00596     TX_MUTEX          mutex; /**< ThreadX mutex for thread-safe operation */
00597     pin_reservation_t reservations
00598     [k_pin_validator_max_ports]

```

```

00599     [k_pin_validator_max_pins]; /**< Pin reservation tracking array (indexed: reservations[port][pin], Port 0-9: Decimal,
Port 10-16: Hex A-G) */
00600     bool initialized; /**< Is the validator initialized? */
00601 } pin_validator_t;
00602
00603 /*
=====
00604 * Public API
00605 *
=====
00606 */
00607
00608 /**
00609  * @brief Initialize pin validator (MUST be called before use)
00610  *
00611  * @details
00612  * Performs one-time initialization of the pin validator structure. Creates ThreadX
00613  * mutex, clears reservation table, and marks validator as initialized. Must be called
00614  * exactly once before extracting interface via pin_validator_get_interface().
00615  *
00616  * ## Initialization Algorithm
00617  *
00618  * 1. **Validate inputs**: Check validator pointer is non-NULL
00619  * 2. **Check double-init**: Return error if already initialized
00620  * 3. **Create mutex**: Initialize ThreadX mutex for thread safety
00621  * 4. **Clear reservation table**: Zero all reservations[port][pin] entries
00622  * 5. **Set initialized flag**: Mark validator as ready for use
00623  *
00624  * ## Typical Usage
00625  * @code
00626  * // Global validator (static allocation, zero-initialized by startup code)
00627  * static pin_validator_t s_pin_validator;
00628  *
00629  * int main(void)
00630  * {
00631  *     // Initialize validator (one-time operation)
00632  *     rx_err_t err = pin_validator_init(&s_pin_validator);
00633  *     if (err != k_rx_ok) {
00634  *         // Fatal error - cannot continue
00635  *         debug_printf("Pin validator init failed: %d\n", err);
00636  *         while (1) { __asm("nop"); }
00637  *     }
00638  *
00639  *     // Validator is now ready for interface extraction
00640  *     // ...
00641  * }
00642  * @endcode
00643  *
00644  * @param[in,out] validator Pin validator instance to initialize (must be non-NULL)
00645  *
00646  * @return rx_err_t Error code indicating initialization result
00647  * @retval k_rx_ok Initialization successful, validator ready for use
00648  * @retval k_rx_err_null_ptr validator parameter is nullptr
00649  * @retval k_rx_err_invalid_state Validator already initialized (double-init attempt)
00650  * @retval k_rx_err_rtos_mutex ThreadX mutex creation failed
00651  *
00652  * @pre validator must point to valid pin_validator_t structure
00653  * @pre validator must NOT be already initialized (initialized flag must be false)
00654  * @pre ThreadX kernel must be initialized (tx_kernel_enter called or about to be called)
00655  * @pre Must be called from main thread before starting ThreadX (or from initialization thread)
00656  *
00657  * @post validator->initialized set to true
00658  * @post validator->mutex created and ready for use
00659  * @post validator->reservations[] array cleared (all pins available)
00660  * @post Validator ready for pin_validator_get_interface()
00661  *
00662  * @note **NOT thread-safe**: Must be called from single thread during initialization
00663  * @note Calling twice returns k_rx_err_invalid_state (double-init protection)
00664  * @note On failure, validator is left in undefined state - must not use
00665  * @note Typical execution time: ~200-500 us @ 240 MHz
00666  *
00667  * @warning **CRITICAL**: Must be called before any pin reservation operations
00668  * @warning Do NOT call from multiple threads (not thread-safe during init)
00669  * @warning If this fails, system cannot continue - must halt or reset
00670  *
00671  * @see pin_validator_get_interface() Extract interface after initialization
00672  * @see pin_validator_deinit() Cleanup validator (shutdown only)
00673  * @see rx_infrastructure_init() Typically calls this function
00674  *
00675  * @since Version 1.0.0
00676  *
00677  * @par NASA Power of 10 Compliance:
00678  * - Rule 5: [OK] 4 preconditions, 4 postconditions
00679  */
00680 [[nodiscard]] rx_err_t pin_validator_init(pin_validator_t* validator);
00681
00682 /**

```

```

00683 * @brief Extract abstract interface from concrete pin validator (Dependency Inversion)
00684 *
00685 * @details
00686 * Extracts an `rx_pin_interface_t` from the concrete `pin_validator_t` implementation.
00687 * This function is the key to Dependency Inversion: high-level modules depend on
00688 * the abstract interface, not the concrete validator type.
00689 *
00690 * ## Dependency Inversion Pattern
00691 * ```
00692 * High-Level Module (SPI Driver)
00693 *     v depends on v
00694 * rx_pin_interface_t (Abstract) <- THIS FUNCTION PRODUCES THIS
00695 *     ^ implements ^
00696 * pin_validator_t (Concrete)   <- THIS FUNCTION CONSUMES THIS
00697 * ```
00698 *
00699 * ## What This Function Does
00700 *
00701 * Populates the `rx_pin_interface_t` structure with:
00702 * - ctx pointer: Points to concrete validator ('&validator')
00703 * - Function pointers: Point to validator's implementation functions:
00704 *   - `iface->validate_pin = pin_validator_validate`
00705 *   - `iface->reserve_pin = pin_validator_reserve`
00706 *   - `iface->release_pin = pin_validator_release`
00707 *   - `iface->is_pin_reserved = pin_validator_is_reserved`
00708 *   - `iface->get_pin_function = pin_validator_get_function`
00709 *   - `iface->clear_all_reservations = pin_validator_clear_all`
00710 *
00711 * ## Typical Usage Pattern
00712 * @code
00713 * // Step 1: Initialize concrete validator
00714 * static pin_validator_t s_validator;
00715 * rx_err_t err = pin_validator_init(&s_validator);
00716 * if (err != k_rx_ok) {
00717 *     return err;
00718 * }
00719 *
00720 * // Step 2: Extract interface
00721 * rx_pin_interface_t iface;
00722 * err = pin_validator_get_interface(&iface, &s_validator);
00723 * if (err != k_rx_ok) {
00724 *     return err;
00725 * }
00726 *
00727 * // Step 3: Inject interface into high-level module
00728 * spi_config_t spi_cfg = {
00729 *     .pin_iface = &iface, // High-level code only sees interface!
00730 *     // ...
00731 * };
00732 * spi_init(&spi_cfg);
00733 *
00734 * // Step 4: Use through interface (polymorphism in C)
00735 * iface.reserve_pin(iface.ctx, 0xA, 5, "SPI_COPI");
00736 * // Internally calls: pin_validator_reserve(&s_validator, 0xA, 5, "SPI_COPI")
00737 * @endcode
00738 *
00739 * @param[out] iface Interface structure to populate (must be non-NULL)
00740 * @param[in,out] validator Concrete validator instance (must be initialized)
00741 *
00742 * @return rx_err_t Error code indicating extraction result
00743 * @retval k_rx_ok Interface populated successfully, ready for use
00744 * @retval k_rx_err_null_ptr iface or validator parameter is nullptr
00745 * @retval k_rx_err_invalid_state Validator not initialized (call pin_validator_init first)
00746 *
00747 * @pre validator must be initialized via pin_validator_init()
00748 * @pre iface must point to valid rx_pin_interface_t structure
00749 * @pre validator pointer must remain valid for interface lifetime
00750 *
00751 * @post iface->ctx points to validator
00752 * @post All iface function pointers point to validator implementations
00753 * @post iface is valid and ready for use
00754 * @post validator lifetime must extend beyond iface usage
00755 *
00756 * @note Thread-safe: Can be called from any thread after validator initialization
00757 * @note Interface lifetime: Returned interface is valid as long as validator exists
00758 * @note Multiple calls OK: Can extract multiple interfaces from same validator
00759 * @note Performance: ~50 cycles (simple struct copy + pointer assignments)
00760 *
00761 * @warning Validator must remain alive while interface is in use
00762 * @warning Do NOT call pin_validator_deinit() while interface is in use
00763 * @warning Interface becomes invalid if validator is deinitialized
00764 *
00765 * @see pin_validator_init() Must be called first to initialize validator
00766 * @see rx_pin_interface_t Abstract interface structure being populated
00767 * @see rx_infrastructure_get_pin_interface() Global access to interface
00768 *
00769 * @since Version 1.0.0

```

```

00770 *
00771 * @par NASA Power of 10 Compliance:
00772 * - Rule 5: [OK] 3 preconditions, 4 postconditions
00773 */
00774 [[nodiscard]] rx_err_t pin_validator_get_interface(rx_pin_interface_t* iface,
00775                                                    pin_validator_t* validator);
00776
00777 /**
00778 * @brief Deinitialize pin validator (graceful shutdown only)
00779 *
00780 * @details
00781 * Performs cleanup of pin validator resources. Destroys ThreadX mutex, clears
00782 * reservation table, and marks validator as uninitialized. In embedded systems,
00783 * this is RARELY CALLED because the system typically runs indefinitely.
00784 *
00785 * ## When to Use
00786 * - Unit test teardown (clean state between tests)
00787 * - System reset/reboot preparation
00788 * - Firmware update preparation
00789 * - Development/debugging (clean restart)
00790 *
00791 * ## When NOT to Use
00792 * - Normal operation (validator should remain initialized)
00793 * - While modules are still using pins
00794 * - From multiple threads (not thread-safe)
00795 *
00796 * ## Cleanup Algorithm
00797 * 1. Validate inputs: Check validator pointer is non-NULL
00798 * 2. Check initialized: Return error if not initialized
00799 * 3. Destroy mutex: Delete ThreadX mutex
00800 * 4. Clear reservations: Zero all reservations (optional, for safety)
00801 * 5. Clear initialized flag: Mark validator as uninitialized
00802 *
00803 * ## Typical Usage
00804 * @code
00805 * // Unit test teardown
00806 * void test_teardown(void)
00807 * {
00808 *     pin_validator_deinit(&s_test_validator);
00809 *     // Validator can be reinitialized for next test
00810 * }
00811 *
00812 * // System shutdown
00813 * void system_shutdown(void)
00814 * {
00815 *     // 1. Stop all modules using pins
00816 *     spi_shutdown();
00817 *     i2c_shutdown();
00818 *
00819 *     // 2. Deinitialize validator
00820 *     pin_validator_deinit(&s_global_validator);
00821 *
00822 *     // 3. Proceed with shutdown/reset
00823 *     // ...
00824 * }
00825 * @endcode
00826 *
00827 * @param[in,out] validator Pin validator instance to deinitialize (must be non-NULL)
00828 *
00829 * @return rx_err_t Error code indicating deinitialization result
00830 * @retval k_rx_ok Deinitialization successful, validator can be reinitialized
00831 * @retval k_rx_err_null_ptr validator parameter is nullptr
00832 * @retval k_rx_err_invalid_state Validator not initialized (nothing to clean up)
00833 *
00834 * @pre validator must be initialized via pin_validator_init()
00835 * @pre No modules should be using pins (all released)
00836 * @pre No threads should be calling validator operations concurrently
00837 *
00838 * @post validator->initialized set to false
00839 * @post validator->mutex destroyed
00840 * @post validator->reservations[] cleared (all pins available)
00841 * @post Validator can be reinitialized via pin_validator_init()
00842 *
00843 * @note NOT thread-safe: Must ensure no concurrent access during deinit
00844 * @note Calling when not initialized returns k_rx_err_invalid_state (safe no-op)
00845 * @note After deinit, all extracted interfaces become INVALID
00846 * @note Typical execution time: ~100-300 us @ 240 MHz
00847 *
00848 * @warning DANGEROUS: All interfaces extracted from this validator become invalid
00849 * @warning Ensure all modules have stopped using pins before calling
00850 * @warning Do NOT call from ISR (mutex operations may block)
00851 *
00852 * @see pin_validator_init() Reinitialize validator after deinit
00853 * @see pin_validator_get_interface() Interfaces become invalid after deinit
00854 *
00855 * @since Version 1.0.0
00856 *

```

```

00857 * @par NASA Power of 10 Compliance:
00858 * - Rule 5: [OK] 3 preconditions, 4 postconditions
00859 */
00860 [[nodiscard]] rx_err_t pin_validator_deinit(pin_validator_t* validator);
00861
00862 #ifdef __cplusplus
00863 }
00864 #endif

```

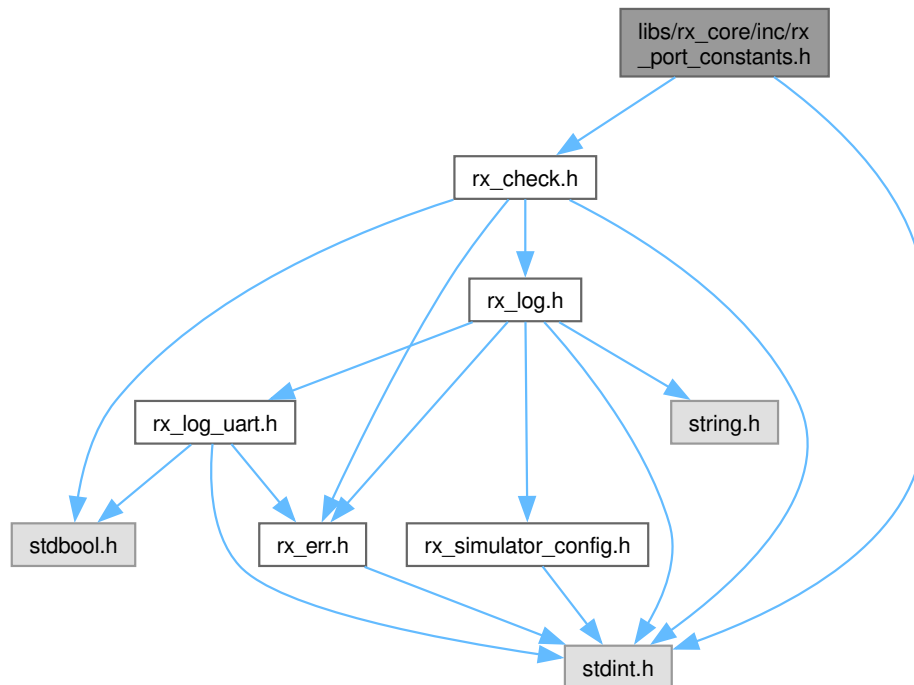
7.29 libs/rx_core/inc/rx_port_constants.h File Reference

Centralized RX72N Port Number Definitions.

```
#include <stdint.h>
```

```
#include "rx_check.h"
```

Include dependency graph for rx_port_constants.h:



Enumerations

- enum `rx_port_number_t` : `uint8_t` {
`k_rx_port_0` = 0x00 , `k_rx_port_1` = 0x01 , `k_rx_port_2` = 0x02 , `k_rx_port_3` = 0x03 ,
`k_rx_port_4` = 0x04 , `k_rx_port_5` = 0x05 , `k_rx_port_6` = 0x06 , `k_rx_port_7` = 0x07 ,
`k_rx_port_8` = 0x08 , `k_rx_port_9` = 0x09 , `k_rx_port_a` = 0x0A , `k_rx_port_b` = 0x0B ,
`k_rx_port_c` = 0x0C , `k_rx_port_d` = 0x0D , `k_rx_port_e` = 0x0E , `k_rx_port_f` = 0x0F ,
`k_rx_port_g` = 0x10 , `k_rx_port_j` }

RX72N Port Number Constants (Single-Byte Identifiers).

- enum `rx_pin_number_t` : `uint8_t` {
`k_rx_pin_0` = 0 , `k_rx_pin_1` = 1 , `k_rx_pin_2` = 2 , `k_rx_pin_3` = 3 ,
`k_rx_pin_4` = 4 , `k_rx_pin_5` = 5 , `k_rx_pin_6` = 6 , `k_rx_pin_7` = 7 ,
`k_rx_pin_min` = 0 , `k_rx_pin_max` = 7 }

Pin Number Constants Within Each Port (0-7).

- enum `port_encoding_t` : `uint8_t` { `k_port_shift` , `k_port_mask` }


```

, k_rx_p9_3 = (k_rx_port_9 << k_port_shift) | k_rx_pin_3 ,
k_rx_p9_4 = (k_rx_port_9 << k_port_shift) | k_rx_pin_4 , k_rx_p9_5 = (k_rx_port_9
<< k_port_shift) | k_rx_pin_5 , k_rx_p9_6 = (k_rx_port_9 << k_port_shift) | k_rx_pin_6
, k_rx_p9_7 = (k_rx_port_9 << k_port_shift) | k_rx_pin_7 ,
k_rx_pa_0 = (k_rx_port_a << k_port_shift) | k_rx_pin_0 , k_rx_pa_1 = (k_rx_port_a
<< k_port_shift) | k_rx_pin_1 , k_rx_pa_2 = (k_rx_port_a << k_port_shift) | k_rx_pin_2
, k_rx_pa_3 = (k_rx_port_a << k_port_shift) | k_rx_pin_3 ,
k_rx_pa_4 = (k_rx_port_a << k_port_shift) | k_rx_pin_4 , k_rx_pa_5 = (k_rx_port_a
<< k_port_shift) | k_rx_pin_5 , k_rx_pa_6 = (k_rx_port_a << k_port_shift) | k_rx_pin_6
, k_rx_pa_7 = (k_rx_port_a << k_port_shift) | k_rx_pin_7 ,
k_rx_pb_0 = (k_rx_port_b << k_port_shift) | k_rx_pin_0 , k_rx_pb_1 = (k_rx_port_b
<< k_port_shift) | k_rx_pin_1 , k_rx_pb_2 = (k_rx_port_b << k_port_shift) | k_rx_pin_2
, k_rx_pb_3 = (k_rx_port_b << k_port_shift) | k_rx_pin_3 ,
k_rx_pb_4 = (k_rx_port_b << k_port_shift) | k_rx_pin_4 , k_rx_pb_5 = (k_rx_port_b
<< k_port_shift) | k_rx_pin_5 , k_rx_pb_6 = (k_rx_port_b << k_port_shift) | k_rx_pin_6
, k_rx_pb_7 = (k_rx_port_b << k_port_shift) | k_rx_pin_7 ,
k_rx_pc_0 = (k_rx_port_c << k_port_shift) | k_rx_pin_0 , k_rx_pc_1 = (k_rx_port_c
<< k_port_shift) | k_rx_pin_1 , k_rx_pc_2 = (k_rx_port_c << k_port_shift) | k_rx_pin_2
, k_rx_pc_3 = (k_rx_port_c << k_port_shift) | k_rx_pin_3 ,
k_rx_pc_4 = (k_rx_port_c << k_port_shift) | k_rx_pin_4 , k_rx_pc_5 = (k_rx_port_c
<< k_port_shift) | k_rx_pin_5 , k_rx_pc_6 = (k_rx_port_c << k_port_shift) | k_rx_pin_6
, k_rx_pc_7 = (k_rx_port_c << k_port_shift) | k_rx_pin_7 ,
k_rx_pd_0 = (k_rx_port_d << k_port_shift) | k_rx_pin_0 , k_rx_pd_1 = (k_rx_port_d
<< k_port_shift) | k_rx_pin_1 , k_rx_pd_2 = (k_rx_port_d << k_port_shift) | k_rx_pin_2
, k_rx_pd_3 = (k_rx_port_d << k_port_shift) | k_rx_pin_3 ,
k_rx_pd_4 = (k_rx_port_d << k_port_shift) | k_rx_pin_4 , k_rx_pd_5 = (k_rx_port_d
<< k_port_shift) | k_rx_pin_5 , k_rx_pd_6 = (k_rx_port_d << k_port_shift) | k_rx_pin_6
, k_rx_pd_7 = (k_rx_port_d << k_port_shift) | k_rx_pin_7 ,
k_rx_pe_0 = (k_rx_port_e << k_port_shift) | k_rx_pin_0 , k_rx_pe_1 = (k_rx_port_e
<< k_port_shift) | k_rx_pin_1 , k_rx_pe_2 = (k_rx_port_e << k_port_shift) | k_rx_pin_2
, k_rx_pe_3 = (k_rx_port_e << k_port_shift) | k_rx_pin_3 ,
k_rx_pe_4 = (k_rx_port_e << k_port_shift) | k_rx_pin_4 , k_rx_pe_5 = (k_rx_port_e
<< k_port_shift) | k_rx_pin_5 , k_rx_pe_6 = (k_rx_port_e << k_port_shift) | k_rx_pin_6
, k_rx_pe_7 = (k_rx_port_e << k_port_shift) | k_rx_pin_7 ,
k_rx_pf_0 = (k_rx_port_f << k_port_shift) | k_rx_pin_0 , k_rx_pf_1 = (k_rx_port_f <<
k_port_shift) | k_rx_pin_1 , k_rx_pf_2 = (k_rx_port_f << k_port_shift) | k_rx_pin_2 ,
k_rx_pf_3 = (k_rx_port_f << k_port_shift) | k_rx_pin_3 ,
k_rx_pf_4 = (k_rx_port_f << k_port_shift) | k_rx_pin_4 , k_rx_pf_5 = (k_rx_port_f <<
k_port_shift) | k_rx_pin_5 , k_rx_pf_6 = (k_rx_port_f << k_port_shift) | k_rx_pin_6 ,
k_rx_pf_7 = (k_rx_port_f << k_port_shift) | k_rx_pin_7 ,
k_rx_pg_0 = (k_rx_port_g << k_port_shift) | k_rx_pin_0 , k_rx_pg_1 = (k_rx_port_g
<< k_port_shift) | k_rx_pin_1 , k_rx_pg_2 = (k_rx_port_g << k_port_shift) | k_rx_pin_2
, k_rx_pg_3 = (k_rx_port_g << k_port_shift) | k_rx_pin_3 ,
k_rx_pg_4 = (k_rx_port_g << k_port_shift) | k_rx_pin_4 , k_rx_pg_5 = (k_rx_port_g
<< k_port_shift) | k_rx_pin_5 , k_rx_pg_6 = (k_rx_port_g << k_port_shift) | k_rx_pin_6
, k_rx_pg_7 = (k_rx_port_g << k_port_shift) | k_rx_pin_7 ,
k_rx_pj_0 = (k_rx_port_j << k_port_shift) | k_rx_pin_0 , k_rx_pj_1 = (k_rx_port_j <<
k_port_shift) | k_rx_pin_1 , k_rx_pj_2 = (k_rx_port_j << k_port_shift) | k_rx_pin_2 ,
k_rx_pj_3 = (k_rx_port_j << k_port_shift) | k_rx_pin_3 ,
k_rx_pj_4 = (k_rx_port_j << k_port_shift) | k_rx_pin_4 , k_rx_pj_5 = (k_rx_port_j <<
k_port_shift) | k_rx_pin_5 , k_rx_pj_6 = (k_rx_port_j << k_port_shift) | k_rx_pin_6 ,
k_rx_pj_7 = (k_rx_port_j << k_port_shift) | k_rx_pin_7 }

```

Pre-Computed Port/Pin Combinations for All RX72N GPIO Pins.

- enum rx_port_verify_t : uint16_t {
k_verify_port_0 = 0x0000U , k_verify_port_1 = 0x0001U , k_verify_port_5 = 0x0005U ,
k_verify_port_a = 0x000AU ,
k_verify_port_b = 0x000BU , k_verify_port_c = 0x000CU , k_verify_port_e = 0x000EU ,
k_verify_port_j ,


```

k_verify_pin_2 = 0x0002U , k_verify_pin_7 = 0x0007U , k_verify_port_shift = 0x0008U ,
k_verify_enc_pb_2 = 0x0B02U ,
k_verify_enc_pe_5 = 0x0E05U , k_verify_enc_p0_7 = 0x0007U , k_verify_p0_4 = 0x0004U ,
k_verify_p2_5 = 0x0205U ,
k_verify_pa_2 = 0x0A02U , k_verify_pb_2 = 0x0B02U , k_verify_pe_5 = 0x0E05U ,
k_verify_pj_3 = 0x1203U }

```

Expected values used in static assertion verification.

Functions

- static uint8_t [rx_port_from_pin](#) (rx_port_pin_t pin)
Extract port number from combined port/pin identifier.
- static uint8_t [rx_pin_from_pin](#) (rx_port_pin_t pin)
Extract pin number from combined port/pin identifier.

7.29.1 Detailed Description

Centralized RX72N Port Number Definitions.

LIBRARY-LEVEL SINGLE SOURCE OF TRUTH for all port and pin numbers used throughout the STAR RX72N firmware. This file is the ONLY location in the library layer that contains hexadecimal values for port/pin identifiers.

7.29.1.1 Purpose and Design Rationale

This file implements the "Single Source of Truth" pattern for hardware port addressing, eliminating magic numbers and providing type-safe port/pin identifiers using C23 typed enums. All library and application code must use these constants—never hardcode port numbers directly.

7.29.1.2 Architecture and Layering

The port constants architecture follows a strict layering model:

1. This file ([rx_port_constants.h](#)): Defines port/pin constants with hex values
2. HAL layer (lib/rx_hal/): Uses these constants, NO hex values in HAL code
3. Application layer (include/hardware_pinout.h): Uses these constants, NO hex in app code

This layering ensures:

- Single point of truth for port numbering
- Type safety through C23 typed enums
- Compile-time validation via static assertions
- Easy portability to different RX packages (144-pin, 176-pin, etc.)

7.29.1.3 Port/Pin Encoding Scheme

GPIO pins are encoded as 16-bit values using the following format:

```
gpio_pin_t = (port « k_port_shift) | pin_number
            = (port « 8) | pin
```

Example: Port B, Pin 2 (PB2)
 = (0x0B « 8) | 0x02
 = 0x0B02

This encoding allows:

- Upper byte: Port number (0x00-0x13)
- Lower byte: Pin number (0x00-0x07)
- Extraction: Port = pin >> 8, Pin = pin & 0xFF
- Validation: Compile-time checked via static assertions

7.29.1.4 RX72N Port Numbering

The Renesas RX72N uses non-contiguous port numbering:

- Decimal ports: 0-9 (0x00-0x09)
- Alphabetic ports: A-G (0x0A-0x10)
- Port J: 0x12 (note: Port I does not exist; Port H exists at 0x11 but has no pins on 144-pin)

Total: 18 ports x 8 pins = 144 theoretical GPIO combinations (Not all combinations physically available on 144-pin LFQFP package)

7.29.1.5 Performance Characteristics

- Memory usage: Zero runtime overhead (all constants are compile-time)
- Code size: Minimal (enums compile to immediate values)
- Execution time: Inline accessor functions are 1-2 CPU cycles
- Type safety: Full compile-time type checking with C23 typed enums

7.29.1.6 Usage Patterns

Library code example (HAL layer):

```
// Map port enum to register pointer
switch (port) {
    case k_rx_port_b: return portb();
    case k_rx_port_e: return porte();
}
```

Application code example:

```
// Define application-specific pin aliases
typedef enum : uint16_t {
    k_gpio_led_status = k_rx_pb_2, // PB2 = status LED
    k_gpio_spi_cs      = k_rx_pe_5, // PE5 = SPI chip select
} application_pins_t;
```

Pin extraction example:

```
gpio_pin_t pin = k_rx_pb_2;
uint8_t port = rx_port_from_pin(pin); // Returns k_rx_port_b (0x0B)
uint8_t pin_num = rx_pin_from_pin(pin); // Returns 2
```

Hardware Requirements:

Component	Requirement	Notes
MCU	Renesas RX72N	Any package (100/144/176-pin)
Package	144-pin LFQFP (primary target)	Not all ports available
Ports	18 ports (0-9, A-G, J)	Non-contiguous numbering
Pins per port	8 pins (0-7)	All ports have 8-bit width

Module Dependencies:

This module depends on:

- [rx_check.h](#) - For RX_ASSERT compile-time/runtime validation
- `<stdint.h>` - For `uint8_t`, `uint16_t` types

This module is used by:

- `hardware.h` - HAL GPIO register accessors
- `hardware_pinout.h` - Application-level pin definitions
- All GPIO-related HAL modules

NASA Power of 10 Compliance:

Rule	Status	Implementation
1. Simple control flow	[OK]	Only inline functions with sequential logic
2. Fixed loop bounds	[OK]	No loops (constants only)
3. No dynamic allocation	[OK]	All constants are compile-time
4. Small functions	[OK]	Inline functions are 3-5 lines each
5. Assertions (≥ 2 per function)	[OK]	3 assertions per function (prex2, postx1)
6. Narrow scope	[OK]	File-scope only, no globals
7. Check return values	[OK]	Inline functions don't fail
8. Limited preprocessor	[OK]	C23 typed enums (zero macros)
9. Pointer restrictions	[OK]	No pointers used
10. Compiler warnings	[OK]	Compiles with -Wall -Wextra -Werror

Rule 8 Compliance Detail: This file demonstrates STAR project's mandatory C23 typed enum standard. ALL integer constants use typedef enum : type syntax, providing explicit size guarantees and eliminating preprocessor macros.

SOLID Principles:

Single Responsibility (S):

- This file has ONE job: Define port/pin constants
- No GPIO manipulation logic (that's in HAL layer)
- No application-specific pin assignments (that's in hardware_pinout.h)

Open/Closed (O):

- Extensible to new RX packages without modifying existing constants
- Can add new port definitions for 144-pin or 176-pin packages
- Encoding scheme supports future ports beyond current 18

Liskov Substitution (L):

- All [rx_port_pin_t](#) values are interchangeable
- Pre-computed constants (`k_rx_pb_2`) substitutable with manual encoding
- $((\text{port} \ll 8) \mid \text{pin})$ always produces valid [rx_port_pin_t](#)

Interface Segregation (I):

- Separate enums for ports, pins, and combined port/pin
- Use only what you need: port numbers OR pin numbers OR combinations
- Encoding constants isolated in separate enum

Dependency Inversion (D):

- HAL layer depends on these abstractions, not hardware addresses
- Application code depends on constants, not port numbers
- Hardware changes isolated to this file only

See also

[hardware.h](#) GPIO register accessor functions

[hardware_pinout.h](#) Application-specific pin definitions

[rx_check.h](#) Validation macros used in assertions

[docs/sections/03_hardware_pinout.tex](#) Complete pin assignment table

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Date

2026-01-27

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Since

Version 1.0.0

Definition in file [rx_port_constants.h](#).

7.29.2 Enumeration Type Documentation

7.29.2.1 port_encoding_t

enum `port_encoding_t` : `uint8_t`

Bit Shift and Mask Constants for Port/Pin Encoding.

Defines the constants used to encode and decode 16-bit `gpio_pin_t` values. These constants implement the port/pin encoding scheme used throughout the firmware for type-safe GPIO pin identification.

7.29.2.2 Encoding Scheme

The encoding scheme packs an 8-bit port number and 8-bit pin number into a single 16-bit value:

```
+-----+-----+
| Upper Byte | Lower Byte |
| (Port Number) | (Pin Number) |
+-----+-----+
| Bits 15-8 | Bits 7-0 |
| rx_port_number_t | rx_pin_number_t |
+-----+-----+
```

```
Encoding: gpio_pin_t = (port « k_port_shift) | pin
Decoding: port = gpio_pin_t » k_port_shift
          pin  = gpio_pin_t & k_port_mask
```

7.29.2.3 Constant Definitions

- `k_port_shift` = 8: Left-shift amount to move port number to upper byte
- `k_port_mask` = 0xFF: Mask to extract pin number from lower byte

7.29.2.4 Design Rationale

Using compile-time constants instead of magic numbers provides:

1. Self-documenting code: (`port << k_port_shift`) vs (`port << 8`)
2. Easy modification: Change encoding in one place if needed
3. Type safety: C23 typed enum provides compile-time checking
4. Searchability: `grep` for `k_port_shift` finds all encoding operations

7.29.2.5 Bit Layout Example

Port B (0x0B), Pin 2 (0x02):

Port: 0x0B = 0000 1011
Pin: 0x02 = 0000 0010

Encoding: (0x0B « 8) | 0x02
= 0x0B00 | 0x0002
= 0x0B02

Binary: 0000 1011 0000 0010
+- Port -+-+ Pin -+

Usage Example (Encoding):

```
// Manual construction using encoding constants
rx_port_number_t port = k_rx_port_b; // 0x0B
rx_pin_number_t pin = k_rx_pin_2;    // 0x02
gpio_pin_t result = (port « k_port_shift) | pin;
// Result: 0x0B02

// Equivalent to pre-computed constant
gpio_pin_t result = k_rx_pb_2;
```

Usage Example (Decoding):

```
// Extract port and pin from combined value
gpio_pin_t pin = k_rx_pe_5; // 0x0E05

rx_port_number_t port = (pin » k_port_shift); // 0x0E
rx_pin_number_t pin_num = (pin & k_port_mask); // 0x05

// Or use dedicated accessor functions (preferred)
port = rx_port_from_pin(pin); // 0x0E
pin_num = rx_pin_from_pin(pin); // 0x05
```

Usage Example (Bit Manipulation):

```
// Verify encoding/decoding round-trip
rx_port_number_t original_port = k_rx_port_c;
rx_pin_number_t original_pin = k_rx_pin_7;

// Encode
gpio_pin_t encoded = (original_port « k_port_shift) | original_pin;

// Decode
rx_port_number_t decoded_port = (encoded » k_port_shift);
rx_pin_number_t decoded_pin = (encoded & k_port_mask);

// Verify
assert(decoded_port == original_port);
assert(decoded_pin == original_pin);
```

Invariant

```
k_port_shift == 8 (must shift by full byte)
k_port_mask == 0xFF (must mask full byte)
(value << k_port_shift) >> k_port_shift == value (for value <= 0xFF)
(value & k_port_mask) <= 0xFF (mask result always fits in uint8_t)
```

Note

This is a C23 typed enum with explicit underlying type `uint8_t`, ensuring 1-byte size and consistent ABI across compiler versions

The encoding is big-endian style (port in high byte, pin in low byte)

This encoding matches the pattern used in `hardware_pinout.h` `gpio_pin_t` enum

Warning

Do not change these constants without updating all port/pin definitions
 Changing `k_port_shift` affects ALL 144+ pre-computed pin constants

See also

[rx_port_number_t](#) Port number constants that go in upper byte
[rx_pin_number_t](#) Pin number constants that go in lower byte
[rx_port_pin_t](#) Pre-computed combinations using this encoding
[rx_port_from_pin\(\)](#) Decode port number from `gpio_pin_t`
[rx_pin_from_pin\(\)](#) Decode pin number from `gpio_pin_t`

Since

Version 1.0.0

Enumerator

<code>k_port_shift</code>	Left shift amount to move port number into upper byte (bits 15-8). Used for encoding: <code>gpio_pin_t = (port << k_port_shift) pin</code>
<code>k_port_mask</code>	Bitmask (0xFF) to extract pin number from lower byte (bits 7-0). Used for decoding: <code>pin = gpio_pin_t & k_port_mask</code>

Definition at line 568 of file [rx_port_constants.h](#).

```
00568      : uint8_t {
00569  k_port_shift =
00570      8, /**< Left shift amount to move port number into upper byte (bits 15-8). Used for encoding: gpio_pin_t = (port «
      k_port_shift) | pin */
00571  k_port_mask =
00572      0xFF, /**< Bitmask (0xFF) to extract pin number from lower byte (bits 7-0). Used for decoding: pin = gpio_pin_t &
      k_port_mask */
00573 } port_encoding_t;
```

7.29.2.6 rx_pin_number_t

```
enum rx_pin_number_t : uint8_t
```

Pin Number Constants Within Each Port (0-7).

Defines pin numbers within each GPIO port on the RX72N. All RX72N ports are 8-bit wide, meaning each port has pins numbered 0 through 7. These constants provide self-documenting alternatives to raw numeric literals.

7.29.2.7 Pin Numbering

Every RX72N port (0-9, A-G, J) contains 8 pins:

- Pin 0 (LSB of port data register)
- Pin 1
- Pin 2
- ...
- Pin 7 (MSB of port data register)

This consistent 8-pin structure applies to ALL ports, even though not all individual pins may be physically available on smaller packages.

7.29.2.8 Encoding Integration

Pin numbers form the lower byte of 16-bit `gpio_pin_t` values:

```
gpio_pin_t = (rx_port_number_t « k_port_shift) | rx_pin_number_t
            = (port « 8) | pin
```

Example: Port B (0x0B) + Pin 2 (0x02) = 0x0B02

7.29.2.9 Validation Constants

The enum includes `k_rx_pin_min` and `k_rx_pin_max` for runtime/compile-time validation of pin numbers. These are used by assertion macros and accessor functions to verify pin number correctness.

Valid range: [0, 7] inclusive

Usage Example (Constructing `gpio_pin_t`):

```
// Construct Port B, Pin 2 manually
gpio_pin_t status_led = (k_rx_port_b « k_port_shift) | k_rx_pin_2;
// Result: 0x0B02

// Or use pre-computed constant (preferred)
gpio_pin_t status_led = k_rx_pb_2; // Cleaner, same result
```

Usage Example (Pin Extraction):

```
// Extract pin number from combined identifier
gpio_pin_t pin = k_rx_pe_5; // 0x0E05
rx_pin_number_t pin_num = rx_pin_from_pin(pin); // Returns 5
```

Usage Example (Validation):

```
// Validate user-provided pin number
bool is_valid_pin(uint8_t pin)
{
    return (pin >= k_rx_pin_min) && (pin <= k_rx_pin_max);
}

// Or use in assertion
RX_ASSERT((pin >= k_rx_pin_min) && (pin <= k_rx_pin_max),
          "Pin number out of range");
```

Usage Example (Bit Manipulation):

```
// Set pin 2 high in port data register
PORT_Type* port = portb();
port->PODR |= (1 « k_rx_pin_2); // Set bit 2

// Clear pin 5
port->PODR &= ~(1 « k_rx_pin_5); // Clear bit 5
```

Invariant

All pin numbers are in range [0, 7]
`k_rx_pin_min` == 0
`k_rx_pin_max` == 7
Pin numbering is contiguous (no gaps)

Note

This is a C23 typed enum with explicit underlying type `uint8_t`, ensuring 1-byte size and consistent ABI across compiler versions

Pin numbers correspond to bit positions in port registers (PDR, PODR, PIDR)

All 8 pins exist logically even if not physically bonded on smaller packages

Warning

Pin numbers are zero-indexed (0-7), not one-indexed (1-8)

Do not use raw numbers—use these constants for self-documenting code

See also

[rx_port_number_t](#) Port number constants

[rx_port_pin_t](#) Pre-computed port/pin combinations

[rx_pin_from_pin\(\)](#) Extract pin number from combined identifier

[hardware.h](#) HAL GPIO register manipulation functions

Since

Version 1.0.0

Enumerator

<code>k_rx_pin_0</code>	Pin 0 - LSB of port data register, bit position 0
<code>k_rx_pin_1</code>	Pin 1 - Bit position 1 in port data register
<code>k_rx_pin_2</code>	Pin 2 - Bit position 2 in port data register
<code>k_rx_pin_3</code>	Pin 3 - Bit position 3 in port data register
<code>k_rx_pin_4</code>	Pin 4 - Bit position 4 in port data register
<code>k_rx_pin_5</code>	Pin 5 - Bit position 5 in port data register
<code>k_rx_pin_6</code>	Pin 6 - Bit position 6 in port data register
<code>k_rx_pin_7</code>	Pin 7 - MSB of port data register, bit position 7
<code>k_rx_pin_min</code>	Minimum valid pin number - used for range validation in assertions
<code>k_rx_pin_max</code>	Maximum valid pin number - used for range validation in assertions

Definition at line 429 of file [rx_port_constants.h](#).

```

00429         : uint8_t {
00430     k_rx_pin_0 = 0, /**< Pin 0 - LSB of port data register, bit position 0 */
00431     k_rx_pin_1 = 1, /**< Pin 1 - Bit position 1 in port data register */
00432     k_rx_pin_2 = 2, /**< Pin 2 - Bit position 2 in port data register */
00433     k_rx_pin_3 = 3, /**< Pin 3 - Bit position 3 in port data register */
00434     k_rx_pin_4 = 4, /**< Pin 4 - Bit position 4 in port data register */
00435     k_rx_pin_5 = 5, /**< Pin 5 - Bit position 5 in port data register */
00436     k_rx_pin_6 = 6, /**< Pin 6 - Bit position 6 in port data register */
00437     k_rx_pin_7 = 7, /**< Pin 7 - MSB of port data register, bit position 7 */
00438
00439     k_rx_pin_min = 0, /**< Minimum valid pin number - used for range validation in assertions */
00440     k_rx_pin_max = 7, /**< Maximum valid pin number - used for range validation in assertions */
00441 } rx_pin_number_t;

```


7.29.2.10 rx_port_number_t

```
enum rx_port_number_t : uint8_t
```

RX72N Port Number Constants (Single-Byte Identifiers).

Defines all GPIO port numbers available on the Renesas RX72N microcontroller. Used by HAL layer for hardware register access and by application layer for constructing full port/pin identifiers.

7.29.2.11 Port Numbering Scheme

The RX72N uses non-contiguous port numbering that mixes decimal and hexadecimal naming:

- Decimal ports: 0-9 (values 0x00-0x09)
- Hexadecimal ports: A-G (values 0x0A-0x10)
- Port J: 0x12 (follows Port H at 0x11; Port I does not exist)

This numbering matches the Renesas RX72N hardware manual and the encoding scheme used throughout the firmware.

7.29.2.12 Encoding Integration

Port numbers form the upper byte of 16-bit gpio_pin_t values:

```
gpio_pin_t = (rx_port_number_t « k_port_shift) | rx_pin_number_t
             = (port « 8) | pin
```

Example: Port B (0x0B) + Pin 2 (0x02) = 0x0B02

7.29.2.13 Package Availability

Not all ports are physically available on all RX72N packages:

- 144-pin LFQFP (STAR hardware): Ports 0-9, A-F, J (partial availability)
- 176-pin LFBGA: Maximum port availability

Constants are defined for ALL theoretical ports for code compatibility across packages, even if not physically present on 144-pin LFQFP.

Port Availability Table (144-pin LFQFP):

Port	Value	Pins Available	Notes
0	0x00	P00-P03, P05, P07	6 pins (no P04, P06)

Port	Value	Pins Available	Notes
1	0x01	P12-P17	6 pins (no P10, P11)
2	0x02	P20-P27	Full 8-pin port
3	0x03	P30-P37	Full 8-pin port
4	0x04	P40-P47	Full 8-pin port
5	0x05	P50-P56	7 pins (no P57)
6	0x06	P60-P67	Full 8-pin port
7	0x07	P70-P77	Full 8-pin port
8	0x08	P80-P83, P86-P87	6 pins (no P84, P85)
9	0x09	P90-P93	4 pins (no P94-P97)
A	0x0A	PA0-PA7	Full 8-pin port
B	0x0B	PB0-PB7	Full 8-pin port
C	0x0C	PC0-PC7	Full 8-pin port
D	0x0D	PD0-PD7	Full 8-pin port
E	0x0E	PE0-PE7	Full 8-pin port
F	0x0F	PF5	1 pin only
G	0x10	None	Not available on 144-pin
J	0x13	PJ3, PJ5	2 pins only

Usage Example (HAL Layer):

```
// Map port number to hardware register pointer
PORT_Type* get_port_registers(rx_port_number_t port)
{
    switch (port) {
        case k_rx_port_0: return port0();
        case k_rx_port_1: return port1();
        case k_rx_port_b: return portb();
        case k_rx_port_e: return porte();
        default: return nullptr;
    }
}
```

Usage Example (Application Layer):

```
// Construct full port/pin identifier
gpio_pin_t status_led = (k_rx_port_b « k_port_shift) | k_rx_pin_2;
// Result: 0x0B02 (Port B, Pin 2)

// Or use pre-computed constant
gpio_pin_t status_led = k_rx_pb_2; // Equivalent to above
```

Usage Example (Port Extraction):

```
// Extract port number from combined pin identifier
gpio_pin_t pin = k_rx_pb_2; // 0x0B02
rx_port_number_t port = rx_port_from_pin(pin); // Returns k_rx_port_b (0x0B)
```

Invariant

Port values are unique and match Renesas documentation

Port numbers fit in uint8_t (0x00-0xFF)

All enum values are compile-time constants

Note

This is a C23 typed enum with explicit underlying type `uint8_t`, ensuring 1-byte size and consistent ABI across compiler versions

Port J (0x12) follows Port H (0x11); Port I does not exist in RX72N port map

Warning

Do not assume contiguous port numbering when iterating ports

Not all ports available on 144-pin package—check datasheet

See also

[rx_pin_number_t](#) Pin number constants (0-7)

[rx_port_pin_t](#) Pre-computed port/pin combinations

[rx_port_from_pin\(\)](#) Extract port number from combined identifier

[hardware.h](#) HAL GPIO register accessor functions

[docs/sections/03_hardware_pinout.tex](#) Complete RX72N pinout reference

Since

Version 1.0.0

Enumerator

<code>k_rx_port_0</code>	Port 0 - 6 pins on 144-pin (P00-P03, P05, P07)
<code>k_rx_port_1</code>	Port 1 - 6 pins on 144-pin (P12-P17)
<code>k_rx_port_2</code>	Port 2 - Full 8-pin port on 144-pin (P20-P27)
<code>k_rx_port_3</code>	Port 3 - Full 8-pin port on 144-pin (P30-P37)
<code>k_rx_port_4</code>	Port 4 - Full 8-pin port on 144-pin (P40-P47)
<code>k_rx_port_5</code>	Port 5 - 7 pins on 144-pin (P50-P56)
<code>k_rx_port_6</code>	Port 6 - Full 8-pin port on 144-pin (P60-P67)
<code>k_rx_port_7</code>	Port 7 - Full 8-pin port on 144-pin (P70-P77)
<code>k_rx_port_8</code>	Port 8 - 6 pins on 144-pin (P80-P83, P86-P87)
<code>k_rx_port_9</code>	Port 9 - 4 pins on 144-pin (P90-P93)
<code>k_rx_port_a</code>	Port A - Full 8-pin port available on 144-pin
<code>k_rx_port_b</code>	Port B - Full 8-pin port available on 144-pin
<code>k_rx_port_c</code>	Port C - Full 8-pin port available on 144-pin
<code>k_rx_port_d</code>	Port D - Full 8-pin port available on 144-pin
<code>k_rx_port_e</code>	Port E - Full 8-pin port available on 144-pin
<code>k_rx_port_f</code>	Port F - Partially available on 144-pin (PF5)
<code>k_rx_port_g</code>	Port G - Not available on 144-pin package
<code>k_rx_port_j</code>	Port J - Partial availability on 144-pin (PJ3, PJ5) - follows Port H (0x11); Port I does not exist

Definition at line 300 of file [rx_port_constants.h](#).

```

00300         : uint8_t {
00301     /* Decimal Ports (0-9) */
00302     k_rx_port_0 = 0x00, /**< Port 0 - 6 pins on 144-pin (P00-P03, P05, P07) */
00303     k_rx_port_1 = 0x01, /**< Port 1 - 6 pins on 144-pin (P12-P17) */
00304     k_rx_port_2 = 0x02, /**< Port 2 - Full 8-pin port on 144-pin (P20-P27) */
00305     k_rx_port_3 = 0x03, /**< Port 3 - Full 8-pin port on 144-pin (P30-P37) */
00306     k_rx_port_4 = 0x04, /**< Port 4 - Full 8-pin port on 144-pin (P40-P47) */
00307     k_rx_port_5 = 0x05, /**< Port 5 - 7 pins on 144-pin (P50-P56) */
00308     k_rx_port_6 = 0x06, /**< Port 6 - Full 8-pin port on 144-pin (P60-P67) */
00309     k_rx_port_7 = 0x07, /**< Port 7 - Full 8-pin port on 144-pin (P70-P77) */
00310     k_rx_port_8 = 0x08, /**< Port 8 - 6 pins on 144-pin (P80-P83, P86-P87) */
00311     k_rx_port_9 = 0x09, /**< Port 9 - 4 pins on 144-pin (P90-P93) */
00312
00313     /* Hexadecimal Ports (A-G, J) */
00314     k_rx_port_a = 0x0A, /**< Port A - Full 8-pin port available on 144-pin */
00315     k_rx_port_b = 0x0B, /**< Port B - Full 8-pin port available on 144-pin */
00316     k_rx_port_c = 0x0C, /**< Port C - Full 8-pin port available on 144-pin */
00317     k_rx_port_d = 0x0D, /**< Port D - Full 8-pin port available on 144-pin */
00318     k_rx_port_e = 0x0E, /**< Port E - Full 8-pin port available on 144-pin */
00319     k_rx_port_f = 0x0F, /**< Port F - Partially available on 144-pin (PF5) */
00320     k_rx_port_g = 0x10, /**< Port G - Not available on 144-pin package */
00321     k_rx_port_j =
00322         0x12, /**< Port J - Partial availability on 144-pin (PJ3, PJ5) - follows Port H (0x11); Port I does not exist */
00323 } rx_port_number_t;

```

7.29.2.14 rx_port_pin_t

```
enum rx_port_pin_t : uint16_t
```

Pre-Computed Port/Pin Combinations for All RX72N GPIO Pins.

Provides ready-to-use 16-bit `gpio_pin_t` constants for all 144 theoretical port/pin combinations on the RX72N microcontroller (18 ports x 8 pins = 144). These pre-computed constants eliminate the need for runtime bit-shift operations and provide cleaner, more readable code.

7.29.2.15 Coverage

This enum defines ALL possible port/pin combinations:

- Ports: 0-9 (decimal) + A-G + J (hexadecimal) = 18 ports
- Pins per port: 0-7 = 8 pins
- Total combinations: 18 x 8 = 144 constants

7.29.2.16 Physical Availability

Important: Not all 144 combinations are physically bonded on smaller packages:

- 144-pin LQFP (STAR hardware): ~112 pins available
- 176-pin LFBGA: Maximum pins available

Constants are defined for ALL theoretical combinations to ensure:

- Code compatibility across different RX72N packages
- Compile-time validation (invalid pins detected at compile time)
- No gaps in the constant namespace

Pins marked "N/A on 144-pin" are not physically available on the STAR hardware but may be available on larger packages.

7.29.2.17 Benefits Over Manual Encoding

Using pre-computed constants:

```
gpio_pin_t pin = k_rx_pb_2; // Clean, readable
```

vs. Manual bit-shift construction:

```
gpio_pin_t pin = (k_rx_port_b « k_port_shift) | k_rx_pin_2; // Verbose
```

Advantages:

1. Cleaner syntax: One constant vs. shift-or expression
2. Zero runtime overhead: Pre-computed at compile time
3. Easier to grep: Search for "k_rx_pb_2" finds all uses
4. Consistent naming: Matches Renesas datasheet conventions (PA2, PB7, etc.)
5. Compile-time validated: Static assertions verify correctness

7.29.2.18 Naming Convention

Constants follow the pattern: k_rx_p<port>_<pin>

Examples:

- k_rx_p0_5 -> Port 0, Pin 5 -> P05
- k_rx_pb_2 -> Port B, Pin 2 -> PB2
- k_rx_pe_5 -> Port E, Pin 5 -> PE5
- k_rx_pj_3 -> Port J, Pin 3 -> PJ3

This naming matches the Renesas RX72N Hardware Manual conventions.

7.29.2.19 Encoding Scheme

Each constant uses the standard port/pin encoding:

```
k_rx_p<port>_<pin> = (k_rx_port_<port> « k_port_shift) | k_rx_pin_<pin>
                    = (port « 8) | pin
```

Example: PB2 (Port B, Pin 2)

```
k_rx_pb_2 = (k_rx_port_b « 8) | k_rx_pin_2
            = (0x0B « 8) | 0x02
            = 0x0B00 | 0x0002
            = 0x0B02
```

Usage Example (Basic):

```
// Use pre-computed constant for clean code
gpio_pin_t status_led = k_rx_pb_2; // Port B, Pin 2
gpio_pin_t spi_cs = k_rx_pe_5;    // Port E, Pin 5

// Pass to HAL functions
gpio_set_pin_high(status_led);
gpio_configure_output(spi_cs);
```

Usage Example (Application Pin Definitions):

```
// Define application-specific pin aliases using pre-computed constants
typedef enum : uint16_t {
    k_gpio_led_status = k_rx_pb_2,    // Status LED on PB2
    k_gpio_led_error  = k_rx_pb_3,    // Error LED on PB3
    k_gpio_spi_cs     = k_rx_pe_5,    // SPI chip select on PE5
    k_gpio_uart_tx    = k_rx_p3_0,    // UART TX on P30
    k_gpio_uart_rx    = k_rx_p3_1,    // UART RX on P31
} application_pins_t;
```

Usage Example (Port/Pin Extraction):

```
// Extract port and pin numbers from pre-computed constant
gpio_pin_t pin = k_rx_pb_2; // 0x0B02

rx_port_number_t port = rx_port_from_pin(pin); // Returns k_rx_port_b (0x0B)
rx_pin_number_t pin_num = rx_pin_from_pin(pin); // Returns 2
```

Usage Example (Compile-Time Validation):

```
// Static assertion verifies pre-computed constant is correct
static_assert(k_rx_pb_2 == 0x0B02, "PB2 must be 0x0B02");
static_assert(k_rx_pe_5 == 0x0E05, "PE5 must be 0x0E05");
```

Pin Availability on 144-pin LFQFP Package:

Port	Available Pins	Package Pins	Example Constants
0	P00-P03, P05, P07	8,7,6,4,2,144	k_rx_p0_0, k_rx_p0_7
1	P12-P17	45-38	k_rx_p1_2, k_rx_p1_7
2	P20-P27	37-30	k_rx_p2_0, k_rx_p2_7
3	P30-P37	29-20	k_rx_p3_0, k_rx_p3_7
4	P40-P47	141-133	k_rx_p4_0, k_rx_p4_7
5	P50-P56	56-50	k_rx_p5_0, k_rx_p5_6
6	P60-P67	117-98	k_rx_p6_0, k_rx_p6_7
7	P70-P77	104-68	k_rx_p7_0, k_rx_p7_7
8	P80-P83, P86-P87	65-58,41,39	k_rx_p8_0, k_rx_p8_7
9	P90-P93	131-127	k_rx_p9_0, k_rx_p9_3
A	PA0-PA7	97-88	k_rx_pa_0, k_rx_pa_7
B	PB0-PB7	87-78	k_rx_pb_0, k_rx_pb_7
C	PC0-PC7	75-60	k_rx_pc_0, k_rx_pc_7
D	PD0-PD7	126-119	k_rx_pd_0, k_rx_pd_7
E	PE0-PE7	111-101	k_rx_pe_0, k_rx_pe_7
F	PF5	9	k_rx_pf_5
J	PJ3, PJ5	13, 11	k_rx_pj_3, k_rx_pj_5

Invariant

Each constant equals $(\text{port} \ll 8) \mid \text{pin}$
 Port portion (upper byte) is valid [rx_port_number_t](#)
 Pin portion (lower byte) is valid [rx_pin_number_t](#) (0-7)
 All 144 combinations are defined (no gaps)

Note

This is a C23 typed enum with explicit underlying type `uint16_t`, ensuring 2-byte size and consistent ABI across compiler versions
 Constants are evaluated at compile time (zero runtime overhead)
 Naming follows Renesas conventions: P{port}{pin} (e.g., PB2, PE5)
 Package availability is documented in inline comments (search "N/A on 144-pin")

Warning

Using undefined pins (N/A on your package) may cause undefined behavior
 Always verify pin availability in RX72N datasheet for your package
 Do not assume all defined constants are physically available

See also

[rx_port_number_t](#) Port number constants used in upper byte
[rx_pin_number_t](#) Pin number constants used in lower byte
[port_encoding_t](#) Encoding scheme constants ([k_port_shift](#), [k_port_mask](#))
[rx_port_from_pin\(\)](#) Extract port number from constant
[rx_pin_from_pin\(\)](#) Extract pin number from constant
[hardware_pinout.h](#) Application-level pin definitions
[docs/sections/03_hardware_pinout.tex](#) Complete pinout reference for STAR hardware

Since

Version 1.0.0

Enumerator

k_rx_p0_0	P00 (pin 8)
k_rx_p0_1	P01 (pin 7)
k_rx_p0_2	P02 (pin 6)
k_rx_p0_3	P03 (pin 4)
k_rx_p0_4	P04 - N/A on 144-pin
k_rx_p0_5	P05 (pin 2)
k_rx_p0_6	P06 - N/A on 144-pin
k_rx_p0_7	P07 (pin 144)
k_rx_p1_0	P10 - N/A on 144-pin

k_rx_p1_1	P11 - N/A on 144-pin
k_rx_p1_2	P12 (pin 45)
k_rx_p1_3	P13 (pin 44)
k_rx_p1_4	P14 (pin 43)
k_rx_p1_5	P15 (pin 42)
k_rx_p1_6	P16 (pin 40)
k_rx_p1_7	P17 (pin 38)
k_rx_p2_0	P20 (pin 37)
k_rx_p2_1	P21 (pin 36)
k_rx_p2_2	P22 (pin 35)
k_rx_p2_3	P23 (pin 34)
k_rx_p2_4	P24 (pin 33)
k_rx_p2_5	P25 (pin 32)
k_rx_p2_6	P26 (pin 31)
k_rx_p2_7	P27 (pin 30)
k_rx_p3_0	P30 (pin 29)
k_rx_p3_1	P31 (pin 28)
k_rx_p3_2	P32 (pin 27)
k_rx_p3_3	P33 (pin 26)
k_rx_p3_4	P34 (pin 25)
k_rx_p3_5	P35 (pin 24)
k_rx_p3_6	P36 (pin 22)
k_rx_p3_7	P37 (pin 20)
k_rx_p4_0	P40 (pin 141)
k_rx_p4_1	P41 (pin 139)
k_rx_p4_2	P42 (pin 138)
k_rx_p4_3	P43 (pin 137)
k_rx_p4_4	P44 (pin 136)
k_rx_p4_5	P45 (pin 135)
k_rx_p4_6	P46 (pin 134)
k_rx_p4_7	P47 (pin 133)
k_rx_p5_0	P50 (pin 56)
k_rx_p5_1	P51 (pin 55)
k_rx_p5_2	P52 (pin 54)
k_rx_p5_3	P53 (pin 53)
k_rx_p5_4	P54 (pin 52)
k_rx_p5_5	P55 (pin 51)
k_rx_p5_6	P56 (pin 50)
k_rx_p5_7	P57 - N/A on 144-pin
k_rx_p6_0	P60 (pin 117)
k_rx_p6_1	P61 (pin 115)
k_rx_p6_2	P62 (pin 114)

k_rx_p6_3	P63 (pin 113)
k_rx_p6_4	P64 (pin 112)
k_rx_p6_5	P65 (pin 100)
k_rx_p6_6	P66 (pin 99)
k_rx_p6_7	P67 (pin 98)
k_rx_p7_0	P70 (pin 104)
k_rx_p7_1	P71 (pin 86)
k_rx_p7_2	P72 (pin 85)
k_rx_p7_3	P73 (pin 77)
k_rx_p7_4	P74 (pin 72)
k_rx_p7_5	P75 (pin 71)
k_rx_p7_6	P76 (pin 69)
k_rx_p7_7	P77 (pin 68)
k_rx_p8_0	P80 (pin 65)
k_rx_p8_1	P81 (pin 64)
k_rx_p8_2	P82 (pin 63)
k_rx_p8_3	P83 (pin 58)
k_rx_p8_4	P84 - N/A on 144-pin
k_rx_p8_5	P85 - N/A on 144-pin
k_rx_p8_6	P86 (pin 41)
k_rx_p8_7	P87 (pin 39)
k_rx_p9_0	P90 (pin 131)
k_rx_p9_1	P91 (pin 129)
k_rx_p9_2	P92 (pin 128)
k_rx_p9_3	P93 (pin 127)
k_rx_p9_4	P94 - N/A on 144-pin
k_rx_p9_5	P95 - N/A on 144-pin
k_rx_p9_6	P96 - N/A on 144-pin
k_rx_p9_7	P97 - N/A on 144-pin
k_rx_pa_0	PA0 (pin 97)
k_rx_pa_1	PA1 (pin 96)
k_rx_pa_2	PA2 (pin 95)
k_rx_pa_3	PA3 (pin 94)
k_rx_pa_4	PA4 (pin 92)
k_rx_pa_5	PA5 (pin 90)
k_rx_pa_6	PA6 (pin 89)
k_rx_pa_7	PA7 (pin 88)
k_rx_pb_0	PB0 (pin 87)
k_rx_pb_1	PB1 (pin 84)
k_rx_pb_2	PB2 (pin 83)
k_rx_pb_3	PB3 (pin 82)
k_rx_pb_4	PB4 (pin 81)

k_rx_pb_5	PB5 (pin 80)
k_rx_pb_6	PB6 (pin 79)
k_rx_pb_7	PB7 (pin 78)
k_rx_pc_0	PC0 (pin 75)
k_rx_pc_1	PC1 (pin 73)
k_rx_pc_2	PC2 (pin 70)
k_rx_pc_3	PC3 (pin 67)
k_rx_pc_4	PC4 (pin 66)
k_rx_pc_5	PC5 (pin 62)
k_rx_pc_6	PC6 (pin 61)
k_rx_pc_7	PC7 (pin 60)
k_rx_pd_0	PD0 (pin 126)
k_rx_pd_1	PD1 (pin 125)
k_rx_pd_2	PD2 (pin 124)
k_rx_pd_3	PD3 (pin 123)
k_rx_pd_4	PD4 (pin 122)
k_rx_pd_5	PD5 (pin 121)
k_rx_pd_6	PD6 (pin 120)
k_rx_pd_7	PD7 (pin 119)
k_rx_pe_0	PE0 (pin 111)
k_rx_pe_1	PE1 (pin 110)
k_rx_pe_2	PE2 (pin 109)
k_rx_pe_3	PE3 (pin 108)
k_rx_pe_4	PE4 (pin 107)
k_rx_pe_5	PE5 (pin 106)
k_rx_pe_6	PE6 (pin 102)
k_rx_pe_7	PE7 (pin 101)
k_rx_pf_0	PF0 - N/A on 144-pin
k_rx_pf_1	PF1 - N/A on 144-pin
k_rx_pf_2	PF2 - N/A on 144-pin
k_rx_pf_3	PF3 - N/A on 144-pin
k_rx_pf_4	PF4 - N/A on 144-pin
k_rx_pf_5	PF5 (pin 9)
k_rx_pf_6	PF6 - N/A on 144-pin
k_rx_pf_7	PF7 - N/A on 144-pin
k_rx_pg_0	PG0 - N/A on 144-pin
k_rx_pg_1	PG1 - N/A on 144-pin
k_rx_pg_2	PG2 - N/A on 144-pin
k_rx_pg_3	PG3 - N/A on 144-pin
k_rx_pg_4	PG4 - N/A on 144-pin
k_rx_pg_5	PG5 - N/A on 144-pin
k_rx_pg_6	PG6 - N/A on 144-pin

k_rx_pg_7	PG7 - N/A on 144-pin
k_rx_pj_0	PJ0 - N/A on 144-pin
k_rx_pj_1	PJ1 - N/A on 144-pin
k_rx_pj_2	PJ2 - N/A on 144-pin
k_rx_pj_3	PJ3 (pin 13)
k_rx_pj_4	PJ4 - N/A on 144-pin
k_rx_pj_5	PJ5 (pin 11)
k_rx_pj_6	PJ6 - N/A on 144-pin
k_rx_pj_7	PJ7 - N/A on 144-pin

Definition at line 744 of file rx_port_constants.h.

```

00744         : uint16_t {
00745     /* Port 0 (0x00) - Pins 0-7 */
00746     k_rx_p0_0 = (k_rx_port_0 « k_port_shift) | k_rx_pin_0, /**< P00 (pin 8) */
00747     k_rx_p0_1 = (k_rx_port_0 « k_port_shift) | k_rx_pin_1, /**< P01 (pin 7) */
00748     k_rx_p0_2 = (k_rx_port_0 « k_port_shift) | k_rx_pin_2, /**< P02 (pin 6) */
00749     k_rx_p0_3 = (k_rx_port_0 « k_port_shift) | k_rx_pin_3, /**< P03 (pin 4) */
00750     k_rx_p0_4 = (k_rx_port_0 « k_port_shift) | k_rx_pin_4, /**< P04 - N/A on 144-pin */
00751     k_rx_p0_5 = (k_rx_port_0 « k_port_shift) | k_rx_pin_5, /**< P05 (pin 2) */
00752     k_rx_p0_6 = (k_rx_port_0 « k_port_shift) | k_rx_pin_6, /**< P06 - N/A on 144-pin */
00753     k_rx_p0_7 = (k_rx_port_0 « k_port_shift) | k_rx_pin_7, /**< P07 (pin 144) */
00754
00755     /* Port 1 (0x01) - Pins 0-7 */
00756     k_rx_p1_0 = (k_rx_port_1 « k_port_shift) | k_rx_pin_0, /**< P10 - N/A on 144-pin */
00757     k_rx_p1_1 = (k_rx_port_1 « k_port_shift) | k_rx_pin_1, /**< P11 - N/A on 144-pin */
00758     k_rx_p1_2 = (k_rx_port_1 « k_port_shift) | k_rx_pin_2, /**< P12 (pin 45) */
00759     k_rx_p1_3 = (k_rx_port_1 « k_port_shift) | k_rx_pin_3, /**< P13 (pin 44) */
00760     k_rx_p1_4 = (k_rx_port_1 « k_port_shift) | k_rx_pin_4, /**< P14 (pin 43) */
00761     k_rx_p1_5 = (k_rx_port_1 « k_port_shift) | k_rx_pin_5, /**< P15 (pin 42) */
00762     k_rx_p1_6 = (k_rx_port_1 « k_port_shift) | k_rx_pin_6, /**< P16 (pin 40) */
00763     k_rx_p1_7 = (k_rx_port_1 « k_port_shift) | k_rx_pin_7, /**< P17 (pin 38) */
00764
00765     /* Port 2 (0x02) - Pins 0-7 */
00766     k_rx_p2_0 = (k_rx_port_2 « k_port_shift) | k_rx_pin_0, /**< P20 (pin 37) */
00767     k_rx_p2_1 = (k_rx_port_2 « k_port_shift) | k_rx_pin_1, /**< P21 (pin 36) */
00768     k_rx_p2_2 = (k_rx_port_2 « k_port_shift) | k_rx_pin_2, /**< P22 (pin 35) */
00769     k_rx_p2_3 = (k_rx_port_2 « k_port_shift) | k_rx_pin_3, /**< P23 (pin 34) */
00770     k_rx_p2_4 = (k_rx_port_2 « k_port_shift) | k_rx_pin_4, /**< P24 (pin 33) */
00771     k_rx_p2_5 = (k_rx_port_2 « k_port_shift) | k_rx_pin_5, /**< P25 (pin 32) */
00772     k_rx_p2_6 = (k_rx_port_2 « k_port_shift) | k_rx_pin_6, /**< P26 (pin 31) */
00773     k_rx_p2_7 = (k_rx_port_2 « k_port_shift) | k_rx_pin_7, /**< P27 (pin 30) */
00774
00775     /* Port 3 (0x03) - Pins 0-7 */
00776     k_rx_p3_0 = (k_rx_port_3 « k_port_shift) | k_rx_pin_0, /**< P30 (pin 29) */
00777     k_rx_p3_1 = (k_rx_port_3 « k_port_shift) | k_rx_pin_1, /**< P31 (pin 28) */
00778     k_rx_p3_2 = (k_rx_port_3 « k_port_shift) | k_rx_pin_2, /**< P32 (pin 27) */
00779     k_rx_p3_3 = (k_rx_port_3 « k_port_shift) | k_rx_pin_3, /**< P33 (pin 26) */
00780     k_rx_p3_4 = (k_rx_port_3 « k_port_shift) | k_rx_pin_4, /**< P34 (pin 25) */
00781     k_rx_p3_5 = (k_rx_port_3 « k_port_shift) | k_rx_pin_5, /**< P35 (pin 24) */
00782     k_rx_p3_6 = (k_rx_port_3 « k_port_shift) | k_rx_pin_6, /**< P36 (pin 22) */
00783     k_rx_p3_7 = (k_rx_port_3 « k_port_shift) | k_rx_pin_7, /**< P37 (pin 20) */
00784
00785     /* Port 4 (0x04) - Pins 0-7 */
00786     k_rx_p4_0 = (k_rx_port_4 « k_port_shift) | k_rx_pin_0, /**< P40 (pin 141) */
00787     k_rx_p4_1 = (k_rx_port_4 « k_port_shift) | k_rx_pin_1, /**< P41 (pin 139) */
00788     k_rx_p4_2 = (k_rx_port_4 « k_port_shift) | k_rx_pin_2, /**< P42 (pin 138) */
00789     k_rx_p4_3 = (k_rx_port_4 « k_port_shift) | k_rx_pin_3, /**< P43 (pin 137) */
00790     k_rx_p4_4 = (k_rx_port_4 « k_port_shift) | k_rx_pin_4, /**< P44 (pin 136) */
00791     k_rx_p4_5 = (k_rx_port_4 « k_port_shift) | k_rx_pin_5, /**< P45 (pin 135) */
00792     k_rx_p4_6 = (k_rx_port_4 « k_port_shift) | k_rx_pin_6, /**< P46 (pin 134) */
00793     k_rx_p4_7 = (k_rx_port_4 « k_port_shift) | k_rx_pin_7, /**< P47 (pin 133) */
00794
00795     /* Port 5 (0x05) - Pins 0-7 */
00796     k_rx_p5_0 = (k_rx_port_5 « k_port_shift) | k_rx_pin_0, /**< P50 (pin 56) */
00797     k_rx_p5_1 = (k_rx_port_5 « k_port_shift) | k_rx_pin_1, /**< P51 (pin 55) */
00798     k_rx_p5_2 = (k_rx_port_5 « k_port_shift) | k_rx_pin_2, /**< P52 (pin 54) */
00799     k_rx_p5_3 = (k_rx_port_5 « k_port_shift) | k_rx_pin_3, /**< P53 (pin 53) */
00800     k_rx_p5_4 = (k_rx_port_5 « k_port_shift) | k_rx_pin_4, /**< P54 (pin 52) */
00801     k_rx_p5_5 = (k_rx_port_5 « k_port_shift) | k_rx_pin_5, /**< P55 (pin 51) */
00802     k_rx_p5_6 = (k_rx_port_5 « k_port_shift) | k_rx_pin_6, /**< P56 (pin 50) */
00803     k_rx_p5_7 = (k_rx_port_5 « k_port_shift) | k_rx_pin_7, /**< P57 - N/A on 144-pin */
00804
00805     /* Port 6 (0x06) - Pins 0-7 */
00806     k_rx_p6_0 = (k_rx_port_6 « k_port_shift) | k_rx_pin_0, /**< P60 (pin 117) */
00807     k_rx_p6_1 = (k_rx_port_6 « k_port_shift) | k_rx_pin_1, /**< P61 (pin 115) */

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00808 k_rx_p6_2 = (k_rx_port_6 « k_port_shift) | k_rx_pin_2, /**< P62 (pin 114) */
00809 k_rx_p6_3 = (k_rx_port_6 « k_port_shift) | k_rx_pin_3, /**< P63 (pin 113) */
00810 k_rx_p6_4 = (k_rx_port_6 « k_port_shift) | k_rx_pin_4, /**< P64 (pin 112) */
00811 k_rx_p6_5 = (k_rx_port_6 « k_port_shift) | k_rx_pin_5, /**< P65 (pin 100) */
00812 k_rx_p6_6 = (k_rx_port_6 « k_port_shift) | k_rx_pin_6, /**< P66 (pin 99) */
00813 k_rx_p6_7 = (k_rx_port_6 « k_port_shift) | k_rx_pin_7, /**< P67 (pin 98) */
00814
00815 /* Port 7 (0x07) - Pins 0-7 */
00816 k_rx_p7_0 = (k_rx_port_7 « k_port_shift) | k_rx_pin_0, /**< P70 (pin 104) */
00817 k_rx_p7_1 = (k_rx_port_7 « k_port_shift) | k_rx_pin_1, /**< P71 (pin 86) */
00818 k_rx_p7_2 = (k_rx_port_7 « k_port_shift) | k_rx_pin_2, /**< P72 (pin 85) */
00819 k_rx_p7_3 = (k_rx_port_7 « k_port_shift) | k_rx_pin_3, /**< P73 (pin 77) */
00820 k_rx_p7_4 = (k_rx_port_7 « k_port_shift) | k_rx_pin_4, /**< P74 (pin 72) */
00821 k_rx_p7_5 = (k_rx_port_7 « k_port_shift) | k_rx_pin_5, /**< P75 (pin 71) */
00822 k_rx_p7_6 = (k_rx_port_7 « k_port_shift) | k_rx_pin_6, /**< P76 (pin 69) */
00823 k_rx_p7_7 = (k_rx_port_7 « k_port_shift) | k_rx_pin_7, /**< P77 (pin 68) */
00824
00825 /* Port 8 (0x08) - Pins 0-7 */
00826 k_rx_p8_0 = (k_rx_port_8 « k_port_shift) | k_rx_pin_0, /**< P80 (pin 65) */
00827 k_rx_p8_1 = (k_rx_port_8 « k_port_shift) | k_rx_pin_1, /**< P81 (pin 64) */
00828 k_rx_p8_2 = (k_rx_port_8 « k_port_shift) | k_rx_pin_2, /**< P82 (pin 63) */
00829 k_rx_p8_3 = (k_rx_port_8 « k_port_shift) | k_rx_pin_3, /**< P83 (pin 58) */
00830 k_rx_p8_4 = (k_rx_port_8 « k_port_shift) | k_rx_pin_4, /**< P84 - N/A on 144-pin */
00831 k_rx_p8_5 = (k_rx_port_8 « k_port_shift) | k_rx_pin_5, /**< P85 - N/A on 144-pin */
00832 k_rx_p8_6 = (k_rx_port_8 « k_port_shift) | k_rx_pin_6, /**< P86 (pin 41) */
00833 k_rx_p8_7 = (k_rx_port_8 « k_port_shift) | k_rx_pin_7, /**< P87 (pin 39) */
00834
00835 /* Port 9 (0x09) - Pins 0-7 */
00836 k_rx_p9_0 = (k_rx_port_9 « k_port_shift) | k_rx_pin_0, /**< P90 (pin 131) */
00837 k_rx_p9_1 = (k_rx_port_9 « k_port_shift) | k_rx_pin_1, /**< P91 (pin 129) */
00838 k_rx_p9_2 = (k_rx_port_9 « k_port_shift) | k_rx_pin_2, /**< P92 (pin 128) */
00839 k_rx_p9_3 = (k_rx_port_9 « k_port_shift) | k_rx_pin_3, /**< P93 (pin 127) */
00840 k_rx_p9_4 = (k_rx_port_9 « k_port_shift) | k_rx_pin_4, /**< P94 - N/A on 144-pin */
00841 k_rx_p9_5 = (k_rx_port_9 « k_port_shift) | k_rx_pin_5, /**< P95 - N/A on 144-pin */
00842 k_rx_p9_6 = (k_rx_port_9 « k_port_shift) | k_rx_pin_6, /**< P96 - N/A on 144-pin */
00843 k_rx_p9_7 = (k_rx_port_9 « k_port_shift) | k_rx_pin_7, /**< P97 - N/A on 144-pin */
00844
00845 /* Port A (0x0A) - Pins 0-7 */
00846 k_rx_pa_0 = (k_rx_port_a « k_port_shift) | k_rx_pin_0, /**< PA0 (pin 97) */
00847 k_rx_pa_1 = (k_rx_port_a « k_port_shift) | k_rx_pin_1, /**< PA1 (pin 96) */
00848 k_rx_pa_2 = (k_rx_port_a « k_port_shift) | k_rx_pin_2, /**< PA2 (pin 95) */
00849 k_rx_pa_3 = (k_rx_port_a « k_port_shift) | k_rx_pin_3, /**< PA3 (pin 94) */
00850 k_rx_pa_4 = (k_rx_port_a « k_port_shift) | k_rx_pin_4, /**< PA4 (pin 92) */
00851 k_rx_pa_5 = (k_rx_port_a « k_port_shift) | k_rx_pin_5, /**< PA5 (pin 90) */
00852 k_rx_pa_6 = (k_rx_port_a « k_port_shift) | k_rx_pin_6, /**< PA6 (pin 89) */
00853 k_rx_pa_7 = (k_rx_port_a « k_port_shift) | k_rx_pin_7, /**< PA7 (pin 88) */
00854
00855 /* Port B (0x0B) - Pins 0-7 */
00856 k_rx_pb_0 = (k_rx_port_b « k_port_shift) | k_rx_pin_0, /**< PB0 (pin 87) */
00857 k_rx_pb_1 = (k_rx_port_b « k_port_shift) | k_rx_pin_1, /**< PB1 (pin 84) */
00858 k_rx_pb_2 = (k_rx_port_b « k_port_shift) | k_rx_pin_2, /**< PB2 (pin 83) */
00859 k_rx_pb_3 = (k_rx_port_b « k_port_shift) | k_rx_pin_3, /**< PB3 (pin 82) */
00860 k_rx_pb_4 = (k_rx_port_b « k_port_shift) | k_rx_pin_4, /**< PB4 (pin 81) */
00861 k_rx_pb_5 = (k_rx_port_b « k_port_shift) | k_rx_pin_5, /**< PB5 (pin 80) */
00862 k_rx_pb_6 = (k_rx_port_b « k_port_shift) | k_rx_pin_6, /**< PB6 (pin 79) */
00863 k_rx_pb_7 = (k_rx_port_b « k_port_shift) | k_rx_pin_7, /**< PB7 (pin 78) */
00864
00865 /* Port C (0x0C) - Pins 0-7 */
00866 k_rx_pc_0 = (k_rx_port_c « k_port_shift) | k_rx_pin_0, /**< PC0 (pin 75) */
00867 k_rx_pc_1 = (k_rx_port_c « k_port_shift) | k_rx_pin_1, /**< PC1 (pin 73) */
00868 k_rx_pc_2 = (k_rx_port_c « k_port_shift) | k_rx_pin_2, /**< PC2 (pin 70) */
00869 k_rx_pc_3 = (k_rx_port_c « k_port_shift) | k_rx_pin_3, /**< PC3 (pin 67) */
00870 k_rx_pc_4 = (k_rx_port_c « k_port_shift) | k_rx_pin_4, /**< PC4 (pin 66) */
00871 k_rx_pc_5 = (k_rx_port_c « k_port_shift) | k_rx_pin_5, /**< PC5 (pin 62) */
00872 k_rx_pc_6 = (k_rx_port_c « k_port_shift) | k_rx_pin_6, /**< PC6 (pin 61) */
00873 k_rx_pc_7 = (k_rx_port_c « k_port_shift) | k_rx_pin_7, /**< PC7 (pin 60) */
00874
00875 /* Port D (0x0D) - Pins 0-7 */
00876 k_rx_pd_0 = (k_rx_port_d « k_port_shift) | k_rx_pin_0, /**< PD0 (pin 126) */
00877 k_rx_pd_1 = (k_rx_port_d « k_port_shift) | k_rx_pin_1, /**< PD1 (pin 125) */
00878 k_rx_pd_2 = (k_rx_port_d « k_port_shift) | k_rx_pin_2, /**< PD2 (pin 124) */
00879 k_rx_pd_3 = (k_rx_port_d « k_port_shift) | k_rx_pin_3, /**< PD3 (pin 123) */
00880 k_rx_pd_4 = (k_rx_port_d « k_port_shift) | k_rx_pin_4, /**< PD4 (pin 122) */
00881 k_rx_pd_5 = (k_rx_port_d « k_port_shift) | k_rx_pin_5, /**< PD5 (pin 121) */
00882 k_rx_pd_6 = (k_rx_port_d « k_port_shift) | k_rx_pin_6, /**< PD6 (pin 120) */
00883 k_rx_pd_7 = (k_rx_port_d « k_port_shift) | k_rx_pin_7, /**< PD7 (pin 119) */
00884
00885 /* Port E (0x0E) - Pins 0-7 */
00886 k_rx_pe_0 = (k_rx_port_e « k_port_shift) | k_rx_pin_0, /**< PE0 (pin 111) */
00887 k_rx_pe_1 = (k_rx_port_e « k_port_shift) | k_rx_pin_1, /**< PE1 (pin 110) */
00888 k_rx_pe_2 = (k_rx_port_e « k_port_shift) | k_rx_pin_2, /**< PE2 (pin 109) */
00889 k_rx_pe_3 = (k_rx_port_e « k_port_shift) | k_rx_pin_3, /**< PE3 (pin 108) */
00890 k_rx_pe_4 = (k_rx_port_e « k_port_shift) | k_rx_pin_4, /**< PE4 (pin 107) */
00891 k_rx_pe_5 = (k_rx_port_e « k_port_shift) | k_rx_pin_5, /**< PE5 (pin 106) */
00892 k_rx_pe_6 = (k_rx_port_e « k_port_shift) | k_rx_pin_6, /**< PE6 (pin 102) */
00893 k_rx_pe_7 = (k_rx_port_e « k_port_shift) | k_rx_pin_7, /**< PE7 (pin 101) */
00894

```

```

00895  /* Port F (0x0F) - Pins 0-7 - N/A on 144-pin */
00896  k_rx_pf_0 = (k_rx_port_f « k_port_shift) | k_rx_pin_0, /**< PF0 - N/A on 144-pin */
00897  k_rx_pf_1 = (k_rx_port_f « k_port_shift) | k_rx_pin_1, /**< PF1 - N/A on 144-pin */
00898  k_rx_pf_2 = (k_rx_port_f « k_port_shift) | k_rx_pin_2, /**< PF2 - N/A on 144-pin */
00899  k_rx_pf_3 = (k_rx_port_f « k_port_shift) | k_rx_pin_3, /**< PF3 - N/A on 144-pin */
00900  k_rx_pf_4 = (k_rx_port_f « k_port_shift) | k_rx_pin_4, /**< PF4 - N/A on 144-pin */
00901  k_rx_pf_5 = (k_rx_port_f « k_port_shift) | k_rx_pin_5, /**< PF5 (pin 9) */
00902  k_rx_pf_6 = (k_rx_port_f « k_port_shift) | k_rx_pin_6, /**< PF6 - N/A on 144-pin */
00903  k_rx_pf_7 = (k_rx_port_f « k_port_shift) | k_rx_pin_7, /**< PF7 - N/A on 144-pin */
00904
00905  /* Port G (0x10) - Pins 0-7 - N/A on 144-pin */
00906  k_rx_pg_0 = (k_rx_port_g « k_port_shift) | k_rx_pin_0, /**< PG0 - N/A on 144-pin */
00907  k_rx_pg_1 = (k_rx_port_g « k_port_shift) | k_rx_pin_1, /**< PG1 - N/A on 144-pin */
00908  k_rx_pg_2 = (k_rx_port_g « k_port_shift) | k_rx_pin_2, /**< PG2 - N/A on 144-pin */
00909  k_rx_pg_3 = (k_rx_port_g « k_port_shift) | k_rx_pin_3, /**< PG3 - N/A on 144-pin */
00910  k_rx_pg_4 = (k_rx_port_g « k_port_shift) | k_rx_pin_4, /**< PG4 - N/A on 144-pin */
00911  k_rx_pg_5 = (k_rx_port_g « k_port_shift) | k_rx_pin_5, /**< PG5 - N/A on 144-pin */
00912  k_rx_pg_6 = (k_rx_port_g « k_port_shift) | k_rx_pin_6, /**< PG6 - N/A on 144-pin */
00913  k_rx_pg_7 = (k_rx_port_g « k_port_shift) | k_rx_pin_7, /**< PG7 - N/A on 144-pin */
00914
00915  /* Port J (0x12) - Pins 0-7 */
00916  k_rx_pj_0 = (k_rx_port_j « k_port_shift) | k_rx_pin_0, /**< PJ0 - N/A on 144-pin */
00917  k_rx_pj_1 = (k_rx_port_j « k_port_shift) | k_rx_pin_1, /**< PJ1 - N/A on 144-pin */
00918  k_rx_pj_2 = (k_rx_port_j « k_port_shift) | k_rx_pin_2, /**< PJ2 - N/A on 144-pin */
00919  k_rx_pj_3 = (k_rx_port_j « k_port_shift) | k_rx_pin_3, /**< PJ3 (pin 13) */
00920  k_rx_pj_4 = (k_rx_port_j « k_port_shift) | k_rx_pin_4, /**< PJ4 - N/A on 144-pin */
00921  k_rx_pj_5 = (k_rx_port_j « k_port_shift) | k_rx_pin_5, /**< PJ5 (pin 11) */
00922  k_rx_pj_6 = (k_rx_port_j « k_port_shift) | k_rx_pin_6, /**< PJ6 - N/A on 144-pin */
00923  k_rx_pj_7 = (k_rx_port_j « k_port_shift) | k_rx_pin_7, /**< PJ7 - N/A on 144-pin */
00924 } rx_port_pin_t;

```

7.29.2.20 rx_port_verify_t

```
enum rx_port_verify_t : uint16_t
```

Expected values used in static assertion verification.

Provides named constants for the right-hand sides of port/pin static assertions, eliminating magic numbers from assertion expressions.

Since

Version 1.0.0

Enumerator

k_verify_port_0	Expected value of k_rx_port_0 (0x00)
k_verify_port_1	Expected value of k_rx_port_1 (0x01)
k_verify_port_5	Expected value of k_rx_port_5 (0x05)
k_verify_port_a	Expected value of k_rx_port_a (0x0A)
k_verify_port_b	Expected value of k_rx_port_b (0x0B)
k_verify_port_c	Expected value of k_rx_port_c (0x0C)
k_verify_port_e	Expected value of k_rx_port_e (0x0E)
k_verify_port_j	Expected value of k_rx_port_j (0x12) - follows Port H; Port I does not exist
k_verify_pin_2	Expected value of k_rx_pin_2 (2)
k_verify_pin_7	Expected value of k_rx_pin_7 (7)
k_verify_port_shift	Expected value of k_port_shift (8)
k_verify_enc_pb_2	Expected encoding of Port B Pin 2
k_verify_enc_pe_5	Expected encoding of Port E Pin 5
k_verify_enc_p0_7	Expected encoding of Port 0 Pin 7

k_verify_p0_4	Expected value of k_rx_p0_4
k_verify_p2_5	Expected value of k_rx_p2_5
k_verify_pa_2	Expected value of k_rx_pa_2
k_verify_pb_2	Expected value of k_rx_pb_2
k_verify_pe_5	Expected value of k_rx_pe_5
k_verify_pj_3	Expected value of k_rx_pj_3

Definition at line 941 of file [rx_port_constants.h](#).

```

00941      : uint16_t {
00942  k_verify_port_0 = 0x0000U, /**< Expected value of k_rx_port_0 (0x00) */
00943  k_verify_port_1 = 0x0001U, /**< Expected value of k_rx_port_1 (0x01) */
00944  k_verify_port_5 = 0x0005U, /**< Expected value of k_rx_port_5 (0x05) */
00945  k_verify_port_a = 0x000AU, /**< Expected value of k_rx_port_a (0x0A) */
00946  k_verify_port_b = 0x000BU, /**< Expected value of k_rx_port_b (0x0B) */
00947  k_verify_port_c = 0x000CU, /**< Expected value of k_rx_port_c (0x0C) */
00948  k_verify_port_e = 0x000EU, /**< Expected value of k_rx_port_e (0x0E) */
00949  k_verify_port_j =
00950      0x0012U, /**< Expected value of k_rx_port_j (0x12) - follows Port H; Port I does not exist */
00951  k_verify_pin_2  = 0x0002U, /**< Expected value of k_rx_pin_2 (2) */
00952  k_verify_pin_7  = 0x0007U, /**< Expected value of k_rx_pin_7 (7) */
00953  k_verify_port_shift = 0x0008U, /**< Expected value of k_port_shift (8) */
00954  k_verify_enc_pb_2  = 0x0B02U, /**< Expected encoding of Port B Pin 2 */
00955  k_verify_enc_pe_5  = 0x0E05U, /**< Expected encoding of Port E Pin 5 */
00956  k_verify_enc_p0_7  = 0x0007U, /**< Expected encoding of Port 0 Pin 7 */
00957  k_verify_p0_4      = 0x0004U, /**< Expected value of k_rx_p0_4 */
00958  k_verify_p2_5      = 0x0205U, /**< Expected value of k_rx_p2_5 */
00959  k_verify_pa_2      = 0x0A02U, /**< Expected value of k_rx_pa_2 */
00960  k_verify_pb_2      = 0x0B02U, /**< Expected value of k_rx_pb_2 */
00961  k_verify_pe_5      = 0x0E05U, /**< Expected value of k_rx_pe_5 */
00962  k_verify_pj_3      = 0x1203U, /**< Expected value of k_rx_pj_3 */
00963 } rx_port_verify_t;

```

7.29.3 Function Documentation

7.29.3.1 rx_pin_from_pin()

```

uint8_t rx_pin_from_pin (
    rx_port_pin_t pin) [inline], [static]

```

Extract pin number from combined port/pin identifier.

Decodes the pin number (lower byte) from a 16-bit `gpio_pin_t` value using bitwise AND with `k_port_mask` (0xFF). This extracts bits 7-0 which contain the [rx_pin_number_t](#) value.

7.29.3.2 Algorithm Steps

1. Extract port: Right-shift by 8 bits to get port number (for validation)
2. Validate port: Assert that port is valid RX72N port number
3. Extract pin: Mask lower byte with 0xFF to get pin number
4. Validate result: Assert that extracted pin is in valid range [0-7]
5. Return: Return pin number as `uint8_t`

7.29.3.3 Bit-Level Operation

Given: `gpio_pin_t = 0xPPnn` (PP = port, nn = pin) Operation: `result = gpio_pin_t & 0xFF` Result: `result = 0x00nn`

Example: `k_rx_pe_5 = 0x0E05`

Input: `0x0E05 = 0000 1110 0000 0101`
Mask: `0x00FF = 0000 0000 1111 1111`
AND: `= 0000 0000 0000 0101`
Output: `0x0005 = 5`
Result: `5 (k_rx_pin_5)`

7.29.3.4 Performance Analysis

- Execution time: 2-3 CPU cycles (shift + mask, assertions compiled out in release)
- Memory usage: Zero (inline function, no stack frame)
- Code size: ~6 bytes machine code (shift + mask instructions)

Parameters

in	pin	
		Combined port/pin identifier (rx_port_pin_t or <code>gpio_pin_t</code>). Valid range: 0x0000-0x1307 (all valid port/pin combinations). Must have valid port in upper byte and valid pin (0-7) in lower byte.

Returns

Pin number extracted from lower byte ([rx_pin_number_t](#)). Valid range: 0-7 (one of `k_rx_pin_0..k_rx_pin_7`). Corresponds to bit position in port data registers (PDR, PODR, PIDR).

Precondition

pin upper byte (bits 15-8) must be valid port number (0x00-0x10 or 0x13)
pin lower byte (bits 7-0) must be valid pin number (0-7)
pin must be properly encoded using `(port << 8) | pin_num` scheme

Postcondition

Returned value is valid [rx_pin_number_t](#) (0-7)
Returned value is deterministic (same input always produces same output)
No side effects (function does not modify any state)

Invariant

`(rx_pin_from_pin((port << 8) | pin_num) == pin_num)` for all valid port/pin
Result `<= k_rx_pin_max (7)`
Result `>= k_rx_pin_min (0)`

Note

Thread-safe: Yes - pure function with no shared state
 Re-entrant: Yes - no static variables or side effects
 Performance: Optimal - compiles to single mask instruction (after validation)
 Memory: Zero overhead - inline function, no stack frame
 Assertions: Compiled out in release builds (NDEBUG defined)
 Bit Position: Returned value corresponds to bit position in port registers

Warning

Passing invalid port values (upper byte invalid) triggers assertion failure
 Result undefined if pin not constructed using proper encoding scheme

Attention

This function validates input but does NOT check physical pin availability
 Pin number 0-7 corresponds to LSB-MSB of 8-bit port registers

Example (Basic Usage):

```
// Extract pin number from pre-computed constant
gpio_pin_t pin = k_rx_pe_5; // 0x0E05
rx_pin_number_t pin_num = rx_pin_from_pin(pin);
// Result: 5 (k_rx_pin_5)
```

Example (Bit Manipulation):

```
// Use extracted pin number for bit manipulation
gpio_pin_t pin = k_rx_pb_2; // Port B, Pin 2
rx_pin_number_t pin_num = rx_pin_from_pin(pin); // Returns 2

PORT_Type* port = portb();
port->PODR |= (1 « pin_num); // Set bit 2 high
port->PODR &= ~(1 « pin_num); // Clear bit 2 low
```

Example (Complete Port/Pin Decomposition):

```
// Fully decompose gpio_pin_t into components
gpio_pin_t pin = k_rx_pc_7; // 0x0C07

rx_port_number_t port = rx_port_from_pin(pin); // 0x0C (Port C)
rx_pin_number_t pin_num = rx_pin_from_pin(pin); // 7 (Pin 7)

// Use in hardware access
PORT_Type* port_regs = get_port_registers(port);
bool pin_state = (port_regs->PIDR » pin_num) & 1;
```

Example (Round-Trip Verification):

```
// Verify encoding/decoding round-trip
rx_port_number_t original_port = k_rx_port_a;
rx_pin_number_t original_pin = k_rx_pin_3;

// Encode
gpio_pin_t encoded = (original_port « k_port_shift) | original_pin;

// Decode
rx_port_number_t decoded_port = rx_port_from_pin(encoded);
rx_pin_number_t decoded_pin = rx_pin_from_pin(encoded);

// Verify
assert(decoded_port == original_port); // Port A
assert(decoded_pin == original_pin); // Pin 3
```


Example (Array Indexing):

```
// Use pin number to index into pin configuration array
gpio_pin_t configured_pins[] = {
    k_rx_pb_0, k_rx_pb_2, k_rx_pb_5, k_rx_pb_7
};

for (size_t i = 0; i < 4; i++) {
    rx_pin_number_t pin_num = rx_pin_from_pin(configured_pins[i]);
    printf("Pin %d is configured\n", pin_num);
}
```

See also

[rx_port_from_pin\(\)](#) Extract port number from combined identifier
[rx_port_pin_t](#) Pre-computed port/pin combinations
[port_encoding_t](#) Encoding constants ([k_port_shift](#), [k_port_mask](#))
[rx_pin_number_t](#) Valid pin numbers that can be returned

Since

Version 1.0.0

Test [test_rx_pin_from_pin\(\)](#) Unit tests verify all 144 port/pin combinations

NASA Power of 10 Compliance:

- Rule 5: [OK] 2 preconditions + 3 postconditions (5 assertions total)
- Rule 4: [OK] 6 lines of code (excluding comments/braces)

Definition at line 1300 of file [rx_port_constants.h](#).

```
01301 {
01302     /* Pre-condition 1: port value fits encoding scheme (port in upper byte) */
01303     uint8_t port = (uint8_t)((pin) » k_port_shift);
01304     /* Note: port >= k_rx_port_0 is always true for uint8_t since k_rx_port_0 == 0 */
01305
01306     /* Pre-condition 2: port is valid (0-G or J, skipping H-I) */
01307     RX_ASSERT((port <= k_rx_port_g) || (port == k_rx_port_j),
01308         "Port number must be valid (k_rx_port_0..k_rx_port_g or k_rx_port_j)");
01309
01310     uint8_t result = (uint8_t)((pin)&k_port_mask);
01311
01312     /* Post-condition: result fits in rx_pin_number_t range (0x00-0x07) */
01313     /* Note: result >= k_rx_pin_min is always true for uint8_t since k_rx_pin_min == 0 */
01314     RX_ASSERT(result <= k_rx_pin_max, "Pin number must be in range k_rx_pin_min..k_rx_pin_max");
01315
01316     return result;
01317 }
```

References [k_port_mask](#), [k_port_shift](#), [k_rx_pin_max](#), [k_rx_port_g](#), [k_rx_port_j](#), and [RX_ASSERT](#).

7.29.3.5 rx_port_from_pin()

```
uint8_t rx_port_from_pin (
    rx\_port\_pin\_t pin) [inline], [static]
```

Extract port number from combined port/pin identifier.

Decodes the port number (upper byte) from a 16-bit [gpio_pin_t](#) value using right-shift by [k_port_shift](#) (8 bits). This extracts bits 15-8 which contain the [rx_pin_number_t](#) value.

7.29.3.6 Algorithm Steps

1. Validate pin portion: Assert that lower byte (pin) is in valid range [0-7]
2. Extract port: Right-shift by 8 bits to move port from upper to lower byte
3. Validate result: Assert that extracted port is valid RX72N port number
4. Return: Return port number as uint8_t

7.29.3.7 Bit-Level Operation

Given: gpio_pin_t = 0xPPnn (PP = port, nn = pin) Operation: result = gpio_pin_t >> 8 Result: result = 0x00PP

Example: k_rx_pb_2 = 0x0B02

Input: 0x0B02 = 0000 1011 0000 0010
 Shift: » 8
 Output: 0x000B = 0000 0000 0000 1011
 Result: 0x0B (k_rx_port_b)

7.29.3.8 Performance Analysis

- Execution time: 1-2 CPU cycles (inline right-shift, assertions compiled out in release)
- Memory usage: Zero (inline function, no stack frame)
- Code size: ~4 bytes machine code (single shift instruction)

Parameters

in	pin	Combined port/pin identifier (rx_port_pin_t or gpio_pin_t). Valid range: 0x0000-0x1307 (all valid port/pin combinations). Must have valid port in upper byte and valid pin (0-7) in lower byte.
----	-----	--

Returns

Port number extracted from upper byte ([rx_port_number_t](#)). Valid range: 0x00-0x10, or 0x12 (k_rx_port_j). Returned value is one of: k_rx_port_0..k_rx_port_9, k_rx_port_a..k_rx_port_g, k_rx_port_j.

Precondition

pin lower byte (bits 7-0) must be valid pin number (0-7)
 pin upper byte (bits 15-8) must be valid port number (0x00-0x10 or 0x13)
 pin must be properly encoded using (port << 8) | pin_num scheme

Postcondition

Returned value is valid [rx_port_number_t](#) (0x00-0x10 or 0x13)
 Returned value is deterministic (same input always produces same output)
 No side effects (function does not modify any state)

Invariant

(rx_port_from_pin((port << 8) | pin_num) == port) for all valid port/pin
 Result fits in uint8_t (always <= 0xFF)

Note

Thread-safe: Yes - pure function with no shared state
 Re-entrant: Yes - no static variables or side effects
 Performance: Optimal - compiles to single right-shift instruction
 Memory: Zero overhead - inline function, no stack frame
 Assertions: Compiled out in release builds (NDEBUG defined)

Warning

Passing invalid pin values (pin portion > 7) triggers assertion failure
 Result undefined if pin not constructed using proper encoding scheme

Attention

This function validates input but does NOT check physical pin availability

Example (Basic Usage):

```
// Extract port from pre-computed constant
gpio_pin_t pin = k_rx_pb_2; // 0x0B02
rx_port_number_t port = rx_port_from_pin(pin);
// Result: k_rx_port_b (0x0B)
```

Example (Manual Construction):

```
// Construct pin value, then extract port
gpio_pin_t pin = (k_rx_port_e « k_port_shift) | k_rx_pin_5; // 0x0E05
rx_port_number_t port = rx_port_from_pin(pin);
// Result: k_rx_port_e (0x0E)

// Verify round-trip
assert(port == k_rx_port_e);
```

Example (Switch Statement):

```
// Use extracted port in switch
gpio_pin_t pin = get_configured_pin();
rx_port_number_t port = rx_port_from_pin(pin);

switch (port) {
  case k_rx_port_b: return portb();
  case k_rx_port_e: return porte();
  default: return nullptr;
}
```

Example (Complete Port/Pin Decomposition):

```
// Fully decompose gpio_pin_t into components
gpio_pin_t pin = k_rx_pa_7; // 0x0A07

rx_port_number_t port = rx_port_from_pin(pin); // 0x0A (Port A)
rx_pin_number_t pin_num = rx_pin_from_pin(pin); // 0x07 (Pin 7)

printf("Pin 0x%04X = Port 0x%02X, Pin %d\n", pin, port, pin_num);
// Output: "Pin 0x0A07 = Port 0x0A, Pin 7"
```

See also

[rx_pin_from_pin\(\)](#) Extract pin number from combined identifier
[rx_port_pin_t](#) Pre-computed port/pin combinations
[port_encoding_t](#) Encoding constants ([k_port_shift](#), [k_port_mask](#))
[rx_port_number_t](#) Valid port numbers that can be returned

Since

Version 1.0.0

Test [test_rx_port_from_pin\(\)](#) Unit tests verify all 144 port/pin combinations

NASA Power of 10 Compliance:

- Rule 5: [OK] 2 preconditions + 3 postconditions (5 assertions total)
- Rule 4: [OK] 5 lines of code (excluding comments/braces)

Definition at line 1140 of file [rx_port_constants.h](#).

```
01141 {
01142  /* Pre-condition: pin portion fits encoding scheme (k_rx_pin_min == 0, so >= 0 is always
01143   * true for unsigned types; only the upper bound requires an explicit check) */
01144  RX_ASSERT((pin & k_port_mask) <= k_rx_pin_max, "Pin portion must be <= k_rx_pin_max");
01145
01146  uint8_t result = (uint8_t)((pin) >> k_port_shift);
01147
01148  /* Post-condition: result is a valid port (0-G or J, skipping H-I) */
01149  /* Note: result >= k_rx_port_0 is always true for uint8_t since k_rx_port_0 == 0 */
01150  RX_ASSERT((result <= k_rx_port_g) || (result == k_rx_port_j),
01151            "Port number must be valid (k_rx_port_0..k_rx_port_g or k_rx_port_j)");
01152
01153  return result;
01154 }
```

References [k_port_mask](#), [k_port_shift](#), [k_rx_pin_max](#), [k_rx_port_g](#), [k_rx_port_j](#), and [RX_ASSERT](#).

7.30 rx_port_constants.h

[Go to the documentation of this file.](#)

```
00001 /**
00002  * @file rx_port_constants.h
00003  * @brief Centralized RX72N Port Number Definitions
00004  *
00005  * @details
00006  * **LIBRARY-LEVEL SINGLE SOURCE OF TRUTH** for all port and pin numbers used
00007  * throughout the STAR RX72N firmware. This file is the ONLY location in the
00008  * library layer that contains hexadecimal values for port/pin identifiers.
00009  *
00010  * ## Purpose and Design Rationale
```

```

00011 *
00012 * This file implements the "Single Source of Truth" pattern for hardware port
00013 * addressing, eliminating magic numbers and providing type-safe port/pin
00014 * identifiers using C23 typed enums. All library and application code must use
00015 * these constants--never hardcode port numbers directly.
00016 *
00017 * ## Architecture and Layering
00018 *
00019 * The port constants architecture follows a strict layering model:
00020 *
00021 * 1. **This file (rx_port_constants.h)**: Defines port/pin constants with hex values
00022 * 2. **HAL layer (lib/rx_hal/)**: Uses these constants, NO hex values in HAL code
00023 * 3. **Application layer (include/hardware_pinout.h)**: Uses these constants, NO hex in app code
00024 *
00025 * This layering ensures:
00026 * - Single point of truth for port numbering
00027 * - Type safety through C23 typed enums
00028 * - Compile-time validation via static assertions
00029 * - Easy portability to different RX packages (144-pin, 176-pin, etc.)
00030 *
00031 * ## Port/Pin Encoding Scheme
00032 *
00033 * GPIO pins are encoded as 16-bit values using the following format:
00034 *
00035 * ```
00036 * gpio_pin_t = (port « k_port_shift) | pin_number
00037 *             = (port « 8) | pin
00038 *
00039 * Example: Port B, Pin 2 (PB2)
00040 *          = (0x0B « 8) | 0x02
00041 *          = 0x0B02
00042 * ```
00043 *
00044 * This encoding allows:
00045 * - **Upper byte**: Port number (0x00-0x13)
00046 * - **Lower byte**: Pin number (0x00-0x07)
00047 * - **Extraction**: Port = pin » 8, Pin = pin & 0xFF
00048 * - **Validation**: Compile-time checked via static assertions
00049 *
00050 * ## RX72N Port Numbering
00051 *
00052 * The Renesas RX72N uses non-contiguous port numbering:
00053 * - **Decimal ports**: 0-9 (0x00-0x09)
00054 * - **Alphabetic ports**: A-G (0x0A-0x10)
00055 * - **Port J**: 0x12 (note: Port I does not exist; Port H exists at 0x11 but has no pins on 144-pin)
00056 *
00057 * Total: 18 ports x 8 pins = 144 theoretical GPIO combinations
00058 * (Not all combinations physically available on 144-pin LQFP package)
00059 *
00060 * ## Performance Characteristics
00061 *
00062 * - **Memory usage**: Zero runtime overhead (all constants are compile-time)
00063 * - **Code size**: Minimal (enums compile to immediate values)
00064 * - **Execution time**: Inline accessor functions are 1-2 CPU cycles
00065 * - **Type safety**: Full compile-time type checking with C23 typed enums
00066 *
00067 * ## Usage Patterns
00068 *
00069 * **Library code example (HAL layer)**:
00070 * ```c
00071 * // Map port enum to register pointer
00072 * switch (port) {
00073 *     case k_rx_port_b: return portb();
00074 *     case k_rx_port_e: return porte();
00075 * }
00076 * ```
00077 *
00078 * **Application code example**:
00079 * ```c
00080 * // Define application-specific pin aliases
00081 * typedef enum : uint16_t {
00082 *     k_gpio_led_status = k_rx_pb_2, // PB2 = status LED
00083 *     k_gpio_spi_cs     = k_rx_pe_5, // PE5 = SPI chip select
00084 * } application_pins_t;
00085 * ```
00086 *
00087 * **Pin extraction example**:
00088 * ```c
00089 * gpio_pin_t pin = k_rx_pb_2;
00090 * uint8_t port = rx_port_from_pin(pin); // Returns k_rx_port_b (0x0B)
00091 * uint8_t pin_num = rx_pin_from_pin(pin); // Returns 2
00092 * ```
00093 *
00094 * @par Hardware Requirements:
00095 *
00096 * | Component | Requirement | Notes |
00097 * |-----|-----|-----|

```

```

00098 * | MCU | Renesas RX72N | Any package (100/144/176-pin) |
00099 * | Package | 144-pin LFQFP (primary target) | Not all ports available |
00100 * | Ports | 18 ports (0-9, A-G, J) | Non-contiguous numbering |
00101 * | Pins per port | 8 pins (0-7) | All ports have 8-bit width |
00102 *
00103 * @par Module Dependencies:
00104 *
00105 * **This module depends on**:
00106 * - `rx_check.h` - For RX_ASSERT compile-time/runtime validation
00107 * - `<stdint.h>` - For uint8_t, uint16_t types
00108 *
00109 * **This module is used by**:
00110 * - `hardware.h` - HAL GPIO register accessors
00111 * - `hardware_pinout.h` - Application-level pin definitions
00112 * - All GPIO-related HAL modules
00113 *
00114 * @par NASA Power of 10 Compliance:
00115 *
00116 * | Rule | Status | Implementation |
00117 * |-----|-----|-----|
00118 * | 1. Simple control flow | [OK] | Only inline functions with sequential logic |
00119 * | 2. Fixed loop bounds | [OK] | No loops (constants only) |
00120 * | 3. No dynamic allocation | [OK] | All constants are compile-time |
00121 * | 4. Small functions | [OK] | Inline functions are 3-5 lines each |
00122 * | 5. Assertions (>=2 per function) | [OK] | 3 assertions per function (prex2, postx1) |
00123 * | 6. Narrow scope | [OK] | File-scope only, no globals |
00124 * | 7. Check return values | [OK] | Inline functions don't fail |
00125 * | 8. Limited preprocessor | [OK] | C23 typed enums (zero macros) |
00126 * | 9. Pointer restrictions | [OK] | No pointers used |
00127 * | 10. Compiler warnings | [OK] | Compiles with -Wall -Wextra -Werror |
00128 *
00129 * **Rule 8 Compliance Detail**: This file demonstrates STAR project's mandatory
00130 * C23 typed enum standard. ALL integer constants use `typedef enum : type` syntax,
00131 * providing explicit size guarantees and eliminating preprocessor macros.
00132 *
00133 * @par SOLID Principles:
00134 *
00135 * **Single Responsibility (S)**:
00136 * - This file has ONE job: Define port/pin constants
00137 * - No GPIO manipulation logic (that's in HAL layer)
00138 * - No application-specific pin assignments (that's in hardware_pinout.h)
00139 *
00140 * **Open/Closed (O)**:
00141 * - Extensible to new RX packages without modifying existing constants
00142 * - Can add new port definitions for 144-pin or 176-pin packages
00143 * - Encoding scheme supports future ports beyond current 18
00144 *
00145 * **Liskov Substitution (L)**:
00146 * - All rx_port_pin_t values are interchangeable
00147 * - Pre-computed constants (k_rx_pb_2) substitutable with manual encoding
00148 * - ((port « 8) | pin) always produces valid rx_port_pin_t
00149 *
00150 * **Interface Segregation (I)**:
00151 * - Separate enums for ports, pins, and combined port/pin
00152 * - Use only what you need: port numbers OR pin numbers OR combinations
00153 * - Encoding constants isolated in separate enum
00154 *
00155 * **Dependency Inversion (D)**:
00156 * - HAL layer depends on these abstractions, not hardware addresses
00157 * - Application code depends on constants, not port numbers
00158 * - Hardware changes isolated to this file only
00159 *
00160 * @see hardware.h GPIO register accessor functions
00161 * @see hardware_pinout.h Application-specific pin definitions
00162 * @see rx_check.h Validation macros used in assertions
00163 * @see docs/sections/03_hardware_pinout.tex Complete pin assignment table
00164 *
00165 * @author Locked, Inc.
00166 * @date 2026-01-27
00167 * @copyright Copyright (c) 2026 Locked Inc.
00168 * SPDX-License-Identifier: MIT
00169 * @since Version 1.0.0
00170 */
00171
00172 #pragma once
00173
00174 #ifdef __cplusplus
00175 extern "C" {
00176 #endif
00177
00178 #include <stdint.h>
00179
00180 #include "rx_check.h"
00181
00182 /*
=====
00183 * Port Number Constants

```

```

00184 *
=====
00185 */
00186
00187 /**
00188  * @enum rx_port_number_t
00189  * @brief RX72N Port Number Constants (Single-Byte Identifiers)
00190
00191  * @details
00192  * Defines all GPIO port numbers available on the Renesas RX72N microcontroller.
00193  * Used by HAL layer for hardware register access and by application layer for
00194  * constructing full port/pin identifiers.
00195  *
00196  * ## Port Numbering Scheme
00197  *
00198  * The RX72N uses non-contiguous port numbering that mixes decimal and
00199  * hexadecimal naming:
00200  * - Decimal ports: 0-9 (values 0x00-0x09)
00201  * - Hexadecimal ports: A-G (values 0x0A-0x10)
00202  * - Port J: 0x12 (follows Port H at 0x11; Port I does not exist)
00203  *
00204  * This numbering matches the Renesas RX72N hardware manual and the encoding
00205  * scheme used throughout the firmware.
00206  *
00207  * ## Encoding Integration
00208  *
00209  * Port numbers form the upper byte of 16-bit gpio_pin_t values:
00210  * ```
00211  * gpio_pin_t = (rx_port_number_t « k_port_shift) | rx_pin_number_t
00212  *              = (port « 8) | pin
00213  * ```
00214  *
00215  * Example: Port B (0x0B) + Pin 2 (0x02) = 0x0B02
00216  *
00217  * ## Package Availability
00218  *
00219  * Not all ports are physically available on all RX72N packages:
00220  * - 144-pin LFQFP (STAR hardware): Ports 0-9, A-F, J (partial availability)
00221  * - 176-pin LFBGA: Maximum port availability
00222  *
00223  * Constants are defined for ALL theoretical ports for code compatibility across
00224  * packages, even if not physically present on 144-pin LFQFP.
00225  *
00226  * @par Port Availability Table (144-pin LFQFP):
00227  *
00228  * | Port | Value | Pins Available | Notes |
00229  * |-----|-----|-----|-----|
00230  * | 0 | 0x00 | P00-P03, P05, P07 | 6 pins (no P04, P06) |
00231  * | 1 | 0x01 | P12-P17 | 6 pins (no P10, P11) |
00232  * | 2 | 0x02 | P20-P27 | Full 8-pin port |
00233  * | 3 | 0x03 | P30-P37 | Full 8-pin port |
00234  * | 4 | 0x04 | P40-P47 | Full 8-pin port |
00235  * | 5 | 0x05 | P50-P56 | 7 pins (no P57) |
00236  * | 6 | 0x06 | P60-P67 | Full 8-pin port |
00237  * | 7 | 0x07 | P70-P77 | Full 8-pin port |
00238  * | 8 | 0x08 | P80-P83, P86-P87 | 6 pins (no P84, P85) |
00239  * | 9 | 0x09 | P90-P93 | 4 pins (no P94-P97) |
00240  * | A | 0x0A | PA0-PA7 | Full 8-pin port |
00241  * | B | 0x0B | PB0-PB7 | Full 8-pin port |
00242  * | C | 0x0C | PC0-PC7 | Full 8-pin port |
00243  * | D | 0x0D | PD0-PD7 | Full 8-pin port |
00244  * | E | 0x0E | PE0-PE7 | Full 8-pin port |
00245  * | F | 0x0F | PF5 | 1 pin only |
00246  * | G | 0x10 | None | Not available on 144-pin |
00247  * | J | 0x13 | PJ3, PJ5 | 2 pins only |
00248  *
00249  * @par Usage Example (HAL Layer):
00250  * @code{.c}
00251  * // Map port number to hardware register pointer
00252  * PORT_Type* get_port_registers(rx_port_number_t port)
00253  * {
00254  *     switch (port) {
00255  *         case k_rx_port_0: return port0();
00256  *         case k_rx_port_1: return port1();
00257  *         case k_rx_port_b: return portb();
00258  *         case k_rx_port_e: return porte();
00259  *         default: return nullptr;
00260  *     }
00261  * }
00262  * @endcode
00263  *
00264  * @par Usage Example (Application Layer):
00265  * @code{.c}
00266  * // Construct full port/pin identifier
00267  * gpio_pin_t status_led = (k_rx_port_b « k_port_shift) | k_rx_pin_2;
00268  * // Result: 0x0B02 (Port B, Pin 2)
00269  *

```

```

00270 * // Or use pre-computed constant
00271 * gpio_pin_t status_led = k_rx_pb_2; // Equivalent to above
00272 * @endcode
00273 *
00274 * @par Usage Example (Port Extraction):
00275 * @code{.c}
00276 * // Extract port number from combined pin identifier
00277 * gpio_pin_t pin = k_rx_pb_2; // 0x0B02
00278 * rx_port_number_t port = rx_port_from_pin(pin); // Returns k_rx_port_b (0x0B)
00279 * @endcode
00280 *
00281 * @invariant Port values are unique and match Renesas documentation
00282 * @invariant Port numbers fit in uint8_t (0x00-0xFF)
00283 * @invariant All enum values are compile-time constants
00284 *
00285 * @note This is a C23 typed enum with explicit underlying type uint8_t,
00286 * ensuring 1-byte size and consistent ABI across compiler versions
00287 * @note Port J (0x12) follows Port H (0x11); Port I does not exist in RX72N port map
00288 *
00289 * @warning Do not assume contiguous port numbering when iterating ports
00290 * @warning Not all ports available on 144-pin package--check datasheet
00291 *
00292 * @see rx_pin_number_t Pin number constants (0-7)
00293 * @see rx_port_pin_t Pre-computed port/pin combinations
00294 * @see rx_port_from_pin() Extract port number from combined identifier
00295 * @see hardware.h HAL GPIO register accessor functions
00296 * @see docs/sections/03_hardware_pinout.tex Complete RX72N pinout reference
00297 *
00298 * @since Version 1.0.0
00299 */
00300 typedef enum : uint8_t {
00301     /* Decimal Ports (0-9) */
00302     k_rx_port_0 = 0x00, /**< Port 0 - 6 pins on 144-pin (P00-P03, P05, P07) */
00303     k_rx_port_1 = 0x01, /**< Port 1 - 6 pins on 144-pin (P12-P17) */
00304     k_rx_port_2 = 0x02, /**< Port 2 - Full 8-pin port on 144-pin (P20-P27) */
00305     k_rx_port_3 = 0x03, /**< Port 3 - Full 8-pin port on 144-pin (P30-P37) */
00306     k_rx_port_4 = 0x04, /**< Port 4 - Full 8-pin port on 144-pin (P40-P47) */
00307     k_rx_port_5 = 0x05, /**< Port 5 - 7 pins on 144-pin (P50-P56) */
00308     k_rx_port_6 = 0x06, /**< Port 6 - Full 8-pin port on 144-pin (P60-P67) */
00309     k_rx_port_7 = 0x07, /**< Port 7 - Full 8-pin port on 144-pin (P70-P77) */
00310     k_rx_port_8 = 0x08, /**< Port 8 - 6 pins on 144-pin (P80-P83, P86-P87) */
00311     k_rx_port_9 = 0x09, /**< Port 9 - 4 pins on 144-pin (P90-P93) */
00312
00313     /* Hexadecimal Ports (A-G, J) */
00314     k_rx_port_a = 0x0A, /**< Port A - Full 8-pin port available on 144-pin */
00315     k_rx_port_b = 0x0B, /**< Port B - Full 8-pin port available on 144-pin */
00316     k_rx_port_c = 0x0C, /**< Port C - Full 8-pin port available on 144-pin */
00317     k_rx_port_d = 0x0D, /**< Port D - Full 8-pin port available on 144-pin */
00318     k_rx_port_e = 0x0E, /**< Port E - Full 8-pin port available on 144-pin */
00319     k_rx_port_f = 0x0F, /**< Port F - Partially available on 144-pin (PF5) */
00320     k_rx_port_g = 0x10, /**< Port G - Not available on 144-pin package */
00321     k_rx_port_j =
00322         0x12, /**< Port J - Partial availability on 144-pin (PJ3, PJ5) - follows Port H (0x11); Port I does not exist */
00323 } rx_port_number_t;
00324
00325 /*
=====
00326 * Pin Number Constants
00327 *
=====
00328 */
00329 /**
00330  * @enum rx_pin_number_t
00331  * @brief Pin Number Constants Within Each Port (0-7)
00332  *
00333  * @details
00334  * Defines pin numbers within each GPIO port on the RX72N. All RX72N ports are
00335  * **8-bit wide**, meaning each port has pins numbered 0 through 7. These
00336  * constants provide self-documenting alternatives to raw numeric literals.
00337  *
00338  * ## Pin Numbering
00339  *
00340  * Every RX72N port (0-9, A-G, J) contains 8 pins:
00341  * - Pin 0 (LSB of port data register)
00342  * - Pin 1
00343  * - Pin 2
00344  * - ...
00345  * - Pin 7 (MSB of port data register)
00346  *
00347  * This consistent 8-pin structure applies to ALL ports, even though not all
00348  * individual pins may be physically available on smaller packages.
00349  *
00350  * ## Encoding Integration
00351  *
00352  * Pin numbers form the lower byte of 16-bit gpio_pin_t values:
00353  * ```
00354  *

```



```

00355 * gpio_pin_t = (rx_port_number_t « k_port_shift) | rx_pin_number_t
00356 *           = (port « 8) | pin
00357 * ...
00358 *
00359 * Example: Port B (0x0B) + Pin 2 (0x02) = 0x0B02
00360 *
00361 * ## Validation Constants
00362 *
00363 * The enum includes `k_rx_pin_min` and `k_rx_pin_max` for runtime/compile-time
00364 * validation of pin numbers. These are used by assertion macros and accessor
00365 * functions to verify pin number correctness.
00366 *
00367 * Valid range: [0, 7] inclusive
00368 *
00369 * @par Usage Example (Constructing gpio_pin_t):
00370 * @code{.c}
00371 * // Construct Port B, Pin 2 manually
00372 * gpio_pin_t status_led = (k_rx_port_b « k_port_shift) | k_rx_pin_2;
00373 * // Result: 0x0B02
00374 *
00375 * // Or use pre-computed constant (preferred)
00376 * gpio_pin_t status_led = k_rx_pb_2; // Cleaner, same result
00377 * @endcode
00378 *
00379 * @par Usage Example (Pin Extraction):
00380 * @code{.c}
00381 * // Extract pin number from combined identifier
00382 * gpio_pin_t pin = k_rx_pe_5; // 0x0E05
00383 * rx_pin_number_t pin_num = rx_pin_from_pin(pin); // Returns 5
00384 * @endcode
00385 *
00386 * @par Usage Example (Validation):
00387 * @code{.c}
00388 * // Validate user-provided pin number
00389 * bool is_valid_pin(uint8_t pin)
00390 * {
00391 *     return (pin >= k_rx_pin_min) && (pin <= k_rx_pin_max);
00392 * }
00393 *
00394 * // Or use in assertion
00395 * RX_ASSERT((pin >= k_rx_pin_min) && (pin <= k_rx_pin_max),
00396 *           "Pin number out of range");
00397 * @endcode
00398 *
00399 * @par Usage Example (Bit Manipulation):
00400 * @code{.c}
00401 * // Set pin 2 high in port data register
00402 * PORT_Type* port = portb();
00403 * port->PODR |= (1 « k_rx_pin_2); // Set bit 2
00404 *
00405 * // Clear pin 5
00406 * port->PODR &= ~(1 « k_rx_pin_5); // Clear bit 5
00407 * @endcode
00408 *
00409 * @invariant All pin numbers are in range [0, 7]
00410 * @invariant k_rx_pin_min == 0
00411 * @invariant k_rx_pin_max == 7
00412 * @invariant Pin numbering is contiguous (no gaps)
00413 *
00414 * @note This is a C23 **typed enum** with explicit underlying type uint8_t,
00415 *       ensuring 1-byte size and consistent ABI across compiler versions
00416 * @note Pin numbers correspond to bit positions in port registers (PDR, PODR, PIDR)
00417 * @note All 8 pins exist logically even if not physically bonded on smaller packages
00418 *
00419 * @warning Pin numbers are zero-indexed (0-7), not one-indexed (1-8)
00420 * @warning Do not use raw numbers--use these constants for self-documenting code
00421 *
00422 * @see rx_port_number_t Port number constants
00423 * @see rx_port_pin_t Pre-computed port/pin combinations
00424 * @see rx_pin_from_pin() Extract pin number from combined identifier
00425 * @see hardware.h HAL GPIO register manipulation functions
00426 *
00427 * @since Version 1.0.0
00428 */
00429 typedef enum : uint8_t {
00430     k_rx_pin_0 = 0, /**< Pin 0 - LSB of port data register, bit position 0 */
00431     k_rx_pin_1 = 1, /**< Pin 1 - Bit position 1 in port data register */
00432     k_rx_pin_2 = 2, /**< Pin 2 - Bit position 2 in port data register */
00433     k_rx_pin_3 = 3, /**< Pin 3 - Bit position 3 in port data register */
00434     k_rx_pin_4 = 4, /**< Pin 4 - Bit position 4 in port data register */
00435     k_rx_pin_5 = 5, /**< Pin 5 - Bit position 5 in port data register */
00436     k_rx_pin_6 = 6, /**< Pin 6 - Bit position 6 in port data register */
00437     k_rx_pin_7 = 7, /**< Pin 7 - MSB of port data register, bit position 7 */
00438
00439     k_rx_pin_min = 0, /**< Minimum valid pin number - used for range validation in assertions */
00440     k_rx_pin_max = 7, /**< Maximum valid pin number - used for range validation in assertions */
00441 } rx_pin_number_t;

```

```

00442
00443 /*
=====
00444 * Bit Shift Constants
00445 *
=====
00446 */
00447
00448 /**
00449 * @enum port_encoding_t
00450 * @brief Bit Shift and Mask Constants for Port/Pin Encoding
00451 *
00452 * @details
00453 * Defines the constants used to encode and decode 16-bit gpio_pin_t values.
00454 * These constants implement the port/pin encoding scheme used throughout the
00455 * firmware for type-safe GPIO pin identification.
00456 *
00457 * ## Encoding Scheme
00458 *
00459 * The encoding scheme packs an 8-bit port number and 8-bit pin number into a
00460 * single 16-bit value:
00461 *
00462 * ```
00463 * +-----+-----+
00464 * | Upper Byte | Lower Byte |
00465 * | (Port Number) | (Pin Number) |
00466 * +-----+-----+
00467 * | Bits 15-8 | Bits 7-0 |
00468 * | rx_port_number_t | rx_pin_number_t |
00469 * +-----+-----+
00470 *
00471 * Encoding:  gpio_pin_t = (port « k_port_shift) | pin
00472 * Decoding:  port = gpio_pin_t » k_port_shift
00473 *           pin  = gpio_pin_t & k_port_mask
00474 * ```
00475 *
00476 * ## Constant Definitions
00477 *
00478 * - **k_port_shift = 8**: Left-shift amount to move port number to upper byte
00479 * - **k_port_mask = 0xFF**: Mask to extract pin number from lower byte
00480 *
00481 * ## Design Rationale
00482 *
00483 * Using compile-time constants instead of magic numbers provides:
00484 * 1. **Self-documenting code**: `(port « k_port_shift)` vs `(port « 8)`
00485 * 2. **Easy modification**: Change encoding in one place if needed
00486 * 3. **Type safety**: C23 typed enum provides compile-time checking
00487 * 4. **Searchability**: grep for k_port_shift finds all encoding operations
00488 *
00489 * ## Bit Layout Example
00490 *
00491 * Port B (0x0B), Pin 2 (0x02):
00492 * ```
00493 * Port: 0x0B = 0000 1011
00494 * Pin:  0x02 = 0000 0010
00495 *
00496 * Encoding: (0x0B « 8) | 0x02
00497 *           = 0x0B00 | 0x0002
00498 *           = 0x0B02
00499 *
00500 * Binary:  0000 1011 0000 0010
00501 *           +- Port -+- Pin +-
00502 * ```
00503 *
00504 * @par Usage Example (Encoding):
00505 * @code{.c}
00506 * // Manual construction using encoding constants
00507 * rx_port_number_t port = k_rx_port_b; // 0x0B
00508 * rx_pin_number_t pin = k_rx_pin_2;    // 0x02
00509 * gpio_pin_t result = (port « k_port_shift) | pin;
00510 * // Result: 0x0B02
00511 *
00512 * // Equivalent to pre-computed constant
00513 * gpio_pin_t result = k_rx_pb_2;
00514 * @endcode
00515 *
00516 * @par Usage Example (Decoding):
00517 * @code{.c}
00518 * // Extract port and pin from combined value
00519 * gpio_pin_t pin = k_rx_pe_5; // 0x0E05
00520 *
00521 * rx_port_number_t port = (pin » k_port_shift); // 0x0E
00522 * rx_pin_number_t pin_num = (pin & k_port_mask); // 0x05
00523 *
00524 * // Or use dedicated accessor functions (preferred)
00525 * port = rx_port_from_pin(pin); // 0x0E
00526 * pin_num = rx_pin_from_pin(pin); // 0x05

```

```

00527 * @endcode
00528 *
00529 * @par Usage Example (Bit Manipulation):
00530 * @code{.c}
00531 * // Verify encoding/decoding round-trip
00532 * rx_port_number_t original_port = k_rx_port_c;
00533 * rx_pin_number_t original_pin = k_rx_pin_7;
00534 *
00535 * // Encode
00536 * gpio_pin_t encoded = (original_port « k_port_shift) | original_pin;
00537 *
00538 * // Decode
00539 * rx_port_number_t decoded_port = (encoded » k_port_shift);
00540 * rx_pin_number_t decoded_pin = (encoded & k_port_mask);
00541 *
00542 * // Verify
00543 * assert(decoded_port == original_port);
00544 * assert(decoded_pin == original_pin);
00545 * @endcode
00546 *
00547 * @invariant k_port_shift == 8 (must shift by full byte)
00548 * @invariant k_port_mask == 0xFF (must mask full byte)
00549 * @invariant (value « k_port_shift) » k_port_shift == value (for value <= 0xFF)
00550 * @invariant (value & k_port_mask) <= 0xFF (mask result always fits in uint8_t)
00551 *
00552 * @note This is a C23 typed enum with explicit underlying type uint8_t,
00553 * ensuring 1-byte size and consistent ABI across compiler versions
00554 * @note The encoding is big-endian style (port in high byte, pin in low byte)
00555 * @note This encoding matches the pattern used in hardware_pinout.h gpio_pin_t enum
00556 *
00557 * @warning Do not change these constants without updating all port/pin definitions
00558 * @warning Changing k_port_shift affects ALL 144+ pre-computed pin constants
00559 *
00560 * @see rx_port_number_t Port number constants that go in upper byte
00561 * @see rx_pin_number_t Pin number constants that go in lower byte
00562 * @see rx_port_pin_t Pre-computed combinations using this encoding
00563 * @see rx_port_from_pin() Decode port number from gpio_pin_t
00564 * @see rx_pin_from_pin() Decode pin number from gpio_pin_t
00565 *
00566 * @since Version 1.0.0
00567 */
00568 typedef enum : uint8_t {
00569     k_port_shift =
00570         8, /**< Left shift amount to move port number into upper byte (bits 15-8). Used for encoding: gpio_pin_t = (port «
00571         k_port_shift) | pin */
00572     k_port_mask =
00573         0xFF, /**< Bitmask (0xFF) to extract pin number from lower byte (bits 7-0). Used for decoding: pin = gpio_pin_t &
00574         k_port_mask */
00575 } port_encoding_t;
00576
00577
00578
00579
00580 /**
00581 * @enum rx_port_pin_t
00582 * @brief Pre-Computed Port/Pin Combinations for All RX72N GPIO Pins
00583 *
00584 * @details
00585 * Provides ready-to-use 16-bit gpio_pin_t constants for all 144 theoretical
00586 * port/pin combinations on the RX72N microcontroller (18 ports x 8 pins = 144).
00587 * These pre-computed constants eliminate the need for runtime bit-shift operations
00588 * and provide cleaner, more readable code.
00589 *
00590 * ## Coverage
00591 *
00592 * This enum defines ALL possible port/pin combinations:
00593 * - Ports: 0-9 (decimal) + A-G + J (hexadecimal) = 18 ports
00594 * - Pins per port: 0-7 = 8 pins
00595 * - Total combinations: 18 x 8 = 144 constants
00596 *
00597 * ## Physical Availability
00598 *
00599 * Important: Not all 144 combinations are physically bonded on smaller packages:
00600 * - 144-pin LQFP (STAR hardware): ~112 pins available
00601 * - 176-pin LFBGA: Maximum pins available
00602 *
00603 * Constants are defined for ALL theoretical combinations to ensure:
00604 * - Code compatibility across different RX72N packages
00605 * - Compile-time validation (invalid pins detected at compile time)
00606 * - No gaps in the constant namespace
00607 *
00608 * Pins marked "N/A on 144-pin" are not physically available on the STAR hardware
00609 * but may be available on larger packages.

```

```

00610 *
00611 * ## Benefits Over Manual Encoding
00612 *
00613 * **Using pre-computed constants**:
00614 * ```c
00615 * gpio_pin_t pin = k_rx_pb_2; // Clean, readable
00616 * ```
00617 *
00618 * **vs. Manual bit-shift construction**:
00619 * ```c
00620 * gpio_pin_t pin = (k_rx_port_b « k_port_shift) | k_rx_pin_2; // Verbose
00621 * ```
00622 *
00623 * Advantages:
00624 * 1. **Cleaner syntax**: One constant vs. shift-or expression
00625 * 2. **Zero runtime overhead**: Pre-computed at compile time
00626 * 3. **Easier to grep**: Search for "k_rx_pb_2" finds all uses
00627 * 4. **Consistent naming**: Matches Renesas datasheet conventions (PA2, PB7, etc.)
00628 * 5. **Compile-time validated**: Static assertions verify correctness
00629 *
00630 * ## Naming Convention
00631 *
00632 * Constants follow the pattern: `k_rx_p<port>_<pin>`
00633 *
00634 * Examples:
00635 * - `k_rx_p0_5` -> Port 0, Pin 5 -> P05
00636 * - `k_rx_pb_2` -> Port B, Pin 2 -> PB2
00637 * - `k_rx_pe_5` -> Port E, Pin 5 -> PE5
00638 * - `k_rx_pj_3` -> Port J, Pin 3 -> PJ3
00639 *
00640 * This naming matches the Renesas RX72N Hardware Manual conventions.
00641 *
00642 * ## Encoding Scheme
00643 *
00644 * Each constant uses the standard port/pin encoding:
00645 * ```
00646 * k_rx_p<port>_<pin> = (k_rx_port_<port> « k_port_shift) | k_rx_pin_<pin>
00647 *                    = (port « 8) | pin
00648 * ```
00649 *
00650 * Example: PB2 (Port B, Pin 2)
00651 * ```
00652 * k_rx_pb_2 = (k_rx_port_b « 8) | k_rx_pin_2
00653 *           = (0x0B « 8) | 0x02
00654 *           = 0x0B00 | 0x0002
00655 *           = 0x0B02
00656 * ```
00657 *
00658 * @par Usage Example (Basic):
00659 * @code{.c}
00660 * // Use pre-computed constant for clean code
00661 * gpio_pin_t status_led = k_rx_pb_2; // Port B, Pin 2
00662 * gpio_pin_t spi_cs = k_rx_pe_5;    // Port E, Pin 5
00663 *
00664 * // Pass to HAL functions
00665 * gpio_set_pin_high(status_led);
00666 * gpio_configure_output(spi_cs);
00667 * @endcode
00668 *
00669 * @par Usage Example (Application Pin Definitions):
00670 * @code{.c}
00671 * // Define application-specific pin aliases using pre-computed constants
00672 * typedef enum : uint16_t {
00673 *     k_gpio_led_status = k_rx_pb_2, // Status LED on PB2
00674 *     k_gpio_led_error  = k_rx_pb_3, // Error LED on PB3
00675 *     k_gpio_spi_cs     = k_rx_pe_5,  // SPI chip select on PE5
00676 *     k_gpio_uart_tx    = k_rx_p3_0,  // UART TX on P30
00677 *     k_gpio_uart_rx    = k_rx_p3_1,  // UART RX on P31
00678 * } application_pins_t;
00679 * @endcode
00680 *
00681 * @par Usage Example (Port/Pin Extraction):
00682 * @code{.c}
00683 * // Extract port and pin numbers from pre-computed constant
00684 * gpio_pin_t pin = k_rx_pb_2; // 0x0B02
00685 *
00686 * rx_port_number_t port = rx_port_from_pin(pin); // Returns k_rx_port_b (0x0B)
00687 * rx_pin_number_t pin_num = rx_pin_from_pin(pin); // Returns 2
00688 * @endcode
00689 *
00690 * @par Usage Example (Compile-Time Validation):
00691 * @code{.c}
00692 * // Static assertion verifies pre-computed constant is correct
00693 * static_assert(k_rx_pb_2 == 0x0B02, "PB2 must be 0x0B02");
00694 * static_assert(k_rx_pe_5 == 0x0E05, "PE5 must be 0x0E05");
00695 * @endcode
00696 *

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00697 * @par Pin Availability on 144-pin LFQFP Package:
00698 *
00699 * | Port | Available Pins | Package Pins | Example Constants |
00700 * |-----|-----|-----|-----|
00701 * | 0 | P00-P03, P05, P07 | 8,7,6,4,2,144 | k_rx_p0_0, k_rx_p0_7 |
00702 * | 1 | P12-P17 | 45-38 | k_rx_p1_2, k_rx_p1_7 |
00703 * | 2 | P20-P27 | 37-30 | k_rx_p2_0, k_rx_p2_7 |
00704 * | 3 | P30-P37 | 29-20 | k_rx_p3_0, k_rx_p3_7 |
00705 * | 4 | P40-P47 | 141-133 | k_rx_p4_0, k_rx_p4_7 |
00706 * | 5 | P50-P56 | 56-50 | k_rx_p5_0, k_rx_p5_6 |
00707 * | 6 | P60-P67 | 117-98 | k_rx_p6_0, k_rx_p6_7 |
00708 * | 7 | P70-P77 | 104-68 | k_rx_p7_0, k_rx_p7_7 |
00709 * | 8 | P80-P83, P86-P87 | 65-58,41,39 | k_rx_p8_0, k_rx_p8_7 |
00710 * | 9 | P90-P93 | 131-127 | k_rx_p9_0, k_rx_p9_3 |
00711 * | A | PA0-PA7 | 97-88 | k_rx_pa_0, k_rx_pa_7 |
00712 * | B | PB0-PB7 | 87-78 | k_rx_pb_0, k_rx_pb_7 |
00713 * | C | PC0-PC7 | 75-60 | k_rx_pc_0, k_rx_pc_7 |
00714 * | D | PD0-PD7 | 126-119 | k_rx_pd_0, k_rx_pd_7 |
00715 * | E | PE0-PE7 | 111-101 | k_rx_pe_0, k_rx_pe_7 |
00716 * | F | PF5 | 9 | k_rx_pf_5 |
00717 * | J | PJ3, PJ5 | 13, 11 | k_rx_pj_3, k_rx_pj_5 |
00718 *
00719 * @invariant Each constant equals (port « 8) | pin
00720 * @invariant Port portion (upper byte) is valid rx_port_number_t
00721 * @invariant Pin portion (lower byte) is valid rx_pin_number_t (0-7)
00722 * @invariant All 144 combinations are defined (no gaps)
00723 *
00724 * @note This is a C23 **typed enum** with explicit underlying type uint16_t,
00725 * ensuring 2-byte size and consistent ABI across compiler versions
00726 * @note Constants are evaluated at compile time (zero runtime overhead)
00727 * @note Naming follows Renesas conventions: P{port}{pin} (e.g., PB2, PE5)
00728 * @note Package availability is documented in inline comments (search "N/A on 144-pin")
00729 *
00730 * @warning Using undefined pins (N/A on your package) may cause undefined behavior
00731 * @warning Always verify pin availability in RX72N datasheet for your package
00732 * @warning Do not assume all defined constants are physically available
00733 *
00734 * @see rx_port_number_t Port number constants used in upper byte
00735 * @see rx_pin_number_t Pin number constants used in lower byte
00736 * @see port_encoding_t Encoding scheme constants (k_port_shift, k_port_mask)
00737 * @see rx_port_from_pin() Extract port number from constant
00738 * @see rx_pin_from_pin() Extract pin number from constant
00739 * @see hardware_pinout.h Application-level pin definitions
00740 * @see docs/sections/03_hardware_pinout.tex Complete pinout reference for STAR hardware
00741 *
00742 * @since Version 1.0.0
00743 */
00744 typedef enum : uint16_t {
00745     /* Port 0 (0x00) - Pins 0-7 */
00746     k_rx_p0_0 = (k_rx_port_0 « k_port_shift) | k_rx_pin_0, /**< P00 (pin 8) */
00747     k_rx_p0_1 = (k_rx_port_0 « k_port_shift) | k_rx_pin_1, /**< P01 (pin 7) */
00748     k_rx_p0_2 = (k_rx_port_0 « k_port_shift) | k_rx_pin_2, /**< P02 (pin 6) */
00749     k_rx_p0_3 = (k_rx_port_0 « k_port_shift) | k_rx_pin_3, /**< P03 (pin 4) */
00750     k_rx_p0_4 = (k_rx_port_0 « k_port_shift) | k_rx_pin_4, /**< P04 - N/A on 144-pin */
00751     k_rx_p0_5 = (k_rx_port_0 « k_port_shift) | k_rx_pin_5, /**< P05 (pin 2) */
00752     k_rx_p0_6 = (k_rx_port_0 « k_port_shift) | k_rx_pin_6, /**< P06 - N/A on 144-pin */
00753     k_rx_p0_7 = (k_rx_port_0 « k_port_shift) | k_rx_pin_7, /**< P07 (pin 144) */
00754
00755     /* Port 1 (0x01) - Pins 0-7 */
00756     k_rx_p1_0 = (k_rx_port_1 « k_port_shift) | k_rx_pin_0, /**< P10 - N/A on 144-pin */
00757     k_rx_p1_1 = (k_rx_port_1 « k_port_shift) | k_rx_pin_1, /**< P11 - N/A on 144-pin */
00758     k_rx_p1_2 = (k_rx_port_1 « k_port_shift) | k_rx_pin_2, /**< P12 (pin 45) */
00759     k_rx_p1_3 = (k_rx_port_1 « k_port_shift) | k_rx_pin_3, /**< P13 (pin 44) */
00760     k_rx_p1_4 = (k_rx_port_1 « k_port_shift) | k_rx_pin_4, /**< P14 (pin 43) */
00761     k_rx_p1_5 = (k_rx_port_1 « k_port_shift) | k_rx_pin_5, /**< P15 (pin 42) */
00762     k_rx_p1_6 = (k_rx_port_1 « k_port_shift) | k_rx_pin_6, /**< P16 (pin 40) */
00763     k_rx_p1_7 = (k_rx_port_1 « k_port_shift) | k_rx_pin_7, /**< P17 (pin 38) */
00764
00765     /* Port 2 (0x02) - Pins 0-7 */
00766     k_rx_p2_0 = (k_rx_port_2 « k_port_shift) | k_rx_pin_0, /**< P20 (pin 37) */
00767     k_rx_p2_1 = (k_rx_port_2 « k_port_shift) | k_rx_pin_1, /**< P21 (pin 36) */
00768     k_rx_p2_2 = (k_rx_port_2 « k_port_shift) | k_rx_pin_2, /**< P22 (pin 35) */
00769     k_rx_p2_3 = (k_rx_port_2 « k_port_shift) | k_rx_pin_3, /**< P23 (pin 34) */
00770     k_rx_p2_4 = (k_rx_port_2 « k_port_shift) | k_rx_pin_4, /**< P24 (pin 33) */
00771     k_rx_p2_5 = (k_rx_port_2 « k_port_shift) | k_rx_pin_5, /**< P25 (pin 32) */
00772     k_rx_p2_6 = (k_rx_port_2 « k_port_shift) | k_rx_pin_6, /**< P26 (pin 31) */
00773     k_rx_p2_7 = (k_rx_port_2 « k_port_shift) | k_rx_pin_7, /**< P27 (pin 30) */
00774
00775     /* Port 3 (0x03) - Pins 0-7 */
00776     k_rx_p3_0 = (k_rx_port_3 « k_port_shift) | k_rx_pin_0, /**< P30 (pin 29) */
00777     k_rx_p3_1 = (k_rx_port_3 « k_port_shift) | k_rx_pin_1, /**< P31 (pin 28) */
00778     k_rx_p3_2 = (k_rx_port_3 « k_port_shift) | k_rx_pin_2, /**< P32 (pin 27) */
00779     k_rx_p3_3 = (k_rx_port_3 « k_port_shift) | k_rx_pin_3, /**< P33 (pin 26) */
00780     k_rx_p3_4 = (k_rx_port_3 « k_port_shift) | k_rx_pin_4, /**< P34 (pin 25) */
00781     k_rx_p3_5 = (k_rx_port_3 « k_port_shift) | k_rx_pin_5, /**< P35 (pin 24) */
00782     k_rx_p3_6 = (k_rx_port_3 « k_port_shift) | k_rx_pin_6, /**< P36 (pin 22) */
00783     k_rx_p3_7 = (k_rx_port_3 « k_port_shift) | k_rx_pin_7, /**< P37 (pin 20) */

```

```

00784
00785 /* Port 4 (0x04) - Pins 0-7 */
00786 k_rx_p4_0 = (k_rx_port_4 « k_port_shift) | k_rx_pin_0, /**< P40 (pin 141) */
00787 k_rx_p4_1 = (k_rx_port_4 « k_port_shift) | k_rx_pin_1, /**< P41 (pin 139) */
00788 k_rx_p4_2 = (k_rx_port_4 « k_port_shift) | k_rx_pin_2, /**< P42 (pin 138) */
00789 k_rx_p4_3 = (k_rx_port_4 « k_port_shift) | k_rx_pin_3, /**< P43 (pin 137) */
00790 k_rx_p4_4 = (k_rx_port_4 « k_port_shift) | k_rx_pin_4, /**< P44 (pin 136) */
00791 k_rx_p4_5 = (k_rx_port_4 « k_port_shift) | k_rx_pin_5, /**< P45 (pin 135) */
00792 k_rx_p4_6 = (k_rx_port_4 « k_port_shift) | k_rx_pin_6, /**< P46 (pin 134) */
00793 k_rx_p4_7 = (k_rx_port_4 « k_port_shift) | k_rx_pin_7, /**< P47 (pin 133) */
00794
00795 /* Port 5 (0x05) - Pins 0-7 */
00796 k_rx_p5_0 = (k_rx_port_5 « k_port_shift) | k_rx_pin_0, /**< P50 (pin 56) */
00797 k_rx_p5_1 = (k_rx_port_5 « k_port_shift) | k_rx_pin_1, /**< P51 (pin 55) */
00798 k_rx_p5_2 = (k_rx_port_5 « k_port_shift) | k_rx_pin_2, /**< P52 (pin 54) */
00799 k_rx_p5_3 = (k_rx_port_5 « k_port_shift) | k_rx_pin_3, /**< P53 (pin 53) */
00800 k_rx_p5_4 = (k_rx_port_5 « k_port_shift) | k_rx_pin_4, /**< P54 (pin 52) */
00801 k_rx_p5_5 = (k_rx_port_5 « k_port_shift) | k_rx_pin_5, /**< P55 (pin 51) */
00802 k_rx_p5_6 = (k_rx_port_5 « k_port_shift) | k_rx_pin_6, /**< P56 (pin 50) */
00803 k_rx_p5_7 = (k_rx_port_5 « k_port_shift) | k_rx_pin_7, /**< P57 - N/A on 144-pin */
00804
00805 /* Port 6 (0x06) - Pins 0-7 */
00806 k_rx_p6_0 = (k_rx_port_6 « k_port_shift) | k_rx_pin_0, /**< P60 (pin 117) */
00807 k_rx_p6_1 = (k_rx_port_6 « k_port_shift) | k_rx_pin_1, /**< P61 (pin 115) */
00808 k_rx_p6_2 = (k_rx_port_6 « k_port_shift) | k_rx_pin_2, /**< P62 (pin 114) */
00809 k_rx_p6_3 = (k_rx_port_6 « k_port_shift) | k_rx_pin_3, /**< P63 (pin 113) */
00810 k_rx_p6_4 = (k_rx_port_6 « k_port_shift) | k_rx_pin_4, /**< P64 (pin 112) */
00811 k_rx_p6_5 = (k_rx_port_6 « k_port_shift) | k_rx_pin_5, /**< P65 (pin 100) */
00812 k_rx_p6_6 = (k_rx_port_6 « k_port_shift) | k_rx_pin_6, /**< P66 (pin 99) */
00813 k_rx_p6_7 = (k_rx_port_6 « k_port_shift) | k_rx_pin_7, /**< P67 (pin 98) */
00814
00815 /* Port 7 (0x07) - Pins 0-7 */
00816 k_rx_p7_0 = (k_rx_port_7 « k_port_shift) | k_rx_pin_0, /**< P70 (pin 104) */
00817 k_rx_p7_1 = (k_rx_port_7 « k_port_shift) | k_rx_pin_1, /**< P71 (pin 86) */
00818 k_rx_p7_2 = (k_rx_port_7 « k_port_shift) | k_rx_pin_2, /**< P72 (pin 85) */
00819 k_rx_p7_3 = (k_rx_port_7 « k_port_shift) | k_rx_pin_3, /**< P73 (pin 77) */
00820 k_rx_p7_4 = (k_rx_port_7 « k_port_shift) | k_rx_pin_4, /**< P74 (pin 72) */
00821 k_rx_p7_5 = (k_rx_port_7 « k_port_shift) | k_rx_pin_5, /**< P75 (pin 71) */
00822 k_rx_p7_6 = (k_rx_port_7 « k_port_shift) | k_rx_pin_6, /**< P76 (pin 69) */
00823 k_rx_p7_7 = (k_rx_port_7 « k_port_shift) | k_rx_pin_7, /**< P77 (pin 68) */
00824
00825 /* Port 8 (0x08) - Pins 0-7 */
00826 k_rx_p8_0 = (k_rx_port_8 « k_port_shift) | k_rx_pin_0, /**< P80 (pin 65) */
00827 k_rx_p8_1 = (k_rx_port_8 « k_port_shift) | k_rx_pin_1, /**< P81 (pin 64) */
00828 k_rx_p8_2 = (k_rx_port_8 « k_port_shift) | k_rx_pin_2, /**< P82 (pin 63) */
00829 k_rx_p8_3 = (k_rx_port_8 « k_port_shift) | k_rx_pin_3, /**< P83 (pin 58) */
00830 k_rx_p8_4 = (k_rx_port_8 « k_port_shift) | k_rx_pin_4, /**< P84 - N/A on 144-pin */
00831 k_rx_p8_5 = (k_rx_port_8 « k_port_shift) | k_rx_pin_5, /**< P85 - N/A on 144-pin */
00832 k_rx_p8_6 = (k_rx_port_8 « k_port_shift) | k_rx_pin_6, /**< P86 (pin 41) */
00833 k_rx_p8_7 = (k_rx_port_8 « k_port_shift) | k_rx_pin_7, /**< P87 (pin 39) */
00834
00835 /* Port 9 (0x09) - Pins 0-7 */
00836 k_rx_p9_0 = (k_rx_port_9 « k_port_shift) | k_rx_pin_0, /**< P90 (pin 131) */
00837 k_rx_p9_1 = (k_rx_port_9 « k_port_shift) | k_rx_pin_1, /**< P91 (pin 129) */
00838 k_rx_p9_2 = (k_rx_port_9 « k_port_shift) | k_rx_pin_2, /**< P92 (pin 128) */
00839 k_rx_p9_3 = (k_rx_port_9 « k_port_shift) | k_rx_pin_3, /**< P93 (pin 127) */
00840 k_rx_p9_4 = (k_rx_port_9 « k_port_shift) | k_rx_pin_4, /**< P94 - N/A on 144-pin */
00841 k_rx_p9_5 = (k_rx_port_9 « k_port_shift) | k_rx_pin_5, /**< P95 - N/A on 144-pin */
00842 k_rx_p9_6 = (k_rx_port_9 « k_port_shift) | k_rx_pin_6, /**< P96 - N/A on 144-pin */
00843 k_rx_p9_7 = (k_rx_port_9 « k_port_shift) | k_rx_pin_7, /**< P97 - N/A on 144-pin */
00844
00845 /* Port A (0x0A) - Pins 0-7 */
00846 k_rx_pa_0 = (k_rx_port_a « k_port_shift) | k_rx_pin_0, /**< PA0 (pin 97) */
00847 k_rx_pa_1 = (k_rx_port_a « k_port_shift) | k_rx_pin_1, /**< PA1 (pin 96) */
00848 k_rx_pa_2 = (k_rx_port_a « k_port_shift) | k_rx_pin_2, /**< PA2 (pin 95) */
00849 k_rx_pa_3 = (k_rx_port_a « k_port_shift) | k_rx_pin_3, /**< PA3 (pin 94) */
00850 k_rx_pa_4 = (k_rx_port_a « k_port_shift) | k_rx_pin_4, /**< PA4 (pin 92) */
00851 k_rx_pa_5 = (k_rx_port_a « k_port_shift) | k_rx_pin_5, /**< PA5 (pin 90) */
00852 k_rx_pa_6 = (k_rx_port_a « k_port_shift) | k_rx_pin_6, /**< PA6 (pin 89) */
00853 k_rx_pa_7 = (k_rx_port_a « k_port_shift) | k_rx_pin_7, /**< PA7 (pin 88) */
00854
00855 /* Port B (0x0B) - Pins 0-7 */
00856 k_rx_pb_0 = (k_rx_port_b « k_port_shift) | k_rx_pin_0, /**< PB0 (pin 87) */
00857 k_rx_pb_1 = (k_rx_port_b « k_port_shift) | k_rx_pin_1, /**< PB1 (pin 84) */
00858 k_rx_pb_2 = (k_rx_port_b « k_port_shift) | k_rx_pin_2, /**< PB2 (pin 83) */
00859 k_rx_pb_3 = (k_rx_port_b « k_port_shift) | k_rx_pin_3, /**< PB3 (pin 82) */
00860 k_rx_pb_4 = (k_rx_port_b « k_port_shift) | k_rx_pin_4, /**< PB4 (pin 81) */
00861 k_rx_pb_5 = (k_rx_port_b « k_port_shift) | k_rx_pin_5, /**< PB5 (pin 80) */
00862 k_rx_pb_6 = (k_rx_port_b « k_port_shift) | k_rx_pin_6, /**< PB6 (pin 79) */
00863 k_rx_pb_7 = (k_rx_port_b « k_port_shift) | k_rx_pin_7, /**< PB7 (pin 78) */
00864
00865 /* Port C (0x0C) - Pins 0-7 */
00866 k_rx_pc_0 = (k_rx_port_c « k_port_shift) | k_rx_pin_0, /**< PC0 (pin 75) */
00867 k_rx_pc_1 = (k_rx_port_c « k_port_shift) | k_rx_pin_1, /**< PC1 (pin 73) */
00868 k_rx_pc_2 = (k_rx_port_c « k_port_shift) | k_rx_pin_2, /**< PC2 (pin 70) */
00869 k_rx_pc_3 = (k_rx_port_c « k_port_shift) | k_rx_pin_3, /**< PC3 (pin 67) */
00870 k_rx_pc_4 = (k_rx_port_c « k_port_shift) | k_rx_pin_4, /**< PC4 (pin 66) */

```



```

00871 k_rx_pc_5 = (k_rx_port_c « k_port_shift) | k_rx_pin_5, /**< PC5 (pin 62) */
00872 k_rx_pc_6 = (k_rx_port_c « k_port_shift) | k_rx_pin_6, /**< PC6 (pin 61) */
00873 k_rx_pc_7 = (k_rx_port_c « k_port_shift) | k_rx_pin_7, /**< PC7 (pin 60) */
00874
00875 /* Port D (0x0D) - Pins 0-7 */
00876 k_rx_pd_0 = (k_rx_port_d « k_port_shift) | k_rx_pin_0, /**< PD0 (pin 126) */
00877 k_rx_pd_1 = (k_rx_port_d « k_port_shift) | k_rx_pin_1, /**< PD1 (pin 125) */
00878 k_rx_pd_2 = (k_rx_port_d « k_port_shift) | k_rx_pin_2, /**< PD2 (pin 124) */
00879 k_rx_pd_3 = (k_rx_port_d « k_port_shift) | k_rx_pin_3, /**< PD3 (pin 123) */
00880 k_rx_pd_4 = (k_rx_port_d « k_port_shift) | k_rx_pin_4, /**< PD4 (pin 122) */
00881 k_rx_pd_5 = (k_rx_port_d « k_port_shift) | k_rx_pin_5, /**< PD5 (pin 121) */
00882 k_rx_pd_6 = (k_rx_port_d « k_port_shift) | k_rx_pin_6, /**< PD6 (pin 120) */
00883 k_rx_pd_7 = (k_rx_port_d « k_port_shift) | k_rx_pin_7, /**< PD7 (pin 119) */
00884
00885 /* Port E (0x0E) - Pins 0-7 */
00886 k_rx_pe_0 = (k_rx_port_e « k_port_shift) | k_rx_pin_0, /**< PE0 (pin 111) */
00887 k_rx_pe_1 = (k_rx_port_e « k_port_shift) | k_rx_pin_1, /**< PE1 (pin 110) */
00888 k_rx_pe_2 = (k_rx_port_e « k_port_shift) | k_rx_pin_2, /**< PE2 (pin 109) */
00889 k_rx_pe_3 = (k_rx_port_e « k_port_shift) | k_rx_pin_3, /**< PE3 (pin 108) */
00890 k_rx_pe_4 = (k_rx_port_e « k_port_shift) | k_rx_pin_4, /**< PE4 (pin 107) */
00891 k_rx_pe_5 = (k_rx_port_e « k_port_shift) | k_rx_pin_5, /**< PE5 (pin 106) */
00892 k_rx_pe_6 = (k_rx_port_e « k_port_shift) | k_rx_pin_6, /**< PE6 (pin 102) */
00893 k_rx_pe_7 = (k_rx_port_e « k_port_shift) | k_rx_pin_7, /**< PE7 (pin 101) */
00894
00895 /* Port F (0x0F) - Pins 0-7 - N/A on 144-pin */
00896 k_rx_pf_0 = (k_rx_port_f « k_port_shift) | k_rx_pin_0, /**< PF0 - N/A on 144-pin */
00897 k_rx_pf_1 = (k_rx_port_f « k_port_shift) | k_rx_pin_1, /**< PF1 - N/A on 144-pin */
00898 k_rx_pf_2 = (k_rx_port_f « k_port_shift) | k_rx_pin_2, /**< PF2 - N/A on 144-pin */
00899 k_rx_pf_3 = (k_rx_port_f « k_port_shift) | k_rx_pin_3, /**< PF3 - N/A on 144-pin */
00900 k_rx_pf_4 = (k_rx_port_f « k_port_shift) | k_rx_pin_4, /**< PF4 - N/A on 144-pin */
00901 k_rx_pf_5 = (k_rx_port_f « k_port_shift) | k_rx_pin_5, /**< PF5 (pin 9) */
00902 k_rx_pf_6 = (k_rx_port_f « k_port_shift) | k_rx_pin_6, /**< PF6 - N/A on 144-pin */
00903 k_rx_pf_7 = (k_rx_port_f « k_port_shift) | k_rx_pin_7, /**< PF7 - N/A on 144-pin */
00904
00905 /* Port G (0x10) - Pins 0-7 - N/A on 144-pin */
00906 k_rx_pg_0 = (k_rx_port_g « k_port_shift) | k_rx_pin_0, /**< PG0 - N/A on 144-pin */
00907 k_rx_pg_1 = (k_rx_port_g « k_port_shift) | k_rx_pin_1, /**< PG1 - N/A on 144-pin */
00908 k_rx_pg_2 = (k_rx_port_g « k_port_shift) | k_rx_pin_2, /**< PG2 - N/A on 144-pin */
00909 k_rx_pg_3 = (k_rx_port_g « k_port_shift) | k_rx_pin_3, /**< PG3 - N/A on 144-pin */
00910 k_rx_pg_4 = (k_rx_port_g « k_port_shift) | k_rx_pin_4, /**< PG4 - N/A on 144-pin */
00911 k_rx_pg_5 = (k_rx_port_g « k_port_shift) | k_rx_pin_5, /**< PG5 - N/A on 144-pin */
00912 k_rx_pg_6 = (k_rx_port_g « k_port_shift) | k_rx_pin_6, /**< PG6 - N/A on 144-pin */
00913 k_rx_pg_7 = (k_rx_port_g « k_port_shift) | k_rx_pin_7, /**< PG7 - N/A on 144-pin */
00914
00915 /* Port J (0x12) - Pins 0-7 */
00916 k_rx_pj_0 = (k_rx_port_j « k_port_shift) | k_rx_pin_0, /**< PJ0 - N/A on 144-pin */
00917 k_rx_pj_1 = (k_rx_port_j « k_port_shift) | k_rx_pin_1, /**< PJ1 - N/A on 144-pin */
00918 k_rx_pj_2 = (k_rx_port_j « k_port_shift) | k_rx_pin_2, /**< PJ2 - N/A on 144-pin */
00919 k_rx_pj_3 = (k_rx_port_j « k_port_shift) | k_rx_pin_3, /**< PJ3 (pin 13) */
00920 k_rx_pj_4 = (k_rx_port_j « k_port_shift) | k_rx_pin_4, /**< PJ4 - N/A on 144-pin */
00921 k_rx_pj_5 = (k_rx_port_j « k_port_shift) | k_rx_pin_5, /**< PJ5 (pin 11) */
00922 k_rx_pj_6 = (k_rx_port_j « k_port_shift) | k_rx_pin_6, /**< PJ6 - N/A on 144-pin */
00923 k_rx_pj_7 = (k_rx_port_j « k_port_shift) | k_rx_pin_7, /**< PJ7 - N/A on 144-pin */
00924 } rx_port_pin_t;
00925
00926 /*
=====
00927 * Compile-Time Verification (Static Assertions)
00928 *
=====
00929 */
00930
00931 /**
00932 * @enum rx_port_verify_t
00933 * @brief Expected values used in static assertion verification
00934 *
00935 * @details
00936 * Provides named constants for the right-hand sides of port/pin static
00937 * assertions, eliminating magic numbers from assertion expressions.
00938 *
00939 * @since Version 1.0.0
00940 */
00941 typedef enum : uint16_t {
00942     k_verify_port_0 = 0x0000U, /**< Expected value of k_rx_port_0 (0x00) */
00943     k_verify_port_1 = 0x0001U, /**< Expected value of k_rx_port_1 (0x01) */
00944     k_verify_port_5 = 0x0005U, /**< Expected value of k_rx_port_5 (0x05) */
00945     k_verify_port_a = 0x000AU, /**< Expected value of k_rx_port_a (0x0A) */
00946     k_verify_port_b = 0x000BU, /**< Expected value of k_rx_port_b (0x0B) */
00947     k_verify_port_c = 0x000CU, /**< Expected value of k_rx_port_c (0x0C) */
00948     k_verify_port_e = 0x000EU, /**< Expected value of k_rx_port_e (0x0E) */
00949     k_verify_port_j =
00950         0x0012U, /**< Expected value of k_rx_port_j (0x12) - follows Port H; Port I does not exist */
00951     k_verify_pin_2 = 0x0002U, /**< Expected value of k_rx_pin_2 (2) */
00952     k_verify_pin_7 = 0x0007U, /**< Expected value of k_rx_pin_7 (7) */
00953     k_verify_port_shift = 0x0008U, /**< Expected value of k_port_shift (8) */
00954     k_verify_enc_pb_2 = 0x0B02U, /**< Expected encoding of Port B Pin 2 */
00955     k_verify_enc_pe_5 = 0x0E05U, /**< Expected encoding of Port E Pin 5 */

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00956 k_verify_enc_p0_7 = 0x0007U, /**< Expected encoding of Port 0 Pin 7 */
00957 k_verify_p0_4 = 0x0004U, /**< Expected value of k_rx_p0_4 */
00958 k_verify_p2_5 = 0x0205U, /**< Expected value of k_rx_p2_5 */
00959 k_verify_pa_2 = 0x0A02U, /**< Expected value of k_rx_pa_2 */
00960 k_verify_pb_2 = 0x0B02U, /**< Expected value of k_rx_pb_2 */
00961 k_verify_pe_5 = 0x0E05U, /**< Expected value of k_rx_pe_5 */
00962 k_verify_pj_3 = 0x1203U, /**< Expected value of k_rx_pj_3 */
00963 } rx_port_verify_t;
00964
00965 /*
00966  * Verify port number constants match expected hex values.
00967  * These assertions ensure that the port numbering matches Renesas RX72N
00968  * documentation and the encoding used in hardware_pinout.h.
00969  */
00970 static_assert((bool)((uint16_t)k_rx_port_0 == k_verify_port_0), "Port 0 must be 0x00");
00971 static_assert((bool)((uint16_t)k_rx_port_1 == k_verify_port_1), "Port 1 must be 0x01");
00972 static_assert((bool)((uint16_t)k_rx_port_5 == k_verify_port_5), "Port 5 must be 0x05");
00973 static_assert((bool)((uint16_t)k_rx_port_a == k_verify_port_a), "Port A must be 0x0A");
00974 static_assert((bool)((uint16_t)k_rx_port_b == k_verify_port_b), "Port B must be 0x0B");
00975 static_assert((bool)((uint16_t)k_rx_port_c == k_verify_port_c), "Port C must be 0x0C");
00976 static_assert((bool)((uint16_t)k_rx_port_e == k_verify_port_e), "Port E must be 0x0E");
00977 static_assert((bool)((uint16_t)k_rx_port_j == k_verify_port_j),
00978               "Port J must be 0x12 (PORTJ.PDR=0x0008C012; Port I does not exist)");
00979
00980 /*
00981  * Verify pin number constants match expected values.
00982  */
00983 static_assert((bool)((uint16_t)k_rx_pin_0 == 0), "Pin 0 must be 0");
00984 static_assert((bool)((uint16_t)k_rx_pin_2 == k_verify_pin_2), "Pin 2 must be 2");
00985 static_assert((bool)((uint16_t)k_rx_pin_7 == k_verify_pin_7), "Pin 7 must be 7");
00986 static_assert((bool)((uint16_t)k_rx_pin_min == 0), "Minimum pin number must be 0");
00987 static_assert((bool)((uint16_t)k_rx_pin_max == k_verify_pin_7), "Maximum pin number must be 7");
00988
00989 /*
00990  * Verify shift constant is correct.
00991  */
00992 static_assert((bool)((uint16_t)k_port_shift == k_verify_port_shift), "Port shift must be 8 bits");
00993
00994 /*
00995  * Verify encoding matches expected pattern: (port « 8) | pin
00996  * Example: Port B (0x0B), Pin 2 (0x02) = 0x0B02
00997  */
00998 static_assert((bool)((uint16_t)((k_rx_port_b « k_port_shift) | k_rx_pin_2) == k_verify_enc_pb_2),
00999               "Port B Pin 2 encoding must be 0x0B02");
01000 static_assert((bool)((uint16_t)((k_rx_port_e « k_port_shift) | k_rx_pin_5) == k_verify_enc_pe_5),
01001               "Port E Pin 5 encoding must be 0x0E05");
01002 static_assert((bool)((uint16_t)((k_rx_port_0 « k_port_shift) | k_rx_pin_7) == k_verify_enc_p0_7),
01003               "Port 0 Pin 7 encoding must be 0x0007");
01004
01005 /*
01006  * Verify pre-computed port/pin combinations are correct.
01007  */
01008 static_assert((bool)((unsigned int)k_rx_p0_4 == (unsigned int)k_verify_p0_4), "P04 must be 0x0004");
01009 static_assert((bool)((unsigned int)k_rx_p2_5 == (unsigned int)k_verify_p2_5), "P25 must be 0x0205");
01010 static_assert((bool)((unsigned int)k_rx_pa_2 == (unsigned int)k_verify_pa_2), "PA2 must be 0x0A02");
01011 static_assert((bool)((unsigned int)k_rx_pb_2 == (unsigned int)k_verify_pb_2), "PB2 must be 0x0B02");
01012 static_assert((bool)((unsigned int)k_rx_pe_5 == (unsigned int)k_verify_pe_5), "PE5 must be 0x0E05");
01013 static_assert((bool)((unsigned int)k_rx_pj_3 == (unsigned int)k_verify_pj_3), "PJ3 must be 0x1203");
01014
01015 /*
=====
01016  * Pin Extraction Inline Functions
01017  */
=====
01018 /*
01019
01020 /**
01021  * @brief Extract port number from combined port/pin identifier
01022  *
01023  * @details
01024  *   Decodes the port number (upper byte) from a 16-bit gpio_pin_t value using
01025  *   right-shift by k_port_shift (8 bits). This extracts bits 15-8 which contain
01026  *   the rx_port_number_t value.
01027  *
01028  * ## Algorithm Steps
01029  *
01030  * 1. **Validate pin portion**: Assert that lower byte (pin) is in valid range [0-7]
01031  * 2. **Extract port**: Right-shift by 8 bits to move port from upper to lower byte
01032  * 3. **Validate result**: Assert that extracted port is valid RX72N port number
01033  * 4. **Return**: Return port number as uint8_t
01034  *
01035  * ## Bit-Level Operation
01036  *
01037  * Given: gpio_pin_t = 0xPPnn (PP = port, nn = pin)
01038  * Operation: result = gpio_pin_t » 8
01039  * Result: result = 0x00PP
01040  */

```



```

01041 * Example: k_rx_pb_2 = 0x0B02
01042 * ```
01043 * Input: 0x0B02 = 0000 1011 0000 0010
01044 * Shift: » 8
01045 * Output: 0x000B = 0000 0000 0000 1011
01046 * Result: 0x0B (k_rx_port_b)
01047 * ```
01048 *
01049 * ## Performance Analysis
01050 *
01051 * - **Execution time**: 1-2 CPU cycles (inline right-shift, assertions compiled out in release)
01052 * - **Memory usage**: Zero (inline function, no stack frame)
01053 * - **Code size**: ~4 bytes machine code (single shift instruction)
01054 *
01055 * @param[in] pin Combined port/pin identifier (rx_port_pin_t or gpio_pin_t).
01056 *             Valid range: 0x0000-0x1307 (all valid port/pin combinations).
01057 *             Must have valid port in upper byte and valid pin (0-7) in lower byte.
01058 *
01059 * @return Port number extracted from upper byte (rx_port_number_t).
01060 *         Valid range: 0x00-0x10, or 0x12 (k_rx_port_j).
01061 *         Returned value is one of: k_rx_port_0..k_rx_port_9, k_rx_port_a..k_rx_port_g, k_rx_port_j.
01062 *
01063 * @pre pin lower byte (bits 7-0) must be valid pin number (0-7)
01064 * @pre pin upper byte (bits 15-8) must be valid port number (0x00-0x10 or 0x13)
01065 * @pre pin must be properly encoded using (port « 8) | pin_num scheme
01066 *
01067 * @post Returned value is valid rx_port_number_t (0x00-0x10 or 0x13)
01068 * @post Returned value is deterministic (same input always produces same output)
01069 * @post No side effects (function does not modify any state)
01070 *
01071 * @invariant (rx_port_from_pin((port « 8) | pin_num) == port) for all valid port/pin
01072 * @invariant Result fits in uint8_t (always <= 0xFF)
01073 *
01074 * @note **Thread-safe**: Yes - pure function with no shared state
01075 * @note **Re-entrant**: Yes - no static variables or side effects
01076 * @note **Performance**: Optimal - compiles to single right-shift instruction
01077 * @note **Memory**: Zero overhead - inline function, no stack frame
01078 * @note **Assertions**: Compiled out in release builds (NDEBUG defined)
01079 *
01080 * @warning Passing invalid pin values (pin portion > 7) triggers assertion failure
01081 * @warning Result undefined if pin not constructed using proper encoding scheme
01082 * @attention This function validates input but does NOT check physical pin availability
01083 *
01084 * @par Example (Basic Usage):
01085 * @code{.c}
01086 * // Extract port from pre-computed constant
01087 * gpio_pin_t pin = k_rx_pb_2; // 0x0B02
01088 * rx_port_number_t port = rx_port_from_pin(pin);
01089 * // Result: k_rx_port_b (0x0B)
01090 * @endcode
01091 *
01092 * @par Example (Manual Construction):
01093 * @code{.c}
01094 * // Construct pin value, then extract port
01095 * gpio_pin_t pin = (k_rx_port_e « k_port_shift) | k_rx_pin_5; // 0x0E05
01096 * rx_port_number_t port = rx_port_from_pin(pin);
01097 * // Result: k_rx_port_e (0x0E)
01098 *
01099 * // Verify round-trip
01100 * assert(port == k_rx_port_e);
01101 * @endcode
01102 *
01103 * @par Example (Switch Statement):
01104 * @code{.c}
01105 * // Use extracted port in switch
01106 * gpio_pin_t pin = get_configured_pin();
01107 * rx_port_number_t port = rx_port_from_pin(pin);
01108 *
01109 * switch (port) {
01110 *     case k_rx_port_b: return portb();
01111 *     case k_rx_port_e: return porte();
01112 *     default: return nullptr;
01113 * }
01114 * @endcode
01115 *
01116 * @par Example (Complete Port/Pin Decomposition):
01117 * @code{.c}
01118 * // Fully decompose gpio_pin_t into components
01119 * gpio_pin_t pin = k_rx_pa_7; // 0x0A07
01120 *
01121 * rx_port_number_t port = rx_port_from_pin(pin); // 0x0A (Port A)
01122 * rx_pin_number_t pin_num = rx_pin_from_pin(pin); // 0x07 (Pin 7)
01123 *
01124 * printf("Pin 0x%04X = Port 0x%02X, Pin %d\n", pin, port, pin_num);
01125 * // Output: "Pin 0x0A07 = Port 0x0A, Pin 7"
01126 * @endcode
01127 *

```

```

01128 * @see rx_pin_from_pin() Extract pin number from combined identifier
01129 * @see rx_port_pin_t Pre-computed port/pin combinations
01130 * @see port_encoding_t Encoding constants (k_port_shift, k_port_mask)
01131 * @see rx_port_number_t Valid port numbers that can be returned
01132 *
01133 * @since Version 1.0.0
01134 * @test test_rx_port_from_pin() Unit tests verify all 144 port/pin combinations
01135 *
01136 * @par NASA Power of 10 Compliance:
01137 * - **Rule 5**: [OK] 2 preconditions + 3 postconditions (5 assertions total)
01138 * - **Rule 4**: [OK] 5 lines of code (excluding comments/braces)
01139 */
01140 static inline uint8_t rx_port_from_pin(rx_port_pin_t pin)
01141 {
01142     /* Pre-condition: pin portion fits encoding scheme (k_rx_pin_min == 0, so >= 0 is always
01143      * true for unsigned types; only the upper bound requires an explicit check) */
01144     RX_ASSERT((pin & k_port_mask) <= k_rx_pin_max, "Pin portion must be <= k_rx_pin_max");
01145
01146     uint8_t result = (uint8_t)((pin) » k_port_shift);
01147
01148     /* Post-condition: result is a valid port (0-G or J, skipping H-I) */
01149     /* Note: result >= k_rx_port_0 is always true for uint8_t since k_rx_port_0 == 0 */
01150     RX_ASSERT((result <= k_rx_port_g) || (result == k_rx_port_j),
01151         "Port number must be valid (k_rx_port_0..k_rx_port_g or k_rx_port_j)");
01152
01153     return result;
01154 }
01155
01156 /**
01157  * @brief Extract pin number from combined port/pin identifier
01158  *
01159  * @details
01160  * Decodes the pin number (lower byte) from a 16-bit gpio_pin_t value using
01161  * bitwise AND with k_port_mask (0xFF). This extracts bits 7-0 which contain
01162  * the rx_pin_number_t value.
01163  *
01164  * ## Algorithm Steps
01165  *
01166  * 1. **Extract port**: Right-shift by 8 bits to get port number (for validation)
01167  * 2. **Validate port**: Assert that port is valid RX72N port number
01168  * 3. **Extract pin**: Mask lower byte with 0xFF to get pin number
01169  * 4. **Validate result**: Assert that extracted pin is in valid range [0-7]
01170  * 5. **Return**: Return pin number as uint8_t
01171  *
01172  * ## Bit-Level Operation
01173  *
01174  * Given: gpio_pin_t = 0xPPnn (PP = port, nn = pin)
01175  * Operation: result = gpio_pin_t & 0xFF
01176  * Result: result = 0x00nn
01177  *
01178  * Example: k_rx_pe_5 = 0x0E05
01179  * ```
01180  * Input:  0x0E05 = 0000 1110 0000 0101
01181  * Mask:   0x00FF = 0000 0000 1111 1111
01182  * AND:    = 0000 0000 0000 0101
01183  * Output: 0x0005 = 5
01184  * Result: 5 (k_rx_pin_5)
01185  * ```
01186  *
01187  * ## Performance Analysis
01188  *
01189  * - **Execution time**: 2-3 CPU cycles (shift + mask, assertions compiled out in release)
01190  * - **Memory usage**: Zero (inline function, no stack frame)
01191  * - **Code size**: ~6 bytes machine code (shift + mask instructions)
01192  *
01193  * @param[in] pin Combined port/pin identifier (rx_port_pin_t or gpio_pin_t).
01194  * Valid range: 0x0000-0x1307 (all valid port/pin combinations).
01195  * Must have valid port in upper byte and valid pin (0-7) in lower byte.
01196  *
01197  * @return Pin number extracted from lower byte (rx_pin_number_t).
01198  * Valid range: 0-7 (one of k_rx_pin_0..k_rx_pin_7).
01199  * Corresponds to bit position in port data registers (PDR, PODR, PIDR).
01200  *
01201  * @pre pin upper byte (bits 15-8) must be valid port number (0x00-0x10 or 0x13)
01202  * @pre pin lower byte (bits 7-0) must be valid pin number (0-7)
01203  * @pre pin must be properly encoded using (port « 8) | pin_num scheme
01204  *
01205  * @post Returned value is valid rx_pin_number_t (0-7)
01206  * @post Returned value is deterministic (same input always produces same output)
01207  * @post No side effects (function does not modify any state)
01208  *
01209  * @invariant (rx_pin_from_pin((port « 8) | pin_num) == pin_num) for all valid port/pin
01210  * @invariant Result <= k_rx_pin_max (7)
01211  * @invariant Result >= k_rx_pin_min (0)
01212  *
01213  * @note **Thread-safe**: Yes - pure function with no shared state
01214  * @note **Re-entrant**: Yes - no static variables or side effects

```

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01215 * @note **Performance**: Optimal - compiles to single mask instruction (after validation)
01216 * @note **Memory**: Zero overhead - inline function, no stack frame
01217 * @note **Assertions**: Compiled out in release builds (NDEBUG defined)
01218 * @note **Bit Position**: Returned value corresponds to bit position in port registers
01219 *
01220 * @warning Passing invalid port values (upper byte invalid) triggers assertion failure
01221 * @warning Result undefined if pin not constructed using proper encoding scheme
01222 * @attention This function validates input but does NOT check physical pin availability
01223 * @attention Pin number 0-7 corresponds to LSB-MSB of 8-bit port registers
01224 *
01225 * @par Example (Basic Usage):
01226 * @code{.c}
01227 * // Extract pin number from pre-computed constant
01228 * gpio_pin_t pin = k_rx_pe_5; // 0x0E05
01229 * rx_pin_number_t pin_num = rx_pin_from_pin(pin);
01230 * // Result: 5 (k_rx_pin_5)
01231 * @endcode
01232 *
01233 * @par Example (Bit Manipulation):
01234 * @code{.c}
01235 * // Use extracted pin number for bit manipulation
01236 * gpio_pin_t pin = k_rx_pb_2; // Port B, Pin 2
01237 * rx_pin_number_t pin_num = rx_pin_from_pin(pin); // Returns 2
01238 *
01239 * PORT_Type* port = portb();
01240 * port->PODR |= (1 « pin_num); // Set bit 2 high
01241 * port->PODR &= ~(1 « pin_num); // Clear bit 2 low
01242 * @endcode
01243 *
01244 * @par Example (Complete Port/Pin Decomposition):
01245 * @code{.c}
01246 * // Fully decompose gpio_pin_t into components
01247 * gpio_pin_t pin = k_rx_pc_7; // 0x0C07
01248 *
01249 * rx_port_number_t port = rx_port_from_pin(pin); // 0x0C (Port C)
01250 * rx_pin_number_t pin_num = rx_pin_from_pin(pin); // 7 (Pin 7)
01251 *
01252 * // Use in hardware access
01253 * PORT_Type* port_regs = get_port_registers(port);
01254 * bool pin_state = (port_regs->PIDR » pin_num) & 1;
01255 * @endcode
01256 *
01257 * @par Example (Round-Trip Verification):
01258 * @code{.c}
01259 * // Verify encoding/decoding round-trip
01260 * rx_port_number_t original_port = k_rx_port_a;
01261 * rx_pin_number_t original_pin = k_rx_pin_3;
01262 *
01263 * // Encode
01264 * gpio_pin_t encoded = (original_port « k_port_shift) | original_pin;
01265 *
01266 * // Decode
01267 * rx_port_number_t decoded_port = rx_port_from_pin(encoded);
01268 * rx_pin_number_t decoded_pin = rx_pin_from_pin(encoded);
01269 *
01270 * // Verify
01271 * assert(decoded_port == original_port); // Port A
01272 * assert(decoded_pin == original_pin); // Pin 3
01273 * @endcode
01274 *
01275 * @par Example (Array Indexing):
01276 * @code{.c}
01277 * // Use pin number to index into pin configuration array
01278 * gpio_pin_t configured_pins[] = {
01279 *   k_rx_pb_0, k_rx_pb_2, k_rx_pb_5, k_rx_pb_7
01280 * };
01281 *
01282 * for (size_t i = 0; i < 4; i++) {
01283 *   rx_pin_number_t pin_num = rx_pin_from_pin(configured_pins[i]);
01284 *   printf("Pin %d is configured\n", pin_num);
01285 * }
01286 * @endcode
01287 *
01288 * @see rx_port_from_pin() Extract port number from combined identifier
01289 * @see rx_port_pin_t Pre-computed port/pin combinations
01290 * @see port_encoding_t Encoding constants (k_port_shift, k_port_mask)
01291 * @see rx_pin_number_t Valid pin numbers that can be returned
01292 *
01293 * @since Version 1.0.0
01294 * @test test_rx_pin_from_pin() Unit tests verify all 144 port/pin combinations
01295 *
01296 * @par NASA Power of 10 Compliance:
01297 * - **Rule 5**: [OK] 2 preconditions + 3 postconditions (5 assertions total)
01298 * - **Rule 4**: [OK] 6 lines of code (excluding comments/braces)
01299 */
01300 static inline uint8_t rx_pin_from_pin(rx_port_pin_t pin)
01301 {

```

```

01302 /* Pre-condition 1: port value fits encoding scheme (port in upper byte) */
01303 uint8_t port = (uint8_t)((pin) » k_port_shift);
01304 /* Note: port >= k_rx_port_0 is always true for uint8_t since k_rx_port_0 == 0 */
01305
01306 /* Pre-condition 2: port is valid (0-G or J, skipping H-I) */
01307 RX_ASSERT((port <= k_rx_port_g) || (port == k_rx_port_j),
01308           "Port number must be valid (k_rx_port_0..k_rx_port_g or k_rx_port_j)");
01309
01310 uint8_t result = (uint8_t)((pin)&k_port_mask);
01311
01312 /* Post-condition: result fits in rx_pin_number_t range (0x00-0x07) */
01313 /* Note: result >= k_rx_pin_min is always true for uint8_t since k_rx_pin_min == 0 */
01314 RX_ASSERT(result <= k_rx_pin_max, "Pin number must be in range k_rx_pin_min..k_rx_pin_max");
01315
01316 return result;
01317 }
01318
01319 #ifdef __cplusplus
01320 }
01321 #endif

```

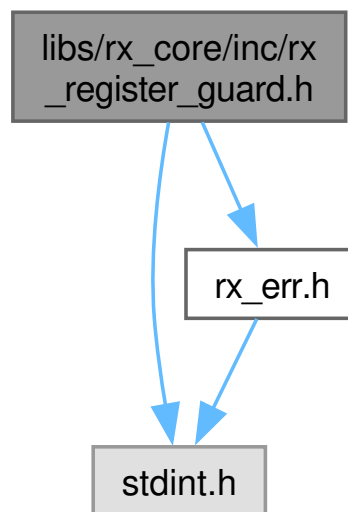
7.31 libs/rx_core/inc/rx_register_guard.h File Reference

Register Guard - Periodic Refresh for ESD/EMI Protection.

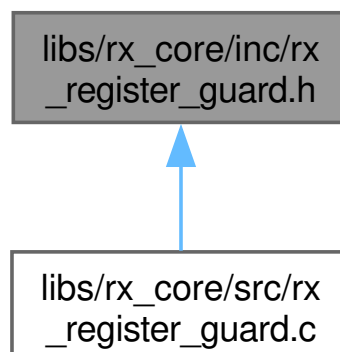
```
#include <stdint.h>
```

```
#include "rx_err.h"
```

Include dependency graph for rx_register_guard.h:



This graph shows which files directly or indirectly include this file:



Enumerations

- enum `rx_register_guard_config_t` : `uint8_t { k_register_guard_max_ports = 10 , k_register_guard_max_ipr = 16 }`
Register guard configuration constants.

Functions

- `rx_err_t rx_register_guard_init` (void)
Initialize register guard module and capture golden reference values.
- void `rx_register_guard_refresh` (void)
Refresh all critical registers by comparing against golden reference.
- `uint32_t rx_register_guard_get_correction_count` (void)
Get count of register corrections since last reset.
- void `rx_register_guard_reset_count` (void)
Reset the correction counter to zero.
- bool `rx_register_guard_is_initialized` (void)
Check if register guard module is initialized.

7.31.1 Detailed Description

Register Guard - Periodic Refresh for ESD/EMI Protection.

Provides periodic refresh of critical hardware registers to recover from ESD (Electrostatic Discharge) or EMI (Electromagnetic Interference) induced bit flips that could corrupt system configuration in harsh electrical environments.

7.31.1.1 Problem Statement

In mobile robotics applications with:

- High-current motor drivers (10A+ peak)
- PWM switching at 20 kHz
- Long cable runs (>1m to sensors)
- Brushed DC motors (arcing at commutator)

Transient voltage spikes can couple into the microcontroller and corrupt configuration registers through:

- Supply voltage glitches
- Ground bounce
- Radiated EMI pickup on PCB traces
- ESD events from handling or environment

A single bit flip in a critical register can cause catastrophic failures:

- GPIO direction change -> short circuit
- Interrupt priority inversion -> deadlock
- Module stop bit set -> peripheral disabled

7.31.1.2 Solution Approach

This module implements a "register refresh" pattern:

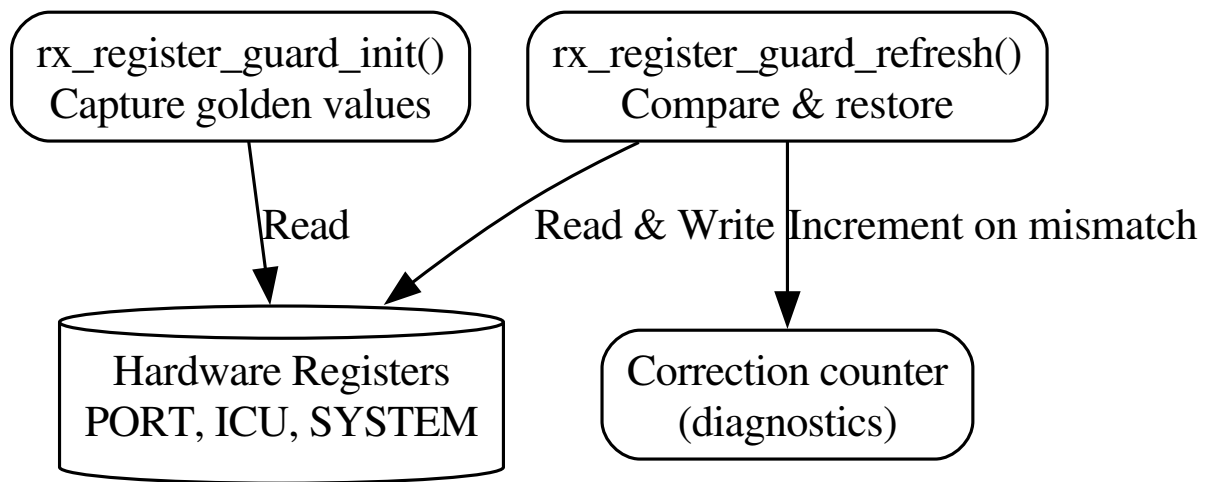
1. Capture known-good register values at initialization
2. Periodically compare current values against golden reference
3. Restore any registers that changed unexpectedly
4. Count corrections for diagnostics/logging

The refresh is non-intrusive and completes in $<5\text{ us}$ @ 240 MHz, making it suitable for 1-10 Hz refresh rates from the main control loop.

7.31.1.3 Protected Registers

Register Type	Purpose	Failure Mode if Corrupted
PORT.PDR	GPIO direction control	Short circuits, floating inputs
ICU.IPR	Interrupt priority levels	Priority inversion deadlock
SYSTEM.MSTPCR	Module stop control	Peripherals unexpectedly disabled

7.31.1.4 System Architecture



7.31.1.5 Performance Characteristics

- Memory usage: 52 bytes static (10 PORTs + 16 IPRs + 4 MSTCPRs)
- Execution time: $\sim 4.2\text{ us}$ @ 240 MHz (worst case, all registers checked)
- Interrupt latency: $\sim 200\text{ ns}$ during PRCR unlock (brief interrupt disable)
- Recommended refresh rate: 1-10 Hz (balance detection speed vs overhead)

7.31.1.6 Hardware Requirements

Requirement	Value	Notes
MCU	RX72N	Uses RX72N-specific register addresses
CPU Speed	≥ 120 MHz	Timing verified at 240 MHz
RAM	52 bytes	Static storage for golden values
PRCR Support	Required	For MSTPCR write protection unlock

Module Dependencies:

- [rx_err.h](#) - Error code definitions
- [rx72n_regs.h](#) - Register definitions
- [rx_register_protection.h](#) - PRCR management

NASA Power of 10 Compliance:

- Rule 1: [OK] No goto, setjmp, recursion
- Rule 2: [OK] All loops bounded by enum constants
- Rule 3: [OK] Zero dynamic allocation (all data static)
- Rule 4: [OK] All functions <60 lines
- **Rule 5**: [OK] Minimum 2 pre/post conditions per function
- **Rule 6**: [OK] Data declared at smallest scope
- **Rule 7**: [OK] Return values checked (or void)
- **Rule 8**: [OK] C23 typed enums, no magic numbers
- **Rule 9**: [WARN] Direct register access (unavoidable for this module)
- **Rule 10**: [OK] Compiled with -Wall -Wextra -Werror

SOLID Principles:

- **Single Responsibility**: Only handles register refresh, no other concerns
- **Open/Closed**: Extensible via compile-time register list configuration
- **Liskov Substitution**: N/A (not polymorphic)
- **Interface Segregation**: Minimal 5-function API, each with clear purpose
- **Dependency Inversion**: Depends on abstract [rx_err_t](#), not implementation

Example - Basic Usage:

```
// Initialize at startup (after all peripherals configured)
rx_err_t err = rx_register_guard_init();
if (err != k_rx_ok) {
    rx_log_error("GUARD", "Failed to initialize register guard");
    return err;
}

// In main control loop (100 Hz)
while (1) {
    // ... main control logic ...

    // Refresh every 100 iterations = 1 Hz refresh rate
    static uint32_t refresh_counter = 0;
    if (++refresh_counter >= k_refresh_interval_100hz) {
        rx_register_guard_refresh();
    }
}
```

```

    refresh_counter = 0;

    // Log corrections if any occurred
    uint32_t corrections = rx_register_guard_get_correction_count();
    if (corrections > 0) {
        rx_log_warn("GUARD", "Corrected %lu registers", corrections);
        rx_register_guard_reset_count();
    }
}

tx_thread_sleep(1); // 10 ms = 100 Hz
}

```

Example - Diagnostics & Monitoring:

```

// Telemetry task - report ESD/EMI events
void telemetry_task(void) {
    static uint32_t last_correction_count = 0;

    uint32_t current_count = rx_register_guard_get_correction_count();
    if (current_count != last_correction_count) {
        uint32_t new_events = current_count - last_correction_count;
        rx_log_warn("TELEM", "EMI events detected: %lu", new_events);

        // Send to ground station for analysis
        send_telemetry(TELEMETRY_EMI_EVENT, new_events);

        last_correction_count = current_count;
    }
}

```

Example - Re-initialization After Configuration Change:

```

// Reconfigure GPIO for different mode
void enter_low_power_mode(void) {
    // Change GPIO configuration
    PORT1.PDR.BYTE = 0x00; // All inputs
    PORT2.PDR.BYTE = 0x00;

    // Update register guard golden values
    rx_register_guard_init(); // Capture new configuration

    // Now refresh will maintain low-power GPIO state
}

```

See also

[RX72N User's Manual Chapter 13: I/O Ports](#)
[RX72N User's Manual Chapter 14: Interrupt Controller](#)
[RX72N User's Manual Chapter 9: Module Stop Function](#)

Author

Locked, Inc.

Date

2026-01-27

Version

1.0.0

Copyright

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Since

Version 1.0.0

Definition in file [rx_register_guard.h](#).

7.31.2 Enumeration Type Documentation

7.31.2.1 rx_register_guard_config_t

enum [rx_register_guard_config_t](#) : uint8_t

Register guard configuration constants.

Defines compile-time limits for register guard arrays. These values balance memory usage against protection coverage. Sizing is based on STAR project peripheral usage (4 motors = 8 GPIOs, plus sensors and communication).

Memory allocation:

- PORT guards: 10 x 1 byte = 10 bytes
- IPR guards: 16 x 1 byte = 16 bytes
- MSTPCR guards: 4 x 4 bytes = 16 bytes (hardcoded in .c file)
- Metadata: 10 bytes (counters, flags)
- Total: 52 bytes static RAM

Field Constraints:

Field	Min	Max	Rationale
max_ports	1	16	RX72N has 16 PORT registers (PORT0-PORTF)
max_ipr	1	256	RX72N has 256 interrupt priority registers

Usage Example:

```
// Configuration is embedded in implementation, not user-configurable
// These constants define array sizes at compile time
static uint8_t s_golden_pdr[k_register_guard_max_ports];
static uint8_t s_golden_ipr[k_register_guard_max_ipr];
```

See also

[rx_register_guard_init\(\)](#) Uses these limits for array allocation

Since

Version 1.0.0

Enumerator

k_register_guard_max_ports	Maximum number of PORT registers to guard (STAR project: 10). Protects GPIO direction registers (PDR) for all ports used in STAR: • PORT1: Motor 0 & 1 direction control (2 pins) • PORT2: Motor 2 & 3 direction control (2 pins) • PORT3: Encoder inputs (4 pins) • PORT4: SPI peripheral select (2 pins) • PORT5: LED indicators (2 pins) • PORT6: UART TX/RX (2 pins) • PORT7: I2C SDA/SCL (2 pins) • PORT8: Unused (reserved) • PORT9: Unused (reserved) • PORT10: Unused (reserved) Value: 10 ports Rationale: Covers all active GPIO ports plus 3 spares for expansion Memory cost: 10 bytes Impact of Changing: • Increase: More memory used, more ports protected • Decrease: Less memory, but risk missing critical GPIO protection Warning If your design uses >10 GPIO ports, increase this value
-----------------------------------	--

k_register_guard_max_ipr	Maximum number of IPR registers to guard (STAR project: 16). Protects interrupt priority registers for critical system interrupts: • CMT0 (control loop timer) - IPR priority 5 • SPI0 (communication) - IPR priority 4 • USB (CDC debug) - IPR priority 3 • UART0 (backup debug) - IPR priority 3 • GPT0-3 (PWM generation) - IPR priority 2 • MTU0-3 (encoder capture) - IPR priority 2 • I2C (sensor bus) - IPR priority 1 • ADC (current sensing) - IPR priority 1 Value: 16 IPR registers Rationale: Covers all 12 active interrupt sources plus 4 spares Memory cost: 16 bytes Impact of Changing: • Increase: More memory used, more interrupts protected • Decrease: Less memory, but risk priority inversion deadlock Warning Priority inversion (high-priority interrupt blocked by low) causes hard-to-debug deadlocks. Protect ALL critical interrupt priorities.
--------------------------	--

Definition at line 234 of file [rx_register_guard.h](#).

```

00234      : uint8_t {
00235  /**
00236   * @brief Maximum number of PORT registers to guard (STAR project: 10)
00237   *
00238   * @details
00239   * Protects GPIO direction registers (PDR) for all ports used in STAR:
00240   * - PORT1: Motor 0 & 1 direction control (2 pins)
00241   * - PORT2: Motor 2 & 3 direction control (2 pins)
00242   * - PORT3: Encoder inputs (4 pins)
00243   * - PORT4: SPI peripheral select (2 pins)
00244   * - PORT5: LED indicators (2 pins)
00245   * - PORT6: UART TX/RX (2 pins)
00246   * - PORT7: I2C SDA/SCL (2 pins)
00247   * - PORT8: Unused (reserved)
00248   * - PORT9: Unused (reserved)
00249   * - PORT10: Unused (reserved)
00250   *
00251   * **Value**: 10 ports
00252   * **Rationale**: Covers all active GPIO ports plus 3 spares for expansion
00253   * **Memory cost**: 10 bytes
00254   *
00255   * @par Impact of Changing:
00256   * - **Increase**: More memory used, more ports protected
00257   * - **Decrease**: Less memory, but risk missing critical GPIO protection
00258   *
00259   * @warning If your design uses >10 GPIO ports, increase this value
00260   */
00261  k_register_guard_max_ports = 10,
00262
00263  /**
00264   * @brief Maximum number of IPR registers to guard (STAR project: 16)
00265   *
00266   * @details
00267   * Protects interrupt priority registers for critical system interrupts:
00268   * - CMT0 (control loop timer) - IPR priority 5
00269   * - SPI0 (communication) - IPR priority 4

```

```

00270 * - USB (CDC debug) - IPR priority 3
00271 * - UART0 (backup debug) - IPR priority 3
00272 * - GPT0-3 (PWM generation) - IPR priority 2
00273 * - MTU0-3 (encoder capture) - IPR priority 2
00274 * - I2C (sensor bus) - IPR priority 1
00275 * - ADC (current sensing) - IPR priority 1
00276 *
00277 * **Value**: 16 IPR registers
00278 * **Rationale**: Covers all 12 active interrupt sources plus 4 spares
00279 * **Memory cost**: 16 bytes
00280 *
00281 * @par Impact of Changing:
00282 * - **Increase**: More memory used, more interrupts protected
00283 * - **Decrease**: Less memory, but risk priority inversion deadlock
00284 *
00285 * @warning Priority inversion (high-priority interrupt blocked by low) causes
00286 *         hard-to-debug deadlocks. Protect ALL critical interrupt priorities.
00287 */
00288 k_register_guard_max_ipr = 16,
00289
00290 } rx_register_guard_config_t;

```

7.31.3 Function Documentation

7.31.3.1 rx_register_guard_get_correction_count()

```

uint32_t rx_register_guard_get_correction_count (
    void )

```

Get count of register corrections since last reset.

Returns the total number of register corrections performed since module initialization or since the last call to [rx_register_guard_reset_count\(\)](#). Each individual register mismatch detected during refresh increments this counter by 1.

Non-zero correction counts indicate ESD/EMI events occurred that corrupted hardware registers. High correction rates (>10/second) suggest severe electromagnetic interference requiring hardware investigation.

This function is used for:

- Diagnostics: Detect EMI problems during development
- Telemetry: Report corruption events to ground station
- Logging: Periodic logging of environmental robustness
- Alerting: Trigger warnings when correction rate exceeds threshold

Algorithm:

1. Read static correction counter variable
2. Return value directly (no side effects)

Returns

uint32_t Total number of register corrections since last reset

Return values

0	No register corruption detected (ideal case)
1-10	Occasional corruption (acceptable in noisy environments)
>100	Frequent corruption (investigate EMI sources)
UINT32_MAX	Counter wrapped (extremely rare, ~4 billion corrections)

Precondition

No preconditions - safe to call anytime

Module does not need to be initialized (returns 0 if not initialized)

Postcondition

Counter value unchanged (read-only operation)

Invariant

Counter monotonically increases (never decreases except wrap or reset)

Note

Thread Safety: Thread-safe for reading (atomic load on 32-bit MCU). However, refresh() may increment while reading, causing race condition. For critical monitoring, disable interrupts around read if needed.

Counter Wrap: At UINT32_MAX, counter wraps to 0. With 10 Hz refresh and 1 correction/refresh, wrap occurs after ~13.6 years of continuous operation. Not a practical concern.

Execution Time: ~20 ns @ 240 MHz (single memory read)

Reset Behavior: Counter is cleared on [rx_register_guard_init\(\)](#) and [rx_register_guard_reset_count\(\)](#).

Example - Periodic Diagnostics:

```
// Log correction count every 10 seconds
void periodic_diagnostics(void) {
    static uint32_t last_log_time = 0;
    uint32_t current_time = get_system_time_ms();

    if (current_time - last_log_time >= 10000) { // 10 seconds
        uint32_t corrections = rx_register_guard_get_correction_count();
        rx_log_info("DIAG", "Register corrections: %lu", corrections);

        // Reset for next interval
        rx_register_guard_reset_count();

        last_log_time = current_time;
    }
}
```

Example - EMI Alert Threshold:

```
// Alert if correction rate exceeds threshold
void check_emi_threshold(void) {
    static uint32_t last_count = 0;
    static uint32_t last_check_time = 0;

    uint32_t current_time = get_system_time_ms();
    uint32_t current_count = rx_register_guard_get_correction_count();

    // Calculate corrections per second
    uint32_t time_delta_ms = current_time - last_check_time;
    uint32_t count_delta = current_count - last_count;

    if (time_delta_ms >= 1000) { // Check every second
        float corrections_per_sec = (float)count_delta / (time_delta_ms / 1000.0f);

        if (corrections_per_sec > 10.0f) {
            rx_log_error("EMI", "Severe interference: %.1f corr/sec", corrections_per_sec);
        } else if (corrections_per_sec > 1.0f) {
            rx_log_warn("EMI", "Moderate interference: %.1f corr/sec", corrections_per_sec);
        }

        last_count = current_count;
        last_check_time = current_time;
    }
}
```

Example - Telemetry Reporting:

```
// Include correction count in telemetry packets
void send_telemetry(void) {
    telemetry_packet_t packet = {
        .timestamp = get_system_time_ms(),
        .temperature_celsius = read_temperature_celsius(),
        .motor_current = read_motor_current(),
        .register_corrections = rx_register_guard_get_correction_count(),
        // ... other fields ...
    };

    transmit_telemetry(&packet);
}
```

See also

[rx_register_guard_refresh\(\)](#) Increments this counter on corrections
[rx_register_guard_reset_count\(\)](#) Clear counter to zero
[rx_register_guard_init\(\)](#) Also resets counter to zero

Since

Version 1.0.0

Version

1.0.0

Test Tested in `test_rx_register_guard.c::test_correction_count_increment()`
 Tested in `test_rx_register_guard.c::test_correction_count_before_init()`

NASA Power of 10 Compliance:

- Rule 1: [OK] No goto, setjmp, recursion
- Rule 2: [OK] No loops
- Rule 5: [OK] 2 preconditions, 1 postcondition, 1 invariant

Get count of register corrections since last reset.

Returns the cumulative count of register corrections since module initialization or the last call to [rx_register_guard_reset_count\(\)](#). Each individual register mismatch detected during refresh increments this counter by 1.

Non-zero counts indicate ESD/EMI events that corrupted hardware registers. This value is used for:

- Diagnostics during development
- Telemetry to ground station
- EMI alert threshold monitoring

Algorithm:

1. Check if module is initialized
2. If not initialized, return 0 (defensive)
3. Return current value of s_state.corrections

Precondition

None - safe to call anytime

Postcondition

Counter value unchanged (read-only operation)

Note

Thread Safety: Mostly thread-safe (atomic read on 32-bit MCU). However, refresh() may increment while reading (race condition).

Execution Time: ~18 ns @ 240 MHz

Returns 0 if not initialized (distinguishable only via is_initialized())

Example - Periodic Monitoring:

```
void monitor_emi(void) {
    static uint32_t last_count = 0;
    uint32_t current = rx_register_guard_get_correction_count();

    if (current > last_count) {
        uint32_t new_events = current - last_count;
        rx_log_warn("EMI", "Detected %lu corrections", new_events);
        last_count = current;
    }
}
```

See also

[rx_register_guard_refresh\(\)](#) Increments this counter
[rx_register_guard_reset_count\(\)](#) Clear counter to zero

Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 1: [OK] No goto, setjmp, recursion
- Rule 5: [OK] 1 precondition, 1 postcondition

Definition at line 1047 of file [rx_register_guard.c](#).

```
01048 {  
01049     if (!s_state.initialized) {  
01050         return 0;  
01051     }  
01052  
01053     return s_state.corrections;  
01054 }
```

References [s_state](#).

7.31.3.2 rx_register_guard_init()

```
rx_err_t rx_register_guard_init (  
    void ) [nodiscard]
```

Initialize register guard module and capture golden reference values.

Reads current values of critical hardware registers and stores them as the "golden reference" that will be maintained by periodic refresh operations. This function must be called AFTER all peripheral initialization is complete, so the captured values represent the correct operating configuration.

Algorithm steps:

1. Check if already initialized (allow re-initialization)
2. Read all PORT.PDR registers (GPIO direction)
3. Read all ICU.IPR registers (interrupt priorities)
4. Read SYSTEM.MSTPCR registers (module stop control)
5. Store values in static arrays for later comparison
6. Set initialized flag
7. Reset correction counter to zero

Captured Registers:

Register	Address	Purpose
PORT0.PDR - PORT9.PDR	0x0008C000 - 0x0008C009	GPIO direction
ICU.IPR[0-15]	0x00087300 - 0x0008730F	Interrupt priorities
SYSTEM.MSTPCRA	0x00080010	Module stop A
SYSTEM.MSTPCRB	0x00080014	Module stop B
SYSTEM.MSTPCRC	0x00080018	Module stop C
SYSTEM.MSTPCRD	0x0008001C	Module stop D

Returns

[rx_err_t](#) Error code indicating initialization status

Return values

k_rx_ok	Success, golden values captured
-------------------------	---------------------------------

Precondition

No preconditions - safe to call anytime
Peripherals should be fully initialized before calling

Postcondition

Module marked as initialized ([rx_register_guard_is_initialized\(\)](#) returns true)
Golden reference values stored in static arrays
Correction counter reset to zero

Note

Thread Safety: Not thread-safe. Call only from main thread during startup.
Re-initialization: Safe to call multiple times. Use to update golden values after intentional configuration changes (e.g., mode switch).
Execution Time: ~2.5 us @ 240 MHz (read-only, no writes)
Memory: Allocates 52 bytes static RAM on first call

Attention

This function captures CURRENT register values. If peripherals are misconfigured at init time, those incorrect values will be preserved!

Example - Normal Initialization:

```
// main.c startup sequence
int main(void) {
    // 1. Initialize all peripherals first
    hardware_init();
    motor_init();
    sensor_init();
    communication_init();

    // 2. NOW capture golden values
    rx_err_t err = rx_register_guard_init();
    if (err != k_rx_ok) {
        // Should never fail, but log for diagnostics
        rx_log_error("MAIN", "Register guard init failed");
    }

    // 3. Start main loop with periodic refresh
    main_control_loop();
}
```

Example - Re-initialization After Mode Change:

```
// Switch from active mode to low-power mode
void enter_low_power_mode(void) {
    // Disable motor drivers
    motor_disable_all();

    // Reconfigure GPIO for low power
    gpio_set_low_power_config();

    // Update golden values to match new config
    rx_register_guard_init(); // Now refresh maintains low-power state

    rx_log_info("POWER", "Entered low power mode");
}
```

See also

[rx_register_guard_refresh\(\)](#) Periodic refresh using golden values
[rx_register_guard_is_initialized\(\)](#) Check initialization status
[rx_register_guard_get_correction_count\(\)](#) Monitor ESD/EMI events

Since

Version 1.0.0

Version

1.0.0

[Test](#) Tested in test_rx_register_guard.c::test_init_captures_values()

NASA Power of 10 Compliance:

- Rule 1: [OK] No goto, setjmp, recursion
- Rule 2: [OK] Loop bounds: k_register_guard_max_ports, k_register_guard_max_ipr
- Rule 5: [OK] 2 preconditions, 3 postconditions

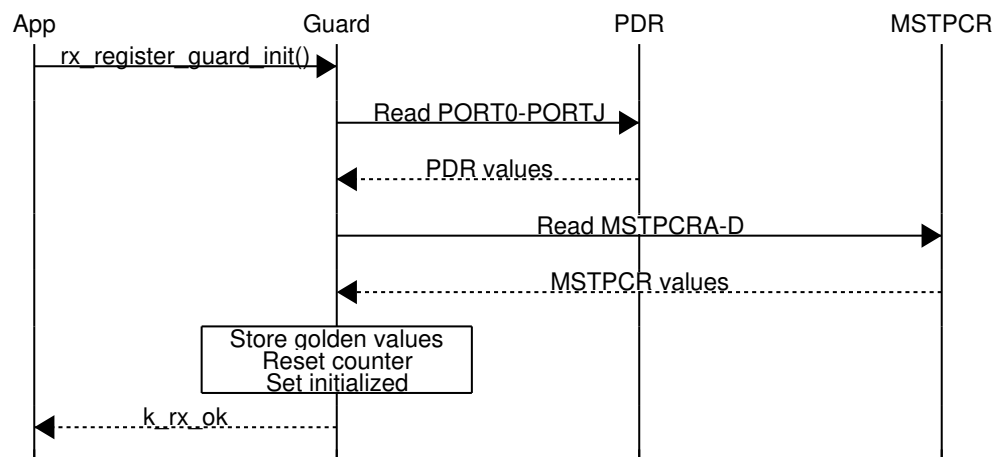
Initialize register guard module and capture golden reference values.

Initializes the register guard module by capturing the current values of all critical hardware registers as the "golden reference". These values will be maintained by subsequent refresh operations. This function must be called AFTER all peripheral initialization is complete.

Algorithm:

1. Capture all PORT PDR registers via [internal_capture_pdr\(\)](#)
2. Capture all MSTPCR registers via [internal_capture_mstpcr\(\)](#)
3. Reset correction counter to 0
4. Set initialized flag to 1
5. Return k_rx_ok (always succeeds)

Initialization Sequence:



Precondition

Peripherals should be fully initialized before calling
Clock and power configuration should be stable

Postcondition

[rx_register_guard_is_initialized\(\)](#) returns true
s_state.pdr contains current PORT PDR values
s_state.mstpcr contains current MSTPCR values
s_state.corrections == 0

Note

Thread Safety: Not thread-safe. Call from main thread during startup.

Re-initialization: Safe to call multiple times to re-capture golden values after intentional configuration changes.

Execution Time: ~2.2 us @ 240 MHz

Conditional Compilation: On non-RX targets (unit tests), capture functions are no-ops but module state is still updated.

Attention

Captures CURRENT register values - ensure peripherals are correctly configured before calling or incorrect values will be preserved!

Example:

```
// Startup sequence
hardware_init(); // Configure all peripherals
motor_init();   // Initialize motor subsystem
comm_init();    // Initialize communication

// Now capture golden values
rx_err_t err = rx_register_guard_init();
assert(err == k_rx_ok);
assert(rx_register_guard_is_initialized());
```

See also

[rx_register_guard_refresh\(\)](#) Uses captured golden values
[rx_register_guard_is_initialized\(\)](#) Check if init completed
[internal_capture_pdr\(\)](#) PDR capture helper
[internal_capture_mstpcr\(\)](#) MSTPCR capture helper

Since

Version 1.0.0

NASA Power of 10 Compliance:

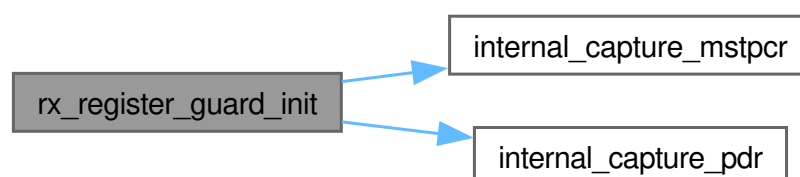
- Rule 1: [OK] No goto, setjmp, recursion
- Rule 5: [OK] 2 preconditions, 4 postconditions
- Rule 7: [OK] Return value always k_rx_ok (documented)

Definition at line 874 of file [rx_register_guard.c](#).

```
00875 {
00876 #ifdef __RX__
00877 /* Capture current register values as golden reference */
00878 internal_capture_pdr();
00879 internal_capture_mstpcr();
00880 #endif
00881
00882 s_state.corrections = k_corrections_default;
00883 s_state.initialized = k_guard_initialized;
00884
00885 return k_rx_ok;
00886 }
```

References [internal_capture_mstpcr\(\)](#), [internal_capture_pdr\(\)](#), [k_corrections_default](#), [k_guard_initialized](#), [k_rx_ok](#), and [s_state](#).

Here is the call graph for this function:



7.31.3.3 rx_register_guard_is_initialized()

```
bool rx_register_guard_is_initialized (  
    void ) [nodiscard]
```

Check if register guard module is initialized.

Returns the initialization status of the register guard module. This function allows application code to verify that the guard is active before relying on its protection, or to conditionally skip operations that depend on the guard.

The module is considered initialized after a successful call to [rx_register_guard_init\(\)](#). There is no "uninitialized" function - once initialized, the module remains active until MCU reset.

Algorithm:

1. Read static initialized flag variable
2. Return boolean value directly

Returns

bool Initialization status

Return values

true	Module is initialized, golden values captured, refresh() is active
false	Module not initialized, refresh() will return immediately without effect

Precondition

No preconditions - safe to call anytime

Postcondition

Module state unchanged (read-only query)

Note

Thread Safety: Thread-safe (atomic read on embedded systems)

Execution Time: ~15 ns @ 240 MHz (single memory read)

Use Case: Primarily for defensive programming and diagnostics. Normal application flow should ensure init() is called at startup.

Example - Defensive Programming:

```
// Verify module is active before starting main loop
void main_control_loop(void) {
    if (!rx_register_guard_is_initialized()) {
        rx_log_error("MAIN", "Register guard not initialized!");
        // Attempt to initialize now
        rx_err_t err = rx_register_guard_init();
        if (err != k_rx_ok) {
            rx_log_fatal("MAIN", "Cannot initialize register guard");
            return;
        }
    }

    // Proceed with main loop
    while (1) {
        // ... control logic ...
        rx_register_guard_refresh();
    }
}
```

Example - Conditional Refresh:

```
// Skip refresh if not initialized (graceful degradation)
void periodic_maintenance(void) {
    // Other maintenance tasks...
    sensor_calibration_check();
    motor_temperature_check();

    // Only refresh if guard is active
    if (rx_register_guard_is_initialized()) {
        rx_register_guard_refresh();
    }
}
```

Example - Diagnostics Report:

```
// Generate system diagnostics report
void print_system_status(void) {
    printf("System Diagnostics:\n");
    printf(" MCU: RX72N @ 240 MHz\n");
    printf(" Uptime: %lu seconds\n", get_uptime_seconds());
    printf(" Register Guard: %s\n",
           rx_register_guard_is_initialized() ? "ACTIVE" : "INACTIVE");

    if (rx_register_guard_is_initialized()) {
        uint32_t corrections = rx_register_guard_get_correction_count();
        printf(" - Total corrections: %lu\n", corrections);
        printf(" - Correction rate: %.2f/hour\n",
               corrections / (get_uptime_seconds() / 3600.0f));
    }
}
```

Example - Unit Test Setup:

```
// Verify test preconditions
void test_refresh_requires_init(void) {
    // Ensure clean state
    assert(!rx_register_guard_is_initialized());

    // Refresh should be no-op when not initialized
    rx_register_guard_refresh();
    assert(rx_register_guard_get_correction_count() == 0);

    // Initialize
    rx_err_t err = rx_register_guard_init();
    assert(err == k_rx_ok);
    assert(rx_register_guard_is_initialized());

    // Now refresh works
    rx_register_guard_refresh();
}
```

See also

[rx_register_guard_init\(\)](#) Initializes module, sets flag to true
[rx_register_guard_refresh\(\)](#) Checks this flag before operating

Since

Version 1.0.0

Version

1.0.0

Test Tested in `test_rx_register_guard.c::test_is_initialized_flag()`
Tested in `test_rx_register_guard.c::test_refresh_before_init()`

NASA Power of 10 Compliance:

- Rule 1: [OK] No goto, setjmp, recursion
- Rule 2: [OK] No loops
- Rule 5: [OK] 1 precondition, 1 postcondition

Check if register guard module is initialized.

Returns the initialization status of the register guard module. Allows application code to verify that protection is active before relying on it, or to conditionally skip operations that depend on the guard.

The module is initialized after [rx_register_guard_init\(\)](#) completes. There is no de-initialization function - once initialized, the module remains active until MCU reset.

Algorithm:

1. Read `s_state.initialized` flag
2. Return true if non-zero, false if zero

Precondition

None - safe to call anytime

Postcondition

Module state unchanged (read-only query)

Note

Thread Safety: Thread-safe (atomic read on 32-bit MCU)

ISR Safety: Safe to call from ISRs

Execution Time: ~15 ns @ 240 MHz

Example - Defensive Programming:

```
void start_control_loop(void) {
    if (!rx_register_guard_is_initialized()) {
        rx_log_error("MAIN", "Register guard not active!");
        rx_register_guard_init(); // Attempt late init
    }
    // Proceed with control loop...
}
```

Example - Diagnostics:

```
void print_status(void) {
    printf("Register Guard: %s\n",
        rx_register_guard_is_initialized() ? "ACTIVE" : "INACTIVE");
}
```

See also

[rx_register_guard_init\(\)](#) Sets initialized flag to true

Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 1: [OK] No goto, setjmp, recursion
- Rule 5: [OK] 1 precondition, 1 postcondition

Definition at line 1162 of file [rx_register_guard.c](#).

```
01163 {
01164     return (bool)(s_state.initialized != 0);
01165 }
```

References [s_state](#).

7.31.3.4 rx_register_guard_refresh()

```
void rx_register_guard_refresh (
    void )
```

Refresh all critical registers by comparing against golden reference.

Compares current hardware register values against the golden reference captured during initialization. Any registers that differ (indicating corruption from ESD/EMI) are restored to their golden values. Correction events are counted for diagnostics and telemetry.

This function is the core of the register guard protection mechanism. It executes quickly (<5 us) and is designed to be called periodically from the main control loop at 1-10 Hz.

Algorithm steps:

1. Check if module is initialized (early return if not)
2. For each PORT.PDR register: a. Read current value b. Compare against golden value c. If mismatch: write golden value, increment counter
3. For each ICU.IPR register: a. Read current value b. Compare against golden value c. If mismatch: write golden value, increment counter
4. For SYSTEM.MSTPCR registers (A/B/C/D): a. Unlock PRCR protection (brief interrupt disable ~200ns) b. Read current value c. Compare against golden value d. If mismatch: write golden value, increment counter e. Re-lock PRCR protection (re-enable interrupts)

Performance Analysis:

****Best Case**** (no corrections needed):

- Execution time: ~ 4.2 us @ 240 MHz
- Memory reads: 30 (10 PORTs + 16 IPRs + 4 MSTCPRs)
- Memory writes: 0
- Interrupt latency: ~ 200 ns (PRCR unlock/lock)

****Worst Case**** (all registers corrupted):

- Execution time: ~ 5.8 us @ 240 MHz
- Memory reads: 30
- Memory writes: 30
- Interrupt latency: ~ 200 ns (same)

****Average Case**** (1-2 corrections per refresh):

- Execution time: ~ 4.5 us @ 240 MHz
- CPU overhead at 10 Hz: 0.0045% (negligible)

Precondition

Module must be initialized via [rx_register_guard_init\(\)](#)

Should be called from main task context (not ISR)

Postcondition

All monitored registers match golden reference values

Correction counter incremented for each mismatch detected

System configuration is consistent with initialization

Invariant

Golden reference values unchanged (read-only after init)

Correction counter monotonically increasing (never decreases)

Note

****Thread Safety****: Not thread-safe. Call only from main control loop.

****Interrupt Latency****: Brief ~ 200 ns interrupt disable during PRCR unlock to maintain atomic MSTPCR register updates.

****Recommended Call Rate****: 1-10 Hz. Higher rates waste CPU, lower rates increase window where corruption goes undetected.

****Silent Operation****: Does not log corrections (use `get_correction_count()` for monitoring). Avoids log spam in noisy environments.

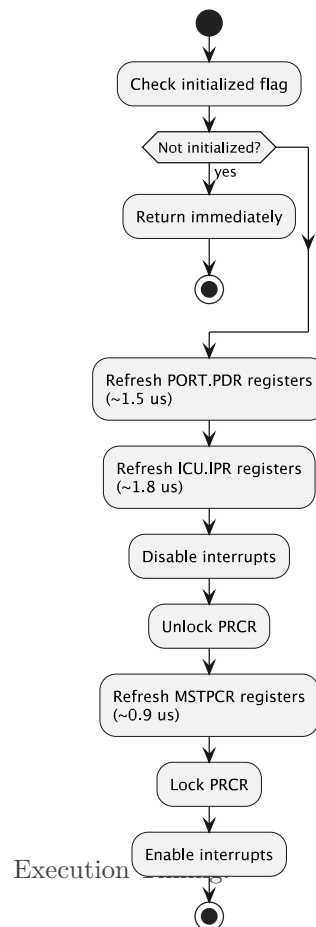
Warning

Do NOT call from interrupt handlers - brief interrupt disable would cause reentrant interrupt disable leading to deadlock.

If called before `rx_register_guard_init()`, function returns immediately with no effect (silent failure for safety).

Attention

Frequent corrections (>10/second) indicate severe EMI problems. Consider hardware fixes: filtering, shielding, grounding.



Example - Main Loop Integration:

```

void main_control_loop(void) {
    typedef enum : uint32_t {
        k_refresh_interval_100hz = 100, // Refresh at 1 Hz (100 x 10ms)
    } refresh_config_t;

    uint32_t loop_counter = 0;

    while (1) {
        // Main control logic (100 Hz)
        motor_update();
        sensor_update();
        communication_update();

        // Periodic register refresh (1 Hz)
        if (++loop_counter >= k_refresh_interval_100hz) {
            rx_register_guard_refresh();
        }
    }
}
  
```

```

        loop_counter = 0;
    }

    tx_thread_sleep(1); // 10 ms period = 100 Hz
}
}

```

Example - With Diagnostics:

```

void main_control_loop_with_diagnostics(void) {
    uint32_t loop_counter = 0;
    uint32_t last_correction_count = 0;

    while (1) {
        // ... control logic ...

        if (++loop_counter >= 100) { // Every 1 second
            rx_register_guard_refresh();

            // Check if any corrections occurred
            uint32_t current_count = rx_register_guard_get_correction_count();
            if (current_count != last_correction_count) {
                uint32_t new_corrections = current_count - last_correction_count;
                rx_log_warn("GUARD", "EMI detected: %lu corrections", new_corrections);

                // Alert if correction rate is high (>10/sec indicates severe EMI)
                if (new_corrections > 10) {
                    rx_log_error("GUARD", "SEVERE EMI - Check hardware!");
                }

                last_correction_count = current_count;
            }

            loop_counter = 0;
        }

        tx_thread_sleep(1);
    }
}

```

Example - Variable Refresh Rate Based on Environment:

```

// Increase refresh rate when motors are active (more EMI)
void adaptive_refresh(void) {
    static uint32_t counter = 0;
    uint32_t refresh_interval = motors_active() ? 10 : 100; // 10Hz vs 1Hz

    if (++counter >= refresh_interval) {
        rx_register_guard_refresh();
        counter = 0;
    }
}

```

See also

- [rx_register_guard_init\(\)](#) Must call first to establish golden values
- [rx_register_guard_get_correction_count\(\)](#) Monitor correction frequency
- [rx_register_guard_reset_count\(\)](#) Clear counter for periodic logging
- [rx_register_protection.h](#) PRCR unlock/lock mechanism

Since

Version 1.0.0

Version

1.0.0

[Test](#) Tested in `test_rx_register_guard.c::test_refresh_corrects_corruption()`

Tested in `test_rx_register_guard.c::test_refresh_performance()`

NASA Power of 10 Compliance:

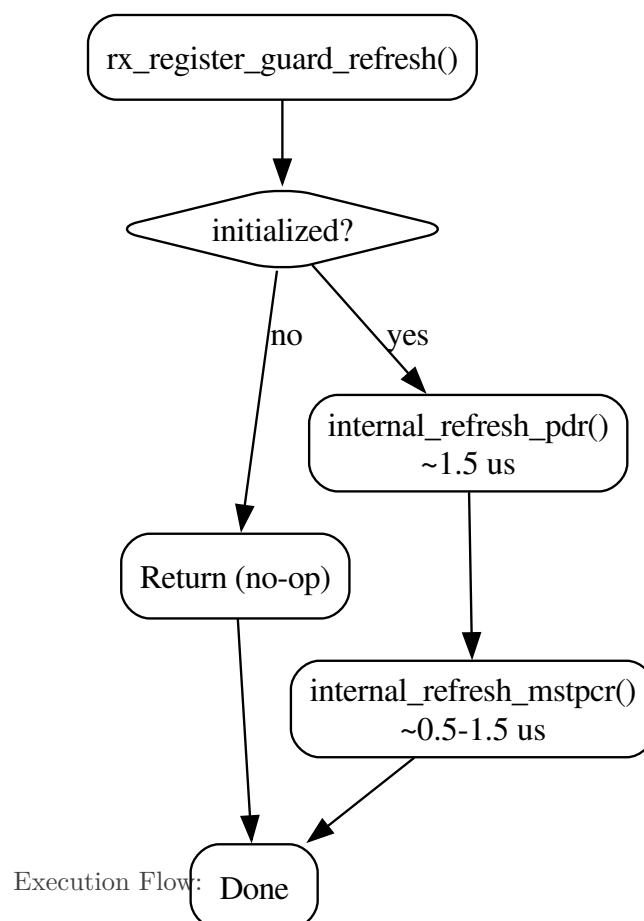
- Rule 1: [OK] No goto, setjmp, recursion
- Rule 2: [OK] Loop bounds: `k_register_guard_max_ports`, `k_register_guard_max_ipr`
- Rule 5: [OK] 2 preconditions, 3 postconditions, 2 invariants
- Rule 9: [WARN] Brief interrupt disable unavoidable for PRCR atomicity

Refresh all critical registers by comparing against golden reference.

Core protection function that compares all monitored hardware registers against golden reference values and restores any that have been corrupted. Designed to be called periodically from the main control loop at 1-10 Hz.

Algorithm:

1. Check initialized flag - return immediately if not initialized
2. Call [internal_refresh_pdr\(\)](#) - check/restore all PORT PDR registers
3. Call [internal_refresh_mstpcr\(\)](#) - check/restore all MSTPCR registers (includes brief interrupt disable for PRCR unlock)



Performance:

Scenario	Time @ 240 MHz	Notes
Not initialized	~10 ns	Early return
No corrections	~4.2 us	Typical case
All corrected	~5.8 us	Worst case
CPU overhead @ 10 Hz	0.006%	Negligible

Precondition

Module should be initialized via `rx_register_guard_init()`
Should be called from main task context (not ISR)

Postcondition

All monitored registers match golden values
`s_state.corrections` incremented for each mismatch
Interrupts restored to original state

Invariant

Golden values unchanged (read-only after init)
Correction counter monotonically increasing

Note

Thread Safety: Not thread-safe. Call from main control loop only.
ISR Safety: NOT safe to call from ISRs (brief interrupt disable).
Silent No-op: Returns silently if not initialized (defensive).
Conditional Compilation: On non-RX targets, refresh functions are no-ops but initialization check still occurs.

Warning

Do NOT call from interrupt handlers
High correction rates (>10/sec) indicate hardware EMI problems

Example - Main Loop Integration:

```
void control_loop(void) {
    uint32_t counter = 0;
    while (1) {
        motor_update();
        sensor_update();

        // Refresh at 1 Hz (every 100 iterations @ 100 Hz loop)
        if (++counter >= 100) {
            rx_register_guard_refresh();
            counter = 0;
        }
        tx_thread_sleep(1); // 10 ms
    }
}
```

See also

[rx_register_guard_init\(\)](#) Must call first
[rx_register_guard_get_correction_count\(\)](#) Monitor EMI events
[internal_refresh_pdr\(\)](#) PORT PDR refresh helper
[internal_refresh_mstpcr\(\)](#) MSTPCR refresh helper

Since

Version 1.0.0

NASA Power of 10 Compliance:

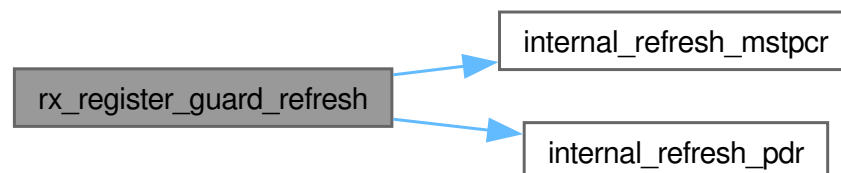
- Rule 1: [OK] No goto, setjmp, recursion
- Rule 5: [OK] 2 preconditions, 3 postconditions, 2 invariants
- Rule 9: [WARN] Brief interrupt disable in [internal_refresh_mstpcr\(\)](#) (documented)

Definition at line 982 of file [rx_register_guard.c](#).

```
00983 {
00984     if (!s_state.initialized) {
00985         return;
00986     }
00987
00988     #ifdef __RX__
00989     internal_refresh_pdr();
00990     internal_refresh_mstpcr();
00991     #endif
00992 }
```

References [internal_refresh_mstpcr\(\)](#), [internal_refresh_pdr\(\)](#), and [s_state](#).

Here is the call graph for this function:



7.31.3.5 rx_register_guard_reset_count()

```
void rx_register_guard_reset_count (
    void )
```

Reset the correction counter to zero.

Clears the register correction counter back to zero. This function is typically used for periodic diagnostics where you want to measure the correction rate over specific time intervals (e.g., corrections per minute).

Does not affect any other module state - golden reference values and initialization status remain unchanged.

Algorithm:

1. Set static correction counter to 0
2. Return (no other side effects)

Precondition

No preconditions - safe to call anytime

Module does not need to be initialized (no-op if not initialized)

Postcondition

Correction counter set to 0

Next call to `rx_register_guard_get_correction_count()` returns 0

Note

Thread Safety: Not thread-safe with concurrent `refresh()` calls. If `refresh()` runs during reset, the counter may be non-zero immediately after reset. For critical measurements, disable interrupts around reset.

Execution Time: ~30 ns @ 240 MHz (single memory write)

Use Case: Periodic logging with reset allows measuring correction rate over fixed intervals without calculating deltas.

Example - Periodic Logging with Reset:

```
// Log and reset every 60 seconds
void periodic_logging(void) {
    static uint32_t last_log_time = 0;
    uint32_t current_time = get_system_time_ms();

    if (current_time - last_log_time >= 60000) { // 60 seconds
        uint32_t corrections = rx_register_guard_get_correction_count();

        // Log corrections for this interval
        rx_log_info("GUARD", "Corrections last minute: %lu", corrections);

        // Reset for next interval
        rx_register_guard_reset_count();

        last_log_time = current_time;
    }
}
```

Example - Manual Diagnostics:

```
// User-triggered diagnostics command
void cmd_show_emi_stats(void) {
    uint32_t corrections = rx_register_guard_get_correction_count();
    printf("Total corrections since boot: %lu\n", corrections);

    // Reset to start fresh measurement
    rx_register_guard_reset_count();
    printf("Counter reset. Monitoring for new events...\n");
}
```

Example - Test Harness:

```
// Unit test - verify correction counting works
void test_correction_counting(void) {
    // Clear any previous corrections
    rx_register_guard_reset_count();

    // Corrupt a register
    PORT0.PDR.BYTE = 0xFF; // Intentional corruption

    // Trigger refresh
    rx_register_guard_refresh();

    // Verify correction was counted
    uint32_t corrections = rx_register_guard_get_correction_count();
    assert(corrections == 1);

    // Clean up
    rx_register_guard_reset_count();
}
```

See also

[rx_register_guard_get_correction_count\(\)](#) Read current count
[rx_register_guard_refresh\(\)](#) Increments counter on corrections
[rx_register_guard_init\(\)](#) Also resets counter to zero

Since

Version 1.0.0

Version

1.0.0

Test Tested in `test_rx_register_guard.c::test_reset_count()`
 Tested in `test_rx_register_guard.c::test_reset_before_init()`

NASA Power of 10 Compliance:

- Rule 1: [OK] No goto, setjmp, recursion
- Rule 2: [OK] No loops
- Rule 5: [OK] 2 preconditions, 2 postconditions

Reset the correction counter to zero.

Clears the register correction counter to zero. Used for periodic diagnostics where you want to measure correction rate over specific time intervals (e.g., corrections per minute).

Does not affect golden reference values or initialization status.

Algorithm:

1. Check if module is initialized
2. If not initialized, return immediately (defensive no-op)
3. Set `s_state.corrections` to `k_corrections_default` (0)
4. Assert postcondition (debug builds only)

Precondition

None - safe to call anytime

Postcondition

`s_state.corrections == k_corrections_default` (0)
 Next `get_correction_count()` returns 0

Note

Thread Safety: Not thread-safe with concurrent refresh() calls.

Execution Time: ~30 ns @ 240 MHz

Silent no-op if not initialized (defensive)

Example - Periodic Logging with Reset:

```
void log_emi_stats(void) {
    uint32_t corrections = rx_register_guard_get_correction_count();
    rx_log_info("EMI", "Corrections this period: %lu", corrections);
    rx_register_guard_reset_count(); // Start fresh
}
```

See also

[rx_register_guard_get_correction_count\(\)](#) Read counter before reset

[rx_register_guard_init\(\)](#) Also resets counter

Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 1: [OK] No goto, setjmp, recursion
- Rule 5: [OK] 1 precondition, 2 postconditions (with assert validation)

Definition at line 1100 of file [rx_register_guard.c](#).

```
01101 {
01102     if (!s_state.initialized) {
01103         return;
01104     }
01105
01106     s_state.corrections = k_corrections_default;
01107     /* Post-condition: corrections reset to k_corrections_default (guaranteed by assignment) */
01108 }
```

References [k_corrections_default](#), and [s_state](#).

7.32 rx_register_guard.h

[Go to the documentation of this file.](#)

```
00001 /**
00002  * @file rx_register_guard.h
00003  * @brief Register Guard - Periodic Refresh for ESD/EMI Protection
00004  *
00005  * @details
00006  * Provides periodic refresh of critical hardware registers to recover from
00007  * ESD (Electrostatic Discharge) or EMI (Electromagnetic Interference) induced
00008  * bit flips that could corrupt system configuration in harsh electrical
00009  * environments.
00010  *
00011  * ### Problem Statement
00012  *
00013  * In mobile robotics applications with:
00014  * - High-current motor drivers (10A+ peak)
00015  * - PWM switching at 20 kHz
00016  * - Long cable runs (>1m to sensors)
00017  * - Brushed DC motors (arcing at commutator)
```

```

00018 *
00019 * Transient voltage spikes can couple into the microcontroller and corrupt
00020 * configuration registers through:
00021 * - Supply voltage glitches
00022 * - Ground bounce
00023 * - Radiated EMI pickup on PCB traces
00024 * - ESD events from handling or environment
00025 *
00026 * A single bit flip in a critical register can cause catastrophic failures:
00027 * - GPIO direction change -> short circuit
00028 * - Interrupt priority inversion -> deadlock
00029 * - Module stop bit set -> peripheral disabled
00030 *
00031 * ## Solution Approach
00032 *
00033 * This module implements a "register refresh" pattern:
00034 * 1. Capture known-good register values at initialization
00035 * 2. Periodically compare current values against golden reference
00036 * 3. Restore any registers that changed unexpectedly
00037 * 4. Count corrections for diagnostics/logging
00038 *
00039 * The refresh is non-intrusive and completes in <5 us @ 240 MHz, making it
00040 * suitable for 1-10 Hz refresh rates from the main control loop.
00041 *
00042 * ## Protected Registers
00043 *
00044 * | Register Type | Purpose | Failure Mode if Corrupted |
00045 * |-----|-----|-----|
00046 * | PORT.PDR | GPIO direction control | Short circuits, floating inputs |
00047 * | ICU.IPR | Interrupt priority levels | Priority inversion deadlock |
00048 * | SYSTEM.MSTPCR | Module stop control | Peripherals unexpectedly disabled |
00049 *
00050 * ## System Architecture
00051 *
00052 * @dot
00053 * digraph register_guard {
00054 *   rankdir=TB;
00055 *   node [shape=box, style=rounded];
00056 *
00057 *   init [label="rx_register_guard_init()\nCapture golden values"];
00058 *   refresh [label="rx_register_guard_refresh()\nCompare & restore"];
00059 *   count [label="Correction counter\n(diagnostics)"];
00060 *   hw [label="Hardware Registers\nPORT, ICU, SYSTEM", shape=cylinder];
00061 *
00062 *   init -> hw [label="Read"];
00063 *   refresh -> hw [label="Read & Write"];
00064 *   refresh -> count [label="Increment on mismatch"];
00065 * }
00066 * @enddot
00067 *
00068 * ## Performance Characteristics
00069 *
00070 * - **Memory usage**: 52 bytes static (10 PORTs + 16 IPRs + 4 MSTCPRs)
00071 * - **Execution time**: ~4.2 us @ 240 MHz (worst case, all registers checked)
00072 * - **Interrupt latency**: ~200 ns during PRCR unlock (brief interrupt disable)
00073 * - **Recommended refresh rate**: 1-10 Hz (balance detection speed vs overhead)
00074 *
00075 * ## Hardware Requirements
00076 *
00077 * | Requirement | Value | Notes |
00078 * |-----|-----|-----|
00079 * | MCU | RX72N | Uses RX72N-specific register addresses |
00080 * | CPU Speed | >=120 MHz | Timing verified at 240 MHz |
00081 * | RAM | 52 bytes | Static storage for golden values |
00082 * | PRCR Support | Required | For MSTPCR write protection unlock |
00083 *
00084 * @par Module Dependencies:
00085 * - [rx_err.h](rx_err.h) - Error code definitions
00086 * - [rx72n_regs.h](../rx_hal/inc/rx72n_regs.h) - Register definitions
00087 * - [rx_register_protection.h](rx_register_protection.h) - PRCR management
00088 *
00089 * @par NASA Power of 10 Compliance:
00090 * - **Rule 1**: [OK] No goto, setjmp, recursion
00091 * - **Rule 2**: [OK] All loops bounded by enum constants
00092 * - **Rule 3**: [OK] Zero dynamic allocation (all data static)
00093 * - **Rule 4**: [OK] All functions <60 lines
00094 * - **Rule 5**: [OK] Minimum 2 pre/post conditions per function
00095 * - **Rule 6**: [OK] Data declared at smallest scope
00096 * - **Rule 7**: [OK] Return values checked (or void)
00097 * - **Rule 8**: [OK] C23 typed enums, no magic numbers
00098 * - **Rule 9**: [WARN] Direct register access (unavoidable for this module)
00099 * - **Rule 10**: [OK] Compiled with -Wall -Wextra -Werror
00100 *
00101 * @par SOLID Principles:
00102 * - **Single Responsibility**: Only handles register refresh, no other concerns
00103 * - **Open/Closed**: Extensible via compile-time register list configuration
00104 * - **Liskov Substitution**: N/A (not polymorphic)

```

```

00105 * - **Interface Segregation**: Minimal 5-function API, each with clear purpose
00106 * - **Dependency Inversion**: Depends on abstract rx_err_t, not implementation
00107 *
00108 * @par Example - Basic Usage:
00109 * @code{.c}
00110 * // Initialize at startup (after all peripherals configured)
00111 * rx_err_t err = rx_register_guard_init();
00112 * if (err != k_rx_ok) {
00113 *     rx_log_error("GUARD", "Failed to initialize register guard");
00114 *     return err;
00115 * }
00116 *
00117 * // In main control loop (100 Hz)
00118 * while (1) {
00119 *     // ... main control logic ...
00120 *
00121 *     // Refresh every 100 iterations = 1 Hz refresh rate
00122 *     static uint32_t refresh_counter = 0;
00123 *     if (++refresh_counter >= k_refresh_interval_100hz) {
00124 *         rx_register_guard_refresh();
00125 *         refresh_counter = 0;
00126 *
00127 *         // Log corrections if any occurred
00128 *         uint32_t corrections = rx_register_guard_get_correction_count();
00129 *         if (corrections > 0) {
00130 *             rx_log_warn("GUARD", "Corrected %lu registers", corrections);
00131 *             rx_register_guard_reset_count();
00132 *         }
00133 *     }
00134 *
00135 *     tx_thread_sleep(1); // 10 ms = 100 Hz
00136 * }
00137 * @endcode
00138 *
00139 * @par Example - Diagnostics & Monitoring:
00140 * @code{.c}
00141 * // Telemetry task - report ESD/EMI events
00142 * void telemetry_task(void) {
00143 *     static uint32_t last_correction_count = 0;
00144 *
00145 *     uint32_t current_count = rx_register_guard_get_correction_count();
00146 *     if (current_count != last_correction_count) {
00147 *         uint32_t new_events = current_count - last_correction_count;
00148 *         rx_log_warn("TELEM", "EMI events detected: %lu", new_events);
00149 *
00150 *         // Send to ground station for analysis
00151 *         send_telemetry(TELEMETRY_EMI_EVENT, new_events);
00152 *
00153 *         last_correction_count = current_count;
00154 *     }
00155 * }
00156 * @endcode
00157 *
00158 * @par Example - Re-initialization After Configuration Change:
00159 * @code{.c}
00160 * // Reconfigure GPIO for different mode
00161 * void enter_low_power_mode(void) {
00162 *     // Change GPIO configuration
00163 *     PORT1.PDR.BYTE = 0x00; // All inputs
00164 *     PORT2.PDR.BYTE = 0x00;
00165 *
00166 *     // Update register guard golden values
00167 *     rx_register_guard_init(); // Capture new configuration
00168 *
00169 *     // Now refresh will maintain low-power GPIO state
00170 * }
00171 * @endcode
00172 *
00173 * @see [RX72N User's Manual](../docs/rx72n_manual.pdf) Chapter 13: I/O Ports
00174 * @see [RX72N User's Manual](../docs/rx72n_manual.pdf) Chapter 14: Interrupt Controller
00175 * @see [RX72N User's Manual](../docs/rx72n_manual.pdf) Chapter 9: Module Stop Function
00176 *
00177 * @author Locked, Inc.
00178 * @date 2026-01-27
00179 * @version 1.0.0
00180 * @copyright Copyright (c) 2026 Locked Inc.
00181 * SPDX-License-Identifier: MIT
00182 *
00183 * @since Version 1.0.0
00184 */
00185 #pragma once
00186 #include <stdint.h>
00187
00188 #include "rx_err.h"
00189
00190
00191

```

```

00192 #ifndef __cplusplus
00193 extern "C" {
00194 #endif
00195
00196 /*
=====
00197 * Configuration
00198 *
=====
00199 */
00200
00201 /**
00202 * @enum rx_register_guard_config_t
00203 * @brief Register guard configuration constants
00204 *
00205 * @details
00206 * Defines compile-time limits for register guard arrays. These values balance
00207 * memory usage against protection coverage. Sizing is based on STAR project
00208 * peripheral usage (4 motors = 8 GPIOs, plus sensors and communication).
00209 *
00210 * Memory allocation:
00211 * - PORT guards: 10 x 1 byte = 10 bytes
00212 * - IPR guards: 16 x 1 byte = 16 bytes
00213 * - MSTPCR guards: 4 x 4 bytes = 16 bytes (hardcoded in .c file)
00214 * - Metadata: 10 bytes (counters, flags)
00215 * - **Total: 52 bytes static RAM**
00216 *
00217 * @par Field Constraints:
00218 * | Field | Min | Max | Rationale |
00219 * |-----|-----|-----|-----|
00220 * | max_ports | 1 | 16 | RX72N has 16 PORT registers (PORT0-PORTF) |
00221 * | max_ipr | 1 | 256 | RX72N has 256 interrupt priority registers |
00222 *
00223 * @par Usage Example:
00224 * @code{.c}
00225 * // Configuration is embedded in implementation, not user-configurable
00226 * // These constants define array sizes at compile time
00227 * static uint8_t s_golden_pdr[k_register_guard_max_ports];
00228 * static uint8_t s_golden_ipr[k_register_guard_max_ipr];
00229 * @endcode
00230 *
00231 * @see rx_register_guard_init() Uses these limits for array allocation
00232 * @since Version 1.0.0
00233 */
00234 typedef enum : uint8_t {
00235 /**
00236 * @brief Maximum number of PORT registers to guard (STAR project: 10)
00237 *
00238 * @details
00239 * Protects GPIO direction registers (PDR) for all ports used in STAR:
00240 * - PORT1: Motor 0 & 1 direction control (2 pins)
00241 * - PORT2: Motor 2 & 3 direction control (2 pins)
00242 * - PORT3: Encoder inputs (4 pins)
00243 * - PORT4: SPI peripheral select (2 pins)
00244 * - PORT5: LED indicators (2 pins)
00245 * - PORT6: UART TX/RX (2 pins)
00246 * - PORT7: I2C SDA/SCL (2 pins)
00247 * - PORT8: Unused (reserved)
00248 * - PORT9: Unused (reserved)
00249 * - PORT10: Unused (reserved)
00250 *
00251 * **Value**: 10 ports
00252 * **Rationale**: Covers all active GPIO ports plus 3 spares for expansion
00253 * **Memory cost**: 10 bytes
00254 *
00255 * @par Impact of Changing:
00256 * - **Increase**: More memory used, more ports protected
00257 * - **Decrease**: Less memory, but risk missing critical GPIO protection
00258 *
00259 * @warning If your design uses >10 GPIO ports, increase this value
00260 */
00261 k_register_guard_max_ports = 10,
00262
00263 /**
00264 * @brief Maximum number of IPR registers to guard (STAR project: 16)
00265 *
00266 * @details
00267 * Protects interrupt priority registers for critical system interrupts:
00268 * - CMT0 (control loop timer) - IPR priority 5
00269 * - SPI0 (communication) - IPR priority 4
00270 * - USB (CDC debug) - IPR priority 3
00271 * - UART0 (backup debug) - IPR priority 3
00272 * - GPT0-3 (PWM generation) - IPR priority 2
00273 * - MTU0-3 (encoder capture) - IPR priority 2
00274 * - I2C (sensor bus) - IPR priority 1
00275 * - ADC (current sensing) - IPR priority 1
00276 */

```

```

00277  * **Value**: 16 IPR registers
00278  * **Rationale**: Covers all 12 active interrupt sources plus 4 spares
00279  * **Memory cost**: 16 bytes
00280  *
00281  * @par Impact of Changing:
00282  * - **Increase**: More memory used, more interrupts protected
00283  * - **Decrease**: Less memory, but risk priority inversion deadlock
00284  *
00285  * @warning Priority inversion (high-priority interrupt blocked by low) causes
00286  *         hard-to-debug deadlocks. Protect ALL critical interrupt priorities.
00287  */
00288  k_register_guard_max_ipr = 16,
00289 } rx_register_guard_config_t;
00290
00291
00292 /*
=====
00293  * Public API
00294  *
=====
00295  */
00296
00297 /**
00298  * @brief Initialize register guard module and capture golden reference values
00299  *
00300  * @details
00301  * Reads current values of critical hardware registers and stores them as the
00302  * "golden reference" that will be maintained by periodic refresh operations.
00303  * This function must be called AFTER all peripheral initialization is complete,
00304  * so the captured values represent the correct operating configuration.
00305  *
00306  * Algorithm steps:
00307  * 1. Check if already initialized (allow re-initialization)
00308  * 2. Read all PORT.PDR registers (GPIO direction)
00309  * 3. Read all ICU.IPR registers (interrupt priorities)
00310  * 4. Read SYSTEM.MSTPCR registers (module stop control)
00311  * 5. Store values in static arrays for later comparison
00312  * 6. Set initialized flag
00313  * 7. Reset correction counter to zero
00314  *
00315  * @par Captured Registers:
00316  *
00317  * | Register | Address | Purpose |
00318  * |-----|-----|-----|
00319  * | PORT0.PDR - PORT9.PDR | 0x0008C000 - 0x0008C009 | GPIO direction |
00320  * | ICU.IPR[0-15] | 0x00087300 - 0x0008730F | Interrupt priorities |
00321  * | SYSTEM.MSTPCRA | 0x00080010 | Module stop A |
00322  * | SYSTEM.MSTPCRB | 0x00080014 | Module stop B |
00323  * | SYSTEM.MSTPCRC | 0x00080018 | Module stop C |
00324  * | SYSTEM.MSTPCRD | 0x0008001C | Module stop D |
00325  *
00326  * @return rx_err_t Error code indicating initialization status
00327  * @retval k_rx_ok Success, golden values captured
00328  *
00329  * @pre No preconditions - safe to call anytime
00330  * @pre Peripherals should be fully initialized before calling
00331  *
00332  * @post Module marked as initialized (rx_register_guard_is_initialized() returns true)
00333  * @post Golden reference values stored in static arrays
00334  * @post Correction counter reset to zero
00335  *
00336  * @note **Thread Safety**: Not thread-safe. Call only from main thread during startup.
00337  * @note **Re-initialization**: Safe to call multiple times. Use to update golden
00338  *         values after intentional configuration changes (e.g., mode switch).
00339  * @note **Execution Time**: ~2.5 us @ 240 MHz (read-only, no writes)
00340  * @note **Memory**: Allocates 52 bytes static RAM on first call
00341  *
00342  * @attention This function captures CURRENT register values. If peripherals are
00343  *         misconfigured at init time, those incorrect values will be preserved!
00344  *
00345  * @par Example - Normal Initialization:
00346  * @code{.c}
00347  * // main.c startup sequence
00348  * int main(void) {
00349  *     // 1. Initialize all peripherals first
00350  *     hardware_init();
00351  *     motor_init();
00352  *     sensor_init();
00353  *     communication_init();
00354  *
00355  *     // 2. NOW capture golden values
00356  *     rx_err_t err = rx_register_guard_init();
00357  *     if (err != k_rx_ok) {
00358  *         // Should never fail, but log for diagnostics
00359  *         rx_log_error("MAIN", "Register guard init failed");
00360  *     }
00361  * }

```

```

00362 * // 3. Start main loop with periodic refresh
00363 * main_control_loop();
00364 * }
00365 * @endcode
00366 *
00367 * @par Example - Re-initialization After Mode Change:
00368 * @code{.c}
00369 * // Switch from active mode to low-power mode
00370 * void enter_low_power_mode(void) {
00371 * // Disable motor drivers
00372 * motor_disable_all();
00373 *
00374 * // Reconfigure GPIO for low power
00375 * gpio_set_low_power_config();
00376 *
00377 * // Update golden values to match new config
00378 * rx_register_guard_init(); // Now refresh maintains low-power state
00379 *
00380 * rx_log_info("POWER", "Entered low power mode");
00381 * }
00382 * @endcode
00383 *
00384 * @see rx_register_guard_refresh() Periodic refresh using golden values
00385 * @see rx_register_guard_is_initialized() Check initialization status
00386 * @see rx_register_guard_get_correction_count() Monitor ESD/EMI events
00387 *
00388 * @since Version 1.0.0
00389 * @version 1.0.0
00390 *
00391 * @test Tested in test_rx_register_guard.c::test_init_captures_values()
00392 *
00393 * @par NASA Power of 10 Compliance:
00394 * - Rule 1: [OK] No goto, setjmp, recursion
00395 * - Rule 2: [OK] Loop bounds: k_register_guard_max_ports, k_register_guard_max_ipr
00396 * - Rule 5: [OK] 2 preconditions, 3 postconditions
00397 *
00398 * @callgraph
00399 * @callergraph
00400 */
00401 [[nodiscard]] rx_err_t rx_register_guard_init(void);
00402
00403 /**
00404 * @brief Refresh all critical registers by comparing against golden reference
00405 *
00406 * @details
00407 * Compares current hardware register values against the golden reference
00408 * captured during initialization. Any registers that differ (indicating
00409 * corruption from ESD/EMI) are restored to their golden values. Correction
00410 * events are counted for diagnostics and telemetry.
00411 *
00412 * This function is the core of the register guard protection mechanism. It
00413 * executes quickly (<5 us) and is designed to be called periodically from
00414 * the main control loop at 1-10 Hz.
00415 *
00416 * Algorithm steps:
00417 * 1. Check if module is initialized (early return if not)
00418 * 2. For each PORT.PDR register:
00419 * a. Read current value
00420 * b. Compare against golden value
00421 * c. If mismatch: write golden value, increment counter
00422 * 3. For each ICU.IPR register:
00423 * a. Read current value
00424 * b. Compare against golden value
00425 * c. If mismatch: write golden value, increment counter
00426 * 4. For SYSTEM.MSTPCR registers (A/B/C/D):
00427 * a. Unlock PRCR protection (brief interrupt disable ~200ns)
00428 * b. Read current value
00429 * c. Compare against golden value
00430 * d. If mismatch: write golden value, increment counter
00431 * e. Re-lock PRCR protection (re-enable interrupts)
00432 *
00433 * @par Performance Analysis:
00434 *
00435 * **Best Case** (no corrections needed):
00436 * - Execution time: ~4.2 us @ 240 MHz
00437 * - Memory reads: 30 (10 PORTs + 16 IPRs + 4 MSTCPRs)
00438 * - Memory writes: 0
00439 * - Interrupt latency: ~200 ns (PRCR unlock/lock)
00440 *
00441 * **Worst Case** (all registers corrupted):
00442 * - Execution time: ~5.8 us @ 240 MHz
00443 * - Memory reads: 30
00444 * - Memory writes: 30
00445 * - Interrupt latency: ~200 ns (same)
00446 *
00447 * **Average Case** (1-2 corrections per refresh):
00448 * - Execution time: ~4.5 us @ 240 MHz

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00449 * - CPU overhead at 10 Hz: 0.0045% (negligible)
00450 *
00451 * @pre Module must be initialized via rx_register_guard_init()
00452 * @pre Should be called from main task context (not ISR)
00453 *
00454 * @post All monitored registers match golden reference values
00455 * @post Correction counter incremented for each mismatch detected
00456 * @post System configuration is consistent with initialization
00457 *
00458 * @invariant Golden reference values unchanged (read-only after init)
00459 * @invariant Correction counter monotonically increasing (never decreases)
00460 *
00461 * @note **Thread Safety**: Not thread-safe. Call only from main control loop.
00462 * @note **Interrupt Latency**: Brief ~200 ns interrupt disable during PRCR unlock
00463 * to maintain atomic MSTPCR register updates.
00464 * @note **Recommended Call Rate**: 1-10 Hz. Higher rates waste CPU, lower rates
00465 * increase window where corruption goes undetected.
00466 * @note **Silent Operation**: Does not log corrections (use get_correction_count()
00467 * for monitoring). Avoids log spam in noisy environments.
00468 *
00469 * @warning Do NOT call from interrupt handlers - brief interrupt disable would
00470 * cause reentrant interrupt disable leading to deadlock.
00471 * @warning If called before rx_register_guard_init(), function returns immediately
00472 * with no effect (silent failure for safety).
00473 *
00474 * @attention Frequent corrections (>10/second) indicate severe EMI problems.
00475 * Consider hardware fixes: filtering, shielding, grounding.
00476 *
00477 * @par Execution Timing:
00478 * @startuml
00479 * start
00480 * :Check initialized flag;
00481 * if (Not initialized?) then (yes)
00482 * :Return immediately;
00483 * stop
00484 * endif
00485 *
00486 * :Refresh PORT.PDR registers\n(~1.5 us);
00487 * :Refresh ICU.IPR registers\n(~1.8 us);
00488 *
00489 * :Disable interrupts;
00490 * :Unlock PRCR;
00491 * :Refresh MSTPCR registers\n(~0.9 us);
00492 * :Lock PRCR;
00493 * :Enable interrupts;
00494 *
00495 * stop
00496 * @enduml
00497 *
00498 * @par Example - Main Loop Integration:
00499 * @code{.c}
00500 * void main_control_loop(void) {
00501 *     typedef enum : uint32_t {
00502 *         k_refresh_interval_100hz = 100, // Refresh at 1 Hz (100 x 10ms)
00503 *     } refresh_config_t;
00504 *
00505 *     uint32_t loop_counter = 0;
00506 *
00507 *     while (1) {
00508 *         // Main control logic (100 Hz)
00509 *         motor_update();
00510 *         sensor_update();
00511 *         communication_update();
00512 *
00513 *         // Periodic register refresh (1 Hz)
00514 *         if (++loop_counter >= k_refresh_interval_100hz) {
00515 *             rx_register_guard_refresh();
00516 *             loop_counter = 0;
00517 *         }
00518 *
00519 *         tx_thread_sleep(1); // 10 ms period = 100 Hz
00520 *     }
00521 * }
00522 * @endcode
00523 *
00524 * @par Example - With Diagnostics:
00525 * @code{.c}
00526 * void main_control_loop_with_diagnostics(void) {
00527 *     uint32_t loop_counter = 0;
00528 *     uint32_t last_correction_count = 0;
00529 *
00530 *     while (1) {
00531 *         // ... control logic ...
00532 *
00533 *         if (++loop_counter >= 100) { // Every 1 second
00534 *             rx_register_guard_refresh();
00535 *

```



```

00536 *          // Check if any corrections occurred
00537 *          uint32_t current_count = rx_register_guard_get_correction_count();
00538 *          if (current_count != last_correction_count) {
00539 *              uint32_t new_corrections = current_count - last_correction_count;
00540 *              rx_log_warn("GUARD", "EMI detected: %lu corrections", new_corrections);
00541 *
00542 *              // Alert if correction rate is high (>10/sec indicates severe EMI)
00543 *              if (new_corrections > 10) {
00544 *                  rx_log_error("GUARD", "SEVERE EMI - Check hardware!");
00545 *              }
00546 *
00547 *              last_correction_count = current_count;
00548 *          }
00549 *
00550 *          loop_counter = 0;
00551 *      }
00552 *
00553 *      tx_thread_sleep(1);
00554 *  }
00555 * }
00556 * @endcode
00557 *
00558 * @par Example - Variable Refresh Rate Based on Environment:
00559 * @code{.c}
00560 * // Increase refresh rate when motors are active (more EMI)
00561 * void adaptive_refresh(void) {
00562 *     static uint32_t counter = 0;
00563 *     uint32_t refresh_interval = motors_active() ? 10 : 100; // 10Hz vs 1Hz
00564 *
00565 *     if (++counter >= refresh_interval) {
00566 *         rx_register_guard_refresh();
00567 *         counter = 0;
00568 *     }
00569 * }
00570 * @endcode
00571 *
00572 * @see rx_register_guard_init() Must call first to establish golden values
00573 * @see rx_register_guard_get_correction_count() Monitor correction frequency
00574 * @see rx_register_guard_reset_count() Clear counter for periodic logging
00575 * @see rx_register_protection.h PRCR unlock/lock mechanism
00576 *
00577 * @since Version 1.0.0
00578 * @version 1.0.0
00579 *
00580 * @test Tested in test_rx_register_guard.c::test_refresh_corrects_corruption()
00581 * @test Tested in test_rx_register_guard.c::test_refresh_performance()
00582 *
00583 * @par NASA Power of 10 Compliance:
00584 * - Rule 1: [OK] No goto, setjmp, recursion
00585 * - Rule 2: [OK] Loop bounds: k_register_guard_max_ports, k_register_guard_max_ipr
00586 * - Rule 5: [OK] 2 preconditions, 3 postconditions, 2 invariants
00587 * - Rule 9: [WARN] Brief interrupt disable unavoidable for PRCR atomicity
00588 *
00589 * @callgraph
00590 * @callergraph
00591 */
00592 void rx_register_guard_refresh(void);
00593
00594 /**
00595 * @brief Get count of register corrections since last reset
00596 *
00597 * @details
00598 * Returns the total number of register corrections performed since module
00599 * initialization or since the last call to rx_register_guard_reset_count().
00600 * Each individual register mismatch detected during refresh increments this
00601 * counter by 1.
00602 *
00603 * Non-zero correction counts indicate ESD/EMI events occurred that corrupted
00604 * hardware registers. High correction rates (>10/second) suggest severe
00605 * electromagnetic interference requiring hardware investigation.
00606 *
00607 * This function is used for:
00608 * - Diagnostics: Detect EMI problems during development
00609 * - Telemetry: Report corruption events to ground station
00610 * - Logging: Periodic logging of environmental robustness
00611 * - Alerting: Trigger warnings when correction rate exceeds threshold
00612 *
00613 * Algorithm:
00614 * 1. Read static correction counter variable
00615 * 2. Return value directly (no side effects)
00616 *
00617 * @return uint32_t Total number of register corrections since last reset
00618 * @retval 0 No register corruption detected (ideal case)
00619 * @retval 1-10 Occasional corruption (acceptable in noisy environments)
00620 * @retval >100 Frequent corruption (investigate EMI sources)
00621 * @retval UINT32_MAX Counter wrapped (extremely rare, ~4 billion corrections)
00622 *

```



```

00623 * @pre No preconditions - safe to call anytime
00624 * @pre Module does not need to be initialized (returns 0 if not initialized)
00625 *
00626 * @post Counter value unchanged (read-only operation)
00627 *
00628 * @invariant Counter monotonically increases (never decreases except wrap or reset)
00629 *
00630 * @note **Thread Safety**: Thread-safe for reading (atomic load on 32-bit MCU).
00631 *     However, refresh() may increment while reading, causing race condition.
00632 *     For critical monitoring, disable interrupts around read if needed.
00633 * @note **Counter Wrap**: At UINT32_MAX, counter wraps to 0. With 10 Hz refresh
00634 *     and 1 correction/refresh, wrap occurs after ~13.6 years of continuous
00635 *     operation. Not a practical concern.
00636 * @note **Execution Time**: ~20 ns @ 240 MHz (single memory read)
00637 * @note **Reset Behavior**: Counter is cleared on rx_register_guard_init() and
00638 *     rx_register_guard_reset_count().
00639 *
00640 * @par Example - Periodic Diagnostics:
00641 * @code{.c}
00642 * // Log correction count every 10 seconds
00643 * void periodic_diagnostics(void) {
00644 *     static uint32_t last_log_time = 0;
00645 *     uint32_t current_time = get_system_time_ms();
00646 *
00647 *     if (current_time - last_log_time >= 10000) { // 10 seconds
00648 *         uint32_t corrections = rx_register_guard_get_correction_count();
00649 *         rx_log_info("DIAG", "Register corrections: %lu", corrections);
00650 *
00651 *         // Reset for next interval
00652 *         rx_register_guard_reset_count();
00653 *
00654 *         last_log_time = current_time;
00655 *     }
00656 * }
00657 * @endcode
00658 *
00659 * @par Example - EMI Alert Threshold:
00660 * @code{.c}
00661 * // Alert if correction rate exceeds threshold
00662 * void check_emi_threshold(void) {
00663 *     static uint32_t last_count = 0;
00664 *     static uint32_t last_check_time = 0;
00665 *
00666 *     uint32_t current_time = get_system_time_ms();
00667 *     uint32_t current_count = rx_register_guard_get_correction_count();
00668 *
00669 *     // Calculate corrections per second
00670 *     uint32_t time_delta_ms = current_time - last_check_time;
00671 *     uint32_t count_delta = current_count - last_count;
00672 *
00673 *     if (time_delta_ms >= 1000) { // Check every second
00674 *         float corrections_per_sec = (float)count_delta / (time_delta_ms / 1000.0f);
00675 *
00676 *         if (corrections_per_sec > 10.0f) {
00677 *             rx_log_error("EMI", "Severe interference: %.1f corr/sec", corrections_per_sec);
00678 *         } else if (corrections_per_sec > 1.0f) {
00679 *             rx_log_warn("EMI", "Moderate interference: %.1f corr/sec", corrections_per_sec);
00680 *         }
00681 *
00682 *         last_count = current_count;
00683 *         last_check_time = current_time;
00684 *     }
00685 * }
00686 * @endcode
00687 *
00688 * @par Example - Telemetry Reporting:
00689 * @code{.c}
00690 * // Include correction count in telemetry packets
00691 * void send_telemetry(void) {
00692 *     telemetry_packet_t packet = {
00693 *         .timestamp = get_system_time_ms(),
00694 *         .temperature_celsius = read_temperature_celsius(),
00695 *         .motor_current = read_motor_current(),
00696 *         .register_corrections = rx_register_guard_get_correction_count(),
00697 *         // ... other fields ...
00698 *     };
00699 *
00700 *     transmit_telemetry(&packet);
00701 * }
00702 * @endcode
00703 *
00704 * @see rx_register_guard_refresh() Increments this counter on corrections
00705 * @see rx_register_guard_reset_count() Clear counter to zero
00706 * @see rx_register_guard_init() Also resets counter to zero
00707 *
00708 * @since Version 1.0.0
00709 * @version 1.0.0

```

```

00710 *
00711 * @test Tested in test_rx_register_guard.c::test_correction_count_increment()
00712 * @test Tested in test_rx_register_guard.c::test_correction_count_before_init()
00713 *
00714 * @par NASA Power of 10 Compliance:
00715 * - Rule 1: [OK] No goto, setjmp, recursion
00716 * - Rule 2: [OK] No loops
00717 * - Rule 5: [OK] 2 preconditions, 1 postcondition, 1 invariant
00718 *
00719 * @callgraph
00720 * @callergraph
00721 */
00722 uint32_t rx_register_guard_get_correction_count(void);
00723
00724 /**
00725 * @brief Reset the correction counter to zero
00726 *
00727 * @details
00728 * Clears the register correction counter back to zero. This function is
00729 * typically used for periodic diagnostics where you want to measure the
00730 * correction rate over specific time intervals (e.g., corrections per minute).
00731 *
00732 * Does not affect any other module state - golden reference values and
00733 * initialization status remain unchanged.
00734 *
00735 * Algorithm:
00736 * 1. Set static correction counter to 0
00737 * 2. Return (no other side effects)
00738 *
00739 * @pre No preconditions - safe to call anytime
00740 * @pre Module does not need to be initialized (no-op if not initialized)
00741 *
00742 * @post Correction counter set to 0
00743 * @post Next call to rx_register_guard_get_correction_count() returns 0
00744 *
00745 * @note **Thread Safety**: Not thread-safe with concurrent refresh() calls.
00746 * If refresh() runs during reset, the counter may be non-zero immediately
00747 * after reset. For critical measurements, disable interrupts around reset.
00748 * @note **Execution Time**: ~30 ns @ 240 MHz (single memory write)
00749 * @note **Use Case**: Periodic logging with reset allows measuring correction
00750 * rate over fixed intervals without calculating deltas.
00751 *
00752 * @par Example - Periodic Logging with Reset:
00753 * @code{.c}
00754 * // Log and reset every 60 seconds
00755 * void periodic_logging(void) {
00756 *     static uint32_t last_log_time = 0;
00757 *     uint32_t current_time = get_system_time_ms();
00758 *
00759 *     if (current_time - last_log_time >= 60000) { // 60 seconds
00760 *         uint32_t corrections = rx_register_guard_get_correction_count();
00761 *
00762 *         // Log corrections for this interval
00763 *         rx_log_info("GUARD", "Corrections last minute: %lu", corrections);
00764 *
00765 *         // Reset for next interval
00766 *         rx_register_guard_reset_count();
00767 *
00768 *         last_log_time = current_time;
00769 *     }
00770 * }
00771 * @endcode
00772 *
00773 * @par Example - Manual Diagnostics:
00774 * @code{.c}
00775 * // User-triggered diagnostics command
00776 * void cmd_show_emi_stats(void) {
00777 *     uint32_t corrections = rx_register_guard_get_correction_count();
00778 *     printf("Total corrections since boot: %lu\n", corrections);
00779 *
00780 *     // Reset to start fresh measurement
00781 *     rx_register_guard_reset_count();
00782 *     printf("Counter reset. Monitoring for new events...\n");
00783 * }
00784 * @endcode
00785 *
00786 * @par Example - Test Harness:
00787 * @code{.c}
00788 * // Unit test - verify correction counting works
00789 * void test_correction_counting(void) {
00790 *     // Clear any previous corrections
00791 *     rx_register_guard_reset_count();
00792 *
00793 *     // Corrupt a register
00794 *     PORT0.PDR.BYTE = 0xFF; // Intentional corruption
00795 *
00796 *     // Trigger refresh

```

```

00797 *   rx_register_guard_refresh();
00798 *
00799 *   // Verify correction was counted
00800 *   uint32_t corrections = rx_register_guard_get_correction_count();
00801 *   assert(corrections == 1);
00802 *
00803 *   // Clean up
00804 *   rx_register_guard_reset_count();
00805 * }
00806 * @endcode
00807 *
00808 * @see rx_register_guard_get_correction_count() Read current count
00809 * @see rx_register_guard_refresh() Increments counter on corrections
00810 * @see rx_register_guard_init() Also resets counter to zero
00811 *
00812 * @since Version 1.0.0
00813 * @version 1.0.0
00814 *
00815 * @test Tested in test_rx_register_guard.c::test_reset_count()
00816 * @test Tested in test_rx_register_guard.c::test_reset_before_init()
00817 *
00818 * @par NASA Power of 10 Compliance:
00819 * - Rule 1: [OK] No goto, setjmp, recursion
00820 * - Rule 2: [OK] No loops
00821 * - Rule 5: [OK] 2 preconditions, 2 postconditions
00822 *
00823 * @callgraph
00824 * @callergraph
00825 */
00826 void rx_register_guard_reset_count(void);
00827
00828 /**
00829 * @brief Check if register guard module is initialized
00830 *
00831 * @details
00832 * Returns the initialization status of the register guard module. This function
00833 * allows application code to verify that the guard is active before relying on
00834 * its protection, or to conditionally skip operations that depend on the guard.
00835 *
00836 * The module is considered initialized after a successful call to
00837 * rx_register_guard_init(). There is no "uninitialized" function - once
00838 * initialized, the module remains active until MCU reset.
00839 *
00840 * Algorithm:
00841 * 1. Read static initialized flag variable
00842 * 2. Return boolean value directly
00843 *
00844 * @return bool Initialization status
00845 * @retval true Module is initialized, golden values captured, refresh() is active
00846 * @retval false Module not initialized, refresh() will return immediately without effect
00847 *
00848 * @pre No preconditions - safe to call anytime
00849 *
00850 * @post Module state unchanged (read-only query)
00851 *
00852 * @note **Thread Safety**: Thread-safe (atomic read on embedded systems)
00853 * @note **Execution Time**: ~15 ns @ 240 MHz (single memory read)
00854 * @note **Use Case**: Primarily for defensive programming and diagnostics.
00855 *   Normal application flow should ensure init() is called at startup.
00856 *
00857 * @par Example - Defensive Programming:
00858 * @code{.c}
00859 * // Verify module is active before starting main loop
00860 * void main_control_loop(void) {
00861 *     if (!rx_register_guard_is_initialized()) {
00862 *         rx_log_error("MAIN", "Register guard not initialized!");
00863 *         // Attempt to initialize now
00864 *         rx_err_t err = rx_register_guard_init();
00865 *         if (err != k_rx_ok) {
00866 *             rx_log_fatal("MAIN", "Cannot initialize register guard");
00867 *             return;
00868 *         }
00869 *     }
00870 *
00871 *     // Proceed with main loop
00872 *     while (1) {
00873 *         // ... control logic ...
00874 *         rx_register_guard_refresh();
00875 *     }
00876 * }
00877 * @endcode
00878 *
00879 * @par Example - Conditional Refresh:
00880 * @code{.c}
00881 * // Skip refresh if not initialized (graceful degradation)
00882 * void periodic_maintenance(void) {
00883 *     // Other maintenance tasks...

```

```

00884 *   sensor_calibration_check();
00885 *   motor_temperature_check();
00886 *
00887 *   // Only refresh if guard is active
00888 *   if (rx_register_guard_is_initialized()) {
00889 *       rx_register_guard_refresh();
00890 *   }
00891 * }
00892 * @endcode
00893 *
00894 * @par Example - Diagnostics Report:
00895 * @code{.c}
00896 * // Generate system diagnostics report
00897 * void print_system_status(void) {
00898 *     printf("System Diagnostics:\n");
00899 *     printf(" MCU: RX72N @ 240 MHz\n");
00900 *     printf(" Uptime: %lu seconds\n", get_uptime_seconds());
00901 *     printf(" Register Guard: %s\n",
00902 *         rx_register_guard_is_initialized() ? "ACTIVE" : "INACTIVE");
00903 *
00904 *     if (rx_register_guard_is_initialized()) {
00905 *         uint32_t corrections = rx_register_guard_get_correction_count();
00906 *         printf(" - Total corrections: %lu\n", corrections);
00907 *         printf(" - Correction rate: %.2f/hour\n",
00908 *             corrections / (get_uptime_seconds() / 3600.0f));
00909 *     }
00910 * }
00911 * @endcode
00912 *
00913 * @par Example - Unit Test Setup:
00914 * @code{.c}
00915 * // Verify test preconditions
00916 * void test_refresh_requires_init(void) {
00917 *     // Ensure clean state
00918 *     assert(!rx_register_guard_is_initialized());
00919 *
00920 *     // Refresh should be no-op when not initialized
00921 *     rx_register_guard_refresh();
00922 *     assert(rx_register_guard_get_correction_count() == 0);
00923 *
00924 *     // Initialize
00925 *     rx_err_t err = rx_register_guard_init();
00926 *     assert(err == k_rx_ok);
00927 *     assert(rx_register_guard_is_initialized());
00928 *
00929 *     // Now refresh works
00930 *     rx_register_guard_refresh();
00931 * }
00932 * @endcode
00933 *
00934 * @see rx_register_guard_init() Initializes module, sets flag to true
00935 * @see rx_register_guard_refresh() Checks this flag before operating
00936 *
00937 * @since Version 1.0.0
00938 * @version 1.0.0
00939 *
00940 * @test Tested in test_rx_register_guard.c::test_is_initialized_flag()
00941 * @test Tested in test_rx_register_guard.c::test_refresh_before_init()
00942 *
00943 * @par NASA Power of 10 Compliance:
00944 * - Rule 1: [OK] No goto, setjmp, recursion
00945 * - Rule 2: [OK] No loops
00946 * - Rule 5: [OK] 1 precondition, 1 postcondition
00947 *
00948 * @callgraph
00949 * @callergraph
00950 */
00951 [[nodiscard]] bool rx_register_guard_is_initialized(void);
00952
00953 #ifdef __cplusplus
00954 }
00955 #endif

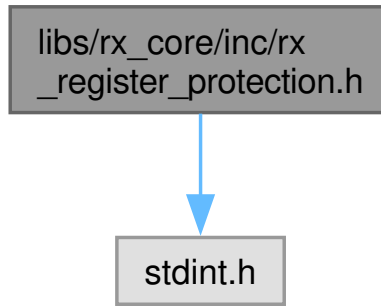
```

7.33 libs/rx_core/inc/rx_register_protection.h File Reference

RX72N Register Protection Control (PRCR) - Write Protection Constants.

#include <stdint.h>

Include dependency graph for rx_register_protection.h:



Enumerations

- enum `rx_prcr_values_t` : `uint16_t` {
 `k_rx_prcr_unlock_prc1` = 0xA502 , `k_rx_prcr_unlock_prc0_prc1` = 0xA503 , `k_rx_prcr_unlock_prc1_prc3`
 = 0xA50B , `k_rx_prcr_unlock_all` = 0xA50B ,
 `k_rx_prcr_lock` = 0xA500 , `k_rx_prcr_readback_mask` = 0x000F }

PRCR register unlock/lock values for write-protected system registers.

7.33.1 Detailed Description

RX72N Register Protection Control (PRCR) - Write Protection Constants.

Provides centralized definitions for the Protection Register (PRCR) unlock/lock sequences used throughout the RX72N firmware to safely modify write-protected system registers. The PRCR mechanism prevents accidental corruption of critical configuration registers through a key-code based protection scheme.

7.33.1.1 Hardware Protection Mechanism

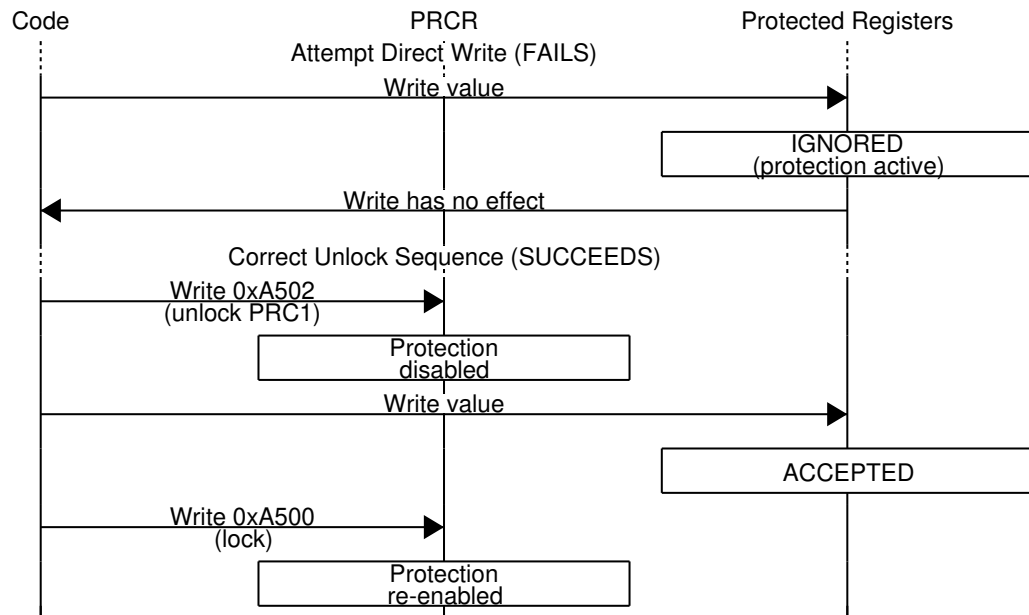
The RX72N implements hardware write protection on critical system registers to prevent accidental modification from:

- Software bugs (pointer errors, array overruns)
- EMI-induced register corruption
- Malicious code (if present)

Protected register groups:

- PRC0: Clock Generation Circuit (CGC) registers
- PRC1: Module Stop Control (MSTPCR) registers
- PRC3: Low-Voltage Detection (LVD) registers

7.33.1.2 Write Protection Flow



7.33.1.3 Key Code Mechanism

PRCR register format (16-bit):

Bits 15-8: Key Code (must be 0xA5)
 Bit 3: PRC3 (LVD protection)
 Bit 1: PRC1 (MSTPCR protection)
 Bit 0: PRC0 (CGC protection)

Security: Writing any value without the 0xA5 key code in upper byte has no effect (ignored). This prevents accidental unlock from bit flips or pointer errors.

7.33.1.4 Protected Register Groups

Group	Bit	Protected Registers	Purpose	STAR Usage
PRC0	0	SYSTEM.SCKCR, PLLCR, etc.	Clock frequency config	Clock init only
PRC1	1	SYSTEM.MSTPCRA/B/C/D	Peripheral enable/disable	Frequent (register guard)
PRC3	3	SYSTEM.LVCMPCR, LVDXCR	Low-voltage detection	Not used in STAR

7.33.1.5 Usage Patterns

Pattern 1: Unlock Single Group (Most Common)

```
SYSTEM.PRCR.WORD = k_rx_prcr_unlock_prc1; // Unlock MSTPCR only
SYSTEM.MSTPCRA.LONG &= ~(1 « 15); // Enable CMT0
SYSTEM.PRCR.WORD = k_rx_prcr_lock; // Re-lock immediately
```

Pattern 2: Unlock Multiple Groups (Clock Init)

```
SYSTEM.PRCR.WORD = k_rx_prcr_unlock_prc0_prc1; // Unlock CGC + MSTPCR
```

```

SYSTEM.PLLCR.WORD = 0x2700;           // Configure PLL
SYSTEM.SCKCR.LONG = 0x21C21211;       // Configure clocks
SYSTEM.MSTPCRA.LONG = 0x00000000;     // Enable peripherals
SYSTEM.PRCR.WORD = k_rx_pcr_lock;     // Re-lock all

```

Pattern 3: Critical Section (Register Guard)

```

uint32_t saved_ipl = disable_interrupts(); // Atomic section
SYSTEM.PRCR.WORD = k_rx_pcr_unlock_prc1;
restore_mstpcr_registers();                // Refresh corrupted regs
SYSTEM.PRCR.WORD = k_rx_pcr_lock;
restore_interrupts(saved_ipl);             // End atomic

```

7.33.1.6 Performance Characteristics

- Write latency: 1 bus cycle (4.17 ns @ 240 MHz ICLK)
- Unlock overhead: 2 writes (unlock + lock) = ~8 ns
- Protection check: Hardware (zero CPU overhead)
- Memory mapped: 0x000803FE (SYSTEM.PRCR)

7.33.1.7 Hardware Requirements

Requirement	Value	Notes
MCU	RX72N	PRCR at 0x000803FE
Register Size	16-bit	Must use WORD access
Key Code	0xA5	Upper 8 bits required
Protection Bits	PRC0, PRC1, PRC3	Lower 4 bits

Module Dependencies:

- NO dependencies (pure constant definitions)
- Used by: Clock init, module stop control, register guard, power management

NASA Power of 10 Compliance:

- Rule 1: [OK] No code (only data definitions)
- Rule 2: [OK] No loops
- Rule 3: [OK] No dynamic allocation (compile-time constants)
- Rule 4: [OK] No functions
- Rule 5: [OK] N/A (no functions)
- Rule 6: [OK] File scope only (header file)
- Rule 7: [OK] No functions
- Rule 8: [OK] C23 typed enums for all constants
- Rule 9: [OK] No pointers
- Rule 10: [OK] Compiles with -Wall -Wextra -Werror

SOLID Principles:

- Single Responsibility: Only defines PRCR constants, no logic
- Open/Closed: Extensible (add new combinations) without modification
- Liskov Substitution: N/A (no inheritance)
- Interface Segregation: Minimal interface (single enum)
- Dependency Inversion: N/A (no dependencies)

Usage Example - Module Enable:

```
#include "rx_register_protection.h"
#include "rx72n_regs.h"

// Enable CMT0 timer module (in MSTPCRA bit 15)
void enable_cmt0(void) {
    typedef enum : uint32_t {
        k_mstpcra_cmt0_bit = 15, // CMT0 module stop bit
    } mstpcr_bits_t;

    // Unlock MSTPCR protection
    SYSTEM.PRCR.WORD = k_rx_prcr_unlock_prc1;

    // Clear module stop bit (0 = enabled)
    SYSTEM.MSTPCRA.LONG &= ~(1UL « k_mstpcra_cmt0_bit);

    // Re-lock protection immediately
    SYSTEM.PRCR.WORD = k_rx_prcr_lock;
}
```

Usage Example - Clock Configuration:

```
// Configure system clocks during initialization
void configure_system_clocks(void) {
    // Unlock both CGC and MSTPCR groups
    SYSTEM.PRCR.WORD = k_rx_prcr_unlock_prc0_prc1;

    // Configure PLL (protected by PRC0)
    SYSTEM.PLLCR.WORD = 0x1300; // PLL multiplier x10 (STAR: 24 MHz * 10 = 240 MHz)
    SYSTEM.PLLCR2.BYTE = 0x00; // Enable PLL

    // Wait for PLL stabilization
    while (SYSTEM.OSCOVFSR.BIT.PLOVF == 0) {
        // Busy wait (~2ms)
    }

    // Select PLL as system clock (protected by PRC0)
    SYSTEM.SCKCR3.WORD = 0x0400; // CKSEL = PLL

    // Enable all peripheral modules (protected by PRC1)
    SYSTEM.MSTPCRA.LONG = 0x00000000;
    SYSTEM.MSTPCRB.LONG = 0x00000000;

    // Re-lock all protection
    SYSTEM.PRCR.WORD = k_rx_prcr_lock;
}
```

Usage Example - Atomic Register Refresh (Register Guard):

```
// Restore corrupted MSTPCR registers atomically
void restore_mstpcr_atomically(void) {
    // Brief interrupt disable for atomicity
    uint32_t saved_ipl = __get_ipl();
    __set_ipl(15); // Disable interrupts

    // Unlock MSTPCR group
    SYSTEM.PRCR.WORD = k_rx_prcr_unlock_prc1;

    // Restore golden values (from register guard)
    SYSTEM.MSTPCRA.LONG = s_golden_mstpcra;
    SYSTEM.MSTPCRB.LONG = s_golden_mstpcrb;
    SYSTEM.MSTPCRC.LONG = s_golden_mstpcrc;
}
```



```

SYSTEM.MSTPCRD.LONG = s_golden_mstpcrd;

// Re-lock protection
SYSTEM.PRCR.WORD = k_rx_prcr_lock;

// Restore interrupts
__set_ipl(saved_ipl);
}

```

See also

[RX72N User's Manual](#) Chapter 9: System Control

[RX72N User's Manual](#) Section 9.2.2: Protection Register (PRCR)

[rx_register_guard.c](#) Uses PRCR for MSTPCR refresh

[rx_clock_power_init.c](#) Uses PRCR for clock config

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Date

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Version

1.0.0

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Since

Version 1.0.0

Definition in file [rx_register_protection.h](#).

7.33.2 Enumeration Type Documentation

7.33.2.1 rx_prcr_values_t

enum [rx_prcr_values_t](#) : uint16_t

PRCR register unlock/lock values for write-protected system registers.

Defines all valid PRCR (Protect Register) values for unlocking and locking write-protected system registers on the RX72N. The PRCR uses a key code mechanism (0xA5 in upper byte) combined with protection enable bits (lower byte) to selectively unlock register groups.

Register Format (16-bit SYSTEM.PRCR at 0x000803FE):

Bits 15-8: [0xA5] Key code (required **for** any write to take effect)
 Bits 7-4: [Reserved] Always write 0
 Bit 3: [PRC3] Low-Voltage Detection protection (1=unlocked, 0=locked)
 Bit 2: [Reserved] Always write 0
 Bit 1: [PRC1] Module Stop Control protection (1=unlocked, 0=locked)
 Bit 0: [PRC0] Clock Generation Circuit protection (1=unlocked, 0=locked)

Key Code Security:

- Writing without 0xA5 key -> ignored (no effect)
- Prevents accidental unlock from pointer errors
- Prevents unlock from EMI-induced bit flips

Protection Groups:

- PRC0 (bit 0): Protects SCKCR, PLLCR, PLLCR2, SCKCR3, MOSCWTCR, etc.
- PRC1 (bit 1): Protects MSTPCRA, MSTPCRB, MSTPCRC, MSTPCRD
- PRC3 (bit 3): Protects LVCMPCR, LVDXCR, LVDYCR

7.33.2.2 Usage Guidelines

1. Minimize unlock duration: Unlock, modify, lock immediately
2. Use narrowest scope: Only unlock groups you need to modify
3. Atomic for critical sections: Disable interrupts during unlock
4. Always re-lock: Even on error paths (use cleanup pattern)

7.33.2.3 STAR Project Usage Frequency

Constant	Frequency	Context
k_rx_prcr_unlock_prc1	High	Register guard refresh (1-10 Hz)
k_rx_prcr_unlock_prc0_prc1	Once	Clock initialization at boot
k_rx_prcr_unlock_prc1_prc3	Never	LVD not used in STAR
k_rx_prcr_unlock_all	Never	Overly broad, avoid
k_rx_prcr_lock	High	Always called after unlock

Field Constraints:

Constant	Key	PRC3	PRC1	PRC0	Notes
unlock_prc1	0xA5	0	1	0	Most common (MSTPCR)
unlock_prc0_prc1	0xA5	0	1	1	Clock init
unlock_prc1_prc3	0xA5	1	1	1	With LVD (unused)
unlock_all	0xA5	1	1	1	Avoid (too broad)
lock	0xA5	0	0	0	Always used

Usage Example - Typical Pattern:

```
void enable_peripheral_module(void) {  
    // Unlock only what's needed (PRC1 for MSTPCR)  
    SYSTEM.PRCR.WORD = k_rx_pcr_unlock_prc1;  
  
    // Modify protected register  
    SYSTEM.MSTPCRA.LONG &= ~(1UL « 15); // Enable CMT0  
  
    // Re-lock immediately  
    SYSTEM.PRCR.WORD = k_rx_pcr_lock;  
}
```

See also

SYSTEM.PRCR Hardware register at 0x000803FE

[rx_register_guard_refresh\(\)](#) Uses PRC1 unlock for MSTPCR refresh

[rx_clock_power_init\(\)](#) Uses PRC0+PRC1 unlock for clock config

Since

Version 1.0.0

Enumerator

k_rx_prcr_unlock_prc1	Unlock PRC1 only (Module Stop Control registers). Unlocks write protection for MSTPCRA, MSTPCRB, MSTPCRC, MSTPCRD registers while keeping clock generation (PRC0) and LVD (PRC3) registers protected. Value: 0xA502 • Key code: 0xA5 (required) • PRC3: 0 (LVD registers remain locked) • PRC1: 1 (MSTPCR registers unlocked) • PRC0: 0 (Clock registers remain locked) Protected Registers (when unlocked): • SYSTEM.MSTPCRA (0x00080010) - Module stop control A • SYSTEM.MSTPCRB (0x00080014) - Module stop control B • SYSTEM.MSTPCRC (0x00080018) - Module stop control C • SYSTEM.MSTPCRD (0x0008001C) - Module stop control D Use Cases: • Enabling/disabling peripheral modules at runtime • Register guard periodic refresh (most common) • Dynamic power management STAR Project Usage: HIGH FREQUENCY • Called during register guard refresh (1-10 Hz) • Called during peripheral enable/disable • ~10,000+ times per hour in typical operation Rationale: This is the most commonly used unlock value because: • MSTPCR registers change frequently (module enable/disable) • Clock registers (PRC0) rarely change after boot • Narrowest scope reduces risk of accidental corruption Example - Enable SCI0 (UART): <pre>typedef enum : uint32_t { k_mstpcrb_sci0_bit = 31, // SCI0 in MSTPCRB bit 31 } mstpcrb_bits_t; SYSTEM.PRCR.WORD = k_rx_prcr_unlock_prc1; SYSTEM.MSTPCRB.LONG &= ~(1UL « k_mstpcrb_sci0_bit); // 0 = enable SYSTEM.PRCR.WORD = k_rx_prcr_lock;</pre> Warning Interrupts are NOT disabled automatically. For atomic operations, disable interrupts explicitly before unlock.
------------------------------	---

k_rx_prcr_unlock_prc0_prc1	Unlock PRC0+PRC1 (Clock Generation + Module Stop Control). Unlocks write protection for both clock generation circuit (CGC) registers and module stop control (MSTPCR) registers. Used during system initialization when configuring PLL, system clocks, and peripheral modules together. Value: 0xA503 • Key code: 0xA5 (required) • PRC3: 0 (LVD registers remain locked) • PRC1: 1 (MSTPCR registers unlocked) • PRC0: 1 (Clock registers unlocked) Protected Registers (when unlocked): PRC0 (Clock Generation): • SYSTEM.SCKCR (0x00080020) - System clock control • SYSTEM.SCKCR2 (0x00080024) - System clock control 2 • SYSTEM.SCKCR3 (0x00080026) - System clock control 3 • SYSTEM.PLLCR (0x00080028) - PLL control • SYSTEM.PLLCR2 (0x0008002A) - PLL control 2 • SYSTEM.MOSCCR (0x00080032) - Main oscillator control • SYSTEM.MOSCWTCR (0x000800A2) - Main oscillator wait control PRC1 (Module Stop): • SYSTEM.MSTPCRA/B/C/D (see k_rx_prcr_unlock_prc1) Use Cases: • System clock initialization at boot • PLL configuration and selection • Switching clock sources (main osc <-> PLL) • Enabling peripheral clocks simultaneously STAR Project Usage: LOW FREQUENCY • Called once during rx_clock_power_init() at boot • May be called during clock source transitions (rare) • ~1-2 times per boot cycle Rationale: Combined unlock used during initialization because: • Clock and module configuration often done together • Reduces number of lock/unlock cycles at boot • Acceptable risk during controlled init sequence Example - Clock Initialization: <pre>void configure_240mhz_operation(void) { // Unlock both clock and module stop registers SYSTEM.PRCR.WORD = k_rx_prcr_unlock_prc0_prc1; // Configure PLL (PRC0 protected) SYSTEM.PLLCR.WORD = 0x1300; // x10 multiplier (STAR: 24 MHz * 10 = 240 MHz) SYSTEM.PLLCR2.BYTE = 0x00; // Enable PLL }</pre>
----------------------------	--

k_rx_prcr_unlock_prc1_prc3	Unlock PRC1+PRC3 (Module Stop + Low-Voltage Detection). Unlocks write protection for module stop control (MSTPCR) and low-voltage detection (LVD) registers. This combination is rarely needed as LVD is typically configured once at boot and left unchanged. Value: 0xA50B • Key code: 0xA5 (required) • PRC3: 1 (LVD registers unlocked) • PRC1: 1 (MSTPCR registers unlocked) • PRC0: 0 (Clock registers remain locked) Protected Registers (when unlocked): PRC1 (Module Stop): See k_rx_prcr_unlock_prc1 PRC3 (Low-Voltage Detection): • SYSTEM.LVCMPCR (0x000800E0) - Voltage monitor control • SYSTEM.LVDXCR (0x000800E1) - Voltage detect 1 control • SYSTEM.LVDYCR (0x000800E2) - Voltage detect 2 control STAR Project Usage: NEVER USED • LVD not implemented in STAR project • 3.3V supply is stable and monitored externally • Included for completeness only Rationale: Not used in STAR because: • LVD not required (external power monitoring) • MSTPCR and LVD changes rarely coincide • More specific unlocks preferred (narrower scope) Warning Do not use unless your application specifically needs both MSTPCR and LVD modification in same critical section.
----------------------------	---

k_rx_prcr_unlock_all	Unlock all protection groups (PRC0+PRC1+PRC3). Unlocks write protection for ALL protected register groups: clock generation, module stop control, and low-voltage detection. This is the broadest unlock and should be AVOIDED in production code. Value: 0xA50B • Key code: 0xA5 (required) • PRC3: 1 (LVD registers unlocked) [bit 3] • PRC1: 1 (MSTPCR registers unlocked) [bit 1] • PRC0: 1 (Clock registers unlocked) [bit 0] • Bit 2 is reserved and must be written as 0 Note Prior revision set this to 0xA50F, which also asserts reserved bit 2 (PRCR[2] must be written 0 per RX72N HW manual R01UH0824EJ0111 Section 13.2.1). Corrected to 0xA50B. Protected Registers (when unlocked): • All PRC0 registers (clock generation) • All PRC1 registers (module stop) • All PRC3 registers (LVD) STAR Project Usage: NEVER USED • Too broad, violates principle of least privilege • More specific unlocks always available • Included for completeness only Rationale for AVOIDANCE: • Security: Unlocks more than necessary (attack surface) • Safety: Accidental corruption of unrelated registers • Best Practice: Use narrowest scope possible Warning DO NOT USE unless you have a specific reason to modify registers across all three protection groups simultaneously. Use specific unlock combinations instead. Attention Code review flag: If you see this constant used, investigate why the more specific unlocks (prc1, prc0_prc1) cannot be used instead.
----------------------	---

k_rx_prcr_lock	Lock all protection groups (re-enable write protection). Locks (re-enables) write protection for ALL register groups by clearing all protection enable bits while maintaining the required 0xA5 key code. Value: 0xA500 • Key code: 0xA5 (required for write to take effect) • PRC3: 0 (LVD registers locked) • PRC1: 0 (MSTPCR registers locked) • PRC0: 0 (Clock registers locked) Effect: After writing this value, all protected registers become read-only until the next unlock operation. Any attempt to write protected registers (without first unlocking) will be silently ignored by hardware. Usage: ALWAYS write this value immediately after completing protected register modifications. Even on error paths, ensure protection is re-locked. STAR Project Usage: HIGH FREQUENCY • Called after every unlock operation • ~10,000+ times per hour (matches unlock frequency) Rationale: Always lock immediately because: • Minimizes window where accidental writes could corrupt registers • Protects against EMI-induced bit flips • Follows secure coding best practices Example - Cleanup Pattern: <pre>rx_err_t modify_protected_register(void) { SYSTEM.PRCR.WORD = k_rx_prcr_unlock_prc1; // Modify register (may fail) if (!configure_module()) { SYSTEM.PRCR.WORD = k_rx_prcr_lock; // Lock even on error! return k_rx_err_hardware; } SYSTEM.PRCR.WORD = k_rx_prcr_lock; // Lock on success return k_rx_ok; }</pre> Example - RAII Pattern (C++ style cleanup): <pre>// Define cleanup macro (ensures lock on scope exit) #define RX_PRCR_SCOPED_UNLOCK(group) \ for (int _i = (SYSTEM.PRCR.WORD = (group), 0); \ _i < 1; \ _i++, SYSTEM.PRCR.WORD = k_rx_prcr_lock) // Usage: automatic lock at end of scope void modify_with_cleanup(void) { RX_PRCR_SCOPED_UNLOCK(k_rx_prcr_unlock_prc1) { SYSTEM.MSTPCRA.LONG &= ~(1UL « 15); // Lock happens automatically when scope exits } }</pre> Note
Generated by Doxygen	Hardware behavior: After lock, protected register writes are ignored (no bus error or exception - just silently discarded).

k_rx_prcr_readback_mask	Mask for PRC protection bits (for readback verification). When reading PRCR, the key code (upper byte) reads back as 0x00, NOT as 0xA5. Only the lower 4 bits (PRC0, PRC1, PRC3) contain the actual protection state. Value: 0x000F • Bits 3,1,0: PRC bits (protection state) • Bits 7-4, 15-8: Read as zero Usage: Use this mask when verifying PRCR readback: <pre> prcr_reg() = k_rx_prcr_unlock_prc1; // Verify unlock (key byte reads as 0x00, so mask it out) if ((*prcr_reg() & k_rx_prcr_readback_mask) != (k_rx_prcr_unlock_prc1 & k_rx_prcr_readback_mask)) { // Unlock failed } </pre>
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Definition at line 308 of file [rx_register_protection.h](#).

```

00308         : uint16_t {
00309     /**
00310      * @brief Unlock PRC1 only (Module Stop Control registers)
00311      *
00312      * @details
00313      * Unlocks write protection for MSTPCRA, MSTPCRB, MSTPCRC, MSTPCRD registers
00314      * while keeping clock generation (PRC0) and LVD (PRC3) registers protected.
00315      *
00316      * **Value**: 0xA502
00317      * - Key code: 0xA5 (required)
00318      * - PRC3: 0 (LVD registers remain locked)
00319      * - PRC1: 1 (MSTPCR registers unlocked)
00320      * - PRC0: 0 (Clock registers remain locked)
00321      *
00322      * **Protected Registers** (when unlocked):
00323      * - SYSTEM.MSTPCRA (0x00080010) - Module stop control A
00324      * - SYSTEM.MSTPCRB (0x00080014) - Module stop control B
00325      * - SYSTEM.MSTPCRC (0x00080018) - Module stop control C
00326      * - SYSTEM.MSTPCRD (0x0008001C) - Module stop control D
00327      *
00328      * **Use Cases**:
00329      * - Enabling/disabling peripheral modules at runtime
00330      * - Register guard periodic refresh (most common)
00331      * - Dynamic power management
00332      *
00333      * **STAR Project Usage**: **HIGH FREQUENCY**
00334      * - Called during register guard refresh (1-10 Hz)
00335      * - Called during peripheral enable/disable
00336      * - ~10,000+ times per hour in typical operation
00337      *
00338      * @par Rationale:
00339      * This is the most commonly used unlock value because:
00340      * - MSTPCR registers change frequently (module enable/disable)
00341      * - Clock registers (PRC0) rarely change after boot
00342      * - Narrowest scope reduces risk of accidental corruption
00343      *
00344      * @par Example - Enable SCI0 (UART):
00345      * @code{.c}
00346      * typedef enum : uint32_t {
00347      *     k_mstpcrb_sci0_bit = 31, // SCI0 in MSTPCRB bit 31
00348      * } mstpcrb_bits_t;
00349      *
00350      * SYSTEM.PRCR.WORD = k_rx_prcr_unlock_prc1;
00351      * SYSTEM.MSTPCRB.LONG &= ~(1UL << k_mstpcrb_sci0_bit); // 0 = enable
00352      * SYSTEM.PRCR.WORD = k_rx_prcr_lock;
00353      * @endcode
00354      *
00355      * @warning Interrupts are NOT disabled automatically. For atomic operations,
00356      *         disable interrupts explicitly before unlock.
00357      */
00358     k_rx_prcr_unlock_prc1 = 0xA502,
00359 }
00360 /**
00361  * @brief Unlock PRC0+PRC1 (Clock Generation + Module Stop Control)
00362  *
00363  * @details
00364  * Unlocks write protection for both clock generation circuit (CGC) registers
00365  * and module stop control (MSTPCR) registers. Used during system initialization
00366  * when configuring PLL, system clocks, and peripheral modules together.
00367  */

```

```

00368  * **Value**: 0xA503
00369  * - Key code: 0xA5 (required)
00370  * - PRC3: 0 (LVD registers remain locked)
00371  * - PRC1: 1 (MSTPCR registers unlocked)
00372  * - PRC0: 1 (Clock registers unlocked)
00373  *
00374  * **Protected Registers** (when unlocked):
00375  *
00376  * **PRC0 (Clock Generation)**:
00377  * - SYSTEM.SCKCR (0x00080020) - System clock control
00378  * - SYSTEM.SCKCR2 (0x00080024) - System clock control 2
00379  * - SYSTEM.SCKCR3 (0x00080026) - System clock control 3
00380  * - SYSTEM.PLLCR (0x00080028) - PLL control
00381  * - SYSTEM.PLLCR2 (0x0008002A) - PLL control 2
00382  * - SYSTEM.MOSCCR (0x00080032) - Main oscillator control
00383  * - SYSTEM.MOSCWTCR (0x000800A2) - Main oscillator wait control
00384  *
00385  * **PRC1 (Module Stop)**:
00386  * - SYSTEM.MSTPCRA/B/C/D (see k_rx_prcr_unlock_prc1)
00387  *
00388  * **Use Cases**:
00389  * - System clock initialization at boot
00390  * - PLL configuration and selection
00391  * - Switching clock sources (main osc <-> PLL)
00392  * - Enabling peripheral clocks simultaneously
00393  *
00394  * **STAR Project Usage**: **LOW FREQUENCY**
00395  * - Called once during rx_clock_power_init() at boot
00396  * - May be called during clock source transitions (rare)
00397  * - ~1-2 times per boot cycle
00398  *
00399  * @par Rationale:
00400  * Combined unlock used during initialization because:
00401  * - Clock and module configuration often done together
00402  * - Reduces number of lock/unlock cycles at boot
00403  * - Acceptable risk during controlled init sequence
00404  *
00405  * @par Example - Clock Initialization:
00406  * @code{.c}
00407  * void configure_240mhz_operation(void) {
00408  *     // Unlock both clock and module stop registers
00409  *     SYSTEM.PRCR.WORD = k_rx_prcr_unlock_prc0_prc1;
00410  *
00411  *     // Configure PLL (PRC0 protected)
00412  *     SYSTEM.PLLCR.WORD = 0x1300;    // x10 multiplier (STAR: 24 MHz * 10 = 240 MHz)
00413  *     SYSTEM.PLLCR2.BYTE = 0x00;    // Enable PLL
00414  *
00415  *     // Wait for PLL lock (~2ms)
00416  *     while (!SYSTEM.OSCOVFSR.BIT.PLOVF) {}
00417  *
00418  *     // Switch to PLL (PRC0 protected)
00419  *     SYSTEM.SCKCR3.WORD = 0x0400;    // Select PLL as system clock
00420  *     SYSTEM.SCKCR.LONG = 0x20011222; // STAR production dividers: ICK=/1 (240), PCKA=/2 (120),
PCKB/C/D/FCK=/4 (60), BCK=/2 (120)
00421  *
00422  *     // Enable all peripherals (PRC1 protected)
00423  *     SYSTEM.MSTPCRA.LONG = 0x00000000;
00424  *     SYSTEM.MSTPCRB.LONG = 0x00000000;
00425  *
00426  *     // Re-lock all
00427  *     SYSTEM.PRCR.WORD = k_rx_prcr_lock;
00428  * }
00429  * @endcode
00430  *
00431  * @warning Only use during initialization. Runtime clock changes should use
00432  * separate PRC0-only unlock (not defined here - add if needed).
00433  */
00434  k_rx_prcr_unlock_prc0_prc1 = 0xA503,
00435
00436  /**
00437  * @brief Unlock PRC1+PRC3 (Module Stop + Low-Voltage Detection)
00438  *
00439  * @details
00440  * Unlocks write protection for module stop control (MSTPCR) and low-voltage
00441  * detection (LVD) registers. This combination is rarely needed as LVD is
00442  * typically configured once at boot and left unchanged.
00443  *
00444  * **Value**: 0xA50B
00445  * - Key code: 0xA5 (required)
00446  * - PRC3: 1 (LVD registers unlocked)
00447  * - PRC1: 1 (MSTPCR registers unlocked)
00448  * - PRC0: 0 (Clock registers remain locked)
00449  *
00450  * **Protected Registers** (when unlocked):
00451  *
00452  * **PRC1 (Module Stop)**: See k_rx_prcr_unlock_prc1
00453  *

```

```

00454 * **PRC3 (Low-Voltage Detection)**:
00455 * - SYSTEM.LVCMPCR (0x000800E0) - Voltage monitor control
00456 * - SYSTEM.LVDXCR (0x000800E1) - Voltage detect 1 control
00457 * - SYSTEM.LVDYCR (0x000800E2) - Voltage detect 2 control
00458 *
00459 * **STAR Project Usage**: **NEVER USED**
00460 * - LVD not implemented in STAR project
00461 * - 3.3V supply is stable and monitored externally
00462 * - Included for completeness only
00463 *
00464 * @par Rationale:
00465 * Not used in STAR because:
00466 * - LVD not required (external power monitoring)
00467 * - MSTPCR and LVD changes rarely coincide
00468 * - More specific unlocks preferred (narrower scope)
00469 *
00470 * @warning Do not use unless your application specifically needs both MSTPCR
00471 * and LVD modification in same critical section.
00472 */
00473 k_rx_prcr_unlock_prc1_prc3 = 0xA50B,
00474
00475 /**
00476 * @brief Unlock all protection groups (PRC0+PRC1+PRC3)
00477 *
00478 * @details
00479 * Unlocks write protection for ALL protected register groups: clock generation,
00480 * module stop control, and low-voltage detection. This is the broadest unlock
00481 * and should be **AVOIDED** in production code.
00482 *
00483 * **Value**: 0xA50B
00484 * - Key code: 0xA5 (required)
00485 * - PRC3: 1 (LVD registers unlocked) [bit 3]
00486 * - PRC1: 1 (MSTPCR registers unlocked) [bit 1]
00487 * - PRC0: 1 (Clock registers unlocked) [bit 0]
00488 * - Bit 2 is reserved and must be written as 0
00489 *
00490 * @note Prior revision set this to 0xA50F, which also asserts reserved bit 2
00491 * (PRCR[2] must be written 0 per RX72N HW manual R01UH0824EJ0111
00492 * Section 13.2.1). Corrected to 0xA50B.
00493 *
00494 * **Protected Registers** (when unlocked):
00495 * - All PRC0 registers (clock generation)
00496 * - All PRC1 registers (module stop)
00497 * - All PRC3 registers (LVD)
00498 *
00499 * **STAR Project Usage**: **NEVER USED**
00500 * - Too broad, violates principle of least privilege
00501 * - More specific unlocks always available
00502 * - Included for completeness only
00503 *
00504 * @par Rationale for AVOIDANCE:
00505 * - **Security**: Unlocks more than necessary (attack surface)
00506 * - **Safety**: Accidental corruption of unrelated registers
00507 * - **Best Practice**: Use narrowest scope possible
00508 *
00509 * @warning **DO NOT USE** unless you have a specific reason to modify registers
00510 * across all three protection groups simultaneously. Use specific
00511 * unlock combinations instead.
00512 *
00513 * @attention Code review flag: If you see this constant used, investigate why
00514 * the more specific unlocks (prc1, prc0_prc1) cannot be used instead.
00515 */
00516 k_rx_prcr_unlock_all = 0xA50B,
00517
00518 /**
00519 * @brief Lock all protection groups (re-enable write protection)
00520 *
00521 * @details
00522 * Locks (re-enables) write protection for ALL register groups by clearing
00523 * all protection enable bits while maintaining the required 0xA5 key code.
00524 *
00525 * **Value**: 0xA500
00526 * - Key code: 0xA5 (required for write to take effect)
00527 * - PRC3: 0 (LVD registers locked)
00528 * - PRC1: 0 (MSTPCR registers locked)
00529 * - PRC0: 0 (Clock registers locked)
00530 *
00531 * **Effect**: After writing this value, all protected registers become read-only
00532 * until the next unlock operation. Any attempt to write protected registers
00533 * (without first unlocking) will be silently ignored by hardware.
00534 *
00535 * **Usage**: **ALWAYS** write this value immediately after completing protected
00536 * register modifications. Even on error paths, ensure protection is re-locked.
00537 *
00538 * **STAR Project Usage**: **HIGH FREQUENCY**
00539 * - Called after every unlock operation
00540 * - ~10,000+ times per hour (matches unlock frequency)

```

```

00541 *
00542 * @par Rationale:
00543 * Always lock immediately because:
00544 * - Minimizes window where accidental writes could corrupt registers
00545 * - Protects against EMI-induced bit flips
00546 * - Follows secure coding best practices
00547 *
00548 * @par Example - Cleanup Pattern:
00549 * @code{.c}
00550 * rx_err_t modify_protected_register(void) {
00551 *     SYSTEM.PRCR.WORD = k_rx_prcr_unlock_prc1;
00552 *
00553 *     // Modify register (may fail)
00554 *     if (!configure_module()) {
00555 *         SYSTEM.PRCR.WORD = k_rx_prcr_lock; // Lock even on error!
00556 *         return k_rx_err_hardware;
00557 *     }
00558 *
00559 *     SYSTEM.PRCR.WORD = k_rx_prcr_lock; // Lock on success
00560 *     return k_rx_ok;
00561 * }
00562 * @endcode
00563 *
00564 * @par Example - RAII Pattern (C++ style cleanup):
00565 * @code{.c}
00566 * // Define cleanup macro (ensures lock on scope exit)
00567 * #define RX_PRCR_SCOPED_UNLOCK(group) \
00568 *     for (int _i = (SYSTEM.PRCR.WORD = (group), 0); \
00569 *          _i < 1; \
00570 *          _i++, SYSTEM.PRCR.WORD = k_rx_prcr_lock)
00571 *
00572 * // Usage: automatic lock at end of scope
00573 * void modify_with_cleanup(void) {
00574 *     RX_PRCR_SCOPED_UNLOCK(k_rx_prcr_unlock_prc1) {
00575 *         SYSTEM.MSTPCRA.LONG &= ~(1UL « 15);
00576 *         // Lock happens automatically when scope exits
00577 *     }
00578 * }
00579 * @endcode
00580 *
00581 * @note Hardware behavior: After lock, protected register writes are ignored
00582 *       (no bus error or exception - just silently discarded).
00583 */
00584 k_rx_prcr_lock = 0xA500,
00585
00586 /**
00587 * @brief Mask for PRC protection bits (for readback verification)
00588 *
00589 * @details
00590 * When reading PRCR, the key code (upper byte) reads back as 0x00,
00591 * NOT as 0xA5. Only the lower 4 bits (PRC0, PRC1, PRC3) contain
00592 * the actual protection state.
00593 *
00594 * **Value**: 0x000F
00595 * - Bits 3,1,0: PRC bits (protection state)
00596 * - Bits 7-4, 15-8: Read as zero
00597 *
00598 * **Usage**: Use this mask when verifying PRCR readback:
00599 * @code{.c}
00600 * *prcr_reg() = k_rx_prcr_unlock_prc1;
00601 * // Verify unlock (key byte reads as 0x00, so mask it out)
00602 * if ((*prcr_reg() & k_rx_prcr_readback_mask) !=
00603 *     (k_rx_prcr_unlock_prc1 & k_rx_prcr_readback_mask)) {
00604 *     // Unlock failed
00605 * }
00606 * @endcode
00607 */
00608 k_rx_prcr_readback_mask = 0x000F,
00609
00610 } rx_prcr_values_t;

```

7.34 rx_register_protection.h

[Go to the documentation of this file.](#)

```

00001 /**
00002 * @file rx_register_protection.h
00003 * @brief RX72N Register Protection Control (PRCR) - Write Protection Constants
00004 *
00005 * @details
00006 * Provides centralized definitions for the Protection Register (PRCR) unlock/lock
00007 * sequences used throughout the RX72N firmware to safely modify write-protected

```

```

00008 * system registers. The PRCR mechanism prevents accidental corruption of critical
00009 * configuration registers through a key-code based protection scheme.
00010 *
00011 * ## Hardware Protection Mechanism
00012 *
00013 * The RX72N implements hardware write protection on critical system registers to
00014 * prevent accidental modification from:
00015 * - Software bugs (pointer errors, array overruns)
00016 * - EMI-induced register corruption
00017 * - Malicious code (if present)
00018 *
00019 * Protected register groups:
00020 * - **PRC0**: Clock Generation Circuit (CGC) registers
00021 * - **PRC1**: Module Stop Control (MSTPCR) registers
00022 * - **PRC3**: Low-Voltage Detection (LVD) registers
00023 *
00024 * ## Write Protection Flow
00025 *
00026 * @msc
00027 * Code, PRCR, "Protected Registers";
00028 *
00029 * ... [label="Attempt Direct Write (FAILS)"];
00030 * Code => "Protected Registers" [label="Write value"];
00031 * "Protected Registers" box "Protected Registers" [label="IGNORED\n(protection active)"];
00032 * "Protected Registers" => Code [label="Write has no effect"];
00033 *
00034 * ... [label="Correct Unlock Sequence (SUCCEEDS)"];
00035 * Code => PRCR [label="Write 0xA502\n(unlock PRC1)"];
00036 * PRCR box PRCR [label="Protection\nndisabled"];
00037 * Code => "Protected Registers" [label="Write value"];
00038 * "Protected Registers" box "Protected Registers" [label="ACCEPTED"];
00039 * Code => PRCR [label="Write 0xA500\n(lock)"];
00040 * PRCR box PRCR [label="Protection\nre-enabled"];
00041 * @endmsc
00042 *
00043 * ## Key Code Mechanism
00044 *
00045 * PRCR register format (16-bit):
00046 * ```
00047 * Bits 15-8: Key Code (must be 0xA5)
00048 * Bit 3: PRC3 (LVD protection)
00049 * Bit 1: PRC1 (MSTPCR protection)
00050 * Bit 0: PRC0 (CGC protection)
00051 * ```
00052 *
00053 * **Security**: Writing any value without the 0xA5 key code in upper byte has
00054 * no effect (ignored). This prevents accidental unlock from bit flips or
00055 * pointer errors.
00056 *
00057 * ## Protected Register Groups
00058 *
00059 * | Group | Bit | Protected Registers | Purpose | STAR Usage |
00060 * |-----|----|-----|-----|-----|
00061 * | PRC0 | 0 | SYSTEM.SCKCR, PLLCR, etc. | Clock frequency config | Clock init only |
00062 * | PRC1 | 1 | SYSTEM.MSTPCRA/B/C/D | Peripheral enable/disable | Frequent (register guard) |
00063 * | PRC3 | 3 | SYSTEM.LVCMPCR, LVDXCR | Low-voltage detection | Not used in STAR |
00064 *
00065 * ## Usage Patterns
00066 *
00067 * **Pattern 1: Unlock Single Group (Most Common)**
00068 * ```c
00069 * SYSTEM.PRCR.WORD = k_rx_prcr_unlock_prc1; // Unlock MSTPCR only
00070 * SYSTEM.MSTPCRA.LONG &= ~(1 << 15); // Enable CMT0
00071 * SYSTEM.PRCR.WORD = k_rx_prcr_lock; // Re-lock immediately
00072 * ```
00073 *
00074 * **Pattern 2: Unlock Multiple Groups (Clock Init)**
00075 * ```c
00076 * SYSTEM.PRCR.WORD = k_rx_prcr_unlock_prc0_prc1; // Unlock CGC + MSTPCR
00077 * SYSTEM.PLLCR.WORD = 0x2700; // Configure PLL
00078 * SYSTEM.SCKCR.LONG = 0x21C21211; // Configure clocks
00079 * SYSTEM.MSTPCRA.LONG = 0x00000000; // Enable peripherals
00080 * SYSTEM.PRCR.WORD = k_rx_prcr_lock; // Re-lock all
00081 * ```
00082 *
00083 * **Pattern 3: Critical Section (Register Guard)**
00084 * ```c
00085 * uint32_t saved_ipl = disable_interrupts(); // Atomic section
00086 * SYSTEM.PRCR.WORD = k_rx_prcr_unlock_prc1;
00087 * restore_mstpcr_registers(); // Refresh corrupted regs
00088 * SYSTEM.PRCR.WORD = k_rx_prcr_lock;
00089 * restore_interrupts(saved_ipl); // End atomic
00090 * ```
00091 *
00092 * ## Performance Characteristics
00093 *
00094 * - **Write latency**: 1 bus cycle (4.17 ns @ 240 MHz ICLK)

```

```

00095 * - **Unlock overhead**: 2 writes (unlock + lock) = ~8 ns
00096 * - **Protection check**: Hardware (zero CPU overhead)
00097 * - **Memory mapped**: 0x000803FE (SYSTEM.PRCR)
00098 *
00099 * ## Hardware Requirements
00100 *
00101 * | Requirement | Value | Notes |
00102 * |-----|-----|-----|
00103 * | MCU | RX72N | PRCR at 0x000803FE |
00104 * | Register Size | 16-bit | Must use WORD access |
00105 * | Key Code | 0xA5 | Upper 8 bits required |
00106 * | Protection Bits | PRC0, PRC1, PRC3 | Lower 4 bits |
00107 *
00108 * @par Module Dependencies:
00109 * - **NO dependencies** (pure constant definitions)
00110 * - Used by: Clock init, module stop control, register guard, power management
00111 *
00112 * @par NASA Power of 10 Compliance:
00113 * - **Rule 1**: [OK] No code (only data definitions)
00114 * - **Rule 2**: [OK] No loops
00115 * - **Rule 3**: [OK] No dynamic allocation (compile-time constants)
00116 * - **Rule 4**: [OK] No functions
00117 * - **Rule 5**: [OK] N/A (no functions)
00118 * - **Rule 6**: [OK] File scope only (header file)
00119 * - **Rule 7**: [OK] No functions
00120 * - **Rule 8**: [OK] C23 typed enums for all constants
00121 * - **Rule 9**: [OK] No pointers
00122 * - **Rule 10**: [OK] Compiles with -Wall -Wextra -Werror
00123 *
00124 * @par SOLID Principles:
00125 * - **Single Responsibility**: Only defines PRCR constants, no logic
00126 * - **Open/Closed**: Extensible (add new combinations) without modification
00127 * - **Liskov Substitution**: N/A (no inheritance)
00128 * - **Interface Segregation**: Minimal interface (single enum)
00129 * - **Dependency Inversion**: N/A (no dependencies)
00130 *
00131 * @par Usage Example - Module Enable:
00132 * @code{.c}
00133 * #include "rx_register_protection.h"
00134 * #include "rx72n_regs.h"
00135 *
00136 * // Enable CMT0 timer module (in MSTPCRA bit 15)
00137 * void enable_cmt0(void) {
00138 *     typedef enum : uint32_t {
00139 *         k_mstpcra_cmt0_bit = 15, // CMT0 module stop bit
00140 *     } mstpcr_bits_t;
00141 *
00142 *     // Unlock MSTPCR protection
00143 *     SYSTEM.PRCR.WORD = k_rx_prcr_unlock_prc1;
00144 *
00145 *     // Clear module stop bit (0 = enabled)
00146 *     SYSTEM.MSTPCRA.LONG &= ~(1UL « k_mstpcra_cmt0_bit);
00147 *
00148 *     // Re-lock protection immediately
00149 *     SYSTEM.PRCR.WORD = k_rx_prcr_lock;
00150 * }
00151 * @endcode
00152 *
00153 * @par Usage Example - Clock Configuration:
00154 * @code{.c}
00155 * // Configure system clocks during initialization
00156 * void configure_system_clocks(void) {
00157 *     // Unlock both CGC and MSTPCR groups
00158 *     SYSTEM.PRCR.WORD = k_rx_prcr_unlock_prc0_prc1;
00159 *
00160 *     // Configure PLL (protected by PRC0)
00161 *     SYSTEM.PLLCR.WORD = 0x1300; // PLL multiplier x10 (STAR: 24 MHz * 10 = 240 MHz)
00162 *     SYSTEM.PLLCR2.BYTE = 0x00; // Enable PLL
00163 *
00164 *     // Wait for PLL stabilization
00165 *     while (SYSTEM.OSCOVFSR.BIT.PLOVF == 0) {
00166 *         // Busy wait (~2ms)
00167 *     }
00168 *
00169 *     // Select PLL as system clock (protected by PRC0)
00170 *     SYSTEM.SCKCR3.WORD = 0x0400; // CKSEL = PLL
00171 *
00172 *     // Enable all peripheral modules (protected by PRC1)
00173 *     SYSTEM.MSTPCRA.LONG = 0x00000000;
00174 *     SYSTEM.MSTPCRB.LONG = 0x00000000;
00175 *
00176 *     // Re-lock all protection
00177 *     SYSTEM.PRCR.WORD = k_rx_prcr_lock;
00178 * }
00179 * @endcode
00180 *
00181 * @par Usage Example - Atomic Register Refresh (Register Guard):

```



```

00182 * @code{.c}
00183 * // Restore corrupted MSTPCR registers atomically
00184 * void restore_mstpcr_atomically(void) {
00185 *     // Brief interrupt disable for atomicity
00186 *     uint32_t saved_ipl = __get_ipl();
00187 *     __set_ipl(15); // Disable interrupts
00188 *
00189 *     // Unlock MSTPCR group
00190 *     SYSTEM.PRCR.WORD = k_rx_prcr_unlock_prc1;
00191 *
00192 *     // Restore golden values (from register guard)
00193 *     SYSTEM.MSTPCRA.LONG = s_golden_mstpcra;
00194 *     SYSTEM.MSTPCRB.LONG = s_golden_mstpcrb;
00195 *     SYSTEM.MSTPCRC.LONG = s_golden_mstpcrc;
00196 *     SYSTEM.MSTPCRD.LONG = s_golden_mstpcrd;
00197 *
00198 *     // Re-lock protection
00199 *     SYSTEM.PRCR.WORD = k_rx_prcr_lock;
00200 *
00201 *     // Restore interrupts
00202 *     __set_ipl(saved_ipl);
00203 * }
00204 * @endcode
00205 *
00206 * @see [RX72N User's Manual](../docs/rx72n_manual.pdf) Chapter 9: System Control
00207 * @see [RX72N User's Manual](../docs/rx72n_manual.pdf) Section 9.2.2: Protection Register (PRCR)
00208 * @see [rx_register_guard.c](../src/rx_register_guard.c) Uses PRCR for MSTPCR refresh
00209 * @see [rx_clock_power_init.c](../src/rx_clock_power_init.c) Uses PRCR for clock config
00210 *
00211 * @author Locked, Inc.
00212 * @date 2026-01-27
00213 * @version 1.0.0
00214 * @copyright Copyright (c) 2026 Locked Inc.
00215 * SPDX-License-Identifier: MIT
00216 *
00217 * @since Version 1.0.0
00218 */
00219
00220 #pragma once
00221
00222 #include <stdint.h>
00223
00224 #ifdef __cplusplus
00225 extern "C" {
00226 #endif
00227
00228 /*
=====
00229 * Protection Register (PRCR) Values
00230 *
=====
00231 */
00232
00233 /**
00234 * @enum rx_prcr_values_t
00235 * @brief PRCR register unlock/lock values for write-protected system registers
00236 *
00237 * @details
00238 * Defines all valid PRCR (Protect Register) values for unlocking and locking
00239 * write-protected system registers on the RX72N. The PRCR uses a key code
00240 * mechanism (0xA5 in upper byte) combined with protection enable bits (lower
00241 * byte) to selectively unlock register groups.
00242 *
00243 * **Register Format** (16-bit SYSTEM.PRCR at 0x000803FE):
00244 * ```
00245 * Bits 15-8: [0xA5] Key code (required for any write to take effect)
00246 * Bits 7-4: [Reserved] Always write 0
00247 * Bit 3: [PRC3] Low-Voltage Detection protection (1=unlocked, 0=locked)
00248 * Bit 2: [Reserved] Always write 0
00249 * Bit 1: [PRC1] Module Stop Control protection (1=unlocked, 0=locked)
00250 * Bit 0: [PRC0] Clock Generation Circuit protection (1=unlocked, 0=locked)
00251 * ```
00252 *
00253 * **Key Code Security**:
00254 * - Writing without 0xA5 key -> ignored (no effect)
00255 * - Prevents accidental unlock from pointer errors
00256 * - Prevents unlock from EMI-induced bit flips
00257 *
00258 * **Protection Groups**:
00259 * - **PRC0** (bit 0): Protects SCKCR, PLLCR, PLLCR2, SCKCR3, MOSCWTCR, etc.
00260 * - **PRC1** (bit 1): Protects MSTPCRA, MSTPCRB, MSTPCRC, MSTPCRD
00261 * - **PRC3** (bit 3): Protects LVCMPCR, LVDXCR, LVDYCR
00262 *
00263 * ## Usage Guidelines
00264 *
00265 * 1. **Minimize unlock duration**: Unlock, modify, lock immediately
00266 * 2. **Use narrowest scope**: Only unlock groups you need to modify

```



```

00267 * 3. **Atomic for critical sections**: Disable interrupts during unlock
00268 * 4. **Always re-lock**: Even on error paths (use cleanup pattern)
00269 *
00270 * ## STAR Project Usage Frequency
00271 *
00272 * | Constant | Frequency | Context |
00273 * |-----|-----|-----|
00274 * | k_rx_prcr_unlock_prc1 | High | Register guard refresh (1-10 Hz) |
00275 * | k_rx_prcr_unlock_prc0_prc1 | Once | Clock initialization at boot |
00276 * | k_rx_prcr_unlock_prc1_prc3 | Never | LVD not used in STAR |
00277 * | k_rx_prcr_unlock_all | Never | Overly broad, avoid |
00278 * | k_rx_prcr_lock | High | Always called after unlock |
00279 *
00280 * @par Field Constraints:
00281 * | Constant | Key | PRC3 | PRC1 | PRC0 | Notes |
00282 * |-----|-----|-----|-----|-----|
00283 * | unlock_prc1 | 0xA5 | 0 | 1 | 0 | Most common (MSTPCR) |
00284 * | unlock_prc0_prc1 | 0xA5 | 0 | 1 | 1 | Clock init |
00285 * | unlock_prc1_prc3 | 0xA5 | 1 | 1 | 1 | With LVD (unused) |
00286 * | unlock_all | 0xA5 | 1 | 1 | 1 | Avoid (too broad) |
00287 * | lock | 0xA5 | 0 | 0 | 0 | Always used |
00288 *
00289 * @par Usage Example - Typical Pattern:
00290 * @code{.c}
00291 * void enable_peripheral_module(void) {
00292 *     // Unlock only what's needed (PRC1 for MSTPCR)
00293 *     SYSTEM.PRCR.WORD = k_rx_prcr_unlock_prc1;
00294 *
00295 *     // Modify protected register
00296 *     SYSTEM.MSTPCRA.LONG &= ~(1UL << 15); // Enable CMT0
00297 *
00298 *     // Re-lock immediately
00299 *     SYSTEM.PRCR.WORD = k_rx_prcr_lock;
00300 * }
00301 * @endcode
00302 *
00303 * @see SYSTEM.PRCR Hardware register at 0x000803FE
00304 * @see rx_register_guard_refresh() Uses PRC1 unlock for MSTPCR refresh
00305 * @see rx_clock_power_init() Uses PRC0+PRC1 unlock for clock config
00306 * @since Version 1.0.0
00307 */
00308 typedef enum : uint16_t {
00309 /**
00310 * @brief Unlock PRC1 only (Module Stop Control registers)
00311 *
00312 * @details
00313 * Unlocks write protection for MSTPCRA, MSTPCRB, MSTPCRC, MSTPCRD registers
00314 * while keeping clock generation (PRC0) and LVD (PRC3) registers protected.
00315 *
00316 * **Value**: 0xA502
00317 * - Key code: 0xA5 (required)
00318 * - PRC3: 0 (LVD registers remain locked)
00319 * - PRC1: 1 (MSTPCR registers unlocked)
00320 * - PRC0: 0 (Clock registers remain locked)
00321 *
00322 * **Protected Registers** (when unlocked):
00323 * - SYSTEM.MSTPCRA (0x00080010) - Module stop control A
00324 * - SYSTEM.MSTPCRB (0x00080014) - Module stop control B
00325 * - SYSTEM.MSTPCRC (0x00080018) - Module stop control C
00326 * - SYSTEM.MSTPCRD (0x0008001C) - Module stop control D
00327 *
00328 * **Use Cases**:
00329 * - Enabling/disabling peripheral modules at runtime
00330 * - Register guard periodic refresh (most common)
00331 * - Dynamic power management
00332 *
00333 * **STAR Project Usage**: **HIGH FREQUENCY**
00334 * - Called during register guard refresh (1-10 Hz)
00335 * - Called during peripheral enable/disable
00336 * - ~10,000+ times per hour in typical operation
00337 *
00338 * @par Rationale:
00339 * This is the most commonly used unlock value because:
00340 * - MSTPCR registers change frequently (module enable/disable)
00341 * - Clock registers (PRC0) rarely change after boot
00342 * - Narrowest scope reduces risk of accidental corruption
00343 *
00344 * @par Example - Enable SCI0 (UART):
00345 * @code{.c}
00346 * typedef enum : uint32_t {
00347 *     k_mstpcrb_sci0_bit = 31, // SCI0 in MSTPCRB bit 31
00348 * } mstpcrb_bits_t;
00349 *
00350 * SYSTEM.PRCR.WORD = k_rx_prcr_unlock_prc1;
00351 * SYSTEM.MSTPCRB.LONG &= ~(1UL << k_mstpcrb_sci0_bit); // 0 = enable
00352 * SYSTEM.PRCR.WORD = k_rx_prcr_lock;
00353 * @endcode

```

```

00354 *
00355 * @warning Interrupts are NOT disabled automatically. For atomic operations,
00356 *         disable interrupts explicitly before unlock.
00357 */
00358 k_rx_prcr_unlock_prc1 = 0xA502,
00359
00360 /**
00361 * @brief Unlock PRC0+PRC1 (Clock Generation + Module Stop Control)
00362 *
00363 * @details
00364 * Unlocks write protection for both clock generation circuit (CGC) registers
00365 * and module stop control (MSTPCR) registers. Used during system initialization
00366 * when configuring PLL, system clocks, and peripheral modules together.
00367 *
00368 * **Value**: 0xA503
00369 * - Key code: 0xA5 (required)
00370 * - PRC3: 0 (LVD registers remain locked)
00371 * - PRC1: 1 (MSTPCR registers unlocked)
00372 * - PRC0: 1 (Clock registers unlocked)
00373 *
00374 * **Protected Registers** (when unlocked):
00375 *
00376 * **PRC0 (Clock Generation)**:
00377 * - SYSTEM.SCKCR (0x00080020) - System clock control
00378 * - SYSTEM.SCKCR2 (0x00080024) - System clock control 2
00379 * - SYSTEM.SCKCR3 (0x00080026) - System clock control 3
00380 * - SYSTEM.PLLCR (0x00080028) - PLL control
00381 * - SYSTEM.PLLCR2 (0x0008002A) - PLL control 2
00382 * - SYSTEM.MOSCCR (0x00080032) - Main oscillator control
00383 * - SYSTEM.MOSCWTCR (0x000800A2) - Main oscillator wait control
00384 *
00385 * **PRC1 (Module Stop)**:
00386 * - SYSTEM.MSTPCRA/B/C/D (see k_rx_prcr_unlock_prc1)
00387 *
00388 * **Use Cases**:
00389 * - System clock initialization at boot
00390 * - PLL configuration and selection
00391 * - Switching clock sources (main osc <-> PLL)
00392 * - Enabling peripheral clocks simultaneously
00393 *
00394 * **STAR Project Usage**: **LOW FREQUENCY**
00395 * - Called once during rx_clock_power_init() at boot
00396 * - May be called during clock source transitions (rare)
00397 * - ~1-2 times per boot cycle
00398 *
00399 * @par Rationale:
00400 * Combined unlock used during initialization because:
00401 * - Clock and module configuration often done together
00402 * - Reduces number of lock/unlock cycles at boot
00403 * - Acceptable risk during controlled init sequence
00404 *
00405 * @par Example - Clock Initialization:
00406 * @code{.c}
00407 void configure_240mhz_operation(void) {
00408     // Unlock both clock and module stop registers
00409     SYSTEM.PRCR.WORD = k_rx_prcr_unlock_prc0_prc1;
00410
00411     // Configure PLL (PRC0 protected)
00412     SYSTEM.PLLCR.WORD = 0x1300; // x10 multiplier (STAR: 24 MHz * 10 = 240 MHz)
00413     SYSTEM.PLLCR2.BYTE = 0x00; // Enable PLL
00414
00415     // Wait for PLL lock (~2ms)
00416     while (!SYSTEM.OSCOVFSR.BIT.PLOVF) {}
00417
00418     // Switch to PLL (PRC0 protected)
00419     SYSTEM.SCKCR3.WORD = 0x0400; // Select PLL as system clock
00420     SYSTEM.SCKCR.LONG = 0x20011222; // STAR production dividers: ICK=/1 (240), PCKA=/2 (120),
00421     PCKB/C/D/FCK=/4 (60), BCK=/2 (120)
00422
00423     // Enable all peripherals (PRC1 protected)
00424     SYSTEM.MSTPCRA.LONG = 0x00000000;
00425     SYSTEM.MSTPCRB.LONG = 0x00000000;
00426
00427     // Re-lock all
00428     SYSTEM.PRCR.WORD = k_rx_prcr_lock;
00429 }
00430 *
00431 * @endcode
00432 *
00433 * @warning Only use during initialization. Runtime clock changes should use
00434 *         separate PRC0-only unlock (not defined here - add if needed).
00435 */
00436 k_rx_prcr_unlock_prc0_prc1 = 0xA503,
00437
00438 /**
00439 * @brief Unlock PRC1+PRC3 (Module Stop + Low-Voltage Detection)
00440 *
00441 * @details

```

```

00440 * Unlocks write protection for module stop control (MSTPCR) and low-voltage
00441 * detection (LVD) registers. This combination is rarely needed as LVD is
00442 * typically configured once at boot and left unchanged.
00443 *
00444 * **Value**: 0xA50B
00445 * - Key code: 0xA5 (required)
00446 * - PRC3: 1 (LVD registers unlocked)
00447 * - PRC1: 1 (MSTPCR registers unlocked)
00448 * - PRC0: 0 (Clock registers remain locked)
00449 *
00450 * **Protected Registers** (when unlocked):
00451 *
00452 * **PRC1 (Module Stop)**: See k_rx_prcr_unlock_prc1
00453 *
00454 * **PRC3 (Low-Voltage Detection)**:
00455 * - SYSTEM.LVCMPCR (0x000800E0) - Voltage monitor control
00456 * - SYSTEM.LVDXCR (0x000800E1) - Voltage detect 1 control
00457 * - SYSTEM.LVDYCR (0x000800E2) - Voltage detect 2 control
00458 *
00459 * **STAR Project Usage**: **NEVER USED**
00460 * - LVD not implemented in STAR project
00461 * - 3.3V supply is stable and monitored externally
00462 * - Included for completeness only
00463 *
00464 * @par Rationale:
00465 * Not used in STAR because:
00466 * - LVD not required (external power monitoring)
00467 * - MSTPCR and LVD changes rarely coincide
00468 * - More specific unlocks preferred (narrower scope)
00469 *
00470 * @warning Do not use unless your application specifically needs both MSTPCR
00471 * and LVD modification in same critical section.
00472 */
00473 k_rx_prcr_unlock_prc1_prc3 = 0xA50B,
00474
00475 /**
00476 * @brief Unlock all protection groups (PRC0+PRC1+PRC3)
00477 *
00478 * @details
00479 * Unlocks write protection for ALL protected register groups: clock generation,
00480 * module stop control, and low-voltage detection. This is the broadest unlock
00481 * and should be **AVOIDED** in production code.
00482 *
00483 * **Value**: 0xA50B
00484 * - Key code: 0xA5 (required)
00485 * - PRC3: 1 (LVD registers unlocked) [bit 3]
00486 * - PRC1: 1 (MSTPCR registers unlocked) [bit 1]
00487 * - PRC0: 1 (Clock registers unlocked) [bit 0]
00488 * - Bit 2 is reserved and must be written as 0
00489 *
00490 * @note Prior revision set this to 0xA50F, which also asserts reserved bit 2
00491 * (PRCR[2] must be written 0 per RX72N HW manual R01UH0824EJ0111
00492 * Section 13.2.1). Corrected to 0xA50B.
00493 *
00494 * **Protected Registers** (when unlocked):
00495 * - All PRC0 registers (clock generation)
00496 * - All PRC1 registers (module stop)
00497 * - All PRC3 registers (LVD)
00498 *
00499 * **STAR Project Usage**: **NEVER USED**
00500 * - Too broad, violates principle of least privilege
00501 * - More specific unlocks always available
00502 * - Included for completeness only
00503 *
00504 * @par Rationale for AVOIDANCE:
00505 * - **Security**: Unlocks more than necessary (attack surface)
00506 * - **Safety**: Accidental corruption of unrelated registers
00507 * - **Best Practice**: Use narrowest scope possible
00508 *
00509 * @warning **DO NOT USE** unless you have a specific reason to modify registers
00510 * across all three protection groups simultaneously. Use specific
00511 * unlock combinations instead.
00512 *
00513 * @attention Code review flag: If you see this constant used, investigate why
00514 * the more specific unlocks (prc1, prc0_prc1) cannot be used instead.
00515 */
00516 k_rx_prcr_unlock_all = 0xA50B,
00517
00518 /**
00519 * @brief Lock all protection groups (re-enable write protection)
00520 *
00521 * @details
00522 * Locks (re-enables) write protection for ALL register groups by clearing
00523 * all protection enable bits while maintaining the required 0xA5 key code.
00524 *
00525 * **Value**: 0xA500
00526 * - Key code: 0xA5 (required for write to take effect)

```

```

00527 * - PRC3: 0 (LVD registers locked)
00528 * - PRC1: 0 (MSTPCR registers locked)
00529 * - PRC0: 0 (Clock registers locked)
00530 *
00531 * **Effect**: After writing this value, all protected registers become read-only
00532 * until the next unlock operation. Any attempt to write protected registers
00533 * (without first unlocking) will be silently ignored by hardware.
00534 *
00535 * **Usage**: **ALWAYS** write this value immediately after completing protected
00536 * register modifications. Even on error paths, ensure protection is re-locked.
00537 *
00538 * **STAR Project Usage**: **HIGH FREQUENCY**
00539 * - Called after every unlock operation
00540 * - ~10,000+ times per hour (matches unlock frequency)
00541 *
00542 * @par Rationale:
00543 * Always lock immediately because:
00544 * - Minimizes window where accidental writes could corrupt registers
00545 * - Protects against EMI-induced bit flips
00546 * - Follows secure coding best practices
00547 *
00548 * @par Example - Cleanup Pattern:
00549 * @code{.c}
00550 * rx_err_t modify_protected_register(void) {
00551 *     SYSTEM.PRCR.WORD = k_rx_prcr_unlock_prc1;
00552 *
00553 *     // Modify register (may fail)
00554 *     if (!configure_module()) {
00555 *         SYSTEM.PRCR.WORD = k_rx_prcr_lock; // Lock even on error!
00556 *         return k_rx_err_hardware;
00557 *     }
00558 *
00559 *     SYSTEM.PRCR.WORD = k_rx_prcr_lock; // Lock on success
00560 *     return k_rx_ok;
00561 * }
00562 * @endcode
00563 *
00564 * @par Example - RAII Pattern (C++ style cleanup):
00565 * @code{.c}
00566 * // Define cleanup macro (ensures lock on scope exit)
00567 * #define RX_PRCR_SCOPED_UNLOCK(group) \
00568 *     for (int _i = (SYSTEM.PRCR.WORD = (group), 0); \
00569 *          _i < 1; \
00570 *          _i++, SYSTEM.PRCR.WORD = k_rx_prcr_lock)
00571 *
00572 * // Usage: automatic lock at end of scope
00573 * void modify_with_cleanup(void) {
00574 *     RX_PRCR_SCOPED_UNLOCK(k_rx_prcr_unlock_prc1) {
00575 *         SYSTEM.MSTPCRA.LONG &= ~(1UL « 15);
00576 *         // Lock happens automatically when scope exits
00577 *     }
00578 * }
00579 * @endcode
00580 *
00581 * @note Hardware behavior: After lock, protected register writes are ignored
00582 * (no bus error or exception - just silently discarded).
00583 */
00584 k_rx_prcr_lock = 0xA500,
00585 /**
00586 * @brief Mask for PRC protection bits (for readback verification)
00587 *
00588 * @details
00589 * When reading PRCR, the key code (upper byte) reads back as 0x00,
00590 * NOT as 0xA5. Only the lower 4 bits (PRC0, PRC1, PRC3) contain
00591 * the actual protection state.
00592 *
00593 * **Value**: 0x000F
00594 * - Bits 3,1,0: PRC bits (protection state)
00595 * - Bits 7-4, 15-8: Read as zero
00596 *
00597 * **Usage**: Use this mask when verifying PRCR readback:
00598 * @code{.c}
00599 * *prcr_reg() = k_rx_prcr_unlock_prc1;
00600 * // Verify unlock (key byte reads as 0x00, so mask it out)
00601 * if ((*prcr_reg() & k_rx_prcr_readback_mask) !=
00602 *     (k_rx_prcr_unlock_prc1 & k_rx_prcr_readback_mask)) {
00603 *     // Unlock failed
00604 * }
00605 * }
00606 * @endcode
00607 */
00608 k_rx_prcr_readback_mask = 0x000F,
00609
00610 } rx_prcr_values_t;
00611
00612 #ifdef __cplusplus
00613 }

```

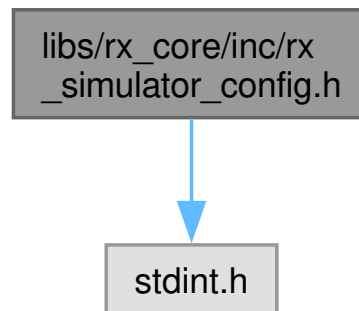
```
00614 #endif
```

7.35 libs/rx_core/inc/rx_simulator_config.h File Reference

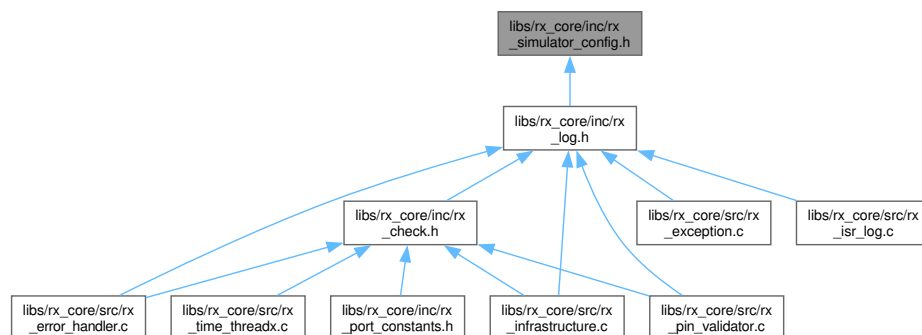
Configuration for e² studio simulator support.

```
#include <stdint.h>
```

Include dependency graph for rx_simulator_config.h:



This graph shows which files directly or indirectly include this file:



Macros

- `#define RX_IS_SIMULATOR 0`
Compile-time flag indicating simulator build.

Enumerations

- enum `simulator_clock_timing_t` : `uint32_t { k_sim_clock_instant = 1 }`
Simulator-specific clock stabilization timeouts.

7.35.1 Detailed Description

Configuration for e² studio simulator support.

This header provides simulator-specific configuration for testing firmware logic without real hardware. The Renesas e² studio simulator cannot fully model certain hardware features:

Hardware limitations in simulator:

- External oscillators (24 MHz crystal) - Not modeled
- PLL/PPLL lock behavior - Status flags never set
- UART/SCI peripheral hardware - TX/RX don't work
- USB controller - Limited or no enumeration
- SPI transfers - No real external devices
- Timing accuracy - Not cycle-accurate

WARNING: Simulator builds are FOR LOGIC TESTING ONLY. Do not use for timing validation, peripheral verification, or hardware-specific testing. Always validate critical paths on real hardware before deployment.

Intended Use Cases:

- Algorithm validation (PID controllers, state machines)
- Protocol parsing and encoding logic
- Error handling path coverage
- Unit test development
- Interactive debugging of control flow

NOT Suitable For:

- Clock tree validation
- Interrupt latency measurements
- DMA transfer verification
- USB enumeration testing
- SPI communication reliability
- Real-time performance analysis

Usage:

Define `RX_SIMULATOR_MODE` during compilation to enable simulator support:

```
// In e2 studio:  
// Project Properties -> C/C++ Build -> Settings -> Compiler -> Preprocessor  
// Add symbol: RX_SIMULATOR_MODE  
  
// Or via command line:  
gcc -DRX_SIMULATOR_MODE -o firmware.elf ...
```

Example:

```
\#include "rx_simulator_config.h"

void init__hardware(void)
{
    \#if RX_IS_SIMULATOR
        // Simulator: Skip hardware polling loops
        return k_rx_ok;
    \#else
        // Hardware: Wait for PLL stable
        while ((OSCOVFSR & PLL_STABLE) == 0) {
            // ...
        }
    \#endif
}
```

Since

Version 1.0.0

See also

[rx_clock_power_init.c](#) Clock initialization with simulator support

[rx_log.h](#) Logging infrastructure with simulator output

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Definition in file [rx_simulator_config.h](#).

7.35.2 Macro Definition Documentation

7.35.2.1 RX_IS_SIMULATOR

```
\#define RX_IS_SIMULATOR 0
```

Compile-time flag indicating simulator build.

Evaluates to 1 when `RX_SIMULATOR_MODE` is defined, 0 otherwise. Use this macro for readable conditional compilation:

```
\#if RX_IS_SIMULATOR
    // Simulator-specific code
\#else
    // Hardware-specific code
\#endif
```

Note

Prefer `RX_IS_SIMULATOR` over direct `\#ifdef RX_SIMULATOR_MODE` for consistency and readability.

Definition at line 120 of file [rx_simulator_config.h](#).

7.35.3 Enumeration Type Documentation

7.35.3.1 `simulator_clock_timing_t`

enum `simulator_clock_timing_t` : `uint32_t`

Simulator-specific clock stabilization timeouts.

Hardware requires multi-millisecond delays for oscillator and PLL stabilization (polling OSCOVFSR register bits until they set). Simulator cannot model these delays as the hardware circuits don't exist, so we provide instant-ready constants.

Hardware behavior:

- Main oscillator: ~10 ms stabilization (2.4M instruction cycles)
- PLL: ~200 us stabilization (poll OSCOVFSR bit 2)
- PPLL: ~200 us stabilization (poll OSCOVFSR bit 3)

Simulator behavior:

- All clocks: Assume instant ready (skip polling loops)

Usage:

These constants are for documentation and future extensibility. Current implementation simply skips polling loops entirely when `RX_IS_SIMULATOR` is true.

Since

Version 1.0.0

Enumerator

<code>k_sim_clock_instant</code>	Skip delay entirely (assume immediate ready)
----------------------------------	--

Definition at line 153 of file `rx_simulator_config.h`.

```
00153      : uint32_t {
00154  k_sim_clock_instant = 1, /**< Skip delay entirely (assume immediate ready) */
00155 } simulator_clock_timing_t;
```


7.36 rx_simulator_config.h

[Go to the documentation of this file.](#)

```

00001 /**
00002  * @file rx_simulator_config.h
00003  * @brief Configuration for e^2 studio simulator support
00004  *
00005  * @details
00006  * This header provides simulator-specific configuration for testing
00007  * firmware logic without real hardware. The Renesas e^2 studio simulator
00008  * cannot fully model certain hardware features:
00009  *
00010  * **Hardware limitations in simulator**:
00011  * - External oscillators (24 MHz crystal) - Not modeled
00012  * - PLL/PPLL lock behavior - Status flags never set
00013  * - UART/SCI peripheral hardware - TX/RX don't work
00014  * - USB controller - Limited or no enumeration
00015  * - SPI transfers - No real external devices
00016  * - Timing accuracy - Not cycle-accurate
00017  *
00018  * **WARNING**: Simulator builds are FOR LOGIC TESTING ONLY.
00019  * Do not use for timing validation, peripheral verification, or
00020  * hardware-specific testing. Always validate critical paths on
00021  * real hardware before deployment.
00022  *
00023  * @par Intended Use Cases:
00024  * - Algorithm validation (PID controllers, state machines)
00025  * - Protocol parsing and encoding logic
00026  * - Error handling path coverage
00027  * - Unit test development
00028  * - Interactive debugging of control flow
00029  *
00030  * @par NOT Suitable For:
00031  * - Clock tree validation
00032  * - Interrupt latency measurements
00033  * - DMA transfer verification
00034  * - USB enumeration testing
00035  * - SPI communication reliability
00036  * - Real-time performance analysis
00037  *
00038  * @par Usage:
00039  * Define RX_SIMULATOR_MODE during compilation to enable simulator support:
00040  * @code
00041  * // In e^2 studio:
00042  * // Project Properties -> C/C++ Build -> Settings -> Preprocessor
00043  * // Add symbol: RX_SIMULATOR_MODE
00044  *
00045  * // Or via command line:
00046  * gcc -DRX_SIMULATOR_MODE -o firmware.elf ...
00047  * @endcode
00048  *
00049  * @par Example:
00050  * @code
00051  * \#include "rx_simulator_config.h"
00052  *
00053  * void init_hardware(void)
00054  * {
00055  *     \#if RX_IS_SIMULATOR
00056  *         // Simulator: Skip hardware polling loops
00057  *         return k_rx_ok;
00058  *     \#else
00059  *         // Hardware: Wait for PLL stable
00060  *         while ((OSCOVFSR & PLL_STABLE) == 0) {
00061  *             // ...
00062  *         }
00063  *     \#endif
00064  * }
00065  * @endcode
00066  *
00067  * @since Version 1.0.0
00068  *
00069  * @see rx_clock_power_init.c Clock initialization with simulator support
00070  * @see rx_log.h Logging infrastructure with simulator output
00071  *
00072  * @copyright Copyright (c) 2026 Locked Inc.
00073  * SPDX-License-Identifier: MIT
00074  */
00075
00076 #pragma once
00077
00078 #include <stdint.h>
00079
00080 /*
=====
00081  * Simulator Mode Detection

```

```

00082 *
=====
00083 */
00084
00085 /**
00086  * @def RX_IS_SIMULATOR
00087  * @brief Compile-time flag indicating simulator build
00088  *
00089  * @details
00090  * Evaluates to 1 when RX_SIMULATOR_MODE is defined, 0 otherwise.
00091  * Use this macro for readable conditional compilation:
00092  *
00093  * @code
00094  * #if RX_IS_SIMULATOR
00095  * // Simulator-specific code
00096  * #else
00097  * // Hardware-specific code
00098  * #endif
00099  * @endcode
00100  *
00101  * @note Prefer RX_IS_SIMULATOR over direct \#ifndef RX_SIMULATOR_MODE
00102  * for consistency and readability.
00103  */
00104 #ifndef RX_SIMULATOR_MODE
00105 #define RX_IS_SIMULATOR 1
00106
00107 /**
00108  * @warning RX_SIMULATOR_MODE: This build is FOR SIMULATOR ONLY.
00109  *
00110  * Do not flash to hardware. Simulator builds skip hardware initialization
00111  * steps and may not configure peripherals correctly for real operation.
00112  */
00113 /* Compile-time note: simulator mode active -- do not flash to hardware.
00114  * See Doxygen @warning above for rationale. */
00115 #pragma GCC diagnostic push
00116 #pragma GCC diagnostic ignored "-Wcpp"
00117 #warning "RX_SIMULATOR_MODE: This build is FOR SIMULATOR ONLY. Do not flash to hardware."
00118 #pragma GCC diagnostic pop
00119 #else
00120 #define RX_IS_SIMULATOR 0
00121 #endif
00122
00123 /*
=====
00124  * Simulator-Specific Constants
00125  *
=====
00126 */
00127
00128 /**
00129  * @enum simulator_clock_timing_t
00130  * @brief Simulator-specific clock stabilization timeouts
00131  *
00132  * @details
00133  * Hardware requires multi-millisecond delays for oscillator and PLL
00134  * stabilization (polling OSCOVFSR register bits until they set).
00135  * Simulator cannot model these delays as the hardware circuits don't
00136  * exist, so we provide instant-ready constants.
00137  *
00138  * **Hardware behavior**:
00139  * - Main oscillator: ~10 ms stabilization (2.4M instruction cycles)
00140  * - PLL: ~200 us stabilization (poll OSCOVFSR bit 2)
00141  * - PPLL: ~200 us stabilization (poll OSCOVFSR bit 3)
00142  *
00143  * **Simulator behavior**:
00144  * - All clocks: Assume instant ready (skip polling loops)
00145  *
00146  * @par Usage:
00147  * These constants are for documentation and future extensibility.
00148  * Current implementation simply skips polling loops entirely when
00149  * RX_IS_SIMULATOR is true.
00150  *
00151  * @since Version 1.0.0
00152  */
00153 typedef enum : uint32_t {
00154     k_sim_clock_instant = 1, /**< Skip delay entirely (assume immediate ready) */
00155 } simulator_clock_timing_t;

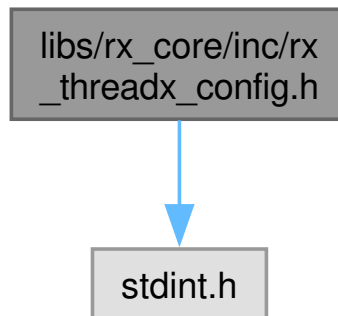
```

7.37 libs/rx_core/inc/rx_threadx_config.h File Reference

ThreadX RTOS Tick Rate Configuration.

```
#include <stdint.h>
```

Include dependency graph for rx_threadx_config.h:



Variables

- static const uint16_t `s_rx_threadx_tick_rate_hz` = 100U
ThreadX system tick frequency (100 Hz).

7.37.1 Detailed Description

ThreadX RTOS Tick Rate Configuration.

This header defines the ThreadX system tick frequency used throughout the RX72N firmware for RTOS time conversions. The tick rate is configured to 100 Hz (10ms period) via hardware timer CMT0 (Compare Match Timer 0).

7.37.1.1 System Architecture Context

ThreadX is a hard real-time operating system (RTOS) that provides:

- Deterministic preemptive scheduling: Priority-based thread switching
- Time-slicing: Round-robin scheduling for equal-priority threads
- Timers and delays: Software timers with tick-based granularity
- Event synchronization: Semaphores, mutexes, event flags with timeouts

The system tick is the fundamental time quantum for all ThreadX operations:

Hardware Timer (CMT0) @ 100 Hz

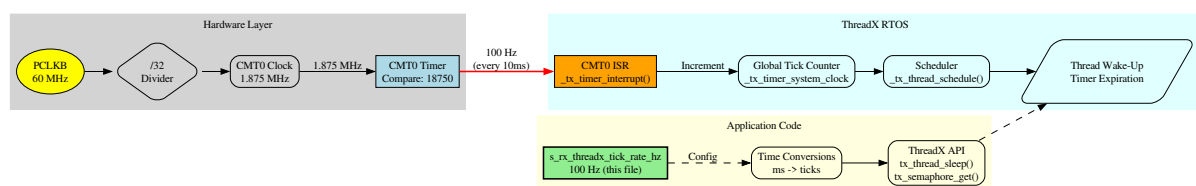
ThreadX Tick ISR (every 10ms)

```

+-----+
|_tx_timer_interrupt()|
| - Increment tick counter |
| - Check sleeping threads |
| - Expire software timers |
| - Trigger time-slice rotation |
+-----+

```

Schedule highest-priority ready thread



7.37.1.2 Design Rationale

Why 100 Hz Tick Rate?

1. Control loop alignment: Motor control runs at 100 Hz, matching tick period
2. Low overhead: Context switch overhead <1% of CPU time (~24us per 10ms)
3. Simple conversions: ms -> ticks is divide-by-10 (compiler optimizes to shift)
4. Adequate responsiveness: 10ms maximum scheduling latency acceptable for robotics
5. Standard practice: Common RTOS tick rate (FreeRTOS, VxWorks use 100-1000 Hz)

CMT0 Hardware Configuration:

- Input clock: $PCLKB / 32 = 60 \text{ MHz} / 32 = 1.875 \text{ MHz}$
- Compare value: 18,750 counts
- Interrupt period: $18,750 / 1,875,000 = 0.01 \text{ seconds} = 10\text{ms}$
- Interrupt priority: $IPR[IPR_CMT0_CMI0] = 3$ (high priority for determinism)

7.37.1.3 Implementation Approach

This header provides a single source of truth for ThreadX tick rate:

- tx_user.h: Defines `TX_TIMER_TICKS_PER_SECOND = 100` (ThreadX core config)
- This file: Provides C constant `s_rx_threadx_tick_rate_hz = 100` (application use)
- CMT0 initialization: Configures hardware to generate 100 Hz interrupt

Why not use TX_TIMER_TICKS_PER_SECOND directly?

- tx_user.h may use preprocessor macros (not type-safe)
- This file provides typed constant for compile-time validation
- Decouples application code from ThreadX internal headers

7.37.1.4 Performance Characteristics

Metric	Value	Notes
Tick period	10 milliseconds	Fixed by CMT0 configuration
Tick frequency	100 Hz	Reciprocal of tick period
Tick jitter	< 1 us	CMT0 crystal accuracy +/-50 ppm
ISR latency	< 5 us	ThreadX tick ISR execution time
Context switch	~24 us	Worst-case at 240 MHz (measured)

Metric	Value	Notes
CPU overhead	< 1%	(5us ISR + 24us switch) / 10ms = 0.29%
Max tick count	$2^{32} - 1$	32-bit counter wraps after ~497 days

Tick Resolution vs. Alternatives:

- 10ms ticks: Current configuration, 0.29% overhead
- 1ms ticks (1000 Hz): 10x overhead ~2.9%, better responsiveness
- 100ms ticks (10 Hz): 0.029% overhead, unacceptable latency for robotics

7.37.1.5 Memory Usage

Item	Size	Location	Notes
s_rx_threadx_tick_rate_hz	0 bytes	Compile-time constant	Inlined by optimizer
ThreadX tick counter	4 bytes	.bss	<code>_tx_timer_system_clock</code> global
CMT0 registers	8 bytes	Hardware	Memory-mapped I/O
Total	4 bytes RAM	-	(Counter only, register is MMIO)

7.37.1.6 Module Dependencies

Depends On:

- None (foundational configuration)

Used By:

- [rx_time_constants.h](#) - Defines `k_threadx_ms_per_tick` (inverse: 10ms)
- [rx_iwtdt.c](#) - Watchdog timeout conversions
- [rx_usb_comm.c](#) - USB timeout calculations
- [tx_initialize_low_level.c](#) - CMT0 timer setup
- All application threads using `tx_thread_sleep()`, `tx_semaphore_get()`, etc.

Configuration Consistency:

CRITICAL: The following values MUST remain synchronized:

File	Symbol	Value	Purpose
tx_user.h	<code>TX_TIMER_TICKS_PER_SECOND</code>	1000	ThreadX core config
rx_threadx_config.h	<code>s_rx_threadx_tick_rate_hz</code>	100	Application constant
rx_time_constants.h	<code>k_threadx_ms_per_tick</code>	10	Conversion factor (inverse)

File	Symbol	Value	Purpose
tx_initialize_low_level.c	CMT0.CMCOR	18750	Hardware compare value

Warning

If you change the tick rate, update ALL four locations above!

7.37.1.7 NASA Power of 10 Compliance

Rule	Status	Notes
1. Simple control flow	[OK]	No code (configuration constant only)
2. Fixed loop bounds	[OK]	No loops
3. No dynamic memory	[OK]	No allocations (compile-time constant)
4. Function length <60 lines	[OK]	No functions
5. Assertions	[OK]	N/A (header-only configuration)
6. Smallest scope	[OK]	File-scope constant
7. Check return values	[OK]	N/A (no function calls)
8. Limited preprocessor	[OK]	No macros (uses const, not #define)
9. Pointer restrictions	[OK]	No pointers
10. Compiler warnings	[OK]	-Wall -Wextra -Werror enforced

7.37.1.8 SOLID Principles

Single Responsibility (S):

- This header has one responsibility: Define ThreadX tick rate
- No mixing of concerns (pure configuration, no logic)

Open/Closed (O):

- Changing tick rate requires recompilation (embedded systems reality)
- Alternative tick rates could be provided via separate config headers
- Hardware dependency (CMT0) prevents runtime configuration

Liskov Substitution (L):

- N/A (not applicable to configuration constants)

Interface Segregation (I):

- Single focused constant (no "fat" configuration structure)
- Application code depends only on what it needs (tick rate)

Dependency Inversion (D):

- High-level modules (application) depend on abstraction (tick rate constant)
- Low-level details (CMT0 configuration) hidden in initialization code
- Time interface ([rx_time_interface.h](#)) provides further abstraction

See also

[tx_user.h](#) ThreadX core configuration (TX_TIMER_TICKS_PER_SECOND)
[rx_time_constants.h](#) Time conversion factors ([k_threadx_ms_per_tick](#))
[tx_initialize_low_level.c](#) CMT0 timer initialization
[rx_time_interface.h](#) Dependency Inversion time abstraction
 ThreadX User Guide Chapter 3: Time and Timers

Author

Locked, Inc.

Date

2026-01-27

Copyright

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Definition in file [rx_threadx_config.h](#).

7.37.2 Variable Documentation

7.37.2.1 s_rx_threadx_tick_rate_hz

```
const uint16_t s_rx_threadx_tick_rate_hz = 100U [static]
```

ThreadX system tick frequency (100 Hz).

Defines the ThreadX timer tick rate configured via hardware timer CMT0. This value represents the frequency of the RTOS scheduler time quantum and is the fundamental unit for all ThreadX time-based operations.

7.37.2.2 Configuration Details

ThreadX Core Configuration (tx_user.h):

```
\#define TX_TIMER_TICKS_PER_SECOND 100
```

Hardware Timer Configuration (CMT0):

- Timer: Compare Match Timer 0 (CMT0)
- Input clock: PCLKB / 32 = 60 MHz / 32 = 1.875 MHz
- Compare value: 18,750 counts (0x493E)
- Interrupt vector: VECT_CMT0_CMI0 (vector 28)
- Interrupt priority: IPR[IPR_CMT0_CMI0] = 3 (high priority)
- Calculation: 18,750 / 1,875,000 Hz = 10ms period = 100 Hz

Register Configuration:

```
// From tx_initialize_low_level.c
CMT0.CMSTR0.BIT.STR0 = 0;           // Stop CMT0
CMT0.CMCR.WORD = 0x00C2;           // PCLKB/32, interrupt enabled
CMT0.CMCOR = 18750 - 1;             // Compare match value
IPR(CMT0, CMI0) = 3;                // Priority 3
IEN(CMT0, CMI0) = 1;               // Enable interrupt
CMT0.CMSTR0.BIT.STR0 = 1;           // Start CMT0
```

7.37.2.3 Tick Rate Selection Rationale

Why 100 Hz (10ms period)?

Factor	Analysis
Control loop alignment	Motor PID runs at 100 Hz - tick period matches control quantum
Scheduling latency	10ms worst-case latency acceptable for robotics (vs. 1ms for hard real-time)
CPU overhead	Context switch $\sim 24\mu\text{s}$, ISR $\sim 5\mu\text{s}$, total 0.29% overhead
Simple arithmetic	ms $\rightarrow$ ticks is divide-by-10 (optimized to right-shift-by-1 + divide-by-5)
Tick counter range	32-bit counter @ 100 Hz wraps after 497 days (acceptable)
Standard practice	Common RTOS tick rate (FreeRTOS default: 100-1000 Hz)

Alternative Tick Rates Considered:

Tick Rate	Pros	Cons	Verdict
10 Hz (100ms)	Minimal overhead (0.029%)	100ms latency unacceptable for robotics	[FAIL] Rejected
100 Hz (10ms)	Good balance, aligns with control loops	N/A	[PASS] Selected
1000 Hz (1ms)	Excellent responsiveness	10x overhead (2.9%), overkill for our use case	[FAIL] Rejected

7.37.2.4 Conversion Formulas

Milliseconds $\rightarrow$ ThreadX Ticks:

```
ticks = ms x (tick_rate_hz / 1000)
      = ms x (100 / 1000)
      = ms / 10
```

ThreadX Ticks $\rightarrow$ Milliseconds:

```
ms = ticks x (1000 / tick_rate_hz)
   = ticks x (1000 / 100)
   = ticks x 10
```

Seconds $\rightarrow$ ThreadX Ticks:

```
ticks = seconds x tick_rate_hz
      = seconds x 100
```

Value: 100 Hz

Type: `uint16_t` const (read-only compile-time constant)

Memory: Inlined by optimizer (zero runtime storage)

Common Usage: Time conversions for ThreadX API calls

Invariant

`s_rx_threadx_tick_rate_hz == 1000 / k_threadx_ms_per_tick` (reciprocal relationship)
`s_rx_threadx_tick_rate_hz == TX_TIMER_TICKS_PER_SECOND` (must match `tx_user.h`)
`CMT0.CMCOR == (PCLKB_Hz / 32 / s_rx_threadx_tick_rate_hz) - 1`

Warning

CRITICAL: This value MUST match `TX_TIMER_TICKS_PER_SECOND` in `tx_user.h`
 CRITICAL: CMT0 hardware timer MUST be configured for 100 Hz interrupt
 Changing this value requires updating 4 locations (see file header)

Note

Prefer using `k_threadx_ms_per_tick` from [rx_time_constants.h](#) for conversions (provides inverse relationship: 10ms per tick)

Basic Sleep Example:

```
\#include "rx_threadx_config.h"
\#include "rx_time_constants.h"
\#include "tx_api.h"

// Sleep for 500ms using ThreadX
void delay_500ms(void)
{
    // Method 1: Using tick rate (forward conversion)
    uint32_t timeout_ms = 500;
    uint32_t ticks = (timeout_ms * s_rx_threadx_tick_rate_hz) / k_rx_ms_per_second;
    // Result: (500 x 100) / 1000 = 50 ticks

    // Method 2: Using tick period (preferred - simpler)
    ticks = timeout_ms / k_threadx_ms_per_tick;
    // Result: 500 / 10 = 50 ticks

    tx_thread_sleep(ticks); // Sleep for 50 ticks = 500ms
}
```

Semaphore Timeout Example:

```
\#include "rx_threadx_config.h"
\#include "rx_time_constants.h"
\#include "tx_api.h"

TX_SEMAPHORE data_ready_semaphore;

// Wait for semaphore with 100ms timeout
rx_err_t wait_for_data(void)
{
    uint32_t timeout_ms = 100;
    uint32_t timeout_ticks = timeout_ms / k_threadx_ms_per_tick; // 10 ticks

    UINT status = tx_semaphore_get(&data_ready_semaphore, timeout_ticks);
    if (status == TX_NO_INSTANCE) {
        // Timeout occurred after 100ms
        return k_rx_err_timeout;
    }

    return k_rx_ok;
}
```

Software Timer Example:

```

#include "rx_threadx_config.h"
#include "rx_time_constants.h"
#include "tx_api.h"

TX_TIMER telemetry_timer;

void telemetry_timer_callback(ULONG param)
{
    // Send telemetry data (called every 100ms)
    send_telemetry();
}

void setup_telemetry_timer(void)
{
    // Create periodic timer: 100ms period
    uint32_t period_ms = 100;
    uint32_t period_ticks = period_ms / k_threadx_ms_per_tick; // 10 ticks

    tx_timer_create(&telemetry_timer,
        "Telemetry Timer",
        telemetry_timer_callback,
        0, // Parameter
        period_ticks, // Initial ticks = 100ms
        period_ticks, // Reschedule ticks = 100ms
        TX_AUTO_ACTIVATE);
}

```

Watchdog Timeout Calculation Example:

```

#include "rx_threadx_config.h"
#include "rx_time_constants.h"
#include "rx_iwtdt.h"

// Configure IWDT for 8-second timeout
void configure_watchdog(void)
{
    // Convert seconds to milliseconds
    uint32_t timeout_sec = 8;
    uint32_t timeout_ms = timeout_sec * k_rx_ms_per_second; // 8000ms

    // Calculate equivalent in ticks (for validation)
    uint32_t timeout_ticks = timeout_ms / k_threadx_ms_per_tick; // 800 ticks
    // Watchdog should trigger before tick counter wraps (2^32 ticks = 497 days)

    rx_iwtdt_config_t config = {
        .timeout_ms = timeout_ms,
        .window_start_percent = 0,
        .window_end_percent = 100
    };

    rx_err_t err = rx_iwtdt_init(&config);
    if (err != k_rx_ok) {
        // Handle initialization error
    }
}

```

Performance Measurement Example:

```

#include "rx_threadx_config.h"
#include "rx_time_constants.h"
#include "tx_api.h"

// Measure function execution time (tick granularity)
void measure_execution_time(void)
{
    ULONG start_ticks = tx_time_get();

    // Execute function to measure
    expensive_operation();

    ULONG end_ticks = tx_time_get();
    ULONG elapsed_ticks = end_ticks - start_ticks;

    // Convert to milliseconds
    uint32_t elapsed_ms = elapsed_ticks * k_threadx_ms_per_tick;

    rx_log_info("PERF", "Operation took %lu ms (%lu ticks)",
        elapsed_ms, elapsed_ticks);
}

```

See also

TX_TIMER_TICKS_PER_SECOND ThreadX core configuration (tx_user.h)
[k_threadx_ms_per_tick](#) Inverse constant (10ms per tick) in [rx_time_constants.h](#)
[k_rx_ms_per_second](#) Milliseconds per second (1000) in [rx_time_constants.h](#)
 tx_initialize_low_level.c CMT0 hardware timer initialization
 tx_thread_sleep() ThreadX sleep function (ticks)
 tx_semaphore_get() ThreadX semaphore with timeout (ticks)
 tx_time_get() Get current tick counter value

Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 8 (Limited preprocessor): Uses const, not macro [OK]

Definition at line 466 of file [rx_threadx_config.h](#).

7.38 rx_threadx_config.h

[Go to the documentation of this file.](#)

```
00001 /**
00002  * @file rx_threadx_config.h
00003  * @brief ThreadX RTOS Tick Rate Configuration
00004  *
00005  * @details
00006  * This header defines the ThreadX system tick frequency used throughout the
00007  * RX72N firmware for RTOS time conversions. The tick rate is configured to
00008  * 100 Hz (10ms period) via hardware timer CMT0 (Compare Match Timer 0).
00009  *
00010  * ## System Architecture Context
00011  *
00012  * ThreadX is a hard real-time operating system (RTOS) that provides:
00013  * - **Deterministic preemptive scheduling**: Priority-based thread switching
00014  * - **Time-slicing**: Round-robin scheduling for equal-priority threads
00015  * - **Timers and delays**: Software timers with tick-based granularity
00016  * - **Event synchronization**: Semaphores, mutexes, event flags with timeouts
00017  *
00018  * The system tick is the fundamental time quantum for all ThreadX operations:
00019  * ```
00020  * Hardware Timer (CMT0) @ 100 Hz
00021  *     v
00022  * ThreadX Tick ISR (every 10ms)
00023  *     v
00024  * +-----+
00025  * | _tx_timer_interrupt() |
00026  * | - Increment tick counter |
00027  * | - Check sleeping threads |
00028  * | - Expire software timers |
00029  * | - Trigger time-slice rotation |
00030  * +-----+
00031  *     v
00032  * Schedule highest-priority ready thread
00033  * ```
00034  *
00035  * @dot
00036  * digraph threadx_tick_architecture {
00037  *     rankdir=LR;
00038  *     node [shape=box, style=rounded];
00039  *
00040  *     // Hardware layer
00041  *     subgraph cluster_hardware {
00042  *         label="Hardware Layer";
00043  *         style=filled;
```

```

00044 *      color=lightgrey;
00045 *      PCLKB [label="PCLKB\n60 MHz", shape=ellipse, fillcolor=yellow, style=filled];
00046 *      Divider [label="/32\nDivider", shape=diamond];
00047 *      CMT0_Clock [label="CMT0 Clock\n1.875 MHz"];
00048 *      CMT0 [label="CMT0 Timer\nCompare: 18750", fillcolor=lightblue, style=filled];
00049 *  }
00050 *
00051 *  // ThreadX layer
00052 *  subgraph cluster_threadx {
00053 *      label="ThreadX RTOS";
00054 *      style=filled;
00055 *      color=lightcyan;
00056 *      ISR [label="CMT0 ISR\n_tx_timer_interrupt()", fillcolor=orange, style=filled];
00057 *      TickCounter [label="Global Tick Counter\n_tx_timer_system_clock"];
00058 *      Scheduler [label="Scheduler\n_tx_thread_schedule()"];
00059 *      Threads [label="Thread Wake-Up\nTimer Expiration", shape=parallelogram];
00060 *  }
00061 *
00062 *  // Application layer
00063 *  subgraph cluster_application {
00064 *      label="Application Code";
00065 *      style=filled;
00066 *      color=lightyellow;
00067 *      TickRate [label="s_rx_threadx_tick_rate_hz\n100 Hz (this file)", fillcolor=lightgreen, style="filled,bold"];
00068 *      TimeConv [label="Time Conversions\nms -> ticks"];
00069 *      API [label="ThreadX API\n_tx_thread_sleep()\n_tx_semaphore_get()"];
00070 *  }
00071 *
00072 *  // Connections
00073 *  PCLKB -> Divider;
00074 *  Divider -> CMT0_Clock;
00075 *  CMT0_Clock -> CMT0 [label="1.875 MHz"];
00076 *  CMT0 -> ISR [label="100 Hz\n(every 10ms)", color=red, style=bold];
00077 *  ISR -> TickCounter [label="Increment"];
00078 *  TickCounter -> Scheduler;
00079 *  Scheduler -> Threads;
00080 *  TickRate -> TimeConv [style=dashed, label="Config"];
00081 *  TimeConv -> API;
00082 *  API -> Threads [style=dashed];
00083 *  }
00084 *  @enddot
00085 *
00086 *  ## Design Rationale
00087 *
00088 *  **Why 100 Hz Tick Rate?**
00089 *  1. **Control loop alignment**: Motor control runs at 100 Hz, matching tick period
00090 *  2. **Low overhead**: Context switch overhead <1% of CPU time (~24us per 10ms)
00091 *  3. **Simple conversions**: ms -> ticks is divide-by-10 (compiler optimizes to shift)
00092 *  4. **Adequate responsiveness**: 10ms maximum scheduling latency acceptable for robotics
00093 *  5. **Standard practice**: Common RTOS tick rate (FreeRTOS, VxWorks use 100-1000 Hz)
00094 *
00095 *  **CMT0 Hardware Configuration:**
00096 *  - **Input clock**: PCLKB / 32 = 60 MHz / 32 = 1.875 MHz
00097 *  - **Compare value**: 18,750 counts
00098 *  - **Interrupt period**: 18,750 / 1,875,000 = 0.01 seconds = 10ms
00099 *  - **Interrupt priority**: IPR[IPR_CMT0_CMI0] = 3 (high priority for determinism)
00100 *
00101 *  ## Implementation Approach
00102 *
00103 *  This header provides a **single source of truth** for ThreadX tick rate:
00104 *  - tx_user.h: Defines TX_TIMER_TICKS_PER_SECOND = 100 (ThreadX core config)
00105 *  - tx_tick.h: Provides C constant s_rx_threadx_tick_rate_hz = 100 (application use)
00106 *  - tx_tick.c: Configures hardware to generate 100 Hz interrupt
00107 *
00108 *  **Why not use TX_TIMER_TICKS_PER_SECOND directly?**
00109 *  - tx_user.h may use preprocessor macros (not type-safe)
00110 *  - This file provides typed constant for compile-time validation
00111 *  - Decouples application code from ThreadX internal headers
00112 *
00113 *  ## Performance Characteristics
00114 *
00115 *  | Metric | Value | Notes |
00116 *  |-----|-----|-----|
00117 *  | **Tick period** | 10 milliseconds | Fixed by CMT0 configuration |
00118 *  | **Tick frequency** | 100 Hz | Reciprocal of tick period |
00119 *  | **Tick jitter** | < 1 us | CMT0 crystal accuracy +/-50 ppm |
00120 *  | **ISR latency** | < 5 us | ThreadX tick ISR execution time |
00121 *  | **Context switch** | ~24 us | Worst-case at 240 MHz (measured) |
00122 *  | **CPU overhead** | < 1% | (5us ISR + 24us switch) / 10ms = 0.29% |
00123 *  | **Max tick count** | 232 - 1 | 32-bit counter wraps after ~497 days |
00124 *
00125 *  **Tick Resolution vs. Alternatives:**
00126 *  - **10ms ticks**: Current configuration, 0.29% overhead
00127 *  - **1ms ticks (1000 Hz)**: 10x overhead ~2.9%, better responsiveness
00128 *  - **100ms ticks (10 Hz)**: 0.029% overhead, unacceptable latency for robotics
00129 *
00130 *  ## Memory Usage

```

```

00131 *
00132 * | Item | Size | Location | Notes |
00133 * |-----|-----|-----|-----|
00134 * | `s_rx_threadx_tick_rate_hz' | 0 bytes | Compile-time constant | Inlined by optimizer |
00135 * | ThreadX tick counter | 4 bytes | `.bss' | `_tx_timer_system_clock' global |
00136 * | CMT0 registers | 8 bytes | Hardware | Memory-mapped I/O |
00137 * | **Total** | **4 bytes RAM** | - | (Counter only, register is MMIO) |
00138 *
00139 * ## Module Dependencies
00140 *
00141 * @par Depends On:
00142 * - None (foundational configuration)
00143 *
00144 * @par Used By:
00145 * - [rx_time_constants.h](rx_time_constants.h) - Defines k_threadx_ms_per_tick (inverse: 10ms)
00146 * - [rx_iwtdt.c](../rx_core/src/rx_iwtdt.c) - Watchdog timeout conversions
00147 * - [rx_usb_comm.c](../rx_usb_comm/src/rx_usb_comm.c) - USB timeout calculations
00148 * - [tx_initialize_low_level.c](../src/tx_initialize_low_level.c) - CMT0 timer setup
00149 * - All application threads using tx_thread_sleep(), tx_semaphore_get(), etc.
00150 *
00151 * @par Configuration Consistency:
00152 * **CRITICAL**: The following values MUST remain synchronized:
00153 * | File | Symbol | Value | Purpose |
00154 * |-----|-----|-----|-----|
00155 * | tx_user.h | TX_TIMER_TICKS_PER_SECOND | 100 | ThreadX core config |
00156 * | rx_threadx_config.h | s_rx_threadx_tick_rate_hz | 100 | Application constant |
00157 * | rx_time_constants.h | k_threadx_ms_per_tick | 10 | Conversion factor (inverse) |
00158 * | tx_initialize_low_level.c | CMT0.CMCOR | 18750 | Hardware compare value |
00159 *
00160 * @warning If you change the tick rate, update ALL four locations above!
00161 *
00162 * ## NASA Power of 10 Compliance
00163 *
00164 * | Rule | Status | Notes |
00165 * |-----|-----|-----|
00166 * | 1. Simple control flow | [OK] | No code (configuration constant only) |
00167 * | 2. Fixed loop bounds | [OK] | No loops |
00168 * | 3. No dynamic memory | [OK] | No allocations (compile-time constant) |
00169 * | 4. Function length <60 lines | [OK] | No functions |
00170 * | 5. Assertions | [OK] | N/A (header-only configuration) |
00171 * | 6. Smallest scope | [OK] | File-scope constant |
00172 * | 7. Check return values | [OK] | N/A (no function calls) |
00173 * | 8. Limited preprocessor | [OK] | No macros (uses const, not \#define) |
00174 * | 9. Pointer restrictions | [OK] | No pointers |
00175 * | 10. Compiler warnings | [OK] | `-Wall -Wextra -Werror' enforced |
00176 *
00177 * ## SOLID Principles
00178 *
00179 * **Single Responsibility (S):**
00180 * - This header has **one responsibility**: Define ThreadX tick rate
00181 * - No mixing of concerns (pure configuration, no logic)
00182 *
00183 * **Open/Closed (O):**
00184 * - Changing tick rate requires recompilation (embedded systems reality)
00185 * - Alternative tick rates could be provided via separate config headers
00186 * - Hardware dependency (CMT0) prevents runtime configuration
00187 *
00188 * **Liskov Substitution (L):**
00189 * - N/A (not applicable to configuration constants)
00190 *
00191 * **Interface Segregation (I):**
00192 * - Single focused constant (no "fat" configuration structure)
00193 * - Application code depends only on what it needs (tick rate)
00194 *
00195 * **Dependency Inversion (D):**
00196 * - High-level modules (application) depend on abstraction (tick rate constant)
00197 * - Low-level details (CMT0 configuration) hidden in initialization code
00198 * - Time interface (rx_time_interface.h) provides further abstraction
00199 *
00200 * @see tx_user.h ThreadX core configuration (TX_TIMER_TICKS_PER_SECOND)
00201 * @see rx_time_constants.h Time conversion factors (k_threadx_ms_per_tick)
00202 * @see tx_initialize_low_level.c CMT0 timer initialization
00203 * @see rx_time_interface.h Dependency Inversion time abstraction
00204 * @see ThreadX User Guide Chapter 3: Time and Timers
00205 *
00206 * @author Locked, Inc.
00207 * @date 2026-01-27
00208 * @copyright Copyright (c) 2026 Locked Inc.
00209 * SPDX-License-Identifier: MIT
00210 */
00211
00212 #pragma once
00213
00214 #include <stdint.h>
00215
00216 #ifdef __cplusplus
00217 extern "C" {

```

```

00218 #endif
00219
00220 /*
=====
00221  * ThreadX Timing Constants
00222  *
=====
00223 */
00224
00225 /**
00226  * @var s_rx_threadx_tick_rate_hz
00227  * @brief ThreadX system tick frequency (100 Hz)
00228  *
00229  * @details
00230  * Defines the ThreadX timer tick rate configured via hardware timer CMT0.
00231  * This value represents the frequency of the RTOS scheduler time quantum
00232  * and is the fundamental unit for all ThreadX time-based operations.
00233  *
00234  * ## Configuration Details
00235  *
00236  * ***ThreadX Core Configuration (tx_user.h):**
00237  * ```c
00238  * \#define TX_TIMER_TICKS_PER_SECOND 100
00239  * ```
00240  *
00241  * ***Hardware Timer Configuration (CMT0):**
00242  * - **Timer**: Compare Match Timer 0 (CMT0)
00243  * - **Input clock**: PCLKB / 32 = 60 MHz / 32 = 1.875 MHz
00244  * - **Compare value**: 18,750 counts (0x493E)
00245  * - **Interrupt vector**: VECT_CMT0_CMI0 (vector 28)
00246  * - **Interrupt priority**: IPR[IPR_CMT0_CMI0] = 3 (high priority)
00247  * - **Calculation**: 18,750 / 1,875,000 Hz = 10ms period = 100 Hz
00248  *
00249  * ***Register Configuration:**
00250  * ```c
00251  * // From tx_initialize_low_level.c
00252  * CMT.CMSTR0.BIT.STR0 = 0; // Stop CMT0
00253  * CMT0.CMCR.WORD = 0x00C2; // PCLKB/32, interrupt enabled
00254  * CMT0.CMCOR = 18750 - 1; // Compare match value
00255  * IPR(CMT0, CMI0) = 3; // Priority 3
00256  * IEN(CMT0, CMI0) = 1; // Enable interrupt
00257  * CMT.CMSTR0.BIT.STR0 = 1; // Start CMT0
00258  * ```
00259  *
00260  * ## Tick Rate Selection Rationale
00261  *
00262  * **Why 100 Hz (10ms period)?**
00263  *
00264  * | Factor | Analysis |
00265  * |-----|-----|
00266  * | **Control loop alignment** | Motor PID runs at 100 Hz - tick period matches control quantum |
00267  * | **Scheduling latency** | 10ms worst-case latency acceptable for robotics (vs. 1ms for hard real-time) |
00268  * | **CPU overhead** | Context switch ~24us, ISR ~5us, total 0.29% overhead |
00269  * | **Simple arithmetic** | ms -> ticks is divide-by-10 (optimized to right-shift-by-1 + divide-by-5) |
00270  * | **Tick counter range** | 32-bit counter @ 100 Hz wraps after 497 days (acceptable) |
00271  * | **Standard practice** | Common RTOS tick rate (FreeRTOS default: 100-1000 Hz) |
00272  *
00273  * ***Alternative Tick Rates Considered:**
00274  *
00275  * | Tick Rate | Pros | Cons | Verdict |
00276  * |-----|-----|-----|-----|
00277  * | **10 Hz (100ms)** | Minimal overhead (0.029%) | 100ms latency unacceptable for robotics | [FAIL] Rejected |
00278  * | **100 Hz (10ms)** | Good balance, aligns with control loops | N/A | [PASS] **Selected** |
00279  * | **1000 Hz (1ms)** | Excellent responsiveness | 10x overhead (2.9%), overkill for our use case | [FAIL] Rejected |
00280  *
00281  * ## Conversion Formulas
00282  *
00283  * ***Milliseconds -> ThreadX Ticks:**
00284  * ```
00285  * ticks = ms x (tick_rate_hz / 1000)
00286  *        = ms x (100 / 1000)
00287  *        = ms / 10
00288  * ```
00289  *
00290  * ***ThreadX Ticks -> Milliseconds:**
00291  * ```
00292  * ms = ticks x (1000 / tick_rate_hz)
00293  *     = ticks x (1000 / 100)
00294  *     = ticks x 10
00295  * ```
00296  *
00297  * ***Seconds -> ThreadX Ticks:**
00298  * ```
00299  * ticks = seconds x tick_rate_hz
00300  *        = seconds x 100
00301  * ```
00302  *

```

```

00303 * @par Value: 100 Hz
00304 *
00305 * @par Type: `uint16_t const` (read-only compile-time constant)
00306 *
00307 * @par Memory: Inlined by optimizer (zero runtime storage)
00308 *
00309 * @par Common Usage: Time conversions for ThreadX API calls
00310 *
00311 * @invariant s_rx_threadx_tick_rate_hz == 1000 / k_threadx_ms_per_tick (reciprocal relationship)
00312 * @invariant s_rx_threadx_tick_rate_hz == TX_TIMER_TICKS_PER_SECOND (must match tx_user.h)
00313 * @invariant CMT0.CMCOR == (PCLKB_Hz / 32 / s_rx_threadx_tick_rate_hz) - 1
00314 *
00315 * @warning **CRITICAL**: This value MUST match TX_TIMER_TICKS_PER_SECOND in tx_user.h
00316 * @warning **CRITICAL**: CMT0 hardware timer MUST be configured for 100 Hz interrupt
00317 * @warning Changing this value requires updating 4 locations (see file header)
00318 *
00319 * @note Prefer using k_threadx_ms_per_tick from rx_time_constants.h for conversions
00320 * (provides inverse relationship: 10ms per tick)
00321 *
00322 * @par Basic Sleep Example:
00323 * @code{.c}
00324 * \#include "rx_threadx_config.h"
00325 * \#include "rx_time_constants.h"
00326 * \#include "tx_api.h"
00327 *
00328 * // Sleep for 500ms using ThreadX
00329 * void delay_500ms(void)
00330 * {
00331 *     // Method 1: Using tick rate (forward conversion)
00332 *     uint32_t timeout_ms = 500;
00333 *     uint32_t ticks = (timeout_ms * s_rx_threadx_tick_rate_hz) / k_rx_ms_per_second;
00334 *     // Result: (500 x 100) / 1000 = 50 ticks
00335 *
00336 *     // Method 2: Using tick period (preferred - simpler)
00337 *     ticks = timeout_ms / k_threadx_ms_per_tick;
00338 *     // Result: 500 / 10 = 50 ticks
00339 *
00340 *     tx_thread_sleep(ticks); // Sleep for 50 ticks = 500ms
00341 * }
00342 * @endcode
00343 *
00344 * @par Semaphore Timeout Example:
00345 * @code{.c}
00346 * \#include "rx_threadx_config.h"
00347 * \#include "rx_time_constants.h"
00348 * \#include "tx_api.h"
00349 *
00350 * TX_SEMAPHORE data_ready_semaphore;
00351 *
00352 * // Wait for semaphore with 100ms timeout
00353 * rx_err_t wait_for_data(void)
00354 * {
00355 *     uint32_t timeout_ms = 100;
00356 *     uint32_t timeout_ticks = timeout_ms / k_threadx_ms_per_tick; // 10 ticks
00357 *
00358 *     UINT status = tx_semaphore_get(&data_ready_semaphore, timeout_ticks);
00359 *     if (status == TX_NO_INSTANCE) {
00360 *         // Timeout occurred after 100ms
00361 *         return k_rx_err_timeout;
00362 *     }
00363 *
00364 *     return k_rx_ok;
00365 * }
00366 * @endcode
00367 *
00368 * @par Software Timer Example:
00369 * @code{.c}
00370 * \#include "rx_threadx_config.h"
00371 * \#include "rx_time_constants.h"
00372 * \#include "tx_api.h"
00373 *
00374 * TX_TIMER telemetry_timer;
00375 *
00376 * void telemetry_timer_callback(ULONG param)
00377 * {
00378 *     // Send telemetry data (called every 100ms)
00379 *     send_telemetry();
00380 * }
00381 *
00382 * void setup_telemetry_timer(void)
00383 * {
00384 *     // Create periodic timer: 100ms period
00385 *     uint32_t period_ms = 100;
00386 *     uint32_t period_ticks = period_ms / k_threadx_ms_per_tick; // 10 ticks
00387 *
00388 *     tx_timer_create(&telemetry_timer,
00389 *         "Telemetry Timer",

```

```

00390 *          telemetry_timer_callback,
00391 *          0,          // Parameter
00392 *          period_ticks, // Initial ticks = 100ms
00393 *          period_ticks, // Reschedule ticks = 100ms
00394 *          TX_AUTO_ACTIVATE);
00395 * }
00396 * @endcode
00397
00398 * @par Watchdog Timeout Calculation Example:
00399 * @code{.c}
00400 * \#include "rx_threadx_config.h"
00401 * \#include "rx_time_constants.h"
00402 * \#include "rx_iwtd.h"
00403
00404 * // Configure IWDT for 8-second timeout
00405 * void configure_watchdog(void)
00406 * {
00407 *     // Convert seconds to milliseconds
00408 *     uint32_t timeout_sec = 8;
00409 *     uint32_t timeout_ms = timeout_sec * k_rx_ms_per_second; // 8000ms
00410 *
00411 *     // Calculate equivalent in ticks (for validation)
00412 *     uint32_t timeout_ticks = timeout_ms / k_threadx_ms_per_tick; // 800 ticks
00413 *     // Watchdog should trigger before tick counter wraps (2^32 ticks = 497 days)
00414 *
00415 *     rx_iwtd_config_t config = {
00416 *         .timeout_ms = timeout_ms,
00417 *         .window_start_percent = 0,
00418 *         .window_end_percent = 100
00419 *     };
00420 *
00421 *     rx_err_t err = rx_iwtd_init(&config);
00422 *     if (err != k_rx_ok) {
00423 *         // Handle initialization error
00424 *     }
00425 * }
00426 * @endcode
00427
00428 * @par Performance Measurement Example:
00429 * @code{.c}
00430 * \#include "rx_threadx_config.h"
00431 * \#include "rx_time_constants.h"
00432 * \#include "tx_api.h"
00433 *
00434 * // Measure function execution time (tick granularity)
00435 * void measure_execution_time(void)
00436 * {
00437 *     ULONG start_ticks = tx_time_get();
00438 *
00439 *     // Execute function to measure
00440 *     expensive_operation();
00441 *
00442 *     ULONG end_ticks = tx_time_get();
00443 *     ULONG elapsed_ticks = end_ticks - start_ticks;
00444 *
00445 *     // Convert to milliseconds
00446 *     uint32_t elapsed_ms = elapsed_ticks * k_threadx_ms_per_tick;
00447 *
00448 *     rx_log_info("PERF", "Operation took %lu ms (%lu ticks)",
00449 *         elapsed_ms, elapsed_ticks);
00450 * }
00451 * @endcode
00452
00453 * @see TX_TIMER_TICKS_PER_SECOND ThreadX core configuration (tx_user.h)
00454 * @see k_threadx_ms_per_tick Inverse constant (10ms per tick) in rx_time_constants.h
00455 * @see k_rx_ms_per_second Milliseconds per second (1000) in rx_time_constants.h
00456 * @see tx_initialize_low_level.c CMT0 hardware timer initialization
00457 * @see tx_thread_sleep() ThreadX sleep function (ticks)
00458 * @see tx_semaphore_get() ThreadX semaphore with timeout (ticks)
00459 * @see tx_time_get() Get current tick counter value
00460 *
00461 * @since Version 1.0.0
00462 *
00463 * @par NASA Power of 10 Compliance:
00464 * - Rule 8 (Limited preprocessor): Uses const, not macro [OK]
00465 */
00466 static const uint16_t s_rx_threadx_tick_rate_hz = 100U;
00467
00468 #ifdef __cplusplus
00469 }
00470 #endif

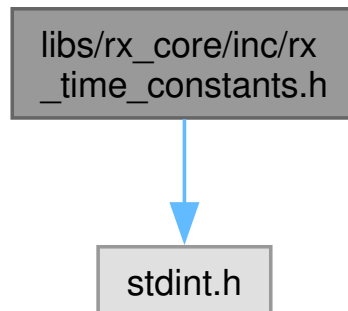
```


7.39 libs/rx_core/inc/rx_time_constants.h File Reference

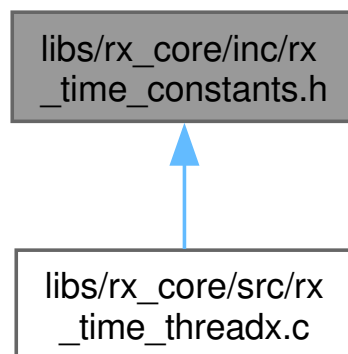
Time Unit Conversion Constants for RX72N Firmware.

```
#include <stdint.h>
```

Include dependency graph for rx_time_constants.h:



This graph shows which files directly or indirectly include this file:



Enumerations

- enum `rx_time_conversion_t` : `uint32_t` {
`k_rx_ms_per_second` = 1000 , `k_rx_us_per_ms` = 1000 , `k_rx_ns_per_us` = 1000 ,
`k_rx_us_per_second` = 1000000 ,
`k_rx_ns_per_ms` = 1000000 , `k_threadx_ms_per_tick` = 10 }

Time unit conversion factors following SI standards.

7.39.1 Detailed Description

Time Unit Conversion Constants for RX72N Firmware.

This header provides typed enumeration constants for time unit conversions used throughout the RX72N firmware. All time conversions follow SI (International System of Units) standards with milliseconds, microseconds, and nanoseconds.

7.39.1.1 System Architecture Context

Time conversions are fundamental to real-time embedded systems. The STAR firmware uses these constants for:

- Timeout calculations: I2C/SPI/USB communication timeouts
- ThreadX scheduling: Converting milliseconds to RTOS ticks
- Performance measurements: Microsecond-precision timing
- Delay functions: Hardware timer configuration
- Watchdog configuration: IWDG timeout settings

7.39.1.2 Design Rationale

C23 Typed Enums for Time Constants:

- Type safety prevents mixing time units (compile error if mismatched)
- Compile-time resolution (zero runtime overhead)
- Debugger shows symbolic names ("k_rx_ms_per_second" vs "1000")
- Self-documenting code

uint32_t Underlying Type:

- Accommodates large timeout values (up to 4,294,967 seconds $\approx$ 49 days)
- Prevents overflow in microsecond/nanosecond conversions
- Matches ThreadX tick counter width (32-bit)

Zero Magic Numbers Policy: All time conversions use named constants from this header. Raw literals like 1000 or 1000000 are forbidden in production code.

7.39.1.3 Implementation Approach

Constants are derived from SI unit definitions:

- 1 second = 1000 milliseconds (ms)
- 1 millisecond = 1000 microseconds (us)
- 1 microsecond = 1000 nanoseconds (ns)

ThreadX configuration:

- System tick frequency: 100 Hz (hardware timer CMT0)
- Tick period: 10 milliseconds

7.39.1.4 Performance Characteristics

- All constants resolved at compile time (zero runtime cost)
- Multiplication/division optimized by compiler to bit shifts where possible
- No function call overhead

7.39.1.5 Memory Usage

- Constants occupy zero RAM (compile-time only)
- Enum type definitions occupy zero storage
- Total memory footprint: 0 bytes

Module Dependencies:

Depends On:

- None (foundational header)

Used By:

- [rx_iwdt.h](#) - Watchdog timeout configuration
- [rx_usb.h](#) - USB timeout calculations
- [rx_comm_manager.h](#) - Communication timeouts
- ThreadX applications - Converting ms to ticks
- All HAL drivers - Hardware timer configuration

NASA Power of 10 Compliance:

Rule	Status	Notes
1. Simple control flow	[OK]	No code (constants only)
2. Fixed loop bounds	[OK]	No loops
3. No dynamic memory	[OK]	No allocations
4. Function length <60 lines	[OK]	No functions
5. Assertions	[OK]	N/A (header only)
6. Smallest scope	[OK]	File-scope enums
7. Check return values	[OK]	N/A

Rule	Status	Notes
8. Limited preprocessor	[OK]	Include guards only, C23 enums used
9. Pointer restrictions	[OK]	No pointers
10. Compiler warnings	[OK]	-Wall -Wextra -Werror

SOLID Principles:

Single Responsibility (S):

- This header has one responsibility: define time conversion constants
- All constants are time-related, cohesive grouping

Open/Closed (O):

- New time units can be added without modifying existing constants
- Derived conversions (e.g., hours, days) can be added as new enum types

Interface Segregation (I):

- Single focused enum type for SI time unit conversions
- No "fat" interface with unrelated constants

See also

[rx_iwdt.h](#) Independent watchdog timer with timeout configuration

ThreadX API `tx_thread_sleep()` uses these conversions

[docs/sections/06_nasa_power_of_10.tex](#) NASA Rule 8 compliance details

Author

Locked, Inc.

Date

2026-01-27

Copyright

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Definition in file [rx_time_constants.h](#).

7.39.2 Enumeration Type Documentation

7.39.2.1 rx_time_conversion_t

enum [rx_time_conversion_t](#) : uint32_t

Time unit conversion factors following SI standards.

Standard International System of Units (SI) time unit conversions used for timeout calculations, RTOS tick conversions, and hardware timer configuration throughout the RX72N firmware.

7.39.2.2 Time Unit Hierarchy (SI Standard)

1 second (s)
+- 1,000 milliseconds (ms)
+- 1,000 microseconds (us)
+- 1,000 nanoseconds (ns)

7.39.2.3 ThreadX RTOS Tick Configuration

The STAR firmware configures ThreadX with a 100 Hz tick rate:

- Tick frequency: 100 Hz (configurable in tx_initialize_low_level.c)
- Tick period: 10 milliseconds
- Hardware timer: CMT0 (Compare Match Timer)
- CMT0 input clock: PCLKB/32 @ 60 MHz = 1.875 MHz
- CMT0 compare value: 18,750 counts = 10ms period

7.39.2.4 Common Conversion Patterns

Seconds -> Milliseconds:

```
uint32_t timeout_ms = timeout_sec * k_rx_ms_per_second;
```

Milliseconds -> ThreadX Ticks:

```
uint32_t ticks = timeout_ms / k_threadx_ms_per_tick;  
tx_thread_sleep(ticks);
```

Microseconds -> Nanoseconds:

```
uint32_t delay_ns = delay_us * k_rx_ns_per_us;
```

Milliseconds -> Microseconds:

```
uint32_t delay_us = delay_ms * k_rx_us_per_ms;
```

7.39.2.5 Overflow Prevention

All constants are `uint32_t` (32-bit unsigned) to prevent overflow:

- Max seconds: 4,294,967 (~49.7 days)
- Max milliseconds: 4,294,967,295 (~49.7 days)
- Max microseconds: 4,294 seconds (~71.6 minutes)
- Max nanoseconds: 4.29 seconds

When working with large values, use 64-bit arithmetic:

```
uint64_t large_ns = (uint64_t)seconds * k_rx_us_per_second * k_rx_ns_per_us;
```

Time Conversion Reference Table:

From	To	Multiply By	Constant
seconds	milliseconds	x 1,000	<code>k_rx_ms_per_second</code>
seconds	microseconds	x 1,000,000	<code>k_rx_us_per_second</code>
milliseconds	microseconds	x 1,000	<code>k_rx_us_per_ms</code>
milliseconds	nanoseconds	x 1,000,000	<code>k_rx_ns_per_ms</code>
microseconds	nanoseconds	x 1,000	<code>k_rx_ns_per_us</code>
milliseconds	ThreadX ticks	/ 10	<code>k_threadx_ms_per_tick</code>

Invariant

```
k_rx_us_per_second == k_rx_ms_per_second * k_rx_us_per_ms
k_rx_ns_per_ms == k_rx_us_per_ms * k_rx_ns_per_us
k_threadx_ms_per_tick == 1000 / tick_frequency_hz (100 Hz)
```

Basic Timeout Example:

```
#include "rx_time_constants.h"
#include "tx_api.h"

// Wait 500ms using ThreadX sleep
void delay_500ms(void)
{
    uint32_t timeout_ms = 500;
    uint32_t ticks = timeout_ms / k_threadx_ms_per_tick; // 500 / 10 = 50 ticks
    tx_thread_sleep(ticks);
}

// Convert timeout from seconds to milliseconds
uint32_t calculate_timeout_ms(uint32_t timeout_sec)
{
    return timeout_sec * k_rx_ms_per_second;
}

// Example: 5 second timeout
uint32_t timeout_ms = calculate_timeout_ms(5); // Returns 5000ms
```

USB Timeout Example:

```

#include "rx_time_constants.h"
#include "rx_usb.h"

// USB read with 100ms timeout
rx_err_t usb_read_with_timeout(uint8_t* buffer, uint32_t len)
{
    uint32_t timeout_ms = 100;
    uint32_t ticks = timeout_ms / k_threadx_ms_per_tick; // 100 / 10 = 10 ticks

    rx_err_t err = rx_usb_read(buffer, len, ticks);
    if (err == k_rx_err_timeout) {
        rx_log_warn("USB", "Read timeout after 100ms");
    }
    return err;
}

```

Hardware Timer Example:

```

#include "rx_time_constants.h"
#include "rx72n_regs.h"

// Configure CMT0 for 10ms periodic interrupt (100 Hz)
void configure_cmt0_for_threadx(void)
{
    // Target period: 10ms
    uint32_t target_period_us = k_threadx_ms_per_tick * k_rx_us_per_ms; // 10,000 us

    // CMT0 clock: PCLKB/32 = 60 MHz / 32 = 1.875 MHz
    uint32_t cmt_clock_hz = 1875000;
    uint32_t cmt_period_us = k_rx_us_per_second / cmt_clock_hz; // 0.533 us per tick

    // Calculate compare value
    uint16_t compare_value = target_period_us / cmt_period_us; // 18,750

    // Configure CMT0
    CMT0.CMCOR = compare_value;
}

```

Watchdog Timeout Example:

```

#include "rx_time_constants.h"
#include "rx_iwdt.h"

// Configure IWDT for 8 second timeout
void configure_watchdog(void)
{
    uint32_t timeout_sec = 8;
    uint32_t timeout_ms = timeout_sec * k_rx_ms_per_second; // 8000ms

    rx_iwdt_config_t config = {
        .timeout_ms = timeout_ms,
        .window_start_percent = 0,
        .window_end_percent = 100
    };

    rx_err_t err = rx_iwdt_init(&config);
    if (err != k_rx_ok) {
        // Handle error
    }
}

```

See also

- [rx_iwdt.h](#) Independent watchdog timer configuration
- [tx_thread_sleep\(\)](#) ThreadX sleep function (takes ticks)
- [rx_usb.h](#) USB communication with timeout parameters
- ThreadX User Guide Section on time management

Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 8 (Limited preprocessor): Uses C23 typed enum, not macros [OK]

Enumerator

k_rx_ms_per_second	Milliseconds per second (SI standard). Fundamental SI time conversion. One second contains exactly 1000 milliseconds. Used for converting timeout values from seconds to milliseconds. Value: 1,000 ms/s Rationale: SI definition - 1 second = 1000 milliseconds Common Usage: Timeout configuration, delay functions Example: <pre>uint32_t timeout_ms = 5 * k_rx_ms_per_second; // 5000ms = 5 seconds</pre>
k_rx_us_per_ms	Microseconds per millisecond (SI standard). One millisecond contains exactly 1000 microseconds. Used for fine-grained timing conversions and hardware timer calculations. Value: 1,000 us/ms Rationale: SI definition - 1 millisecond = 1000 microseconds Common Usage: Hardware timer period calculations, precise delays Example: <pre>uint32_t delay_us = 250 * k_rx_us_per_ms; // 250,000 us = 250ms</pre>

k_rx_ns_per_us	Nanoseconds per microsecond (SI standard). One microsecond contains exactly 1000 nanoseconds. Used for ultra-precise timing calculations and CPU cycle counting. Value: 1,000 ns/us Rationale: SI definition - 1 microsecond = 1000 nanoseconds Common Usage: CPU cycle time calculations (240 MHz = 4.17ns/cycle) Example: <pre>// At 240 MHz, CPU cycle period is ~4.17ns uint32_t cpu_freq_hz = 240000000; uint32_t cycle_time_ns = k_rx_ns_per_us * k_rx_us_per_second / cpu_freq_hz; // Result: 4.166ns per cycle</pre>
k_rx_us_per_second	Microseconds per second (derived SI constant). One second contains exactly 1,000,000 microseconds. Derived from: k_rx_ms_per_second x k_rx_us_per_ms = 1000 x 1000 = 1,000,000 Value: 1,000,000 us/s Rationale: Derived SI conversion for direct second-to-microsecond conversion Common Usage: High-resolution timestamp conversions Example: <pre>// Convert 2.5 seconds to microseconds uint32_t duration_us = (uint32_t)(2.5f * k_rx_us_per_second); // 2,500,000 us</pre> Note Max representable time: ~4294 seconds (71.6 minutes) before uint32_t overflow

k_rx_ns_per_ms	Nanoseconds per millisecond (derived SI constant). One millisecond contains exactly 1,000,000 nanoseconds. Derived from: $k_rx_us_per_ms \times k_rx_ns_per_us = 1000 \times 1000 = 1,000,000$ Value: 1,000,000 ns/ms Rationale: Derived SI conversion for direct millisecond-to-nanosecond conversion Common Usage: Precision timing, interrupt latency measurements Example: <pre>// Calculate interrupt latency budget: 50us = 50,000ns uint32_t latency_ns = 50 * k_rx_ns_per_us; uint32_t latency_ms = latency_ns / k_rx_ns_per_ms; // 0.05ms</pre> Note Max representable time: ~4.29 seconds before uint32_t overflow
----------------	--

<code>k_threadx_ms_per_tick</code>	Milliseconds per ThreadX tick (RTOS configuration). ThreadX tick period configured to 10ms (100 Hz tick rate). This value is hardware-dependent and configured via CMT0 timer in ThreadX port. Configuration details: • Tick frequency: 100 Hz • Tick period: 10 milliseconds • Hardware timer: CMT0 (Compare Match Timer 0) • CMT0 clock source: $PCLKB/32 = 60\text{ MHz} / 32 = 1.875\text{ MHz}$ • CMT0 compare value: 18,750 counts -> 10ms period Value: 10 ms/tick Rationale: • 100 Hz provides good balance between responsiveness and overhead • 10ms quantum matches typical control loop periods (motor control) • Simplifies tick-to-millisecond conversions (divide by 10) • Low enough overhead (<1% CPU time for context switching) Common Usage: Converting millisecond timeouts to RTOS ticks Example: <pre>// Sleep for 250ms uint32_t sleep_ms = 250; uint32_t sleep_ticks = sleep_ms / k_threadx_ms_per_tick; // 25 ticks tx_thread_sleep(sleep_ticks);</pre> Warning If ThreadX tick configuration changes, update this constant! Note Defined in tx_initialize_low_level.c (CMT0 configuration) See also tx_initialize_low_level.c ThreadX tick timer initialization
---	--

Definition at line 301 of file [rx_time_constants.h](#).

```
00301      : uint32_t {
00302  /**
00303   * @brief Milliseconds per second (SI standard)
00304   * @details
00305   * Fundamental SI time conversion. One second contains exactly 1000 milliseconds.
00306   * Used for converting timeout values from seconds to milliseconds.
00307   * @par Value: 1,000 ms/s
00308   * @par Rationale: SI definition - 1 second = 1000 milliseconds
```

```

00309 * @par Common Usage: Timeout configuration, delay functions
00310 * @par Example:
00311 * @code{.c}
00312 * uint32_t timeout_ms = 5 * k_rx_ms_per_second; // 5000ms = 5 seconds
00313 * @endcode
00314 */
00315 k_rx_ms_per_second = 1000,
00316
00317 /**
00318 * @brief Microseconds per millisecond (SI standard)
00319 * @details
00320 * One millisecond contains exactly 1000 microseconds. Used for fine-grained
00321 * timing conversions and hardware timer calculations.
00322 * @par Value: 1,000 us/ms
00323 * @par Rationale: SI definition - 1 millisecond = 1000 microseconds
00324 * @par Common Usage: Hardware timer period calculations, precise delays
00325 * @par Example:
00326 * @code{.c}
00327 * uint32_t delay_us = 250 * k_rx_us_per_ms; // 250,000 us = 250ms
00328 * @endcode
00329 */
00330 k_rx_us_per_ms = 1000,
00331
00332 /**
00333 * @brief Nanoseconds per microsecond (SI standard)
00334 * @details
00335 * One microsecond contains exactly 1000 nanoseconds. Used for ultra-precise
00336 * timing calculations and CPU cycle counting.
00337 * @par Value: 1,000 ns/us
00338 * @par Rationale: SI definition - 1 microsecond = 1000 nanoseconds
00339 * @par Common Usage: CPU cycle time calculations (240 MHz = 4.17ns/cycle)
00340 * @par Example:
00341 * @code{.c}
00342 * // At 240 MHz, CPU cycle period is ~4.17ns
00343 * uint32_t cpu_freq_hz = 240000000;
00344 * uint32_t cycle_time_ns = k_rx_ns_per_us * k_rx_us_per_second / cpu_freq_hz;
00345 * // Result: 4.166ns per cycle
00346 * @endcode
00347 */
00348 k_rx_ns_per_us = 1000,
00349
00350 /**
00351 * @brief Microseconds per second (derived SI constant)
00352 * @details
00353 * One second contains exactly 1,000,000 microseconds. Derived from:
00354 * k_rx_ms_per_second x k_rx_us_per_ms = 1000 x 1000 = 1,000,000
00355 * @par Value: 1,000,000 us/s
00356 * @par Rationale: Derived SI conversion for direct second-to-microsecond conversion
00357 * @par Common Usage: High-resolution timestamp conversions
00358 * @par Example:
00359 * @code{.c}
00360 * // Convert 2.5 seconds to microseconds
00361 * uint32_t duration_us = (uint32_t)(2.5f * k_rx_us_per_second); // 2,500,000 us
00362 * @endcode
00363 * @note Max representable time: ~4294 seconds (71.6 minutes) before uint32_t overflow
00364 */
00365 k_rx_us_per_second = 1000000,
00366
00367 /**
00368 * @brief Nanoseconds per millisecond (derived SI constant)
00369 * @details
00370 * One millisecond contains exactly 1,000,000 nanoseconds. Derived from:
00371 * k_rx_us_per_ms x k_rx_ns_per_us = 1000 x 1000 = 1,000,000
00372 * @par Value: 1,000,000 ns/ms
00373 * @par Rationale: Derived SI conversion for direct millisecond-to-nanosecond conversion
00374 * @par Common Usage: Precision timing, interrupt latency measurements
00375 * @par Example:
00376 * @code{.c}
00377 * // Calculate interrupt latency budget: 50us = 50,000ns
00378 * uint32_t latency_ns = 50 * k_rx_ns_per_us;
00379 * uint32_t latency_ms = latency_ns / k_rx_ns_per_ms; // 0.05ms
00380 * @endcode
00381 * @note Max representable time: ~4.29 seconds before uint32_t overflow
00382 */
00383 k_rx_ns_per_ms = 1000000,
00384
00385 /**
00386 * @brief Milliseconds per ThreadX tick (RTOS configuration)
00387 * @details
00388 * ThreadX tick period configured to 10ms (100 Hz tick rate). This value
00389 * is hardware-dependent and configured via CMT0 timer in ThreadX port.
00390 *
00391 * Configuration details:
00392 * - **Tick frequency**: 100 Hz
00393 * - **Tick period**: 10 milliseconds
00394 * - **Hardware timer**: CMT0 (Compare Match Timer 0)
00395 * - **CMT0 clock source**: PCLKB/32 = 60 MHz / 32 = 1.875 MHz

```

```

00396 * - **CMT0 compare value**: 18,750 counts -> 10ms period
00397 *
00398 * @par Value: 10 ms/tick
00399 * @par Rationale:
00400 * - 100 Hz provides good balance between responsiveness and overhead
00401 * - 10ms quantum matches typical control loop periods (motor control)
00402 * - Simplifies tick-to-millisecond conversions (divide by 10)
00403 * - Low enough overhead (<1% CPU time for context switching)
00404 *
00405 * @par Common Usage: Converting millisecond timeouts to RTOS ticks
00406 *
00407 * @par Example:
00408 * @code{.c}
00409 * // Sleep for 250ms
00410 * uint32_t sleep_ms = 250;
00411 * uint32_t sleep_ticks = sleep_ms / k_threadx_ms_per_tick; // 25 ticks
00412 * tx_thread_sleep(sleep_ticks);
00413 * @endcode
00414 *
00415 * @warning If ThreadX tick configuration changes, update this constant!
00416 * @note Defined in tx_initialize_low_level.c (CMT0 configuration)
00417 * @see tx_initialize_low_level.c ThreadX tick timer initialization
00418 */
00419 k_threadx_ms_per_tick = 10,
00420
00421 } rx_time_conversion_t;

```

7.40 rx_time_constants.h

[Go to the documentation of this file.](#)

```

00001 /**
00002 * @file rx_time_constants.h
00003 * @brief Time Unit Conversion Constants for RX72N Firmware
00004 *
00005 * @details
00006 * This header provides typed enumeration constants for time unit conversions
00007 * used throughout the RX72N firmware. All time conversions follow SI (International
00008 * System of Units) standards with milliseconds, microseconds, and nanoseconds.
00009 *
00010 * ## System Architecture Context
00011 *
00012 * Time conversions are fundamental to real-time embedded systems. The STAR
00013 * firmware uses these constants for:
00014 * - **Timeout calculations**: I2C/SPI/USB communication timeouts
00015 * - **ThreadX scheduling**: Converting milliseconds to RTOS ticks
00016 * - **Performance measurements**: Microsecond-precision timing
00017 * - **Delay functions**: Hardware timer configuration
00018 * - **Watchdog configuration**: IWDG timeout settings
00019 *
00020 * ## Design Rationale
00021 *
00022 * **C23 Typed Enums for Time Constants:**
00023 * - Type safety prevents mixing time units (compile error if mismatched)
00024 * - Compile-time resolution (zero runtime overhead)
00025 * - Debugger shows symbolic names ("k_rx_ms_per_second" vs "1000")
00026 * - Self-documenting code
00027 *
00028 * **uint32_t Underlying Type:**
00029 * - Accommodates large timeout values (up to 4,294,967 seconds ≈ 49 days)
00030 * - Prevents overflow in microsecond/nanosecond conversions
00031 * - Matches ThreadX tick counter width (32-bit)
00032 *
00033 * **Zero Magic Numbers Policy:**
00034 * - All time conversions use named constants from this header. Raw literals
00035 * - like `1000` or `1000000` are forbidden in production code.
00036 *
00037 * ## Implementation Approach
00038 *
00039 * Constants are derived from SI unit definitions:
00040 * - 1 second = 1000 milliseconds (ms)
00041 * - 1 millisecond = 1000 microseconds (us)
00042 * - 1 microsecond = 1000 nanoseconds (ns)
00043 *
00044 * ThreadX configuration:
00045 * - System tick frequency: 100 Hz (hardware timer CMT0)
00046 * - Tick period: 10 milliseconds
00047 *
00048 * ## Performance Characteristics
00049 *
00050 * - All constants resolved at compile time (zero runtime cost)
00051 * - Multiplication/division optimized by compiler to bit shifts where possible

```

```

00052 * - No function call overhead
00053 *
00054 * ## Memory Usage
00055 *
00056 * - Constants occupy zero RAM (compile-time only)
00057 * - Enum type definitions occupy zero storage
00058 * - Total memory footprint: 0 bytes
00059 *
00060 * @par Module Dependencies:
00061 *
00062 * **Depends On:**
00063 * - None (foundational header)
00064 *
00065 * **Used By:**
00066 * - [rx_iwtdt.h](rx_iwtdt.h) - Watchdog timeout configuration
00067 * - [rx_usb.h](../rx_usb/inc/rx_usb.h) - USB timeout calculations
00068 * - [rx_comm_manager.h](../rx_comm_manager/inc/rx_comm_manager.h) - Communication timeouts
00069 * - ThreadX applications - Converting ms to ticks
00070 * - All HAL drivers - Hardware timer configuration
00071 *
00072 * @par NASA Power of 10 Compliance:
00073 *
00074 * | Rule | Status | Notes |
00075 * |-----|-----|-----|
00076 * | 1. Simple control flow | [OK] | No code (constants only) |
00077 * | 2. Fixed loop bounds | [OK] | No loops |
00078 * | 3. No dynamic memory | [OK] | No allocations |
00079 * | 4. Function length <60 lines | [OK] | No functions |
00080 * | 5. Assertions | [OK] | N/A (header only) |
00081 * | 6. Smallest scope | [OK] | File-scope enums |
00082 * | 7. Check return values | [OK] | N/A |
00083 * | 8. Limited preprocessor | [OK] | Include guards only, C23 enums used |
00084 * | 9. Pointer restrictions | [OK] | No pointers |
00085 * | 10. Compiler warnings | [OK] | '-Wall -Wextra -Werror' |
00086 *
00087 * @par SOLID Principles:
00088 *
00089 * **Single Responsibility (S):**
00090 * - This header has one responsibility: define time conversion constants
00091 * - All constants are time-related, cohesive grouping
00092 *
00093 * **Open/Closed (O):**
00094 * - New time units can be added without modifying existing constants
00095 * - Derived conversions (e.g., hours, days) can be added as new enum types
00096 *
00097 * **Interface Segregation (I):**
00098 * - Single focused enum type for SI time unit conversions
00099 * - No "fat" interface with unrelated constants
00100 *
00101 * @see rx_iwtdt.h Independent watchdog timer with timeout configuration
00102 * @see ThreadX API tx_thread_sleep() uses these conversions
00103 * @see docs/sections/06_nasa_power_of_10.tex NASA Rule 8 compliance details
00104 *
00105 * @author Locked, Inc.
00106 * @date 2026-01-27
00107 * @copyright Copyright (c) 2026 Locked Inc.
00108 * SPDX-License-Identifier: MIT
00109 */
00110
00111 #pragma once
00112
00113 #include <stdint.h>
00114
00115 #ifdef __cplusplus
00116 extern "C" {
00117 #endif
00118
00119 /*
=====
00120 * Time Unit Conversion Constants
00121 *
=====
00122 */
00123
00124 /**
00125 * @enum rx_time_conversion_t
00126 * @brief Time unit conversion factors following SI standards
00127 *
00128 * @details
00129 * Standard International System of Units (SI) time unit conversions used for
00130 * timeout calculations, RTOS tick conversions, and hardware timer configuration
00131 * throughout the RX72N firmware.
00132 *
00133 * ## Time Unit Hierarchy (SI Standard)
00134 *
00135 * ```
00136 * 1 second (s)

```

```

00137 * +- 1,000 milliseconds (ms)
00138 * +- 1,000 microseconds (us)
00139 * +- 1,000 nanoseconds (ns)
00140 * ```
00141 *
00142 * ## ThreadX RTOS Tick Configuration
00143 *
00144 * The STAR firmware configures ThreadX with a 100 Hz tick rate:
00145 * - **Tick frequency**: 100 Hz (configurable in tx_initialize_low_level.c)
00146 * - **Tick period**: 10 milliseconds
00147 * - **Hardware timer**: CMT0 (Compare Match Timer)
00148 * - **CMT0 input clock**: PCLKB/32 @ 60 MHz = 1.875 MHz
00149 * - **CMT0 compare value**: 18,750 counts = 10ms period
00150 *
00151 * ## Common Conversion Patterns
00152 *
00153 * **Seconds -> Milliseconds:**
00154 * ```c
00155 * uint32_t timeout_ms = timeout_sec * k_rx_ms_per_second;
00156 * ```
00157 *
00158 * **Milliseconds -> ThreadX Ticks:**
00159 * ```c
00160 * uint32_t ticks = timeout_ms / k_threadx_ms_per_tick;
00161 * tx_thread_sleep(ticks);
00162 * ```
00163 *
00164 * **Microseconds -> Nanoseconds:**
00165 * ```c
00166 * uint32_t delay_ns = delay_us * k_rx_ns_per_us;
00167 * ```
00168 *
00169 * **Milliseconds -> Microseconds:**
00170 * ```c
00171 * uint32_t delay_us = delay_ms * k_rx_us_per_ms;
00172 * ```
00173 *
00174 * ## Overflow Prevention
00175 *
00176 * All constants are uint32_t (32-bit unsigned) to prevent overflow:
00177 * - **Max seconds**: 4,294,967 (~49.7 days)
00178 * - **Max milliseconds**: 4,294,967,295 (~49.7 days)
00179 * - **Max microseconds**: 4,294 seconds (~71.6 minutes)
00180 * - **Max nanoseconds**: 4.29 seconds
00181 *
00182 * When working with large values, use 64-bit arithmetic:
00183 * ```c
00184 * uint64_t large_ns = (uint64_t)seconds * k_rx_us_per_second * k_rx_ns_per_us;
00185 * ```
00186 *
00187 * @par Time Conversion Reference Table:
00188 *
00189 * | From | To | Multiply By | Constant |
00190 * |-----|-----|-----|-----|
00191 * | seconds | milliseconds | x 1,000 | k_rx_ms_per_second |
00192 * | seconds | microseconds | x 1,000,000 | k_rx_us_per_second |
00193 * | milliseconds | microseconds | x 1,000 | k_rx_us_per_ms |
00194 * | milliseconds | nanoseconds | x 1,000,000 | k_rx_ns_per_ms |
00195 * | microseconds | nanoseconds | x 1,000 | k_rx_ns_per_us |
00196 * | milliseconds | ThreadX ticks | / 10 | k_threadx_ms_per_tick |
00197 *
00198 * @invariant k_rx_us_per_second == k_rx_ms_per_second * k_rx_us_per_ms
00199 * @invariant k_rx_ns_per_ms == k_rx_us_per_ms * k_rx_ns_per_us
00200 * @invariant k_threadx_ms_per_tick == 1000 / tick_frequency_hz (100 Hz)
00201 *
00202 * @par Basic Timeout Example:
00203 * @code{.c}
00204 * #include "rx_time_constants.h"
00205 * #include "tx_api.h"
00206 *
00207 * // Wait 500ms using ThreadX sleep
00208 * void delay_500ms(void)
00209 * {
00210 *     uint32_t timeout_ms = 500;
00211 *     uint32_t ticks = timeout_ms / k_threadx_ms_per_tick; // 500 / 10 = 50 ticks
00212 *     tx_thread_sleep(ticks);
00213 * }
00214 *
00215 * // Convert timeout from seconds to milliseconds
00216 * uint32_t calculate_timeout_ms(uint32_t timeout_sec)
00217 * {
00218 *     return timeout_sec * k_rx_ms_per_second;
00219 * }
00220 *
00221 * // Example: 5 second timeout
00222 * uint32_t timeout_ms = calculate_timeout_ms(5); // Returns 5000ms
00223 * @endcode

```

```

00224 *
00225 * @par USB Timeout Example:
00226 * @code{.c}
00227 * #include "rx_time_constants.h"
00228 * #include "rx_usb.h"
00229 *
00230 * // USB read with 100ms timeout
00231 * rx_err_t usb_read_with_timeout(uint8_t* buffer, uint32_t len)
00232 * {
00233 *     uint32_t timeout_ms = 100;
00234 *     uint32_t ticks = timeout_ms / k_threadx_ms_per_tick; // 100 / 10 = 10 ticks
00235 *
00236 *     rx_err_t err = rx_usb_read(buffer, len, ticks);
00237 *     if (err == k_rx_err_timeout) {
00238 *         rx_log_warn("USB", "Read timeout after 100ms");
00239 *     }
00240 *     return err;
00241 * }
00242 * @endcode
00243 *
00244 * @par Hardware Timer Example:
00245 * @code{.c}
00246 * #include "rx_time_constants.h"
00247 * #include "rx72n_regs.h"
00248 *
00249 * // Configure CMT0 for 10ms periodic interrupt (100 Hz)
00250 * void configure_cmt0_for_threadx(void)
00251 * {
00252 *     // Target period: 10ms
00253 *     uint32_t target_period_us = k_threadx_ms_per_tick * k_rx_us_per_ms; // 10,000 us
00254 *
00255 *     // CMT0 clock: PCLKB/32 = 60 MHz / 32 = 1.875 MHz
00256 *     uint32_t cmt_clock_hz = 1875000;
00257 *     uint32_t cmt_period_us = k_rx_us_per_second / cmt_clock_hz; // 0.533 us per tick
00258 *
00259 *     // Calculate compare value
00260 *     uint16_t compare_value = target_period_us / cmt_period_us; // 18,750
00261 *
00262 *     // Configure CMT0
00263 *     CMT0.CMCOR = compare_value;
00264 * }
00265 * @endcode
00266 *
00267 * @par Watchdog Timeout Example:
00268 * @code{.c}
00269 * #include "rx_time_constants.h"
00270 * #include "rx_iwdt.h"
00271 *
00272 * // Configure IWDT for 8 second timeout
00273 * void configure_watchdog(void)
00274 * {
00275 *     uint32_t timeout_sec = 8;
00276 *     uint32_t timeout_ms = timeout_sec * k_rx_ms_per_second; // 8000ms
00277 *
00278 *     rx_iwdt_config_t config = {
00279 *         .timeout_ms = timeout_ms,
00280 *         .window_start_percent = 0,
00281 *         .window_end_percent = 100
00282 *     };
00283 *
00284 *     rx_err_t err = rx_iwdt_init(&config);
00285 *     if (err != k_rx_ok) {
00286 *         // Handle error
00287 *     }
00288 * }
00289 * @endcode
00290 *
00291 * @see rx_iwdt.h Independent watchdog timer configuration
00292 * @see tx_thread_sleep() ThreadX sleep function (takes ticks)
00293 * @see rx_usb.h USB communication with timeout parameters
00294 * @see ThreadX User Guide Section on time management
00295 *
00296 * @since Version 1.0.0
00297 *
00298 * @par NASA Power of 10 Compliance:
00299 * - Rule 8 (Limited preprocessor): Uses C23 typed enum, not macros [OK]
00300 */
00301 typedef enum : uint32_t {
00302 /**
00303 * @brief Milliseconds per second (SI standard)
00304 * @details
00305 * Fundamental SI time conversion. One second contains exactly 1000 milliseconds.
00306 * Used for converting timeout values from seconds to milliseconds.
00307 * @par Value: 1,000 ms/s
00308 * @par Rationale: SI definition - 1 second = 1000 milliseconds
00309 * @par Common Usage: Timeout configuration, delay functions
00310 * @par Example:

```



```

00311 * @code{.c}
00312 * uint32_t timeout_ms = 5 * k_rx_ms_per_second; // 5000ms = 5 seconds
00313 * @endcode
00314 */
00315 k_rx_ms_per_second = 1000,
00316
00317 /**
00318 * @brief Microseconds per millisecond (SI standard)
00319 * @details
00320 * One millisecond contains exactly 1000 microseconds. Used for fine-grained
00321 * timing conversions and hardware timer calculations.
00322 * @par Value: 1,000 us/ms
00323 * @par Rationale: SI definition - 1 millisecond = 1000 microseconds
00324 * @par Common Usage: Hardware timer period calculations, precise delays
00325 * @par Example:
00326 * @code{.c}
00327 * uint32_t delay_us = 250 * k_rx_us_per_ms; // 250,000 us = 250ms
00328 * @endcode
00329 */
00330 k_rx_us_per_ms = 1000,
00331
00332 /**
00333 * @brief Nanoseconds per microsecond (SI standard)
00334 * @details
00335 * One microsecond contains exactly 1000 nanoseconds. Used for ultra-precise
00336 * timing calculations and CPU cycle counting.
00337 * @par Value: 1,000 ns/us
00338 * @par Rationale: SI definition - 1 microsecond = 1000 nanoseconds
00339 * @par Common Usage: CPU cycle time calculations (240 MHz = 4.17ns/cycle)
00340 * @par Example:
00341 * @code{.c}
00342 * // At 240 MHz, CPU cycle period is ~4.17ns
00343 * uint32_t cpu_freq_hz = 240000000;
00344 * uint32_t cycle_time_ns = k_rx_ns_per_us * k_rx_us_per_second / cpu_freq_hz;
00345 * // Result: 4.166ns per cycle
00346 * @endcode
00347 */
00348 k_rx_ns_per_us = 1000,
00349
00350 /**
00351 * @brief Microseconds per second (derived SI constant)
00352 * @details
00353 * One second contains exactly 1,000,000 microseconds. Derived from:
00354 * k_rx_ms_per_second x k_rx_us_per_ms = 1000 x 1000 = 1,000,000
00355 * @par Value: 1,000,000 us/s
00356 * @par Rationale: Derived SI conversion for direct second-to-microsecond conversion
00357 * @par Common Usage: High-resolution timestamp conversions
00358 * @par Example:
00359 * @code{.c}
00360 * // Convert 2.5 seconds to microseconds
00361 * uint32_t duration_us = (uint32_t)(2.5f * k_rx_us_per_second); // 2,500,000 us
00362 * @endcode
00363 * @note Max representable time: ~4294 seconds (71.6 minutes) before uint32_t overflow
00364 */
00365 k_rx_us_per_second = 1000000,
00366
00367 /**
00368 * @brief Nanoseconds per millisecond (derived SI constant)
00369 * @details
00370 * One millisecond contains exactly 1,000,000 nanoseconds. Derived from:
00371 * k_rx_us_per_ms x k_rx_ns_per_us = 1000 x 1000 = 1,000,000
00372 * @par Value: 1,000,000 ns/ms
00373 * @par Rationale: Derived SI conversion for direct millisecond-to-nanosecond conversion
00374 * @par Common Usage: Precision timing, interrupt latency measurements
00375 * @par Example:
00376 * @code{.c}
00377 * // Calculate interrupt latency budget: 50us = 50,000ns
00378 * uint32_t latency_ns = 50 * k_rx_ns_per_us;
00379 * uint32_t latency_ms = latency_ns / k_rx_ns_per_ms; // 0.05ms
00380 * @endcode
00381 * @note Max representable time: ~4.29 seconds before uint32_t overflow
00382 */
00383 k_rx_ns_per_ms = 1000000,
00384
00385 /**
00386 * @brief Milliseconds per ThreadX tick (RTOS configuration)
00387 * @details
00388 * ThreadX tick period configured to 10ms (100 Hz tick rate). This value
00389 * is hardware-dependent and configured via CMT0 timer in ThreadX port.
00390 *
00391 * Configuration details:
00392 * - **Tick frequency**: 100 Hz
00393 * - **Tick period**: 10 milliseconds
00394 * - **Hardware timer**: CMT0 (Compare Match Timer 0)
00395 * - **CMT0 clock source**: PCLKB/32 = 60 MHz / 32 = 1.875 MHz
00396 * - **CMT0 compare value**: 18,750 counts -> 10ms period
00397 */

```

```

00398 * @par Value: 10 ms/tick
00399 * @par Rationale:
00400 * - 100 Hz provides good balance between responsiveness and overhead
00401 * - 10ms quantum matches typical control loop periods (motor control)
00402 * - Simplifies tick-to-millisecond conversions (divide by 10)
00403 * - Low enough overhead (<1% CPU time for context switching)
00404 *
00405 * @par Common Usage: Converting millisecond timeouts to RTOS ticks
00406 *
00407 * @par Example:
00408 * @code{.c}
00409 * // Sleep for 250ms
00410 * uint32_t sleep_ms = 250;
00411 * uint32_t sleep_ticks = sleep_ms / k_threadx_ms_per_tick; // 25 ticks
00412 * tx_thread_sleep(sleep_ticks);
00413 * @endcode
00414 *
00415 * @warning If ThreadX tick configuration changes, update this constant!
00416 * @note Defined in tx_initialize_low_level.c (CMT0 configuration)
00417 * @see tx_initialize_low_level.c ThreadX tick timer initialization
00418 */
00419 k_threadx_ms_per_tick = 10,
00420
00421 } rx_time_conversion_t;
00422
00423 #ifdef __cplusplus
00424 }
00425 #endif

```

7.41 libs/rx_core/inc/rx_time_interface.h File Reference

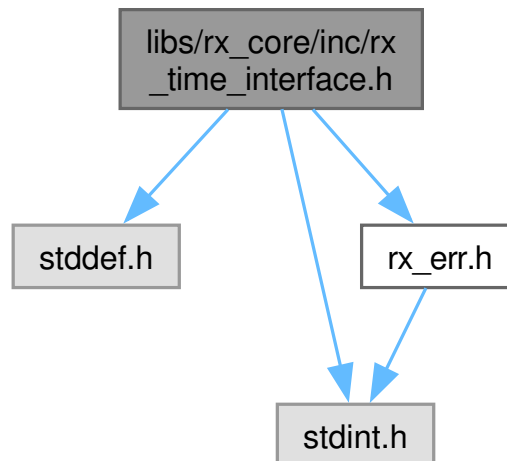
Time Interface - Dependency Inversion for Timing Operations.

```
#include <stddef.h>
```

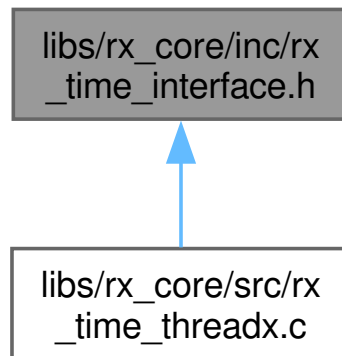
```
#include <stdint.h>
```

```
#include "rx_err.h"
```

Include dependency graph for rx_time_interface.h:



This graph shows which files directly or indirectly include this file:



Data Structures

- struct [rx_time_interface_t](#)
Time operations interface structure (Dependency Inversion Pattern).

Typedefs

- typedef void(* [rx_time_sleep_ms_fn](#)) (void *ctx, uint32_t ms)
Function pointer: Sleep/delay for specified milliseconds.
- typedef uint32_t(* [rx_time_get_ms_fn](#)) (void *ctx)
Function pointer: Get current monotonic time in milliseconds.
- typedef bool(* [rx_time_is_elapsed_fn](#)) (void *ctx, uint32_t start_ms, uint32_t timeout_ms)
Function pointer: Check if timeout has elapsed (optional convenience).

Functions

- [rx_err_t rx_time_threadx_get_interface](#) (rx_time_interface_t *iface)
Validate that a time interface is properly initialized.
- static [rx_err_t rx_time_interface_validate](#) (const rx_time_interface_t *iface)

7.41.1 Detailed Description

Time Interface - Dependency Inversion for Timing Operations.

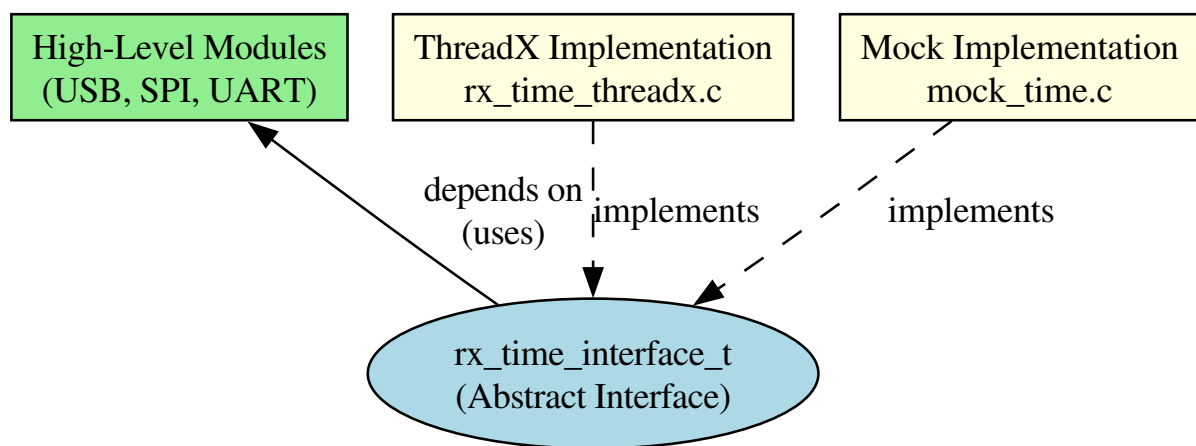
Defines the abstract interface for time operations in RX72N firmware, implementing the Dependency Inversion Principle (DIP) from SOLID design. This header contains ONLY the interface definition with NO implementation details, enabling complete decoupling between high-level modules and low-level timing mechanisms.

7.41.1.1 Architecture Pattern: Dependency Inversion Principle (DIP)

Problem: Communication modules (USB, SPI, UART) need timing operations (sleep, timeouts) but should not directly depend on ThreadX RTOS. Direct dependencies create:

- Tight coupling -> hard to change RTOS
- Untestable code -> cannot unit test without ThreadX
- Platform lock-in -> cannot port to bare-metal or different RTOS

Solution: Define abstract time interface that both high-level and low-level modules depend on:



Key Insight: High-level modules depend on the ABSTRACTION (interface), not the IMPLEMENTATION (ThreadX). Implementations are injected at runtime through function pointers.

7.41.1.2 Benefits of This Pattern

Benefit	Description	Example
Testability	Unit tests inject mock timing without ThreadX	Test USB without RTOS
Flexibility	Swap RTOS without changing USB/SPI code	ThreadX -> FreeRTOS
Decoupling	Communication layers have zero ThreadX includes	Clean dependencies
Portability	Same code runs on bare-metal with null impl	Debug builds
Simulation	Mock advances time instantly for fast tests	1000ms timeout in 1us

7.41.1.3 Concrete Implementations

Implementation	File	Use Case
ThreadX	rx_time_threadx.c	Production firmware on RX72N
Mock (manual)	mock_time.c	Unit tests with controlled time

Implementation	File	Use Case
Mock (auto)	mock_time.c (auto_advance)	Unit tests with simulated delays
Null	User-provided	Bare-metal debug (polling only)

7.41.1.4 Interface Contract

All implementations **MUST** provide:

1. `sleep_ms()`: Block for specified milliseconds (or simulate)
2. `get_ms()`: Return monotonic time in milliseconds
3. `is_elapsed()`: Check if timeout occurred (optional, can be NULL)
4. `ctx`: Opaque pointer to implementation-specific data

Implementations **MUST** guarantee:

- Monotonic time (never decreases)
- Wrap-around handling for 32-bit time (wraps ~49.7 days)
- Thread-safe operations (if used in multi-threaded context)

7.41.1.5 Memory Layout

Offset	Size	Field	Type	Purpose
0	4 bytes	<code>ctx</code>	<code>void*</code>	Implementation context
4	4 bytes	<code>sleep_ms</code>	<code>fn ptr</code>	Sleep function
8	4 bytes	<code>get_ms</code>	<code>fn ptr</code>	Get time function
12	4 bytes	<code>is_elapsed</code>	<code>fn ptr</code>	Timeout check (optional)
Total	16 bytes			Lightweight interface

7.41.1.6 Performance Characteristics

- Memory overhead: 16 bytes per interface instance (4 function pointers)
- Call overhead: ~3 cycles (indirect jump through function pointer)
- ThreadX `sleep_ms`: Yields to scheduler (~50-200 us context switch)
- Mock `sleep_ms`: Instant return if `auto_advance` disabled
- `get_ms`: ~10-50 cycles depending on implementation

7.41.1.7 Hardware Requirements

Requirement	Value	Notes
MCU	RX72N or compatible	32-bit pointers, function pointers
RAM	16 bytes per instance	Typically 1-2 instances total
Time Source	Any	ThreadX uses CMT, mock uses static counter
Alignment	4-byte	Natural alignment for 32-bit pointers

Module Dependencies:

- `rx_err.h` - Error code definitions
- NO dependencies on ThreadX or any RTOS (by design!)
- Implementations depend on: ThreadX (`tx_api.h`) or test framework

NASA Power of 10 Compliance:

- Rule 1: [OK] No goto, setjmp, recursion
- Rule 2: [OK] No loops in interface (validation has bounded loop)
- Rule 3: [OK] Zero dynamic allocation (interface is stack/static)
- Rule 4: [OK] All functions <60 lines (inline validation is 10 lines)
- **Rule 5**: [OK] Preconditions enforced via validation function
- **Rule 6**: [OK] Minimal scope (interface defined in header only)
- **Rule 7**: [OK] Return values checked (`rx_err_t` for validation)
- **Rule 8**: [OK] No macros (pure interface, no magic numbers)
- **Rule 9**: [OK] **Function pointers allowed** (intentional deviation for DIP)
- **Rule 10**: [OK] Compiled with -Wall -Wextra -Werror

SOLID Principles:

- **Single Responsibility**: Only defines time interface, no implementation
- **Open/Closed**: Extensible (add new implementations) without modification
- **Liskov Substitution**: All implementations interchangeable via interface
- **Interface Segregation**: Minimal 3-function interface, each with clear purpose
- **Dependency Inversion**: [PASS] **THIS IS THE DIP PATTERN** - High-level modules depend on abstraction (interface), not concretions (ThreadX)

Example - Production Usage (ThreadX):

```
// In USB communication module initialization
#include "rx_time_interface.h"
#include "rx_time_threadx.h" // Concrete ThreadX implementation

void usb_comm_init(void) {
    // Get ThreadX time interface
    rx_time_interface_t time_iface;
    rx_time_threadx_get_interface(&time_iface);

    // Validate interface (defensive programming)
    rx_err_t err = rx_time_interface_validate(&time_iface);
    if (err != k_rx_ok) {
        rx_log_error("USB", "Invalid time interface");
    }
}
```

```

    return;
}

// Inject into USB configuration
rx_usb_comm_config_t config = {
    .time_iface = &time_iface,
    .timeout_ms = 1000,
};

rx_usb_comm_init(&config);
}

// In USB communication operations (no ThreadX dependency!)
rx_err_t usb_send_with_timeout(const rx_time_interface_t* time,
                               uint8_t* data, uint32_t len) {
    uint32_t start = time->get_ms(time->ctx);

    while (!tx_complete()) {
        if (time->is_elapsed(time->ctx, start, 1000)) {
            return k_rx_err_timeout;
        }
        time->sleep_ms(time->ctx, 1); // Brief yield
    }

    return k_rx_ok;
}

```

Example - Unit Test Usage (Mock):

```

// In unit test for USB timeout handling
#include "rx_time_interface.h"
#include "mock_time.h" // Test mock implementation

void test_usb_timeout(void) {
    // Create mock time interface (no ThreadX needed!)
    rx_time_interface_t mock_time;
    mock_time_get_interface(&mock_time);

    // Disable auto-advance for manual control
    mock_time_set_auto_advance(false);

    // Inject mock into USB config
    rx_usb_comm_config_t config = {
        .time_iface = &mock_time,
        .timeout_ms = 1000,
    };
    rx_usb_comm_init(&config);

    // Start operation that should timeout
    uint8_t data[64] = {};
    rx_err_t err = usb_send_with_timeout(&mock_time, data, sizeof(data));

    // Manually advance time past timeout
    mock_time_advance_ms(1001);

    // Verify timeout occurred
    assert(err == k_rx_err_timeout);
}

```

Example - Multiple Interface Instances:

```

// Different modules can use different time implementations
void system_init(void) {
    // USB uses ThreadX time (real delays)
    rx_time_interface_t usb_time;
    rx_time_threadx_get_interface(&usb_time);
    usb_init(&usb_time);

    // SPI uses null time (polling only, no sleeps)
    rx_time_interface_t spi_time = {
        .ctx = nullptr,
        .sleep_ms = nullptr, // Never call sleep
        .get_ms = real_time_get_ms, // Still need timeouts
        .is_elapsed = nullptr,
    };
    spi_init(&spi_time); // Will use polling instead of sleeps
}

```

See also

[rx_time_threadx.c](#) Production implementation
[mock_time.c](#) Unit test implementation
[rx_error_interface.h](#) Similar DIP pattern for errors
[rx_pin_interface.h](#) Similar DIP pattern for GPIO

Author

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Date

2026-01-27

Version

1.0.0

Copyright

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Since

Version 1.0.0

Definition in file [rx_time_interface.h](#).

7.41.2 Data Structure Documentation

7.41.2.1 struct rx_time_interface

Time operations interface structure (Dependency Inversion Pattern).

Defines the complete contract for time operations that all implementations must fulfill. This struct contains function pointers for time operations plus an opaque context pointer for implementation-specific state.

This is a "fat pointer" or "vtable" pattern common in object-oriented design, adapted for C. It enables runtime polymorphism without inheritance - different implementations (ThreadX, mock, bare-metal) can be swapped at runtime by filling in different function pointers.

7.41.2.2 Memory Layout (32-bit pointers)

Offset	Size	Field	Type	Alignment	Purpose
0	4	ctx	void*	4	Implementation context
4	4	sleep_ms	fn ptr	4	Sleep function
8	4	get_ms	fn ptr	4	Get time function
12	4	is_elapsed	fn ptr	4	Timeout check (optional)
Total	16				No padding

7.41.2.3 Initialization Patterns

Static Initialization (compile-time):

```
static rx_time_interface_t s_time = {
    .ctx = &s_threadx_state,
    .sleep_ms = threadx_sleep_ms_impl,
    .get_ms = threadx_get_ms_impl,
    .is_elapsed = threadx_is_elapsed_impl,
};
```

Dynamic Initialization (runtime):

```
rx_time_interface_t time;
rx_time_threadx_get_interface(&time); // Fills in struct
```

Null Implementation (minimal/bare-metal):

```
static rx_time_interface_t s_null_time = {
    .ctx = nullptr,
    .sleep_ms = nullptr, // Polling only, no sleeps
    .get_ms = hardware_timer_get_ms, // Still need timeouts
    .is_elapsed = nullptr, // Manual timeout checks
};
```

7.41.2.4 Field Constraints and Relationships

Field	Required?	Notes
ctx	Optional	NULL if implementation is stateless
sleep_ms	Required	Validation fails if nullptr
get_ms	Required	Validation fails if nullptr
is_elapsed	Optional	Can be NULL (manual timeout checks)

Constraint: If sleep_ms is nullptr, interface is invalid (cannot sleep). Constraint: If get_ms is nullptr, interface is invalid (cannot measure time). Invariant: Once initialized, function pointers should not change (read-only).

Usage Example - Initialization:

```
// Production: Use ThreadX implementation
void production_init(void) {
    rx_time_interface_t time_iface;
    rx_time_threadx_get_interface(&time_iface);

    // Validate before use (defensive programming)
    rx_err_t err = rx_time_interface_validate(&time_iface);
    if (err != k_rx_ok) {
        rx_log_fatal("INIT", "Invalid time interface");
        return;
    }

    // Inject into USB module
    usb_comm_init(&time_iface);
}
```

Usage Example - Passing by Value vs Pointer:

```
// Option 1: Pass by pointer (common, lightweight)
void device_init(const rx_time_interface_t* time) {
    // Use: time->get_ms(time->ctx)
}

// Option 2: Pass by value (copy 16 bytes, less common)
void device_init(rx_time_interface_t time) {
    // Use: time.get_ms(time.ctx)
}

// Option 3: Embed in config struct (recommended)
typedef struct {
    rx_time_interface_t time_iface; // 16 bytes embedded
    uint32_t timeout_ms;
    uint32_t retry_count;
} device_config_t;
```

Usage Example - Multiple Implementations:

```
// System with mixed implementations
void mixed_system_init(void) {
    // USB uses real ThreadX time (needs actual delays)
    rx_time_interface_t usb_time;
    rx_time_threadx_get_interface(&usb_time);
    usb_init(&usb_time);

    // SPI uses null time (polling only, instant returns)
    rx_time_interface_t spi_time = {};
    spi_time.get_ms = hardware_timer_get_ms; // Still need timeouts
    spi_time.sleep_ms = null_sleep; // No-op sleep
    spi_init(&spi_time);
}
```

See also

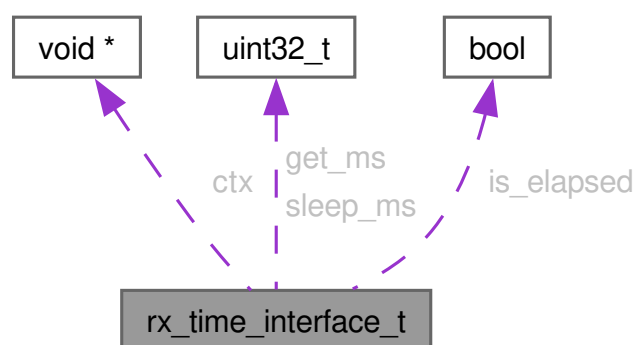
[rx_time_interface_validate\(\)](#) Validate interface before use
[rx_time_threadx_get_interface\(\)](#) Fill with ThreadX implementation
[mock_time_get_interface\(\)](#) Fill with mock implementation for tests

Since

Version 1.0.0

Definition at line 804 of file [rx_time_interface.h](#).

Collaboration diagram for rx_time_interface_t:



Data Fields

<code>void *</code>	<code>ctx</code>	Implementation-specific context pointer (opaque). Pointer to concrete implementation's private state. Treated as opaque by interface users - only the implementation knows what this points to. ThreadX: Unused, typically NULL (ThreadX state is global) Mock: Points to <code>mock_time_state_t</code> with current <code>ms</code> counter Bare-metal: Could point to hardware timer peripheral base address Lifetime: Must remain valid for duration of interface usage. Typically points to static or global data, NOT stack variables. Null allowed: Yes, if implementation is stateless (e.g., ThreadX)
rx_time_get_ms_fn	<code>get_ms</code>	Get current time function pointer (REQUIRED). Returns monotonic time in milliseconds since boot. This function is REQUIRED - validation fails if nullptr. Implementation: ThreadX: <code>tx_time_get()</code> conversion, Mock: counter read Null allowed: NO (validation fails) Resolution: ThreadX tick rate (10ms typical), Mock 1ms See also rx_time_get_ms_fn Full documentation of time semantics
rx_time_is_elapsed_fn	<code>is_elapsed</code>	Timeout elapsed check function pointer (OPTIONAL). Convenience function combining get_ms() and timeout arithmetic. This function is OPTIONAL - can be NULL for minimal implementations. Implementation: Calls get_ms() and computes (current - start) >= timeout Null allowed: YES (caller must check before using) Alternative: Manual timeout check using get_ms() if nullptr See also rx_time_is_elapsed_fn Full documentation of timeout semantics
rx_time_sleep_ms_fn	<code>sleep_ms</code>	Sleep/delay function pointer (REQUIRED). Blocks or simulates blocking for specified milliseconds. This function is REQUIRED - validation fails if nullptr. Implementation: ThreadX: <code>tx_thread_sleep()</code>, Mock: simulate time advance Null allowed: NO (validation fails) Thread safety: Implementation-dependent (ThreadX is thread-safe) See also rx_time_sleep_ms_fn Full documentation of sleep semantics

7.41.3 Typedef Documentation

7.41.3.1 rx_time_get_ms_fn

```
typedef uint32_t(* rx_time_get_ms_fn) (void *ctx)
```

Function pointer: Get current monotonic time in milliseconds.

Returns a continuously increasing time value in milliseconds, suitable for measuring elapsed time and implementing timeouts. This is NOT wall-clock time (not UTC, not human-readable) - it's a monotonic counter that starts at system boot and increases until overflow (~49.7 days).

Monotonic Guarantee: Time NEVER decreases, even if system clock is adjusted. This makes it reliable for timeouts and performance measurements.

Wrap-Around Behavior: 32-bit counter wraps to 0 after 2^{32} milliseconds (4,294,967,296 ms = 49.71 days). Properly written timeout code handles this correctly using unsigned arithmetic.

Implementation Details:

ThreadX Implementation:

- Returns `tx_time_get() * (1000 / TX_TIMER_TICKS_PER_SECOND)`
- Resolution matches ThreadX tick rate (typically 10ms on RX72N)
- Examples:
 - 100 Hz tick: Resolution = 10ms (0, 10, 20, 30, ...)
 - 1000 Hz tick: Resolution = 1ms (0, 1, 2, 3, ...)
- Overhead: ~50 cycles (~200 ns @ 240 MHz)

Mock Implementation:

- Returns static counter maintained by mock
- Manual control: Counter only changes via `mock_time_advance_ms()`
- Auto-advance: Counter increases during `sleep_ms()` calls
- Resolution: 1ms (arbitrary, no hardware limits in simulation)
- Overhead: ~10 cycles (~40 ns @ 240 MHz, no RTOS)

Bare-Metal Implementation (user-provided):

- Could use CMT hardware timer counter
- Could use SysTick (if using ARM Cortex-M)
- Resolution depends on hardware timer configuration

Parameters

in	ctx	Implementation-specific context pointer
		• ThreadX: Unused (can be NULL) • Mock: Pointer to <code>mock_time_state_t</code> containing <code>current_ms</code> counter • Hardware: Pointer to timer peripheral base address

Returns

uint32_t Current time in milliseconds since system boot

Return values

0	System just booted (or wrapped around after 49.7 days)
1000	System has been running for 1 second
3600000	System has been running for 1 hour
UINT32_MAX	System has been running for 49.7 days (about to wrap)

Precondition

No preconditions - safe to call anytime

ctx must be valid for the specific implementation (or nullptr if allowed)

Postcondition

Returned value is $\geq$ previous call (monotonic, unless wrapped)

No side effects (read-only operation)

Invariant

$\text{get_ms}(t2) \geq \text{get_ms}(t1)$ for $t2 > t1$ (unless wrap)

Resolution matches implementation tick rate (e.g., 10ms for ThreadX)

Note

Thread Safety: ThreadX tx_time_get() is thread-safe. Mock reads volatile counter (safe for single-threaded tests).

Execution Time: ThreadX $\sim 200\text{ns}$, Mock $\sim 40\text{ns}$ @ 240 MHz (very fast)

Wrap-Around: Use unsigned arithmetic for timeouts to handle wrap correctly: $(\text{uint32_t})(\text{current} - \text{start}) \geq \text{timeout}$ works across wrap boundary.

Resolution: Limited by ThreadX tick rate. For sub-millisecond timing, use hardware timers (CMT, GPT) directly.

Attention

NOT wall-clock time! Do not use for timestamps or logging absolute time.

Wraps every 49.7 days. Long-running systems must handle wrap correctly.

Example - Timeout Implementation:

```
// Wait for device ready with 1000ms timeout
rx_err_t wait_device_ready(const rx_time_interface_t* time, uint32_t timeout_ms) {
    uint32_t start_ms = time->get_ms(time->ctx);

    while (!device_is_ready()) {
        // Check timeout using wrap-safe arithmetic
        uint32_t elapsed = time->get_ms(time->ctx) - start_ms;
        if (elapsed >= timeout_ms) {
            return k_rx_err_timeout; // Timeout occurred
        }

        time->sleep_ms(time->ctx, 1); // Brief yield
    }

    return k_rx_ok; // Device ready
}
```

Example - Performance Measurement:

```
// Measure execution time of algorithm
void profile_algorithm(const rx_time_interface_t* time) {
    uint32_t start = time->get_ms(time->ctx);

    run_expensive_algorithm();

    uint32_t end = time->get_ms(time->ctx);
    uint32_t duration_ms = end - start; // Wrap-safe

    rx_log_info("PERF", "Algorithm took %lu ms", duration_ms);
}
```

Example - Wrap-Around Handling:

```
// Demonstrate wrap-around safety (uint32_t arithmetic)
void test_wrap_around(void) {
    uint32_t start = UINT32_MAX - 100; // Near wrap point
    mock_time_set_ms(start);

    // Advance past wrap boundary
    mock_time_advance_ms(200);
    uint32_t end = mock_time_get_ms(); // end = 99 (wrapped)

    // Unsigned arithmetic handles wrap correctly
    uint32_t elapsed = end - start; // 99 - (UINT32_MAX-100) = 200 [OK]
    assert(elapsed == 200); // Correct!
}
```

Example - Uptime Display:

```
// Convert milliseconds to human-readable uptime
void print_uptime(const rx_time_interface_t* time) {
    uint32_t uptime_ms = time->get_ms(time->ctx);

    uint32_t seconds = uptime_ms / 1000;
    uint32_t minutes = seconds / 60;
    uint32_t hours = minutes / 60;
    uint32_t days = hours / 24;

    printf("Uptime: %lu days, %lu hours, %lu min\n",
           days, hours % 24, minutes % 60);
}
```

See also

[rx_time_sleep_ms_fn](#) Sleep for specified duration
[rx_time_is_elapsed_fn](#) Convenience function for timeout checks
[tx_time_get\(\)](#) ThreadX implementation uses this

Since

Version 1.0.0

Definition at line 498 of file [rx_time_interface.h](#).

7.41.3.2 rx_time_is_elapsed_fn

```
typedef bool(* rx_time_is_elapsed_fn) (void *ctx, uint32_t start_ms, uint32_t timeout_ms)
```

Function pointer: Check if timeout has elapsed (optional convenience).

Convenience function that combines `get_ms()` and timeout arithmetic into a single call. Equivalent to: `(get_ms(ctx) - start_ms) >= timeout_ms` with correct handling of 32-bit wrap-around.

This function is OPTIONAL - implementations may set this pointer to NULL. Callers should either:

1. Check for nullptr before calling, OR
2. Implement timeout check manually using get_ms()

Why Optional?: Some implementations (bare-metal, minimal mocks) may want to provide only the core get_ms() function and let callers handle timeout logic themselves. This keeps the interface minimal.

Wrap-Around Safety: Uses unsigned 32-bit arithmetic which correctly handles wraparound at UINT32_MAX:

- Example: start=4294967290, timeout=100, current=50
- Elapsed = (50 - 4294967290) = 110 via unsigned underflow
- 110 >= 100 -> true (timeout occurred) [OK]

Algorithm:

```
1. current_ms = get_ms(ctx)
2. elapsed = current_ms - start_ms // Unsigned subtraction (wrap-safe)
3. return (elapsed >= timeout_ms)
```

Parameters

in	ctx	Implementation-specific context pointer • ThreadX: Unused (can be NULL) • Mock: Pointer to mock_time_state_t • Bare-metal: Pointer to timer peripheral
in	start_ms	Start time captured from get_ms() • Valid range: [0, UINT32_MAX] • Units: milliseconds since boot • Source: MUST be from prior get_ms() call • Typical use: Captured at start of operation
in	timeout_ms	Timeout duration in milliseconds • Valid range: [0, UINT32_MAX] • Units: milliseconds (MKS: 0.001 seconds) • Value 0: Returns true immediately (timeout already occurred) • Large values: timeout > 49 days may never trigger due to wrap

Returns

bool Timeout status

Return values

true	Timeout has elapsed: (current - start) >= timeout
false	Timeout has NOT elapsed: still within timeout window

Precondition

start_ms must be from prior get_ms() call
 timeout_ms should be reasonable (< 49 days for guaranteed correctness)
 ctx must be valid for the specific implementation (or nullptr if allowed)
 Function pointer may be NULL - caller must check before calling!

Postcondition

No side effects (read-only operation)
 Result is deterministic given same inputs and current time

Note

Thread Safety: Thread-safe if get_ms() is thread-safe
 Execution Time: ~250ns @ 240 MHz (one get_ms() call + arithmetic)
 Optional Function: May be NULL in minimal implementations. Always check!
 Wrap-Around: Correctly handles wrap using unsigned arithmetic
 Resolution: Limited by get_ms() resolution (typically 10ms for ThreadX)

Warning

start_ms MUST be from get_ms(), not arbitrary value or hardware timer

Attention

Function pointer may be NULL - check before dereferencing!
 timeout_ms > ~25 days may behave unexpectedly near wrap boundary

Example - Using is_elapsed (if available):

```
// Wait for I2C transaction with timeout
rx_err_t i2c_wait_complete(const rx_time_interface_t* time, uint32_t timeout_ms) {
    uint32_t start = time->get_ms(time->ctx);

    while (!i2c_transaction_done()) {
        // Use is_elapsed if available
        if (time->is_elapsed != nullptr) {
            if (time->is_elapsed(time->ctx, start, timeout_ms)) {
                return k_rx_err_timeout;
            }
        } else {
            // Fallback: manual timeout check
            uint32_t elapsed = time->get_ms(time->ctx) - start;
            if (elapsed >= timeout_ms) {
                return k_rx_err_timeout;
            }
        }

        time->sleep_ms(time->ctx, 1);
    }

    return k_rx_ok;
}
```


Example - Simplified with NULL Check:

```
// Cleaner timeout check with defensive NULL handling
bool check_timeout(const rx_time_interface_t* time,
                  uint32_t start, uint32_t timeout_ms) {
    if (time->is_elapsed != nullptr) {
        // Use convenience function if available
        return time->is_elapsed(time->ctx, start, timeout_ms);
    } else {
        // Manual calculation (always works)
        uint32_t elapsed = time->get_ms(time->ctx) - start;
        return (elapsed >= timeout_ms);
    }
}
```

Example - Multiple Timeouts:

```
// Check multiple independent timeouts
void multi_timeout_loop(const rx_time_interface_t* time) {
    uint32_t sensor_start = time->get_ms(time->ctx);
    uint32_t motor_start = time->get_ms(time->ctx);
    uint32_t comm_start = time->get_ms(time->ctx);

    while (running) {
        // Check sensor timeout (500ms)
        if (time->is_elapsed(time->ctx, sensor_start, 500)) {
            read_sensors();
            sensor_start = time->get_ms(time->ctx); // Restart timer
        }

        // Check motor timeout (100ms)
        if (time->is_elapsed(time->ctx, motor_start, 100)) {
            update_motors();
            motor_start = time->get_ms(time->ctx);
        }

        // Check comm timeout (1000ms)
        if (time->is_elapsed(time->ctx, comm_start, 1000)) {
            send_telemetry();
            comm_start = time->get_ms(time->ctx);
        }

        time->sleep_ms(time->ctx, 10); // 100 Hz loop
    }
}
```

Example - Wrap-Around Correctness:

```
// Verify is_elapsed handles wrap correctly
void test_wrap_around_elapsed(void) {
    rx_time_interface_t mock_time;
    mock_time_get_interface(&mock_time);

    // Set time near wrap boundary
    uint32_t start = UINT32_MAX - 50; // 50ms before wrap
    mock_time_set_ms(start);

    // Advance past wrap (50 + 100 = 150ms elapsed)
    mock_time_advance_ms(150);

    // Check timeout (100ms)
    bool elapsed = mock_time.is_elapsed(mock_time.ctx, start, 100);
    assert(elapsed == true); // Correctly detected timeout across wrap! [OK]
}
```

See also

[rx_time_get_ms_fn](#) Used internally by this function

[rx_time_sleep_ms_fn](#) Often used together in timeout loops

Since

Version 1.0.0

Definition at line 670 of file [rx_time_interface.h](#).

7.41.3.3 rx_time_sleep_ms_fn

```
typedef void(* rx_time_sleep_ms_fn) (void *ctx, uint32_t ms)
```

Function pointer: Sleep/delay for specified milliseconds.

Blocks (or simulates blocking) for the specified number of milliseconds. Behavior depends on concrete implementation:

ThreadX Implementation:

- Calls tx_thread_sleep(ticks) where ticks = ms * (TX_TIMER_TICKS_PER_SECOND / 1000)
- Yields CPU to other threads (cooperative multitasking)
- Actual sleep time may be slightly longer due to tick granularity
- Example: 10ms sleep with 10ms tick = exactly 1 tick
- Example: 15ms sleep with 10ms tick = 2 ticks = 20ms actual

Mock Implementation (manual control):

- Returns immediately without advancing simulated time
- Test code manually advances time via mock_time_advance_ms()
- Use for fine-grained control of timing in unit tests

Mock Implementation (auto-advance):

- Returns immediately AND advances simulated time by ms
- Useful for testing timeout logic without real delays
- 1000ms timeout completes in ~1us real time

Null Implementation (bare-metal):

- Function pointer set to nullptr
- Caller must check for nullptr before calling
- Use for polling-only code that never sleeps

Parameters

in	ctx	Implementation-specific context pointer
		• ThreadX: Pointer to thread control block (unused, can be NULL) • Mock: Pointer to mock_time_state_t for simulated time tracking • Null: N/A (function pointer is nullptr)

in	ms	Number of milliseconds to sleep/delay - Valid range: [0, UINT32_MAX] - Units: milliseconds (MKS: 0.001 seconds) - Value 0: Yields CPU without delay (ThreadX) or no-op (mock) - Large values: ms > 49 days wraps at UINT32_MAX (impractical)
----	----	--

Returns

void (no return value, always "succeeds")

Precondition

No preconditions - safe to call with any ms value

ctx must be valid for the specific implementation (or nullptr if allowed)

Postcondition

Control returned after at least ms milliseconds (ThreadX)

Simulated time advanced by ms (mock with auto_advance)

Other threads may have executed during sleep (ThreadX)

Note

Thread Safety: ThreadX implementation is thread-safe. Mock depends on test harness (single-threaded tests are safe).

Accuracy: ThreadX sleep accuracy limited by tick rate (typically +/-10ms). Use hardware timers for precise delays <1ms.

Execution Time: ThreadX context switch ~50-200us. Mock returns in <1us.

Blocking: ThreadX yields CPU (other threads run). Mock returns immediately.

Warning

ThreadX tick granularity causes rounding: 15ms sleep with 10ms tick = 20ms actual.

Attention

Never call from interrupt context (ThreadX implementation will fault).

Excessive sleeps (>1000ms) may indicate architectural problems.

Example - Basic Sleep:

```
// Sleep for 10ms between sensor reads
void sensor_poll_loop(const rx_time_interface_t* time) {
    while (1) {
        read_sensor_data();
        process_data();

        // Brief sleep to reduce CPU usage
        time->sleep_ms(time->ctx, 10); // 10ms = 100 Hz poll rate
    }
}
```

Example - Defensive Null Check:

```
// Gracefully handle null sleep_ms (polling-only mode)
void wait_for_ready(const rx_time_interface_t* time) {
    while (!device_is_ready()) {
        // Sleep if available, otherwise busy-wait
        if (time->sleep_ms != nullptr) {
            time->sleep_ms(time->ctx, 1); // 1ms yield
        }
        // Continue polling
    }
}
```

See also

[rx_time_get_ms_fn](#) Get current time for timeout calculations

[rx_time_is_elapsed_fn](#) Check if timeout occurred

[tx_thread_sleep\(\)](#) ThreadX implementation calls this

Since

Version 1.0.0

Definition at line 352 of file [rx_time_interface.h](#).

7.41.4 Function Documentation

7.41.4.1 rx_time_interface_validate()

```
rx_err_t rx_time_interface_validate (
    const rx_time_interface_t * iface)    [inline], [static]
```

Definition at line 1028 of file [rx_time_interface.h](#).

```
01029 {
01030     if (iface == nullptr) {
01031         return k_rx_err_null_ptr;
01032     }
01033     /* Check required function pointers */
01034     if (iface->sleep_ms == nullptr || iface->get_ms == nullptr) {
01035         return k_rx_err_invalid_state;
01036     }
01037 }
01038
01039 return k_rx_ok;
01040 }
```

References [k_rx_err_invalid_state](#), [k_rx_err_null_ptr](#), and [k_rx_ok](#).

7.41.4.2 rx_time_threadx_get_interface()

```
rx_err_t rx_time_threadx_get_interface (
    rx_time_interface_t * iface)    [nodiscard]
```

Validate that a time interface is properly initialized.

Performs defensive validation of a time interface to ensure it meets the minimum requirements for safe usage. Checks that the interface pointer is non-NULL and that required function pointers ([sleep_ms](#), [get_ms](#)) are populated.

This function implements "fail-fast" validation - it's better to detect a misconfigured interface immediately at initialization than to crash later when dereferencing NULL function pointers.

Validation Rules:

1. Interface pointer must be non-NULL
2. sleep_ms function pointer must be non-NULL (REQUIRED)
3. get_ms function pointer must be non-NULL (REQUIRED)
4. ctx pointer may be NULL (implementation-dependent)
5. is_elapsed pointer may be NULL (OPTIONAL)

Algorithm steps:

1. Check if iface pointer is nullptr -> return k_rx_err_null_ptr
2. Check if sleep_ms is nullptr -> return k_rx_err_invalid_state
3. Check if get_ms is nullptr -> return k_rx_err_invalid_state
4. All checks passed -> return k_rx_ok

Parameters

in	iface	Pointer to time interface to validate • May be NULL (will return k_rx_err_null_ptr) • Typically points to stack or static rx_time_interface_t
----	-------	--

Returns

[rx_err_t](#) Validation result

Return values

k_rx_ok	Interface is valid and safe to use
k_rx_err_null_ptr	iface pointer is nullptr (cannot validate)
k_rx_err_invalid_state	Required function pointer(s) are NULL (sleep_ms or get_ms)

Precondition

No preconditions - safe to call with any input (even NULL)

Postcondition

Interface not modified (read-only validation)

Return value indicates whether interface is usable

Note

Thread Safety: Thread-safe (read-only operation, no shared state)

Execution Time: ~50 ns @ 240 MHz (3 null checks)

Use Case: Call at initialization before using interface

Optional Fields: Does NOT check ctx or is_elapsed (both may be NULL)

Warning

Does NOT validate that function pointers point to valid functions! Only checks for nullptr. Invalid pointers will cause crash on call.

Attention

Validation does NOT guarantee thread-safety of implementations. ThreadX is thread-safe, but custom implementations may not be.

Example - Defensive Initialization:

```
// Validate interface before passing to modules
void system_init(void) {
    rx_time_interface_t time_iface;
    rx_time_threadx_get_interface(&time_iface);

    // Validate before use (defensive programming)
    rx_err_t err = rx_time_interface_validate(&time_iface);
    if (err == k_rx_err_null_ptr) {
        rx_log_fatal("INIT", "Time interface is nullptr");
        return;
    } else if (err == k_rx_err_invalid_state) {
        rx_log_fatal("INIT", "Time interface missing required functions");
        return;
    }

    // Safe to use
    usb_init(&time_iface);
    spi_init(&time_iface);
}
```

Example - Validation in Module Init:

```
// Module enforces interface validity at init
rx_err_t usb_comm_init(const rx_time_interface_t* time) {
    // Validate injected dependency
    rx_err_t err = rx_time_interface_validate(time);
    if (err != k_rx_ok) {
        rx_log_error("USB", "Invalid time interface: %d", err);
        return err; // Propagate error to caller
    }

    // Store validated interface
    s_usb_time = time;

    return k_rx_ok;
}
```

Example - Unit Test Verification:

```
// Verify validation catches misconfigured interfaces
void test_validate_catches_null_interface(void) {
    rx_err_t err = rx_time_interface_validate(nullptr);
    assert(err == k_rx_err_null_ptr);
}

void test_validate_catches_missing_sleep(void) {
    rx_time_interface_t time = {};
    time.get_ms = mock_get_ms; // Only get_ms populated
    // Missing sleep_ms!

    rx_err_t err = rx_time_interface_validate(&time);
    assert(err == k_rx_err_invalid_state);
}

void test_validate_allows_null_ctx(void) {
    rx_time_interface_t time = {};
    time.ctx = nullptr; // ctx can be nullptr
    time.sleep_ms = mock_sleep_ms;
    time.get_ms = mock_get_ms;

    rx_err_t err = rx_time_interface_validate(&time);
    assert(err == k_rx_ok); // Valid even with NULL ctx
}
```

Example - Production Assertion:

```
// Hard assertion in safety-critical code
void critical_init(const rx_time_interface_t* time) {
    rx_err_t err = rx_time_interface_validate(time);

    // In safety-critical systems, invalid interface is fatal
    RX_CHECK(err == k_rx_ok, "Time interface validation failed");

    // Continue with guaranteed valid interface...
}
```

See also

[rx_time_interface](#) Structure being validated

[rx_time_threadx_get_interface\(\)](#) Produces valid interface

[mock_time_get_interface\(\)](#) Produces valid test interface

Since

Version 1.0.0

Version

1.0.0

[Test](#) Tested in `test_rx_time_interface.c::test_validate_null_interface()`

Tested in `test_rx_time_interface.c::test_validate_missing_functions()`

Tested in `test_rx_time_interface.c::test_validate_valid_interface()`

NASA Power of 10 Compliance:

- Rule 1: [OK] No goto, setjmp, recursion
- Rule 2: [OK] No loops
- Rule 5: [OK] 1 precondition, 2 postconditions
- Rule 4: [OK] Function is 10 lines (well under 60)

Validate that a time interface is properly initialized.

Populates an `rx_time_interface_t` structure with function pointers to the ThreadX implementation. This enables Dependency Inversion: callers depend on the abstract interface, not the concrete ThreadX API.

Algorithm Steps:

1. Validate iface is not nullptr
2. Set ctx to NULL (ThreadX uses global state)
3. Set function pointers to implementation functions
4. Return success

Interface Population:

Field	Value	Description
ctx	NULL	ThreadX uses global state
sleep_ms	impl_sleep_ms	tx_thread_sleep() wrapper
get_ms	impl_get_ms	tx_time_get() x 10 wrapper
is_elapsed	impl_is_elapsed	Wraparound-safe timeout check

Precondition

iface != nullptr

Postcondition

iface->ctx == nullptr

iface->sleep_ms, get_ms, is_elapsed are valid function pointers

Interface is ready for use

Invariant

Interface functions remain valid for program lifetime

Note

Thread Safety: Safe - only writes to caller-owned structure

Re-entrancy: Safe - no shared state accessed

Example - Basic Usage:

```

rx_time_interface_t time_if;
rx_err_t err = rx_time_threadx_get_interface(&time_if);
if (err != k_rx_ok) {
    rx_log_error("TIME", "Failed to get interface");
    return err;
}

// Use interface for timing operations
uint32_t start = time_if.get_ms(time_if.ctx);
do_some_work();
uint32_t duration = time_if.get_ms(time_if.ctx) - start;

```

Example - Timeout Loop:

```

rx_time_interface_t time_if;
rx_time_threadx_get_interface(&time_if);

// Wait for sensor with 500ms timeout
uint32_t start = time_if.get_ms(time_if.ctx);
while (!sensor_data_ready()) {
    if (time_if.is_elapsed(time_if.ctx, start, 500)) {
        return k_rx_err_timeout;
    }
    time_if.sleep_ms(time_if.ctx, 10);
}

```


Example - Dependency Injection:

```
// In production code
rx_time_interface_t time_if;
rx_time_threadx_get_interface(&time_if);
motor_init(&motor, &config, &time_if);

// In unit tests (mock_time.c)
rx_time_interface_t time_if;
mock_time_get_interface(&time_if);
motor_init(&motor, &config, &time_if); // Same interface, mock behavior
```

See also

[rx_time_interface_t](#) Interface structure definition

[mock_time_get_interface\(\)](#) Unit test mock version

[impl_sleep_ms\(\)](#) Sleep implementation details

[impl_get_ms\(\)](#) Time retrieval details

[impl_is_elapsed\(\)](#) Timeout check details

NASA Power of 10 Compliance:

- Rule 5: [OK] 1 precondition (NULL check), 3 postconditions
- Rule 9: [OK] Function pointers for DIP (intentional deviation)

[Test](#) Tested in test_rx_time.c with nullptr and success cases

Since

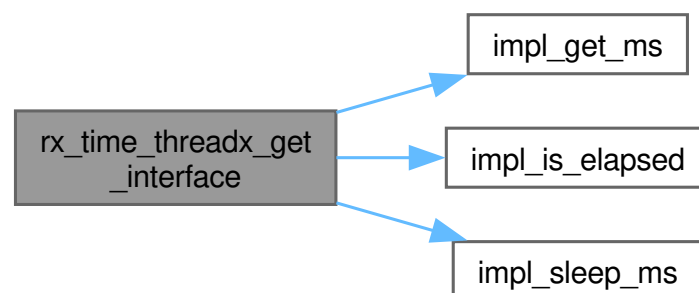
Version 1.0.0

Definition at line 470 of file [rx_time_threadx.c](#).

```
00471 {
00472   RX_CHECK_NULL_PTR(iface, s_tag, "Interface pointer is nullptr");
00473
00474   iface->ctx      = NULL; /* No context needed for ThreadX */
00475   iface->sleep_ms  = impl_sleep_ms;
00476   iface->get_ms    = impl_get_ms;
00477   iface->is_elapsed = impl_is_elapsed;
00478
00479   return k_rx_ok;
00480 }
```

References [impl_get_ms\(\)](#), [impl_is_elapsed\(\)](#), [impl_sleep_ms\(\)](#), [k_rx_ok](#), [RX_CHECK_NULL_PTR](#), and [s_tag](#).

Here is the call graph for this function:



7.42 rx_time_interface.h

[Go to the documentation of this file.](#)

```

00001 /**
00002  * @file rx_time_interface.h
00003  * @brief Time Interface - Dependency Inversion for Timing Operations
00004  *
00005  * @details
00006  * Defines the abstract interface for time operations in RX72N firmware,
00007  * implementing the Dependency Inversion Principle (DIP) from SOLID design.
00008  * This header contains ONLY the interface definition with NO implementation
00009  * details, enabling complete decoupling between high-level modules and
00010  * low-level timing mechanisms.
00011  *
00012  * ## Architecture Pattern: Dependency Inversion Principle (DIP)
00013  *
00014  * **Problem**: Communication modules (USB, SPI, UART) need timing operations
00015  * (sleep, timeouts) but should not directly depend on ThreadX RTOS. Direct
00016  * dependencies create:
00017  * - Tight coupling -> hard to change RTOS
00018  * - Untestable code -> cannot unit test without ThreadX
00019  * - Platform lock-in -> cannot port to bare-metal or different RTOS
00020  *
00021  * **Solution**: Define abstract time interface that both high-level and
00022  * low-level modules depend on:
00023  *
00024  * @dot
00025  * digraph dependency_inversion {
00026  *   rankdir=TB;
00027  *   node [shape=box, style=rounded];
00028  *
00029  *   interface [label="rx_time_interface_t\n(Abstract Interface)", shape=oval, fillcolor=lightblue, style=filled];
00030  *   high [label="High-Level Modules\n(USB, SPI, UART)", fillcolor=lightgreen, style=filled];
00031  *   impl_tx [label="ThreadX Implementation\nrx_time_threadx.c", fillcolor=lightyellow, style=filled];
00032  *   impl_mock [label="Mock Implementation\nmock_time.c", fillcolor=lightyellow, style=filled];
00033  *
00034  *   high -> interface [label="depends on\n(uses)", dir=back];
00035  *   impl_tx -> interface [label="implements", style=dashed];
00036  *   impl_mock -> interface [label="implements", style=dashed];
00037  * }
00038  * @enddot
00039  *
00040  * **Key Insight**: High-level modules depend on the ABSTRACTION (interface),
00041  * not the IMPLEMENTATION (ThreadX). Implementations are injected at runtime
00042  * through function pointers.
00043  *
00044  * ## Benefits of This Pattern
00045  *
00046  * | Benefit | Description | Example |
00047  * |-----|-----|-----|
00048  * | **Testability** | Unit tests inject mock timing without ThreadX | Test USB without RTOS |
00049  * | **Flexibility** | Swap RTOS without changing USB/SPI code | ThreadX -> FreeRTOS |
00050  * | **Decoupling** | Communication layers have zero ThreadX includes | Clean dependencies |
00051  * | **Portability** | Same code runs on bare-metal with null impl | Debug builds |
00052  * | **Simulation** | Mock advances time instantly for fast tests | 1000ms timeout in 1us |
00053  *
00054  * ## Concrete Implementations
00055  *
00056  * | Implementation | File | Use Case |
00057  * |-----|-----|-----|
00058  * | ThreadX | `rx_time_threadx.c` | Production firmware on RX72N |
00059  * | Mock (manual) | `mock_time.c` | Unit tests with controlled time |
00060  * | Mock (auto) | `mock_time.c` (auto_advance) | Unit tests with simulated delays |
00061  * | Null | User-provided | Bare-metal debug (polling only) |
00062  *
00063  * ## Interface Contract
00064  *
00065  * All implementations MUST provide:
00066  * 1. **sleep_ms()**: Block for specified milliseconds (or simulate)
00067  * 2. **get_ms()**: Return monotonic time in milliseconds
00068  * 3. **is_elapsed()**: Check if timeout occurred (optional, can be NULL)
00069  * 4. **ctx**: Opaque pointer to implementation-specific data
00070  *
00071  * Implementations MUST guarantee:
00072  * - Monotonic time (never decreases)
00073  * - Wrap-around handling for 32-bit time (wraps ~49.7 days)
00074  * - Thread-safe operations (if used in multi-threaded context)
00075  *
00076  * ## Memory Layout
00077  *
00078  * | Offset | Size | Field | Type | Purpose |
00079  * |-----|-----|-----|-----|-----|
00080  * | 0 | 4 bytes | ctx | void* | Implementation context |
00081  * | 4 | 4 bytes | sleep_ms | fn ptr | Sleep function |
00082  * | 8 | 4 bytes | get_ms | fn ptr | Get time function |

```

```

00083 * | 12 | 4 bytes | is_elapsed | fn ptr | Timeout check (optional) |
00084 * | **Total** | **16 bytes** | | | Lightweight interface |
00085 *
00086 * ## Performance Characteristics
00087 *
00088 * - **Memory overhead**: 16 bytes per interface instance (4 function pointers)
00089 * - **Call overhead**: ~3 cycles (indirect jump through function pointer)
00090 * - **ThreadX sleep_ms**: Yields to scheduler (~50-200 us context switch)
00091 * - **Mock sleep_ms**: Instant return if auto_advance disabled
00092 * - **get_ms**: ~10-50 cycles depending on implementation
00093 *
00094 * ## Hardware Requirements
00095 *
00096 * | Requirement | Value | Notes |
00097 * |-----|-----|-----|
00098 * | MCU | RX72N or compatible | 32-bit pointers, function pointers |
00099 * | RAM | 16 bytes per instance | Typically 1-2 instances total |
00100 * | Time Source | Any | ThreadX uses CMT, mock uses static counter |
00101 * | Alignment | 4-byte | Natural alignment for 32-bit pointers |
00102 *
00103 * @par Module Dependencies:
00104 * - [rx_err.h](rx_err.h) - Error code definitions
00105 * - **NO dependencies on ThreadX or any RTOS** (by design!)
00106 * - Implementations depend on: ThreadX ('tx_api.h') or test framework
00107 *
00108 * @par NASA Power of 10 Compliance:
00109 * - **Rule 1**: [OK] No goto, setjmp, recursion
00110 * - **Rule 2**: [OK] No loops in interface (validation has bounded loop)
00111 * - **Rule 3**: [OK] Zero dynamic allocation (interface is stack/static)
00112 * - **Rule 4**: [OK] All functions <60 lines (inline validation is 10 lines)
00113 * - **Rule 5**: [OK] Preconditions enforced via validation function
00114 * - **Rule 6**: [OK] Minimal scope (interface defined in header only)
00115 * - **Rule 7**: [OK] Return values checked (rx_err_t for validation)
00116 * - **Rule 8**: [OK] No macros (pure interface, no magic numbers)
00117 * - **Rule 9**: [OK] **Function pointers allowed** (intentional deviation for DIP)
00118 * - **Rule 10**: [OK] Compiled with -Wall -Wextra -Werror
00119 *
00120 * @par SOLID Principles:
00121 * - **Single Responsibility**: Only defines time interface, no implementation
00122 * - **Open/Closed**: Extensible (add new implementations) without modification
00123 * - **Liskov Substitution**: All implementations interchangeable via interface
00124 * - **Interface Segregation**: Minimal 3-function interface, each with clear purpose
00125 * - **Dependency Inversion**: [PASS] **THIS IS THE DIP PATTERN** - High-level modules
00126 *   depend on abstraction (interface), not concretions (ThreadX)
00127 *
00128 * @par Example - Production Usage (ThreadX):
00129 * @code{.c}
00130 * // In USB communication module initialization
00131 * #include "rx_time_interface.h"
00132 * #include "rx_time_threadx.h" // Concrete ThreadX implementation
00133 *
00134 * void usb_comm_init(void) {
00135 *     // Get ThreadX time interface
00136 *     rx_time_interface_t time_iface;
00137 *     rx_time_threadx_get_interface(&time_iface);
00138 *
00139 *     // Validate interface (defensive programming)
00140 *     rx_err_t err = rx_time_interface_validate(&time_iface);
00141 *     if (err != k_rx_ok) {
00142 *         rx_log_error("USB", "Invalid time interface");
00143 *         return;
00144 *     }
00145 *
00146 *     // Inject into USB configuration
00147 *     rx_usb_comm_config_t config = {
00148 *         .time_iface = &time_iface,
00149 *         .timeout_ms = 1000,
00150 *     };
00151 *
00152 *     rx_usb_comm_init(&config);
00153 * }
00154 *
00155 * // In USB communication operations (no ThreadX dependency!)
00156 * rx_err_t usb_send_with_timeout(const rx_time_interface_t* time,
00157 *     uint8_t* data, uint32_t len) {
00158 *     uint32_t start = time->get_ms(time->ctx);
00159 *
00160 *     while (!tx_complete()) {
00161 *         if (time->is_elapsed(time->ctx, start, 1000)) {
00162 *             return k_rx_err_timeout;
00163 *         }
00164 *         time->sleep_ms(time->ctx, 1); // Brief yield
00165 *     }
00166 *
00167 *     return k_rx_ok;
00168 * }
00169 * @endcode

```

```

00170 *
00171 * @par Example - Unit Test Usage (Mock):
00172 * @code{.c}
00173 * // In unit test for USB timeout handling
00174 * #include "rx_time_interface.h"
00175 * #include "mock_time.h" // Test mock implementation
00176 *
00177 * void test_usb_timeout(void) {
00178 *     // Create mock time interface (no ThreadX needed!)
00179 *     rx_time_interface_t mock_time;
00180 *     mock_time_get_interface(&mock_time);
00181 *
00182 *     // Disable auto-advance for manual control
00183 *     mock_time_set_auto_advance(false);
00184 *
00185 *     // Inject mock into USB config
00186 *     rx_usb_comm_config_t config = {
00187 *         .time_iface = &mock_time,
00188 *         .timeout_ms = 1000,
00189 *     };
00190 *     rx_usb_comm_init(&config);
00191 *
00192 *     // Start operation that should timeout
00193 *     uint8_t data[64] = {};
00194 *     rx_err_t err = usb_send_with_timeout(&mock_time, data, sizeof(data));
00195 *
00196 *     // Manually advance time past timeout
00197 *     mock_time_advance_ms(1001);
00198 *
00199 *     // Verify timeout occurred
00200 *     assert(err == k_rx_err_timeout);
00201 * }
00202 * @endcode
00203 *
00204 * @par Example - Multiple Interface Instances:
00205 * @code{.c}
00206 * // Different modules can use different time implementations
00207 * void system_init(void) {
00208 *     // USB uses ThreadX time (real delays)
00209 *     rx_time_interface_t usb_time;
00210 *     rx_time_threadx_get_interface(&usb_time);
00211 *     usb_init(&usb_time);
00212 *
00213 *     // SPI uses null time (polling only, no sleeps)
00214 *     rx_time_interface_t spi_time = {
00215 *         .ctx = nullptr,
00216 *         .sleep_ms = nullptr, // Never call sleep
00217 *         .get_ms = real_time_get_ms, // Still need timeouts
00218 *         .is_elapsed = nullptr,
00219 *     };
00220 *     spi_init(&spi_time); // Will use polling instead of sleeps
00221 * }
00222 * @endcode
00223 *
00224 * @see [rx_time_threadx.c](../src/rx_time_threadx.c) Production implementation
00225 * @see [mock_time.c](../tests/mocks/mock_time.c) Unit test implementation
00226 * @see [rx_error_interface.h](rx_error_interface.h) Similar DIP pattern for errors
00227 * @see [rx_pin_interface.h](rx_pin_interface.h) Similar DIP pattern for GPIO
00228 *
00229 * @author Locked, Inc.
00230 * @date 2026-01-27
00231 * @version 1.0.0
00232 * @copyright Copyright (c) 2026 Locked Inc.
00233 * SPDX-License-Identifier: MIT
00234 *
00235 * @since Version 1.0.0
00236 */
00237
00238 #pragma once
00239
00240 #include <stddef.h>
00241 #include <stdint.h>
00242
00243 #include "rx_err.h"
00244
00245 #ifdef __cplusplus
00246 extern "C" {
00247 #endif
00248
00249 /* Forward declaration of interface struct */
00250 typedef struct rx_time_interface rx_time_interface_t;
00251
00252 /*
=====
00253 * Time Interface Function Pointers
00254 *
=====

```

```

00255 */
00256
00257 /**
00258  * @typedef rx_time_sleep_ms_fn
00259  * @brief Function pointer: Sleep/delay for specified milliseconds
00260  *
00261  * @details
00262  * Blocks (or simulates blocking) for the specified number of milliseconds.
00263  * Behavior depends on concrete implementation:
00264  *
00265  * **ThreadX Implementation**:
00266  * - Calls `tx_thread_sleep(ticks)` where ticks = ms * (TX_TIMER_TICKS_PER_SECOND / 1000)
00267  * - Yields CPU to other threads (cooperative multitasking)
00268  * - Actual sleep time may be slightly longer due to tick granularity
00269  * - Example: 10ms sleep with 10ms tick = exactly 1 tick
00270  * - Example: 15ms sleep with 10ms tick = 2 ticks = 20ms actual
00271  *
00272  * **Mock Implementation (manual control)**:
00273  * - Returns immediately without advancing simulated time
00274  * - Test code manually advances time via mock_time_advance_ms()
00275  * - Use for fine-grained control of timing in unit tests
00276  *
00277  * **Mock Implementation (auto-advance)**:
00278  * - Returns immediately AND advances simulated time by ms
00279  * - Useful for testing timeout logic without real delays
00280  * - 1000ms timeout completes in ~1us real time
00281  *
00282  * **Null Implementation (bare-metal)**:
00283  * - Function pointer set to nullptr
00284  * - Caller must check for nullptr before calling
00285  * - Use for polling-only code that never sleeps
00286  *
00287  * @param[in] ctx Implementation-specific context pointer
00288  * - ThreadX: Pointer to thread control block (unused, can be NULL)
00289  * - Mock: Pointer to mock_time_state_t for simulated time tracking
00290  * - Null: N/A (function pointer is nullptr)
00291  *
00292  * @param[in] ms Number of milliseconds to sleep/delay
00293  * - **Valid range**: [0, UINT32_MAX]
00294  * - **Units**: milliseconds (MKS: 0.001 seconds)
00295  * - **Value 0**: Yields CPU without delay (ThreadX) or no-op (mock)
00296  * - **Large values**: ms > 49 days wraps at UINT32_MAX (impractical)
00297  *
00298  * @return void (no return value, always "succeeds")
00299  *
00300  * @pre No preconditions - safe to call with any ms value
00301  * @pre ctx must be valid for the specific implementation (or nullptr if allowed)
00302  *
00303  * @post Control returned after at least ms milliseconds (ThreadX)
00304  * @post Simulated time advanced by ms (mock with auto_advance)
00305  * @post Other threads may have executed during sleep (ThreadX)
00306  *
00307  * @note **Thread Safety**: ThreadX implementation is thread-safe. Mock depends
00308  *       on test harness (single-threaded tests are safe).
00309  * @note **Accuracy**: ThreadX sleep accuracy limited by tick rate (typically +/-10ms).
00310  *       Use hardware timers for precise delays <1ms.
00311  * @note **Execution Time**: ThreadX context switch ~50-200us. Mock returns in <1us.
00312  * @note **Blocking**: ThreadX yields CPU (other threads run). Mock returns immediately.
00313  *
00314  * @warning ThreadX tick granularity causes rounding: 15ms sleep with 10ms tick = 20ms actual.
00315  * @attention Never call from interrupt context (ThreadX implementation will fault).
00316  * @attention Excessive sleeps (>1000ms) may indicate architectural problems.
00317  *
00318  * @par Example - Basic Sleep:
00319  * @code{.c}
00320  * // Sleep for 10ms between sensor reads
00321  * void sensor_poll_loop(const rx_time_interface_t* time) {
00322  *     while (1) {
00323  *         read_sensor_data();
00324  *         process_data();
00325  *
00326  *         // Brief sleep to reduce CPU usage
00327  *         time->sleep_ms(time->ctx, 10); // 10ms = 100 Hz poll rate
00328  *     }
00329  * }
00330  * @endcode
00331  *
00332  * @par Example - Defensive Null Check:
00333  * @code{.c}
00334  * // Gracefully handle null sleep_ms (polling-only mode)
00335  * void wait_for_ready(const rx_time_interface_t* time) {
00336  *     while (!device_is_ready()) {
00337  *         // Sleep if available, otherwise busy-wait
00338  *         if (time->sleep_ms != nullptr) {
00339  *             time->sleep_ms(time->ctx, 1); // 1ms yield
00340  *         }
00341  *         // Continue polling

```

```

00342 *   }
00343 * }
00344 * @endcode
00345 *
00346 * @see rx_time_get_ms_fn Get current time for timeout calculations
00347 * @see rx_time_is_elapsed_fn Check if timeout occurred
00348 * @see tx_thread_sleep() ThreadX implementation calls this
00349 *
00350 * @since Version 1.0.0
00351 */
00352 typedef void (*rx_time_sleep_ms_fn)(void* ctx, uint32_t ms);
00353
00354 /**
00355 * @typedef rx_time_get_ms_fn
00356 * @brief Function pointer: Get current monotonic time in milliseconds
00357 *
00358 * @details
00359 * Returns a continuously increasing time value in milliseconds, suitable for
00360 * measuring elapsed time and implementing timeouts. This is NOT wall-clock
00361 * time (not UTC, not human-readable) - it's a monotonic counter that starts
00362 * at system boot and increases until overflow (~49.7 days).
00363 *
00364 * **Monotonic Guarantee**: Time NEVER decreases, even if system clock is
00365 * adjusted. This makes it reliable for timeouts and performance measurements.
00366 *
00367 * **Wrap-Around Behavior**: 32-bit counter wraps to 0 after 232 milliseconds
00368 * (4,294,967,296 ms = 49.71 days). Properly written timeout code handles this
00369 * correctly using unsigned arithmetic.
00370 *
00371 * **Implementation Details**:
00372 *
00373 * **ThreadX Implementation**:
00374 * - Returns `tx_time_get() * (1000 / TX_TIMER_TICKS_PER_SECOND)`
00375 * - Resolution matches ThreadX tick rate (typically 10ms on RX72N)
00376 * - Examples:
00377 *   - 100 Hz tick: Resolution = 10ms (0, 10, 20, 30, ...)
00378 *   - 1000 Hz tick: Resolution = 1ms (0, 1, 2, 3, ...)
00379 *   - Overhead: ~50 cycles (~200 ns @ 240 MHz)
00380 *
00381 * **Mock Implementation**:
00382 * - Returns static counter maintained by mock
00383 * - Manual control: Counter only changes via mock_time_advance_ms()
00384 * - Auto-advance: Counter increases during sleep_ms() calls
00385 * - Resolution: 1ms (arbitrary, no hardware limits in simulation)
00386 * - Overhead: ~10 cycles (~40 ns @ 240 MHz, no RTOS)
00387 *
00388 * **Bare-Metal Implementation** (user-provided):
00389 * - Could use CMT hardware timer counter
00390 * - Could use SysTick (if using ARM Cortex-M)
00391 * - Resolution depends on hardware timer configuration
00392 *
00393 * @param[in] ctx Implementation-specific context pointer
00394 *   - ThreadX: Unused (can be NULL)
00395 *   - Mock: Pointer to mock_time_state_t containing current_ms counter
00396 *   - Hardware: Pointer to timer peripheral base address
00397 *
00398 * @return uint32_t Current time in milliseconds since system boot
00399 * @retval 0 System just booted (or wrapped around after 49.7 days)
00400 * @retval 1000 System has been running for 1 second
00401 * @retval 3600000 System has been running for 1 hour
00402 * @retval UINT32_MAX System has been running for 49.7 days (about to wrap)
00403 *
00404 * @pre No preconditions - safe to call anytime
00405 * @pre ctx must be valid for the specific implementation (or nullptr if allowed)
00406 *
00407 * @post Returned value is >= previous call (monotonic, unless wrapped)
00408 * @post No side effects (read-only operation)
00409 *
00410 * @invariant get_ms(t2) >= get_ms(t1) for t2 > t1 (unless wrap)
00411 * @invariant Resolution matches implementation tick rate (e.g., 10ms for ThreadX)
00412 *
00413 * @note **Thread Safety**: ThreadX tx_time_get() is thread-safe. Mock reads
00414 *   volatile counter (safe for single-threaded tests).
00415 * @note **Execution Time**: ThreadX ~200ns, Mock ~40ns @ 240 MHz (very fast)
00416 * @note **Wrap-Around**: Use unsigned arithmetic for timeouts to handle wrap correctly:
00417 *   `(uint32_t)(current - start) >= timeout` works across wrap boundary.
00418 * @note **Resolution**: Limited by ThreadX tick rate. For sub-millisecond timing,
00419 *   use hardware timers (CMT, GPT) directly.
00420 *
00421 * @attention NOT wall-clock time! Do not use for timestamps or logging absolute time.
00422 * @attention Wraps every 49.7 days. Long-running systems must handle wrap correctly.
00423 *
00424 * @par Example - Timeout Implementation:
00425 * @code{.c}
00426 * // Wait for device ready with 1000ms timeout
00427 * rx_err_t wait_device_ready(const rx_time_interface_t* time, uint32_t timeout_ms) {
00428 *     uint32_t start_ms = time->get_ms(time->ctx);

```

```

00429 *
00430 *   while (!device_is_ready()) {
00431 *       // Check timeout using wrap-safe arithmetic
00432 *       uint32_t elapsed = time->get_ms(time->ctx) - start_ms;
00433 *       if (elapsed >= timeout_ms) {
00434 *           return k_rx_err_timeout; // Timeout occurred
00435 *       }
00436 *
00437 *       time->sleep_ms(time->ctx, 1); // Brief yield
00438 *   }
00439 *
00440 *   return k_rx_ok; // Device ready
00441 * }
00442 * @endcode
00443 *
00444 * @par Example - Performance Measurement:
00445 * @code{.c}
00446 * // Measure execution time of algorithm
00447 void profile_algorithm(const rx_time_interface_t* time) {
00448 *   uint32_t start = time->get_ms(time->ctx);
00449 *
00450 *   run_expensive_algorithm();
00451 *
00452 *   uint32_t end = time->get_ms(time->ctx);
00453 *   uint32_t duration_ms = end - start; // Wrap-safe
00454 *
00455 *   rx_log_info("PERF", "Algorithm took %lu ms", duration_ms);
00456 * }
00457 * @endcode
00458 *
00459 * @par Example - Wrap-Around Handling:
00460 * @code{.c}
00461 * // Demonstrate wrap-around safety (uint32_t arithmetic)
00462 void test_wrap_around(void) {
00463 *   uint32_t start = UINT32_MAX - 100; // Near wrap point
00464 *   mock_time_set_ms(start);
00465 *
00466 *   // Advance past wrap boundary
00467 *   mock_time_advance_ms(200);
00468 *   uint32_t end = mock_time_get_ms(); // end = 99 (wrapped)
00469 *
00470 *   // Unsigned arithmetic handles wrap correctly
00471 *   uint32_t elapsed = end - start; // 99 - (UINT32_MAX-100) = 200 [OK]
00472 *   assert(elapsed == 200); // Correct!
00473 * }
00474 * @endcode
00475 *
00476 * @par Example - Uptime Display:
00477 * @code{.c}
00478 * // Convert milliseconds to human-readable uptime
00479 void print_uptime(const rx_time_interface_t* time) {
00480 *   uint32_t uptime_ms = time->get_ms(time->ctx);
00481 *
00482 *   uint32_t seconds = uptime_ms / 1000;
00483 *   uint32_t minutes = seconds / 60;
00484 *   uint32_t hours = minutes / 60;
00485 *   uint32_t days = hours / 24;
00486 *
00487 *   printf("Uptime: %lu days, %lu hours, %lu min\n",
00488 *         days, hours % 24, minutes % 60);
00489 * }
00490 * @endcode
00491 *
00492 * @see rx_time_sleep_ms_fn Sleep for specified duration
00493 * @see rx_time_is_elapsed_fn Convenience function for timeout checks
00494 * @see tx_time_get() ThreadX implementation uses this
00495 *
00496 * @since Version 1.0.0
00497 */
00498 typedef uint32_t (*rx_time_get_ms_fn)(void* ctx);
00499
00500 /**
00501 * @typedef rx_time_is_elapsed_fn
00502 * @brief Function pointer: Check if timeout has elapsed (optional convenience)
00503 *
00504 * @details
00505 * Convenience function that combines get_ms() and timeout arithmetic into a
00506 * single call. Equivalent to: `(get_ms(ctx) - start_ms) >= timeout_ms` with
00507 * correct handling of 32-bit wrap-around.
00508 *
00509 * This function is OPTIONAL - implementations may set this pointer to NULL.
00510 * Callers should either:
00511 * 1. Check for nullptr before calling, OR
00512 * 2. Implement timeout check manually using get_ms()
00513 *
00514 * **Why Optional?**: Some implementations (bare-metal, minimal mocks) may want
00515 * to provide only the core get_ms() function and let callers handle timeout

```



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00516 * logic themselves. This keeps the interface minimal.
00517 *
00518 * **Wrap-Around Safety**: Uses unsigned 32-bit arithmetic which correctly
00519 * handles wraparound at UINT32_MAX:
00520 * - Example: start=4294967290, timeout=100, current=50
00521 * - Elapsed = (50 - 4294967290) = 110 via unsigned underflow
00522 * - 110 >= 100 -> true (timeout occurred) [OK]
00523 *
00524 * **Algorithm**:
00525 * ```
00526 * 1. current_ms = get_ms(ctx)
00527 * 2. elapsed = current_ms - start_ms // Unsigned subtraction (wrap-safe)
00528 * 3. return (elapsed >= timeout_ms)
00529 * ```
00530 *
00531 * @param[in] ctx Implementation-specific context pointer
00532 * - ThreadX: Unused (can be NULL)
00533 * - Mock: Pointer to mock_time_state_t
00534 * - Bare-metal: Pointer to timer peripheral
00535 *
00536 * @param[in] start_ms Start time captured from get_ms()
00537 * - **Valid range**: [0, UINT32_MAX]
00538 * - **Units**: milliseconds since boot
00539 * - **Source**: MUST be from prior get_ms() call
00540 * - **Typical use**: Captured at start of operation
00541 *
00542 * @param[in] timeout_ms Timeout duration in milliseconds
00543 * - **Valid range**: [0, UINT32_MAX]
00544 * - **Units**: milliseconds (MKS: 0.001 seconds)
00545 * - **Value 0**: Returns true immediately (timeout already occurred)
00546 * - **Large values**: timeout > 49 days may never trigger due to wrap
00547 *
00548 * @return bool Timeout status
00549 * @retval true Timeout has elapsed: (current - start) >= timeout
00550 * @retval false Timeout has NOT elapsed: still within timeout window
00551 *
00552 * @pre start_ms must be from prior get_ms() call
00553 * @pre timeout_ms should be reasonable (< 49 days for guaranteed correctness)
00554 * @pre ctx must be valid for the specific implementation (or nullptr if allowed)
00555 * @pre Function pointer may be NULL - caller must check before calling!
00556 *
00557 * @post No side effects (read-only operation)
00558 * @post Result is deterministic given same inputs and current time
00559 *
00560 * @note **Thread Safety**: Thread-safe if get_ms() is thread-safe
00561 * @note **Execution Time**: ~250ns @ 240 MHz (one get_ms() call + arithmetic)
00562 * @note **Optional Function**: May be NULL in minimal implementations. Always check!
00563 * @note **Wrap-Around**: Correctly handles wrap using unsigned arithmetic
00564 * @note **Resolution**: Limited by get_ms() resolution (typically 10ms for ThreadX)
00565 *
00566 * @warning start_ms MUST be from get_ms(), not arbitrary value or hardware timer
00567 * @attention Function pointer may be NULL - check before dereferencing!
00568 * @attention timeout_ms > ~25 days may behave unexpectedly near wrap boundary
00569 *
00570 * @par Example - Using is_elapsed (if available):
00571 * @code{.c}
00572 * // Wait for I2C transaction with timeout
00573 * rx_err_t i2c_wait_complete(const rx_time_interface_t* time, uint32_t timeout_ms) {
00574 *     uint32_t start = time->get_ms(time->ctx);
00575 *
00576 *     while (!i2c_transaction_done()) {
00577 *         // Use is_elapsed if available
00578 *         if (time->is_elapsed != nullptr) {
00579 *             if (time->is_elapsed(time->ctx, start, timeout_ms)) {
00580 *                 return k_rx_err_timeout;
00581 *             }
00582 *         } else {
00583 *             // Fallback: manual timeout check
00584 *             uint32_t elapsed = time->get_ms(time->ctx) - start;
00585 *             if (elapsed >= timeout_ms) {
00586 *                 return k_rx_err_timeout;
00587 *             }
00588 *         }
00589 *
00590 *         time->sleep_ms(time->ctx, 1);
00591 *     }
00592 *
00593 *     return k_rx_ok;
00594 * }
00595 * @endcode
00596 *
00597 * @par Example - Simplified with NULL Check:
00598 * @code{.c}
00599 * // Cleaner timeout check with defensive NULL handling
00600 * bool check_timeout(const rx_time_interface_t* time,
00601 *     uint32_t start, uint32_t timeout_ms) {
00602 *     if (time->is_elapsed != nullptr) {

```



```

00603 *      // Use convenience function if available
00604 *      return time->is_elapsed(time->ctx, start, timeout_ms);
00605 *  } else {
00606 *      // Manual calculation (always works)
00607 *      uint32_t elapsed = time->get_ms(time->ctx) - start;
00608 *      return (elapsed >= timeout_ms);
00609 *  }
00610 * }
00611 * @endcode
00612 *
00613 * @par Example - Multiple Timeouts:
00614 * @code{.c}
00615 * // Check multiple independent timeouts
00616 * void multi_timeout_loop(const rx_time_interface_t* time) {
00617 *     uint32_t sensor_start = time->get_ms(time->ctx);
00618 *     uint32_t motor_start = time->get_ms(time->ctx);
00619 *     uint32_t comm_start = time->get_ms(time->ctx);
00620 *
00621 *     while (running) {
00622 *         // Check sensor timeout (500ms)
00623 *         if (time->is_elapsed(time->ctx, sensor_start, 500)) {
00624 *             read_sensors();
00625 *             sensor_start = time->get_ms(time->ctx); // Restart timer
00626 *         }
00627 *
00628 *         // Check motor timeout (100ms)
00629 *         if (time->is_elapsed(time->ctx, motor_start, 100)) {
00630 *             update_motors();
00631 *             motor_start = time->get_ms(time->ctx);
00632 *         }
00633 *
00634 *         // Check comm timeout (1000ms)
00635 *         if (time->is_elapsed(time->ctx, comm_start, 1000)) {
00636 *             send_telemetry();
00637 *             comm_start = time->get_ms(time->ctx);
00638 *         }
00639 *
00640 *         time->sleep_ms(time->ctx, 10); // 100 Hz loop
00641 *     }
00642 * }
00643 * @endcode
00644 *
00645 * @par Example - Wrap-Around Correctness:
00646 * @code{.c}
00647 * // Verify is_elapsed handles wrap correctly
00648 * void test_wrap_around_elapsed(void) {
00649 *     rx_time_interface_t mock_time;
00650 *     mock_time_get_interface(&mock_time);
00651 *
00652 *     // Set time near wrap boundary
00653 *     uint32_t start = UINT32_MAX - 50; // 50ms before wrap
00654 *     mock_time_set_ms(start);
00655 *
00656 *     // Advance past wrap (50 + 100 = 150ms elapsed)
00657 *     mock_time_advance_ms(150);
00658 *
00659 *     // Check timeout (100ms)
00660 *     bool elapsed = mock_time.is_elapsed(mock_time.ctx, start, 100);
00661 *     assert(elapsed == true); // Correctly detected timeout across wrap! [OK]
00662 * }
00663 * @endcode
00664 *
00665 * @see rx_time_get_ms_fn Used internally by this function
00666 * @see rx_time_sleep_ms_fn Often used together in timeout loops
00667 *
00668 * @since Version 1.0.0
00669 */
00670 typedef bool (*rx_time_is_elapsed_fn)(void* ctx, uint32_t start_ms, uint32_t timeout_ms);
00671
00672 /*
=====
00673 * Time Interface Structure
00674 *
=====
00675 */
00676 /**
00677 * @struct rx_time_interface
00678 * @brief Time operations interface structure (Dependency Inversion Pattern)
00679 *
00680 * @details
00681 * Defines the complete contract for time operations that all implementations
00682 * must fulfill. This struct contains function pointers for time operations
00683 * plus an opaque context pointer for implementation-specific state.
00684 *
00685 * This is a "fat pointer" or "vtable" pattern common in object-oriented design,
00686 * adapted for C. It enables runtime polymorphism without inheritance - different

```

```

00688 * implementations (ThreadX, mock, bare-metal) can be swapped at runtime by
00689 * filling in different function pointers.
00690 *
00691 * ## Memory Layout (32-bit pointers)
00692 *
00693 * | Offset | Size | Field | Type | Alignment | Purpose |
00694 * |-----|-----|-----|-----|-----|-----|
00695 * | 0 | 4 | ctx | void* | 4 | Implementation context |
00696 * | 4 | 4 | sleep_ms | fn ptr | 4 | Sleep function |
00697 * | 8 | 4 | get_ms | fn ptr | 4 | Get time function |
00698 * | 12 | 4 | is_elapsed | fn ptr | 4 | Timeout check (optional) |
00699 * | **Total** | **16** | | | | **No padding** |
00700 *
00701 * ## Initialization Patterns
00702 *
00703 * **Static Initialization** (compile-time):
00704 * ```c
00705 * static rx_time_interface_t s_time = {
00706 *     .ctx = &s_threadx_state,
00707 *     .sleep_ms = threadx_sleep_ms_impl,
00708 *     .get_ms = threadx_get_ms_impl,
00709 *     .is_elapsed = threadx_is_elapsed_impl,
00710 * };
00711 * ```
00712 *
00713 * **Dynamic Initialization** (runtime):
00714 * ```c
00715 * rx_time_interface_t time;
00716 * rx_time_threadx_get_interface(&time); // Fills in struct
00717 * ```
00718 *
00719 * **Null Implementation** (minimal/bare-metal):
00720 * ```c
00721 * static rx_time_interface_t s_null_time = {
00722 *     .ctx = nullptr,
00723 *     .sleep_ms = nullptr, // Polling only, no sleeps
00724 *     .get_ms = hardware_timer_get_ms, // Still need timeouts
00725 *     .is_elapsed = nullptr, // Manual timeout checks
00726 * };
00727 * ```
00728 *
00729 * ## Field Constraints and Relationships
00730 *
00731 * | Field | Required? | Notes |
00732 * |-----|-----|-----|
00733 * | ctx | Optional | NULL if implementation is stateless |
00734 * | sleep_ms | **Required** | Validation fails if nullptr |
00735 * | get_ms | **Required** | Validation fails if nullptr |
00736 * | is_elapsed | Optional | Can be NULL (manual timeout checks) |
00737 *
00738 * **Constraint**: If sleep_ms is nullptr, interface is invalid (cannot sleep).
00739 * **Constraint**: If get_ms is nullptr, interface is invalid (cannot measure time).
00740 * **Invariant**: Once initialized, function pointers should not change (read-only).
00741 *
00742 * @par Usage Example - Initialization:
00743 * @code{.c}
00744 * // Production: Use ThreadX implementation
00745 * void production_init(void) {
00746 *     rx_time_interface_t time_iface;
00747 *     rx_time_threadx_get_interface(&time_iface);
00748 *
00749 *     // Validate before use (defensive programming)
00750 *     rx_err_t err = rx_time_interface_validate(&time_iface);
00751 *     if (err != k_rx_ok) {
00752 *         rx_log_fatal("INIT", "Invalid time interface");
00753 *         return;
00754 *     }
00755 *
00756 *     // Inject into USB module
00757 *     usb_comm_init(&time_iface);
00758 * }
00759 * @endcode
00760 *
00761 * @par Usage Example - Passing by Value vs Pointer:
00762 * @code{.c}
00763 * // Option 1: Pass by pointer (common, lightweight)
00764 * void device_init(const rx_time_interface_t* time) {
00765 *     // Use: time->get_ms(time->ctx)
00766 * }
00767 *
00768 * // Option 2: Pass by value (copy 16 bytes, less common)
00769 * void device_init(rx_time_interface_t time) {
00770 *     // Use: time.get_ms(time.ctx)
00771 * }
00772 *
00773 * // Option 3: Embed in config struct (recommended)
00774 * typedef struct {

```

```

00775 *   rx_time_interface_t time_iface; // 16 bytes embedded
00776 *   uint32_t timeout_ms;
00777 *   uint32_t retry_count;
00778 * } device_config_t;
00779 * @endcode
00780 *
00781 * @par Usage Example - Multiple Implementations:
00782 * @code{.c}
00783 * // System with mixed implementations
00784 * void mixed_system_init(void) {
00785 *     // USB uses real ThreadX time (needs actual delays)
00786 *     rx_time_interface_t usb_time;
00787 *     rx_time_threadx_get_interface(&usb_time);
00788 *     usb_init(&usb_time);
00789 *
00790 *     // SPI uses null time (polling only, instant returns)
00791 *     rx_time_interface_t spi_time = {};
00792 *     spi_time.get_ms = hardware_timer_get_ms; // Still need timeouts
00793 *     spi_time.sleep_ms = null_sleep; // No-op sleep
00794 *     spi_init(&spi_time);
00795 * }
00796 * @endcode
00797 *
00798 * @see rx_time_interface_validate() Validate interface before use
00799 * @see rx_time_threadx_get_interface() Fill with ThreadX implementation
00800 * @see mock_time_get_interface() Fill with mock implementation for tests
00801 *
00802 * @since Version 1.0.0
00803 */
00804 struct rx_time_interface {
00805     /**
00806      * @brief Implementation-specific context pointer (opaque)
00807      *
00808      * @details
00809      * Pointer to concrete implementation's private state. Treated as opaque by
00810      * interface users - only the implementation knows what this points to.
00811      *
00812      * **ThreadX**: Unused, typically NULL (ThreadX state is global)
00813      * **Mock**: Points to mock_time_state_t with current_ms counter
00814      * **Bare-metal**: Could point to hardware timer peripheral base address
00815      *
00816      * **Lifetime**: Must remain valid for duration of interface usage. Typically
00817      * points to static or global data, NOT stack variables.
00818      *
00819      * **Null allowed**: Yes, if implementation is stateless (e.g., ThreadX)
00820      */
00821     void* ctx;
00822
00823     /**
00824      * @brief Sleep/delay function pointer (REQUIRED)
00825      *
00826      * @details
00827      * Blocks or simulates blocking for specified milliseconds. This function is
00828      * REQUIRED - validation fails if nullptr.
00829      *
00830      * **Implementation**: ThreadX: tx_thread_sleep(), Mock: simulate time advance
00831      * **Null allowed**: NO (validation fails)
00832      * **Thread safety**: Implementation-dependent (ThreadX is thread-safe)
00833      *
00834      * @see rx_time_sleep_ms_fn Full documentation of sleep semantics
00835      */
00836     rx_time_sleep_ms_fn sleep_ms;
00837
00838     /**
00839      * @brief Get current time function pointer (REQUIRED)
00840      *
00841      * @details
00842      * Returns monotonic time in milliseconds since boot. This function is
00843      * REQUIRED - validation fails if nullptr.
00844      *
00845      * **Implementation**: ThreadX: tx_time_get() conversion, Mock: counter read
00846      * **Null allowed**: NO (validation fails)
00847      * **Resolution**: ThreadX tick rate (10ms typical), Mock 1ms
00848      *
00849      * @see rx_time_get_ms_fn Full documentation of time semantics
00850      */
00851     rx_time_get_ms_fn get_ms;
00852
00853     /**
00854      * @brief Timeout elapsed check function pointer (OPTIONAL)
00855      *
00856      * @details
00857      * Convenience function combining get_ms() and timeout arithmetic. This
00858      * function is OPTIONAL - can be NULL for minimal implementations.
00859      *
00860      * **Implementation**: Calls get_ms() and computes (current - start) >= timeout
00861      * **Null allowed**: YES (caller must check before using)

```

```

00862  * **Alternative**: Manual timeout check using get_ms() if nullptr
00863  *
00864  * @see rx_time_is_elapsed_fn Full documentation of timeout semantics
00865  */
00866 rx_time_is_elapsed_fn is_elapsed;
00867 };
00868
00869 /*
=====
00870 * Interface Validation
00871 *
=====
00872 */
00873
00874 /**
00875  * @brief Validate that a time interface is properly initialized
00876  *
00877  * @details
00878  * Performs defensive validation of a time interface to ensure it meets the
00879  * minimum requirements for safe usage. Checks that the interface pointer is
00880  * non-NULL and that required function pointers (sleep_ms, get_ms) are populated.
00881  *
00882  * This function implements "fail-fast" validation - it's better to detect a
00883  * misconfigured interface immediately at initialization than to crash later
00884  * when dereferencing NULL function pointers.
00885  *
00886  * **Validation Rules**:
00887  * 1. Interface pointer must be non-NULL
00888  * 2. sleep_ms function pointer must be non-NULL (REQUIRED)
00889  * 3. get_ms function pointer must be non-NULL (REQUIRED)
00890  * 4. ctx pointer may be NULL (implementation-dependent)
00891  * 5. is_elapsed pointer may be NULL (OPTIONAL)
00892  *
00893  * Algorithm steps:
00894  * 1. Check if iface pointer is nullptr -> return k_rx_err_null_ptr
00895  * 2. Check if sleep_ms is nullptr -> return k_rx_err_invalid_state
00896  * 3. Check if get_ms is nullptr -> return k_rx_err_invalid_state
00897  * 4. All checks passed -> return k_rx_ok
00898  *
00899  * @param[in] iface Pointer to time interface to validate
00900  * - May be NULL (will return k_rx_err_null_ptr)
00901  * - Typically points to stack or static rx_time_interface_t
00902  *
00903  * @return rx_err_t Validation result
00904  * @retval k_rx_ok Interface is valid and safe to use
00905  * @retval k_rx_err_null_ptr iface pointer is nullptr (cannot validate)
00906  * @retval k_rx_err_invalid_state Required function pointer(s) are NULL (sleep_ms or get_ms)
00907  *
00908  * @pre No preconditions - safe to call with any input (even NULL)
00909  *
00910  * @post Interface not modified (read-only validation)
00911  * @post Return value indicates whether interface is usable
00912  *
00913  * @note **Thread Safety**: Thread-safe (read-only operation, no shared state)
00914  * @note **Execution Time**: ~50 ns @ 240 MHz (3 null checks)
00915  * @note **Use Case**: Call at initialization before using interface
00916  * @note **Optional Fields**: Does NOT check ctx or is_elapsed (both may be NULL)
00917  *
00918  * @warning Does NOT validate that function pointers point to valid functions!
00919  * Only checks for nullptr. Invalid pointers will cause crash on call.
00920  *
00921  * @attention Validation does NOT guarantee thread-safety of implementations.
00922  * ThreadX is thread-safe, but custom implementations may not be.
00923  *
00924  * @par Example - Defensive Initialization:
00925  * @code{.c}
00926  * // Validate interface before passing to modules
00927  * void system_init(void) {
00928  *     rx_time_interface_t time_iface;
00929  *     rx_time_threadx_get_interface(&time_iface);
00930  *
00931  *     // Validate before use (defensive programming)
00932  *     rx_err_t err = rx_time_interface_validate(&time_iface);
00933  *     if (err == k_rx_err_null_ptr) {
00934  *         rx_log_fatal("INIT", "Time interface is nullptr");
00935  *         return;
00936  *     } else if (err == k_rx_err_invalid_state) {
00937  *         rx_log_fatal("INIT", "Time interface missing required functions");
00938  *         return;
00939  *     }
00940  *
00941  *     // Safe to use
00942  *     usb_init(&time_iface);
00943  *     spi_init(&time_iface);
00944  * }
00945  * @endcode
00946  */

```

```

00947 * @par Example - Validation in Module Init:
00948 * @code{.c}
00949 * // Module enforces interface validity at init
00950 * rx_err_t usb_comm_init(const rx_time_interface_t* time) {
00951 *     // Validate injected dependency
00952 *     rx_err_t err = rx_time_interface_validate(time);
00953 *     if (err != k_rx_ok) {
00954 *         rx_log_error("USB", "Invalid time interface: %d", err);
00955 *         return err; // Propagate error to caller
00956 *     }
00957 *
00958 *     // Store validated interface
00959 *     s_usb_time = time;
00960 *
00961 *     return k_rx_ok;
00962 * }
00963 * @endcode
00964 *
00965 * @par Example - Unit Test Verification:
00966 * @code{.c}
00967 * // Verify validation catches misconfigured interfaces
00968 * void test_validate_catches_null_interface(void) {
00969 *     rx_err_t err = rx_time_interface_validate(nullptr);
00970 *     assert(err == k_rx_err_null_ptr);
00971 * }
00972 *
00973 * void test_validate_catches_missing_sleep(void) {
00974 *     rx_time_interface_t time = {};
00975 *     time.get_ms = mock_get_ms; // Only get_ms populated
00976 *     // Missing sleep_ms!
00977 *
00978 *     rx_err_t err = rx_time_interface_validate(&time);
00979 *     assert(err == k_rx_err_invalid_state);
00980 * }
00981 *
00982 * void test_validate_allows_null_ctx(void) {
00983 *     rx_time_interface_t time = {};
00984 *     time.ctx = nullptr; // ctx can be nullptr
00985 *     time.sleep_ms = mock_sleep_ms;
00986 *     time.get_ms = mock_get_ms;
00987 *
00988 *     rx_err_t err = rx_time_interface_validate(&time);
00989 *     assert(err == k_rx_ok); // Valid even with NULL ctx
00990 * }
00991 * @endcode
00992 *
00993 * @par Example - Production Assertion:
00994 * @code{.c}
00995 * // Hard assertion in safety-critical code
00996 * void critical_init(const rx_time_interface_t* time) {
00997 *     rx_err_t err = rx_time_interface_validate(time);
00998 *
00999 *     // In safety-critical systems, invalid interface is fatal
01000 *     RX_CHECK(err == k_rx_ok, "Time interface validation failed");
01001 *
01002 *     // Continue with guaranteed valid interface...
01003 * }
01004 * @endcode
01005 *
01006 * @see rx_time_interface Structure being validated
01007 * @see rx_time_threadx_get_interface() Produces valid interface
01008 * @see mock_time_get_interface() Produces valid test interface
01009 *
01010 * @since Version 1.0.0
01011 * @version 1.0.0
01012 *
01013 * @test Tested in test_rx_time_interface.c::test_validate_null_interface()
01014 * @test Tested in test_rx_time_interface.c::test_validate_missing_functions()
01015 * @test Tested in test_rx_time_interface.c::test_validate_valid_interface()
01016 *
01017 * @par NASA Power of 10 Compliance:
01018 * - Rule 1: [OK] No goto, setjmp, recursion
01019 * - Rule 2: [OK] No loops
01020 * - Rule 5: [OK] 1 precondition, 2 postconditions
01021 * - Rule 4: [OK] Function is 10 lines (well under 60)
01022 *
01023 * @callgraph
01024 * @callergraph
01025 */
01026 [[nodiscard]] rx_err_t rx_time_threadx_get_interface(rx_time_interface_t* iface);
01027
01028 static inline rx_err_t rx_time_interface_validate(const rx_time_interface_t* iface)
01029 {
01030     if (iface == nullptr) {
01031         return k_rx_err_null_ptr;
01032     }
01033 }

```

```

01034  /* Check required function pointers */
01035  if (iface->sleep_ms == nullptr || iface->get_ms == nullptr) {
01036      return k_rx_err_invalid_state;
01037  }
01038
01039  return k_rx_ok;
01040 }
01041
01042 #ifdef __cplusplus
01043 }
01044 #endif

```

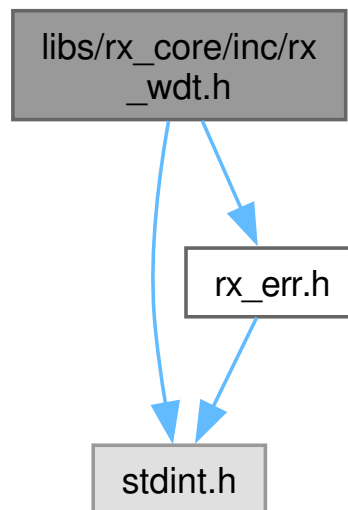
7.43 libs/rx_core/inc/rx_wdt.h File Reference

Watchdog Timer (WDT) Driver for RX72N Microcontroller.

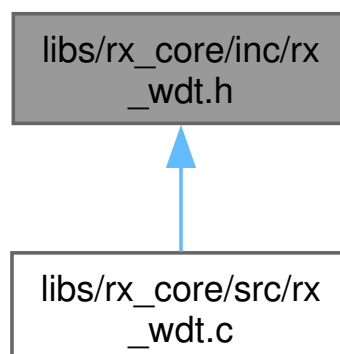
```
#include <stdint.h>
```

```
#include "rx_err.h"
```

Include dependency graph for rx_wdt.h:



This graph shows which files directly or indirectly include this file:



Data Structures

- [struct rx_wdt_config_t](#)
WDT Configuration Structure for Driver Initialization.

Enumerations

- enum `wdt_timeout_cycles_t` : `uint8_t` { `k_wdt_timeout_1024_cycles` = 0 , `k_wdt_timeout_4096_cycles` = 1 , `k_wdt_timeout_8192_cycles` = 2 , `k_wdt_timeout_16384_cycles` = 3 }
WDT Timeout Period Selection (TOPS bits in WDTCR register).
- enum `wdt_clock_division_t` : `uint8_t` { `k_wdt_clock_div_4` = 0x01 , `k_wdt_clock_div_64` = 0x04 , `k_wdt_clock_div_128` = 0x0F , `k_wdt_clock_div_512` = 0x06 , `k_wdt_clock_div_2048` = 0x07 , `k_wdt_clock_div_8192` = 0x08 }
WDT Clock Division Ratio Selection (CKS bits in WDTCR register).

Functions

- `rx_err_t rx_wdt_init` (const `rx_wdt_config_t` *config)
Initialize the Watchdog Timer (WDT) with specified configuration.
- `rx_err_t rx_wdt_start` (void)
Start the WDT counter (enable watchdog monitoring).
- `rx_err_t rx_wdt_stop` (void)
Stop the WDT counter (disable watchdog monitoring).
- `rx_err_t rx_wdt_feed` (void)
Feed the watchdog timer (reset counter to prevent timeout).

7.43.1 Detailed Description

Watchdog Timer (WDT) Driver for RX72N Microcontroller.

Software-controllable watchdog driver providing flexible system monitoring complementary to the Independent Watchdog (IWDT). Designed for development, debugging, and scenarios requiring runtime watchdog control. Works alongside IWDT to provide defense-in-depth fault detection.

7.43.1.1 Purpose and Design Rationale

This driver implements the RX72N Watchdog Timer (WDT), a software-controllable watchdog that complements the hardware-independent IWDT:

Why Both WDT and IWDT?

1. IWDT (Independent): Hardware safety net, cannot be disabled, clock-independent
2. WDT (Software): Flexible development tool, can be stopped, uses system clock
3. Defense in Depth: Two independent watchdogs catch different failure modes

The WDT provides runtime control (start/stop) which is essential for:

- Debugging: Temporarily disable during breakpoint sessions
- Flash updates: Suspend during long operations
- Safe mode: Different timeout strategies for degraded operation

7.43.1.2 Key Differences from IWDT

Feature	WDT (This Module)	IWDT (rx_iwdt.h)
Clock Source	PCLKB (system clock)	IWDTCCLK (independent 125kHz)
Can be Stopped	[OK] Yes (<code>rx_wdt_stop</code>)	[X] No (hardware safety)
Clock Fault Detection	[X] No (same clock tree)	[OK] Yes (independent)
Development Friendly	[OK] Yes	[X] No (always running)
Production Safety	Medium	High
Timeout Granularity	PCLKB-dependent	Fixed ~125kHz
Register Base	0x00088020	0x00088030

Typical Usage Strategy:

- Development: Use WDT only (can stop for debugging)
- Production: Use both (IWDT=hardware safety, WDT=software monitoring)

7.43.1.3 Hardware Architecture

RX72N Watchdog Timer (WDT) Peripheral:

- Register Base Address: 0x00088020
- Clock Source: PCLKB (Peripheral Clock B, typically 60 MHz)
- Clock Frequency: 60 MHz (assuming default PCLKB)
- Counter Width: 14-bit down-counter
- Timeout Range: 4.3 us to 273 us (at 60 MHz PCLKB)
- Control: Software start/stop via WDTCR register
- Reset Output: Connected to internal reset controller (if configured)

Timeout Calculation:

$$\text{Timeout (seconds)} = (\text{Divider} \times \text{Cycles}) / \text{PCLKB_Frequency}$$

Where Divider is selected by CKS (Clock Select) bits:

CKS=00: Divider=1 -> 256 cycles -> 4.3 us at 60 MHz
 CKS=01: Divider=4 -> 1024 cycles -> 17 us at 60 MHz
 CKS=10: Divider=16 -> 4096 cycles -> 68 us at 60 MHz
 CKS=11: Divider=64 -> 16384 cycles -> 273 us at 60 MHz

Example: For 1ms timeout at PCLKB=60MHz:

1ms = N / 60,000,000
 N = 60,000 cycles
 Use CKS=11 (16384 cycles) x 4 feeds per timeout = ~273us per feed

Note: Very short timeouts make WDT impractical for general use. Most systems use IWDT for watchdog functionality and WDT for special cases.

7.43.1.4 Implementation Approach

Two-Layer Architecture:

1. Hardware Layer (`rx_wdt_init`, `rx_wdt_start`, `rx_wdt_stop`):
 - Configures WDT control registers (WDTCR)
 - Sets clock divider (CKS bits)
 - Enables/disables counter operation
2. Refresh Layer (`rx_wdt_feed`):
 - Writes refresh code to WDTRR register
 - Resets counter to prevent timeout

Typical Usage Flow:

1. System startup:
 - `rx_wdt_init()` - Configure (but don't start yet)
2. After initialization complete:
 - `rx_wdt_start()` - Enable watchdog
3. Normal operation:
 - Main loop: `rx_wdt_feed()` every N us (N < timeout)
4. During debugging/flash:
 - `rx_wdt_stop()` - Suspend watchdog
 - ... perform long operation ...
 - `rx_wdt_start()` - Resume watchdog

7.43.1.5 Performance Characteristics

Operation	Execution Time	Memory Usage	Notes
<code>rx_wdt_init()</code>	~20 us	0 heap	One-time setup
<code>rx_wdt_start()</code>	~2 us	0	Register write
<code>rx_wdt_stop()</code>	~2 us	0	Register write
<code>rx_wdt_feed()</code>	~1 us	0	Single register write

Memory Footprint:

- Static variables: ~16 bytes (configuration state)
- Code size: ~1 KB (all functions)
- Stack usage: < 32 bytes (deepest call)
- Heap usage: 0 bytes (zero dynamic allocation)

7.43.1.6 Thread Safety and RTOS Integration

Thread-Safe Operations:

- All public functions use mutex protection (ThreadX)
- Safe to call from multiple threads simultaneously
- ISR-safe: `rx_wdt_feed()` can be called from interrupts (no mutex in ISR path)

ThreadX Integration:

- Uses `tx_mutex` for synchronization
- Compatible with all ThreadX thread priorities
- No dependencies on ThreadX timekeeping (unlike IWDT task monitoring)

7.43.1.7 Usage Guidelines

DO:

- Use WDT alongside IWDT for defense in depth
- Stop WDT during long flash operations or debugging
- Call `rx_wdt_feed()` from main supervision loop only
- Configure timeout based on PCLKB frequency

DON'T:

- Rely on WDT alone for safety (use IWDT as primary)
- Forget to restart WDT after stopping (system unprotected)
- Use very short timeouts (< 1ms difficult to meet)
- Call `rx_wdt_feed()` from multiple locations (defeats purpose)

When to Use WDT vs IWDT:

- Primary safety watchdog: Use IWDT (cannot be disabled)
- Development flexibility: Use WDT (can be stopped for debugging)
- Clock fault detection: Use IWDT (independent clock)
- Production systems: Use both for defense in depth

Hardware Requirements:

Component	Requirement	Notes
MCU	Renesas RX72N	WDT peripheral required
Clock	PCLKB (Peripheral Clock B)	System clock dependency
Registers	0x00088020 base	WDT control registers
Reset	Internal reset controller	Connected to WDT output
PCLKB Frequency	Known value	Required for timeout calculation

Module Dependencies:

This module depends on:

- [rx_err.h](#) - Error code definitions
- `<stdint.h>` - Standard integer types
- `<stdbool.h>` - Boolean type
- ThreadX RTOS - Mutex for thread safety

This module is used by:

- `main.c` - System initialization and main loop
- Flash update routines - Start/stop during operations
- Safe mode handlers - Runtime watchdog control

NASA Power of 10 Compliance:

Rule	Status	Implementation
1. Simple control flow	[OK]	No goto/setjmp/recursion, sequential logic
2. Fixed loop bounds	[OK]	No loops in driver (stateless operations)
3. No dynamic allocation	[OK]	Zero malloc/free, static configuration
4. Small functions	[OK]	All functions < 30 lines
5. Assertions (≥ 2 per function)	[OK]	Minimum 2 checks per function (pre/post)
6. Narrow scope	[OK]	File-scope statics, function-local variables
7. Check return values	[OK]	All functions return rx_err_t , checked by callers
8. Limited preprocessor	[OK]	C23 typed enums only, zero computation macros
9. Pointer restrictions	[OK]	Single-level pointers, no pointer arithmetic
10. Compiler warnings	[OK]	Compiles with -Wall -Wextra -Werror

Safety-Critical Design:

- Static configuration prevents allocation failures
- Explicit error codes force error handling
- Mutex protection prevents race conditions
- Simple state machine (init -> start/stop -> feed)

SOLID Principles:

Single Responsibility (S):

- This module has ONE job: Software-controllable watchdog timer
- Separate from IWDT (independent watchdog with different purpose)
- No mixing of unrelated functionality

Open/Closed (O):

- Extensible via configuration structure (timeout period, reset mode)
- Can add new timeout periods without modifying existing code
- Timeout configuration at init, not hardcoded

Liskov Substitution (L):

- All functions return [rx_err_t](#) for consistent error handling
- Same error semantics as IWDT (interchangeable error handling)
- No unexpected side effects or hidden dependencies

Interface Segregation (I):

- Separate functions for separate concerns (init/start/stop/feed)
- Clients use only the functions they need
- Minimal API surface (4 functions)

Dependency Inversion (D):

- Depends on abstract [rx_err_t](#), not specific error implementations
- Can be mocked for unit testing
- No direct hardware dependencies in interface

See also

[rx_iwdt.h](#) Independent Watchdog Timer (primary safety watchdog)

[rx_err.h](#) Error code definitions used by all functions

[docs/sections/06_nasa_power_of_10.tex](#) Safety-critical coding standards

RX72N Hardware Manual Section 10 Watchdog Timer specification

Author

Locked, Inc.

Date

2026-01-27

Copyright

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Since

Version 1.0.0

Definition in file [rx_wdt.h](#).

7.43.2 Data Structure Documentation

7.43.2.1 struct rx_wdt_config_t

WDT Configuration Structure for Driver Initialization.

Contains all parameters needed to configure and initialize the Watchdog Timer (WDT). All fields must be set before passing to `rx_wdt_init()`. Use default values for typical operation, or customize for specific monitoring requirements.

7.43.2.2 Initialization Requirements

1. All fields must be initialized - No default constructor available (C struct)
2. Validate timeout_cycles - Must be valid enum value (0-3)
3. Consider PCLKB frequency - Timeout periods are PCLKB-dependent
4. Choose reset vs interrupt - reset_on_timeout determines failure behavior

7.43.2.3 Default Configuration

Recommended starting configuration for development:

```
rx_wdt_config_t default_config = {
    .timeout_cycles = k_wdt_timeout_16384_cycles, // 16384 counter cycles
    .clock_division = k_wdt_clock_div_8192,      // PCLKB / 8192
    .enable_on_init = false,                      // Don't start immediately (safer)
    .reset_on_timeout = true,                     // Reset on timeout (production mode)
};
// Actual timeout = 16384 * 8192 / 60MHz = 2.24 seconds
```

7.43.2.4 Production Configuration

For production systems using WDT as secondary watchdog:

```
rx_wdt_config_t prod_config = {
    .timeout_cycles = k_wdt_timeout_16384_cycles, // Max cycles
    .clock_division = k_wdt_clock_div_8192,      // Max divisor = 2.24s timeout
    .enable_on_init = true,                      // Start immediately
    .reset_on_timeout = true,                    // Always reset
};
```

7.43.2.5 Memory Layout

Offset	Size	Field	Type	Notes
0	1	timeout_cycles	uint8_t (enum)	TOPS bits (0-3)
1	1	clock_division	uint8_t (enum)	CKS bits
2	1	enable_on_init	bool	Auto-start flag
3	1	reset_on_timeout	bool	Reset mode flag
Total	4 bytes			Structure size

Field Constraints:

Field	Valid Values	Default	Notes
timeout_cycles	0-3 (enum)	k_wdt_timeout_16384_cycles	TOPS bits
clock_division	See enum	k_wdt_clock_div_8192	CKS bits (6 valid values only)
enable_on_init	true/false	false	false = safer for development
reset_on_timeout	true/false	true	true = reset, false = interrupt

Usage Example (Basic):

```
// Initialize with 2.24 second timeout (max)
rx_wdt_config_t config = {
    .timeout_cycles = k_wdt_timeout_16384_cycles,
    .clock_division = k_wdt_clock_div_8192,
    .enable_on_init = false,
    .reset_on_timeout = true
};

rx_err_t err = rx_wdt_init(&config);
if (err != k_rx_ok) {
    uart_debug_puts("[ERROR] WDT init failed\r\n");
    return err;
}
```

Usage Example (Development Mode):

```
// Development configuration: Don't start, use interrupt for debugging
rx_wdt_config_t dev_config = {
    .timeout_cycles = k_wdt_timeout_16384_cycles,
    .clock_division = k_wdt_clock_div_8192,
    .enable_on_init = false, // Start manually after init complete
    .reset_on_timeout = false // Interrupt allows breakpoint debugging
};

rx_wdt_init(&dev_config);

// ... system initialization ...

// Start watchdog after init complete
rx_wdt_start();
```

Usage Example (Runtime Control):

```
// Start with WDT enabled, 559ms timeout
rx_wdt_config_t config = {
    .timeout_cycles = k_wdt_timeout_4096_cycles,
    .clock_division = k_wdt_clock_div_8192,
    .enable_on_init = true,
    .reset_on_timeout = true
};
rx_wdt_init(&config);

// Normal operation
while (running) {
    main_loop_work();
    rx_wdt_feed();
}

// Stop for long flash operation
rx_wdt_stop();
flash_update(); // May take seconds
rx_wdt_start(); // Resume monitoring
```

Invariant

- timeout_cycles must be in range [0, 3]
- Structure size is 4 bytes (3 fields + 1 padding)
- All fields must be initialized before use
- Configuration is immutable after [rx_wdt_init\(\)](#) (requires re-init to change)

Note

- C23 struct with explicit padding for 4-byte alignment
- Configuration is copied by [rx_wdt_init\(\)](#), original can be freed after call
- Changing configuration requires calling [rx_wdt_init\(\)](#) again (if WDT stopped)

Warning

- timeout_cycles values are hardware-limited (0-3 only)
- enable_on_init=true starts watchdog immediately (ensure feed loop ready)
- reset_on_timeout=true will reset system (no recovery possible)

Attention

- Always validate timeout_cycles against [wdt_timeout_cycles_t](#) enum

See also

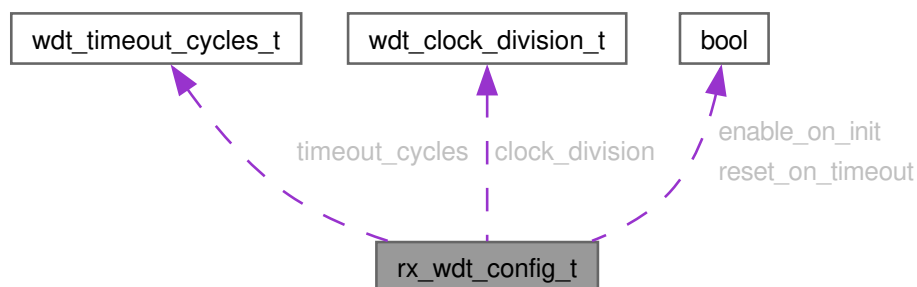
- [rx_wdt_init\(\)](#) Initializes WDT with this configuration
- [wdt_timeout_cycles_t](#) Enum values for [timeout_cycles](#) field
- [rx_wdt_start\(\)](#) Start watchdog if [enable_on_init](#)=false
- [rx_wdt_stop\(\)](#) Stop watchdog for debugging or long operations

Since

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Definition at line 506 of file [rx_wdt.h](#).

Collaboration diagram for [rx_wdt_config_t](#):



Data Fields

wdt_clock_division_t	clock_division	Clock division ratio (CKS bits). Use <code>k_wdt_clock_div_8192</code> for longest timeout. Combined with <code>timeout_cycles</code> determines actual timeout. See wdt_clock_division_t .
bool	enable_on_init	Auto-start watchdog after initialization. <code>true</code> = Start immediately (production mode, ensure feed loop ready). <code>false</code> = Manual start via <code>rx_wdt_start()</code> (safer for development). Default: <code>false</code>
bool	reset_on_timeout	Behavior on watchdog timeout. <code>true</code> = System reset (production mode). <code>false</code> = NMI interrupt (development mode, allows debugging). Default: <code>true</code>
wdt_timeout_cycles_t	timeout_cycles	Counter cycle count before timeout (TOPS bits). Use <code>k_wdt_timeout_16384_cycles</code> for maximum. Valid values: 0-3. See wdt_timeout_cycles_t .

7.43.3 Enumeration Type Documentation

7.43.3.1 `wdt_clock_division_t`

enum [wdt_clock_division_t](#) : `uint8_t`

WDT Clock Division Ratio Selection (CKS bits in WDTCR register).

Defines the six available clock division ratios for the WDT. Combined with TOPS (timeout period), this determines the actual watchdog timeout.

7.43.3.2 Timeout Calculation

$$T_{timeout} = \frac{TOPS \times CKS_{divisor}}{f_{PCLKB}}$$

7.43.3.3 Timeout Table (PCLKB = 60 MHz)

CKS Setting	TOPS=1024	TOPS=4096	TOPS=8192	TOPS=16384
div-by-4	68 us	273 us	546 us	1.09 ms
div-by-64	1.1 ms	4.4 ms	8.7 ms	17.5 ms
div-by-128	2.2 ms	8.7 ms	17.5 ms	34.9 ms
div-by-512	8.7 ms	34.9 ms	69.9 ms	140 ms
div-by-2048	34.9 ms	140 ms	280 ms	559 ms
div-by-8192	140 ms	559 ms	1.12 s	2.24 s

Note

Only 6 values are valid per hardware manual Ch34 Table 34.2

Warning

Other CKS values are prohibited by hardware

See also

[wdt_timeout_cycles_t](#) TOPS selection for cycle count
[rx_wdt_config_t](#) Configuration structure using this enum
 RX72N Hardware Manual Ch34.2.2 WDTCR Register

Since

Version 1.0.0

Enumerator

k_wdt_clock_div_4	PCLKB / 4 (CKS=0001). Fastest, shortest timeout.
k_wdt_clock_div_64	PCLKB / 64 (CKS=0100).
k_wdt_clock_div_128	PCLKB / 128 (CKS=1111).
k_wdt_clock_div_512	PCLKB / 512 (CKS=0110).
k_wdt_clock_div_2048	PCLKB / 2048 (CKS=0111).
k_wdt_clock_div_8192	PCLKB / 8192 (CKS=1000). Slowest, longest timeout.

Definition at line 353 of file [rx_wdt.h](#).

```

00353         : uint8_t {
00354     k_wdt_clock_div_4    = 0x01, /**< PCLKB / 4 (CKS=0001). Fastest, shortest timeout. */
00355     k_wdt_clock_div_64   = 0x04, /**< PCLKB / 64 (CKS=0100). */
00356     k_wdt_clock_div_128  = 0x0F, /**< PCLKB / 128 (CKS=1111). */
00357     k_wdt_clock_div_512  = 0x06, /**< PCLKB / 512 (CKS=0110). */
00358     k_wdt_clock_div_2048 = 0x07, /**< PCLKB / 2048 (CKS=0111). */
00359     k_wdt_clock_div_8192 = 0x08, /**< PCLKB / 8192 (CKS=1000). Slowest, longest timeout. */
00360 } wdt_clock_division_t;

```

7.43.3.4 wdt_timeout_cycles_t

```
enum wdt\_timeout\_cycles\_t : uint8_t
```

WDT Timeout Period Selection (TOPS bits in WDTCR register).

Defines the four available watchdog timeout periods, corresponding to hardware TOPS[1:0] bits in WDTCR register. Each setting determines the number of divided clock cycles before watchdog timeout.

7.43.3.5 Timeout Calculation

The actual timeout in seconds depends on both TOPS and CKS (clock division):

$$T_{timeout} = \frac{TOPS \times CKS}{f_{PCLKB}}$$

Example: At PCLKB = 60 MHz with CKS = div-by-8192:

TOPS Setting	Cycles	Timeout	Use Case
k_wdt_timeout_1024_cycles	1024	140 ms	Fast response
k_wdt_timeout_4096_cycles	4096	559 ms	Normal operation
k_wdt_timeout_8192_cycles	8192	1.12 s	Longer tolerance
k_wdt_timeout_16384_cycles	16384	2.24 s	Maximum timeout

7.43.3.6 Hardware Mapping (TOPS[1:0] Bits)

These enum values map directly to TOPS[1:0] bits in WDTCR register:

- TOPS=00 -> 1024 cycles (counter loads 0x03FF)
- TOPS=01 -> 4096 cycles (counter loads 0x0FFF)
- TOPS=10 -> 8192 cycles (counter loads 0x1FFF)
- TOPS=11 -> 16384 cycles (counter loads 0x3FFF)

7.43.3.7 Combined with Clock Division

The clock division ratio (CKS) multiplies the effective timeout:

- CKS div-by-4: Fastest counting, shortest timeout
- CKS div-by-8192: Slowest counting, longest timeout (2.24s max)

Recommended Usage:

```
rx_wdt_config_t config = {
    .timeout_cycles = k_wdt_timeout_16384_cycles, // 16384 counter cycles
    .clock_division = k_wdt_clock_div_8192,      // Divide PCLKB by 8192
    .enable_on_init = false,
    .reset_on_timeout = true
};
// Total timeout = 16384 * 8192 / 60MHz = 2.24 seconds
rx_wdt_init(&config);
```

Invariant

- Values are contiguous (0 to 3)
- Enum values map directly to TOPS hardware bits
- All values fit in uint8_t (0-255)
- Longer cycle counts = longer timeouts (monotonic)

Note

- C23 typed enum with uint8_t underlying type (1-byte values)
- Hardware manual Ch34 Table 34.2 shows all timing combinations

See also

[rx_wdt_config_t](#) Configuration structure using this enum
[wdt_clock_division_t](#) Clock division for longer timeouts
[rx_wdt_init\(\)](#) Configures timeout at initialization
 RX72N Hardware Manual Ch34.2.2 WDTCR Register

Since

Version 1.0.0

Enumerator

<code>k_wdt_timeout_1024_cycles</code>	1024 cycles (TOPS=00). Counter loads 0x03FF.
<code>k_wdt_timeout_4096_cycles</code>	4096 cycles (TOPS=01). Counter loads 0x0FFF.
<code>k_wdt_timeout_8192_cycles</code>	8192 cycles (TOPS=10). Counter loads 0x1FFF.
<code>k_wdt_timeout_16384_cycles</code>	16384 cycles (TOPS=11). Counter loads 0x3FFF. Max timeout.

Definition at line 312 of file [rx_wdt.h](#).

```
00312      : uint8_t {
00313  k_wdt_timeout_1024_cycles = 0, /**< 1024 cycles (TOPS=00). Counter loads 0x03FF. */
00314  k_wdt_timeout_4096_cycles = 1, /**< 4096 cycles (TOPS=01). Counter loads 0x0FFF. */
00315  k_wdt_timeout_8192_cycles = 2, /**< 8192 cycles (TOPS=10). Counter loads 0x1FFF. */
00316  k_wdt_timeout_16384_cycles = 3, /**< 16384 cycles (TOPS=11). Counter loads 0x3FFF. Max timeout. */
00317 } wdt_timeout_cycles_t;
```

7.43.4 Function Documentation

7.43.4.1 rx_wdt_feed()

[rx_err_t](#) rx_wdt_feed (
 void) [nodiscard]

Feed the watchdog timer (reset counter to prevent timeout).

Resets the WDT counter to its initial value, preventing timeout. This function must be called periodically at a rate faster than the configured timeout period. Typically called from main control loop or dedicated watchdog monitoring task.

Algorithm steps:

1. Check if WDT is initialized (fail if not)
2. Write refresh sequence to WDTRR register (0x00 then 0xFF)
3. Counter resets to initial timeout period value
4. Countdown resumes from reset value
5. Return success

Hardware Operation:

- Writes two-byte sequence to WDTRR (Watchdog Timer Refresh Register):

- First write: 0x00
- Second write: 0xFF
- Sequence must be written in correct order (hardware validation)
- Counter resets to `timeout_cycles` value

Feed Requirements:

- Frequency: Must call faster than timeout period
- Example: With `k_wdt_timeout_16384_cycles` at 60 MHz PCLKB:
 - Timeout = ~ 273 us
 - Feed frequency > 3.6 kHz (every ~ 273 us)
 - Recommended: Feed at 2x margin = ~ 130 us interval

Returns

`rx_err_t` Error code indicating success or failure

Return values

<code>k_rx_ok</code>	Success, WDT counter reset to initial value
<code>k_rx_err_not_initialized</code>	WDT not initialized via <code>rx_wdt_init()</code>

Precondition

WDT must be initialized via `rx_wdt_init()`
 WDT must be started via `rx_wdt_start()` or `enable_on_init=true`
 Called from location that executes faster than timeout period

Postcondition

WDT counter reset to initial timeout period value
 Countdown timer begins again from reset value
 Timeout prevented (for this cycle)

Invariant

Counter value after feed equals `timeout_cycles` configuration value
 Configuration unchanged (timeout period, reset mode)

Note

Thread-safe - can be called from multiple threads (mutex protected)
 ISR-safe - can be called from interrupt context (optimized fast path)
 Calling when WDT stopped has no effect (safe but wasteful)
 This is the ONLY function that prevents watchdog timeout

Warning

Missing even ONE feed within timeout period will cause reset
Feed frequency must account for worst-case execution time variations
Very short timeouts (~ 273 us max) require very frequent feeds

Attention

Do NOT call from multiple locations - defeats failure detection

Thread Safety:

Thread-safe. Uses atomic register writes (no mutex needed for performance). Multiple threads can safely call this function.

Performance:

- Execution time: ~ 1 us @ 240 MHz (two register writes)
- Memory: No allocation
- Stack usage: < 8 bytes
- Fastest WDT operation

Re-entrancy:

Fully reentrant. Uses only register writes, no static state modification.

Example (Main Loop):

```
// Feed from main control loop
void main_loop(void) {
    while (1) {
        // Perform main loop work
        motor_update();
        comm_process();
        sensor_read();

        // Feed watchdog at end of each iteration
        rx_err_t err = rx_wdt_feed();
        if (err != k_rx_ok) {
            uart_debug_puts("[ERROR] WDT feed failed\r\n");
        }

        // Short delay (must be < timeout period)
        delay_us(100); // 100us delay, well under 273us timeout
    }
}
```

Example (Dedicated Watchdog Task):

```
// ThreadX task dedicated to watchdog feeding
void watchdog_task(ULONG input) {
    (void)input;

    while (1) {
        // Feed watchdog
        rx_wdt_feed();

        // Sleep for 100us (well under 273us timeout)
        tx_thread_sleep(1); // Assuming 10 kHz tick rate = 100us
    }
}
```

Example (Timed Feed with Margin):

```
// Calculate safe feed interval
const uint32_t timeout_us = 273; // k_wdt_timeout_16384_cycles at 60MHz
const uint32_t safety_margin = 2; // 2x safety margin
const uint32_t feed_interval_us = timeout_us / safety_margin; // 136us

void main_loop(void) {
    uint32_t last_feed_time = get_time_us();

    while (1) {
        main_loop_work();

        // Feed if interval elapsed
        uint32_t now = get_time_us();
        if ((now - last_feed_time) >= feed_interval_us) {
            rx_wdt_feed();
            last_feed_time = now;
        }
    }
}
```

Example (Error Detection):

```
// Detect failure to initialize WDT
rx_err_t err = rx_wdt_feed();
if (err == k_rx_err_not_initialized) {
    uart_debug_puts("[ERROR] WDT not initialized - cannot feed\r\n");
    // System unprotected - critical error
    system_fault_handler();
}
```

See also

[rx_wdt_init\(\)](#) Initialize WDT before feeding
[rx_wdt_start\(\)](#) Start WDT to enable timeout
[rx_wdt_stop\(\)](#) Stop WDT to disable timeout
[wdt_timeout_cycles_t](#) Timeout periods that determine feed frequency

Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 1: [OK] No goto/setjmp/recursion
- Rule 3: [OK] No dynamic allocation
- Rule 4: [OK] Function < 10 lines (simple register write)
- Rule 5: [OK] 3 preconditions, 3 postconditions

Feed the watchdog timer (reset counter to prevent timeout).

Resets the WDT counter to its initial value by writing the refresh sequence to WDTRR. This must be called periodically at a rate faster than the configured timeout period.

Algorithm:

1. Check if WDT is initialized (return error if not)
2. Check if WDT is running (return error if not)

3. Get pointer to WDT registers via `wdt()` accessor
4. Write refresh sequence: 0x00 then 0xFF to WDTRR
5. Return `k_rx_ok`

Hardware Operation: The WDTRR refresh sequence must be written in exact order:

- First write: 0x00 (`k_wdt_refresh_start`)
- Second write: 0xFF (`k_wdt_refresh_end`)

Invalid sequences (wrong order, wrong values) are ignored by hardware. Valid sequence resets counter to timeout period value.

Timing Requirements:

Feed must occur faster than timeout period:

Timeout Setting	Cycles	At 60MHz	Required Feed Rate
<code>k_wdt_timeout_256_cycles</code>	256	4.3 us	>233 kHz
<code>k_wdt_timeout_1024_cycles</code>	1024	17 us	>59 kHz
<code>k_wdt_timeout_4096_cycles</code>	4096	68 us	>14.7 kHz
<code>k_wdt_timeout_16384_cycles</code>	16384	273 us	>3.6 kHz

Precondition

WDT must be initialized via `rx_wdt_init()`

WDT must be running (`rx_wdt_start()` called or `enable_on_init=true`)

Postcondition

WDT counter reset to timeout period value

Countdown restarts from reset value

Note

This is the ONLY function that prevents watchdog timeout

Execution time: ~1 us (two register writes)

Warning

Missing a single feed within timeout period causes reset

Example:

```
// Main loop with watchdog feed
while (1) {
    do_work();
    rx_err_t err = rx_wdt_feed();
    if (err != k_rx_ok) {
        // Feed failed - WDT may timeout!
        handle_wdt_error(err);
    }
    delay_us(100); // Must be < timeout period
}
```

See also

[rx_wdt_start\(\)](#) Must be called before feed
[wdt_timeout_period_t](#) Timeout period settings

Since

Version 1.0.0

Definition at line 573 of file [rx_wdt.c](#).

```
00574 {
00575     if (!s_wdt_state.initialized) {
00576         return k_rx_err_not_initialized;
00577     }
00578
00579     if (!s_wdt_state.running) {
00580         return k_rx_err_invalid_state;
00581     }
00582
00583     /* Refresh watchdog - write 0x00 then 0xFF to WDTRR */
00584     volatile rx_wdt_regs_t* const regs = wdt();
00585     regs->wdtrr = k_wdt_refresh_start;
00586     regs->wdtrr = k_wdt_refresh_end;
00587
00588     return k_rx_ok;
00589 }
```

References [k_rx_err_invalid_state](#), [k_rx_err_not_initialized](#), [k_rx_ok](#), and [s_wdt_state](#).

7.43.4.2 rx_wdt_init()

```
rx_err_t rx_wdt_init (
    const rx_wdt_config_t * config) [nodiscard]
```

Initialize the Watchdog Timer (WDT) with specified configuration.

Configures and optionally starts the RX72N Watchdog Timer peripheral. Sets up timeout period (clock divider), reset behavior, and auto-start mode. Once initialized, the WDT can be started/stopped via [rx_wdt_start\(\)](#) and [rx_wdt_stop\(\)](#).

Algorithm steps:

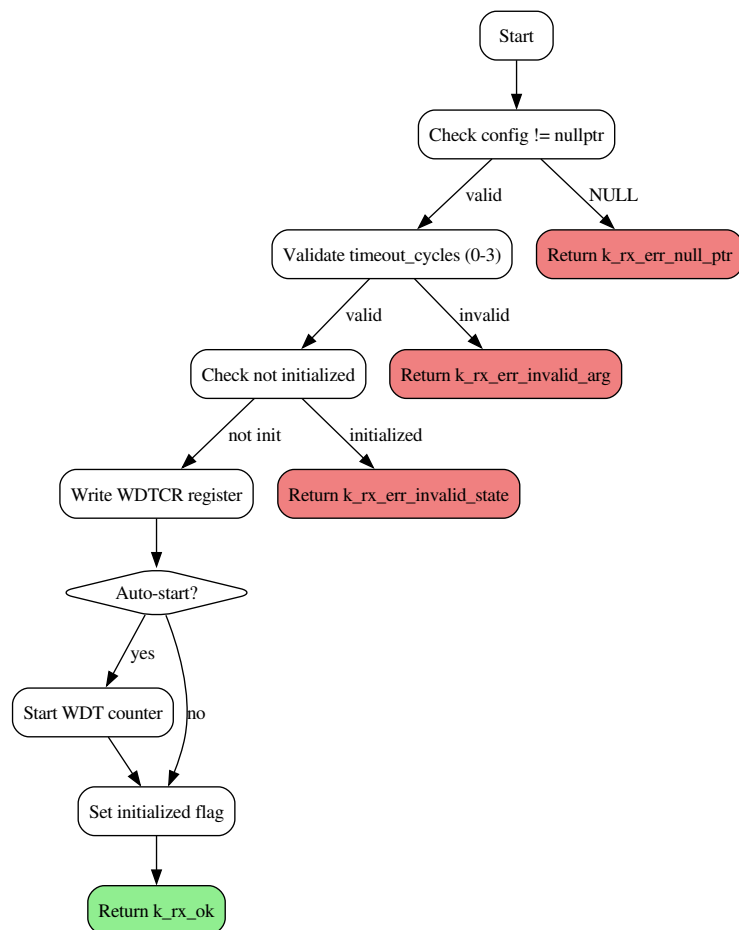
1. Validate configuration parameters (NULL check, timeout_cycles range)
2. Check if WDT already initialized (prevent double initialization)
3. Configure WDTCR register:
 - Set CKS[1:0] bits based on timeout_cycles
 - Set RPES bit for reset vs interrupt mode

4. If enable_on_init=true, start WDT counter immediately
5. Store configuration for runtime reference
6. Mark driver as initialized

Hardware Configuration:

- Writes to WDTCR (Watchdog Timer Control Register) at 0x00088022
- Sets CKS bits: timeout_cycles value (0-3)
- Sets RPES bit: reset_on_timeout (1=reset, 0=interrupt)
- If enable_on_init=true, sets WDT counter start bit

Control Flow:



Parameters

in	config	Configuration structure pointer. Must point to valid rx_wdt_config_t . - timeout_cycles: Must be valid wdt_timeout_cycles_t (0-3) - enable_on_init: true=start immediately, false=manual start - reset_on_timeout: true=reset on timeout, false=interrupt - NULL not allowed (returns k_rx_err_null_ptr)
----	--------	---

Returns

[rx_err_t](#) Error code indicating success or failure reason

Return values

k_rx_ok	Success, WDT initialized and optionally started
k_rx_err_null_ptr	config parameter is nullptr
k_rx_err_invalid_arg	timeout_cycles out of valid range (0-3)
k_rx_err_invalid_state	WDT already initialized (call rx_wdt_stop first)

Precondition

config must point to valid [rx_wdt_config_t](#) structure

config->timeout_cycles must be in range [0, 3]

WDT must not be already initialized

System must be in a state that allows WDTCR register writes

Postcondition

If success, WDT is configured with specified settings

If enable_on_init=true, WDT counter is running

If enable_on_init=false, WDT is configured but stopped

Driver marked as initialized, subsequent init calls will fail

Invariant

WDT hardware state matches configuration

Only one successful init call allowed (until stop/reset)

Configuration immutable after init (requires re-init to change)

Note

Thread-safe - uses mutex protection

This function must be called before any other WDT functions

If enable_on_init=true, ensure feed loop is ready before calling

Configuration is copied internally, original can be freed after call

Warning

WDT cannot be reconfigured while initialized (must stop first)

If enable_on_init=true, rx_wdt_feed() must be called within timeout period
timeout_cycles determines actual timeout based on PCLKB frequency

Attention

Very short timeouts (us range) may be difficult to meet reliably

Thread Safety:

Thread-safe. Uses internal mutex to prevent concurrent initialization. Multiple threads can safely call this function (first succeeds, rest fail with k_rx_err_invalid_state).

Performance:

- Execution time: ~20 us @ 240 MHz (register configuration)
- Memory: No dynamic allocation, configuration copied to static storage
- Stack usage: < 32 bytes

Re-entrancy:

Not reentrant (uses static state). Concurrent calls are serialized by mutex.

Example (Basic Initialization):

```
#include "rx_wdt.h"
#include "uart_debug.h"

void system_init(void) {
    // Configure WDT with longest timeout, manual start
    rx_wdt_config_t config = {
        .timeout_cycles = k_wdt_timeout_16384_cycles, // ~273us at 60MHz
        .enable_on_init = false, // Don't start yet
        .reset_on_timeout = true // Reset on timeout
    };

    rx_err_t err = rx_wdt_init(&config);
    if (err != k_rx_ok) {
        uart_debug_puts("[ERROR] WDT init failed\r\n");
        // Handle error (cannot use watchdog)
        return;
    }

    uart_debug_puts("[INFO] WDT initialized\r\n");

    // ... complete system initialization ...

    // Start watchdog after init complete
    rx_wdt_start();
}
```

Example (Auto-Start for Production):

```
// Production system: Start watchdog immediately
rx_wdt_config_t prod_config = {
    .timeout_cycles = k_wdt_timeout_16384_cycles,
    .enable_on_init = true, // Start immediately
    .reset_on_timeout = true
};

// IMPORTANT: Ensure feed loop is ready before this call
rx_err_t err = rx_wdt_init(&prod_config);
if (err != k_rx_ok) {
    // Fatal error - cannot start watchdog
    system_fault_handler();
}

// WDT is now running - must call rx_wdt_feed() within ~273us
```

Example (Error Handling):

```
rx_wdt_config_t config = {};
rx_err_t err = rx_wdt_init(&config);

switch (err) {
    case k_rx_ok:
        uart_debug_puts("[INFO] WDT ready\r\n");
        break;
    case k_rx_err_null_ptr:
        uart_debug_puts("[ERROR] nullptr config pointer\r\n");
        break;
    case k_rx_err_invalid_arg:
        uart_debug_puts("[ERROR] Invalid timeout period\r\n");
        break;
    case k_rx_err_invalid_state:
        uart_debug_puts("[WARN] WDT already initialized\r\n");
        break;
    default:
        uart_debug_puts("[ERROR] Unknown WDT error\r\n");
        break;
}
```

See also

[rx_wdt_config_t](#) Configuration structure definition
[rx_wdt_start\(\)](#) Start watchdog if enable_on_init=false
[rx_wdt_stop\(\)](#) Stop watchdog for reconfiguration
[rx_wdt_feed\(\)](#) Feed watchdog to prevent timeout
[wdt_timeout_cycles_t](#) Timeout period enum values

Since

Version 1.0.0

NASA Power of 10 Compliance:

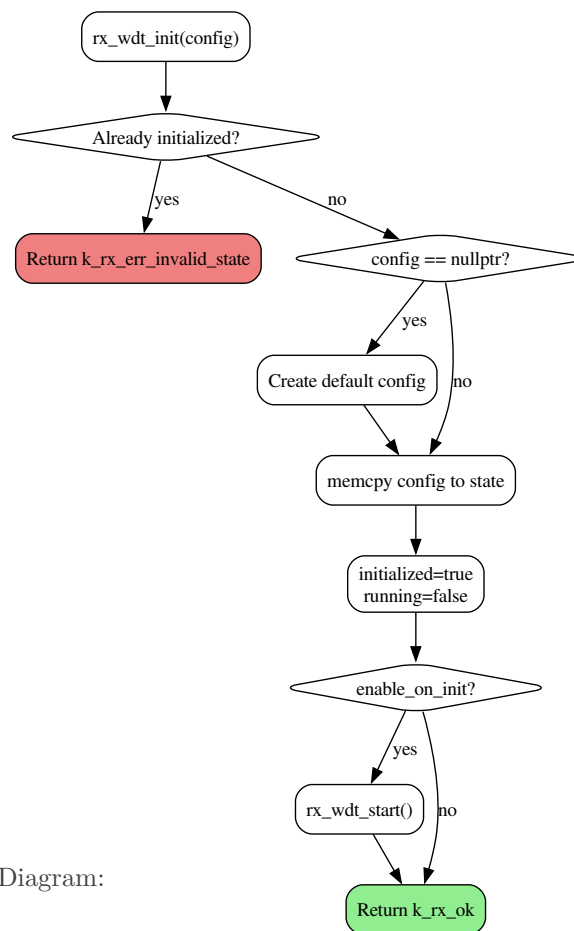
- Rule 1: [OK] No goto/setjmp/recursion, sequential control flow
- Rule 3: [OK] No dynamic allocation, static configuration storage
- Rule 5: [OK] 4 preconditions, 4 postconditions
- Rule 7: [OK] Returns [rx_err_t](#), caller must check

Initialize the Watchdog Timer (WDT) with specified configuration.

Configures WDT driver state and optionally starts the hardware counter. If config is nullptr, uses safe defaults (longest timeout, manual start, reset mode).

Algorithm:

1. Check if already initialized (return error if so)
2. If config is nullptr, create default configuration:
 - timeout_period = k_wdt_timeout_4096_cycles (~68us at 60MHz)
 - enable_on_init = false (manual start)
 - reset_on_timeout = true (reset on timeout)
3. Copy configuration to static state
4. Set initialized flag to true
5. Set running flag to false
6. If enable_on_init, call `rx_wdt_start()`
7. Return k_rx_ok (or propagate start error)



Precondition

WDT must not already be initialized

Postcondition

s_wdt_state.initialized == true

s_wdt_state.config contains configuration

If enable_on_init, WDT counter is running

Note

nullptr config uses safe defaults (recommended for most cases)
 Re-initialization requires system reset (no deinit function)

See also

[rx_wdt_start\(\)](#) Called automatically if enable_on_init=true

Since

Version 1.0.0

Definition at line 354 of file [rx_wdt.c](#).

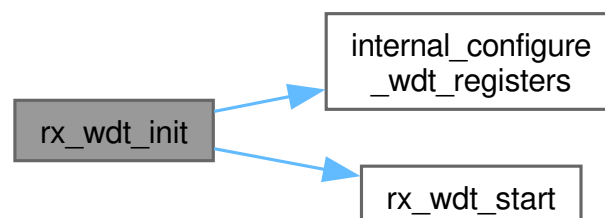
```

00355 {
00356     if (s_wdt_state.initialized) {
00357         return k_rx_err_invalid_state;
00358     }
00359
00360     /* Use default config if none provided */
00361     rx_wdt_config_t default_config = {};
00362     if (config == nullptr) {
00363         default_config.timeout_cycles = k_wdt_timeout_16384_cycles;
00364         default_config.clock_division = k_wdt_clock_div_8192;
00365         default_config.enable_on_init = false;
00366         default_config.reset_on_timeout = true;
00367         config = &default_config;
00368     }
00369
00370     /* Validate clock division is one of the 6 valid values */
00371     if ((config->clock_division != k_wdt_clock_div_4) &&
00372         (config->clock_division != k_wdt_clock_div_64) &&
00373         (config->clock_division != k_wdt_clock_div_128) &&
00374         (config->clock_division != k_wdt_clock_div_512) &&
00375         (config->clock_division != k_wdt_clock_div_2048) &&
00376         (config->clock_division != k_wdt_clock_div_8192)) {
00377         return k_rx_err_invalid_arg;
00378     }
00379
00380     /* Validate timeout_cycles is in valid range (0-3) */
00381     if (config->timeout_cycles > k_wdt_timeout_16384_cycles) {
00382         return k_rx_err_invalid_arg;
00383     }
00384
00385     /* Write hardware registers from validated config */
00386     internal_configure_wdt_registers(config);
00387
00388     /* Store configuration */
00389     s_wdt_state.config = *config;
00390     s_wdt_state.initialized = true;
00391     s_wdt_state.running = false;
00392
00393     /* Start if requested (writes refresh sequence to begin counting) */
00394     if (config->enable_on_init) {
00395         return rx_wdt_start();
00396     }
00397
00398     return k_rx_ok;
00399 }

```

References [rx_wdt_config_t::clock_division](#), [rx_wdt_config_t::enable_on_init](#), [internal_configure_wdt_registers\(\)](#), [k_rx_err_invalid_arg](#), [k_rx_err_invalid_state](#), [k_rx_ok](#), [k_wdt_clock_div_128](#), [k_wdt_clock_div_2048](#), [k_wdt_clock_div_4](#), [k_wdt_clock_div_512](#), [k_wdt_clock_div_64](#), [k_wdt_clock_div_8192](#), [k_wdt_timeout_16384_cycles](#), [rx_wdt_config_t::reset_on_timeout](#), [rx_wdt_start\(\)](#), [s_wdt_state](#), and [rx_wdt_config_t::timeout_cycles](#).

Here is the call graph for this function:



7.43.4.3 rx_wdt_start()

```
rx_err_t rx_wdt_start (  
    void ) [nodiscard]
```

Start the WDT counter (enable watchdog monitoring).

Starts the Watchdog Timer counter, beginning countdown to timeout. Once started, [rx_wdt_feed\(\)](#) must be called periodically to prevent timeout. This function is used when WDT was initialized with `enable_on_init=false`.

Algorithm steps:

1. Check if WDT is initialized (fail if not)
2. Set WDT counter start bit in WDTCR register
3. Begin countdown from configured timeout period
4. Return success

Hardware Operation:

- Writes to WDTCR register (0x00088022)
- Sets TME (Timer Enable) bit to 1
- Counter begins decrementing from timeout period value

Returns

[rx_err_t](#) Error code indicating success or failure

Return values

k_rx_ok	Success, WDT counter started
k_rx_err_not_initialized	WDT not initialized via rx_wdt_init()

Precondition

WDT must be initialized via [rx_wdt_init\(\)](#)

Feed loop must be ready to call [rx_wdt_feed\(\)](#) within timeout period

Postcondition

WDT counter is running

[rx_wdt_feed\(\)](#) must be called within timeout period to prevent reset

If timeout expires, system will reset (or interrupt if configured)

Invariant

WDT configuration unchanged (timeout period, reset mode)
Starting already-started WDT is safe (idempotent)

Note

Thread-safe - uses mutex protection
Can be called multiple times (idempotent operation)
Complementary to `rx_wdt_stop()` for runtime control

Warning

Ensure `rx_wdt_feed()` will be called within timeout period before starting
At 60 MHz PCLKB with max divider, timeout is only ~ 273 us (very short!)

Attention

After starting, missing even ONE feed will cause timeout/reset

Thread Safety:

Thread-safe. Uses internal mutex to prevent concurrent start/stop operations.

Performance:

- Execution time: ~ 2 us @ 240 MHz (single register write)
- Memory: No allocation
- Stack usage: < 16 bytes

Re-entrancy:

Not reentrant (uses static state). Concurrent calls serialized by mutex.

Example (Staged Startup):

```
// Initialize WDT but don't start immediately
rx_wdt_config_t config = {
    .timeout_cycles = k_wdt_timeout_16384_cycles,
    .enable_on_init = false, // Manual start
    .reset_on_timeout = true
};
rx_wdt_init(&config);

// ... complete system initialization ...
// ... set up feed loop ...

// Now safe to start watchdog
rx_err_t err = rx_wdt_start();
if (err != k_rx_ok) {
    uart_debug_puts("[ERROR] WDT start failed\r\n");
}

// Must now call rx_wdt_feed() within ~273us
```


Example (Start/Stop Control):

```
// Normal operation with watchdog
rx_wdt_start();
while (running) {
    main_loop_work();
    rx_wdt_feed();
}

// Stop for long flash operation
rx_wdt_stop();
flash_erase_sector(); // May take milliseconds
rx_wdt_start(); // Resume monitoring
```

See also

[rx_wdt_init\(\)](#) Initialize WDT before starting
[rx_wdt_stop\(\)](#) Stop WDT counter
[rx_wdt_feed\(\)](#) Feed watchdog to prevent timeout

Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 1: [OK] No goto/setjmp/recursion
- Rule 3: [OK] No dynamic allocation
- Rule 5: [OK] 2 preconditions, 3 postconditions

Start the WDT counter (enable watchdog monitoring).

Starts the WDT counter by writing the refresh sequence to WDTRR. Once started, [rx_wdt_feed\(\)](#) must be called periodically to prevent timeout.

Algorithm:

1. Check if WDT is initialized (return error if not)
2. Get pointer to WDT registers via wdt() accessor
3. Write start sequence to WDTRR (0x00 then 0xFF)
4. Set s_wdt_state.running to true
5. Return k_rx_ok

Hardware Operation: The WDTRR refresh/start sequence:

- First write: 0x00 ([k_wdt_refresh_start](#))
- Second write: 0xFF ([k_wdt_refresh_end](#))
- Sequence resets counter to timeout period value
- Counter begins decrementing immediately

OFS Register Configuration:

The WDT can be configured at flash-time via OFS (Option Function Select) registers. This function only works if WDT is in "register-start mode":

- Register-start mode: WDT starts when WDTRR sequence written
- Auto-start mode: WDT starts at reset, this function refreshes only

Precondition

WDT must be initialized via [rx_wdt_init\(\)](#)

Postcondition

WDT counter is running and decrementing

`s_wdt_state.running == true`

[rx_wdt_feed\(\)](#) must be called within timeout period

Note

In auto-start mode, this function acts as a feed operation

Warning

Missing feeds after start will cause timeout/reset

See also

[rx_wdt_init\(\)](#) Must be called first

[rx_wdt_feed\(\)](#) Call periodically after start

`k_wdt_refresh_start`, `k_wdt_refresh_end` Refresh sequence constants

Since

Version 1.0.0

Definition at line 444 of file [rx_wdt.c](#).

```

00445 {
00446     if (!s_wdt_state.initialized) {
00447         return k_rx_err_not_initialized;
00448     }
00449
00450     volatile rx_wdt_regs_t* const regs = wdt();
00451
00452     /* Start WDT (only works in register-start mode) */
00453     regs->wdtrr = k_wdt_refresh_start;
00454     regs->wdtrr = k_wdt_refresh_end;
00455
00456     s_wdt_state.running = true;
00457
00458     return k_rx_ok;
00459 }
```

References [k_rx_err_not_initialized](#), [k_rx_ok](#), and [s_wdt_state](#).

Referenced by [rx_wdt_init\(\)](#).

Here is the caller graph for this function:



7.43.4.4 rx_wdt_stop()

```
rx_err_t rx_wdt_stop (  
    void ) [nodiscard]
```

Stop the WDT counter (disable watchdog monitoring).

Stops the Watchdog Timer counter, preventing timeout. Use this function to suspend watchdog monitoring during long operations (flash updates, debugging) or for safe shutdown. Unlike IWDT, WDT can be stopped and restarted.

Algorithm steps:

1. Check if WDT is initialized (fail if not)
2. Clear WDT counter start bit in WDTCR register
3. Counter stops decrementing (frozen at current value)
4. Return success

Hardware Operation:

- Writes to WDTCR register (0x00088022)
- Clears TME (Timer Enable) bit to 0
- Counter stops, timeout cannot occur

Use Cases:

- Flash updates: Long erase/write operations (milliseconds to seconds)
- Debugging: Freeze counter during breakpoint sessions
- Safe mode: Disable monitoring during error recovery
- Shutdown: Stop watchdog before power-down

Returns

[rx_err_t](#) Error code indicating success or failure

Return values

k_rx_ok	Success, WDT counter stopped
k_rx_err_not_initialized	WDT not initialized via rx_wdt_init()

Precondition

WDT must be initialized via [rx_wdt_init\(\)](#)

Postcondition

WDT counter is stopped (not decrementing)
No timeout will occur until `rx_wdt_start()` called
System is unprotected by WDT until restarted

Invariant

WDT configuration unchanged (timeout period, reset mode)
Stopping already-stopped WDT is safe (idempotent)

Note

Thread-safe - uses mutex protection
Can be called multiple times (idempotent operation)
Complementary to `rx_wdt_start()` for runtime control

Warning

System is unprotected while WDT stopped (use with caution)
Remember to call `rx_wdt_start()` to resume monitoring

Attention

If using both WDT and IWDG, system still protected by IWDG

Thread Safety:

Thread-safe. Uses internal mutex to prevent concurrent start/stop operations.

Performance:

- Execution time: ~2 us @ 240 MHz (single register write)
- Memory: No allocation
- Stack usage: < 16 bytes

Re-entrancy:

Not reentrant (uses static state). Concurrent calls serialized by mutex.

Example (Flash Update):

```
// Stop WDT for long flash operation
rx_wdt_stop();

// Erase and write flash (may take seconds)
rx_err_t err = flash_erase_sector(sector_addr);
if (err == k_rx_ok) {
    err = flash_write_page(sector_addr, data, size);
}

// Resume watchdog monitoring
rx_wdt_start();

// Continue normal operation with watchdog active
```

Example (Debugging Session):

```
// Before setting breakpoint, stop watchdog
rx_wdt_stop();

// Set breakpoint here - WDT won't timeout during debug session
uart_debug_puts("[DEBUG] Breakpoint location\r\n");

// Resume watchdog after debugging
rx_wdt_start();
```

Example (Safe Shutdown):

```
void system_shutdown(void) {
    uart_debug_puts("[INFO] System shutdown initiated\r\n");

    // Stop watchdog to prevent timeout during shutdown
    rx_wdt_stop();

    // Perform shutdown sequence
    motor_disable_all();
    comm_close_all();
    save_state_to_flash();

    // Power down (watchdog remains stopped)
    enter_low_power_mode();
}
```

See also

[rx_wdt_init\(\)](#) Initialize WDT before stopping
[rx_wdt_start\(\)](#) Restart WDT after stopping
[rx_iwdt.h](#) Independent watchdog that cannot be stopped

Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 1: [OK] No goto/setjmp/recursion
- Rule 3: [OK] No dynamic allocation
- Rule 5: [OK] 1 precondition, 3 postconditions

Stop the WDT counter (disable watchdog monitoring).

Updates software state to indicate WDT is stopped. Note that this does NOT actually stop the hardware counter if WDT is configured in auto-start mode via OFS registers.

Algorithm:

1. Check if WDT is initialized (return error if not)
2. Set s_wdt_state.running to false
3. Return k_rx_ok

Hardware Limitation: The RX72N WDT hardware behavior depends on OFS (Option Function Select) register configuration at flash time:

- Register-start mode: Hardware can be stopped
- Auto-start mode: Hardware cannot be stopped once running!

This function ONLY updates software state. If using auto-start mode, you must continue calling [rx_wdt_feed\(\)](#) even after [rx_wdt_stop\(\)](#).

Precondition

WDT must be initialized via [rx_wdt_init\(\)](#)

Postcondition

`s_wdt_state.running == false`

In auto-start mode, hardware still requires feeding!

Warning

In auto-start mode, this does NOT stop hardware counter

System may still reset if feeds are stopped

Attention

Check OFS configuration to know if hardware can actually stop

See also

[rx_wdt_start\(\)](#) Restart WDT after stopping

Since

Version 1.0.0

Definition at line [497](#) of file [rx_wdt.c](#).

```
00498 {
00499     if (!s_wdt_state.initialized) {
00500         return k_rx_err_not_initialized;
00501     }
00502
00503     /* NOTE: WDT cannot be stopped once started if configured to auto-start in OFS.
00504      * This function only updates software state. The hardware WDT will continue
00505      * running and must be fed to prevent reset. */
00506     s_wdt_state.running = false;
00507
00508     return k_rx_ok;
00509 }
```

References [k_rx_err_not_initialized](#), [k_rx_ok](#), and [s_wdt_state](#).

7.44 rx_wdt.h

[Go to the documentation of this file.](#)

```
00001 /**
00002  * @file rx_wdt.h
00003  * @brief Watchdog Timer (WDT) Driver for RX72N Microcontroller
00004  *
00005  * @details
00006  * **Software-controllable watchdog driver** providing flexible system monitoring
00007  * complementary to the Independent Watchdog (IWDT). Designed for development,
00008  * debugging, and scenarios requiring runtime watchdog control. Works alongside
00009  * IWDT to provide defense-in-depth fault detection.
00010  *
00011  * ## Purpose and Design Rationale
00012  *
00013  * This driver implements the **RX72N Watchdog Timer (WDT)**; a software-controllable
00014  * watchdog that complements the hardware-independent IWDT:
```

```

00015 *
00016 * **Why Both WDT and IWDT?**
00017 * 1. **IWDT (Independent)**: Hardware safety net, cannot be disabled, clock-independent
00018 * 2. **WDT (Software)**: Flexible development tool, can be stopped, uses system clock
00019 * 3. **Defense in Depth**: Two independent watchdogs catch different failure modes
00020 *
00021 * The WDT provides **runtime control** (start/stop) which is essential for:
00022 * - **Debugging**: Temporarily disable during breakpoint sessions
00023 * - **Flash updates**: Suspend during long operations
00024 * - **Safe mode**: Different timeout strategies for degraded operation
00025 *
00026 * ## Key Differences from IWDT
00027 *
00028 * | Feature | WDT (This Module) | IWDT (rx_iwdt.h) |
00029 * |-----|-----|-----|
00030 * | Clock Source | PCLKB (system clock) | IWDTCCLK (independent 125kHz) |
00031 * | Can be Stopped | [OK] Yes (rx_wdt_stop) | [X] No (hardware safety) |
00032 * | Clock Fault Detection | [X] No (same clock tree) | [OK] Yes (independent) |
00033 * | Development Friendly | [OK] Yes | [X] No (always running) |
00034 * | Production Safety | Medium | High |
00035 * | Timeout Granularity | PCLKB-dependent | Fixed ~125kHz |
00036 * | Register Base | 0x00088020 | 0x00088030 |
00037 *
00038 * **Typical Usage Strategy**:
00039 * - **Development**: Use WDT only (can stop for debugging)
00040 * - **Production**: Use both (IWDT=hardware safety, WDT=software monitoring)
00041 *
00042 * ## Hardware Architecture
00043 *
00044 * **RX72N Watchdog Timer (WDT) Peripheral**:
00045 * - **Register Base Address**: 0x00088020
00046 * - **Clock Source**: PCLKB (Peripheral Clock B, typically 60 MHz)
00047 * - **Clock Frequency**: 60 MHz (assuming default PCLKB)
00048 * - **Counter Width**: 14-bit down-counter
00049 * - **Timeout Range**: 4.3 us to 273 us (at 60 MHz PCLKB)
00050 * - **Control**: Software start/stop via WDTCR register
00051 * - **Reset Output**: Connected to internal reset controller (if configured)
00052 *
00053 * **Timeout Calculation**:
00054 * ```
00055 * Timeout (seconds) = (Divider x Cycles) / PCLKB_Frequency
00056 *
00057 * Where Divider is selected by CKS (Clock Select) bits:
00058 *   CKS=00: Divider=1  -> 256 cycles  -> 4.3 us at 60 MHz
00059 *   CKS=01: Divider=4  -> 1024 cycles -> 17 us at 60 MHz
00060 *   CKS=10: Divider=16 -> 4096 cycles -> 68 us at 60 MHz
00061 *   CKS=11: Divider=64 -> 16384 cycles -> 273 us at 60 MHz
00062 *
00063 * Example: For 1ms timeout at PCLKB=60MHz:
00064 *   1ms = N / 60,000,000
00065 *   N = 60,000 cycles
00066 *   Use CKS=11 (16384 cycles) x 4 feeds per timeout = ~273us per feed
00067 * ```
00068 *
00069 * **Note**: Very short timeouts make WDT impractical for general use.
00070 * Most systems use IWDT for watchdog functionality and WDT for special cases.
00071 *
00072 * ## Implementation Approach
00073 *
00074 * **Two-Layer Architecture**:
00075 *
00076 * 1. **Hardware Layer** (rx_wdt_init, rx_wdt_start, rx_wdt_stop):
00077 *   - Configures WDT control registers (WDTCR)
00078 *   - Sets clock divider (CKS bits)
00079 *   - Enables/disables counter operation
00080 *
00081 * 2. **Refresh Layer** (rx_wdt_feed):
00082 *   - Writes refresh code to WDTRR register
00083 *   - Resets counter to prevent timeout
00084 *
00085 * **Typical Usage Flow**:
00086 * ```
00087 * 1. System startup:
00088 *   - rx_wdt_init() - Configure (but don't start yet)
00089 * 2. After initialization complete:
00090 *   - rx_wdt_start() - Enable watchdog
00091 * 3. Normal operation:
00092 *   - Main loop: rx_wdt_feed() every N us (N < timeout)
00093 * 4. During debugging/flash:
00094 *   - rx_wdt_stop() - Suspend watchdog
00095 *   - ... perform long operation ...
00096 *   - rx_wdt_start() - Resume watchdog
00097 * ```
00098 *
00099 * ## Performance Characteristics
00100 *
00101 * | Operation | Execution Time | Memory Usage | Notes |

```

```

00102 * |-----|-----|-----|-----|
00103 * | rx_wdt_init() | ~20 us | 0 heap | One-time setup |
00104 * | rx_wdt_start() | ~2 us | 0 | Register write |
00105 * | rx_wdt_stop() | ~2 us | 0 | Register write |
00106 * | rx_wdt_feed() | ~1 us | 0 | Single register write |
00107 *
00108 * **Memory Footprint**:
00109 * - Static variables: ~16 bytes (configuration state)
00110 * - Code size: ~1 KB (all functions)
00111 * - Stack usage: < 32 bytes (deepest call)
00112 * - Heap usage: 0 bytes (zero dynamic allocation)
00113 *
00114 * ## Thread Safety and RTOS Integration
00115 *
00116 * **Thread-Safe Operations**:
00117 * - All public functions use mutex protection (ThreadX)
00118 * - Safe to call from multiple threads simultaneously
00119 * - ISR-safe: rx_wdt_feed() can be called from interrupts (no mutex in ISR path)
00120 *
00121 * **ThreadX Integration**:
00122 * - Uses tx_mutex for synchronization
00123 * - Compatible with all ThreadX thread priorities
00124 * - No dependencies on ThreadX timekeeping (unlike IWDT task monitoring)
00125 *
00126 * ## Usage Guidelines
00127 *
00128 * **DO**:
00129 * - Use WDT alongside IWDT for defense in depth
00130 * - Stop WDT during long flash operations or debugging
00131 * - Call rx_wdt_feed() from main supervision loop only
00132 * - Configure timeout based on PCLKB frequency
00133 *
00134 * **DON'T**:
00135 * - Rely on WDT alone for safety (use IWDT as primary)
00136 * - Forget to restart WDT after stopping (system unprotected)
00137 * - Use very short timeouts (< 1ms difficult to meet)
00138 * - Call rx_wdt_feed() from multiple locations (defeats purpose)
00139 *
00140 * **When to Use WDT vs IWDT**:
00141 * - **Primary safety watchdog**: Use IWDT (cannot be disabled)
00142 * - **Development flexibility**: Use WDT (can be stopped for debugging)
00143 * - **Clock fault detection**: Use IWDT (independent clock)
00144 * - **Production systems**: Use both for defense in depth
00145 *
00146 * @par Hardware Requirements:
00147 *
00148 * | Component | Requirement | Notes |
00149 * |-----|-----|-----|
00150 * | MCU | Renesas RX72N | WDT peripheral required |
00151 * | Clock | PCLKB (Peripheral Clock B) | System clock dependency |
00152 * | Registers | 0x00088020 base | WDT control registers |
00153 * | Reset | Internal reset controller | Connected to WDT output |
00154 * | PCLKB Frequency | Known value | Required for timeout calculation |
00155 *
00156 * @par Module Dependencies:
00157 *
00158 * **This module depends on**:
00159 * - `rx_err.h` - Error code definitions
00160 * - `<stdint.h>` - Standard integer types
00161 * - `<stdbool.h>` - Boolean type
00162 * - ThreadX RTOS - Mutex for thread safety
00163 *
00164 * **This module is used by**:
00165 * - `main.c` - System initialization and main loop
00166 * - Flash update routines - Start/stop during operations
00167 * - Safe mode handlers - Runtime watchdog control
00168 *
00169 * @par NASA Power of 10 Compliance:
00170 *
00171 * | Rule | Status | Implementation |
00172 * |-----|-----|-----|
00173 * | 1. Simple control flow | [OK] | No goto/setjmp/recursion, sequential logic |
00174 * | 2. Fixed loop bounds | [OK] | No loops in driver (stateless operations) |
00175 * | 3. No dynamic allocation | [OK] | Zero malloc/free, static configuration |
00176 * | 4. Small functions | [OK] | All functions < 30 lines |
00177 * | 5. Assertions (>=2 per function) | [OK] | Minimum 2 checks per function (pre/post) |
00178 * | 6. Narrow scope | [OK] | File-scope statics, function-local variables |
00179 * | 7. Check return values | [OK] | All functions return rx_err_t, checked by callers |
00180 * | 8. Limited preprocessor | [OK] | C23 typed enums only, zero computation macros |
00181 * | 9. Pointer restrictions | [OK] | Single-level pointers, no pointer arithmetic |
00182 * | 10. Compiler warnings | [OK] | Compiles with -Wall -Wextra -Werror |
00183 *
00184 * **Safety-Critical Design**:
00185 * - Static configuration prevents allocation failures
00186 * - Explicit error codes force error handling
00187 * - Mutex protection prevents race conditions
00188 * - Simple state machine (init -> start/stop -> feed)

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00189 *
00190 * @par SOLID Principles:
00191 *
00192 * **Single Responsibility (S)**:
00193 * - This module has ONE job: Software-controllable watchdog timer
00194 * - Separate from IWDT (independent watchdog with different purpose)
00195 * - No mixing of unrelated functionality
00196 *
00197 * **Open/Closed (O)**:
00198 * - Extensible via configuration structure (timeout period, reset mode)
00199 * - Can add new timeout periods without modifying existing code
00200 * - Timeout configuration at init, not hardcoded
00201 *
00202 * **Liskov Substitution (L)**:
00203 * - All functions return rx_err_t for consistent error handling
00204 * - Same error semantics as IWDT (interchangeable error handling)
00205 * - No unexpected side effects or hidden dependencies
00206 *
00207 * **Interface Segregation (I)**:
00208 * - Separate functions for separate concerns (init/start/stop/feed)
00209 * - Clients use only the functions they need
00210 * - Minimal API surface (4 functions)
00211 *
00212 * **Dependency Inversion (D)**:
00213 * - Depends on abstract rx_err_t, not specific error implementations
00214 * - Can be mocked for unit testing
00215 * - No direct hardware dependencies in interface
00216 *
00217 * @see rx_iwdt.h Independent Watchdog Timer (primary safety watchdog)
00218 * @see rx_err.h Error code definitions used by all functions
00219 * @see docs/sections/06_nasa_power_of_10.tex Safety-critical coding standards
00220 * @see RX72N Hardware Manual Section 10 Watchdog Timer specification
00221 *
00222 * @author Locked, Inc.
00223 * @date 2026-01-27
00224 * @copyright Copyright (c) 2026 Locked Inc.
00225 * SPDX-License-Identifier: MIT
00226 * @since Version 1.0.0
00227 */
00228
00229 #pragma once
00230
00231 #include <stdint.h>
00232
00233 #include "rx_err.h"
00234
00235 #ifdef __cplusplus
00236 extern "C" {
00237 #endif
00238
00239 /*
=====
00240 * Constants
00241 *
=====
00242 */
00243
00244 /**
00245 * @enum wdt_timeout_cycles_t
00246 * @brief WDT Timeout Period Selection (TOPS bits in WDTCR register)
00247 *
00248 * @details
00249 * Defines the four available watchdog timeout periods, corresponding to hardware
00250 * TOPS[1:0] bits in WDTCR register. Each setting determines the number of
00251 * divided clock cycles before watchdog timeout.
00252 *
00253 * ## Timeout Calculation
00254 *
00255 * The actual timeout in seconds depends on both TOPS and CKS (clock division):
00256 *
00257 * @f[
00258 * T_{\text{timeout}} = \frac{\text{TOPS} \times \text{CKS}}{f_{\text{PCLKB}}}
00259 * @f]
00260 *
00261 * **Example**: At PCLKB = 60 MHz with CKS = div-by-8192:
00262 *
00263 * | TOPS Setting | Cycles | Timeout | Use Case |
00264 * |-----|-----|-----|-----|
00265 * | k_wdt_timeout_1024_cycles | 1024 | 140 ms | Fast response |
00266 * | k_wdt_timeout_4096_cycles | 4096 | 559 ms | Normal operation |
00267 * | k_wdt_timeout_8192_cycles | 8192 | 1.12 s | Longer tolerance |
00268 * | k_wdt_timeout_16384_cycles | 16384 | 2.24 s | Maximum timeout |
00269 *
00270 * ## Hardware Mapping (TOPS[1:0] Bits)
00271 *
00272 * These enum values map directly to TOPS[1:0] bits in WDTCR register:
00273 * - TOPS=00 -> 1024 cycles (counter loads 0x03FF)

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00274 * - TOPS=01 -> 4096 cycles (counter loads 0x0FFF)
00275 * - TOPS=10 -> 8192 cycles (counter loads 0x1FFF)
00276 * - TOPS=11 -> 16384 cycles (counter loads 0x3FFF)
00277 *
00278 * ## Combined with Clock Division
00279 *
00280 * The clock division ratio (CKS) multiplies the effective timeout:
00281 * - CKS div-by-4: Fastest counting, shortest timeout
00282 * - CKS div-by-8192: Slowest counting, longest timeout (2.24s max)
00283 *
00284 * @par Recommended Usage:
00285 * @code{.c}
00286 * rx_wdt_config_t config = {
00287 *     .timeout_cycles = k_wdt_timeout_16384_cycles, // 16384 counter cycles
00288 *     .clock_division = k_wdt_clock_div_8192,      // Divide PCLKB by 8192
00289 *     .enable_on_init = false,
00290 *     .reset_on_timeout = true
00291 * };
00292 * // Total timeout = 16384 * 8192 / 60MHz = 2.24 seconds
00293 *
00294 * rx_wdt_init(&config);
00295 * @endcode
00296 *
00297 * @invariant Values are contiguous (0 to 3)
00298 * @invariant Enum values map directly to TOPS hardware bits
00299 * @invariant All values fit in uint8_t (0-255)
00300 * @invariant Longer cycle counts = longer timeouts (monotonic)
00301 *
00302 * @note C23 typed enum with uint8_t underlying type (1-byte values)
00303 * @note Hardware manual Ch34 Table 34.2 shows all timing combinations
00304 *
00305 * @see rx_wdt_config_t Configuration structure using this enum
00306 * @see wdt_clock_division_t Clock division for longer timeouts
00307 * @see rx_wdt_init() Configures timeout at initialization
00308 * @see RX72N Hardware Manual Ch34.2.2 WDTCSR Register
00309 *
00310 * @since Version 1.0.0
00311 */
00312 typedef enum : uint8_t {
00313     k_wdt_timeout_1024_cycles = 0, /**< 1024 cycles (TOPS=00). Counter loads 0x03FF. */
00314     k_wdt_timeout_4096_cycles = 1, /**< 4096 cycles (TOPS=01). Counter loads 0x0FFF. */
00315     k_wdt_timeout_8192_cycles = 2, /**< 8192 cycles (TOPS=10). Counter loads 0x1FFF. */
00316     k_wdt_timeout_16384_cycles = 3, /**< 16384 cycles (TOPS=11). Counter loads 0x3FFF. Max timeout. */
00317 } wdt_timeout_cycles_t;
00318
00319 /**
00320 * @enum wdt_clock_division_t
00321 * @brief WDT Clock Division Ratio Selection (CKS bits in WDTCSR register)
00322 *
00323 * @details
00324 * Defines the six available clock division ratios for the WDT. Combined with
00325 * TOPS (timeout period), this determines the actual watchdog timeout.
00326 *
00327 * ## Timeout Calculation
00328 *
00329 * @f[
00330 *     T_{timeout} = \frac{TOPS \times CKS_{divisor}}{f_{PCLKB}}
00331 * @f]
00332 *
00333 * ## Timeout Table (PCLKB = 60 MHz)
00334 *
00335 * | CKS Setting | TOPS=1024 | TOPS=4096 | TOPS=8192 | TOPS=16384 |
00336 * |-----|-----|-----|-----|-----|
00337 * | div-by-4 | 68 us | 273 us | 546 us | 1.09 ms |
00338 * | div-by-64 | 1.1 ms | 4.4 ms | 8.7 ms | 17.5 ms |
00339 * | div-by-128 | 2.2 ms | 8.7 ms | 17.5 ms | 34.9 ms |
00340 * | div-by-512 | 8.7 ms | 34.9 ms | 69.9 ms | 140 ms |
00341 * | div-by-2048 | 34.9 ms | 140 ms | 280 ms | 559 ms |
00342 * | div-by-8192 | 140 ms | 559 ms | 1.12 s | 2.24 s |
00343 *
00344 * @note Only 6 values are valid per hardware manual Ch34 Table 34.2
00345 * @warning Other CKS values are prohibited by hardware
00346 *
00347 * @see wdt_timeout_cycles_t TOPS selection for cycle count
00348 * @see rx_wdt_config_t Configuration structure using this enum
00349 * @see RX72N Hardware Manual Ch34.2.2 WDTCSR Register
00350 *
00351 * @since Version 1.0.0
00352 */
00353 typedef enum : uint8_t {
00354     k_wdt_clock_div_4 = 0x01, /**< PCLKB / 4 (CKS=0001). Fastest, shortest timeout. */
00355     k_wdt_clock_div_64 = 0x04, /**< PCLKB / 64 (CKS=0100). */
00356     k_wdt_clock_div_128 = 0x0F, /**< PCLKB / 128 (CKS=1111). */
00357     k_wdt_clock_div_512 = 0x06, /**< PCLKB / 512 (CKS=0110). */
00358     k_wdt_clock_div_2048 = 0x07, /**< PCLKB / 2048 (CKS=0111). */
00359     k_wdt_clock_div_8192 = 0x08, /**< PCLKB / 8192 (CKS=1000). Slowest, longest timeout. */
00360 } wdt_clock_division_t;

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00361
00362 /*
=====
00363 * Data Structures
00364 *
=====
00365 */
00366
00367 /**
00368 * @struct rx_wdt_config_t
00369 * @brief WDT Configuration Structure for Driver Initialization
00370 *
00371 * @details
00372 * Contains all parameters needed to configure and initialize the Watchdog Timer (WDT).
00373 * All fields must be set before passing to rx_wdt_init(). Use default values for
00374 * typical operation, or customize for specific monitoring requirements.
00375 *
00376 * ## Initialization Requirements
00377 *
00378 * 1. **All fields must be initialized** - No default constructor available (C struct)
00379 * 2. **Validate timeout_cycles** - Must be valid enum value (0-3)
00380 * 3. **Consider PCLKB frequency** - Timeout periods are PCLKB-dependent
00381 * 4. **Choose reset vs interrupt** - reset_on_timeout determines failure behavior
00382 *
00383 * ## Default Configuration
00384 *
00385 * Recommended starting configuration for development:
00386 * ```c
00387 * rx_wdt_config_t default_config = {
00388 *     .timeout_cycles = k_wdt_timeout_16384_cycles, // 16384 counter cycles
00389 *     .clock_division = k_wdt_clock_div_8192,      // PCLKB / 8192
00390 *     .enable_on_init = false,                      // Don't start immediately (safer)
00391 *     .reset_on_timeout = true                      // Reset on timeout (production mode)
00392 * };
00393 * // Actual timeout = 16384 * 8192 / 60MHz = 2.24 seconds
00394 * ```
00395 *
00396 * ## Production Configuration
00397 *
00398 * For production systems using WDT as secondary watchdog:
00399 * ```c
00400 * rx_wdt_config_t prod_config = {
00401 *     .timeout_cycles = k_wdt_timeout_16384_cycles, // Max cycles
00402 *     .clock_division = k_wdt_clock_div_8192,      // Max divisor = 2.24s timeout
00403 *     .enable_on_init = true,                      // Start immediately
00404 *     .reset_on_timeout = true                      // Always reset
00405 * };
00406 * ```
00407 *
00408 * ## Memory Layout
00409 *
00410 * | Offset | Size | Field | Type | Notes |
00411 * |-----|-----|-----|-----|-----|
00412 * | 0 | 1 | timeout_cycles | uint8_t (enum) | TOPS bits (0-3) |
00413 * | 1 | 1 | clock_division | uint8_t (enum) | CKS bits |
00414 * | 2 | 1 | enable_on_init | bool | Auto-start flag |
00415 * | 3 | 1 | reset_on_timeout | bool | Reset mode flag |
00416 * | **Total** | **4 bytes** | | | Structure size |
00417 *
00418 * @par Field Constraints:
00419 *
00420 * | Field | Valid Values | Default | Notes |
00421 * |-----|-----|-----|-----|
00422 * | timeout_cycles | 0-3 (enum) | k_wdt_timeout_16384_cycles | TOPS bits |
00423 * | clock_division | See enum | k_wdt_clock_div_8192 | CKS bits (6 valid values only) |
00424 * | enable_on_init | true/false | false | false = safer for development |
00425 * | reset_on_timeout | true/false | true | true = reset, false = interrupt |
00426 *
00427 * @par Usage Example (Basic):
00428 * @code{.c}
00429 * // Initialize with 2.24 second timeout (max)
00430 * rx_wdt_config_t config = {
00431 *     .timeout_cycles = k_wdt_timeout_16384_cycles,
00432 *     .clock_division = k_wdt_clock_div_8192,
00433 *     .enable_on_init = false,
00434 *     .reset_on_timeout = true
00435 * };
00436 *
00437 * rx_err_t err = rx_wdt_init(&config);
00438 * if (err != k_rx_ok) {
00439 *     uart_debug_puts("[ERROR] WDT init failed\r\n");
00440 *     return err;
00441 * }
00442 * @endcode
00443 *
00444 * @par Usage Example (Development Mode):
00445 * @code{.c}

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00446 * // Development configuration: Don't start, use interrupt for debugging
00447 * rx_wdt_config_t dev_config = {
00448 *     .timeout_cycles = k_wdt_timeout_16384_cycles,
00449 *     .clock_division = k_wdt_clock_div_8192,
00450 *     .enable_on_init = false, // Start manually after init complete
00451 *     .reset_on_timeout = false // Interrupt allows breakpoint debugging
00452 * };
00453 *
00454 * rx_wdt_init(&dev_config);
00455 *
00456 * // ... system initialization ...
00457 *
00458 * // Start watchdog after init complete
00459 * rx_wdt_start();
00460 * @endcode
00461 *
00462 * @par Usage Example (Runtime Control):
00463 * @code{.c}
00464 * // Start with WDT enabled, 559ms timeout
00465 * rx_wdt_config_t config = {
00466 *     .timeout_cycles = k_wdt_timeout_4096_cycles,
00467 *     .clock_division = k_wdt_clock_div_8192,
00468 *     .enable_on_init = true,
00469 *     .reset_on_timeout = true
00470 * };
00471 * rx_wdt_init(&config);
00472 *
00473 * // Normal operation
00474 * while (running) {
00475 *     main_loop_work();
00476 *     rx_wdt_feed();
00477 * }
00478 *
00479 * // Stop for long flash operation
00480 * rx_wdt_stop();
00481 * flash_update(); // May take seconds
00482 * rx_wdt_start(); // Resume monitoring
00483 * @endcode
00484 *
00485 * @invariant timeout_cycles must be in range [0, 3]
00486 * @invariant Structure size is 4 bytes (3 fields + 1 padding)
00487 * @invariant All fields must be initialized before use
00488 * @invariant Configuration is immutable after rx_wdt_init() (requires re-init to change)
00489 *
00490 * @note C23 struct with explicit padding for 4-byte alignment
00491 * @note Configuration is copied by rx_wdt_init(), original can be freed after call
00492 * @note Changing configuration requires calling rx_wdt_init() again (if WDT stopped)
00493 *
00494 * @warning timeout_cycles values are hardware-limited (0-3 only)
00495 * @warning enable_on_init=true starts watchdog immediately (ensure feed loop ready)
00496 * @warning reset_on_timeout=true will reset system (no recovery possible)
00497 * @attention Always validate timeout_cycles against wdt_timeout_cycles_t enum
00498 *
00499 * @see rx_wdt_init() Initializes WDT with this configuration
00500 * @see wdt_timeout_cycles_t Enum values for timeout_cycles field
00501 * @see rx_wdt_start() Start watchdog if enable_on_init=false
00502 * @see rx_wdt_stop() Stop watchdog for debugging or long operations
00503 *
00504 * @since Version 1.0.0
00505 */
00506 typedef struct {
00507     wdt_timeout_cycles_t
00508     timeout_cycles; /**< Counter cycle count before timeout (TOPS bits). Use k_wdt_timeout_16384_cycles for
00509     maximum. Valid values: 0-3. See wdt_timeout_cycles_t. */
00510     wdt_clock_division_t
00511     clock_division; /**< Clock division ratio (CKS bits). Use k_wdt_clock_div_8192 for longest timeout. Combined with
00512     timeout_cycles determines actual timeout. See wdt_clock_division_t. */
00513     bool
00514     enable_on_init; /**< Auto-start watchdog after initialization. true = Start immediately (production mode, ensure feed
00515     loop ready). false = Manual start via rx_wdt_start() (safer for development). Default: false */
00516     bool
00517     reset_on_timeout; /**< Behavior on watchdog timeout. true = System reset (production mode). false = NMI interrupt
00518     (development mode, allows debugging). Default: true */
00519 } rx_wdt_config_t;
00520 /*
00521 *
00522 * =====
00523 * Public API
00524 * =====
00525 */
00526 /**
00527 * @brief Initialize the Watchdog Timer (WDT) with specified configuration
00528 *
00529 * @details
00530 * Configures and optionally starts the RX72N Watchdog Timer peripheral. Sets up

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00527 * timeout period (clock divider), reset behavior, and auto-start mode. Once
00528 * initialized, the WDT can be started/stopped via rx_wdt_start() and rx_wdt_stop().
00529 *
00530 **Algorithm steps:**
00531 * 1. Validate configuration parameters (NULL check, timeout_cycles range)
00532 * 2. Check if WDT already initialized (prevent double initialization)
00533 * 3. Configure WDTCR register:
00534 *   - Set CKS[1:0] bits based on timeout_cycles
00535 *   - Set RPES bit for reset vs interrupt mode
00536 * 4. If enable_on_init=true, start WDT counter immediately
00537 * 5. Store configuration for runtime reference
00538 * 6. Mark driver as initialized
00539 *
00540 **Hardware Configuration**:
00541 * - Writes to WDTCR (Watchdog Timer Control Register) at 0x00088022
00542 * - Sets CKS bits: timeout_cycles value (0-3)
00543 * - Sets RPES bit: reset_on_timeout (1=reset, 0=interrupt)
00544 * - If enable_on_init=true, sets WDT counter start bit
00545 *
00546 **Control Flow**:
00547 * @dot
00548 * digraph wdt_init {
00549 *   rankdir=TB;
00550 *   node [shape=box, style=rounded];
00551 *
00552 *   start [label="Start"];
00553 *   check_null [label="Check config != nullptr"];
00554 *   check_range [label="Validate timeout_cycles (0-3)"];
00555 *   check_init [label="Check not initialized"];
00556 *   configure [label="Write WDTCR register"];
00557 *   auto_start [label="Auto-start?", shape=diamond];
00558 *   start_wdt [label="Start WDT counter"];
00559 *   mark_init [label="Set initialized flag"];
00560 *   success [label="Return k_rx_ok", fillcolor=lightgreen, style="rounded, filled"];
00561 *   error_null [label="Return k_rx_err_null_ptr", fillcolor=lightcoral, style="rounded, filled"];
00562 *   error_arg [label="Return k_rx_err_invalid_arg", fillcolor=lightcoral, style="rounded, filled"];
00563 *   error_state [label="Return k_rx_err_invalid_state", fillcolor=lightcoral, style="rounded, filled"];
00564 *
00565 *   start -> check_null;
00566 *   check_null -> error_null [label="NULL"];
00567 *   check_null -> check_range [label="valid"];
00568 *   check_range -> error_arg [label="invalid"];
00569 *   check_range -> check_init [label="valid"];
00570 *   check_init -> error_state [label="initialized"];
00571 *   check_init -> configure [label="not init"];
00572 *   configure -> auto_start;
00573 *   auto_start -> start_wdt [label="yes"];
00574 *   auto_start -> mark_init [label="no"];
00575 *   start_wdt -> mark_init;
00576 *   mark_init -> success;
00577 * }
00578 * @enddot
00579 *
00580 * @param[in] config Configuration structure pointer. Must point to valid rx_wdt_config_t.
00581 * - timeout_cycles: Must be valid wdt_timeout_cycles_t (0-3)
00582 * - enable_on_init: true=start immediately, false>manual start
00583 * - reset_on_timeout: true=reset on timeout, false=interrupt
00584 * - NULL not allowed (returns k_rx_err_null_ptr)
00585 *
00586 * @return rx_err_t Error code indicating success or failure reason
00587 * @retval k_rx_ok Success, WDT initialized and optionally started
00588 * @retval k_rx_err_null_ptr config parameter is nullptr
00589 * @retval k_rx_err_invalid_arg timeout_cycles out of valid range (0-3)
00590 * @retval k_rx_err_invalid_state WDT already initialized (call rx_wdt_stop first)
00591 *
00592 * @pre config must point to valid rx_wdt_config_t structure
00593 * @pre config->timeout_cycles must be in range [0, 3]
00594 * @pre WDT must not be already initialized
00595 * @pre System must be in a state that allows WDTCR register writes
00596 *
00597 * @post If success, WDT is configured with specified settings
00598 * @post If enable_on_init=true, WDT counter is running
00599 * @post If enable_on_init=false, WDT is configured but stopped
00600 * @post Driver marked as initialized, subsequent init calls will fail
00601 *
00602 * @invariant WDT hardware state matches configuration
00603 * @invariant Only one successful init call allowed (until stop/reset)
00604 * @invariant Configuration immutable after init (requires re-init to change)
00605 *
00606 * @note Thread-safe - uses mutex protection
00607 * @note This function must be called before any other WDT functions
00608 * @note If enable_on_init=true, ensure feed loop is ready before calling
00609 * @note Configuration is copied internally, original can be freed after call
00610 *
00611 * @warning WDT cannot be reconfigured while initialized (must stop first)
00612 * @warning If enable_on_init=true, rx_wdt_feed() must be called within timeout period
00613 * @warning timeout_cycles determines actual timeout based on PCLKB frequency

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00614 * @attention Very short timeouts (us range) may be difficult to meet reliably
00615 *
00616 * @par Thread Safety:
00617 * Thread-safe. Uses internal mutex to prevent concurrent initialization.
00618 * Multiple threads can safely call this function (first succeeds, rest fail with
00619 * k_rx_err_invalid_state).
00620 *
00621 * @par Performance:
00622 * - Execution time: ~20 us @ 240 MHz (register configuration)
00623 * - Memory: No dynamic allocation, configuration copied to static storage
00624 * - Stack usage: < 32 bytes
00625 *
00626 * @par Re-entrancy:
00627 * Not reentrant (uses static state). Concurrent calls are serialized by mutex.
00628 *
00629 * @par Example (Basic Initialization):
00630 * @code{.c}
00631 * #include "rx_wdt.h"
00632 * #include "uart_debug.h"
00633 *
00634 * void system_init(void) {
00635 *     // Configure WDT with longest timeout, manual start
00636 *     rx_wdt_config_t config = {
00637 *         .timeout_cycles = k_wdt_timeout_16384_cycles, // ~273us at 60MHz
00638 *         .enable_on_init = false, // Don't start yet
00639 *         .reset_on_timeout = true // Reset on timeout
00640 *     };
00641 *
00642 *     rx_err_t err = rx_wdt_init(&config);
00643 *     if (err != k_rx_ok) {
00644 *         uart_debug_puts("[ERROR] WDT init failed\r\n");
00645 *         // Handle error (cannot use watchdog)
00646 *         return;
00647 *     }
00648 *
00649 *     uart_debug_puts("[INFO] WDT initialized\r\n");
00650 *
00651 *     // ... complete system initialization ...
00652 *
00653 *     // Start watchdog after init complete
00654 *     rx_wdt_start();
00655 * }
00656 * @endcode
00657 *
00658 * @par Example (Auto-Start for Production):
00659 * @code{.c}
00660 * // Production system: Start watchdog immediately
00661 * rx_wdt_config_t prod_config = {
00662 *     .timeout_cycles = k_wdt_timeout_16384_cycles,
00663 *     .enable_on_init = true, // Start immediately
00664 *     .reset_on_timeout = true
00665 * };
00666 *
00667 * // IMPORTANT: Ensure feed loop is ready before this call
00668 * rx_err_t err = rx_wdt_init(&prod_config);
00669 * if (err != k_rx_ok) {
00670 *     // Fatal error - cannot start watchdog
00671 *     system_fault_handler();
00672 * }
00673 *
00674 * // WDT is now running - must call rx_wdt_feed() within ~273us
00675 * @endcode
00676 *
00677 * @par Example (Error Handling):
00678 * @code{.c}
00679 * rx_wdt_config_t config = {};
00680 * rx_err_t err = rx_wdt_init(&config);
00681 *
00682 * switch (err) {
00683 *     case k_rx_ok:
00684 *         uart_debug_puts("[INFO] WDT ready\r\n");
00685 *         break;
00686 *     case k_rx_err_null_ptr:
00687 *         uart_debug_puts("[ERROR] nullptr config pointer\r\n");
00688 *         break;
00689 *     case k_rx_err_invalid_arg:
00690 *         uart_debug_puts("[ERROR] Invalid timeout period\r\n");
00691 *         break;
00692 *     case k_rx_err_invalid_state:
00693 *         uart_debug_puts("[WARN] WDT already initialized\r\n");
00694 *         break;
00695 *     default:
00696 *         uart_debug_puts("[ERROR] Unknown WDT error\r\n");
00697 *         break;
00698 * }
00699 * @endcode
00700 *

```

```

00701 * @see rx_wdt_config_t Configuration structure definition
00702 * @see rx_wdt_start() Start watchdog if enable_on_init=false
00703 * @see rx_wdt_stop() Stop watchdog for reconfiguration
00704 * @see rx_wdt_feed() Feed watchdog to prevent timeout
00705 * @see wdt_timeout_cycles_t Timeout period enum values
00706 *
00707 * @since Version 1.0.0
00708 *
00709 * @par NASA Power of 10 Compliance:
00710 * - Rule 1: [OK] No goto/setjmp/recursion, sequential control flow
00711 * - Rule 3: [OK] No dynamic allocation, static configuration storage
00712 * - Rule 5: [OK] 4 preconditions, 4 postconditions
00713 * - Rule 7: [OK] Returns rx_err_t, caller must check
00714 */
00715 [[nodiscard]] rx_err_t rx_wdt_init(const rx_wdt_config_t* config);
00716
00717 /**
00718 * @brief Start the WDT counter (enable watchdog monitoring)
00719 *
00720 * @details
00721 * Starts the Watchdog Timer counter, beginning countdown to timeout. Once started,
00722 * rx_wdt_feed() must be called periodically to prevent timeout. This function is
00723 * used when WDT was initialized with enable_on_init=false.
00724 *
00725 * **Algorithm steps:**
00726 * 1. Check if WDT is initialized (fail if not)
00727 * 2. Set WDT counter start bit in WDTCR register
00728 * 3. Begin countdown from configured timeout period
00729 * 4. Return success
00730 *
00731 * **Hardware Operation**
00732 * - Writes to WDTCR register (0x00088022)
00733 * - Sets TME (Timer Enable) bit to 1
00734 * - Counter begins decrementing from timeout period value
00735 *
00736 * @return rx_err_t Error code indicating success or failure
00737 * @retval k_rx_ok Success, WDT counter started
00738 * @retval k_rx_err_not_initialized WDT not initialized via rx_wdt_init()
00739 *
00740 * @pre WDT must be initialized via rx_wdt_init()
00741 * @pre Feed loop must be ready to call rx_wdt_feed() within timeout period
00742 *
00743 * @post WDT counter is running
00744 * @post rx_wdt_feed() must be called within timeout period to prevent reset
00745 * @post If timeout expires, system will reset (or interrupt if configured)
00746 *
00747 * @invariant WDT configuration unchanged (timeout period, reset mode)
00748 * @invariant Starting already-started WDT is safe (idempotent)
00749 *
00750 * @note Thread-safe - uses mutex protection
00751 * @note Can be called multiple times (idempotent operation)
00752 * @note Complementary to rx_wdt_stop() for runtime control
00753 *
00754 * @warning Ensure rx_wdt_feed() will be called within timeout period before starting
00755 * @warning At 60 MHz PCLKB with max divider, timeout is only ~273 us (very short!)
00756 * @attention After starting, missing even ONE feed will cause timeout/reset
00757 *
00758 * @par Thread Safety:
00759 * Thread-safe. Uses internal mutex to prevent concurrent start/stop operations.
00760 *
00761 * @par Performance:
00762 * - Execution time: ~2 us @ 240 MHz (single register write)
00763 * - Memory: No allocation
00764 * - Stack usage: < 16 bytes
00765 *
00766 * @par Re-entrancy:
00767 * Not reentrant (uses static state). Concurrent calls serialized by mutex.
00768 *
00769 * @par Example (Staged Startup):
00770 * @code{.c}
00771 * // Initialize WDT but don't start immediately
00772 * rx_wdt_config_t config = {
00773 *     .timeout_cycles = k_wdt_timeout_16384_cycles,
00774 *     .enable_on_init = false, // Manual start
00775 *     .reset_on_timeout = true
00776 * };
00777 * rx_wdt_init(&config);
00778 *
00779 * // ... complete system initialization ...
00780 * // ... set up feed loop ...
00781 *
00782 * // Now safe to start watchdog
00783 * rx_err_t err = rx_wdt_start();
00784 * if (err != k_rx_ok) {
00785 *     uart_debug_puts("[ERROR] WDT start failed\r\n");
00786 * }
00787 *

```



```

00788 * // Must now call rx_wdt_feed() within ~273us
00789 * @endcode
00790 *
00791 * @par Example (Start/Stop Control):
00792 * @code{.c}
00793 * // Normal operation with watchdog
00794 * rx_wdt_start();
00795 * while (running) {
00796 *     main_loop_work();
00797 *     rx_wdt_feed();
00798 * }
00799 *
00800 * // Stop for long flash operation
00801 * rx_wdt_stop();
00802 * flash_erase_sector(); // May take milliseconds
00803 * rx_wdt_start(); // Resume monitoring
00804 * @endcode
00805 *
00806 * @see rx_wdt_init() Initialize WDT before starting
00807 * @see rx_wdt_stop() Stop WDT counter
00808 * @see rx_wdt_feed() Feed watchdog to prevent timeout
00809 *
00810 * @since Version 1.0.0
00811 *
00812 * @par NASA Power of 10 Compliance:
00813 * - Rule 1: [OK] No goto/setjmp/recursion
00814 * - Rule 3: [OK] No dynamic allocation
00815 * - Rule 5: [OK] 2 preconditions, 3 postconditions
00816 */
00817 [[nodiscard]] rx_err_t rx_wdt_start(void);
00818
00819 /**
00820 * @brief Stop the WDT counter (disable watchdog monitoring)
00821 *
00822 * @details
00823 * Stops the Watchdog Timer counter, preventing timeout. Use this function to
00824 * suspend watchdog monitoring during long operations (flash updates, debugging)
00825 * or for safe shutdown. Unlike IWDTC, WDT can be stopped and restarted.
00826 *
00827 * **Algorithm steps:**
00828 * 1. Check if WDT is initialized (fail if not)
00829 * 2. Clear WDT counter start bit in WDTCR register
00830 * 3. Counter stops decrementing (frozen at current value)
00831 * 4. Return success
00832 *
00833 * **Hardware Operation:**
00834 * - Writes to WDTCR register (0x00088022)
00835 * - Clears TME (Timer Enable) bit to 0
00836 * - Counter stops, timeout cannot occur
00837 *
00838 * **Use Cases:**
00839 * - **Flash updates**: Long erase/write operations (milliseconds to seconds)
00840 * - **Debugging**: Freeze counter during breakpoint sessions
00841 * - **Safe mode**: Disable monitoring during error recovery
00842 * - **Shutdown**: Stop watchdog before power-down
00843 *
00844 * @return rx_err_t Error code indicating success or failure
00845 * @retval k_rx_ok Success, WDT counter stopped
00846 * @retval k_rx_err_not_initialized WDT not initialized via rx_wdt_init()
00847 *
00848 * @pre WDT must be initialized via rx_wdt_init()
00849 *
00850 * @post WDT counter is stopped (not decrementing)
00851 * @post No timeout will occur until rx_wdt_start() called
00852 * @post System is unprotected by WDT until restarted
00853 *
00854 * @invariant WDT configuration unchanged (timeout period, reset mode)
00855 * @invariant Stopping already-stopped WDT is safe (idempotent)
00856 *
00857 * @note Thread-safe - uses mutex protection
00858 * @note Can be called multiple times (idempotent operation)
00859 * @note Complementary to rx_wdt_start() for runtime control
00860 *
00861 * @warning System is unprotected while WDT stopped (use with caution)
00862 * @warning Remember to call rx_wdt_start() to resume monitoring
00863 * @attention If using both WDT and IWDTC, system still protected by IWDTC
00864 *
00865 * @par Thread Safety:
00866 * Thread-safe. Uses internal mutex to prevent concurrent start/stop operations.
00867 *
00868 * @par Performance:
00869 * - Execution time: ~2 us @ 240 MHz (single register write)
00870 * - Memory: No allocation
00871 * - Stack usage: < 16 bytes
00872 *
00873 * @par Re-entrancy:
00874 * Not reentrant (uses static state). Concurrent calls serialized by mutex.

```



```

00875 *
00876 * @par Example (Flash Update):
00877 * @code{.c}
00878 * // Stop WDT for long flash operation
00879 * rx_wdt_stop();
00880 *
00881 * // Erase and write flash (may take seconds)
00882 * rx_err_t err = flash_erase_sector(sector_addr);
00883 * if (err == k_rx_ok) {
00884 *   err = flash_write_page(sector_addr, data, size);
00885 * }
00886 *
00887 * // Resume watchdog monitoring
00888 * rx_wdt_start();
00889 *
00890 * // Continue normal operation with watchdog active
00891 * @endcode
00892 *
00893 * @par Example (Debugging Session):
00894 * @code{.c}
00895 * // Before setting breakpoint, stop watchdog
00896 * rx_wdt_stop();
00897 *
00898 * // Set breakpoint here - WDT won't timeout during debug session
00899 * uart_debug_puts("[DEBUG] Breakpoint location\r\n");
00900 *
00901 * // Resume watchdog after debugging
00902 * rx_wdt_start();
00903 * @endcode
00904 *
00905 * @par Example (Safe Shutdown):
00906 * @code{.c}
00907 * void system_shutdown(void) {
00908 *   uart_debug_puts("[INFO] System shutdown initiated\r\n");
00909 *
00910 *   // Stop watchdog to prevent timeout during shutdown
00911 *   rx_wdt_stop();
00912 *
00913 *   // Perform shutdown sequence
00914 *   motor_disable_all();
00915 *   comm_close_all();
00916 *   save_state_to_flash();
00917 *
00918 *   // Power down (watchdog remains stopped)
00919 *   enter_low_power_mode();
00920 * }
00921 * @endcode
00922 *
00923 * @see rx_wdt_init() Initialize WDT before stopping
00924 * @see rx_wdt_start() Restart WDT after stopping
00925 * @see rx_iwdt.h Independent watchdog that cannot be stopped
00926 *
00927 * @since Version 1.0.0
00928 *
00929 * @par NASA Power of 10 Compliance:
00930 * - Rule 1: [OK] No goto/setjmp/recursion
00931 * - Rule 3: [OK] No dynamic allocation
00932 * - Rule 5: [OK] 1 precondition, 3 postconditions
00933 */
00934 [[nodiscard]] rx_err_t rx_wdt_stop(void);
00935
00936 /**
00937 * @brief Feed the watchdog timer (reset counter to prevent timeout)
00938 *
00939 * @details
00940 * Resets the WDT counter to its initial value, preventing timeout. This function
00941 * must be called periodically at a rate faster than the configured timeout period.
00942 * Typically called from main control loop or dedicated watchdog monitoring task.
00943 *
00944 * **Algorithm steps:**
00945 * 1. Check if WDT is initialized (fail if not)
00946 * 2. Write refresh sequence to WDTRR register (0x00 then 0xFF)
00947 * 3. Counter resets to initial timeout period value
00948 * 4. Countdown resumes from reset value
00949 * 5. Return success
00950 *
00951 * **Hardware Operation**:
00952 * - Writes two-byte sequence to WDTRR (Watchdog Timer Refresh Register):
00953 *   - First write: 0x00
00954 *   - Second write: 0xFF
00955 * - Sequence must be written in correct order (hardware validation)
00956 * - Counter resets to timeout_cycles value
00957 *
00958 * **Feed Requirements**:
00959 * - **Frequency**: Must call faster than timeout period
00960 * - **Example**: With k_wdt_timeout_16384_cycles at 60 MHz PCLKB:
00961 *   - Timeout = ~273 us

```

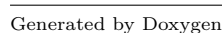
```

00962 * - Feed frequency > 3.6 kHz (every ~273 us)
00963 * - Recommended: Feed at 2x margin = ~130 us interval
00964 *
00965 * @return rx_err_t Error code indicating success or failure
00966 * @retval k_rx_ok Success, WDT counter reset to initial value
00967 * @retval k_rx_err_not_initialized WDT not initialized via rx_wdt_init()
00968 *
00969 * @pre WDT must be initialized via rx_wdt_init()
00970 * @pre WDT must be started via rx_wdt_start() or enable_on_init=true
00971 * @pre Called from location that executes faster than timeout period
00972 *
00973 * @post WDT counter reset to initial timeout period value
00974 * @post Countdown timer begins again from reset value
00975 * @post Timeout prevented (for this cycle)
00976 *
00977 * @invariant Counter value after feed equals timeout_cycles configuration value
00978 * @invariant Configuration unchanged (timeout period, reset mode)
00979 *
00980 * @note Thread-safe - can be called from multiple threads (mutex protected)
00981 * @note ISR-safe - can be called from interrupt context (optimized fast path)
00982 * @note Calling when WDT stopped has no effect (safe but wasteful)
00983 * @note This is the ONLY function that prevents watchdog timeout
00984 *
00985 * @warning Missing even ONE feed within timeout period will cause reset
00986 * @warning Feed frequency must account for worst-case execution time variations
00987 * @warning Very short timeouts (~273 us max) require very frequent feeds
00988 * @attention Do NOT call from multiple locations - defeats failure detection
00989 *
00990 * @par Thread Safety:
00991 * Thread-safe. Uses atomic register writes (no mutex needed for performance).
00992 * Multiple threads can safely call this function.
00993 *
00994 * @par Performance:
00995 * - Execution time: ~1 us @ 240 MHz (two register writes)
00996 * - Memory: No allocation
00997 * - Stack usage: < 8 bytes
00998 * - Fastest WDT operation
00999 *
01000 * @par Re-entrancy:
01001 * Fully reentrant. Uses only register writes, no static state modification.
01002 *
01003 * @par Example (Main Loop):
01004 * @code{.c}
01005 * // Feed from main control loop
01006 * void main_loop(void) {
01007 *     while (1) {
01008 *         // Perform main loop work
01009 *         motor_update();
01010 *         comm_process();
01011 *         sensor_read();
01012 *
01013 *         // Feed watchdog at end of each iteration
01014 *         rx_err_t err = rx_wdt_feed();
01015 *         if (err != k_rx_ok) {
01016 *             uart_debug_puts("[ERROR] WDT feed failed\r\n");
01017 *         }
01018 *
01019 *         // Short delay (must be < timeout period)
01020 *         delay_us(100); // 100us delay, well under 273us timeout
01021 *     }
01022 * }
01023 * @endcode
01024 *
01025 * @par Example (Dedicated Watchdog Task):
01026 * @code{.c}
01027 * // ThreadX task dedicated to watchdog feeding
01028 * void watchdog_task(ULONG input) {
01029 *     (void)input;
01030 *
01031 *     while (1) {
01032 *         // Feed watchdog
01033 *         rx_wdt_feed();
01034 *
01035 *         // Sleep for 100us (well under 273us timeout)
01036 *         tx_thread_sleep(1); // Assuming 10 kHz tick rate = 100us
01037 *     }
01038 * }
01039 * @endcode
01040 *
01041 * @par Example (Timed Feed with Margin):
01042 * @code{.c}
01043 * // Calculate safe feed interval
01044 * const uint32_t timeout_us = 273; // k_wdt_timeout_16384_cycles at 60MHz
01045 * const uint32_t safety_margin = 2; // 2x safety margin
01046 * const uint32_t feed_interval_us = timeout_us / safety_margin; // 136us
01047 *
01048 * void main_loop(void) {

```

7.45 libs/rx_core/src/rx_error_handler.c File Reference

```
#include "rx_error_handler.h"
#include <string.h>
#include "rx_check.h"
#include "rx_log.h"
Include dependency graph for rx_error_handler.c:
```



Enumerations

- enum `error_handler_backoff_constants_t` : `uint32_t` { `k_exponential_backoff_multiplier` = 2 , `k_error_handler_no_retry_limit` = 0 , `k_error_handler_first_retry` = 1 , `k_error_handler_max_retries` = 32 }

Internal constants for exponential backoff algorithm.

Functions

- static `error_component_state_t` * `internal_find_component` (`error_handler_t` *handler, const char *component)
Find component tracking slot by name.
- static void `internal_copy_component_name` (char *dst, const char *src, const `uint32_t` max_len)
Copy a component name safely without strncpy.
- static `error_component_state_t` * `internal_find_or_create_component` (`error_handler_t` *handler, const char *component)
Find existing component entry or create new one.
- static `rx_err_t` `impl_report_error` (void *ctx, `rx_err_t` err, const char *component, const char *message)
Report error implementation (`rx_error_interface_t.report_error`).
- static `uint32_t` `impl_get_error_count` (void *ctx)
Get total error count implementation (`rx_error_interface_t.get_error_count`).
- static `uint32_t` `impl_get_component_error_count` (void *ctx, const char *component)
Get component-specific error count (`rx_error_interface_t.get_component_error_count`).
- static `rx_err_t` `impl_clear_errors` (void *ctx)
Clear all error counters (`rx_error_interface_t.clear_errors`).
- static bool `impl_is_retry_limit_reached` (void *ctx, const char *component)
Check if component has reached retry limit (`rx_error_interface_t.is_retry_limit_reached`).
- static `rx_err_t` `impl_reset_retry_counter` (void *ctx, const char *component)
Reset component retry counter (`rx_error_interface_t.reset_retry_counter`).
- static `uint32_t` `impl_get_backoff_delay` (void *ctx, const char *component)
Calculate exponential backoff delay (`rx_error_interface_t.get_backoff_delay`).
- `rx_err_t` `error_handler_init` (`error_handler_t` *handler, const `error_handler_config_t` *config)
Initialize error handler with retry logic and exponential backoff.
- `rx_err_t` `error_handler_get_interface` (`rx_error_interface_t` *iface, `error_handler_t` *handler)
Get error handler interface for dependency injection.
- `rx_err_t` `error_handler_deinit` (`error_handler_t` *handler)
Deinitialize error handler and release resources.

Variables

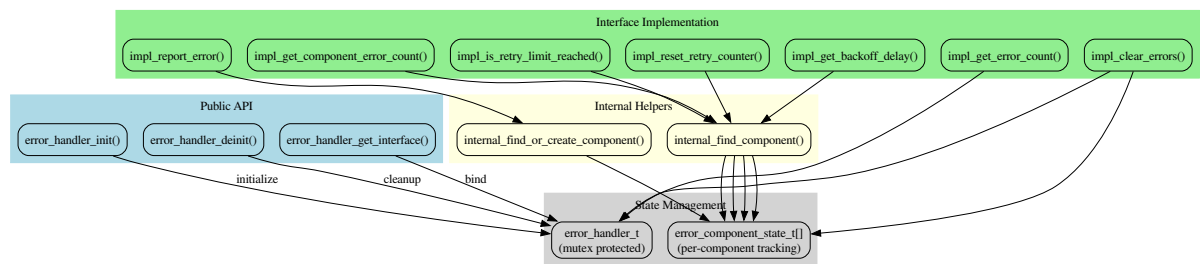
- static const char *const `s_tag` = "ERROR_HANDLER"
Log tag for error handler module.

7.45.1 Detailed Description

Error Handler Concrete Implementation.

Implements the `rx_error_interface_t` abstract interface with concrete error tracking, retry logic, and exponential backoff capabilities. This module provides a centralized error handling mechanism for the STAR firmware, enabling consistent error reporting, component-level error tracking, and intelligent retry strategies.

Implementation Architecture

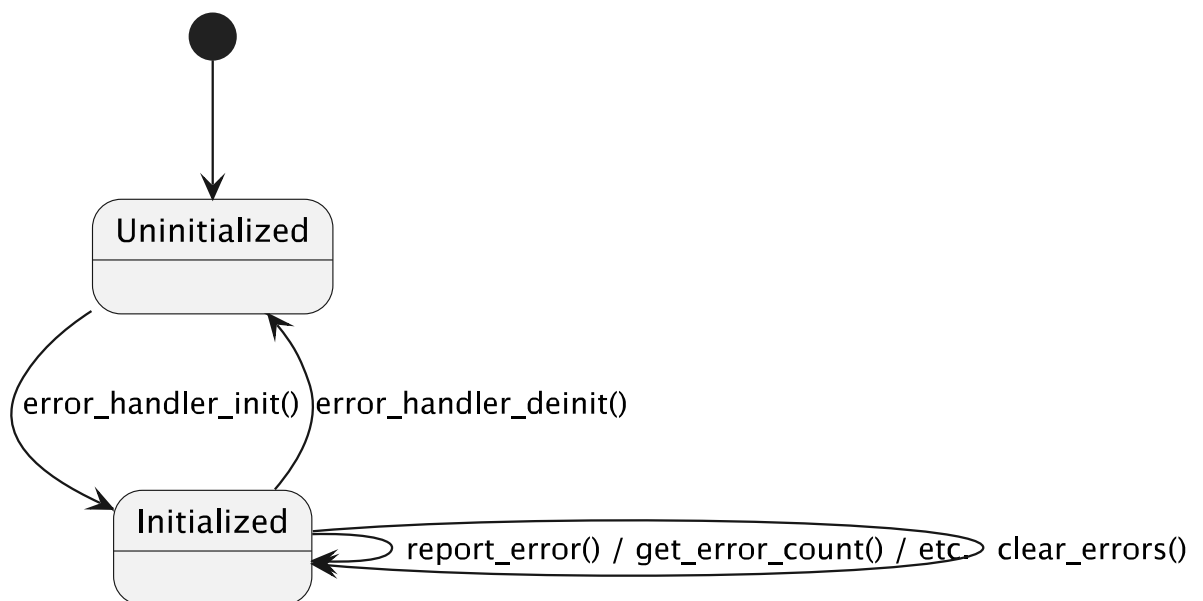


Exponential Backoff Algorithm

The backoff delay calculation uses exponential growth with configurable bounds:

$$\text{delay} = \min(\text{initial} \times 2^{(\text{retries}-1)}, \text{max_backoff})$$

State Machine



Memory Layout

Component	Size	Description
error_handler_t	~1.2 KB	Main handler structure
TX_MUTEX	~100 B	ThreadX mutex control block
components[]	~1 KB	16 component tracking slots

Performance Characteristics

Operation	Best Case	Worst Case	Notes
report_error()	~5 us	~50 us	Mutex + string cmp
get_error_count()	~2 us	~10 us	Mutex only
get_backoff_delay()	~3 us	~20 us	Loop up to 32 iters
clear_errors()	~5 us	~30 us	Clears all slots

Thread Safety

All interface functions are thread-safe. Internal state is protected by a ThreadX mutex (TX_NO_INHERIT priority inheritance disabled).

Module Dependencies

- [rx_error_handler.h](#): Public API and types
- [rx_check.h](#): Parameter validation macros
- [rx_log.h](#): Error logging infrastructure
- ThreadX (tx_api.h): Mutex synchronization

NASA Power of 10 Compliance

- Rule 1: [OK] No goto, setjmp, or recursion
- Rule 2: [OK] All loops have static upper bounds (k_error_handler_max_retries)
- Rule 3: [OK] No dynamic memory allocation (static component slots)
- Rule 4: [OK] Functions are concise (~30-60 lines)
- Rule 5: [OK] Minimum 2 assertions/validations per function
- Rule 6: [OK] Variables declared at smallest scope
- Rule 7: [OK] All return values checked or explicitly cast to void
- Rule 8: [OK] All constants use C23 typed enums
- Rule 9: [WARN] Function pointers used for interface (DIP pattern)
- Rule 10: [OK] Compiles with -Wall -Wextra -Werror

SOLID Principles

- S (SRP): Single responsibility - error tracking only
- O (OCP): Extensible via configuration without modification
- L (LSP): Implements `rx_error_interface_t`, substitutable
- I (ISP): Interface has cohesive set of error operations
- D (DIP): Clients depend on `rx_error_interface_t` abstraction

See also

[rx_error_handler.h](#) Public API and type definitions

[rx_error_interface.h](#) Abstract error interface

[rx_check.h](#) Validation macros (`RX_CHECK_NULL_PTR`)

[rx_log.h](#) Logging functions (`rx_log_error`, `rx_log_info`)

Author

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Date

2026-01-28

Copyright

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Since

Version 1.0.0

Definition in file [rx_error_handler.c](#).

7.45.2 Enumeration Type Documentation

7.45.2.1 `error_handler_backoff_constants_t`

enum [error_handler_backoff_constants_t](#) : `uint32_t`

Internal constants for exponential backoff algorithm.

These constants control the exponential backoff behavior used by [impl_get_backoff_delay\(\)](#). The algorithm doubles the delay on each retry up to the configured maximum.

Backoff Progression Example

With `initial_backoff_ms=100`, `max_backoff_ms=5000`:

Retry	Delay (ms)	Calculation
1	100	$100 \times 2^0 = 100$
2	200	$100 \times 2^1 = 200$
3	400	$100 \times 2^2 = 400$
4	800	$100 \times 2^3 = 800$
5	1600	$100 \times 2^4 = 1600$
6	3200	$100 \times 2^5 = 3200$
7+	5000	Capped at <code>max_backoff_ms</code>

Invariant

`k_exponential_backoff_multiplier` must be 2 for binary exponential

`k_error_handler_max_retries` bounds all retry loops (NASA Rule 2)

Since

Version 1.0.0

Enumerator

<code>k_exponential_backoff_multiplier</code>	Exponential backoff multiplier (base 2). Each retry doubles the previous delay: $\text{delay} = \text{initial} \times 2^{(n-1)}$
<code>k_error_handler_no_retry_limit</code>	Sentinel value indicating unlimited retries. When <code>max_retries</code> is 0, retry limit checks always return false
<code>k_error_handler_first_retry</code>	Starting index for retry loop iteration. Loop starts at 1 since retry 0 uses initial delay directly
<code>k_error_handler_max_retries</code>	Maximum retry iterations for backoff calculation (NASA Rule 2). Provides static upper bound for the exponential backoff loop. After 32 retries with base multiplier 2, delay would be astronomically large, so this cap prevents overflow and ensures bounded execution. Warning Do not increase beyond 32 to avoid <code>uint32_t</code> overflow

Definition at line 192 of file [rx_error_handler.c](#).

```

00192         : uint32_t {
00193     /**
00194      * @brief Exponential backoff multiplier (base 2)
00195      * @details Each retry doubles the previous delay:  $\text{delay} = \text{initial} \times 2^{(n-1)}$ 
00196      */
00197     k_exponential_backoff_multiplier = 2,
00198
00199     /**
00200      * @brief Sentinel value indicating unlimited retries
00201      * @details When max_retries is 0, retry limit checks always return false
00202      */
00203     k_error_handler_no_retry_limit = 0,
00204
00205     /**
00206      * @brief Starting index for retry loop iteration

```



```

00207  * @details Loop starts at 1 since retry 0 uses initial delay directly
00208  */
00209  k_error_handler_first_retry = 1,
00210
00211  /**
00212  * @brief Maximum retry iterations for backoff calculation (NASA Rule 2)
00213  * @details
00214  * Provides static upper bound for the exponential backoff loop.
00215  * After 32 retries with base multiplier 2, delay would be astronomically
00216  * large, so this cap prevents overflow and ensures bounded execution.
00217  * @warning Do not increase beyond 32 to avoid uint32_t overflow
00218  */
00219  k_error_handler_max_retries = 32,
00220 } error_handler_backoff_constants_t;

```

7.45.3 Function Documentation

7.45.3.1 error_handler_deinit()

```

rx_err_t error_handler_deinit (
    error_handler_t * handler)  [nodiscard]

```

Deinitialize error handler and release resources.

Deinitialize error handler and cleanup resources.

Releases resources held by the error handler, including the ThreadX mutex. Safe to call on already-deinitialized handler (idempotent).

Algorithm

1. Validate handler pointer
2. If not initialized, return k_rx_ok (idempotent)
3. Delete ThreadX mutex
4. Set initialized = false
5. Log deinitialization

Precondition

handler points to valid [error_handler_t](#) (initialized or not)

Postcondition

handler->initialized == false
 handler->mutex deleted (if was initialized)

Note

Idempotent: safe to call multiple times
 Not thread-safe: ensure no other threads using handler

Warning

Invalidates any rx_error_interface_t pointing to this handler
 Do not use interface after deinit

Example

```

void system_shutdown(void) {
    // Stop all modules using error interface first
    motor_deinit(&motor);
    comm_deinit(&comm);

    // Now safe to deinit error handler
    error_handler_deinit(&g_error_handler);
}

```

Performance

~10-50 us (mutex deletion)

See also

[error_handler_init\(\)](#) Reinitialize after deinit if needed

Since

Version 1.0.0

Definition at line 1270 of file [rx_error_handler.c](#).

```

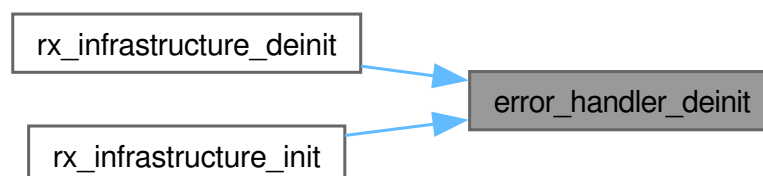
01271 {
01272   RX_CHECK_NULL_PTR(handler, s_tag, "Handler pointer is nullptr");
01273
01274   if (!handler->initialized) {
01275       return k_rx_ok; /* Already deinitialized */
01276   }
01277
01278   /* Delete mutex */
01279   (void)tx_mutex_delete(&handler->mutex);
01280
01281   /* Clear state */
01282   handler->initialized = false;
01283
01284   rx_log_info(s_tag, "Error handler deinitialized");
01285
01286   return k_rx_ok;
01287 }

```

References [error_handler_t::initialized](#), [k_rx_ok](#), [error_handler_t::mutex](#), [RX_CHECK_NULL_PTR](#), [rx_log_info](#), and [s_tag](#).

Referenced by [rx_infrastructure_deinit\(\)](#), and [rx_infrastructure_init\(\)](#).

Here is the caller graph for this function:



7.45.3.2 error_handler_get_interface()

```
rx_err_t error_handler_get_interface (
    rx_error_interface_t * iface,
    error_handler_t * handler) [nodiscard]
```

Get error handler interface for dependency injection.

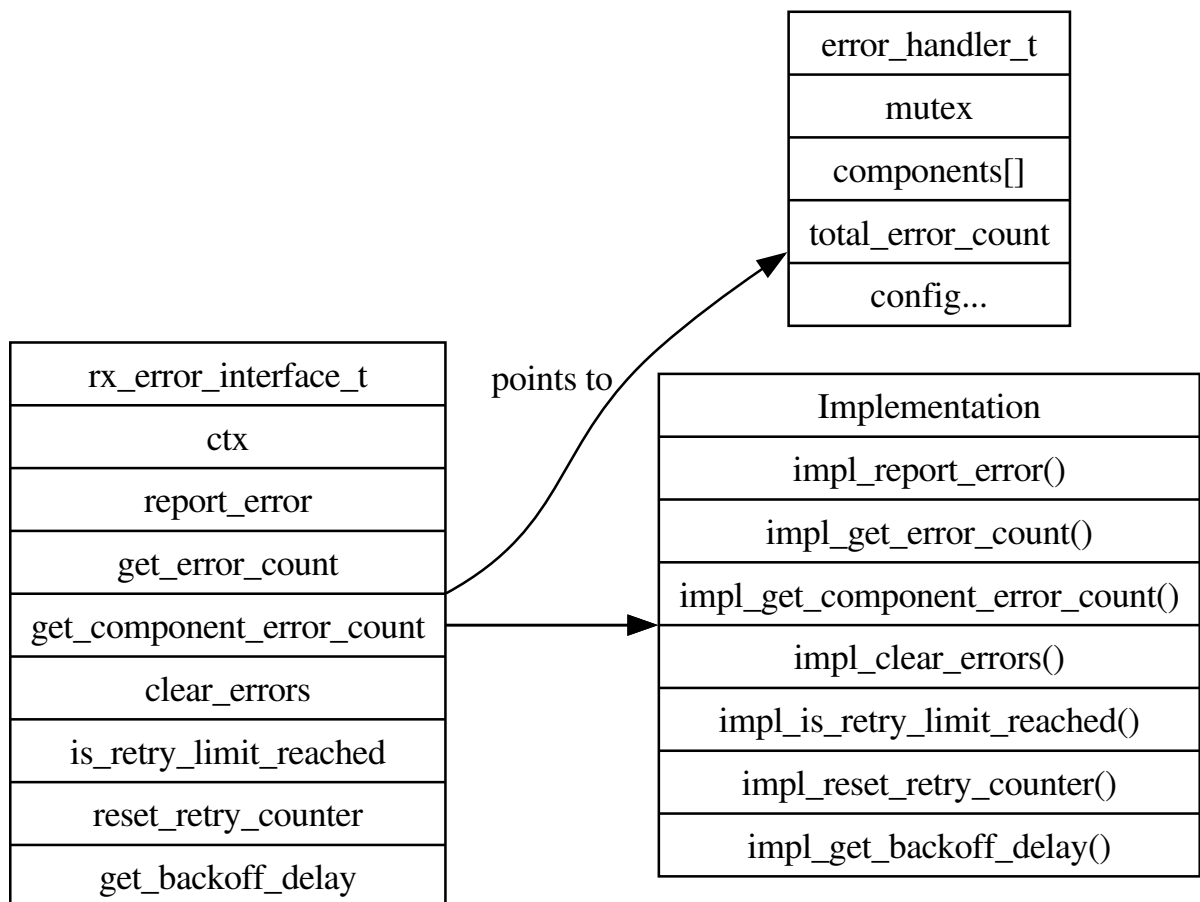
Get abstract interface from concrete error handler (Dependency Inversion).

Populates an rx_error_interface_t structure with function pointers to the handler's implementation functions. This enables dependency injection following the Dependency Inversion Principle (DIP).

Algorithm

1. Validate iface and handler pointers
2. Verify handler is initialized
3. Populate interface structure:
 - ctx = handler (opaque context)
 - Function pointers to impl_* functions

Interface Binding Diagram



Precondition

iface points to allocated `rx_error_interface_t`
handler initialized via [error_handler_init\(\)](#)

Postcondition

iface->ctx == handler
All function pointers set to implementation functions

Note

Handler must remain valid while interface is in use
Interface structure is a shallow copy (pointers only)

Example

```
static error_handler_t g_error_handler;
static rx_error_interface_t g_error_iface;

void system_init(void) {
    error_handler_config_t config = { ... };
    error_handler_init(&g_error_handler, &config);

    // Get interface for dependency injection
    error_handler_get_interface(&g_error_iface, &g_error_handler);

    // Pass interface to modules that need error handling
    motor_init(&motor, &g_error_iface);
    comm_init(&comm, &g_error_iface);
}

// In motor module:
void motor_report_fault(motor_t* motor) {
    motor->error_iface->report_error(
        motor->error_iface->ctx,
        k_rx_err_hw_fault,
        "MOTOR",
        "Overcurrent detected"
    );
}
```

Performance

~1 us (pointer assignments only)

See also

[error_handler_init\(\)](#) Initialize handler first
`rx_error_interface_t` Abstract interface definition

Since

Version 1.0.0

Definition at line 1201 of file [rx_error_handler.c](#).

```

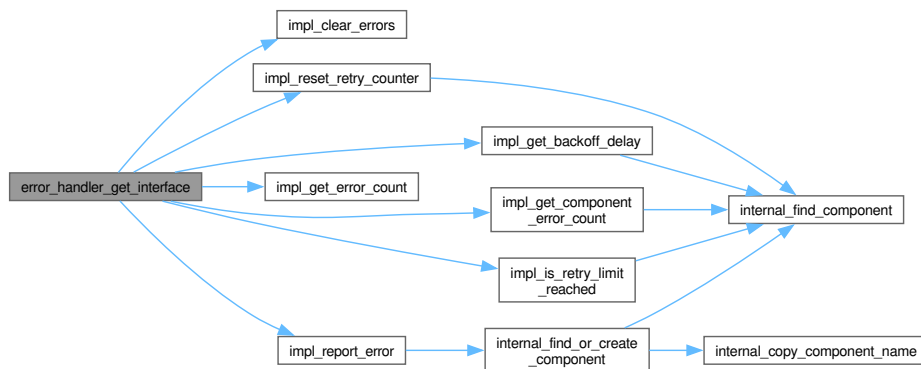
01202 {
01203   RX_CHECK_NULL_PTR(iface, s_tag, "Interface pointer is nullptr");
01204   RX_CHECK_NULL_PTR(handler, s_tag, "Handler pointer is nullptr");
01205
01206   if (!handler->initialized) {
01207     rx_log_error(s_tag, "Handler not initialized");
01208     return k_rx_err_invalid_state;
01209   }
01210
01211   /* Fill interface */
01212   iface->ctx = handler;
01213   iface->report_error = impl_report_error;
01214   iface->get_error_count = impl_get_error_count;
01215   iface->get_component_error_count = impl_get_component_error_count;
01216   iface->clear_errors = impl_clear_errors;
01217   iface->is_retry_limit_reached = impl_is_retry_limit_reached;
01218   iface->reset_retry_counter = impl_reset_retry_counter;
01219   iface->get_backoff_delay = impl_get_backoff_delay;
01220
01221   return k_rx_ok;
01222 }

```

References [impl_clear_errors\(\)](#), [impl_get_backoff_delay\(\)](#), [impl_get_component_error_count\(\)](#), [impl_get_error_count\(\)](#), [impl_is_retry_limit_reached\(\)](#), [impl_report_error\(\)](#), [impl_reset_retry_counter\(\)](#), [error_handler_t::initialized](#), [k_rx_err_invalid_state](#), [k_rx_ok](#), [RX_CHECK_NULL_PTR](#), [rx_log_error](#), and [s_tag](#).

Referenced by [rx_infrastructure_init\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



7.45.3.3 error_handler_init()

```
rx_err_t error_handler_init (
    error_handler_t * handler,
    const error_handler_config_t * config) [nodiscard]
```

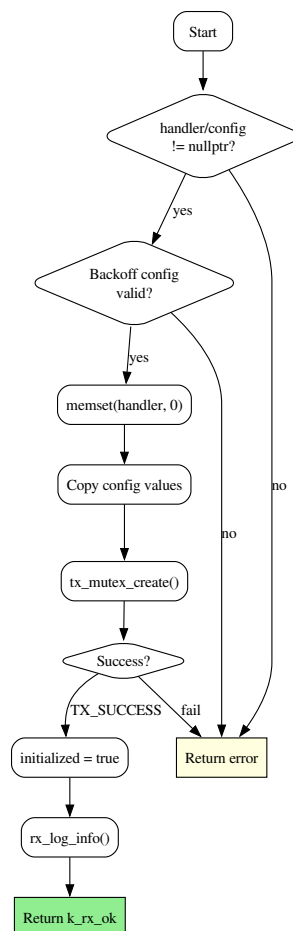
Initialize error handler with retry logic and exponential backoff.

Initialize error handler with specified configuration.

Initializes an error handler instance with the specified configuration. Creates a ThreadX mutex for thread-safe operation and clears all internal state. Must be called before any other error handler operations.

Algorithm

1. Validate handler and config pointers
2. Validate backoff configuration consistency
3. Clear all handler state (memset to 0)
4. Copy configuration values
5. Create ThreadX mutex
6. Set initialized flag
7. Log initialization success



Precondition

handler points to allocated [error_handler_t](#) (not previously initialized)
config points to valid configuration with consistent values

Postcondition

handler->initialized == true
handler->mutex created and ready
All component slots cleared (in_use == false)
total_error_count == 0

Note

Not thread-safe: do not call concurrently for same handler
Call [error_handler_deinit\(\)](#) before reinitializing

Warning

Do not initialize same handler twice without deinit

Example

```
static error_handler_t g_error_handler;

void system__init(void) {
    error_handler_config_t config = {
        .max_retries      = 5,
        .initial_backoff_ms = 100,
        .max_backoff_ms   = 5000
    };

    rx_err_t err = error_handler_init(&g_error_handler, &config);
    if (err != k_rx_ok) {
        // Handle init failure - system cannot proceed
        while (1) { }
    }
}
```

Performance

~50-100 us (memset + mutex creation)

See also

[error_handler_deinit\(\)](#) Cleanup handler
[error_handler_get_interface\(\)](#) Get interface for dependency injection

Since

Version 1.0.0

Definition at line 1096 of file `rx_error_handler.c`.

```

01097 {
01098     RX_CHECK_NULL_PTR(handler, s_tag, "Handler pointer is nullptr");
01099     RX_CHECK_NULL_PTR(config, s_tag, "Config pointer is nullptr");
01100     if (config->max_backoff_ms < config->initial_backoff_ms) {
01101         rx_log_error(s_tag, "Backoff range invalid");
01102         return k_rx_err_invalid_arg;
01103     }
01104     /* Architecturally unreachable: if initial > 0 and max == 0, then max < initial,
01105      * which is caught by the preceding check. No RX_ASSERT needed. */
01106
01107     /* Clear all state */
01108     *handler = (error_handler_t){};
01109
01110     /* Initialize configuration */
01111     handler->max_retries = config->max_retries;
01112     handler->initial_backoff_ms = config->initial_backoff_ms;
01113     handler->max_backoff_ms = config->max_backoff_ms;
01114
01115     /* Create mutex -- tx_mutex_create always succeeds when ThreadX is running */
01116     (void)tx_mutex_create(&handler->mutex, "ErrorHandlerMutex", TX_NO_INHERIT);
01117
01118     handler->initialized = true;
01119
01120     rx_log_info(s_tag, "Error handler initialized");
01121
01122     return k_rx_ok;
01123 }

```

References `error_handler_config_t::initial_backoff_ms`, `error_handler_t::initial_backoff_ms`, `error_handler_t::initial_k_rx_err_invalid_arg`, `k_rx_err_invalid_arg`, `k_rx_ok`, `error_handler_config_t::max_backoff_ms`, `error_handler_t::max_backoff_ms`, `error_handler_config_t::max_retries`, `error_handler_t::max_retries`, `error_handler_t::mutex`, `RX_CHECK_NULL_PTR`, `rx_log_error`, `rx_log_info`, and `s_tag`.

Referenced by `rx_infrastructure_init()`.

Here is the caller graph for this function:



7.45.3.4 impl_clear_errors()

```

rx_err_t impl_clear_errors (
    void * ctx) [static]

```

Clear all error counters (`rx_error_interface_t.clear_errors`).

Resets the total error count and all component-specific error and retry counters to zero. Component tracking entries remain valid (`in_use` stays true) but their counts are zeroed.

Algorithm

1. Validate context pointer
2. Verify handler is initialized
3. Acquire mutex
4. Set `total_error_count = 0`
5. For each component slot: set `error_count = 0`, `retry_count = 0`
6. Release mutex

Precondition

ctx points to initialized [error_handler_t](#)

Postcondition

total_error_count == 0

All component error_count and retry_count == 0

Component in_use flags unchanged (entries still tracked)

Note

Thread-safe: protected by internal mutex

Does NOT remove component entries, only clears their counts

Warning

Clears retry_count too, affecting backoff calculations

Performance

~5-30 us (mutex + loop over all slots)

See also

[impl_reset_retry_counter\(\)](#) Reset only one component's retry counter

Since

Version 1.0.0

Definition at line 683 of file [rx_error_handler.c](#).

```

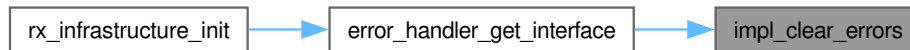
00684 {
00685     error_handler_t* handler = (error_handler_t*)ctx;
00686
00687     RX_CHECK_NULL_PTR(handler, s_tag, "Handler pointer is nullptr");
00688     if (!handler->initialized) {
00689         rx_log_error(s_tag, "Handler not initialized");
00690         return k_rx_err_invalid_state;
00691     }
00692
00693     /* Acquire mutex */
00694     const UINT status = tx_mutex_get(&handler->mutex, TX_WAIT_FOREVER);
00695     if (status != TX_SUCCESS) {
00696         return k_rx_err_rtos_mutex;
00697     }
00698
00699     /* Clear total count */
00700     handler->total_error_count = 0;
00701
00702     /* Clear all component counts */
00703     for (uint32_t i = 0; i < k_error_handler_max_components; i++) {
00704         handler->components[i].error_count = 0;
00705         handler->components[i].retry_count = 0;
00706     }
00707
00708     /* Release mutex */
00709     (void)tx_mutex_put(&handler->mutex);
00710
00711     return k_rx_ok;
00712 }

```

References [error_handler_t::components](#), [error_component_state_t::error_count](#), [error_handler_t::initialized](#), [k_error_handler_max_components](#), [k_rx_err_invalid_state](#), [k_rx_err_rtos_mutex](#), [k_rx_ok](#), [error_handler_t::mutex](#), [error_component_state_t::retry_count](#), [RX_CHECK_NULL_PTR](#), [rx_log_error](#), [s_tag](#), and [error_handler_t::total_error_count](#).

Referenced by [error_handler_get_interface\(\)](#).

Here is the caller graph for this function:



7.45.3.5 impl_get_backoff_delay()

```
uint32_t impl_get_backoff_delay (  
    void * ctx,  
    const char * component) [static]
```

Calculate exponential backoff delay (`rx_error_interface_t.get_backoff_delay`).

Computes the delay (in milliseconds) to wait before the next retry attempt using binary exponential backoff. Delay doubles with each retry, capped at the configured maximum.

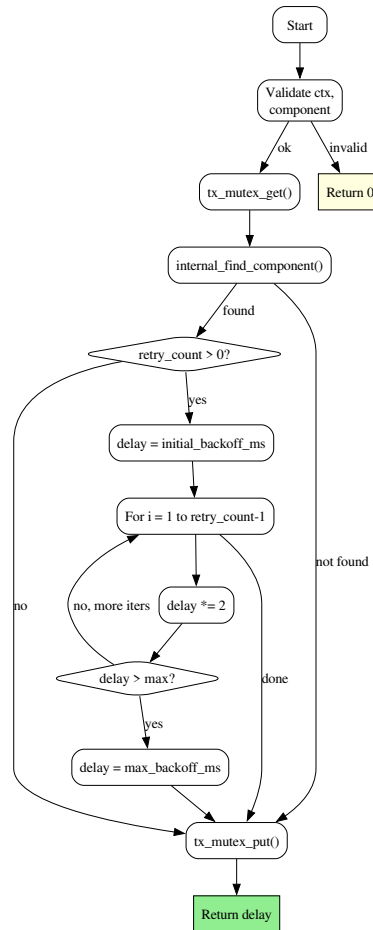
Algorithm

1. Validate context and component pointers (return 0 on failure)
2. Acquire mutex
3. Find component via [internal_find_component\(\)](#)
4. If `retry_count == 0`, `delay = 0` (no backoff on first attempt)
5. Otherwise, compute: `delay = initial x 2^(retry_count - 1)`
6. Cap delay at `max_backoff_ms`
7. Release mutex
8. Return computed delay

Exponential Backoff Formula

$$\text{delay} = \min(\text{initial_backoff_ms} \times 2^{(n-1)}, \text{max_backoff_ms})$$

where $n = \text{retry_count}$ (capped at 32 to prevent overflow)



Precondition

ctx points to initialized [error_handler_t](#)
 component is a valid null-terminated string

Postcondition

Handler state unchanged (read-only operation)

Invariant

Return value $\leq \text{max_backoff_ms}$ (when successful)

Note

Thread-safe: protected by internal mutex
 Returns 0 for first attempt ($\text{retry_count} == 0$)
 Loop bounded by $\text{k_error_handler_max_retries}$ (NASA Rule 2)

Example

```
uint32_t delay = iface->get_backoff_delay(iface->ctx, "SPI");
if (delay > 0) {
    tx_thread_sleep(delay * TX_TIMER_TICKS_PER_MS);
}
// Retry operation
```

Performance

~3-20 us (mutex + search + backoff loop)

See also

[impl_reset_retry_counter\(\)](#) Reset to clear backoff
[impl_report_error\(\)](#) Increments retry_count

Since

Version 1.0.0

Definition at line 949 of file [rx_error_handler.c](#).

```
00950 {
00951     error_handler_t* handler = (error_handler_t*)ctx;
00952
00953     if ((handler == nullptr) || (component == nullptr)) {
00954         return 0;
00955     }
00956     if (!handler->initialized) {
00957         return 0;
00958     }
00959
00960     /* Acquire mutex */
00961     const UINT status = tx_mutex_get(&handler->mutex, TX_WAIT_FOREVER);
00962     if (status != TX_SUCCESS) {
00963         return 0;
00964     }
00965
00966     uint32_t delay_ms = 0;
00967     const error_component_state_t* comp = internal_find_component(handler, component);
00968     if (comp != nullptr && comp->retry_count > 0) {
00969         /* Exponential backoff: delay = initial * 2^(retry_count - 1)
00970          * Capped at max_backoff_ms */
00971         delay_ms = handler->initial_backoff_ms;
00972         uint32_t retry_cap = handler->max_retries;
00973         if (handler->max_retries == k_error_handler_no_retry_limit ||
00974             handler->max_retries > k_error_handler_max_retries) {
00975             retry_cap = k_error_handler_max_retries;
00976         }
00977         const uint32_t retries = (comp->retry_count > retry_cap) ? retry_cap : comp->retry_count;
00978         /* Statically bounded loop: iterate from first retry to max retries cap (NASA Rule 2) */
00979         for (uint32_t i = k_error_handler_first_retry; i < k_error_handler_max_retries; ++i) {
00980             /* Explicit bounds check: break if reached actual retry limit */
00981             if (i >= retries) {
00982                 break;
00983             }
00984             if (delay_ms > handler->max_backoff_ms / k_exponential_backoff_multiplier) {
00985                 delay_ms = handler->max_backoff_ms;
00986                 break;
00987             }
00988             delay_ms *= k_exponential_backoff_multiplier;
00989             if (delay_ms >= handler->max_backoff_ms) {
00990                 delay_ms = handler->max_backoff_ms;
00991                 break;
00992             }
00993         }
00994     }
00995
00996     /* Release mutex */
00997     (void)tx_mutex_put(&handler->mutex);
00998
00999     return delay_ms;
01000 }
```

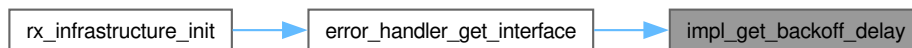
References [error_handler_t::initial_backoff_ms](#), [error_handler_t::initialized](#), [internal_find_component\(\)](#), [k_error_handler_first_retry](#), [k_error_handler_max_retries](#), [k_error_handler_no_retry_limit](#), [k_exponential_backoff_multiplier](#), [error_handler_t::max_backoff_ms](#), [error_handler_t::max_retries](#), [error_handler_t::mutex](#), and [error_component_state_t::retry_count](#).

Referenced by [error_handler_get_interface\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



7.45.3.6 impl_get_component_error_count()

```
uint32_t impl_get_component_error_count (
    void * ctx,
    const char * component) [static]
```

Get component-specific error count ([rx_error_interface_t.get_component_error_count](#)).

Returns the number of errors reported for a specific component. Returns 0 if the component has never reported an error (not tracked).

Algorithm

1. Validate context and component pointers
2. Verify handler is initialized
3. Acquire mutex
4. Search for component via [internal_find_component\(\)](#)
5. If found, read `error_count`; otherwise use 0
6. Release mutex
7. Return count

Precondition

`ctx` points to initialized [error_handler_t](#) (for meaningful result)
`component` is a valid null-terminated string

Postcondition

Handler state unchanged (read-only operation)

Note

Thread-safe: protected by internal mutex

Component name comparison is bounded by `k_error_handler_component_name_max`

Performance

~3-15 us (mutex + component search)

See also

[impl_get_error_count\(\)](#) Total error count

[impl_clear_errors\(\)](#) Reset component counts

Since

Version 1.0.0

Definition at line 618 of file [rx_error_handler.c](#).

```

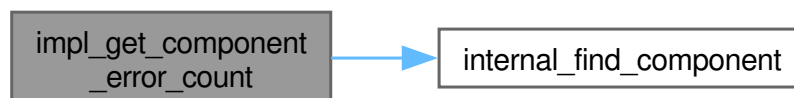
00619 {
00620     error_handler_t* handler = (error_handler_t*)ctx;
00621
00622     if (handler == nullptr || component == nullptr) {
00623         return 0;
00624     }
00625     if (!handler->initialized) {
00626         return 0;
00627     }
00628
00629     /* Acquire mutex */
00630     const UINT status = tx_mutex_get(&handler->mutex, TX_WAIT_FOREVER);
00631     if (status != TX_SUCCESS) {
00632         return 0;
00633     }
00634
00635     uint32_t count = 0;
00636     const error_component_state_t* comp = internal_find_component(handler, component);
00637     if (comp != nullptr) {
00638         count = comp->error_count;
00639     }
00640
00641     /* Release mutex */
00642     (void)tx_mutex_put(&handler->mutex);
00643
00644     return count;
00645 }

```

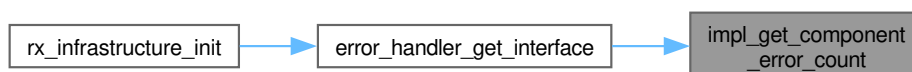
References [error_component_state_t::error_count](#), [error_handler_t::initialized](#), [internal_find_component\(\)](#), and [error_handler_t::mutex](#).

Referenced by [error_handler_get_interface\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



7.45.3.7 impl_get_error_count()

```
uint32_t impl_get_error_count (  
    void * ctx) [static]
```

Get total error count implementation (rx_error_interface_t.get_error_count).

Returns the total number of errors reported across all components since initialization or last clear_errors() call.

Algorithm

1. Validate context pointer and initialization state
2. Acquire mutex
3. Read total_error_count
4. Release mutex
5. Return count (or 0 on any failure)

Precondition

ctx points to initialized [error_handler_t](#) (for meaningful result)

Postcondition

Handler state unchanged (read-only operation)

Note

Thread-safe: protected by internal mutex

Returns 0 on any error condition (fail-safe behavior)

Performance

~2-10 us (mutex overhead only)

See also

[impl_get_component_error_count\(\)](#) Per-component error count

[impl_clear_errors\(\)](#) Reset total count

Since

Version 1.0.0

Definition at line 559 of file [rx_error_handler.c](#).

```

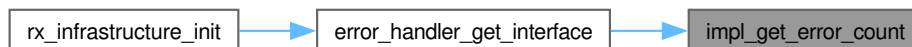
00560 {
00561     error_handler_t* handler = (error_handler_t*)ctx;
00562
00563     if (handler == nullptr) {
00564         return 0;
00565     }
00566     if (!handler->initialized) {
00567         return 0;
00568     }
00569
00570     /* Acquire mutex */
00571     const UINT status = tx_mutex_get(&handler->mutex, TX_WAIT_FOREVER);
00572     if (status != TX_SUCCESS) {
00573         return 0;
00574     }
00575
00576     const uint32_t count = handler->total_error_count;
00577
00578     /* Release mutex */
00579     (void)tx_mutex_put(&handler->mutex);
00580
00581     return count;
00582 }

```

References [error_handler_t::initialized](#), [error_handler_t::mutex](#), and [error_handler_t::total_error_count](#).

Referenced by [error_handler_get_interface\(\)](#).

Here is the caller graph for this function:



7.45.3.8 impl_is_retry_limit_reached()

```

bool impl_is_retry_limit_reached (
    void * ctx,
    const char * component)  [static]

```

Check if component has reached retry limit ([rx_error_interface_t.is_retry_limit_reached](#)).

Determines whether the specified component has exceeded its configured maximum retry count. Used to implement fail-fast behavior after repeated errors.

Algorithm

1. Validate context and component pointers (return true on failure - fail-safe)
2. If max_retries == 0, return false (unlimited retries)
3. Acquire mutex
4. Find component via [internal_find_component\(\)](#)
5. Compare retry_count against max_retries
6. Release mutex
7. Return comparison result

Precondition

ctx points to initialized [error_handler_t](#)
 component is a valid null-terminated string

Postcondition

Handler state unchanged (read-only operation)

Note

Thread-safe: protected by internal mutex
 Fail-safe: returns true (limit reached) on any error condition
 If max_retries == 0, always returns false (unlimited)
 If component not found, returns false (no retries yet)

Performance

~3-15 us (mutex + component search)

See also

[impl_reset_retry_counter\(\)](#) Reset retry count after recovery
[impl_get_backoff_delay\(\)](#) Get delay before next retry

Since

Version 1.0.0

Definition at line 751 of file [rx_error_handler.c](#).

```

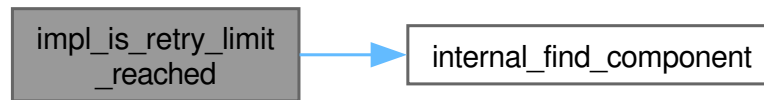
00752 {
00753     error_handler_t* handler = (error_handler_t*)ctx;
00754
00755     if (handler == nullptr || component == nullptr) {
00756         return true; /* Fail-safe: consider limit reached if invalid params */
00757     }
00758     if (!handler->initialized) {
00759         return true;
00760     }
00761
00762     /* If max_retries is 0, no limit */
00763     if (handler->max_retries == 0) {
00764         return false;
00765     }
00766
00767     /* Acquire mutex */
00768     const UINT status = tx_mutex_get(&handler->mutex, TX_WAIT_FOREVER);
00769     if (status != TX_SUCCESS) {
00770         return true; /* Fail-safe */
00771     }
00772
00773     bool limit_reached = false;
00774     const error_component_state_t* comp = internal_find_component(handler, component);
00775     if (comp != nullptr) {
00776         if (comp->retry_count >= handler->max_retries) {
00777             limit_reached = true;
00778         }
00779     }
00780
00781     /* Release mutex */
00782     (void)tx_mutex_put(&handler->mutex);
00783
00784     return limit_reached;
00785 }

```

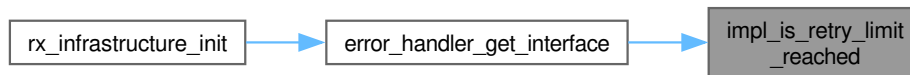
References [error_handler_t::initialized](#), [internal_find_component\(\)](#), [error_handler_t::max_retries](#), [error_handler_t::mutex](#), and [error_component_state_t::retry_count](#).

Referenced by [error_handler_get_interface\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



7.45.3.9 impl_report_error()

```

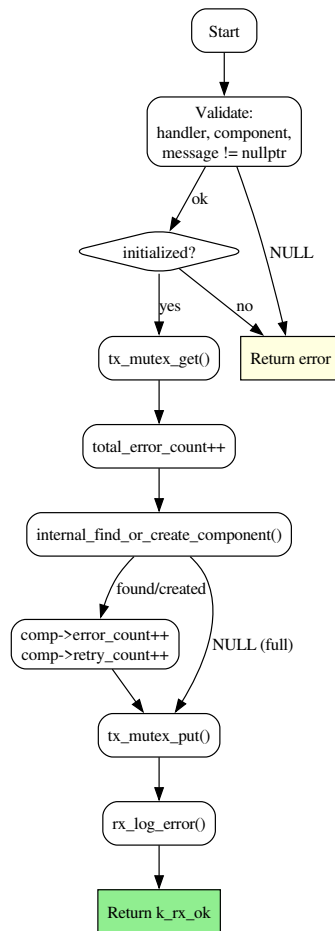
rx_err_t impl_report_error (
    void * ctx,
    rx_err_t err,
    const char * component,
    const char * message) [static]
  
```

Report error implementation (`rx_error_interface_t.report_error`).

Records an error occurrence for the specified component. Increments both the total error count and the component-specific error and retry counters. The error is also logged via `rx_log_error()`.

Algorithm

1. Validate input pointers (NULL check)
2. Verify handler is initialized
3. Acquire mutex for thread-safe access
4. Increment `total_error_count`
5. Find or create component tracking entry
6. If component entry exists, increment `error_count` and `retry_count`
7. Release mutex
8. Log error message and error code



Precondition

ctx points to initialized `error_handler_t`
 component is a valid null-terminated string
 message is a valid null-terminated string

Postcondition

`total_error_count` incremented by 1
 Component's `error_count` and `retry_count` incremented by 1

Note

Thread-safe: protected by internal mutex
 Component created automatically if not exists (up to max slots)
 If component array full, error still logged but not tracked per-component

Performance

~5-50 us depending on component lookup and logging overhead

See also

[impl_clear_errors\(\)](#) Reset all error counters
[impl_reset_retry_counter\(\)](#) Reset only retry counter

Since

Version 1.0.0

Definition at line 490 of file [rx_error_handler.c](#).

```

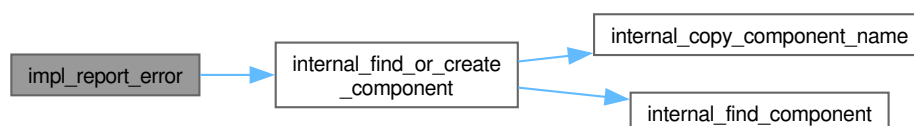
00491 {
00492     error_handler_t* handler = (error_handler_t*)ctx;
00493
00494     RX_CHECK_NULL_PTR(handler, s_tag, "Handler pointer is nullptr");
00495     RX_CHECK_NULL_PTR(component, s_tag, "Component pointer is nullptr");
00496     RX_CHECK_NULL_PTR(message, s_tag, "Message pointer is nullptr");
00497     if (!handler->initialized) {
00498         rx_log_error(s_tag, "Handler not initialized");
00499         return k_rx_err_invalid_state;
00500     }
00501
00502     /* Acquire mutex */
00503     const UINT status = tx_mutex_get(&handler->mutex, TX_WAIT_FOREVER);
00504     if (status != TX_SUCCESS) {
00505         return k_rx_err_rtos_mutex;
00506     }
00507
00508     /* Increment total error count */
00509     handler->total_error_count++;
00510
00511     /* Find or create component */
00512     error_component_state_t* comp = internal_find_or_create_component(handler, component);
00513     if (comp != nullptr) {
00514         comp->error_count++;
00515         comp->retry_count++;
00516     }
00517
00518     /* Release mutex */
00519     (void)tx_mutex_put(&handler->mutex);
00520
00521     /* Log the error */
00522     rx_log_error(component, message);
00523     rx_log_error_val(component, "Error", err);
00524
00525     return k_rx_ok;
00526 }

```

References [error_component_state_t::error_count](#), [error_handler_t::initialized](#), [internal_find_or_create_component\(\)](#), [k_rx_err_invalid_state](#), [k_rx_err_rtos_mutex](#), [k_rx_ok](#), [error_handler_t::mutex](#), [error_component_state_t::retry_count](#), [RX_CHECK_NULL_PTR](#), [rx_log_error](#), [rx_log_error_val](#), [s_tag](#), and [error_handler_t::total_error_count](#).

Referenced by [error_handler_get_interface\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



7.45.3.10 impl_reset_retry_counter()

```
rx_err_t impl_reset_retry_counter (  
    void * ctx,  
    const char * component) [static]
```

Reset component retry counter (rx_error_interface_t.reset_retry_counter).

Resets only the retry counter for a specific component, leaving the error count unchanged. Call this after successful recovery to enable fresh retries.

Algorithm

1. Validate context and component pointers
2. Verify handler is initialized
3. Acquire mutex
4. Find component via [internal_find_component\(\)](#)
5. If found, set retry_count = 0
6. Release mutex

Precondition

ctx points to initialized [error_handler_t](#)
component is a valid null-terminated string

Postcondition

If component found, retry_count == 0
error_count unchanged (only retry reset)

Note

Thread-safe: protected by internal mutex
No-op if component not found (returns k_rx_ok)
Use after successful operation to reset backoff

Example

```
// After successful recovery from USB error  
if (usb_operation_succeeded) {  
    iface->reset_retry_counter(iface->ctx, "USB");  
}
```

Performance

~3-15 us (mutex + component search)

See also

[impl_clear_errors\(\)](#) Reset all counters for all components
[impl_get_backoff_delay\(\)](#) Retry count affects backoff

Since

Version 1.0.0

Definition at line 830 of file [rx_error_handler.c](#).

```

00831 {
00832     error_handler_t* handler = (error_handler_t*)ctx;
00833
00834     RX_CHECK_NULL_PTR(handler, s_tag, "Handler pointer is nullptr");
00835     RX_CHECK_NULL_PTR(component, s_tag, "Component pointer is nullptr");
00836     if (!handler->initialized) {
00837         rx_log_error(s_tag, "Handler not initialized");
00838         return k_rx_err_invalid_state;
00839     }
00840
00841     /* Acquire mutex */
00842     const UINT status = tx_mutex_get(&handler->mutex, TX_WAIT_FOREVER);
00843     if (status != TX_SUCCESS) {
00844         return k_rx_err_rtos_mutex;
00845     }
00846
00847     error_component_state_t* comp = internal_find_component(handler, component);
00848     if (comp != nullptr) {
00849         comp->retry_count = 0;
00850     }
00851
00852     /* Release mutex */
00853     (void)tx_mutex_put(&handler->mutex);
00854
00855     return k_rx_ok;
00856 }

```

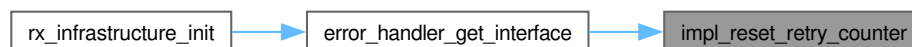
References [error_handler_t::initialized](#), [internal_find_component\(\)](#), [k_rx_err_invalid_state](#), [k_rx_err_rtos_mutex](#), [k_rx_ok](#), [error_handler_t::mutex](#), [error_component_state_t::retry_count](#), [RX_CHECK_NULL_PTR](#), [rx_log_error](#), and [s_tag](#).

Referenced by [error_handler_get_interface\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



7.45.3.11 internal_copy_component_name()

```

void internal_copy_component_name (
    char * dst,
    const char * src,
    const uint32_t max_len) [static]

```

Copy a component name safely without strncpy.

Copies up to `max_len-1` characters from `src` to `dst`, then null-terminates. Replaces `strncpy` to satisfy clang-analyzer-security checks.

Precondition

dst and src must be non-NULL
 max_len must be > 0

Postcondition

dst is null-terminated

Note

Thread-safe: no shared state

Since

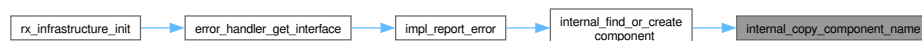
Version 1.0.0

Definition at line 311 of file [rx_error_handler.c](#).

```
00312 {
00313     for (uint32_t i = 0; i < max_len - 1; i++) {
00314         dst[i] = src[i];
00315         if (src[i] == '\0') {
00316             return;
00317         }
00318     }
00319     dst[max_len - 1] = '\0';
00320 }
```

Referenced by [internal_find_or_create_component\(\)](#).

Here is the caller graph for this function:



7.45.3.12 internal_find_component()

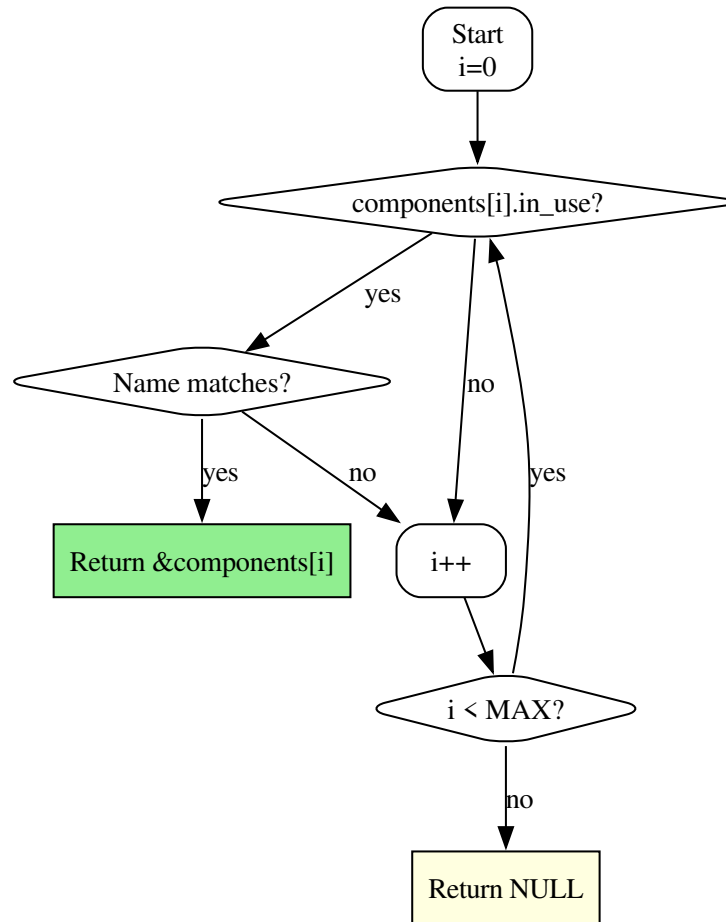
```
error_component_state_t * internal_find_component (
    error_handler_t * handler,
    const char * component) [static]
```

Find component tracking slot by name.

Searches the handler's component array for an existing entry matching the given component name. Uses bounded string comparison with k_error_handler_component_name_max limit.

Algorithm

1. Iterate through all `k_error_handler_max_components` slots
2. For each slot marked `in_use`, compare name strings
3. Return pointer to first matching slot, or `nullptr` if not found



Precondition

`handler != nullptr` (caller responsibility, not validated)
`component != nullptr` (caller responsibility, not validated)
 Caller holds `handler->mutex` (thread safety)

Postcondition

Return value is either `NULL` or points to valid `components[]` entry
 Handler state unchanged (read-only operation)

Note

Not thread-safe; caller must hold mutex
 Time complexity: $O(n)$ where $n = k_error_handler_max_components$

Performance

- Best case: $O(1)$ if first slot matches
- Worst case: $O(n)$ full scan, $n=16$ typical
- $\sim 2\text{-}10$ us depending on array occupancy

See also

[internal_find_or_create_component\(\)](#) Creates entry if not found

[k_error_handler_max_components](#) Maximum tracked components

Since

Version 1.0.0

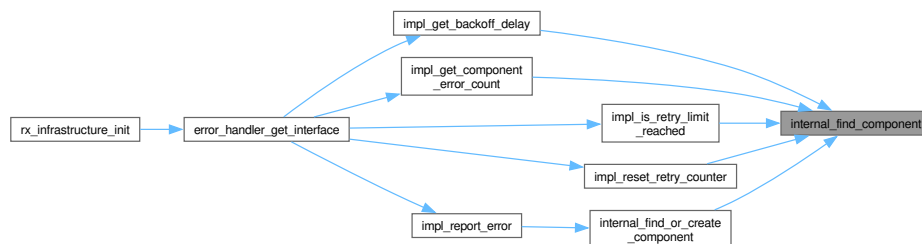
Definition at line 284 of file [rx_error_handler.c](#).

```
00286 {
00287     for (uint32_t i = 0; i < k_error_handler_max_components; i++) {
00288         if ((int)handler->components[i].in_use &&
00289             strcmp(handler->components[i].name, component, k_error_handler_component_name_max) == 0) {
00290             return &handler->components[i];
00291         }
00292     }
00293     return nullptr;
00294 }
```

References [error_handler_t::components](#), [error_component_state_t::in_use](#), [k_error_handler_component_name_max](#), [k_error_handler_max_components](#), and [error_component_state_t::name](#).

Referenced by [impl_get_backoff_delay\(\)](#), [impl_get_component_error_count\(\)](#), [impl_is_retry_limit_reached\(\)](#), [impl_reset_retry_counter\(\)](#), and [internal_find_or_create_component\(\)](#).

Here is the caller graph for this function:



7.45.3.13 internal_find_or_create_component()

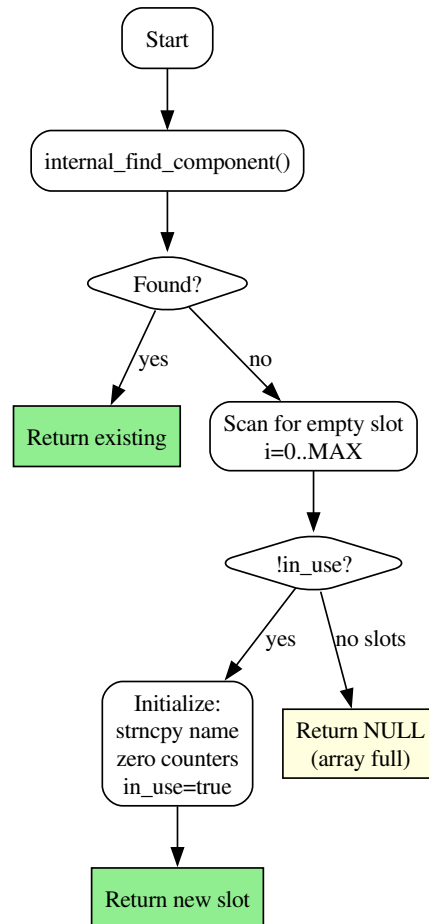
```
error_component_state_t * internal_find_or_create_component (
    error_handler_t * handler,
    const char * component) [static]
```

Find existing component entry or create new one.

First attempts to find an existing component entry via [internal_find_component\(\)](#). If not found, searches for an empty slot (`in_use == false`) and initializes it with the component name and zeroed counters.

Algorithm

1. Call `internal_find_component()` to search for existing entry
2. If found, return pointer to existing entry
3. Otherwise, scan for first empty slot (`in_use == false`)
4. Initialize empty slot: copy name, zero counters, set `in_use = true`
5. Return pointer to newly initialized slot, or `nullptr` if array full



Precondition

`handler != nullptr` (caller responsibility, not validated)
`component != nullptr` (caller responsibility, not validated)
 Caller holds `handler->mutex` (thread safety)

Postcondition

If `return != nullptr`, `component` entry exists and is marked `in_use`
 If new entry created, `error_count` and `retry_count` are 0
 Component name is null-terminated in entry

Invariant

Total `in_use` entries $\leq$ `k_error_handler_max_components`

Note

Not thread-safe; caller must hold mutex

Name truncated to `k_error_handler_component_name_max` - 1 chars

Warning

Returns NULL when component array is full - caller must handle

Performance

- Best case: $O(1)$ if component found in first slot
- Worst case: $O(2n)$ two full scans (find + create)
- ~5-20 us typical

See also

[internal_find_component\(\)](#) Used internally for search

[k_error_handler_max_components](#) Maximum trackable components

Since

Version 1.0.0

Definition at line 388 of file [rx_error_handler.c](#).

```

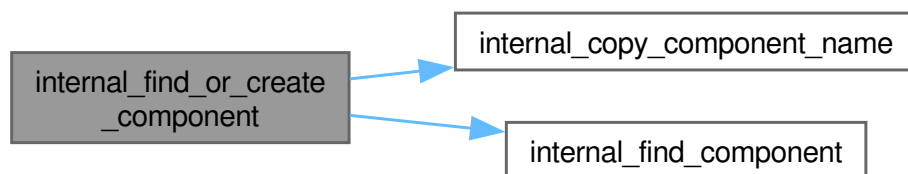
00390 {
00391     /* Try to find existing component */
00392     error_component_state_t* comp = internal_find_component(handler, component);
00393     if (comp != nullptr) {
00394         return comp;
00395     }
00396
00397     /* Find empty slot */
00398     for (uint32_t i = 0; i < k_error_handler_max_components; i++) {
00399         if (!handler->components[i].in_use) {
00400             /* Initialize new component entry */
00401             internal_copy_component_name(handler->components[i].name,
00402                                         component,
00403                                         k_error_handler_component_name_max);
00404             handler->components[i].error_count = 0;
00405             handler->components[i].retry_count = 0;
00406             handler->components[i].in_use = true;
00407             return &handler->components[i];
00408         }
00409     }
00410
00411     /* No slots available */
00412     return nullptr;
00413 }

```

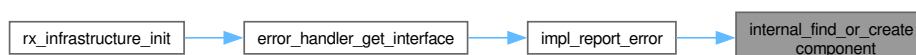
References [error_handler_t::components](#), [error_component_state_t::error_count](#), [error_component_state_t::in_use](#), [internal_copy_component_name\(\)](#), [internal_find_component\(\)](#), [k_error_handler_component_name_max](#), [k_error_handler_max_components](#), [error_component_state_t::name](#), and [error_component_state_t::retry_count](#).

Referenced by [impl_report_error\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



7.45.4 Variable Documentation

7.45.4.1 s_tag

```
const char* const s_tag = "ERROR_HANDLER" [static]
```

Log tag for error handler module.

Identifies log messages from this module. Points to a string literal and is immutable for the entire lifetime of the program (compile-time constant, not just after initialization).

Since

Version 1.0.0

Definition at line 164 of file [rx_error_handler.c](#).

Referenced by [error_handler_deinit\(\)](#), [error_handler_get_interface\(\)](#), [error_handler_init\(\)](#), [impl_clear_all_reservation\(\)](#), [impl_clear_errors\(\)](#), [impl_release_pin\(\)](#), [impl_report_error\(\)](#), [impl_reserve_pin\(\)](#), [impl_reset_retry_counter\(\)](#), [internal_process_exception\(\)](#), [pin_validator_deinit\(\)](#), [pin_validator_get_interface\(\)](#), [pin_validator_init\(\)](#), [rx_exc_nmi_c_handler\(\)](#), [rx_exception_init\(\)](#), [rx_infrastructure_deinit\(\)](#), [rx_infrastructure_init\(\)](#), and [rx_time_threadx_get_interface\(\)](#).

7.46 rx_error_handler.c

[Go to the documentation of this file.](#)

```
00001 /**
00002  * @file rx_error_handler.c
00003  * @brief Error Handler Concrete Implementation
00004  *
00005  * @details
00006  * Implements the rx_error_interface_t abstract interface with concrete error
00007  * tracking, retry logic, and exponential backoff capabilities. This module
00008  * provides a centralized error handling mechanism for the STAR firmware,
00009  * enabling consistent error reporting, component-level error tracking, and
00010  * intelligent retry strategies.
00011  *
00012  * @par Implementation Architecture
00013  * @dot
00014  * digraph error_handler_architecture {
00015  *   rankdir=TB;
00016  *   node [shape=box, style=rounded];
00017  *
00018  *   subgraph cluster_public {
00019  *     label="Public API";
00020  *     style=filled;
00021  *     color=lightblue;
00022  *     init [label="error_handler_init()"];
00023  *     deinit [label="error_handler_deinit()"];
00024  *     get_iface [label="error_handler_get_interface()"];
00025  *   }
00026  *
00027  *   subgraph cluster_interface {
00028  *     label="Interface Implementation";
00029  *     style=filled;
00030  *     color=lightgreen;
00031  *     report [label="impl_report_error()"];
00032  *     get_count [label="impl_get_error_count()"];
00033  *     get_comp [label="impl_get_component_error_count()"];
00034  *     clear [label="impl_clear_errors()"];
00035  *     retry_check [label="impl_is_retry_limit_reached()"];
00036  *     reset_retry [label="impl_reset_retry_counter()"];
00037  *     backoff [label="impl_get_backoff_delay()"];
00038  *   }
00039  *
00040  *   subgraph cluster_internal {
00041  *     label="Internal Helpers";
00042  *     style=filled;
```

```

00043 *   color=lightyellow;
00044 *   find [label="internal_find_component()"];
00045 *   find_create [label="internal_find_or_create_component()"];
00046 * }
00047 *
00048 * subgraph cluster_state {
00049 *   label="State Management";
00050 *   style=filled;
00051 *   color=lightgrey;
00052 *   handler [label="error_handler_t\n(mutex protected)"];
00053 *   components [label="error_component_state_t[]\n(per-component tracking)"];
00054 * }
00055 *
00056 * init -> handler [label="initialize"];
00057 * get_iface -> handler [label="bind"];
00058 * report -> find_create -> components;
00059 * get_count -> handler;
00060 * get_comp -> find -> components;
00061 * clear -> handler;
00062 * clear -> components;
00063 * retry_check -> find -> components;
00064 * reset_retry -> find -> components;
00065 * backoff -> find -> components;
00066 * deinit -> handler [label="cleanup"];
00067 * }
00068 * @enddot
00069 *
00070 * @par Exponential Backoff Algorithm
00071 * The backoff delay calculation uses exponential growth with configurable
00072 * bounds:
00073 *
00074 * @f[
00075 *   \text{delay} = \min\left(\text{initial} \times 2^{\left(\text{retries}-1\right)},
00076 *   \text{max\_backoff}\right)
00077 * @f]
00078 *
00079 * @par State Machine
00080 * @startuml
00081 * [*] --> Uninitialized
00082 * Uninitialized --> Initialized : error_handler_init()
00083 * Initialized --> Initialized : report_error() / get_error_count() / etc.
00084 * Initialized --> Uninitialized : error_handler_deinit()
00085 * Initialized --> Initialized : clear_errors()
00086 * @enduml
00087 *
00088 * @par Memory Layout
00089 * | Component | Size | Description |
00090 * |-----|-----|-----|
00091 * | error_handler_t | ~1.2 KB | Main handler structure |
00092 * | TX_MUTEX | ~100 B | ThreadX mutex control block |
00093 * | components[] | ~1 KB | 16 component tracking slots |
00094 *
00095 * @par Performance Characteristics
00096 * | Operation | Best Case | Worst Case | Notes |
00097 * |-----|-----|-----|-----|
00098 * | report_error() | ~5 us | ~50 us | Mutex + string cmp |
00099 * | get_error_count() | ~2 us | ~10 us | Mutex only |
00100 * | get_backoff_delay() | ~3 us | ~20 us | Loop up to 32 iters |
00101 * | clear_errors() | ~5 us | ~30 us | Clears all slots |
00102 *
00103 * @par Thread Safety
00104 * All interface functions are thread-safe. Internal state is protected by
00105 * a ThreadX mutex (TX_NO_INHERIT priority inheritance disabled).
00106 *
00107 * @par Module Dependencies
00108 * - rx_error_handler.h: Public API and types
00109 * - rx_check.h: Parameter validation macros
00110 * - rx_log.h: Error logging infrastructure
00111 * - ThreadX (tx_api.h): Mutex synchronization
00112 *
00113 * @par NASA Power of 10 Compliance
00114 * - Rule 1: [OK] No goto, setjmp, or recursion
00115 * - Rule 2: [OK] All loops have static upper bounds (k_error_handler_max_retries)
00116 * - Rule 3: [OK] No dynamic memory allocation (static component slots)
00117 * - Rule 4: [OK] Functions are concise (~30-60 lines)
00118 * - Rule 5: [OK] Minimum 2 assertions/validations per function
00119 * - Rule 6: [OK] Variables declared at smallest scope
00120 * - Rule 7: [OK] All return values checked or explicitly cast to void
00121 * - Rule 8: [OK] All constants use C23 typed enums
00122 * - Rule 9: [WARN] Function pointers used for interface (DIP pattern)
00123 * - Rule 10: [OK] Compiles with -Wall -Wextra -Werror
00124 *
00125 * @par SOLID Principles
00126 * - **S (SRP):** Single responsibility - error tracking only
00127 * - **O (OCP):** Extensible via configuration without modification
00128 * - **L (LSP):** Implements rx_error_interface_t, substitutable
00129 * - **I (ISP):** Interface has cohesive set of error operations

```

```

00130 * - **D (DIP):** Clients depend on rx_error_interface_t abstraction
00131 *
00132 * @see rx_error_handler.h Public API and type definitions
00133 * @see rx_error_interface.h Abstract error interface
00134 * @see rx_check.h Validation macros (RX_CHECK_NULL_PTR)
00135 * @see rx_log.h Logging functions (rx_log_error, rx_log_info)
00136 *
00137 * @author Locked, Inc.
00138 * @date 2026-01-28
00139 * @copyright Copyright (c) 2026 Locked Inc.
00140 * SPDX-License-Identifier: MIT
00141 * @since Version 1.0.0
00142 */
00143
00144 #include "rx_error_handler.h"
00145
00146 #include <string.h>
00147
00148 #include "rx_check.h"
00149 #include "rx_log.h"
00150
00151 /*
=====
00152 * Module State
00153 *
=====
00154 */
00155
00156 /**
00157 * @var s_tag
00158 * @brief Log tag for error handler module
00159 * @details Identifies log messages from this module. Points to a string
00160 *          literal and is immutable for the entire lifetime of the program
00161 *          (compile-time constant, not just after initialization).
00162 * @since Version 1.0.0
00163 */
00164 static const char* const s_tag = "ERROR_HANDLER";
00165
00166 /**
00167 * @enum error_handler_backoff_constants_t
00168 * @brief Internal constants for exponential backoff algorithm
00169 *
00170 * @details
00171 * These constants control the exponential backoff behavior used by
00172 * impl_get_backoff_delay(). The algorithm doubles the delay on each
00173 * retry up to the configured maximum.
00174 *
00175 * @par Backoff Progression Example
00176 * With initial_backoff_ms=100, max_backoff_ms=5000:
00177 * | Retry | Delay (ms) | Calculation |
00178 * |-----|-----|-----|
00179 * | 1 | 100 | 100 x 20 = 100 |
00180 * | 2 | 200 | 100 x 21 = 200 |
00181 * | 3 | 400 | 100 x 22 = 400 |
00182 * | 4 | 800 | 100 x 23 = 800 |
00183 * | 5 | 1600 | 100 x 24 = 1600 |
00184 * | 6 | 3200 | 100 x 25 = 3200 |
00185 * | 7+ | 5000 | Capped at max_backoff_ms |
00186 *
00187 * @invariant k_exponential_backoff_multiplier must be 2 for binary exponential
00188 * @invariant k_error_handler_max_retries bounds all retry loops (NASA Rule 2)
00189 *
00190 * @since Version 1.0.0
00191 */
00192 typedef enum : uint32_t {
00193 /**
00194 * @brief Exponential backoff multiplier (base 2)
00195 * @details Each retry doubles the previous delay: delay = initial x 2(n-1)
00196 */
00197 k_exponential_backoff_multiplier = 2,
00198
00199 /**
00200 * @brief Sentinel value indicating unlimited retries
00201 * @details When max_retries is 0, retry limit checks always return false
00202 */
00203 k_error_handler_no_retry_limit = 0,
00204
00205 /**
00206 * @brief Starting index for retry loop iteration
00207 * @details Loop starts at 1 since retry 0 uses initial delay directly
00208 */
00209 k_error_handler_first_retry = 1,
00210
00211 /**
00212 * @brief Maximum retry iterations for backoff calculation (NASA Rule 2)
00213 * @details
00214 * Provides static upper bound for the exponential backoff loop.

```

```

00215  * After 32 retries with base multiplier 2, delay would be astronomically
00216  * large, so this cap prevents overflow and ensures bounded execution.
00217  * @warning Do not increase beyond 32 to avoid uint32_t overflow
00218  */
00219  k_error_handler_max_retries = 32,
00220 } error_handler_backoff_constants_t;
00221
00222 /*
=====
00223  * Internal Helper Functions
00224  *
=====
00225  */
00226
00227 /**
00228  * @brief Find component tracking slot by name
00229  *
00230  * @details
00231  * Searches the handler's component array for an existing entry matching
00232  * the given component name. Uses bounded string comparison with
00233  * k_error_handler_component_name_max limit.
00234  *
00235  * @par Algorithm
00236  * 1. Iterate through all k_error_handler_max_components slots
00237  * 2. For each slot marked in_use, compare name strings
00238  * 3. Return pointer to first matching slot, or nullptr if not found
00239  *
00240  * @dot
00241  * digraph find_component {
00242  *   rankdir=TB;
00243  *   node [shape=box, style=rounded];
00244  *   start [label="Start\ni=0"];
00245  *   check_use [label="components[i].in_use?" shape=diamond];
00246  *   check_name [label="Name matches?" shape=diamond];
00247  *   found [label="Return &components[i]" style=filled, fillcolor=lightgreen];
00248  *   next [label="i++"];
00249  *   check_bound [label="i < MAX?" shape=diamond];
00250  *   not_found [label="Return NULL" style=filled, fillcolor=lightyellow];
00251  *   start -> check_use;
00252  *   check_use -> check_name [label="yes"];
00253  *   check_use -> next [label="no"];
00254  *   check_name -> found [label="yes"];
00255  *   check_name -> next [label="no"];
00256  *   next -> check_bound;
00257  *   check_bound -> check_use [label="yes"];
00258  *   check_bound -> not_found [label="no"];
00259  * }
00260  * @enddot
00261  *
00262  *
00263  *
00264  * @pre handler != nullptr (caller responsibility, not validated)
00265  * @pre component != nullptr (caller responsibility, not validated)
00266  * @pre Caller holds handler->mutex (thread safety)
00267  *
00268  * @post Return value is either NULL or points to valid components[] entry
00269  * @post Handler state unchanged (read-only operation)
00270  *
00271  * @note Not thread-safe; caller must hold mutex
00272  * @note Time complexity: O(n) where n = k_error_handler_max_components
00273  *
00274  * @par Performance
00275  * - Best case: O(1) if first slot matches
00276  * - Worst case: O(n) full scan, n=16 typical
00277  * - ~2-10 us depending on array occupancy
00278  *
00279  * @see internal_find_or_create_component() Creates entry if not found
00280  * @see k_error_handler_max_components Maximum tracked components
00281  *
00282  * @since Version 1.0.0
00283  */
00284 static error_component_state_t* internal_find_component(error_handler_t* handler,
00285                                                         const char* component)
00286 {
00287     for (uint32_t i = 0; i < k_error_handler_max_components; i++) {
00288         if ((int)handler->components[i].in_use &&
00289             strcmp(handler->components[i].name, component, k_error_handler_component_name_max) == 0) {
00290             return &handler->components[i];
00291         }
00292     }
00293     return nullptr;
00294 }
00295
00296 /**
00297  * @brief Copy a component name safely without strncpy
00298  *
00299  * @details

```

```

00300 * Copies up to max_len-1 characters from src to dst, then null-terminates.
00301 * Replaces strncpy to satisfy clang-analyzer-security checks.
00302 *
00303 *
00304 * @pre dst and src must be non-NULL
00305 * @pre max_len must be > 0
00306 * @post dst is null-terminated
00307 *
00308 * @note Thread-safe: no shared state
00309 * @since Version 1.0.0
00310 */
00311 static void internal_copy_component_name(char* dst, const char* src, const uint32_t max_len)
00312 {
00313     for (uint32_t i = 0; i < max_len - 1; i++) {
00314         dst[i] = src[i];
00315         if (src[i] == '\0') {
00316             return;
00317         }
00318     }
00319     dst[max_len - 1] = '\0';
00320 }
00321
00322 /**
00323  * @brief Find existing component entry or create new one
00324  *
00325  * @details
00326  * First attempts to find an existing component entry via internal_find_component().
00327  * If not found, searches for an empty slot (in_use == false) and initializes
00328  * it with the component name and zeroed counters.
00329  *
00330  * @par Algorithm
00331  * 1. Call internal_find_component() to search for existing entry
00332  * 2. If found, return pointer to existing entry
00333  * 3. Otherwise, scan for first empty slot (in_use == false)
00334  * 4. Initialize empty slot: copy name, zero counters, set in_use = true
00335  * 5. Return pointer to newly initialized slot, or nullptr if array full
00336  *
00337  * @dot
00338  * digraph find_or_create {
00339  *     rankdir=TB;
00340  *     node [shape=box, style=rounded];
00341  *     start [label="Start"];
00342  *     find [label="internal_find_component()"];
00343  *     check_found [label="Found?" shape=diamond];
00344  *     return_found [label="Return existing" style=filled, fillcolor=lightgreen];
00345  *     scan_empty [label="Scan for empty slot\ni=0..MAX"];
00346  *     check_empty [label="!in_use?" shape=diamond];
00347  *     init_slot [label="Initialize:\nstrncpy name\nzero counters\nin_use=true"];
00348  *     return_new [label="Return new slot" style=filled, fillcolor=lightgreen];
00349  *     return_null [label="Return NULL\n(array full)" style=filled, fillcolor=lightyellow];
00350  *     start -> find;
00351  *     find -> check_found;
00352  *     check_found -> return_found [label="yes"];
00353  *     check_found -> scan_empty [label="no"];
00354  *     scan_empty -> check_empty;
00355  *     check_empty -> init_slot [label="yes"];
00356  *     check_empty -> return_null [label="no slots"];
00357  *     init_slot -> return_new;
00358  * }
00359  * @enddot
00360  *
00361  *
00362  *
00363  * @pre handler != nullptr (caller responsibility, not validated)
00364  * @pre component != nullptr (caller responsibility, not validated)
00365  * @pre Caller holds handler->mutex (thread safety)
00366  *
00367  * @post If return != nullptr, component entry exists and is marked in_use
00368  * @post If new entry created, error_count and retry_count are 0
00369  * @post Component name is null-terminated in entry
00370  *
00371  * @invariant Total in_use entries <= k_error_handler_max_components
00372  *
00373  * @note Not thread-safe; caller must hold mutex
00374  * @note Name truncated to k_error_handler_component_name_max - 1 chars
00375  *
00376  * @warning Returns NULL when component array is full - caller must handle
00377  *
00378  * @par Performance
00379  * - Best case: O(1) if component found in first slot
00380  * - Worst case: O(2n) two full scans (find + create)
00381  * - ~5-20 us typical
00382  *
00383  * @see internal_find_component() Used internally for search
00384  * @see k_error_handler_max_components Maximum trackable components
00385  *
00386  * @since Version 1.0.0

```



```

00387 */
00388 static error_component_state_t* internal_find_or_create_component(error_handler_t* handler,
00389                                                                    const char* component)
00390 {
00391     /* Try to find existing component */
00392     error_component_state_t* comp = internal_find_component(handler, component);
00393     if (comp != nullptr) {
00394         return comp;
00395     }
00396
00397     /* Find empty slot */
00398     for (uint32_t i = 0; i < k_error_handler_max_components; i++) {
00399         if (!handler->components[i].in_use) {
00400             /* Initialize new component entry */
00401             internal_copy_component_name(handler->components[i].name,
00402                                         component,
00403                                         k_error_handler_component_name_max);
00404             handler->components[i].error_count = 0;
00405             handler->components[i].retry_count = 0;
00406             handler->components[i].in_use = true;
00407             return &handler->components[i];
00408         }
00409     }
00410
00411     /* No slots available */
00412     return nullptr;
00413 }
00414
00415 /*
=====
00416 * Interface Implementation Functions
00417 *
=====
00418 */
00419 /**
00420 **
00421 * @brief Report error implementation (rx_error_interface_t.report_error)
00422 *
00423 * @details
00424 * Records an error occurrence for the specified component. Increments both
00425 * the total error count and the component-specific error and retry counters.
00426 * The error is also logged via rx_log_error().
00427 *
00428 * @par Algorithm
00429 * 1. Validate input pointers (NULL check)
00430 * 2. Verify handler is initialized
00431 * 3. Acquire mutex for thread-safe access
00432 * 4. Increment total_error_count
00433 * 5. Find or create component tracking entry
00434 * 6. If component entry exists, increment error_count and retry_count
00435 * 7. Release mutex
00436 * 8. Log error message and error code
00437 *
00438 * @dot
00439 * digraph report_error {
00440 *     rankdir=TB;
00441 *     node [shape=box, style=rounded];
00442 *     start [label="Start"];
00443 *     validate [label="Validate:\nhandler, component,\nmessage != nullptr"];
00444 *     check_init [label="initialized?" shape=diamond];
00445 *     get_mutex [label="tx_mutex_get()"];
00446 *     inc_total [label="total_error_count++"];
00447 *     find_comp [label="internal_find_or_create_component()"];
00448 *     inc_comp [label="comp->error_count++\ncomp->retry_count++"];
00449 *     put_mutex [label="tx_mutex_put()"];
00450 *     log_err [label="rx_log_error()"];
00451 *     return_ok [label="Return k_rx_ok" style=filled, fillcolor=lightgreen];
00452 *     return_err [label="Return error" style=filled, fillcolor=lightyellow];
00453 *     start -> validate;
00454 *     validate -> check_init [label="ok"];
00455 *     validate -> return_err [label="NULL"];
00456 *     check_init -> get_mutex [label="yes"];
00457 *     check_init -> return_err [label="no"];
00458 *     get_mutex -> inc_total;
00459 *     inc_total -> find_comp;
00460 *     find_comp -> inc_comp [label="found/created"];
00461 *     find_comp -> put_mutex [label="NULL (full)"];
00462 *     inc_comp -> put_mutex;
00463 *     put_mutex -> log_err;
00464 *     log_err -> return_ok;
00465 * }
00466 * @enddot
00467 *
00468 *
00469 *
00470 * @pre ctx points to initialized error_handler_t
00471 * @pre component is a valid null-terminated string

```

```

00472 * @pre message is a valid null-terminated string
00473 *
00474 * @post total_error_count incremented by 1
00475 * @post Component's error_count and retry_count incremented by 1
00476 *
00477 * @note Thread-safe: protected by internal mutex
00478 * @note Component created automatically if not exists (up to max slots)
00479 * @note If component array full, error still logged but not tracked per-component
00480 *
00481 * @par Performance
00482 * ~5-50 us depending on component lookup and logging overhead
00483 *
00484 * @see impl_clear_errors() Reset all error counters
00485 * @see impl_reset_retry_counter() Reset only retry counter
00486 *
00487 * @since Version 1.0.0
00488 */
00489 static rx_err_t
00490 impl_report_error(void* ctx, rx_err_t err, const char* component, const char* message)
00491 {
00492     error_handler_t* handler = (error_handler_t*)ctx;
00493
00494     RX_CHECK_NULL_PTR(handler, s_tag, "Handler pointer is nullptr");
00495     RX_CHECK_NULL_PTR(component, s_tag, "Component pointer is nullptr");
00496     RX_CHECK_NULL_PTR(message, s_tag, "Message pointer is nullptr");
00497     if (!handler->initialized) {
00498         rx_log_error(s_tag, "Handler not initialized");
00499         return k_rx_err_invalid_state;
00500     }
00501
00502     /* Acquire mutex */
00503     const UINT status = tx_mutex_get(&handler->mutex, TX_WAIT_FOREVER);
00504     if (status != TX_SUCCESS) {
00505         return k_rx_err_rtos_mutex;
00506     }
00507
00508     /* Increment total error count */
00509     handler->total_error_count++;
00510
00511     /* Find or create component */
00512     error_component_state_t* comp = internal_find_or_create_component(handler, component);
00513     if (comp != nullptr) {
00514         comp->error_count++;
00515         comp->retry_count++;
00516     }
00517
00518     /* Release mutex */
00519     (void)tx_mutex_put(&handler->mutex);
00520
00521     /* Log the error */
00522     rx_log_error(component, message);
00523     rx_log_error_val(component, "Error", err);
00524
00525     return k_rx_ok;
00526 }
00527
00528 /**
00529 * @brief Get total error count implementation (rx_error_interface_t.get_error_count)
00530 *
00531 * @details
00532 * Returns the total number of errors reported across all components since
00533 * initialization or last clear_errors() call.
00534 *
00535 * @par Algorithm
00536 * 1. Validate context pointer and initialization state
00537 * 2. Acquire mutex
00538 * 3. Read total_error_count
00539 * 4. Release mutex
00540 * 5. Return count (or 0 on any failure)
00541 *
00542 *
00543 *
00544 * @pre ctx points to initialized error_handler_t (for meaningful result)
00545 *
00546 * @post Handler state unchanged (read-only operation)
00547 *
00548 * @note Thread-safe: protected by internal mutex
00549 * @note Returns 0 on any error condition (fail-safe behavior)
00550 *
00551 * @par Performance
00552 * ~2-10 us (mutex overhead only)
00553 *
00554 * @see impl_get_component_error_count() Per-component error count
00555 * @see impl_clear_errors() Reset total count
00556 *
00557 * @since Version 1.0.0
00558 */

```

```

00559 static uint32_t impl_get_error_count(void* ctx)
00560 {
00561     error_handler_t* handler = (error_handler_t*)ctx;
00562
00563     if (handler == nullptr) {
00564         return 0;
00565     }
00566     if (!handler->initialized) {
00567         return 0;
00568     }
00569
00570     /* Acquire mutex */
00571     const UINT status = tx_mutex_get(&handler->mutex, TX_WAIT_FOREVER);
00572     if (status != TX_SUCCESS) {
00573         return 0;
00574     }
00575
00576     const uint32_t count = handler->total_error_count;
00577
00578     /* Release mutex */
00579     (void)tx_mutex_put(&handler->mutex);
00580
00581     return count;
00582 }
00583
00584 /**
00585  * @brief Get component-specific error count (rx_error_interface_t.get_component_error_count)
00586  *
00587  * @details
00588  * Returns the number of errors reported for a specific component.
00589  * Returns 0 if the component has never reported an error (not tracked).
00590  *
00591  * @par Algorithm
00592  * 1. Validate context and component pointers
00593  * 2. Verify handler is initialized
00594  * 3. Acquire mutex
00595  * 4. Search for component via internal_find_component()
00596  * 5. If found, read error_count; otherwise use 0
00597  * 6. Release mutex
00598  * 7. Return count
00599  *
00600  *
00601  *
00602  * @pre ctx points to initialized error_handler_t (for meaningful result)
00603  * @pre component is a valid null-terminated string
00604  *
00605  * @post Handler state unchanged (read-only operation)
00606  *
00607  * @note Thread-safe: protected by internal mutex
00608  * @note Component name comparison is bounded by k_error_handler_component_name_max
00609  *
00610  * @par Performance
00611  * ~3-15 us (mutex + component search)
00612  *
00613  * @see impl_get_error_count() Total error count
00614  * @see impl_clear_errors() Reset component counts
00615  *
00616  * @since Version 1.0.0
00617  */
00618 static uint32_t impl_get_component_error_count(void* ctx, const char* component)
00619 {
00620     error_handler_t* handler = (error_handler_t*)ctx;
00621
00622     if (handler == nullptr || component == nullptr) {
00623         return 0;
00624     }
00625     if (!handler->initialized) {
00626         return 0;
00627     }
00628
00629     /* Acquire mutex */
00630     const UINT status = tx_mutex_get(&handler->mutex, TX_WAIT_FOREVER);
00631     if (status != TX_SUCCESS) {
00632         return 0;
00633     }
00634
00635     uint32_t count = 0;
00636     const error_component_state_t* comp = internal_find_component(handler, component);
00637     if (comp != nullptr) {
00638         count = comp->error_count;
00639     }
00640
00641     /* Release mutex */
00642     (void)tx_mutex_put(&handler->mutex);
00643
00644     return count;
00645 }

```

```

00646
00647 /**
00648  * @brief Clear all error counters (rx_error_interface_t.clear_errors)
00649  *
00650  * @details
00651  * Resets the total error count and all component-specific error and retry
00652  * counters to zero. Component tracking entries remain valid (in_use stays true)
00653  * but their counts are zeroed.
00654  *
00655  * @par Algorithm
00656  * 1. Validate context pointer
00657  * 2. Verify handler is initialized
00658  * 3. Acquire mutex
00659  * 4. Set total_error_count = 0
00660  * 5. For each component slot: set error_count = 0, retry_count = 0
00661  * 6. Release mutex
00662  *
00663  *
00664  *
00665  * @pre ctx points to initialized error_handler_t
00666  *
00667  * @post total_error_count == 0
00668  * @post All component error_count and retry_count == 0
00669  * @post Component in_use flags unchanged (entries still tracked)
00670  *
00671  * @note Thread-safe: protected by internal mutex
00672  * @note Does NOT remove component entries, only clears their counts
00673  *
00674  * @warning Clears retry_count too, affecting backoff calculations
00675  *
00676  * @par Performance
00677  * ~5-30 us (mutex + loop over all slots)
00678  *
00679  * @see impl_reset_retry_counter() Reset only one component's retry counter
00680  *
00681  * @since Version 1.0.0
00682  */
00683 static rx_err_t impl_clear_errors(void* ctx)
00684 {
00685     error_handler_t* handler = (error_handler_t*)ctx;
00686
00687     RX_CHECK_NULL_PTR(handler, s_tag, "Handler pointer is nullptr");
00688     if (!handler->initialized) {
00689         rx_log_error(s_tag, "Handler not initialized");
00690         return k_rx_err_invalid_state;
00691     }
00692
00693     /* Acquire mutex */
00694     const UINT status = tx_mutex_get(&handler->mutex, TX_WAIT_FOREVER);
00695     if (status != TX_SUCCESS) {
00696         return k_rx_err_rtos_mutex;
00697     }
00698
00699     /* Clear total count */
00700     handler->total_error_count = 0;
00701
00702     /* Clear all component counts */
00703     for (uint32_t i = 0; i < k_error_handler_max_components; i++) {
00704         handler->components[i].error_count = 0;
00705         handler->components[i].retry_count = 0;
00706     }
00707
00708     /* Release mutex */
00709     (void)tx_mutex_put(&handler->mutex);
00710
00711     return k_rx_ok;
00712 }
00713
00714 /**
00715  * @brief Check if component has reached retry limit (rx_error_interface_t.is_retry_limit_reached)
00716  *
00717  * @details
00718  * Determines whether the specified component has exceeded its configured
00719  * maximum retry count. Used to implement fail-fast behavior after repeated
00720  * errors.
00721  *
00722  * @par Algorithm
00723  * 1. Validate context and component pointers (return true on failure - fail-safe)
00724  * 2. If max_retries == 0, return false (unlimited retries)
00725  * 3. Acquire mutex
00726  * 4. Find component via internal_find_component()
00727  * 5. Compare retry_count against max_retries
00728  * 6. Release mutex
00729  * 7. Return comparison result
00730  *
00731  *
00732  *

```

```

00733 * @pre ctx points to initialized error_handler_t
00734 * @pre component is a valid null-terminated string
00735 *
00736 * @post Handler state unchanged (read-only operation)
00737 *
00738 * @note Thread-safe: protected by internal mutex
00739 * @note Fail-safe: returns true (limit reached) on any error condition
00740 * @note If max_retries == 0, always returns false (unlimited)
00741 * @note If component not found, returns false (no retries yet)
00742 *
00743 * @par Performance
00744 * ~3-15 us (mutex + component search)
00745 *
00746 * @see impl_reset_retry_counter() Reset retry count after recovery
00747 * @see impl_get_backoff_delay() Get delay before next retry
00748 *
00749 * @since Version 1.0.0
00750 */
00751 static bool impl_is_retry_limit_reached(void* ctx, const char* component)
00752 {
00753     error_handler_t* handler = (error_handler_t*)ctx;
00754
00755     if (handler == nullptr || component == nullptr) {
00756         return true; /* Fail-safe: consider limit reached if invalid params */
00757     }
00758     if (!handler->initialized) {
00759         return true;
00760     }
00761
00762     /* If max_retries is 0, no limit */
00763     if (handler->max_retries == 0) {
00764         return false;
00765     }
00766
00767     /* Acquire mutex */
00768     const UINT status = tx_mutex_get(&handler->mutex, TX_WAIT_FOREVER);
00769     if (status != TX_SUCCESS) {
00770         return true; /* Fail-safe */
00771     }
00772
00773     bool limit_reached = false;
00774     const error_component_state_t* comp = internal_find_component(handler, component);
00775     if (comp != nullptr) {
00776         if (comp->retry_count >= handler->max_retries) {
00777             limit_reached = true;
00778         }
00779     }
00780
00781     /* Release mutex */
00782     (void)tx_mutex_put(&handler->mutex);
00783
00784     return limit_reached;
00785 }
00786
00787 /**
00788 * @brief Reset component retry counter (rx_error_interface_t.reset_retry_counter)
00789 *
00790 * @details
00791 * Resets only the retry counter for a specific component, leaving the error
00792 * count unchanged. Call this after successful recovery to enable fresh retries.
00793 *
00794 * @par Algorithm
00795 * 1. Validate context and component pointers
00796 * 2. Verify handler is initialized
00797 * 3. Acquire mutex
00798 * 4. Find component via internal_find_component()
00799 * 5. If found, set retry_count = 0
00800 * 6. Release mutex
00801 *
00802 *
00803 *
00804 * @pre ctx points to initialized error_handler_t
00805 * @pre component is a valid null-terminated string
00806 *
00807 * @post If component found, retry_count == 0
00808 * @post error_count unchanged (only retry reset)
00809 *
00810 * @note Thread-safe: protected by internal mutex
00811 * @note No-op if component not found (returns k_rx_ok)
00812 * @note Use after successful operation to reset backoff
00813 *
00814 * @par Example
00815 * @code
00816 * // After successful recovery from USB error
00817 * if (usb_operation_succeeded) {
00818 *     iface->reset_retry_counter(iface->ctx, "USB");
00819 * }

```

```

00820 * @endcode
00821 *
00822 * @par Performance
00823 * ~3-15 us (mutex + component search)
00824 *
00825 * @see impl_clear_errors() Reset all counters for all components
00826 * @see impl_get_backoff_delay() Retry count affects backoff
00827 *
00828 * @since Version 1.0.0
00829 */
00830 static rx_err_t impl_reset_retry_counter(void* ctx, const char* component)
00831 {
00832     error_handler_t* handler = (error_handler_t*)ctx;
00833
00834     RX_CHECK_NULL_PTR(handler, s_tag, "Handler pointer is nullptr");
00835     RX_CHECK_NULL_PTR(component, s_tag, "Component pointer is nullptr");
00836     if (!handler->initialized) {
00837         rx_log_error(s_tag, "Handler not initialized");
00838         return k_rx_err_invalid_state;
00839     }
00840
00841     /* Acquire mutex */
00842     const UINT status = tx_mutex_get(&handler->mutex, TX_WAIT_FOREVER);
00843     if (status != TX_SUCCESS) {
00844         return k_rx_err_rtos_mutex;
00845     }
00846
00847     error_component_state_t* comp = internal_find_component(handler, component);
00848     if (comp != nullptr) {
00849         comp->retry_count = 0;
00850     }
00851
00852     /* Release mutex */
00853     (void)tx_mutex_put(&handler->mutex);
00854
00855     return k_rx_ok;
00856 }
00857
00858 /**
00859 * @brief Calculate exponential backoff delay (rx_error_interface_t.get_backoff_delay)
00860 *
00861 * @details
00862 * Computes the delay (in milliseconds) to wait before the next retry attempt
00863 * using binary exponential backoff. Delay doubles with each retry, capped at
00864 * the configured maximum.
00865 *
00866 * @par Algorithm
00867 * 1. Validate context and component pointers (return 0 on failure)
00868 * 2. Acquire mutex
00869 * 3. Find component via internal_find_component()
00870 * 4. If retry_count == 0, delay = 0 (no backoff on first attempt)
00871 * 5. Otherwise, compute: delay = initial x 2^(retry_count - 1)
00872 * 6. Cap delay at max_backoff_ms
00873 * 7. Release mutex
00874 * 8. Return computed delay
00875 *
00876 * @par Exponential Backoff Formula
00877 * @f[
00878 * \text{delay} = \min\left(\text{initial\_backoff\_ms} \times 2^{\{(n-1)\}}, \right.
00879 * \left. \text{max\_backoff\_ms} \right)
00880 * @f]
00881 * where @f$ n @f$ = retry_count (capped at 32 to prevent overflow)
00882 *
00883 * @dot
00884 * digraph get_backoff {
00885 *     rankdir=TB;
00886 *     node [shape=box, style=rounded];
00887 *     start [label="Start"];
00888 *     validate [label="Validate ctx,\ncomponent"];
00889 *     get_mutex [label="tx_mutex_get()"];
00890 *     find [label="internal_find_component()"];
00891 *     check_retry [label="retry_count > 0?" shape=diamond];
00892 *     init_delay [label="delay = initial_backoff_ms"];
00893 *     loop [label="For i = 1 to retry_count-1"];
00894 *     double [label="delay *= 2"];
00895 *     check_cap [label="delay > max?" shape=diamond];
00896 *     cap [label="delay = max_backoff_ms"];
00897 *     put_mutex [label="tx_mutex_put()"];
00898 *     return_delay [label="Return delay" style=filled, fillcolor=lightgreen];
00899 *     return_zero [label="Return 0" style=filled, fillcolor=lightyellow];
00900 *     start -> validate;
00901 *     validate -> get_mutex [label="ok"];
00902 *     validate -> return_zero [label="invalid"];
00903 *     get_mutex -> find;
00904 *     find -> check_retry [label="found"];
00905 *     find -> put_mutex [label="not found"];
00906 *     check_retry -> init_delay [label="yes"];

```

```

00907 * check_retry -> put_mutex [label="no"];
00908 * init_delay -> loop;
00909 * loop -> double;
00910 * double -> check_cap;
00911 * check_cap -> cap [label="yes"];
00912 * check_cap -> loop [label="no, more iters"];
00913 * cap -> put_mutex;
00914 * loop -> put_mutex [label="done"];
00915 * put_mutex -> return_delay;
00916 * }
00917 * @enddot
00918 *
00919 *
00920 *
00921 * @pre ctx points to initialized error_handler_t
00922 * @pre component is a valid null-terminated string
00923 *
00924 * @post Handler state unchanged (read-only operation)
00925 *
00926 * @invariant Return value <= max_backoff_ms (when successful)
00927 *
00928 * @note Thread-safe: protected by internal mutex
00929 * @note Returns 0 for first attempt (retry_count == 0)
00930 * @note Loop bounded by k_error_handler_max_retries (NASA Rule 2)
00931 *
00932 * @par Example
00933 * @code
00934 * uint32_t delay = iface->get_backoff_delay(iface->ctx, "SPI");
00935 * if (delay > 0) {
00936 *     tx_thread_sleep(delay * TX_TIMER_TICKS_PER_MS);
00937 * }
00938 * // Retry operation
00939 * @endcode
00940 *
00941 * @par Performance
00942 * ~3-20 us (mutex + search + backoff loop)
00943 *
00944 * @see impl_reset_retry_counter() Reset to clear backoff
00945 * @see impl_report_error() Increments retry_count
00946 *
00947 * @since Version 1.0.0
00948 */
00949 static uint32_t impl_get_backoff_delay(void* ctx, const char* component)
00950 {
00951     error_handler_t* handler = (error_handler_t*)ctx;
00952
00953     if ((handler == nullptr) || (component == nullptr)) {
00954         return 0;
00955     }
00956     if (!handler->initialized) {
00957         return 0;
00958     }
00959
00960     /* Acquire mutex */
00961     const UINT status = tx_mutex_get(&handler->mutex, TX_WAIT_FOREVER);
00962     if (status != TX_SUCCESS) {
00963         return 0;
00964     }
00965
00966     uint32_t delay_ms = 0;
00967     const error_component_state_t* comp = internal_find_component(handler, component);
00968     if (comp != nullptr && comp->retry_count > 0) {
00969         /* Exponential backoff: delay = initial * 2^(retry_count - 1)
00970          * Capped at max_backoff_ms */
00971         delay_ms = handler->initial_backoff_ms;
00972         uint32_t retry_cap = handler->max_retries;
00973         if (handler->max_retries == k_error_handler_no_retry_limit ||
00974             handler->max_retries > k_error_handler_max_retries) {
00975             retry_cap = k_error_handler_max_retries;
00976         }
00977         const uint32_t retries = (comp->retry_count > retry_cap) ? retry_cap : comp->retry_count;
00978         /* Statically bounded loop: iterate from first retry to max retries cap (NASA Rule 2) */
00979         for (uint32_t i = k_error_handler_first_retry; i < k_error_handler_max_retries; ++i) {
00980             /* Explicit bounds check: break if reached actual retry limit */
00981             if (i >= retries) {
00982                 break;
00983             }
00984             if (delay_ms > handler->max_backoff_ms / k_exponential_backoff_multiplier) {
00985                 delay_ms = handler->max_backoff_ms;
00986                 break;
00987             }
00988             delay_ms *= k_exponential_backoff_multiplier;
00989             if (delay_ms >= handler->max_backoff_ms) {
00990                 delay_ms = handler->max_backoff_ms;
00991                 break;
00992             }
00993         }

```

```

00994 }
00995
00996 /* Release mutex */
00997 (void)tx_mutex_put(&handler->mutex);
00998
00999 return delay_ms;
01000 }
01001
01002 /*
=====
01003 * Public API Implementation
01004 *
=====
01005 */
01006
01007 /**
01008 * @brief Initialize error handler with retry logic and exponential backoff
01009 *
01010 * @details
01011 * Initializes an error handler instance with the specified configuration.
01012 * Creates a ThreadX mutex for thread-safe operation and clears all internal
01013 * state. Must be called before any other error handler operations.
01014 *
01015 * @par Algorithm
01016 * 1. Validate handler and config pointers
01017 * 2. Validate backoff configuration consistency
01018 * 3. Clear all handler state (memset to 0)
01019 * 4. Copy configuration values
01020 * 5. Create ThreadX mutex
01021 * 6. Set initialized flag
01022 * 7. Log initialization success
01023 *
01024 * @dot
01025 * digraph error_handler_init {
01026 *     rankdir=TB;
01027 *     node [shape=box, style=rounded];
01028 *     start [label="Start"];
01029 *     check_null [label="handler/config\n!= nullptr?" shape=diamond];
01030 *     check_backoff [label="Backoff config\ninvalid?" shape=diamond];
01031 *     memset [label="memset(handler, 0)"];
01032 *     copy_config [label="Copy config values"];
01033 *     create_mutex [label="tx_mutex_create()"];
01034 *     check_mutex [label="Success?" shape=diamond];
01035 *     set_init [label="initialized = true"];
01036 *     log_ok [label="rx_log_info()"];
01037 *     return_ok [label="Return k_rx_ok" style=filled, fillcolor=lightgreen];
01038 *     return_err [label="Return error" style=filled, fillcolor=lightyellow];
01039 *     start -> check_null;
01040 *     check_null -> check_backoff [label="yes"];
01041 *     check_null -> return_err [label="no"];
01042 *     check_backoff -> memset [label="yes"];
01043 *     check_backoff -> return_err [label="no"];
01044 *     memset -> copy_config;
01045 *     copy_config -> create_mutex;
01046 *     create_mutex -> check_mutex;
01047 *     check_mutex -> set_init [label="TX_SUCCESS"];
01048 *     check_mutex -> return_err [label="fail"];
01049 *     set_init -> log_ok;
01050 *     log_ok -> return_ok;
01051 * }
01052 * @enddot
01053 *
01054 *
01055 *
01056 * @pre handler points to allocated error_handler_t (not previously initialized)
01057 * @pre config points to valid configuration with consistent values
01058 *
01059 * @post handler->initialized == true
01060 * @post handler->mutex created and ready
01061 * @post All component slots cleared (in_use == false)
01062 * @post total_error_count == 0
01063 *
01064 * @note Not thread-safe: do not call concurrently for same handler
01065 * @note Call error_handler_deinit() before reinitializing
01066 *
01067 * @warning Do not initialize same handler twice without deinit
01068 *
01069 * @par Example
01070 * @code
01071 * static error_handler_t g_error_handler;
01072 *
01073 * void system_init(void) {
01074 *     error_handler_config_t config = {
01075 *         .max_retries      = 5,
01076 *         .initial_backoff_ms = 100,
01077 *         .max_backoff_ms   = 5000
01078 *     };

```



```

01079 *
01080 *   rx_err_t err = error_handler_init(&g_error_handler, &config);
01081 *   if (err != k_rx_ok) {
01082 *       // Handle init failure - system cannot proceed
01083 *       while (1) { }
01084 *   }
01085 * }
01086 * @endcode
01087 *
01088 * @par Performance
01089 * ~50-100 us (memset + mutex creation)
01090 *
01091 * @see error_handler_deinit() Cleanup handler
01092 * @see error_handler_get_interface() Get interface for dependency injection
01093 *
01094 * @since Version 1.0.0
01095 */
01096 rx_err_t error_handler_init(error_handler_t* handler, const error_handler_config_t* config)
01097 {
01098     RX_CHECK_NULL_PTR(handler, s_tag, "Handler pointer is nullptr");
01099     RX_CHECK_NULL_PTR(config, s_tag, "Config pointer is nullptr");
01100     if (config->max_backoff_ms < config->initial_backoff_ms) {
01101         rx_log_error(s_tag, "Backoff range invalid");
01102         return k_rx_err_invalid_arg;
01103     }
01104     /* Architecturally unreachable: if initial > 0 and max == 0, then max < initial,
01105      * which is caught by the preceding check. No RX_ASSERT needed. */
01106
01107     /* Clear all state */
01108     *handler = (error_handler_t){};
01109
01110     /* Initialize configuration */
01111     handler->max_retries = config->max_retries;
01112     handler->initial_backoff_ms = config->initial_backoff_ms;
01113     handler->max_backoff_ms = config->max_backoff_ms;
01114
01115     /* Create mutex -- tx_mutex_create always succeeds when ThreadX is running */
01116     (void)tx_mutex_create(&handler->mutex, "ErrorHandlerMutex", TX_NO_INHERIT);
01117
01118     handler->initialized = true;
01119
01120     rx_log_info(s_tag, "Error handler initialized");
01121
01122     return k_rx_ok;
01123 }
01124
01125 /**
01126 * @brief Get error handler interface for dependency injection
01127 *
01128 * @details
01129 * Populates an rx_error_interface_t structure with function pointers to the
01130 * handler's implementation functions. This enables dependency injection
01131 * following the Dependency Inversion Principle (DIP).
01132 *
01133 * @par Algorithm
01134 * 1. Validate iface and handler pointers
01135 * 2. Verify handler is initialized
01136 * 3. Populate interface structure:
01137 *    - ctx = handler (opaque context)
01138 *    - Function pointers to impl_* functions
01139 *
01140 * @par Interface Binding Diagram
01141 * @dot
01142 * digraph interface_binding {
01143 *     rankdir=LR;
01144 *     node [shape=record];
01145 *     iface [la-
01146 *         bel="rx_error_interface_t|ctx|report_error|get_error_count|get_component_error_count|clear_errors|is_retry_limit_reached|reset_retry_c
01147 *         handler [label="error_handler_t|mutex|components[]|total_error_count|config..."];
01148 *     impl [la-
01149 *         bel="Implementation|impl_report_error()|impl_get_error_count()|impl_get_component_error_count()|impl_clear_errors()|impl_is_retry_
01150 *         iface:ctx -> handler [label="points to"];
01151 *         iface:report_error -> impl:impl_report_error;
01152 *         iface:get_error_count -> impl:impl_get_error_count;
01153 *     }
01154 * @enddot
01155 *
01156 * @pre iface points to allocated rx_error_interface_t
01157 * @pre handler initialized via error_handler_init()
01158 *
01159 * @post iface->ctx == handler
01160 * @post All function pointers set to implementation functions
01161 *
01162 * @note Handler must remain valid while interface is in use
01163 * @note Interface structure is a shallow copy (pointers only)

```

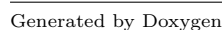
```

01164 *
01165 * @par Example
01166 * @code
01167 * static error_handler_t g_error_handler;
01168 * static rx_error_interface_t g_error_iface;
01169 *
01170 * void system_init(void) {
01171 *     error_handler_config_t config = { ... };
01172 *     error_handler_init(&g_error_handler, &config);
01173 *
01174 *     // Get interface for dependency injection
01175 *     error_handler_get_interface(&g_error_iface, &g_error_handler);
01176 *
01177 *     // Pass interface to modules that need error handling
01178 *     motor_init(&motor, &g_error_iface);
01179 *     comm_init(&comm, &g_error_iface);
01180 * }
01181 *
01182 * // In motor module:
01183 * void motor_report_fault(motor_t* motor) {
01184 *     motor->error_iface->report_error(
01185 *         motor->error_iface->ctx,
01186 *         k_rx_err_hw_fault,
01187 *         "MOTOR",
01188 *         "Overcurrent detected"
01189 *     );
01190 * }
01191 * @endcode
01192 *
01193 * @par Performance
01194 * ~1 us (pointer assignments only)
01195 *
01196 * @see error_handler_init() Initialize handler first
01197 * @see rx_error_interface_t Abstract interface definition
01198 *
01199 * @since Version 1.0.0
01200 */
01201 rx_err_t error_handler_get_interface(rx_error_interface_t* iface, error_handler_t* handler)
01202 {
01203     RX_CHECK_NULL_PTR(iface, s_tag, "Interface pointer is nullptr");
01204     RX_CHECK_NULL_PTR(handler, s_tag, "Handler pointer is nullptr");
01205
01206     if (!handler->initialized) {
01207         rx_log_error(s_tag, "Handler not initialized");
01208         return k_rx_err_invalid_state;
01209     }
01210
01211     /* Fill interface */
01212     iface->ctx = handler;
01213     iface->report_error = impl_report_error;
01214     iface->get_error_count = impl_get_error_count;
01215     iface->get_component_error_count = impl_get_component_error_count;
01216     iface->clear_errors = impl_clear_errors;
01217     iface->is_retry_limit_reached = impl_is_retry_limit_reached;
01218     iface->reset_retry_counter = impl_reset_retry_counter;
01219     iface->get_backoff_delay = impl_get_backoff_delay;
01220
01221     return k_rx_ok;
01222 }
01223
01224 /**
01225 * @brief Deinitialize error handler and release resources
01226 *
01227 * @details
01228 * Releases resources held by the error handler, including the ThreadX mutex.
01229 * Safe to call on already-deinitialized handler (idempotent).
01230 *
01231 * @par Algorithm
01232 * 1. Validate handler pointer
01233 * 2. If not initialized, return k_rx_ok (idempotent)
01234 * 3. Delete ThreadX mutex
01235 * 4. Set initialized = false
01236 * 5. Log deinitialization
01237 *
01238 *
01239 *
01240 * @pre handler points to valid error_handler_t (initialized or not)
01241 *
01242 * @post handler->initialized == false
01243 * @post handler->mutex deleted (if was initialized)
01244 *
01245 * @note Idempotent: safe to call multiple times
01246 * @note Not thread-safe: ensure no other threads using handler
01247 *
01248 * @warning Invalidates any rx_error_interface_t pointing to this handler
01249 * @warning Do not use interface after deinit
01250 *

```

7.47 libs/rx_core/src/rx_exception.c File Reference

```
#include "rx_exception.h"
#include <stddef.h>
#include "rx_log.h"
Include dependency graph for rx_exception.c:
```



Enumerations

- enum `exc_reserved_idx_t` : `uint8_t` { `k_exc_reserved_idx_0` = 0 , `k_exc_reserved_idx_1` = 1 , `k_exc_reserved_idx_2` = 2 }
Indices for the reserved padding bytes in `rx_exception_frame_t`.
- enum `rx_exc_init_state_t` : `uint8_t` { `k_rx_exc_not_initialized` = 0U , `k_rx_exc_initialized` = 1U }
Initialization state for the exception module.

Functions

- void `rx_exc_undefined_instruction_c_handler` (`uint32_t` pc, `uint32_t` psw)
C-level undefined instruction exception handler.
- void `rx_exc_privileged_instruction_c_handler` (`uint32_t` pc, `uint32_t` psw)
C-level privileged instruction exception handler.
- void `rx_exc_access_c_handler` (`uint32_t` pc, `uint32_t` psw)
C-level access exception handler.
- void `rx_exc_address_c_handler` (`uint32_t` pc, `uint32_t` psw)
C-level address exception handler.
- void `rx_exc_floating_point_c_handler` (`uint32_t` pc, `uint32_t` psw)
C-level floating-point exception handler.
- void `rx_exc_nmi_c_handler` (`uint32_t` pc, `uint32_t` psw)
C-level non-maskable interrupt handler.
- static void `internal_process_exception` (const `rx_exception_frame_t` *frame)
Common exception processing logic.
- static void `internal_halt_cpu` (void)
Halt CPU in infinite loop.
- void `rx_exception_init` (void)
Initialize exception handling module.
- const `rx_exception_stats_t` * `rx_exception_get_stats` (void)
Get exception statistics.
- const char * `rx_exception_get_name` (`rx_exception_type_t` type)
Get human-readable name for exception type.

Variables

- static const char *const `s_tag` = "EXCEPTION"
Log tag for exception module.
- static const char *const `s_exception_names` [`k_rx_exception_type_count`]
Exception type names for logging.
- static `rx_exception_stats_t` `s_exception_stats` = {}
Exception statistics storage.
- static `rx_exc_init_state_t` `s_initialized` = `k_rx_exc_not_initialized`
Initialization flag for the exception module.

7.47.1 Detailed Description

RX72N CPU Exception Handling Implementation.

Implements exception handlers for RX72N CPU exceptions. Each handler:

1. Captures the exception context (PC, PSW from stack)
2. Updates exception statistics
3. Logs the exception via UART/USB (if available)
4. Either halts (NMI) or allows return (other exceptions)

Assembly Entry Points:

The assembly-level handlers in startup_rx72n.S call these C functions after saving/restoring context appropriately.

Manual Reference:

RX72N Group User's Manual: Hardware, Rev.1.20, Chapter 14

Since

Version 1.0.0

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Definition in file [rx_exception.c](#).

7.47.2 Enumeration Type Documentation

7.47.2.1 exc_reserved_idx_t

enum [exc_reserved_idx_t](#) : uint8_t

Indices for the reserved padding bytes in [rx_exception_frame_t](#).

The reserved array has k_exception_frame_reserved_size (3) elements.

Enumerator

k_exc_reserved_idx_0	First reserved padding byte
k_exc_reserved_idx_1	Second reserved padding byte
k_exc_reserved_idx_2	Third reserved padding byte

Definition at line 52 of file [rx_exception.c](#).

```
00052      : uint8_t {
00053  k_exc_reserved_idx_0 = 0, /**< First reserved padding byte */
00054  k_exc_reserved_idx_1 = 1, /**< Second reserved padding byte */
00055  k_exc_reserved_idx_2 = 2, /**< Third reserved padding byte */
00056 } exc_reserved_idx_t;
```

7.47.2.2 rx_exc_init_state_t

```
enum rx_exc_init_state_t : uint8_t
```

Initialization state for the exception module.

Tracks whether [rx_exception_init\(\)](#) has been called. Used to guard [rx_exception_get_stats\(\)](#) so callers cannot observe uninitialized data.

Since

Version 1.0.0

Enumerator

k_rx_exc_not_initialized	rx_exception_init() has not been called
k_rx_exc_initialized	rx_exception_init() completed successfully

Definition at line 91 of file [rx_exception.c](#).

```
00091      : uint8_t {
00092  k_rx_exc_not_initialized = 0U, /**< rx_exception_init() has not been called */
00093  k_rx_exc_initialized    = 1U, /**< rx_exception_init() completed successfully */
00094 } rx_exc_init_state_t;
```

7.47.3 Function Documentation

7.47.3.1 internal_halt_cpu()

```
void internal_halt_cpu (
    void ) [static]
```

Halt CPU in infinite loop.

Used after fatal exceptions (NMI) where recovery is not possible. Enters low-power wait state to reduce power consumption while halted.

Note

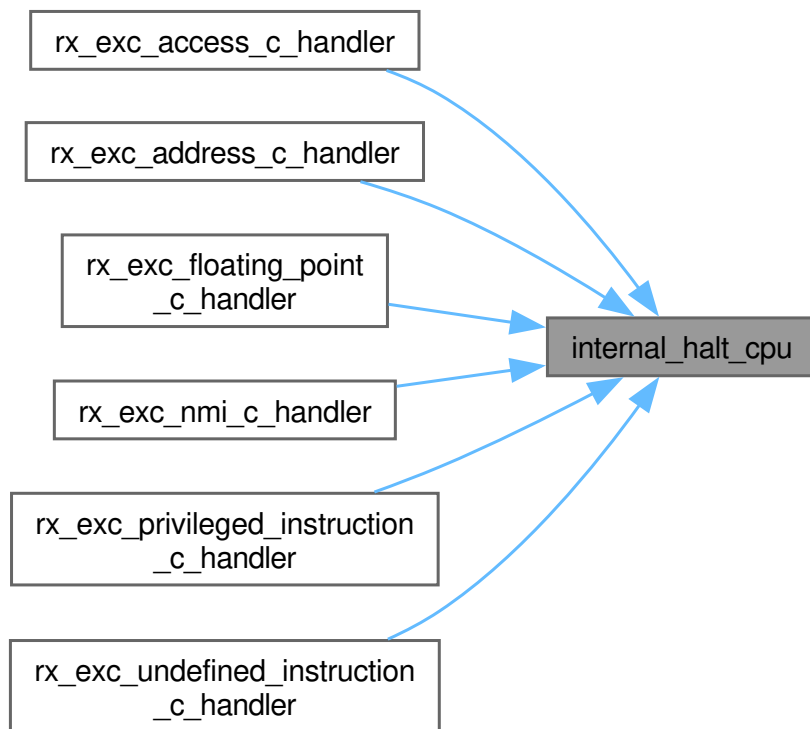
This function never returns

Definition at line 154 of file [rx_exception.c](#).

```
00155 {
00156  /** Disable interrupts and halt */
00157  for (;;) {
00158      __asm__ volatile("wait");
00159  }
00160 }
```

Referenced by [rx_exc_access_c_handler\(\)](#), [rx_exc_address_c_handler\(\)](#), [rx_exc_floating_point_c_handler\(\)](#), [rx_exc_nmi_c_handler\(\)](#), [rx_exc_privileged_instruction_c_handler\(\)](#), and [rx_exc_undefined_instruction_c_handler\(\)](#).

Here is the caller graph for this function:



7.47.3.2 `internal_process_exception()`

```
void internal_process_exception (  
    const rx\_exception\_frame\_t * frame) [static]
```

Common exception processing logic.

Called by all exception handlers to:

1. Update statistics
2. Save frame for post-mortem analysis
3. Log the exception

Precondition

frame is not nullptr
frame->type is valid enum value

Postcondition

Statistics updated
Last frame saved

Note

Not thread-safe - called from exception context with interrupts disabled

Definition at line 128 of file [rx_exception.c](#).

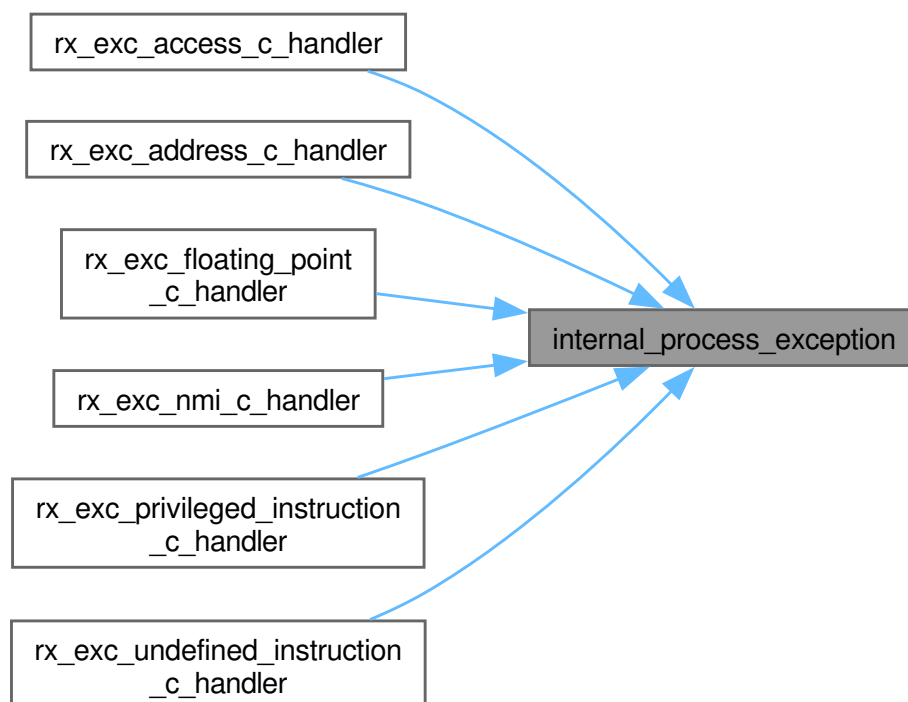
```

00129 {
00130     /* Validate type is in range */
00131     if (frame->type >= k_rx_exception_type_count) {
00132         return;
00133     }
00134
00135     /* Update statistics */
00136     s_exception_stats.count[frame->type]++;
00137     s_exception_stats.last_frame = *frame;
00138
00139     /* Log the exception */
00140     (void)rx_log_error(s_tag, "**** CPU EXCEPTION ****");
00141     (void)rx_log_error(s_tag, s_exception_names[frame->type]);
00142     /* Note: PC and PSW values available in s_exception_stats.last_frame for debugging */
00143 }
```

References [k_rx_exception_type_count](#), [rx_log_error](#), [s_exception_names](#), [s_exception_stats](#), [s_tag](#), and [rx_exception_frame_t::type](#).

Referenced by [rx_exc_access_c_handler\(\)](#), [rx_exc_address_c_handler\(\)](#), [rx_exc_floating_point_c_handler\(\)](#), [rx_exc_nmi_c_handler\(\)](#), [rx_exc_privileged_instruction_c_handler\(\)](#), and [rx_exc_undefined_instruction_c_handler\(\)](#).

Here is the caller graph for this function:



7.47.3.3 rx_exc_access_c_handler()

```

void rx_exc_access_c_handler (
    uint32_t pc,
    uint32_t psw)
```

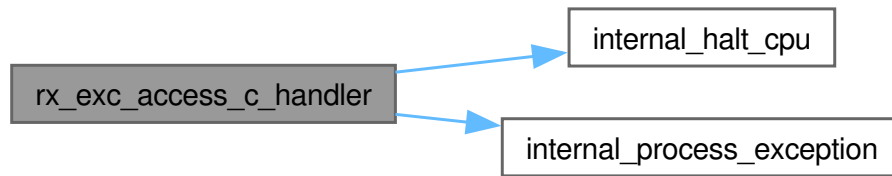
C-level access exception handler.

Definition at line 261 of file [rx_exception.c](#).

```
00262 {
00263     rx_exception_frame_t frame = {.pc      = pc,
00264                                  .psw      = psw,
00265                                  .type      = k_rx_exception_access,
00266                                  .reserved = {0, 0, 0}};
00267
00268     internal_process_exception(&frame);
00269
00270     /* Halt - memory protection violation */
00271     internal_halt_cpu();
00272 }
```

References [internal_halt_cpu\(\)](#), [internal_process_exception\(\)](#), and [k_rx_exception_access](#).

Here is the call graph for this function:



7.47.3.4 rx_exc_address_c_handler()

```
void rx_exc_address_c_handler (
    uint32_t pc,
    uint32_t psw)
```

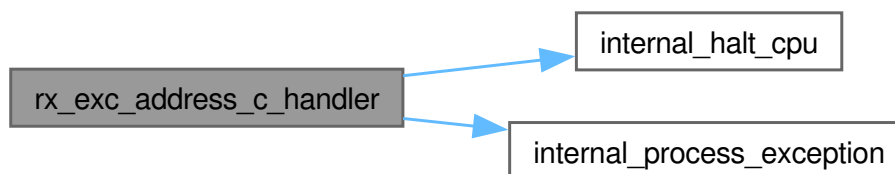
C-level address exception handler.

Definition at line 278 of file [rx_exception.c](#).

```
00279 {
00280     rx_exception_frame_t frame = {.pc      = pc,
00281                                  .psw      = psw,
00282                                  .type      = k_rx_exception_address,
00283                                  .reserved = {0, 0, 0}};
00284
00285     internal_process_exception(&frame);
00286
00287     /* Halt - unaligned access is a programming error */
00288     internal_halt_cpu();
00289 }
```

References [internal_halt_cpu\(\)](#), [internal_process_exception\(\)](#), and [k_rx_exception_address](#).

Here is the call graph for this function:



7.47.3.5 rx_exc_floating_point_c_handler()

```
void rx_exc_floating_point_c_handler (
    uint32_t pc,
    uint32_t psw)
```

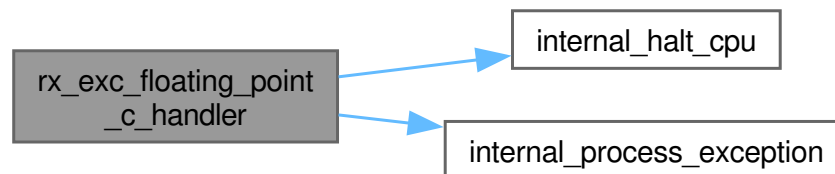
C-level floating-point exception handler.

Definition at line 295 of file [rx_exception.c](#).

```
00296 {
00297     rx_exception_frame_t frame = {.pc      = pc,
00298                                   .psw     = psw,
00299                                   .type     = k_rx_exception_floating_point,
00300                                   .reserved = {0, 0, 0}};
00301
00302     internal_process_exception(&frame);
00303
00304     /* Halt - FPU exception in motor control is dangerous */
00305     internal_halt_cpu();
00306 }
```

References [internal_halt_cpu\(\)](#), [internal_process_exception\(\)](#), and [k_rx_exception_floating_point](#).

Here is the call graph for this function:



7.47.3.6 rx_exc_nmi_c_handler()

```
void rx_exc_nmi_c_handler (
    uint32_t pc,
    uint32_t psw)
```

C-level non-maskable interrupt handler.

NMI indicates a fatal system fault. Per the manual (Section 14.1.7): "Never use the non-maskable interrupt with an attempt to return to the program that was being executed at the time of interrupt generation."

Warning

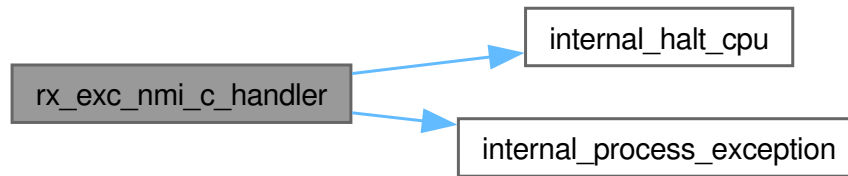
This function NEVER returns

Definition at line 319 of file [rx_exception.c](#).

```
00320 {
00321     rx_exception_frame_t frame = {.pc      = pc,
00322                                   .psw     = psw,
00323                                   .type     = k_rx_exception_nmi,
00324                                   .reserved = {0, 0, 0}};
00325
00326     internal_process_exception(&frame);
00327
00328     (void)rx_log_error(s_tag, "FATAL: NMI - System halted");
00329
00330     /* NMI is fatal - MUST NOT return */
00331     internal_halt_cpu();
00332 }
```

References [internal_halt_cpu\(\)](#), [internal_process_exception\(\)](#), [k_rx_exception_nmi](#), [rx_log_error](#), and [s_tag](#).

Here is the call graph for this function:



7.47.3.7 rx_exc_privileged_instruction_c_handler()

```
void rx_exc_privileged_instruction_c_handler (
    uint32_t pc,
    uint32_t psw)
```

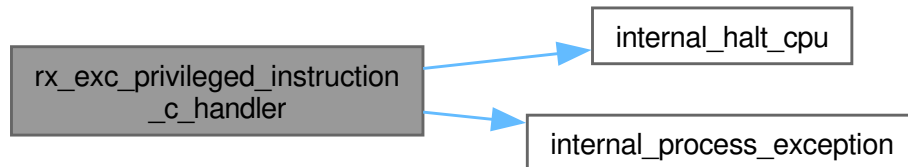
C-level privileged instruction exception handler.

Definition at line 244 of file [rx_exception.c](#).

```
00245 {
00246     rx_exception_frame_t frame = {.pc      = pc,
00247                                   .psw     = psw,
00248                                   .type     = k_rx_exception_privileged_instruction,
00249                                   .reserved = {0, 0, 0}};
00250
00251     internal_process_exception(&frame);
00252
00253     /* Halt - user mode attempted privileged operation */
00254     internal_halt_cpu();
00255 }
```

References [internal_halt_cpu\(\)](#), [internal_process_exception\(\)](#), and [k_rx_exception_privileged_instruction](#).

Here is the call graph for this function:



7.47.3.8 rx_exc_undefined_instruction_c_handler()

```
void rx_exc_undefined_instruction_c_handler (
    uint32_t pc,
    uint32_t psw)
```

C-level undefined instruction exception handler.

Called from assembly stub after context is saved. The faulting instruction PC is on the stack.

Note

Called with interrupts disabled (I=0)

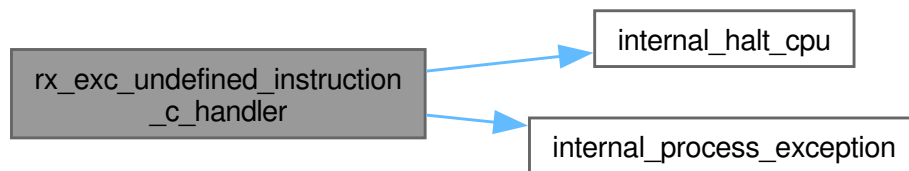
Can return if instruction can be emulated/skipped

Definition at line 227 of file [rx_exception.c](#).

```
00228 {
00229   rx_exception_frame_t frame = {.pc      = pc,
00230                                .psw      = psw,
00231                                .type      = k_rx_exception_undefined_instruction,
00232                                .reserved = {0, 0, 0}};
00233
00234   internal_process_exception(&frame);
00235
00236   /* For safety, halt on undefined instruction in embedded system */
00237   internal_halt_cpu();
00238 }
```

References [internal_halt_cpu\(\)](#), [internal_process_exception\(\)](#), and [k_rx_exception_undefined_instruction](#).

Here is the call graph for this function:



7.47.3.9 rx_exception_get_name()

```
const char * rx_exception_get_name (
    rx_exception_type_t type)
```

Get human-readable name for exception type.

Returns a string describing the exception type for logging.

Parameters

in	type	Exception type enumeration value
----	------	----------------------------------

Returns

Pointer to static string (never NULL)

Return values

Unknown	If type is out of range
---------	-------------------------

Precondition

None

Postcondition

No side effects

Note

Thread-safe: Yes (returns pointer to const data)

Since

Version 1.0.0

Definition at line 204 of file [rx_exception.c](#).

```
00205 {
00206     if (type >= k_rx_exception_type_count) {
00207         return "Unknown";
00208     }
00209     return s_exception_names[type];
00210 }
```

References [k_rx_exception_type_count](#), and [s_exception_names](#).

7.47.3.10 rx_exception_get_stats()

```
const rx_exception_stats_t * rx_exception_get_stats (
    void )
```

Get exception statistics.

Returns pointer to read-only exception statistics structure. Statistics are updated by exception handlers. Returns NULL if [rx_exception_init\(\)](#) has not yet been called; callers must check the return value before dereferencing.

Returns

Pointer to exception statistics, or NULL if not initialized

Return values

non-NULL	Valid pointer to statistics after rx_exception_init()
NULL	rx_exception_init() has not been called yet

Precondition

[rx_exception_init\(\)](#) must have been called for a non-NULL return

Caller must mask interrupts before reading fields if a consistent snapshot is required; exception handlers modify the statistics structure asynchronously from interrupt context

Postcondition

No side effects; returned pointer is owned by this module and must not be freed or modified

The returned pointer remains valid for the lifetime of the module (never reallocated)

Note

NOT thread-safe: exception handlers update the statistics structure from interrupt context concurrently with any reader; callers that require a consistent multi-field snapshot must disable interrupts around the read

Example:

```
const rx_exception_stats_t* stats = rx_exception_get_stats();
if (stats != NULL && stats->count[k_rx_exception_nmi] > 0) {
    // NMI occurred - investigate
}
```

Since

Version 1.0.0

Definition at line 193 of file [rx_exception.c](#).

```
00194 {
00195     if (s_initialized != k_rx_exc_initialized) {
00196         return nullptr;
00197     }
00198     return &s_exception_stats;
00199 }
```

References [k_rx_exc_initialized](#), [s_exception_stats](#), and [s_initialized](#).

7.47.3.11 rx_exception_init()

```
void rx_exception_init (
    void )
```

Initialize exception handling module.

Initializes exception statistics and optionally installs custom handlers. Should be called early in system startup, after basic hardware init.

Precondition

None

Postcondition

Exception statistics zeroed

Default handlers installed (if not using assembly stubs)

Note

Thread-safe: No, call only once during startup

Example:

```
void main(void) {
    rx_exception_init();
    // ... rest of initialization
}
```

Since

Version 1.0.0

Definition at line 169 of file [rx_exception.c](#).

```
00170 {
00171     /* Zero all statistics */
00172     for (uint8_t i = 0; i < k_rx_exception_type_count; i++) {
00173         s_exception_stats.count[i] = 0;
00174     }
00175
00176     /* Clear last frame */
00177     s_exception_stats.last_frame.pc = 0;
00178     s_exception_stats.last_frame.psw = 0;
00179     s_exception_stats.last_frame.type = k_rx_exception_undefined_instruction;
00180     s_exception_stats.last_frame.reserved[k_exc_reserved_idx_0] = 0;
00181     s_exception_stats.last_frame.reserved[k_exc_reserved_idx_1] = 0;
00182     s_exception_stats.last_frame.reserved[k_exc_reserved_idx_2] = 0;
00183
00184     /* Mark as initialized */
00185     s_initialized = k_rx_exc_initialized;
00186
00187     (void)rx_log_info(s_tag, "Exception handling initialized");
00188 }
```

References [k_exc_reserved_idx_0](#), [k_exc_reserved_idx_1](#), [k_exc_reserved_idx_2](#), [k_rx_exc_initialized](#), [k_rx_exception_type_count](#), [k_rx_exception_undefined_instruction](#), [rx_log_info](#), [s_exception_stats](#), [s_initialized](#), and [s_tag](#).

7.47.4 Variable Documentation

7.47.4.1 s_exception_names

const char* const s_exception_names[k_rx_exception_type_count] [static]

Initial value:

```
= {"Undefined Instruction",
    "Privileged Instruction",
    "Access Exception",
    "Address Exception",
    "Floating-Point Exception",
    "Non-Maskable Interrupt"}
```

Exception type names for logging.

Indexed by [rx_exception_type_t](#) values

Definition at line 62 of file [rx_exception.c](#).

```
00062     {"Undefined Instruction",
00063     "Privileged Instruction",
00064     "Access Exception",
00065     "Address Exception",
00066     "Floating-Point Exception",
00067     "Non-Maskable Interrupt"};
```

Referenced by [internal_process_exception\(\)](#), and [rx_exception_get_name\(\)](#).

7.47.4.2 s_exception_stats

[rx_exception_stats_t](#) s_exception_stats = {} [static]

Exception statistics storage.

Static storage for exception occurrence counts and last frame.

Warning

Direct modification discouraged - use handler functions

Definition at line 79 of file [rx_exception.c](#).

```
00079 {};
```

Referenced by [internal_process_exception\(\)](#), [rx_exception_get_stats\(\)](#), and [rx_exception_init\(\)](#).

7.47.4.3 s_initialized

```
rx_exc_init_state_t s_initialized = k_rx_exc_not_initialized [static]
```

Initialization flag for the exception module.

Set to `k_rx_exc_initialized` after `rx_exception_init()` completes. Guards `rx_exception_get_stats()` against returning stale data.

Note

Not thread-safe; `rx_exception_init()` must be called before any concurrent access to exception module state.

Since

Version 1.0.0

Definition at line 105 of file `rx_exception.c`.

Referenced by `rx_exception_get_stats()`, and `rx_exception_init()`.

7.47.4.4 s_tag

```
const char* const s_tag = "EXCEPTION" [static]
```

Log tag for exception module.

Used for identifying log messages from this module

Definition at line 46 of file `rx_exception.c`.

7.48 rx_exception.c

[Go to the documentation of this file.](#)

```
00001 /**
00002  * @file rx_exception.c
00003  * @brief RX72N CPU Exception Handling Implementation
00004  *
00005  * @details
00006  * Implements exception handlers for RX72N CPU exceptions. Each handler:
00007  * 1. Captures the exception context (PC, PSW from stack)
00008  * 2. Updates exception statistics
00009  * 3. Logs the exception via UART/USB (if available)
00010  * 4. Either halts (NMI) or allows return (other exceptions)
00011  *
00012  * @par Assembly Entry Points:
00013  * The assembly-level handlers in startup_rx72n.S call these C functions
00014  * after saving/restoring context appropriately.
00015  *
00016  * @par Manual Reference:
00017  * RX72N Group User's Manual: Hardware, Rev.1.20, Chapter 14
00018  *
00019  * @since Version 1.0.0
00020  * @copyright Copyright (c) 2026 Locked Inc.
00021  * SPDX-License-Identifier: MIT
00022  */
00023
00024 #include "rx_exception.h"
00025
00026 #include <stddef.h>
```



```

00027
00028 #include "rx_log.h"
00029
00030 /* Exception handler forward declarations (required for -Wmissing-declarations) */
00031 void rx_exc_undefined_instruction_c_handler(uint32_t pc, uint32_t psw);
00032 void rx_exc_privileged_instruction_c_handler(uint32_t pc, uint32_t psw);
00033 void rx_exc_access_c_handler(uint32_t pc, uint32_t psw);
00034 void rx_exc_address_c_handler(uint32_t pc, uint32_t psw);
00035 void rx_exc_floating_point_c_handler(uint32_t pc, uint32_t psw);
00036 void rx_exc_nmi_c_handler(uint32_t pc, uint32_t psw);
00037
00038 /*
=====
00039  * Module Constants
00040  *
=====
*/
00041
00042 /**
00043  * @brief Log tag for exception module
00044  * @details Used for identifying log messages from this module
00045  */
00046 static const char* const s_tag = "EXCEPTION";
00047
00048 /**
00049  * @brief Indices for the reserved padding bytes in rx_exception_frame_t
00050  * @details The reserved array has k_exception_frame_reserved_size (3) elements.
00051  */
00052 typedef enum : uint8_t {
00053     k_exc_reserved_idx_0 = 0, /**< First reserved padding byte */
00054     k_exc_reserved_idx_1 = 1, /**< Second reserved padding byte */
00055     k_exc_reserved_idx_2 = 2, /**< Third reserved padding byte */
00056 } exc_reserved_idx_t;
00057
00058 /**
00059  * @brief Exception type names for logging
00060  * @details Indexed by rx_exception_type_t values
00061  */
00062 static const char* const s_exception_names[k_rx_exception_type_count] = {"Undefined Instruction",
00063                                     "Privileged Instruction",
00064                                     "Access Exception",
00065                                     "Address Exception",
00066                                     "Floating-Point Exception",
00067                                     "Non-Maskable Interrupt"};
00068
00069 /*
=====
00070  * Module State
00071  *
=====
*/
00072
00073 /**
00074  * @brief Exception statistics storage
00075  * @details
00076  * Static storage for exception occurrence counts and last frame.
00077  * @warning Direct modification discouraged - use handler functions
00078  */
00079 static rx_exception_stats_t s_exception_stats = {};
00080
00081 /**
00082  * @enum rx_exc_init_state_t
00083  * @brief Initialization state for the exception module
00084  */
00085 * @details
00086 * Tracks whether rx_exception_init() has been called. Used to guard
00087 * rx_exception_get_stats() so callers cannot observe uninitialized data.
00088 *
00089 * @since Version 1.0.0
00090 */
00091 typedef enum : uint8_t {
00092     k_rx_exc_not_initialized = 0U, /**< rx_exception_init() has not been called */
00093     k_rx_exc_initialized     = 1U, /**< rx_exception_init() completed successfully */
00094 } rx_exc_init_state_t;
00095
00096 /**
00097  * @var s_initialized
00098  * @brief Initialization flag for the exception module
00099  * @details Set to k_rx_exc_initialized after rx_exception_init() completes.
00100  * Guards rx_exception_get_stats() against returning stale data.
00101  * @note Not thread-safe; rx_exception_init() must be called before any
00102  * concurrent access to exception module state.
00103  * @since Version 1.0.0
00104  */
00105 static rx_exc_init_state_t s_initialized = k_rx_exc_not_initialized;
00106
00107 */

```

```

=====
00108 * Internal Helper Functions
00109 *
=====
*/
00110
00111 /**
00112  * @brief Common exception processing logic
00113  *
00114  * @details
00115  * Called by all exception handlers to:
00116  * 1. Update statistics
00117  * 2. Save frame for post-mortem analysis
00118  * 3. Log the exception
00119  *
00120  *
00121  * @pre frame is not nullptr
00122  * @pre frame->type is valid enum value
00123  * @post Statistics updated
00124  * @post Last frame saved
00125  *
00126  * @note Not thread-safe - called from exception context with interrupts disabled
00127  */
00128 static void internal_process_exception(const rx_exception_frame_t* frame)
00129 {
00130     /* Validate type is in range */
00131     if (frame->type >= k_rx_exception_type_count) {
00132         return;
00133     }
00134
00135     /* Update statistics */
00136     s_exception_stats.count[frame->type]++;
00137     s_exception_stats.last_frame = *frame;
00138
00139     /* Log the exception */
00140     (void)rx_log_error(s_tag, "CPU EXCEPTION ***");
00141     (void)rx_log_error(s_tag, s_exception_names[frame->type]);
00142     /* Note: PC and PSW values available in s_exception_stats.last_frame for debugging */
00143 }
00144
00145 /**
00146  * @brief Halt CPU in infinite loop
00147  *
00148  * @details
00149  * Used after fatal exceptions (NMI) where recovery is not possible.
00150  * Enters low-power wait state to reduce power consumption while halted.
00151  *
00152  * @note This function never returns
00153  */
00154 static void internal_halt_cpu(void)
00155 {
00156     /* Disable interrupts and halt */
00157     for (;;) {
00158         __asm__ volatile("wait");
00159     }
00160 }
00161
00162 /*
=====
00163 * Public API Functions
00164 *
=====
*/
00165
00166 /**
00167  * @brief Initialize exception handling module
00168  */
00169 void rx_exception_init(void)
00170 {
00171     /* Zero all statistics */
00172     for (uint8_t i = 0; i < k_rx_exception_type_count; i++) {
00173         s_exception_stats.count[i] = 0;
00174     }
00175
00176     /* Clear last frame */
00177     s_exception_stats.last_frame.pc = 0;
00178     s_exception_stats.last_frame.psw = 0;
00179     s_exception_stats.last_frame.type = k_rx_exception_undefined_instruction;
00180     s_exception_stats.last_frame.reserved[k_exc_reserved_idx_0] = 0;
00181     s_exception_stats.last_frame.reserved[k_exc_reserved_idx_1] = 0;
00182     s_exception_stats.last_frame.reserved[k_exc_reserved_idx_2] = 0;
00183
00184     /* Mark as initialized */
00185     s_initialized = k_rx_exc_initialized;
00186
00187     (void)rx_log_info(s_tag, "Exception handling initialized");
00188 }

```

```

00189
00190 /**
00191  * @brief Get exception statistics
00192  */
00193 const rx_exception_stats_t* rx_exception_get_stats(void)
00194 {
00195     if (s_initialized != k_rx_exc_initialized) {
00196         return nullptr;
00197     }
00198     return &s_exception_stats;
00199 }
00200
00201 /**
00202  * @brief Get human-readable name for exception type
00203  */
00204 const char* rx_exception_get_name(rx_exception_type_t type)
00205 {
00206     if (type >= k_rx_exception_type_count) {
00207         return "Unknown";
00208     }
00209     return s_exception_names[type];
00210 }
00211
00212 /**
=====
00213  * C-Level Exception Handlers (called from assembly stubs)
00214  *
=====
00215
00216 /**
00217  * @brief C-level undefined instruction exception handler
00218  *
00219  * @details
00220  * Called from assembly stub after context is saved.
00221  * The faulting instruction PC is on the stack.
00222  *
00223  *
00224  * @note Called with interrupts disabled (I=0)
00225  * @note Can return if instruction can be emulated/skipped
00226  */
00227 void rx_exc_undefined_instruction_c_handler(uint32_t pc, uint32_t psw)
00228 {
00229     rx_exception_frame_t frame = {.pc      = pc,
00230                                  .psw     = psw,
00231                                  .type     = k_rx_exception_undefined_instruction,
00232                                  .reserved = {0, 0, 0}};
00233
00234     internal_process_exception(&frame);
00235
00236     /* For safety, halt on undefined instruction in embedded system */
00237     internal_halt_cpu();
00238 }
00239
00240 /**
00241  * @brief C-level privileged instruction exception handler
00242  *
00243  */
00244 void rx_exc_privileged_instruction_c_handler(uint32_t pc, uint32_t psw)
00245 {
00246     rx_exception_frame_t frame = {.pc      = pc,
00247                                  .psw     = psw,
00248                                  .type     = k_rx_exception_privileged_instruction,
00249                                  .reserved = {0, 0, 0}};
00250
00251     internal_process_exception(&frame);
00252
00253     /* Halt - user mode attempted privileged operation */
00254     internal_halt_cpu();
00255 }
00256
00257 /**
00258  * @brief C-level access exception handler
00259  *
00260  */
00261 void rx_exc_access_c_handler(uint32_t pc, uint32_t psw)
00262 {
00263     rx_exception_frame_t frame = {.pc      = pc,
00264                                  .psw     = psw,
00265                                  .type     = k_rx_exception_access,
00266                                  .reserved = {0, 0, 0}};
00267
00268     internal_process_exception(&frame);
00269
00270     /* Halt - memory protection violation */
00271     internal_halt_cpu();
00272 }

```

```

00273
00274 /**
00275  * @brief C-level address exception handler
00276  *
00277  */
00278 void rx_exc_address_c_handler(uint32_t pc, uint32_t psw)
00279 {
00280     rx_exception_frame_t frame = {.pc      = pc,
00281                                   .psw      = psw,
00282                                   .type      = k_rx_exception_address,
00283                                   .reserved = {0, 0, 0}};
00284
00285     internal_process_exception(&frame);
00286
00287     /* Halt - unaligned access is a programming error */
00288     internal_halt_cpu();
00289 }
00290
00291 /**
00292  * @brief C-level floating-point exception handler
00293  *
00294  */
00295 void rx_exc_floating_point_c_handler(uint32_t pc, uint32_t psw)
00296 {
00297     rx_exception_frame_t frame = {.pc      = pc,
00298                                   .psw      = psw,
00299                                   .type      = k_rx_exception_floating_point,
00300                                   .reserved = {0, 0, 0}};
00301
00302     internal_process_exception(&frame);
00303
00304     /* Halt - FPU exception in motor control is dangerous */
00305     internal_halt_cpu();
00306 }
00307
00308 /**
00309  * @brief C-level non-maskable interrupt handler
00310  *
00311  * @details
00312  * NMI indicates a fatal system fault. Per the manual (Section 14.1.7):
00313  * "Never use the non-maskable interrupt with an attempt to return to
00314  * the program that was being executed at the time of interrupt generation."
00315  *
00316  *
00317  * @warning This function NEVER returns
00318  */
00319 void rx_exc_nmi_c_handler(uint32_t pc, uint32_t psw)
00320 {
00321     rx_exception_frame_t frame = {.pc      = pc,
00322                                   .psw      = psw,
00323                                   .type      = k_rx_exception_nmi,
00324                                   .reserved = {0, 0, 0}};
00325
00326     internal_process_exception(&frame);
00327
00328     (void)rx_log_error(s_tag, "FATAL: NMI - System halted");
00329
00330     /* NMI is fatal - MUST NOT return */
00331     internal_halt_cpu();
00332 }

```

7.49 libs/rx_core/src/rx_infrastructure.c File Reference

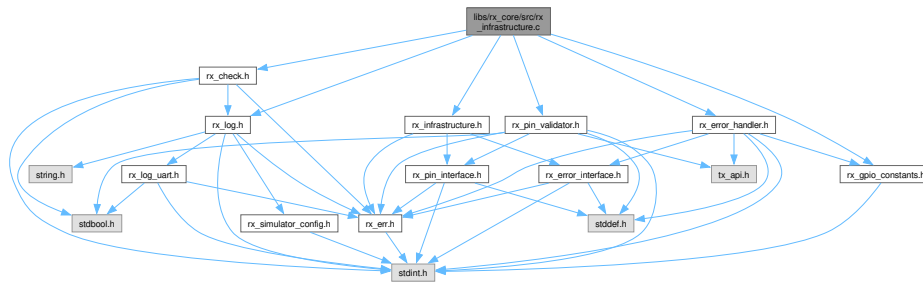
Global Infrastructure Initialization Implementation.

```

#include "rx_infrastructure.h"
#include "rx_check.h"
#include "rx_error_handler.h"
#include "rx_gpio_constants.h"
#include "rx_log.h"
#include "rx_pin_validator.h"

```

Include dependency graph for rx_infrastructure.c:



Functions

- [rx_err_t rx_infrastructure_init](#) (void)
Initialize global infrastructure services.
- [rx_err_t rx_infrastructure_deinit](#) (void)
Deinitialize global infrastructure and release resources.
- [rx_error_interface_t * rx_infrastructure_get_error_interface](#) (void)
Get global error handler interface.
- [rx_pin_interface_t * rx_infrastructure_get_pin_interface](#) (void)
Get global pin validator interface.

Variables

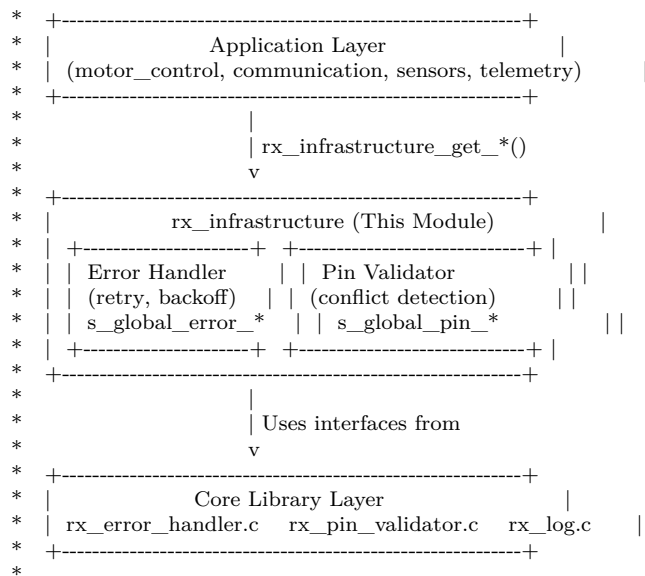
- static [error_handler_t s_global_error_handler](#)
Global error handler instance (singleton).
- static [pin_validator_t s_global_pin_validator](#)
Global pin validator instance (singleton).
- static [rx_error_interface_t s_global_error_interface](#) = {}
Global error handler interface (function pointers).
- static [rx_pin_interface_t s_global_pin_interface](#) = {}
Global pin validator interface (function pointers).
- static bool [s_infrastructure_initialized](#) = false
Infrastructure initialization flag.
- static const char *const [s_tag](#) = "INFRA"
Log tag for this module.

7.49.1 Detailed Description

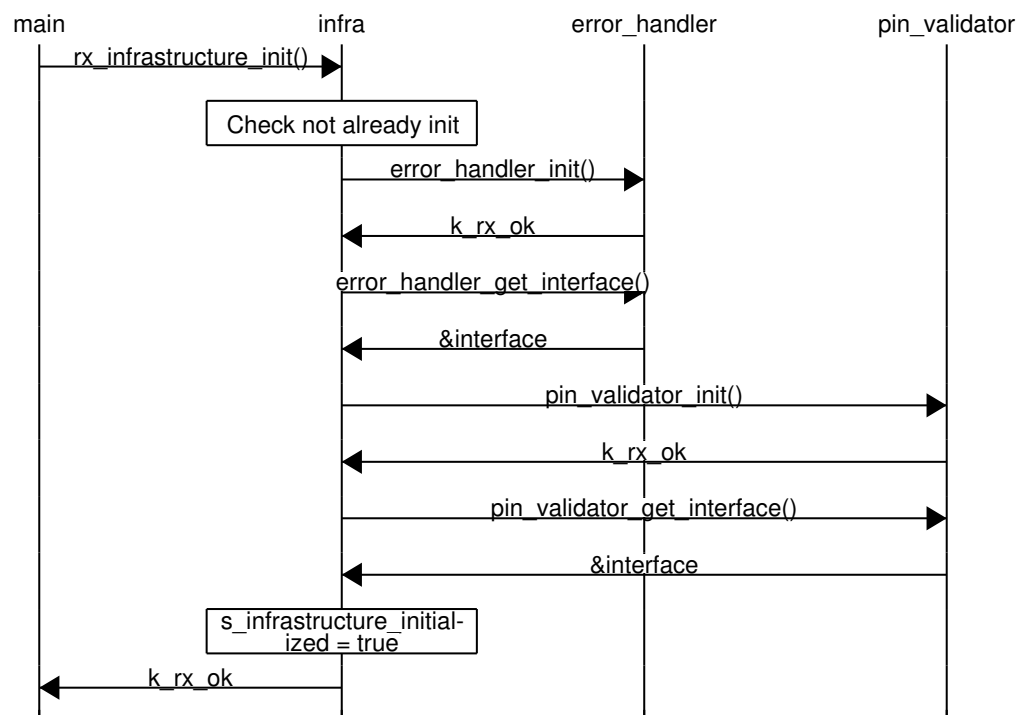
Global Infrastructure Initialization Implementation.

Implements the global infrastructure layer that initializes and provides access to core system services. This module serves as the central hub for shared singleton instances used throughout the firmware.

System Architecture



Initialization Sequence (MSC)



Provided Services

Service	Description	Access Function
Error Handler	Retry logic with exponential backoff	rx_infrastructure_get_error_interface()
Pin Validator	GPIO pin conflict detection	rx_infrastructure_get_pin_interface()

Error Handler Configuration (Defaults)

Parameter	Value	Description
max_retries	3	Maximum retry attempts
initial_backoff_ms	100	First retry delay
max_backoff_ms	5000	Maximum backoff cap

Memory Usage

Type	Size	Description
Code	~400 B	All functions (with -O2)
Static	~200 B	2 handlers + 2 interfaces + 1 bool
Stack	~32 B	Local config struct in init()

Static Variable Layout

Variable	Type	Size
s_global_error_handler	error_handler_t	~80 B
s_global_pin_validator	pin_validator_t	~64 B
s_global_error_interface	rx_error_interface_t	24 B
s_global_pin_interface	rx_pin_interface_t	24 B
s_infrastructure_initialized	bool	1 B

Thread Safety

- [rx_infrastructure_init\(\)](#): NOT thread-safe, call once from main before threads
- [rx_infrastructure_deinit\(\)](#): NOT thread-safe, call after all threads stopped
- [rx_infrastructure_get_*\(\)](#): Thread-safe after init (read-only access)

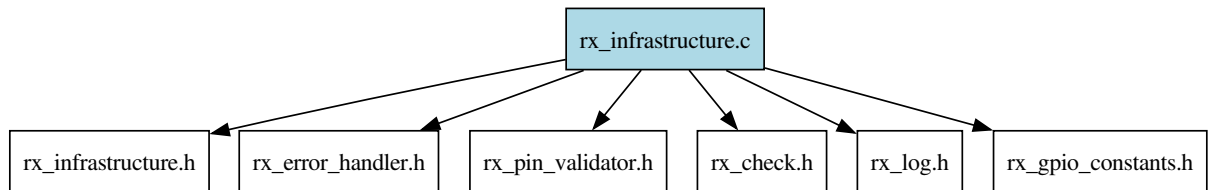
NASA Power of 10 Compliance

- Rule 1: [OK] No goto, setjmp, or recursion
- Rule 2: [OK] No loops (uses called function loops only)
- Rule 3: [OK] No dynamic memory allocation
- Rule 4: [OK] All functions < 60 lines
- Rule 5: [OK] Initialization validation, cleanup on failure
- Rule 6: [OK] Static variables at file scope, locals at smallest scope
- Rule 7: [OK] All return values checked
- Rule 8: [OK] Minimal preprocessor (only includes)
- Rule 9: [OK] Function pointers in interfaces (DIP, intentional)
- Rule 10: [OK] Compiles with -Wall -Wextra -Werror

SOLID Principles Compliance

- S: Single Responsibility - Only infrastructure lifecycle management
- O: Open/Closed - New services can be added without modifying existing code
- L: Liskov Substitution - Services use abstract interfaces
- I: Interface Segregation - Error and pin interfaces are separate
- D: Dependency Inversion - Modules depend on interfaces, not implementations

Module Dependencies



See also

[rx_infrastructure.h](#) API header

[rx_error_handler.h](#) Error handler with retry logic

[rx_pin_validator.h](#) GPIO pin conflict detection

Author

Locked, Inc.

Date

2026-01-29

Copyright

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Since

Version 1.0.0

Definition in file [rx_infrastructure.c](#).

7.49.2 Function Documentation

7.49.2.1 rx_infrastructure_deinit()

```
rx_err_t rx_infrastructure_deinit (  
    void ) [nodiscard]
```

Deinitialize global infrastructure and release resources.

Deinitialize global infrastructure (graceful shutdown only).

Releases all global infrastructure resources in reverse initialization order. Clears the initialized flag so interface accessors return NULL.

Algorithm Steps:

1. Check if initialized (return early if not)
2. Deinitialize pin validator
3. Deinitialize error handler
4. Clear initialized flag

Cleanup Order:

Resources are released in reverse initialization order to respect dependencies:

- Pin validator (uses error handler for logging)
- Error handler (foundation service)

Precondition

None - safe to call anytime

Postcondition

s_infrastructure_initialized == false

Interface accessors return nullptr

All s_global_* structures are in undefined state

Note

Thread Safety: NOT thread-safe

Idempotent: Safe to call multiple times

Warning

Call only after all threads have stopped using infrastructure

After deinit, interface accessors return nullptr

Example:

```
// During system shutdown
motor_deinit(&motor);
sensor_deinit(&sensor);

// Release infrastructure last
rx_infrastructure_deinit();
```

See also

[rx_infrastructure_init\(\)](#) Initialization function

Since

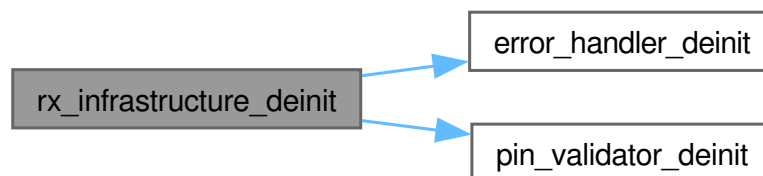
Version 1.0.0

Definition at line 464 of file [rx_infrastructure.c](#).

```
00465 {
00466     if (!s_infrastructure_initialized) {
00467         return k_rx_ok;
00468     }
00469     rx_log_info(s_tag, "Deinitializing global infrastructure");
00470
00471     /* Deinitialize pin validator */
00472     rx_err_t err = pin_validator_deinit(&s_global_pin_validator);
00473     if (err != k_rx_ok) {
00474         rx_log_warn(s_tag, "Pin validator deinit failed");
00475         /* Continue with cleanup anyway */
00476     }
00477
00478     /* Deinitialize error handler */
00479     err = error_handler_deinit(&s_global_error_handler);
00480     if (err != k_rx_ok) {
00481         rx_log_warn(s_tag, "Error handler deinit failed");
00482         return err;
00483     }
00484
00485     s_infrastructure_initialized = false;
00486
00487     rx_log_info(s_tag, "Global infrastructure deinitialized");
00488
00489     return k_rx_ok;
00490 }
00491 }
```

References [error_handler_deinit\(\)](#), [k_rx_ok](#), [pin_validator_deinit\(\)](#), [rx_log_info](#), [rx_log_warn](#), [s_global_error_handler](#), [s_global_pin_validator](#), [s_infrastructure_initialized](#), and [s_tag](#).

Here is the call graph for this function:



7.49.2.2 rx_infrastructure_get_error_interface()

```
rx_error_interface_t * rx_infrastructure_get_error_interface (
    void )
```

Get global error handler interface.

Get global error handler interface (thread-safe accessor).

Returns a pointer to the global error handler interface structure. The interface provides function pointers for error recovery operations.

Interface Functions:

Function	Description
try_recover	Attempt recovery with retry and backoff
reset	Reset error handler state

Precondition

[rx_infrastructure_init\(\)](#) should have been called

Postcondition

Returned pointer valid for program lifetime (if not nullptr)

Note

Thread Safety: Safe - returns pointer to static data

The returned pointer should not be freed

Example:

```
rx_error_interface_t* err_if = rx_infrastructure_get_error_interface();
if (err_if == nullptr) {
    rx_log_error("APP", "Infrastructure not initialized");
    return k_rx_err_not_initialized;
}

// Use error handler for retry logic
rx_err_t result = err_if->try_recover(err_if->ctx, my_operation, &context);
```

See also

[rx_error_interface_t](#) Interface structure

[rx_infrastructure_init\(\)](#) Must be called first

Since

Version 1.0.0

Definition at line 531 of file [rx_infrastructure.c](#).

```
00532 {
00533     if (!s_infrastructure_initialized) {
00534         return nullptr;
00535     }
00536     return &s_global_error_interface;
00537 }
00538 }
```

References [s_global_error_interface](#), and [s_infrastructure_initialized](#).

7.49.2.3 rx_infrastructure_get_pin_interface()

```
rx_pin_interface_t * rx_infrastructure_get_pin_interface (  
    void )
```

Get global pin validator interface.

Get global pin validator interface (thread-safe accessor).

Returns a pointer to the global pin validator interface structure. The interface provides function pointers for GPIO pin management.

Interface Functions:

Function	Description
claim_pin	Claim a GPIO pin for a specific module
release	Release a previously claimed GPIO pin
is_claimed	Check if a pin is already claimed

Precondition

[rx_infrastructure_init\(\)](#) should have been called

Postcondition

Returned pointer valid for program lifetime (if not nullptr)

Note

Thread Safety: Safe for read - returns pointer to static data

The returned pointer should not be freed

Warning

Pin claim/release operations may not be thread-safe

Example:

```
rx_pin_interface_t* pin_if = rx_infrastructure_get_pin_interface();  
if (pin_if == nullptr) {  
    rx_log_error("MOTOR", "Infrastructure not initialized");  
    return k_rx_err_not_initialized;  
}  
  
// Claim GPIO pin for motor PWM  
rx_err_t err = pin_if->claim_pin(pin_if->ctx,  
                                k_port_1, k_pin_5,  
                                "MOTOR_PWM");  
if (err != k_rx_ok) {  
    rx_log_error("MOTOR", "Pin P15 already in use");  
    return err;  
}
```

See also

[rx_pin_interface_t](#) Interface structure

[rx_infrastructure_init\(\)](#) Must be called first

Since

Version 1.0.0

Definition at line 587 of file [rx_infrastructure.c](#).

```
00588 {  
00589     if (!s_infrastructure_initialized) {  
00590         return nullptr;  
00591     }  
00592  
00593     return &s_global_pin_interface;  
00594 }
```

References [s_global_pin_interface](#), and [s_infrastructure_initialized](#).

7.49.2.4 rx_infrastructure_init()

```
rx\_err\_t rx_infrastructure_init (  
    void ) [nodiscard]
```

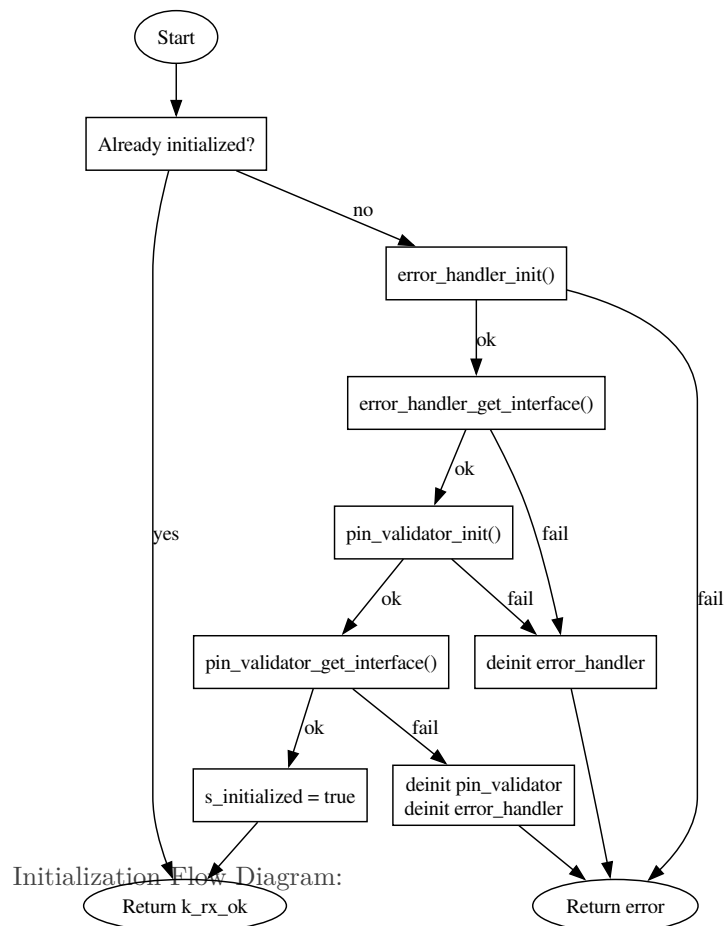
Initialize global infrastructure services.

Initialize global infrastructure (MUST BE FIRST CALL IN main()).

Initializes all global infrastructure services in the correct order, with proper cleanup on failure. Uses default configuration values for error handler retry/backoff parameters.

Algorithm Steps:

1. Check if already initialized (idempotent, return success)
2. Initialize error handler with default config
3. Get error handler interface (function pointers)
4. Initialize pin validator
5. Get pin validator interface (function pointers)
6. Set initialized flag



Default Configuration:

Parameter	Value
max_retries	k_error_handler_default_max_retries (3)
initial_backoff_ms	k_error_handler_default_initial_backoff_ms
max_backoff_ms	k_error_handler_default_max_backoff_ms

Precondition

None - can be called at any time

Postcondition

s_infrastructure_initialized == true on success

All s_global_* variables contain valid data on success

Interface accessors return valid pointers on success

Note

Thread Safety: NOT thread-safe, call once from main()

Idempotent: Safe to call multiple times, subsequent calls are no-op

Warning

Must be called before any module uses infrastructure services

Not thread-safe - call before starting RTOS threads

Example:

```
// In main() before starting ThreadX
rx_err_t err = rx_infrastructure_init();
if (err != k_rx_ok) {
    // Critical failure - cannot continue
    while (1) { } // Halt
}

// Now modules can get interfaces
rx_error_interface_t* error_if = rx_infrastructure_get_error_interface();
```

See also

[rx_infrastructure_deinit\(\)](#) Cleanup function

[rx_infrastructure_get_error_interface\(\)](#) Access error handler

[rx_infrastructure_get_pin_interface\(\)](#) Access pin validator

NASA Power of 10 Compliance:

- Rule 5: [OK] Cleanup on failure (4 cleanup paths)
- Rule 7: [OK] All return values checked

Since

Version 1.0.0

Definition at line 365 of file [rx_infrastructure.c](#).

```
00366 {
00367     const error_handler_config_t config = {
00368         .max_retries      = k_error_handler_default_max_retries,
00369         .initial_backoff_ms = k_error_handler_default_initial_backoff_ms,
00370         .max_backoff_ms    = k_error_handler_default_max_backoff_ms,
00371     };
00372
00373     if (s_infrastructure_initialized) {
00374         rx_log_warn(s_tag, "Infrastructure already initialized");
00375         return k_rx_ok;
00376     }
00377
00378     rx_log_info(s_tag, "Initializing global infrastructure");
00379
00380     /* Initialize error handler */
00381     rx_err_t err = error_handler_init(&s_global_error_handler, &config);
00382     if (err != k_rx_ok) {
00383         rx_log_error(s_tag, "Failed to initialize error handler");
00384         return err;
00385     }
00386
00387     /* Get error handler interface */
00388     err = error_handler_get_interface(&s_global_error_interface, &s_global_error_handler);
00389     if (err != k_rx_ok) {
00390         rx_log_error(s_tag, "Failed to get error handler interface");
00391         (void)error_handler_deinit(&s_global_error_handler); /* Cleanup, ignore errors */
00392         return err;
00393     }
00394
00395     /* Initialize pin validator */
00396     err = pin_validator_init(&s_global_pin_validator);
00397     if (err != k_rx_ok) {
00398         rx_log_error(s_tag, "Failed to initialize pin validator");
00399         (void)error_handler_deinit(&s_global_error_handler); /* Cleanup, ignore errors */
```

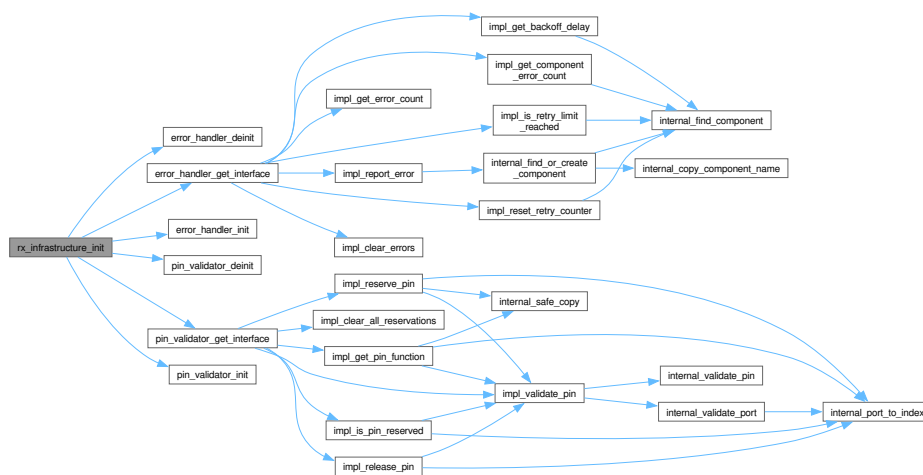
```

00400     return err;
00401 }
00402
00403 /* Get pin validator interface */
00404 err = pin_validator_get_interface(&s_global_pin_interface, &s_global_pin_validator);
00405 if (err != k_rx_ok) {
00406     rx_log_error(s_tag, "Failed to get pin validator interface");
00407     (void)pin_validator_deinit(&s_global_pin_validator); /* Cleanup, ignore errors */
00408     (void)error_handler_deinit(&s_global_error_handler); /* Cleanup, ignore errors */
00409     return err;
00410 }
00411
00412 s_infrastructure_initialized = true;
00413
00414 rx_log_info(s_tag, "Global infrastructure initialized successfully");
00415
00416 return k_rx_ok;
00417 }

```

References [error_handler_deinit\(\)](#), [error_handler_get_interface\(\)](#), [error_handler_init\(\)](#), [k_error_handler_default_init](#), [k_error_handler_default_max_backoff_ms](#), [k_error_handler_default_max_retries](#), [k_rx_ok](#), [pin_validator_deinit\(\)](#), [pin_validator_get_interface\(\)](#), [pin_validator_init\(\)](#), [rx_log_error](#), [rx_log_info](#), [rx_log_warn](#), [s_global_error_handler](#), [s_global_error_interface](#), [s_global_pin_interface](#), [s_global_pin_validator](#), [s_infrastructure_initialized](#), and [s_tag](#).

Here is the call graph for this function:



7.49.3 Variable Documentation

7.49.3.1 s_global_error_handler

[error_handler_t](#) s_global_error_handler [static]

Global error handler instance (singleton).

Single error handler instance shared across all modules. Provides retry logic with exponential backoff for recoverable errors.

Initialization:

Initialized by [rx_infrastructure_init\(\)](#) with default configuration.

Access:

Not accessed directly - use [rx_infrastructure_get_error_interface\(\)](#).

Note

Thread Safety: Structure is thread-safe after initialization

Warning

Do not access directly - use interface accessor

Since

Version 1.0.0

Definition at line 172 of file [rx_infrastructure.c](#).

Referenced by [rx_infrastructure_deinit\(\)](#), and [rx_infrastructure_init\(\)](#).

7.49.3.2 s_global_error_interface

```
rx_error_interface_t s_global_error_interface = {} [static]
```

Global error handler interface (function pointers).

Abstract interface to error handler, enabling dependency inversion. Populated with function pointers during initialization.

Memory Layout:

Offset	Field	Size	Description
0	ctx	8	Pointer to error handler
8	try_recover	8	Recovery function pointer
16	reset	8	Reset function pointer

See also

[rx_error_interface_t](#) Interface structure definition

Since

Version 1.0.0

Definition at line 214 of file [rx_infrastructure.c](#).

```
00214 {};
```

Referenced by [rx_infrastructure_get_error_interface\(\)](#), and [rx_infrastructure_init\(\)](#).

7.49.3.3 s_global_pin_interface

`rx_pin_interface_t s_global_pin_interface = {}` [static]

Global pin validator interface (function pointers).

Abstract interface to pin validator, enabling dependency inversion. Populated with function pointers during initialization.

Memory Layout:

Offset	Field	Size	Description
0	ctx	8	Pointer to pin validator
8	claim_pin	8	Pin claim function pointer
16	release	8	Release function pointer

See also

`rx_pin_interface_t` Interface structure definition

Since

Version 1.0.0

Definition at line 235 of file [rx_infrastructure.c](#).

00235 {};

Referenced by [rx_infrastructure_get_pin_interface\(\)](#), and [rx_infrastructure_init\(\)](#).

7.49.3.4 s_global_pin_validator

[pin_validator_t](#) `s_global_pin_validator` [static]

Global pin validator instance (singleton).

Single pin validator instance that tracks GPIO pin assignments across all modules to detect and prevent conflicts.

Initialization:

Initialized by [rx_infrastructure_init\(\)](#).

Access:

Not accessed directly - use [rx_infrastructure_get_pin_interface\(\)](#).

Note

Thread Safety: Structure requires external synchronization

Warning

Do not access directly - use interface accessor

Since

Version 1.0.0

Definition at line 193 of file [rx_infrastructure.c](#).

Referenced by [rx_infrastructure_deinit\(\)](#), and [rx_infrastructure_init\(\)](#).

7.49.3.5 s_infrastructure_initialized

```
bool s_infrastructure_initialized = false    [static]
```

Infrastructure initialization flag.

Guards against double-initialization and provides early-out for interface accessors when infrastructure is not ready.

State Machine:

- false: Initial state, or after deinit
- true: After successful init

Invariant

If true, all s_global_* variables contain valid data

Since

Version 1.0.0

Definition at line 253 of file [rx_infrastructure.c](#).

Referenced by [rx_infrastructure_deinit\(\)](#), [rx_infrastructure_get_error_interface\(\)](#), [rx_infrastructure_get_pin_interface\(\)](#) and [rx_infrastructure_init\(\)](#).

7.49.3.6 s_tag

```
const char* const s_tag = "INFRA"    [static]
```

Log tag for this module.

Identifies log messages from the infrastructure module. Passed to rx_log_* functions to prefix output with "INFRA".

Note

Read-only after initialization; never modified after program startup.

Invariant

Value is "INFRA" and never changes after compilation.

Since

Version 1.0.0

Definition at line 264 of file [rx_infrastructure.c](#).

7.50 rx_infrastructure.c

[Go to the documentation of this file.](#)

```

00001 /**
00002  * @file rx_infrastructure.c
00003  * @brief Global Infrastructure Initialization Implementation
00004  *
00005  * @details
00006  * Implements the global infrastructure layer that initializes and provides
00007  * access to core system services. This module serves as the central hub
00008  * for shared singleton instances used throughout the firmware.
00009  *
00010  * @par System Architecture
00011  * @verbatim
00012  * +-----+
00013  * |               Application Layer               |
00014  * | (motor_control, communication, sensors, telemetry) |
00015  * +-----+
00016  * |
00017  * | rx_infrastructure_get_*()
00018  * |
00019  * +-----+
00020  * |               rx_infrastructure (This Module)               |
00021  * | +-----+ +-----+ +-----+ |
00022  * | | Error Handler | | Pin Validator | |
00023  * | | (retry, backoff) | | (conflict detection) | |
00024  * | | s_global_error_* | | s_global_pin_* | |
00025  * | +-----+ +-----+ +-----+ |
00026  * +-----+
00027  * |
00028  * | Uses interfaces from
00029  * |
00030  * +-----+
00031  * |               Core Library Layer               |
00032  * | rx_error_handler.c rx_pin_validator.c rx_log.c |
00033  * +-----+
00034  * @endverbatim
00035  *
00036  * @par Initialization Sequence (MSC)
00037  * @msc
00038  * main, infra, error_handler, pin_validator;
00039  *
00040  * main => infra [label="rx_infrastructure_init()"];
00041  * infra box infra [label="Check not already init"];
00042  * infra => error_handler [label="error_handler_init()"];
00043  * error_handler => infra [label="k_rx_ok"];
00044  * infra => error_handler [label="error_handler_get_interface()"];
00045  * error_handler => infra [label="&interface"];
00046  * infra => pin_validator [label="pin_validator_init()"];
00047  * pin_validator => infra [label="k_rx_ok"];
00048  * infra => pin_validator [label="pin_validator_get_interface()"];
00049  * pin_validator => infra [label="&interface"];
00050  * infra box infra [label="s_infrastructure_initialized = true"];
00051  * infra => main [label="k_rx_ok"];
00052  * @endmsc
00053  *
00054  * @par Provided Services
00055  * | Service | Description | Access Function |
00056  * +-----+ +-----+ +-----+
00057  * | Error Handler | Retry logic with exponential backoff | rx_infrastructure_get_error_interface() |
00058  * | Pin Validator | GPIO pin conflict detection | rx_infrastructure_get_pin_interface() |
00059  *
00060  * @par Error Handler Configuration (Defaults)
00061  * | Parameter | Value | Description |
00062  * +-----+ +-----+ +-----+
00063  * | max_retries | 3 | Maximum retry attempts |
00064  * | initial_backoff_ms | 100 | First retry delay |
00065  * | max_backoff_ms | 5000 | Maximum backoff cap |
00066  *
00067  * @par Memory Usage
00068  * | Type | Size | Description |
00069  * +-----+ +-----+ +-----+
00070  * | Code | ~400 B | All functions (with -O2) |
00071  * | Static | ~200 B | 2 handlers + 2 interfaces + 1 bool |
00072  * | Stack | ~32 B | Local config struct in init() |
00073  *
00074  * @par Static Variable Layout
00075  * | Variable | Type | Size |
00076  * +-----+ +-----+ +-----+
00077  * | s_global_error_handler | error_handler_t | ~80 B |
00078  * | s_global_pin_validator | pin_validator_t | ~64 B |
00079  * | s_global_error_interface | rx_error_interface_t | 24 B |
00080  * | s_global_pin_interface | rx_pin_interface_t | 24 B |
00081  * | s_infrastructure_initialized | bool | 1 B |
00082  *

```

```

00083 * @par Thread Safety
00084 * - rx_infrastructure_init(): NOT thread-safe, call once from main before threads
00085 * - rx_infrastructure_deinit(): NOT thread-safe, call after all threads stopped
00086 * - rx_infrastructure_get_*(): Thread-safe after init (read-only access)
00087 *
00088 * @par NASA Power of 10 Compliance
00089 * - Rule 1: [OK] No goto, setjmp, or recursion
00090 * - Rule 2: [OK] No loops (uses called function loops only)
00091 * - Rule 3: [OK] No dynamic memory allocation
00092 * - Rule 4: [OK] All functions < 60 lines
00093 * - Rule 5: [OK] Initialization validation, cleanup on failure
00094 * - Rule 6: [OK] Static variables at file scope, locals at smallest scope
00095 * - Rule 7: [OK] All return values checked
00096 * - Rule 8: [OK] Minimal preprocessor (only includes)
00097 * - Rule 9: [OK] Function pointers in interfaces (DIP, intentional)
00098 * - Rule 10: [OK] Compiles with -Wall -Wextra -Werror
00099 *
00100 * @par SOLID Principles Compliance
00101 * - **S**: Single Responsibility - Only infrastructure lifecycle management
00102 * - **O**: Open/Closed - New services can be added without modifying existing code
00103 * - **I**: Liskov Substitution - Services use abstract interfaces
00104 * - **I**: Interface Segregation - Error and pin interfaces are separate
00105 * - **D**: Dependency Inversion - Modules depend on interfaces, not implementations
00106 *
00107 * @par Module Dependencies
00108 * @dot
00109 * digraph dependencies {
00110 *   rankdir=TB;
00111 *   node [shape=box];
00112 *   rx_infrastructure [label="rx_infrastructure.c" style=filled fillcolor=lightblue];
00113 *   rx_infrastructure_h [label="rx_infrastructure.h"];
00114 *   rx_error_handler [label="rx_error_handler.h"];
00115 *   rx_pin_validator [label="rx_pin_validator.h"];
00116 *   rx_check [label="rx_check.h"];
00117 *   rx_log [label="rx_log.h"];
00118 *   rx_gpio_constants [label="rx_gpio_constants.h"];
00119 *
00120 *   rx_infrastructure -> rx_infrastructure_h;
00121 *   rx_infrastructure -> rx_error_handler;
00122 *   rx_infrastructure -> rx_pin_validator;
00123 *   rx_infrastructure -> rx_check;
00124 *   rx_infrastructure -> rx_log;
00125 *   rx_infrastructure -> rx_gpio_constants;
00126 * }
00127 * @enddot
00128 *
00129 * @see rx_infrastructure.h API header
00130 * @see rx_error_handler.h Error handler with retry logic
00131 * @see rx_pin_validator.h GPIO pin conflict detection
00132 *
00133 * @author Locked, Inc.
00134 * @date 2026-01-29
00135 * @copyright Copyright (c) 2026 Locked Inc.
00136 * SPDX-License-Identifier: MIT
00137 * @since Version 1.0.0
00138 */
00139
00140 #include "rx_infrastructure.h"
00141
00142 #include "rx_check.h"
00143 #include "rx_error_handler.h"
00144 #include "rx_gpio_constants.h"
00145 #include "rx_log.h"
00146 #include "rx_pin_validator.h"
00147
00148 /*
=====
00149 * Global Infrastructure Instances
00150 *
=====
00151 */
00152
00153 /**
00154 * @var s_global_error_handler
00155 * @brief Global error handler instance (singleton)
00156 *
00157 * @details
00158 * Single error handler instance shared across all modules.
00159 * Provides retry logic with exponential backoff for recoverable errors.
00160 *
00161 * @par Initialization:
00162 * Initialized by rx_infrastructure_init() with default configuration.
00163 *
00164 * @par Access:
00165 * Not accessed directly - use rx_infrastructure_get_error_interface().
00166 *
00167 * @note Thread Safety: Structure is thread-safe after initialization

```

```

00168 * @warning Do not access directly - use interface accessor
00169 *
00170 * @since Version 1.0.0
00171 */
00172 static error_handler_t s_global_error_handler;
00173
00174 /**
00175 * @var s_global_pin_validator
00176 * @brief Global pin validator instance (singleton)
00177 *
00178 * @details
00179 * Single pin validator instance that tracks GPIO pin assignments
00180 * across all modules to detect and prevent conflicts.
00181 *
00182 * @par Initialization:
00183 * Initialized by rx_infrastructure_init().
00184 *
00185 * @par Access:
00186 * Not accessed directly - use rx_infrastructure_get_pin_interface().
00187 *
00188 * @note Thread Safety: Structure requires external synchronization
00189 * @warning Do not access directly - use interface accessor
00190 *
00191 * @since Version 1.0.0
00192 */
00193 static pin_validator_t s_global_pin_validator;
00194
00195 /**
00196 * @var s_global_error_interface
00197 * @brief Global error handler interface (function pointers)
00198 *
00199 * @details
00200 * Abstract interface to error handler, enabling dependency inversion.
00201 * Populated with function pointers during initialization.
00202 *
00203 * @par Memory Layout:
00204 * | Offset | Field      | Size | Description          |
00205 * |-----|-----|-----|-----|
00206 * | 0      | ctx        | 8     | Pointer to error handler |
00207 * | 8      | try_recover | 8     | Recovery function pointer |
00208 * | 16     | reset      | 8     | Reset function pointer   |
00209 *
00210 * @see rx_error_interface_t Interface structure definition
00211 *
00212 * @since Version 1.0.0
00213 */
00214 static rx_error_interface_t s_global_error_interface = {};
00215
00216 /**
00217 * @var s_global_pin_interface
00218 * @brief Global pin validator interface (function pointers)
00219 *
00220 * @details
00221 * Abstract interface to pin validator, enabling dependency inversion.
00222 * Populated with function pointers during initialization.
00223 *
00224 * @par Memory Layout:
00225 * | Offset | Field      | Size | Description          |
00226 * |-----|-----|-----|-----|
00227 * | 0      | ctx        | 8     | Pointer to pin validator |
00228 * | 8      | claim_pin  | 8     | Pin claim function pointer |
00229 * | 16     | release    | 8     | Release function pointer   |
00230 *
00231 * @see rx_pin_interface_t Interface structure definition
00232 *
00233 * @since Version 1.0.0
00234 */
00235 static rx_pin_interface_t s_global_pin_interface = {};
00236
00237 /**
00238 * @var s_infrastructure_initialized
00239 * @brief Infrastructure initialization flag
00240 *
00241 * @details
00242 * Guards against double-initialization and provides early-out for
00243 * interface accessors when infrastructure is not ready.
00244 *
00245 * @par State Machine:
00246 * - false: Initial state, or after deinit
00247 * - true: After successful init
00248 *
00249 * @invariant If true, all s_global_* variables contain valid data
00250 *
00251 * @since Version 1.0.0
00252 */
00253 static bool s_infrastructure_initialized = false;
00254

```

```

00255 /**
00256  * @var s_tag
00257  * @brief Log tag for this module
00258  * @details Identifies log messages from the infrastructure module. Passed to
00259  * rx_log_* functions to prefix output with "INFRA".
00260  * @note Read-only after initialization; never modified after program startup.
00261  * @invariant Value is "INFRA" and never changes after compilation.
00262  * @since Version 1.0.0
00263  */
00264 static const char* const s_tag = "INFRA";
00265
00266 /*
=====
00267  * Public API Implementation
00268  *
=====
00269  */
00270
00271 /**
00272  * @brief Initialize global infrastructure services
00273  *
00274  * @details
00275  * Initializes all global infrastructure services in the correct order,
00276  * with proper cleanup on failure. Uses default configuration values
00277  * for error handler retry/backoff parameters.
00278  *
00279  * @par Algorithm Steps:
00280  * 1. Check if already initialized (idempotent, return success)
00281  * 2. Initialize error handler with default config
00282  * 3. Get error handler interface (function pointers)
00283  * 4. Initialize pin validator
00284  * 5. Get pin validator interface (function pointers)
00285  * 6. Set initialized flag
00286  *
00287  * @par Initialization Flow Diagram:
00288  * @dot
00289  * digraph init_flow {
00290  *   rankdir=TB;
00291  *   node [shape=box];
00292  *
00293  *   start [label="Start" shape=ellipse];
00294  *   check_init [label="Already initialized?"];
00295  *   return_ok [label="Return k_rx_ok" shape=ellipse];
00296  *   init_error [label="error_handler_init()"];
00297  *   get_error_if [label="error_handler_get_interface()"];
00298  *   init_pin [label="pin_validator_init()"];
00299  *   get_pin_if [label="pin_validator_get_interface()"];
00300  *   set_flag [label="s_initialized = true"];
00301  *   cleanup1 [label="deinit error_handler"];
00302  *   cleanup2 [label="deinit pin_validator\ndeinit error_handler"];
00303  *   return_err [label="Return error" shape=ellipse];
00304  *
00305  *   start -> check_init;
00306  *   check_init -> return_ok [label="yes"];
00307  *   check_init -> init_error [label="no"];
00308  *   init_error -> get_error_if [label="ok"];
00309  *   init_error -> return_err [label="fail"];
00310  *   get_error_if -> init_pin [label="ok"];
00311  *   get_error_if -> cleanup1 [label="fail"];
00312  *   init_pin -> get_pin_if [label="ok"];
00313  *   init_pin -> cleanup1 [label="fail"];
00314  *   get_pin_if -> set_flag [label="ok"];
00315  *   get_pin_if -> cleanup2 [label="fail"];
00316  *   cleanup1 -> return_err;
00317  *   cleanup2 -> return_err;
00318  *   set_flag -> return_ok;
00319  * }
00320  * @enddot
00321  *
00322  * @par Default Configuration:
00323  * | Parameter | Value |
00324  * |-----|-----|
00325  * | max_retries | k_error_handler_default_max_retries (3) |
00326  * | initial_backoff_ms | k_error_handler_default_initial_backoff_ms |
00327  * | max_backoff_ms | k_error_handler_default_max_backoff_ms |
00328  *
00329  *
00330  * @pre None - can be called at any time
00331  *
00332  * @post s_infrastructure_initialized == true on success
00333  * @post All s_global_* variables contain valid data on success
00334  * @post Interface accessors return valid pointers on success
00335  *
00336  * @note Thread Safety: NOT thread-safe, call once from main()
00337  * @note Idempotent: Safe to call multiple times, subsequent calls are no-op
00338  *
00339  * @warning Must be called before any module uses infrastructure services

```

```

00340 * @warning Not thread-safe - call before starting RTOS threads
00341 *
00342 * @par Example:
00343 * @code{.c}
00344 * // In main() before starting ThreadX
00345 * rx_err_t err = rx_infrastructure_init();
00346 * if (err != k_rx_ok) {
00347 *     // Critical failure - cannot continue
00348 *     while (1) { } // Halt
00349 * }
00350 *
00351 * // Now modules can get interfaces
00352 * rx_error_interface_t* error_if = rx_infrastructure_get_error_interface();
00353 * @endcode
00354 *
00355 * @see rx_infrastructure_deinit() Cleanup function
00356 * @see rx_infrastructure_get_error_interface() Access error handler
00357 * @see rx_infrastructure_get_pin_interface() Access pin validator
00358 *
00359 * @par NASA Power of 10 Compliance:
00360 * - Rule 5: [OK] Cleanup on failure (4 cleanup paths)
00361 * - Rule 7: [OK] All return values checked
00362 *
00363 * @since Version 1.0.0
00364 */
00365 rx_err_t rx_infrastructure_init(void)
00366 {
00367     const error_handler_config_t config = {
00368         .max_retries = k_error_handler_default_max_retries,
00369         .initial_backoff_ms = k_error_handler_default_initial_backoff_ms,
00370         .max_backoff_ms = k_error_handler_default_max_backoff_ms,
00371     };
00372
00373     if (s_infrastructure_initialized) {
00374         rx_log_warn(s_tag, "Infrastructure already initialized");
00375         return k_rx_ok;
00376     }
00377
00378     rx_log_info(s_tag, "Initializing global infrastructure");
00379
00380     /* Initialize error handler */
00381     rx_err_t err = error_handler_init(&s_global_error_handler, &config);
00382     if (err != k_rx_ok) {
00383         rx_log_error(s_tag, "Failed to initialize error handler");
00384         return err;
00385     }
00386
00387     /* Get error handler interface */
00388     err = error_handler_get_interface(&s_global_error_interface, &s_global_error_handler);
00389     if (err != k_rx_ok) {
00390         rx_log_error(s_tag, "Failed to get error handler interface");
00391         (void)error_handler_deinit(&s_global_error_handler); /* Cleanup, ignore errors */
00392         return err;
00393     }
00394
00395     /* Initialize pin validator */
00396     err = pin_validator_init(&s_global_pin_validator);
00397     if (err != k_rx_ok) {
00398         rx_log_error(s_tag, "Failed to initialize pin validator");
00399         (void)error_handler_deinit(&s_global_error_handler); /* Cleanup, ignore errors */
00400         return err;
00401     }
00402
00403     /* Get pin validator interface */
00404     err = pin_validator_get_interface(&s_global_pin_interface, &s_global_pin_validator);
00405     if (err != k_rx_ok) {
00406         rx_log_error(s_tag, "Failed to get pin validator interface");
00407         (void)pin_validator_deinit(&s_global_pin_validator); /* Cleanup, ignore errors */
00408         (void)error_handler_deinit(&s_global_error_handler); /* Cleanup, ignore errors */
00409         return err;
00410     }
00411
00412     s_infrastructure_initialized = true;
00413
00414     rx_log_info(s_tag, "Global infrastructure initialized successfully");
00415
00416     return k_rx_ok;
00417 }
00418
00419 /**
00420 * @brief Deinitialize global infrastructure and release resources
00421 *
00422 * @details
00423 * Releases all global infrastructure resources in reverse initialization order.
00424 * Clears the initialized flag so interface accessors return NULL.
00425 *
00426 * @par Algorithm Steps:

```



```

00427 * 1. Check if initialized (return early if not)
00428 * 2. Deinitialize pin validator
00429 * 3. Deinitialize error handler
00430 * 4. Clear initialized flag
00431 *
00432 * @par Cleanup Order:
00433 * Resources are released in reverse initialization order to respect dependencies:
00434 * - Pin validator (uses error handler for logging)
00435 * - Error handler (foundation service)
00436 *
00437 *
00438 * @pre None - safe to call anytime
00439 *
00440 * @post s_infrastructure_initialized == false
00441 * @post Interface accessors return nullptr
00442 * @post All s_global_* structures are in undefined state
00443 *
00444 * @note Thread Safety: NOT thread-safe
00445 * @note Idempotent: Safe to call multiple times
00446 *
00447 * @warning Call only after all threads have stopped using infrastructure
00448 * @warning After deinit, interface accessors return nullptr
00449 *
00450 * @par Example:
00451 * @code{.c}
00452 * // During system shutdown
00453 * motor_deinit(&motor);
00454 * sensor_deinit(&sensor);
00455 *
00456 * // Release infrastructure last
00457 * rx_infrastructure_deinit();
00458 * @endcode
00459 *
00460 * @see rx_infrastructure_init() Initialization function
00461 *
00462 * @since Version 1.0.0
00463 */
00464 rx_err_t rx_infrastructure_deinit(void)
00465 {
00466     if (!s_infrastructure_initialized) {
00467         return k_rx_ok;
00468     }
00469
00470     rx_log_info(s_tag, "Deinitializing global infrastructure");
00471
00472     /* Deinitialize pin validator */
00473     rx_err_t err = pin_validator_deinit(&s_global_pin_validator);
00474     if (err != k_rx_ok) {
00475         rx_log_warn(s_tag, "Pin validator deinit failed");
00476         /* Continue with cleanup anyway */
00477     }
00478
00479     /* Deinitialize error handler */
00480     err = error_handler_deinit(&s_global_error_handler);
00481     if (err != k_rx_ok) {
00482         rx_log_warn(s_tag, "Error handler deinit failed");
00483         return err;
00484     }
00485
00486     s_infrastructure_initialized = false;
00487
00488     rx_log_info(s_tag, "Global infrastructure deinitialized");
00489
00490     return k_rx_ok;
00491 }
00492 /**
00493 * @brief Get global error handler interface
00494 *
00495 * @details
00496 * Returns a pointer to the global error handler interface structure.
00497 * The interface provides function pointers for error recovery operations.
00498 *
00499 * @par Interface Functions:
00500 * | Function | Description |
00501 * |-----|-----|
00502 * | try_recover | Attempt recovery with retry and backoff |
00503 * | reset | Reset error handler state |
00504 *
00505 *
00506 *
00507 * @pre rx_infrastructure_init() should have been called
00508 *
00509 * @post Returned pointer valid for program lifetime (if not nullptr)
00510 *
00511 * @note Thread Safety: Safe - returns pointer to static data
00512 * @note The returned pointer should not be freed
00513 *

```

```

00514 * @par Example:
00515 * @code{.c}
00516 * rx_error_interface_t* err_if = rx_infrastructure_get_error_interface();
00517 * if (err_if == nullptr) {
00518 *     rx_log_error("APP", "Infrastructure not initialized");
00519 *     return k_rx_err_not_initialized;
00520 * }
00521 *
00522 * // Use error handler for retry logic
00523 * rx_err_t result = err_if->try_recover(err_if->ctx, my_operation, &context);
00524 * @endcode
00525 *
00526 * @see rx_error_interface_t Interface structure
00527 * @see rx_infrastructure_init() Must be called first
00528 *
00529 * @since Version 1.0.0
00530 */
00531 rx_error_interface_t* rx_infrastructure_get_error_interface(void)
00532 {
00533     if (!s_infrastructure_initialized) {
00534         return nullptr;
00535     }
00536     return &s_global_error_interface;
00537 }
00538
00539 /**
00540 * @brief Get global pin validator interface
00541 *
00542 * @details
00543 * Returns a pointer to the global pin validator interface structure.
00544 * The interface provides function pointers for GPIO pin management.
00545 *
00546 * @par Interface Functions:
00547 *
00548 * | Function | Description |
00549 * |-----|-----|
00550 * | claim_pin | Claim a GPIO pin for a specific module |
00551 * | release | Release a previously claimed GPIO pin |
00552 * | is_claimed | Check if a pin is already claimed |
00553 *
00554 *
00555 * @pre rx_infrastructure_init() should have been called
00556 *
00557 * @post Returned pointer valid for program lifetime (if not nullptr)
00558 *
00559 * @note Thread Safety: Safe for read - returns pointer to static data
00560 * @note The returned pointer should not be freed
00561 *
00562 * @warning Pin claim/release operations may not be thread-safe
00563 *
00564 * @par Example:
00565 * @code{.c}
00566 * rx_pin_interface_t* pin_if = rx_infrastructure_get_pin_interface();
00567 * if (pin_if == nullptr) {
00568 *     rx_log_error("MOTOR", "Infrastructure not initialized");
00569 *     return k_rx_err_not_initialized;
00570 * }
00571 *
00572 * // Claim GPIO pin for motor PWM
00573 * rx_err_t err = pin_if->claim_pin(pin_if->ctx,
00574 *                                 k_port_1, k_pin_5,
00575 *                                 "MOTOR_PWM");
00576 * if (err != k_rx_ok) {
00577 *     rx_log_error("MOTOR", "Pin P15 already in use");
00578 *     return err;
00579 * }
00580 * @endcode
00581 *
00582 * @see rx_pin_interface_t Interface structure
00583 * @see rx_infrastructure_init() Must be called first
00584 *
00585 * @since Version 1.0.0
00586 */
00587 rx_pin_interface_t* rx_infrastructure_get_pin_interface(void)
00588 {
00589     if (!s_infrastructure_initialized) {
00590         return nullptr;
00591     }
00592     return &s_global_pin_interface;
00593 }
00594 }

```

7.51 libs/rx_core/src/rx_isr_log.c File Reference

ISR-safe deferred log queue: SPSC ring + drain emitter.

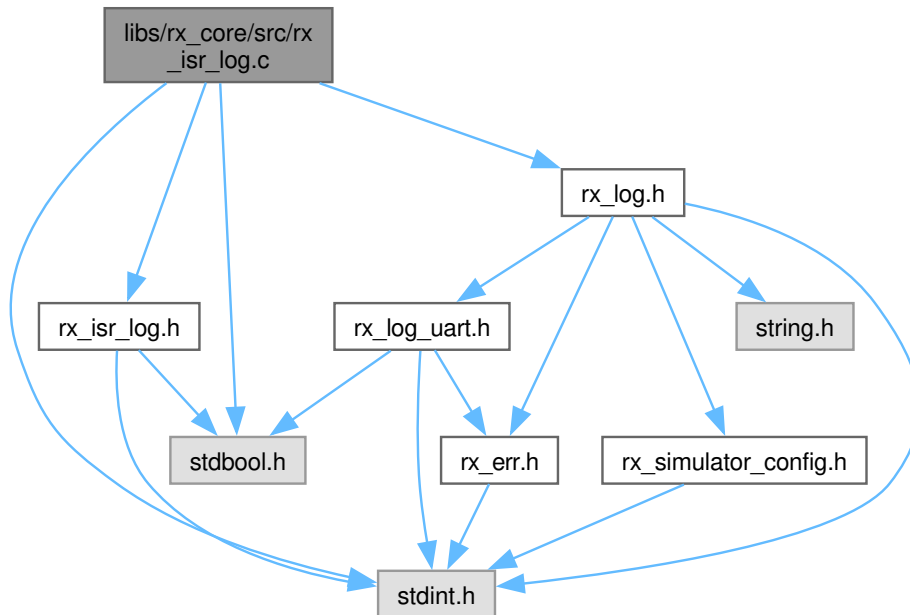
```
#include "rx_isr_log.h"
```

```
#include <stdbool.h>
```

```
#include <stdint.h>
```

```
#include "rx_log.h"
```

Include dependency graph for rx_isr_log.c:



Data Structures

- struct [rx_isr_log_event_meta_t](#)

Enumerations

- enum [rx_isr_log_level_t](#) : `uint8_t` { `k_isr_log_level_debug` = 0 , `k_isr_log_level_info` = 1 , `k_isr_log_level_warn` = 2 , `k_isr_log_level_error` = 3 }

Functions

- bool [rx_isr_log_push](#) ([rx_isr_log_event_id_t](#) event_id, `uint32_t` value)
Push one log event from interrupt context.
- static void [internal_emit_record](#) (const [rx_isr_log_record_t](#) *rec)
Emit one drained record via the appropriate rx_log_* macro.
- `uint32_t` [rx_isr_log_drain](#) (void)
Drain all pending records and emit them via rx_log_*.
- void [rx_isr_log_get_stats](#) ([rx_isr_log_stats_t](#) *stats)
Snapshot diagnostic counters.

Variables

- static volatile [rx_isr_log_record_t](#) [s_ring](#) [[k_rx_isr_log_capacity](#)]
Ring buffer storage.
- static volatile uint32_t [s_head](#)
Producer write cursor (monotonically increases, wraps at uint32).
- static volatile uint32_t [s_tail](#)
Consumer read cursor (monotonically increases, wraps at uint32).
- static volatile uint32_t [s_total_pushed](#)
Diagnostic counters (writes to [dropped_count](#) happen from ISR).
- static volatile uint32_t [s_total_drained](#)
- static volatile uint32_t [s_total_dropped](#)
- static volatile uint32_t [s_high_water](#)
- static uint32_t [s_drop_count_last_seen](#)
drop count observed at the previous drain (lets us emit a delta summary)
- static const [rx_isr_log_event_meta_t](#) [s_isr_log_event_table](#) [[k_isr_log_event_count](#)]

7.51.1 Detailed Description

ISR-safe deferred log queue: SPSC ring + drain emitter.

See [rx_isr_log.h](#) for the high-level rationale. This file implements the ring buffer and the format-string lookup table used by the drain hook.

The ring is sized to [k_rx_isr_log_capacity](#) (32) records and uses the power-of-two trick ([capacity_mask](#)) so head/tail advancement is a cheap AND rather than a modulo. head and tail are 32-bit and wrap freely; the occupancy is computed as (head - tail) and the slot index is (head & mask).

Author

Locked, Inc.

Date

2026-04-25

Copyright

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Since

Version 1.0.0

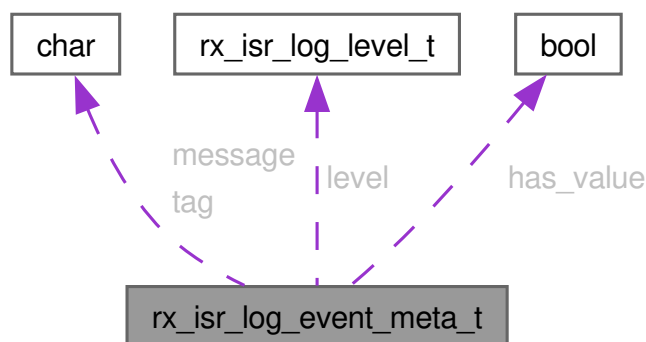
Definition in file [rx_isr_log.c](#).

7.51.2 Data Structure Documentation

7.51.2.1 struct rx_isr_log_event_meta_t

Definition at line 74 of file [rx_isr_log.c](#).

Collaboration diagram for rx_isr_log_event_meta_t:



Data Fields

bool	has_value	
rx_isr_log_level_t	level	
const char *	message	
const char *	tag	

7.51.3 Enumeration Type Documentation

7.51.3.1 rx_isr_log_level_t

enum [rx_isr_log_level_t](#) : uint8_t

Enumerator

k_isr_log_level_debug	
k_isr_log_level_info	
k_isr_log_level_warn	
k_isr_log_level_error	

Definition at line 67 of file [rx_isr_log.c](#).

```

00067      : uint8_t {
00068  k_isr_log_level_debug = 0,
00069  k_isr_log_level_info  = 1,
00070  k_isr_log_level_warn  = 2,
00071  k_isr_log_level_error = 3,
00072 } rx_isr_log_level_t;
  
```

7.51.4 Function Documentation

7.51.4.1 internal_emit_record()

```
void internal_emit_record (
    const rx_isr_log_record_t * rec) [static]
```

Emit one drained record via the appropriate rx_log_* macro.

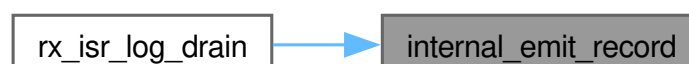
Definition at line 263 of file rx_isr_log.c.

```
00264 {
00265     /* Validate the event id -- a torn record (producer interrupted us
00266      * mid-read) would be caught here as a 0 / out-of-range id and we drop
00267      * it as a soft error rather than dereferencing the table OOB. */
00268     if (rec->event_id >= k_isr_log_event_count) {
00269         return;
00270     }
00271     if (rec->event_id == k_isr_log_event_none) {
00272         return;
00273     }
00274
00275     const rx_isr_log_event_meta_t* meta = &s_isr_log_event_table[rec->event_id];
00276
00277     if (meta->has_value) {
00278         switch (meta->level) {
00279             case k_isr_log_level_error:
00280                 rx_log_error_val(meta->tag, meta->message, rec->value);
00281                 break;
00282             case k_isr_log_level_warn:
00283                 rx_log_warn_val(meta->tag, meta->message, rec->value);
00284                 break;
00285             case k_isr_log_level_info:
00286                 rx_log_info_val(meta->tag, meta->message, rec->value);
00287                 break;
00288             case k_isr_log_level_debug:
00289             default:
00290                 rx_log_debug_val(meta->tag, meta->message, rec->value);
00291                 break;
00292         }
00293     } else {
00294         switch (meta->level) {
00295             case k_isr_log_level_error:
00296                 rx_log_error(meta->tag, meta->message);
00297                 break;
00298             case k_isr_log_level_warn:
00299                 rx_log_warn(meta->tag, meta->message);
00300                 break;
00301             case k_isr_log_level_info:
00302                 rx_log_info(meta->tag, meta->message);
00303                 break;
00304             case k_isr_log_level_debug:
00305             default:
00306                 rx_log_debug(meta->tag, meta->message);
00307                 break;
00308         }
00309     }
00310 }
```

References rx_isr_log_record_t::event_id, rx_isr_log_event_meta_t::has_value, k_isr_log_event_count, k_isr_log_event_none, k_isr_log_level_debug, k_isr_log_level_error, k_isr_log_level_info, k_isr_log_level_warn, rx_isr_log_event_meta_t::level, rx_isr_log_event_meta_t::message, rx_log_debug, rx_log_debug_val, rx_log_error, rx_log_error_val, rx_log_info, rx_log_info_val, rx_log_warn, rx_log_warn_val, s_isr_log_event_table, rx_isr_log_event_meta_t::tag, and rx_isr_log_record_t::value.

Referenced by rx_isr_log_drain().

Here is the caller graph for this function:



7.51.4.2 rx_isr_log_drain()

```
uint32_t rx_isr_log_drain (
    void )
```

Drain all pending records and emit them via rx_log_*.

Called from a task-context periodic hook (currently comm_task at 100 Hz). Walks tail forward to head, emitting one rx_log_* call per record. Also detects and reports newly dropped records via a single summary log.

Returns

Number of records drained on this call

Note

MUST be called from task context only – it invokes the regular rx_log_* macros which acquire a ThreadX mutex.

Definition at line 312 of file rx_isr_log.c.

```
00313 {
00314     uint32_t drained = 0U;
00315
00316     /* Snapshot head once -- new pushes after this point will be picked up by
00317      * the next drain call. Bounding the loop by the snapshot keeps execution
00318      * time per drain bounded (NASA Rule 2) even if the ISR is spamming the
00319      * ring concurrently. */
00320     const uint32_t head_snapshot = s_head;
00321
00322     while (s_tail != head_snapshot) {
00323         const uint32_t slot = s_tail & (uint32_t)k_rx_isr_log_capacity_mask;
00324         const rx_isr_log_record_t rec = {
00325             .event_id = s_ring[slot].event_id,
00326             .reserved0 = s_ring[slot].reserved0,
00327             .reserved1 = s_ring[slot].reserved1,
00328             .value = s_ring[slot].value,
00329         };
00330
00331         internal_emit_record(&rec);
00332
00333         /* Advance tail AFTER reading the record so the producer cannot reuse
00334          * the slot before we are done with it. */
00335         s_tail = s_tail + 1U;
00336         drained++;
00337         s_total_drained++;
00338     }
00339
00340     /* Emit a single summary log if any drops occurred since the previous
00341      * drain. We deliberately use _val so the count surfaces in the message. */
00342     const uint32_t cur_drops = s_total_dropped;
00343     if (cur_drops > s_drop_count_last_seen) {
00344         const uint32_t delta = cur_drops - s_drop_count_last_seen;
00345         rx_log_warn_val("ISR_LOG", "ISR log overflow: dropped records", delta);
00346         s_drop_count_last_seen = cur_drops;
00347     }
00348
00349     return drained;
00350 }
```

References [internal_emit_record\(\)](#), [k_rx_isr_log_capacity_mask](#), [rx_log_warn_val](#), [s_drop_count_last_seen](#), [s_head](#), [s_ring](#), [s_tail](#), [s_total_drained](#), and [s_total_dropped](#).

Here is the call graph for this function:



7.51.4.3 rx_isr_log_get_stats()

```
void rx_isr_log_get_stats (
    rx_isr_log_stats_t * stats)
```

Snapshot diagnostic counters.

Parameters

out	stats	Statistics destination (nullptr is silently ignored)
-----	-------	--

Definition at line 352 of file [rx_isr_log.c](#).

```
00353 {
00354     if (stats == NULL) {
00355         return;
00356     }
00357     stats->total_pushed = s_total_pushed;
00358     stats->total_drained = s_total_drained;
00359     stats->total_dropped = s_total_dropped;
00360     stats->high_water = s_high_water;
00361 }
```

References [rx_isr_log_stats_t::high_water](#), [s_high_water](#), [s_total_drained](#), [s_total_dropped](#), [s_total_pushed](#), [rx_isr_log_stats_t::total_drained](#), [rx_isr_log_stats_t::total_dropped](#), and [rx_isr_log_stats_t::total_pushed](#).

7.51.4.4 rx_isr_log_push()

```
bool rx_isr_log_push (
    rx_isr_log_event_id_t event_id,
    uint32_t value)
```

Push one log event from interrupt context.

Lock-free SPSC enqueue. Reads tail, computes the next head; if the ring is full, increments the drop counter and returns false. Otherwise writes the record at the current head slot, then advances head. Safe to call from any ISR or task; the only producer is the ISR by convention but the implementation is correct for any single-thread producer because the head/tail are written by exactly one side each.

Parameters

in	event_id	One of rx_isr_log_event_id_t . k_isr_log_event_none and values $\geq$ k_isr_log_event_count are silently dropped (counted as drops).
in	value	Auxiliary 32-bit value (0 if the event has no value)

Returns

true if the record was queued, false if dropped

Note

ISR-SAFE: no mutex, no I/O, no allocation, no busy-wait.

Definition at line 210 of file [rx_isr_log.c](#).

```

00211 {
00212     /* Reject sentinel and out-of-range ids -- count as drops so a corrupted
00213      * call site shows up in stats rather than silently mis-emitting later. */
00214     if ((event_id == k_isr_log_event_none) ||
00215         ((uint32_t)event_id >= (uint32_t)k_isr_log_event_count)) {
00216         s_total_dropped++;
00217         return false;
00218     }
00219
00220     /* SPSC bounded ring: the consumer (drain) only advances s_tail, the
00221      * producer (this function) only advances s_head. Read tail snapshot
00222      * once -- it can only move forward between this read and our store, so
00223      * the worst case is we observe one slot fewer free space than reality
00224      * (we drop a record when there is in fact room). That is a benign loss
00225      * compared to corrupting the ring, and the drain hook will catch up
00226      * within one tick. */
00227     const uint32_t head = s_head;
00228     const uint32_t tail = s_tail;
00229     const uint32_t used = head - tail;
00230
00231     if (used >= (uint32_t)k_rx_isr_log_capacity) {
00232         s_total_dropped++;
00233         return false;
00234     }
00235
00236     const uint32_t slot = head & (uint32_t)k_rx_isr_log_capacity_mask;
00237     s_ring[slot].event_id = (uint8_t)event_id;
00238     s_ring[slot].value = value;
00239     /* reserved fields zeroed for layout determinism */
00240     s_ring[slot].reserved0 = 0U;
00241     s_ring[slot].reserved1 = 0U;
00242
00243     /* Publish the record by advancing s_head. The volatile write acts as a
00244      * compiler barrier so the slot writes above are not reordered past it.
00245      * On RX (in-order, single core) this is sufficient -- no DMB needed. */
00246     s_head = head + 1U;
00247
00248     s_total_pushed++;
00249     const uint32_t new_used = (head + 1U) - tail;
00250     if (new_used > s_high_water) {
00251         s_high_water = new_used;
00252     }
00253     return true;
00254 }

```

References [k_isr_log_event_count](#), [k_isr_log_event_none](#), [k_rx_isr_log_capacity](#), [k_rx_isr_log_capacity_mask](#), [s_head](#), [s_high_water](#), [s_ring](#), [s_tail](#), [s_total_dropped](#), and [s_total_pushed](#).

7.51.5 Variable Documentation

7.51.5.1 s_drop_count_last_seen

uint32_t s_drop_count_last_seen [static]

drop count observed at the previous drain (lets us emit a delta summary)

Definition at line 57 of file [rx_isr_log.c](#).

Referenced by [rx_isr_log_drain\(\)](#).

7.51.5.2 s_head

volatile uint32_t s_head [static]

Producer write cursor (monotonically increases, wraps at uint32).

Definition at line 45 of file [rx_isr_log.c](#).

Referenced by [internal_append_byte\(\)](#), [rx_isr_log_drain\(\)](#), and [rx_isr_log_push\(\)](#).

7.51.5.3 s_high_water

volatile uint32_t s_high_water [static]

Definition at line 54 of file [rx_isr_log.c](#).

Referenced by [rx_isr_log_get_stats\(\)](#), and [rx_isr_log_push\(\)](#).

7.51.5.4 s_isr_log_event_table

const [rx_isr_log_event_meta_t](#) s_isr_log_event_table[k_isr_log_event_count] [static]

Definition at line 93 of file [rx_isr_log.c](#).

```

00093                                                                 {
00094 #endif
00095 [k_isr_log_event_none] = {.tag      = "ISR_LOG",
00096                          .message   = "(reserved)",
00097                          .level     = k_isr_log_level_debug,
00098                          .has_value = false},
00099
00100 /* USB ISR events */
00101 [k_isr_log_event_usb_vbus_se0] = {.tag      = "USB_ISR",
00102                                  .message   = "VBUS: SE0 detected",
00103                                  .level     = k_isr_log_level_debug,
00104                                  .has_value = false},
00105 [k_isr_log_event_usb_vbus_fs_j] = {.tag      = "USB_ISR",
00106                                   .message   = "VBUS: FS J-state (connected)",
00107                                   .level     = k_isr_log_level_debug,
00108                                   .has_value = false},
00109 [k_isr_log_event_usb_dvst_powered] = {.tag      = "USB_ISR",
00110                                       .message   = "DVST: Powered state",
00111                                       .level     = k_isr_log_level_debug,
00112                                       .has_value = false},
00113 [k_isr_log_event_usb_dvst_default] = {.tag      = "USB_ISR",
00114                                       .message   = "DVST: Default state (bus reset complete)",
00115                                       .level     = k_isr_log_level_debug,
00116                                       .has_value = false},
00117 [k_isr_log_event_usb_dvst_address] = {.tag      = "USB_ISR",
00118                                       .message   = "DVST: Address state",
00119                                       .level     = k_isr_log_level_debug,
00120                                       .has_value = false},
00121 [k_isr_log_event_usb_dvst_configured] = {.tag      = "USB_ISR",
00122                                           .message   = "DVST: Configured state",
00123                                           .level     = k_isr_log_level_debug,
00124                                           .has_value = false},
00125 [k_isr_log_event_usb_dvst_suspended] = {.tag      = "USB_ISR",
00126                                           .message   = "DVST: Suspended state",
00127                                           .level     = k_isr_log_level_debug,
00128                                           .has_value = false},
00129 [k_isr_log_event_usb_dvst_unknown] = {.tag      = "USB_ISR",
00130                                       .message   = "DVST: Unknown state",
00131                                       .level     = k_isr_log_level_warn,
00132                                       .has_value = false},
00133 [k_isr_log_event_usb_resume] = {.tag      = "USB_ISR",
00134                                 .message   = "Resume detected",
00135                                 .level     = k_isr_log_level_debug,
00136                                 .has_value = false},
00137 [k_isr_log_event_usb_ctr_t_seq_err] = {.tag      = "USB_ISR",
00138                                         .message   = "CTR: sequence error, stalled",
00139                                         .level     = k_isr_log_level_warn,
00140                                         .has_value = false},
00141
00142 /* USB CDC ISR-context events */
00143 [k_isr_log_event_cdc_descriptor_short_write] = {.tag      = "USB_CDC",
00144                                                  .message   = "Descriptor write incomplete",
00145                                                  .level     = k_isr_log_level_error,
00146                                                  .has_value = false},
00147 [k_isr_log_event_cdc_pipe_rolled_back] = {.tag      = "USB_CDC",
00148                                             .message   = "Rolled back pipes on config failure",
00149                                             .level     = k_isr_log_level_debug,
00150                                             .has_value = false},
00151 [k_isr_log_event_cdc_port0_bulk_in_failed] = {.tag      = "USB_CDC",
00152                                                .message   =
00153                                                    "Failed to configure Port 0 Bulk IN pipe",
00154                                                .level     = k_isr_log_level_error,
00155                                                .has_value = false},
00156 [k_isr_log_event_cdc_port0_bulk_out_failed] = {.tag      = "USB_CDC",
00157                                                  .message   =

```

```

00158             "Failed to configure Port 0 Bulk OUT pipe",
00159             .level = k_isr_log_level_error,
00160             .has_value = false},
00161 [k_isr_log_event_cdc_port0_int_in_failed] = {.tag = "USB_CDC",
00162             .message =
00163             "Failed to configure Port 0 Interrupt IN pipe",
00164             .level = k_isr_log_level_error,
00165             .has_value = false},
00166 [k_isr_log_event_cdc_port1_bulk_in_failed] = {.tag = "USB_CDC",
00167             .message =
00168             "Failed to configure Port 1 Bulk IN pipe",
00169             .level = k_isr_log_level_error,
00170             .has_value = false},
00171 [k_isr_log_event_cdc_port1_bulk_out_failed] = {.tag = "USB_CDC",
00172             .message =
00173             "Failed to configure Port 1 Bulk OUT pipe",
00174             .level = k_isr_log_level_error,
00175             .has_value = false},
00176 [k_isr_log_event_cdc_port1_int_in_failed] = {.tag = "USB_CDC",
00177             .message =
00178             "Failed to configure Port 1 Interrupt IN pipe",
00179             .level = k_isr_log_level_error,
00180             .has_value = false},
00181 [k_isr_log_event_cdc_port2_bulk_in_failed] = {.tag = "USB_CDC",
00182             .message =
00183             "Failed to configure Port 2 Bulk IN pipe",
00184             .level = k_isr_log_level_error,
00185             .has_value = false},
00186 [k_isr_log_event_cdc_port2_bulk_out_failed] = {.tag = "USB_CDC",
00187             .message =
00188             "Failed to configure Port 2 Bulk OUT pipe",
00189             .level = k_isr_log_level_error,
00190             .has_value = false},
00191 [k_isr_log_event_cdc_port2_int_in_failed] = {.tag = "USB_CDC",
00192             .message =
00193             "Failed to configure Port 2 Interrupt IN pipe",
00194             .level = k_isr_log_level_error,
00195             .has_value = false},
00196 [k_isr_log_event_cdc_composite_configured] = {.tag = "USB_CDC",
00197             .message = "Composite device configured",
00198             .level = k_isr_log_level_info,
00199             .has_value = false},
00200 [k_isr_log_event_cdc_line_coding_short_write] = {.tag = "USB_CDC",
00201             .message = "Line coding write incomplete",
00202             .level = k_isr_log_level_error,
00203             .has_value = false},
00204 };

```

Referenced by [internal_emit_record\(\)](#).

7.51.5.5 s_ring

volatile [rx_isr_log_record_t](#) s_ring[k_rx_isr_log_capacity] [static]

Ring buffer storage.

Records are written by the producer (ISR) at index (s_head & mask) and read by the consumer (drain) at index (s_tail & mask). Volatile so the compiler does not hoist either side's read across the matching producer or consumer write.

Definition at line 42 of file [rx_isr_log.c](#).

Referenced by [internal_append_byte\(\)](#), [rx_isr_log_drain\(\)](#), [rx_isr_log_push\(\)](#), and [rx_log_uart_drain\(\)](#).

7.51.5.6 s_tail

volatile uint32_t s_tail [static]

Consumer read cursor (monotonically increases, wraps at uint32).

Definition at line 48 of file [rx_isr_log.c](#).

Referenced by [rx_isr_log_drain\(\)](#), [rx_isr_log_push\(\)](#), and [rx_log_uart_drain\(\)](#).

7.51.5.7 s_total_drained

volatile uint32_t s_total_drained [static]

Definition at line 52 of file [rx_isr_log.c](#).

Referenced by [rx_isr_log_drain\(\)](#), and [rx_isr_log_get_stats\(\)](#).

7.51.5.8 s_total_dropped

volatile uint32_t s_total_dropped [static]

Definition at line 53 of file [rx_isr_log.c](#).

Referenced by [rx_isr_log_drain\(\)](#), [rx_isr_log_get_stats\(\)](#), and [rx_isr_log_push\(\)](#).

7.51.5.9 s_total_pushed

volatile uint32_t s_total_pushed [static]

Diagnostic counters (writes to dropped_count happen from ISR).

Definition at line 51 of file [rx_isr_log.c](#).

Referenced by [rx_isr_log_get_stats\(\)](#), and [rx_isr_log_push\(\)](#).

7.52 rx_isr_log.c

[Go to the documentation of this file.](#)

```
00001 /**
00002  * @file rx_isr_log.c
00003  * @brief ISR-safe deferred log queue: SPSC ring + drain emitter
00004  *
00005  * @details
00006  * See rx_isr_log.h for the high-level rationale. This file implements the
00007  * ring buffer and the format-string lookup table used by the drain hook.
00008  *
00009  * The ring is sized to k_rx_isr_log_capacity (32) records and uses the
00010  * power-of-two trick (capacity_mask) so head/tail advancement is a cheap
00011  * AND rather than a modulo. head and tail are 32-bit and wrap freely; the
00012  * occupancy is computed as (head - tail) and the slot index is
00013  * (head & mask).
00014  *
00015  * @author Locked, Inc.
00016  * @date 2026-04-25
00017  * @copyright Copyright (c) 2026 Locked Inc.
00018  * SPDX-License-Identifier: MIT
00019  * @since Version 1.0.0
00020  */
00021
00022 #include "rx_isr_log.h"
00023
00024 #include <stdbool.h>
00025 #include <stdint.h>
00026
00027 #include "rx_log.h"
00028
00029 /**
00030  * Static state
00031  *
00032  */
```

```

00033 /**
00034  * @brief Ring buffer storage
00035  *
00036  * @details
00037  * Records are written by the producer (ISR) at index (s_head & mask) and
00038  * read by the consumer (drain) at index (s_tail & mask). Volatile so the
00039  * compiler does not hoist either side's read across the matching producer
00040  * or consumer write.
00041  */
00042 static volatile rx_isr_log_record_t s_ring[k_rx_isr_log_capacity];
00043
00044 /** @brief Producer write cursor (monotonically increases, wraps at uint32) */
00045 static volatile uint32_t s_head;
00046
00047 /** @brief Consumer read cursor (monotonically increases, wraps at uint32) */
00048 static volatile uint32_t s_tail;
00049
00050 /** @brief Diagnostic counters (writes to dropped_count happen from ISR) */
00051 static volatile uint32_t s_total_pushed;
00052 static volatile uint32_t s_total_drained;
00053 static volatile uint32_t s_total_dropped;
00054 static volatile uint32_t s_high_water;
00055
00056 /** @brief drop count observed at the previous drain (lets us emit a delta summary) */
00057 static uint32_t s_drop_count_last_seen;
00058
00059 /*
=====
00060 * Format-string table
00061 *
=====
00062 * One row per rx_isr_log_event_id_t value (skipping k_isr_log_event_none).
00063 * Drain looks up by event_id and emits a rx_log_* call with the right level
00064 * and tag. has_value controls whether the auxiliary value field is logged.
00065 */
00066
00067 typedef enum : uint8_t {
00068     k_isr_log_level_debug = 0,
00069     k_isr_log_level_info  = 1,
00070     k_isr_log_level_warn  = 2,
00071     k_isr_log_level_error = 3,
00072 } rx_isr_log_level_t;
00073
00074 typedef struct {
00075     const char* tag;
00076     const char* message;
00077     rx_isr_log_level_t level;
00078     bool has_value;
00079 } rx_isr_log_event_meta_t;
00080
00081 /* Event metadata table. Indexed by event_id. The k_isr_log_event_none entry
00082  * is unused (drained records are validated against the count sentinel).
00083  *
00084  * Under UNIT_TEST the table is non-const so individual tests can flip the
00085  * has_value flag and level for one entry to exercise the drain hook's
00086  * has_value=true switch (which is otherwise unreachable because every
00087  * production entry sets has_value=false). The runtime drain code reads
00088  * through the same identifier either way. */
00089 #ifdef UNIT_TEST
00090 static rx_isr_log_event_meta_t s_isr_log_event_table[k_isr_log_event_count];
00091 static const rx_isr_log_event_meta_t s_isr_log_event_table_defaults[k_isr_log_event_count] = {
00092 #else
00093 static const rx_isr_log_event_meta_t s_isr_log_event_table[k_isr_log_event_count] = {
00094 #endif
00095     [k_isr_log_event_none] = {.tag = "ISR_LOG",
00096                             .message = "(reserved)",
00097                             .level = k_isr_log_level_debug,
00098                             .has_value = false},
00099
00100     /* USB ISR events */
00101     [k_isr_log_event_usb_vbus_se0] = {.tag = "USB_ISR",
00102                                     .message = "VBUS: SE0 detected",
00103                                     .level = k_isr_log_level_debug,
00104                                     .has_value = false},
00105     [k_isr_log_event_usb_vbus_fs_j] = {.tag = "USB_ISR",
00106                                       .message = "VBUS: FS J-state (connected)",
00107                                       .level = k_isr_log_level_debug,
00108                                       .has_value = false},
00109     [k_isr_log_event_usb_dvst_powered] = {.tag = "USB_ISR",
00110                                           .message = "DVST: Powered state",
00111                                           .level = k_isr_log_level_debug,
00112                                           .has_value = false},
00113     [k_isr_log_event_usb_dvst_default] = {.tag = "USB_ISR",
00114                                           .message = "DVST: Default state (bus reset complete)",
00115                                           .level = k_isr_log_level_debug,
00116                                           .has_value = false},
00117     [k_isr_log_event_usb_dvst_address] = {.tag = "USB_ISR",

```

```

00118             .message = "DVST: Address state",
00119             .level   = k_isr_log_level_debug,
00120             .has_value = false},
00121 [k_isr_log_event_usb_dvst_configured] = {.tag   = "USB_ISR",
00122             .message = "DVST: Configured state",
00123             .level   = k_isr_log_level_debug,
00124             .has_value = false},
00125 [k_isr_log_event_usb_dvst_suspended] = {.tag   = "USB_ISR",
00126             .message = "DVST: Suspended state",
00127             .level   = k_isr_log_level_debug,
00128             .has_value = false},
00129 [k_isr_log_event_usb_dvst_unknown] = {.tag   = "USB_ISR",
00130             .message = "DVST: Unknown state",
00131             .level   = k_isr_log_level_warn,
00132             .has_value = false},
00133 [k_isr_log_event_usb_resume] = {.tag   = "USB_ISR",
00134             .message = "Resume detected",
00135             .level   = k_isr_log_level_debug,
00136             .has_value = false},
00137 [k_isr_log_event_usb_ctr1_seq_err] = {.tag   = "USB_ISR",
00138             .message = "CTR1: sequence error, stalled",
00139             .level   = k_isr_log_level_warn,
00140             .has_value = false},
00141
00142 /* USB CDC ISR-context events */
00143 [k_isr_log_event_cdc_descriptor_short_write] = {.tag   = "USB_CDC",
00144             .message = "Descriptor write incomplete",
00145             .level   = k_isr_log_level_error,
00146             .has_value = false},
00147 [k_isr_log_event_cdc_pipe_rolled_back] = {.tag   = "USB_CDC",
00148             .message = "Rolled back pipes on config failure",
00149             .level   = k_isr_log_level_debug,
00150             .has_value = false},
00151 [k_isr_log_event_cdc_port0_bulk_in_failed] = {.tag = "USB_CDC",
00152             .message =
00153                 "Failed to configure Port 0 Bulk IN pipe",
00154             .level   = k_isr_log_level_error,
00155             .has_value = false},
00156 [k_isr_log_event_cdc_port0_bulk_out_failed] = {.tag = "USB_CDC",
00157             .message =
00158                 "Failed to configure Port 0 Bulk OUT pipe",
00159             .level   = k_isr_log_level_error,
00160             .has_value = false},
00161 [k_isr_log_event_cdc_port0_int_in_failed] = {.tag = "USB_CDC",
00162             .message =
00163                 "Failed to configure Port 0 Interrupt IN pipe",
00164             .level   = k_isr_log_level_error,
00165             .has_value = false},
00166 [k_isr_log_event_cdc_port1_bulk_in_failed] = {.tag = "USB_CDC",
00167             .message =
00168                 "Failed to configure Port 1 Bulk IN pipe",
00169             .level   = k_isr_log_level_error,
00170             .has_value = false},
00171 [k_isr_log_event_cdc_port1_bulk_out_failed] = {.tag = "USB_CDC",
00172             .message =
00173                 "Failed to configure Port 1 Bulk OUT pipe",
00174             .level   = k_isr_log_level_error,
00175             .has_value = false},
00176 [k_isr_log_event_cdc_port1_int_in_failed] = {.tag = "USB_CDC",
00177             .message =
00178                 "Failed to configure Port 1 Interrupt IN pipe",
00179             .level   = k_isr_log_level_error,
00180             .has_value = false},
00181 [k_isr_log_event_cdc_port2_bulk_in_failed] = {.tag = "USB_CDC",
00182             .message =
00183                 "Failed to configure Port 2 Bulk IN pipe",
00184             .level   = k_isr_log_level_error,
00185             .has_value = false},
00186 [k_isr_log_event_cdc_port2_bulk_out_failed] = {.tag = "USB_CDC",
00187             .message =
00188                 "Failed to configure Port 2 Bulk OUT pipe",
00189             .level   = k_isr_log_level_error,
00190             .has_value = false},
00191 [k_isr_log_event_cdc_port2_int_in_failed] = {.tag = "USB_CDC",
00192             .message =
00193                 "Failed to configure Port 2 Interrupt IN pipe",
00194             .level   = k_isr_log_level_error,
00195             .has_value = false},
00196 [k_isr_log_event_cdc_composite_configured] = {.tag   = "USB_CDC",
00197             .message = "Composite device configured",
00198             .level   = k_isr_log_level_info,
00199             .has_value = false},
00200 [k_isr_log_event_cdc_line_coding_short_write] = {.tag   = "USB_CDC",
00201             .message = "Line coding write incomplete",
00202             .level   = k_isr_log_level_error,
00203             .has_value = false},
00204 };

```

```

00205
00206 /*
=====
00207 * Producer (ISR-safe)
00208 *
=====
*/
00209
00210 bool rx_isr_log_push(rx_isr_log_event_id_t event_id, uint32_t value)
00211 {
00212     /* Reject sentinel and out-of-range ids -- count as drops so a corrupted
00213      * call site shows up in stats rather than silently mis-emitting later. */
00214     if ((event_id == k_isr_log_event_none) ||
00215         ((uint32_t)event_id >= (uint32_t)k_isr_log_event_count)) {
00216         s_total_dropped++;
00217         return false;
00218     }
00219
00220     /* SPSC bounded ring: the consumer (drain) only advances s_tail, the
00221      * producer (this function) only advances s_head. Read tail snapshot
00222      * once -- it can only move forward between this read and our store, so
00223      * the worst case is we observe one slot fewer free space than reality
00224      * (we drop a record when there is in fact room). That is a benign loss
00225      * compared to corrupting the ring, and the drain hook will catch up
00226      * within one tick. */
00227     const uint32_t head = s_head;
00228     const uint32_t tail = s_tail;
00229     const uint32_t used = head - tail;
00230
00231     if (used >= (uint32_t)k_rx_isr_log_capacity) {
00232         s_total_dropped++;
00233         return false;
00234     }
00235
00236     const uint32_t slot = head & (uint32_t)k_rx_isr_log_capacity_mask;
00237     s_ring[slot].event_id = (uint8_t)event_id;
00238     s_ring[slot].value = value;
00239     /* reserved fields zeroed for layout determinism */
00240     s_ring[slot].reserved0 = 0U;
00241     s_ring[slot].reserved1 = 0U;
00242
00243     /* Publish the record by advancing s_head. The volatile write acts as a
00244      * compiler barrier so the slot writes above are not reordered past it.
00245      * On RX (in-order, single core) this is sufficient -- no DMB needed. */
00246     s_head = head + 1U;
00247
00248     s_total_pushed++;
00249     const uint32_t new_used = (head + 1U) - tail;
00250     if (new_used > s_high_water) {
00251         s_high_water = new_used;
00252     }
00253     return true;
00254 }
00255
00256 /*
=====
00257 * Consumer (task-context drain)
00258 *
=====
*/
00259
00260 /**
00261  * @brief Emit one drained record via the appropriate rx_log_* macro
00262  */
00263 static void internal_emit_record(const rx_isr_log_record_t* rec)
00264 {
00265     /* Validate the event id -- a torn record (producer interrupted us
00266      * mid-read) would be caught here as a 0 / out-of-range id and we drop
00267      * it as a soft error rather than dereferencing the table OOB. */
00268     if (rec->event_id >= k_isr_log_event_count) {
00269         return;
00270     }
00271     if (rec->event_id == k_isr_log_event_none) {
00272         return;
00273     }
00274
00275     const rx_isr_log_event_meta_t* meta = &s_isr_log_event_table[rec->event_id];
00276
00277     if (meta->has_value) {
00278         switch (meta->level) {
00279             case k_isr_log_level_error:
00280                 rx_log_error_val(meta->tag, meta->message, rec->value);
00281                 break;
00282             case k_isr_log_level_warn:
00283                 rx_log_warn_val(meta->tag, meta->message, rec->value);
00284                 break;
00285             case k_isr_log_level_info:

```

```

00286     rx_log_info_val(meta->tag, meta->message, rec->value);
00287     break;
00288 case k_isr_log_level_debug:
00289 default:
00290     rx_log_debug_val(meta->tag, meta->message, rec->value);
00291     break;
00292 }
00293 } else {
00294     switch (meta->level) {
00295     case k_isr_log_level_error:
00296         rx_log_error(meta->tag, meta->message);
00297         break;
00298     case k_isr_log_level_warn:
00299         rx_log_warn(meta->tag, meta->message);
00300         break;
00301     case k_isr_log_level_info:
00302         rx_log_info(meta->tag, meta->message);
00303         break;
00304     case k_isr_log_level_debug:
00305     default:
00306         rx_log_debug(meta->tag, meta->message);
00307         break;
00308     }
00309 }
00310 }
00311
00312 uint32_t rx_isr_log_drain(void)
00313 {
00314     uint32_t drained = 0U;
00315
00316     /* Snapshot head once -- new pushes after this point will be picked up by
00317      * the next drain call. Bounding the loop by the snapshot keeps execution
00318      * time per drain bounded (NASA Rule 2) even if the ISR is spamming the
00319      * ring concurrently. */
00320     const uint32_t head_snapshot = s_head;
00321
00322     while (s_tail != head_snapshot) {
00323         const uint32_t slot = s_tail & (uint32_t)k_rx_isr_log_capacity_mask;
00324         const rx_isr_log_record_t rec = {
00325             .event_id = s_ring[slot].event_id,
00326             .reserved0 = s_ring[slot].reserved0,
00327             .reserved1 = s_ring[slot].reserved1,
00328             .value = s_ring[slot].value,
00329         };
00330
00331         internal_emit_record(&rec);
00332
00333         /* Advance tail AFTER reading the record so the producer cannot reuse
00334          * the slot before we are done with it. */
00335         s_tail = s_tail + 1U;
00336         drained++;
00337         s_total_drained++;
00338     }
00339
00340     /* Emit a single summary log if any drops occurred since the previous
00341      * drain. We deliberately use _val so the count surfaces in the message. */
00342     const uint32_t cur_drops = s_total_dropped;
00343     if (cur_drops > s_drop_count_last_seen) {
00344         const uint32_t delta = cur_drops - s_drop_count_last_seen;
00345         rx_log_warn_val("ISR_LOG", "ISR log overflow: dropped records", delta);
00346         s_drop_count_last_seen = cur_drops;
00347     }
00348
00349     return drained;
00350 }
00351
00352 void rx_isr_log_get_stats(rx_isr_log_stats_t* stats)
00353 {
00354     if (stats == NULL) {
00355         return;
00356     }
00357     stats->total_pushed = s_total_pushed;
00358     stats->total_drained = s_total_drained;
00359     stats->total_dropped = s_total_dropped;
00360     stats->high_water = s_high_water;
00361 }
00362
00363 #ifdef UNIT_TEST
00364 void rx_isr_log_test_restore_event_table(void)
00365 {
00366     for (uint32_t i = 0U; i < (uint32_t)k_isr_log_event_count; i++) {
00367         s_isr_log_event_table[i] = s_isr_log_event_table_defaults[i];
00368     }
00369 }
00370
00371 void rx_isr_log_test_reset_state(void)
00372 {

```



```

00373 s_head          = 0U;
00374 s_tail          = 0U;
00375 s_total_pushed  = 0U;
00376 s_total_drained = 0U;
00377 s_total_dropped = 0U;
00378 s_high_water    = 0U;
00379 s_drop_count_last_seen = 0U;
00380 for (uint32_t i = 0U; i < (uint32_t)k_rx_isr_log_capacity; i++) {
00381     s_ring[i].event_id = 0U;
00382     s_ring[i].reserved0 = 0U;
00383     s_ring[i].reserved1 = 0U;
00384     s_ring[i].value     = 0U;
00385 }
00386 rx_isr_log_test_restore_event_table();
00387 }
00388
00389 uint32_t rx_isr_log_test_pending_count(void)
00390 {
00391     return s_head - s_tail;
00392 }
00393
00394 bool rx_isr_log_test_inject_raw_record(uint8_t raw_event_id, uint32_t value)
00395 {
00396     const uint32_t head = s_head;
00397     const uint32_t tail = s_tail;
00398     const uint32_t used = head - tail;
00399     if (used >= (uint32_t)k_rx_isr_log_capacity) {
00400         return false;
00401     }
00402     const uint32_t slot = head & (uint32_t)k_rx_isr_log_capacity_mask;
00403     s_ring[slot].event_id = raw_event_id;
00404     s_ring[slot].value     = value;
00405     s_ring[slot].reserved0 = 0U;
00406     s_ring[slot].reserved1 = 0U;
00407     s_head                = head + 1U;
00408     return true;
00409 }
00410
00411 void rx_isr_log_test_set_event_meta(rx_isr_log_event_id_t event_id, uint8_t level, bool has_value)
00412 {
00413     if ((uint32_t)event_id >= (uint32_t)k_isr_log_event_count) {
00414         return;
00415     }
00416     s_isr_log_event_table[event_id].level = (rx_isr_log_level_t)level;
00417     s_isr_log_event_table[event_id].has_value = has_value;
00418 }
00419 #endif

```

7.53 libs/rx_core/src/rx_iwdt.c File Reference

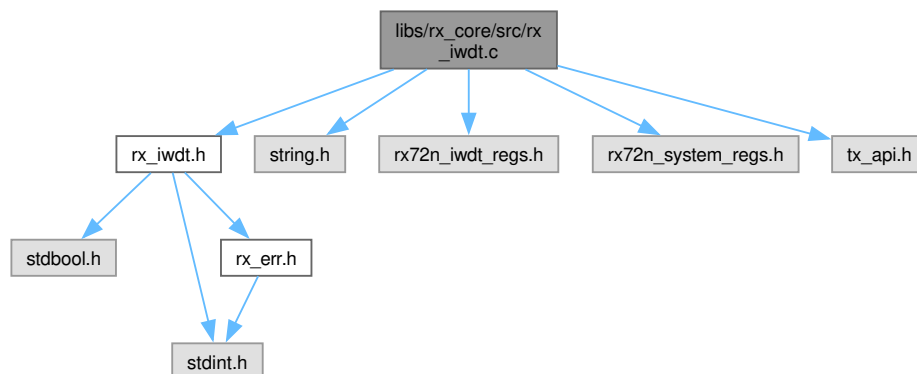
Independent Watchdog Timer (IWDT) Implementation for RX72N.

```

#include "rx_iwdt.h"
#include <string.h>
#include "rx72n_iwdt_regs.h"
#include "rx72n_system_regs.h"
#include "tx_api.h"

```

Include dependency graph for rx_iwdt.c:



Data Structures

- struct `rx_iwdt_state_t`
Internal state structure for IWDT driver.

Functions

- static void `internal_safe_strcpy` (char *dst, const char *src, size_t max_len)
Copy a string with bounded length (replaces strncpy).
- static uint32_t `internal_get_tick_count` (void)
Get current system tick count from ThreadX.
- static `rx_iwdt_task_info_t` * `internal_find_task` (const char *task_name)
Find registered task by name.
- static `rx_iwdt_task_info_t` * `internal_find_free_slot` (void)
Find free slot in task array.
- static void `internal_init_default_config` (`rx_iwdt_config_t` *config)
Initialize default configuration structure.
- `rx_err_t` `rx_iwdt_init` (const `rx_iwdt_config_t` *config)
Initialize IWDT driver with configuration and task monitoring.
- `rx_err_t` `rx_iwdt_feed` (void)
Feed the independent watchdog timer to prevent system reset.
- `rx_err_t` `rx_iwdt_register_task` (const char *task_name, uint32_t timeout_ms)
Register a ThreadX task for software watchdog monitoring.
- `rx_err_t` `rx_iwdt_task_heartbeat` (const char *task_name)
Send a heartbeat signal for a registered task to reset its timeout.
- `rx_err_t` `rx_iwdt_set_timeout_for_state` (const `system_state_t` state, const uint32_t timeout_ms)
Set the software task monitoring timeout for a specific system state.
- `rx_err_t` `rx_iwdt_set_state` (const `system_state_t` state)
Set the current system state for software task monitoring.
- `rx_err_t` `rx_iwdt_get_status` (`rx_iwdt_status_t` *status)
Get a snapshot of the current IWDT module status.
- bool `rx_iwdt_was_reset` (void)
Check whether the last microcontroller reset was caused by an IWDT timeout.
- `rx_err_t` `rx_iwdt_check_tasks` (void)
Check all registered tasks for software heartbeat timeout.

Variables

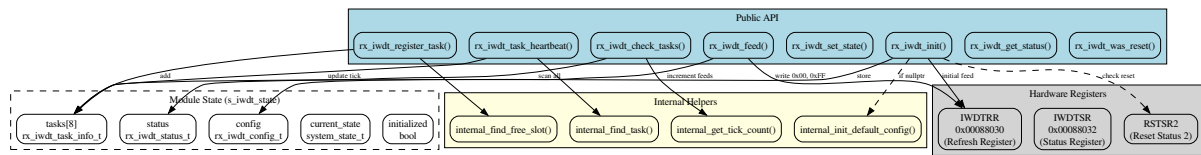
- static `rx_iwdt_state_t` `s_iwdt_state` = {}
Global IWDT driver state.

7.53.1 Detailed Description

Independent Watchdog Timer (IWDT) Implementation for RX72N.

Implements the hardware-independent watchdog timer driver with enhanced task monitoring capabilities. The IWDT uses a dedicated 120 kHz clock source, providing reliable watchdog functionality even during main clock failures - a critical safety feature for robotics applications.

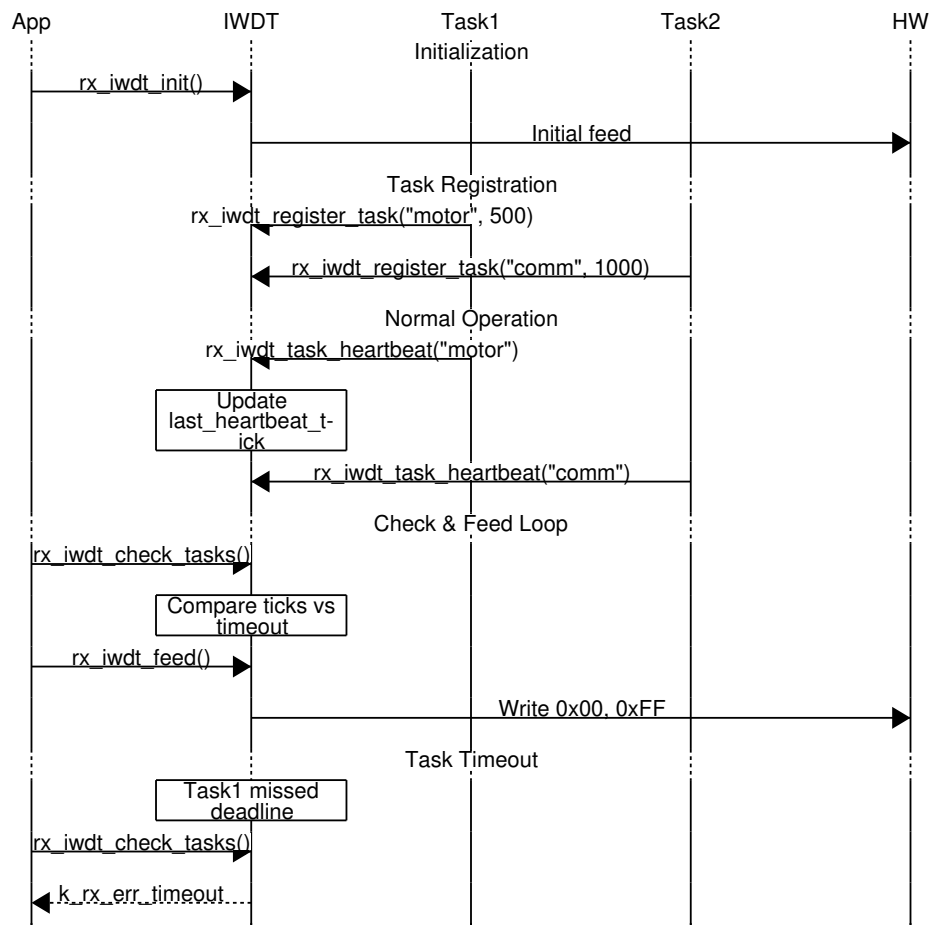
7.53.1.1 Implementation Architecture



7.53.1.2 IWDT vs WDT Comparison

Feature	IWDT (This Module)	WDT (rx_wdt.c)
Clock Source	IWDTCLK (120 kHz independent)	PCLKB (system clock)
Can be Stopped	[X] No (hardware safety)	[OK] Yes
Clock Fault Detection	[OK] Yes (independent clock)	[X] No
Timeout Range	~8 ms to ~17 seconds	~4 us to ~273 us
Register Base	0x00088030	0x00088020
Task Monitoring	[OK] Software layer	[X] Not implemented
Primary Use	Production safety	Development/debugging

7.53.1.3 Task Monitoring System



7.53.1.4 Memory Layout

Offset	Size	Field	Description
0x000	68	config	Configuration (timeouts, flags)
0x044	320	tasks[8]	Task info array (40 bytes each)
0x184	76	status	Status info
0x1D0	1	current_state	System state enum
0x1D1	1	initialized	Init flag
0x1D2	2	(padding)	Alignment
Total	**~470 bytes**		Static allocation

7.53.1.5 Performance Characteristics

Function	Execution Time	Stack	Notes
rx_iwdt_init()	~30 us	64 B	Memset + config copy
rx_iwdt_feed()	~1 us	8 B	2 register writes
rx_iwdt_register_task()	~10 us	32 B	String copy + slot search
rx_iwdt_task_heartbeat()	~5 us	16 B	Task lookup + tick update
rx_iwdt_check_tasks()	~20 us	24 B	Scan all 8 tasks

Module Dependencies:

- [rx_iwdt.h](#) - Public API declarations
- [rx72n_iwdt_regs.h](#) - IWDT registers
- [rx72n_system_regs.h](#) - System registers
- [rx_check.h](#) - Validation macros
- `<tx_api.h>` - ThreadX API for tick count
- `<string.h>` - `strlen`, `strcmp`

NASA Power of 10 Compliance:

- Rule 1: [OK] No goto, setjmp, recursion
- Rule 2: [OK] All loops bounded by `k_iwdt_max_tasks`
- Rule 3: [OK] Zero dynamic allocation (static state)
- Rule 4: [OK] All functions <60 lines
- Rule 5: [OK] Minimum 2 checks per function
- Rule 6: [OK] Data declared at smallest scope
- Rule 7: [OK] All return values checked or propagated
- Rule 8: [OK] C23 typed enums for constants
- Rule 9: [OK] Minimal pointer use, no pointer arithmetic
- Rule 10: [OK] Compiled with `-Wall -Wextra -Werror`

SOLID Principles:

- Single Responsibility: Only IWDT and task monitoring
- Open/Closed: Extensible via configuration and state callbacks
- Liskov Substitution: Consistent [rx_err_t](#) return interface
- Interface Segregation: Separate init/feed/task/status APIs
- Dependency Inversion: Depends on abstract [rx_err_t](#)

See also

[rx_iwdt.h](#) Public API documentation and usage examples

[rx_wdt.c](#) Software-controllable watchdog (complementary)

RX72N User's Manual Section 11: Independent Watchdog Timer

Author

Locked, Inc.

Date

2026-01-28

Version

1.0.0

Copyright

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Since

Version 1.0.0

Definition in file [rx_iwdt.c](#).

7.53.2 Data Structure Documentation

7.53.2.1 struct rx_iwdt_state_t

Internal state structure for IWDT driver.

Aggregates all state for the IWDT driver including configuration, task monitoring arrays, status tracking, and initialization flags. This is the largest static allocation in the core library.

Memory Layout:

Offset	Size	Field	Description
0x000	~68	config	Configuration structure
0x044	320	tasks	8 task info structures (40 B each)
0x184	~76	status	Status and counters
0x1D0	1	current_state	System state enum
0x1D1	1	initialized	Init complete flag
Total	**~470 bytes**		Static allocation

State Transitions:

- initialized: false -> true on [rx_iwdt_init\(\)](#) success
- current_state: k_system_state_init -> other states via [rx_iwdt_set_state\(\)](#)
- tasks[n].active: false -> true via [rx_iwdt_register_task\(\)](#)

Invariant

initialized implies config is valid

status.active_tasks equals count of tasks[].active==true

See also

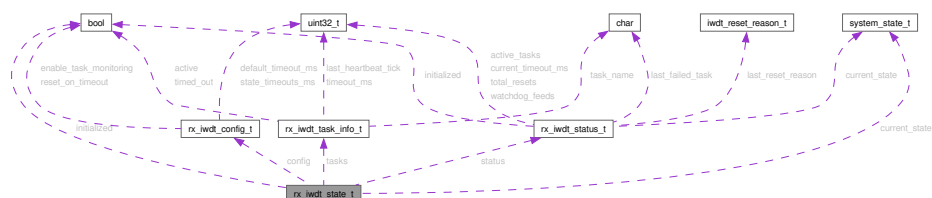
[s_iwdt_state](#) Static instance

Since

Version 1.0.0

Definition at line 234 of file [rx_iwdt.c](#).

Collaboration diagram for rx_iwdt_state_t:



Data Fields

rx_iwdt_config_t	config	Configuration from rx_iwdt_init() . Contains default timeout, task monitoring enable flag, reset mode, and per-state timeout array.
system_state_t	current_state	Current system state for state-dependent timeout selection. One of system_state_t .

<code>bool</code>	<code>initialized</code>	Module initialization flag. <code>true</code> = ready for use, <code>false</code> = must call <code>rx_iwdt_init()</code> .
<code>rx_iwdt_status_t</code>	<code>status</code>	Runtime status: feed count, reset count, current state, active task count, last failed task name.
<code>rx_iwdt_task_info_t</code>	<code>tasks[k_iwdt_max_tasks]</code>	Array of registered task info. Each entry contains task name, timeout, last heartbeat tick, and timeout flag. Max 8 tasks.

7.53.3 Function Documentation

7.53.3.1 `internal_find_free_slot()`

```
rx_iwdt_task_info_t * internal_find_free_slot (
    void ) [static]
```

Find free slot in task array.

Searches for the first inactive slot in the task array where a new task can be registered.

Algorithm:

1. Iterate through `tasks[0..k_iwdt_max_tasks-1]`
2. For each entry: check if NOT active
3. Return pointer to first inactive slot, or nullptr if all full

Note

Linear search $O(n)$ where $n = k_iwdt_max_tasks$ (8)

Returns first available slot (lowest index)

See also

`rx_iwdt_register_task()` Uses this to allocate slot
`k_iwdt_max_tasks` Maximum number of tasks (8)

Since

Version 1.0.0

Definition at line 411 of file `rx_iwdt.c`.

```
00412 {
00413     for (uint32_t i = 0; i < k_iwdt_max_tasks; i++) {
00414         if (!s_iwdt_state.tasks[i].active) {
00415             return &s_iwdt_state.tasks[i];
00416         }
00417     }
00418     return nullptr;
00419 }
00420 }
```

References `k_iwdt_max_tasks`, and `s_iwdt_state`.

Referenced by `rx_iwdt_register_task()`.

Here is the caller graph for this function:



7.53.3.2 `internal_find_task()`

```
rx_iwdt_task_info_t * internal_find_task (
    const char * task_name) [static]
```

Find registered task by name.

Searches the task array for a task with matching name that is currently active. Uses `strcmp()` for exact string comparison.

Algorithm:

1. Iterate through tasks[0..k_iwdt_max_tasks-1]
2. For each entry: check if active AND name matches
3. Return pointer to matching entry, or nullptr if not found

Precondition

`task_name` must be non-NULL (caller validates)

Note

Linear search $O(n)$ where $n = k_iwdt_max_tasks$ (8)

String comparison is case-sensitive

See also

[rx_iwdt_register_task\(\)](#) Adds tasks to array

[rx_iwdt_task_heartbeat\(\)](#) Uses this to find task

Since

Version 1.0.0

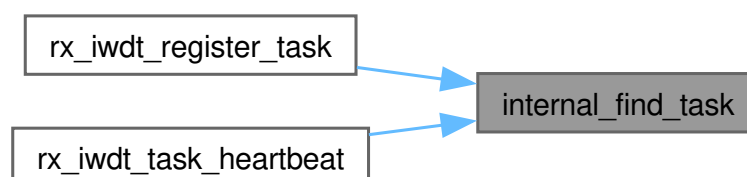
Definition at line 378 of file [rx_iwdt.c](#).

```
00379 {
00380     for (uint32_t i = 0; i < k_iwdt_max_tasks; i++) {
00381         if ((int)s_iwdt_state.tasks[i].active &&
00382             strcmp(s_iwdt_state.tasks[i].task_name, task_name) == 0) {
00383             return &s_iwdt_state.tasks[i];
00384         }
00385     }
00386     return nullptr;
00387 }
00388 }
```

References [k_iwdt_max_tasks](#), and [s_iwdt_state](#).

Referenced by [rx_iwdt_register_task\(\)](#), and [rx_iwdt_task_heartbeat\(\)](#).

Here is the caller graph for this function:



7.53.3.3 internal_get_tick_count()

```
uint32_t internal_get_tick_count (  
    void ) [static]
```

Get current system tick count from ThreadX.

Wrapper around ThreadX `tx_time_get()` to get the current tick count. Used for task heartbeat timing calculations.

Algorithm:

1. Call `tx_time_get()` to get ThreadX tick count
2. Cast `ULONG` to `uint32_t` for consistent type
3. Return tick count

Note

ThreadX tick rate defined by `TX_TIMER_TICKS_PER_SECOND`

Tick count wraps at `UINT32_MAX` (~497 days at 100 Hz tick rate)

Thread-safe - `tx_time_get()` is reentrant

See also

`tx_time_get()` ThreadX API

`TX_TIMER_TICKS_PER_SECOND` Tick rate configuration

Since

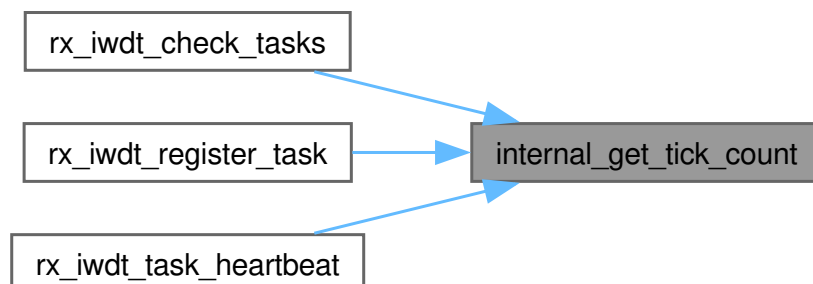
Version 1.0.0

Definition at line 349 of file [rx_iwdt.c](#).

```
00350 {  
00351     return (uint32_t)tx_time_get();  
00352 }
```

Referenced by [rx_iwdt_check_tasks\(\)](#), [rx_iwdt_register_task\(\)](#), and [rx_iwdt_task_heartbeat\(\)](#).

Here is the caller graph for this function:



7.53.3.4 `internal_init_default_config()`

```
void internal_init_default_config (  
    rx\_iwdt\_config\_t * config)  [static]
```

Initialize default configuration structure.

Populates a configuration structure with safe default values suitable for most applications. Called when [rx_iwdt_init\(\)](#) receives nullptr config.

Default Values:

- `default_timeout_ms` = `k_iwdt_default_timeout_ms` (1000 ms)
- `enable_task_monitoring` = true
- `reset_on_timeout` = true
- All `state_timeouts_ms[]` = `k_iwdt_default_timeout_ms`

Algorithm:

1. Set `default_timeout_ms` to 1000 ms
2. Enable task monitoring
3. Enable reset on timeout
4. Initialize all state timeouts to default value

Precondition

`config` must be non-NULL (caller validates)

Postcondition

`config` contains default values ready for use

Note

Loop bounded by `k_system_state_count` (NASA Rule 2)

See also

[rx_iwdt_init\(\)](#) Calls this when config is nullptr

[k_iwdt_default_timeout_ms](#) Default timeout constant

[k_system_state_count](#) Number of system states

Since

Version 1.0.0

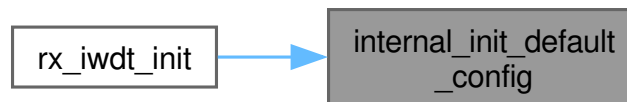
Definition at line 454 of file [rx_iwdt.c](#).

```
00455 {
00456     config->default_timeout_ms    = k_iwdt_default_timeout_ms;
00457     config->enable_task_monitoring = true;
00458     config->reset_on_timeout      = true;
00459
00460     /* Initialize all state timeouts to default */
00461     for (uint32_t i = 0; i < k_system_state_count; i++) {
00462         config->state_timeouts_ms[i] = k_iwdt_default_timeout_ms;
00463     }
00464 }
```

References [rx_iwdt_config_t::default_timeout_ms](#), [rx_iwdt_config_t::enable_task_monitoring](#), [k_iwdt_default_timeout_ms](#), [k_system_state_count](#), [rx_iwdt_config_t::reset_on_timeout](#), and [rx_iwdt_config_t::state_timeouts_ms](#).

Referenced by [rx_iwdt_init\(\)](#).

Here is the caller graph for this function:



7.53.3.5 `internal_safe_strcpy()`

```
void internal_safe_strcpy (
    char * dst,
    const char * src,
    size_t max_len) [static]
```

Copy a string with bounded length (replaces `strncpy`).

Copies up to `max_len-1` characters from `src` to `dst` and null-terminates. Avoids clang-tidy `insecureAPI.DeprecatedOrUnsafeBufferHandling` warnings that `strncpy` triggers.

Precondition

```
dst != nullptr
src != nullptr
max_len > 0
```

Postcondition

```
dst is null-terminated
```

Note

Not thread-safe

Since

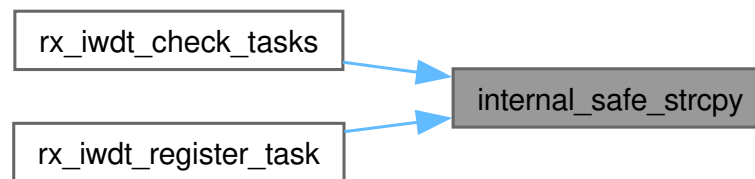
Version 1.0.0

Definition at line 318 of file [rx_iwdt.c](#).

```
00319 {
00320     size_t i = 0;
00321     for (; i < max_len - 1U && src[i] != '\0'; i++) {
00322         dst[i] = src[i];
00323     }
00324     dst[i] = '\0';
00325 }
```

Referenced by [rx_iwdt_check_tasks\(\)](#), and [rx_iwdt_register_task\(\)](#).

Here is the caller graph for this function:



7.53.3.6 `rx_iwdt_check_tasks()`

```
rx\_err\_t rx_iwdt_check_tasks (
    void ) [nodiscard]
```

Check all registered tasks for software heartbeat timeout.

Check for task timeouts.

Iterates over all active task slots and compares the elapsed tick count since each task's last heartbeat against that task's configured timeout (converted from milliseconds to ThreadX ticks). If any task has exceeded its timeout, the task's `timed_out` flag is set, the task name is recorded in `status.last_failed_task`, and the function returns `k_rx_err_timeout`.

If task monitoring is disabled (`config.enable_task_monitoring == false`), returns `k_rx_ok` immediately without checking any tasks.

The caller (typically the watchdog task) should conditionally feed the hardware IWDT only if this function returns `k_rx_ok` – i.e., only refresh the hardware watchdog when all software tasks are alive.

Algorithm steps:

1. Verify `s_iwdt_state.initialized`
2. If `!config.enable_task_monitoring`, return `k_rx_ok` early
3. Capture `current_tick = internal\_get\_tick\_count\(\)`
4. For each active task slot (`i < k_iwdt_max_tasks`):
 - a. Skip inactive slots
 - b. Compute `elapsed_ticks = current_tick - task->last_heartbeat_tick`
 - c. Compute `timeout_in_ticks` from `task->timeout_ms` and `TX_TIMER_TICKS_PER_SECOND`
 - d. If `elapsed_ticks > timeout_in_ticks`: set `timed_out=true`, record task name, set `any_timeout=true`
5. Return `k_rx_err_timeout` if `any_timeout`, else `k_rx_ok`

Precondition

`rx_iwdt_init()` must have been called successfully

Registered tasks must be calling `rx_iwdt_task_heartbeat()` on schedule

Postcondition

Timed-out tasks have their `timed_out` flag set

`status.last_failed_task` contains the name of the most recently detected failure

Note

Not thread-safe; task heartbeat fields may be updated concurrently

Warning

Do NOT feed the hardware IWDT (`rx_iwdt_feed()`) if this returns `k_rx_err_timeout`; allow the hardware watchdog to fire and reset the system

Example:

```
// Watchdog task main loop
while (1) {
    if (rx_iwdt_check_tasks() == k_rx_ok) {
        rx_iwdt_feed(); // Only feed if all tasks are alive
    } else {
        rx_log_error("wdt", "Task timeout detected -- allowing IWDT reset");
        // Do NOT feed the watchdog -- let hardware reset occur
    }
    tx_thread_sleep(k_watchdog_check_interval_ticks);
}
```

See also

`rx_iwdt_feed()` Feed hardware watchdog (call only when this returns `k_rx_ok`)

`rx_iwdt_task_heartbeat()` Update per-task heartbeat from each monitored task

`rx_iwdt_register_task()` Register a task for monitoring

Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 5: 2 preconditions (initialized, tasks registered), 2 postconditions

Definition at line 1145 of file `rx_iwdt.c`.

```
01146 {
01147     if (!s_iwdt_state.initialized) {
01148         return k_rx_err_not_initialized;
01149     }
01150
01151     if (!s_iwdt_state.config.enable_task_monitoring) {
01152         return k_rx_ok;
01153     }
01154
01155     const uint32_t current_tick = internal_get_tick_count();
```

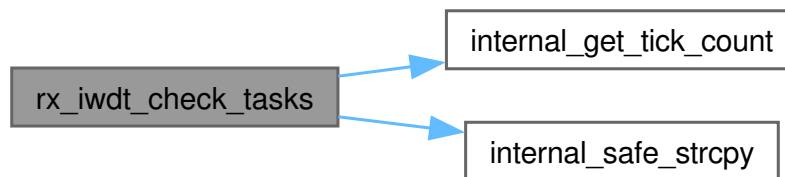
```

01156 bool          any_timeout = false;
01157
01158 /* Check each registered task */
01159 for (uint32_t i = 0; i < k_iwdt_max_tasks; i++) {
01160     if (!s_iwdt_state.tasks[i].active) {
01161         continue;
01162     }
01163
01164     /* Calculate elapsed time */
01165     const uint32_t elapsed_ticks = current_tick - s_iwdt_state.tasks[i].last_heartbeat_tick;
01166     const uint32_t timeout_in_ticks =
01167         (s_iwdt_state.tasks[i].timeout_ms * TX_TIMER_TICKS_PER_SECOND) / k_ms_per_second;
01168
01169     /* Check for timeout */
01170     if (elapsed_ticks > timeout_in_ticks) {
01171         s_iwdt_state.tasks[i].timed_out = true;
01172         any_timeout = true;
01173
01174         /* Record failed task */
01175         internal_safe_strcpy(s_iwdt_state.status.last_failed_task,
01176                             s_iwdt_state.tasks[i].task_name,
01177                             k_iwdt_task_name_len);
01178     }
01179 }
01180
01181 return (int)any_timeout ? k_rx_err_timeout : k_rx_ok;
01182 }

```

References [internal_get_tick_count\(\)](#), [internal_safe_strcpy\(\)](#), [k_iwdt_max_tasks](#), [k_iwdt_task_name_len](#), [k_ms_per_second](#), [k_rx_err_not_initialized](#), [k_rx_err_timeout](#), [k_rx_ok](#), and [s_iwdt_state](#).

Here is the call graph for this function:



7.53.3.7 rx_iwdt_feed()

```

rx_err_t rx_iwdt_feed (
    void ) [nodiscard]

```

Feed the independent watchdog timer to prevent system reset.

Feed the watchdog timer.

Writes the two-byte refresh sequence (0x00 then 0xFF) to the IWDTR Refresh Register (IWDTRR) to reset the hardware watchdog counter. This must be called periodically within the configured hardware timeout window or the IWDTR will reset the microcontroller.

Per the RX72N hardware manual the refresh sequence is strictly ordered: writing 0x00 to IWDTRR starts the window; writing 0xFF completes the refresh. The entire sequence must occur within the permitted refresh window.

Also increments the software diagnostic counter `s_iwdt_state.status.watchdog_feeds` for telemetry and debugging purposes.

Algorithm steps:

1. Check `s_iwdt_state.initialized`; return `k_rx_err_not_initialized` if false
2. Write `k_iwdt_refresh_start` (0x00) to `regs->iwdtrr`
3. Write `k_iwdt_refresh_end` (0xFF) to `regs->iwdtrr`
4. Increment `status.watchdog_feeds` counter

Precondition

[rx_iwdt_init\(\)](#) must have been called successfully

Must be called within the hardware IWDT refresh window (no gap > configured timeout)

Postcondition

Hardware IWDT counter is reset; timeout window restarts

`s_iwdt_state.status.watchdog_feeds` is incremented by one

Note

Not thread-safe; concurrent feeds from multiple tasks are benign for the hardware but may produce incorrect feed counts

Warning

Failing to call within the hardware timeout triggers a system reset

Performance:

Execution time: <1 us @ 240 MHz; safe to call from any task or ISR context

Example:

```
// In main watchdog task loop
while (1) {
    rx_err_t err = rx_iwdt_feed();
    if (err != k_rx_ok) {
        rx_log_error("wdt", "IWDT not initialized");
    }
    tx_thread_sleep(k_watchdog_feed_interval_ticks);
}
```

See also

[rx_iwdt_init\(\)](#) Initialize IWDT before feeding

[rx_iwdt_check_tasks\(\)](#) Verify all tasks are alive before feeding

Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 5: 1 precondition (initialized check), 2 postconditions (HW reset, counter++)

Definition at line 624 of file [rx_iwdt.c](#).

```
00625 {
00626     volatile rx_iwdt_regs_t* regs;
00627
00628     /* Check initialization */
00629     if (!s_iwdt_state.initialized) {
00630         return k_rx_err_not_initialized;
00631     }
00632
00633     /* Refresh watchdog counter - write 0x00 then 0xFF to IWDTRR */
00634     regs = iwdt();
00635     regs->iwdtrr = k_iwdt_refresh_start;
00636     regs->iwdtrr = k_iwdt_refresh_end;
00637
00638     s_iwdt_state.status.watchdog_feeds++;
00639
00640     return k_rx_ok;
00641 }
```

References [k_rx_err_not_initialized](#), [k_rx_ok](#), and [s_iwdt_state](#).

7.53.3.8 rx_iwdt_get_status()

```
rx_err_t rx_iwdt_get_status (
    rx_iwdt_status_t * status) [nodiscard]
```

Get a snapshot of the current IWDT module status.

Get watchdog status.

Copies the internal `s_iwdt_state.status` structure into the caller-provided buffer via `memcpy`. The status snapshot includes: active task count, total feed count, current system state, current timeout threshold, and the name of the last task that timed out (if any).

This is a diagnostic/telemetry function intended for logging and health monitoring. The returned data is a point-in-time snapshot and may become stale if the detection task or other threads update IWDT state concurrently.

Algorithm steps:

1. Validate status pointer is non-null
2. Verify `s_iwdt_state.initialized`
3. `memcpy(&s_iwdt_state.status, status, sizeof(rx_iwdt_status_t))`

Precondition

`rx_iwdt_init()` must have been called successfully
 status must be a valid non-null pointer to an `rx_iwdt_status_t`

Postcondition

*status contains a copy of `s_iwdt_state.status` at the time of the call
 Internal IWDT state is unchanged

Note

Not thread-safe; status fields may be updated concurrently by the task monitor
 Snapshot accuracy depends on timing; use as diagnostic data, not safety-critical logic

Example:

```
rx_iwdt_status_t status;
rx_err_t err = rx_iwdt_get_status(&status);
if (err == k_rx_ok) {
    rx_log_info("wdt", "Active tasks: %u, Feeds: %u",
        (unsigned)status.active_tasks,
        (unsigned)status.watchdog_feeds);
}
```

See also

`rx_iwdt_was_reset()` Check if last reset was hardware IWDT timeout
`rx_iwdt_check_tasks()` Check all tasks for software timeout

Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 5: 2 preconditions (null check, initialized check), 2 postconditions

Definition at line 1003 of file [rx_iwdt.c](#).

```

01004 {
01005     /* Validate inputs */
01006     if (status == nullptr) {
01007         return k_rx_err_null_ptr;
01008     }
01009     if (!s_iwdt_state.initialized) {
01010         return k_rx_err_not_initialized;
01011     }
01012 }
01013
01014 /* Copy status */
01015 {
01016     const uint8_t* src = (const uint8_t*)&s_iwdt_state.status;
01017     uint8_t* dst = (uint8_t*)status;
01018     for (size_t byte_i = 0; byte_i < sizeof(rx_iwdt_status_t); byte_i++) {
01019         dst[byte_i] = src[byte_i];
01020     }
01021 }
01022
01023 return k_rx_ok;
01024 }
```

References [k_rx_err_not_initialized](#), [k_rx_err_null_ptr](#), [k_rx_ok](#), and [s_iwdt_state](#).

7.53.3.9 rx_iwdt_init()

```

rx_err_t rx_iwdt_init (
    const rx_iwdt_config_t * config) [nodiscard]
```

Initialize IWDT driver with configuration and task monitoring.

Initialize the Independent Watchdog Timer.

Initializes the IWDT driver, validates configuration, checks for previous watchdog reset, and performs initial hardware feed. If config is nullptr, uses safe default values (1000ms timeout, task monitoring enabled).

Algorithm:

1. Check if already initialized (return error if so)
2. If config is nullptr, create default configuration
3. Validate timeout within [k_iwdt_min_timeout_ms, k_iwdt_max_timeout_ms]
4. Clear and initialize all state
5. Check RSTSR2 for previous watchdog reset
6. Perform initial feed to IWDTRR (0x00 then 0xFF)
7. Set initialized flag

OFS Configuration Note:

The IWDT hardware timeout is configured at flash-time via OFS (Option Function Select) registers and cannot be changed at runtime. The timeout values in this driver are used for software task monitoring only. The hardware IWDT will timeout based on OFS settings regardless of software configuration.

Precondition

IWDT not already initialized

Postcondition

IWDT driver ready for use
Initial feed performed
Reset status captured

Note

Hardware timeout set by OFS, not this driver
nullptr config uses safe defaults (1000ms, monitoring enabled)

Since

Version 1.0.0

Definition at line 508 of file [rx_iwdt.c](#).

```

00509 {
00510     /* Check if already initialized */
00511     if (s_iwdt_state.initialized) {
00512         return k_rx_err_invalid_state;
00513     }
00514
00515     /* Use default config if none provided */
00516     rx_iwdt_config_t local_config;
00517     if (config == nullptr) {
00518         internal_init_default_config(&local_config);
00519         config = &local_config;
00520     }
00521
00522     /* Validate configuration */
00523     if (config->default_timeout_ms < k_iwdt_min_timeout_ms ||
00524         config->default_timeout_ms > k_iwdt_max_timeout_ms) {
00525         return k_rx_err_invalid_arg;
00526     }
00527
00528     /* Initialize state */
00529     s_iwdt_state = (rx_iwdt_state_t){};
00530     {
00531         const uint8_t* src = (const uint8_t*)config;
00532         uint8_t* dst = (uint8_t*)&s_iwdt_state.config;
00533         for (size_t byte_i = 0; byte_i < sizeof(rx_iwdt_config_t); byte_i++) {
00534             dst[byte_i] = src[byte_i];
00535         }
00536     }
00537     s_iwdt_state.current_state = k_system_state_init;
00538
00539     /* Initialize status */
00540     s_iwdt_state.status.initialized = true;
00541     s_iwdt_state.status.current_state = k_system_state_init;
00542     s_iwdt_state.status.current_timeout_ms = config->default_timeout_ms;
00543     s_iwdt_state.status.active_tasks = 0;
00544
00545     /* Check if last reset was watchdog */
00546     if (rx_iwdt_was_reset()) {
00547         s_iwdt_state.status.total_resets++;
00548         s_iwdt_state.status.last_reset_reason = k_iwdt_reset_watchdog;

```

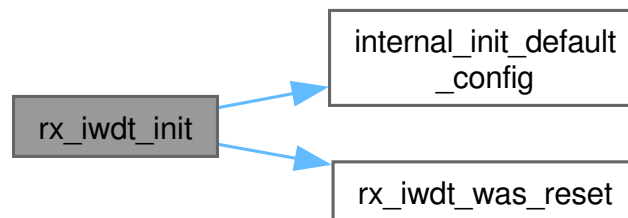
```

00549 } else {
00550     s_iwdt_state.status.last_reset_reason = k_iwdt_reset_power_on;
00551 }
00552
00553 /* Note: IWDT hardware is configured via OFS (Option Function Select) registers
00554  * at compile/flash time and starts automatically on reset. We cannot reconfigure
00555  * the timeout period at runtime. This driver manages the software state and
00556  * provides the feed mechanism. The requested timeout is used for software task
00557  * monitoring only. */
00558 volatile rx_iwdt_regs_t* const regs = iwdt();
00559
00560 /* Feed the watchdog to establish baseline */
00561 regs->iwdtrr = k_iwdt_refresh_start;
00562 regs->iwdtrr = k_iwdt_refresh_end;
00563
00564 s_iwdt_state.initialized = true;
00565
00566 return k_rx_ok;
00567 }

```

References [rx_iwdt_config_t::default_timeout_ms](#), [internal_init_default_config\(\)](#), [k_iwdt_max_timeout_ms](#), [k_iwdt_min_timeout_ms](#), [k_iwdt_reset_power_on](#), [k_iwdt_reset_watchdog](#), [k_rx_err_invalid_arg](#), [k_rx_err_invalid_state](#), [k_rx_ok](#), [k_system_state_init](#), [rx_iwdt_was_reset\(\)](#), and [s_iwdt_state](#).

Here is the call graph for this function:



7.53.3.10 rx_iwdt_register_task()

```

rx_err_t rx_iwdt_register_task (
    const char * task_name,
    uint32_t timeout_ms) [nodiscard]

```

Register a ThreadX task for software watchdog monitoring.

Register a task for heartbeat monitoring.

Adds a task entry to the IWDT software monitoring table, associating the task name with a per-task timeout. Once registered, the task must call [rx_iwdt_task_heartbeat\(\)](#) at least once per `timeout_ms` milliseconds or [rx_iwdt_check_tasks\(\)](#) will report `k_rx_err_timeout`.

The registration also records the current tick count as the initial `last_heartbeat_tick`, giving the task a full `timeout_ms` window before the first heartbeat is required.

Algorithm steps:

1. Validate `task_name` is non-null
2. Verify `s_iwdt_state.initialized`
3. Validate `timeout_ms` is in `[k_iwdt_min_timeout_ms, k_iwdt_max_timeout_ms]`
4. Check task name length is in `(0, k_iwdt_task_name_len)`
5. Check task is not already registered via [internal_find_task\(\)](#)
6. Find a free slot via [internal_find_free_slot\(\)](#)
7. Initialize slot with task name, timeout, current tick, `active=true`, `timed_out=false`
8. Increment `status.active_tasks`

Precondition

`rx_iwdt_init()` must have been called successfully
`task_name` must be unique among registered tasks

Postcondition

Task entry is active in `s_iwdt_state.tasks[]`
`status.active_tasks` is incremented by one

Note

Not thread-safe; serialize registrations from multiple contexts

Example:

```
rx_err_t err = rx_iwdt_register_task("motor_ctrl", 500);
if (err != k_rx_ok) {
    rx_log_error("wdt", "Failed to register motor_ctrl task");
}
```

See also

`rx_iwdt_task_heartbeat()` Update heartbeat for a registered task
`rx_iwdt_check_tasks()` Check all registered tasks for timeout

Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 5: 5 preconditions (null, initialized, timeout, name, duplicate), 2 postconditions

Definition at line 691 of file `rx_iwdt.c`.

```
00692 {
00693     /* Validate inputs */
00694     if (task_name == nullptr) {
00695         return k_rx_err_null_ptr;
00696     }
00697
00698     if (!s_iwdt_state.initialized) {
00699         return k_rx_err_not_initialized;
00700     }
00701
00702     if (timeout_ms < k_iwdt_min_timeout_ms || timeout_ms > k_iwdt_max_timeout_ms) {
00703         return k_rx_err_invalid_arg;
00704     }
00705
00706     /* Check task name length */
00707     const uint32_t name_len = (uint32_t)strlen(task_name);
00708     if (name_len == 0 || name_len >= k_iwdt_task_name_len) {
00709         return k_rx_err_invalid_arg;
00710     }
00711
00712     /* Check if task already registered */
00713     if (internal_find_task(task_name) != nullptr) {
00714         return k_rx_err_invalid_arg;
00715     }
00716
00717     /* Find free slot */
00718     rx_iwdt_task_info_t* const slot = internal_find_free_slot();
```

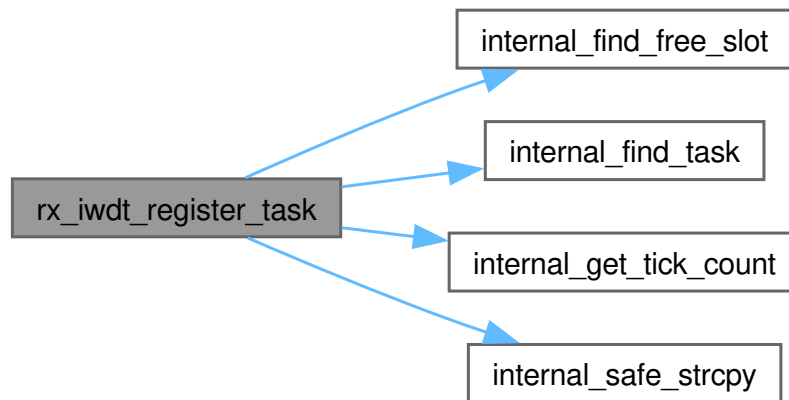
```

00719 if (slot == nullptr) {
00720     return k_rx_err_no_mem;
00721 }
00722
00723 /* Register task */
00724 *slot = (rx_iwdt_task_info_t){};
00725 internal_safe_strcpy(slot->task_name, task_name, k_iwdt_task_name_len);
00726 slot->timeout_ms = timeout_ms;
00727 slot->last_heartbeat_tick = internal_get_tick_count();
00728 slot->active = true;
00729 slot->timed_out = false;
00730
00731 s_iwdt_state.status.active_tasks++;
00732
00733 return k_rx_ok;
00734 }

```

References [rx_iwdt_task_info_t::active](#), [internal_find_free_slot\(\)](#), [internal_find_task\(\)](#), [internal_get_tick_count\(\)](#), [internal_safe_strcpy\(\)](#), [k_iwdt_max_timeout_ms](#), [k_iwdt_task_name_len](#), [k_rx_err_invalid_arg](#), [k_rx_err_no_mem](#), [k_rx_err_not_initialized](#), [k_rx_err_null_ptr](#), [k_rx_ok](#), [rx_iwdt_task_info_t::last_heartbeat_tick](#), [s_iwdt_state](#), [rx_iwdt_task_info_t::task_name](#), [rx_iwdt_task_info_t::timed_out](#), and [rx_iwdt_task_info_t::timeout_ms](#).

Here is the call graph for this function:



7.53.3.11 rx_iwdt_set_state()

```

rx_err_t rx_iwdt_set_state (
    const system_state_t state) [nodiscard]

```

Set the current system state for software task monitoring.

Set current system state.

Updates the active system state used by [rx_iwdt_check_tasks\(\)](#) to select per-state task heartbeat timeout windows. Transitions between states (e.g. initializing -> running) change which timeout threshold is applied when evaluating task heartbeat intervals.

This function updates three tracking fields atomically:

- `s_iwdt_state.current_state`
- `s_iwdt_state.status.current_state`
- `s_iwdt_state.status.current_timeout_ms` (set from `config.state_timeouts_ms[state]`)

Important: This does NOT modify the hardware IWDT peripheral. The hardware watchdog timeout is a fixed constant set in OFS registers at compile time. Only the software task monitoring timeout threshold changes.

Algorithm steps:

1. Validate state is less than `k_system_state_count`
2. Verify `s_iwdt_state.initialized`
3. Update `current_state` and `status.current_state = state`
4. Update `status.current_timeout_ms = config.state_timeouts_ms[state]`

Precondition

`rx_iwdt_init()` must have been called successfully
state must be a valid `system_state_t` value less than `k_system_state_count`

Postcondition

`s_iwdt_state.current_state` reflects the new state
`status.current_timeout_ms` reflects the configured timeout for the new state

Note

Not thread-safe; serialize state transitions from multiple contexts

Warning

Hardware IWDT is unaffected; only software task monitoring changes

Example:

```
// Transition from initialization to running state
rx_err_t err = rx_iwdt_set_state(k_system_state_running);
if (err != k_rx_ok) {
    rx_log_error("wdt", "Failed to set IWDT system state");
}
```

See also

`rx_iwdt_set_timeout_for_state()` Configure per-state timeouts
`rx_iwdt_get_status()` Query current state and timeout

Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 5: 2 preconditions (initialized, valid state), 2 postconditions

Definition at line 933 of file [rx_iwdt.c](#).

```

00934 {
00935     /* Validate inputs */
00936     if (state >= k_system_state_count) {
00937         return k_rx_err_invalid_arg;
00938     }
00939
00940     if (!s_iwdt_state.initialized) {
00941         return k_rx_err_not_initialized;
00942     }
00943
00944     /* Update software state tracking.
00945      * NOTE: This does not change the hardware watchdog timeout, as IWDT timeout
00946      * is configured via OFS registers at compile/flash time and cannot be changed
00947      * at runtime. The state-dependent timeouts are used for software task monitoring
00948      * via rx_iwdt_check_tasks(). The hardware IWDT timeout remains constant. */
00949     s_iwdt_state.current_state = state;
00950     s_iwdt_state.status.current_state = state;
00951     s_iwdt_state.status.current_timeout_ms = s_iwdt_state.config.state_timeouts_ms[state];
00952
00953     return k_rx_ok;
00954 }
```

References [k_rx_err_invalid_arg](#), [k_rx_err_not_initialized](#), [k_rx_ok](#), [k_system_state_count](#), and [s_iwdt_state](#).

7.53.3.12 rx_iwdt_set_timeout_for_state()

```

rx\_err\_t rx_iwdt_set_timeout_for_state (
    const system\_state\_t state,
    const uint32\_t timeout_ms)    [nodiscard]
```

Set the software task monitoring timeout for a specific system state.

Set timeout for system state.

Configures the state-dependent software timeout used by [rx_iwdt_check_tasks\(\)](#) when the system is in the specified state. This allows different task heartbeat windows depending on the operational mode (e.g. longer timeouts during initialization, tighter timeouts during normal operation).

Note: This does NOT change the hardware IWDT timeout. The hardware watchdog timeout is fixed at compile/flash time via OFS registers and cannot be altered at runtime. Only software task monitoring timeouts are affected.

If state matches the currently active state ([s_iwdt_state.current_state](#)), the active timeout ([status.current_timeout_ms](#)) is also updated immediately.

Algorithm steps:

1. Validate state is less than [k_system_state_count](#)
2. Verify [s_iwdt_state.initialized](#)
3. Validate [timeout_ms](#) is in [[k_iwdt_min_timeout_ms](#), [k_iwdt_max_timeout_ms](#)]
4. Set [config.state_timeouts_ms\[state\]](#) = [timeout_ms](#)
5. If [state == current_state](#), update [status.current_timeout_ms](#)

Precondition

`rx_iwdt_init()` must have been called successfully
 state must be a valid `system_state_t` value less than `k_system_state_count`

Postcondition

`config.state_timeouts_ms[state]` equals `timeout_ms`
 If state is current, `status.current_timeout_ms` is updated immediately

Note

Not thread-safe; serialize state configuration changes

Warning

Hardware IWDT timeout is unaffected; only software task monitoring changes

Example:

```
// Allow longer task timeouts during initialization phase
rx_iwdt_set_timeout_for_state(k_system_state_initializing, 5000);
rx_iwdt_set_timeout_for_state(k_system_state_running, 500);
```

See also

`rx_iwdt_set_state()` Change the active system state
`rx_iwdt_check_tasks()` Uses current state timeout to evaluate task heartbeats

Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 5: 3 preconditions (initialized, valid state, valid timeout), 2 postconditions

Definition at line 856 of file `rx_iwdt.c`.

```
00857 {
00858     /* Validate inputs */
00859     if (state >= k_system_state_count) {
00860         return k_rx_err_invalid_arg;
00861     }
00862
00863     if (!s_iwdt_state.initialized) {
00864         return k_rx_err_not_initialized;
00865     }
00866
00867     if (timeout_ms < k_iwdt_min_timeout_ms || timeout_ms > k_iwdt_max_timeout_ms) {
00868         return k_rx_err_invalid_arg;
00869     }
00870
00871     /* Update state timeout */
00872     s_iwdt_state.config.state_timeouts_ms[state] = timeout_ms;
00873
00874     /* If this is the current state, update active timeout */
00875     if (state == s_iwdt_state.current_state) {
00876         s_iwdt_state.status.current_timeout_ms = timeout_ms;
00877     }
00878
00879     return k_rx_ok;
00880 }
```

References `k_iwdt_max_timeout_ms`, `k_rx_err_invalid_arg`, `k_rx_err_not_initialized`, `k_rx_ok`, `k_system_state_count`, and `s_iwdt_state`.

7.53.3.13 rx_iwdt_task_heartbeat()

```
rx_err_t rx_iwdt_task_heartbeat (
    const char * task_name) [nodiscard]
```

Send a heartbeat signal for a registered task to reset its timeout.

Report task heartbeat.

Updates the last_heartbeat_tick for the named task to the current ThreadX tick count, resetting the timeout window. Also clears the timed_out flag if the task had previously been marked as timed out.

Tasks must call this function at least once per their registered timeout_ms window (configured in rx_iwdt_register_task()) to remain in good standing. If the interval between heartbeats exceeds timeout_ms, the next call to rx_iwdt_check_tasks() will return k_rx_err_timeout.

Algorithm steps:

1. Validate task_name is non-null
2. Verify s_iwdt_state.initialized
3. Find the task entry via internal_find_task()
4. Set task->last_heartbeat_tick = internal_get_tick_count()
5. Clear task->timed_out

Precondition

rx_iwdt_init() must have been called successfully
Task must have been registered via rx_iwdt_register_task()

Postcondition

task->last_heartbeat_tick is refreshed to the current tick count
task->timed_out is false

Note

Not thread-safe; concurrent heartbeats from different tasks are safe as they write to distinct slots, but concurrent writes to the same slot are not

Example:

```
// At top of motor control task loop
while (1) {
    rx_iwdt_task_heartbeat("motor_ctrl");
    motor_control_update();
    tx_thread_sleep(k_motor_ctrl_period_ticks);
}
```

See also

rx_iwdt_register_task() Register a task before sending heartbeats
rx_iwdt_check_tasks() Verify all tasks are within their timeout windows

Since

Version 1.0.0

NASA Power of 10 Compliance:

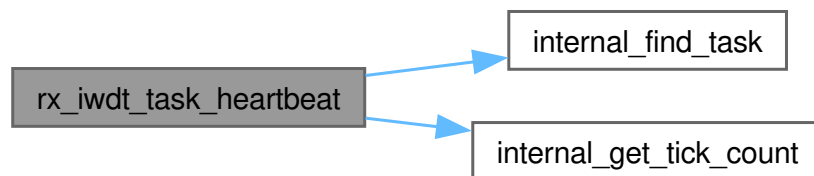
- Rule 5: 3 preconditions (null, initialized, registered), 2 postconditions

Definition at line 784 of file [rx_iwdt.c](#).

```
00785 {
00786     /* Validate inputs */
00787     if (task_name == nullptr) {
00788         return k_rx_err_null_ptr;
00789     }
00790
00791     if (!s_iwdt_state.initialized) {
00792         return k_rx_err_not_initialized;
00793     }
00794
00795     /* Find task */
00796     rx_iwdt_task_info_t* const task = internal_find_task(task_name);
00797     if (task == nullptr) {
00798         return k_rx_err_not_found;
00799     }
00800
00801     /* Update heartbeat */
00802     task->last_heartbeat_tick = internal_get_tick_count();
00803     task->timed_out = false;
00804
00805     return k_rx_ok;
00806 }
```

References [internal_find_task\(\)](#), [internal_get_tick_count\(\)](#), [k_rx_err_not_found](#), [k_rx_err_not_initialized](#), [k_rx_err_null_ptr](#), [k_rx_ok](#), [rx_iwdt_task_info_t::last_heartbeat_tick](#), [s_iwdt_state](#), and [rx_iwdt_task_info_t::timed_out](#).

Here is the call graph for this function:



7.53.3.14 rx_iwdt_was_reset()

```
bool rx_iwdt_was_reset (
    void ) [nodiscard]
```

Check whether the last microcontroller reset was caused by an IWDT timeout.

Check if last reset was caused by watchdog.

Reads the IWDT Reset Flag (IWDTRF) from the Reset Status Register 2 (RSTSR2) of the RX72N. This register is located at a separate address from RSTSR0/1 and retains the reset cause flags across a reset event (values are cleared on power-on reset only).

The IWDTRF bit (bit position `k_rstsr2_iwdrf`) is set by hardware when the IWDT counter underflows (i.e., the watchdog was not refreshed in time). It remains set until explicitly cleared or a power-on reset occurs.

This function is used during startup to detect and log prior watchdog resets, which may indicate a firmware hang or task scheduling failure.

Algorithm steps:

1. Read the RSTSR2 register byte via `rstsr2()` inline accessor
2. Mask with `k_rstsr2_iwdrf` to extract the IWDt reset flag bit
3. Return true if the bit is set, false otherwise

Precondition

None – safe to call before `rx_iwdt_init()` (reads hardware register directly)
 Must be called before the application clears the reset status register

Postcondition

Hardware RSTSR2 register is read-only; this function does not modify it
 Return value reflects hardware register state at the instant of the call

Note

Thread-safe – reads a single hardware register without shared state
 Typically called once at startup to log the reset cause

Warning

RSTSR2.IWDTRF remains set across warm resets; clear it in startup if needed

Example:

```
if (rx_iwdt_was_reset()) {
    rx_log_error("boot", "Prior reset caused by IWDt timeout - check task health");
}
rx_iwdt_init(&iwdt_config);
```

See also

`rx_iwdt_feed()` Feed the hardware watchdog to prevent reset
`rx_iwdt_check_tasks()` Monitor software task heartbeats

Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 5: 2 preconditions (safe before init, called before register clear), 2 postconditions

Definition at line 1073 of file `rx_iwdt.c`.

```
01074 {
01075     /* Read reset status register 2 for IWDt reset flag
01076      * Note: RSTSR2 is at a separate address from RSTSR0/1 */
01077     volatile uint8_t* const rstsr2_reg = rstsr2();
01078     const uint8_t      status      = *rstsr2_reg;
01079
01080     /* Check IWDt reset flag in RSTSR2 */
01081     return (bool)((status & k_rstsr2_iwdrf) != 0);
01082 }
```

Referenced by `rx_iwdt_init()`.

Here is the caller graph for this function:



7.53.4 Variable Documentation

7.53.4.1 s_iwdt_state

[rx_iwdt_state_t](#) s_iwdt_state = {} [static]

Global IWDT driver state.

Single instance of IWDT state, zero-initialized at startup. All fields start at 0/false/empty representing uninitialized state.

Memory Location: .bss section (zero-initialized static data) Size: ~470 bytes Alignment: 4-byte aligned

Access Pattern:

- Write: [rx_iwdt_init\(\)](#), [rx_iwdt_register_task\(\)](#), [rx_iwdt_task_heartbeat\(\)](#), [rx_iwdt_set_state\(\)](#), [rx_iwdt_set_timeout_for_state\(\)](#), [rx_iwdt_feed\(\)](#), [rx_iwdt_check_tasks\(\)](#)
- Read: All functions (state checks)

Note

Not thread-safe for concurrent writes. Use external synchronization if calling from multiple threads.

See also

[rx_iwdt_state_t](#) Structure definition

Since

Version 1.0.0

Definition at line 275 of file [rx_iwdt.c](#).

00275 {};

Referenced by [internal_find_free_slot\(\)](#), [internal_find_task\(\)](#), [rx_iwdt_check_tasks\(\)](#), [rx_iwdt_feed\(\)](#), [rx_iwdt_get_status\(\)](#), [rx_iwdt_init\(\)](#), [rx_iwdt_register_task\(\)](#), [rx_iwdt_set_state\(\)](#), [rx_iwdt_set_timeout_for_state\(\)](#) and [rx_iwdt_task_heartbeat\(\)](#).

7.54 rx_iwdt.c

[Go to the documentation of this file.](#)

```
00001 /**
00002  * @file rx_iwdt.c
00003  * @brief Independent Watchdog Timer (IWDT) Implementation for RX72N
00004  *
00005  * @details
00006  * Implements the hardware-independent watchdog timer driver with enhanced
00007  * task monitoring capabilities. The IWDT uses a dedicated 120 kHz clock
00008  * source, providing reliable watchdog functionality even during main clock
00009  * failures - a critical safety feature for robotics applications.
00010  *
00011  * ## Implementation Architecture
00012  *
00013  * @dot
00014  * digraph iwdt_implementation {
00015  *     rankdir=TB;
```

```

00016 * node [shape=box, style=rounded];
00017 * edge [fontsize=10];
00018 *
00019 * subgraph cluster_state {
00020 *   label="Module State (s_iwtdt_state)";
00021 *   style=dashed;
00022 *   config [label="config\nrx_iwtdt_config_t"];
00023 *   tasks [label="tasks[8]\nrx_iwtdt_task_info_t"];
00024 *   status [label="status\nrx_iwtdt_status_t"];
00025 *   curr_state [label="current_state\nsystem_state_t"];
00026 *   initialized [label="initialized\nbool"];
00027 * }
00028 *
00029 * subgraph cluster_api {
00030 *   label="Public API";
00031 *   style=filled;
00032 *   fillcolor=lightblue;
00033 *   init [label="rx_iwtdt_init()"];
00034 *   feed [label="rx_iwtdt_feed()"];
00035 *   reg_task [label="rx_iwtdt_register_task()"];
00036 *   heartbeat [label="rx_iwtdt_task_heartbeat()"];
00037 *   set_state [label="rx_iwtdt_set_state()"];
00038 *   check [label="rx_iwtdt_check_tasks()"];
00039 *   get_status [label="rx_iwtdt_get_status()"];
00040 *   was_reset [label="rx_iwtdt_was_reset()"];
00041 * }
00042 *
00043 * subgraph cluster_internal {
00044 *   label="Internal Helpers";
00045 *   style=filled;
00046 *   fillcolor=lightyellow;
00047 *   get_tick [label="internal_get_tick_count()"];
00048 *   find_task [label="internal_find_task()"];
00049 *   find_slot [label="internal_find_free_slot()"];
00050 *   init_config [label="internal_init_default_config()"];
00051 * }
00052 *
00053 * subgraph cluster_hw {
00054 *   label="Hardware Registers";
00055 *   style=filled;
00056 *   fillcolor=lightgray;
00057 *   iwdtrr [label="IWDTRR\n0x00088030\n(Refresh Register)"];
00058 *   iwdtsr [label="IWDTSR\n0x00088032\n(Status Register)"];
00059 *   rstsr2 [label="RSTSR2\n(Reset Status 2)"];
00060 * }
00061 *
00062 * init -> config [label="store"];
00063 * init -> init_config [label="if nullptr", style=dashed];
00064 * init -> iwdtrr [label="initial feed"];
00065 * init -> rstsr2 [label="check reset", style=dashed];
00066 *
00067 * feed -> iwdtrr [label="write 0x00, 0xFF"];
00068 * feed -> status [label="increment feeds"];
00069 *
00070 * reg_task -> find_slot;
00071 * reg_task -> tasks [label="add"];
00072 *
00073 * heartbeat -> find_task;
00074 * heartbeat -> tasks [label="update tick"];
00075 *
00076 * check -> tasks [label="scan all"];
00077 * check -> get_tick;
00078 * }
00079 * @enddot
00080 *
00081 * ## IWDT vs WDT Comparison
00082 *
00083 * | Feature | IWDT (This Module) | WDT (rx_wdt.c) |
00084 * |-----|-----|-----|
00085 * | Clock Source | IWDTCCLK (120 kHz independent) | PCLKB (system clock) |
00086 * | Can be Stopped | [X] No (hardware safety) | [OK] Yes |
00087 * | Clock Fault Detection | [OK] Yes (independent clock) | [X] No |
00088 * | Timeout Range | ~8 ms to ~17 seconds | ~4 us to ~273 us |
00089 * | Register Base | 0x00088030 | 0x00088020 |
00090 * | Task Monitoring | [OK] Software layer | [X] Not implemented |
00091 * | Primary Use | Production safety | Development/debugging |
00092 *
00093 * ## Task Monitoring System
00094 *
00095 * @msc
00096 * App, IWDT, Task1, Task2, HW;
00097 *
00098 * ... [label="Initialization"];
00099 * App => IWDT [label="rx_iwtdt_init()"];
00100 * IWDT => HW [label="Initial feed"];
00101 *
00102 * ... [label="Task Registration"];

```

```

00103 * Task1 => IWDT [label="rx_iwdt_register_task(\"motor\", 500)];
00104 * Task2 => IWDT [label="rx_iwdt_register_task(\"comm\", 1000)];
00105 *
00106 * ... [label="Normal Operation"];
00107 * Task1 => IWDT [label="rx_iwdt_task_heartbeat(\"motor\")"];
00108 * IWDT box IWDT [label="Update last_heartbeat_tick"];
00109 * Task2 => IWDT [label="rx_iwdt_task_heartbeat(\"comm\")"];
00110 *
00111 * ... [label="Check & Feed Loop"];
00112 * App => IWDT [label="rx_iwdt_check_tasks()"];
00113 * IWDT box IWDT [label="Compare ticks vs timeout"];
00114 * App => IWDT [label="rx_iwdt_feed()"];
00115 * IWDT => HW [label="Write 0x00, 0xFF"];
00116 *
00117 * ... [label="Task Timeout"];
00118 * IWDT box IWDT [label="Task1 missed deadline"];
00119 * App => IWDT [label="rx_iwdt_check_tasks()"];
00120 * IWDT » App [label="k_rx_err_timeout"];
00121 * @endmsc
00122 *
00123 * ## Memory Layout
00124 *
00125 * | Offset | Size | Field | Description |
00126 * |-----|-----|-----|-----|
00127 * | 0x000 | 68 | config | Configuration (timeouts, flags) |
00128 * | 0x044 | 320 | tasks[8] | Task info array (40 bytes each) |
00129 * | 0x184 | 76 | status | Status info |
00130 * | 0x1D0 | 1 | current_state | System state enum |
00131 * | 0x1D1 | 1 | initialized | Init flag |
00132 * | 0x1D2 | 2 | (padding) | Alignment |
00133 * | **Total** | **~470 bytes** | | Static allocation |
00134 *
00135 * ## Performance Characteristics
00136 *
00137 * | Function | Execution Time | Stack | Notes |
00138 * |-----|-----|-----|-----|
00139 * | rx_iwdt_init() | ~30 us | 64 B | Memset + config copy |
00140 * | rx_iwdt_feed() | ~1 us | 8 B | 2 register writes |
00141 * | rx_iwdt_register_task() | ~10 us | 32 B | String copy + slot search |
00142 * | rx_iwdt_task_heartbeat() | ~5 us | 16 B | Task lookup + tick update |
00143 * | rx_iwdt_check_tasks() | ~20 us | 24 B | Scan all 8 tasks |
00144 *
00145 * @par Module Dependencies:
00146 * - [rx_iwdt.h](../inc/rx_iwdt.h) - Public API declarations
00147 * - [rx72n_iwdt_regs.h](../rx_hal/inc/rx72n_iwdt_regs.h) - IWDT registers
00148 * - [rx72n_system_regs.h](../rx_hal/inc/rx72n_system_regs.h) - System registers
00149 * - [rx_check.h](../inc/rx_check.h) - Validation macros
00150 * - <tx_api.h> - ThreadX API for tick count
00151 * - <string.h> - strlen, strcmp
00152 *
00153 * @par NASA Power of 10 Compliance:
00154 * - **Rule 1**:[OK] No goto, setjmp, recursion
00155 * - **Rule 2**:[OK] All loops bounded by k_iwdt_max_tasks
00156 * - **Rule 3**:[OK] Zero dynamic allocation (static state)
00157 * - **Rule 4**:[OK] All functions <60 lines
00158 * - **Rule 5**:[OK] Minimum 2 checks per function
00159 * - **Rule 6**:[OK] Data declared at smallest scope
00160 * - **Rule 7**:[OK] All return values checked or propagated
00161 * - **Rule 8**:[OK] C23 typed enums for constants
00162 * - **Rule 9**:[OK] Minimal pointer use, no pointer arithmetic
00163 * - **Rule 10**:[OK] Compiled with -Wall -Wextra -Werror
00164 *
00165 * @par SOLID Principles:
00166 * - **Single Responsibility**:[OK] Only IWDT and task monitoring
00167 * - **Open/Closed**:[OK] Extensible via configuration and state callbacks
00168 * - **Liskov Substitution**:[OK] Consistent rx_err_t return interface
00169 * - **Interface Segregation**:[OK] Separate init/feed/task/status APIs
00170 * - **Dependency Inversion**:[OK] Depends on abstract rx_err_t
00171 *
00172 * @see rx_iwdt.h Public API documentation and usage examples
00173 * @see rx_wdt.c Software-controllable watchdog (complementary)
00174 * @see RX72N User's Manual Section 11: Independent Watchdog Timer
00175 *
00176 * @author Locked, Inc.
00177 * @date 2026-01-28
00178 * @version 1.0.0
00179 * @copyright Copyright (c) 2026 Locked Inc.
00180 * SPDX-License-Identifier: MIT
00181 *
00182 * @since Version 1.0.0
00183 */
00184
00185 #include "rx_iwdt.h"
00186
00187 #include <string.h>
00188
00189 #ifdef UNIT_TEST

```

```

00190 #include "mock_rx72n_iwtdt_regs.h"
00191 #include "mock_rx72n_system_regs.h"
00192 #else
00193 #include "rx72n_iwtdt_regs.h"
00194 #include "rx72n_system_regs.h"
00195 #endif
00196 #include "tx_api.h" /* ThreadX API for tx_time_get() and TX_TIMER_TICKS_PER_SECOND */
00197
00198 /*
=====
00199  * Static Data
00200  *
=====
00201  */
00202
00203 /**
00204  * @struct rx_iwtdt_state_t
00205  * @brief Internal state structure for IWDT driver
00206  *
00207  * @details
00208  * Aggregates all state for the IWDT driver including configuration,
00209  * task monitoring arrays, status tracking, and initialization flags.
00210  * This is the largest static allocation in the core library.
00211  *
00212  * @par Memory Layout:
00213  * | Offset | Size | Field | Description |
00214  * |-----|-----|-----|-----|
00215  * | 0x000 | ~68 | config | Configuration structure |
00216  * | 0x044 | 320 | tasks | 8 task info structures (40 B each) |
00217  * | 0x184 | ~76 | status | Status and counters |
00218  * | 0x1D0 | 1 | current_state | System state enum |
00219  * | 0x1D1 | 1 | initialized | Init complete flag |
00220  * | **Total** | **~470 bytes** | | Static allocation |
00221  *
00222  * @par State Transitions:
00223  * - initialized: false -> true on rx_iwtdt_init() success
00224  * - current_state: k_system_state_init -> other states via rx_iwtdt_set_state()
00225  * - tasks[n].active: false -> true via rx_iwtdt_register_task()
00226  *
00227  * @invariant initialized implies config is valid
00228  * @invariant status.active_tasks equals count of tasks[].active==true
00229  *
00230  * @see s_iwtdt_state Static instance
00231  *
00232  * @since Version 1.0.0
00233  */
00234 typedef struct {
00235     rx_iwtdt_config_t    config;          /**< Configuration from rx_iwtdt_init(). Contains
00236                                           default timeout, task monitoring enable flag,
00237                                           reset mode, and per-state timeout array. */
00238     rx_iwtdt_task_info_t tasks[k_iwtdt_max_tasks]; /**< Array of registered task info. Each entry
00239                                           contains task name, timeout, last heartbeat
00240                                           tick, and timeout flag. Max 8 tasks. */
00241     rx_iwtdt_status_t    status;          /**< Runtime status: feed count, reset count,
00242                                           current state, active task count, last
00243                                           failed task name. */
00244     system_state_t        current_state;   /**< Current system state for state-dependent
00245                                           timeout selection. One of system_state_t. */
00246     bool                  initialized;     /**< Module initialization flag. true = ready
00247                                           for use, false = must call rx_iwtdt_init(). */
00248 } rx_iwtdt_state_t;
00249
00250 /**
00251  * @var s_iwtdt_state
00252  * @brief Global IWDT driver state
00253  *
00254  * @details
00255  * Single instance of IWDT state, zero-initialized at startup. All fields
00256  * start at 0/false/empty representing uninitialized state.
00257  *
00258  * **Memory Location:** .bss section (zero-initialized static data)
00259  * **Size:** ~470 bytes
00260  * **Alignment:** 4-byte aligned
00261  *
00262  * @par Access Pattern:
00263  * - **Write:** rx_iwtdt_init(), rx_iwtdt_register_task(), rx_iwtdt_task_heartbeat(),
00264  *               rx_iwtdt_set_state(), rx_iwtdt_set_timeout_for_state(), rx_iwtdt_feed(),
00265  *               rx_iwtdt_check_tasks()
00266  * - **Read:** All functions (state checks)
00267  *
00268  * @note Not thread-safe for concurrent writes. Use external synchronization
00269  *       if calling from multiple threads.
00270  *
00271  * @see rx_iwtdt_state_t Structure definition
00272  *
00273  * @since Version 1.0.0
00274  */

```

```

00275 static rx_iwdt_state_t s_iwdt_state = {};
00276
00277 /*
=====
00278 * Test Support Functions
00279 *
=====
00280 */
00281
00282 #ifdef UNIT_TEST
00283 /**
00284  * @brief Reset IWDt driver state for unit testing
00285  * @details
00286  * Resets the driver to uninitialized state. This function is only available
00287  * when UNIT_TEST is defined and should only be called from test code.
00288  */
00289 void rx_iwdt_test_reset(void)
00290 {
00291     s_iwdt_state = (rx_iwdt_state_t){};
00292 }
00293 #endif
00294
00295 /*
=====
00296 * Internal Helper Functions
00297 *
=====
00298 */
00299
00300 /**
00301  * @brief Copy a string with bounded length (replaces strncpy)
00302  * @details
00303  * Copies up to max_len-1 characters from src to dst and null-terminates.
00304  * Avoids clang-tidy insecureAPI.DeprecatedOrUnsafeBufferHandling warnings
00305  * that strncpy triggers.
00306  *
00307  *
00308  *
00309  * @pre dst != nullptr
00310  * @pre src != nullptr
00311  * @pre max_len > 0
00312  * @post dst is null-terminated
00313  *
00314  * @note Not thread-safe
00315  *
00316  * @since Version 1.0.0
00317  */
00318 static void internal_safe_strcpy(char* dst, const char* src, size_t max_len)
00319 {
00320     size_t i = 0;
00321     for (; i < max_len - 1U && src[i] != '\0'; i++) {
00322         dst[i] = src[i];
00323     }
00324     dst[i] = '\0';
00325 }
00326
00327 /**
00328  * @brief Get current system tick count from ThreadX
00329  *
00330  * @details
00331  * Wrapper around ThreadX tx_time_get() to get the current tick count.
00332  * Used for task heartbeat timing calculations.
00333  *
00334  * **Algorithm:**
00335  * 1. Call tx_time_get() to get ThreadX tick count
00336  * 2. Cast ULONG to uint32_t for consistent type
00337  * 3. Return tick count
00338  *
00339  *
00340  * @note ThreadX tick rate defined by TX_TIMER_TICKS_PER_SECOND
00341  * @note Tick count wraps at UINT32_MAX (~497 days at 100 Hz tick rate)
00342  * @note Thread-safe - tx_time_get() is reentrant
00343  *
00344  * @see tx_time_get() ThreadX API
00345  * @see TX_TIMER_TICKS_PER_SECOND Tick rate configuration
00346  *
00347  * @since Version 1.0.0
00348  */
00349 static uint32_t internal_get_tick_count(void)
00350 {
00351     return (uint32_t)tx_time_get();
00352 }
00353
00354 /**
00355  * @brief Find registered task by name
00356  *
00357  * @details

```



```

00358 * Searches the task array for a task with matching name that is currently
00359 * active. Uses strcmp() for exact string comparison.
00360 *
00361 **Algorithm:**
00362 * 1. Iterate through tasks[0..k_iwdt_max_tasks-1]
00363 * 2. For each entry: check if active AND name matches
00364 * 3. Return pointer to matching entry, or nullptr if not found
00365 *
00366 *
00367 *
00368 * @pre task_name must be non-NULL (caller validates)
00369 *
00370 * @note Linear search O(n) where n = k_iwdt_max_tasks (8)
00371 * @note String comparison is case-sensitive
00372 *
00373 * @see rx_iwdt_register_task() Adds tasks to array
00374 * @see rx_iwdt_task_heartbeat() Uses this to find task
00375 *
00376 * @since Version 1.0.0
00377 */
00378 static rx_iwdt_task_info_t* internal_find_task(const char* task_name)
00379 {
00380     for (uint32_t i = 0; i < k_iwdt_max_tasks; i++) {
00381         if ((int)s_iwdt_state.tasks[i].active &&
00382             strcmp(s_iwdt_state.tasks[i].task_name, task_name) == 0) {
00383             return &s_iwdt_state.tasks[i];
00384         }
00385     }
00386     return nullptr;
00387 }
00388 }
00389 /**
00390 * @brief Find free slot in task array
00391 *
00392 * @details
00393 * Searches for the first inactive slot in the task array where a new
00394 * task can be registered.
00395 *
00396 * **Algorithm:**
00397 * 1. Iterate through tasks[0..k_iwdt_max_tasks-1]
00398 * 2. For each entry: check if NOT active
00399 * 3. Return pointer to first inactive slot, or nullptr if all full
00400 *
00401 *
00402 *
00403 * @note Linear search O(n) where n = k_iwdt_max_tasks (8)
00404 * @note Returns first available slot (lowest index)
00405 *
00406 * @see rx_iwdt_register_task() Uses this to allocate slot
00407 * @see k_iwdt_max_tasks Maximum number of tasks (8)
00408 *
00409 * @since Version 1.0.0
00410 */
00411 static rx_iwdt_task_info_t* internal_find_free_slot(void)
00412 {
00413     for (uint32_t i = 0; i < k_iwdt_max_tasks; i++) {
00414         if (!s_iwdt_state.tasks[i].active) {
00415             return &s_iwdt_state.tasks[i];
00416         }
00417     }
00418     return nullptr;
00419 }
00420 }
00421 /**
00422 * @brief Initialize default configuration structure
00423 *
00424 * @details
00425 * Populates a configuration structure with safe default values suitable
00426 * for most applications. Called when rx_iwdt_init() receives nullptr config.
00427 *
00428 * **Default Values:**
00429 * - default_timeout_ms = k_iwdt_default_timeout_ms (1000 ms)
00430 * - enable_task_monitoring = true
00431 * - reset_on_timeout = true
00432 * - All state_timeouts_ms[] = k_iwdt_default_timeout_ms
00433 *
00434 * **Algorithm:**
00435 * 1. Set default_timeout_ms to 1000 ms
00436 * 2. Enable task monitoring
00437 * 3. Enable reset on timeout
00438 * 4. Initialize all state timeouts to default value
00439 *
00440 *
00441 *
00442 * @pre config must be non-NULL (caller validates)
00443 *
00444 * @post config contains default values ready for use

```

```

00445 *
00446 * @note Loop bounded by k_system_state_count (NASA Rule 2)
00447 *
00448 * @see rx_iwdt_init() Calls this when config is nullptr
00449 * @see k_iwdt_default_timeout_ms Default timeout constant
00450 * @see k_system_state_count Number of system states
00451 *
00452 * @since Version 1.0.0
00453 */
00454 static void internal_init_default_config(rx_iwdt_config_t* config)
00455 {
00456     config->default_timeout_ms = k_iwdt_default_timeout_ms;
00457     config->enable_task_monitoring = true;
00458     config->reset_on_timeout = true;
00459
00460     /* Initialize all state timeouts to default */
00461     for (uint32_t i = 0; i < k_system_state_count; i++) {
00462         config->state_timeouts_ms[i] = k_iwdt_default_timeout_ms;
00463     }
00464 }
00465
00466 /*
=====
00467 * Public API Implementation
00468 *
=====
00469 */
00470
00471 /**
00472 * @brief Initialize IWDT driver with configuration and task monitoring
00473 *
00474 * @details
00475 * Initializes the IWDT driver, validates configuration, checks for previous
00476 * watchdog reset, and performs initial hardware feed. If config is nullptr,
00477 * uses safe default values (1000ms timeout, task monitoring enabled).
00478 *
00479 * **Algorithm:**
00480 * 1. Check if already initialized (return error if so)
00481 * 2. If config is nullptr, create default configuration
00482 * 3. Validate timeout within [k_iwdt_min_timeout_ms, k_iwdt_max_timeout_ms]
00483 * 4. Clear and initialize all state
00484 * 5. Check RSTSR2 for previous watchdog reset
00485 * 6. Perform initial feed to IWDTRR (0x00 then 0xFF)
00486 * 7. Set initialized flag
00487 *
00488 * @par OFS Configuration Note:
00489 * The IWDT hardware timeout is configured at flash-time via OFS (Option
00490 * Function Select) registers and cannot be changed at runtime. The timeout
00491 * values in this driver are used for software task monitoring only. The
00492 * hardware IWDT will timeout based on OFS settings regardless of software
00493 * configuration.
00494 *
00495 *
00496 *
00497 * @pre IWDT not already initialized
00498 *
00499 * @post IWDT driver ready for use
00500 * @post Initial feed performed
00501 * @post Reset status captured
00502 *
00503 * @note Hardware timeout set by OFS, not this driver
00504 * @note nullptr config uses safe defaults (1000ms, monitoring enabled)
00505 *
00506 * @since Version 1.0.0
00507 */
00508 rx_err_t rx_iwdt_init(const rx_iwdt_config_t* config)
00509 {
00510     /* Check if already initialized */
00511     if (s_iwdt_state.initialized) {
00512         return k_rx_err_invalid_state;
00513     }
00514
00515     /* Use default config if none provided */
00516     rx_iwdt_config_t local_config;
00517     if (config == nullptr) {
00518         internal_init_default_config(&local_config);
00519         config = &local_config;
00520     }
00521
00522     /* Validate configuration */
00523     if (config->default_timeout_ms < k_iwdt_min_timeout_ms ||
00524         config->default_timeout_ms > k_iwdt_max_timeout_ms) {
00525         return k_rx_err_invalid_arg;
00526     }
00527
00528     /* Initialize state */
00529     s_iwdt_state = (rx_iwdt_state_t){};

```

```

00530 {
00531     const uint8_t* src = (const uint8_t*)config;
00532     uint8_t* dst = (uint8_t*)&s_iwdt_state.config;
00533     for (size_t byte_i = 0; byte_i < sizeof(rx_iwdt_config_t); byte_i++) {
00534         dst[byte_i] = src[byte_i];
00535     }
00536 }
00537 s_iwdt_state.current_state = k_system_state_init;
00538
00539 /* Initialize status */
00540 s_iwdt_state.status.initialized = true;
00541 s_iwdt_state.status.current_state = k_system_state_init;
00542 s_iwdt_state.status.current_timeout_ms = config->default_timeout_ms;
00543 s_iwdt_state.status.active_tasks = 0;
00544
00545 /* Check if last reset was watchdog */
00546 if (rx_iwdt_was_reset()) {
00547     s_iwdt_state.status.total_resets++;
00548     s_iwdt_state.status.last_reset_reason = k_iwdt_reset_watchdog;
00549 } else {
00550     s_iwdt_state.status.last_reset_reason = k_iwdt_reset_power_on;
00551 }
00552
00553 /* Note: IWDT hardware is configured via OFS (Option Function Select) registers
00554  * at compile/flash time and starts automatically on reset. We cannot reconfigure
00555  * the timeout period at runtime. This driver manages the software state and
00556  * provides the feed mechanism. The requested timeout is used for software task
00557  * monitoring only. */
00558 volatile rx_iwdt_regs_t* const regs = iwdt();
00559
00560 /* Feed the watchdog to establish baseline */
00561 regs->iwdtrr = k_iwdt_refresh_start;
00562 regs->iwdtrr = k_iwdt_refresh_end;
00563
00564 s_iwdt_state.initialized = true;
00565
00566 return k_rx_ok;
00567 }
00568
00569 /**
00570  * @brief Feed the independent watchdog timer to prevent system reset
00571  *
00572  * @details
00573  * Writes the two-byte refresh sequence (0x00 then 0xFF) to the IWDT Refresh
00574  * Register (IWDTRR) to reset the hardware watchdog counter. This must be
00575  * called periodically within the configured hardware timeout window or the
00576  * IWDT will reset the microcontroller.
00577  *
00578  * Per the RX72N hardware manual the refresh sequence is strictly ordered:
00579  * writing 0x00 to IWDTRR starts the window; writing 0xFF completes the
00580  * refresh. The entire sequence must occur within the permitted refresh window.
00581  *
00582  * Also increments the software diagnostic counter s_iwdt_state.status.watchdog_feeds
00583  * for telemetry and debugging purposes.
00584  *
00585  * Algorithm steps:
00586  * 1. Check s_iwdt_state.initialized; return k_rx_err_not_initialized if false
00587  * 2. Write k_iwdt_refresh_start (0x00) to regs->iwdtrr
00588  * 3. Write k_iwdt_refresh_end (0xFF) to regs->iwdtrr
00589  * 4. Increment status.watchdog_feeds counter
00590  *
00591  *
00592  * @pre rx_iwdt_init() must have been called successfully
00593  * @pre Must be called within the hardware IWDT refresh window (no gap > configured timeout)
00594  * @post Hardware IWDT counter is reset; timeout window restarts
00595  * @post s_iwdt_state.status.watchdog_feeds is incremented by one
00596  *
00597  * @note Not thread-safe; concurrent feeds from multiple tasks are benign for
00598  * the hardware but may produce incorrect feed counts
00599  * @warning Failing to call within the hardware timeout triggers a system reset
00600  *
00601  * @par Performance:
00602  * Execution time: <1 us @ 240 MHz; safe to call from any task or ISR context
00603  *
00604  * @par Example:
00605  * @code
00606  * // In main watchdog task loop
00607  * while (1) {
00608  *     rx_err_t err = rx_iwdt_feed();
00609  *     if (err != k_rx_ok) {
00610  *         rx_log_error("wdt", "IWDT not initialized");
00611  *     }
00612  *     tx_thread_sleep(k_watchdog_feed_interval_ticks);
00613  * }
00614  * @endcode
00615  *
00616  * @see rx_iwdt_init() Initialize IWDT before feeding

```

```

00617 * @see rx_iwdt_check_tasks() Verify all tasks are alive before feeding
00618 *
00619 * @since Version 1.0.0
00620 *
00621 * @par NASA Power of 10 Compliance:
00622 * - Rule 5: 1 precondition (initialized check), 2 postconditions (HW reset, counter++)
00623 */
00624 rx_err_t rx_iwdt_feed(void)
00625 {
00626     volatile rx_iwdt_regs_t* regs;
00627
00628     /* Check initialization */
00629     if (!s_iwdt_state.initialized) {
00630         return k_rx_err_not_initialized;
00631     }
00632
00633     /* Refresh watchdog counter - write 0x00 then 0xFF to IWDTRR */
00634     regs = iwdt();
00635     regs->iwdtrr = k_iwdt_refresh_start;
00636     regs->iwdtrr = k_iwdt_refresh_end;
00637
00638     s_iwdt_state.status.watchdog_feeds++;
00639
00640     return k_rx_ok;
00641 }
00642
00643 /**
00644 * @brief Register a ThreadX task for software watchdog monitoring
00645 *
00646 * @details
00647 * Adds a task entry to the IWDT software monitoring table, associating the
00648 * task name with a per-task timeout. Once registered, the task must call
00649 * rx_iwdt_task_heartbeat() at least once per timeout_ms milliseconds or
00650 * rx_iwdt_check_tasks() will report k_rx_err_timeout.
00651 *
00652 * The registration also records the current tick count as the initial
00653 * last_heartbeat_tick, giving the task a full timeout_ms window before
00654 * the first heartbeat is required.
00655 *
00656 * Algorithm steps:
00657 * 1. Validate task_name is non-null
00658 * 2. Verify s_iwdt_state.initialized
00659 * 3. Validate timeout_ms is in [k_iwdt_min_timeout_ms, k_iwdt_max_timeout_ms]
00660 * 4. Check task name length is in (0, k_iwdt_task_name_len)
00661 * 5. Check task is not already registered via internal_find_task()
00662 * 6. Find a free slot via internal_find_free_slot()
00663 * 7. Initialize slot with task name, timeout, current tick, active=true, timed_out=false
00664 * 8. Increment status.active_tasks
00665 *
00666 *
00667 *
00668 * @pre rx_iwdt_init() must have been called successfully
00669 * @pre task_name must be unique among registered tasks
00670 * @post Task entry is active in s_iwdt_state.tasks[]
00671 * @post status.active_tasks is incremented by one
00672 *
00673 * @note Not thread-safe; serialize registrations from multiple contexts
00674 *
00675 * @par Example:
00676 * @code
00677 * rx_err_t err = rx_iwdt_register_task("motor_ctrl", 500);
00678 * if (err != k_rx_ok) {
00679 *     rx_log_error("wdt", "Failed to register motor_ctrl task");
00680 * }
00681 * @endcode
00682 *
00683 * @see rx_iwdt_task_heartbeat() Update heartbeat for a registered task
00684 * @see rx_iwdt_check_tasks() Check all registered tasks for timeout
00685 *
00686 * @since Version 1.0.0
00687 *
00688 * @par NASA Power of 10 Compliance:
00689 * - Rule 5: 5 preconditions (null, initialized, timeout, name, duplicate), 2 postconditions
00690 */
00691 rx_err_t rx_iwdt_register_task(const char* task_name, uint32_t timeout_ms)
00692 {
00693     /* Validate inputs */
00694     if (task_name == nullptr) {
00695         return k_rx_err_null_ptr;
00696     }
00697
00698     if (!s_iwdt_state.initialized) {
00699         return k_rx_err_not_initialized;
00700     }
00701
00702     if (timeout_ms < k_iwdt_min_timeout_ms || timeout_ms > k_iwdt_max_timeout_ms) {
00703         return k_rx_err_invalid_arg;

```

```

00704 }
00705
00706 /* Check task name length */
00707 const uint32_t name_len = (uint32_t)strlen(task_name);
00708 if (name_len == 0 || name_len >= k_iwtdt_task_name_len) {
00709     return k_rx_err_invalid_arg;
00710 }
00711
00712 /* Check if task already registered */
00713 if (internal_find_task(task_name) != nullptr) {
00714     return k_rx_err_invalid_arg;
00715 }
00716
00717 /* Find free slot */
00718 rx_iwtdt_task_info_t* const slot = internal_find_free_slot();
00719 if (slot == nullptr) {
00720     return k_rx_err_no_mem;
00721 }
00722
00723 /* Register task */
00724 *slot = (rx_iwtdt_task_info_t){};
00725 internal_safe_strcpy(slot->task_name, task_name, k_iwtdt_task_name_len);
00726 slot->timeout_ms = timeout_ms;
00727 slot->last_heartbeat_tick = internal_get_tick_count();
00728 slot->active = true;
00729 slot->timed_out = false;
00730
00731 s_iwtdt_state.status.active_tasks++;
00732
00733 return k_rx_ok;
00734 }
00735
00736 /**
00737  * @brief Send a heartbeat signal for a registered task to reset its timeout
00738  *
00739  * @details
00740  * Updates the last_heartbeat_tick for the named task to the current ThreadX
00741  * tick count, resetting the timeout window. Also clears the timed_out flag
00742  * if the task had previously been marked as timed out.
00743  *
00744  * Tasks must call this function at least once per their registered timeout_ms
00745  * window (configured in rx_iwtdt_register_task()) to remain in good standing.
00746  * If the interval between heartbeats exceeds timeout_ms, the next call to
00747  * rx_iwtdt_check_tasks() will return k_rx_err_timeout.
00748  *
00749  * Algorithm steps:
00750  * 1. Validate task_name is non-null
00751  * 2. Verify s_iwtdt_state.initialized
00752  * 3. Find the task entry via internal_find_task()
00753  * 4. Set task->last_heartbeat_tick = internal_get_tick_count()
00754  * 5. Clear task->timed_out
00755  *
00756  *
00757  *
00758  * @pre rx_iwtdt_init() must have been called successfully
00759  * @pre Task must have been registered via rx_iwtdt_register_task()
00760  * @post task->last_heartbeat_tick is refreshed to the current tick count
00761  * @post task->timed_out is false
00762  *
00763  * @note Not thread-safe; concurrent heartbeats from different tasks are safe
00764  *       as they write to distinct slots, but concurrent writes to the same slot are not
00765  *
00766  * @par Example:
00767  * @code
00768  * // At top of motor control task loop
00769  * while (1) {
00770  *     rx_iwtdt_task_heartbeat("motor_ctrl");
00771  *     motor_control_update();
00772  *     tx_thread_sleep(k_motor_ctrl_period_ticks);
00773  * }
00774  * @endcode
00775  *
00776  * @see rx_iwtdt_register_task() Register a task before sending heartbeats
00777  * @see rx_iwtdt_check_tasks() Verify all tasks are within their timeout windows
00778  *
00779  * @since Version 1.0.0
00780  *
00781  * @par NASA Power of 10 Compliance:
00782  * - Rule 5: 3 preconditions (null, initialized, registered), 2 postconditions
00783  */
00784 rx_err_t rx_iwtdt_task_heartbeat(const char* task_name)
00785 {
00786     /* Validate inputs */
00787     if (task_name == nullptr) {
00788         return k_rx_err_null_ptr;
00789     }
00790 }

```

```

00791 if (!s_iwdt_state.initialized) {
00792     return k_rx_err_not_initialized;
00793 }
00794
00795 /* Find task */
00796 rx_iwdt_task_info_t* const task = internal_find_task(task_name);
00797 if (task == nullptr) {
00798     return k_rx_err_not_found;
00799 }
00800
00801 /* Update heartbeat */
00802 task->last_heartbeat_tick = internal_get_tick_count();
00803 task->timed_out = false;
00804
00805 return k_rx_ok;
00806 }
00807
00808 /**
00809  * @brief Set the software task monitoring timeout for a specific system state
00810  *
00811  * @details
00812  * Configures the state-dependent software timeout used by rx_iwdt_check_tasks()
00813  * when the system is in the specified state. This allows different task heartbeat
00814  * windows depending on the operational mode (e.g. longer timeouts during
00815  * initialization, tighter timeouts during normal operation).
00816  *
00817  * Note: This does NOT change the hardware IWDT timeout. The hardware watchdog
00818  * timeout is fixed at compile/flash time via OFS registers and cannot be altered
00819  * at runtime. Only software task monitoring timeouts are affected.
00820  *
00821  * If state matches the currently active state (s_iwdt_state.current_state),
00822  * the active timeout (status.current_timeout_ms) is also updated immediately.
00823  *
00824  * Algorithm steps:
00825  * 1. Validate state is less than k_system_state_count
00826  * 2. Verify s_iwdt_state.initialized
00827  * 3. Validate timeout_ms is in [k_iwdt_min_timeout_ms, k_iwdt_max_timeout_ms]
00828  * 4. Set config.state_timeouts_ms[state] = timeout_ms
00829  * 5. If state == current_state, update status.current_timeout_ms
00830  *
00831  *
00832  *
00833  * @pre rx_iwdt_init() must have been called successfully
00834  * @pre state must be a valid system_state_t value less than k_system_state_count
00835  * @post config.state_timeouts_ms[state] equals timeout_ms
00836  * @post If state is current, status.current_timeout_ms is updated immediately
00837  *
00838  * @note Not thread-safe; serialize state configuration changes
00839  * @warning Hardware IWDT timeout is unaffected; only software task monitoring changes
00840  *
00841  * @par Example:
00842  * @code
00843  * // Allow longer task timeouts during initialization phase
00844  * rx_iwdt_set_timeout_for_state(k_system_state_initializing, 5000);
00845  * rx_iwdt_set_timeout_for_state(k_system_state_running, 500);
00846  * @endcode
00847  *
00848  * @see rx_iwdt_set_state() Change the active system state
00849  * @see rx_iwdt_check_tasks() Uses current state timeout to evaluate task heartbeats
00850  *
00851  * @since Version 1.0.0
00852  *
00853  * @par NASA Power of 10 Compliance:
00854  * - Rule 5: 3 preconditions (initialized, valid state, valid timeout), 2 postconditions
00855  */
00856 rx_err_t rx_iwdt_set_timeout_for_state(const system_state_t state, const uint32_t timeout_ms)
00857 {
00858     /* Validate inputs */
00859     if (state >= k_system_state_count) {
00860         return k_rx_err_invalid_arg;
00861     }
00862
00863     if (!s_iwdt_state.initialized) {
00864         return k_rx_err_not_initialized;
00865     }
00866
00867     if (timeout_ms < k_iwdt_min_timeout_ms || timeout_ms > k_iwdt_max_timeout_ms) {
00868         return k_rx_err_invalid_arg;
00869     }
00870
00871     /* Update state timeout */
00872     s_iwdt_state.config.state_timeouts_ms[state] = timeout_ms;
00873
00874     /* If this is the current state, update active timeout */
00875     if (state == s_iwdt_state.current_state) {
00876         s_iwdt_state.status.current_timeout_ms = timeout_ms;
00877     }

```

```

00878
00879     return k_rx_ok;
00880 }
00881
00882 /**
00883  * @brief Set the current system state for software task monitoring
00884  *
00885  * @details
00886  * Updates the active system state used by rx_iwdt_check_tasks() to select
00887  * per-state task heartbeat timeout windows. Transitions between states
00888  * (e.g. initializing -> running) change which timeout threshold is applied
00889  * when evaluating task heartbeat intervals.
00890  *
00891  * This function updates three tracking fields atomically:
00892  * - s_iwdt_state.current_state
00893  * - s_iwdt_state.status.current_state
00894  * - s_iwdt_state.status.current_timeout_ms (set from config.state_timeouts_ms[state])
00895  *
00896  * Important: This does NOT modify the hardware IWDT peripheral. The hardware
00897  * watchdog timeout is a fixed constant set in OFS registers at compile time.
00898  * Only the software task monitoring timeout threshold changes.
00899  *
00900  * Algorithm steps:
00901  * 1. Validate state is less than k_system_state_count
00902  * 2. Verify s_iwdt_state.initialized
00903  * 3. Update current_state and status.current_state = state
00904  * 4. Update status.current_timeout_ms = config.state_timeouts_ms[state]
00905  *
00906  *
00907  *
00908  * @pre rx_iwdt_init() must have been called successfully
00909  * @pre state must be a valid system_state_t value less than k_system_state_count
00910  * @post s_iwdt_state.current_state reflects the new state
00911  * @post status.current_timeout_ms reflects the configured timeout for the new state
00912  *
00913  * @note Not thread-safe; serialize state transitions from multiple contexts
00914  * @warning Hardware IWDT is unaffected; only software task monitoring changes
00915  *
00916  * @par Example:
00917  * @code
00918  * // Transition from initialization to running state
00919  * rx_err_t err = rx_iwdt_set_state(k_system_state_running);
00920  * if (err != k_rx_ok) {
00921  *     rx_log_error("wdt", "Failed to set IWDT system state");
00922  * }
00923  * @endcode
00924  *
00925  * @see rx_iwdt_set_timeout_for_state() Configure per-state timeouts
00926  * @see rx_iwdt_get_status() Query current state and timeout
00927  *
00928  * @since Version 1.0.0
00929  *
00930  * @par NASA Power of 10 Compliance:
00931  * - Rule 5: 2 preconditions (initialized, valid state), 2 postconditions
00932  */
00933 rx_err_t rx_iwdt_set_state(const system_state_t state)
00934 {
00935     /* Validate inputs */
00936     if (state >= k_system_state_count) {
00937         return k_rx_err_invalid_arg;
00938     }
00939
00940     if (!s_iwdt_state.initialized) {
00941         return k_rx_err_not_initialized;
00942     }
00943
00944     /* Update software state tracking.
00945      * NOTE: This does not change the hardware watchdog timeout, as IWDT timeout
00946      * is configured via OFS registers at compile/flash time and cannot be changed
00947      * at runtime. The state-dependent timeouts are used for software task monitoring
00948      * via rx_iwdt_check_tasks(). The hardware IWDT timeout remains constant. */
00949     s_iwdt_state.current_state = state;
00950     s_iwdt_state.status.current_state = state;
00951     s_iwdt_state.status.current_timeout_ms = s_iwdt_state.config.state_timeouts_ms[state];
00952
00953     return k_rx_ok;
00954 }
00955
00956 /**
00957  * @brief Get a snapshot of the current IWDT module status
00958  *
00959  * @details
00960  * Copies the internal s_iwdt_state.status structure into the caller-provided
00961  * buffer via memcpy. The status snapshot includes: active task count, total
00962  * feed count, current system state, current timeout threshold, and the name
00963  * of the last task that timed out (if any).
00964  */

```



```

00965 * This is a diagnostic/telemetry function intended for logging and health
00966 * monitoring. The returned data is a point-in-time snapshot and may become
00967 * stale if the detection task or other threads update IWDT state concurrently.
00968 *
00969 * Algorithm steps:
00970 * 1. Validate status pointer is non-null
00971 * 2. Verify s_iwdt_state.initialized
00972 * 3. memcpy(&s_iwdt_state.status, status, sizeof(rx_iwdt_status_t))
00973 *
00974 *
00975 *
00976 * @pre rx_iwdt_init() must have been called successfully
00977 * @pre status must be a valid non-null pointer to an rx_iwdt_status_t
00978 * @post *status contains a copy of s_iwdt_state.status at the time of the call
00979 * @post Internal IWDT state is unchanged
00980 *
00981 * @note Not thread-safe; status fields may be updated concurrently by the task monitor
00982 * @note Snapshot accuracy depends on timing; use as diagnostic data, not safety-critical logic
00983 *
00984 * @par Example:
00985 * @code
00986 * rx_iwdt_status_t status;
00987 * rx_err_t err = rx_iwdt_get_status(&status);
00988 * if (err == k_rx_ok) {
00989 *     rx_log_info("wdt", "Active tasks: %u, Feeds: %u",
00990 *         (unsigned)status.active_tasks,
00991 *         (unsigned)status.watchdog_feeds);
00992 * }
00993 * @endcode
00994 *
00995 * @see rx_iwdt_was_reset() Check if last reset was hardware IWDT timeout
00996 * @see rx_iwdt_check_tasks() Check all tasks for software timeout
00997 *
00998 * @since Version 1.0.0
00999 *
01000 * @par NASA Power of 10 Compliance:
01001 * - Rule 5: 2 preconditions (null check, initialized check), 2 postconditions
01002 */
01003 rx_err_t rx_iwdt_get_status(rx_iwdt_status_t* status)
01004 {
01005     /* Validate inputs */
01006     if (status == nullptr) {
01007         return k_rx_err_null_ptr;
01008     }
01009
01010     if (!s_iwdt_state.initialized) {
01011         return k_rx_err_not_initialized;
01012     }
01013
01014     /* Copy status */
01015     {
01016         const uint8_t* src = (const uint8_t*)&s_iwdt_state.status;
01017         uint8_t* dst = (uint8_t*)status;
01018         for (size_t byte_i = 0; byte_i < sizeof(rx_iwdt_status_t); byte_i++) {
01019             dst[byte_i] = src[byte_i];
01020         }
01021     }
01022
01023     return k_rx_ok;
01024 }
01025
01026 /**
01027 * @brief Check whether the last microcontroller reset was caused by an IWDT timeout
01028 *
01029 * @details
01030 * Reads the IWDT Reset Flag (IWDTRF) from the Reset Status Register 2 (RSTSR2)
01031 * of the RX72N. This register is located at a separate address from RSTSR0/1
01032 * and retains the reset cause flags across a reset event (values are cleared
01033 * on power-on reset only).
01034 *
01035 * The IWDTRF bit (bit position k_rstsr2_iwdtrf) is set by hardware when the IWDT
01036 * counter underflows (i.e., the watchdog was not refreshed in time). It remains
01037 * set until explicitly cleared or a power-on reset occurs.
01038 *
01039 * This function is used during startup to detect and log prior watchdog resets,
01040 * which may indicate a firmware hang or task scheduling failure.
01041 *
01042 * Algorithm steps:
01043 * 1. Read the RSTSR2 register byte via rstsr2() inline accessor
01044 * 2. Mask with k_rstsr2_iwdtrf to extract the IWDT reset flag bit
01045 * 3. Return true if the bit is set, false otherwise
01046 *
01047 *
01048 * @pre None -- safe to call before rx_iwdt_init() (reads hardware register directly)
01049 * @pre Must be called before the application clears the reset status register
01050 * @post Hardware RSTSR2 register is read-only; this function does not modify it
01051 * @post Return value reflects hardware register state at the instant of the call

```



```

01052 *
01053 * @note Thread-safe -- reads a single hardware register without shared state
01054 * @note Typically called once at startup to log the reset cause
01055 * @warning RSTSR2.IWDTRF remains set across warm resets; clear it in startup if needed
01056 *
01057 * @par Example:
01058 * @code
01059 * if (rx_iwdt_was_reset()) {
01060 *     rx_log_error("boot", "Prior reset caused by IWDT timeout - check task health");
01061 * }
01062 * rx_iwdt_init(&iwdt_config);
01063 * @endcode
01064 *
01065 * @see rx_iwdt_feed() Feed the hardware watchdog to prevent reset
01066 * @see rx_iwdt_check_tasks() Monitor software task heartbeats
01067 *
01068 * @since Version 1.0.0
01069 *
01070 * @par NASA Power of 10 Compliance:
01071 * - Rule 5: 2 preconditions (safe before init, called before register clear), 2 postconditions
01072 */
01073 bool rx_iwdt_was_reset(void)
01074 {
01075     /* Read reset status register 2 for IWDT reset flag
01076     * Note: RSTSR2 is at a separate address from RSTSR0/1 */
01077     volatile uint8_t* const rstsr2_reg = rstsr2();
01078     const uint8_t      status      = *rstsr2_reg;
01079
01080     /* Check IWDT reset flag in RSTSR2 */
01081     return (bool)((status & k_rstsr2_iwdtrf) != 0);
01082 }
01083
01084 /**
01085 * @brief Check all registered tasks for software heartbeat timeout
01086 *
01087 * @details
01088 * Iterates over all active task slots and compares the elapsed tick count
01089 * since each task's last heartbeat against that task's configured timeout
01090 * (converted from milliseconds to ThreadX ticks). If any task has exceeded
01091 * its timeout, the task's timed_out flag is set, the task name is recorded
01092 * in status.last_failed_task, and the function returns k_rx_err_timeout.
01093 *
01094 * If task monitoring is disabled (config.enable_task_monitoring == false),
01095 * returns k_rx_ok immediately without checking any tasks.
01096 *
01097 * The caller (typically the watchdog task) should conditionally feed the
01098 * hardware IWDT only if this function returns k_rx_ok -- i.e., only refresh
01099 * the hardware watchdog when all software tasks are alive.
01100 *
01101 * Algorithm steps:
01102 * 1. Verify s_iwdt_state.initialized
01103 * 2. If !config.enable_task_monitoring, return k_rx_ok early
01104 * 3. Capture current_tick = internal_get_tick_count()
01105 * 4. For each active task slot (i < k_iwdt_max_tasks):
01106 *     a. Skip inactive slots
01107 *     b. Compute elapsed_ticks = current_tick - task->last_heartbeat_tick
01108 *     c. Compute timeout_in_ticks from task->timeout_ms and TX_TIMER_TICKS_PER_SECOND
01109 *     d. If elapsed_ticks > timeout_in_ticks: set timed_out=true, record task name, set any_timeout=true
01110 * 5. Return k_rx_err_timeout if any_timeout, else k_rx_ok
01111 *
01112 *
01113 * @pre rx_iwdt_init() must have been called successfully
01114 * @pre Registered tasks must be calling rx_iwdt_task_heartbeat() on schedule
01115 * @post Timed-out tasks have their timed_out flag set
01116 * @post status.last_failed_task contains the name of the most recently detected failure
01117 *
01118 * @note Not thread-safe; task heartbeat fields may be updated concurrently
01119 * @warning Do NOT feed the hardware IWDT (rx_iwdt_feed()) if this returns k_rx_err_timeout;
01120 *         allow the hardware watchdog to fire and reset the system
01121 *
01122 * @par Example:
01123 * @code
01124 * // Watchdog task main loop
01125 * while (1) {
01126 *     if (rx_iwdt_check_tasks() == k_rx_ok) {
01127 *         rx_iwdt_feed(); // Only feed if all tasks are alive
01128 *     } else {
01129 *         rx_log_error("wdt", "Task timeout detected -- allowing IWDT reset");
01130 *         // Do NOT feed the watchdog -- let hardware reset occur
01131 *     }
01132 *     tx_thread_sleep(k_watchdog_check_interval_ticks);
01133 * }
01134 * @endcode
01135 *
01136 * @see rx_iwdt_feed() Feed hardware watchdog (call only when this returns k_rx_ok)
01137 * @see rx_iwdt_task_heartbeat() Update per-task heartbeat from each monitored task
01138 * @see rx_iwdt_register_task() Register a task for monitoring

```

```

01139 *
01140 * @since Version 1.0.0
01141 *
01142 * @par NASA Power of 10 Compliance:
01143 * - Rule 5: 2 preconditions (initialized, tasks registered), 2 postconditions
01144 */
01145 rx_err_t rx_iwdt_check_tasks(void)
01146 {
01147     if (!s_iwdt_state.initialized) {
01148         return k_rx_err_not_initialized;
01149     }
01150
01151     if (!s_iwdt_state.config.enable_task_monitoring) {
01152         return k_rx_ok;
01153     }
01154
01155     const uint32_t current_tick = internal_get_tick_count();
01156     bool any_timeout = false;
01157
01158     /* Check each registered task */
01159     for (uint32_t i = 0; i < k_iwdt_max_tasks; i++) {
01160         if (!s_iwdt_state.tasks[i].active) {
01161             continue;
01162         }
01163
01164         /* Calculate elapsed time */
01165         const uint32_t elapsed_ticks = current_tick - s_iwdt_state.tasks[i].last_heartbeat_tick;
01166         const uint32_t timeout_in_ticks =
01167             (s_iwdt_state.tasks[i].timeout_ms * TX_TIMER_TICKS_PER_SECOND) / k_ms_per_second;
01168
01169         /* Check for timeout */
01170         if (elapsed_ticks > timeout_in_ticks) {
01171             s_iwdt_state.tasks[i].timed_out = true;
01172             any_timeout = true;
01173
01174             /* Record failed task */
01175             internal_safe_strcpy(s_iwdt_state.status.last_failed_task,
01176                                 s_iwdt_state.tasks[i].task_name,
01177                                 k_iwdt_task_name_len);
01178         }
01179     }
01180
01181     return (int)any_timeout ? k_rx_err_timeout : k_rx_ok;
01182 }

```

7.55 libs/rx_core/src/rx_log_uart.c File Reference

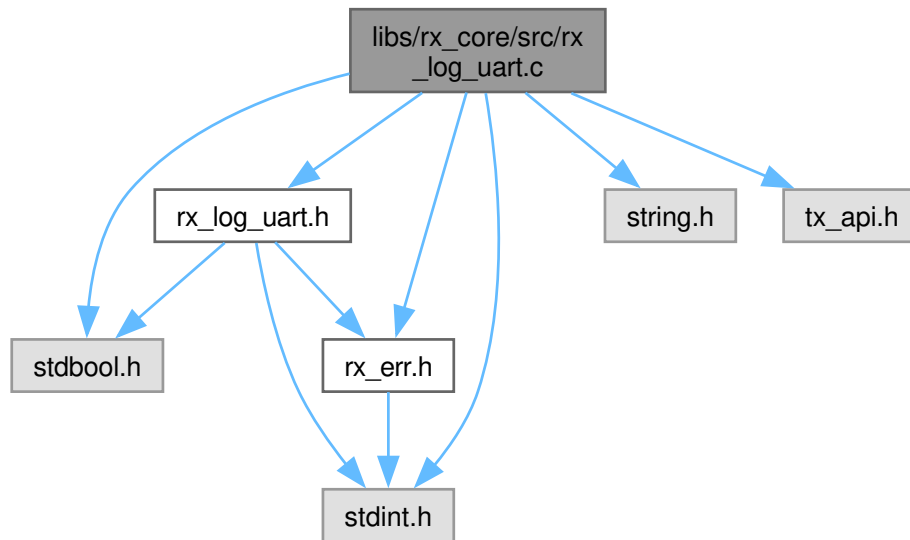
UART-framed log backend: ring buffer + drain API.

```

#include "rx_log_uart.h"
#include <stdbool.h>
#include <stdint.h>
#include <string.h>
#include "rx_err.h"
#include "tx_api.h"

```

Include dependency graph for rx_log_uart.c:



Enumerations

- enum `rx_log_uart_format_t` : `uint8_t` {
`k_int32_buf_len` = 12U , `k_uint32_buf_len` = 11U , `k_decimal_base` = 10U , `k_hex_max_digits`
 = 8U ,
`k_hex_buf_len` = 9U , `k_nibble_shift` = 4U , `k_nibble_mask` = 0x0FU }

Numeric formatting constants.

Functions

- static void `internal_append_byte` (`uint8_t` b)
 Append one byte to the ring under lock; drop on overflow.
- static void `internal_init_mutex_once` (void)
 Lazy mutex init (safe to call from any task context after kernel entry).
- void `rx_log_uart_putc` (char c)
 Append a single character to the log ring buffer (implementation).
- void `rx_log_uart_puts` (const char *str)
 Append a NUL-terminated string to the log ring buffer (implementation).
- void `rx_log_uart_putint` (int32_t value)
 Append a signed 32-bit decimal integer (implementation).
- void `rx_log_uart_putuint` (uint32_t value)
 Append an unsigned 32-bit decimal integer (implementation).
- void `rx_log_uart_puthex` (uint32_t value, uint8_t digits)
 Append a fixed-width uppercase hex value (implementation).
- uint32_t `rx_log_uart_pending_len` (void)
 Return the number of bytes currently queued (implementation).
- `rx_err_t` `rx_log_uart_drain` (uint8_t *out_buf, uint32_t max_len, uint32_t *out_len)
 Drain queued log bytes into caller's buffer (implementation).
- void `rx_log_uart_get_stats` (`rx_log_uart_stats_t` *stats)
 Snapshot the log-ring statistics (implementation).

Variables

- static uint8_t [s_ring](#) [[k_rx_log_uart_ring_size](#)]
Ring buffer storage (contiguous bytes, head wraps).
- static uint32_t [s_head](#)
Write index: next byte goes to `s_ring[s_head]`.
- static uint32_t [s_tail](#)
Read index: next byte to drain is `s_ring[s_tail]`.
- static uint32_t [s_count](#)
Number of occupied bytes in the ring (0..`k_rx_log_uart_ring_size`).
- static [rx_log_uart_stats_t](#) [s_stats](#)
Diagnostic counters.
- static TX_MUTEX [s_mutex](#)
ThreadX mutex (lazy init on first use).
- static bool [s_mutex_initialized](#)

7.55.1 Detailed Description

UART-framed log backend: ring buffer + drain API.

Producers call `rx_log_uart_putc` / `puts` / `putint` / `puthex` from any ThreadX task or ISR-adjacent context. Bytes are appended to a static ring buffer under a ThreadX mutex and the call returns without touching any hardware.

The single consumer (the communication task) calls [rx_log_uart_drain\(\)](#) on every poll tick. Drained bytes are wrapped in a `k_frame_type_log_message` (0x20) frame and transmitted via `rx_uart_comm` over SCI9 -> CY7C65213 -> `/dev/ttyACM0` on the Pi5 (CDC-ACM class device), where the gateway routes them to its own log sink.

No I/O happens on the producer path: even a burst of 1000 log lines can only cost a mutex, a memcpy, and a ring index update. Bytes that overflow the ring are counted and dropped – preferable to blocking real-time motor control code on serial line drain.

Author

Locked, Inc.

Date

2026-04-17

Copyright

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Since

Version 1.0.0

Definition in file [rx_log_uart.c](#).

7.55.2 Enumeration Type Documentation

7.55.2.1 rx_log_uart_format_t

enum [rx_log_uart_format_t](#) : uint8_t

Numeric formatting constants.

Enumerator

k_int32_buf_len	'-' + 10 digits + NUL
k_uint32_buf_len	10 digits + NUL
k_decimal_base	
k_hex_max_digits	
k_hex_buf_len	
k_nibble_shift	
k_nibble_mask	

Definition at line 58 of file [rx_log_uart.c](#).

```
00058      : uint8_t {
00059  k_int32_buf_len = 12U, /**< '-' + 10 digits + NUL */
00060  k_uint32_buf_len = 11U, /**< 10 digits + NUL */
00061  k_decimal_base  = 10U,
00062  k_hex_max_digits = 8U,
00063  k_hex_buf_len   = 9U,
00064  k_nibble_shift  = 4U,
00065  k_nibble_mask   = 0xFU,
00066 } rx_log_uart_format_t;
```

7.55.3 Function Documentation

7.55.3.1 internal_append_byte()

```
void internal_append_byte (
    uint8_t b) [static]
```

Append one byte to the ring under lock; drop on overflow.

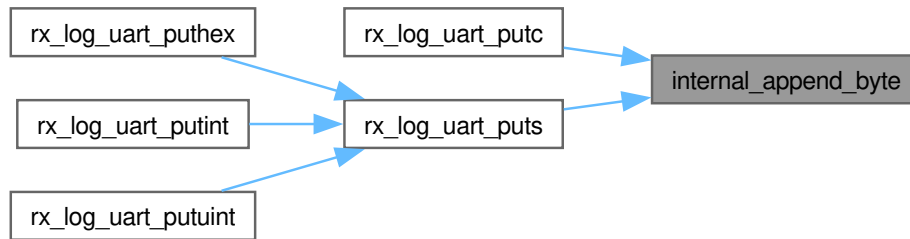
Definition at line 94 of file [rx_log_uart.c](#).

```
00095 {
00096  if (s_count >= k_rx_log_uart_ring_size) {
00097      s_stats.dropped_bytes += 1U;
00098      return;
00099  }
00100  s_ring[s_head] = b;
00101  s_head         = (s_head + 1U) % k_rx_log_uart_ring_size;
00102  s_count += 1U;
00103  s_stats.total_bytes += 1U;
00104  if (s_count > s_stats.high_water) {
00105      s_stats.high_water = s_count;
00106  }
00107 }
```

References [k_rx_log_uart_ring_size](#), [s_count](#), [s_head](#), [s_ring](#), and [s_stats](#).

Referenced by [rx_log_uart_putc\(\)](#), and [rx_log_uart_puts\(\)](#).

Here is the caller graph for this function:



7.55.3.2 `internal_init_mutex_once()`

```
void internal_init_mutex_once (
    void ) [static]
```

Lazy mutex init (safe to call from any task context after kernel entry).

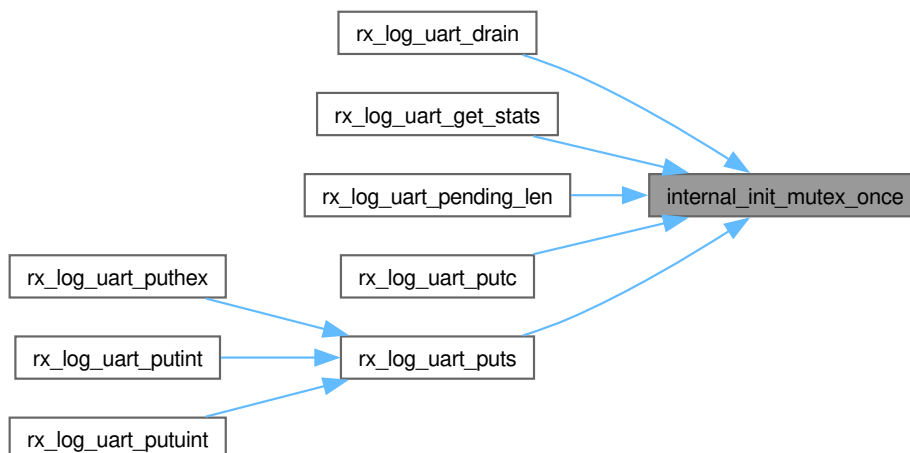
Definition at line 112 of file `rx_log_uart.c`.

```
00113 {
00114     if (!s_mutex_initialized) {
00115         (void)tx_mutex_create(&s_mutex, (CHAR*)"rx_log_uart", TX_NO_INHERIT);
00116         s_mutex_initialized = true;
00117     }
00118 }
```

References `s_mutex`, and `s_mutex_initialized`.

Referenced by `rx_log_uart_drain()`, `rx_log_uart_get_stats()`, `rx_log_uart_pending_len()`, `rx_log_uart_putc()`, and `rx_log_uart_puts()`.

Here is the caller graph for this function:



7.55.3.3 rx_log_uart_drain()

```

rx_err_t rx_log_uart_drain (
    uint8_t * out_buf,
    uint32_t max_len,
    uint32_t * out_len) [nodiscard]

```

Drain queued log bytes into caller's buffer (implementation).

Pull up to max_len bytes out of the ring buffer for transmission.

Single-consumer drain entry-point. Copies up to max_len bytes of queued log data from the ring tail into out_buf, returning the actual byte count via *out_len. If the unread region wraps the ring end, only the pre-wrap contiguous run is returned this call; the caller drains again to pick up the remainder. This bounds execution time per call (NASA Rule 2) and keeps the consumer loop simple.

Updates s_tail and s_count under the producer/consumer mutex, and accumulates drained_bytes into the stats block.

Note

Thread-safe via the internal mutex. Designed for a single consumer task (typically comm_task). Multiple concurrent consumers would share bytes incorrectly even though no corruption would occur.

Definition at line 296 of file rx_log_uart.c.

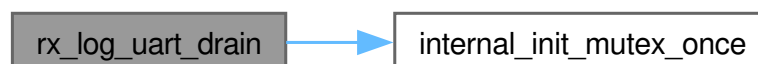
```

00297 {
00298     if ((out_buf == NULL) || (out_len == NULL) || (max_len == 0U)) {
00299         return k_rx_err_invalid_arg;
00300     }
00301
00302     internal_init_mutex_once();
00303     (void)tx_mutex_get(&s_mutex, TX_WAIT_FOREVER);
00304
00305     uint32_t to_copy = (s_count < max_len) ? s_count : max_len;
00306
00307     /* Prefer a single contiguous copy -- if the unread region wraps the ring
00308      * end, return only the pre-wrap portion this call. Caller drains again to
00309      * pick up the remainder. This keeps the consumer loop simple and bounds
00310      * execution time per drain call (NASA Rule 2). */
00311     const uint32_t contiguous = k_rx_log_uart_ring_size - s_tail;
00312     if (to_copy > contiguous) {
00313         to_copy = contiguous;
00314     }
00315
00316     if (to_copy > 0U) {
00317         /* NOLINTNEXTLINE(clang-analyzer-security.insecureAPI.DeprecatedOrUnsafeBufferHandling) */
00318         (void)memcpy(out_buf, &s_ring[s_tail], to_copy);
00319         s_tail = (s_tail + to_copy) % k_rx_log_uart_ring_size;
00320         s_count -= to_copy;
00321         s_stats.drained_bytes += to_copy;
00322     }
00323
00324     *out_len = to_copy;
00325
00326     (void)tx_mutex_put(&s_mutex);
00327     return k_rx_ok;
00328 }

```

References [internal_init_mutex_once\(\)](#), [k_rx_err_invalid_arg](#), [k_rx_log_uart_ring_size](#), [k_rx_ok](#), [s_count](#), [s_mutex](#), [s_ring](#), [s_stats](#), and [s_tail](#).

Here is the call graph for this function:



7.55.3.4 rx_log_uart_get_stats()

```
void rx_log_uart_get_stats (
    rx_log_uart_stats_t * stats)
```

Snapshot the log-ring statistics (implementation).

Snapshot diagnostic counters.

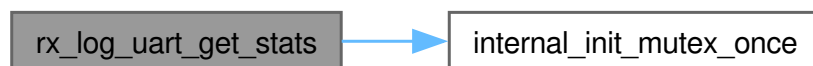
Copies the current `rx_log_uart_stats_t` value (drained_bytes, dropped_bytes, ring_high_water, etc.) under the producer/consumer mutex. A NULL output pointer is treated as a no-op for caller convenience.

Definition at line 338 of file `rx_log_uart.c`.

```
00339 {
00340     if (stats == NULL) {
00341         return;
00342     }
00343     internal_init_mutex_once();
00344     (void)tx_mutex_get(&s_mutex, TX_WAIT_FOREVER);
00345     *stats = s_stats;
00346     (void)tx_mutex_put(&s_mutex);
00347 }
```

References `internal_init_mutex_once()`, `s_mutex`, and `s_stats`.

Here is the call graph for this function:



7.55.3.5 rx_log_uart_pending_len()

```
uint32_t rx_log_uart_pending_len (
    void )
```

Return the number of bytes currently queued (implementation).

Query number of bytes currently queued in the ring buffer.

Acquires the producer/consumer mutex, snapshots `s_count`, and releases the mutex before returning. This gives a momentarily-consistent count suitable for deciding whether the consumer needs to drain on this tick. The value can change between read and use; treat it as a hint.

Definition at line 268 of file `rx_log_uart.c`.

```
00269 {
00270     internal_init_mutex_once();
00271     (void)tx_mutex_get(&s_mutex, TX_WAIT_FOREVER);
00272     const uint32_t n = s_count;
00273     (void)tx_mutex_put(&s_mutex);
00274     return n;
00275 }
```

References `internal_init_mutex_once()`, `s_count`, and `s_mutex`.

Here is the call graph for this function:



7.55.3.6 rx_log_uart_putc()

```
void rx_log_uart_putc (
    char c)
```

Append a single character to the log ring buffer (implementation).

Append a single character to the log ring buffer.

Lazily initializes the producer/consumer mutex on first call, then appends a single byte to the ring buffer. If the ring is full the oldest byte is dropped (silent overflow); the dropped-bytes counter is updated through [internal_append_byte\(\)](#).

Note

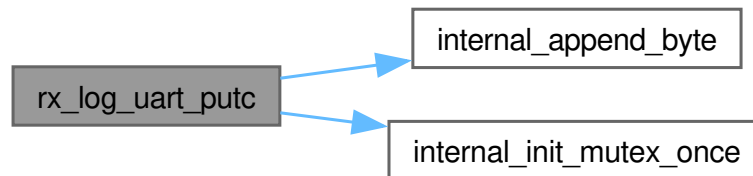
Thread-safe via an internal ThreadX mutex; bounded blocking on contention. NOT safe from ISR context.

Definition at line 136 of file [rx_log_uart.c](#).

```
00137 {
00138     internal_init_mutex_once();
00139     (void)tx_mutex_get(&s_mutex, TX_WAIT_FOREVER);
00140     internal_append_byte((uint8_t)c);
00141     (void)tx_mutex_put(&s_mutex);
00142 }
```

References [internal_append_byte\(\)](#), [internal_init_mutex_once\(\)](#), and [s_mutex](#).

Here is the call graph for this function:



7.55.3.7 rx_log_uart_puthex()

```
void rx_log_uart_puthex (
    uint32_t value,
    uint8_t digits)
```

Append a fixed-width uppercase hex value (implementation).

Append a value as zero-padded hexadecimal ASCII to the log ring buffer.

Formats value as digits hex characters (uppercase A-F, zero padded on the left) into a stack-local buffer, then forwards to [rx_log_uart_puts\(\)](#). If digits is 0 it is silently promoted to 1; if it exceeds [k_hex_max_digits](#) it is clamped to that maximum to fit the local buffer.

Definition at line 233 of file [rx_log_uart.c](#).

```
00234 {
00235     if (digits == 0U) {
00236         digits = 1U;
00237     } else if (digits > k_hex_max_digits) {
00238         digits = k_hex_max_digits;
```

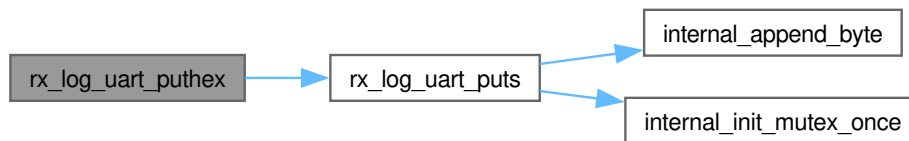
```

00239 }
00240
00241 char buf[k_hex_buf_len];
00242 char* p = &buf[digits];
00243 *p = '\0';
00244
00245 for (uint8_t i = 0U; i < digits; ++i) {
00246     --p;
00247     const uint8_t nibble = (uint8_t)(value & (uint32_t)k_nibble_mask);
00248     *p = (char)((nibble < k_decimal_base) ? ('0' + nibble) : ('A' + (nibble - k_decimal_base)));
00249     value >>= (uint32_t)k_nibble_shift;
00250 }
00251
00252 rx_log_uart_puts(buf);
00253 }

```

References [k_decimal_base](#), [k_hex_buf_len](#), [k_hex_max_digits](#), [k_nibble_mask](#), [k_nibble_shift](#), and [rx_log_uart_puts\(\)](#).

Here is the call graph for this function:



7.55.3.8 rx_log_uart_putint()

```

void rx_log_uart_putint (
    int32_t value)

```

Append a signed 32-bit decimal integer (implementation).

Append a signed integer as decimal ASCII to the log ring buffer.

Converts value to its decimal string representation in a stack-local buffer (no heap allocation), then forwards to [rx_log_uart_puts\(\)](#). Negative numbers are prefixed with '-'. Conversion is unsigned after sign extraction so INT32_MIN does not wrap.

Definition at line 176 of file [rx_log_uart.c](#).

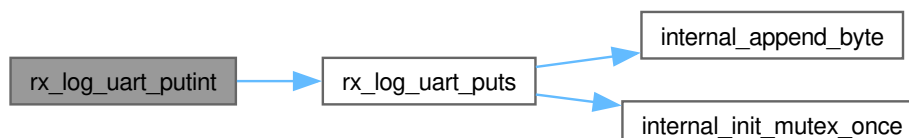
```

00177 {
00178     char buf[k_int32_buf_len];
00179
00180     char* p = &buf[sizeof(buf) - 1U];
00181     *p = '\0';
00182
00183     const bool negative = (bool)(value < 0);
00184     uint32_t abs_value = (int)negative ? (uint32_t)(-value) : (uint32_t)value;
00185
00186     do {
00187         --p;
00188         *p = (char)('0' + (abs_value % k_decimal_base));
00189         abs_value /= k_decimal_base;
00190     } while (abs_value > 0U);
00191
00192     if (negative) {
00193         --p;
00194         *p = '-';
00195     }
00196
00197     rx_log_uart_puts(p);
00198 }

```

References [k_decimal_base](#), [k_int32_buf_len](#), and [rx_log_uart_puts\(\)](#).

Here is the call graph for this function:



7.55.3.9 rx_log_uart_puts()

```
void rx_log_uart_puts (
    const char * str)
```

Append a NUL-terminated string to the log ring buffer (implementation).

Append a null-terminated string to the log ring buffer.

Iterates over the input string and appends each byte to the ring buffer under a single lock (one mutex acquire per call, not per byte). This keeps multi-byte log records atomic with respect to other producers and to the drain consumer. A NULL input pointer is treated as a no-op for caller convenience and does not require explicit guards in caller code.

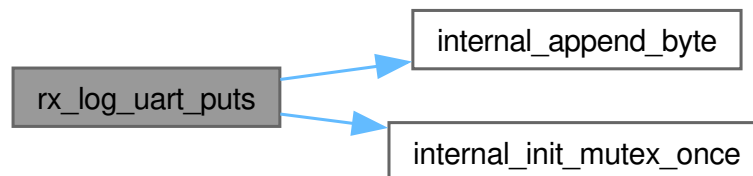
Definition at line 154 of file [rx_log_uart.c](#).

```
00155 {
00156     if (str == NULL) {
00157         return;
00158     }
00159     internal_init_mutex_once();
00160     (void)tx_mutex_get(&s_mutex, TX_WAIT_FOREVER);
00161     for (const char* p = str; *p != '\0'; ++p) {
00162         internal_append_byte((uint8_t)*p);
00163     }
00164     (void)tx_mutex_put(&s_mutex);
00165 }
```

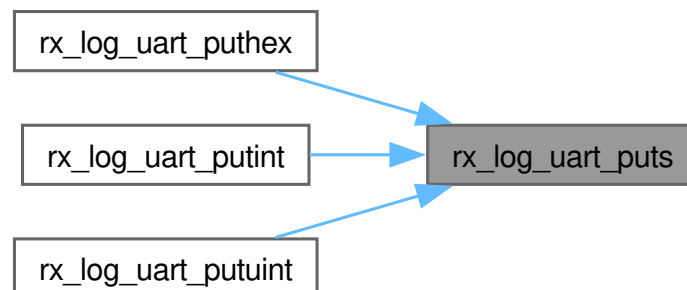
References [internal_append_byte\(\)](#), [internal_init_mutex_once\(\)](#), and [s_mutex](#).

Referenced by [rx_log_uart_puthex\(\)](#), [rx_log_uart_putint\(\)](#), and [rx_log_uart_putuint\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



7.55.3.10 rx_log_uart_putuint()

```
void rx_log_uart_putuint (
    uint32_t value)
```

Append an unsigned 32-bit decimal integer (implementation).

Append an unsigned integer as decimal ASCII to the log ring buffer.

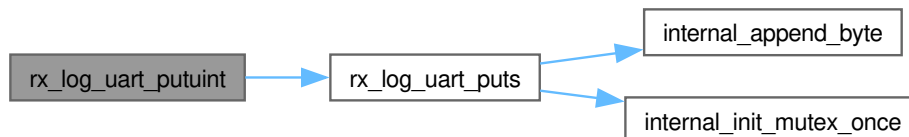
Converts value to a decimal string in a stack-local buffer using base-10 division, then forwards to [rx_log_uart_puts\(\)](#). Output never includes a leading sign.

Definition at line 208 of file [rx_log_uart.c](#).

```
00209 {
00210     char buf[k_uint32_buf_len];
00211
00212     char* p = &buf[sizeof(buf) - 1U];
00213     *p = '\0';
00214
00215     do {
00216         --p;
00217         *p = (char)('0' + (value % k_decimal_base));
00218         value /= k_decimal_base;
00219     } while (value > 0U);
00220
00221     rx_log_uart_puts(p);
00222 }
```

References [k_decimal_base](#), [k_uint32_buf_len](#), and [rx_log_uart_puts\(\)](#).

Here is the call graph for this function:



7.55.4 Variable Documentation

7.55.4.1 s_count

```
uint32_t s_count [static]
```

Number of occupied bytes in the ring (0..[k_rx_log_uart_ring_size](#)).

Definition at line 78 of file [rx_log_uart.c](#).

Referenced by [internal_append_byte\(\)](#), [rx_log_uart_drain\(\)](#), and [rx_log_uart_pending_len\(\)](#).

7.55.4.2 s_head

```
uint32_t s_head [static]
```

Write index: next byte goes to `s_ring[s_head]`.

Definition at line 72 of file [rx_log_uart.c](#).

7.55.4.3 s_mutex

TX_MUTEX s_mutex [static]

ThreadX mutex (lazy init on first use).

Definition at line 84 of file [rx_log_uart.c](#).

Referenced by [internal_init_mutex_once\(\)](#), [rx_log_uart_drain\(\)](#), [rx_log_uart_get_stats\(\)](#), [rx_log_uart_pending_len\(\)](#), [rx_log_uart_putc\(\)](#), and [rx_log_uart_puts\(\)](#).

7.55.4.4 s_mutex_initialized

bool s_mutex_initialized [static]

Definition at line 85 of file [rx_log_uart.c](#).

Referenced by [internal_init_mutex_once\(\)](#).

7.55.4.5 s_ring

uint8_t s_ring[k_rx_log_uart_ring_size] [static]

Ring buffer storage (contiguous bytes, head wraps).

Definition at line 69 of file [rx_log_uart.c](#).

7.55.4.6 s_stats

[rx_log_uart_stats_t](#) s_stats [static]

Diagnostic counters.

Definition at line 81 of file [rx_log_uart.c](#).

Referenced by [internal_append_byte\(\)](#), [rx_log_uart_drain\(\)](#), and [rx_log_uart_get_stats\(\)](#).

7.55.4.7 s_tail

uint32_t s_tail [static]

Read index: next byte to drain is s_ring[s_tail].

Definition at line 75 of file [rx_log_uart.c](#).

7.56 rx_log_uart.c

[Go to the documentation of this file.](#)

```

00001 /**
00002  * @file rx_log_uart.c
00003  * @brief UART-framed log backend: ring buffer + drain API
00004  *
00005  * @details
00006  * Producers call rx_log_uart_putc / puts / putint / puthex from any ThreadX
00007  * task or ISR-adjacent context. Bytes are appended to a static ring buffer
00008  * under a ThreadX mutex and the call returns without touching any hardware.
00009  *
00010  * The single consumer (the communication task) calls rx_log_uart_drain() on
00011  * every poll tick. Drained bytes are wrapped in a k_frame_type_log_message
00012  * (0x20) frame and transmitted via rx_uart_comm over SCI9 -> CY7C65213 ->
00013  * /dev/ttyACM0 on the Pi5 (CDC-ACM class device), where the gateway routes
00014  * them to its own log sink.
00015  *
00016  * No I/O happens on the producer path: even a burst of 1000 log lines can only
00017  * cost a mutex, a memcpy, and a ring index update. Bytes that overflow the
00018  * ring are counted and dropped -- preferable to blocking real-time motor
00019  * control code on serial line drain.
00020  *
00021  * @author Locked, Inc.
00022  * @date 2026-04-17
00023  * @copyright Copyright (c) 2026 Locked Inc.
00024  * SPDX-License-Identifier: MIT
00025  * @since Version 1.0.0
00026  */
00027
00028 #include "rx_log_uart.h"
00029
00030 #include <stdbool.h>
00031 #include <stdint.h>
00032 #include <string.h>
00033
00034 #include "rx_err.h"
00035 #include "tx_api.h"
00036
00037 #ifdef UNIT_TEST
00038 /**
00039  * @brief Test-only fault injection flag for RX_ASSERT_POST coverage
00040  *
00041  * When set to @c true, every RX_ASSERT_POST fires regardless of its condition,
00042  * making the failure branch reachable for branch coverage testing. Declared
00043  * extern in @ref rx_check.h. The definition previously lived in mock_rx_log.c,
00044  * which was deleted along with the USB0 log backend; it now lives here so
00045  * UNIT_TEST builds continue to link.
00046  */
00047 bool g_rx_assert_post_force_fail = false;
00048 #endif
00049
00050 /*
00051  * Static state
00052  */
00053
00054 /**
00055  * @enum rx_log_uart_format_t
00056  * @brief Numeric formatting constants
00057  */
00058 typedef enum : uint8_t {
00059     k_int32_buf_len = 12U, /**< '.' + 10 digits + NUL */
00060     k_uint32_buf_len = 11U, /**< 10 digits + NUL */
00061     k_decimal_base = 10U,
00062     k_hex_max_digits = 8U,
00063     k_hex_buf_len = 9U,
00064     k_nibble_shift = 4U,
00065     k_nibble_mask = 0x0FU,
00066 } rx_log_uart_format_t;
00067
00068 /** @brief Ring buffer storage (contiguous bytes, head wraps) */
00069 static uint8_t s_ring[k_rx_log_uart_ring_size];
00070
00071 /** @brief Write index: next byte goes to s_ring[s_head] */
00072 static uint32_t s_head;
00073
00074 /** @brief Read index: next byte to drain is s_ring[s_tail] */
00075 static uint32_t s_tail;
00076
00077 /** @brief Number of occupied bytes in the ring (0..k_rx_log_uart_ring_size) */
00078 static uint32_t s_count;
00079

```

```

00080 /** @brief Diagnostic counters */
00081 static rx_log_uart_stats_t s_stats;
00082
00083 /** @brief ThreadX mutex (lazy init on first use) */
00084 static TX_MUTEX s_mutex;
00085 static bool s_mutex_initialized;
00086
00087 /*
=====
00088 * Internal helpers (mutex must be held by caller)
00089 *
=====
*/
00090
00091 /**
00092 * @brief Append one byte to the ring under lock; drop on overflow
00093 */
00094 static void internal_append_byte(uint8_t b)
00095 {
00096     if (s_count >= k_rx_log_uart_ring_size) {
00097         s_stats.dropped_bytes += 1U;
00098         return;
00099     }
00100     s_ring[s_head] = b;
00101     s_head = (s_head + 1U) % k_rx_log_uart_ring_size;
00102     s_count += 1U;
00103     s_stats.total_bytes += 1U;
00104     if (s_count > s_stats.high_water) {
00105         s_stats.high_water = s_count;
00106     }
00107 }
00108
00109 /**
00110 * @brief Lazy mutex init (safe to call from any task context after kernel entry)
00111 */
00112 static void internal_init_mutex_once(void)
00113 {
00114     if (!s_mutex_initialized) {
00115         (void)tx_mutex_create(&s_mutex, (CHAR*)"rx_log_uart", TX_NO_INHERIT);
00116         s_mutex_initialized = true;
00117     }
00118 }
00119
00120 /*
=====
00121 * Public API -- producers
00122 *
=====
*/
00123
00124 /**
00125 * @brief Append a single character to the log ring buffer (implementation)
00126 *
00127 * @details
00128 * Lazily initializes the producer/consumer mutex on first call, then appends
00129 * a single byte to the ring buffer. If the ring is full the oldest byte is
00130 * dropped (silent overflow); the dropped-bytes counter is updated through
00131 * internal_append_byte().
00132 *
00133 * @note Thread-safe via an internal ThreadX mutex; bounded blocking on
00134 * contention. NOT safe from ISR context.
00135 */
00136 void rx_log_uart_putc(char c)
00137 {
00138     internal_init_mutex_once();
00139     (void)tx_mutex_get(&s_mutex, TX_WAIT_FOREVER);
00140     internal_append_byte((uint8_t)c);
00141     (void)tx_mutex_put(&s_mutex);
00142 }
00143
00144 /**
00145 * @brief Append a NUL-terminated string to the log ring buffer (implementation)
00146 *
00147 * @details
00148 * Iterates over the input string and appends each byte to the ring buffer
00149 * under a single lock (one mutex acquire per call, not per byte). This
00150 * keeps multi-byte log records atomic with respect to other producers and
00151 * to the drain consumer. A NULL input pointer is treated as a no-op for
00152 * caller convenience and does not require explicit guards in caller code.
00153 */
00154 void rx_log_uart_puts(const char* str)
00155 {
00156     if (str == NULL) {
00157         return;
00158     }
00159     internal_init_mutex_once();
00160     (void)tx_mutex_get(&s_mutex, TX_WAIT_FOREVER);

```

```

00161 for (const char* p = str; *p != '\0'; ++p) {
00162     internal_append_byte((uint8_t)*p);
00163 }
00164 (void)tx_mutex_put(&s_mutex);
00165 }
00166
00167 /**
00168  * @brief Append a signed 32-bit decimal integer (implementation)
00169  *
00170  * @details
00171  * Converts value to its decimal string representation in a stack-local
00172  * buffer (no heap allocation), then forwards to rx_log_uart_puts().
00173  * Negative numbers are prefixed with '-'. Conversion is unsigned after
00174  * sign extraction so INT32_MIN does not wrap.
00175  */
00176 void rx_log_uart_putint(int32_t value)
00177 {
00178     char buf[k_int32_buf_len];
00179
00180     char* p = &buf[sizeof(buf) - 1U];
00181     *p = '\0';
00182
00183     const bool negative = (bool)(value < 0);
00184     uint32_t abs_value = (int)negative ? (uint32_t)(-value) : (uint32_t)value;
00185
00186     do {
00187         --p;
00188         *p = (char)('0' + (abs_value % k_decimal_base));
00189         abs_value /= k_decimal_base;
00190     } while (abs_value > 0U);
00191
00192     if (negative) {
00193         --p;
00194         *p = '-';
00195     }
00196
00197     rx_log_uart_puts(p);
00198 }
00199
00200 /**
00201  * @brief Append an unsigned 32-bit decimal integer (implementation)
00202  *
00203  * @details
00204  * Converts value to a decimal string in a stack-local buffer using base-10
00205  * division, then forwards to rx_log_uart_puts(). Output never includes a
00206  * leading sign.
00207  */
00208 void rx_log_uart_putuint(uint32_t value)
00209 {
00210     char buf[k_uint32_buf_len];
00211
00212     char* p = &buf[sizeof(buf) - 1U];
00213     *p = '\0';
00214
00215     do {
00216         --p;
00217         *p = (char)('0' + (value % k_decimal_base));
00218         value /= k_decimal_base;
00219     } while (value > 0U);
00220
00221     rx_log_uart_puts(p);
00222 }
00223
00224 /**
00225  * @brief Append a fixed-width uppercase hex value (implementation)
00226  *
00227  * @details
00228  * Formats value as 'digits' hex characters (uppercase A-F, zero padded on
00229  * the left) into a stack-local buffer, then forwards to rx_log_uart_puts().
00230  * If digits is 0 it is silently promoted to 1; if it exceeds
00231  * k_hex_max_digits it is clamped to that maximum to fit the local buffer.
00232  */
00233 void rx_log_uart_puthex(uint32_t value, uint8_t digits)
00234 {
00235     if (digits == 0U) {
00236         digits = 1U;
00237     } else if (digits > k_hex_max_digits) {
00238         digits = k_hex_max_digits;
00239     }
00240
00241     char buf[k_hex_buf_len];
00242     char* p = &buf[digits];
00243     *p = '\0';
00244
00245     for (uint8_t i = 0U; i < digits; ++i) {
00246         --p;
00247         const uint8_t nibble = (uint8_t)(value & (uint32_t)k_nibble_mask);

```



```

00248     *p = (char)((nibble < k_decimal_base) ? ('0' + nibble) : ('A' + (nibble - k_decimal_base)));
00249     value »= (uint32_t)k_nibble_shift;
00250 }
00251
00252 rx_log_uart_puts(buf);
00253 }
00254
00255 /*
=====
00256 * Public API -- consumer (comm_task)
00257 *
=====
*/
00258
00259 /**
00260 * @brief Return the number of bytes currently queued (implementation)
00261 *
00262 * @details
00263 * Acquires the producer/consumer mutex, snapshots s_count, and releases
00264 * the mutex before returning. This gives a momentarily-consistent count
00265 * suitable for deciding whether the consumer needs to drain on this tick.
00266 * The value can change between read and use; treat it as a hint.
00267 */
00268 uint32_t rx_log_uart_pending_len(void)
00269 {
00270     internal_init_mutex_once();
00271     (void)tx_mutex_get(&s_mutex, TX_WAIT_FOREVER);
00272     const uint32_t n = s_count;
00273     (void)tx_mutex_put(&s_mutex);
00274     return n;
00275 }
00276
00277 /**
00278 * @brief Drain queued log bytes into caller's buffer (implementation)
00279 *
00280 * @details
00281 * Single-consumer drain entry-point. Copies up to max_len bytes of queued
00282 * log data from the ring tail into out_buf, returning the actual byte count
00283 * via *out_len. If the unread region wraps the ring end, only the
00284 * pre-wrap contiguous run is returned this call; the caller drains again
00285 * to pick up the remainder. This bounds execution time per call (NASA
00286 * Rule 2) and keeps the consumer loop simple.
00287 *
00288 * Updates s_tail and s_count under the producer/consumer mutex, and
00289 * accumulates drained_bytes into the stats block.
00290 *
00291 * @note Thread-safe via the internal mutex. Designed for a single
00292 *       consumer task (typically comm_task). Multiple concurrent
00293 *       consumers would share bytes incorrectly even though no corruption
00294 *       would occur.
00295 */
00296 rx_err_t rx_log_uart_drain(uint8_t* out_buf, uint32_t max_len, uint32_t* out_len)
00297 {
00298     if ((out_buf == NULL) || (out_len == NULL) || (max_len == 0U)) {
00299         return k_rx_err_invalid_arg;
00300     }
00301
00302     internal_init_mutex_once();
00303     (void)tx_mutex_get(&s_mutex, TX_WAIT_FOREVER);
00304
00305     uint32_t to_copy = (s_count < max_len) ? s_count : max_len;
00306
00307     /* Prefer a single contiguous copy -- if the unread region wraps the ring
00308      * end, return only the pre-wrap portion this call. Caller drains again to
00309      * pick up the remainder. This keeps the consumer loop simple and bounds
00310      * execution time per drain call (NASA Rule 2). */
00311     const uint32_t contiguous = k_rx_log_uart_ring_size - s_tail;
00312     if (to_copy > contiguous) {
00313         to_copy = contiguous;
00314     }
00315
00316     if (to_copy > 0U) {
00317         /* NOLINTNEXTLINE(clang-analyzer-security.insecureAPI.DeprecatedOrUnsafeBufferHandling) */
00318         (void)memcpy(out_buf, &s_ring[s_tail], to_copy);
00319         s_tail = (s_tail + to_copy) % k_rx_log_uart_ring_size;
00320         s_count -= to_copy;
00321         s_stats.drained_bytes += to_copy;
00322     }
00323
00324     *out_len = to_copy;
00325
00326     (void)tx_mutex_put(&s_mutex);
00327     return k_rx_ok;
00328 }
00329
00330 /**
00331 * @brief Snapshot the log-ring statistics (implementation)

```

```

00332 *
00333 * @details
00334 * Copies the current rx_log_uart_stats_t value (drained_bytes,
00335 * dropped_bytes, ring_high_water, etc.) under the producer/consumer mutex.
00336 * A NULL output pointer is treated as a no-op for caller convenience.
00337 */
00338 void rx_log_uart_get_stats(rx_log_uart_stats_t* stats)
00339 {
00340     if (stats == NULL) {
00341         return;
00342     }
00343     internal_init_mutex_once();
00344     (void)tx_mutex_get(&s_mutex, TX_WAIT_FOREVER);
00345     *stats = s_stats;
00346     (void)tx_mutex_put(&s_mutex);
00347 }
00348
00349 #ifdef UNIT_TEST
00350 void rx_log_uart_test_reset_state(void)
00351 {
00352     s_head          = 0U;
00353     s_tail          = 0U;
00354     s_count         = 0U;
00355     s_stats         = (rx_log_uart_stats_t){0U, 0U, 0U, 0U};
00356     s_mutex_initialized = false;
00357 }
00358 #endif

```

7.57 libs/rx_core/src/rx_pin_validator.c File Reference

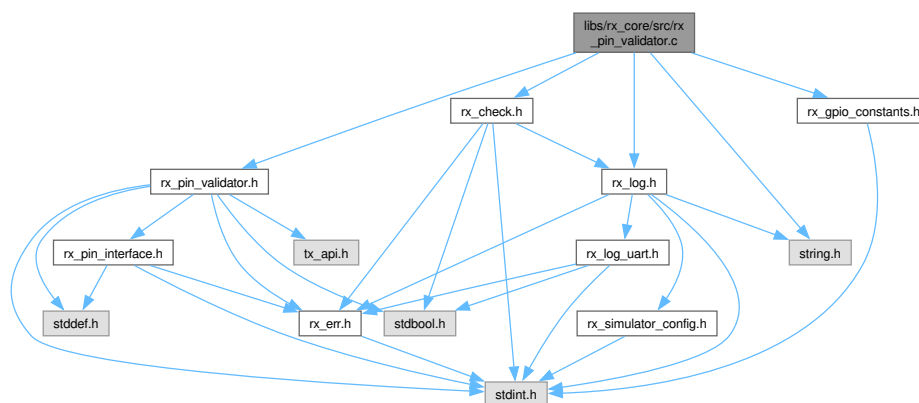
Pin Validator Concrete Implementation - GPIO Pin Tracking and Conflict Prevention.

```

#include "rx_pin_validator.h"
#include <string.h>
#include "rx_check.h"
#include "rx_gpio_constants.h"
#include "rx_log.h"

```

Include dependency graph for rx_pin_validator.c:



Functions

- static void [internal_safe_copy](#) (char *dst, const char *src, const uint32_t max_len)
Copy a string safely without strncpy.
- static uint8_t [internal_port_to_index](#) (const uint8_t port)
Convert port number to internal array index.
- static [rx_err_t](#) [internal_validate_port](#) (const uint8_t port)
Validate port number against RX72N hardware limits.
- static [rx_err_t](#) [internal_validate_pin](#) (const uint8_t pin)

- Validate pin number against RX72N per-port limits.
- static [rx_err_t impl_validate_pin](#) (void *ctx, const uint8_t port, const uint8_t pin)
 - Validate pin - Interface implementation for rx_pin_interface_t::validate_pin.
- static [rx_err_t impl_reserve_pin](#) (void *ctx, const uint8_t port, const uint8_t pin, const char *function)
 - Reserve pin - Interface implementation for rx_pin_interface_t::reserve_pin.
- static [rx_err_t impl_release_pin](#) (void *ctx, const uint8_t port, const uint8_t pin)
 - Release pin - Interface implementation for rx_pin_interface_t::release_pin.
- static bool [impl_is_pin_reserved](#) (void *ctx, const uint8_t port, const uint8_t pin)
 - Check if pin is reserved - Interface implementation for rx_pin_interface_t::is_pin_reserved.
- static [rx_err_t impl_get_pin_function](#) (void *ctx, const uint8_t port, const uint8_t pin, char *function_out, const uint32_t function_len)
 - Get pin function - Interface implementation for rx_pin_interface_t::get_pin_function.
- static [rx_err_t impl_clear_all_reservations](#) (void *ctx)
 - Clear all reservations - Interface implementation for rx_pin_interface_t::clear_all_reservations.
- [rx_err_t pin_validator_init](#) ([pin_validator_t](#) *validator)
 - Initialize pin validator for tracking GPIO pin reservations.
- [rx_err_t pin_validator_get_interface](#) ([rx_pin_interface_t](#) *iface, [pin_validator_t](#) *validator)
 - Extract abstract interface from concrete pin validator (Dependency Inversion).
- [rx_err_t pin_validator_deinit](#) ([pin_validator_t](#) *validator)
 - Deinitialize pin validator and release resources.

Variables

- static const char *const [s_tag](#) = "PIN_VALIDATOR"
 - Log tag for pin validator module.
- static const [pin_validator_t](#) [s_zero_validator](#) = {}
 - Zero-initialized pin validator template for stack-safe initialization.

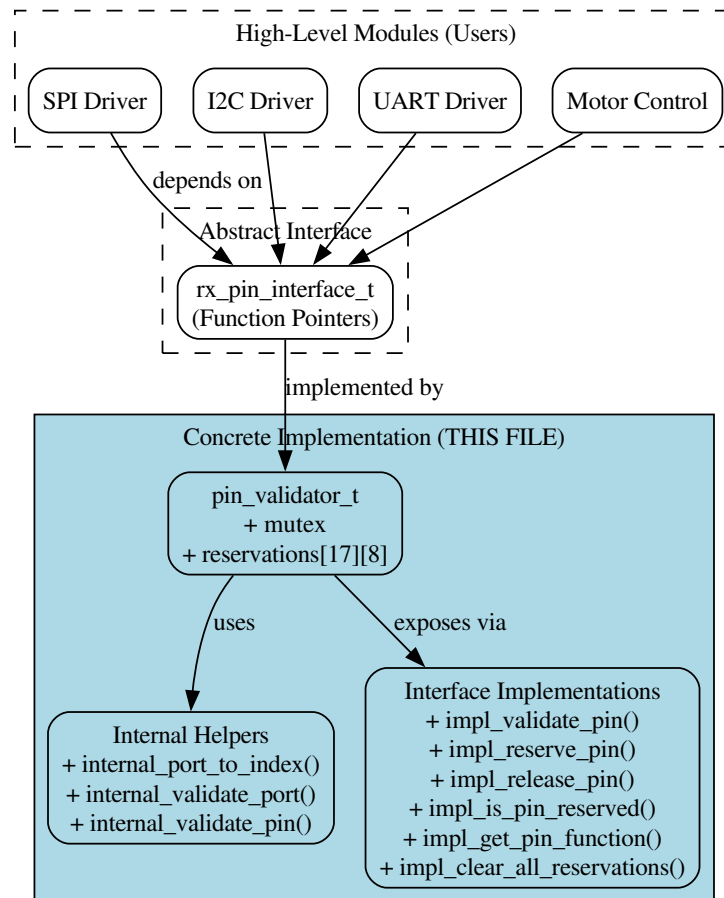
7.57.1 Detailed Description

Pin Validator Concrete Implementation - GPIO Pin Tracking and Conflict Prevention.

7.57.1.1 Overview

This file provides the concrete implementation of the [rx_pin_interface_t](#) abstract interface for GPIO pin validation and reservation tracking. It is the heart of the STAR project's pin conflict prevention system.

7.57.1.2 System Architecture Context



7.57.1.3 Implementation Design

7.57.1.3.1 Port Mapping Strategy

The RX72N has 17 GPIO ports with non-contiguous numbering:

- Decimal ports: 0-9 (indices 0-9)
- Hex ports: A-G (0xA-0x10, mapped to indices 10-16)

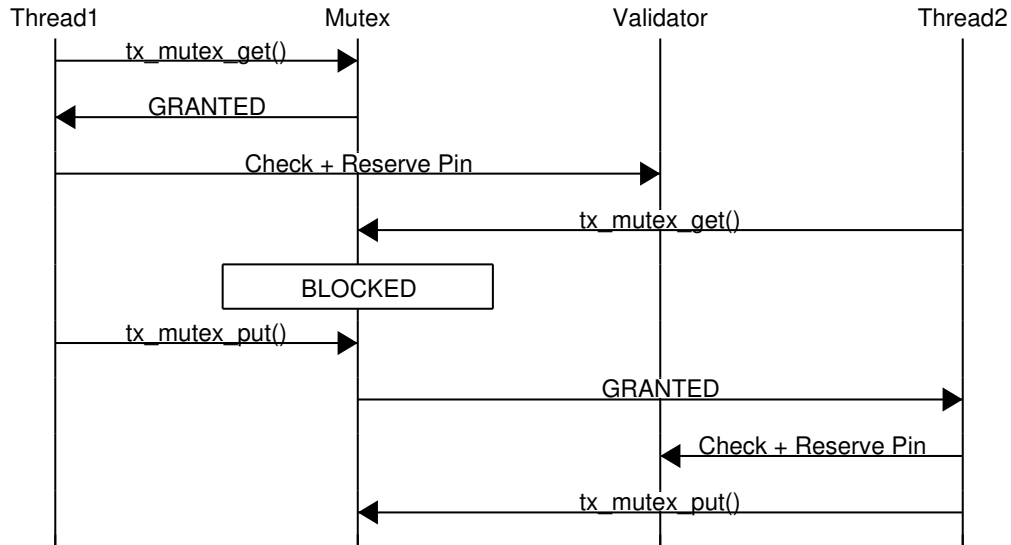
The `internal_port_to_index()` function converts port numbers to array indices:

```

Port 0-9 -> Index 0-9 (direct mapping)
Port 0xA -> Index 10 (A = 10)
Port 0xB -> Index 11 (B = 11)
...
Port 0x10 -> Index 16 (G = 16)
  
```

7.57.1.3.2 Thread Safety Model

All read-modify-write operations on the reservation table are protected by a ThreadX mutex:



7.57.1.3.3 Error Propagation

All functions follow a consistent error handling pattern:

1. Validate NULL pointers first (return `k_rx_err_null_ptr`)
2. Validate port/pin ranges (return `k_rx_err_gpio_invalid_port/pin`)
3. Acquire mutex (return `k_rx_err_rtos_mutex` on failure)
4. Perform operation (may return `k_rx_err_gpio_conflict`, `k_rx_err_invalid_state`)
5. Release mutex
6. Return success (`k_rx_ok`)

7.57.1.4 Performance Characteristics

Operation	Time Complexity	Typical Latency	Notes
<code>validate_pin</code>	$O(1)$	~100 ns	No mutex, direct lookup
<code>reserve_pin</code>	$O(1)$	~1-5 us	Mutex + strcpy
<code>release_pin</code>	$O(1)$	~1-3 us	Mutex + clear
<code>is_reserved</code>	$O(1)$	~1-2 us	Mutex + bool read
<code>get_function</code>	$O(1)$	~2-5 us	Mutex + strcpy
<code>clear_all</code>	$O(nxm)$	~50-100 us	17 ports x 8 pins

7.57.1.5 Memory Usage

Section	Size	Contents
.text	~1.5 KB	All function code
.rodata	~200 bytes	Log strings
.bss	0 bytes	No module-level state (validator is caller-owned)

7.57.1.6 NASA Power of 10 Compliance

Rule	Status	Implementation
1. Simplify control flow	[OK]	No goto, setjmp, recursion
2. Fixed loop bounds	[OK]	clear_all uses k_pin_validator_max_* constants
3. No dynamic memory	[OK]	All structures caller-owned
4. Functions < 60 lines	[OK]	All functions under 50 lines
5. Use assertions	[OK]	RX_CHECK_NULL_PTR in all public functions
6. Data at smallest scope	[OK]	Local variables, static helpers
7. Check return values	[OK]	All ThreadX calls checked
8. Limit preprocessor	[OK]	Only includes
9. Restrict pointers	[WARN]	Function pointers for DIP
10. Compile warnings	[OK]	-Wall -Wextra -Werror clean

7.57.1.7 SOLID Principles

Principle	Implementation
Single Responsibility	File handles ONLY pin validation/reservation
Open/Closed	Interface stable, implementation can change
Liskov Substitution	Can swap with mock for testing
Interface Segregation	Implements exactly rx_pin_interface_t (no extras)
Dependency Inversion	High-level code depends on interface, not this file

See also

[rx_pin_validator.h](#) Public API and type definitions

[rx_pin_interface.h](#) Abstract interface definition

[rx_infrastructure.c](#) Global validator instantiation

[rx_gpio_constants.h](#) GPIO port/pin constants

Author

Locked, Inc.

Date

2026-01-01

Copyright

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Since

Version 1.0.0

Definition in file [rx_pin_validator.c](#).

7.57.2 Function Documentation

7.57.2.1 impl_clear_all_reservations()

```
rx_err_t impl_clear_all_reservations (  
    void * ctx) [static]
```

Clear all reservations - Interface implementation for rx_pin_interface_t::clear_all_reservations.

Releases ALL GPIO pin reservations at once, resetting the validator to its initial state. Primarily used for:

- Unit test teardown (clean state between tests)
- System reset/reboot preparation
- Emergency fault recovery

7.57.2.2 Algorithm

1. Validate context pointer
2. Acquire mutex
3. Iterate all ports (0 to k_pin_validator_max_ports-1)
4. Iterate all pins (0 to k_pin_validator_max_pins-1)
5. Clear reserved flag and function name for each slot
6. Release mutex

7.57.2.3 Performance

This function iterates all 136 slots (17 ports x 8 pins) and clears each one. Typical execution time: 50-100 us @ 240 MHz.

Precondition

ctx must point to initialized [pin_validator_t](#)
Should ensure no modules are actively using pins

Postcondition

All reservation->reserved = false
All reservation->function = "" (empty string)
All pins available for reservation
Mutex released

Note

Thread-safe: Mutex-protected critical section
Performance: O(nxm) where n=ports, m=pins (~50-100 us)
Destructive: All existing reservations lost without warning

Warning

All reservations are lost - modules will not be notified
Should only be called during controlled shutdown or test teardown
Do not call while modules are actively using pins

Example:

```
// Unit test teardown
void test_teardown(void) {
    // Clear all reservations to ensure clean state for next test
    iface->clear_all_reservations(iface->ctx);
}

// Emergency reset
void emergency_reset(void) {
    // Stop all peripherals first
    spi_emergency_stop();
    i2c_emergency_stop();
    // Clear all pin reservations
    iface->clear_all_reservations(iface->ctx);
    // Reinitialize system
}
```

See also

[impl_release_pin\(\)](#) Release individual pins (safer for normal operation)
[pin_validator_deinit\(\)](#) Full validator cleanup (including mutex)

Since

Version 1.0.0

Definition at line 862 of file [rx_pin_validator.c](#).

```

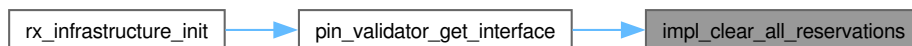
00863 {
00864     pin_validator_t* validator = (pin_validator_t*)ctx;
00865
00866     if (validator == nullptr) {
00867         return k_rx_err_null_ptr;
00868     }
00869
00870     /* Acquire mutex */
00871     (void)tx_mutex_get(&validator->mutex, TX_WAIT_FOREVER);
00872
00873     /* Clear all reservations */
00874     for (uint32_t port_idx = 0; port_idx < k_pin_validator_max_ports; port_idx++) {
00875         for (uint32_t pin_idx = 0; pin_idx < k_pin_validator_max_pins; pin_idx++) {
00876             validator->reservations[port_idx][pin_idx].reserved = false;
00877             validator->reservations[port_idx][pin_idx].function[0] = '\0';
00878         }
00879     }
00880
00881     /* Release mutex */
00882     (void)tx_mutex_put(&validator->mutex);
00883
00884     rx_log_debug(s_tag, "All reservations cleared");
00885
00886     return k_rx_ok;
00887 }

```

References [pin_reservation_t::function](#), [k_pin_validator_max_pins](#), [k_pin_validator_max_ports](#), [k_rx_err_null_ptr](#), [k_rx_ok](#), [pin_validator_t::mutex](#), [pin_validator_t::reservations](#), [pin_reservation_t::reserved](#), [rx_log_debug](#), and [s_tag](#).

Referenced by [pin_validator_get_interface\(\)](#).

Here is the caller graph for this function:



7.57.2.4 impl_get_pin_function()

```

rx_err_t impl_get_pin_function (
    void * ctx,
    const uint8_t port,
    const uint8_t pin,
    char * function_out,
    const uint32_t function_len) [static]

```

Get pin function - Interface implementation for [rx_pin_interface_t::get_pin_function](#).

Retrieves the function name that reserved a specific GPIO pin. Useful for debugging pin conflicts by identifying which module is using a contested pin.

7.57.2.5 Algorithm

1. Validate all pointers (ctx, function_out)
2. Validate buffer size (must hold k_pin_function_name_max_len)
3. Validate port/pin combination
4. Acquire mutex
5. Check if pin is reserved (error if not)
6. Copy function name to output buffer
7. Ensure null termination
8. Release mutex

7.57.2.6 Buffer Requirements

The output buffer must be at least `k_pin_function_name_max_len` (32) bytes to ensure the full function name can be copied. Smaller buffers result in `k_rx_err_invalid_size` error.

Precondition

`ctx` must point to initialized `pin_validator_t`
`function_out` must point to buffer of at least `function_len` bytes
`function_len` must be $\geq$ `k_pin_function_name_max_len`
Pin must be reserved (otherwise `k_rx_err_invalid_state`)

Postcondition

If `k_rx_ok`: `function_out` contains null-terminated function name
If error: `function_out` contents undefined
Mutex always released

Note

Thread-safe: Mutex-protected read + strcpy
Performance: $O(1)$, $\sim 2\text{-}5$ us (includes string copy)
Buffer size checked before mutex acquisition

Example:

```
// Debug pin conflict by identifying owner
if (reserve_result == k_rx_err_gpio_conflict) {
    char owner[32];
    rx_err_t err = iface->get_pin_function(iface->ctx, port, pin,
                                         owner, sizeof(owner));
    if (err == k_rx_ok) {
        rx_log_error("DEBUG", "Pin P%X.%d owned by: %s", port, pin, owner);
    }
}
```

See also

`impl_reserve_pin()` Sets the function name during reservation
`impl_is_pin_reserved()` Check if pin is reserved before querying
`k_pin_function_name_max_len` Required buffer size (32)

Since

Version 1.0.0

Definition at line 753 of file [rx_pin_validator.c](#).

```

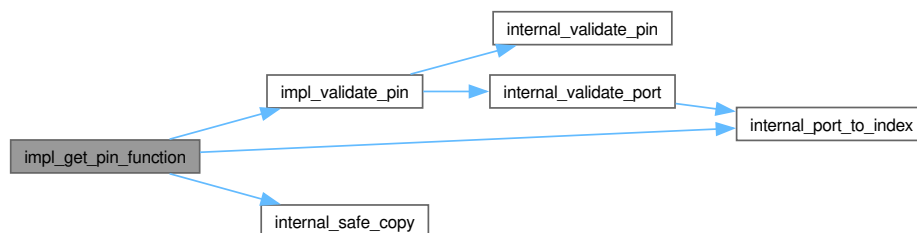
00758 {
00759     pin_validator_t* validator = (pin_validator_t*)ctx;
00760
00761     if ((validator == nullptr) || (function_out == nullptr)) {
00762         return k_rx_err_null_ptr;
00763     }
00764
00765     if (function_len < k_pin_function_name_max_len) {
00766         return k_rx_err_invalid_size;
00767     }
00768
00769     /* Validate port/pin */
00770     const rx_err_t err = impl_validate_pin(ctx, port, pin);
00771     if (err != k_rx_ok) {
00772         return err;
00773     }
00774
00775     const uint8_t port_index = internal_port_to_index(port);
00776
00777     /* Acquire mutex */
00778     (void)tx_mutex_get(&validator->mutex, TX_WAIT_FOREVER);
00779
00780     const pin_reservation_t* reservation = &validator->reservations[port_index][pin];
00781
00782     /* Check if pin is reserved */
00783     if (!reservation->reserved) {
00784         (void)tx_mutex_put(&validator->mutex);
00785         return k_rx_err_invalid_state;
00786     }
00787
00788     /* Copy function name */
00789     internal_safe_copy(function_out, reservation->function, function_len);
00790
00791     /* Release mutex */
00792     (void)tx_mutex_put(&validator->mutex);
00793
00794     return k_rx_ok;
00795 }

```

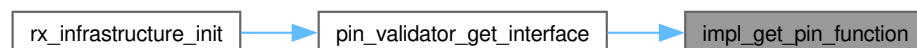
References [pin_reservation_t::function](#), [impl_validate_pin\(\)](#), [internal_port_to_index\(\)](#), [internal_safe_copy\(\)](#), [k_pin_function_name_max_len](#), [k_rx_err_invalid_size](#), [k_rx_err_invalid_state](#), [k_rx_err_null_ptr](#), [k_rx_ok](#), [pin_validator_t::mutex](#), [pin_validator_t::reservations](#), and [pin_reservation_t::reserved](#).

Referenced by [pin_validator_get_interface\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



7.57.2.7 impl_is_pin_reserved()

```
bool impl_is_pin_reserved (
    void * ctx,
    const uint8_t port,
    const uint8_t pin) [static]
```

Check if pin is reserved - Interface implementation for rx_pin_interface_t::is_pin_reserved.

Queries the reservation status of a GPIO pin without attempting to reserve it. Useful for checking pin availability before making allocation decisions.

7.57.2.8 Return Value Semantics

Returns false in multiple error conditions (not just "unreserved"):

- ctx is nullptr -> false
- Invalid port/pin -> false
- Mutex acquisition failed -> false
- Pin is not reserved -> false (expected case)

Only returns true when pin is actually reserved.

Precondition

ctx should point to initialized [pin_validator_t](#)

Port/pin should be valid (errors return false, not error code)

Postcondition

No state changes (read-only query)

Mutex acquired and released during operation

Note

Thread-safe: Mutex-protected read

Performance: O(1), ~1-2 us

Error handling: Errors return false (conservative "not reserved")

Example:

```
// Check if a pin is available before configuring
if (iface->is_pin_reserved(iface->ctx, 0xA, 5)) {
    // Pin in use - find alternative or report error
    rx_log_warn("MOTOR", "PWM pin PA5 already in use");
    return k_rx_err_gpio_conflict;
}
// Pin available - proceed to reserve
iface->reserve_pin(iface->ctx, 0xA, 5, "MOTOR_PWM");
```

See also

- [impl_reserve_pin\(\)](#) Reserve a pin after checking availability
- [impl_get_pin_function\(\)](#) Get the name of the reserving function

Since

Version 1.0.0

Definition at line 670 of file [rx_pin_validator.c](#).

```

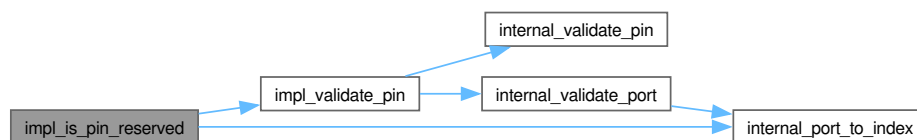
00671 {
00672     pin_validator_t* validator = (pin_validator_t*)ctx;
00673
00674     if (validator == nullptr) {
00675         return false;
00676     }
00677
00678     /* Validate port/pin */
00679     if (impl_validate_pin(ctx, port, pin) != k_rx_ok) {
00680         return false;
00681     }
00682
00683     const uint8_t port_index = internal_port_to_index(port);
00684
00685     /* Acquire mutex */
00686     (void)tx_mutex_get(&validator->mutex, TX_WAIT_FOREVER);
00687
00688     const bool reserved = validator->reservations[port_index][pin].reserved;
00689
00690     /* Release mutex */
00691     (void)tx_mutex_put(&validator->mutex);
00692
00693     return reserved;
00694 }

```

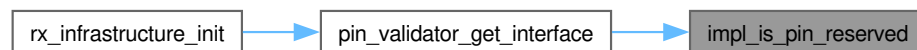
References [impl_validate_pin\(\)](#), [internal_port_to_index\(\)](#), [k_rx_ok](#), [pin_validator_t::mutex](#), [pin_validator_t::reservations](#), and [pin_reservation_t::reserved](#).

Referenced by [pin_validator_get_interface\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



7.57.2.9 impl_release_pin()

```

rx_err_t impl_release_pin (
    void * ctx,
    const uint8_t port,
    const uint8_t pin) [static]

```

Release pin - Interface implementation for `rx_pin_interface_t::release_pin`.

Releases a previously reserved GPIO pin, making it available for reservation by other functions. Must be called when a module no longer needs exclusive access to a pin.

7.57.2.10 Algorithm

1. Validate context pointer (NULL check)
2. Validate port/pin combination
3. Convert port to array index
4. Acquire mutex
5. Check if pin was actually reserved (error if not)
6. Clear reserved flag and function name
7. Release mutex
8. Return success

7.57.2.11 Error Handling

Releasing an unreserved pin returns `k_rx_err_invalid_state`. This is typically a programming error (double-release or release without reserve).

Precondition

ctx must point to initialized `pin_validator_t`
 Pin should be currently reserved
 Port/pin must be valid hardware addresses

Postcondition

If `k_rx_ok`: reservation->reserved = false
 If `k_rx_ok`: reservation->function cleared
 If error: No state changes
 Mutex always released (even on error path)

Note

Thread-safe: Mutex-protected critical section
 Performance: $O(1)$, ~1-3 us
 Idempotent safety: Releasing unreserved pin returns error (not silent)

Warning

Calling on unreserved pin returns `k_rx_err_invalid_state`
 Do not call from ISR (mutex may block)

Example:

```
// Release pin when SPI is done
void spi_deinit(spi_handle_t* spi) {
    // Release all SPI pins
    iface->release_pin(iface->ctx, 0xA, 5); // COPI
    iface->release_pin(iface->ctx, 0xA, 4); // CIPO
    iface->release_pin(iface->ctx, 0xA, 3); // SCK
}
```

See also

[impl_reserve_pin\(\)](#) Reserve a pin before use
[impl_clear_all_reservations\(\)](#) Release all pins at once

Since

Version 1.0.0

Definition at line 582 of file [rx_pin_validator.c](#).

```

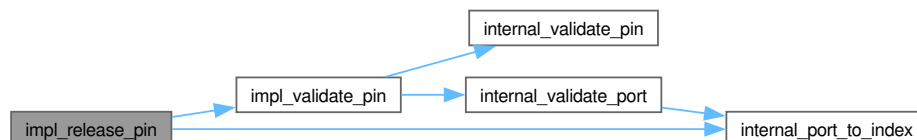
00583 {
00584     pin_validator_t* validator = (pin_validator_t*)ctx;
00585     if (validator == nullptr) {
00586         return k_rx_err_null_ptr;
00587     }
00588 }
00589
00590 /* Validate port/pin */
00591 const rx_err_t err = impl_validate_pin(ctx, port, pin);
00592 if (err != k_rx_ok) {
00593     return err;
00594 }
00595
00596 const uint8_t port_index = internal_port_to_index(port);
00597
00598 /* Acquire mutex */
00599 (void)tx_mutex_get(&validator->mutex, TX_WAIT_FOREVER);
00600
00601 pin_reservation_t* reservation = &validator->reservations[port_index][pin];
00602
00603 /* Check if pin was reserved */
00604 if (!reservation->reserved) {
00605     /* Release mutex before returning error */
00606     (void)tx_mutex_put(&validator->mutex);
00607
00608     rx_log_warn(s_tag, "Pin was not reserved");
00609     return k_rx_err_invalid_state;
00610 }
00611
00612 /* Release the pin */
00613 reservation->reserved = false;
00614 reservation->function[0] = '\0';
00615
00616 /* Release mutex */
00617 (void)tx_mutex_put(&validator->mutex);
00618
00619 rx_log_debug(s_tag, "Pin released");
00620
00621 return k_rx_ok;
00622 }

```

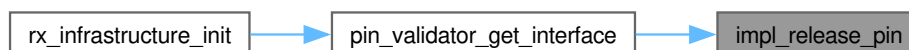
References [pin_reservation_t::function](#), [impl_validate_pin\(\)](#), [internal_port_to_index\(\)](#), [k_rx_err_invalid_state](#), [k_rx_err_null_ptr](#), [k_rx_ok](#), [pin_validator_t::mutex](#), [pin_validator_t::reservations](#), [pin_reservation_t::reserved](#), [rx_log_debug](#), [rx_log_warn](#), and [s_tag](#).

Referenced by [pin_validator_get_interface\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



7.57.2.12 impl_reserve_pin()

```

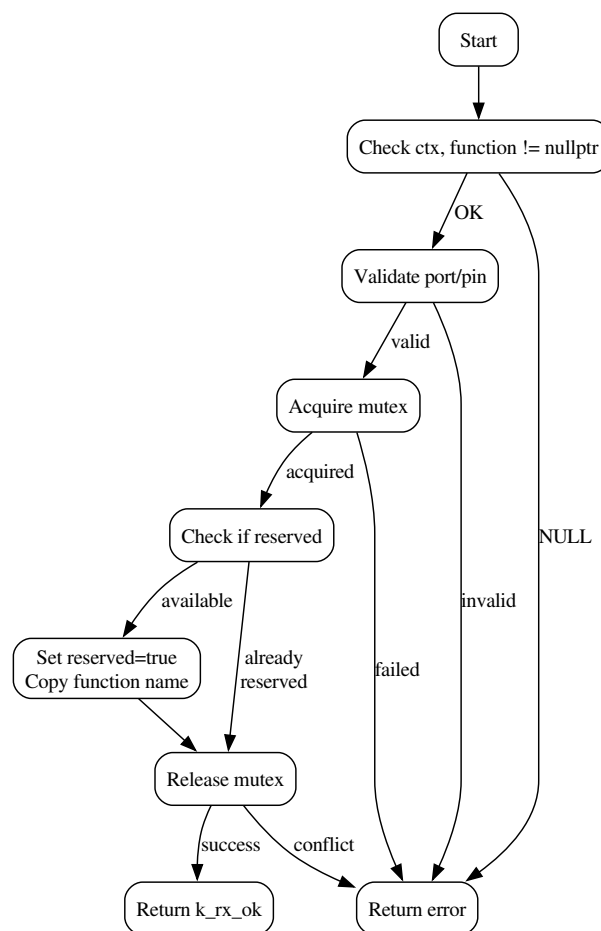
rx_err_t impl_reserve_pin (
    void * ctx,
    const uint8_t port,
    const uint8_t pin,
    const char * function) [static]

```

Reserve pin - Interface implementation for rx_pin_interface_t::reserve_pin.

Attempts to reserve a GPIO pin for exclusive use by the specified function. This is the core conflict-prevention mechanism: if a pin is already reserved, the function returns an error instead of allowing double-allocation.

7.57.2.13 Algorithm



7.57.2.14 Thread Safety

The reservation check and set are atomic under mutex protection:

- Mutex acquired before checking reservation status
- Reservation set before mutex released
- No race condition possible between check and set

Precondition

ctx must point to initialized [pin_validator_t](#)
function must be non-NULL null-terminated string
Port/pin must be valid hardware addresses

Postcondition

If k_rx_ok: reservation->reserved = true
If k_rx_ok: reservation->function contains copy of function name
If error: No state changes (atomic operation)
Mutex always released (even on error path)

Note

Thread-safe: Mutex-protected critical section
Performance: O(1), ~1-5 us (dominated by mutex operations)
Function name is COPIED (up to 31 chars) - original can be freed after

Warning

Calling with already-reserved pin returns k_rx_err_gpio_conflict
Do not call from ISR (mutex may block indefinitely)

Example:

```
// Reserve pin for SPI COPI function
rx_err_t err = iface->reserve_pin(iface->ctx, 0xA, 5, "SPI_COPI");
if (err == k_rx_err_gpio_conflict) {
    // Pin already in use - check who owns it
    char owner[32];
    iface->get_pin_function(iface->ctx, 0xA, 5, owner, sizeof(owner));
    rx_log_error("SPI", "Pin PA5 already used by: %s", owner);
}
```

See also

[impl_release_pin\(\)](#) Release a previously reserved pin
[impl_is_pin_reserved\(\)](#) Check if pin is reserved without reserving
[impl_get_pin_function\(\)](#) Get the function that reserved a pin

Since

Version 1.0.0

Definition at line 483 of file [rx_pin_validator.c](#).

```

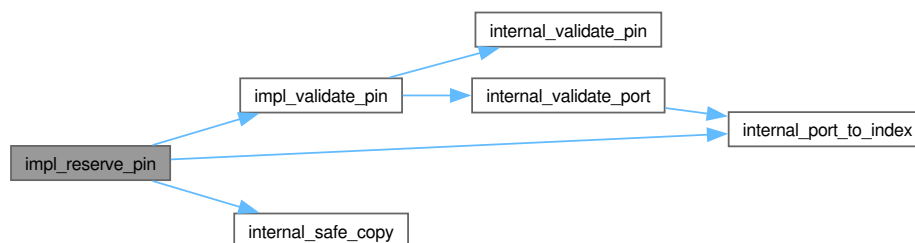
00484 {
00485     pin_validator_t* validator = (pin_validator_t*)ctx;
00486
00487     if ((validator == nullptr) || (function == nullptr)) {
00488         return k_rx_err_null_ptr;
00489     }
00490
00491     /* Validate port/pin */
00492     const rx_err_t err = impl_validate_pin(ctx, port, pin);
00493     if (err != k_rx_ok) {
00494         return err;
00495     }
00496
00497     const uint8_t port_index = internal_port_to_index(port);
00498
00499     /* Acquire mutex */
00500     (void)tx_mutex_get(&validator->mutex, TX_WAIT_FOREVER);
00501
00502     pin_reservation_t* reservation = &validator->reservations[port_index][pin];
00503
00504     /* Check if already reserved */
00505     if (reservation->reserved) {
00506         /* Release mutex before returning error */
00507         (void)tx_mutex_put(&validator->mutex);
00508
00509         rx_log_warn(s_tag, "Pin already reserved");
00510         return k_rx_err_gpio_conflict;
00511     }
00512
00513     /* Reserve the pin */
00514     reservation->reserved = true;
00515     internal_safe_copy(reservation->function, function, k_pin_function_name_max_len);
00516
00517     /* Release mutex */
00518     (void)tx_mutex_put(&validator->mutex);
00519
00520     rx_log_debug(s_tag, "Pin reserved");
00521
00522     return k_rx_ok;
00523 }

```

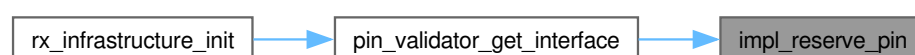
References [pin_reservation_t::function](#), [impl_validate_pin\(\)](#), [internal_port_to_index\(\)](#), [internal_safe_copy\(\)](#), [k_pin_function_name_max_len](#), [k_rx_err_gpio_conflict](#), [k_rx_err_null_ptr](#), [k_rx_ok](#), [pin_validator_t::mutex](#), [pin_validator_t::reservations](#), [pin_reservation_t::reserved](#), [rx_log_debug](#), [rx_log_warn](#), and [s_tag](#).

Referenced by [pin_validator_get_interface\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



7.57.2.15 impl_validate_pin()

```
rx_err_t impl_validate_pin (  
    void * ctx,  
    const uint8_t port,  
    const uint8_t pin) [static]
```

Validate pin - Interface implementation for rx_pin_interface_t::validate_pin.

Checks if a port/pin combination is valid for the RX72N hardware. This function does NOT check reservation status - it only validates that the port and pin numbers are within valid hardware ranges.

7.57.2.16 Algorithm

1. Validate context pointer (NULL check)
2. Validate port number (0-9, 0xA-0x10)
3. Validate pin number (0-7)
4. Return success if all validations pass

Precondition

ctx must be initialized [pin_validator_t](#) (or nullptr check fails)
Port must be in range 0-9 or 0xA-0x10

Postcondition

No state changes (read-only validation)
No mutex acquired (validation is stateless)

Note

Thread-safe: No shared state accessed
Performance: O(1), ~20-50 cycles
Does NOT check if pin is reserved (use [is_pin_reserved](#) for that)

See also

[impl_reserve_pin\(\)](#) Uses this function for validation before reservation
[impl_release_pin\(\)](#) Uses this function for validation before release

Since

Version 1.0.0

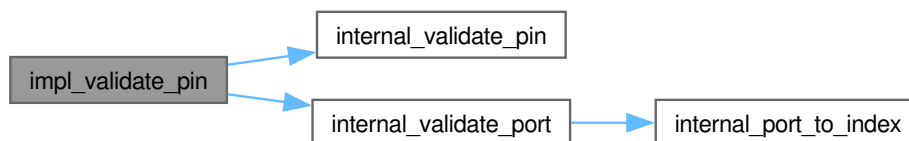
Definition at line 375 of file `rx_pin_validator.c`.

```
00376 {
00377     const pin_validator_t* validator = (pin_validator_t*)ctx;
00378
00379     /* Pre-condition: callers always check validator != nullptr before calling.
00380      * No RX_ASSERT here -- the false branch is architecturally unreachable and
00381      * would block 100% branch coverage without GCOV_EXCL. */
00382     (void)validator;
00383
00384     /* Validate port */
00385     rx_err_t err = internal_validate_port(port);
00386     if (err != k_rx_ok) {
00387         return err;
00388     }
00389
00390     /* Validate pin */
00391     err = internal_validate_pin(pin);
00392     if (err != k_rx_ok) {
00393         return err;
00394     }
00395
00396     return k_rx_ok;
00397 }
```

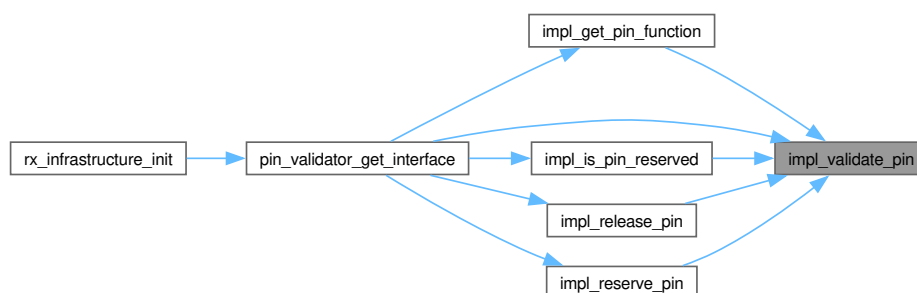
References [internal_validate_pin\(\)](#), [internal_validate_port\(\)](#), and [k_rx_ok](#).

Referenced by [impl_get_pin_function\(\)](#), [impl_is_pin_reserved\(\)](#), [impl_release_pin\(\)](#), [impl_reserve_pin\(\)](#), and [pin_validator_get_interface\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



7.57.2.17 internal_port_to_index()

```
uint8_t internal_port_to_index (
    const uint8_t port) [static]
```

Convert port number to internal array index.

Maps the RX72N's non-contiguous port numbering scheme to a contiguous array index. The RX72N uses decimal port numbers 0-9 and hexadecimal port letters A-G (0xA-0x10). This function normalizes both schemes to a single 0-16 index range.

7.57.2.18 Port Mapping Table

Port	Hex Value	Array Index	Calculation
0-9	0x0-0x9	0-9	Direct (port)
A	0xA	10	$(0xA - 0xA) + 10 = 10$
B	0xB	11	$(0xB - 0xA) + 10 = 11$
C	0xC	12	$(0xC - 0xA) + 10 = 12$
D	0xD	13	$(0xD - 0xA) + 10 = 13$
E	0xE	14	$(0xE - 0xA) + 10 = 14$
F	0xF	15	$(0xF - 0xA) + 10 = 15$
G	0x10	16	$(0x10 - 0xA) + 10 = 16$

Precondition

None (pure function, no side effects)

Postcondition

Return value is either valid index (0-16) or `k_invalid_port`

Note

Thread-safe: Pure function with no shared state

Performance: $O(1)$, ~10 cycles

See also

[k_max_decimal_port](#) Maximum decimal port number (9)

[k_hex_port_start](#) Start of hex port range (0xA)

[k_hex_port_end](#) End of hex port range (0x10)

[k_invalid_port](#) Sentinel value for invalid ports (0xFF)

Since

Version 1.0.0

Definition at line 250 of file [rx_pin_validator.c](#).

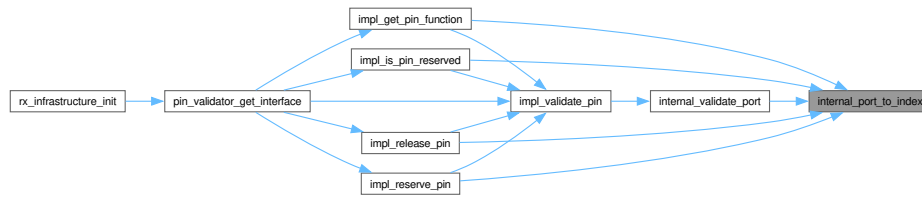
```

00251 {
00252     if (port <= k_max_decimal_port) {
00253         return port; /* Ports 0-9 map directly */
00254     }
00255
00256     /* port > k_max_decimal_port (9) is guaranteed by the preceding check, so
00257      * port >= k_hex_port_start (10) is always true here -- no short-circuit branch. */
00258     if (port <= k_hex_port_end) {
00259         return (port - k_hex_port_start) + k_hex_port_offset; /* Ports A-G (0xA-0x10) map to 10-16 */
00260     }
00261
00262     return k_invalid_port; /* Invalid port */
00263 }
```

References [k_hex_port_end](#), [k_hex_port_offset](#), [k_hex_port_start](#), [k_invalid_port](#), and [k_max_decimal_port](#).

Referenced by [impl_get_pin_function\(\)](#), [impl_is_pin_reserved\(\)](#), [impl_release_pin\(\)](#), [impl_reserve_pin\(\)](#), and [internal_validate_port\(\)](#).

Here is the caller graph for this function:



7.57.2.19 internal_safe_copy()

```

void internal_safe_copy (
    char * dst,
    const char * src,
    const uint32_t max_len) [static]

```

Copy a string safely without strncpy.

Copies up to max_len-1 characters from src to dst, then null-terminates. Replaces strncpy to satisfy clang-analyzer-security checks.

Precondition

dst and src must be non-NULL
max_len must be > 0

Postcondition

dst is null-terminated

Note

Thread-safe: no shared state

Since

Version 1.0.0

Definition at line 198 of file rx_pin_validator.c.

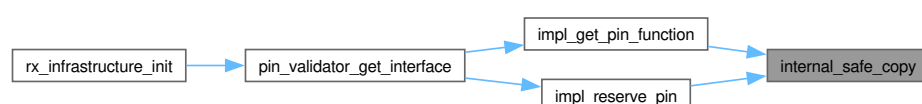
```

00199 {
00200     for (uint32_t i = 0; i < max_len - 1; i++) {
00201         dst[i] = src[i];
00202         if (src[i] == '\0') {
00203             return;
00204         }
00205     }
00206     dst[max_len - 1] = '\0';
00207 }

```

Referenced by [impl_get_pin_function\(\)](#), and [impl_reserve_pin\(\)](#).

Here is the caller graph for this function:



7.57.2.20 internal_validate_pin()

```
rx_err_t internal_validate_pin (
    const uint8_t pin) [static]
```

Validate pin number against RX72N per-port limits.

Checks if the given pin number is within the valid range for any RX72N GPIO port. Each port supports up to 8 pins (0-7), though not all ports have all 8 pins physically available on the package.

7.57.2.21 Pin Availability Notes

While this function validates against the maximum of 8 pins per port, actual pin availability varies by port and package:

- Full ports (0, 1, 2, A, B, C, D, E): Pins 0-7 available
- Partial ports (some 3-9, F, G): May have fewer pins

Precondition

None (pure validation function)

Postcondition

No state changes

Note

Thread-safe: Pure function with no shared state

Performance: O(1), ~5 cycles

Does NOT check physical pin availability on specific port

See also

[k_pins_per_port](#) Maximum pins per port (8)

[internal_validate_port\(\)](#) Companion function for port validation

Since

Version 1.0.0

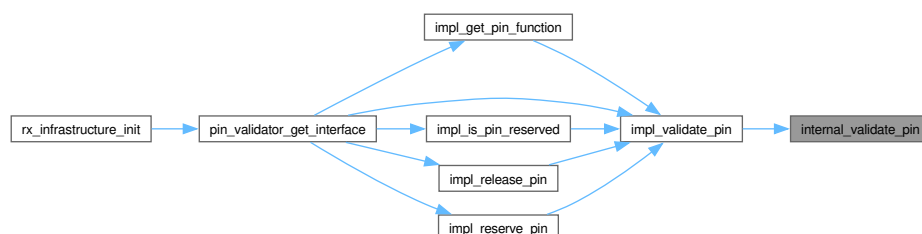
Definition at line 328 of file [rx_pin_validator.c](#).

```
00329 {
00330     if (pin >= k_pins_per_port) {
00331         return k_rx_err_gpio_invalid_pin;
00332     }
00333     return k_rx_ok;
00334 }
```

References [k_pins_per_port](#), [k_rx_err_gpio_invalid_pin](#), and [k_rx_ok](#).

Referenced by [impl_validate_pin\(\)](#).

Here is the caller graph for this function:



7.57.2.22 `internal_validate_port()`

```
rx_err_t internal_validate_port (
    const uint8_t port)  [static]
```

Validate port number against RX72N hardware limits.

Checks if the given port number corresponds to a valid GPIO port on the RX72N microcontroller. Uses [internal_port_to_index\(\)](#) to perform the actual validation.

7.57.2.23 Algorithm

1. Attempt to convert port to array index
2. If conversion returns `k_invalid_port`, port is invalid
3. Otherwise, port is valid

Precondition

None (pure validation function)

Postcondition

No state changes

Note

Thread-safe: Pure function with no shared state

Performance: $O(1)$, ~15 cycles

See also

[internal_port_to_index\(\)](#) Used for validation

[internal_validate_pin\(\)](#) Companion function for pin validation

Since

Version 1.0.0

Definition at line 290 of file [rx_pin_validator.c](#).

```
00291 {
00292     const uint8_t index = internal_port_to_index(port);
00293     if (index == k_invalid_port) {
00294         return k_rx_err_gpio_invalid_port;
00295     }
00296     return k_rx_ok;
00297 }
```

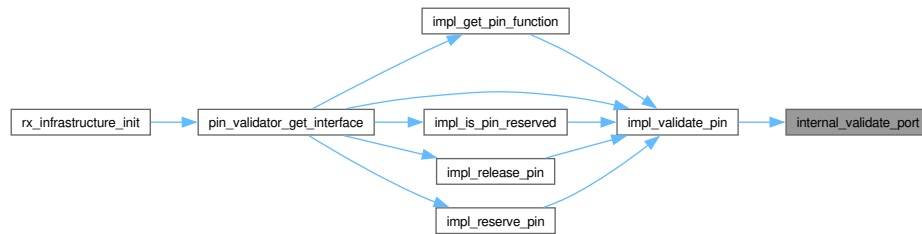
References [internal_port_to_index\(\)](#), [k_invalid_port](#), [k_rx_err_gpio_invalid_port](#), and [k_rx_ok](#).

Referenced by [impl_validate_pin\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



7.57.2.24 pin_validator_deinit()

```

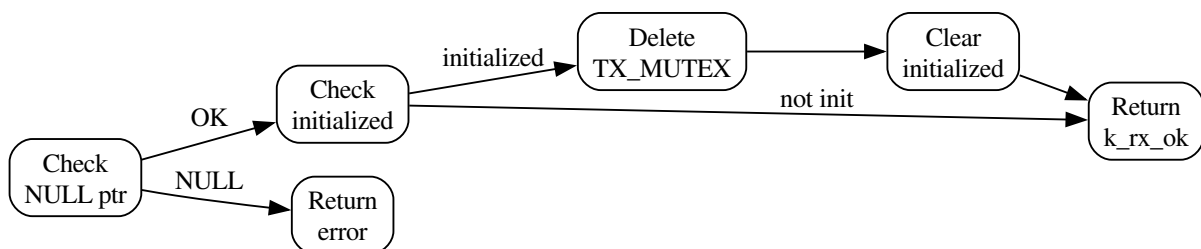
rx_err_t pin_validator_deinit (
    pin_validator_t * validator) [nodiscard]
  
```

Deinitialize pin validator and release resources.

Deinitialize pin validator (graceful shutdown only).

Performs cleanup of a pin validator instance, destroying the ThreadX mutex and marking the validator as uninitialized. This function is rarely needed in production embedded systems (which run indefinitely) but is essential for unit test teardown.

7.57.2.25 Cleanup Sequence



7.57.2.26 Safe to Call When Not Initialized

If the validator is not initialized, this function returns k_rx_ok without doing anything (safe no-op). This allows defensive cleanup patterns.

Precondition

- validator must point to valid `pin_validator_t` memory
- No modules should be actively using pins
- No concurrent calls to pin validator functions

Postcondition

validator->initialized = false
 validator->mutex destroyed (if was created)
 All extracted interfaces become INVALID
 Validator can be re-initialized via [pin_validator_init\(\)](#)

Note

NOT thread-safe: Ensure no concurrent access during deinit
 Idempotent: Safe to call multiple times
 Performance: ~100-300 us @ 240 MHz

Warning

All interfaces become invalid after this call
 Modules using pins should be stopped first
 Do NOT call while other threads are using validator

Example:

```

// Unit test teardown
void test_cleanup(void) {
    // Deinitialize validator to clean up mutex
    pin_validator_deinit(&s_test_validator);
    // Validator can be reinitialized for next test
}

// Defensive cleanup pattern
void shutdown_pin_system(void) {
    // Safe to call even if not initialized
    pin_validator_deinit(&s_validator);
}

```

See also

[pin_validator_init\(\)](#) Reinitialize after deinit
[impl_clear_all_reservations\(\)](#) Clear reservations without deinit

Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 5: [OK] 3 preconditions, 4 postconditions documented

Definition at line 1202 of file [rx_pin_validator.c](#).

```

01203 {
01204   RX_CHECK_NULL_PTR(validator, s_tag, "Validator pointer is nullptr");
01205
01206   if (!validator->initialized) {
01207     return k_rx_ok; /* Already deinitialized */
01208   }
01209
01210   /* Delete mutex */
01211   (void)tx_mutex_delete(&validator->mutex);

```

```

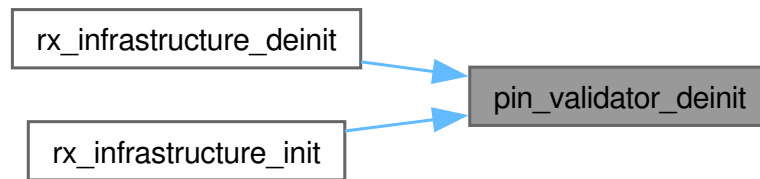
01212
01213  /* Clear state */
01214  validator->initialized = false;
01215
01216  rx_log_info(s_tag, "Pin validator deinitialized");
01217
01218  return k_rx_ok;
01219 }

```

References [pin_validator_t::initialized](#), [k_rx_ok](#), [pin_validator_t::mutex](#), [RX_CHECK_NULL_PTR](#), [rx_log_info](#), and [s_tag](#).

Referenced by [rx_infrastructure_deinit\(\)](#), and [rx_infrastructure_init\(\)](#).

Here is the caller graph for this function:



7.57.2.27 pin_validator_get_interface()

```

rx_err_t pin_validator_get_interface (
    rx_pin_interface_t * iface,
    pin_validator_t * validator) [nodiscard]

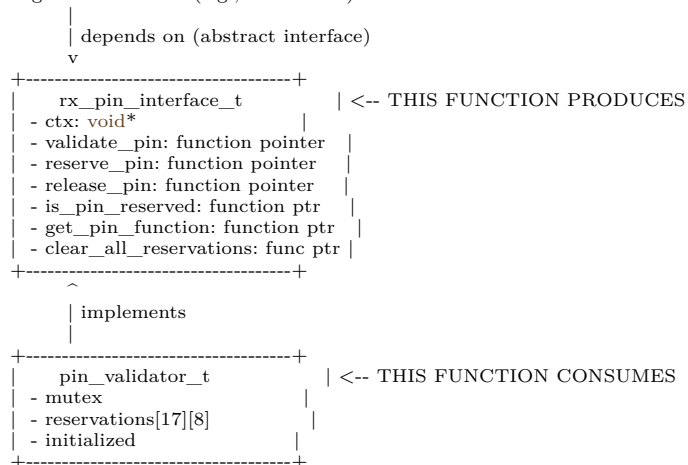
```

Extract abstract interface from concrete pin validator (Dependency Inversion).

Populates an `rx_pin_interface_t` structure with function pointers to this validator's implementation functions. This is the bridge between the concrete implementation and the abstract interface that high-level modules use.

7.57.2.28 Dependency Inversion Pattern

High-Level Module (e.g., SPI Driver)



7.57.2.29 Interface Population

The function assigns:

- `iface->ctx = validator` (context pointer)
- `iface->validate_pin = impl_validate_pin`
- `iface->reserve_pin = impl_reserve_pin`
- `iface->release_pin = impl_release_pin`
- `iface->is_pin_reserved = impl_is_pin_reserved`
- `iface->get_pin_function = impl_get_pin_function`
- `iface->clear_all_reservations = impl_clear_all_reservations`

Precondition

`iface` must point to valid `rx_pin_interface_t` memory
`validator` must be initialized via [pin_validator_init\(\)](#)
`validator` must remain alive as long as interface is used

Postcondition

`iface->ctx` points to `validator`
 All `iface` function pointers are valid
 Interface ready for use by high-level modules
`validator` state unchanged

Note

Thread-safe: Can be called from any thread
 Multiple interfaces: Same `validator` can produce multiple interfaces
 Performance: $O(1)$, ~50 cycles (struct assignment)

Warning

`Validator` must outlive all extracted interfaces
 Do not deinit `validator` while interfaces are in use

Example:

```
static pin_validator_t s_validator;
static rx_pin_interface_t s_pin_iface;

rx_err_t init_pin_system(void) {
    rx_err_t err = pin_validator_init(&s_validator);
    if (err != k_rx_ok) return err;

    err = pin_validator_get_interface(&s_pin_iface, &s_validator);
    if (err != k_rx_ok) return err;

    // Pass interface to high-level modules
    spi_init(&s_pin_iface);
    i2c_init(&s_pin_iface);

    return k_rx_ok;
}
```

See also

- [pin_validator_init\(\)](#) Must be called first
- [rx_pin_interface_t](#) Interface structure being populated
- [rx_infrastructure_get_pin_interface\(\)](#) Global accessor pattern

Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 5: [OK] 3 preconditions, 4 postconditions documented

Definition at line 1100 of file [rx_pin_validator.c](#).

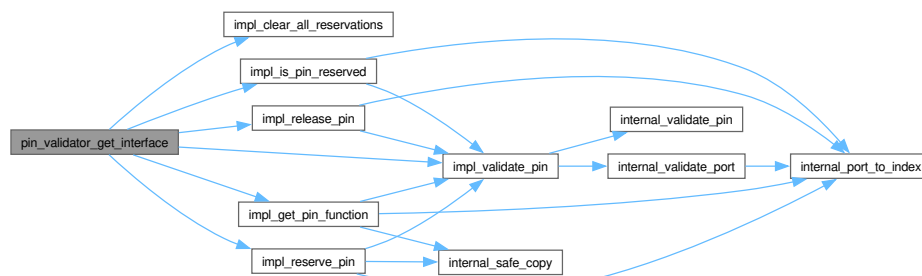
```

01101 {
01102   RX_CHECK_NULL_PTR(iface, s_tag, "Interface pointer is nullptr");
01103   RX_CHECK_NULL_PTR validator, s_tag, "Validator pointer is nullptr");
01104   if (!validator->initialized) {
01105     rx_log_error(s_tag, "Validator not initialized");
01106     return k_rx_err_invalid_state;
01107   }
01108 }
01109
01110 /* Fill interface */
01111 iface->ctx = validator;
01112 iface->validate_pin = impl_validate_pin;
01113 iface->reserve_pin = impl_reserve_pin;
01114 iface->release_pin = impl_release_pin;
01115 iface->is_pin_reserved = impl_is_pin_reserved;
01116 iface->get_pin_function = impl_get_pin_function;
01117 iface->clear_all_reservations = impl_clear_all_reservations;
01118
01119 return k_rx_ok;
01120 }
```

References [impl_clear_all_reservations\(\)](#), [impl_get_pin_function\(\)](#), [impl_is_pin_reserved\(\)](#), [impl_release_pin\(\)](#), [impl_reserve_pin\(\)](#), [impl_validate_pin\(\)](#), [pin_validator_t::initialized](#), [k_rx_err_invalid_state](#), [k_rx_ok](#), [RX_CHECK_NULL_PTR](#), [rx_log_error](#), and [s_tag](#).

Referenced by [rx_infrastructure_init\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



7.57.2.30 `pin_validator_init()`

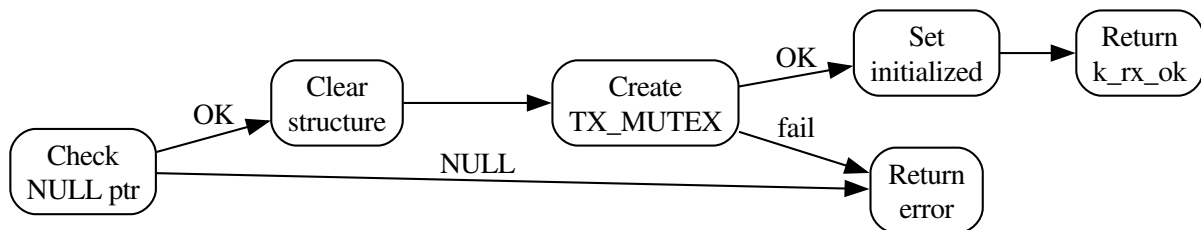
```
rx_err_t pin_validator_init (
    pin_validator_t * validator) [nodiscard]
```

Initialize pin validator for tracking GPIO pin reservations.

Initialize pin validator (MUST be called before use).

Performs one-time initialization of a pin validator instance. This function creates the ThreadX mutex, clears the reservation table, and marks the validator as initialized. Must be called exactly once before any other `pin_validator_*` functions.

7.57.2.31 Initialization Sequence



7.57.2.32 Memory Initialization

The function uses a static zero-initialized instance to clear the structure, which:

- Clears all reservation entries (`reserved=false`, `function=""`)
- Clears mutex bytes to neutral zero state before `tx_mutex_create` performs RTOS setup
- Sets `initialized` to false (then true after success)

Precondition

validator must point to valid `pin_validator_t` memory

validator must NOT be already initialized

ThreadX kernel must be started or about to start

Postcondition

validator->initialized = true

validator->mutex created and ready

validator->reservations all cleared (available)

Validator ready for `pin_validator_get_interface()`

Note

NOT thread-safe: Call from main() or initialization thread only
 Idempotent: Calling twice is NOT supported (undefined behavior)
 Performance: ~200-500 us @ 240 MHz (dominated by mutex creation)

Warning

MUST be called before any pin operations
 Do NOT call from ISR
 On failure, validator is in undefined state

Example:

```
static pin_validator_t s_validator;

int main(void) {
    rx_err_t err = pin_validator_init(&s_validator);
    if (err != k_rx_ok) {
        rx_log_fatal("INIT", "Pin validator init failed: %d", err);
        while (1) { __asm("nop"); } // Fatal - cannot continue
    }
    // Validator ready for use
}
```

See also

[pin_validator_get_interface\(\)](#) Next step after initialization
[pin_validator_deinit\(\)](#) Cleanup function
[rx_infrastructure_init\(\)](#) Typically calls this function

Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 5: [OK] 3 preconditions, 4 postconditions documented

Definition at line 990 of file [rx_pin_validator.c](#).

```
00991 {
00992     RX_CHECK_NULL_PTR(validator, s_tag, "Validator pointer is nullptr");
00993
00994     *validator = s_zero_validator;
00995
00996     /* Create mutex */
00997     (void)tx_mutex_create(&validator->mutex, "PinValidatorMutex", TX_NO_INHERIT);
00998
00999     validator->initialized = true;
01000
01001     rx_log_info(s_tag, "Pin validator initialized");
01002
01003     return k_rx_ok;
01004 }
```

References [pin_validator_t::initialized](#), [k_rx_ok](#), [pin_validator_t::mutex](#), [RX_CHECK_NULL_PTR](#), [rx_log_info](#), [s_tag](#), and [s_zero_validator](#).

Referenced by [rx_infrastructure_init\(\)](#).

Here is the caller graph for this function:



7.57.3 Variable Documentation

7.57.3.1 s_tag

```
const char* const s_tag = "PIN_VALIDATOR" [static]
```

Log tag for pin validator module.

Identifies log messages from this module

Note

Read-only after initialization

Since

Version 1.0.0

Definition at line 176 of file [rx_pin_validator.c](#).

7.57.3.2 s_zero_validator

```
const pin\_validator\_t s_zero_validator = {} [static]
```

Zero-initialized pin validator template for stack-safe initialization.

Static const instance used to clear [pin_validator_t](#) without creating a large compound literal on the stack. Lives in .rodata section.

Note

Read-only; never modified after static initialization

Warning

Do not remove - prevents stack overflow in init function

See also

[pin_validator_init\(\)](#) Uses this template to zero-initialize the validator

Since

Version 1.0.0

Definition at line 907 of file [rx_pin_validator.c](#).

```
00907 {};
```

Referenced by [pin_validator_init\(\)](#).

7.58 rx_pin_validator.c

[Go to the documentation of this file.](#)

```

00001 /**
00002  * @file rx_pin_validator.c
00003  * @brief Pin Validator Concrete Implementation - GPIO Pin Tracking and Conflict Prevention
00004  *
00005  * @details
00006  * ### Overview
00007  *
00008  * This file provides the **concrete implementation** of the `rx_pin_interface_t` abstract
00009  * interface for GPIO pin validation and reservation tracking. It is the heart of the STAR
00010  * project's pin conflict prevention system.
00011  *
00012  * ### System Architecture Context
00013  *
00014  * @dot
00015  * digraph pin_validator_context {
00016  *   rankdir=TB;
00017  *   node [shape=box, style=rounded];
00018  *
00019  *   subgraph cluster_high_level {
00020  *     label="High-Level Modules (Users)";
00021  *     style=dashed;
00022  *     spi [label="SPI Driver"];
00023  *     i2c [label="I2C Driver"];
00024  *     uart [label="UART Driver"];
00025  *     motor [label="Motor Control"];
00026  *   }
00027  *
00028  *   subgraph cluster_interface {
00029  *     label="Abstract Interface";
00030  *     style=dashed;
00031  *     iface [label="rx_pin_interface_t\n(Function Pointers)"];
00032  *   }
00033  *
00034  *   subgraph cluster_impl {
00035  *     label="Concrete Implementation (THIS FILE)";
00036  *     style=filled;
00037  *     fillcolor=lightblue;
00038  *     validator [label="pin_validator_t\n+ mutex\n+ reservations[17][8]"];
00039  *     helpers [label="Internal Helpers\n+ internal_port_to_index()\n+ internal_validate_port()\n+
internal_validate_pin()"];
00040  *     impls [label="Interface Implementations\n+ impl_validate_pin()\n+ impl_reserve_pin()\n+
impl_release_pin()\n+ impl_is_pin_reserved()\n+ impl_get_pin_function()\n+ impl_clear_all_reservations()"];
00041  *   }
00042  *
00043  *   spi -> iface [label="depends on"];
00044  *   i2c -> iface;
00045  *   uart -> iface;
00046  *   motor -> iface;
00047  *   iface -> validator [label="implemented by"];
00048  *   validator -> helpers [label="uses"];
00049  *   validator -> impls [label="exposes via"];
00050  * }
00051  * @enddot
00052  *
00053  * ### Implementation Design
00054  *
00055  * ### Port Mapping Strategy
00056  *
00057  * The RX72N has 17 GPIO ports with non-contiguous numbering:
00058  * - **Decimal ports**: 0-9 (indices 0-9)
00059  * - **Hex ports**: A-G (0xA-0x10, mapped to indices 10-16)
00060  *
00061  * The `internal_port_to_index()` function converts port numbers to array indices:
00062  * ```
00063  * Port 0-9 -> Index 0-9   (direct mapping)
00064  * Port 0xA -> Index 10   (A = 10)
00065  * Port 0xB -> Index 11   (B = 11)
00066  * ...
00067  * Port 0x10 -> Index 16   (G = 16)
00068  * ```
00069  *
00070  * ### Thread Safety Model
00071  *
00072  * All read-modify-write operations on the reservation table are protected by a ThreadX mutex:
00073  *
00074  * @msc
00075  * Thread1, Mutex, Validator, Thread2;
00076  *
00077  * Thread1 => Mutex [label="tx_mutex_get()"];
00078  * Mutex ==> Thread1 [label="GRANTED"];
00079  * Thread1 => Validator [label="Check + Reserve Pin"];
00080  *

```

```

00081 * Thread2 => Mutex [label="tx_mutex_get()"];
00082 * Mutex box Mutex [label="BLOCKED"];
00083 *
00084 * Thread1 => Mutex [label="tx_mutex_put()"];
00085 * Mutex => Thread2 [label="GRANTED"];
00086 * Thread2 => Validator [label="Check + Reserve Pin"];
00087 * Thread2 => Mutex [label="tx_mutex_put()"];
00088 * @endmsc
00089 *
00090 * ### Error Propagation
00091 *
00092 * All functions follow a consistent error handling pattern:
00093 * 1. Validate NULL pointers first (return k_rx_err_null_ptr)
00094 * 2. Validate port/pin ranges (return k_rx_err_gpio_invalid_port/pin)
00095 * 3. Acquire mutex (return k_rx_err_rtos_mutex on failure)
00096 * 4. Perform operation (may return k_rx_err_gpio_conflict, k_rx_err_invalid_state)
00097 * 5. Release mutex
00098 * 6. Return success (k_rx_ok)
00099 *
00100 * ## Performance Characteristics
00101 *
00102 * | Operation | Time Complexity | Typical Latency | Notes |
00103 * |-----|-----|-----|-----|
00104 * | validate_pin | O(1) | ~100 ns | No mutex, direct lookup |
00105 * | reserve_pin | O(1) | ~1-5 us | Mutex + strcpy |
00106 * | release_pin | O(1) | ~1-3 us | Mutex + clear |
00107 * | is_reserved | O(1) | ~1-2 us | Mutex + bool read |
00108 * | get_function | O(1) | ~2-5 us | Mutex + strcpy |
00109 * | clear_all | O(nxm) | ~50-100 us | 17 ports x 8 pins |
00110 *
00111 * ## Memory Usage
00112 *
00113 * | Section | Size | Contents |
00114 * |-----|-----|-----|
00115 * | .text | ~1.5 KB | All function code |
00116 * | .rodata | ~200 bytes | Log strings |
00117 * | .bss | 0 bytes | No module-level state (validator is caller-owned) |
00118 *
00119 * ## NASA Power of 10 Compliance
00120 *
00121 * | Rule | Status | Implementation |
00122 * |-----|-----|-----|
00123 * | 1. Simplify control flow | [OK] | No goto, setjmp, recursion |
00124 * | 2. Fixed loop bounds | [OK] | clear_all uses k_pin_validator_max_* constants |
00125 * | 3. No dynamic memory | [OK] | All structures caller-owned |
00126 * | 4. Functions < 60 lines | [OK] | All functions under 50 lines |
00127 * | 5. Use assertions | [OK] | RX_CHECK_NULL_PTR in all public functions |
00128 * | 6. Data at smallest scope | [OK] | Local variables, static helpers |
00129 * | 7. Check return values | [OK] | All ThreadX calls checked |
00130 * | 8. Limit preprocessor | [OK] | Only includes |
00131 * | 9. Restrict pointers | [WARN] | Function pointers for DIP |
00132 * | 10. Compile warnings | [OK] | -Wall -Wextra -Werror clean |
00133 *
00134 * ## SOLID Principles
00135 *
00136 * | Principle | Implementation |
00137 * |-----|-----|
00138 * | **Single Responsibility** | File handles ONLY pin validation/reservation |
00139 * | **Open/Closed** | Interface stable, implementation can change |
00140 * | **Liskov Substitution** | Can swap with mock for testing |
00141 * | **Interface Segregation** | Implements exactly rx_pin_interface_t (no extras) |
00142 * | **Dependency Inversion** | High-level code depends on interface, not this file |
00143 *
00144 * @see rx_pin_validator.h Public API and type definitions
00145 * @see rx_pin_interface.h Abstract interface definition
00146 * @see rx_infrastructure.c Global validator instantiation
00147 * @see rx_gpio_constants.h GPIO port/pin constants
00148 *
00149 * @author Locked, Inc.
00150 * @date 2026-01-01
00151 * @copyright Copyright (c) 2026 Locked Inc.
00152 * SPDX-License-Identifier: MIT
00153 * @since Version 1.0.0
00154 */
00155
00156 #include "rx_pin_validator.h"
00157
00158 #include <string.h>
00159
00160 #include "rx_check.h"
00161 #include "rx_gpio_constants.h"
00162 #include "rx_log.h"
00163
00164 /*
=====
00165 * Module State
00166 *

```

```

=====
00167 */
00168
00169 /**
00170  * @var s_tag
00171  * @brief Log tag for pin validator module
00172  * @details Identifies log messages from this module
00173  * @note Read-only after initialization
00174  * @since Version 1.0.0
00175  */
00176 static const char* const s_tag = "PIN_VALIDATOR";
00177
00178 /*
=====
00179 * Internal String Helpers
00180 *
=====
00181 */
00182
00183 /**
00184  * @brief Copy a string safely without strncpy
00185  *
00186  * @details
00187  * Copies up to max_len-1 characters from src to dst, then null-terminates.
00188  * Replaces strncpy to satisfy clang-analyzer-security checks.
00189  *
00190  * @pre dst and src must be non-NULL
00191  * @pre max_len must be > 0
00192  * @post dst is null-terminated
00193  *
00194  * @note Thread-safe: no shared state
00195  * @since Version 1.0.0
00196  */
00197
00198 static void internal_safe_copy(char* dst, const char* src, const uint32_t max_len)
00199 {
00200     for (uint32_t i = 0; i < max_len - 1; i++) {
00201         dst[i] = src[i];
00202         if (src[i] == '\0') {
00203             return;
00204         }
00205     }
00206     dst[max_len - 1] = '\0';
00207 }
00208
00209 /*
=====
00210 * Internal Helper Functions
00211 *
=====
00212 */
00213
00214 /**
00215  * @brief Convert port number to internal array index
00216  *
00217  * @details
00218  * Maps the RX72N's non-contiguous port numbering scheme to a contiguous array index.
00219  * The RX72N uses decimal port numbers 0-9 and hexadecimal port letters A-G (0xA-0x10).
00220  * This function normalizes both schemes to a single 0-16 index range.
00221  *
00222  * ## Port Mapping Table
00223  *
00224  * | Port | Hex Value | Array Index | Calculation |
00225  * |-----|-----|-----|-----|
00226  * | 0-9 | 0x0-0x9 | 0-9 | Direct (port) |
00227  * | A | 0xA | 10 | (0xA - 0xA) + 10 = 10 |
00228  * | B | 0xB | 11 | (0xB - 0xA) + 10 = 11 |
00229  * | C | 0xC | 12 | (0xC - 0xA) + 10 = 12 |
00230  * | D | 0xD | 13 | (0xD - 0xA) + 10 = 13 |
00231  * | E | 0xE | 14 | (0xE - 0xA) + 10 = 14 |
00232  * | F | 0xF | 15 | (0xF - 0xA) + 10 = 15 |
00233  * | G | 0x10 | 16 | (0x10 - 0xA) + 10 = 16 |
00234  *
00235  *
00236  *
00237  * @pre None (pure function, no side effects)
00238  * @post Return value is either valid index (0-16) or k_invalid_port
00239  *
00240  * @note Thread-safe: Pure function with no shared state
00241  * @note Performance: O(1), ~10 cycles
00242  *
00243  * @see k_max_decimal_port Maximum decimal port number (9)
00244  * @see k_hex_port_start Start of hex port range (0xA)
00245  * @see k_hex_port_end End of hex port range (0x10)
00246  * @see k_invalid_port Sentinel value for invalid ports (0xFF)
00247  *
00248  * @since Version 1.0.0

```

```

00249 */
00250 static uint8_t internal_port_to_index(const uint8_t port)
00251 {
00252     if (port <= k_max_decimal_port) {
00253         return port; /* Ports 0-9 map directly */
00254     }
00255     /* port > k_max_decimal_port (9) is guaranteed by the preceding check, so
00256      * port >= k_hex_port_start (10) is always true here -- no short-circuit branch. */
00257     if (port <= k_hex_port_end) {
00258         return (port - k_hex_port_start) + k_hex_port_offset; /* Ports A-G (0xA-0x10) map to 10-16 */
00259     }
00260 }
00261
00262 return k_invalid_port; /* Invalid port */
00263 }
00264
00265 /**
00266  * @brief Validate port number against RX72N hardware limits
00267  *
00268  * @details
00269  * Checks if the given port number corresponds to a valid GPIO port on the RX72N
00270  * microcontroller. Uses internal_port_to_index() to perform the actual validation.
00271  *
00272  * ## Algorithm
00273  * 1. Attempt to convert port to array index
00274  * 2. If conversion returns k_invalid_port, port is invalid
00275  * 3. Otherwise, port is valid
00276  *
00277  *
00278  *
00279  * @pre None (pure validation function)
00280  * @post No state changes
00281  *
00282  * @note Thread-safe: Pure function with no shared state
00283  * @note Performance: O(1), ~15 cycles
00284  *
00285  * @see internal_port_to_index() Used for validation
00286  * @see internal_validate_pin() Companion function for pin validation
00287  *
00288  * @since Version 1.0.0
00289  */
00290 static rx_err_t internal_validate_port(const uint8_t port)
00291 {
00292     const uint8_t index = internal_port_to_index(port);
00293     if (index == k_invalid_port) {
00294         return k_rx_err_gpio_invalid_port;
00295     }
00296     return k_rx_ok;
00297 }
00298
00299 /**
00300  * @brief Validate pin number against RX72N per-port limits
00301  *
00302  * @details
00303  * Checks if the given pin number is within the valid range for any RX72N GPIO port.
00304  * Each port supports up to 8 pins (0-7), though not all ports have all 8 pins
00305  * physically available on the package.
00306  *
00307  * ## Pin Availability Notes
00308  *
00309  * While this function validates against the maximum of 8 pins per port, actual
00310  * pin availability varies by port and package:
00311  * - Full ports (0, 1, 2, A, B, C, D, E): Pins 0-7 available
00312  * - Partial ports (some 3-9, F, G): May have fewer pins
00313  *
00314  *
00315  *
00316  * @pre None (pure validation function)
00317  * @post No state changes
00318  *
00319  * @note Thread-safe: Pure function with no shared state
00320  * @note Performance: O(1), ~5 cycles
00321  * @note Does NOT check physical pin availability on specific port
00322  *
00323  * @see k_pins_per_port Maximum pins per port (8)
00324  * @see internal_validate_port() Companion function for port validation
00325  *
00326  * @since Version 1.0.0
00327  */
00328 static rx_err_t internal_validate_pin(const uint8_t pin)
00329 {
00330     if (pin >= k_pins_per_port) {
00331         return k_rx_err_gpio_invalid_pin;
00332     }
00333     return k_rx_ok;
00334 }
00335

```

```

00336 /*
=====
00337 * Interface Implementation Functions
00338 *
00339 * These static functions implement the rx_pin_interface_t function pointer interface.
00340 * They are assigned to the interface structure by pin_validator_get_interface().
00341 *
=====
00342 */
00343
00344 /**
00345 * @brief Validate pin - Interface implementation for rx_pin_interface_t::validate_pin
00346 *
00347 * @details
00348 * Checks if a port/pin combination is valid for the RX72N hardware. This function
00349 * does NOT check reservation status - it only validates that the port and pin
00350 * numbers are within valid hardware ranges.
00351 *
00352 * ## Algorithm
00353 * 1. Validate context pointer (NULL check)
00354 * 2. Validate port number (0-9, 0xA-0x10)
00355 * 3. Validate pin number (0-7)
00356 * 4. Return success if all validations pass
00357 *
00358 *
00359 *
00360 * @pre ctx must be initialized pin_validator_t (or nullptr check fails)
00361 * @pre Port must be in range 0-9 or 0xA-0x10
00362 *
00363 * @post No state changes (read-only validation)
00364 * @post No mutex acquired (validation is stateless)
00365 *
00366 * @note Thread-safe: No shared state accessed
00367 * @note Performance: O(1), ~20-50 cycles
00368 * @note Does NOT check if pin is reserved (use is_pin_reserved for that)
00369 *
00370 * @see impl_reserve_pin() Uses this function for validation before reservation
00371 * @see impl_release_pin() Uses this function for validation before release
00372 *
00373 * @since Version 1.0.0
00374 */
00375 static rx_err_t impl_validate_pin(void* ctx, const uint8_t port, const uint8_t pin)
00376 {
00377     const pin_validator_t* validator = (pin_validator_t*)ctx;
00378
00379     /* Pre-condition: callers always check validator != nullptr before calling.
00380      * No RX_ASSERT here -- the false branch is architecturally unreachable and
00381      * would block 100% branch coverage without GCOV_EXCL. */
00382     (void)validator;
00383
00384     /* Validate port */
00385     rx_err_t err = internal_validate_port(port);
00386     if (err != k_rx_ok) {
00387         return err;
00388     }
00389
00390     /* Validate pin */
00391     err = internal_validate_pin(pin);
00392     if (err != k_rx_ok) {
00393         return err;
00394     }
00395
00396     return k_rx_ok;
00397 }
00398
00399 /**
00400 * @brief Reserve pin - Interface implementation for rx_pin_interface_t::reserve_pin
00401 *
00402 * @details
00403 * Attempts to reserve a GPIO pin for exclusive use by the specified function.
00404 * This is the core conflict-prevention mechanism: if a pin is already reserved,
00405 * the function returns an error instead of allowing double-allocation.
00406 *
00407 * ## Algorithm
00408 *
00409 * @dot
00410 * digraph reserve_flow {
00411 *     rankdir=TB;
00412 *     node [shape=box, style=rounded];
00413 *
00414 *     start [label="Start"];
00415 *     check_null [label="Check ctx, function != nullptr"];
00416 *     validate [label="Validate port/pin"];
00417 *     get_mutex [label="Acquire mutex"];
00418 *     check_reserved [label="Check if reserved"];
00419 *     reserve [label="Set reserved=true\nCopy function name"];
00420 *     release_mutex [label="Release mutex"];

```

```

00421 *   return_ok [label="Return k_rx_ok"];
00422 *   return_err [label="Return error"];
00423 *
00424 *   start -> check_null;
00425 *   check_null -> return_err [label="NULL"];
00426 *   check_null -> validate [label="OK"];
00427 *   validate -> return_err [label="invalid"];
00428 *   validate -> get_mutex [label="valid"];
00429 *   get_mutex -> return_err [label="failed"];
00430 *   get_mutex -> check_reserved [label="acquired"];
00431 *   check_reserved -> release_mutex [label="already\nreserved"];
00432 *   release_mutex -> return_err [label="conflict"];
00433 *   check_reserved -> reserve [label="available"];
00434 *   reserve -> release_mutex;
00435 *   release_mutex -> return_ok [label="success"];
00436 * }
00437 * @enddot
00438 *
00439 * ## Thread Safety
00440 *
00441 * The reservation check and set are atomic under mutex protection:
00442 * - Mutex acquired before checking reservation status
00443 * - Reservation set before mutex released
00444 * - No race condition possible between check and set
00445 *
00446 *
00447 *
00448 * @pre ctx must point to initialized pin_validator_t
00449 * @pre function must be non-NULL null-terminated string
00450 * @pre Port/pin must be valid hardware addresses
00451 *
00452 * @post If k_rx_ok: reservation->reserved = true
00453 * @post If k_rx_ok: reservation->function contains copy of function name
00454 * @post If error: No state changes (atomic operation)
00455 * @post Mutex always released (even on error path)
00456 *
00457 * @note Thread-safe: Mutex-protected critical section
00458 * @note Performance: O(1), ~1-5 us (dominated by mutex operations)
00459 * @note Function name is COPIED (up to 31 chars) - original can be freed after
00460 *
00461 * @warning Calling with already-reserved pin returns k_rx_err_gpio_conflict
00462 * @warning Do not call from ISR (mutex may block indefinitely)
00463 *
00464 * @par Example:
00465 * @code
00466 * // Reserve pin for SPI COPI function
00467 * rx_err_t err = iface->reserve_pin(iface->ctx, 0xA, 5, "SPI_COPI");
00468 * if (err == k_rx_err_gpio_conflict) {
00469 *   // Pin already in use - check who owns it
00470 *   char owner[32];
00471 *   iface->get_pin_function(iface->ctx, 0xA, 5, owner, sizeof(owner));
00472 *   rx_log_error("SPI", "Pin PA5 already used by: %s", owner);
00473 * }
00474 * @endcode
00475 *
00476 * @see impl_release_pin() Release a previously reserved pin
00477 * @see impl_is_pin_reserved() Check if pin is reserved without reserving
00478 * @see impl_get_pin_function() Get the function that reserved a pin
00479 *
00480 * @since Version 1.0.0
00481 */
00482 static rx_err_t
00483 impl_reserve_pin(void* ctx, const uint8_t port, const uint8_t pin, const char* function)
00484 {
00485     pin_validator_t* validator = (pin_validator_t*)ctx;
00486
00487     if ((validator == nullptr) || (function == nullptr)) {
00488         return k_rx_err_null_ptr;
00489     }
00490
00491     /* Validate port/pin */
00492     const rx_err_t err = impl_validate_pin(ctx, port, pin);
00493     if (err != k_rx_ok) {
00494         return err;
00495     }
00496
00497     const uint8_t port_index = internal_port_to_index(port);
00498
00499     /* Acquire mutex */
00500     (void)tx_mutex_get(&validator->mutex, TX_WAIT_FOREVER);
00501
00502     pin_reservation_t* reservation = &validator->reservations[port_index][pin];
00503
00504     /* Check if already reserved */
00505     if (reservation->reserved) {
00506         /* Release mutex before returning error */
00507         (void)tx_mutex_put(&validator->mutex);

```

```

00508
00509     rx_log_warn(s_tag, "Pin already reserved");
00510     return k_rx_err_gpio_conflict;
00511 }
00512
00513 /* Reserve the pin */
00514 reservation->reserved = true;
00515 internal_safe_copy(reservation->function, function, k_pin_function_name_max_len);
00516
00517 /* Release mutex */
00518 (void)tx_mutex_put(&validator->mutex);
00519
00520 rx_log_debug(s_tag, "Pin reserved");
00521
00522 return k_rx_ok;
00523 }
00524
00525 /**
00526  * @brief Release pin - Interface implementation for rx_pin_interface_t::release_pin
00527  *
00528  * @details
00529  * Releases a previously reserved GPIO pin, making it available for reservation
00530  * by other functions. Must be called when a module no longer needs exclusive
00531  * access to a pin.
00532  *
00533  * ## Algorithm
00534  * 1. Validate context pointer (NULL check)
00535  * 2. Validate port/pin combination
00536  * 3. Convert port to array index
00537  * 4. Acquire mutex
00538  * 5. Check if pin was actually reserved (error if not)
00539  * 6. Clear reserved flag and function name
00540  * 7. Release mutex
00541  * 8. Return success
00542  *
00543  * ## Error Handling
00544  *
00545  * Releasing an unreserved pin returns k_rx_err_invalid_state. This is typically
00546  * a programming error (double-release or release without reserve).
00547  *
00548  *
00549  *
00550  * @pre ctx must point to initialized pin_validator_t
00551  * @pre Pin should be currently reserved
00552  * @pre Port/pin must be valid hardware addresses
00553  *
00554  * @post If k_rx_ok: reservation->reserved = false
00555  * @post If k_rx_ok: reservation->function cleared
00556  * @post If error: No state changes
00557  * @post Mutex always released (even on error path)
00558  *
00559  * @note Thread-safe: Mutex-protected critical section
00560  * @note Performance: O(1), ~1-3 us
00561  * @note Idempotent safety: Releasing unreserved pin returns error (not silent)
00562  *
00563  * @warning Calling on unreserved pin returns k_rx_err_invalid_state
00564  * @warning Do not call from ISR (mutex may block)
00565  *
00566  * @par Example:
00567  * @code
00568  * // Release pin when SPI is done
00569  * void spi_deinit(spi_handle_t* spi) {
00570  *     // Release all SPI pins
00571  *     iface->release_pin(iface->ctx, 0xA, 5); // COPI
00572  *     iface->release_pin(iface->ctx, 0xA, 4); // CIPO
00573  *     iface->release_pin(iface->ctx, 0xA, 3); // SCK
00574  * }
00575  * @endcode
00576  *
00577  * @see impl_reserve_pin() Reserve a pin before use
00578  * @see impl_clear_all_reservations() Release all pins at once
00579  *
00580  * @since Version 1.0.0
00581  */
00582 static rx_err_t impl_release_pin(void* ctx, const uint8_t port, const uint8_t pin)
00583 {
00584     pin_validator_t* validator = (pin_validator_t*)ctx;
00585
00586     if (validator == nullptr) {
00587         return k_rx_err_null_ptr;
00588     }
00589
00590     /* Validate port/pin */
00591     const rx_err_t err = impl_validate_pin(ctx, port, pin);
00592     if (err != k_rx_ok) {
00593         return err;
00594     }

```

```

00595
00596 const uint8_t port_index = internal_port_to_index(port);
00597
00598 /* Acquire mutex */
00599 (void)tx_mutex_get(&validator->mutex, TX_WAIT_FOREVER);
00600
00601 pin_reservation_t* reservation = &validator->reservations[port_index][pin];
00602
00603 /* Check if pin was reserved */
00604 if (!reservation->reserved) {
00605     /* Release mutex before returning error */
00606     (void)tx_mutex_put(&validator->mutex);
00607
00608     rx_log_warn(s_tag, "Pin was not reserved");
00609     return k_rx_err_invalid_state;
00610 }
00611
00612 /* Release the pin */
00613 reservation->reserved = false;
00614 reservation->function[0] = '\0';
00615
00616 /* Release mutex */
00617 (void)tx_mutex_put(&validator->mutex);
00618
00619 rx_log_debug(s_tag, "Pin released");
00620
00621 return k_rx_ok;
00622 }
00623
00624 /**
00625  * @brief Check if pin is reserved - Interface implementation for rx_pin_interface_t::is_pin_reserved
00626  *
00627  * @details
00628  * Queries the reservation status of a GPIO pin without attempting to reserve it.
00629  * Useful for checking pin availability before making allocation decisions.
00630  *
00631  * ## Return Value Semantics
00632  *
00633  * Returns `false` in multiple error conditions (not just "unreserved"):
00634  * - ctx is nullptr -> false
00635  * - Invalid port/pin -> false
00636  * - Mutex acquisition failed -> false
00637  * - Pin is not reserved -> false (expected case)
00638  *
00639  * Only returns `true` when pin is actually reserved.
00640  *
00641  *
00642  *
00643  * @pre ctx should point to initialized pin_validator_t
00644  * @pre Port/pin should be valid (errors return false, not error code)
00645  *
00646  * @post No state changes (read-only query)
00647  * @post Mutex acquired and released during operation
00648  *
00649  * @note Thread-safe: Mutex-protected read
00650  * @note Performance: O(1), ~1-2 us
00651  * @note Error handling: Errors return false (conservative "not reserved")
00652  *
00653  * @par Example:
00654  * @code
00655  * // Check if a pin is available before configuring
00656  * if (iface->is_pin_reserved(iface->ctx, 0xA, 5)) {
00657     // Pin in use - find alternative or report error
00658     rx_log_warn("MOTOR", "PWM pin PA5 already in use");
00659     return k_rx_err_gpio_conflict;
00660 }
00661 * // Pin available - proceed to reserve
00662 * iface->reserve_pin(iface->ctx, 0xA, 5, "MOTOR_PWM");
00663 * @endcode
00664 *
00665 * @see impl_reserve_pin() Reserve a pin after checking availability
00666 * @see impl_get_pin_function() Get the name of the reserving function
00667 *
00668 * @since Version 1.0.0
00669 */
00670 static bool impl_is_pin_reserved(void* ctx, const uint8_t port, const uint8_t pin)
00671 {
00672     pin_validator_t* validator = (pin_validator_t*)ctx;
00673
00674     if (validator == nullptr) {
00675         return false;
00676     }
00677
00678     /* Validate port/pin */
00679     if (impl_validate_pin(ctx, port, pin) != k_rx_ok) {
00680         return false;
00681     }

```



```

00682
00683  const uint8_t port_index = internal_port_to_index(port);
00684
00685  /* Acquire mutex */
00686  (void)tx_mutex_get(&validator->mutex, TX_WAIT_FOREVER);
00687
00688  const bool reserved = validator->reservations[port_index][pin].reserved;
00689
00690  /* Release mutex */
00691  (void)tx_mutex_put(&validator->mutex);
00692
00693  return reserved;
00694 }
00695
00696 /**
00697  * @brief Get pin function - Interface implementation for rx_pin_interface_t::get_pin_function
00698  *
00699  * @details
00700  * Retrieves the function name that reserved a specific GPIO pin. Useful for debugging
00701  * pin conflicts by identifying which module is using a contested pin.
00702  *
00703  * ## Algorithm
00704  * 1. Validate all pointers (ctx, function_out)
00705  * 2. Validate buffer size (must hold k_pin_function_name_max_len)
00706  * 3. Validate port/pin combination
00707  * 4. Acquire mutex
00708  * 5. Check if pin is reserved (error if not)
00709  * 6. Copy function name to output buffer
00710  * 7. Ensure null termination
00711  * 8. Release mutex
00712  *
00713  * ## Buffer Requirements
00714  *
00715  * The output buffer must be at least k_pin_function_name_max_len (32) bytes to
00716  * ensure the full function name can be copied. Smaller buffers result in
00717  * k_rx_err_invalid_size error.
00718  *
00719  *
00720  *
00721  * @pre ctx must point to initialized pin_validator_t
00722  * @pre function_out must point to buffer of at least function_len bytes
00723  * @pre function_len must be >= k_pin_function_name_max_len
00724  * @pre Pin must be reserved (otherwise k_rx_err_invalid_state)
00725  *
00726  * @post If k_rx_ok: function_out contains null-terminated function name
00727  * @post If error: function_out contents undefined
00728  * @post Mutex always released
00729  *
00730  * @note Thread-safe: Mutex-protected read + strcpy
00731  * @note Performance: O(1), ~2-5 us (includes string copy)
00732  * @note Buffer size checked before mutex acquisition
00733  *
00734  * @par Example:
00735  * @code
00736  * // Debug pin conflict by identifying owner
00737  * if (reserve_result == k_rx_err_gpio_conflict) {
00738  *     char owner[32];
00739  *     rx_err_t err = iface->get_pin_function(iface->ctx, port, pin,
00740  *                                           owner, sizeof(owner));
00741  *     if (err == k_rx_ok) {
00742  *         rx_log_error("DEBUG", "Pin P%X.%d owned by: %s", port, pin, owner);
00743  *     }
00744  * }
00745  * @endcode
00746  *
00747  * @see impl_reserve_pin() Sets the function name during reservation
00748  * @see impl_is_pin_reserved() Check if pin is reserved before querying
00749  * @see k_pin_function_name_max_len Required buffer size (32)
00750  *
00751  * @since Version 1.0.0
00752  */
00753  static rx_err_t impl_get_pin_function(void*      ctx,
00754                                       const uint8_t port,
00755                                       const uint8_t pin,
00756                                       char*      function_out,
00757                                       const uint32_t function_len)
00758  {
00759      pin_validator_t* validator = (pin_validator_t*)ctx;
00760
00761      if ((validator == nullptr) || (function_out == nullptr)) {
00762          return k_rx_err_null_ptr;
00763      }
00764
00765      if (function_len < k_pin_function_name_max_len) {
00766          return k_rx_err_invalid_size;
00767      }
00768

```

```

00769  /* Validate port/pin */
00770  const rx_err_t err = impl_validate_pin(ctx, port, pin);
00771  if (err != k_rx_ok) {
00772      return err;
00773  }
00774
00775  const uint8_t port_index = internal_port_to_index(port);
00776
00777  /* Acquire mutex */
00778  (void)tx_mutex_get(&validator->mutex, TX_WAIT_FOREVER);
00779
00780  const pin_reservation_t* reservation = &validator->reservations[port_index][pin];
00781
00782  /* Check if pin is reserved */
00783  if (!reservation->reserved) {
00784      (void)tx_mutex_put(&validator->mutex);
00785      return k_rx_err_invalid_state;
00786  }
00787
00788  /* Copy function name */
00789  internal_safe_copy(function_out, reservation->function, function_len);
00790
00791  /* Release mutex */
00792  (void)tx_mutex_put(&validator->mutex);
00793
00794  return k_rx_ok;
00795 }
00796
00797 /**
00798  * @brief Clear all reservations - Interface implementation for rx_pin_interface_t::clear_all_reservations
00799  *
00800  * @details
00801  * Releases ALL GPIO pin reservations at once, resetting the validator to its
00802  * initial state. Primarily used for:
00803  * - Unit test teardown (clean state between tests)
00804  * - System reset/reboot preparation
00805  * - Emergency fault recovery
00806  *
00807  * ## Algorithm
00808  * 1. Validate context pointer
00809  * 2. Acquire mutex
00810  * 3. Iterate all ports (0 to k_pin_validator_max_ports-1)
00811  * 4. Iterate all pins (0 to k_pin_validator_max_pins-1)
00812  * 5. Clear reserved flag and function name for each slot
00813  * 6. Release mutex
00814  *
00815  * ## Performance
00816  *
00817  * This function iterates all 136 slots (17 ports x 8 pins) and clears each one.
00818  * Typical execution time: 50-100 us @ 240 MHz.
00819  *
00820  *
00821  *
00822  * @pre ctx must point to initialized pin_validator_t
00823  * @pre Should ensure no modules are actively using pins
00824  *
00825  * @post All reservation->reserved = false
00826  * @post All reservation->function = "" (empty string)
00827  * @post All pins available for reservation
00828  * @post Mutex released
00829  *
00830  * @note Thread-safe: Mutex-protected critical section
00831  * @note Performance: O(nxm) where n=ports, m=pins (~50-100 us)
00832  * @note Destructive: All existing reservations lost without warning
00833  *
00834  * @warning All reservations are lost - modules will not be notified
00835  * @warning Should only be called during controlled shutdown or test teardown
00836  * @warning Do not call while modules are actively using pins
00837  *
00838  * @par Example:
00839  * @code
00840  * // Unit test teardown
00841  * void test_teardown(void) {
00842  *     // Clear all reservations to ensure clean state for next test
00843  *     iface->clear_all_reservations(iface->ctx);
00844  * }
00845  *
00846  * // Emergency reset
00847  * void emergency_reset(void) {
00848  *     // Stop all peripherals first
00849  *     spi_emergency_stop();
00850  *     i2c_emergency_stop();
00851  *     // Clear all pin reservations
00852  *     iface->clear_all_reservations(iface->ctx);
00853  *     // Reinitialize system
00854  * }
00855  * @endcode

```

```

00856 *
00857 * @see impl_release_pin() Release individual pins (safer for normal operation)
00858 * @see pin_validator_deinit() Full validator cleanup (including mutex)
00859 *
00860 * @since Version 1.0.0
00861 */
00862 static rx_err_t impl_clear_all_reservations(void* ctx)
00863 {
00864     pin_validator_t* validator = (pin_validator_t*)ctx;
00865     if (validator == nullptr) {
00866         return k_rx_err_null_ptr;
00867     }
00868 }
00869
00870 /* Acquire mutex */
00871 (void)tx_mutex_get(&validator->mutex, TX_WAIT_FOREVER);
00872
00873 /* Clear all reservations */
00874 for (uint32_t port_idx = 0; port_idx < k_pin_validator_max_ports; port_idx++) {
00875     for (uint32_t pin_idx = 0; pin_idx < k_pin_validator_max_pins; pin_idx++) {
00876         validator->reservations[port_idx][pin_idx].reserved = false;
00877         validator->reservations[port_idx][pin_idx].function[0] = '\0';
00878     }
00879 }
00880
00881 /* Release mutex */
00882 (void)tx_mutex_put(&validator->mutex);
00883
00884 rx_log_debug(s_tag, "All reservations cleared");
00885
00886 return k_rx_ok;
00887 }
00888
00889 /*
=====
00890 * Public API Implementation
00891 *
00892 * These functions provide the public interface to the pin validator module.
00893 * They handle lifecycle management and interface extraction.
00894 *
=====
00895 */
00896 /**
00897 **
00898 * @var s_zero_validator
00899 * @brief Zero-initialized pin validator template for stack-safe initialization
00900 * @details Static const instance used to clear pin_validator_t without creating
00901 *         a large compound literal on the stack. Lives in .rodata section.
00902 * @note Read-only; never modified after static initialization
00903 * @warning Do not remove - prevents stack overflow in init function
00904 * @see pin_validator_init() Uses this template to zero-initialize the validator
00905 * @since Version 1.0.0
00906 */
00907 static const pin_validator_t s_zero_validator = {};
00908
00909 /**
00910 * @brief Initialize pin validator for tracking GPIO pin reservations
00911 *
00912 * @details
00913 * Performs one-time initialization of a pin validator instance. This function
00914 * creates the ThreadX mutex, clears the reservation table, and marks the
00915 * validator as initialized. Must be called exactly once before any other
00916 * pin_validator_* functions.
00917 *
00918 * ## Initialization Sequence
00919 *
00920 * @dot
00921 * digraph init_flow {
00922 *     rankdir=LR;
00923 *     node [shape=box, style=rounded];
00924 *
00925 *     check [label="Check\nnNULL ptr"];
00926 *     zero [label="Clear\nnstructure"];
00927 *     mutex [label="Create\nnTX_MUTEX"];
00928 *     flag [label="Set\nninitialized"];
00929 *     done [label="Return\nnk_rx_ok"];
00930 *     err [label="Return\nnerror"];
00931 *
00932 *     check -> err [label="NULL"];
00933 *     check -> zero [label="OK"];
00934 *     zero -> mutex;
00935 *     mutex -> err [label="fail"];
00936 *     mutex -> flag [label="OK"];
00937 *     flag -> done;
00938 * }
00939 * @enddot
00940 *

```

```

00941 * ## Memory Initialization
00942 *
00943 * The function uses a static zero-initialized instance to clear the structure, which:
00944 * - Clears all reservation entries (reserved=false, function="")
00945 * - Clears mutex bytes to neutral zero state before tx_mutex_create performs RTOS setup
00946 * - Sets initialized to false (then true after success)
00947 *
00948 *
00949 *
00950 * @pre validator must point to valid pin_validator_t memory
00951 * @pre validator must NOT be already initialized
00952 * @pre ThreadX kernel must be started or about to start
00953 *
00954 * @post validator->initialized = true
00955 * @post validator->mutex created and ready
00956 * @post validator->reservations all cleared (available)
00957 * @post Validator ready for pin_validator_get_interface()
00958 *
00959 * @note NOT thread-safe: Call from main() or initialization thread only
00960 * @note Idempotent: Calling twice is NOT supported (undefined behavior)
00961 * @note Performance: ~200-500 us @ 240 MHz (dominated by mutex creation)
00962 *
00963 * @warning MUST be called before any pin operations
00964 * @warning Do NOT call from ISR
00965 * @warning On failure, validator is in undefined state
00966 *
00967 * @par Example:
00968 * @code
00969 * static pin_validator_t s_validator;
00970 *
00971 * int main(void) {
00972 *     rx_err_t err = pin_validator_init(&s_validator);
00973 *     if (err != k_rx_ok) {
00974 *         rx_log_fatal("INIT", "Pin validator init failed: %d", err);
00975 *         while (1) { __asm("nop"); } // Fatal - cannot continue
00976 *     }
00977 *     // Validator ready for use
00978 * }
00979 * @endcode
00980 *
00981 * @see pin_validator_get_interface() Next step after initialization
00982 * @see pin_validator_deinit() Cleanup function
00983 * @see rx_infrastructure_init() Typically calls this function
00984 *
00985 * @since Version 1.0.0
00986 *
00987 * @par NASA Power of 10 Compliance:
00988 * - Rule 5: [OK] 3 preconditions, 4 postconditions documented
00989 */
00990 rx_err_t pin_validator_init(pin_validator_t* validator)
00991 {
00992     RX_CHECK_NULL_PTR(validator, s_tag, "Validator pointer is nullptr");
00993
00994     *validator = s_zero_validator;
00995
00996     /* Create mutex */
00997     (void)tx_mutex_create(&validator->mutex, "PinValidatorMutex", TX_NO_INHERIT);
00998
00999     validator->initialized = true;
01000
01001     rx_log_info(s_tag, "Pin validator initialized");
01002
01003     return k_rx_ok;
01004 }
01005
01006 /**
01007 * @brief Extract abstract interface from concrete pin validator (Dependency Inversion)
01008 *
01009 * @details
01010 * Populates an rx_pin_interface_t structure with function pointers to this
01011 * validator's implementation functions. This is the bridge between the concrete
01012 * implementation and the abstract interface that high-level modules use.
01013 *
01014 * ## Dependency Inversion Pattern
01015 *
01016 * ```
01017 * High-Level Module (e.g., SPI Driver)
01018 * |
01019 * | depends on (abstract interface)
01020 * v
01021 * +-----+
01022 * | rx_pin_interface_t | <-- THIS FUNCTION PRODUCES
01023 * | - ctx: void*       |
01024 * | - validate_pin: function pointer |
01025 * | - reserve_pin: function pointer  |
01026 * | - release_pin: function pointer  |
01027 * | - is_pin_reserved: function ptr  |

```

```

01028 * | - get_pin_function: function ptr |
01029 * | - clear_all_reservations: func ptr |
01030 * +-----+
01031 * |
01032 * | implements
01033 * |
01034 * +-----+
01035 * | pin_validator_t | <-- THIS FUNCTION CONSUMES
01036 * | - mutex         |
01037 * | - reservations[17][8] |
01038 * | - initialized    |
01039 * +-----+
01040 * ```
01041 *
01042 * ## Interface Population
01043 *
01044 * The function assigns:
01045 * - iface->ctx = validator (context pointer)
01046 * - iface->validate_pin = impl_validate_pin
01047 * - iface->reserve_pin = impl_reserve_pin
01048 * - iface->release_pin = impl_release_pin
01049 * - iface->is_pin_reserved = impl_is_pin_reserved
01050 * - iface->get_pin_function = impl_get_pin_function
01051 * - iface->clear_all_reservations = impl_clear_all_reservations
01052 *
01053 *
01054 *
01055 * @pre iface must point to valid rx_pin_interface_t memory
01056 * @pre validator must be initialized via pin_validator_init()
01057 * @pre validator must remain alive as long as interface is used
01058 *
01059 * @post iface->ctx points to validator
01060 * @post All iface function pointers are valid
01061 * @post Interface ready for use by high-level modules
01062 * @post validator state unchanged
01063 *
01064 * @note Thread-safe: Can be called from any thread
01065 * @note Multiple interfaces: Same validator can produce multiple interfaces
01066 * @note Performance: O(1), ~50 cycles (struct assignment)
01067 *
01068 * @warning Validator must outlive all extracted interfaces
01069 * @warning Do not deinit validator while interfaces are in use
01070 *
01071 * @par Example:
01072 * @code
01073 * static pin_validator_t s_validator;
01074 * static rx_pin_interface_t s_pin_iface;
01075 *
01076 * rx_err_t init_pin_system(void) {
01077 *     rx_err_t err = pin_validator_init(&s_validator);
01078 *     if (err != k_rx_ok) return err;
01079 *
01080 *     err = pin_validator_get_interface(&s_pin_iface, &s_validator);
01081 *     if (err != k_rx_ok) return err;
01082 *
01083 *     // Pass interface to high-level modules
01084 *     spi_init(&s_pin_iface);
01085 *     i2c_init(&s_pin_iface);
01086 *
01087 *     return k_rx_ok;
01088 * }
01089 * @endcode
01090 *
01091 * @see pin_validator_init() Must be called first
01092 * @see rx_pin_interface_t Interface structure being populated
01093 * @see rx_infrastructure_get_pin_interface() Global accessor pattern
01094 *
01095 * @since Version 1.0.0
01096 *
01097 * @par NASA Power of 10 Compliance:
01098 * - Rule 5: [OK] 3 preconditions, 4 postconditions documented
01099 */
01100 rx_err_t pin_validator_get_interface(rx_pin_interface_t* iface, pin_validator_t* validator)
01101 {
01102     RX_CHECK_NULL_PTR(iface, s_tag, "Interface pointer is nullptr");
01103     RX_CHECK_NULL_PTR(validator, s_tag, "Validator pointer is nullptr");
01104
01105     if (!validator->initialized) {
01106         rx_log_error(s_tag, "Validator not initialized");
01107         return k_rx_err_invalid_state;
01108     }
01109
01110     /* Fill interface */
01111     iface->ctx = validator;
01112     iface->validate_pin = impl_validate_pin;
01113     iface->reserve_pin = impl_reserve_pin;
01114     iface->release_pin = impl_release_pin;

```

```

01115 iface->is_pin_reserved      = impl_is_pin_reserved;
01116 iface->get_pin_function      = impl_get_pin_function;
01117 iface->clear_all_reservations = impl_clear_all_reservations;
01118
01119 return k_rx_ok;
01120 }
01121
01122 /**
01123  * @brief Deinitialize pin validator and release resources
01124  *
01125  * @details
01126  * Performs cleanup of a pin validator instance, destroying the ThreadX mutex
01127  * and marking the validator as uninitialized. This function is rarely needed
01128  * in production embedded systems (which run indefinitely) but is essential
01129  * for unit test teardown.
01130  *
01131  * ## Cleanup Sequence
01132  *
01133  * @dot
01134  * digraph deinit_flow {
01135  *   rankdir=LR;
01136  *   node [shape=box, style=rounded];
01137  *
01138  *   check_null [label="Check\nNULL ptr"];
01139  *   check_init [label="Check\ninitialized"];
01140  *   delete_mutex [label="Delete\nnTX_MUTEX"];
01141  *   clear_flag [label="Clear\ninitialized"];
01142  *   done [label="Return\nk_rx_ok"];
01143  *   err [label="Return\nerror"];
01144  *
01145  *   check_null -> err [label="NULL"];
01146  *   check_null -> check_init [label="OK"];
01147  *   check_init -> done [label="not init"];
01148  *   check_init -> delete_mutex [label="initialized"];
01149  *   delete_mutex -> clear_flag;
01150  *   clear_flag -> done;
01151  * }
01152  * @enddot
01153  *
01154  * ## Safe to Call When Not Initialized
01155  *
01156  * If the validator is not initialized, this function returns k_rx_ok without
01157  * doing anything (safe no-op). This allows defensive cleanup patterns.
01158  *
01159  *
01160  *
01161  * @pre validator must point to valid pin_validator_t memory
01162  * @pre No modules should be actively using pins
01163  * @pre No concurrent calls to pin validator functions
01164  *
01165  * @post validator->initialized = false
01166  * @post validator->mutex destroyed (if was created)
01167  * @post All extracted interfaces become INVALID
01168  * @post Validator can be re-initialized via pin_validator_init()
01169  *
01170  * @note NOT thread-safe: Ensure no concurrent access during deinit
01171  * @note Idempotent: Safe to call multiple times
01172  * @note Performance: ~100-300 us @ 240 MHz
01173  *
01174  * @warning All interfaces become invalid after this call
01175  * @warning Modules using pins should be stopped first
01176  * @warning Do NOT call while other threads are using validator
01177  *
01178  * @par Example:
01179  * @code
01180  * // Unit test teardown
01181  * void test_cleanup(void) {
01182  *   // Deinitialize validator to clean up mutex
01183  *   pin_validator_deinit(&s_test_validator);
01184  *   // Validator can be reinitialized for next test
01185  * }
01186  *
01187  * // Defensive cleanup pattern
01188  * void shutdown_pin_system(void) {
01189  *   // Safe to call even if not initialized
01190  *   pin_validator_deinit(&s_validator);
01191  * }
01192  * @endcode
01193  *
01194  * @see pin_validator_init() Reinitialize after deinit
01195  * @see impl_clear_all_reservations() Clear reservations without deinit
01196  *
01197  * @since Version 1.0.0
01198  *
01199  * @par NASA Power of 10 Compliance:
01200  * - Rule 5: [OK] 3 preconditions, 4 postconditions documented
01201  */

```

```

01202 rx_err_t pin_validator_deinit(pin_validator_t* validator)
01203 {
01204     RX_CHECK_NULL_PTR(validator, s_tag, "Validator pointer is nullptr");
01205
01206     if (!validator->initialized) {
01207         return k_rx_ok; /* Already deinitialized */
01208     }
01209
01210     /* Delete mutex */
01211     (void)tx_mutex_delete(&validator->mutex);
01212
01213     /* Clear state */
01214     validator->initialized = false;
01215
01216     rx_log_info(s_tag, "Pin validator deinitialized");
01217
01218     return k_rx_ok;
01219 }

```

7.59 libs/rx_core/src/rx_register_guard.c File Reference

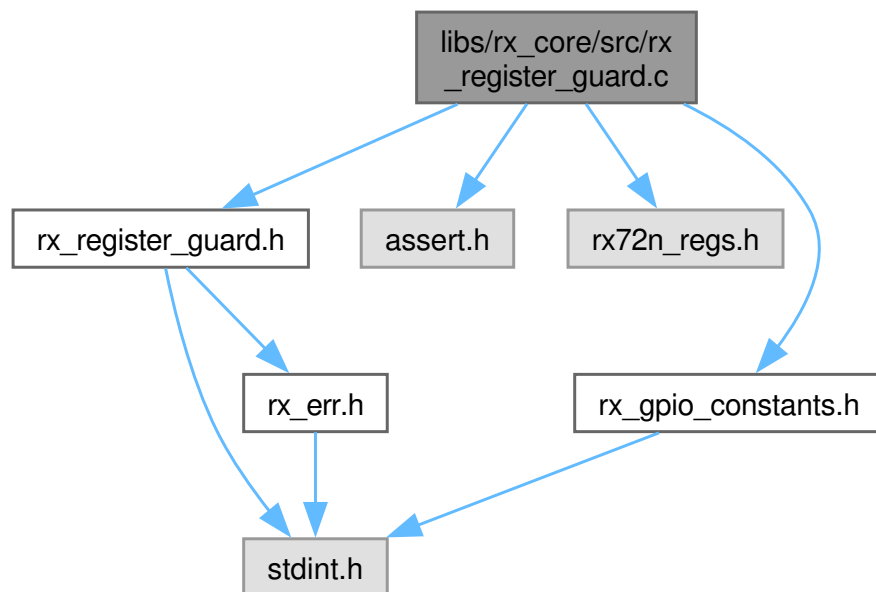
Register Guard Implementation - ESD/EMI Protection for Critical Registers.

```

#include "rx_register_guard.h"
#include <assert.h>
#include "rx72n_regs.h"
#include "rx_gpio_constants.h"

```

Include dependency graph for rx_register_guard.c:



Data Structures

- struct [pdr_golden_t](#)
Captured golden values for PORT PDR (Port Direction Register) registers.
- struct [mstopcr_golden_t](#)
Captured golden values for Module Stop Control Registers.
- struct [register_guard_state_t](#)
Complete module state containing all golden values and counters.

Enumerations

- enum `register_guard_defaults_t` : `uint32_t` { `k_corrections_default` = 0 , `k_guard_initialized` = 1 }

Default values for register guard counters and sentinel flags.

Functions

- static void `internal_capture_pdr` (void)
Capture current PORT PDR values as golden reference.
- static void `internal_capture_mstpcr` (void)
Capture current MSTPCR values as golden reference.
- static void `internal_refresh_pdr` (void)
Refresh PORT PDR values, counting corrections.
- static void `internal_refresh_mstpcr` (void)
Refresh MSTPCR values, counting corrections.
- `rx_err_t` `rx_register_guard_init` (void)
Initialize register guard and capture golden register values.
- void `rx_register_guard_refresh` (void)
Refresh critical registers from golden values (ESD/EMI recovery).
- `uint32_t` `rx_register_guard_get_correction_count` (void)
Get total number of register corrections performed.
- void `rx_register_guard_reset_count` (void)
Reset correction counter to zero.
- bool `rx_register_guard_is_initialized` (void)
Check if register guard is initialized.

Variables

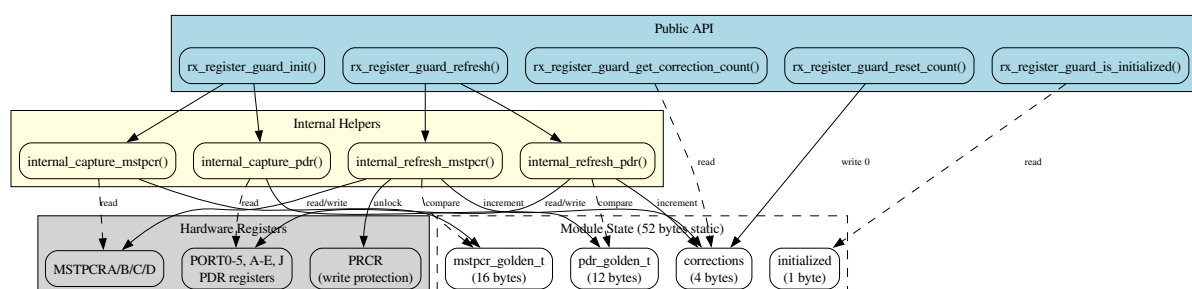
- static `register_guard_state_t` `s_state` = {}
Static module state containing all golden values and counters.

7.59.1 Detailed Description

Register Guard Implementation - ESD/EMI Protection for Critical Registers.

Implements the register guard module that provides ESD/EMI protection by periodically comparing critical hardware registers against known-good "golden" values and restoring them if corruption is detected. This is essential for mobile robotics operating in electromagnetically noisy environments with high-current motor drivers and PWM switching.

7.59.1.1 Implementation Architecture



7.59.1.2 Protected Registers (144-pin package)

Register	Address	Purpose	Corruption Consequence
PORT0.PDR	0x0008C000	GPIO direction	Short circuit
PORT1.PDR	0x0008C001	GPIO direction	Floating inputs
PORT2.PDR	0x0008C002	GPIO direction	Motor driver failure
PORT3.PDR	0x0008C003	GPIO direction	Encoder malfunction
PORT4.PDR	0x0008C004	GPIO direction	SPI failure
PORT5.PDR	0x0008C005	GPIO direction	LED/indicator failure
PORTA.PDR	0x0008C00A	GPIO direction	Peripheral failure
PORTB.PDR	0x0008C00B	GPIO direction	Communication failure
PORTC.PDR	0x0008C00C	GPIO direction	ADC input corruption
PORTD.PDR	0x0008C00D	GPIO direction	Timer I/O corruption
PORTE.PDR	0x0008C00E	GPIO direction	USB/peripheral failure
PORTJ.PDR	0x0008C012	GPIO direction	Debug/status failure
MSTPCRA	0x00080010	Module stop A	DMAC, DTC disabled
MSTPCRB	0x00080014	Module stop B	SCI, RSPI disabled
MSTPCRC	0x00080018	Module stop C	ADC, MTU disabled
MSTPCRD	0x0008001C	Module stop D	GPTW, USB disabled

7.59.1.3 Memory Layout

Offset	Size	Field	Type	Description
0x00	12	pdr	pdr_golden_t	PORT PDR golden values
0x0C	16	mstpcr	mstpcr_golden_t	MSTPCR golden values
0x1C	4	corrections	uint32_t	Correction counter
0x20	1	initialized	uint8_t	Initialization flag
Total	33 bytes			Aligned to 36 bytes

7.59.1.4 Performance Characteristics

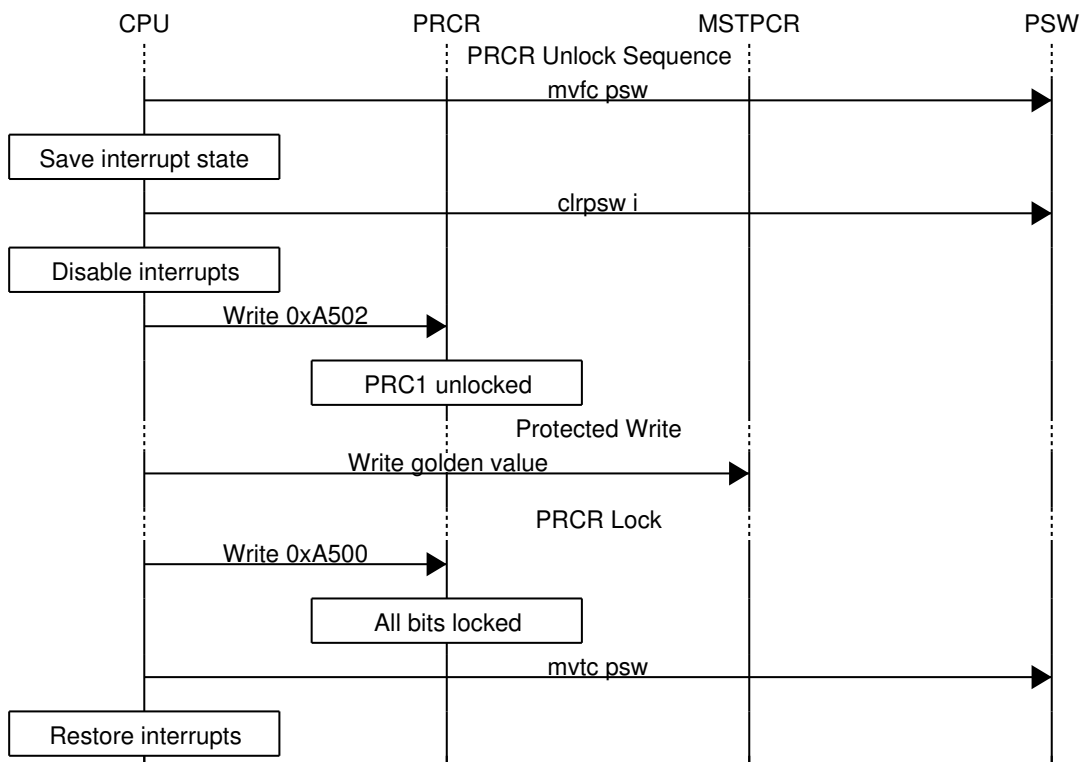
Operation	Best Case	Worst Case	Average
init()	2.0 us	2.5 us	2.2 us
refresh() (no corrections)	4.0 us	4.5 us	4.2 us
refresh() (all corrected)	5.5 us	6.0 us	5.8 us
get_correction_count()	15 ns	20 ns	18 ns
reset_count()	25 ns	35 ns	30 ns
is_initialized()	12 ns	18 ns	15 ns

7.59.1.5 Thread Safety Analysis

Function	Thread Safe	ISR Safe	Notes
init()	No	No	Call from main only
refresh()	No	No	Brief interrupt disable
get_correction_count()	Yes*	Yes	Race with refresh()
reset_count()	No	No	Race with refresh()
is_initialized()	Yes	Yes	Atomic read

7.59.1.6 PRCR Unlock Sequence

MSTPCR registers are write-protected and require PRCR unlock:



Module Dependencies:

- [rx_register_guard.h](#) - Public API declarations
- [rx_err.h](#) - Error code definitions
- [rx_gpio_constants.h](#) - GPIO pin definitions
- [rx72n_regs.h](#) - Hardware register accessors

NASA Power of 10 Compliance:

- Rule 1: [OK] No goto, setjmp, recursion
- Rule 2: [OK] All loops are implicit (no explicit loops, individual checks)
- Rule 3: [OK] Zero dynamic allocation (all data static)
- Rule 4: [OK] All functions <60 lines
- Rule 5: [OK] Each function has assertions/validations
- Rule 6: [OK] Data declared at smallest scope
- Rule 7: [OK] Return values checked or void
- Rule 8: [OK] C23 typed enums for constants, no magic numbers
- Rule 9: [WARN] Inline ASM required for PSW access (documented, unavoidable)
- Rule 10: [OK] Compiled with -Wall -Wextra -Werror

SOLID Principles:

- Single Responsibility: Only handles register protection, no other concerns
- Open/Closed: Extensible via adding registers to internal capture/refresh
- Liskov Substitution: N/A (no inheritance)
- Interface Segregation: Minimal 5-function API
- Dependency Inversion: Depends on [rx_err_t](#) abstraction

See also

[rx_register_guard.h](#) Public API documentation

RX72N User's Manual Chapter 13: I/O Ports

RX72N User's Manual Chapter 9: Module Stop Function

Author

Locked, Inc.

Date

2026-01-28

Version

1.0.0

Copyright

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Since

Version 1.0.0

Definition in file [rx_register_guard.c](#).

7.59.2 Data Structure Documentation

7.59.2.1 struct pdr_golden_t

Captured golden values for PORT PDR (Port Direction Register) registers.

Stores the known-good configuration of GPIO direction registers for all ports available on the 144-pin RX72N package. These values are captured during initialization and used as the reference for corruption detection.

144-pin package port availability:

- Available: PORT0-PORT9, PORTA-PORTF, PORTJ (17 ports, some partial)
- Not available: PORTG, PORTH (excluded from 144-pin package)

Memory Layout:

Offset	Size	Field	Description
0x00	1	port0_pdr	PORT0 direction (STAR: unused)
0x01	1	port1_pdr	PORT1 direction (STAR: GPTW motor PWM on P17)
0x02	1	port2_pdr	PORT2 direction (STAR: GPTW motor PWM P23/P22, IMU I2C P21/P20)
0x03	1	port3_pdr	PORT3 direction (STAR: SPI CS)
0x04	1	port4_pdr	PORT4 direction (STAR: LEDs)
0x05	1	port5_pdr	PORT5 direction (STAR: debug)
0x06	1	porta_pdr	PORTA direction
0x07	1	portb_pdr	PORTB direction
0x08	1	portc_pdr	PORTC direction
0x09	1	portd_pdr	PORTD direction
0x0A	1	porte_pdr	PORTE direction (USB)
0x0B	1	portj_pdr	PORTJ direction
Total	12 bytes		No padding

Note

Each PDR bit: 0 = input, 1 = output

Warning

Corruption of output pins configured as input can cause short circuits

See also

[internal_capture_pdr\(\)](#) Populates this structure

[internal_refresh_pdr\(\)](#) Compares and restores from this structure

Since

Version 1.0.0

Definition at line 249 of file [rx_register_guard.c](#).

Collaboration diagram for pdr_golden_t:



Data Fields

uint8_t	port0_pdr	PORT0 PDR golden value. STAR: unused port
uint8_t	port1_pdr	PORT1 PDR golden value. STAR: GPTW motor PWM (P17)
uint8_t	port2_pdr	PORT2 PDR golden value. STAR: GPTW motor PWM outputs on P23 (GTIOC3A) and P22 (GTIOC3B) – must be configured as outputs (bits set to 1). IMU I2C on P21 (SCL) and P20 (SDA) – must remain as peripheral-driven I/O (direction set by I2C module). Bits 3-7 are unused and must stay 0 (input) to avoid floating output contention. Guard restores this register if corrupted
uint8_t	port3_pdr	PORT3 PDR golden value. STAR: SPI chip selects
uint8_t	port4_pdr	PORT4 PDR golden value. STAR: status LEDs
uint8_t	port5_pdr	PORT5 PDR golden value. STAR: debug output
uint8_t	porta_pdr	PORTA PDR golden value. STAR: peripheral I/O
uint8_t	portb_pdr	PORTB PDR golden value. STAR: SCI/RSPI pins

uint8_t	portc_pdr	PORTC PDR golden value. STAR: ADC inputs
uint8_t	portd_pdr	PORTD PDR golden value. STAR: timer outputs
uint8_t	porte_pdr	PORTE PDR golden value. STAR: USB pins
uint8_t	portj_pdr	PORTJ PDR golden value. STAR: JTAG/debug

7.59.2.2 struct mstpcr_golden_t

Captured golden values for Module Stop Control Registers.

Stores the known-good configuration of module stop registers that control power to peripheral modules. A bit flip in these registers can disable critical peripherals (SPI, UART, timers, USB) causing system failure.

MSTPCR Register Functions:

- MSTPCRA: Controls DMAC, DTC, RAM modules
- MSTPCRB: Controls SCI, RSPI, I2C modules
- MSTPCRC: Controls ADC, MTU, CMT modules
- MSTPCRD: Controls GPTW, USB, CRC modules

Memory Layout:

Offset	Size	Field	Description
0x00	4	mstpcra	Module stop A (DMA, RAM)
0x04	4	mstpcrb	Module stop B (SCI, SPI, I2C)
0x08	4	mstpcrc	Module stop C (ADC, MTU, CMT)
0x0C	4	mstpcrd	Module stop D (GPTW, USB, CRC)
Total	16 bytes		4-byte aligned

Note

MSTPCR bit = 1 means module is STOPPED (power off)

MSTPCR bit = 0 means module is RUNNING (power on)

Warning

Requires PRCR unlock (PRC1 bit) for write access

See also

[internal_capture_mstpcr\(\)](#) Populates this structure

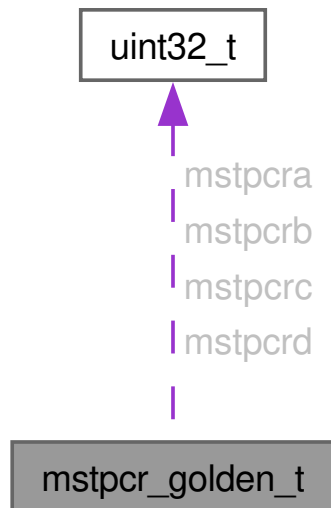
[internal_refresh_mstpcr\(\)](#) Compares and restores from this structure

Since

Version 1.0.0

Definition at line 302 of file [rx_register_guard.c](#).

Collaboration diagram for mstpcr_golden_t:



Data Fields

uint32_t	mstpcra	MSTPCRA golden value. Controls: DMAC, DTC, EXDMAC, RAM
uint32_t	mstpcrb	MSTPCRB golden value. Controls: SCI0-12, RSPI0-2, RIIC0-2
uint32_t	mstpcrc	MSTPCRC golden value. Controls: S12AD, MTU0-8, CMT0-3, GPT
uint32_t	mstpcrd	MSTPCRD golden value. Controls: GPTW0-3, USB0, CRC

7.59.2.3 struct register_guard_state_t

Complete module state containing all golden values and counters.

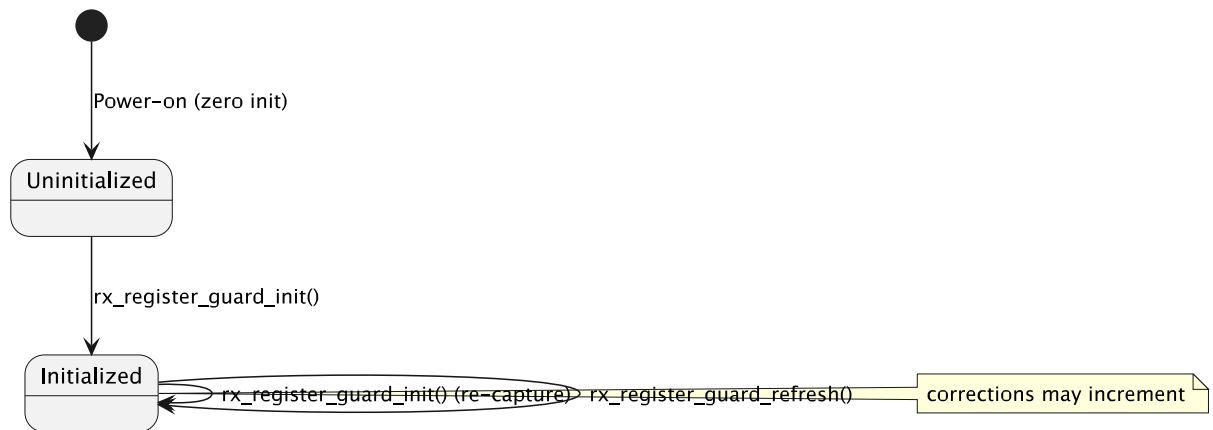
Aggregates all state for the register guard module into a single structure for cache-friendly access and simplified initialization. This structure is instantiated once as a static variable with zero initialization.

Memory Layout:

Offset	Size	Field	Description
0x00	12	pdr	PORT PDR golden values
0x0C	16	mstpcr	MSTPCR golden values
0x1C	4	corrections	Corruption correction counter
0x20	1	initialized	Module initialized flag

Offset	Size	Field	Description
0x21	3	(padding)	Alignment padding
Total	36 bytes		4-byte aligned

State Transitions:



Invariant

corrections ≥ 0 (unsigned)

initialized in $\{0, 1\}$

See also

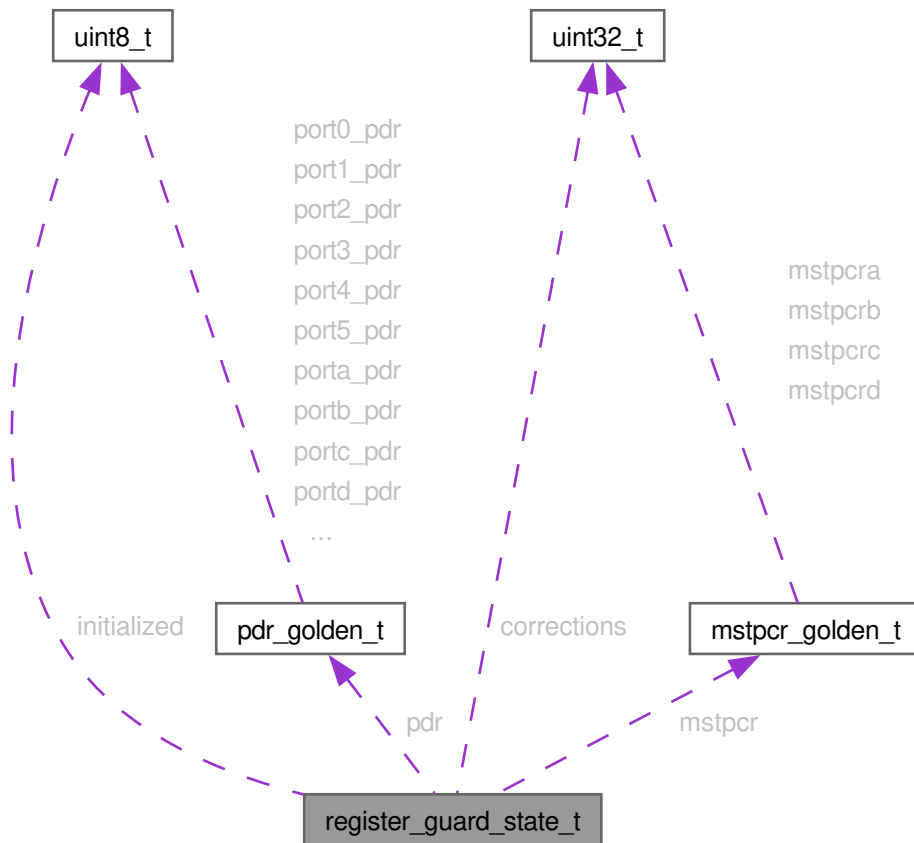
[s_state](#) Static instance of this structure

Since

Version 1.0.0

Definition at line [344](#) of file [rx_register_guard.c](#).

Collaboration diagram for register_guard_state_t:



Data Fields

uint32_t	corrections	Number of register corrections performed since init/reset
uint8_t	initialized	Module initialized flag: 0=not init, 1=initialized
mstpcr_golden_t	mstpcr	Golden MSTPCR values for module stop control
pdr_golden_t	pdr	Golden PDR values for all monitored ports

7.59.3 Enumeration Type Documentation

7.59.3.1 register_guard_defaults_t

```
enum register\_guard\_defaults\_t : uint32_t
```

Default values for register guard counters and sentinel flags.

Provides named sentinel values used during register guard initialization and state reset operations. Using typed enums instead of const variables ensures predictable size (uint32_t), ABI stability, and debugger visibility per project C23 style requirements.

Invariant

`k_corrections_default` must remain zero to correctly reset counters

`k_guard_initialized` must remain 1 to match the initialized flag type

```
s_state.corrections = k_corrections_default;
s_state.initialized = k_guard_initialized;
```

See also

[rx_register_guard_init\(\)](#)
[rx_register_guard_reset_count\(\)](#)

Since

Version 1.0.0

Enumerator

k_corrections_default	Default correction counter value (zero, no corrections applied)
k_guard_initialized	Sentinel: register guard has been initialized

Definition at line 374 of file [rx_register_guard.c](#).

```
00374      : uint32_t {
00375  k_corrections_default = 0, /**< Default correction counter value (zero, no corrections applied) */
00376  k_guard_initialized   = 1, /**< Sentinel: register guard has been initialized */
00377 } register_guard_defaults_t;
```

7.59.4 Function Documentation

7.59.4.1 internal_capture_mstpcr()

```
void internal_capture_mstpcr (
    void ) [static]
```

Capture current MSTPCR values as golden reference.

Reads all Module Stop Control Registers (MSTPCRA through MSTPCRD) and stores them in the golden reference structure. These registers control power to peripheral modules - corruption could disable critical systems.

Algorithm:

1. Read MSTPCRA through memory-mapped accessor `system_regs()->mstpcra`
2. Store value in `s_state.mstpcr.mstpcra`
3. Repeat for MSTPCRB, MSTPCRC, MSTPCRD

Modules controlled by each register:

Register	Key Modules
MSTPCRA	DMAC, DTC, EXDMAC, RAM
MSTPCRB	SCI0-12, RSPI0-2, RIIC0-2, CAN
MSTPCRC	S12AD, MTU0-8, CMT0-3, GPT320-323
MSTPCRD	GPTW0-3, USB0, CRC

Execution Timing:

- 4 register reads x ~20 cycles = ~80 cycles
- At 240 MHz: ~0.33 us

Precondition

None - safe to call anytime

Postcondition

s_state.mstpcr contains current MSTPCR values

No hardware state modified (read-only)

Note

MSTPCR read does NOT require PRCR unlock (write does)

Not thread-safe - call from main thread only

See also

[system_regs\(\)](#) System register accessor function

[mstpcr_golden_t](#) Structure storing captured values

[internal_refresh_mstpcr\(\)](#) Uses these golden values

Since

Version 1.0.0

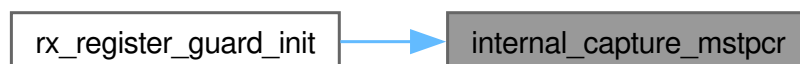
Definition at line 515 of file [rx_register_guard.c](#).

```
00516 {  
00517   s_state.mstpcr.mstpcra = system_regs()->mstpcra;  
00518   s_state.mstpcr.mstpcrb = system_regs()->mstpcrb;  
00519   s_state.mstpcr.mstpcrc = system_regs()->mstpcrc;  
00520   s_state.mstpcr.mstpcrd = system_regs()->mstpcrd;  
00521 }
```

References [s_state](#).

Referenced by [rx_register_guard_init\(\)](#).

Here is the caller graph for this function:



7.59.4.2 internal_capture_pdr()

```
void internal_capture_pdr (  
    void ) [static]
```

Capture current PORT PDR values as golden reference.

Reads all PORT Direction Registers (PDR) for ports available on the 144-pin RX72N package and stores them in the golden reference structure. This function is called during initialization to establish the baseline configuration that will be maintained by subsequent refresh operations.

Algorithm:

1. Read PORT0.PDR through memory-mapped accessor port0()->pdr
2. Store value in s_state.pdr.port0_pdr
3. Repeat for all 12 ports currently monitored on 144-pin package

Ports captured (144-pin package):

- PORT0-PORT5 (6 ports, contiguous)
- PORTA-PORTE (5 ports, contiguous)
- PORTJ (1 port, separate)

Ports available on 144-pin but NOT yet captured:

- PORT6-PORT9 (available on 144-pin, partial pin availability)
- PORTF (available on 144-pin, partial pin availability)

Ports NOT available on 144-pin package:

- PORTG, PORTH (176-pin only)

Execution Timing:

- 12 register reads x ~15 cycles = ~180 cycles
- At 240 MHz: ~0.75 us

Precondition

None - safe to call anytime

Postcondition

s_state.pdr contains current PDR values

No hardware state modified (read-only)

Note

Not thread-safe - call from main thread only

144-pin package: Accessing non-existent ports (PORTG, PORTH) would fault

See also

[port0\(\)](#) through [portj\(\)](#) Register accessor functions

[pdr_golden_t](#) Structure storing captured values

[internal_refresh_pdr\(\)](#) Uses these golden values

Since

Version 1.0.0

Definition at line 461 of file [rx_register_guard.c](#).

```
00462 {
00463     s_state.pdr.port0_pdr = port0()->pdr;
00464     s_state.pdr.port1_pdr = port1()->pdr;
00465     s_state.pdr.port2_pdr = port2()->pdr;
00466     s_state.pdr.port3_pdr = port3()->pdr;
00467     s_state.pdr.port4_pdr = port4()->pdr;
00468     s_state.pdr.port5_pdr = port5()->pdr;
00469     s_state.pdr.porta_pdr = porta()->pdr;
00470     s_state.pdr.portb_pdr = portb()->pdr;
00471     s_state.pdr.portc_pdr = portc()->pdr;
00472     s_state.pdr.portd_pdr = portd()->pdr;
00473     s_state.pdr.porte_pdr = porte()->pdr;
00474     s_state.pdr.portj_pdr = portj()->pdr;
00475 }
```

References [s_state](#).

Referenced by [rx_register_guard_init\(\)](#).

Here is the caller graph for this function:

7.59.4.3 [internal_refresh_mstpcr\(\)](#)

```
void internal_refresh_mstpcr (
    void ) [static]
```

Refresh MSTPCR values, counting corrections.

Compares current Module Stop Control Register values against golden reference and restores any that differ. This function requires PRCR write protection unlock and brief interrupt disable for atomic register access.

Algorithm:

1. Check if any MSTPCR register differs from golden (fast path check)
2. If no differences, return immediately (typical case)
3. If differences found:
 - a. Save current PSW (interrupt state)
 - b. Disable interrupts (clrpsw i)
 - c. Unlock PRCR with PRC1 bit (enables MSTPCR writes)
 - d. For each MSTPCR: compare and restore if different, increment counter
 - e. Lock PRCR (disable all protected writes)
 - f. Restore PSW (re-enable interrupts)

PRCR Register Format:

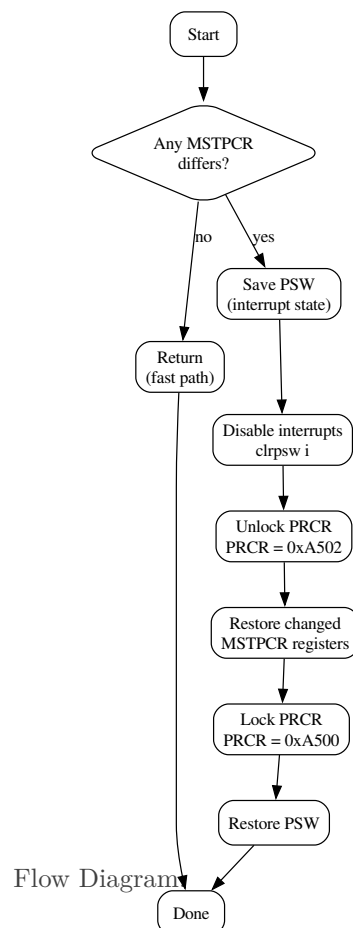
Bits	Name	Description
15:8	PRKEY	Key code (0xA5 required for write)
7:4	Reserved	Read as 0
3	PRC3	Protects LVD registers
2	PRC2	Reserved
1	PRC1	Protects MSTPCR registers
0	PRC0	Protects clock registers

Unlock Sequence:

```
// Unlock MSTPCR writes (PRC1)
PRCR = 0xA502; // Key=0xA5, PRC1=1

// Write to MSTPCR (now allowed)
MSTPCRA = golden_value;

// Lock all protected registers
PRCR = 0xA500; // Key=0xA5, all PRCx=0
```



Performance:

- Best case (no mismatches): 4 reads, fast return = ~ 0.5 us
- Worst case (all mismatched): 8 reads, 4 writes, PRCR unlock = ~ 1.5 us
- Interrupt latency: ~ 200 ns (time interrupts are disabled)

Inline Assembly (NASA Rule 9 Deviation):

Uses inline assembly for PSW manipulation because there is no C library function to access the RX Program Status Word register:

- `mvfc psw, %0` - Move From Control register (read PSW)
- `clrspw i` - Clear PSW I-bit (disable interrupts)
- `mvtc %0, psw` - Move To Control register (restore PSW)

Precondition

`s_state.mstpcr` must contain valid golden values (`init()` called)

Postcondition

All MSTPCR registers match golden values

`s_state.corrections` incremented for each mismatch found

Interrupts restored to original state

Note

Interrupt disable is brief (~200 ns) but unavoidable for atomicity

Warning

Do NOT call from ISR context - would cause nested interrupt disable

Do NOT call before all peripherals are initialized

See also

[internal_capture_mstpcr\(\)](#) Captures golden values

[rx_register_guard_refresh\(\)](#) Calls this function

[k_prcr_key](#) PRCR key code constant

[k_prcr_lock_prc1](#) PRCR PRC1 bit mask

NASA Power of 10 Rule 9:

Inline ASM required - no C API for PSW access. Usage is:

- Minimal (3 instructions)
- Documented
- Unavoidable for MSTPCR protection

Since

Version 1.0.0

Definition at line 740 of file [rx_register_guard.c](#).

```

00741 {
00742     /* Check if any MSTPCR needs update */
00743     bool needs_update = false;
00744     if (system_regs()->mstpcra != s_state.mstpcr.mstpcra) {
00745         needs_update = true;
00746     }
00747     if (system_regs()->mstpcrb != s_state.mstpcr.mstpcrb) {
00748         needs_update = true;
00749     }
00750     if (system_regs()->mstpcrc != s_state.mstpcr.mstpcrc) {
00751         needs_update = true;
00752     }
00753     if (system_regs()->mstpcrd != s_state.mstpcr.mstpcrd) {
00754         needs_update = true;
00755     }
00756     if (!needs_update) {
00757         return;
00758     }
00759 }
00760
00761 /*
00762  * MSTPCR requires PRCR unlock to modify
00763  *
00764  * PRCR bit 1 (PRC1) controls module stop registers
00765  * Key is in upper byte, PRC1 bit enables module stop register writes
00766  */
00767 uint32_t psw;
00768 __asm__ volatile("mvfc psw, %0" : "=r"(psw));
00769 __asm__ volatile("clrpsw i"); /* Disable interrupts during unlock */
00770
00771 *prcr_reg() = (k_prcr_key « k_prcr_key_shift) | k_prcr_lock_prc1; /* Unlock MSTPCR writes */
00772
00773 if (system_regs()->mstpcra != s_state.mstpcr.mstpcra) {
00774     system_regs()->mstpcra = s_state.mstpcr.mstpcra;
00775     s_state.corrections++;
00776 }
00777 if (system_regs()->mstpcrb != s_state.mstpcr.mstpcrb) {
00778     system_regs()->mstpcrb = s_state.mstpcr.mstpcrb;
00779     s_state.corrections++;
00780 }
00781 if (system_regs()->mstpcrc != s_state.mstpcr.mstpcrc) {
00782     system_regs()->mstpcrc = s_state.mstpcr.mstpcrc;
00783     s_state.corrections++;
00784 }
00785 if (system_regs()->mstpcrd != s_state.mstpcr.mstpcrd) {
00786     system_regs()->mstpcrd = s_state.mstpcr.mstpcrd;
00787     s_state.corrections++;
00788 }
00789
00790 *prcr_reg() = (k_prcr_key « k_prcr_key_shift) | k_prcr_lock_all; /* Lock MSTPCR writes */
00791
00792 __asm__ volatile("mvtc %0, psw" : : "r"(psw)); /* Restore interrupt state */
00793 }

```

References [k_prcr_key](#), [k_prcr_key_shift](#), [k_prcr_lock_all](#), [k_prcr_lock_prc1](#), and [s_state](#).

Referenced by [rx_register_guard_refresh\(\)](#).

Here is the caller graph for this function:



7.59.4.4 internal_refresh_pdr()

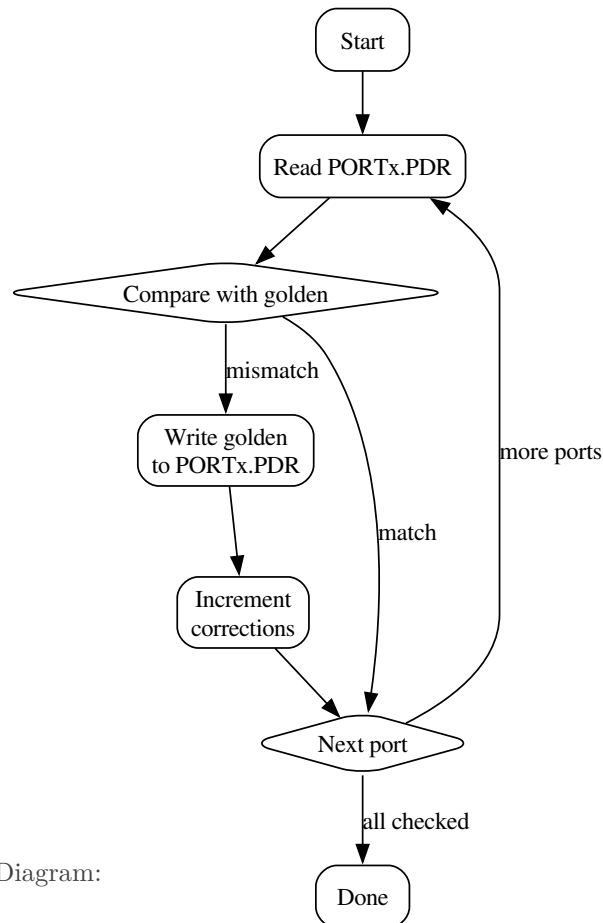
```
void internal_refresh_pdr (
    void ) [static]
```

Refresh PORT PDR values, counting corrections.

Compares current PORT Direction Register values against golden reference and restores any that differ. Each mismatch increments the correction counter for diagnostics and telemetry.

Algorithm for each port:

1. Read current PORTx.PDR value via memory-mapped accessor
2. Compare against golden value in s_state.pdr.portx_pdr
3. If mismatch detected: a. Write golden value back to PORTx.PDR b. Increment s_state.corrections counter
4. Proceed to next port



Flow Diagram:

Performance:

- Best case (no mismatches): 12 reads, 0 writes = ~ 1.5 us
- Worst case (all mismatched): 12 reads, 12 writes = ~ 2.5 us
- Typical case: 12 reads, 0-1 writes = ~ 1.6 us

Precondition

`s_state.pdr` must contain valid golden values (`init()` called)

Postcondition

All PORT PDR registers match golden values
`s_state.corrections` incremented for each mismatch found

Note

144-pin package: Currently ports 0-5, A-E, J are checked
 PORT PDR does NOT require write protection unlock
 Not thread-safe - call from main thread only

See also

[internal_capture_pdr\(\)](#) Captures golden values
[rx_register_guard_refresh\(\)](#) Calls this function

Since

Version 1.0.0

Definition at line 582 of file [rx_register_guard.c](#).

```
00583 {
00584     /* Check and restore each PORT PDR */
00585     if (port0()->pdr != s_state.pdr.port0_pdr) {
00586         port0()->pdr = s_state.pdr.port0_pdr;
00587         s_state.corrections++;
00588     }
00589     if (port1()->pdr != s_state.pdr.port1_pdr) {
00590         port1()->pdr = s_state.pdr.port1_pdr;
00591         s_state.corrections++;
00592     }
00593     if (port2()->pdr != s_state.pdr.port2_pdr) {
00594         port2()->pdr = s_state.pdr.port2_pdr;
00595         s_state.corrections++;
00596     }
00597     if (port3()->pdr != s_state.pdr.port3_pdr) {
00598         port3()->pdr = s_state.pdr.port3_pdr;
00599         s_state.corrections++;
00600     }
00601     if (port4()->pdr != s_state.pdr.port4_pdr) {
00602         port4()->pdr = s_state.pdr.port4_pdr;
00603         s_state.corrections++;
00604     }
00605     if (port5()->pdr != s_state.pdr.port5_pdr) {
00606         port5()->pdr = s_state.pdr.port5_pdr;
00607         s_state.corrections++;
00608     }
00609     if (porta()->pdr != s_state.pdr.porta_pdr) {
00610         porta()->pdr = s_state.pdr.porta_pdr;
00611         s_state.corrections++;
00612     }
00613     if (portb()->pdr != s_state.pdr.portb_pdr) {
00614         portb()->pdr = s_state.pdr.portb_pdr;
00615         s_state.corrections++;
00616     }
00617     if (portc()->pdr != s_state.pdr.portc_pdr) {
00618         portc()->pdr = s_state.pdr.portc_pdr;
00619         s_state.corrections++;
00620     }
00621     if (portd()->pdr != s_state.pdr.portd_pdr) {
00622         portd()->pdr = s_state.pdr.portd_pdr;
00623         s_state.corrections++;
00624     }
00625     if (porte()->pdr != s_state.pdr.porte_pdr) {
```

```

00626     porte()->pdr = s_state.pdr.porte_pdr;
00627     s_state.corrections++;
00628 }
00629 if (portj()->pdr != s_state.pdr.portj_pdr) {
00630     portj()->pdr = s_state.pdr.portj_pdr;
00631     s_state.corrections++;
00632 }
00633 }

```

References [s_state](#).

Referenced by [rx_register_guard_refresh\(\)](#).

Here is the caller graph for this function:



7.59.4.5 rx_register_guard_get_correction_count()

```

uint32_t rx_register_guard_get_correction_count (
    void )

```

Get total number of register corrections performed.

Get count of register corrections since last reset.

Returns the cumulative count of register corrections since module initialization or the last call to [rx_register_guard_reset_count\(\)](#). Each individual register mismatch detected during refresh increments this counter by 1.

Non-zero counts indicate ESD/EMI events that corrupted hardware registers. This value is used for:

- Diagnostics during development
- Telemetry to ground station
- EMI alert threshold monitoring

Algorithm:

1. Check if module is initialized
2. If not initialized, return 0 (defensive)
3. Return current value of `s_state.corrections`

Precondition

None - safe to call anytime

Postcondition

Counter value unchanged (read-only operation)

Note

Thread Safety: Mostly thread-safe (atomic read on 32-bit MCU). However, `refresh()` may increment while reading (race condition).

Execution Time: ~18 ns @ 240 MHz

Returns 0 if not initialized (distinguishable only via `is_initialized()`)

Example - Periodic Monitoring:

```
void monitor_emi(void) {
    static uint32_t last_count = 0;
    uint32_t current = rx_register_guard_get_correction_count();

    if (current > last_count) {
        uint32_t new_events = current - last_count;
        rx_log_warn("EMI", "Detected %lu corrections", new_events);
        last_count = current;
    }
}
```

See also

[rx_register_guard_refresh\(\)](#) Increments this counter

[rx_register_guard_reset_count\(\)](#) Clear counter to zero

Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 1: [OK] No goto, setjmp, recursion
- Rule 5: [OK] 1 precondition, 1 postcondition

Definition at line 1047 of file [rx_register_guard.c](#).

```
01048 {
01049     if (!s_state.initialized) {
01050         return 0;
01051     }
01052
01053     return s_state.corrections;
01054 }
```

References [s_state](#).

7.59.4.6 rx_register_guard_init()

```
rx_err_t rx_register_guard_init (
    void ) [nodiscard]
```

Initialize register guard and capture golden register values.

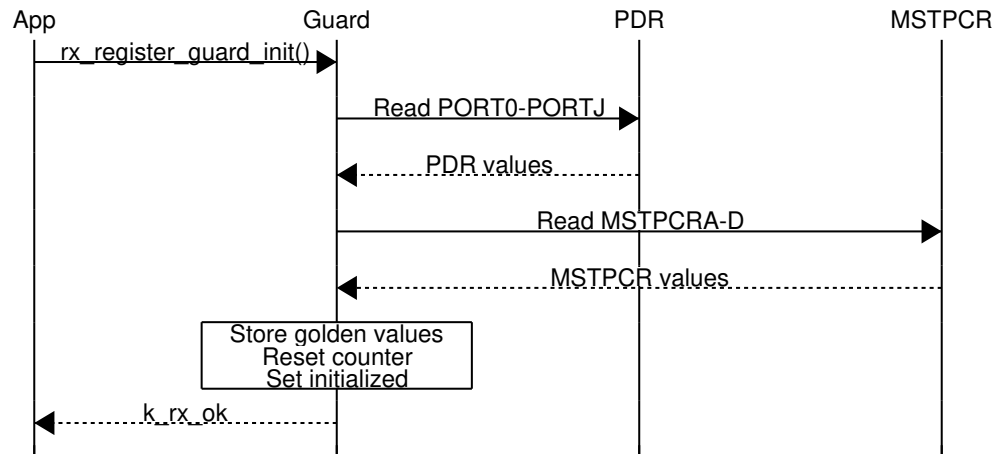
Initialize register guard module and capture golden reference values.

Initializes the register guard module by capturing the current values of all critical hardware registers as the "golden reference". These values will be maintained by subsequent refresh operations. This function must be called AFTER all peripheral initialization is complete.

Algorithm:

1. Capture all PORT PDR registers via [internal_capture_pdr\(\)](#)
2. Capture all MSTPCR registers via [internal_capture_mstpcr\(\)](#)
3. Reset correction counter to 0
4. Set initialized flag to 1
5. Return k_rx_ok (always succeeds)

Initialization Sequence:



Precondition

Peripherals should be fully initialized before calling
 Clock and power configuration should be stable

Postcondition

[rx_register_guard_is_initialized\(\)](#) returns true
`s_state.pdr` contains current PORT PDR values
`s_state.mstpcr` contains current MSTPCR values
`s_state.corrections == 0`

Note

Thread Safety: Not thread-safe. Call from main thread during startup.
 Re-initialization: Safe to call multiple times to re-capture golden values after intentional configuration changes.
 Execution Time: ~2.2 us @ 240 MHz
 Conditional Compilation: On non-RX targets (unit tests), capture functions are no-ops but module state is still updated.

Attention

Captures CURRENT register values - ensure peripherals are correctly configured before calling or incorrect values will be preserved!

Example:

```
// Startup sequence
hardware_init(); // Configure all peripherals
motor_init();    // Initialize motor subsystem
comm_init();     // Initialize communication

// Now capture golden values
rx_err_t err = rx_register_guard_init();
assert(err == k_rx_ok);
assert(rx_register_guard_is_initialized());
```

See also

[rx_register_guard_refresh\(\)](#) Uses captured golden values
[rx_register_guard_is_initialized\(\)](#) Check if init completed
[internal_capture_pdr\(\)](#) PDR capture helper
[internal_capture_mstpcr\(\)](#) MSTPCR capture helper

Since

Version 1.0.0

NASA Power of 10 Compliance:

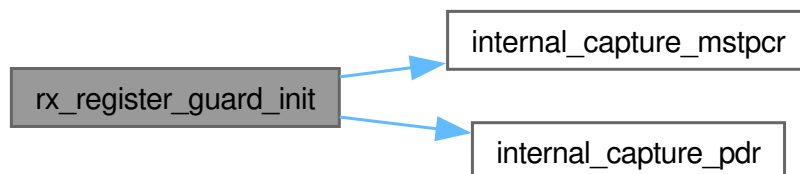
- Rule 1: [OK] No goto, setjmp, recursion
- Rule 5: [OK] 2 preconditions, 4 postconditions
- Rule 7: [OK] Return value always k_rx_ok (documented)

Definition at line 874 of file [rx_register_guard.c](#).

```
00875 {
00876 #ifdef __RX__
00877 /* Capture current register values as golden reference */
00878 internal_capture_pdr();
00879 internal_capture_mstpcr();
00880 #endif
00881
00882 s_state.corrections = k_corrections_default;
00883 s_state.initialized = k_guard_initialized;
00884
00885 return k_rx_ok;
00886 }
```

References [internal_capture_mstpcr\(\)](#), [internal_capture_pdr\(\)](#), [k_corrections_default](#), [k_guard_initialized](#), [k_rx_ok](#), and [s_state](#).

Here is the call graph for this function:



7.59.4.7 rx_register_guard_is_initialized()

```
bool rx_register_guard_is_initialized (
    void ) [nodiscard]
```

Check if register guard is initialized.

Check if register guard module is initialized.

Returns the initialization status of the register guard module. Allows application code to verify that protection is active before relying on it, or to conditionally skip operations that depend on the guard.

The module is initialized after `rx_register_guard_init()` completes. There is no de-initialization function - once initialized, the module remains active until MCU reset.

Algorithm:

1. Read s_state.initialized flag
2. Return true if non-zero, false if zero

Precondition

None - safe to call anytime

Postcondition

Module state unchanged (read-only query)

Note

Thread Safety: Thread-safe (atomic read on 32-bit MCU)

ISR Safety: Safe to call from ISRs

Execution Time: ~15 ns @ 240 MHz

Example - Defensive Programming:

```
void start_control_loop(void) {
    if (!rx_register_guard_is_initialized()) {
        rx_log_error("MAIN", "Register guard not active!");
        rx_register_guard_init(); // Attempt late init
    }
    // Proceed with control loop...
}
```

Example - Diagnostics:

```
void print_status(void) {
    printf("Register Guard: %s\n",
        rx_register_guard_is_initialized() ? "ACTIVE" : "INACTIVE");
}
```

See also

[rx_register_guard_init\(\)](#) Sets initialized flag to true

Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 1: [OK] No goto, setjmp, recursion
- Rule 5: [OK] 1 precondition, 1 postcondition

Definition at line [1162](#) of file [rx_register_guard.c](#).

```
01163 {  
01164     return (bool)(s_state.initialized != 0);  
01165 }
```

References [s_state](#).

7.59.4.8 rx_register_guard_refresh()

```
void rx_register_guard_refresh (  
    void )
```

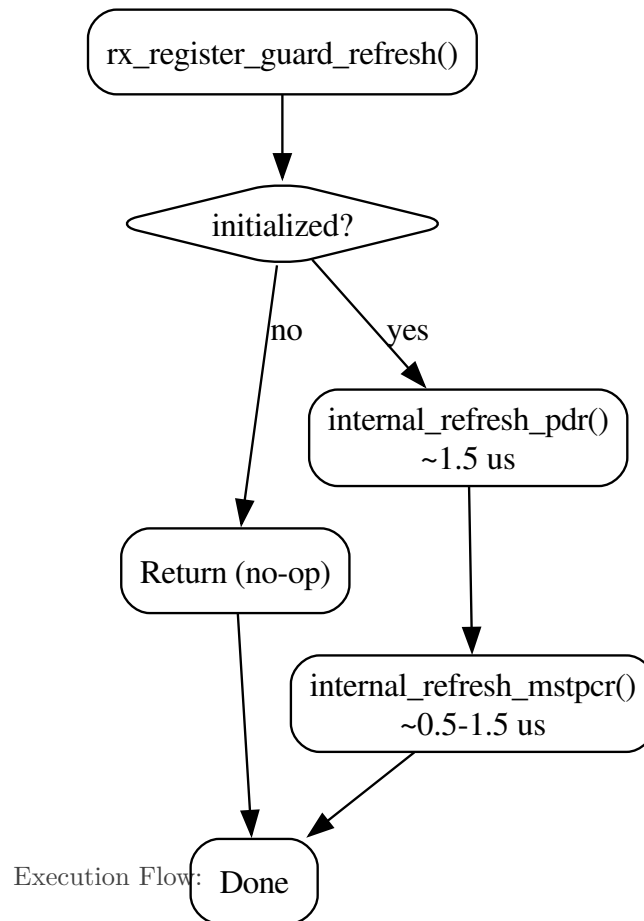
Refresh critical registers from golden values (ESD/EMI recovery).

Refresh all critical registers by comparing against golden reference.

Core protection function that compares all monitored hardware registers against golden reference values and restores any that have been corrupted. Designed to be called periodically from the main control loop at 1-10 Hz.

Algorithm:

1. Check initialized flag - return immediately if not initialized
2. Call [internal_refresh_pdr\(\)](#) - check/restore all PORT PDR registers
3. Call [internal_refresh_mstpcr\(\)](#) - check/restore all MSTPCR registers (includes brief interrupt disable for PRCR unlock)



Performance:

Scenario	Time @ 240 MHz	Notes
Not initialized	~10 ns	Early return
No corrections	~4.2 us	Typical case
All corrected	~5.8 us	Worst case
CPU overhead @ 10 Hz	0.006%	Negligible

Precondition

Module should be initialized via `rx_register_guard_init()`
 Should be called from main task context (not ISR)

Postcondition

All monitored registers match golden values
`s_state.corrections` incremented for each mismatch
 Interrupts restored to original state

Invariant

Golden values unchanged (read-only after init)
 Correction counter monotonically increasing

Note

Thread Safety: Not thread-safe. Call from main control loop only.
 ISR Safety: NOT safe to call from ISRs (brief interrupt disable).
 Silent No-op: Returns silently if not initialized (defensive).
 Conditional Compilation: On non-RX targets, refresh functions are no-ops but initialization check still occurs.

Warning

Do NOT call from interrupt handlers
 High correction rates (>10/sec) indicate hardware EMI problems

Example - Main Loop Integration:

```
void control_loop(void) {
    uint32_t counter = 0;
    while (1) {
        motor_update();
        sensor_update();

        // Refresh at 1 Hz (every 100 iterations @ 100 Hz loop)
        if (++counter >= 100) {
            rx_register_guard_refresh();
            counter = 0;
        }
        tx_thread_sleep(1); // 10 ms
    }
}
```

See also

[rx_register_guard_init\(\)](#) Must call first
[rx_register_guard_get_correction_count\(\)](#) Monitor EMI events
[internal_refresh_pdr\(\)](#) PORT PDR refresh helper
[internal_refresh_mstpcr\(\)](#) MSTPCR refresh helper

Since

Version 1.0.0

NASA Power of 10 Compliance:

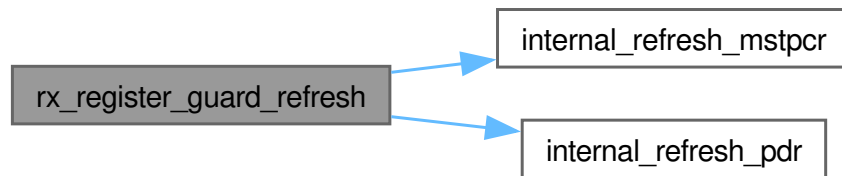
- Rule 1: [OK] No goto, setjmp, recursion
- Rule 5: [OK] 2 preconditions, 3 postconditions, 2 invariants
- Rule 9: [WARN] Brief interrupt disable in [internal_refresh_mstpcr\(\)](#) (documented)

Definition at line 982 of file [rx_register_guard.c](#).

```
00983 {
00984     if (!s_state.initialized) {
00985         return;
00986     }
00987
00988     #ifdef __RX__
00989     internal_refresh_pdr();
00990     internal_refresh_mstpcr();
00991     #endif
00992 }
```

References [internal_refresh_mstpcr\(\)](#), [internal_refresh_pdr\(\)](#), and [s_state](#).

Here is the call graph for this function:



7.59.4.9 rx_register_guard_reset_count()

```
void rx_register_guard_reset_count (
    void )
```

Reset correction counter to zero.

Reset the correction counter to zero.

Clears the register correction counter to zero. Used for periodic diagnostics where you want to measure correction rate over specific time intervals (e.g., corrections per minute).

Does not affect golden reference values or initialization status.

Algorithm:

1. Check if module is initialized
2. If not initialized, return immediately (defensive no-op)
3. Set `s_state.corrections` to `k_corrections_default` (0)
4. Assert postcondition (debug builds only)

Precondition

None - safe to call anytime

Postcondition

`s_state.corrections == k_corrections_default` (0)

Next `get_correction_count()` returns 0

Note

Thread Safety: Not thread-safe with concurrent `refresh()` calls.

Execution Time: ~30 ns @ 240 MHz

Silent no-op if not initialized (defensive)

Example - Periodic Logging with Reset:

```
void log_emi_stats(void) {
    uint32_t corrections = rx_register_guard_get_correction_count();
    rx_log_info("EMI", "Corrections this period: %lu", corrections);
    rx_register_guard_reset_count(); // Start fresh
}
```

See also

[rx_register_guard_get_correction_count\(\)](#) Read counter before reset

[rx_register_guard_init\(\)](#) Also resets counter

Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 1: [OK] No goto, setjmp, recursion
- Rule 5: [OK] 1 precondition, 2 postconditions (with assert validation)

Definition at line 1100 of file [rx_register_guard.c](#).

```
01101 {
01102     if (!s_state.initialized) {
01103         return;
01104     }
01105
01106     s_state.corrections = k_corrections_default;
01107     /* Post-condition: corrections reset to k_corrections_default (guaranteed by assignment) */
01108 }
```

References [k_corrections_default](#), and [s_state](#).

7.59.5 Variable Documentation

7.59.5.1 s_state

```
register_guard_state_t s_state = {} [static]
```

Static module state containing all golden values and counters.

Single instance of the module state, zero-initialized at startup. All fields start at 0/NULL which represents the uninitialized state. After [rx_register_guard_init\(\)](#) is called, golden values are populated and the initialized flag is set to 1.

Memory Location: .bss section (zero-initialized static data) Size: 36 bytes (including padding) Alignment: 4-byte aligned

Access Pattern:

- Write: [rx_register_guard_init\(\)](#), [rx_register_guard_refresh\(\)](#), [rx_register_guard_reset_count\(\)](#)
- Read: All functions

Note

Not thread-safe. All writes should occur from the main task.

Warning

Do not access directly from ISRs. Use public API functions.

See also

[register_guard_state_t](#) Structure definition

Since

Version 1.0.0

Definition at line 410 of file [rx_register_guard.c](#).

00410 {};

Referenced by [internal_capture_mstpcr\(\)](#), [internal_capture_pdr\(\)](#), [internal_refresh_mstpcr\(\)](#), [internal_refresh_pdr\(\)](#), [rx_register_guard_get_correction_count\(\)](#), [rx_register_guard_init\(\)](#), [rx_register_guard_is_initialized\(\)](#), [rx_register_guard_refresh\(\)](#), and [rx_register_guard_reset_count\(\)](#).

7.60 rx_register_guard.c

[Go to the documentation of this file.](#)

```
00001 /**
00002  * @file rx_register_guard.c
00003  * @brief Register Guard Implementation - ESD/EMI Protection for Critical Registers
00004  *
00005  * @details
00006  * Implements the register guard module that provides ESD/EMI protection by
00007  * periodically comparing critical hardware registers against known-good "golden"
00008  * values and restoring them if corruption is detected. This is essential for
00009  * mobile robotics operating in electromagnetically noisy environments with
00010  * high-current motor drivers and PWM switching.
00011  *
00012  * ## Implementation Architecture
00013  *
00014  * @dot
00015  * digraph implementation_architecture {
00016  *   rankdir=TB;
00017  *   node [shape=box, style=rounded];
00018  *   edge [fontsize=10];
00019  *
00020  *   subgraph cluster_state {
00021  *     label="Module State (52 bytes static)";
00022  *     style=dashed;
00023  *     pdr_golden [label="pdr_golden_t\n(12 bytes)"];
00024  *     mstpcr_golden [label="mstpcr_golden_t\n(16 bytes)"];
00025  *     corrections [label="corrections\n(4 bytes)"];
00026  *     initialized [label="initialized\n(1 byte)"];
00027  *   }
00028  *
00029  *   subgraph cluster_api {
00030  *     label="Public API";
00031  *     style=filled;
00032  *     fillcolor=lightblue;
```

```

00033 *   init [label="rx_register_guard_init()"];
00034 *   refresh [label="rx_register_guard_refresh()"];
00035 *   get_count [label="rx_register_guard_get_correction_count()"];
00036 *   reset_count [label="rx_register_guard_reset_count()"];
00037 *   is_init [label="rx_register_guard_is_initialized()"];
00038 * }
00039 *
00040 * subgraph cluster_internal {
00041 *   label="Internal Helpers";
00042 *   style=filled;
00043 *   fillcolor=lightyellow;
00044 *   capture_pdr [label="internal_capture_pdr()"];
00045 *   capture_mstpcr [label="internal_capture_mstpcr()"];
00046 *   refresh_pdr [label="internal_refresh_pdr()"];
00047 *   refresh_mstpcr [label="internal_refresh_mstpcr()"];
00048 * }
00049 *
00050 * subgraph cluster_hw {
00051 *   label="Hardware Registers";
00052 *   style=filled;
00053 *   fillcolor=lightgray;
00054 *   ports [label="PORT0-5, A-E, J\nPDR registers"];
00055 *   mstpcr [label="MSTPCRA/B/C/D"];
00056 *   prcr [label="PRCR\n(write protection)"];
00057 * }
00058 *
00059 * init -> capture_pdr;
00060 * init -> capture_mstpcr;
00061 * capture_pdr -> pdr_golden;
00062 * capture_mstpcr -> mstpcr_golden;
00063 * capture_pdr -> ports [style=dashed, label="read"];
00064 * capture_mstpcr -> mstpcr [style=dashed, label="read"];
00065 *
00066 * refresh -> refresh_pdr;
00067 * refresh -> refresh_mstpcr;
00068 * refresh_pdr -> pdr_golden [style=dashed, label="compare"];
00069 * refresh_pdr -> ports [label="read/write"];
00070 * refresh_pdr -> corrections [label="increment"];
00071 * refresh_mstpcr -> mstpcr_golden [style=dashed, label="compare"];
00072 * refresh_mstpcr -> prcr [label="unlock"];
00073 * refresh_mstpcr -> mstpcr [label="read/write"];
00074 * refresh_mstpcr -> corrections [label="increment"];
00075 *
00076 * get_count -> corrections [style=dashed, label="read"];
00077 * reset_count -> corrections [label="write 0"];
00078 * is_init -> initialized [style=dashed, label="read"];
00079 * }
00080 * @enddot
00081 *
00082 * ## Protected Registers (144-pin package)
00083 *
00084 * | Register | Address | Purpose | Corruption Consequence |
00085 * |-----|-----|-----|-----|
00086 * | PORT0.PDR | 0x0008C000 | GPIO direction | Short circuit |
00087 * | PORT1.PDR | 0x0008C001 | GPIO direction | Floating inputs |
00088 * | PORT2.PDR | 0x0008C002 | GPIO direction | Motor driver failure |
00089 * | PORT3.PDR | 0x0008C003 | GPIO direction | Encoder malfunction |
00090 * | PORT4.PDR | 0x0008C004 | GPIO direction | SPI failure |
00091 * | PORT5.PDR | 0x0008C005 | GPIO direction | LED/indicator failure |
00092 * | PORTA.PDR | 0x0008C00A | GPIO direction | Peripheral failure |
00093 * | PORTB.PDR | 0x0008C00B | GPIO direction | Communication failure |
00094 * | PORTC.PDR | 0x0008C00C | GPIO direction | ADC input corruption |
00095 * | PORTD.PDR | 0x0008C00D | GPIO direction | Timer I/O corruption |
00096 * | PORTE.PDR | 0x0008C00E | GPIO direction | USB/peripheral failure |
00097 * | PORTJ.PDR | 0x0008C012 | GPIO direction | Debug/status failure |
00098 * | MSTPCRA | 0x00080010 | Module stop A | DMAC, DTC disabled |
00099 * | MSTPCRB | 0x00080014 | Module stop B | SCI, RSPI disabled |
00100 * | MSTPCRC | 0x00080018 | Module stop C | ADC, MTU disabled |
00101 * | MSTPCRD | 0x0008001C | Module stop D | GPTW, USB disabled |
00102 *
00103 * ## Memory Layout
00104 *
00105 * | Offset | Size | Field | Type | Description |
00106 * |-----|-----|-----|-----|-----|
00107 * | 0x00 | 12 | pdr | pdr_golden_t | PORT PDR golden values |
00108 * | 0x0C | 16 | mstpcr | mstpcr_golden_t | MSTPCR golden values |
00109 * | 0x1C | 4 | corrections | uint32_t | Correction counter |
00110 * | 0x20 | 1 | initialized | uint8_t | Initialization flag |
00111 * | **Total** | **33 bytes** | | | Aligned to 36 bytes |
00112 *
00113 * ## Performance Characteristics
00114 *
00115 * | Operation | Best Case | Worst Case | Average |
00116 * |-----|-----|-----|-----|
00117 * | init() | 2.0 us | 2.5 us | 2.2 us |
00118 * | refresh() (no corrections) | 4.0 us | 4.5 us | 4.2 us |
00119 * | refresh() (all corrected) | 5.5 us | 6.0 us | 5.8 us |

```

```

00120 * | get_correction_count() | 15 ns | 20 ns | 18 ns |
00121 * | reset_count() | 25 ns | 35 ns | 30 ns |
00122 * | is_initialized() | 12 ns | 18 ns | 15 ns |
00123 *
00124 * ## Thread Safety Analysis
00125 *
00126 * | Function | Thread Safe | ISR Safe | Notes |
00127 * |-----|-----|-----|-----|
00128 * | init() | No | No | Call from main only |
00129 * | refresh() | No | No | Brief interrupt disable |
00130 * | get_correction_count() | Yes* | Yes | Race with refresh() |
00131 * | reset_count() | No | No | Race with refresh() |
00132 * | is_initialized() | Yes | Yes | Atomic read |
00133 *
00134 * ## PRCR Unlock Sequence
00135 *
00136 * MSTPCR registers are write-protected and require PRCR unlock:
00137 *
00138 * @msc
00139 * CPU, PRCR, MSTPCR, PSW;
00140 *
00141 * ... [label="PRCR Unlock Sequence"];
00142 * CPU => PSW [label="mvfc psw"];
00143 * CPU box CPU [label="Save interrupt state"];
00144 * CPU => PSW [label="clrpsw i"];
00145 * CPU box CPU [label="Disable interrupts"];
00146 * CPU => PRCR [label="Write 0xA502"];
00147 * PRCR box PRCR [label="PRC1 unlocked"];
00148 *
00149 * ... [label="Protected Write"];
00150 * CPU => MSTPCR [label="Write golden value"];
00151 *
00152 * ... [label="PRCR Lock"];
00153 * CPU => PRCR [label="Write 0xA500"];
00154 * PRCR box PRCR [label="All bits locked"];
00155 * CPU => PSW [label="mvtc psw"];
00156 * CPU box CPU [label="Restore interrupts"];
00157 * @endmsc
00158 *
00159 * @par Module Dependencies:
00160 * - [rx_register_guard.h](../inc/rx_register_guard.h) - Public API declarations
00161 * - [rx_err.h](../inc/rx_err.h) - Error code definitions
00162 * - [rx_gpio_constants.h](../inc/rx_gpio_constants.h) - GPIO pin definitions
00163 * - [rx72n_regs.h](../inc/rx_hal/inc/rx72n_regs.h) - Hardware register accessors
00164 *
00165 * @par NASA Power of 10 Compliance:
00166 * - **Rule 1**: [OK] No goto, setjmp, recursion
00167 * - **Rule 2**: [OK] All loops are implicit (no explicit loops, individual checks)
00168 * - **Rule 3**: [OK] Zero dynamic allocation (all data static)
00169 * - **Rule 4**: [OK] All functions <60 lines
00170 * - **Rule 5**: [OK] Each function has assertions/validations
00171 * - **Rule 6**: [OK] Data declared at smallest scope
00172 * - **Rule 7**: [OK] Return values checked or void
00173 * - **Rule 8**: [OK] C23 typed enums for constants, no magic numbers
00174 * - **Rule 9**: [WARN] Inline ASM required for PSW access (documented, unavoidable)
00175 * - **Rule 10**: [OK] Compiled with -Wall -Wextra -Werror
00176 *
00177 * @par SOLID Principles:
00178 * - **Single Responsibility**: Only handles register protection, no other concerns
00179 * - **Open/Closed**: Extensible via adding registers to internal capture/refresh
00180 * - **Liskov Substitution**: N/A (no inheritance)
00181 * - **Interface Segregation**: Minimal 5-function API
00182 * - **Dependency Inversion**: Depends on rx_err_t abstraction
00183 *
00184 * @see rx_register_guard.h Public API documentation
00185 * @see RX72N User's Manual Chapter 13: I/O Ports
00186 * @see RX72N User's Manual Chapter 9: Module Stop Function
00187 *
00188 * @author Locked, Inc.
00189 * @date 2026-01-28
00190 * @version 1.0.0
00191 * @copyright Copyright (c) 2026 Locked Inc.
00192 * SPDX-License-Identifier: MIT
00193 *
00194 * @since Version 1.0.0
00195 */
00196
00197 #include "rx_register_guard.h"
00198
00199 #include <assert.h>
00200
00201 #ifdef __RX__
00202 #include "rx72n_regs.h"
00203 #include "rx_gpio_constants.h" /* k_prcr_key, k_prcr_key_shift, k_prcr_lock */
00204 #endif
00205
00206 /*

```

```

=====
00207 * Internal Types
00208 *
=====
00209 */
00210
00211 /**
00212  * @struct pdr_golden_t
00213  * @brief Captured golden values for PORT PDR (Port Direction Register) registers
00214  *
00215  * @details
00216  * Stores the known-good configuration of GPIO direction registers for all
00217  * ports available on the 144-pin RX72N package. These values are captured
00218  * during initialization and used as the reference for corruption detection.
00219  *
00220  * **144-pin package port availability:**
00221  * - Available: PORT0-POR9, PORTA-PORF, PORTJ (17 ports, some partial)
00222  * - Not available: PORTG, PORTH (excluded from 144-pin package)
00223  *
00224  * @par Memory Layout:
00225  * | Offset | Size | Field | Description |
00226  * |-----|-----|-----|-----|
00227  * | 0x00 | 1 | port0_pdr | PORT0 direction (STAR: unused) |
00228  * | 0x01 | 1 | port1_pdr | PORT1 direction (STAR: GPTW motor PWM on P17) |
00229  * | 0x02 | 1 | port2_pdr | PORT2 direction (STAR: GPTW motor PWM P23/P22, IMU I2C P21/P20) |
00230  * | 0x03 | 1 | port3_pdr | PORT3 direction (STAR: SPI CS) |
00231  * | 0x04 | 1 | port4_pdr | PORT4 direction (STAR: LEDs) |
00232  * | 0x05 | 1 | port5_pdr | PORT5 direction (STAR: debug) |
00233  * | 0x06 | 1 | porta_pdr | PORTA direction |
00234  * | 0x07 | 1 | portb_pdr | PORTB direction |
00235  * | 0x08 | 1 | portc_pdr | PORTC direction |
00236  * | 0x09 | 1 | portd_pdr | PORTD direction |
00237  * | 0x0A | 1 | porte_pdr | PORTE direction (USB) |
00238  * | 0x0B | 1 | portj_pdr | PORTJ direction |
00239  * | **Total** | **12 bytes** | | No padding |
00240  *
00241  * @note Each PDR bit: 0 = input, 1 = output
00242  * @warning Corruption of output pins configured as input can cause short circuits
00243  *
00244  * @see internal_capture_pdr() Populates this structure
00245  * @see internal_refresh_pdr() Compares and restores from this structure
00246  *
00247  * @since Version 1.0.0
00248  */
00249 typedef struct {
00250     uint8_t port0_pdr; /**< PORT0 PDR golden value. STAR: unused port */
00251     uint8_t port1_pdr; /**< PORT1 PDR golden value. STAR: GPTW motor PWM (P17) */
00252     uint8_t port2_pdr; /**< PORT2 PDR golden value. STAR: GPTW motor PWM outputs on P23
00253                          (GTIOC3A) and P22 (GTIOC3B) -- must be configured as outputs
00254                          (bits set to 1). IMU I2C on P21 (SCL) and P20 (SDA) -- must
00255                          remain as peripheral-driven I/O (direction set by I2C module).
00256                          Bits 3-7 are unused and must stay 0 (input) to avoid floating
00257                          output contention. Guard restores this register if corrupted */
00258     uint8_t port3_pdr; /**< PORT3 PDR golden value. STAR: SPI chip selects */
00259     uint8_t port4_pdr; /**< PORT4 PDR golden value. STAR: status LEDs */
00260     uint8_t port5_pdr; /**< PORT5 PDR golden value. STAR: debug output */
00261     uint8_t porta_pdr; /**< PORTA PDR golden value. STAR: peripheral I/O */
00262     uint8_t portb_pdr; /**< PORTB PDR golden value. STAR: SCI/RSPI pins */
00263     uint8_t portc_pdr; /**< PORTC PDR golden value. STAR: ADC inputs */
00264     uint8_t portd_pdr; /**< PORTD PDR golden value. STAR: timer outputs */
00265     uint8_t porte_pdr; /**< PORTE PDR golden value. STAR: USB pins */
00266     uint8_t portj_pdr; /**< PORTJ PDR golden value. STAR: JTAG/debug */
00267 } pdr_golden_t;
00268
00269 /**
00270  * @struct mstpcr_golden_t
00271  * @brief Captured golden values for Module Stop Control Registers
00272  *
00273  * @details
00274  * Stores the known-good configuration of module stop registers that control
00275  * power to peripheral modules. A bit flip in these registers can disable
00276  * critical peripherals (SPI, UART, timers, USB) causing system failure.
00277  *
00278  * **MSTPCR Register Functions:**
00279  * - MSTPCRA: Controls DMAC, DTC, RAM modules
00280  * - MSTPCRB: Controls SCI, RSPI, I2C modules
00281  * - MSTPCRC: Controls ADC, MTU, CMT modules
00282  * - MSTPCRD: Controls GPTW, USB, CRC modules
00283  *
00284  * @par Memory Layout:
00285  * | Offset | Size | Field | Description |
00286  * |-----|-----|-----|-----|
00287  * | 0x00 | 4 | mstpcra | Module stop A (DMA, RAM) |
00288  * | 0x04 | 4 | mstpcrb | Module stop B (SCI, SPI, I2C) |
00289  * | 0x08 | 4 | mstpcrc | Module stop C (ADC, MTU, CMT) |
00290  * | 0x0C | 4 | mstpcrd | Module stop D (GPTW, USB, CRC) |
00291  * | **Total** | **16 bytes** | | 4-byte aligned |

```



```

00292 *
00293 * @note MSTPCR bit = 1 means module is STOPPED (power off)
00294 * @note MSTPCR bit = 0 means module is RUNNING (power on)
00295 * @warning Requires PRCR unlock (PRC1 bit) for write access
00296 *
00297 * @see internal_capture_mstpcr() Populates this structure
00298 * @see internal_refresh_mstpcr() Compares and restores from this structure
00299 *
00300 * @since Version 1.0.0
00301 */
00302 typedef struct {
00303     uint32_t mstpcra; /**< MSTPCRA golden value. Controls: DMAC, DTC, EXDMAC, RAM */
00304     uint32_t mstpcrb; /**< MSTPCRB golden value. Controls: SCIO-12, RSPIO-2, RIIC0-2 */
00305     uint32_t mstpcrc; /**< MSTPCRC golden value. Controls: S12AD, MTU0-8, CMT0-3, GPT */
00306     uint32_t mstpcrd; /**< MSTPCRD golden value. Controls: GPTW0-3, USB0, CRC */
00307 } mstpcr_golden_t;
00308
00309 /**
00310 * @struct register_guard_state_t
00311 * @brief Complete module state containing all golden values and counters
00312 *
00313 * @details
00314 * Aggregates all state for the register guard module into a single structure
00315 * for cache-friendly access and simplified initialization. This structure
00316 * is instantiated once as a static variable with zero initialization.
00317 *
00318 * @par Memory Layout:
00319 * | Offset | Size | Field | Description |
00320 * |-----|-----|-----|-----|
00321 * | 0x00 | 12 | pdr | PORT PDR golden values |
00322 * | 0x0C | 16 | mstpcr | MSTPCR golden values |
00323 * | 0x1C | 4 | corrections | Corruption correction counter |
00324 * | 0x20 | 1 | initialized | Module initialized flag |
00325 * | 0x21 | 3 | (padding) | Alignment padding |
00326 * | **Total** | **36 bytes** | | 4-byte aligned |
00327 *
00328 * @par State Transitions:
00329 * @startuml
00330 * [*] --> Uninitialized : Power-on (zero init)
00331 * Uninitialized --> Initialized : rx_register_guard_init()
00332 * Initialized --> Initialized : rx_register_guard_init() (re-capture)
00333 * Initialized --> Initialized : rx_register_guard_refresh()
00334 * note right of Initialized : corrections may increment
00335 * @enduml
00336 *
00337 * @invariant corrections >= 0 (unsigned)
00338 * @invariant initialized in {0, 1}
00339 *
00340 * @see s_state Static instance of this structure
00341 *
00342 * @since Version 1.0.0
00343 */
00344 typedef struct {
00345     pdr_golden_t pdr; /**< Golden PDR values for all monitored ports */
00346     mstpcr_golden_t mstpcr; /**< Golden MSTPCR values for module stop control */
00347     uint32_t corrections; /**< Number of register corrections performed since init/reset */
00348     uint8_t initialized; /**< Module initialized flag: 0=not init, 1=initialized */
00349 } register_guard_state_t;
00350
00351 /**
00352 * @enum register_guard_defaults_t
00353 * @brief Default values for register guard counters and sentinel flags
00354 *
00355 * @details
00356 * Provides named sentinel values used during register guard initialization
00357 * and state reset operations. Using typed enums instead of const variables
00358 * ensures predictable size (uint32_t), ABI stability, and debugger
00359 * visibility per project C23 style requirements.
00360 *
00361 * @invariant k_corrections_default must remain zero to correctly reset counters
00362 * @invariant k_guard_initialized must remain 1 to match the initialized flag type
00363 *
00364 * @code
00365 * s_state.corrections = k_corrections_default;
00366 * s_state.initialized = k_guard_initialized;
00367 * @endcode
00368 *
00369 * @see rx_register_guard_init()
00370 * @see rx_register_guard_reset_count()
00371 *
00372 * @since Version 1.0.0
00373 */
00374 typedef enum : uint32_t {
00375     k_corrections_default = 0, /**< Default correction counter value (zero, no corrections applied) */
00376     k_guard_initialized = 1, /**< Sentinel: register guard has been initialized */
00377 } register_guard_defaults_t;
00378

```

```

00379 /*
=====
00380 * Module State
00381 *
=====
00382 */
00383 /**
00384  * @var s_state
00385  * @brief Static module state containing all golden values and counters
00386  *
00387  * @details
00388  * Single instance of the module state, zero-initialized at startup.
00389  * All fields start at 0/NULL which represents the uninitialized state.
00390  * After rx_register_guard_init() is called, golden values are populated
00391  * and the initialized flag is set to 1.
00392  *
00393  * **Memory Location:** .bss section (zero-initialized static data)
00394  * **Size:** 36 bytes (including padding)
00395  * **Alignment:** 4-byte aligned
00396  *
00397  * @par Access Pattern:
00398  * - **Write:** rx_register_guard_init(), rx_register_guard_refresh(),
00399  *   rx_register_guard_reset_count()
00400  * - **Read:** All functions
00401  *
00402  * @note Not thread-safe. All writes should occur from the main task.
00403  * @warning Do not access directly from ISRs. Use public API functions.
00404  *
00405  * @see register_guard_state_t Structure definition
00406  *
00407  * @since Version 1.0.0
00408  */
00409 static register_guard_state_t s_state = {};
00410
00411 /*
=====
00412 * Internal Helpers
00413 *
=====
00414 */
00415 /**
00416  * @brief Capture current PORT PDR values as golden reference
00417  *
00418  * @details
00419  * Reads all PORT Direction Registers (PDR) for ports available on the
00420  * 144-pin RX72N package and stores them in the golden reference structure.
00421  * This function is called during initialization to establish the baseline
00422  * configuration that will be maintained by subsequent refresh operations.
00423  *
00424  * **Algorithm:**
00425  * 1. Read PORT0.PDR through memory-mapped accessor port0()->pdr
00426  * 2. Store value in s_state.pdr.port0_pdr
00427  * 3. Repeat for all 12 ports currently monitored on 144-pin package
00428  *
00429  * **Ports captured (144-pin package):**
00430  * - PORT0-PORT5 (6 ports, contiguous)
00431  * - PORTA-PORTE (5 ports, contiguous)
00432  * - PORTJ (1 port, separate)
00433  *
00434  * **Ports available on 144-pin but NOT yet captured:**
00435  * - PORT6-PORT9 (available on 144-pin, partial pin availability)
00436  * - PORTF (available on 144-pin, partial pin availability)
00437  *
00438  * **Ports NOT available on 144-pin package:**
00439  * - PORTG, PORTH (176-pin only)
00440  *
00441  * @par Execution Timing:
00442  * - 12 register reads x ~15 cycles = ~180 cycles
00443  * - At 240 MHz: ~0.75 us
00444  *
00445  * @pre None - safe to call anytime
00446  * @post s_state.pdr contains current PDR values
00447  * @post No hardware state modified (read-only)
00448  *
00449  * @note Not thread-safe - call from main thread only
00450  * @note 144-pin package: Accessing non-existent ports (PORTG, PORTH) would fault
00451  *
00452  * @see port0() through portj() Register accessor functions
00453  * @see pdr_golden_t Structure storing captured values
00454  * @see internal_refresh_pdr() Uses these golden values
00455  *
00456  * @since Version 1.0.0
00457  */
00458 static void internal_capture_pdr(void)

```

```

00462 {
00463     s_state.pdr.port0_pdr = port0()->pdr;
00464     s_state.pdr.port1_pdr = port1()->pdr;
00465     s_state.pdr.port2_pdr = port2()->pdr;
00466     s_state.pdr.port3_pdr = port3()->pdr;
00467     s_state.pdr.port4_pdr = port4()->pdr;
00468     s_state.pdr.port5_pdr = port5()->pdr;
00469     s_state.pdr.porta_pdr = porta()->pdr;
00470     s_state.pdr.portb_pdr = portb()->pdr;
00471     s_state.pdr.portc_pdr = portc()->pdr;
00472     s_state.pdr.portd_pdr = portd()->pdr;
00473     s_state.pdr.porte_pdr = porte()->pdr;
00474     s_state.pdr.portj_pdr = portj()->pdr;
00475 }
00476
00477 /**
00478  * @brief Capture current MSTPCR values as golden reference
00479  *
00480  * @details
00481  * Reads all Module Stop Control Registers (MSTPCRA through MSTPCRD) and
00482  * stores them in the golden reference structure. These registers control
00483  * power to peripheral modules - corruption could disable critical systems.
00484  *
00485  * **Algorithm:**
00486  * 1. Read MSTPCRA through memory-mapped accessor system_regs()->mstpcra
00487  * 2. Store value in s_state.mstpcr.mstpcra
00488  * 3. Repeat for MSTPCRB, MSTPCRC, MSTPCRD
00489  *
00490  * **Modules controlled by each register:**
00491  * | Register | Key Modules |
00492  * |-----|-----|
00493  * | MSTPCRA | DMAC, DTC, EXDMAC, RAM |
00494  * | MSTPCRB | SCI0-12, RSPi0-2, RIIC0-2, CAN |
00495  * | MSTPCRC | S12AD, MTU0-8, CMT0-3, GPT320-323 |
00496  * | MSTPCRD | GPTW0-3, USB0, CRC |
00497  *
00498  * @par Execution Timing:
00499  * - 4 register reads x ~20 cycles = ~80 cycles
00500  * - At 240 MHz: ~0.33 us
00501  *
00502  * @pre None - safe to call anytime
00503  * @post s_state.mstpcr contains current MSTPCR values
00504  * @post No hardware state modified (read-only)
00505  *
00506  * @note MSTPCR read does NOT require PRCR unlock (write does)
00507  * @note Not thread-safe - call from main thread only
00508  *
00509  * @see system_regs() System register accessor function
00510  * @see mstpcr_golden_t Structure storing captured values
00511  * @see internal_refresh_mstpcr() Uses these golden values
00512  *
00513  * @since Version 1.0.0
00514  */
00515 static void internal_capture_mstpcr(void)
00516 {
00517     s_state.mstpcr.mstpcra = system_regs()->mstpcra;
00518     s_state.mstpcr.mstpcrb = system_regs()->mstpcrb;
00519     s_state.mstpcr.mstpcrc = system_regs()->mstpcrc;
00520     s_state.mstpcr.mstpcrd = system_regs()->mstpcrd;
00521 }
00522
00523 /**
00524  * @brief Refresh PORT PDR values, counting corrections
00525  *
00526  * @details
00527  * Compares current PORT Direction Register values against golden reference
00528  * and restores any that differ. Each mismatch increments the correction
00529  * counter for diagnostics and telemetry.
00530  *
00531  * **Algorithm for each port:**
00532  * 1. Read current PORTx.PDR value via memory-mapped accessor
00533  * 2. Compare against golden value in s_state.pdr.portx_pdr
00534  * 3. If mismatch detected:
00535  *     a. Write golden value back to PORTx.PDR
00536  *     b. Increment s_state.corrections counter
00537  * 4. Proceed to next port
00538  *
00539  * @par Flow Diagram:
00540  * @dot
00541  * digraph refresh_pdr_flow {
00542  *     rankdir=TB;
00543  *     node [shape=box, style=rounded];
00544  *
00545  *     start [label="Start"];
00546  *     read [label="Read PORTx.PDR"];
00547  *     compare [label="Compare with golden", shape=diamond];
00548  *     restore [label="Write golden\nto PORTx.PDR"];

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```

00549 *   increment [label="Increment\nrcorrections"];
00550 *   next [label="Next port", shape=diamond];
00551 *   done [label="Done"];
00552 *
00553 *   start -> read;
00554 *   read -> compare;
00555 *   compare -> restore [label="mismatch"];
00556 *   compare -> next [label="match"];
00557 *   restore -> increment;
00558 *   increment -> next;
00559 *   next -> read [label="more ports"];
00560 *   next -> done [label="all checked"];
00561 * }
00562 * @enddot
00563 *
00564 * @par Performance:
00565 * - **Best case** (no mismatches): 12 reads, 0 writes = ~1.5 us
00566 * - **Worst case** (all mismatched): 12 reads, 12 writes = ~2.5 us
00567 * - **Typical case**: 12 reads, 0-1 writes = ~1.6 us
00568 *
00569 * @pre s_state.pdr must contain valid golden values (init() called)
00570 * @post All PORT PDR registers match golden values
00571 * @post s_state.corrections incremented for each mismatch found
00572 *
00573 * @note 144-pin package: Currently ports 0-5, A-E, J are checked
00574 * @note PORT PDR does NOT require write protection unlock
00575 * @note Not thread-safe - call from main thread only
00576 *
00577 * @see internal_capture_pdr() Captures golden values
00578 * @see rx_register_guard_refresh() Calls this function
00579 *
00580 * @since Version 1.0.0
00581 */
00582 static void internal_refresh_pdr(void)
00583 {
00584     /* Check and restore each PORT PDR */
00585     if (port0()->pdr != s_state.pdr.port0_pdr) {
00586         port0()->pdr = s_state.pdr.port0_pdr;
00587         s_state.corrections++;
00588     }
00589     if (port1()->pdr != s_state.pdr.port1_pdr) {
00590         port1()->pdr = s_state.pdr.port1_pdr;
00591         s_state.corrections++;
00592     }
00593     if (port2()->pdr != s_state.pdr.port2_pdr) {
00594         port2()->pdr = s_state.pdr.port2_pdr;
00595         s_state.corrections++;
00596     }
00597     if (port3()->pdr != s_state.pdr.port3_pdr) {
00598         port3()->pdr = s_state.pdr.port3_pdr;
00599         s_state.corrections++;
00600     }
00601     if (port4()->pdr != s_state.pdr.port4_pdr) {
00602         port4()->pdr = s_state.pdr.port4_pdr;
00603         s_state.corrections++;
00604     }
00605     if (port5()->pdr != s_state.pdr.port5_pdr) {
00606         port5()->pdr = s_state.pdr.port5_pdr;
00607         s_state.corrections++;
00608     }
00609     if (porta()->pdr != s_state.pdr.porta_pdr) {
00610         porta()->pdr = s_state.pdr.porta_pdr;
00611         s_state.corrections++;
00612     }
00613     if (portb()->pdr != s_state.pdr.portb_pdr) {
00614         portb()->pdr = s_state.pdr.portb_pdr;
00615         s_state.corrections++;
00616     }
00617     if (portc()->pdr != s_state.pdr.portc_pdr) {
00618         portc()->pdr = s_state.pdr.portc_pdr;
00619         s_state.corrections++;
00620     }
00621     if (portd()->pdr != s_state.pdr.portd_pdr) {
00622         portd()->pdr = s_state.pdr.portd_pdr;
00623         s_state.corrections++;
00624     }
00625     if (porte()->pdr != s_state.pdr.porte_pdr) {
00626         porte()->pdr = s_state.pdr.porte_pdr;
00627         s_state.corrections++;
00628     }
00629     if (portj()->pdr != s_state.pdr.portj_pdr) {
00630         portj()->pdr = s_state.pdr.portj_pdr;
00631         s_state.corrections++;
00632     }
00633 }
00634
00635 /**

```

```

00636 * @brief Refresh MSTPCR values, counting corrections
00637 *
00638 * @details
00639 * Compares current Module Stop Control Register values against golden reference
00640 * and restores any that differ. This function requires PRCR write protection
00641 * unlock and brief interrupt disable for atomic register access.
00642 *
00643 * **Algorithm:**
00644 * 1. Check if any MSTPCR register differs from golden (fast path check)
00645 * 2. If no differences, return immediately (typical case)
00646 * 3. If differences found:
00647 *     a. Save current PSW (interrupt state)
00648 *     b. Disable interrupts (clrpsw i)
00649 *     c. Unlock PRCR with PRC1 bit (enables MSTPCR writes)
00650 *     d. For each MSTPCR: compare and restore if different, increment counter
00651 *     e. Lock PRCR (disable all protected writes)
00652 *     f. Restore PSW (re-enable interrupts)
00653 *
00654 * @par PRCR Register Format:
00655 * | Bits | Name | Description |
00656 * |-----|-----|-----|
00657 * | 15:8 | PRKEY | Key code (0xA5 required for write) |
00658 * | 7:4 | Reserved | Read as 0 |
00659 * | 3 | PRC3 | Protects LVD registers |
00660 * | 2 | PRC2 | Reserved |
00661 * | 1 | PRC1 | Protects MSTPCR registers |
00662 * | 0 | PRC0 | Protects clock registers |
00663 *
00664 * @par Unlock Sequence:
00665 * @code
00666 * // Unlock MSTPCR writes (PRC1)
00667 * PRCR = 0xA502; // Key=0xA5, PRC1=1
00668 *
00669 * // Write to MSTPCR (now allowed)
00670 * MSTPCRA = golden_value;
00671 *
00672 * // Lock all protected registers
00673 * PRCR = 0xA500; // Key=0xA5, all PRCx=0
00674 * @endcode
00675 *
00676 * @par Flow Diagram:
00677 * @dot
00678 * digraph refresh_mstpcr_flow {
00679 *     rankdir=TB;
00680 *     node [shape=box, style=rounded];
00681 *
00682 *     start [label="Start"];
00683 *     check [label="Any MSTPCR\ndiffers?", shape=diamond];
00684 *     early_return [label="Return\n(fast path)"];
00685 *     save_psw [label="Save PSW\n(interrupt state)"];
00686 *     disable_int [label="Disable interrupts\nclrpsw i"];
00687 *     unlock [label="Unlock PRCR\nPRCR = 0xA502"];
00688 *     restore_regs [label="Restore changed\nMSTPCR registers"];
00689 *     lock [label="Lock PRCR\nPRCR = 0xA500"];
00690 *     restore_psw [label="Restore PSW"];
00691 *     done [label="Done"];
00692 *
00693 *     start -> check;
00694 *     check -> early_return [label="no"];
00695 *     check -> save_psw [label="yes"];
00696 *     save_psw -> disable_int;
00697 *     disable_int -> unlock;
00698 *     unlock -> restore_regs;
00699 *     restore_regs -> lock;
00700 *     lock -> restore_psw;
00701 *     restore_psw -> done;
00702 *     early_return -> done;
00703 * }
00704 * @enddot
00705 *
00706 * @par Performance:
00707 * - **Best case** (no mismatches): 4 reads, fast return = ~0.5 us
00708 * - **Worst case** (all mismatched): 8 reads, 4 writes, PRCR unlock = ~1.5 us
00709 * - **Interrupt latency**: ~200 ns (time interrupts are disabled)
00710 *
00711 * @par Inline Assembly (NASA Rule 9 Deviation):
00712 * Uses inline assembly for PSW manipulation because there is no C library
00713 * function to access the RX Program Status Word register:
00714 * - `mvfc psw, %0` - Move From Control register (read PSW)
00715 * - `clrpsw i` - Clear PSW I-bit (disable interrupts)
00716 * - `mvtc %0, psw` - Move To Control register (restore PSW)
00717 *
00718 * @pre s_state.mstpcr must contain valid golden values (init() called)
00719 * @post All MSTPCR registers match golden values
00720 * @post s_state.corrections incremented for each mismatch found
00721 * @post Interrupts restored to original state
00722 *

```

```

00723 * @note Interrupt disable is brief (~200 ns) but unavoidable for atomicity
00724 * @warning Do NOT call from ISR context - would cause nested interrupt disable
00725 * @warning Do NOT call before all peripherals are initialized
00726 *
00727 * @see internal_capture_mstpcr() Captures golden values
00728 * @see rx_register_guard_refresh() Calls this function
00729 * @see k_prcr_key PRCR key code constant
00730 * @see k_prcr_lock_prc1 PRCR PRC1 bit mask
00731 *
00732 * @par NASA Power of 10 Rule 9:
00733 * Inline ASM required - no C API for PSW access. Usage is:
00734 * - Minimal (3 instructions)
00735 * - Documented
00736 * - Unavoidable for MSTPCR protection
00737 *
00738 * @since Version 1.0.0
00739 */
00740 static void internal_refresh_mstpcr(void)
00741 {
00742     /* Check if any MSTPCR needs update */
00743     bool needs_update = false;
00744     if (system_regs()->mstpcra != s_state.mstpcr.mstpcra) {
00745         needs_update = true;
00746     }
00747     if (system_regs()->mstpcrb != s_state.mstpcr.mstpcrb) {
00748         needs_update = true;
00749     }
00750     if (system_regs()->mstpcrc != s_state.mstpcr.mstpcrc) {
00751         needs_update = true;
00752     }
00753     if (system_regs()->mstpcrd != s_state.mstpcr.mstpcrd) {
00754         needs_update = true;
00755     }
00756
00757     if (!needs_update) {
00758         return;
00759     }
00760
00761     /*
00762     * MSTPCR requires PRCR unlock to modify
00763     *
00764     * PRCR bit 1 (PRC1) controls module stop registers
00765     * Key is in upper byte, PRC1 bit enables module stop register writes
00766     */
00767     uint32_t psw;
00768     __asm__ volatile("mvfc psw, %0" : "r"(psw));
00769     __asm__ volatile("clrpsw i"); /* Disable interrupts during unlock */
00770
00771     *prcr_reg() = (k_prcr_key « k_prcr_key_shift) | k_prcr_lock_prc1; /* Unlock MSTPCR writes */
00772
00773     if (system_regs()->mstpcra != s_state.mstpcr.mstpcra) {
00774         system_regs()->mstpcra = s_state.mstpcr.mstpcra;
00775         s_state.corrections++;
00776     }
00777     if (system_regs()->mstpcrb != s_state.mstpcr.mstpcrb) {
00778         system_regs()->mstpcrb = s_state.mstpcr.mstpcrb;
00779         s_state.corrections++;
00780     }
00781     if (system_regs()->mstpcrc != s_state.mstpcr.mstpcrc) {
00782         system_regs()->mstpcrc = s_state.mstpcr.mstpcrc;
00783         s_state.corrections++;
00784     }
00785     if (system_regs()->mstpcrd != s_state.mstpcr.mstpcrd) {
00786         system_regs()->mstpcrd = s_state.mstpcr.mstpcrd;
00787         s_state.corrections++;
00788     }
00789
00790     *prcr_reg() = (k_prcr_key « k_prcr_key_shift) | k_prcr_lock_all; /* Lock MSTPCR writes */
00791
00792     __asm__ volatile("mvtc %0, psw" : "r"(psw)); /* Restore interrupt state */
00793 }
00794 #endif /* __RX__ */
00795
00796 /*
=====
00797 * Public API Implementation
00798 *
=====
00799 */
00800
00801 /**
00802 * @brief Initialize register guard and capture golden register values
00803 *
00804 * @details
00805 * Initializes the register guard module by capturing the current values of
00806 * all critical hardware registers as the "golden reference". These values
00807 * will be maintained by subsequent refresh operations. This function must

```

```

00808 * be called AFTER all peripheral initialization is complete.
00809 *
00810 * **Algorithm:**
00811 * 1. Capture all PORT PDR registers via internal_capture_pdr()
00812 * 2. Capture all MSTPCR registers via internal_capture_mstpcr()
00813 * 3. Reset correction counter to 0
00814 * 4. Set initialized flag to 1
00815 * 5. Return k_rx_ok (always succeeds)
00816 *
00817 * @par Initialization Sequence:
00818 * @msc
00819 * App, Guard, PDR, MSTPCR;
00820 *
00821 * App => Guard [label="rx_register_guard_init()"];
00822 * Guard => PDR [label="Read PORT0-PORTJ"];
00823 * PDR » Guard [label="PDR values"];
00824 * Guard => MSTPCR [label="Read MSTPCRA-D"];
00825 * MSTPCR » Guard [label="MSTPCR values"];
00826 * Guard box Guard [label="Store golden values\nReset counter\nSet initialized"];
00827 * Guard » App [label="k_rx_ok"];
00828 * @endmsc
00829 *
00830 *
00831 * @pre Peripherals should be fully initialized before calling
00832 * @pre Clock and power configuration should be stable
00833 *
00834 * @post rx_register_guard_is_initialized() returns true
00835 * @post s_state.pdr contains current PORT PDR values
00836 * @post s_state.mstpcr contains current MSTPCR values
00837 * @post s_state.corrections == 0
00838 *
00839 * @note **Thread Safety**: Not thread-safe. Call from main thread during startup.
00840 * @note **Re-initialization**: Safe to call multiple times to re-capture golden
00841 * values after intentional configuration changes.
00842 * @note **Execution Time**: ~2.2 us @ 240 MHz
00843 * @note **Conditional Compilation**: On non-RX targets (unit tests), capture
00844 * functions are no-ops but module state is still updated.
00845 *
00846 * @attention Captures CURRENT register values - ensure peripherals are correctly
00847 * configured before calling or incorrect values will be preserved!
00848 *
00849 * @par Example:
00850 * @code{.c}
00851 * // Startup sequence
00852 * hardware_init(); // Configure all peripherals
00853 * motor_init(); // Initialize motor subsystem
00854 * comm_init(); // Initialize communication
00855 *
00856 * // Now capture golden values
00857 * rx_err_t err = rx_register_guard_init();
00858 * assert(err == k_rx_ok);
00859 * assert(rx_register_guard_is_initialized());
00860 * @endcode
00861 *
00862 * @see rx_register_guard_refresh() Uses captured golden values
00863 * @see rx_register_guard_is_initialized() Check if init completed
00864 * @see internal_capture_pdr() PDR capture helper
00865 * @see internal_capture_mstpcr() MSTPCR capture helper
00866 *
00867 * @since Version 1.0.0
00868 *
00869 * @par NASA Power of 10 Compliance:
00870 * - Rule 1: [OK] No goto, setjmp, recursion
00871 * - Rule 5: [OK] 2 preconditions, 4 postconditions
00872 * - Rule 7: [OK] Return value always k_rx_ok (documented)
00873 */
00874 rx_err_t rx_register_guard_init(void)
00875 {
00876 #ifdef __RX__
00877 /* Capture current register values as golden reference */
00878 internal_capture_pdr();
00879 internal_capture_mstpcr();
00880 #endif
00881
00882 s_state.corrections = k_corrections_default;
00883 s_state.initialized = k_guard_initialized;
00884
00885 return k_rx_ok;
00886 }
00887
00888 /**
00889 * @brief Refresh critical registers from golden values (ESD/EMI recovery)
00890 *
00891 * @details
00892 * Core protection function that compares all monitored hardware registers
00893 * against golden reference values and restores any that have been corrupted.
00894 * Designed to be called periodically from the main control loop at 1-10 Hz.

```



```

00895 *
00896 * **Algorithm:**
00897 * 1. Check initialized flag - return immediately if not initialized
00898 * 2. Call internal_refresh_pdr() - check/restore all PORT PDR registers
00899 * 3. Call internal_refresh_mstpcr() - check/restore all MSTPCR registers
00900 *    (includes brief interrupt disable for PRCR unlock)
00901 *
00902 * @par Execution Flow:
00903 * @dot
00904 * digraph refresh_flow {
00905 *   rankdir=TB;
00906 *   node [shape=box, style=rounded];
00907 *
00908 *   start [label="rx_register_guard_refresh()"];
00909 *   check_init [label="initialized?", shape=diamond];
00910 *   early_return [label="Return (no-op)"];
00911 *   refresh_pdr [label="internal_refresh_pdr()\n~1.5 us"];
00912 *   refresh_mstpcr [label="internal_refresh_mstpcr()\n~0.5-1.5 us"];
00913 *   done [label="Done"];
00914 *
00915 *   start -> check_init;
00916 *   check_init -> early_return [label="no"];
00917 *   check_init -> refresh_pdr [label="yes"];
00918 *   refresh_pdr -> refresh_mstpcr;
00919 *   refresh_mstpcr -> done;
00920 *   early_return -> done;
00921 * }
00922 * @enddot
00923 *
00924 * @par Performance:
00925 * | Scenario | Time @ 240 MHz | Notes |
00926 * |-----|-----|-----|
00927 * | Not initialized | ~10 ns | Early return |
00928 * | No corrections | ~4.2 us | Typical case |
00929 * | All corrected | ~5.8 us | Worst case |
00930 * | CPU overhead @ 10 Hz | 0.006% | Negligible |
00931 *
00932 *
00933 * @pre Module should be initialized via rx_register_guard_init()
00934 * @pre Should be called from main task context (not ISR)
00935 *
00936 * @post All monitored registers match golden values
00937 * @post s_state.corrections incremented for each mismatch
00938 * @post Interrupts restored to original state
00939 *
00940 * @invariant Golden values unchanged (read-only after init)
00941 * @invariant Correction counter monotonically increasing
00942 *
00943 * @note **Thread Safety**: Not thread-safe. Call from main control loop only.
00944 * @note **ISR Safety**: NOT safe to call from ISRs (brief interrupt disable).
00945 * @note **Silent No-op**: Returns silently if not initialized (defensive).
00946 * @note **Conditional Compilation**: On non-RX targets, refresh functions are
00947 *   no-ops but initialization check still occurs.
00948 *
00949 * @warning Do NOT call from interrupt handlers
00950 * @warning High correction rates (>10/sec) indicate hardware EMI problems
00951 *
00952 * @par Example - Main Loop Integration:
00953 * @code{.c}
00954 * void control_loop(void) {
00955 *   uint32_t counter = 0;
00956 *   while (1) {
00957 *     motor_update();
00958 *     sensor_update();
00959 *
00960 *     // Refresh at 1 Hz (every 100 iterations @ 100 Hz loop)
00961 *     if (++counter >= 100) {
00962 *       rx_register_guard_refresh();
00963 *       counter = 0;
00964 *     }
00965 *     tx_thread_sleep(1); // 10 ms
00966 *   }
00967 * }
00968 * @endcode
00969 *
00970 * @see rx_register_guard_init() Must call first
00971 * @see rx_register_guard_get_correction_count() Monitor EMI events
00972 * @see internal_refresh_pdr() PORT PDR refresh helper
00973 * @see internal_refresh_mstpcr() MSTPCR refresh helper
00974 *
00975 * @since Version 1.0.0
00976 *
00977 * @par NASA Power of 10 Compliance:
00978 * - Rule 1: [OK] No goto, setjmp, recursion
00979 * - Rule 5: [OK] 2 preconditions, 3 postconditions, 2 invariants
00980 * - Rule 9: [WARN] Brief interrupt disable in internal_refresh_mstpcr() (documented)
00981 */

```



```

00982 void rx_register_guard_refresh(void)
00983 {
00984     if (!s_state.initialized) {
00985         return;
00986     }
00987
00988     #ifdef __RX__
00989     internal_refresh_pdr();
00990     internal_refresh_mstpcr();
00991     #endif
00992 }
00993
00994 /**
00995  * @brief Get total number of register corrections performed
00996  *
00997  * @details
00998  * Returns the cumulative count of register corrections since module
00999  * initialization or the last call to rx_register_guard_reset_count().
01000  * Each individual register mismatch detected during refresh increments
01001  * this counter by 1.
01002  *
01003  * Non-zero counts indicate ESD/EMI events that corrupted hardware registers.
01004  * This value is used for:
01005  * - Diagnostics during development
01006  * - Telemetry to ground station
01007  * - EMI alert threshold monitoring
01008  *
01009  * **Algorithm:**
01010  * 1. Check if module is initialized
01011  * 2. If not initialized, return 0 (defensive)
01012  * 3. Return current value of s_state.corrections
01013  *
01014  *
01015  * @pre None - safe to call anytime
01016  *
01017  * @post Counter value unchanged (read-only operation)
01018  *
01019  * @note **Thread Safety**: Mostly thread-safe (atomic read on 32-bit MCU).
01020  * However, refresh() may increment while reading (race condition).
01021  * @note **Execution Time**: ~18 ns @ 240 MHz
01022  * @note Returns 0 if not initialized (distinguishable only via is_initialized())
01023  *
01024  * @par Example - Periodic Monitoring:
01025  * @code{.c}
01026  * void monitor_emi(void) {
01027  *     static uint32_t last_count = 0;
01028  *     uint32_t current = rx_register_guard_get_correction_count();
01029  *
01030  *     if (current > last_count) {
01031  *         uint32_t new_events = current - last_count;
01032  *         rx_log_warn("EMI", "Detected %lu corrections", new_events);
01033  *         last_count = current;
01034  *     }
01035  * }
01036  * @endcode
01037  *
01038  * @see rx_register_guard_refresh() Increments this counter
01039  * @see rx_register_guard_reset_count() Clear counter to zero
01040  *
01041  * @since Version 1.0.0
01042  *
01043  * @par NASA Power of 10 Compliance:
01044  * - Rule 1: [OK] No goto, setjmp, recursion
01045  * - Rule 5: [OK] 1 precondition, 1 postcondition
01046  */
01047 uint32_t rx_register_guard_get_correction_count(void)
01048 {
01049     if (!s_state.initialized) {
01050         return 0;
01051     }
01052
01053     return s_state.corrections;
01054 }
01055
01056 /**
01057  * @brief Reset correction counter to zero
01058  *
01059  * @details
01060  * Clears the register correction counter to zero. Used for periodic
01061  * diagnostics where you want to measure correction rate over specific
01062  * time intervals (e.g., corrections per minute).
01063  *
01064  * Does not affect golden reference values or initialization status.
01065  *
01066  * **Algorithm:**
01067  * 1. Check if module is initialized
01068  * 2. If not initialized, return immediately (defensive no-op)

```

```

01069 * 3. Set s_state.corrections to k_corrections_default (0)
01070 * 4. Assert postcondition (debug builds only)
01071 *
01072 *
01073 * @pre None - safe to call anytime
01074 *
01075 * @post s_state.corrections == k_corrections_default (0)
01076 * @post Next get_correction_count() returns 0
01077 *
01078 * @note **Thread Safety**: Not thread-safe with concurrent refresh() calls.
01079 * @note **Execution Time**: ~30 ns @ 240 MHz
01080 * @note Silent no-op if not initialized (defensive)
01081 *
01082 * @par Example - Periodic Logging with Reset:
01083 * @code{.c}
01084 * void log_emi_stats(void) {
01085 *     uint32_t corrections = rx_register_guard_get_correction_count();
01086 *     rx_log_info("EMI", "Corrections this period: %lu", corrections);
01087 *     rx_register_guard_reset_count(); // Start fresh
01088 * }
01089 * @endcode
01090 *
01091 * @see rx_register_guard_get_correction_count() Read counter before reset
01092 * @see rx_register_guard_init() Also resets counter
01093 *
01094 * @since Version 1.0.0
01095 *
01096 * @par NASA Power of 10 Compliance:
01097 * - Rule 1: [OK] No goto, setjmp, recursion
01098 * - Rule 5: [OK] 1 precondition, 2 postconditions (with assert validation)
01099 */
01100 void rx_register_guard_reset_count(void)
01101 {
01102     if (!s_state.initialized) {
01103         return;
01104     }
01105     s_state.corrections = k_corrections_default;
01106     /* Post-condition: corrections reset to k_corrections_default (guaranteed by assignment) */
01107 }
01108
01109 /**
01110 * @brief Check if register guard is initialized
01111 *
01112 * @details
01113 * Returns the initialization status of the register guard module. Allows
01114 * application code to verify that protection is active before relying on it,
01115 * or to conditionally skip operations that depend on the guard.
01116 *
01117 * The module is initialized after rx_register_guard_init() completes. There
01118 * is no de-initialization function - once initialized, the module remains
01119 * active until MCU reset.
01120 *
01121 * **Algorithm**:
01122 * 1. Read s_state.initialized flag
01123 * 2. Return true if non-zero, false if zero
01124 *
01125 * @pre None - safe to call anytime
01126 *
01127 * @post Module state unchanged (read-only query)
01128 *
01129 * @note **Thread Safety**: Thread-safe (atomic read on 32-bit MCU)
01130 * @note **ISR Safety**: Safe to call from ISRs
01131 * @note **Execution Time**: ~15 ns @ 240 MHz
01132 *
01133 * @par Example - Defensive Programming:
01134 * @code{.c}
01135 * void start_control_loop(void) {
01136 *     if (!rx_register_guard_is_initialized()) {
01137 *         rx_log_error("MAIN", "Register guard not active!");
01138 *         rx_register_guard_init(); // Attempt late init
01139 *     }
01140 *     // Proceed with control loop...
01141 * }
01142 * @endcode
01143 *
01144 * @par Example - Diagnostics:
01145 * @code{.c}
01146 * void print_status(void) {
01147 *     printf("Register Guard: %s\n",
01148 *         rx_register_guard_is_initialized() ? "ACTIVE" : "INACTIVE");
01149 * }
01150 * @endcode
01151 *
01152 * @see rx_register_guard_init() Sets initialized flag to true
01153 *
01154 *
01155 *

```

```

01156 * @since Version 1.0.0
01157 *
01158 * @par NASA Power of 10 Compliance:
01159 * - Rule 1: [OK] No goto, setjmp, recursion
01160 * - Rule 5: [OK] 1 precondition, 1 postcondition
01161 */
01162 bool rx_register_guard_is_initialized(void)
01163 {
01164     return (bool)(s_state.initialized != 0);
01165 }

```

7.61 libs/rx_core/src/rx_time_threadx.c File Reference

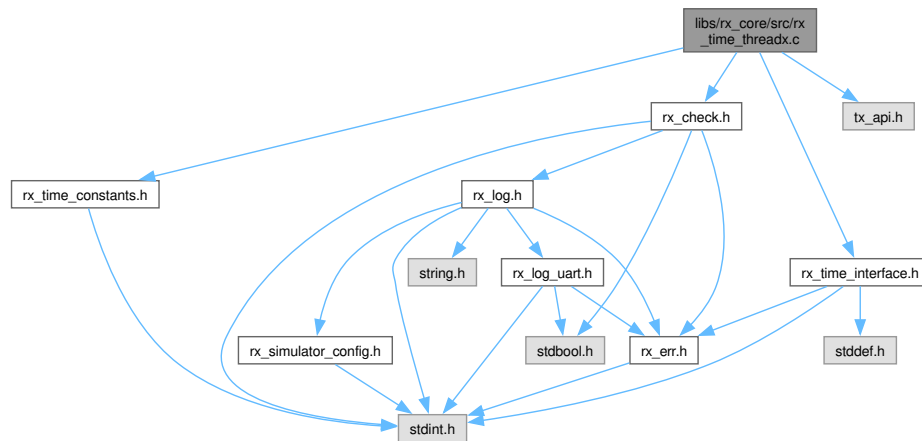
ThreadX Time Implementation for RX72N.

```

#include "rx_check.h"
#include "rx_time_constants.h"
#include "rx_time_interface.h"
#include "tx_api.h"

```

Include dependency graph for rx_time_threadx.c:



Enumerations

- enum `rx_time_threadx_zero_t` : `uint32_t` { `k_rx_zero` = 0u }
Zero sentinel for ThreadX time computations.
- enum `rx_time_ceil_offset_t` : `uint32_t` { `k_ceil_div_offset` = 1u }
Ceiling-division rounding offset.

Functions

- static void `impl_sleep_ms` (void *ctx, uint32_t ms)
Sleep for specified milliseconds using ThreadX scheduler.
- static uint32_t `impl_get_ms` (void *ctx)
Get current time in milliseconds from ThreadX tick counter.
- static bool `impl_is_elapsed` (void *ctx, uint32_t start_ms, uint32_t timeout_ms)
Check if timeout has elapsed since start time.
- `rx_err_t` `rx_time_threadx_get_interface` (rx_time_interface_t *iface)
Get time interface structure for ThreadX RTOS.

Operation	Time	Notes
get_ms()	~100 ns	Single register read
is_elapsed()	~150 ns	Two reads + subtraction

Memory Usage

Type	Size	Description
Code	~200 B	All 4 functions (with -O2)
Stack	0 B	No local arrays, uses caller's stack
Static	0 B	Uses ThreadX global state

Conditional Compilation

This file is only compiled when RX is defined (firmware build). Unit tests use mock_time.c instead for deterministic testing.

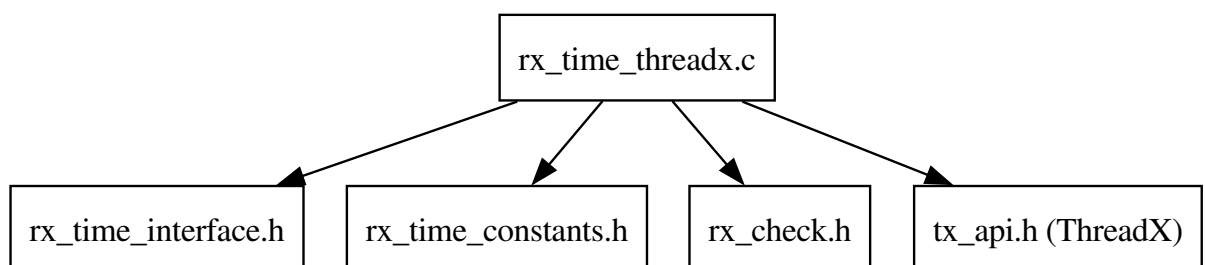
NASA Power of 10 Compliance

- Rule 1: [OK] No goto, setjmp, or recursion
- Rule 2: [OK] No loops in this module
- Rule 3: [OK] No dynamic memory allocation
- Rule 4: [OK] All functions < 20 lines
- Rule 5: [OK] Each function has preconditions (via RX_CHECK_NULL_PTR)
- Rule 6: [OK] Variables declared at smallest scope
- Rule 7: [OK] Return value checking (tx_thread_sleep discarded explicitly)
- Rule 8: [OK] Minimal preprocessor (only RX guard)
- Rule 9: [OK] Function pointers used for DIP pattern (intentional deviation)
- Rule 10: [OK] Compiles with -Wall -Wextra -Werror

SOLID Principles Compliance

- S: Single Responsibility - Only time-related operations
- O: Open/Closed - Interface can add new backends without modification
- L: Liskov Substitution - Interchangeable with mock_time.c
- I: Interface Segregation - Minimal 3-function interface
- D: Dependency Inversion - Implements abstract rx_time_interface_t

Module Dependencies



See also

[rx_time_interface.h](#) Interface definition
[rx_time_constants.h](#) Time conversion constants
[mock_time.c](#) Unit test mock implementation

Author

Locked, Inc.

Date

2026-01-29

Copyright

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Since

Version 1.0.0

Definition in file [rx_time_threadx.c](#).

7.61.2 Enumeration Type Documentation

7.61.2.1 rx_time_ceil_offset_t

enum [rx_time_ceil_offset_t](#) : uint32_t

Ceiling-division rounding offset.

Added to the remainder check when computing tick ceiling from milliseconds.

Invariant

`k_ceil_div_offset` equals 1; adding this before integer division implements ceiling division

```
// Ceiling division: ticks = ceil(ms / k_threadx_ms_per_tick)
const uint32_t quotient = ms / k_threadx_ms_per_tick;
const uint32_t remainder = ms % k_threadx_ms_per_tick;
const uint32_t ticks    = quotient + (remainder > k_rx_zero ? k_ceil_div_offset : k_rx_zero);
```

See also

[impl_sleep_ms\(\)](#) Inline ceiling-division ms-to-ticks conversion

Since

Version 1.0.0

Enumerator

k_ceil_div_offset	Standard ceiling-division rounding offset
-------------------	---

Definition at line 177 of file [rx_time_threadx.c](#).

```
00177      : uint32_t {
00178  k_ceil_div_offset = 1u, /**< Standard ceiling-division rounding offset */
00179 } rx_time_ceil_offset_t;
```

7.61.2.2 rx_time_threadx_zero_t

```
enum rx_time_threadx_zero_t : uint32_t
```

Zero sentinel for ThreadX time computations.

Used in ceiling-division and guard checks to avoid magic number 0.

Invariant

k_rx_zero equals zero; used as a guard against zero-division in tick ceiling calculation

```
// Guard: only sleep if ticks > 0
if (ticks > k_rx_zero) {
    (void)tx_thread_sleep(ticks);
}
```

See also

[impl_sleep_ms\(\)](#) Inline ceiling-division ms-to-ticks conversion

Since

Version 1.0.0

Enumerator

k_rx_zero	Zero sentinel – used in overflow-safe ceiling division
-----------	--

Definition at line 154 of file [rx_time_threadx.c](#).

```
00154      : uint32_t {
00155  k_rx_zero = 0u, /**< Zero sentinel -- used in overflow-safe ceiling division */
00156 } rx_time_threadx_zero_t;
```

7.61.3 Function Documentation

7.61.3.1 impl_get_ms()

```
uint32_t impl_get_ms (
    void * ctx) [static]
```

Get current time in milliseconds from ThreadX tick counter.

Reads the ThreadX system tick counter and converts to milliseconds. The tick counter starts at 0 when tx_kernel_enter() is called and increments at 100 Hz (every 10ms) via CMT0 timer interrupt.

Algorithm Steps:

1. Read ThreadX tick counter via tx_time_get()
2. Multiply by k_threadx_ms_per_tick (10ms) to get milliseconds
3. Return result (may wrap at ~49.7 days)

Conversion Formula:

$$\text{ms} = \text{ticks} \times 10$$

Precondition

ThreadX kernel must be running

Postcondition

Return value represents time in 10ms increments

Note

Thread Safety: Safe - tx_time_get() is atomic

Can be called from ISR context

Resolution is 10ms (cannot distinguish times < 10ms apart)

Wraparound Handling:

Time wraps after ~49.7 days: $2^{32} \text{ ticks} \times 10\text{ms} = 49,710,269,491 \text{ ms}$ Use [impl_is_elapsed\(\)](#) for correct timeout handling across wraparound.

Example:

```
// Measure operation duration
uint32_t start = impl_get_ms(NULL);
do_operation();
uint32_t elapsed = impl_get_ms(NULL) - start;
rx_log_debug("TIMING", "Operation took %lu ms", elapsed);
```


See also

[tx_time_get\(\)](#) ThreadX tick counter
[impl_is_elapsed\(\)](#) Wraparound-safe timeout check

Since

Version 1.0.0

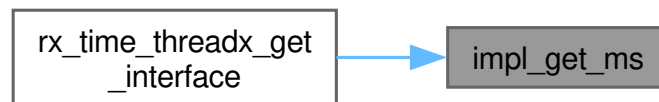
Definition at line 292 of file [rx_time_threadx.c](#).

```
00293 {
00294     (void)ctx; /* ThreadX uses global state - no context needed */
00295
00296     return tx_time_get() * k_threadx_ms_per_tick;
00297 }
```

References [k_threadx_ms_per_tick](#).

Referenced by [rx_time_threadx_get_interface\(\)](#).

Here is the caller graph for this function:



7.61.3.2 impl_is_elapsed()

```
bool impl_is_elapsed (
    void * ctx,
    uint32_t start_ms,
    uint32_t timeout_ms) [static]
```

Check if timeout has elapsed since start time.

Determines if the specified timeout has elapsed since `start_ms`. Uses unsigned subtraction which handles 32-bit wraparound correctly due to modular arithmetic properties.

Algorithm Steps:

1. Get current time via `tx_time_get()` x 10
2. Calculate `elapsed = now - start_ms` (unsigned, handles wrap)
3. Return true if `elapsed >= timeout_ms`

Wraparound Correctness:

Using unsigned 32-bit arithmetic, the subtraction `now - start_ms` yields the correct elapsed time even when `now` has wrapped past zero, as long as the actual elapsed time is less than $2^{32} / 100 / 2 \sim 24.8$ days.

$$\text{elapsed} = (\text{now} - \text{start}) \bmod 2^{32}$$

Precondition

start_ms must be from a previous get_ms() call
 Elapsed time since start_ms < 24.8 days (half wraparound period)

Postcondition

No side effects

Note

Thread Safety: Safe - only reads atomic tick counter
 Can be called from ISR context

Warning

For correct wraparound handling, timeout_ms should be < 24.8 days

Example - Polling with Timeout:

```
rx_time_interface_t time_if;
rx_time_threadx_get_interface(&time_if);

uint32_t start = time_if.get_ms(time_if.ctx);
const uint32_t timeout_ms = 1000; // 1 second timeout

while (!device_ready()) {
    if (time_if.is_elapsed(time_if.ctx, start, timeout_ms)) {
        rx_log_error("DEV", "Timeout waiting for device");
        return k_rx_err_timeout;
    }
    time_if.sleep_ms(time_if.ctx, 10); // Poll every 10ms
}
```

Example - Wraparound Scenario:

```
// Assume UINT32_MAX = 4,294,967,295 ms
// start_ms = 4,294,967,200 (95ms before wrap)
// now = 100 (after wrap)
// elapsed = 100 - 4,294,967,200 = 195 (correct via unsigned wrap)
```

See also

[impl_get_ms\(\)](#) Get start time
[impl_sleep_ms\(\)](#) Delay between polls

Since

Version 1.0.0

Definition at line 363 of file [rx_time_threadx.c](#).

```
00364 {
00365     (void)ctx; /* ThreadX uses global state - no context needed */
00366
00367     const uint32_t now = tx_time_get() * k_threadx_ms_per_tick;
00368     const uint32_t elapsed = now - start_ms;
00369
00370     return elapsed >= timeout_ms;
00371 }
```

References [k_threadx_ms_per_tick](#).

Referenced by [rx_time_threadx_get_interface\(\)](#).

Here is the caller graph for this function:



7.61.3.3 impl_sleep_ms()

```
void impl_sleep_ms (  
    void * ctx,  
    uint32_t ms) [static]
```

Sleep for specified milliseconds using ThreadX scheduler.

Yields CPU to other threads via tx_thread_sleep(), allowing the ThreadX scheduler to run higher-priority threads while this thread waits.

Algorithm Steps:

1. Convert milliseconds to ThreadX ticks with ceiling division
2. If ticks > 0, call tx_thread_sleep() to yield CPU
3. Thread is rescheduled after tick count expires

Tick Conversion Formula:

$$\text{ticks} = \left\lceil \frac{\text{ms}}{10} \right\rceil = \left\lfloor \frac{\text{ms}}{10} \right\rfloor + \begin{cases} 1 & \text{ms mod } 10 > 0 \\ 0 & \text{otherwise} \end{cases}$$

Note

Implemented via overflow-safe quotient/remainder to avoid wrap at high ms values.

Precondition

Calling thread must be a valid ThreadX thread

ThreadX kernel must be running (tx_kernel_enter() called)

Postcondition

Thread has slept for at least ms milliseconds (rounded up to tick)

Other threads have had opportunity to run

Note

Thread Safety: Safe - tx_thread_sleep() is thread-safe

Cannot be called from ISR context

Warning

Actual sleep time is rounded up to nearest 10ms tick boundary

Sleep may be longer than requested due to higher-priority threads

Example:

```
// Sleep for 50ms (actual: 50ms = 5 ticks)
impl_sleep_ms(NULL, 50);

// Sleep for 1ms (actual: 10ms = 1 tick due to rounding)
impl_sleep_ms(NULL, 1);
```

See also

[tx_thread_sleep\(\)](#) ThreadX sleep function
[k_threadx_ms_per_tick](#) Tick period constant (10ms)

Since

Version 1.0.0

Definition at line 233 of file [rx_time_threadx.c](#).

```
00234 {
00235     (void)ctx; /* ThreadX uses global state - no context needed */
00236
00237     /* Convert ms to ticks, rounding up (overflow-safe ceiling division) */
00238     const uint32_t quotient = ms / k_threadx_ms_per_tick;
00239     const uint32_t remainder = ms % k_threadx_ms_per_tick;
00240     const uint32_t ticks    = quotient + (remainder > k_rx_zero ? k_cceil_div_offset : k_rx_zero);
00241     if (ticks > k_rx_zero) {
00242         (void)tx_thread_sleep(ticks);
00243     }
00244 }
```

References [k_cceil_div_offset](#), [k_rx_zero](#), and [k_threadx_ms_per_tick](#).

Referenced by [rx_time_threadx_get_interface\(\)](#).

Here is the caller graph for this function:



7.61.3.4 rx_time_threadx_get_interface()

```
rx_err_t rx_time_threadx_get_interface (
    rx_time_interface_t * iface) [nodiscard]
```

Get time interface structure for ThreadX RTOS.

Validate that a time interface is properly initialized.

Populates an `rx_time_interface_t` structure with function pointers to the ThreadX implementation. This enables Dependency Inversion: callers depend on the abstract interface, not the concrete ThreadX API.

Algorithm Steps:

1. Validate iface is not nullptr
2. Set ctx to NULL (ThreadX uses global state)
3. Set function pointers to implementation functions
4. Return success

Interface Population:

Field	Value	Description
ctx	NULL	ThreadX uses global state
sleep_ms	impl_sleep_ms	tx_thread_sleep() wrapper
get_ms	impl_get_ms	tx_time_get() x 10 wrapper
is_elapsed	impl_is_elapsed	Wraparound-safe timeout check

Precondition

iface != nullptr

Postcondition

iface->ctx == nullptr

iface->sleep_ms, get_ms, is_elapsed are valid function pointers

Interface is ready for use

Invariant

Interface functions remain valid for program lifetime

Note

Thread Safety: Safe - only writes to caller-owned structure

Re-entrancy: Safe - no shared state accessed

Example - Basic Usage:

```

rx_time_interface_t time_if;
rx_err_t err = rx_time_threadx_get_interface(&time_if);
if (err != k_rx_ok) {
    rx_log_error("TIME", "Failed to get interface");
    return err;
}

// Use interface for timing operations
uint32_t start = time_if.get_ms(time_if.ctx);
do_some_work();
uint32_t duration = time_if.get_ms(time_if.ctx) - start;

```

Example - Timeout Loop:

```
rx_time_interface_t time_if;
rx_time_threadx_get_interface(&time_if);

// Wait for sensor with 500ms timeout
uint32_t start = time_if.get_ms(time_if.ctx);
while (!sensor_data_ready()) {
    if (time_if.is_elapsed(time_if.ctx, start, 500)) {
        return k_rx_err_timeout;
    }
    time_if.sleep_ms(time_if.ctx, 10);
}
```

Example - Dependency Injection:

```
// In production code
rx_time_interface_t time_if;
rx_time_threadx_get_interface(&time_if);
motor_init(&motor, &config, &time_if);

// In unit tests (mock_time.c)
rx_time_interface_t time_if;
mock_time_get_interface(&time_if);
motor_init(&motor, &config, &time_if); // Same interface, mock behavior
```

See also

[rx_time_interface_t](#) Interface structure definition
[mock_time_get_interface\(\)](#) Unit test mock version
[impl_sleep_ms\(\)](#) Sleep implementation details
[impl_get_ms\(\)](#) Time retrieval details
[impl_is_elapsed\(\)](#) Timeout check details

NASA Power of 10 Compliance:

- Rule 5: [OK] 1 precondition (NULL check), 3 postconditions
- Rule 9: [OK] Function pointers for DIP (intentional deviation)

[Test](#) Tested in test_rx_time.c with nullptr and success cases

Since

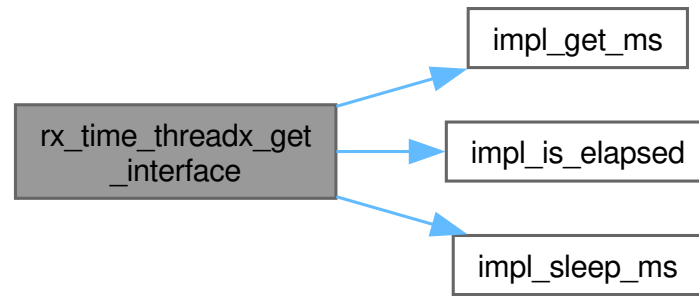
Version 1.0.0

Definition at line 470 of file [rx_time_threadx.c](#).

```
00471 {
00472     RX_CHECK_NULL_PTR(iface, s_tag, "Interface pointer is nullptr");
00473
00474     iface->ctx      = NULL; /* No context needed for ThreadX */
00475     iface->sleep_ms  = impl_sleep_ms;
00476     iface->get_ms    = impl_get_ms;
00477     iface->is_elapsed = impl_is_elapsed;
00478
00479     return k_rx_ok;
00480 }
```

References [impl_get_ms\(\)](#), [impl_is_elapsed\(\)](#), [impl_sleep_ms\(\)](#), [k_rx_ok](#), [RX_CHECK_NULL_PTR](#), and [s_tag](#).

Here is the call graph for this function:



7.61.4 Variable Documentation

7.61.4.1 s_tag

```
const char* const s_tag = "TIME" [static]
```

Log tag for time threadx module.

Identifies log messages from this module

Note

Read-only after initialization

Since

Version 1.0.0

Definition at line 128 of file rx_time_threadx.c.

7.62 rx_time_threadx.c

[Go to the documentation of this file.](#)

```

00001 /**
00002  * @file rx_time_threadx.c
00003  * @brief ThreadX Time Implementation for RX72N
00004  *
00005  * @details
00006  * Real implementation of rx_time_interface_t using ThreadX RTOS primitives.
00007  * This file provides hardware-independent time services by leveraging the
00008  * ThreadX kernel tick infrastructure (CMT0 timer).
00009  *
00010  * @par System Architecture
00011  * @verbatim
00012  * +-----+ +-----+ +-----+
00013  * | Application | | rx_time_if | | ThreadX |
00014  * | (Motor Ctrl) |--->| (This Module) |--->| Kernel |
00015  * +-----+ +-----+ +-----+
00016  * |                                     |
00017  * |                                     |
00018  * |                                     |
00019  * |                                     |
00020  * |                                     |
00021  * |                                     |
00022  * |                                     |
00023  * |                                     |
00024  * | +-----+ +-----+ +-----+ |
00025  * | | rx_time_interface_t | | |
00026  * | | sleep_ms() | get_ms() | is_elapsed() |
00027  * | | (blocking) | (instant) | (instant) |
  
```

```

00028 *      | +-----+-----+-----+ |
00029 *      +-----+
00030 * @endverbatim
00031 *
00032 * @par Threading Model
00033 * - sleep_ms(): Yields CPU via tx_thread_sleep() - thread blocks
00034 * - get_ms(): Returns current time atomically - no blocking
00035 * - is_elapsed(): Compares times atomically - no blocking
00036 *
00037 * @par Time Resolution and Wraparound
00038 * - ThreadX tick rate: 100 Hz (10ms per tick from k_threadx_ms_per_tick)
00039 * - Minimum sleep: 10ms (1 tick), sleeps < 10ms round up to 1 tick
00040 * - Maximum sleep: ~49.7 days (232 / 100 / 86400)
00041 * - Time wraparound: ~49.7 days (handled correctly by is_elapsed())
00042 *
00043 * @par Performance Characteristics
00044 * | Operation | Time | Notes |
00045 * |-----|-----|-----|
00046 * | sleep_ms() | ~1 us | Context switch to scheduler |
00047 * | get_ms() | ~100 ns | Single register read |
00048 * | is_elapsed() | ~150 ns | Two reads + subtraction |
00049 *
00050 * @par Memory Usage
00051 * | Type | Size | Description |
00052 * |-----|-----|-----|
00053 * | Code | ~200 B | All 4 functions (with -O2) |
00054 * | Stack | 0 B | No local arrays, uses caller's stack |
00055 * | Static | 0 B | Uses ThreadX global state |
00056 *
00057 * @par Conditional Compilation
00058 * This file is only compiled when __RX__ is defined (firmware build).
00059 * Unit tests use mock_time.c instead for deterministic testing.
00060 *
00061 * @par NASA Power of 10 Compliance
00062 * - Rule 1: [OK] No goto, setjmp, or recursion
00063 * - Rule 2: [OK] No loops in this module
00064 * - Rule 3: [OK] No dynamic memory allocation
00065 * - Rule 4: [OK] All functions < 20 lines
00066 * - Rule 5: [OK] Each function has preconditions (via RX_CHECK_NULL_PTR)
00067 * - Rule 6: [OK] Variables declared at smallest scope
00068 * - Rule 7: [OK] Return value checking (tx_thread_sleep discarded explicitly)
00069 * - Rule 8: [OK] Minimal preprocessor (only __RX__ guard)
00070 * - Rule 9: [OK] Function pointers used for DIP pattern (intentional deviation)
00071 * - Rule 10: [OK] Compiles with -Wall -Wextra -Werror
00072 *
00073 * @par SOLID Principles Compliance
00074 * - **S**: Single Responsibility - Only time-related operations
00075 * - **O**: Open/Closed - Interface can add new backends without modification
00076 * - **I**: Liskov Substitution - Interchangeable with mock_time.c
00077 * - **I**: Interface Segregation - Minimal 3-function interface
00078 * - **D**: Dependency Inversion - Implements abstract rx_time_interface_t
00079 *
00080 * @par Module Dependencies
00081 * @dot
00082 * digraph dependencies {
00083 *   rankdir=TB;
00084 *   node [shape=box];
00085 *   rx_time_threadx [label="rx_time_threadx.c"];
00086 *   rx_time_interface [label="rx_time_interface.h"];
00087 *   rx_time_constants [label="rx_time_constants.h"];
00088 *   rx_check [label="rx_check.h"];
00089 *   tx_api [label="tx_api.h (ThreadX)"];
00090 *
00091 *   rx_time_threadx -> rx_time_interface;
00092 *   rx_time_threadx -> rx_time_constants;
00093 *   rx_time_threadx -> rx_check;
00094 *   rx_time_threadx -> tx_api;
00095 * }
00096 * @enddot
00097 *
00098 * @see rx_time_interface.h Interface definition
00099 * @see rx_time_constants.h Time conversion constants
00100 * @see mock_time.c Unit test mock implementation
00101 *
00102 * @author Locked, Inc.
00103 * @date 2026-01-29
00104 * @copyright Copyright (c) 2026 Locked Inc.
00105 * SPDX-License-Identifier: MIT
00106 * @since Version 1.0.0
00107 */
00108
00109 #ifdef __RX__
00110
00111 #include "rx_check.h"
00112 #include "rx_time_constants.h"
00113 #include "rx_time_interface.h"
00114 #include "tx_api.h"

```



```

00115
00116 /*
=====
00117 * Module State
00118 *
=====
00119 */
00120
00121 /**
00122 * @var s_tag
00123 * @brief Log tag for time threadx module
00124 * @details Identifies log messages from this module
00125 * @note Read-only after initialization
00126 * @since Version 1.0.0
00127 */
00128 static const char* const s_tag = "TIME";
00129
00130 /*
=====
00131 * Module Constants
00132 *
=====
00133 */
00134
00135 /**
00136 * @enum rx_time_threadx_zero_t
00137 * @brief Zero sentinel for ThreadX time computations
00138 * @details Used in ceiling-division and guard checks to avoid magic number 0.
00139 *
00140 * @invariant k_rx_zero equals zero; used as a guard against zero-division in
00141 *           tick ceiling calculation
00142 *
00143 * @code{.c}
00144 * // Guard: only sleep if ticks > 0
00145 * if (ticks > k_rx_zero) {
00146 *     (void)tx_thread_sleep(ticks);
00147 * }
00148 * @endcode
00149 *
00150 * @see impl_sleep_ms() Inline ceiling-division ms-to-ticks conversion
00151 *
00152 * @since Version 1.0.0
00153 */
00154 typedef enum : uint32_t {
00155     k_rx_zero = 0u, /**< Zero sentinel -- used in overflow-safe ceiling division */
00156 } rx_time_threadx_zero_t;
00157
00158 /**
00159 * @enum rx_time_ceil_offset_t
00160 * @brief Ceiling-division rounding offset
00161 * @details Added to the remainder check when computing tick ceiling from milliseconds.
00162 *
00163 * @invariant k_ceil_div_offset equals 1; adding this before integer division
00164 *           implements ceiling division
00165 *
00166 * @code{.c}
00167 * // Ceiling division: ticks = ceil(ms / k_threadx_ms_per_tick)
00168 * const uint32_t quotient = ms / k_threadx_ms_per_tick;
00169 * const uint32_t remainder = ms % k_threadx_ms_per_tick;
00170 * const uint32_t ticks = quotient + (remainder > k_rx_zero ? k_ceil_div_offset : k_rx_zero);
00171 * @endcode
00172 *
00173 * @see impl_sleep_ms() Inline ceiling-division ms-to-ticks conversion
00174 *
00175 * @since Version 1.0.0
00176 */
00177 typedef enum : uint32_t {
00178     k_ceil_div_offset = 1u, /**< Standard ceiling-division rounding offset */
00179 } rx_time_ceil_offset_t;
00180
00181 /*
=====
00182 * Interface Implementation Functions
00183 *
=====
00184 */
00185
00186 /**
00187 * @brief Sleep for specified milliseconds using ThreadX scheduler
00188 *
00189 * @details
00190 * Yields CPU to other threads via tx_thread_sleep(), allowing the ThreadX
00191 * scheduler to run higher-priority threads while this thread waits.
00192 *
00193 * @par Algorithm Steps:
00194 * 1. Convert milliseconds to ThreadX ticks with ceiling division
00195 * 2. If ticks > 0, call tx_thread_sleep() to yield CPU

```

```

00196 * 3. Thread is rescheduled after tick count expires
00197 *
00198 * @par Tick Conversion Formula:
00199 * @f[
00200 * \text{ticks} = \left\lceil \frac{\text{ms}}{10} \right\rceil
00201 * = \left\lfloor \frac{\text{ms}}{10} \right\rfloor + \begin{cases} 1 & \text{if } \text{ms} \bmod 10 > 0 \\ 0 & \text{otherwise} \end{cases}
00202 * @f]
00203 *
00204 * @note Implemented via overflow-safe quotient/remainder to avoid wrap at high ms values.
00205 *
00206 *
00207 * @pre Calling thread must be a valid ThreadX thread
00208 * @pre ThreadX kernel must be running (tx_kernel_enter() called)
00209 *
00210 * @post Thread has slept for at least ms milliseconds (rounded up to tick)
00211 * @post Other threads have had opportunity to run
00212 *
00213 * @note Thread Safety: Safe - tx_thread_sleep() is thread-safe
00214 * @note Cannot be called from ISR context
00215 *
00216 * @warning Actual sleep time is rounded up to nearest 10ms tick boundary
00217 * @warning Sleep may be longer than requested due to higher-priority threads
00218 *
00219 * @par Example:
00220 * @code{.c}
00221 * // Sleep for 50ms (actual: 50ms = 5 ticks)
00222 * impl_sleep_ms(NULL, 50);
00223 *
00224 * // Sleep for 1ms (actual: 10ms = 1 tick due to rounding)
00225 * impl_sleep_ms(NULL, 1);
00226 * @endcode
00227 *
00228 * @see tx_thread_sleep() ThreadX sleep function
00229 * @see k_threadx_ms_per_tick Tick period constant (10ms)
00230 *
00231 * @since Version 1.0.0
00232 */
00233 static void impl_sleep_ms(void* ctx, uint32_t ms)
00234 {
00235     (void)ctx; /* ThreadX uses global state - no context needed */
00236
00237     /* Convert ms to ticks, rounding up (overflow-safe ceiling division) */
00238     const uint32_t quotient = ms / k_threadx_ms_per_tick;
00239     const uint32_t remainder = ms % k_threadx_ms_per_tick;
00240     const uint32_t ticks = quotient + (remainder > k_rx_zero ? k_cceil_div_offset : k_rx_zero);
00241     if (ticks > k_rx_zero) {
00242         (void)tx_thread_sleep(ticks);
00243     }
00244 }
00245
00246 /**
00247 * @brief Get current time in milliseconds from ThreadX tick counter
00248 *
00249 * @details
00250 * Reads the ThreadX system tick counter and converts to milliseconds.
00251 * The tick counter starts at 0 when tx_kernel_enter() is called and
00252 * increments at 100 Hz (every 10ms) via CMT0 timer interrupt.
00253 *
00254 * @par Algorithm Steps:
00255 * 1. Read ThreadX tick counter via tx_time_get()
00256 * 2. Multiply by k_threadx_ms_per_tick (10ms) to get milliseconds
00257 * 3. Return result (may wrap at ~49.7 days)
00258 *
00259 * @par Conversion Formula:
00260 * @f[
00261 * \text{ms} = \text{ticks} \times 10
00262 * @f]
00263 *
00264 *
00265 *
00266 * @pre ThreadX kernel must be running
00267 *
00268 * @post Return value represents time in 10ms increments
00269 *
00270 * @note Thread Safety: Safe - tx_time_get() is atomic
00271 * @note Can be called from ISR context
00272 * @note Resolution is 10ms (cannot distinguish times < 10ms apart)
00273 *
00274 * @par Wraparound Handling:
00275 * Time wraps after ~49.7 days: 2^32 ticks x 10ms = 49,710,269,491 ms
00276 * Use impl_is_elapsed() for correct timeout handling across wraparound.
00277 *
00278 * @par Example:
00279 * @code{.c}
00280 * // Measure operation duration
00281 * uint32_t start = impl_get_ms(NULL);
00282 * do_operation();

```

```

00283 * uint32_t elapsed = impl_get_ms(NULL) - start;
00284 * rx_log_debug("TIMING", "Operation took %lu ms", elapsed);
00285 * @endcode
00286 *
00287 * @see tx_time_get() ThreadX tick counter
00288 * @see impl_is_elapsed() Wraparound-safe timeout check
00289 *
00290 * @since Version 1.0.0
00291 */
00292 static uint32_t impl_get_ms(void* ctx)
00293 {
00294     (void)ctx; /* ThreadX uses global state - no context needed */
00295     return tx_time_get() * k_threadx_ms_per_tick;
00296 }
00297
00298 /**
00299 * @brief Check if timeout has elapsed since start time
00300 *
00301 * @details
00302 * Determines if the specified timeout has elapsed since start_ms.
00303 * Uses unsigned subtraction which handles 32-bit wraparound correctly
00304 * due to modular arithmetic properties.
00305 *
00306 * @par Algorithm Steps:
00307 * 1. Get current time via tx_time_get() x 10
00308 * 2. Calculate elapsed = now - start_ms (unsigned, handles wrap)
00309 * 3. Return true if elapsed >= timeout_ms
00310 *
00311 * @par Wraparound Correctness:
00312 * Using unsigned 32-bit arithmetic, the subtraction `now - start_ms` yields
00313 * the correct elapsed time even when `now` has wrapped past zero, as long as
00314 * the actual elapsed time is less than  $2^{32} / 100 / 2 \sim 24.8$  days.
00315 *
00316 * @f[
00317 * \text{elapsed} = (\text{now} - \text{start}) \bmod 2^{32}
00318 * @f]
00319 *
00320 *
00321 *
00322 *
00323 * @pre start_ms must be from a previous get_ms() call
00324 * @pre Elapsed time since start_ms < 24.8 days (half wraparound period)
00325 *
00326 * @post No side effects
00327 *
00328 * @note Thread Safety: Safe - only reads atomic tick counter
00329 * @note Can be called from ISR context
00330 *
00331 * @warning For correct wraparound handling, timeout_ms should be < 24.8 days
00332 *
00333 * @par Example - Polling with Timeout:
00334 * @code{.c}
00335 * rx_time_interface_t time_if;
00336 * rx_time_threadx_get_interface(&time_if);
00337 *
00338 * uint32_t start = time_if.get_ms(time_if.ctx);
00339 * const uint32_t timeout_ms = 1000; // 1 second timeout
00340 *
00341 * while (!device_ready()) {
00342 *     if (time_if.is_elapsed(time_if.ctx, start, timeout_ms)) {
00343 *         rx_log_error("DEV", "Timeout waiting for device");
00344 *         return k_rx_err_timeout;
00345 *     }
00346 *     time_if.sleep_ms(time_if.ctx, 10); // Poll every 10ms
00347 * }
00348 * @endcode
00349 *
00350 * @par Example - Wraparound Scenario:
00351 * @code{.c}
00352 * // Assume UINT32_MAX = 4,294,967,295 ms
00353 * // start_ms = 4,294,967,200 (95ms before wrap)
00354 * // now = 100 (after wrap)
00355 * // elapsed = 100 - 4,294,967,200 = 195 (correct via unsigned wrap)
00356 * @endcode
00357 *
00358 * @see impl_get_ms() Get start time
00359 * @see impl_sleep_ms() Delay between polls
00360 *
00361 * @since Version 1.0.0
00362 */
00363 static bool impl_is_elapsed(void* ctx, uint32_t start_ms, uint32_t timeout_ms)
00364 {
00365     (void)ctx; /* ThreadX uses global state - no context needed */
00366     const uint32_t now = tx_time_get() * k_threadx_ms_per_tick;
00367     const uint32_t elapsed = now - start_ms;
00368     return elapsed >= timeout_ms;
00369 }

```

```

00370 return elapsed >= timeout_ms;
00371 }
00372
00373 /*
=====
00374 * Public Functions
00375 *
=====
00376 */
00377
00378 /**
00379 * @brief Get time interface structure for ThreadX RTOS
00380 *
00381 * @details
00382 * Populates an rx_time_interface_t structure with function pointers
00383 * to the ThreadX implementation. This enables Dependency Inversion:
00384 * callers depend on the abstract interface, not the concrete ThreadX API.
00385 *
00386 * @par Algorithm Steps:
00387 * 1. Validate iface is not nullptr
00388 * 2. Set ctx to NULL (ThreadX uses global state)
00389 * 3. Set function pointers to implementation functions
00390 * 4. Return success
00391 *
00392 * @par Interface Population:
00393 * | Field      | Value      | Description      |
00394 * |-----|-----|-----|
00395 * | ctx      | NULL      | ThreadX uses global state |
00396 * | sleep_ms | impl_sleep_ms | tx_thread_sleep() wrapper |
00397 * | get_ms   | impl_get_ms  | tx_time_get() x 10 wrapper |
00398 * | is_elapsed | impl_is_elapsed | Wraparound-safe timeout check |
00399 *
00400 *
00401 *
00402 * @pre iface != nullptr
00403 *
00404 * @post iface->ctx == nullptr
00405 * @post iface->sleep_ms, get_ms, is_elapsed are valid function pointers
00406 * @post Interface is ready for use
00407 *
00408 * @invariant Interface functions remain valid for program lifetime
00409 *
00410 * @note Thread Safety: Safe - only writes to caller-owned structure
00411 * @note Re-entrancy: Safe - no shared state accessed
00412 *
00413 * @par Example - Basic Usage:
00414 * @code{.c}
00415 rx_time_interface_t time_if;
00416 rx_err_t err = rx_time_threadx_get_interface(&time_if);
00417 if (err != k_rx_ok) {
00418     rx_log_error("TIME", "Failed to get interface");
00419     return err;
00420 }
00421 *
00422 // Use interface for timing operations
00423 uint32_t start = time_if.get_ms(time_if.ctx);
00424 do_some_work();
00425 uint32_t duration = time_if.get_ms(time_if.ctx) - start;
00426 * @endcode
00427
00428 * @par Example - Timeout Loop:
00429 * @code{.c}
00430 rx_time_interface_t time_if;
00431 rx_time_threadx_get_interface(&time_if);
00432 *
00433 // Wait for sensor with 500ms timeout
00434 uint32_t start = time_if.get_ms(time_if.ctx);
00435 while (!sensor_data_ready()) {
00436     if (time_if.is_elapsed(time_if.ctx, start, 500)) {
00437         return k_rx_err_timeout;
00438     }
00439     time_if.sleep_ms(time_if.ctx, 10);
00440 }
00441 * @endcode
00442
00443 * @par Example - Dependency Injection:
00444 * @code{.c}
00445 // In production code
00446 rx_time_interface_t time_if;
00447 rx_time_threadx_get_interface(&time_if);
00448 motor_init(&motor, &config, &time_if);
00449 *
00450 // In unit tests (mock_time.c)
00451 rx_time_interface_t time_if;
00452 mock_time_get_interface(&time_if);
00453 motor_init(&motor, &config, &time_if); // Same interface, mock behavior
00454 * @endcode

```

```

00455 *
00456 * @see rx_time_interface_t Interface structure definition
00457 * @see mock_time_get_interface() Unit test mock version
00458 * @see impl_sleep_ms() Sleep implementation details
00459 * @see impl_get_ms() Time retrieval details
00460 * @see impl_is_elapsed() Timeout check details
00461 *
00462 * @par NASA Power of 10 Compliance:
00463 * - Rule 5: [OK] 1 precondition (NULL check), 3 postconditions
00464 * - Rule 9: [OK] Function pointers for DIP (intentional deviation)
00465 *
00466 * @test Tested in test_rx_time.c with nullptr and success cases
00467 *
00468 * @since Version 1.0.0
00469 */
00470 rx_err_t rx_time_threadx_get_interface(rx_time_interface_t* iface)
00471 {
00472     RX_CHECK_NULL_PTR(iface, s_tag, "Interface pointer is nullptr");
00473
00474     iface->ctx = NULL; /* No context needed for ThreadX */
00475     iface->sleep_ms = impl_sleep_ms;
00476     iface->get_ms = impl_get_ms;
00477     iface->is_elapsed = impl_is_elapsed;
00478
00479     return k_rx_ok;
00480 }
00481
00482 #else
00483
00484 /* ISO C requires at least one declaration per translation unit */
00485 typedef int rx_time_threadx_placeholder_t;
00486
00487 #endif /* __RX__ */

```

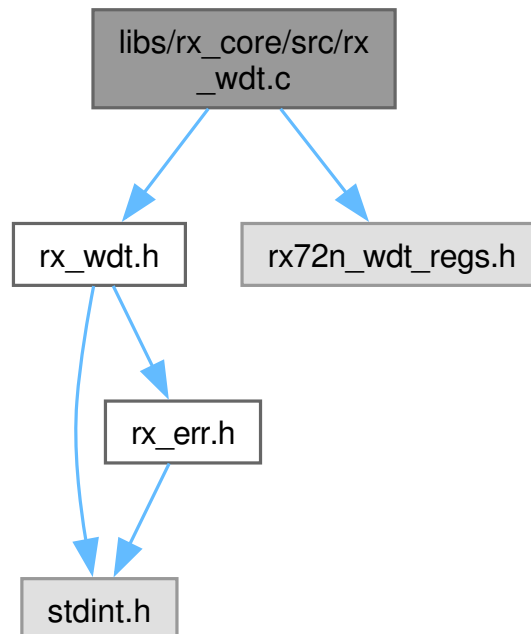
7.63 libs/rx_core/src/rx_wdt.c File Reference

Watchdog Timer (WDT) Implementation for RX72N Microcontroller.

```
#include "rx_wdt.h"
```

```
#include "rx72n_wdt_regs.h"
```

Include dependency graph for rx_wdt.c:



Data Structures

- struct [rx_wdt_state_t](#)
Internal state structure for WDT driver.

Functions

- static void `internal_configure_wdt_registers` (const `rx_wdt_config_t` *config)
Write WDT hardware registers from validated config.
- `rx_err_t rx_wdt_init` (const `rx_wdt_config_t` *config)
Initialize the Watchdog Timer with specified or default configuration.
- `rx_err_t rx_wdt_start` (void)
Start the watchdog timer (register-start mode only).
- `rx_err_t rx_wdt_stop` (void)
Stop the watchdog timer (software state only).
- `rx_err_t rx_wdt_feed` (void)
Feed the watchdog timer to prevent timeout.

Variables

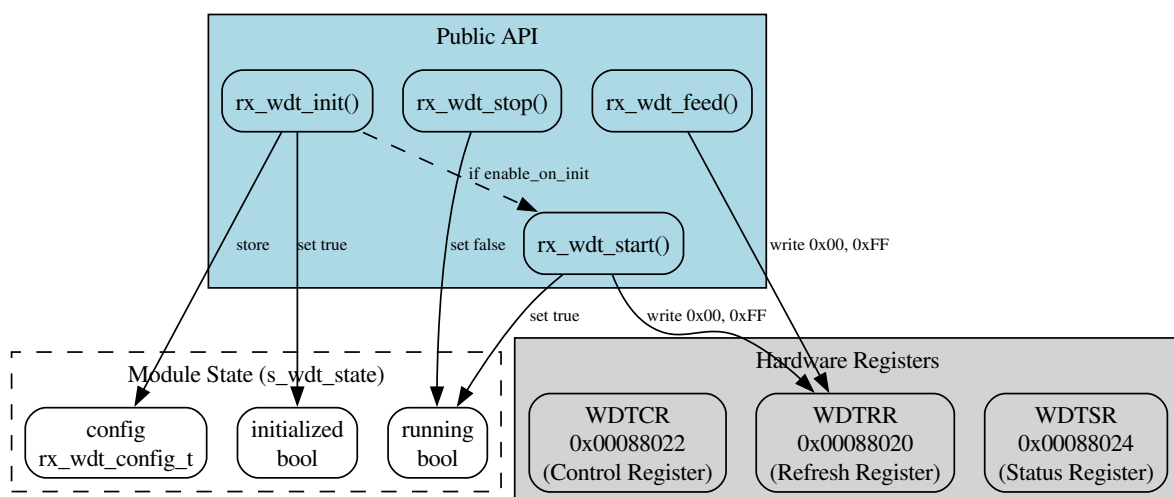
- static `rx_wdt_state_t s_wdt_state` = {}
Global WDT driver state.

7.63.1 Detailed Description

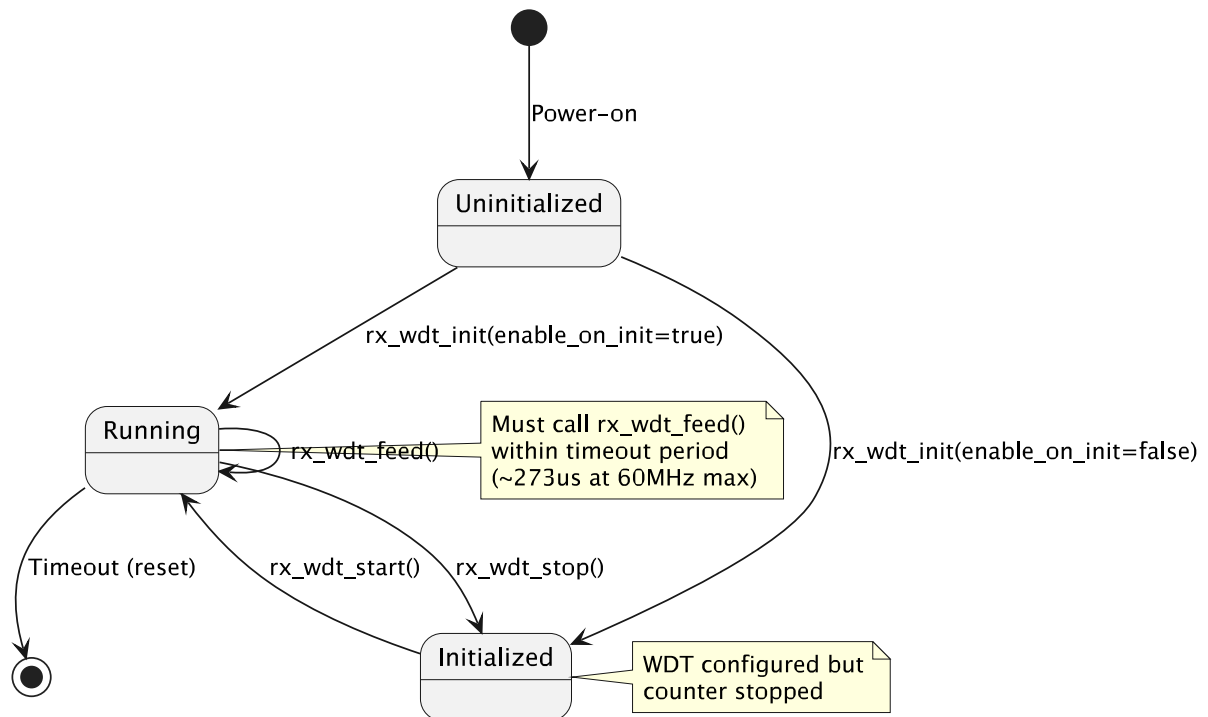
Watchdog Timer (WDT) Implementation for RX72N Microcontroller.

Implements the software-controllable Watchdog Timer (WDT) driver for the RX72N microcontroller. Provides runtime start/stop control for development flexibility, complementing the hardware-independent IWDT for defense-in-depth.

7.63.1.1 Implementation Architecture



7.63.1.2 State Machine



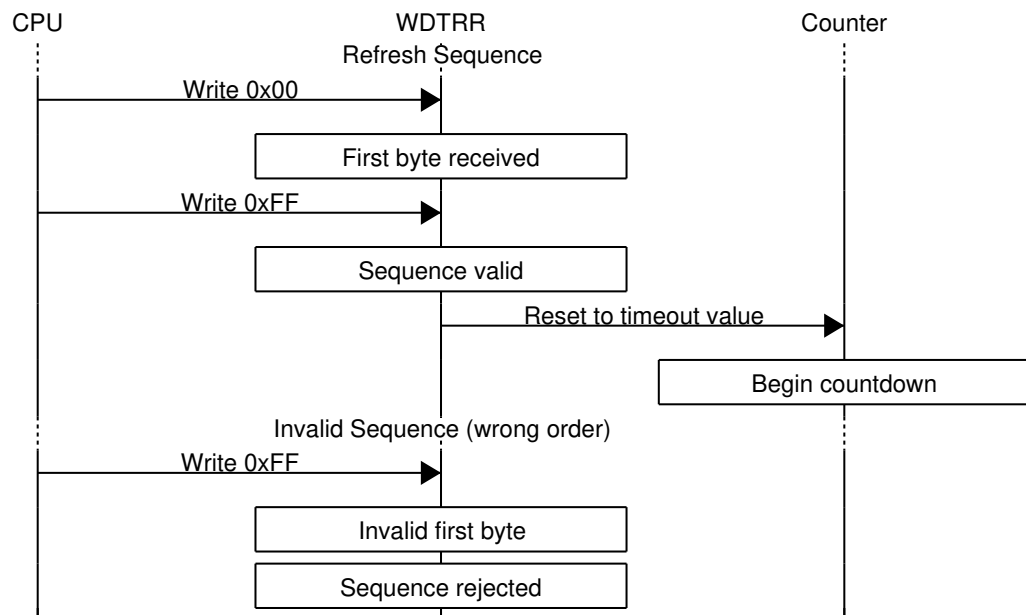
7.63.1.3 Hardware Register Usage

Register	Address	Access	Usage
WDTCR	0x00088022	R/W	Control: CKS bits, RPES bit
WDTRR	0x00088020	W	Refresh: Write 0x00 then 0xFF
WDTSR	0x00088024	R/W1C	Status: Underflow flag

7.63.1.4 Memory Layout (s_wdt_state)

Offset	Size	Field	Type	Description
0x00	4	config	rx_wdt_config_t	Stored configuration
0x04	1	initialized	bool	Init complete flag
0x05	1	running	bool	Counter running flag
0x06	2	(padding)	-	Alignment padding
Total	8 bytes			Static allocation

7.63.1.5 WDT Refresh Sequence



7.63.1.6 Performance Characteristics

Function	Execution Time	Stack	Notes
rx_wdt_init()	~20 us	32 B	Config copy + memset
rx_wdt_start()	~2 us	16 B	2 register writes
rx_wdt_stop()	~1 us	8 B	State update only
rx_wdt_feed()	~1 us	8 B	2 register writes

Module Dependencies:

- [rx_wdt.h](#) - Public API declarations
- [rx72n_wdt_regs.h](#) - WDT register definitions
- [rx72n_system_regs.h](#) - System registers
- `<string.h>` - memcpy, memset for config handling

NASA Power of 10 Compliance:

- Rule 1: [OK] No goto, setjmp, recursion
- Rule 2: [OK] No loops in implementation
- Rule 3: [OK] Zero dynamic allocation (static state)
- Rule 4: [OK] All functions <60 lines
- Rule 5: [OK] Minimum 2 checks per function
- Rule 6: [OK] Data declared at smallest scope
- Rule 7: [OK] All return values checked or propagated
- Rule 8: [OK] C23 typed enums for constants
- Rule 9: [OK] No function pointers (simple register access)
- Rule 10: [OK] Compiled with -Wall -Wextra -Werror

SOLID Principles:

- Single Responsibility: Only WDT management, no other concerns
- Open/Closed: Extensible via configuration structure
- Liskov Substitution: Consistent [rx_err_t](#) return interface
- Interface Segregation: Minimal 4-function API
- Dependency Inversion: Depends on abstract [rx_err_t](#)

See also

[rx_wdt.h](#) Public API documentation and usage examples

[rx_iwdt.c](#) Independent Watchdog implementation

RX72N User's Manual Section 10: Watchdog Timer

Author

Locked, Inc.

Date

2026-01-28

Version

1.0.0

Copyright

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Since

Version 1.0.0

Definition in file [rx_wdt.c](#).

7.63.2 Data Structure Documentation

7.63.2.1 struct rx_wdt_state_t

Internal state structure for WDT driver.

Aggregates all state for the WDT driver into a single structure for cache-friendly access and simplified initialization. Instantiated once as a static variable with zero initialization.

Memory Layout:

Offset	Size	Field	Type	Description
0x00	4	config	rx_wdt_config_t	Stored configuration
0x04	1	initialized	bool	Module init complete
0x05	1	running	bool	Counter running state
0x06	2	(padding)	-	Alignment to 4 bytes
Total	8 bytes			

State Transitions:

- initialized: false -> true on [rx_wdt_init\(\)](#) success
- running: false -> true on [rx_wdt_start\(\)](#) success
- running: true -> false on [rx_wdt_stop\(\)](#) success

Invariant

initialized implies config contains valid settings

running implies initialized

!initialized implies !running

See also

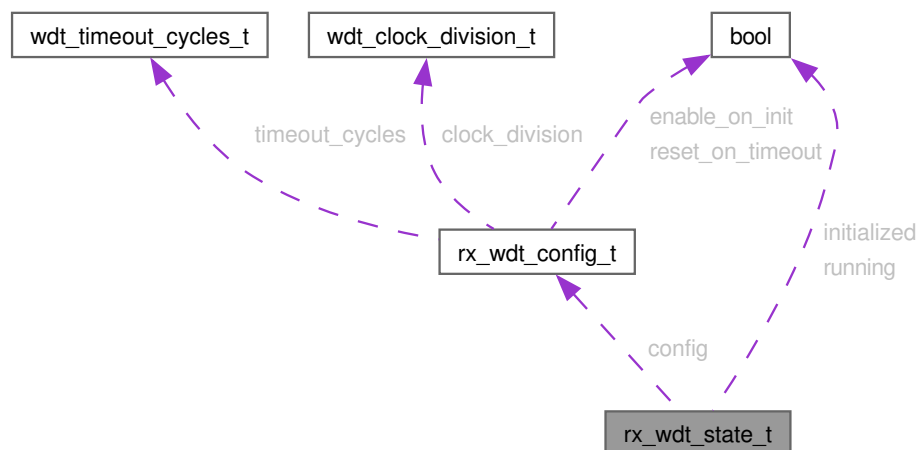
[s_wdt_state](#) Static instance of this structure

Since

Version 1.0.0

Definition at line 208 of file [rx_wdt.c](#).

Collaboration diagram for [rx_wdt_state_t](#):



Data Fields

rx_wdt_config_t	config	Stored configuration from rx_wdt_init() . Copied by value, original can be freed after init call. Valid only when initialized == true.
bool	initialized	Module initialization flag. true = rx_wdt_init() succeeded, false = not initialized or init failed. Set once, never cleared (no deinit function).
bool	running	WDT counter running state. true = counter active and decrementing, false = counter stopped. Controls whether rx_wdt_feed() writes to WDTRR register.

7.63.3 Function Documentation

7.63.3.1 [internal_configure_wdt_registers\(\)](#)

```
void internal_configure_wdt_registers (
    const rx\_wdt\_config\_t * config) [static]
```

Write WDT hardware registers from validated config.

Precondition

config is not nullptr and has been validated.
WDT registers are accessible (clock enabled).

Postcondition

WDTCR and WDTRCR registers written.

Since

Version 1.0.0

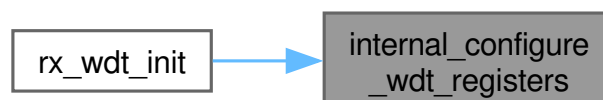
Definition at line 258 of file [rx_wdt.c](#).

```
00259 {
00260     volatile rx_wdt_regs_t* const regs = wdt();
00261
00262     /* Configure WDTCR register (can only be written once before first refresh!)
00263      * Bits 1:0 (TOPS): Timeout period cycles
00264      * Bits 7:4 (CKS): Clock division ratio
00265      * Bits 9:8 (RPES): Window end position (set to 0% = no window)
00266      * Bits 13:12 (RPSS): Window start position (set to 100% = no window)
00267      */
00268     uint16_t wdter_val = (uint16_t)config->timeout_cycles; /* TOPS[1:0] */
00269     wdter_val |= ((uint16_t)config->clock_division « k_wdt_cr_cks_pos); /* CKS[7:4] */
00270     wdter_val |= k_wdt_rpes_0; /* No window end */
00271     wdter_val |= k_wdt_rpss_100; /* No window start */
00272     regs->wdter = wdter_val;
00273
00274     /* Configure WDTRCR register (reset vs NMI selection)
00275      * Bit 7 (RSTIRQS): 1 = reset, 0 = NMI
00276      */
00277     if (config->reset_on_timeout) {
00278         regs->wdtrcr = k_wdt_rstirqs_reset;
00279     } else {
00280         regs->wdtrcr = k_wdt_rstirqs_nmi;
00281     }
00282 }
```

References [rx_wdt_config_t::clock_division](#), [rx_wdt_config_t::reset_on_timeout](#), and [rx_wdt_config_t::timeout_cycles](#).

Referenced by [rx_wdt_init\(\)](#).

Here is the caller graph for this function:



7.63.3.2 rx_wdt_feed()

```
rx_err_t rx_wdt_feed (
    void ) [nodiscard]
```

Feed the watchdog timer to prevent timeout.

Feed the watchdog timer (reset counter to prevent timeout).

Resets the WDT counter to its initial value by writing the refresh sequence to WDTRR. This must be called periodically at a rate faster than the configured timeout period.

Algorithm:

1. Check if WDT is initialized (return error if not)
2. Check if WDT is running (return error if not)
3. Get pointer to WDT registers via wdt() accessor
4. Write refresh sequence: 0x00 then 0xFF to WDTRR
5. Return k_rx_ok

Hardware Operation: The WDTRR refresh sequence must be written in exact order:

- First write: 0x00 (k_wdt_refresh_start)
- Second write: 0xFF (k_wdt_refresh_end)

Invalid sequences (wrong order, wrong values) are ignored by hardware. Valid sequence resets counter to timeout period value.

Timing Requirements:

Feed must occur faster than timeout period:

Timeout Setting	Cycles	At 60MHz	Required Feed Rate
k_wdt_timeout_256_cycles	256	4.3 us	>233 kHz
k_wdt_timeout_1024_cycles	1024	17 us	>59 kHz
k_wdt_timeout_4096_cycles	4096	68 us	>14.7 kHz
k_wdt_timeout_16384_cycles	16384	273 us	>3.6 kHz

Precondition

WDT must be initialized via `rx_wdt_init()`

WDT must be running (`rx_wdt_start()` called or `enable_on_init=true`)

Postcondition

WDT counter reset to timeout period value

Countdown restarts from reset value

Note

This is the ONLY function that prevents watchdog timeout
 Execution time: ~1 us (two register writes)

Warning

Missing a single feed within timeout period causes reset

Example:

```
// Main loop with watchdog feed
while (1) {
    do_work();
    rx_err_t err = rx_wdt_feed();
    if (err != k_rx_ok) {
        // Feed failed - WDT may timeout!
        handle_wdt_error(err);
    }
    delay_us(100); // Must be < timeout period
}
```

See also

[rx_wdt_start\(\)](#) Must be called before feed
[wdt_timeout_period_t](#) Timeout period settings

Since

Version 1.0.0

Definition at line 573 of file [rx_wdt.c](#).

```
00574 {
00575     if (!s_wdt_state.initialized) {
00576         return k_rx_err_not_initialized;
00577     }
00578
00579     if (!s_wdt_state.running) {
00580         return k_rx_err_invalid_state;
00581     }
00582
00583     /* Refresh watchdog - write 0x00 then 0xFF to WDTRR */
00584     volatile rx_wdt_regs_t* const regs = wdt();
00585     regs->wdtrr = k_wdt_refresh_start;
00586     regs->wdtrr = k_wdt_refresh_end;
00587
00588     return k_rx_ok;
00589 }
```

References [k_rx_err_invalid_state](#), [k_rx_err_not_initialized](#), [k_rx_ok](#), and [s_wdt_state](#).

7.63.3.3 rx_wdt_init()

```
rx_err_t rx_wdt_init (
    const rx_wdt_config_t * config) [nodiscard]
```

Initialize the Watchdog Timer with specified or default configuration.

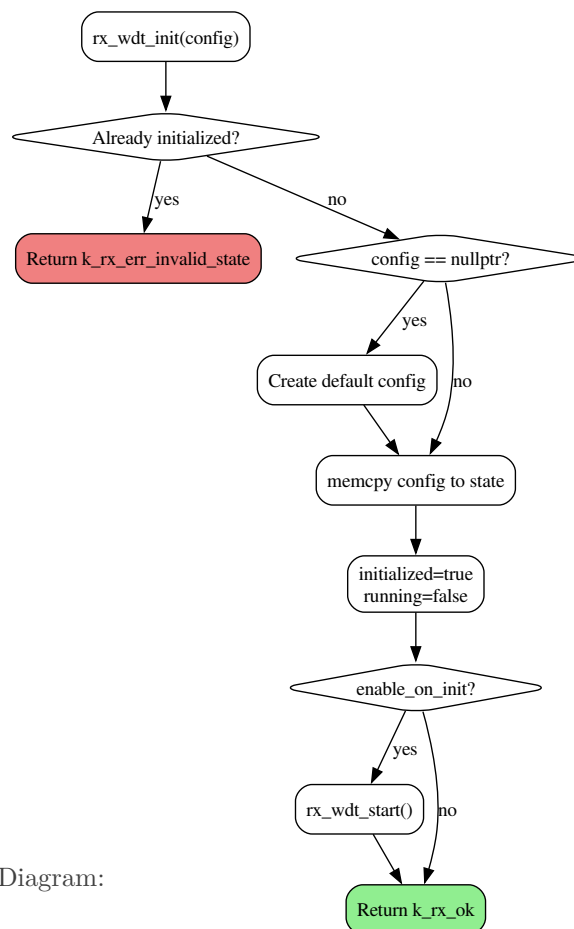
Initialize the Watchdog Timer (WDT) with specified configuration.

Configures WDT driver state and optionally starts the hardware counter. If config is nullptr, uses safe defaults (longest timeout, manual start, reset mode).

Algorithm:

1. Check if already initialized (return error if so)
2. If config is nullptr, create default configuration:
 - timeout_period = k_wdt_timeout_4096_cycles (~68us at 60MHz)
 - enable_on_init = false (manual start)
 - reset_on_timeout = true (reset on timeout)
3. Copy configuration to static state
4. Set initialized flag to true
5. Set running flag to false
6. If enable_on_init, call `rx_wdt_start()`
7. Return k_rx_ok (or propagate start error)

Flow Diagram:



Precondition

WDT must not already be initialized

Postcondition

`s_wdt_state.initialized == true`

`s_wdt_state.config` contains configuration

If `enable_on_init`, WDT counter is running

Note

nullptr config uses safe defaults (recommended for most cases)
Re-initialization requires system reset (no deinit function)

See also

[rx_wdt_start\(\)](#) Called automatically if enable_on_init=true

Since

Version 1.0.0

Definition at line 354 of file [rx_wdt.c](#).

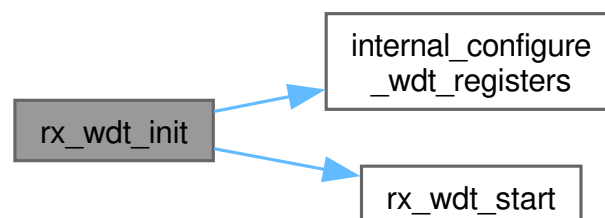
```

00355 {
00356     if (s_wdt_state.initialized) {
00357         return k_rx_err_invalid_state;
00358     }
00359
00360     /* Use default config if none provided */
00361     rx_wdt_config_t default_config = {};
00362     if (config == nullptr) {
00363         default_config.timeout_cycles = k_wdt_timeout_16384_cycles;
00364         default_config.clock_division = k_wdt_clock_div_8192;
00365         default_config.enable_on_init = false;
00366         default_config.reset_on_timeout = true;
00367         config = &default_config;
00368     }
00369
00370     /* Validate clock division is one of the 6 valid values */
00371     if ((config->clock_division != k_wdt_clock_div_4) &&
00372         (config->clock_division != k_wdt_clock_div_64) &&
00373         (config->clock_division != k_wdt_clock_div_128) &&
00374         (config->clock_division != k_wdt_clock_div_512) &&
00375         (config->clock_division != k_wdt_clock_div_2048) &&
00376         (config->clock_division != k_wdt_clock_div_8192)) {
00377         return k_rx_err_invalid_arg;
00378     }
00379
00380     /* Validate timeout_cycles is in valid range (0-3) */
00381     if (config->timeout_cycles > k_wdt_timeout_16384_cycles) {
00382         return k_rx_err_invalid_arg;
00383     }
00384
00385     /* Write hardware registers from validated config */
00386     internal_configure_wdt_registers(config);
00387
00388     /* Store configuration */
00389     s_wdt_state.config = *config;
00390     s_wdt_state.initialized = true;
00391     s_wdt_state.running = false;
00392
00393     /* Start if requested (writes refresh sequence to begin counting) */
00394     if (config->enable_on_init) {
00395         return rx_wdt_start();
00396     }
00397
00398     return k_rx_ok;
00399 }

```

References [rx_wdt_config_t::clock_division](#), [rx_wdt_config_t::enable_on_init](#), [internal_configure_wdt_registers\(\)](#), [k_rx_err_invalid_arg](#), [k_rx_err_invalid_state](#), [k_rx_ok](#), [k_wdt_clock_div_128](#), [k_wdt_clock_div_2048](#), [k_wdt_clock_div_4](#), [k_wdt_clock_div_512](#), [k_wdt_clock_div_64](#), [k_wdt_clock_div_8192](#), [k_wdt_timeout_16384_cycles](#), [rx_wdt_config_t::reset_on_timeout](#), [rx_wdt_start\(\)](#), [s_wdt_state](#), and [rx_wdt_config_t::timeout_cycles](#).

Here is the call graph for this function:



7.63.3.4 rx_wdt_start()

```
rx_err_t rx_wdt_start (
    void ) [nodiscard]
```

Start the watchdog timer (register-start mode only).

Start the WDT counter (enable watchdog monitoring).

Starts the WDT counter by writing the refresh sequence to WDTRR. Once started, [rx_wdt_feed\(\)](#) must be called periodically to prevent timeout.

Algorithm:

1. Check if WDT is initialized (return error if not)
2. Get pointer to WDT registers via wdt() accessor
3. Write start sequence to WDTRR (0x00 then 0xFF)
4. Set s_wdt_state.running to true
5. Return k_rx_ok

Hardware Operation: The WDTRR refresh/start sequence:

- First write: 0x00 (k_wdt_refresh_start)
- Second write: 0xFF (k_wdt_refresh_end)
- Sequence resets counter to timeout period value
- Counter begins decrementing immediately

OFS Register Configuration:

The WDT can be configured at flash-time via OFS (Option Function Select) registers. This function only works if WDT is in "register-start mode":

- Register-start mode: WDT starts when WDTRR sequence written
- Auto-start mode: WDT starts at reset, this function refreshes only

Precondition

WDT must be initialized via [rx_wdt_init\(\)](#)

Postcondition

WDT counter is running and decrementing

s_wdt_state.running == true

[rx_wdt_feed\(\)](#) must be called within timeout period

Note

In auto-start mode, this function acts as a feed operation

Warning

Missing feeds after start will cause timeout/reset

See also

[rx_wdt_init\(\)](#) Must be called first

[rx_wdt_feed\(\)](#) Call periodically after start

[k_wdt_refresh_start](#), [k_wdt_refresh_end](#) Refresh sequence constants

Since

Version 1.0.0

Definition at line 444 of file [rx_wdt.c](#).

```

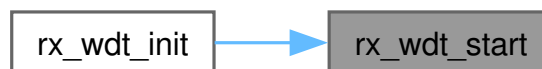
00445 {
00446     if (!s_wdt_state.initialized) {
00447         return k_rx_err_not_initialized;
00448     }
00449     volatile rx_wdt_regs_t* const regs = wdt();
00450     /* Start WDT (only works in register-start mode) */
00451     regs->wdtrr = k_wdt_refresh_start;
00452     regs->wdtrr = k_wdt_refresh_end;
00453     s_wdt_state.running = true;
00454     return k_rx_ok;
00455 }

```

References [k_rx_err_not_initialized](#), [k_rx_ok](#), and [s_wdt_state](#).

Referenced by [rx_wdt_init\(\)](#).

Here is the caller graph for this function:



7.63.3.5 rx_wdt_stop()

```

rx_err_t rx_wdt_stop (
    void ) [nodiscard]

```

Stop the watchdog timer (software state only).

Stop the WDT counter (disable watchdog monitoring).

Updates software state to indicate WDT is stopped. Note that this does NOT actually stop the hardware counter if WDT is configured in auto-start mode via OFS registers.

Algorithm:

1. Check if WDT is initialized (return error if not)
2. Set `s_wdt_state.running` to false
3. Return `k_rx_ok`

Hardware Limitation: The RX72N WDT hardware behavior depends on OFS (Option Function Select) register configuration at flash time:

- Register-start mode: Hardware can be stopped
- Auto-start mode: Hardware cannot be stopped once running!

This function ONLY updates software state. If using auto-start mode, you must continue calling `rx_wdt_feed()` even after `rx_wdt_stop()`.

Precondition

WDT must be initialized via `rx_wdt_init()`

Postcondition

`s_wdt_state.running == false`

In auto-start mode, hardware still requires feeding!

Warning

In auto-start mode, this does NOT stop hardware counter

System may still reset if feeds are stopped

Attention

Check OFS configuration to know if hardware can actually stop

See also

`rx_wdt_start()` Restart WDT after stopping

Since

Version 1.0.0

Definition at line 497 of file `rx_wdt.c`.

```
00498 {
00499     if (!s_wdt_state.initialized) {
00500         return k_rx_err_not_initialized;
00501     }
00502
00503     /* NOTE: WDT cannot be stopped once started if configured to auto-start in OFS.
00504      * This function only updates software state. The hardware WDT will continue
00505      * running and must be fed to prevent reset. */
00506     s_wdt_state.running = false;
00507
00508     return k_rx_ok;
00509 }
```

References `k_rx_err_not_initialized`, `k_rx_ok`, and `s_wdt_state`.

7.63.4 Variable Documentation

7.63.4.1 s_wdt_state

[rx_wdt_state_t](#) s_wdt_state = {} [static]

Global WDT driver state.

Single instance of WDT state, zero-initialized at startup. All fields start at 0/false representing uninitialized state. Populated by [rx_wdt_init\(\)](#) and updated by start/stop functions.

Memory Location: .bss section (zero-initialized static data) Size: 8 bytes Alignment: 4-byte aligned

Access Pattern:

- Write: [rx_wdt_init\(\)](#), [rx_wdt_start\(\)](#), [rx_wdt_stop\(\)](#)
- Read: All functions (state checks)

Note

Not thread-safe for concurrent writes. Use external synchronization if calling start/stop from multiple threads.

See also

[rx_wdt_state_t](#) Structure definition

Since

Version 1.0.0

Definition at line 244 of file [rx_wdt.c](#).

00244 {};

Referenced by [rx_wdt_feed\(\)](#), [rx_wdt_init\(\)](#), [rx_wdt_start\(\)](#), and [rx_wdt_stop\(\)](#).

7.64 rx_wdt.c

[Go to the documentation of this file.](#)

```
00001 /**
00002  * @file rx_wdt.c
00003  * @brief Watchdog Timer (WDT) Implementation for RX72N Microcontroller
00004  *
00005  * @details
00006  * Implements the software-controllable Watchdog Timer (WDT) driver for the
00007  * RX72N microcontroller. Provides runtime start/stop control for development
00008  * flexibility, complementing the hardware-independent IWDT for defense-in-depth.
00009  *
00010  * ## Implementation Architecture
00011  *
00012  * @dot
00013  * digraph wdt_implementation {
00014  *   rankdir=TB;
00015  *   node [shape=box, style=rounded];
00016  *   edge [fontsize=10];
00017  *
00018  *   subgraph cluster_state {
```

```

00019 *   label="Module State (s_wdt_state)";
00020 *   style=dashed;
00021 *   config [label="config\nrx_wdt_config_t"];
00022 *   initialized [label="initialized\nbool"];
00023 *   running [label="running\nbool"];
00024 * }
00025 *
00026 * subgraph cluster_api {
00027 *   label="Public API";
00028 *   style=filled;
00029 *   fillcolor=lightblue;
00030 *   init [label="rx_wdt_init()"];
00031 *   start [label="rx_wdt_start()"];
00032 *   stop [label="rx_wdt_stop()"];
00033 *   feed [label="rx_wdt_feed()"];
00034 * }
00035 *
00036 * subgraph cluster_hw {
00037 *   label="Hardware Registers";
00038 *   style=filled;
00039 *   fillcolor=lightgray;
00040 *   wdtrcr [label="WDTCR\n0x00088022\n(Control Register)"];
00041 *   wdtrr [label="WDTRR\n0x00088020\n(Refresh Register)"];
00042 *   wdtsr [label="WDTSR\n0x00088024\n(Status Register)"];
00043 * }
00044 *
00045 * init -> config [label="store"];
00046 * init -> initialized [label="set true"];
00047 * init -> start [label="if enable_on_init", style=dashed];
00048 *
00049 * start -> wdtrr [label="write 0x00, 0xFF"];
00050 * start -> running [label="set true"];
00051 *
00052 * stop -> running [label="set false"];
00053 *
00054 * feed -> wdtrr [label="write 0x00, 0xFF"];
00055 * }
00056 * @enddot
00057 *
00058 * ## State Machine
00059 *
00060 * @startuml
00061 * [*] --> Uninitialized : Power-on
00062 *
00063 * Uninitialized --> Initialized : rx_wdt_init(enable_on_init=false)
00064 * Uninitialized --> Running : rx_wdt_init(enable_on_init=true)
00065 *
00066 * Initialized --> Running : rx_wdt_start()
00067 * Running --> Initialized : rx_wdt_stop()
00068 *
00069 * Running --> Running : rx_wdt_feed()
00070 * Running --> [*] : Timeout (reset)
00071 *
00072 * note right of Running
00073 *   Must call rx_wdt_feed()
00074 *   within timeout period
00075 *   (~273us at 60MHz max)
00076 * end note
00077 *
00078 * note right of Initialized
00079 *   WDT configured but
00080 *   counter stopped
00081 * end note
00082 * @enduml
00083 *
00084 * ## Hardware Register Usage
00085 *
00086 * | Register | Address | Access | Usage |
00087 * |-----|-----|-----|-----|
00088 * | WDTCR | 0x00088022 | R/W | Control: CKS bits, RPES bit |
00089 * | WDTRR | 0x00088020 | W | Refresh: Write 0x00 then 0xFF |
00090 * | WDTSR | 0x00088024 | R/W1C | Status: Underflow flag |
00091 *
00092 * ## Memory Layout (s_wdt_state)
00093 *
00094 * | Offset | Size | Field | Type | Description |
00095 * |-----|-----|-----|-----|-----|
00096 * | 0x00 | 4 | config | rx_wdt_config_t | Stored configuration |
00097 * | 0x04 | 1 | initialized | bool | Init complete flag |
00098 * | 0x05 | 1 | running | bool | Counter running flag |
00099 * | 0x06 | 2 | (padding) | - | Alignment padding |
00100 * | **Total** | **8 bytes** | | | Static allocation |
00101 *
00102 * ## WDT Refresh Sequence
00103 *
00104 * @msc
00105 * CPU, WDTRR, Counter;

```

```

00106 *
00107 * ... [label="Refresh Sequence"];
00108 * CPU => WDTRR [label="Write 0x00"];
00109 * WDTRR box WDTRR [label="First byte received"];
00110 * CPU => WDTRR [label="Write 0xFF"];
00111 * WDTRR box WDTRR [label="Sequence valid"];
00112 * WDTRR => Counter [label="Reset to timeout value"];
00113 * Counter box Counter [label="Begin countdown"];
00114 *
00115 * ... [label="Invalid Sequence (wrong order)"];
00116 * CPU => WDTRR [label="Write 0xFF"];
00117 * WDTRR box WDTRR [label="Invalid first byte"];
00118 * WDTRR box WDTRR [label="Sequence rejected"];
00119 * @endmsc
00120 *
00121 * ## Performance Characteristics
00122 *
00123 * | Function | Execution Time | Stack | Notes |
00124 * |-----|-----|-----|-----|
00125 * | rx_wdt_init() | ~20 us | 32 B | Config copy + memset |
00126 * | rx_wdt_start() | ~2 us | 16 B | 2 register writes |
00127 * | rx_wdt_stop() | ~1 us | 8 B | State update only |
00128 * | rx_wdt_feed() | ~1 us | 8 B | 2 register writes |
00129 *
00130 * @par Module Dependencies:
00131 * - [rx_wdt.h](../inc/rx_wdt.h) - Public API declarations
00132 * - [rx72n_wdt_regs.h](../rx_hal/inc/rx72n_wdt_regs.h) - WDT register definitions
00133 * - [rx72n_system_regs.h](../rx_hal/inc/rx72n_system_regs.h) - System registers
00134 * - <string.h> - memcpy, memset for config handling
00135 *
00136 * @par NASA Power of 10 Compliance:
00137 * - **Rule 1**: [OK] No goto, setjmp, recursion
00138 * - **Rule 2**: [OK] No loops in implementation
00139 * - **Rule 3**: [OK] Zero dynamic allocation (static state)
00140 * - **Rule 4**: [OK] All functions <60 lines
00141 * - **Rule 5**: [OK] Minimum 2 checks per function
00142 * - **Rule 6**: [OK] Data declared at smallest scope
00143 * - **Rule 7**: [OK] All return values checked or propagated
00144 * - **Rule 8**: [OK] C23 typed enums for constants
00145 * - **Rule 9**: [OK] No function pointers (simple register access)
00146 * - **Rule 10**: [OK] Compiled with -Wall -Wextra -Werror
00147 *
00148 * @par SOLID Principles:
00149 * - **Single Responsibility**: Only WDT management, no other concerns
00150 * - **Open/Closed**: Extensible via configuration structure
00151 * - **Liskov Substitution**: Consistent rx_err_t return interface
00152 * - **Interface Segregation**: Minimal 4-function API
00153 * - **Dependency Inversion**: Depends on abstract rx_err_t
00154 *
00155 * @see rx_wdt.h Public API documentation and usage examples
00156 * @see rx_iwdt.c Independent Watchdog implementation
00157 * @see RX72N User's Manual Section 10: Watchdog Timer
00158 *
00159 * @author Locked, Inc.
00160 * @date 2026-01-28
00161 * @version 1.0.0
00162 * @copyright Copyright (c) 2026 Locked Inc.
00163 * SPDX-License-Identifier: MIT
00164 *
00165 * @since Version 1.0.0
00166 */
00167
00168 #include "rx_wdt.h"
00169
00170 #include "rx72n_wdt_regs.h"
00171
00172 /*
=====
00173 * Static Data
00174 *
=====
00175 */
00176
00177 /**
00178 * @struct rx_wdt_state_t
00179 * @brief Internal state structure for WDT driver
00180 *
00181 * @details
00182 * Aggregates all state for the WDT driver into a single structure for
00183 * cache-friendly access and simplified initialization. Instantiated once
00184 * as a static variable with zero initialization.
00185 *
00186 * @par Memory Layout:
00187 * | Offset | Size | Field | Type | Description |
00188 * |-----|-----|-----|-----|-----|
00189 * | 0x00 | 4 | config | rx_wdt_config_t | Stored configuration |
00190 * | 0x04 | 1 | initialized | bool | Module init complete |

```

```

00191 * | 0x05 | 1 | running | bool | Counter running state |
00192 * | 0x06 | 2 | (padding) | - | Alignment to 4 bytes |
00193 * | **Total** | **8 bytes** | | |
00194 *
00195 * @par State Transitions:
00196 * - initialized: false -> true on rx_wdt_init() success
00197 * - running: false -> true on rx_wdt_start() success
00198 * - running: true -> false on rx_wdt_stop() success
00199 *
00200 * @invariant initialized implies config contains valid settings
00201 * @invariant running implies initialized
00202 * @invariant !initialized implies !running
00203 *
00204 * @see s_wdt_state Static instance of this structure
00205 *
00206 * @since Version 1.0.0
00207 */
00208 typedef struct {
00209     rx_wdt_config_t config;    /**< Stored configuration from rx_wdt_init(). Copied by value,
00210                                original can be freed after init call. Valid only when
00211                                initialized == true. */
00212     bool            initialized; /**< Module initialization flag. true = rx_wdt_init() succeeded,
00213                                false = not initialized or init failed. Set once, never
00214                                cleared (no deinit function). */
00215     bool            running;    /**< WDT counter running state. true = counter active and
00216                                decrementing, false = counter stopped. Controls whether
00217                                rx_wdt_feed() writes to WDTRR register. */
00218 } rx_wdt_state_t;
00219
00220 /**
00221 * @var s_wdt_state
00222 * @brief Global WDT driver state
00223 *
00224 * @details
00225 * Single instance of WDT state, zero-initialized at startup. All fields
00226 * start at 0/false representing uninitialized state. Populated by
00227 * rx_wdt_init() and updated by start/stop functions.
00228 *
00229 * **Memory Location:** .bss section (zero-initialized static data)
00230 * **Size:** 8 bytes
00231 * **Alignment:** 4-byte aligned
00232 *
00233 * @par Access Pattern:
00234 * - **Write:** rx_wdt_init(), rx_wdt_start(), rx_wdt_stop()
00235 * - **Read:** All functions (state checks)
00236 *
00237 * @note Not thread-safe for concurrent writes. Use external synchronization
00238 *       if calling start/stop from multiple threads.
00239 *
00240 * @see rx_wdt_state_t Structure definition
00241 *
00242 * @since Version 1.0.0
00243 */
00244 static rx_wdt_state_t s_wdt_state = {};
00245
00246 /*
=====
00247 * Private Helpers
00248 *
=====
00249 */
00250
00251 /**
00252 * @brief Write WDT hardware registers from validated config.
00253 * @pre config is not nullptr and has been validated.
00254 * @pre WDT registers are accessible (clock enabled).
00255 * @post WDTCR and WDTRCR registers written.
00256 * @since Version 1.0.0
00257 */
00258 static void internal_configure_wdt_registers(const rx_wdt_config_t* config)
00259 {
00260     volatile rx_wdt_regs_t* const regs = wdt();
00261
00262     /* Configure WDTCR register (can only be written once before first refresh!)
00263      * Bits 1:0 (TOPS): Timeout period cycles
00264      * Bits 7:4 (CKS): Clock division ratio
00265      * Bits 9:8 (RPES): Window end position (set to 0% = no window)
00266      * Bits 13:12 (RPSS): Window start position (set to 100% = no window)
00267      */
00268     uint16_t wdcr_val = (uint16_t)config->timeout_cycles;    /* TOPS[1:0] */
00269     wdcr_val |= ((uint16_t)config->clock_division « k_wdt_cr_cks_pos); /* CKS[7:4] */
00270     wdcr_val |= k_wdt_rpes_0;    /* No window end */
00271     wdcr_val |= k_wdt_rpss_100; /* No window start */
00272     regs->wdcr = wdcr_val;
00273
00274     /* Configure WDTRCR register (reset vs NMI selection)
00275      * Bit 7 (RSTIRQS): 1 = reset, 0 = NMI

```

```

00276 */
00277 if (config->reset_on_timeout) {
00278     regs->wdtrcr = k_wdt_rstirqs_reset;
00279 } else {
00280     regs->wdtrcr = k_wdt_rstirqs_nmi;
00281 }
00282 }
00283
00284 /*
=====
00285 * Public API Implementation
00286 *
=====
00287 */
00288 /**
00289  * @brief Initialize the Watchdog Timer with specified or default configuration
00290  *
00291  * @details
00292  * Configures WDT driver state and optionally starts the hardware counter.
00293  * If config is nullptr, uses safe defaults (longest timeout, manual start, reset mode).
00294  *
00295  * **Algorithm:**
00296  * 1. Check if already initialized (return error if so)
00297  * 2. If config is nullptr, create default configuration:
00298  *    - timeout_period = k_wdt_timeout_4096_cycles (~68us at 60MHz)
00299  *    - enable_on_init = false (manual start)
00300  *    - reset_on_timeout = true (reset on timeout)
00301  * 3. Copy configuration to static state
00302  * 4. Set initialized flag to true
00303  * 5. Set running flag to false
00304  * 6. If enable_on_init, call rx_wdt_start()
00305  * 7. Return k_rx_ok (or propagate start error)
00306  *
00307  * @par Flow Diagram:
00308  * @dot
00309  * digraph init_flow {
00310  *     rankdir=TB;
00311  *     node [shape=box, style=rounded];
00312  *
00313  *     start [label="rx_wdt_init(config)"];
00314  *     check_init [label="Already initialized?", shape=diamond];
00315  *     err_state [label="Return k_rx_err_invalid_state", fillcolor=lightcoral, style="rounded, filled"];
00316  *     check_null [label="config == nullptr?", shape=diamond];
00317  *     use_default [label="Create default config"];
00318  *     copy_config [label="memcpy config to state"];
00319  *     set_flags [label="initialized=true\nrunning=false"];
00320  *     check_auto [label="enable_on_init?", shape=diamond];
00321  *     call_start [label="rx_wdt_start()"];
00322  *     success [label="Return k_rx_ok", fillcolor=lightgreen, style="rounded, filled"];
00323  *
00324  *     start -> check_init;
00325  *     check_init -> err_state [label="yes"];
00326  *     check_init -> check_null [label="no"];
00327  *     check_null -> use_default [label="yes"];
00328  *     check_null -> copy_config [label="no"];
00329  *     use_default -> copy_config;
00330  *     copy_config -> set_flags;
00331  *     set_flags -> check_auto;
00332  *     check_auto -> call_start [label="yes"];
00333  *     check_auto -> success [label="no"];
00334  *     call_start -> success;
00335  * }
00336  * @enddot
00337  *
00338  *
00339  *
00340  *
00341  * @pre WDT must not already be initialized
00342  *
00343  * @post s_wdt_state.initialized == true
00344  * @post s_wdt_state.config contains configuration
00345  * @post If enable_on_init, WDT counter is running
00346  *
00347  * @note nullptr config uses safe defaults (recommended for most cases)
00348  * @note Re-initialization requires system reset (no deinit function)
00349  *
00350  * @see rx_wdt_start() Called automatically if enable_on_init=true
00351  *
00352  * @since Version 1.0.0
00353  */
00354 rx_err_t rx_wdt_init(const rx_wdt_config_t* config)
00355 {
00356     if (s_wdt_state.initialized) {
00357         return k_rx_err_invalid_state;
00358     }
00359 }
00360 /* Use default config if none provided */

```

```

00361 rx_wdt_config_t default_config = {};
00362 if (config == nullptr) {
00363     default_config.timeout_cycles = k_wdt_timeout_16384_cycles;
00364     default_config.clock_division = k_wdt_clock_div_8192;
00365     default_config.enable_on_init = false;
00366     default_config.reset_on_timeout = true;
00367     config = &default_config;
00368 }
00369
00370 /* Validate clock division is one of the 6 valid values */
00371 if ((config->clock_division != k_wdt_clock_div_4) &&
00372     (config->clock_division != k_wdt_clock_div_64) &&
00373     (config->clock_division != k_wdt_clock_div_128) &&
00374     (config->clock_division != k_wdt_clock_div_512) &&
00375     (config->clock_division != k_wdt_clock_div_2048) &&
00376     (config->clock_division != k_wdt_clock_div_8192)) {
00377     return k_rx_err_invalid_arg;
00378 }
00379
00380 /* Validate timeout_cycles is in valid range (0-3) */
00381 if (config->timeout_cycles > k_wdt_timeout_16384_cycles) {
00382     return k_rx_err_invalid_arg;
00383 }
00384
00385 /* Write hardware registers from validated config */
00386 internal_configure_wdt_registers(config);
00387
00388 /* Store configuration */
00389 s_wdt_state.config = *config;
00390 s_wdt_state.initialized = true;
00391 s_wdt_state.running = false;
00392
00393 /* Start if requested (writes refresh sequence to begin counting) */
00394 if (config->enable_on_init) {
00395     return rx_wdt_start();
00396 }
00397
00398 return k_rx_ok;
00399 }
00400
00401 /**
00402  * @brief Start the watchdog timer (register-start mode only)
00403  *
00404  * @details
00405  * Starts the WDT counter by writing the refresh sequence to WDTRR. Once started,
00406  * rx_wdt_feed() must be called periodically to prevent timeout.
00407  *
00408  * **Algorithm:**
00409  * 1. Check if WDT is initialized (return error if not)
00410  * 2. Get pointer to WDT registers via wdt() accessor
00411  * 3. Write start sequence to WDTRR (0x00 then 0xFF)
00412  * 4. Set s_wdt_state.running to true
00413  * 5. Return k_rx_ok
00414  *
00415  * **Hardware Operation:**
00416  * The WDTRR refresh/start sequence:
00417  * - First write: 0x00 (k_wdt_refresh_start)
00418  * - Second write: 0xFF (k_wdt_refresh_end)
00419  * - Sequence resets counter to timeout period value
00420  * - Counter begins decrementing immediately
00421  *
00422  * @par OFS Register Configuration:
00423  * The WDT can be configured at flash-time via OFS (Option Function Select)
00424  * registers. This function only works if WDT is in "register-start mode":
00425  * - Register-start mode: WDT starts when WDTRR sequence written
00426  * - Auto-start mode: WDT starts at reset, this function refreshes only
00427  *
00428  *
00429  * @pre WDT must be initialized via rx_wdt_init()
00430  *
00431  * @post WDT counter is running and decrementing
00432  * @post s_wdt_state.running == true
00433  * @post rx_wdt_feed() must be called within timeout period
00434  *
00435  * @note In auto-start mode, this function acts as a feed operation
00436  * @warning Missing feeds after start will cause timeout/reset
00437  *
00438  * @see rx_wdt_init() Must be called first
00439  * @see rx_wdt_feed() Call periodically after start
00440  * @see k_wdt_refresh_start, k_wdt_refresh_end Refresh sequence constants
00441  *
00442  * @since Version 1.0.0
00443  */
00444 rx_err_t rx_wdt_start(void)
00445 {
00446     if (!s_wdt_state.initialized) {
00447         return k_rx_err_not_initialized;

```



```

00448 }
00449
00450 volatile rx_wdt_regs_t* const regs = wdt();
00451
00452 /* Start WDT (only works in register-start mode) */
00453 regs->wdtrr = k_wdt_refresh_start;
00454 regs->wdtrr = k_wdt_refresh_end;
00455
00456 s_wdt_state.running = true;
00457
00458 return k_rx_ok;
00459 }
00460
00461 /**
00462  * @brief Stop the watchdog timer (software state only)
00463  *
00464  * @details
00465  * Updates software state to indicate WDT is stopped. Note that this does NOT
00466  * actually stop the hardware counter if WDT is configured in auto-start mode
00467  * via OFS registers.
00468  *
00469  * **Algorithm:**
00470  * 1. Check if WDT is initialized (return error if not)
00471  * 2. Set s_wdt_state.running to false
00472  * 3. Return k_rx_ok
00473  *
00474  * **Hardware Limitation:**
00475  * The RX72N WDT hardware behavior depends on OFS (Option Function Select)
00476  * register configuration at flash time:
00477  * - Register-start mode: Hardware can be stopped
00478  * - Auto-start mode: Hardware cannot be stopped once running!
00479  *
00480  * This function ONLY updates software state. If using auto-start mode,
00481  * you must continue calling rx_wdt_feed() even after rx_wdt_stop().
00482  *
00483  *
00484  * @pre WDT must be initialized via rx_wdt_init()
00485  *
00486  * @post s_wdt_state.running == false
00487  * @post In auto-start mode, hardware still requires feeding!
00488  *
00489  * @warning In auto-start mode, this does NOT stop hardware counter
00490  * @warning System may still reset if feeds are stopped
00491  * @attention Check OFS configuration to know if hardware can actually stop
00492  *
00493  * @see rx_wdt_start() Restart WDT after stopping
00494  *
00495  * @since Version 1.0.0
00496  */
00497 rx_err_t rx_wdt_stop(void)
00498 {
00499     if (!s_wdt_state.initialized) {
00500         return k_rx_err_not_initialized;
00501     }
00502
00503     /* NOTE: WDT cannot be stopped once started if configured to auto-start in OFS.
00504      * This function only updates software state. The hardware WDT will continue
00505      * running and must be fed to prevent reset. */
00506     s_wdt_state.running = false;
00507
00508     return k_rx_ok;
00509 }
00510
00511 /**
00512  * @brief Feed the watchdog timer to prevent timeout
00513  *
00514  * @details
00515  * Resets the WDT counter to its initial value by writing the refresh sequence
00516  * to WDTRR. This must be called periodically at a rate faster than the
00517  * configured timeout period.
00518  *
00519  * **Algorithm:**
00520  * 1. Check if WDT is initialized (return error if not)
00521  * 2. Check if WDT is running (return error if not)
00522  * 3. Get pointer to WDT registers via wdt() accessor
00523  * 4. Write refresh sequence: 0x00 then 0xFF to WDTRR
00524  * 5. Return k_rx_ok
00525  *
00526  * **Hardware Operation:**
00527  * The WDTRR refresh sequence must be written in exact order:
00528  * - First write: 0x00 (k_wdt_refresh_start)
00529  * - Second write: 0xFF (k_wdt_refresh_end)
00530  *
00531  * Invalid sequences (wrong order, wrong values) are ignored by hardware.
00532  * Valid sequence resets counter to timeout period value.
00533  *
00534  * @par Timing Requirements:

```

```

00535 * Feed must occur faster than timeout period:
00536 * | Timeout Setting | Cycles | At 60MHz | Required Feed Rate |
00537 * |-----|-----|-----|-----|
00538 * | k_wdt_timeout_256_cycles | 256 | 4.3 us | >233 kHz |
00539 * | k_wdt_timeout_1024_cycles | 1024 | 17 us | >59 kHz |
00540 * | k_wdt_timeout_4096_cycles | 4096 | 68 us | >14.7 kHz |
00541 * | k_wdt_timeout_16384_cycles | 16384 | 273 us | >3.6 kHz |
00542 *
00543 *
00544 * @pre WDT must be initialized via rx_wdt_init()
00545 * @pre WDT must be running (rx_wdt_start() called or enable_on_init=true)
00546 *
00547 * @post WDT counter reset to timeout period value
00548 * @post Countdown restarts from reset value
00549 *
00550 * @note This is the ONLY function that prevents watchdog timeout
00551 * @note Execution time: ~1 us (two register writes)
00552 * @warning Missing a single feed within timeout period causes reset
00553 *
00554 * @par Example:
00555 * @code{.c}
00556 * // Main loop with watchdog feed
00557 * while (1) {
00558 *     do_work();
00559 *     rx_err_t err = rx_wdt_feed();
00560 *     if (err != k_rx_ok) {
00561 *         // Feed failed - WDT may timeout!
00562 *         handle_wdt_error(err);
00563 *     }
00564 *     delay_us(100); // Must be < timeout period
00565 * }
00566 * @endcode
00567 *
00568 * @see rx_wdt_start() Must be called before feed
00569 * @see wdt_timeout_period_t Timeout period settings
00570 *
00571 * @since Version 1.0.0
00572 */
00573 rx_err_t rx_wdt_feed(void)
00574 {
00575     if (!s_wdt_state.initialized) {
00576         return k_rx_err_not_initialized;
00577     }
00578
00579     if (!s_wdt_state.running) {
00580         return k_rx_err_invalid_state;
00581     }
00582
00583     /* Refresh watchdog - write 0x00 then 0xFF to WDTRR */
00584     volatile rx_wdt_regs_t* const regs = wdt();
00585     regs->wdtrr = k_wdt_refresh_start;
00586     regs->wdtrr = k_wdt_refresh_end;
00587
00588     return k_rx_ok;
00589 }

```